

MODERN APPROACH TO CHEMISTRY

Grade XII

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Preface

A dire need of a text book in Chemistry, which can provide an insight of various aspects of Chemistry in a comprehensive and interactive way, has been felt for a long time. Hence, a noble strategy for presenting the text in a simple and interesting manner becomes the true motive of the book for the students to achieve clear chemical concepts and principles. Our dream is to fulfill the need of students to cover all the curriculum of Chemistry for Higher Secondary Level and medical and engineering entrance examinations. Having long years of experience in teaching chemistry in various levels (+2, I. Sc., B. Sc., B. E., Masters and Ph.D. level), it's our immense pleasure in bringing out **Modern Approach to Chemistry** for the students of grade XII of Higher Secondary Education Board (HSEB), Nepal. This is a text book of chemistry that provides a firm foundation primarily aimed at delivering concepts of chemistry critically.

Salient features of the book are:

- Simple, systematic and well-organized texts in lucid and comprehensive manner
- Eye catchy photograph in the beginning of each chapter emphasizing its practical application
- Logically explained concepts
- Illustration of concepts with the help of carefully drawn and well- labeled diagrams
- Well-chosen examples and numerical problems with the solutions
- Proper presentation of reactions with mechanisms for their better understanding
- Exercises along with short and long answer questions from the latest examination question papers of grade XII of HSEB, Nepal
- Objective questions for different levels of competitive entrance exams
- Self-test questions to evaluate the level of understanding of students after the completion of each section of different units.

We hope this book will serve its purpose and will be beneficial for chemistry instructors and students. However, we shall appreciate and acknowledge any valuable suggestions and constructive criticisms for the betterment of the text book in the best interest of the students.

July 2015

Authors

Syllabus

Full marks: 100 (75T + 25 P)

Pass Marks: 27T + 10P

Teaching Hours: 150T+50P

Course Contents

General & Physical Chemistry (Section A)

Unit 1: Chemical Bonding and Shape of Molecules

- 3 teaching hours

• Hybridization and concept of sigma and pi bond • Valence shell Electron Pair Repulsion (VSEPR) theory • Prediction of molecular geometry (Shape of molecules) on the basis of VSEPR and hybridization. (BeF₂, BF₃, NH₃, H₂O, CH₄, H₂O, C₂H₂, C₂H₄, H₂S)

Unit 2: Volumetric Analysis

- 8 teaching hours

• Different ways of expressing the concentration of solutions: (i) Molarity, (ii) Normality (iii) Molality (iv) Gram /Litre (v) Percentage • Titration: (i) Acid-base titration (ii) Redox titration • Primary standard substances, primary standard solution, secondary standard solution, end point, equivalence point, neutral point, indicators • Derivation of normality equation • Relation between normality and molarity • Selection of indicators in acid-base titration and p^H curve • Solving related numerical problems

Unit 3: Ionic Equilibrium

- 12 teaching hours

• Introduction • Ionization of weak electrolyte (Ostwald's dilution law) • Degree of ionization and ionization constant • Strength of acids and base in term of K_a, K_b and pK_a and pK_b values • Acid-base concept: (i) Arrhenius concept of acids and bases. (ii) Bronsted Lowry concept of acids and bases (iii) Lewis concept of acids and bases. • Ionization of water, pH and pOH scale. • Hydrolysis of salts. (qualitative concept) • Solubility product principle and its application • Common ion effects and its application • Application of solubility product principle in qualitative analysis • Buffer Solution (Solving numerical problems related with solubility, solubility product, pH and pOH)

Unit 4: Electrochemistry

- 10 teaching hours

• Introduction • Electrolysis; strong and weak electrolyte • Arrhenius theory of ionization • Faraday's laws of electrolysis • Criteria of product formation during electrolysis • Electrolytic conduction, equivalent and molar conductivities • Variation of conductivity with concentration • Electrode potential, standard electrode potential, standard hydrogen electrode and its applications • Electrochemical series and its use to predict the feasibility of redox reactions • Electrochemical cell (Galvanic cell) • EMF of electrochemical cell in the standard state (Solving related numerical problems)

Unit 5: Energetics of Chemical Reactions

- 8 teaching hours

• Introduction, unit of energy • Some thermodynamical terms: system, surrounding, boundary, universe different types of system, state function, state variables and internal energy • Exchange of energy between the system and surrounding • Different types of thermodynamic process • The first law of thermodynamics • Sign convention of heat and work • Enthalpy, enthalpy change in chemical reactions • Hess's law of constant heat summation • Heat of neutralization, heat of solution, heat of combustion, heat of vaporization, heat of formation and bond energy (Solving related numerical problems)

Unit 6: Chemical Thermodynamics

- 6 teaching hours

- Spontaneous process • Second law of thermodynamics • Entropy and its physical concept • Entropy change in phase transformation • Entropy and spontaneity • Entropy changes and their calculation
- Gibbs free energy and prediction for the feasibility of reaction • Standard free energy change and equilibrium constant • Influence of temperature on spontaneous process (Calculation involving in standard free energy change and equilibrium constant)

Unit 7: Chemical Kinetics

- 10 Teaching hours

- Concept of reaction rate • Average rate and instantaneous rate of a reaction • Factors that influences the rate of reaction • Rate law equation, rate constant and its units • 1st order, 2nd order, 3rd order and zero order reactions • Order and molecularity of a reaction • Integrated rate law for a first order reaction
- Half-life of a reaction (first order) • Explaining the increase in reaction rate with temperature or collision theory (qualitative concept only) • Concept of activation energy as the energy barrier, activated complex and effect of catalyst on the rate of reaction (Solving related numerical problems)

Organic Chemistry (Section B)

Unit 8: Aromatic Hydrocarbon

- 3 teaching hours

- Definition, characteristics of aromatic compounds, Huckel's rule, structure of benzene, isomerism and orientation of benzene derivatives • Preparation of benzene from (i) decarboxylation; (ii) phenol; (iii) ethyne; (iv) chlorobenzene • Physical properties of benzene • Chemical properties of benzene (i) Addition reaction: hydrogen, halogen and ozone (ii) Electrophilic substitution reactions: nitration, halogenation, sulphonation, Friedel Crafts alkylation and acylation (iii) Combustion of benzene and uses

Unit 9: Haloalkanes and Haloarenes

- 8 teaching hours

Haloalkanes: • Introduction, classification and isomerism • Preparation of monohaloalkanes from alkanes, alkenes and alcohols • Physical properties of monohaloalkanes • Chemical properties -Substitution reactions - Elimination reaction (dehydrohalogenation) - Grignard's reactions -Reduction reactions - Wurtz's reaction • Polyhaloalkane; - Laboratory preparation of trichloromethane from ethanol and propanone - Physical properties of trichloromethane - Chemical properties: oxidation reduction, action on Silver Powder, con. nitric acid, propanone, aqueous alkali, Carbylamine reaction, Remer Tiemann reaction, Iodoform reaction, etc.

Haloarenes: • Preparation of chlorobenzene from (i) benzene (ii) benzene diazonium chloride • Physical properties • Chemical properties - Low reactivity of haloarene as compared to haloalkane in term of nucleophilic substitution reaction - Reduction of chlorobenzene - Electrophilic substitution reactions - Action with Na, Mg and chloral, etc. - uses

Unit 10: Alcohols and Phenols

- 10 teaching hours

Alcohols: • Introduction, classification, nomenclature and isomerism • Distinction of primary, secondary and tertiary alcohol by Victor Meyer's Method • Preparation of monohydric alcohols from (i) haloalkane (ii) Grignard's reagents using aldehydes and ketones (iii) primary amines (iv) Ester • Industrial preparation ethanol from: (i) Oxo process ii. Fermentation of sugar (iii) hydroboration of ethane • Physical properties monohydric alcohols • Chemical properties of monohydric alcohols - Reaction with HX, PX_3 , $SOCl_2$ - Action with reactive metals like Na, K, Li - Esterification process - Dehydration of alcohols. - Oxidation of primary, secondary and tertiary alcohol with oxidizing agents. - Reduction of alcohols (Catalytic dehydrogenation) - Laboratory test of ethanol - Absolute alcohol, methylated spirit, rectified spirit; alcoholic beverage. • Preparation and uses of ethane- 1, 2. diol (glycol) • Preparation and uses of Propane - 1, 2, 3 triol (glycerol)

Phenols: • Introduction to phenol • Preparation of phenol from (i) chlorobenzene (ii) Diazonium salt and (iii) benzene sulphonic acid • Physical properties of phenol • Chemical properties - Acidic nature of phenol - Action with PCl_5 , PX_3 , NH_3 , Zn, Na benzene diazonium chloride and phthalic anhydride - Acylation reaction, Kolbe's reaction, Reimer Tiemann's reaction - **Electrophilic substitution:** halogenation, nitration, sulphonation, bromination and Friedel Craft's alkylation - Laboratory test of phenol - Uses of phenol

Unit 11: Ethers

- 4 teaching hours

Aliphatic Ethers: • Introduction, nomenclature classification, isomerism in ether • Preparation of ethers from (i) alcohol (ii) Williamson's etherification process • Laboratory preparation of ethoxyethane from ethanol • Physical properties of ether • Chemical properties of ethoxyethane - action with HI, PCl₅, conc. HCl, Conc. H₂SO₄, air and Cl₂ - Uses of ethoxyethane 11.2 Aromatic Ether:

Aromatic Ether: - Preparation of methoxybenzene (anisole) - Halogenation, nitration and sulphonation reactions

Unit 12: Aldehydes and Ketones

- 11 teaching hours

Aliphatic Aldehydes and Ketones: • Introduction, structure of carbonyl group, nomenclature and isomerism in carbonyl compound • Preparation of aldehydes and ketones from (i) Dehydrogenation and oxidation of alcohol (ii) Ozonolysis of alkenes (iii) Acid chloride (iv) Gem dihaloalkane (v) Catalytic distillation of fatty acid (vi) Distillation of salt of fatty acid (vii) Catalytic hydration of alkynes • Physical properties • Chemical properties (i) Addition reaction: addition of H₂, HCN, NaHSO₃ and Grignard's reagents (ii) Action with ammonia derivatives; NH₂OH, NH₂-NH₂, phenyl hydrazine, semicarbazides and 2, 4-DNP (iii) Reduction properties of aldehydes - Oxidation with Tollen's reagent, Fehling's solution (iv) Aldol condensation reaction; Clemmensen's reduction Wolff-Kishner reduction, Action with PCl₅, action with LiAlH₄ (v) Special reaction of methanal; Cannizzaro's reaction, action with ammonia, action with phenol; formalin and its uses

Aromatic Aldehydes and Ketones: • Preparation of benzaldehyde from toluene • Properties of benzaldehyde • Important reaction benzaldehyde different from aliphatic aldehydes: - Perkin condensation - Benzoin condensation - Electrophilic substitution reaction - Cannizzaro's reaction • Preparation of acetophenone by Friedel Crafts acylation

Unit 13: Carboxylic Acids

- 10 teaching hours

Aliphatic Carboxylic Acids: • Introduction, nomenclature, examples • Preparation of monocarboxylic acids from (i) aldehydes (ii) nitriles (iii) Grignard's reagents (iv) dicarboxylic acid (v) sodium alkoxide. (vi) trihaloalkanes • Physical properties of monocarboxylic acids • Chemical properties: Action with alkalis metal oxides, metal carbonates, metal bicarbonates, PCl₃, LiAlH₄ and dehydration of carboxylic acid, esterification, halogenation • Effect of constituents on the acidic strength of carboxylic acid • Laboratory preparation of methanoic acid • Abnormal behaviour of methanoic acid • Uses of carboxylic acid

Derivatives of Carboxylic Acid: • Nomenclature, preparation and properties of (i) Acid halides (ii) Acid amides (iii) Acid anhydrides and (iv) Esters

Aromatic Carboxylic Acids: • Preparation of benzoic acid • Physical and chemical properties • Uses of benzoic acid

Unit 14: Nitrocompounds:

- 4 teaching hours

Aliphatic Nitrocompounds (Nitroalkanes): • Introduction and nomenclature • Preparation from haloalkane and alkane • Physical properties • Reduction of nitroalkane • Uses

Aromatic Nitrocompounds: • Laboratory preparation of nitrobenzene • Physical properties • Chemical properties - Reduction in different media - Electrophilic substitution reactions - Uses of nitrobenzene

Unit 15: Amino Compounds (Amines and Aniline)

- 7 teaching hours

Aliphatic Amines: • Introduction, nomenclature and classification • Separation of primary, secondary and tertiary amines by Hoffmann's method • Preparation of primary amines from haloalkane, nitriles, nitroalkanes and amides • Physical properties • **Chemical Properties:** basicity of amines, comparative study of basic nature of 1°, 2° and 3° amines. Reaction of Primary amines with chloroform, conc. HCl, R-X, RCOX and nitrous acid (NaNO₂ / HCl) • Test of 1°, 2° and 3° amines. (nitrous acid test)

Aromatic Amine (Aniline): • Laboratory preparation of aniline • Physical properties • **Chemical properties:** basicity of aniline, comparison of basic nature of aniline with aliphatic amines; alkylation, acylation, diazotization, carbylamine and coupling reaction • **Electrophilic substitution:** Nitration sulphonation and bromination • Uses of aniline

Unit 16: Molecules of Life

- 8 teaching hours

• **Carbohydrates:** definition, classification of carbohydrates, various examples of carbohydrate of different class. structure and glucose and fructose, function of carbohydrates, sugar and non-sugar • **Protein:** definition, amino acid, essential and non-essential amino acids, peptide linkage, hydrolysis of amino acids, denaturation of protein, zwitter ions, functions of amino acids • **Nucleic acid:** definition, basic components of nucleic acid; double helix, difference between RNA and DNA; biological function of nucleic acid • **Lipid:** definition, fatty acids, fat as ester of fatty acid and difference between fats and oils, function of lipid • **Enzymes** and their functions

Unit 17: Chemistry in Service to Mankind

- 10 teaching hours

• **Polymer:** definition, natural and synthetic polymers, homopolymers and co-polymer, Preparation of some polymers; PVC, polyethylene polystyrene, Teflon, Nylon-66, Bakelite and their uses • **Dyes:** definition, natural and synthetic dyes, names and structure of some common drug, drug addiction • **Fertilizer:** definition, chemical and organic fertilizers, nitrogen fertilizer, phosphatic fertilizer; fertilizer as pollution • **Pesticides:** insecticides, herbicides. weedicides and fungicides (examples and their uses)

Inorganic Chemistry Section C

Unit 18: Heavy Metals

-18 teaching hours

General Characteristics of Transition Metals: **Copper:** • Position in periodic table • Occurrence and extraction of copper from copper pyrites • Properties and uses • Chemistry of (i) blue vitriol (ii) black oxide of copper (iii) red oxide of copper. **Zinc:** • Position in periodic table • Occurrence and extraction of zinc from zinc blende • Properties and uses of copper • Preparation properties and uses of zinc white and white vitriol • Galvanization. **Mercury:** • Occurrence and extraction of Hg from Cinnabar • Properties of mercury • Mercury poisoning and uses of Hg • Preparation, properties and uses of (i) Calomel (ii) Corrosive Sublimate. **Iron:** • Occurrence and extraction • Varieties of Iron • Properties of Iron • Manufacture of Steel by (i) Bessemer process (ii) Open hearth process • Heat treatment of steel • Stainless steel • Rusting of iron and its prevention • Uses and biological importance of iron • Structure and uses of green vitriol, Ferric chloride Mohr's salt. **Silver:** • Extraction of Silver by cyanide process and its uses • Preparation and uses of (i) Silver chloride (ii) Silver nitrate

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17. Chemistry in the Service of Mankind

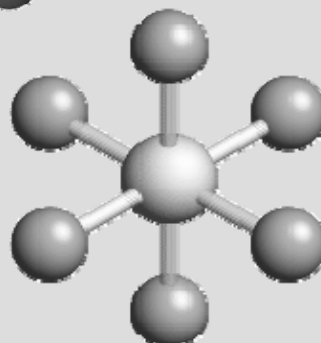
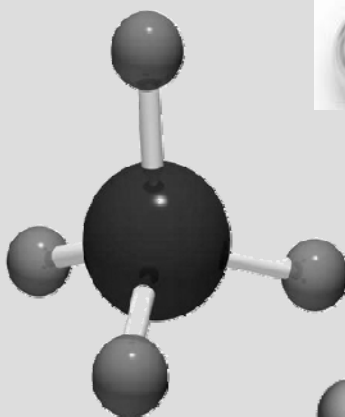
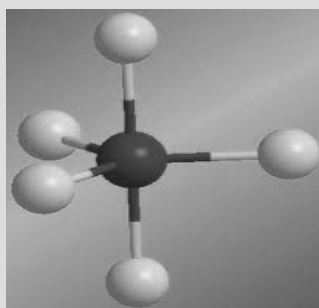
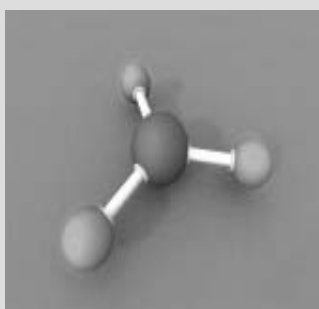
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Chemical Bonding and Shape of Molecules

1



Ball and stick models showing geometries of different molecules.

1.1 Introduction

In molecules, atoms are held together by a force of attraction called chemical bond. Ionic bonds are non-directional i.e., an ion can attract oppositely charged ion in any direction. However, covalent bonds are directional. Hence, covalent molecules have particular bond angles and shapes. In the case of diatomic covalent compounds, the molecules are always linear. **The molecule having three or more atoms possesses particular shape. The shape of a particular molecule is determined by specific arrangement of atoms in it and the bond angles. The shape of molecule depending on the arrangement of atoms within the molecule in space is called geometry of the molecule. Depending upon the different orientations of atoms with respect to each other and central atom, molecules have different geometries.**

The modern techniques like X-ray diffraction (XRD), electron diffraction, spectroscopy etc. are used to determine the geometry of molecules. In order to predict the geometry of molecules and various aspects of bonds, the following theories have been proposed.

- Valence Shell Electron Pair Repulsion (VSEPR) Theory
- Valance Bond Theory (VBT)
- Molecular Orbital Theory (MOT)

1.2 Valence Shell Electron Pair Repulsion (VSEPR) Theory

1.2.1 Introduction

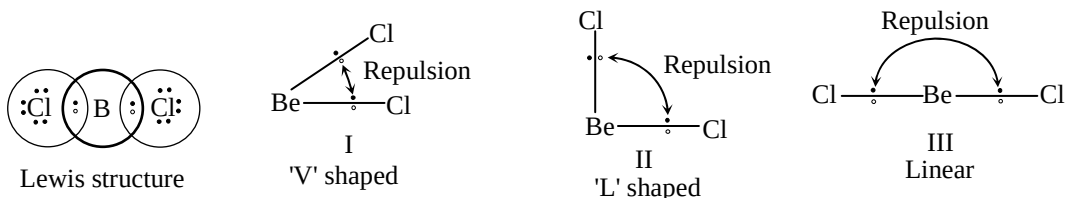
Valence Shell Electron Pair Repulsion (VSEPR) theory is a simple theory to predict and explain the shape and bond angles in a number of simple covalent molecules and ions. This theory was proposed by Sedgwick and Powell in 1940 initially and developed by Gillespie and Nyholm in 1957 as VSEPR theory. The main postulates of this theory may be summarized as follows;

1.2.2 Postulates of VSEPR Theory

- The shape of a covalent molecule or ion is determined by the total number of electron pairs present in the valence shell of the central atom in the molecule. The electron pairs remain arranged around the central atom in space so as to get the minimum electrostatic repulsion hence, determines the shape.

Examples:

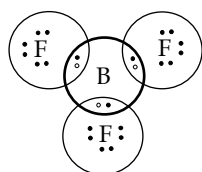
- Beryllium chloride (BeCl_2):** Here, the central atom Be has two bond pairs in its valence shell. For this molecule, there may a number of possible structures. Some of them are;



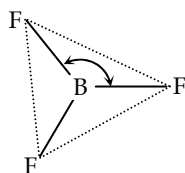
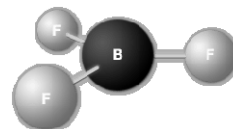
Repulsion of bond pairs: I > II > III

The actual structure is the one in which the repulsion of electron pairs is minimum. This is in structure (III). Hence, BeCl_2 is linear molecule.

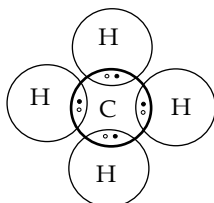
- ii. **Boron trifluoride (BF_3):** In BF_3 , the central atom B has three bond pairs in its valence shell. The repulsion of three bond pairs is minimum in trigonal planar geometry with bond angle 120° . Hence, BF_3 is trigonal planar.



Lewis structure

Trigonal planar shape of BF_3 .

- iii. **Methane (CH_4):** In CH_4 , the central atom carbon has four electron pairs. All are bond pairs. The repulsion of four bond pairs is minimum in tetrahedral geometry with bond angle $109^\circ 28'$. Hence, CH_4 has tetrahedral geometry.



Lewis structure

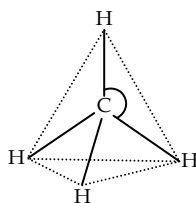
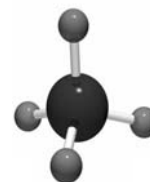
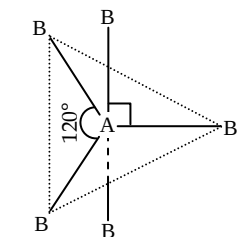
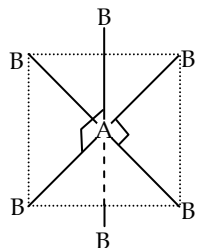
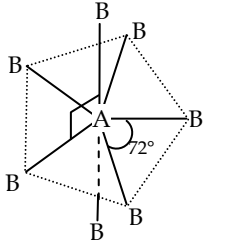
Tetrahedral geometry of CH_4 

Table 1.1: Molecular geometry (ideal shape) on the basis of VSEPR theory if central atom has only bond pairs. Here, A = central atom and B = atom bonded to central atom.

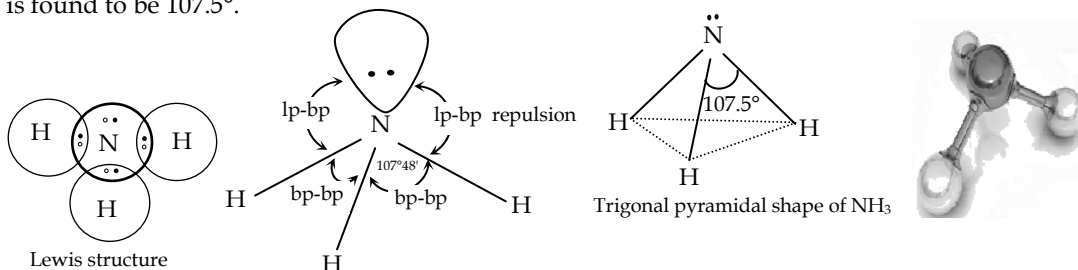
No. of e- pairs in the valence shell	Ideal shape of molecule	Bond angle	Examples
2	 Linear	180°	BeF_2 , BeCl_2
3	 Trigonal planar	120°	BF_3 , BCl_3
4	 Tetrahedral	$109^\circ 28'$ (109.5°)	CH_4 , CCl_4 , NH_4^+

5	 Trigonal bipyramidal	$90^\circ, 120^\circ$	PCl_5
6	 Octahedral	90°	SF_6
7	 Pentagonal bipyramidal	$72^\circ, 90^\circ$	IF_7

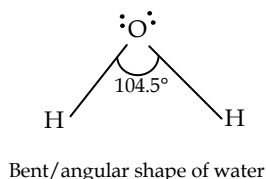
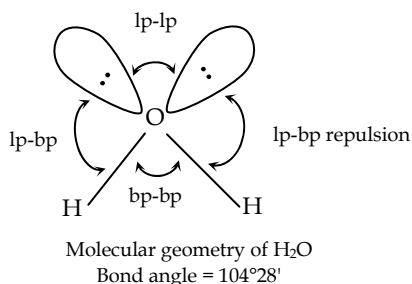
2. If the central atom has lone pair(s), along with bond pairs, the ideal shape is distorted to give different bond angle than expected. This is because a lone pair (*lp*) occupies more space around the central atom than the bond pair (*bp*) since lone pair is attracted by one nucleus and the bond pair is shared between two nuclei. Hence, repulsion of electron pairs around the central atom is in the order: $lp-lp > lp-bp > bp-bp$.

Examples:

- i. **Ammonia (NH_3):** In NH_3 , the central atom contains four effective electron pairs but only three of them are bonded pairs. Since one of the electron pair is the lone pair, the molecule will have trigonal pyramidal geometry. Since, lone pair exerts greater repulsion than the bonded pair, the bond angle is less than expected in regular tetrahedral bond angle ($109^\circ 28'$). Actual bond angle is found to be 107.5° .

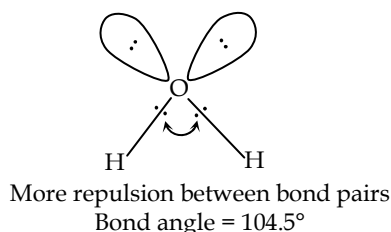
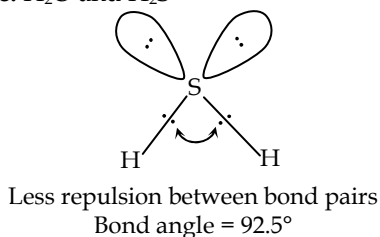


- ii. **Water (H_2O):** In H_2O , the central atom oxygen has four electron pairs hence, expected geometry is tetrahedral but there are two lone pairs and two bond pairs. Since there are two lone pairs of electrons around central atom, the actual geometry is bent or angular. Since lone pair exerts greater repulsion than the bonded pairs, the bond angle is less than the regular tetrahedral bond angle ($109^\circ 28'$). Actual bond angle is found to be 104.5° .



3. The electronegativity of atoms in the molecule also affects the bond angle of the molecule, in the iso-structural molecules (molecules having same number of bonded and lone pairs around the central atom).
- a. As the electronegativity of the central atom increases, the bond pairs of electrons are pulled more towards the central atom due to which repulsion between bond pairs increases and bond angle increases.

Example: H_2O and H_2S



Electronegativity of oxygen is greater than that of sulphur.

Table 1.2: Variation of bond angle with change in electronegativity (EN) of central atom

Group 15	Bond angle	EN of central atom	Group 16	Bond angle
NH_3	$107^\circ 48'$	Decreases ↓	H_2O	$104^\circ 28'$
PH_3	$93^\circ 36'$		H_2S	92°
AsH_3	$91^\circ 48'$		H_2Se	91°
SbH_3	$91^\circ 18'$		H_2Te	90°

- b. As the electronegativity (EN) of the atoms bonded to the central atom increases, the bond pairs of electrons are pulled towards themselves by electronegative atoms due to which repulsion between bond pairs decreases. As a result, bond angle decreases.

Example: H_2O and OF_2

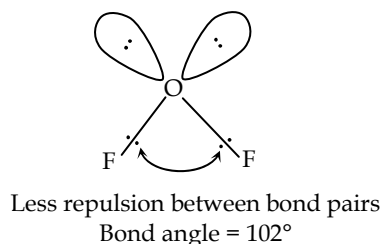
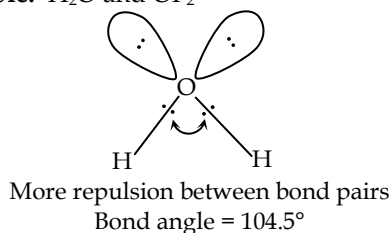


Table 1.3: Variation of bond angle with change in electronegativity of atom bonded to central atom

Group 17	Bond angle	EN of surrounding atoms
PF ₃	97°	↓ Decreases
PCl ₃	100°	
PBr ₃	101.5°	
PI ₃	102°	

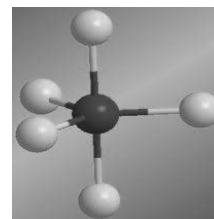
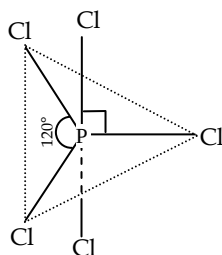
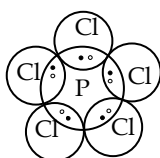
4. The double or triple bond is taken as one effective electron pair, however, the repulsion is in the order: triple bond > double bond > single bond

Self Test

- 1.1 Predict the shape of following molecules on the basis of VSEPR theory.
 - i. BF₃ ii. CH₄ iii. CCl₄ iv. PCl₅ v. SF₆
- 1.2 Predict the shape of following molecules on the basis of VSEPR theory.
 - i. PCl₃ ii. NH₃ iii. PH₃ iv. H₂O v. H₂S vi. OF₂
- 1.3 BF₃ is trigonal planar but PF₃ is trigonal pyramidal, explain.
- 1.4 Arrange CH₄, NH₃ and H₂O in the decreasing order of their bond angle, on the basis of VSEPR theory.
- 1.5 How do you account for the fact that bond angle in H₂O is greater than that in H₂S?
- 1.6 How do you account for the fact that bond angle in PF₃ is smaller than that in PCl₃?
- 1.7 Draw the molecular geometries of NH₃ and NF₃. Which one is expected to have smaller bond angle?

1.2.3 Some More Examples

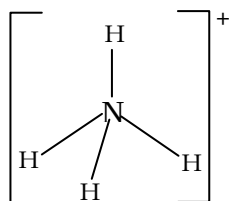
- i. **Shape of Phosphorous Pentachloride (PCl₅):** In PCl₅, the central atom phosphorous has five electron pairs. The repulsion of five electron pairs is minimum when they are arranged in trigonal bipyramidal shape. Hence, PCl₅ is trigonal bipyramidal in shape. However, trigonal bipyramid is not a regular structure. Here, two axial bonds are longer than three equatorial bonds.



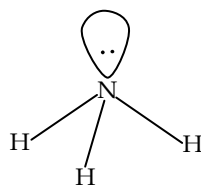
- ii. **Shapes of NH₄⁺, NH₃ and NH₂⁻:** NH₄⁺ is regarded as being formed by adding H⁺ to NH₃ by coordinate covalent bond. Hence, the central atom nitrogen has four bond pairs in its valence shell. The repulsion of four bond pairs is minimum when bond angle is 109°28' with tetrahedral geometry. Hence, NH₄⁺ is tetrahedral.

In NH₃, the central atom nitrogen has four electron pairs; three bond pairs and one lone pair. The repulsion between lp and bp is greater than bp and bp. Hence, bond angle in NH₃ is smaller than the normal tetrahedral value the shape is trigonal pyramidal.

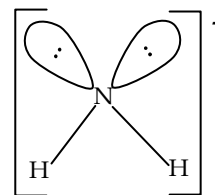
NH₂⁻ is regarded as being formed by removing H⁺ from NH₃. Hence, the central atom nitrogen has two lone pairs and two bond pairs in its valence shell. The repulsion of electron pairs around the central atom is in the order of lp-lp > lp-bp > bp-bp. Hence, bond angle in NH₂⁻ is still smaller than that in NH₃ and it is bent or angular shaped. Hence, bond angle is in the order



Shape of NH_4^+
Tetrahedral



Shape of NH_3
Trigonal pyramidal



Shape of NH_2^-
Bent or angular

- iii. **Examples involving four electron pairs in the valence shell of central atom:** If the central atom has four electron pairs in its valence shell, three different shapes are possible; **tetrahedral**, **trigonal pyramidal** and **bent or angular** depending upon number of lone pair as in table 1.4.

Table 1.4: Shapes of molecules having four electron pairs.

Molecular Geometry			
Total electron pairs	4	4	4
Bond pairs	4	3	2
Lone pairs	0	1	2
Bond angle	$109^\circ 28'$	Less than $109^\circ 28'$	Less than $109^\circ 28'$
Shape	<i>Tetrahedral</i>	<i>Trigonal pyramidal</i>	<i>Bent or angular</i>
Type of molecule	AB_4	AB_3L	AB_2L_2
Examples	$\text{CH}_4, \text{CHCl}_3, \text{CCl}_4, \text{SiF}_4$	$\text{NH}_3, \text{NF}_3, \text{PH}_3, \text{PX}_3$	$\text{H}_2\text{O}, \text{H}_2\text{S}, \text{OF}_2, \text{SCl}_2$

- iv. **Examples involving five electron pairs in the valence shell of central atom:** If the central atom has five electron pairs in its valence shell, the structure is *trigonal bipyramidal*. If lone pairs occur in a trigonal bipyramid, they will be located in the equatorial positions rather than the apical positions, since this arrangement minimizes the electrostatic repulsions

Table 1.5: Shapes of molecules having five electron pairs.

Molecular Geometry				
Bond pairs	5	4	3	2
Lone pairs	0	1	2	3
shape	Trigonal bipyramidal	Seesaw	T shaped	linear
Type of molecule	AB_5	AB_4L	AB_3L_2	AB_2L_3
Examples	PX_5	$\text{SF}_4, \text{TeCl}_4$	ClF_3	$\text{XeF}_2, \text{I}_3^-, \text{ICl}_2^-$

- v. **Examples involving six electron pairs in the valence shell of central atom:** If the central atom has six electron pairs in its valence shell, the structure is octahedral.

Table 1.6: Shapes of molecules having six electron pairs.

Bond pairs	6	5	4
Lone pairs	0	1	2
shape	Octahedral	Square pyramidal	Square planar
Type of molecule	AB_6	AB_5L	AB_4L_2
Examples	SF_6	IF_5, ClF_5, BrF_5	XeF_4, ICl_4^-

Self Test

1.8 Predict the geometry of the following molecules on the basis of VSEPR theory.

- i. ClF_3 ii. XeF_2 iii. XeF_4 iv. SF_4

1.9 Why do lone pair(s) occupy equatorial positions in case of five electron pairs containing molecules?

1.3 Valence Bond Theory

Valence bond theory (VBT) was proposed by Heitler and London in 1927 and improved by Linus Pauling to explain the nature of bonding in covalent molecules. This theory is based on wave mechanical model of atom. According to this theory, a covalent bond is formed by overlapping of atomic orbitals having unpaired electrons. The main postulates of this theory are as follows;

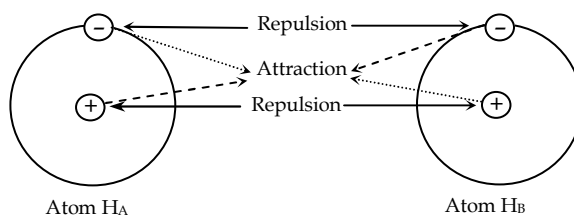
1.3.1 Postulates

1. Half filled atomic orbital of one atom overlaps with half filled atomic orbital of another atom to form a covalent bond.
2. Atomic orbitals undergoing overlap must be sufficiently close to each other with proper alignment.
3. The strength of bond formed depends upon the extent of overlapping of atomic orbitals. Greater the overlapping of atomic orbitals, stronger the covalent bond formed.
4. Overlapping lowers the energy of the molecule and excess energy is released. The energy released per mole is called stabilization energy or bond energy and the equilibrium distance between two nuclei of bonding atoms is **bond length**.
5. Number of unpaired electrons in an atom can increase at the time of reaction due to the excitation of electron from one orbital to the orbital of slightly higher energy.
6. Number of unpaired electrons in ground state or excited state of an atom is called covalency of the element.

Formation of Hydrogen Molecule

In valence bond theory, the stability of a molecule is explained in terms of following interactions.

- Electrons-nucleus attractive interaction
- Electron-electron repulsive interaction
- Nucleus-nucleus repulsive interaction



The attractive and repulsive interactions oppose each other. When attractive interactions are stronger than repulsive interactions, certain amount of energy is released. Due to lowering of energy, the molecule becomes stable. The qualitative treatment of valence bond theory can be understood by describing the formation of hydrogen molecule.

Consider two hydrogen atoms; H_A and H_B at large separation before formation of bond to each other. There is no interaction between them. The total energy is equal to sum of energies of two H atoms (location A in the figure)

When two H atoms approach each other, the attractive and repulsive force begin to operate; electron in an atom H_A is attracted by the nucleus of another atom H_B and electron in an atom H_B is attracted by the nucleus of another atom H_A . Similarly, there is repulsion between the two nuclei. The attractive interaction decreases energy. Therefore, when two atoms approach each other, energy of system goes on decreasing.

When two H atoms H_A and H_B come still closer there exist electron-electron and nucleus-nucleus repulsions which tend to increase energy. As large as the attractive interaction is stronger than repulsion, the energy of the system continues to decrease.

At certain distance between the two atoms, the attractive and repulsive interaction balance each other and the energy of system attains a minimum value (location C). At this state, the two H atoms have a fixed distance between them and form a stable H_2 molecule. This inter-nuclear distance when the energy of the system is minimum is called bond length which is $0.74\text{\AA}$ in H_2 .

If the two H atoms in a H_2 molecule further come closer then the bond length ($0.74\text{\AA}$), the repulsive force predominates and energy of system increases sharply. At point C, H_2 has 104 kcal less energy than uncombined state at point A.

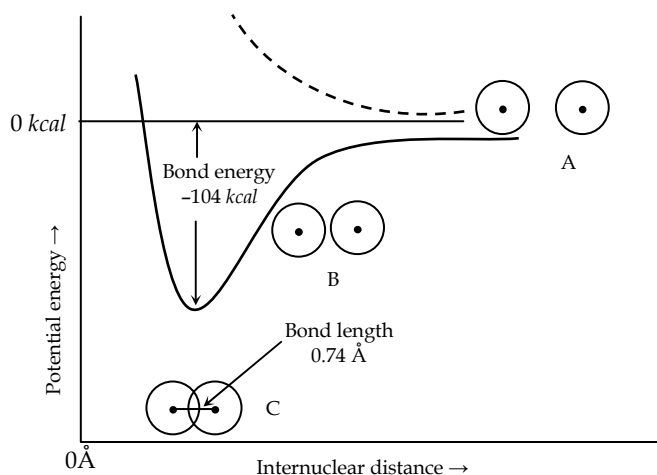


Fig 1.1: Energy profile diagram for formation of H_2 molecule

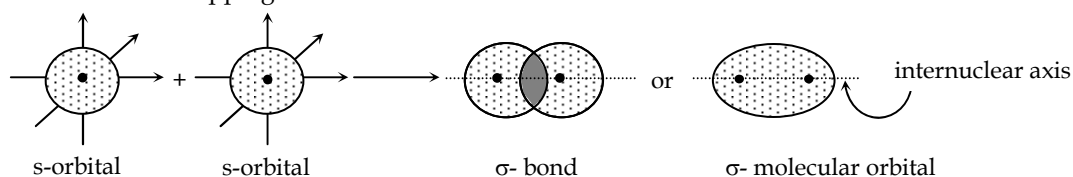
Hence, 104 kcal energy is required to separate H atoms from H_2 . This energy is called bond energy of H_2 .

1.3.2 Types of Overlapping: Types of Covalent Bonds

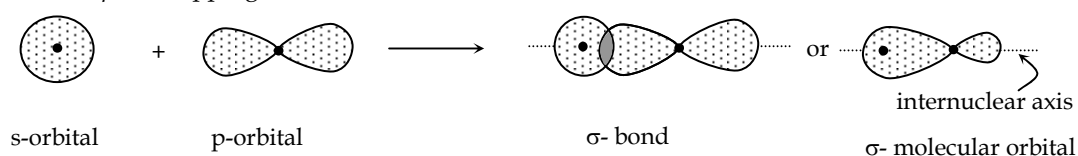
There are two types of covalent bonds depending upon type of overlapping of orbitals; sigma bond and pi bond.

- a. **Sigma bond (σ - bond):** It is a covalent bond formed by end to end (head on or axial) overlapping of half filled atomic orbitals. Overlapping takes place along the internuclear axis. The extent of overlapping is large and the bond formed is strong. It is formed by following types of overlapping;

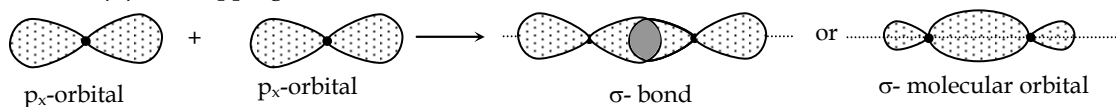
i. s-s overlapping



ii. s-p overlapping

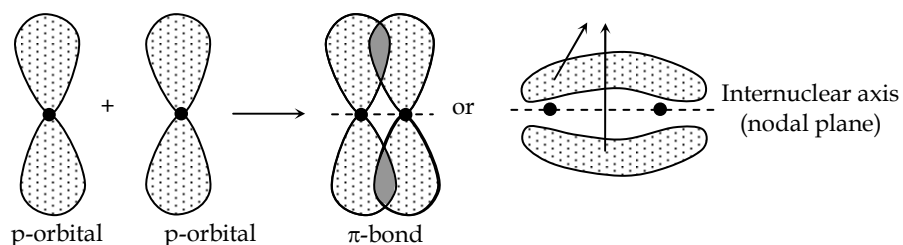


iii. p-p overlapping



where, x is taken as inter-nuclear axis.

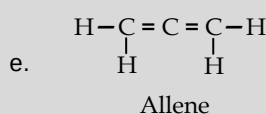
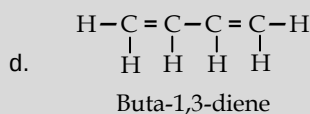
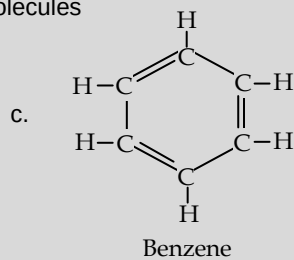
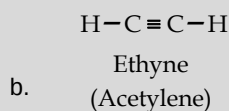
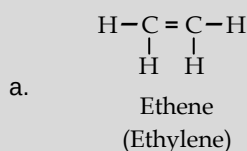
- b. **Pi bond (π - bond):** It is a covalent bond formed by sideways or lateral overlapping of half filled atomic orbitals. Overlapping takes place perpendicular to the internuclear axis. The extent of overlapping is small and the bond formed is weak. It is formed by following types of overlapping;



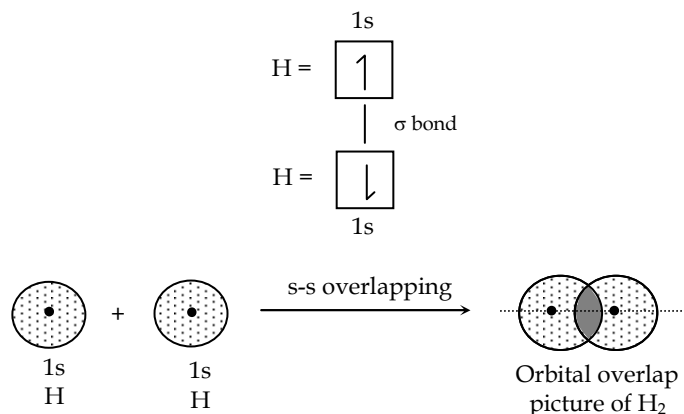
Differences between σ -bond and π bond

Sigma (σ) bond	Pi (π) bond
i. It is formed by end to end, head on or axial overlapping of half filled orbitals.	i. It is formed by sideways or lateral overlapping of half filled orbitals.
ii. Overlapping of orbitals takes place along the internuclear axis.	ii. Overlapping of orbitals takes place perpendicular to the internuclear axis.
iii. The extent of overlapping is large so the bond formed is strong.	iii. The extent of overlapping is small so the bond formed is weak.
iv. It is symmetrical along the internuclear axis so free rotation about sigma is possible.	iv. It consists two lobes on either side of internuclear axis and free rotation about pi bond is restricted.
v. It may be present alone or along with pi bonds.	v. π bond cannot be formed alone. It is formed when σ bond is present.
vi. It is formed by overlapping of all types of orbitals; <i>s-s, s-p, p-p orbitals</i> .	vi. It is formed by other than s orbitals.

Self Test

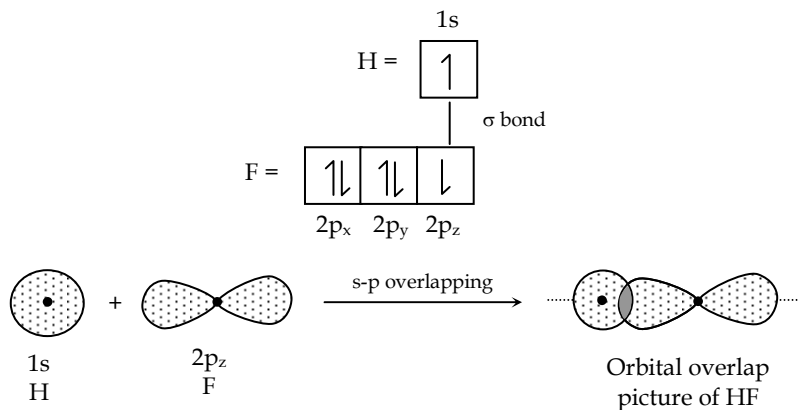
1.10 Calculate the number of σ and π bonds in the following molecules1.3.3 Bonding in H_2 , HF , F_2 , O_2 and N_2 on the Basis of Valence Bond Theory

1. **Hydrogen (H_2) molecule:** Hydrogen atom has one half filled orbital ($1s^1$). Therefore, it can form a single covalent bond. In the formation of H_2 , half filled $1s$ orbital of one hydrogen atom overlaps with half filled $1s$ orbital of another hydrogen atom forming a sigma bond.

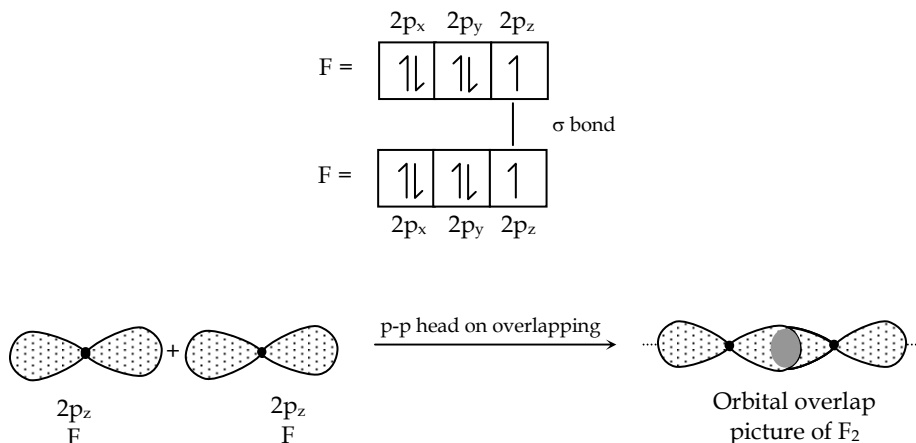


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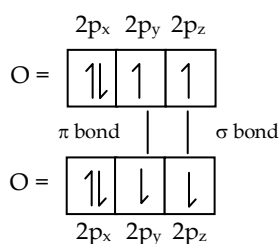
2. **Hydrogen fluoride (HF) molecule:** Hydrogen atom has one half filled orbital ($1s^1$). Fluorine atom has one half filled orbital ($1s^2 2s^2 2p_x^2 p_y^2 p_z^1$). So, each can form a single covalent bond. In the formation of HF, half filled $1s$ orbital of one hydrogen atom overlaps with half filled $2p_z$ orbital of fluorine atom forming a sigma bond (here, Z-axis is considered as inter-nuclear axis).



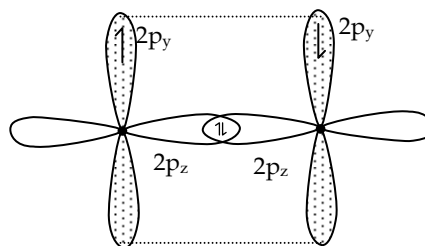
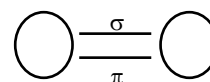
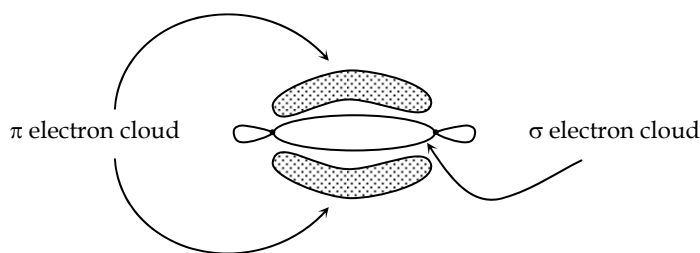
3. **Fluorine (F_2) molecule:** Fluorine atom has one half filled orbital ($1s^2 2s^2 2p_x^2 p_y^2 p_z^1$). So, it can form a single covalent bond. In the formation of F_2 , half filled $2p_z$ orbital of one fluorine atom overlaps with half filled $2p_z$ orbital of another fluorine atom *end to end* forming a sigma bond (here, Z-axis is considered as inter-nuclear axis).



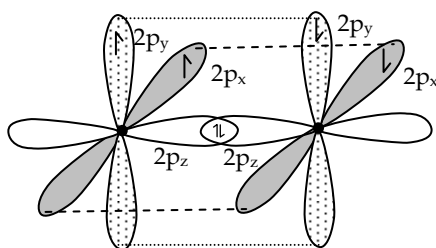
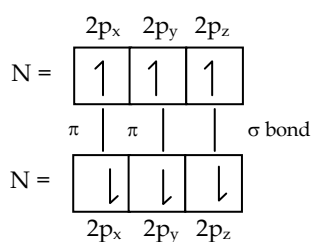
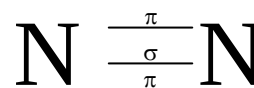
4. **Oxygen (O_2) molecule:** Oxygen atom has two half filled orbitals ($1s^2 2s^2 2p_x^2 p_y^1 p_z^1$). So, it can form two covalent bonds or a double bond. In the formation of O_2 , half filled $2p_z$ orbital of one oxygen atom overlaps with half filled $2p_z$ orbital of another oxygen atom *end to end* forming a sigma bond. The remaining half filled $2p_y$ orbital of oxygen atom which is perpendicular to its $2p_z$ orbital overlaps with that of another oxygen atom *laterally* forming a pi bond. Hence, O_2 has a double bond, a sigma bond and a pi bond. Sigma bond is strong since overlapping of orbitals is more and pi bond is weak since overlapping of orbitals is less.



(here, z is taken as internuclear axis)

Orbital picture of O₂

5. **Nitrogen (N₂) molecule:** Nitrogen atom has three half filled orbital ($1s^2 2s^2 2p_x^1 p_y^1 p_z^1$). So, it can form three covalent bonds or a triple bond. In the formation of N₂, half filled $2p_z$ orbital of one nitrogen atom overlaps with half filled $2p_z$ orbital of another nitrogen atom *end to end* forming a sigma bond. The remaining half filled $2p_y$ orbital which is perpendicular to $2p_z$ orbital overlaps with that of another nitrogen atom *laterally* forming a pi bond. The remaining half filled $2p_x$ orbital which is perpendicular to $2p_z$ orbital overlaps with that of another nitrogen atom *laterally* forming another pi bond. Hence, N₂ has a triple bond, a sigma bond and two pi bond. Sigma bond is strong since overlapping of orbitals is more and pi bond is weak since overlapping of orbitals is less.

Orbital picture of N₂

Self Test

- 1.11 What are the differences between a sigma bond and a pi bond?
- 1.12 Draw the orbital picture of following molecules and indicate the number of unpaired electrons according to VBT.
- i. HCl ii. Cl₂ iii. O₂ iv. N₂

1.3.4 Hybridization

The concept of hybridization was introduced to explain the nature of bonding and geometry of molecules especially in polyatomic molecules. According to this concept, atomic orbitals or orbitals wave function, undergo mixing to form new orbitals of equivalent energy and identical shape and size called hybrid orbitals which then participate in bonding. The process is called **hybridization**.

Characteristics of hybridization

1. Atomic orbitals only in an atom undergo hybridization. It is not mixing of atomic orbitals of different atoms.
2. Atomic orbitals undergoing hybridization must have nearly equal energy. They should not have very large difference in their energy.
3. Full filled, half filled as well as empty atomic orbitals undergo hybridization.
4. The number of hybrid orbitals formed is equal to total number of orbitals undergoing hybridization.
5. Hybrid orbitals formed in a hybridization have new character and are equivalent in energy and identical in shape and size and take the geometry so as to get minimum electrostatic repulsion.
6. Hybrid orbital has one lobe enlarged, so it overlaps more effectively. Hence, a hybrid orbital forms stronger bond than the original atomic orbitals.

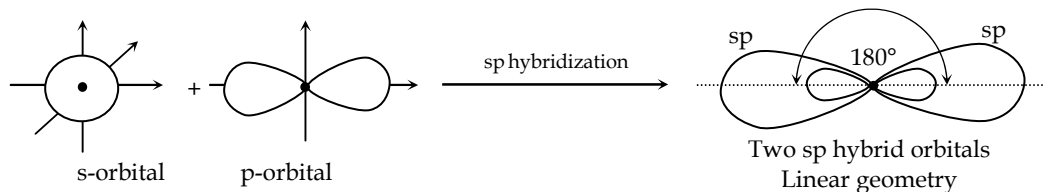
Hence, **hybridization is the process of mixing atomic orbitals of nearly equal energy in an atom to form new orbitals of equivalent energy and identical shape and size called hybrid orbitals.**

Types of hybridization

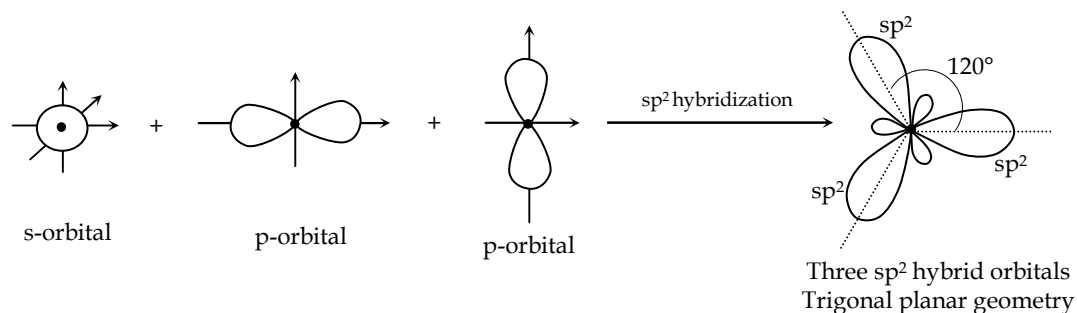
There are various types of hybridizations depending on nature and number of atomic orbitals involved. The s and p orbitals undergo three types of hybridization. Only these types of hybridizations (sp , sp^2 and sp^3) will be discussed here through d orbitals can also take part in hybridization.

1. sp hybridization
2. sp^2 hybridization
3. sp^3 hybridization

1. **sp hybridization (Linear or diagonal hybridization):** It is the process of mixing one s and one p orbital of nearly equal energy in an atom to form two new orbitals of equivalent energy and identical shape. Each orbital formed is called sp hybrid orbital. Two sp hybrid orbitals take linear geometry with bond angle 180° . Hence, this type of hybridization is also called linear hybridization. Each sp hybrid orbital has 50% s -character and 50% p -character.



2. **sp^2 hybridization (Trigonal planar hybridization):** It is the process of mixing one s and two p orbitals of nearly equal energy in an atom to form three new orbitals of equivalent energy and identical shape. Each orbital formed is called sp^2 hybrid orbital. Three sp^2 hybrid orbitals takes trigonal planar geometry with bond angle 120° . Hence, this type of hybridization is also called **trigonal planar hybridization**. Each sp^2 hybrid orbital has 33.33% s -character and 66.67% p -character.



3. **sp^3 hybridization (Tetrahedral hybridization):** It is the process of mixing one s and three p orbitals of nearly equal energy in an atom to form four new orbitals of equivalent energy and identical shape and size. Each orbital formed is called sp^3 hybrid orbital. Four sp^3 hybrid orbitals take tetrahedral geometry with bond angle $109^\circ 28'$. Hence, this type of hybridization is also called tetrahedral hybridization. Each sp^3 hybrid orbital has 25% s -character and 75% p -character. An sp^3 hybrid orbitals has one lobe enlarged so it can overlap more effectively than an s or a p orbital. Hence, sp^3 hybrid orbitals form stronger bonds than the original s and p orbitals.

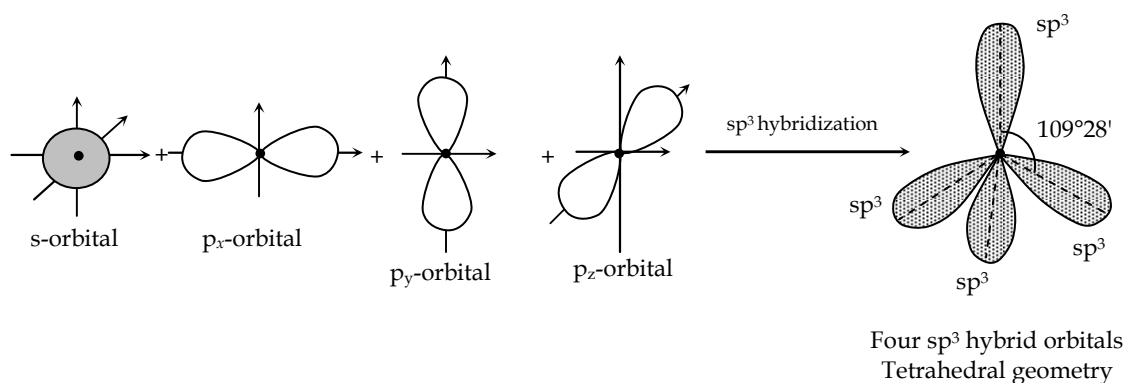


Table 1.7: Approximate strength of bonds formed by various orbitals

Orbital	Relative bond strengths
s	1
p	1.73
sp	1.93
sp^2	1.99
sp^3	2.00

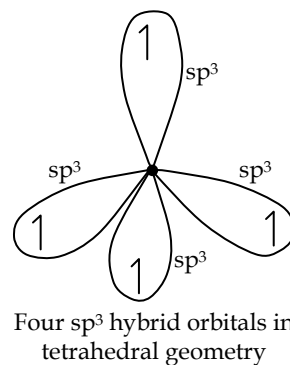
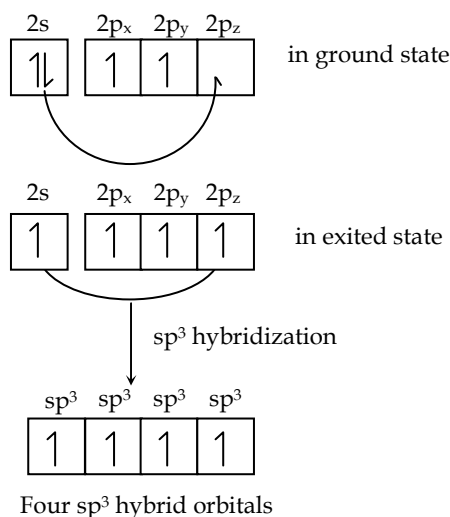
Table 1.8: Various bond properties based upon the mode of hybridization

	sp	sp^2	sp^3
% of s -character	50	33.33	25
Electronegativity	—————> Decreases		
Acidic character	—————> Decreases		
Bond length	—————> Increases		

Examples involving sp^3 , sp^2 and sp hybridization

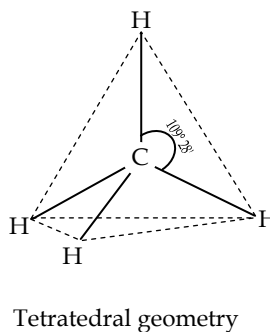
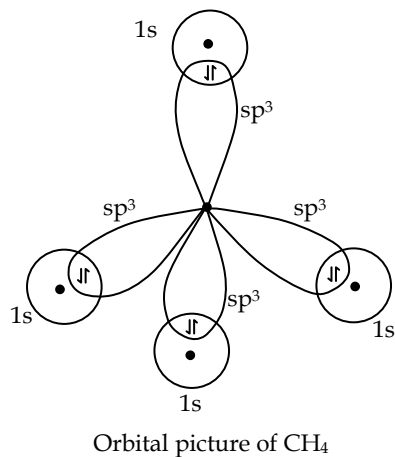
1. Examples involving sp^3 hybridization

- i. **Methane (CH_4):** Carbon has two half filled orbitals in ground state ($1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^0$). It has four half filled orbitals in excited state since one electron is promoted from $2s$ to empty $2p$ orbital during chemical combination ($1s^1, 2s^1, 2p_x^1, 2p_y^1, 2p_z^1$) i.e.



Now one $2s$ and three $2p$ orbitals undergo sp^3 hybridization to form four sp^3 hybrid orbitals of equivalent energy and identical shape and size which take tetrahedral geometry with bond angle $109^\circ 28'$.

Each half filled sp^3 hybrid orbital of carbon atom overlaps with half filled $1s$ orbital of four hydrogen atoms forming four carbon to hydrogen sigma bonds. Hence, all four bonds in methane are identical with bond angle $109^\circ 28'$ and methane has a tetrahedral geometry.

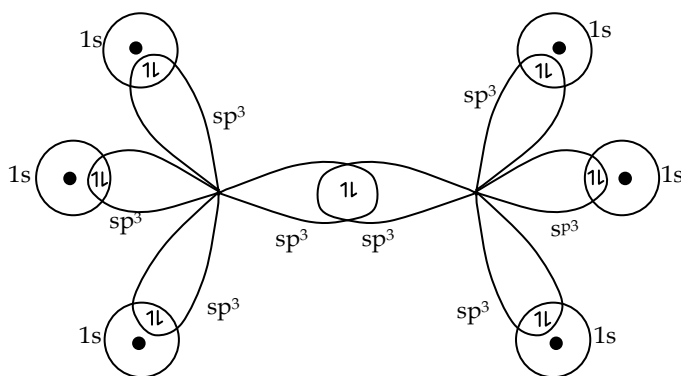


Why is the concept of hybridization needed in CH_4 ?

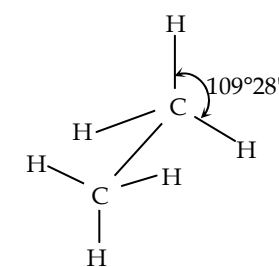
The answer is to explain the nature of bond and bond angle. In CH_4 , all four bonds are identical (same bond length ($1.10\text{\AA}$) and bond dissociation energy) with bond angle $109^\circ 28'$. This means all four half filled orbitals in carbon are equivalent. This is possible only when $2s$ and three $2p$ orbitals undergo mixing forming four new orbitals of equivalent energy during bond formation. If the concept of hybridization is not introduced, there should have two types of bonds in CH_4 ; one type of bond due to $2s$ orbital and another type of bonds due to three $2p$ orbitals. Similarly, bond angle in methane should have 90° since three p orbitals are perpendicular to each other.

- ii. **Ethane (C_2H_6):** Each carbon has four half filled orbitals in excited state since one electron is promoted from $2s$ to empty $2p$ orbital during chemical combination.

Now, one $2s$ and three $2p$ orbitals undergo sp^3 hybridization to form four sp^3 hybrid orbitals of equivalent energy and identical shape which take tetrahedral geometry with bond angle $109^\circ 28'$.



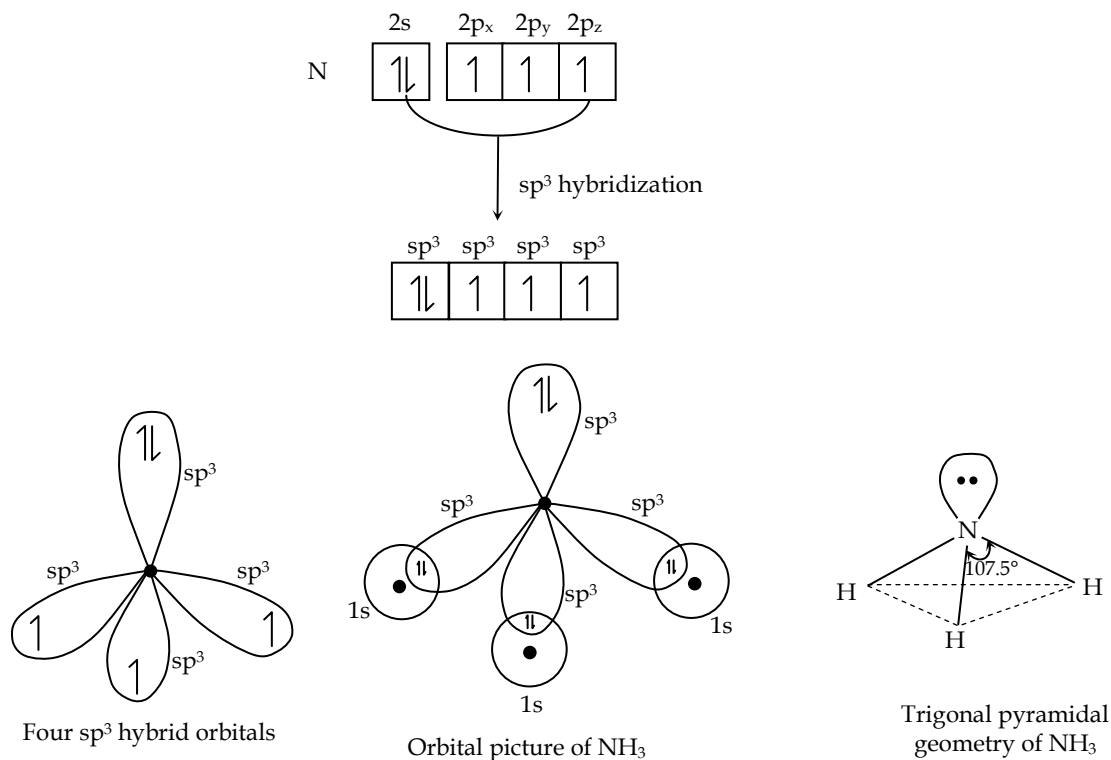
Orbital picture of ethane molecule



Geometry of Ethane

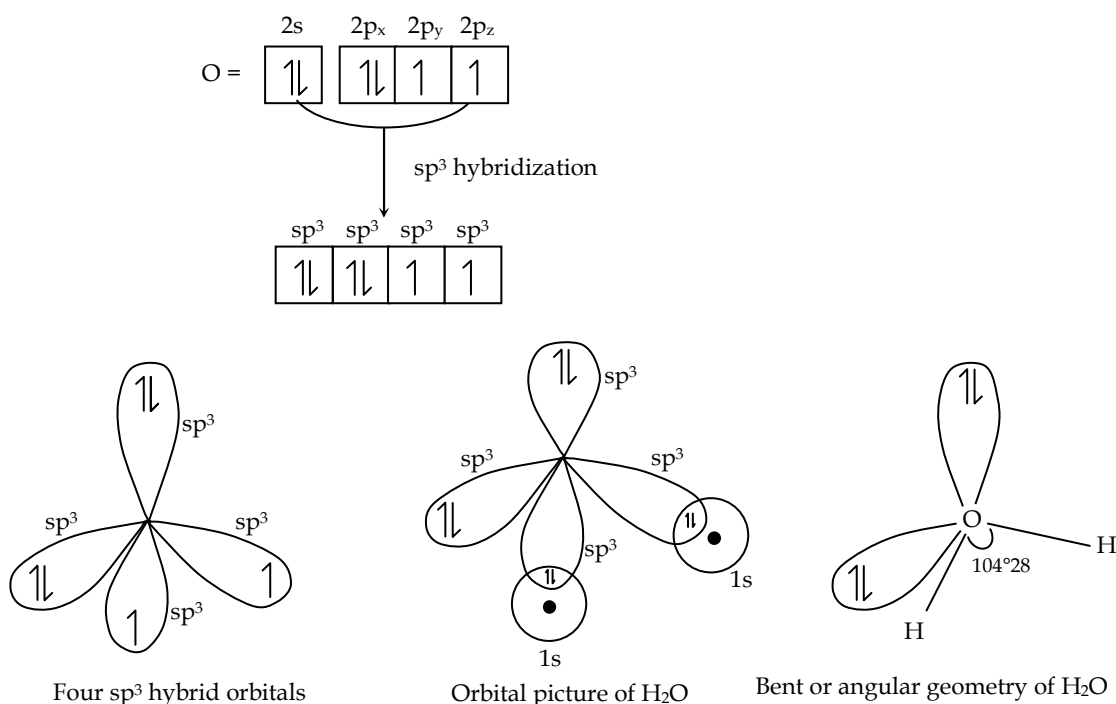
One half filled sp^3 hybrid orbital of one carbon atom overlaps with half filled sp^3 orbital of another carbon atom head to head forming a carbon to carbon sigma bond. The remaining three half filled sp^3 orbitals of each carbon atom overlap with half filled $1s$ orbital of six hydrogen atoms forming six carbon to hydrogen sigma bonds. Hence, all carbon to hydrogen bonds in ethane are identical with bond angle $109^\circ 28'$.

- iii. **Ammonia (NH_3):** Nitrogen has three half filled orbitals ($1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$). Now, one $2s$ and three $2p$ orbitals undergo sp^3 hybridization to form four sp^3 hybrid orbitals of equivalent energy and identical shape which take tetrahedral geometry with bond angle $109^\circ 28'$.



Three sp^3 hybrid orbitals are half filled and one sp^3 orbital is completely filled. Each half filled sp^3 hybrid orbital overlaps with half filled $1s$ orbital of three hydrogen atoms forming three sigma bonds. Now, there are three bond pairs and one lone pair around the central atom. The repulsion between lone pair and bond pair is greater than the repulsion between bond pairs. Hence, bond angle in NH_3 is lower than the normal tetrahedral value. NH_3 is trigonal pyramidal in shape with bond angle $107^\circ 48'$.

- iv. **Water (H_2O):** Oxygen has two half filled orbitals ($1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$). Now one $2s$ and three $2p$ orbitals undergo sp^3 hybridization to form four sp^3 hybrid orbitals of equivalent energy and identical shape which take tetrahedral geometry with bond angle $109^\circ 28'$.

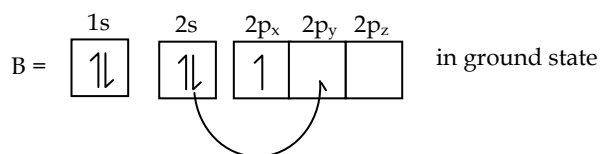


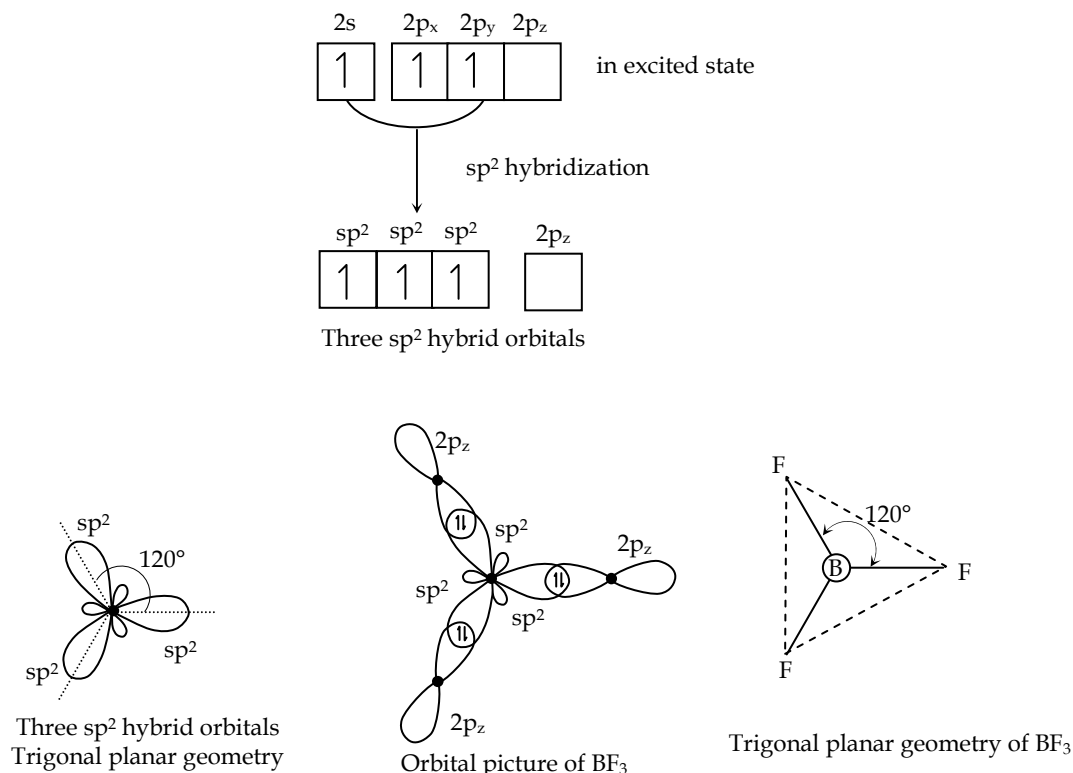
Two sp^3 hybrid orbitals are half filled and two sp^3 orbital are completely filled. Each half filled sp^3 hybrid orbital overlaps with half filled $1s$ orbital of two hydrogen atoms forming two sigma bonds. Now there are two bond pairs and two lone pairs around the central atom. The repulsion between lone pair and lone pair is greater than lone pair and bond pair which in turn is greater than the repulsion between bond pairs. Hence, bond angle in H_2O is lower than the normal tetrahedral value with bond angle $104^\circ 28'$. The geometry of H_2O molecular is "angular" or "bent".

2. Examples involving sp^2 hybridization

- i. **Boron trifluoride (BF_3):** Boron has one half filled orbital in ground state ($1s^2 2s^2 2p_x^1 2p_y^0 2p_z^0$). It has three half filled orbitals in excited state since one electron is promoted from $2s$ to $2p$ orbital during chemical combination.

Now, one $2s$ and two $2p$ orbitals undergo sp^2 hybridization to form three sp^2 hybrid orbitals of equivalent energy and identical shape and size which take trigonal planar geometry with bond angle 120° .





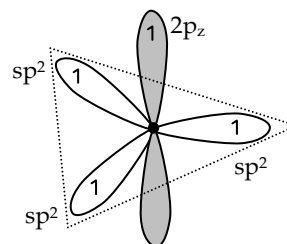
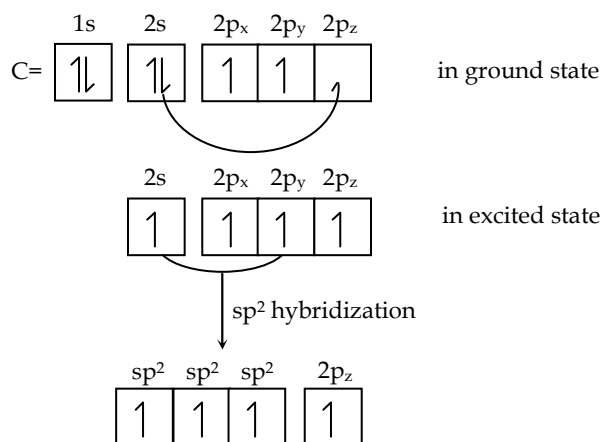
Each half filled sp^2 hybrid orbital of carbon atom overlaps with half filled $2p_z$ orbital of three fluorine atoms forming three boron to fluorine sigma bonds. Hence, all three bonds in BF_3 are identical with bond angle 120° and BF_3 is trigonal planar molecule.

Why is the concept of hybridization needed in BF_3 ?

The answer is to explain the nature of bond and bond angle. In BF_3 , all three bonds are identical (same bond length and bond energy) with bond angle 120° . This means all three half filled orbitals in boron are equivalent. This is possible only when $2s$ and two $2p$ orbitals undergo mixing forming three new orbitals of equivalent energy during bond formation. If the concept of hybridization is not introduced, there should have two types of bonds in BF_3 ; one type of bond due to $2s$ orbital and another type of bonds due to two $2p$ orbitals. Similarly, bond angle in BF_3 should have 90° since two p orbitals are perpendicular to each other.

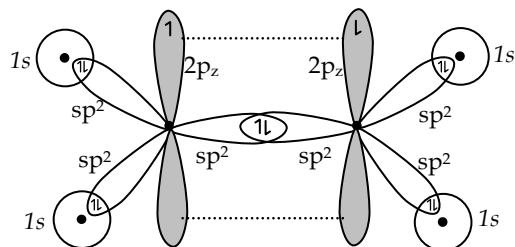
- ii. **Ethylene (C_2H_4):** Each carbon has two half filled orbitals in ground state ($1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^0$). It has four half filled orbitals in excited state since one electron is promoted from $2s$ to empty $2p$ orbital during chemical combination ($1s^2, 2s^1, 2p_x^1, 2p_y^1, 2p_z^1$).

Now, one $2s$ and two $2p$ orbitals undergo sp^2 hybridization to form three sp^2 hybrid orbitals of equivalent energy and identical shape and size which take trigonal planar geometry with bond angle 120° . Unhybridized $2p_z$ half filled orbital remains perpendicular to the plane of hybrid orbitals.

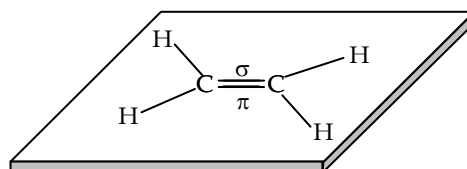


Three sp^2 hybrid orbitals Unhybridized orbital

One half filled sp^2 orbital of one carbon atom overlaps with half filled sp^2 orbital of another carbon atom forming carbon to carbon sigma bond. Each half filled sp^2 hybrid orbitals of both carbons atoms overlap with half filled $1s$ orbital of four hydrogen atoms forming four carbon to hydrogen sigma bonds. The remaining half filled $2p_z$ orbital of one carbon which is perpendicular to the plane of hybrid orbitals overlaps with that of another carbon atom *laterally* forming carbon to carbon pi bond. Hence, ethene is flat molecule (all carbon and hydrogen atoms lie in a plane) with pi electron cloud above and below the plane of the molecule. The geometry is trigonal planar with respect to each carbon.



Orbital picture of ethylene



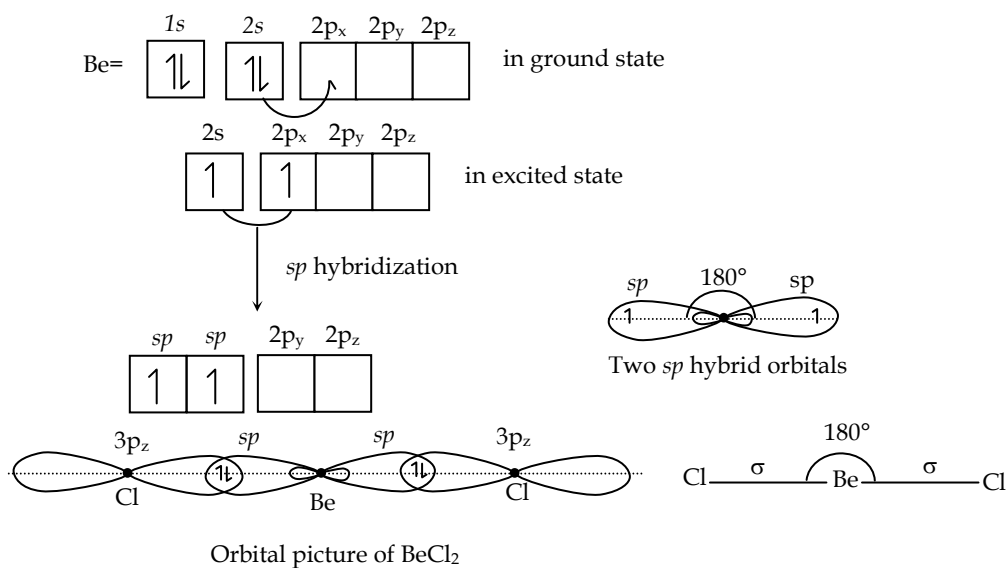
Trigonal planar geometry of ethylene

3. Examples involving sp hybridization

- i. **Beryllium chloride (BeCl_2):** Be has no half filled orbitals in ground state ($1s^2, 2s^2, 2p_x^0, 2p_y^0, 2p_z^0$). It has two half filled orbitals in excited state since one electron is promoted from $2s$ to empty $2p$ orbital during chemical combination ($1s^2, 2s^1, 2p_x^1, 2p_y^0, 2p_z^0$).

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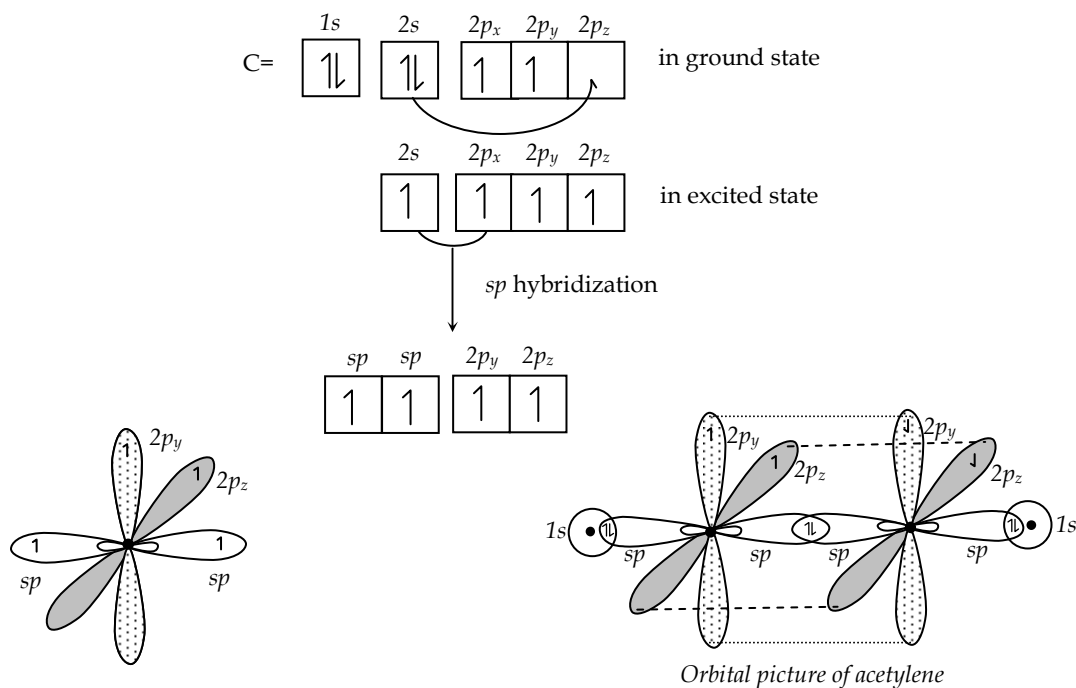
i.e.

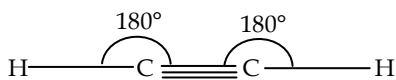


Now one $2s$ and one $2p$ orbitals undergo sp hybridization to form two sp hybrid orbitals of equivalent energy and identical shape and size which take linear geometry with bond angle 180° .

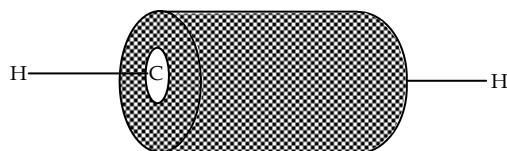
Each half filled sp hybrid orbital of Be atom overlaps with half filled $2p_z$ orbital of two chlorine atoms forming two sigma bonds. Hence, all two bonds in BeCl_2 are identical with bond angle 180° and BeCl_2 is linear molecule.

- ii. **Acetylene (C_2H_2):** Each carbon has two half filled orbitals in ground state ($1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^0$). It has four unpaired electrons in excited state since one electron is promoted from $2s$ to empty $2p$ orbital during chemical reaction ($1s^2, 2s^1, 2p_x^1, 2p_y^1, 2p_z^1$). i. e.





Linear shape of acetylene



Cylindrical electron cloud in acetylene

Now, one $2s$ and one $2p$ orbitals undergo sp hybridization to form two sp hybrid orbitals of equivalent energy and identical shape and size which take linear geometry with bond angle 180° . Unhybridized $2p_y$ and $2p_z$ orbitals remain perpendicular to hybrid orbitals.

One half filled sp hybrid orbital of carbon atom overlaps with half filled sp orbital of another carbon atom forming carbon to carbon sigma bond. The remaining half filled sp orbitals of each carbon overlaps with half filled $1s$ orbital of two hydrogen atoms forming two carbon to hydrogen sigma bonds. Unhybridized and half filled $2p_y$ and $2p_z$ orbitals of one carbon overlap with respective orbitals of another carbon *laterally* forming two carbon to carbon pi bonds.

iii. Carbon dioxide (CO_2):

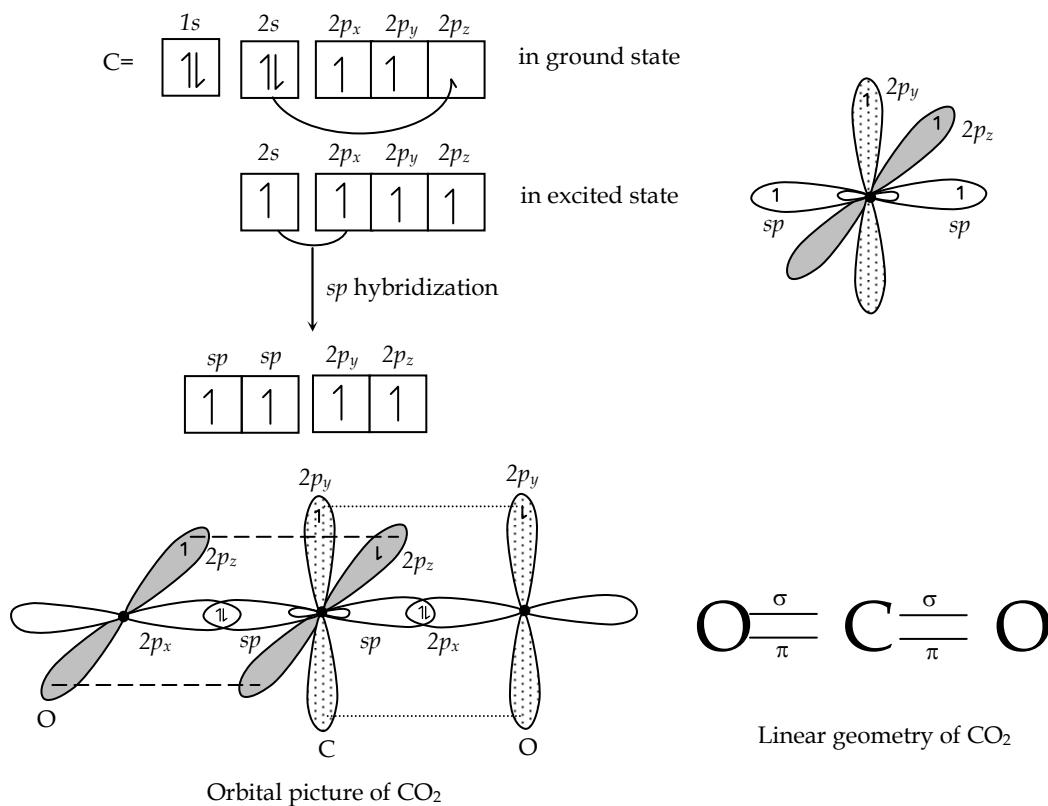


Table 1.9: A summary of hybridization

Hybridization	Geometry	Bond angle	Examples
sp	Linear	180°	Acetylenes, CO_2 , BeF_2 , BeCl_2 , etc.
sp^2	Trigonal planar	120°	Alkenes, benzene, BF_3 , etc.
sp^3	Tetrahedral	$109^\circ 28'$	Alkanes, CH_3X , etc.
sp^3d	Trigonal bipyramidal	90° , 120°	PX_5 ,
sp^3d^2	Octahedral	90°	SF_6
sp^3d^3	Pentagonal bipyramidal	72° , 90°	IF_7
dsp^2	Square planar	90°	$[\text{Cu}(\text{NH}_3)_4]^{2+}$

1.3.5 Limitations of Valence Bond Theory (VBT)

1. According to VBT, a covalent bond is formed by overlapping half filled orbitals. It can not explain bonding in electron deficient compounds.
2. It fails to explain the paramagnetic behavior of O_2 . Paramagnetic behavior is due to unpaired electrons but according to VBT, O_2 has no unpaired electrons.
3. Presence of other nuclei in the molecule should affect the whole electronic arrangement of atoms in the molecule. This has been neglected in VBT.

A simple way to predict the type of hybridization

The type of hybridization and hence, the shape of covalent molecules and ions can be predicted with the help of general formula

$$H = \frac{1}{2} (V + M + A - C)$$

where, H = Hybridization

V = Number of valence electrons of the central atom

M = Number of monovalent atoms

A = Charge on anion

C = Charge on cation.

It is very easy to find number of valence electrons (V) of an element if its group or family is known.

Group 14 Carbon family	Group 15 Nitrogen family	Group 16 Chalcogens Oxygen family	Group 17 Halogens	Group 18 Noble gases
				He^* (exception)
C	N	O	F	Ne
Si	P	S	Cl	Ar
Ge	As	Se	Br	Kr
	Sb	Te	I	Xe
	Bi	Po		Rn
No. of Valence e = 4	No. of Valence e = 5	No. of Valence e = 6	No. of Valence e = 7	No. of Valence e = 8

Examples:

Example i: NCl_3

$$H = \frac{1}{2} (V + M + A - C)$$

Here, $V = 5$ ($\text{N} = 2, 5$), $M = 3$ (three Cl atoms), $A = 0$ (no negative charge), $C = 0$ (no positive charge).

$$H = \frac{1}{2} (5 + 3 + 0 - 0) = 4$$

Hence, hybridization of N in NCl_3 is sp^3 .

Example ii: PCl_5

$$H = \frac{1}{2} (V + M + A - C)$$

Here, $V = 5$ (It belongs to nitrogen family),

$$M = 5 \text{ (five Cl atoms), } H = \frac{1}{2} (5 + 5 + 0 - 0) = 5$$

Hence, hybridization of P in PCl_5 is sp^3d .

Example iii: SO_3

$$H = \frac{1}{2} (V + M + A - C)$$

Here, $V = 6$ (It belongs to oxygen family),

$M = 0$ (There is no monovalent atoms, oxygen is divalent)

$$H = \frac{1}{2} (6 + 0 + 0 - 0) = 3$$

Hence, hybridization of S in SO_2 is sp^2 .

Example iv: ClF_3

$$H = \frac{1}{2} (V + M + A - C)$$

Here, $V = 7$ (It belongs to halogen family), $M = 3$ (three F atoms)

$$H = \frac{1}{2} (7 + 3 + 0 - 0) = 5$$

Hence, hybridization of Cl in ClF_3 is sp^3d .

Example v: XeF_4

$$H = \frac{1}{2} (V + M + A - C)$$

Here, $V = 8$ (It belongs to group 18), $M = 4$ (four F atoms)

$$H = \frac{1}{2} (8 + 4 + 0 - 0) = 6$$

Hence, hybridization of Xe in XeF_2 is sp^3d^2 .

Example vi: CH_3^+

$$H = \frac{1}{2} (V + M + A - C)$$

Here, $V = 4$ ($C = 2, 4$), $M = 3$ (three H atoms), $A = 0$ (no negative charge), $C = 1$

$$H = \frac{1}{2} (4 + 3 + 0 - 1) = 3$$

Hence, hybridization of C in CH_3^+ is sp^2 .

Example vii: SO_4^{2-}

$$H = \frac{1}{2} (V + M + A - C)$$

Here, $V = 6$ ($N = 2, 5$), $M = 0$ (no monovalent atoms), $A = 2$, $C = 0$ (no positive charge).

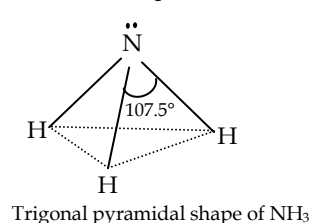
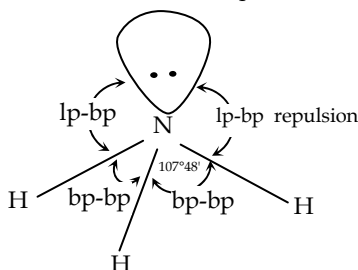
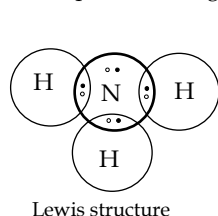
$$H = \frac{1}{2} (6 + 0 + 2 - 0) = 4$$

Hence, hybridization of S in SO_4^{2-} is sp^3 .

■ Solved Questions

1. Predict the molecular geometry of ammonia on the basis of VSEPR theory.

- The shape of a covalent molecule or ion is determined by the total number of electron pairs present in the valence shell of the central atom in the molecule. The electron pairs remain arranged around the central atom in space so as to get the minimum electrostatic repulsion hence determines the shape.



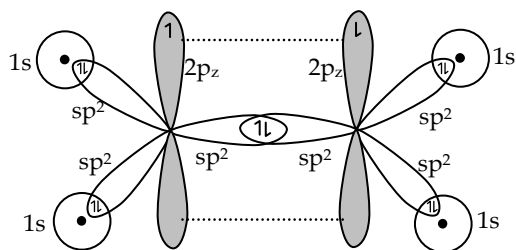
In NH_3 , the central atom nitrogen has four electron pairs hence, expected geometry is tetrahedral but there are three bond pairs (bp) only. So, actual shape of ammonia is trigonal pyramidal. Repulsion between lone pairs (lp) is more than that between a lone pair and a bond pair. Hence, the bond angle in NH_3 is less than normal tetrahedral value ($109^\circ 28'$). Experimentally the bond angle has been found to be $107^\circ 28'$.

2. Write differences between a sigma and a pi bond.

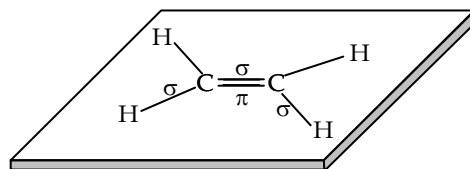
σ -bond	π -bond
i. It is formed by end to end, head on or axial overlapping of half filled orbitals.	i. It is formed by sideways or lateral overlapping of half filled orbitals.
ii. Overlapping of orbitals takes place along the internuclear axis.	ii. Overlapping of orbitals takes place perpendicular to the internuclear axis.
iii. The extent of overlapping is large so the bond formed is strong.	iii. The extent of overlapping is small so the bond formed is weak.
iv. It is symmetrical along the internuclear axis so free rotation about sigma is possible.	iv. It consists of two lobes on either side of internuclear axis and free rotation about pi bond is restricted.
v. It may be present alone or along with pi bonds.	v. It can not present alone. It is always present in addition to sigma bond.
vi. It is formed by overlapping of all types of orbitals; $s-s$, $s-p$, $p-p$ orbitals.	6. It is formed by the overlapping of other than s orbitals.

3. Define hybridization. Draw the orbital picture of ethylene.

- Hybridization is the process of mixing atomic orbitals of nearly equal energy in an atom to form new orbitals of equivalent energy and identical shape and size called hybrid orbitals. For example: in sp^3 hybridization, one s -orbital and three p -orbitals hybridize together giving four sp^3 hybrid orbitals.



Orbital picture of ethylene



Planar geometry of ethylene

Questions

Very Short Answer Questions:

- BCl_3 is trigonal planar whereas PCl_3 is trigonal pyramidal, why?
- NH_3 is trigonal pyramidal whereas NH_4^+ is tetrahedral. Why?
- The bond angle in NF_3 is 103° whereas in BF_3 is 120° . What factor account for this difference in bond angle?
- Why is H-O-H bond angle in water molecule comparatively higher than H-S-H bond angle in H_2S molecule?
- What are the differences between a σ -bond and a π -bond?
- Define hybridization. What are the characteristic of hybridization?
- Mention the type of hybridization of underlined atom in the following molecules.
 - $\underline{C}H_4$
 - $\underline{C}_2H_4$
 - $\underline{C}_2H_2$
 - $\underline{B}F_3$
 - $\underline{Be}F_2$
 - $\underline{N}H_3$
 - $H_2\underline{O}$
 - $\underline{C}O_2$
- Nitrogen of ammonia has sp^3 hybridization but ammonia molecule has trigonal pyramidal geometry. Why?
- Mention with an example of
 - Diagonal hybridization
 - Trigonal hybridization
 - Tetrahedral hybridization
- Draw the orbital picture of following compounds indicating σ and π bond.
 - Ethane
 - Ethylene
 - Acetylene
 - Ammonia
- Write any two features of sp^3 hybrid orbitals with examples.
- Why do NH_3 and BF_3 have dissimilar geometries?
- What is the mode of hybridization in BF_3 ? Write any two important features of this hybridization.
- What is the mode of hybridization of the central atom whose molecular geometry is tetrahedral? Give an example.
- Draw the orbital picture of a hydrocarbon showing (a) tetrahedral structure (b) Trigonal planar structure.
- Draw orbital picture of ethyne and mention σ and π bonds.

Short /Long Answer Questions:

- Predict the geometry of the following species on the basis of VSEPR model?
 - BeF_2
 - BF_3
 - NH_3
 - NH_4^+
 - PCl_5
 - SO_2
 - SF_6
 - ClF_3
 - NCl_3
 - NO_3^-
 - XeF_4
 - I_3^-
- How would you interpret that all four C-H bonds of methane are identical?

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- Predict the molecular geometry of the following species on the basis of hybridization?
 - BeF_2
 - BF_3
 - NH_3
 - CH_4
 - C_2H_4
 - C_2H_2
 - CH_3Cl
 - CCl_4
- What do you understand by sp^2 hybridization? Explain the molecular geometry involved with an example for the same.

Multiple Choice Questions

- VSEPR theory is developed by
 - Lewis and Kossel
 - Sidgwick and Powell
 - Gillespie and Nyholm
 - Linus Pauling
- VSEPR theory is useful to explain
 - Nature of bond
 - Number of π electron pairs
 - Geometry of molecules
 - Number of π and σ electron pairs
- The order of bond angle is
 - $\text{PH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{SbH}_3$
 - $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$
 - $\text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{NH}_3$
 - $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$
- The order of bond angle is
 - $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Te} > \text{H}_2\text{Se}$
 - $\text{H}_2\text{S} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{O}$
 - $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$
 - $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$
- The order of bond angle is
 - $\text{PF}_3 > \text{PBr}_3 > \text{PCl}_3 > \text{PI}_3$
 - $\text{PF}_3 > \text{PCl}_3 > \text{PBr}_3 > \text{PI}_3$
 - $\text{PCl}_3 > \text{PBr}_3 > \text{PI}_3 > \text{PF}_3$
 - $\text{PF}_3 < \text{PCl}_3 < \text{PBr}_3 < \text{PI}_3$
- Which of the followings has highest value of bond angle?
 - BF_3
 - NH_3
 - PCl_3
 - NCl_3
- The lowest value of bond angle is in
 - NH_3
 - NH_4^+
 - NH_2^-
 - NCl_3
- If the central atom has three bond pairs and one lone pair, the expected shape is
 - Tetrahedral
 - Square planar
 - Pyramidal
 - Trigonal planar
- If the central atom has three bond pairs and two lone pairs, the expected shape of the molecule is
 - Pentagonal bipyramidal
 - Trigonal bipyramidal
 - T shaped
 - Trigonal planar
- The central atom has three lone pairs and two bond pairs, the shape of molecule is
 - Linear
 - 'V' shaped
 - Trigonal bipyramidal
 - 'L' shaped
- The shape of ClF_3 is
 - T shaped
 - Trigonal planar
 - Pyramidal
 - Tetrahedral
- The geometry of BF_4^- is similar to
 - BF_3
 - NF_3
 - CH_4
 - SF_4
- Which of the followings is trigonal planar?
 - AlCl_3
 - PCl_3
 - ClF_3
 - ICl_2^-
- Shape of XeF_2 is similar to
 - OF_2
 - I_3^-
 - SCl_2
 - None
- Which of the following pairs has identical geometry?
 - BCl_3 and PCl_3
 - CH_3Cl and CHCl_3
 - CH_4 and CCl_4
 - XeF_4 and SiF_4
- CO_2 is isostructural with
 - C_2H_2
 - NO_2
 - SO_2
 - C_2H_6
- Which of the followings has unequal bond length?
 - NF_3
 - SF_4
 - PF_5
 - SF_6
- According to valence bond theory, number of unpaired electron in O_2 is
 - 0
 - 1
 - 2
 - 4
- The number of unpaired electron in O_2 is
 - 0
 - 1
 - 2
 - 4
- A pi bond is formed by
 - sideways overlapping of sp^2 hybrid orbitals
 - axial overlapping of p orbitals
 - lateral overlapping of p orbitals
 - head on overlapping of p orbitals
- pi bonds in benzene are formed by
 - sp^2 -s overlapping
 - sp^2 - sp^2 overlapping
 - sideways p-p overlapping
 - head on p-p overlapping
- The number of pi bond in acetylene is
 - 1
 - 2
 - 3
 - 4

Answer Key

1	2	3	4	5	6	7	8	9	10	11
c	c	b	c	d	a	c	c	c	a	a
12	13	14	15	16	17	18	19	20	21	22
c	a	b	c	a	c	a	c	c	c	b

Volumetric Analysis

2



Titration is a very simple and efficient method of determining the concentration of solutions. Nowadays, instrumental methods are being mostly used in titration owing their accuracy and versatility.

2.1 Introduction

Chemical analysis is one of the most important tasks in chemistry. It involves qualitative analysis and quantitative analysis. Qualitative analysis involves the determination or identification of chemical substances or radicals present there in. Qualitative salt analysis is an example of this category. Quantitative analysis involves the estimation of amount of substance present. There are different methods of quantitative analysis. Some of these methods are volumetric analysis (e.g., titrimetric analysis, gravimetric analysis, instrumental analysis, etc.).

Volumetric analysis involves the determination of concentration of one solution comparing with another standard solution, by measuring their reacting volumes. If concentration of base and acid are measured with respect to standard acid and standard alkali, these analyses are called acidimetry and alkalimetry, respectively. This topic is mainly related to acidimetry and alkalimetry. Before proceeding to this, first we shall discuss about equivalent weight, knowledge of which is the prerequisite to understand the acidimetry and alkalimetry.

2.2 Concept of Equivalent Weight

Equivalent weight is the combining capacity of a chemical species by weight. Equivalent weight of any compound is defined as the number of parts by mass (in gram) of it which reacts completely with one gram equivalent of another compound. Equivalent weight of compound can be studied under different headings.

1. Equivalent weight of acid

Equivalent weight of an acid is defined as the number of parts by weight of it which contains 1.008 parts by weight of replaceable hydrogen atom.

In HCl: 36.5 parts by weight of HCl consists 1.008 parts by weight of replaceable hydrogen. Hence equivalent weight of HCl is 36.5.

$$\text{Equivalent weight of acid} = \frac{\text{Molecular mass of acid}}{\text{Number of replaceable hydrogen atoms}}$$

The number of replaceable hydrogen atom(s) present in a molecule of an acid is called its **basicity**.

$$\therefore \text{Equivalent weight of acid} = \frac{\text{Molecular mass of acid}}{\text{Basicity}}$$

Self Test

2.1 Calculate the equivalent weight of carbonic acid.

Table 2.1: Some acids with their molecular mass (Mol.mass) and equivalent weight (Eq.Wt.).

S.N.	Name of acid	Molecular formula	Molecular mass (amu)	Basicity	Eq.wt. = $\frac{\text{Mol. mass}}{\text{Basicity}}$
1.	Hydrochloric acid	HCl	36.5	1	36.5
2.	Nitric acid	HNO ₃	63.0	1	63.0
3.	Acetic acid	CH ₃ COOH	60	1	60
4.	Formic acid	HCOOH	46	1	46
5.	Sulphuric acid	H ₂ SO ₄	98	2	49
6.	Oxalic acid (anhydrous)	$\begin{array}{c} \text{COOH} \\ \\ \text{COOH} \end{array}$	90	2	45
7.	Oxalic acid (hydrated)	$\begin{array}{c} \text{COOH.H}_2\text{O} \\ \\ \text{COOH.H}_2\text{O} \end{array}$	126	2	63
8.	Phosphoric acid	H ₃ PO ₄	98	3	32.6
9.	Phosphorous acid	H ₃ PO ₃	82	2	41.0
10.	Pyrophosphoric acid	H ₄ P ₂ O ₇	178	4	44.5
11.	Metaphosphoric acid	HPO ₃	80	1	80

2. Equivalent weight of base or alkali

Equivalent weight of base is defined as the number of parts by weight of it which is exactly neutralized by one gram equivalent of acid.

$$\text{Equivalent weight of base} = \frac{\text{Molecular mass of base}}{\text{Number of replaceable OH}^- \text{ ions}}$$

The number of replaceable OH[−] ions present in a molecule is called **acidity**.

$$\therefore \text{Equivalent weight of base} = \frac{\text{Molecular mass of base}}{\text{Acidity}}$$

Let us find the equivalent weight of Ca(OH)₂

(Molecular mass = 74 amu)

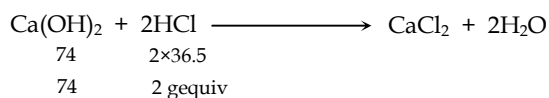
Method 1:

2 replaceable OH[−] ions are present in 74 parts by wt. of Ca(OH)₂

1 replaceable OH[−] ion is present in 37 parts by wt. of Ca(OH)₂.

Method 2:

Let us take an acid-base reaction



We know 1 gram equivalent (gequiv) of HCl = 36.5 g

2 gequiv of HCl is neutralized by 74 g of Ca(OH)₂

1 gequiv of HCl is neutralized by $\frac{74}{2}$ g of Ca(OH)₂ = 37 g of Ca(OH)₂

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∴ 1 gram equivalent of HCl is neutralized by 37 g of Ca(OH)₂

1 gram equivalent of Ca(OH)₂ = 37 g

Hence, Equivalent weight of Ca(OH)₂ is 37.

Table 2.2: Some bases with their Molecular masses (Mol.mass) and Equivalent weights (Eq.Wt.).

S.N.	Name of base	Molecular formula	Mol. mass (amu)	Acidity	Eq.wt. = $\frac{\text{Mol. mass}}{\text{Acidity}}$
1.	Sodium hydroxide	NaOH	40	1	40
2.	Ammonium hydroxide	NH ₄ OH	35	1	35
3.	Potassium hydroxide	KOH	56	1	56
4.	Calcium hydroxide	Ca(OH) ₂	74	2	37
5.	Magnesium hydroxide	Mg(OH) ₂	58	2	29
6.	Barium hydroxide (anhydrous)	Ba(OH) ₂	171	2	85.50
7.	Barium hydroxide (hydrated)	Ba(OH) ₂ .8H ₂ O	315.50	2	157.75

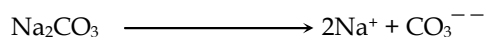
3. Equivalent weight of salts

Equivalent weight of a salt¹ is the number of parts by weight of it which is formed by complete neutralization of 1 gram equivalent of acid with 1 gram equivalent of base.

$$\text{Equivalent weight of salt} = \frac{\text{Molecular mass or formula unit mass of salt}}{\text{Total number of cationic or anionic charge}}$$

Let us find the equivalent weight of Na₂CO₃ [Mol. mass = 106 amu]

Method 1:

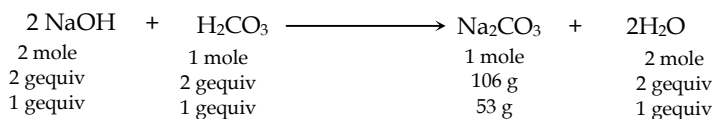


Total number of cationic charge = Total number of anionic charge = 2

$$\therefore \text{Equivalent weight of Na}_2\text{CO}_3 = \frac{106}{2} = 53$$

Method 2:

Let us take a reaction of formation of Na₂CO₃



When one gram equivalent of NaOH reacts with one gram equivalent of H₂CO₃, 1 gram equivalent of Na₂CO₃ is formed.

1 gequiv of Na₂CO₃ = 53 g

$$\therefore \text{Equivalent weight of Na}_2\text{CO}_3 = 53$$

¹ Ionic compounds are composed of cations and anions. Some prefer to use the term formula unit rather than molecule. The latter is used to denote covalent compound. In this sense, formula unit weight is used in preference to molecular mass especially for ionic compound.

Table 2.3: Some salts with their molecular masses (Mol. mass) and Equivalent weights (Eq. Wt.)

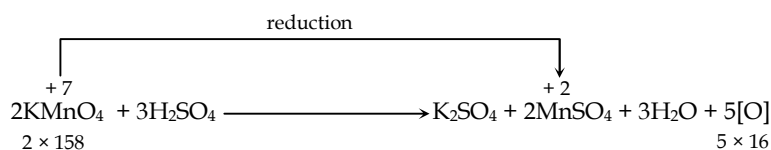
S.N.	Name of salt	Molecular formula	Mol. mass or formula unit mass (amu)	Total number of cationic or anionic charge	Eq.wt. = $\frac{\text{Formula unit mass}}{\text{Total no. of cationic or anionic charge}}$
1.	Sodium chloride	NaCl	58.5	1	58.5
2.	Potassium chloride	KCl	74.5	1	74.5
3.	Sodium carbonate	Na ₂ CO ₃	106	2	53
4.	Sodium bicarbonate	NaHCO ₃	84	1	84
5.	Silver nitrate	AgNO ₃	169.8	1	169.8
6.	Potassium nitrate	KNO ₃	101.0	1	101.0
7.	Sodium Sulphate	Na ₂ SO ₄	142	2	71

4. Equivalent weight of oxidizing agent and reducing agent

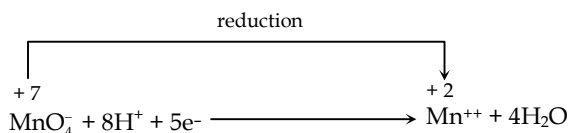
- a. **Equivalent weight of KMnO₄:** KMnO₄ is oxidizing agent having variable equivalent weight. Its equivalent weight is different in different media.

$$[\text{Mol. mass of KMnO}_4 = 39 + 55 + 4 \times 16 = 158 \text{ amu}]$$

i. In Acidic medium



Or



Method 1:

5 × 16 parts by weight of oxygen is released from 2 × 158 parts by wt. of KMnO₄.

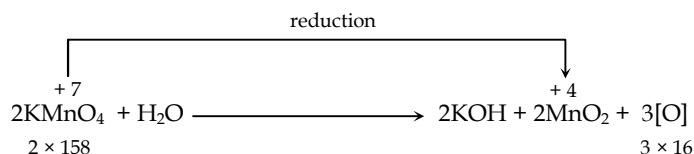
8 parts by weight of oxygen is released from $\frac{2 \times 158}{5 \times 16} \times 8 = 31.6$ parts by weight of KMnO₄.

So, equivalent weight of KMnO₄ in acidic medium = 31.6.

Method 2:

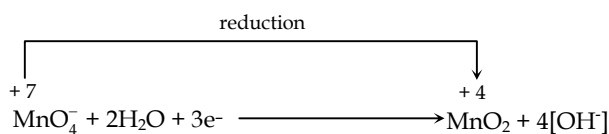
$$\begin{aligned}
 \text{Equivalent weight of KMnO}_4 &= \frac{\text{Mol. mass of KMnO}_4}{\text{Total change of oxidation number (O.N.) per molecule}} \\
 &= \frac{158}{5} = 31.6
 \end{aligned}$$

ii. In Neutral medium



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Or



Method 1:

3 × 16 parts by wt. of oxygen is released from 2 × 158 parts by wt. of KMnO₄

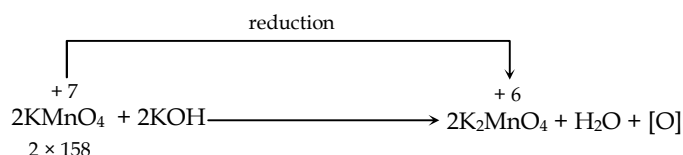
8 parts by wt. of oxygen is released from $\frac{2 \times 158}{3 \times 16} \times 8$ parts by wt. of KMnO₄.

$$= \frac{158}{3} = 52.6$$

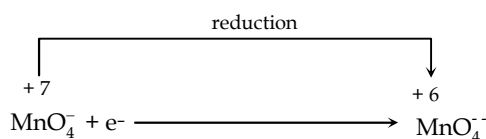
Method 2:

$$\text{Equivalent weight of KMnO}_4 = \frac{\text{Molecular mass of KMnO}_4}{\text{Change in oxidation number per molecule}} = \frac{158}{3} = 52.6$$

iii. In Alkaline medium



Or



Method 1:

16 parts by weight of oxygen is released from 2 × 158 parts by weight of KMnO₄

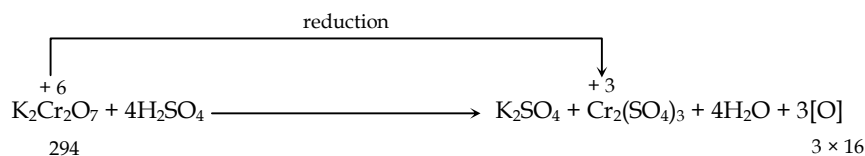
8 parts by weight of oxygen is released from $\frac{2 \times 158}{16} \times 8$ parts by weight of KMnO₄ = 158

Method 2:

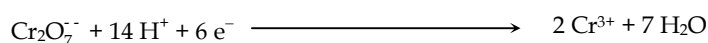
$$\text{Equivalent weight of KMnO}_4 = \frac{\text{Mol. mass of KMnO}_4}{\text{Change in O.N. per molecule}} = \frac{158}{1} = 158$$

b. Equivalent weight of potassium dichromate

[Mol. mass of K₂Cr₂O₇ = 2 × 39 + 2 × 52 + 7 × 16 = 294 amu]



Or



Method 1:

3 × 16 parts by weight of oxygen is produced from 294 parts by weight of K₂Cr₂O₇

8 parts by weight of oxygen is produced from $\frac{294}{3 \times 16} \times 8$ parts by weight of $K_2Cr_2O_7$

$$= \frac{294}{6} = 49$$

Method 2:

$$\text{Equivalent weight of } K_2Cr_2O_7 = \frac{\text{Molecular mass of } K_2Cr_2O_7}{\text{Change in O.N. per molecule}} = \frac{294}{2 \times 3} = 49$$

Here, one molecule of $K_2Cr_2O_7$ consists 2 atoms of chromium.

Change in oxidation number in one molecule of $K_2Cr_2O_7$

$$= 2 \times \text{change in O.N. of Cr.} = 2 \times 3 = 6$$

Self Test

2.2 Calculate the equivalent weight of $K_2Cr_2O_7$ in the reaction,

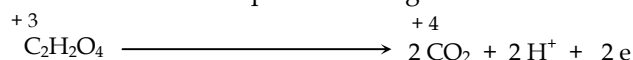


Ans: [M/2]

Table 2.4: Some compounds with their molecular masses and equivalent weights

S.N.	Name of compounds	Molecular formula	Mol. mass	Change in O.N. per molecule	Eq. wt.
1.	Potassium permanganate	KMnO ₄	158	5 (in acidic medium) 3 (in neutral medium) 1 (in alkaline medium)	31.6 52.6 158
2.	Potassium dichromate	K ₂ Cr ₂ O ₇	294.2	6	49.03
3.	Potassium iodate	KIO ₃	214	1	214
4.	Mohr's salt	FeSO ₄ (NH ₄) ₂ SO ₄ ·6H ₂ O	392	1	392

- c. **Equivalent weight of oxalic acid:** Oxalic acid is reducing agent and molecular mass of anhydrous oxalic acid is 90 amu. Its equivalent weight can be calculated as



$$\therefore \text{Equivalent weight} = \frac{\text{Molecular mass}}{1 \times 2} = \frac{90}{2} = 45.$$

Self Test

2.3 Find out the equivalent weight of (a) H_2O_2 (b) hydrated oxalic acid, $(COOH)_2 \cdot 2H_2O$.

Ans: (a) [M/2] (b) [M/2]

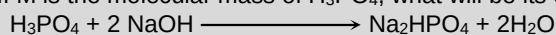
5. Equivalent weight of ions

$$\text{Equivalent weight of ion} = \frac{\text{Ionic mass}}{\text{Charge on ion}}$$

$$\text{e.g. Eq. wt. of } CO_3^{2-} \text{ ion} = \frac{60}{2} = 30$$

Self Test

2.4 If M is the molecular mass of H_3PO_4 , what will be its equivalent weight in



[Ans: M/2]

Table 2.5: Equivalent weights of some ions

S.N.	Ions	Ionic Mass	Charge of ions	Eq. wt. = $\frac{\text{Ionic mass}}{\text{Charge on ion}}$
1.	Na^+	23	1	23
2.	Ca^{++}	40	2	20
3.	SO_4^{--}	96	2	48
4.	HCO_3^-	61	1	61

If a compound is formed by radicals then the equivalent weight of the compound is equal to the sum of equivalent weight of radicals. For example, Na_2CO_3 is composed of Na^+ and CO_3^{--} .
 Eqvt. wt. of Na_2CO_3 = Eqvt. wt of Na^+ + Eqvt. wt. of CO_3^{--} = 23 + 30 = 53.

2.3 Different Ways of Expressing the Concentration of Solution

Amount of solute present in the given amount of solution is concentration.

$$\text{Concentration} = \frac{\text{Amount of solute}}{\text{Amount of solution}}$$

It can be expressed in different ways:

Depending upon the relative amount of solute and solution, solution can be classified into diluted solution and concentrated solution.

- Diluted solution:** The solution which consists relatively low amount of solute dissolved in a given amount of a solution is called diluted solution.
- Concentrated solution:** The solution which consists relatively more amount of solute dissolved in a given amount of solution is called concentrated solution.

The terms dilute and concentrate do not express the actual amount of solute dissolved in solution. These express only relative amount. Hence actual concentrations of solutions are expressed different ways as follows:

1. Percentage Concentration

Percentage concentration of a solution can be expressed in following three terms.

- Percentage concentration in mass/volume:** It is defined as the mass of a solute in gram present in 100 mL of the solution.

$$\% \text{ (w/v)} = \frac{\text{Mass of solute in gram}}{\text{Volume of solution in mL}} \times 100.$$

For example: when 4 g of sodium chloride is dissolved in water to prepare 100 mL solution then the strength of solution is said to be 4% (w/v) NaCl.

Self Test

2.5 100 mL of oxalic acid solution consists 2.45 gram of oxalic acid. Calculate the percentage strength.

- Percentage by mass or % (w/w):** It is defined as the mass of a solute in gram present per 100 g of solution. Mathematically,

$\% (w/w) = \frac{\text{Mass of a solute in gram}}{\text{Mass of solution in gram}} \times 100 = \frac{w \times 100}{w_{\text{soln}}} = \frac{w \times 100}{V \times \rho_{\text{soln}}} = \frac{\% (w/v)}{\rho_{\text{soln}}}$, where ρ_{soln} is the density of the solution in g/mL (or specific gravity).

When 4 g of NaCl is dissolved to prepare 100 g of its solution, then the strength of NaCl is said to be 4% (w/w). In other words 4g of NaCl is dissolved in 96 g of water to prepare 100 g solution.

- c. **Percentage concentration in volume/volume:** If solute and solvent both are liquid then percentage strength can be expressed in volume/volume. It is defined as the volume of solution present per 100 volume of the solution.

$$\% (v/v) = \frac{\text{Volume of solute in mL}}{\text{Volume of solution in mL}} \times 100$$

When 10 mL of ethyl alcohol is mixed with 90 mL of water to prepare 100 mL of solution, then the strength of solution is 10% V/V.

Self Test

2.6 What do you mean by 90% (v/v) methanol solution in water?

2. Gram per litre

It is defined as the mass of solute in gram present per litre of its solution.

$$\text{Strength in gram/lit (g/L)} = \frac{\text{Mass of solute in gram}}{\text{Volume of solution in litre}}$$

$$\text{or, Strength in g/L} = \frac{\text{Mass of solute in gram}}{\text{Volume of solution in mL}} \times 1000 = \frac{w \times 1000}{v}$$

When 4.9 g of H_2SO_4 is dissolved to prepare 1 litre solution then the strength of solution is said to be 4.9 g/L.

Again,

$$\text{Strength in g/L} = \frac{\text{Mass of solute in gram}}{\text{Volume of solution in mL}} \times 100 \times 10$$

$$\text{So, g/L} = \% (w/v) \times 10$$

Self Test

2.7 Find the strength of 4.5% (w/v) Na_2CO_3 in terms of g/L?

3. Molarity (M)

It is defined as the number of mole of solute dissolved per litre solution.

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litre}}$$

If volume of solution is expressed in mL then the above expression becomes.

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in mL}} \times 1000 = \frac{w \times 1000}{M_w \times v} = \frac{\text{g/L}}{M_w}$$

where M_w = Molecular mass of the solute.

For example, when 1 mole of anhydrous sodium carbonate (i.e. 106 gram) is dissolved to prepare one litre solution then the strength of solution is said to one molar (written as 1 M or M).

Semi-molar solution (M/2)

When half mole of solute is dissolved to prepare one litre solution then the strength of solution is said to be semi-molar. If 49 gram of sulphuric acid is dissolved to prepare one litre solution then the strength of solution is semi-molar.

Decimolar solution (M/10)

When one-tenth mole of solute is dissolved to prepare one litre solution then the strength of solution is said to be decimolar. If 9.8 gram of sulphuric acid is dissolved to prepare one litre solution then the strength of solution is decimolar.

Centimolar solution (M/100)

When 0.01 mole of solute is dissolved to prepare one litre of solution then the strength of solution is called centimolar. If 0.98 gram of sulphuric acid is dissolved to prepare one litre solution, then the strength of solution is centimolar.

Different ways of expression of molarity

$$\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litre}} \quad \dots \text{(i)}$$

$$= \frac{\text{Number of millimoles of solute}}{\text{Volume of solution in mL}} \quad \dots \text{(ii)}$$

$$= \frac{\text{Number of moles of solute}}{\text{Volume of solution in mL}} \times 1000 \quad \dots \text{(iii)}$$

We know,

$$\text{Number of mole} = \frac{\text{Mass of solute in gram}}{\text{Molar mass (in g/mol)}}$$

Now relation (i) becomes,

$$\text{Molarity} = \frac{\text{Mass of solute in gram}}{\text{Molecular mass} \times \text{Volume of solution in litre}} \quad \dots \text{(iv)}$$

$$= \frac{\text{Gram per litre}}{\text{Molecular mass}} \quad \dots \text{(v)}$$

$$= \frac{\text{Percentage strength (w/v)} \times 10}{\text{Molecular mass}} \quad [\because \text{Gram per litre} = \% (w/v) \times 10] \dots \text{(vi)}$$

If volume is expressed in mL then equation (iv) can be expressed as,

$$\text{Molarity} = \frac{\text{Mass of solute in gram} \times 1000}{\text{Molecular mass} \times \text{Volume of solution in mL}} = \frac{w \times 1000}{M_w \times v} \quad \dots \text{(vii)}$$

$$\text{or, } M = \frac{w \times 1000}{M_w \times v}$$

Example 2.1. Calculate the molarity of 250 mL solution prepared by dissolving 1.375 g of anhydrous sodium carbonate.

Solution

Molecular mass of anhydrous $\text{Na}_2\text{CO}_3 = 106 \text{ amu}$

Method 1:

106 g of Na_2CO_3 in 1000 mL solution = 1M

1.375 g of Na_2CO_3 in 1000 mL solution = $\frac{1}{106} \times 1.375 \text{ M}$

$$1.375 \text{ g of Na}_2\text{CO}_3 \text{ in } 250 \text{ mL solution} = \frac{1}{106} \times 1.375 \times \frac{1000}{250} = 0.0518 \text{ M}$$

Method 2:

$$\text{Number of moles of Na}_2\text{CO}_3 = \frac{1.375}{106} = 0.01297 \text{ mol}$$

$$\text{Volume of solution in litre} = \frac{250}{1000} = 0.25 \text{ L}$$

$$\text{Molarity} = \frac{\text{Number of mole}}{\text{Volume in litre}} = \frac{0.01297}{0.25} = 0.0518 \text{ M}$$

Method 3:

$$\text{Molarity} = \frac{\text{Mass of solute} \times 1000}{\text{Volume in mL} \times \text{Molecular mass}} = \frac{1.375 \times 1000}{250 \times 106} = 0.0518 \text{ M}$$

Method 4:

250 mL solution consists 1.375 g

$$1000 \text{ mL solution consist } \frac{1.375}{250} \times 1000 \text{ g} = 5.5 \text{ g}$$

Strength = 5.5 g/L

$$\text{Molarity} = \frac{\text{g/L}}{\text{Molecular mass}} = \frac{5.5}{106} = 0.0518 \text{ M}$$

Example 2.2 Calculate the amount of NaHCO₃ in 500 mL of 0.25 M solution.

Solution

Molecular mass of NaHCO₃ = 84

Method 1:

$$\text{Mass of solute} = \frac{\text{Volume in mL} \times \text{Mol. wt} \times \text{Molarity}}{1000} = \frac{500 \times 84 \times 0.25}{1000} = 10.5 \text{ g}$$

Method 2:

$$\text{Strength in g/L} = \text{Molarity} \times \text{Molecular mass} = 0.25 \times 84 = 21 \text{ g/L}$$

i.e. 1000 mL solution consists 21g

500 mL solution consists 10.5g

(Practice other methods also)

Example 2.3 Calculate the molarity of pure water.

Solution

If density of water is 1 g/mL,

1 litre of water = 1000 mL = 1000 g

For 1 litre of water,

$$\text{Number of moles} = \frac{\text{Mass of 1 litre water}}{\text{Molecular mass of water}} = \frac{1000}{18} = 55.55 \text{ mol}$$

∴ Strength of pure water is 55.55 mol/L (55.55 M)

Example 2.4 25 millimole of glucose is present in 250 mL solution. Calculate the molarity of solution.

Solution

$$\text{Molarity} = \frac{\text{Number of millimole of solute}}{\text{Volume of solution in mL}} = \frac{25}{250} = \frac{1}{10} = 0.1 \text{ M}$$

4. Formality (F)

Ionic compounds exist in the form the ions rather than in the form of molecules. So the term formula mass is used instead of molecular mass. That is why the term formality is used for ionic compound in place of molarity. As the molecular mass is the sum of all atoms present in the molecule, formula

mass is also the sum of all atoms present in the formula. So, mathematically, molarity and formality will have same values, for a given solution.

Formality of a solution is defined as the number of gram formula mass of solute dissolved per litre of solution. It is denoted by F.

$$\text{Formality (F)} = \frac{\text{Number of formula mass in gram}}{\text{Volume of solution in litre}}$$

5. Molality (m)

Molality is defined as the number of moles of solute dissolved in 1 kg of solvent.

$$\text{Molality (m)} = \frac{\text{Number of moles of solute}}{\text{mass of solvent in kg}}$$

Different ways of expression of molality

$$\text{Molality (m)} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent in kg}} \quad \dots \text{(i)}$$

$$= \frac{\text{Number of millimoles of solute}}{\text{Mass of solvent in gram}} \quad \dots \text{(ii)}$$

If mass of solvent is expressed in gram,

$$\text{Molality (m)} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent in gram}} \times 1000 \quad \dots \text{(iii)}$$

We know,

$$\text{Number of mole} = \frac{\text{Mass of solute}}{\text{Molar mass}}$$

Now, relation (i) becomes

$$\text{Molality} = \frac{\text{Mass of solute in gram}}{\text{Molecular Mass} \times \text{Mass of solvent in kg}} \quad \dots \text{(iv)}$$

$$= \frac{\text{Strength in gram kg}^{-1} \text{ solvent}}{\text{Molecular mass}} \quad \dots \text{(v)}$$

If weight is expressed in gram, relation (iii) changes to

$$\text{Molality} = \frac{\text{Mass of solute in gram} \times 1000}{\text{Molecular mass} \times \text{Mass of solvent in gram}} \quad \dots \text{(vi)}$$

Relation between molarity (M) and molality (m)

$$M = \frac{w \times 1000}{M_w \times v} \text{ and } m = \frac{w \times 1000}{M_w \times w_{\text{solv}}} \text{ . So, } \frac{m}{M} = \frac{v}{w_{\text{solv}}} \quad \dots \text{(vii)}$$

Self Test

2.8 Calculate the molarity of 3M NaOH solution having a density 1.1 g/mL.

Ans: 2.95M

Example 2.5 29.25 g of sodium chloride is dissolved in 500 g water. What is the molality of solution?

Solution

Molecular mass or formula mass of NaCl = 58.5 amu

Method 1:

58.5 g of NaCl in 1000 g water makes 1 molal solution

29.25 g of NaCl in 1000 g water makes $\frac{1}{58.5} \times 29.25$ molal solution

29.25 g of NaCl in 500 g water makes $\frac{1}{58.5} \times 29.25 \times \frac{1000}{500} = 1$ molal.

Method 2:

$$\text{Number of moles of solute} = \frac{29.25}{58.5} = 0.5 \text{ mol}$$

$$\text{Mass of solvent in kg} = \frac{500}{1000} = 0.5 \text{ kg}$$

$$\therefore \text{ Molality} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent in kg}} = \frac{0.5}{0.5} = 1 \text{ m}$$

Method 3:

$$\text{Molality} = \frac{\text{Mass of solute in gram} \times 1000}{\text{Molecular mass of solute} \times \text{mass of solvent in gram}} = \frac{29.25 \times 1000}{58.5 \times 500} = 1 \text{ m}$$

Method 4:

500 g solvent dissolves 29.25 g solute.

$$1000 \text{ g solvent dissolves } \frac{29.25}{500} \times 1000 \text{ g} = 58.5 \text{ g}$$

$$\text{Strength in molality (m)} = \frac{\text{Strength in gram kg}^{-1} \text{ solvent}}{\text{Molecular mass}} = \frac{58.5}{58.5} = 1 \text{ m}$$

Example 2.6 A sulphuric acid solution is 9.8% by mass. Calculate its molality.

Solution

9.8% by mass of sulphuric acid solution means, 9.8 g of sulphuric acid is dissolved to prepare 100 g of its solution.

$$\therefore \text{ Mass of water} = 100 - 9.8 = 90.2 \text{ g}$$

90.2 g of water dissolves 9.8 g of H_2SO_4

$$1000 \text{ g of water dissolves } \frac{9.8}{90.2} \times 1000 \text{ g of } \text{H}_2\text{SO}_4 = 108.64 \text{ g}$$

$$\therefore \text{ Molality} = \frac{\text{Strength in gram kg}^{-1} \text{ solvent}}{\text{Molecular mass}} = \frac{108.64}{98} = 1.10 \text{ m}$$

Example 2.7 The density of 15% (w/w) H_2SO_4 solution is 1.1 g/mL.

- Calculate % (w/v)
- Calculate molarity
- Calculate molality

Solution**(a) Method 1:**

15% (w/w) H_2SO_4 means

100 g H_2SO_4 solution has 15 g H_2SO_4

Mass of solution = 100 g

Mass of H_2SO_4 = 15 g

Mass of H_2O = 100 - 15 = 85 g

$$\text{Volume of solution} = \frac{\text{Mass of solution}}{\text{Density of solution}} = \frac{100}{1.1} = 90.9 \text{ mL}$$

$$\text{So, \% (w/v)} = \frac{w \times 100}{v} = \frac{15}{90.9} \times 100 = 16.5\% \text{ (w/v)}$$

Method 2:

$$\% \text{ (w/v)} = \% \text{ (w/w)} \times \text{Density of solution} = 15 \times 1.1 = 16.5\% \text{ (w/v)}$$

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(b) 90.5 mL solution has 15 g H_2SO_4

Method 1:

$$\text{g/L} = \% (\text{w/v}) \times 10$$

$$M = \frac{\text{g/L}}{\text{Molecular mass}}$$

$$\text{So, Molarity} = \frac{16.5 \times 10}{98} = 1.69 \text{ M}$$

$$\text{Molality} = \frac{w \times 1000}{\text{Molecular mass} \times \text{Weight of solution}} = \frac{15 \times 1000}{98 \times 85} = 1.80 \text{ m}$$

Method 2:

$$\frac{m}{M} = \frac{\text{Volume of solution}}{\text{Weight of solvent}}$$

$$\text{or, } \frac{m}{1.69} = \frac{1000 + (1.69 \times 98)}{1000 \times 1.1}$$

$$\text{So, } m = 1.80 \text{ m}$$

6. Normality (N)

Normality of a solution is defined as the number of gram equivalent (g equiv) of solute dissolved per litre of solution.

Mathematically,

$$\text{Normality (N)} = \frac{\text{Number of gram equivalent of solute}}{\text{Volume of solution in litre}}$$

If volume of solution is expressed in mL then above expression becomes,

$$\text{Normality} = \frac{\text{Number of gram equivalent of solute}}{\text{Volume of solution in mL}} \times 1000$$

When 49 g of sulphuric acid is dissolved to prepare 1 litre solution, then the strength of solution is one normal (1 N or N).

Bi-normal solution (2N solution)

The solution in which two gram equivalent solute is dissolved to prepare one litre solution is said to be bi-normal solution. When 80 g of NaOH is dissolved to prepare 1 litre solution then the strength of solution is bi-normal.

Semi-normal solution (N/2 solution)

The solution in which half gram equivalent solute is dissolved to prepare one litre solution is said to be semi-normal solution. When 20 g of NaOH is dissolved to prepare 1 litre solution then the strength of solution is semi-normal.

Deci-normal solution: (N/10 solution)

The solution in which one tenth gram equivalent of solute is dissolved to prepare one litre solution is said to be deci-normal solution

When 4 g of NaOH is dissolved to prepare 1 litre of solution then the strength of solution is deci-normal.

Centi-normal solution (N/100 solution)

The solution in which one-hundredth gram equivalent solute is dissolved to prepare one litre solution is said to be centi-normal solution.

When 0.4 g of NaOH is dissolved to prepare 1 litre of solution then the strength of solution is centi-normal.

Self Test

2.9 Define

- | | |
|-----------------------------------|----------------------------------|
| a. Normal solution (N) | b. Binormal solution (2N) |
| c. Penta-normal solution (5N) | d. Deca-normal solution (10N) |
| e. Semi-normal solution (N/2) | f. Penti-normal solution (N/5) |
| g. Deci-normal solution (N/10) | h. Centi-normal solution (N/100) |
| i. Milli-normal solution (N/1000) | |

Different ways of expressing of normality

$$\text{Normality} = \frac{\text{Number of gram equivalent of solute}}{\text{Volume of solution in litre}} \quad \dots \text{(i)}$$

$$= \frac{\text{Number of milligram equivalent of solute}}{\text{Volume of solution in mL}} \quad \dots \text{(ii)}$$

$$= \frac{\text{Number of gram equivalent of solute}}{\text{Volume of solution in mL}} \times 1000 \quad \dots \text{(iii)}$$

We know,

$$\text{Number of gram equivalent} = \frac{\text{Mass of solute in gram}}{\text{Equivalent weight of solute}}$$

Now, relation (i) becomes,

$$\text{Normality (N)} = \frac{\text{Mass of solute in gram (w)}}{\text{Equivalent weight of solute (E)} \times \text{Volume of solution in litre (V)}} \quad \dots \text{(iv)}$$

$$N = \frac{\text{Strength in g/L}}{\text{Equivalent weight}} \quad \dots \text{(v)}$$

$$= \frac{\% (w/v) \times 10}{\text{Equivalent weight}} \quad [\because \text{g/L} = \% (w/v) \times 10] \quad \dots \text{(vi)}$$

If volume is expressed in mL then equation (iv) changes to,

$$\text{Normality (N)} = \frac{\text{Weight of solute in gram (w)} \times 1000}{\text{Equivalent weight of solute (E)} \times \text{Volume in mL (v)}} \quad \dots \text{(vii)}$$

$$\text{or } w = \frac{VEN}{1000}$$

Example 2.8 What amount of anhydrous sodium carbonate should be dissolved to prepare 250 mL of decinormal solution?

SolutionMolecular mass of anhydrous $\text{Na}_2\text{CO}_3 = 106 \text{ amu}$

$$\text{Equivalent weight} = \frac{106}{2} = 53$$

Method 1:To prepare 1000 mL of 1N Na_2CO_3 solution 53 g Na_2CO_3 should be dissolved.To prepare 250 mL of 1N Na_2CO_3 solution $\frac{53}{1000} \times 250 \text{ g}$ should be dissolved.

To prepare 250 mL of 0.1N Na_2CO_3 solution $\frac{53 \times 250 \times 0.1}{1000} \text{ g}$ should be dissolved.
 $= 1.325 \text{ g}$

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Method 2:

$$\begin{aligned}\text{Normality} &= \frac{\text{Number of gram equivalent of solute}}{\text{Volume in litre}} \\ 0.1 &= \frac{\text{Number of gram equivalent of solute}}{0.250}\end{aligned}$$

$$\therefore \text{Number of gram equivalent} = 0.1 \times 0.250 = 0.025 \text{ g equiv.}$$

$$\text{Mass in gram} = \text{Number of gram equivalent} \times \text{Equivalent weight} = 0.025 \times 53 = 1.325 \text{ g}$$

Method 3:

$$\begin{aligned}\text{Normality} &= \frac{\text{Gram/lit}}{\text{Eq. wt.}} \\ \text{Gram/litre} &= \text{Normality} \times \text{Equivalent weight} = 0.1 \times 53 = 5.3 \text{ g/L} \\ 1000 \text{ mL solution} &\text{ consists } 5.3 \text{ g of Na}_2\text{CO}_3 \\ 250 \text{ mL solution} &\text{ consists } \frac{5.3}{1000} \times 250 \text{ g of Na}_2\text{CO}_3 = 1.325 \text{ g}\end{aligned}$$

Method 4:

$$w = \frac{VEN}{1000} = \frac{250 \times 53 \times 0.1}{1000} = 1.325 \text{ g}$$

Example 2.9 Calculate the number of milli-equivalent of oxalic acid in 50 mL of 0.2 N oxalic acid solution.

Solution

$$\text{Number of milli-equivalent} = \text{Normality} \times \text{volume in mL} = 0.2 \times 50 = 10.0 \text{ milli-equivalent (10 m equiv)}$$

Example 2.10 Commercially available concentrated hydrochloric acid contains 36.5% HCl by mass and has density 1.25 g cm⁻³. Calculate the normality of solution.

Solution

36.5% HCl means, 100 g of solution consist 36.5 g of dissolved HCl.

Method 1:

$$\begin{aligned}100 \text{ g of solution} &\text{ consists } 36.5 \text{ g of HCl} \\ \frac{100}{1.25} \text{ mL of solution} &\text{ consists } 36.5 \text{ g of HCl} \quad \left[\because V = \frac{\text{Mass}}{\text{Density}} \right] \\ 1000 \text{ mL of solution} &\text{ consists } \frac{36.5 \times 1.25 \times 1000}{100} \text{ g of HCl} = 456.25 \text{ g} \\ \text{Strength is } &456.25 \text{ g/L} \\ \text{Normality} &= \frac{\text{Gram/litre}}{\text{Eq. wt.}} = \frac{456.25}{36.5} = 12.5 \text{ N}\end{aligned}$$

Method 2:

$$\text{Normality} = \frac{\% (w/w) \text{ strength} \times \text{density (sp. gravity)} \times 10}{\text{Eqvt. wt.}} = \frac{36.5 \times 1.25 \times 10}{36.5} = 12.5 \text{ N}$$

Relation between molarity and normality

We know the relation,

$$\text{Normality} = \frac{\text{Gram/litre}}{\text{Eqvt. wt.}}$$

$$\text{or, Gram/litre} = \text{Normality} \times \text{Eq. wt.} \quad \dots \text{ (i)}$$

$$\text{Molarity} = \frac{\text{Gram/litre}}{\text{Mol. mass}}$$

$$\text{or, Gram/litre} = \text{Molarity} \times \text{Mol. mass} \quad \dots \text{ (ii)}$$

From relation (i) and (ii),

$$\text{Normality} \times \text{Eqvt. wt.} = \text{Molarity} \times \text{Mol. mass}$$

$$\text{Normality} = \text{Molarity} \times \frac{\text{Mol. mass}}{\text{Eqvt. wt.}}$$

$$\text{or, } N = M \times \left(\frac{M_w}{E_w} \right) \quad \dots \text{(iii)}$$

Note:

$\left(\frac{\text{Mol. mass}}{\text{Eqvt. wt.}} \right)$ represents acidity for base or basicity for acid or total change in oxidation number for oxidizing or reducing agents, or total +ve charge on cation or total -ve charge on anion for a salt.

7. Mole fraction

Mole fraction is the ratio of number of moles of solute or solvent to the total number of moles of solute and solvent.

If n_1 be the number of moles of solute and n_2 be the number of moles of solvent, then

$$\text{Mole fraction of solute } X_{\text{solute}} = \frac{n_1}{n_1 + n_2}$$

$$\text{Mole fraction of solvent } X_{\text{solvent}} = \frac{n_2}{n_1 + n_2}$$

$$\text{Also, } X_{\text{solute}} + X_{\text{solvent}} = 1$$

Example 2.11 5.0g of CaCO_3 is mixed with 8.4 g of MgCO_3 .

- Calculate the number of moles of CaCO_3 in the mixture.
- Calculate the number of moles of MgCO_3 in the mixture.
- Calculate the total number of moles of components.
- Calculate the percentage mass of CaCO_3 in mixture.
- Calculate the percentage mass of MgCO_3 in mixture.
- Calculate the mole fraction of CaCO_3 .
- Calculate the mole fraction of MgCO_3 .
- Calculate the sum of mole fraction of components.
- What amount of which component should be added to the mixture so that each component of mixture will have same value of mole fraction?

Solution

- $\text{Molecular mass of } \text{CaCO}_3 = 40 + 12 + 3 \times 16 = 100 \text{ amu}$
 $\text{Molecular mass of } \text{MgCO}_3 = 24 + 12 + 3 \times 16 = 84 \text{ amu}$
 $\text{No. of mole of } \text{CaCO}_3 (n_1) = \frac{\text{Mass of } \text{CaCO}_3}{\text{Mol. mass of } \text{CaCO}_3} = \frac{5}{100} = 0.05 \text{ mol}$
- $\text{No. of mole of } \text{MgCO}_3 (n_2) = \frac{\text{Mass of } \text{MgCO}_3}{\text{Mol. mass of } \text{MgCO}_3} = \frac{8.4}{84} = 0.10 \text{ mol}$
- $\text{Total no. of mole of components } (n_t) = n_1 + n_2 = 0.05 + 0.10 = 0.15 \text{ mol}$
- $\% \text{ mass of } \text{CaCO}_3 = \frac{\text{Wt. of } \text{CaCO}_3}{\text{Total mass}} \times 100\% = \frac{5}{5 + 8.4} \times 100\% = 37.31\%$
- $\% \text{ mass of } \text{MgCO}_3 = \frac{\text{Mass of } \text{MgCO}_3}{\text{Total mass}} \times 100\% = \frac{8.4}{5 + 8.4} \times 100\% = 62.69\%$
- $\text{Mole fraction of } \text{CaCO}_3 (x_1) = \frac{n_1}{n_1 + n_2} = \frac{0.05}{0.05 + 0.10} = 0.33$
- $\text{Mole fraction of } \text{MgCO}_3 (x_2) = \frac{n_2}{n_1 + n_2} = \frac{0.10}{0.05 + 0.10} = 0.67$
- $\text{Sum of mole fraction} = x_1 + x_2 = 0.33 + 0.67 = 1$
- To have equal mole fraction the mole fraction of each component should be 0.5.

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i.e. mole fraction of CaCO_3 should be 0.5

let x g of CaCO_3 should be added to the mixture.

$$\text{No. of mole of } \text{CaCO}_3 = \frac{5+x}{100}$$

$$\text{Mole fraction of } \text{CaCO}_3 (x_1) = \frac{\text{No. of mole of } \text{CaCO}_3}{\text{Total no. of mole}}$$

$$\text{or } \frac{1}{2} = \frac{(5+x)/100}{\left(\frac{5+x}{100} + 0.10\right)}$$

$$\text{or } \frac{1}{2} = \frac{(5+x)/100}{(5+x+10)/100}$$

$$\text{or } \frac{1}{2} = \frac{5+x}{15+x}$$

$$\text{or } 10 + 2x = 15 + x$$

$$\text{or } 2x - x = 15 - 10$$

$$\therefore x = 5 \text{ g}$$

8. Part per million (ppm)

When very small quantity of solute is present in large quantity of solution then it is less significant to express the concentration in terms of normality or molarity or gram per litre. For very dilute solution, the concentration is often expressed in terms of part per million. Concentration of ions such as Hg^{++} , Pb^{++} etc in solution, and presence of traces amount of gas in atmosphere are also expressed in terms of ppm.

Concentration of ppm is defined as the weight of solute present in one million parts by weight of solution.

$$\text{Concentration in ppm} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6 = \frac{w \times 10^6}{w_{\text{soln}}}$$

But for very dilute aqueous solution, the amount of solute is very low so $v \cong w_{\text{soln}}$

$$\text{ppm} = \frac{\text{Mass of solute (in g)}}{\text{vol. of solution (in mL)}} \times 10^6 = \frac{w \times 10^6}{v}$$

$$\text{and ppm} = \text{g/L} \times 10^3$$

Example 2.12 The iodide salt provided by salt trading corporation consists 50 ppm iodine in 1 kg packet. Calculate the amount of iodine in 1 kg packet salt.

Solution

Method 1:

50 ppm means, 50 parts by mass of iodine is present in 10^6 parts by mass of salt.

10^6 kg of salt consists 50 kg iodine

$$1 \text{ kg of salt consist } \frac{50}{10^6} \text{ kg of iodine} = 5 \times 10^{-5} \text{ kg} = 5 \times 10^{-5} \times 1000 \text{ g} = 0.05 \text{ g} = 50 \text{ mg}$$

Method 2:

$$\text{ppm} = \frac{\text{Mass of solute}}{\text{Mass of solution or mixture}} \times 10^6$$

$$\text{Mass of solute} = \frac{\text{ppm} \times \text{mass of solution or mixture}}{10^6} = \frac{50 \times 1 \text{ kg}}{10^6} = 5 \times 10^{-5} \text{ kg} = 50 \text{ mg}.$$

Note: Temperature dependence in concentration units

Normality, molarity, g/L and % (w/v) include volume of solution. If temperature is increased, volume of solution is also increased (due to expansion), but amount of solute remains constant. So, these concentration units are temperature dependent units. Their values decrease with increasing temperature. But molality, mole fraction and % (w/w) include masses or number of moles only in their units. Values of these quantities do not change with the change in temperature. So, these concentration units are temperature independent units.

2.4 Primary and Secondary Standard Substances

In volumetric analysis, concentration of one solution is determined with the help of another solution of known concentration by measuring their reacting volumes. Volumetric analysis is likely to be confused with the measurement of volume of gas or solid. Hence, a better term titrimetric analysis is used which involves the measurement of volume of solution only used in the titration.

A solution having known concentration is called **standard solution**. The process of determination of concentration of a solution with the help of standard solution is called **standardization** (or titration) and the solution is called standardized solution.

Standard solution is prepared by dissolving a known mass of pure substance in water and making a definite volume of solution. These types of solute substances are called **primary standard substance**. However, all solutions of the exact concentrations cannot be prepared directly because some substances interact with air or moisture or may be present in impure form. In such conditions, once these prepared solutions are required to be standardized and these types of solute substances are called **secondary standard substances**.

Primary standard substance is that pure chemical substance which can be used for the direct preparation of solution of known concentration.

To be a primary standard, a substance should have following requisites.

1. The substance must be easy to obtain and purify. It should be present in pure state.
2. The substance should not be hygroscopic or efflorescent or deliquescent.
3. The substance shouldn't be affected by carbon-dioxide and should not be oxidized by air. The composition should not change during storage or weighing.
4. It is better to have a high relative molecular mass so that weighing error can be minimized.
5. The substance should be readily soluble under the employed condition.
6. The composition of substance should not change in its solid state or in solution form for sufficiently long period.

Some examples of primary standard substances include

- | | |
|--|---|
| i. Anhydrous sodium carbonate (Na_2CO_3) | ii. Oxalic acid ($(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$) |
| iii. Potassium chloride (KCl) | iv. Arseneous oxide (As_2O_3) |
| v. Mohr's salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ | vi. Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) |
| vii. Potassium iodate (KIO_3) | viii. Potassium bromate (KBrO_3) |
| ix. Sodium chloride (NaCl) | |

The solution which is directly prepared by dissolving primary standard substance and composition of solution does not vary on prolong standing is called **primary standard solution**.

Secondary standard substance is defined as that chemical substance whose composition may change either in solid phase or in its solution phase on long standing and cannot be used to prepare primary standard solution directly. Secondary standard solutions are also prepared as the primary standard solutions. But after preparing the solutions, it must be titrated with a primary standard solution to get its accurate strength. The accurate strength is then utilized for the further calculations.

Hydrated salt as a rule do not make good standard solution (except few like hydrated oxalic acid) as it is difficult to dry these salt efficiently. The following salts are found to be secondary standard substances.

- | | |
|---|---|
| i. Copper sulphate ($\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$) | ii. Sodium tetra borate ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10 \text{H}_2\text{O}$) |
| iii. Sodium hydroxide (NaOH) | iv. Acids such as HCl , HNO_3 , H_2SO_4 |
| v. Potassium permanganate (KMnO_4) | vi. Ferrous sulphate (FeSO_4) |

So, the solution whose strength is to be redetermined after its preparation with the help of a primary standard solution is called **secondary standard solution**. In such type of solution exact strength can not be determined on the basis of amount of substance dissolved to prepare solution because this type of substance may not be present in pure and dry state. Their composition may change while storing.

Preparation of Primary Standard Solution (Say Na_2CO_3 Solution)

To prepare primary standard solution of a substance, first we have to determine volume and strength of solution to be prepared. The mass of the substance required is calculated. The calculated mass of primary standard substance is taken in volumetric flask and water is added upto mark. The solution is shaken well. Let us take an example for the preparation of 100 mL of decinormal oxalic acid solution. The solution is prepared in the following steps.

Calculation

$$\text{Mass required (w)} = \frac{\text{Volume (v)} \times \text{Eqvt. mass (E)} \times \text{Normality (N)}}{1000} = \frac{100 \times 63 \times 0.1}{1000} = 0.63 \text{ g}$$

- ☛ Weigh out exactly 0.63 g of pure oxalic acid crystals in watch glass.
- ☛ Pour the weighted oxalic acid into 100 mL volumetric flask using funnel.
- ☛ Wash the watch glass by sprinkling water drops through wash bottle into volumetric flask.
- ☛ Wash the funnel upto the brim and allow to fall the wash into volumetric flask.
- ☛ Dissolve the content within volumetric flask by adding little quantity of water.
- ☛ Then add water to make the volume of solution upto the mark.
- ☛ Shake the solution properly to make the solution homogeneous.

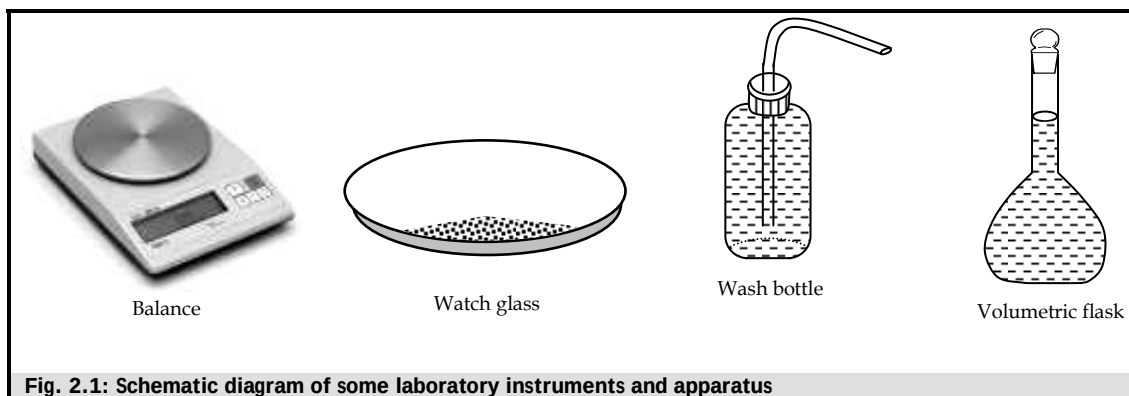


Fig. 2.1: Schematic diagram of some laboratory instruments and apparatus

Preparation of decinormal NaOH solution (Secondary Standard Solution)

Lets suppose, we have to prepare decinormal solution of NaOH (which may be approximately decinormal solution). Weight out $\frac{100 \times 40 \times 0.1}{1000}$ g = 0.4 g NaOH in a watch glass and transfer and dissolve it in 100 mL volumetric flask. Add water to make the volume upto the mark. The solution is approximately decinormal.

Preparation of secondary standard solution (say exactly Decinormal NaOH Solution)

Let us take an example for the preparation of 100 mL of decinormal NaOH solution. The solution is prepared in the following steps.

Calculation:

$$\begin{aligned}\text{Weight required of NaOH (W)} &= \frac{\text{Volume (V)} \times \text{Eqvt. mass (E)} \times \text{Normality (N)}}{1000} \\ &= \frac{100 \times 40 \times 0.1}{1000} = 0.4 \text{ g}\end{aligned}$$

Step 1: Preparation of approximate 0.1 N NaOH solution.

- Weigh out slightly more than 0.4 g (say 0.5-0.6 g) of NaOH in watch glass.
- Transfer the weighed NaOH into 100 mL volumetric flask using funnel.
- Wash the watch glass by sprinkling water drops through wash bottle into volumetric flasks.
- Dissolve the content within volumetric flask by adding little quantity of water.
- Then add water to make the volume of solution upto the mark.
- Shake the solution properly to make the solution homogeneous.

Step 2: Standardization of NaOH solution

- Take NaOH solution in burette and 10 mL of standard oxalic acid solution in conical flask. Then add about 2 drops of phenolphthalein.
- Titrate the solution to obtain end point.

Calculation**Oxalic acid**

$$V_1 = 10 \text{ mL (from pipette)}$$

$$N_1 = \frac{N}{10} \text{ (say)}$$

Sodium hydroxide

$$V_2 = 8 \text{ mL (say, from burette reading)}$$

$$N_2 = ?$$

Applying normality equation, $N_1 V_1 = N_2 V_2$

$$N_2 = \frac{V_1 N_1}{V_2} = \frac{10}{8} \times \frac{1}{10} = \frac{1}{8} \text{ N}$$

Step 3: Dilution of NaOH solution to make it exactly decinormal.**NaOH (Prepared above)**

$$V_1 = ?$$

$$N_1 = \frac{1}{8}$$

NaOH (To be prepared)

$$V_2 = 100 \text{ mL}$$

$$N_2 = 0.1 \text{ N}$$

Applying Law of Dilution ($C_1V_1 = C_2V_2$)

$$V_1 = \frac{V_2N_2}{N_1} = \frac{100 \times 0.1}{1/8} = 100 \times 0.1 \times 8 = 80 \text{ mL}$$

Take exactly 80 mL of thus prepared NaOH solution in 100 mL measuring flask and add water to make it exactly 100 mL. Shake the solution to make homogeneous. The solution can now be used as a decinormal solution.

2.5 Law of Chemical Equivalence: Normality Equation

According to the **law of chemical equivalence**, one gram equivalent of one chemical substance reacts exactly with one gram equivalent of another chemical substance or in any chemical reaction, reactants must react in equal number of gram equivalents. Accordingly, x gram equivalent of particular compound exactly reacts with x gram equivalent of another compound. For two reacting substance A and B to have complete reaction,

$$\text{Number of gram equivalent for A (}g_1\text{)} = \text{Number of gram equivalent for B (}g_2\text{)} \quad \dots \text{(i)}$$

We know,

$$\text{Normality (}N\text{)} = \frac{\text{Number of gram equivalent (}n\text{)}}{\text{Volume of solution in litre (}V\text{)}}$$

$$\text{or, Number of gram equivalent (}g\text{)} = \text{Normality (}N\text{)} \times \text{Volume in litre (}V\text{)}$$

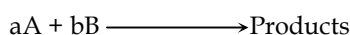
$$\text{So, } g_1 = N_1V_1 \text{ and } g_2 = N_2V_2$$

Now relation (i) becomes,

$$N_1V_1 = N_2V_2 \quad \dots \text{(ii)}$$

The relation (ii) is called **normality equation**. This relation is always true because chemical substance always react in equivalent proportions,

For molarity equation, the balance chemical equation is required. Example:



$$\frac{\text{No. of moles of A}}{\text{No. of moles of B}} = \frac{a}{b}$$

$$\text{or } \frac{M_1V_1}{M_2V_2} = \frac{a}{b}$$

So, the molarity equation is

$$a(M_2V_2) = b(M_1V_1) \quad \dots \text{(iii)}$$

Note:

$M_1V_1 = M_2V_2$ holds true only when reactants react in equal number of moles.

Normality factor

To understand the normality factor, let's take an example of preparation of decinormal 250 mL solution of anhydrous sodium carbonate 1.325 g should be weighted. If 1.325g of anhydrous Na_2CO_3 is weighted and dissolved to prepare 250 mL solution then the strength is exactly $\frac{N}{10}$. If weight slightly greater or less is taken, the strength of solution is not exactly $\frac{N}{10}$. In such condition,

normality factor is calculated. Normality factor (f) is defined as the ratio of mass taken to the mass to be taken.

$$\text{Normality factor } (f) = \frac{\text{Mass taken}}{\text{Mass to be taken}}$$

If normality factor is greater than one then the strength of solution is greater than that to be required. If normality factor is one then the strength is exactly same as that is required. If normality factor is less than one then the strength of solution is less than that to be required. To understand more clearly, let us discuss few cases here

a. If weight taken is 1.350 g.

$$\text{Normality factor } (f) = \frac{1.350}{1.325} = 1.01$$

The concentration of solution is expressed as $\frac{N}{10}$ ($f = 1.01$)

b. If weight taken is 1.325 g.

$$\text{Normality factor } (f) = \frac{1.325}{1.325} = 1$$

In such case, it is not necessary to write normality factor.

The concentration of solution is expressed as $\frac{N}{10}$ ($f = 1$) or simply $\frac{N}{10}$

c. If weight taken is 1.320 g.

$$\text{Normality factor } (f) = \frac{1.320}{1.325} = 0.99$$

The concentration of solution is expressed as $\frac{N}{10}$ ($f = 0.99$)

In calculation, the normality factor is multiplied

Examples:

$$\frac{N}{10} (f = 1.01) = \frac{1}{10} \times 1.01N = 0.101N$$

$$\frac{N}{10} (f = 1) = \frac{1}{10} \times 1 N = 0.1N$$

$$\frac{N}{10} (f = 0.99) = \frac{1}{10} \times 0.99 N = 0.099N$$

In the days where weight of chemical substance were used to be taken by using, non-digital balance, the normality factors were mostly used. Now a day's digital balances are used to weigh out the chemical substance. Hence, normality factors are less commonly used.

2.6 Theoretical Aspects of Volumetric Analysis

2.6.1 Introduction

Volumetric analysis is that analysis in which strength of a solution is determined by reacting it with standard solution by measuring their reacting volumes. Volumetric analysis is based on the fact that chemical substances react in their equivalent proportions. Volumetric analysis is carried out by titration. The principle of volumetric analysis can be outlined as follows.

- i. Chemical substances always react in their equivalent proportions. It leads to the normality equation $N_1V_1 = N_2V_2$. This law is called **law of equivalence**.
- ii. During dilution, if volume of solution is multiplied by x , strength of solution must be divided by x . For the dilution of a particular solution, the product of $C \times V$ remains constant.

This is called **law of dilution**. The general relation for dilution is $C_1V_1 = C_2V_2$. Where C_1 and V_1 are the concentration and volume of solution before dilution, C_2 and V_2 are the concentration and volume of solution after dilution. Here C is the concentration of the solution in any unit.

$$1 \text{ mL of } 1N \text{ NaOH} \equiv 10 \text{ mL of } \frac{1}{10} N \text{ NaOH} \equiv 20 \text{ mL of } \frac{N}{20} \text{ NaOH}$$

$$\equiv 5 \text{ mL of } \frac{N}{5} \text{ NaOH} \equiv 0.1 \text{ mL of } \frac{N}{0.1} \text{ NaOH and so on.}$$

- iii. If solutions of reactants have same normality, these react with each other in equal volume. If acid and base neutralize by equal volume, these should have same normality.

$$9 \text{ mL of } 0.1 N \text{ HCl} \equiv 9 \text{ mL of } 0.1N \text{ NaOH} \equiv 9 \text{ mL of } 0.1N \text{ Na}_2\text{CO}_3 \equiv 9 \text{ mL of } 0.1N \text{ KOH}$$

- iv. When chemical substances of same kind are added then number of gram equivalent increases.

If three different acid solutions having normality N_1, N_2, N_3 and volume V_1, V_2, V_3 , respectively are mixed then

$$NV = N_1V_1 + N_2V_2 + N_3V_3$$

$$\text{or, } N = \frac{N_1V_1 + N_2V_2 + N_3V_3}{(V_1 + V_2 + V_3)}$$

Where N is normality and V is volume of resulting solution. The same is true for base.

- v. If acids and alkalis are mixed, the chemical substance having minimum number of gram equivalent is completely neutralized and solution possesses the acidic or basic character due to excess reagent.

If N_1 and V_1 are normality and volume of acid

N_2 and V_2 are normality and volume of base.

N and V are normality and volume of resulting solution,

then $NV = N_1V_1 - N_2V_2$ if $N_1V_1 > N_2V_2$

$$\text{or } N = \frac{N_1V_1 - N_2V_2}{V_1 + V_2}$$

and, $NV = N_2V_2 - N_1V_1$ if $N_2V_2 > N_1V_1$

$$\text{or } N = \frac{N_2V_2 - N_1V_1}{V_1 + V_2}$$

Example 2.13 0.715 g of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ required 20 mL of seminormal hydrochloric acid solution for complete reaction. Find the value of x .

Solution

Method 1:

1 gequiv of HCl $\equiv$ 1 gequiv of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$.

1000 mL of 1N HCl $\equiv$ E g of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$, where E = Equivalent weight of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$

20 mL of 1N HCl $\equiv \frac{E}{1000} \times 20$ g $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$

20 mL of $\frac{N}{2}$ HCl $\equiv \frac{E}{1000} \times 20 \times \frac{1}{2} \text{ g} \equiv \frac{E}{100} \text{ g}$.

$$\therefore \frac{E}{100} \text{ g} = 0.715 \text{ g}$$

$$\therefore E = 71.5$$

$$\therefore \text{Equivalent wt. of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = 71.5$$

$$\therefore \text{Mol. mass of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = 71.5 \times 2 \text{ [acidity of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} \text{ is } 2] = 143 \text{ amu.}$$

$$\text{Mol. mass of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = 2 \times 23 + 12 + 3 \times 16 + x \times 18$$

$$143 = 46 + 12 + 48 + x \times 18$$

$$143 - 106 = 18x$$

$$37 = 18x$$

$$x = 2.05$$

$$\therefore \text{Value of } x = 2.$$

Method 2:

$$\text{Equivalent wt. of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = \frac{w \times 1000}{v \times N} = \frac{0.715 \times 1000}{20 \times 0.5} = 71.5$$

$$\therefore \text{Mol. mass} = 71.5 \times 2 = 143$$

$$\text{Mol. mass of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = 106 + x \times 18$$

$$\text{or, } 143 = 106 + 18x$$

$$\therefore x = \frac{143 - 106}{18} = 2.05 \approx 2$$

Method 3:

No. of gram equivalents of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ = No. of gram equivalents of HCl

$$\text{or, } \frac{0.715}{E} = \frac{1}{2} \times \frac{20}{1000}, \text{ since number of g equiv} = N \times V \text{ (in L)} = \frac{w \text{ (in g)}}{E}$$

$$\text{or, } E = 71.5$$

$$\text{Molecular mass of } \text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = 71.5 \times 2 = 143 \text{ amu}$$

$$\text{Now, } 143 = 106 + 18x$$

$$\therefore x = 2$$

Example 2.14 7.35 g of a dibasic acid was dissolved in water and diluted to 250 mL. 25 mL of this solution was neutralized by 15 mL of N NaOH solution. What is equivalent weight and molecular mass of the acid?

Solution

Method 1:

25 mL of acid solution was neutralized by 15 mL of 1N NaOH

Acid solution

$$v_1 = 25 \text{ mL}$$

$$N_1 = ?$$

NaOH solution

$$v_2 = 15 \text{ mL}$$

$$N_2 = 1\text{N}$$

$$\therefore N_1 = \frac{V_2 N_2}{V_1} = \frac{15 \times 1}{25} = 0.6 \text{ N, where } N_1 \text{ is the normality of the diluted solution.}$$

$$\therefore \text{Strength of 25 mL acid solution} = \text{strength of 250 mL of acid solution} = 0.6\text{N.}$$

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0.6 N of 250 mL solution consists 7.35 g dibasic acid

1N of 250 mL solution consists $\frac{7.35}{0.6}$ g of dibasic acid

1N of 1000 mL solution consists $\frac{7.35 \times 1000}{0.6 \times 250}$ g = 49 g dibasic acid

So, Eq. wt of dibasic acid = 49.

Since acid is dibasic

Molecular mass = Eq. wt $\times$ basicity = $49 \times 2 = 98$ amu

Alternatively,

$$\text{Eq. wt of dibasic acid} = \frac{\frac{7.35 \times 1000}{\frac{15 \times 1}{25} \times 250}}{\frac{7.35 \times 1000 \times 25}{15 \times 250}} = 49$$

$\therefore$ Mol. mass of dibasic acid = $49 \times 2 = 98$ amu

Example 2.15 x g of CaCO_3 required 40 mL of $\frac{N}{2}$ HCl for complete neutralization. Calculate x.

Solution

Note:

$$\text{No. of gequiv of solute} = N \times V (\text{in L}) = \frac{W (\text{in g})}{E}$$

$$\text{No. of mequiv of solute} = N \times v (\text{in mL}) = \frac{w (\text{in mg})}{E}$$

Method 1:

$$1000 \text{ mL of } 1 \text{ N HCl} = 50 \text{ g CaCO}_3 \text{ (as } E_{\text{CaCO}_3} = \frac{100}{2} = 50)$$

$$40 \text{ mL of } \frac{N}{2} \text{ HCl} = \frac{50 \times 40 \times 1}{1000 \times 2} \text{ g CaCO}_3 = 1 \text{ g}$$

Method 2:

No. of g equiv of CaCO_3 = No. of g equiv of HCl

$$\text{So, } \frac{x}{50} = \frac{1}{2} \times \frac{40}{1000}$$

or, $x = 1 \text{ g}$

Method 3

No. of mequiv CaCO_3 = No. of mequiv HCl

Suppose, mass of CaCO_3 = w mg

$$\frac{w}{50} = \frac{1}{2} \times 40$$

$$w = 1000 \text{ mg} = 1 \text{ g}$$

Example 2.16 1.0 g of CaCO_3 was dissolved in 50 mL at $\frac{N}{2}$ HCl solution. The solution was diluted 100 mL. Calculate the normality and pH of the diluted acid.

Solution

$$\text{No. of m equiv of CaCO}_3 = \frac{1000}{50} = 20 \text{ mequiv}$$

$$\text{No. of m equiv of HCl taken} = \frac{1}{2} \times 50 = 25 \text{ mequiv.}$$

No. of mequiv of HCl that reacted with CaCO_3 = 20 mequiv

No. of mequiv HCl left unreacted = $25 - 20 = 5$ mequiv in 50 mL of solution

$$\text{Normality of residual and before dilution (N}_1\text{)} = \frac{5}{50} = 0.1 \text{ N}$$

Let,

So, normality of diluted acid (to 100 mL) = N_2

Using law of dilution, $N_1V_1 = N_2V_2$

$$0.1 \times 50 = N_2 \times 100$$

$$N_2 = \frac{0.1 \times 50}{100} = \frac{5}{100} = 0.05 \text{ N}$$

$$\text{pH of diluted acid} = -\log(N_{\text{acid}}) = -\log(0.05) = 1.7$$

Example 2.17 1.5 g of calcium carbonate sample required 40 mL of $\frac{N}{2}$ HCl for complete neutralization. Calculate percentage purity of the calcium carbonate in the sample.

Solution

Lets suppose: Mass of CaCO_3 in the sample = x g

No. of g equiv. of CaCO_3 = No. of g equiv of HCl

$$\text{or, } \frac{x}{50} = \frac{1}{2} \times \frac{40}{1000}$$

$$x = 1 \text{ g}$$

$$\text{So, \% purity of } \text{CaCO}_3 = \frac{1}{1.5} \times 100 = 66.67\%$$

Example 2.18 50 mL of 1M acetic acid is mixed with 50 mL of 0.2 M NaOH. If k_a for CH_3COOH is 1.8×10^{-5} , calculate

(a) pH of resulting solution

(b) Concentration of salt formed

Solution

Number of millimole of CH_3COOH = volume in mL $\times$ molarity = $50 \times 1 = 50$ mmol

Number of millimole of NaOH = $V(\text{mL}) \times \text{molarity} = 50 \times 0.2 = 10$ mmol

The reaction takes place as follows:

	CH_3COOH	NaOH	$\longrightarrow$	CH_3COONa	H_2O
Initially present	50 mmol	10 mmol		-	-
Changed quantity	10 mmol	10 mmol		10 mmol	10 mmol
Quantity after reaction	40 mmol	-		10 mmol	10 mmol

Total volume of solution = $50 + 50 = 100$ mL

(a) **Calculation of pH of resulting solution**

$$\text{Molarity of acetic acid in resulting solution (M)} = \frac{\text{number of mmol}}{\text{volume in mL}} = \frac{40}{100} = 0.4 \text{ M}$$

The solution is equivalent to 100 mL of 0.4M H_3COOH . The acetic acid ionizes as follows:

	CH_3COOH	$\rightleftharpoons$	CH_3COO^-	$+$	H^+
Initially,	$C \text{ mol lit}^{-1}$		0		0
At time t	$c - c\alpha$		$c\alpha$		$c\alpha$

From Ostwald's dilution law, $k_a = \alpha^2 c$

$$\text{or, } \alpha = \sqrt{\frac{k_a}{c}}$$

$$[\text{H}^+] = c\alpha = c \cdot \sqrt{\frac{k_a}{c}} = \sqrt{k_a \cdot c} = \sqrt{1.8 \times 10^{-5} \times 0.4} = 2.68 \times 10^{-3}$$

$$\text{pH} = -\log [\text{H}^+] = -\log (2.68 \times 10^{-3}) = 2.57$$

(b) **Concentration of salt formed**

$$\text{Molar concentration of salt} = \frac{\text{number of mmol of salt}}{\text{volume of mL}} = \frac{10}{100} = 0.1 \text{ M}$$

2.6.2 Titration

In the previous unit we have discussed that volumetric analysis involves the quantitative determination of volume of standard solution required to react with the measured volume of solution of unknown strength. The term volumetric analysis is likely to be confused with the measurement of volume of gas. Therefore, a clear term **titrimetric analysis** is introduced which expresses the process of titration rather than volume measurement.

Titration is the method of quantitative chemical analysis which involves the determination of volume of standard solution required to react with measured volume of solution of unknown strength. The solution of known strength is called **standard solution**. The solution of known strength is also called **titrant**. In titration, the standard solution (titrant) is usually added from a burette. The substance being titrated is called **titrand**. Titrand is usually taken in conical flask. However, it is not the affix rule. There are other factors used to determine which solution is to be kept in burette or in conical flask.

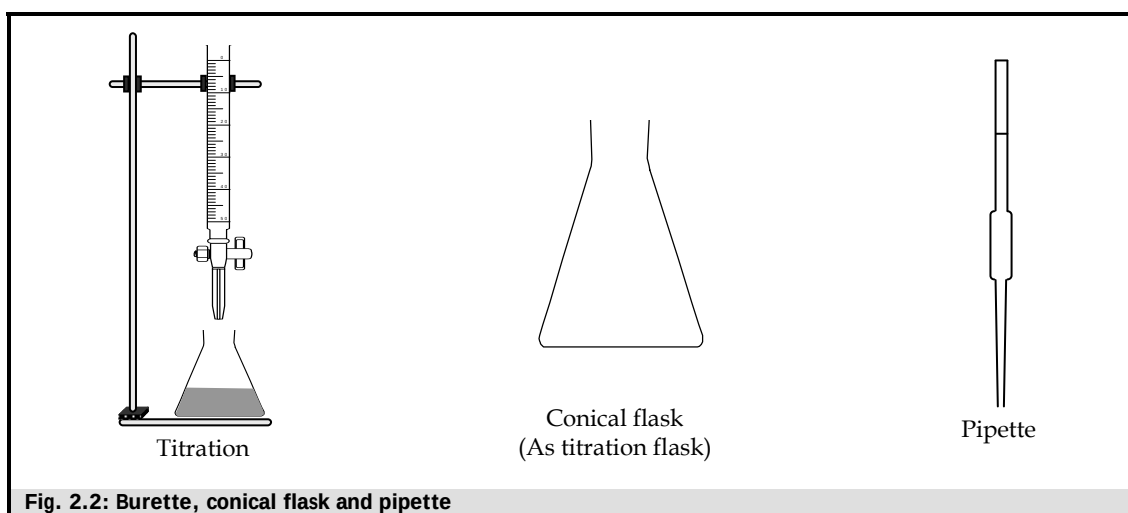


Fig. 2.2: Burette, conical flask and pipette

In titration process, a fixed volume of solution (usually, the solution whose strength is to be determined) is taken in a conical flask. Few drops of suitable indicator are added. Then another solution (usually standard solution) is added from burette into the conical flask. The conical flask is shaken with swirling motion. During titration, a point will appear at which equivalent quantity of titrand is neutralized by titrant which is called **equivalence point** or **theoretical end point** or **stoichiometric end point**. However, this point is not visible in acid base titration. To indicate the completion of reaction between titrant and titrand usually an auxiliary chemical substance is added which is called **indicator**. The completion of titration is detected by some physical change produced by one of the solution itself or by indicator. Most of the indicators show their colour change either in acidic medium or in alkaline medium. Therefore, at the equivalence point the indicators are unable to show their colour change. The point at which indicator show the completion of reaction between titrant and titrand by the sharp visual change in colour or the appearance of turbidity is called **end point** of the titration. In the ideal titration, end point coincides with equivalence point but in practice the value of end point is slightly different than the value of stoichiometric end point or equivalence point. The difference between end point and equivalence point of titration is called **titration error**. In titration, indicator should be chosen in such a manner that it would avoid or minimize the titration error.

While conducting the titrimetric analysis the following points are to be considered.

- The chemical substances react completely in equivalent proportion.
- The reaction should be relatively fast. If rate of reaction is slow, a suitable catalyst should be added or solution can be heated.
- There must be some sort of physical change at the end point. The change may be in colour or the formation of precipitate.
- If indicator fails to indicate the end point, other alternative methods are adopted to find out the equivalence point. The alternative methods may be conductometric titration, (measuring the change in conductance of ions in solution), potentiometric titration (measuring the potential between indicator electrode and reference electrode), coulometric titration (developing the titrant by electrolysis) or amperometric titration (measurement of current under constant applied voltage). These methods of titration are not discussed here. This is beyond the scope of this text.

Differences between equivalence point and end point

Equivalence point		End point	
1.	The point in titration at which equivalent quantity of titrand is exactly neutralized by titrant is called equivalence point	1.	The point in titration at which neutralization of titrand by titrant is shown by sharp change in visual colour of solution is called end point.
2.	Equivalence point is not visible in acid-base titration involving burette and pipette system.	2.	End point is visible in acid base titration using burette and pipette system.
3.	It is determined by using (conductometer or potentiometer) instrumental method.	3.	It is determined by observing the colour change of indicator in solution.
4.	It is exact point where the reactants are completely reacted.	4.	It is practical point where the completion of the reaction is indicated by the change in colour.

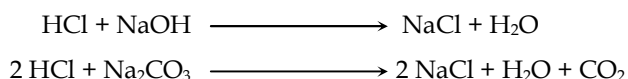
Classification of Titration Reactions in Titrimetric Analysis

The titration involved in titrimetric analysis can be classified into following types:

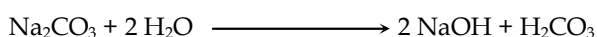
- Acid-base titration or neutralization titration.
- Redox titration or oxidation-reduction titration.
- Complexometric titration or complex formation titration.
- Precipitation titration.

a. Acid-base titration

The titration which involves the reaction of free acid and free base or base formed from salt of weak acids by hydrolysis (e.g. Na_2CO_3) is called **acid-base titration**. Some of the acid-base reactions involved in acid-base titration are as follows.



The reaction of HCl and Na_2CO_3 is also regarded as acid-base reaction although Na_2CO_3 is salt because Na_2CO_3 undergoes hydrolysis to give alkali NaOH.



In this type of titration, end point is detected by the use of acid-base indicator or pH indicator. It will be discussed in next unit.

b. Redox titration

The chemical reaction which involves change in oxidation number or transfer of electron among the reacting chemical species is called redox reaction or oxidation-reduction reaction. In this type of reaction standard solution is either oxidising or reducing agent. It means if titrant is oxidising agent, titrand should be reducing agent and vice-versa. Titration involving such type of reaction is called **redox titration**. The commonly used oxidising agents are KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, I_2 , KIO_3 , KBrO_3 , etc. The commonly used reducing agents are iron (II) compounds such as Mohr's salt ($\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$), sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$), arseneous oxide (As_2O_3), oxalic acid (COOH)₂ etc.

The reaction between acidified KMnO_4 solution and oxalic acid is discussed below.

KMnO_4 is strong oxidising agent and it oxidises oxalic acid to carbondioxide. KMnO_4 is reduced while oxalic acid is oxidised.

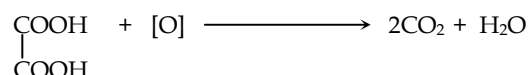
The oxidation half reaction of KMnO_4 in acidic medium is,



Ionic reaction,



The reduction half reaction of oxalic acid is,

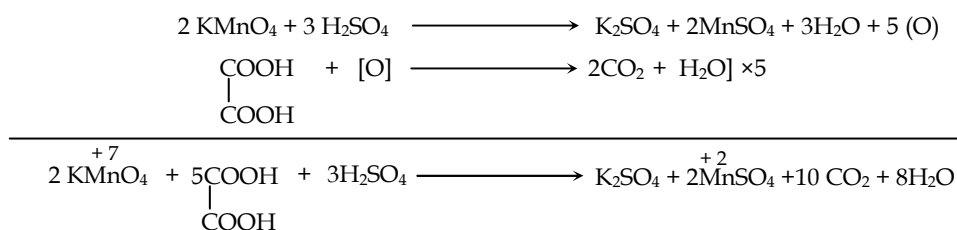


Ionic reaction is,

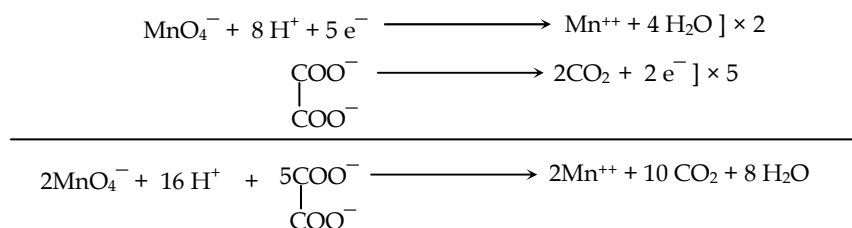


To obtain overall reaction, oxidation half and reduction half reaction are added.

The molecular reaction is



The ionic reaction is,

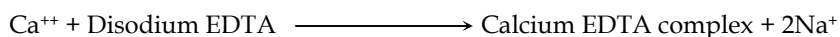


Note:

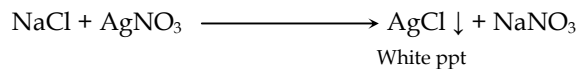
1. The titration involving potassium permanganate is called *permanganometric titration*.
2. Potassium permanganate is not primary standard substance because it is not obtained in pure state. It may consist MnO_2 as impurity. The KMnO_4 solution is not so soluble in aqueous solution. The aqueous solution of KMnO_4 is boiled for some time, cooled and filtered. Then it is standardised by either oxalic acid or by Mohr's salt. In this titration KMnO_4 is taken in burette.
3. KMnO_4 is acidified by dilute H_2SO_4 but not by dilute HCl or HNO_3 because KMnO_4 may oxidise HCl . Whatever may be the strength, HNO_3 acts as oxidising agent.
4. In the titration of KMnO_4 , no external or internal indicator is required because disappearance of pink colour indicates the completion of reaction. So KMnO_4 acts as self indicator.
5. When KMnO_4 is reduced to Mn^{++} , the Mn^{++} acts as catalyst and increases the rate of reaction. For example, in the titration of acidified KMnO_4 and oxalic acid, initially the rate of disappearance of pink colour of KMnO_4 is relatively slow. After few readings Mn^{++} is sufficient in solution to catalyse and rate of disappearance of pink colour of KMnO_4 is fast in the same pace of titration.
6. Since the reaction between KMnO_4 and oxalic acid is slow, so the acidified oxalic acid must be heated to about 60°C , before running down the KMnO_4 from burette.

c. Complexometric titration

The titration which involves the formation of complex compound is called **complexometric titration**. The most commonly used reagent in this type of titration is disodium salt of EDTA (Ethylenediamine tetra-acetic acid). In such case, the equivalence point is determined by the use of metal ion indicator such as *Eriochrome black-T*. This type of titration can be used in the estimation of hardness of water.

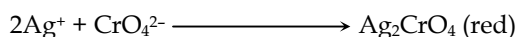
**d. Precipitation titration**

This type of titration involves the formation of precipitate by the combination of ions of titrant and titrand.

Example:

The formation of precipitate is guided by solubility product principle. (This is discussed in ionic equilibria). In the titration of AgNO_3 and NaCl , *potassium chromate* solution can be used as indicator.

At end point:

**Indicators in acid-base titration**

The auxiliary chemical substance which is used to indicate the completion of reaction by the sharp change in colour of solution or of precipitate is called indicator. The indicator used in acid-base reaction is called acid-base indicator. There are different theories about the action of indicator. Acid-base indicators are either organic weak acids or organic weak bases. These show one colour in certain pH range and show another type of colour in another **pH range**. Every acid-base indicator has its own colour changing pH range called pH transition. If equivalence point of reaction falls in this pH range, equivalence point coincides with colour changing pH range of indicator and end point is attained accurately.

The choice of indicator depends upon the nature of titrant and titrand. However the basic principle and common feature is that the indicator should be chosen in such a way that end point should coincide with equivalence point and titration error should be zero. This condition may not be achieved exactly and very small difference usually occurs. The indicator and experimental condition should be chosen in such a way that the difference between end point and theoretical end point is as small as possible.

In case of acid base titration, acid-base indicator is used. Every acid-base indicator has a certain effective pH range within which it shows sharp change in colour. The pH value at the end point of titration depends upon the relative strength of acid-base under titration. The pH value at the end point of titration and pH range of indicator should have proper matching. The suitable indicator and effective pH range for acid-base titration are as follows:

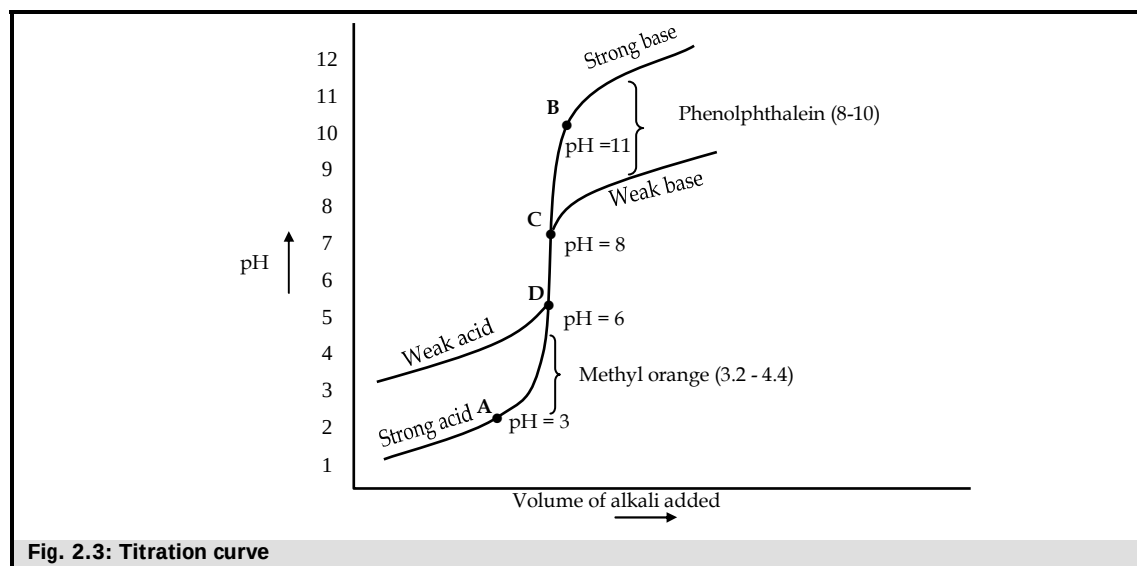
Table 2.6: Choice of indicators in acid-base titration

Acid	Base	Approx. p^H range near equivalence point	Indicator	Approximate p^H transition of indicator
Strong	Strong	3 - 11	Methyl orange phenolphthalein	3.2 - 4.4 8 - 10
Strong	Weak	3 - 8	Methyl orange	3.2 - 4.4
Weak	Strong	6 to 11	phenolphthalein	8 - 10
Weak	Weak	about 6 to 8	None	----

Table 2.7: Colour of indicators in different media

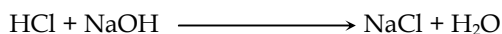
Indicator	Colour in	
	Acidic medium	Basic medium
Methyl orange	Pink or red	Light yellow
Litmus paper	Red	Blue
Phenolphthalein	Colourless	Pink

Following titration curve is useful to choose acid base indicator.

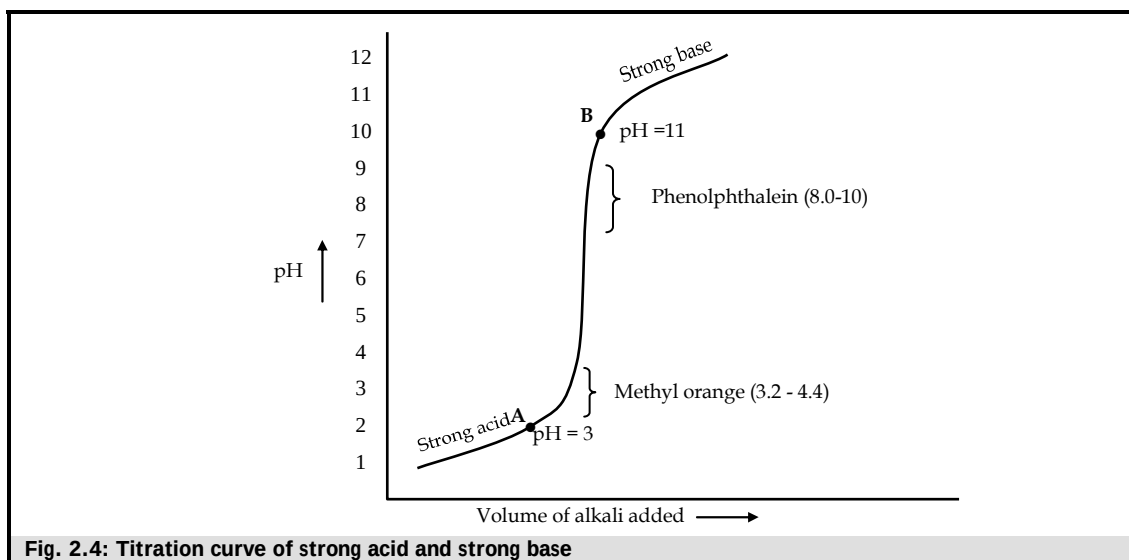
**Fig. 2.3: Titration curve**

A curve obtained by plotting a graph of pH of solution versus volume of alkali added is called **titration curve**. When acid is taken in a conical flask with few drops of indicator and alkali is added gradually from burette, H^+ ions of acid is neutralized by OH^- ion of base. In this condition there is smooth increase in pH value of resulting solution. Near the end point or equivalence point there is sudden rise in pH value. After this, again there is smooth rise in pH value. The sudden rise or steep rise in pH value during titration indicates the neutralization in acid-base reaction. The nature of titration curve depends upon the nature of acid and base used.

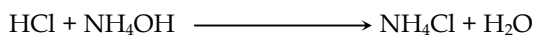
- a. **Titration of strong acid and strong base:** In the titration of strong acid such as HCl and strong base such as NaOH, the steep rise of curve occurs at pH 3 to 11. Any indicator capable of changing colour within this range is a suitable indicator.



The colour changing pH range of methyl orange is 3.2 to 4.4 and that of phenolphthalein is 8 to 10. So either methyl orange or phenolphthalein can be used as indicator.

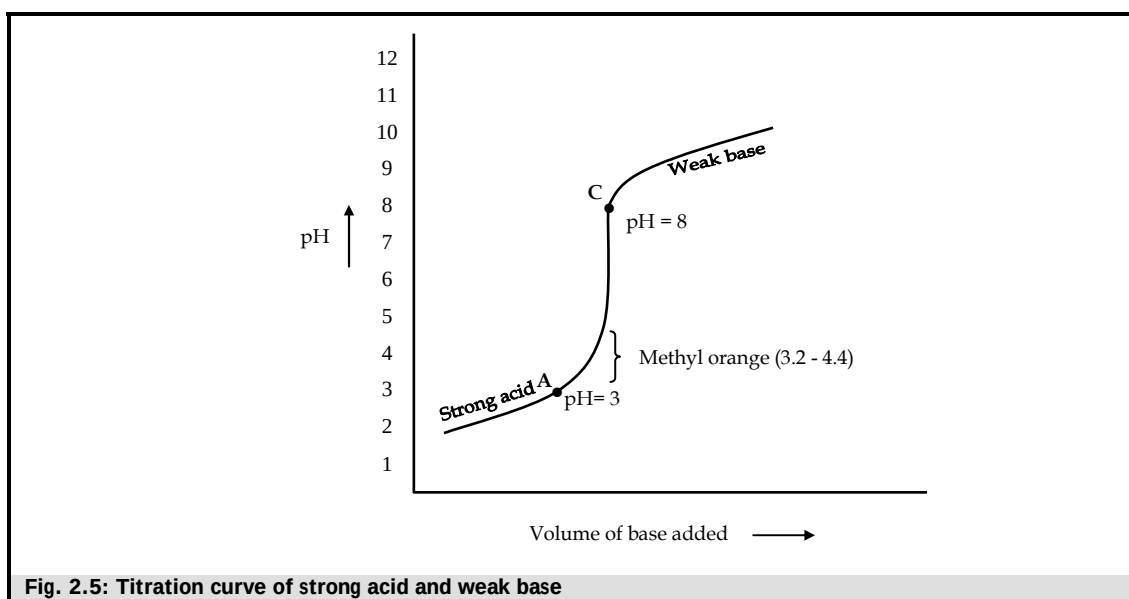


- b. **Titration of strong acid and weak base:** In the titration of strong acid such as HCl and weak base such as NH_4OH , the steep rise of curve occurs at pH 3 to 8.



Any indicator capable of changing its colour within this pH range is a suitable indicator.

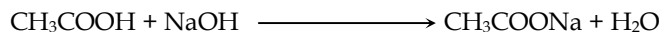
The colour changing pH range of methyl orange is 3.2 to 4.4. So methyl orange is a suitable indicator in the titration of strong acid and weak base.



Self Test

2.10 Why is phenolphthalein not used as indicator in the titration of strong acid and weak base?

- c. **Titration of weak acid and strong base:** In the titration of weak acid such as CH_3COOH and strong base such as NaOH the steep rise of curve occurs at pH 6 to 11.



Any indicator capable of changing its colour within this pH range is a suitable indicator. The colour changing pH range of phenolphthalein is 8.0 to 10.0. So phenolphthalein is a suitable indicator in the titration of weak acid and strong base.

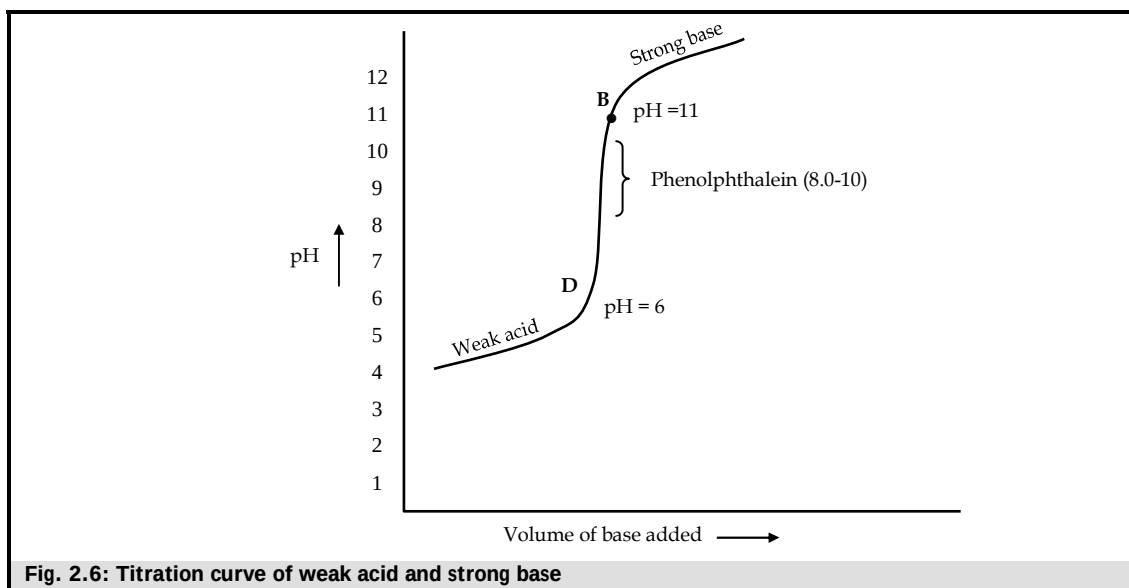


Fig. 2.6: Titration curve of weak acid and strong base

- d. **Titration of weak acid and weak base:** In the titration of weak acid such as CH_3COOH and weak base such as NH_4OH there is no enough steep rises in the curve. The tapering of curve takes place within pH value of 6 to 8.

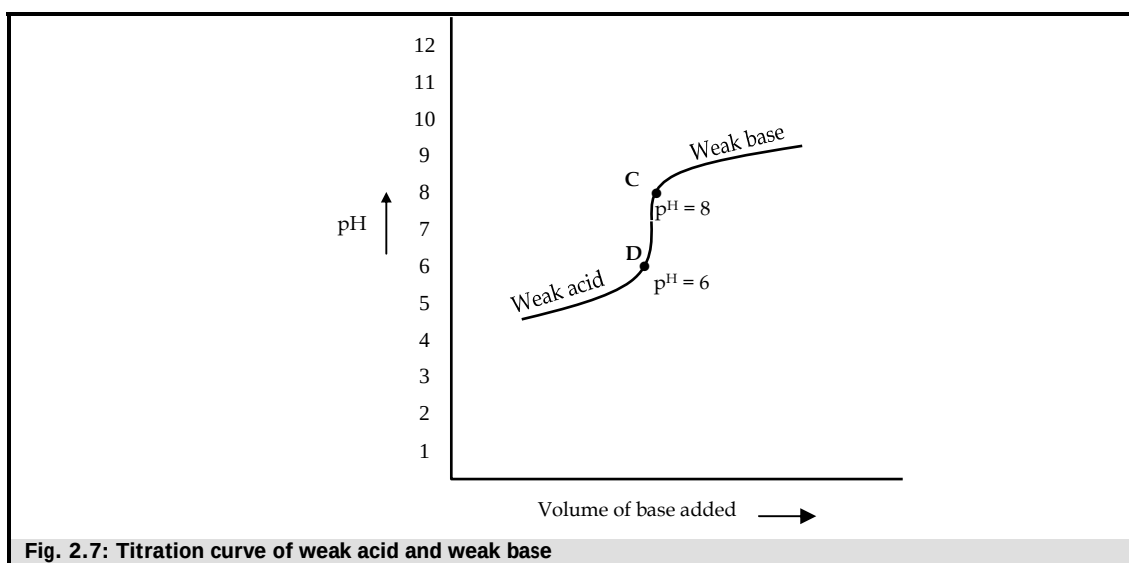
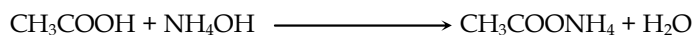


Fig. 2.7: Titration curve of weak acid and weak base

The colour changing pH range of methyl orange is 3.2 to 4.4 and that of phenolphthalein is 8.0 to 10.0. Therefore, neither methyl orange nor phenolphthalein is suitable indicator. Hence end point of weak acid-weak base titration is not determined by acid-base titration which may be determined by other methods such as conductometric titration.

■ Solved Questions

1. Define decinormal solution.

- ↪ A solution containing one tenth gram equivalent of dissolved solute per litre solution is said to be decinormal solution (N/10). Example: 5.3 g Na_2CO_3 dissolved in 1 L solution gives decinormal Na_2CO_3 solution.

2. What do you mean by equivalent weight of an element?

- ↪ Equivalent weight of an element may be defined as the number of parts by weight of it which combines with or displaces directly or indirectly 1.008 parts by weight of hydrogen, 8 parts by weight of oxygen and 35.5 parts by weight of chlorine or its equivalent. For example: Equivalent wt. of Mg = 12

3. What is primary standard solution?

- ↪ The solution which can be prepared directly by dissolving calculated amount of solute to prepare specified volume of solution and can be stored for a sufficiently long period without any change in concentration, is called primary standard solution. Primary standard solution is prepared by dissolving primary standard substance. e.g., Oxalic acid, anhydrous sodium carbonate, Mohr's salt, etc. can be used to prepare primary standard solution.

4. What is secondary standard solution?

- ↪ The standard solution, whose strength is to be redetermined after its preparation is secondary standard solution. The solution whose strength is determined first by titrating it with standard solution then afterward it can be used for titration purpose is called secondary standard solution. Solution of potassium permanganate is an example of secondary standard solution.

5. What is an acid-base indicator?

- ↪ The organic compound either weak acid or weak base which acts as auxiliary reagent and shows the completion of acid-base reaction by the sharp change in visual colour of solution or of precipitate is called acid-base indicator. Example: Methyl orange and phenolphthalein are the examples of acid-base indicators.

■ Questions

■ Very Short Answer Questions:

1. What is normality? How is it related with molarity?
2. Why are the concentration units like g/L, M, N etc. temperature dependent?
3. Define acidimetry and alkalimetry.
4. Write an example of redox titration. Why is it called so?
5. Differentiate primary and secondary standard solutions.
6. What are the requisites for a substance to be a primary standard?
7. Define secondary standard solution with a suitable example
8. What is normality factor? How does it arise during the preparation of standard solution?
9. Why are the following substances not taken as primary standard substances (a) NaOH (b) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (c) FeSO_4 (d) conc. H_2SO_4 (e) KMnO_4
10. Write down the differences between equivalence point and end point.
11. Why is chemical indicator method not suitable for the titration between weak acid and weak base?
12. What is normality equation? Derive it.

■ Short /Long Answer Questions:

1. Define: (a) normality factor (b) binormal solution (c) semi-molar solution (d) centimolar solution (e) molarity (f) decinormal solution (g) decimolar solution (h) end point (i) equivalence point (j) indicator.
2. What is an indicator? How is a suitable indicator selected for a particular acid-base titration?
3. Write short notes on (a) choice of indicators in acid base titration (b) law of equivalence.

■ Numerical Problems

1. What volume of water should be added to 500 mL of 2 N ($f = 0.98$) Na_2CO_3 to make it exactly N/10?
Ans: 9.3 litres
2. X cc of 5 N HCl was diluted to one litre to make normal solution. Calculate the value of x.
Ans: 200cc
3. What mass of Na_2CO_3 is required to make 50cc of its seminormal solution?
Ans: 1.325 g
4. What volume of 5% NaOH is required to neutralize 2 litres of decinormal H_2SO_4 ?
Ans: 160 mL
5. 0.715 g of $\text{Na}_2\text{CO}_3 \cdot x \text{H}_2\text{O}$ required 20 mL of seminormal hydrochloric acid solution for complete reaction. Find the value of x.
Ans: $x = 2$
6. 7.35 g of a dibasic acid was dissolved in water and diluted to 250 mL. 25 mL of this solution was neutralized by 15 mL of N NaOH solution. What is equivalent weight and molecular mass of the acid?
Ans: Eqvt. wt. = 49, Molecular wt. = 98
7. x g of a metal (eq. wt = 12) was completely dissolved in 100 cc of N/2 HCl. The volume was then made up to 500 cc. 25 cc of this diluted acid required 17.5 cc N/10 NaOH for complete neutralization. Find the value of x.
Ans: $x = 0.18\text{g}$
8. 100 mL of 0.1 M HCl is mixed with 50 mL of 0.1M KOH. Calculate the concentration of acid in terms of g/L in the resulting solution.
Ans: Gram/Litre = 1.21
9. If 20 mL of 0.5 N NaOH is mixed with 30 mL of 0.3 N HCl, is the resulting solution acidic or basic? Calculate the normality with respect to the acidic or basic final solution.
Ans: Basic, 0.02 N
10. What is the normality of 50 mL of 3 M H_3PO_4 solution?
11. What volume of 10 M and 5 M NaOH solutions must be mixed to prepare 6L of 6M NaOH solution?
Ans: 1.2 L and 4.8 L
12. What mass of 50% pure CaCO_3 is required to neutralize 40 mL of $\frac{\text{N}}{2}$ HCl?
Ans: 2.0 g
13. Calculate the molarity and normality of 36.5% (w/w) HCl solution having a density 1.15 g/mL.
Ans: 11.5 N, 11.5 M
14. How many moles of H_2SO_4 are required to prepare 4 L of 2N NaOH solution?
Ans: 4 mol
15. 0.8 g of a divalent metal was dissolved in 100 cc of 1.28 N HCl and the solution was diluted to

200 cc. 50 cc of this dilute solution required 54.6 cc of 0.22 N NaOH for neutralization. Calculate the atomic mass of the metal.

Ans: 20

16. 4 g of NaOH was added to 20 cc of 2N H_2SO_4 solution and solution was diluted to 1 L. Predict whether the solution is acidic, basic or neutral. Calculate the normality of the resulting solution.

Ans: Basic, 3 N

17. 30 g of NaOH was dissolved in 500 mL of its solution, Determine the concentration of the solution in terms of (a) g/L (b) Normality (c) Molarity (d) % (w/v).

Ans: 60 g/L, 1.5 N, 1.5 M, 6%

18. 5 g of diacidic base is completely neutralized by 50 mL of 2N HCl. Find the molecular mass of the base.

Ans: 100 amu.

19. 100 mL of 0.1 N HCl and of 0.3 N NaOH are mixed. Calculate (a) normality of NaCl formed (b) Mass of NaOH (in g).

Ans: 0.067 N, 0.585 g

Multiple Choice Questions

- The number of milli-moles of HCl required to neutralize 10 mL of 0.2 M Na_2CO_3 is
a. 2.0 mmol b. 4.0 mmol
c. 0.2 mmol d. 0.4 mmol
- 100 mL of 0.5 M H_2SO_4 solution and 0.1 litre of 1 M HCl were mixed. The normality of the resulting solution will be,
a. 1.5 N b. 0.75 N
c. 2.0 N d. 1 N
- The molarity of a solution obtained by mixing 50 mL of 24 N sulphuric acid with 50 mL of water is
a. 12 M b. 18 M
c. 6 M d. 9 M
- 20 mL of 1N HCl reacted with 30 mL of 0.5 N NaOH. e normality of resulting solution is
a. 0.1 N HCl b. 0.1 N NaOH
c. 0.5 N HCl d. 0.5 N NaOH
- The equivalent weight of divalent metal is 12. Find the molecular mass of the metal oxide,
a. 40 b. 16
c. 28 d. 32
- The normality of 7.3% solution of HCl is
a. 0.2 N b. 2.0 N
c. 7.3 N d. 0.1 N
- The normality of 1% solution of sulphuric acid is nearly
a. 1.0 N b. 0.1 N
c. 0.2 N d. 1.5 N
- What is the molarity of a solution containing 34.2 g of cane sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in 1 litre of its aqueous solution?
a. 1.0 M b. 3.42 M
c. 0.1 M d. 34.2 M
- 2 g of CaCO_3 sample is required for the complete neutralization of 10 mL of 2N HCl. The % purity of CaCO_3 is
a. 50.00% b. 100.00%
c. 66.67% d. 33.33%
- 1.5g of mixture of NaCl and CaCO_3 requires 40 mL of N/2 HCl for neutralization. The % NaCl in the mixture is
a. 66.67% b. 50.00%
c. 33.33% d. 80.10%

Answer Key

1	2	3	4	5	6	7	8	9	10
b	d	c	a	a	b	c	c	a	c

Ionic Equilibrium

3



pH meters are very useful devices to determine the acidic and basic nature of various solutions.

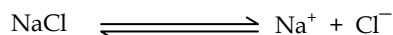
3.1 Introduction

The equilibrium state existing in reversible chemical reaction is discussed under the chapter Chemical Equilibrium (chapter 9 of modern approach to chemistry I). Such phenomenon existing in aqueous solutions of electrolytes are studied under ionic equilibrium. Since, some of the most important processes in biological and industrial systems consist of ionic species of acids, bases or salts in aqueous solutions, the concept of ionic equilibrium helps us to understand the nature of different electrolytes in aqueous solution for different applications. The equilibrium among ions and undissociated or unionised electrolytes is **ionic equilibrium**.

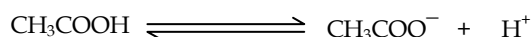
3.2 Electrolytes and Non-Electrolytes

Any substances which are ionized in water and can conduct electricity through their aqueous solution are known as **electrolytes** whereas those which are not ionized in aqueous solution and cannot conduct electricity are known as **non-electrolytes**. The electrolytes are further categorized into two types according to the extent of ionization in aqueous solution; (i) strong electrolytes and (ii) weak electrolytes.

Strong electrolytes are those electrolytes which are almost completely ionized in any concentration of solution. Mineral acids (HCl, HBr, HI, HNO₃, H₂SO₄), alkalies (LiOH, NaOH, KOH), and water soluble ionic compounds (NaCl, KCl, KNO₃, K₂SO₄, Na₂SO₄) are the common examples of strong electrolytes. The ionization or dissociation of strong electrolyte is given as



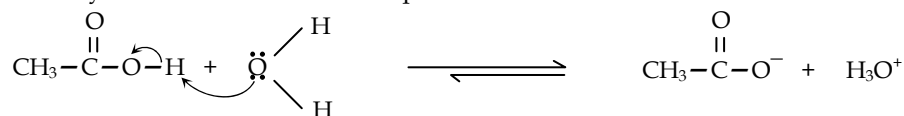
The reversibility sign is necessary for strong electrolytes even as the ionization is not exactly 100%. Weak electrolytes are those which are feebly (partially) ionized in a certain concentration of solution. Almost all organic acids (e.g. HCOOH, CH₃COOH), some inorganic acids (HCN, HClO, H₂CO₃) and some bases (NH₄OH, Al(OH)₃, Fe(OH)₃) are the common example of weak electrolytes. The ionization of weak electrolyte is given as



The ionization or dissociation of electrolyte also depends upon the nature of solvent. If solvent is changed, the extent of ionization may be changed and the concept of strong electrolyte and weak electrolyte is also changed. That is why a weak electrolyte in one solvent system may behave as strong electrolyte in another solvent system. The behaviour of strong electrolyte can be explained in the same way.

Moreover, electrolytes can be classified into two types according to their bonding nature. They are; (i) true electrolytes and (ii) potential electrolytes. A **true electrolyte** is a substance which consists of ions in its purest form. They dissociate to give free ions when dissolved in water. The ionic compounds such as Na⁺Cl⁻, K⁺Cl⁻, Ba⁺⁺SO₄^{- -} etc., are true electrolytes.

A **potential electrolyte** is a highly polar covalent compound which gets dissolved in water and gives ions in solution. In their purest form, these electrolytes do not contain any ions. However, they can easily form ions after dissolution in water. For example HCl and CH₃COOH are covalent compounds. They can interact with water to produce ions as in this reaction.



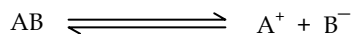
Note:

True electrolytes dissociate in aqueous solution whereas potential electrolytes ionize in aqueous solution.

3.3 Arrhenius Theory of Ionization

Svante Arrhenius postulated the theory of ionization in 1887. The basic postulates of this theory are:

1. When an electrolyte is dissolved in water, it gives two types of species in which one carries positive charge and another carries negative charge. Such charged particles are called ions and process is called ionization.
2. The ions which carry positive charges are called **cations** and those which carry negative charges are called **anions**.
3. In any electrolytic solution, the total number of positive charges is equal to the total number of negative charges. Therefore, given electrolytic solution is electrically neutral.
4. The ions present in the solution constantly reunite to form neutral molecules and thus there is a state of dynamic equilibrium between the dissociated and undissociated molecules.



5. The properties of an electrolyte in the solution are due to the ions given by it in the solution.
6. When an electric current is passed through the solution of an electrolyte, the positive ions (cations) move towards the cathode and the negative ions (anions) move towards the anode.
7. The electrical conductivity of an electrolyte depends on the number of ions present in the solution which is determined by the term degree of ionization (α) which is defined as

$$\alpha = \frac{\text{Number of moles of electrolyte ionised/dissociated}}{\text{Total number of moles of electrolyte taken}}$$

The degree of ionization depends upon temperature, concentration, and nature of the electrolyte and solvent.

Limitations of Arrhenius Theory

1. Arrhenius assumed that ions are formed when electrolytes are dissolved in water but ionic electrolytes are previously in ionized form. They only dissociate when dissolved in water.
2. According to Arrhenius, water or solvent is must for ionization but X-ray diffraction studies have revealed that ionization occurs even in molten state.
3. According to Arrhenius, ionization increases due to dilution but strong electrolytes are almost completely ionized even in concentrated solution.
4. The increase in equivalent conductance and molar conductance of strong electrolyte is not due to increase in dissociation but it is due to increase in ionic mobility with dilution.

3.4 Factors Affecting the Degree of Ionization

$$\text{Degree of ionization } (\alpha) = \frac{\text{Number of moles of electrolyte ionised or dissociated}}{\text{Total number of moles of electrolyte taken}}$$

The important factors that affect the degree of ionization are:

1. **Nature of electrolyte:** Nature of electrolyte is the most important factor that affects the degree of ionization. For example, ionic compounds such as NaCl, KI, etc. dissociate almost completely even in moderately concentrated solution. Such compounds are called strong electrolytes. However, organic acids such as CH_3COOH , HCOOH can ionize partially, and are called weak electrolytes. For strong electrolytes, $\alpha \approx 1$ and for weak electrolytes, $\alpha \ll 1$.

- Nature of solvent:** The degree of ionization of an electrolyte is directly proportional to the dielectric constant of the solvent. For example, dielectric constant of water is greater than that of dimethyl sulfoxide (DMSO). Therefore, degree of ionization of given electrolyte is higher in water than in DMSO.
- Temperature:** The degree of ionization increases with increase in temperature because at higher temperature the kinetic energy of solute molecules increases which helps in increasing the effective collisions between molecules, increasing the degree of ionization.
- Concentration of electrolyte:** The degree of ionization increases with increase in dilution (or decrease in concentration). This is strikingly found in weak electrolytes because they can almost completely ionize at infinite dilution which is explained by Ostwald dilution law.
- Common ion effect:** The presence of a strong electrolyte in the solution of a weak electrolyte having a common ion greatly suppresses the ionization of weak electrolyte. For example, ionization of CH_3COOH (weak acid) is decreased in presence of CH_3COONa (strong electrolyte). This effect is simply called common ion effect and its detail explanation is given in section 3.12.

3.5 Degree of Ionization (α) and Ionization Constant (K): Ostwald Dilution Law

Ostwald dilution law relates the degree of ionisation (α) with concentration of a weak electrolyte. Let us consider the ionization of a binary electrolyte AB in the solution of concentration $C \text{ mol/L}$. Suppose the degree of ionization of electrolyte is α .

	$AB \rightleftharpoons A^+ + B^-$	
Initial concentration	C	0
Change in concentration	$-C\alpha$	$+C\alpha$
Concentration at equilibrium	$(C - C\alpha)$	$C\alpha$

Applying the law of mass action at equilibrium state of this ionization, the equilibrium constant of the electrolyte is given by,

$$K = \frac{[A^+][B^-]}{[AB]} \quad \dots (i)$$

Here, K = ionisation or dissociation constant of the electrolyte.

Substituting their concentration, we get,

$$K = \frac{\alpha C \times \alpha C}{C(1 - \alpha)}$$

$$\text{or, } K = \frac{\alpha^2 C}{(1 - \alpha)} \quad \dots (ii)$$

Equation (iii) is the mathematical form of Ostwald dilution law. For weak electrolyte, the degree of ionization is very small. Since, $1 - \alpha \approx 1$. Now equation (iii) becomes

$$K = \alpha^2 C \quad \dots (iii)$$

$$\text{or, } \alpha^2 = \frac{K}{C}$$

$$\text{or, } \alpha = \sqrt{\frac{K}{C}} \quad \dots \text{ (iv)}$$

We know,

$$\text{Molar concentration (C)} = \frac{\text{Number of mole of solute (n)}}{\text{Volume of solution in litre (V)}}$$

If one mole of solute is used to prepare V liter of solution then, $C = \frac{1}{V}$.

$$\text{Now equation (iii) becomes, } K = \frac{\alpha^2}{V(1 - \alpha)} \quad \dots \text{ (v)}$$

$$\text{Taking } 1 - \alpha = 1, \text{ equation (iv) becomes, } K = \frac{\alpha^2}{V} \quad \dots \text{ (vi)}$$

$$\text{or, } \alpha = \sqrt{KV} \quad \dots \text{ (vii)}$$

Equation (vii) reveals that the degree of ionization is directly proportional to the square root of dilution.

Therefore, according to Ostwald dilution law:

"Degree of ionisation of an electrolyte is inversely proportional to square root of the concentration of the electrolyte." Or

"Degree of ionisation of an electrolyte is directly proportional to the square root of dilution."

$$\alpha = \sqrt{\frac{K}{C}} \text{ or } \alpha = \sqrt{KV}$$

Applications of Ostwald Dilution Law

1. To calculate degree of ionisation (α): The above mentioned equation can be utilized as:

* For weak acids: $\alpha = \sqrt{\frac{K_a}{C}}$, where K_a = ionisation constant of an acid.

* For weak bases: $\alpha = \sqrt{\frac{K_b}{C}}$, where K_b = ionization/dissociation constant of a base.

* For salt hydrolysis: $h = \sqrt{\frac{K_h}{C}}$, where h is degree of hydrolysis and K_h = hydrolysis constant

2. To determine equilibrium molar concentration of ions like H^+ , OH^- etc.: Molar concentration of ions formed by the ionisation of an electrolyte is $C\alpha$. For example, $[H^+] = C\alpha$, $[OH^-] = C\alpha$ etc.

3. To calculate pH and pOH of weak acids and bases: The molar concentrations of H^+ and OH^- can be determined as given in the example above. Thus, pH and pOH of weak acid and bases can be determined as:

$$pH = -\log [H^+], pOH = -\log [OH^-]$$

$$\text{Also, } pH = 14 - pOH \text{ (for } 25^\circ\text{C)}$$

(iv) To determine the relative acidic or basic nature of acids or bases: The relation is:

$$\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_1}{K_2}}$$

Limitation of Ostwald's dilution law

Ostwald dilution law is only applicable for weak electrolytes. Strong electrolyte ionise/dissociate almost completely. For strong electrolyte, $1 - \alpha$ cannot be approximated as 1. So, the equation number (iii) as given above becomes $K = \frac{\alpha^2 C}{0}$ = Undetermined. This means, Ostwald's dilution law is not applicable for the strong electrolytes.

Example 3.1 At 25°C, ionisation constant of CH_3COOH is 1.8×10^{-5} . For 2 M CH_3COOH solution, calculate:

(a) degree of ionisation (b) % ionisation, and (c) molar concentration of H^+ ions in equilibrium. d) pH of the solution

Solution

- Using Ostwald dilution law: $\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{1.8 \times 10^{-5}}{2}} = \sqrt{9 \times 10^{-6}} = 3 \times 10^{-3}$
- % ionisation = $\alpha \times 100 = 3 \times 10^{-3} \times 100 = 0.3\%$
- $[\text{H}^+] = C\alpha = 2 \times 3 \times 10^{-3} = 6 \times 10^{-3} \text{ mol/L}$
- $\text{pH} = -\log[\text{H}^+] = -\log(6 \times 10^{-3}) = 2.2$

Note:

For weak acid, $[\text{H}^+] = C\alpha$ and for weak base, $[\text{OH}^-] = C\alpha$. This is because, out of 1 mol electrolyte, α mol of acid or base are ionized. So for C mol/L C α mol of acid or base are ionised giving C α mol $[\text{H}^+]$ or $[\text{OH}^-]$

Self Test

3.1 How many times an acid with $K_a = 10^{-4}$ is stronger than an acid with $K_a = 10^{-6}$? (Ans. 10)

3.6 Ionization Constant and Relative Strength of Acids and Bases

The relative strengths of weak acids can be compared in terms of their ionization constant (K_a or K_b). The smaller the value of K_a , the weaker will be the acidic strength. This fact can be explained with the help of Ostwald's dilution law.

We have, $K_a = \alpha^2 C$

Stronger the weak acid, higher will be the value of degree of ionization for a given concentration and therefore, higher is the value of K_a . For example, K_a values for acetic acid, formic acid, and hypochlorous acid at 25°C are 1.8×10^{-5} , 2.0×10^{-4} , and 3.5×10^{-8} , respectively. Therefore, the order of acidic strength should be formic acid > acetic acid > hypochlorous acid.

Table 3.1: Ionization constants of some weak acids at 25°C

Name of acid	Formula	K_a
Formic acid	HCOOH	1.7×10^{-4}
Hydrofluoric acid	HF	7.1×10^{-4}
Nitrous acid	HNO_2	4.5×10^{-4}
Acetic acid	CH_3COOH	1.8×10^{-5}
Benzoic acid	$\text{C}_6\text{H}_5\text{COOH}$	6.5×10^{-5}
Asorbic acid	$\text{C}_6\text{H}_8\text{O}_6$	8×10^{-10}
Hydrocyanic acid	HCN	4.9×10^{-10}
Phenol	$\text{C}_6\text{H}_5\text{OH}$	1.3×10^{-10}

Similarly, higher the value of K_b , higher is the degree of ionization and hence, stronger will be the base. For example, K_b values for weak bases ammonia, aniline, and methylamine at 25°C are 1.8×10^{-5} , 5×10^{-10} and 5.1×10^{-4} , respectively. Therefore, the order of basic strength should be methylamine > ammonia > aniline.

Table: 3.2: Ionization constants of some weak bases at 25°C

Name or base	Formula	K_b
Caffeine	$\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$	4.1×10^{-4}
Methylamine	CH_3NH_2	4.4×10^{-4}
Ethylamine	$\text{C}_2\text{H}_5\text{NH}_2$	5.6×10^{-4}
Ammonia	NH_3	1.8×10^{-5}
Pyridine	$\text{C}_5\text{H}_5\text{N}$	1.7×10^{-9}
Aniline	$\text{C}_6\text{H}_5\text{NH}_2$	3.8×10^{-10}
Urea	$\text{N}_2\text{H}_4\text{CO}$	1.5×10^{-14}

Since, the value of K_a and K_b for weak acid and base is very small, the term $\text{p}K_a$ and $\text{p}K_b$ is used to compare the acidic and basic strength. The terms $\text{p}K_a$ and $\text{p}K_b$ are given by

$$\text{p}K_a = -\log K_a, \text{ and } \text{p}K_b = -\log K_b$$

We have seen that the magnitude of K_a indicates the strength of an acid. Another measure of the strength of an acid is its percent ionization, which is defined as

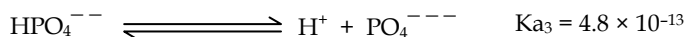
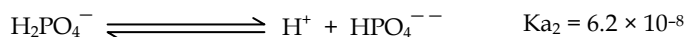
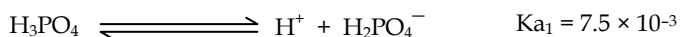
$$\text{Percent ionization} = \alpha \times 100\%.$$

Stronger the acid, greater will be the percent ionization. For a monoprotic acid HA, the concentration of the acid that undergoes ionization is equal to the concentration of the H^+ ions or the concentration of the A^- ions at equilibrium. Therefore, we can write the percent ionization as

$$\text{Percent ionization} = \frac{[\text{H}^+]}{[\text{HA}]^0} \times 100\%$$

Where, $[\text{H}^+]$ is the molar concentration of H^+ at equilibrium and $[\text{HA}]^0$ is the initial concentration of the acid taken.

Polybasic acid undergoes ionization in different successive steps. Each step has its own ionization constant. The first ionization constant is always greater than the second ionization constant and so on. For example, orthophosphoric acid undergoes ionization in three steps as follows



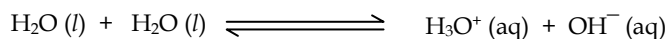
Here, it is to be noted that $K_{a1} > K_{a2} > K_{a3}$

Self Test

3.2 First ionization constant of an acid is always greater than second ionization constant. Do you agree this? Give reason for your answer.

3.7 Autoionization of Water

Pure water ionizes to a very small extent. Its autoionization can be expressed as,



By applying the law of mass action:

$$K_{\text{eq}} = \frac{[\text{H}_3\text{O}^+] [\text{OH}^-]}{[\text{H}_2\text{O}]^2}$$

Here, K_{eq} is called ionization constant of water.

Concentration of water can be taken to be constant.

$$\text{or, } K_{\text{eq}} \times [\text{H}_2\text{O}]^2 = [\text{H}_3\text{O}^+] [\text{OH}^-]$$

$$\text{or, } K_w = [\text{H}_3\text{O}^+] [\text{OH}^-]$$

$$\text{or, Simply: } K_w = [\text{H}^+] [\text{OH}^-]$$

Here, K_w is a constant called ionic product of water which depends upon temperature only.

At 25°C, 10^{-7} mol/L water ionises producing 10^{-7} mol/L of H^+ ions and 10^{-7} mol/L of OH^- ions.

For pure water, $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ ions are formed in equal numbers. Therefore, at 25°C,

$$[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1.0 \times 10^{-7} \text{ mol/L.}$$

$$\text{Therefore, } K_w = (1.0 \times 10^{-7}) \times (1.0 \times 10^{-7})$$

$$\text{or, } K_w = 1.0 \times 10^{-14}$$

$$\text{Since, } \text{p}K_w = -\log (K_w)$$

$$\text{or, } \text{p}K_w = -\log (10^{-14})$$

$$\text{or, } \text{p}K_w = 14$$

Therefore, at 25°C, $\text{pH} + \text{pOH} = 14$

Effect of temperature on auto-ionization of water

Increase in temperature increases the degree of ionization of water consequently concentration of H^+ and OH^- increases and ionic product of water also increases. As result of this $\text{p}K_w$ value decreases. At different temperatures the values of K_w and $\text{p}K_w$ are given below. This means, on increasing temperature, there is decrease in pH , pOH and $\text{p}K_w$.

Example:

For pure water at 25°C: $\text{pH}=7$, $\text{pOH}=7$, $\text{p}K_w=14$ and $\text{p}K_w= 14$. Here, $\text{pH} + \text{pOH}=14$.

For pure water at 50°C: $\text{pH}=6.63$, $\text{pOH}=6.63$ and $\text{p}K_w=13.26$. Here, $\text{pH}+\text{pOH}=13.26$.

Temperature	0°C	25°C	50°C	60°C
K_w	1.14×10^{-15}	1.00×10^{-14}	5.47×10^{-14}	9.60×10^{-14}
$\text{p}K_w$	14.94	14	13.26	13.01

3.8 pH and pH Scale

Because the concentration of H^+ and OH^- ions in aqueous solution are very small numbers which makes inconvenient to work with, Sorensen in 1909 proposed a more practical measure, called pH . Simply, **pH is defined as the negative common logarithm of molar concentration of H^+ ions.**

$$\text{pH} = -\log_{10} [\text{H}^+]$$

In aqueous solution, H^+ ion mostly exists in hydrated form. More correctly, $\text{pH} = -\log_{10} [\text{H}_3\text{O}^+]$

Similarly, for basic solution

$$\text{pOH} = -\log_{10}[\text{OH}^-]$$

The relationship between pH and pOH can be established from autoionization of water.

We know, $[\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ at 25°C

Taking log of both sides

$$\log[\text{H}_3\text{O}^+] + \log[\text{OH}^-] = \log(1 \times 10^{-14})$$

$$\text{or, } -\log[\text{H}_3\text{O}^+] - \log[\text{OH}^-] = -\log(1 \times 10^{-14})$$

$$\text{pH} + \text{pOH} = 14$$

Because pH is simply a way to express hydrogen ion concentration, nature of the solution (acidic, basic or neutral) can be distinguished by their pH values expressed in pH scale.

For acidic solutions at 25°C

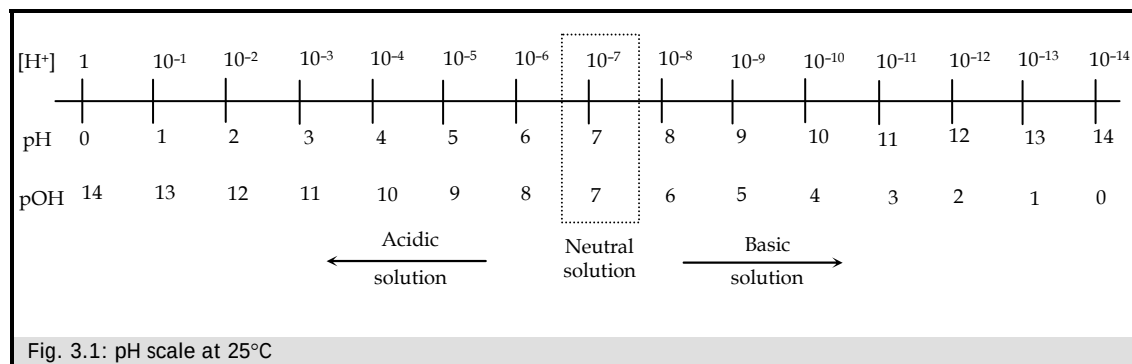
The concentration of H^+ ion is greater than 10^{-7} M and hence the pH value is less than 7.

For basic solution at 25°C

The concentration of H^+ ion is less than 10^{-7} M and hence the pH value is more than 7.

For neutral solution at 25°C

The concentration of H^+ ion is equal to 10^{-7} M and hence the pH value is 7.



Example 3.2 Calculate the pH of 0.04N H_2SO_4 solution (Assume complete ionization)

Solution

Method 1:

Since basicity of H_2SO_4 is 2, its molarity

$M = N/2 = 0.04/2 = 0.02 \text{ M}$ and it is completely ionized

	H_2SO_4	$\longrightarrow$	2H^+	+	SO_4^{2-}
Initial concentration	0.02 M		0		0
Change after ionization	-0.02 M		2×0.02		0.02
Concentration at equilibrium	≈ 0		0.04 M		0.02 M

Now,

$$\text{pH} = -\log[\text{H}^+] = -\log(0.04) = 1.398$$

Method 2: In case of complete ionization of acid or base,

$$\text{pH} = -\log(N_{\text{acid}}) \text{ and } \text{pOH} = -\log(N_{\text{base}})$$

$$\text{So, pH} = -\log(0.04) = 1.398$$

Example 3.3 Calculate the pH of 0.01M solution of ammonia ($K_b = 1.8 \times 10^{-5}$)

Solution

Since ammonia is weak base, it is partially ionized. The degree of ionization of ammonia is calculated by Ostwald's dilution law.

Method 1:

	NH_4OH	$\rightleftharpoons$	NH_4^+	+	OH^-
Initial concentration	0.01 M		0		0
Change at equilibrium	-0.01 α		+0.01 α		+0.01 α
Concentration at equilibrium	(0.01-0.01 α) M		+ 0.01 α M		+0.01 α M

Now,

$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} = \frac{0.01 \alpha \times 0.01 \alpha}{(0.01 - 0.01 \alpha)} = \frac{0.01 \alpha^2}{(1 - \alpha)}$$

For weak electrolyte $1 - \alpha \approx 1$

So, $K_b = 0.01 \alpha^2$

$$\text{or } \alpha = \sqrt{\frac{K_b}{0.01}} = \sqrt{\frac{1.8 \times 10^{-5}}{0.01}} = 0.042$$

and $[\text{OH}^-] = 0.01 \times 0.042 = 0.00042 \text{ M}$

$\text{pOH} = -\log[\text{OH}^-] = -\log(0.00042) = 3.377$

$\text{pH} = 14 - \text{pOH} = 14 - 3.377 = 10.623$

Method 2:

$$\text{Using Ostwald's dilution law: } \alpha = \sqrt{\frac{K_b}{C}} = \sqrt{\frac{1.8 \times 10^{-5}}{0.01}} = 0.042$$

$[\text{H}^+] = C\alpha = 0.01 \times 0.042 = 0.00042 \text{ M}$

$\text{pOH} = -\log[\text{OH}^-] = -\log(0.00042) = 3.377$

$\text{pH} = 14 - \text{pOH} = 14 - 3.377 = 10.623$

3.9 Buffer Solution or Buffered Solution

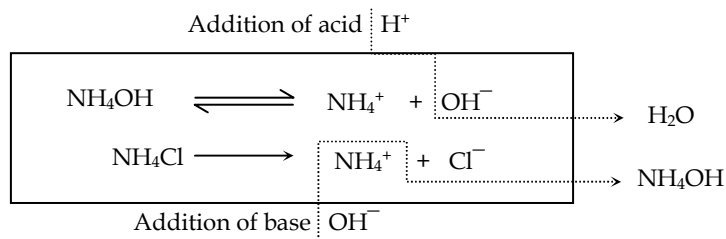
The literal meaning of buffer is "to resist". A buffer solution resists the change in pH value on addition of small amount of acid or base. Generally, a buffer solution is a mixture of a weak acid and its salt with strong base or a mixture of weak base and its salt with strong acid. For example, if we mix CH_3COOH solution (solution 1) and its salt CH_3COONa solution (solution 2), the resultant mixture is called **acidic buffer**. The solution has the ability to resist changes in pH upon the addition of small amounts of either strong acid or base. Buffers are very important to chemical and biological systems. The pH in the human body varies greatly from one fluid to another; for example, the pH of blood is about 7.4, whereas the gastric juice in our stomachs has a pH of about 1.5. A buffer solution must contain a relatively large concentration of acid to react with any OH^- ions that are added to it, and it must contain a similar concentration of base to react with any added H^+ ions. Furthermore, the acid and the base components of the buffer must not consume each other in a neutralization reaction. There are two types of buffer solutions;

- Acidic buffer:** It is the mixture of weak acid and its salt with strong base. Example: Mixture of CH_3COOH and CH_3COONa solutions.
- Basic buffer:** It is the mixture of weak base and its salt with strong acid. Example: mixture of NH_4OH and NH_4Cl solutions.

Besides these two types of buffered solutions, solution of the salt of weak acid and weak base can also act like a buffered solution, which can be termed as a **single buffer**. For example, the solution of ammonium acetate ($\text{CH}_3\text{COONH}_4$) can act as a single buffer. In the text, we shall only be discussing about the first two types only.

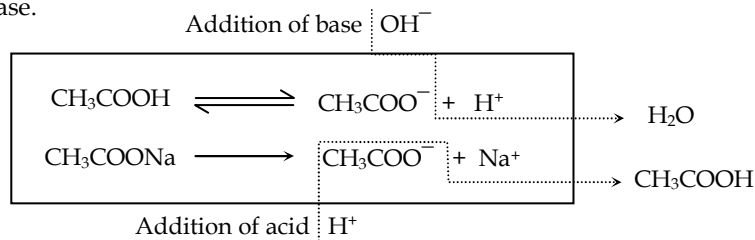
Buffer action can be understood with the help of common ion effect and Le-Chatelier principle. The following diagram illustrates how basic buffer of ammonia and ammonium chloride can maintain the pH almost constant upon the addition of small amount of strong acid (HCl) or strong base (NaOH).

When small quantity of acid (H^+) is added, it combines with OH^- ion of buffer system to give H_2O molecules, which is feebly ionized. Therefore, OH^- ion concentration is decreased which causes the disturbance of equilibrium state of ionization of NH_4OH . However, according to Le-Chatelier's principle, this disturbance is minimized by shifting this equilibrium towards forward direction i.e. more NH_4OH molecule will be ionized to fulfill the, amount of OH^- ion and therefore pH remains constants. In this way addition of small quantity of acid does not change the pH of solution appreciably.



When small quantity of base (OH^-) is added, it combines with NH_4^+ ion of buffer solution to furnish NH_4OH , which is feebly ionized. Due to this addition of small quantity of base does not change the pH of solution appreciably.

Now, let us take an example of acid buffer to study how pH is maintained on addition of small quantity of acid or base.



When small quantity of base is added, OH^- ion of base is neutralized by H^+ of buffer solution to give H_2O , which is feebly ionized. Therefore, H^+ ion concentration is decreased which causes the disturbance of equilibrium state of ionization of CH_3COOH . However, according to Le-Chatelier's principle, this disturbance is minimized by shifting this equilibrium towards forward direction i.e. more CH_3COOH molecule will be ionized to fulfill the, amount of H^+ ion and therefore pH remains constant.

When small quantity of acid (H^+) is added, H^+ ion combines with CH_3COO^- ion of solution to give CH_3COOH , which is feebly ionized. Hence there is no appreciable increase in H^+ ion concentration in buffer solution.

pH of buffer solution can be calculated by Henderson equation, which is

For acidic buffer: $\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$

For basic buffer: $\text{pOH} = \text{pK}_b + \log \frac{[\text{salt}]}{[\text{base}]}$

Self test

3.2 Give two examples of basic buffer solution.

Example 3.4 Calculate the pH of mixture that contains 0.03 M NH_4OH and 0.04M of NH_4Cl . (K_b for ammonia is 1.8×10^{-5})

Solution

Since mixture contains weak base and its salt, it is basic buffer and its pH can be calculated by using Henderson equation.

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]} = -\log K_b + \log \frac{[\text{salt}]}{[\text{base}]} = 4.74 + \log \frac{0.04}{0.03} = 4.865$$

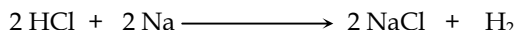
$$\text{pH} = 14 - \text{pOH} = 14 - 4.865 = 9.135$$

3.10 Concept of Acids, Bases and Salts

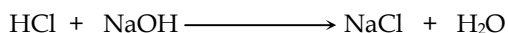
Classically, a compound containing replaceable hydrogen atom (s), which has sour taste, has a capacity to neutralize base and turns blue litmus paper red, is called an acid.

For example HCl is acid because;

- i. it contains hydrogen which can be replaced by treating with active metal like sodium



- ii. it can be neutralized by base like as NaOH to give salt and water



- iii. Its aqueous solution turns blue litmus paper to red, methyl orange to pink and it has sour taste.

Similarly, the oxides or hydroxides of metals, which have a capacity to neutralize acid, turn red litmus paper blue and have bitter taste, are called bases. Water soluble bases are called alkalis. All alkalis are bases but all bases are not alkalis. $\text{Fe}(\text{OH})_3$, $\text{Cu}(\text{OH})_2$, etc. are bases but are not alkalis.

For example, NaOH is alkali because;

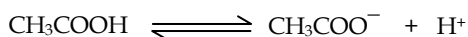
- i. It is water soluble hydroxide of metal
- ii. It neutralize acid to give salt and water
- iii. Its aqueous solution turns red litmus paper into blue and phenolphthalein to pink.
- iv. It is bitter in taste and soapy to touch (**such types of bases should never be tasted !**).

The important acid-base concepts are explained here.

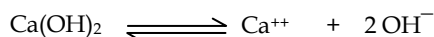
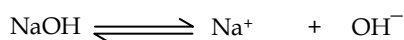
3.10.1 Arrhenius Concept of Acid-Base

Depending upon the Arrhenius theory of ionization, clear definition of acids, bases and salt have been given.

An acid is a compound which ionizes to give hydrogen ion as only the positive ion in aqueous solution. For example, hydrochloric acid and acetic acid are acids because they furnish hydrogen ion when dissolved in water.



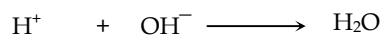
A base is a compound which ionizes to give hydroxide ion as only the negative ion in aqueous system. For example, sodium hydroxide and calcium hydroxide are bases (alkalies) because they furnishes OH^- ion when dissolved in water.



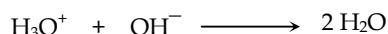
The acidic or basic properties explained above are due to the presence of H^+ or OH^- ions of acid or base, respectively.

A salt is a compound which gives positive ion excepting hydrogen ion and negative ion, excepting hydroxide ion, when taken in aqueous solution.

Neutralization is a process through which H^+ of acid combines with OH^- of the base to give neutral water molecule.



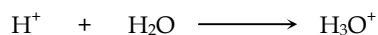
or more correctly



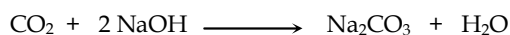
Limitations of Arrhenius Concept

Arrhenius theory suffers from many serious defects, which are given below;

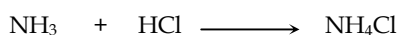
1. Arrhenius theory does not explain the nature of hydrogen ion in aqueous solution. Actually, H^+ ion does not exist independently in solution. It combines with water molecule to give hydronium ion.



2. This theory does not explain the acidic or basic properties of those compounds which do not consist H^+ or OH^- ions. For example, CO_2 does not contain H^+ ion but behaves as an acid.



Similarly, NH_3 is base even though it does not contain OH^- ion in its molecule.



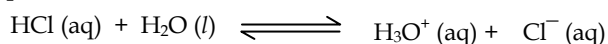
3. Arrhenius theory explains the acids and bases in terms of their aqueous solution (i.e., this concept is limited to water soluble compounds only) but it does not say anything about the acidic or basic properties of substance themselves.
4. Some salts such as AlCl_3 , FeCl_3 , etc, show acidic behaviour. But this theory does not explain about their acidic nature.

3.10.2 Bronsted-Lowry Concept of Acid-Base

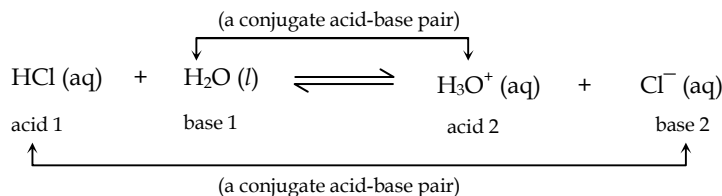
Arrhenius definitions of acids and bases are limited in aqueous solutions. Broader definitions were proposed by **Johannes Bronsted** and **T. M. Lowry** in 1932; a Bronsted-Lowry acid is a proton donor and a Bronsted-Lowry base is a proton acceptor.

Therefore, **any hydrogen containing species (molecule, cation or anion) which is capable of donating one or more protons (H^+) to any substance is called an acid and that which is capable of accepting proton from an acid is called a base.** Note that this definition of acid-base does not require acids and bases to be in aqueous solution. Moreover, an important feature of this concept is interdependent acid-base pair. This means, for a substance to act as an acid, there should be another substance to act as a base such that there is a proton transfer from the acid to base.

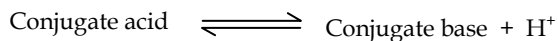
This concept of acid and base is illustrated below. Hydrochloric acid is a Bronsted acid since it donates a proton in water:



Here, HCl donates a proton and becomes Cl^- whereas H_2O accepts this proton to become H_3O^+ . Thus, HCl is an acid and H_2O is a base. If we consider the reverse reaction, H_3O^+ can donate its proton to Cl^- . Hence H_3O^+ is also a Bronsted acid and Cl^- is a Bronsted base. It is clear that every acid must form a base on donating its proton, and every base must form an acid on accepting a proton. Therefore, we can write,

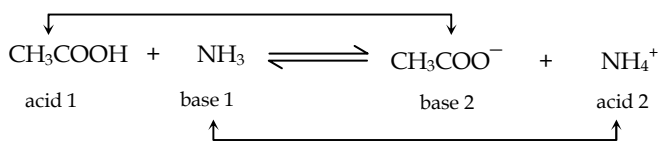


Here, HCl donates a proton to water, and water accepts a proton, hence HCl is an acid and water is a base. In reverse process, the Cl^- ion accepts a proton from H_3O^+ , hence Cl^- ion acts as a base and H_3O^+ acts as an acid. The pair of substances which can be formed from each other by gain or loss of proton is called a conjugate acid-base pair. Therefore, acid1 and base-2 (i.e., HCl and Cl^-) is one conjugate acid-base pair and base-1 and acid-2 (i.e., H_2O and H_3O^+) is another conjugate acid-base pair. Simply, acid after loss of proton becomes **conjugate base** while a base after gaining a proton becomes a **conjugate acid**.



Therefore, a pair of an acid and base which differ from one another by a proton is called a **conjugate acid-base pair**.

Interestingly, according to this concept neutralization means the formation of corresponding conjugate acid-base pair. For example, neutralization of acetic acid by ammonia as in this reaction



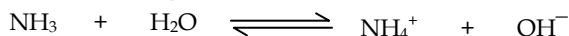
The strength of acid or base is related to the tendency of donating or accepting of proton. A strong acid has a greater tendency to donate its proton, while a strong base has a greater tendency to accept the proton. From conjugate acid-base pair, if given acid is strong, its conjugate base must be weak and vice versa. In above example, acetic acid is weak acid and therefore its conjugate base (CH_3COO^- ion) should be strong base.

Relative strength of conjugate acid-base pairs

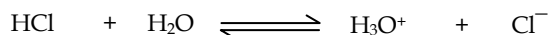
Acid strength increases ↑	Perchloric acid (HClO_4)	Strong acid }	ClO_4^-	Base strength increases ↓
	Sulphuric acid (H_2SO_4)		HSO_4^-	
	Hydroiodic acid (HI)		I^-	
	Hydrobromic acid (HBr)		Br^-	
	Hydrochloric acid (HCl)		Cl^-	
	Nitric acid (HNO_3)		NO_3^-	
	Hydrofluoric acid (HF)		F^-	
	Nitrous acid (HNO_2)	Weak acids }	NO_2^-	
	Formic acid (HCOOH)		HCOO^-	
	Acetic acid (CH_3COOH)		CH_3COO^-	
	Hydrocyanic acid (HCN)		CN^-	
	Water (H_2O)		OH^-	
	Ammonia (NH_3)		NH_2^-	
	Acids		Conjugate bases	

This theory is much broader than Arrhenius theory because it is found that all Arrhenius acids and bases are also Bronsted-Lowry acids and bases, but the reverse may not be true. For example,

ammonia has no OH group and is not Arrhenius base. However, it can accept proton from water to give NH_4^+ ion and therefore is Bronsted base. Moreover, NH_4^+ ion is Bronsted acid as it can donate proton in reverse direction of given reaction.



In this reaction water acts as acid. However, if we consider the dissolution of hydrochloric acid in water, water acts as base as it accepts proton from HCl as:



Therefore, H_2O in first reaction acts as an acid, while in second reaction it acts as a base. It is concluded that Bronsted acid or base can be identified from the given reaction where they donate or accept a proton. As it is clear from above, with NH_3 , water acts as an acid, while with HCl water acts as a base. Water is therefore an amphoteric substance.

The substances which can act as proton donor as well as proton acceptor are called **amphoteric** substances

Self Test

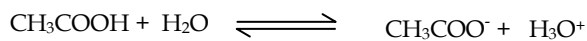
3.3 HCO_3^- is amphoteric. Explain

3.4 Explain acid-base properties of water based upon Bronsted-Lowry concept.

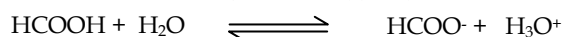
Significance of Bronsted-Lowry concept of acid and base

This concept is capable of explaining the acidic and basic natures of some compounds (like NH_3 , HCl, etc.) in non-aqueous medium also. Besides this, it provides an excellent way of comparing the relative acidic strengths considering their ionizations in water and determining their ionization constant values.

Example:



For this ionization: $K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}_3\text{O}^+]}{[\text{CH}_3\text{COOH}][\text{H}_2\text{O}]} = 1.8 \times 10^{-5}$, at 25°C

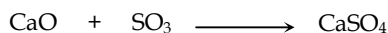
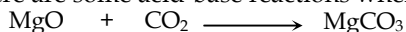


For this ionization: $K_a = \frac{[\text{HCOO}^-][\text{H}_3\text{O}^+]}{[\text{HCOOH}][\text{H}_2\text{O}]} = 1.7 \times 10^{-4}$, at 25°C

The above data shows that HCOOH is stronger acid than CH_3COOH .

Limitation of Bronsted Lowry concept of acid and base

1. The Bronsted-Lowry concept is not applicable for non-protonic chemical species. For example, SO_2 , CO_2 , NO_2 , etc. are acidic in nature but these do not have proton. Similarly, CaO , MgO , BaO , etc. are basic in nature but these do not accept proton.
2. There are some acid-base reactions where there is no transfer of proton. The example includes,



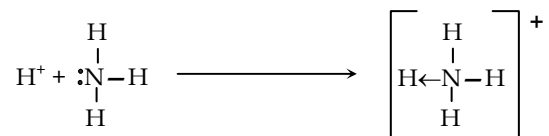
These types of reactions can not be explained by Bronsted-Lowry concept.

3. Some salts such as BF_3 , AlCl_3 , FeCl_3 , etc. are acidic in nature but these do not have proton.

3.10.3 Lewis Concept of Acid-Base

G. N. Lewis proposed a more general theory about acids and bases. This concept is based on electron rather than proton. A Lewis base is a substance which can donate a pair of electrons. A Lewis acid is a substance which can accept a pair of electrons. For example, in the protonation of ammonia, NH_3

acts as a Lewis base because it donates a pair of electrons to the proton H^+ , which acts as a Lewis acid by accepting the pair of electrons. A Lewis acid-base reaction, therefore, is one that involves the donation of a pair of electrons from one species to another. Such a reaction does not produce a salt and water.



Here, a lone pair of electron is donated by ammonia to hydrogen ion, hence ammonia is a Lewis base, and hydrogen ion that accepts the lone pair of electron is a Lewis acid.

Examples of Lewis acid:

- Cations (NO_2^+ , Cl^+ , Ag^+ , Cu^{2+} etc.)
- Compounds having incomplete octet in their molecules (e.g. BCl_3 , BF_3 , AlCl_3 , FeCl_3 etc.)
- Some molecules with multiple bonds (e.g., CO_2 , SO_3 , $\text{H}-\text{C} \equiv \text{N}$ etc.)

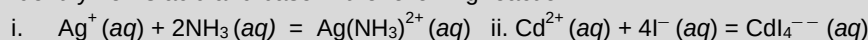
Examples of Lewis bases:

- All anions (e.g. Cl^- , I^- , OH^- , CN^- , etc.)
- Molecules having at least one lone pair electrons. (e.g. H_2O , NH_3 , ROH , ROR etc.)

Lewis acids have an electron deficient centre and it attract electron pair where as Lewis base has the electron rich centre and it looks for electron deficient species. Hence, Lewis acids are electrophiles and Lewis bases are nucleophiles.

Self Test

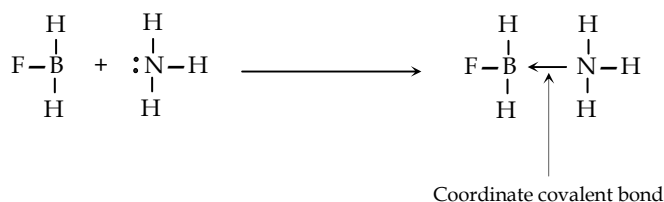
3.5 Identify Lewis acid and base in the following reaction



Limitations of Lewis concept of acid and base

Even though this is the advanced concept of acid-base, it still has following limitations.

- In Lewis acid-base reaction, there is formation of coordinate covalent bond which is very slow process. But true acid-base reactions are very fast.



- This concept cannot tell anything about the strength of acid and base.
- This concept does not consider about ionization process of acid and base.

Self Test

3.6 With the help of Lewis dot formula how can you show that CO_2 in water acts as Lewis acid.

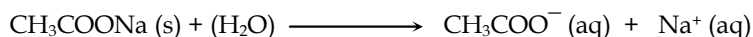
3.11 Salt Hydrolysis

A salt is an ionic compound formed by the combination of basic radical and acid radical. Most of the salts are strong electrolytes that completely dissociate into ions in water. The term **salt hydrolysis** describes the reaction of an anion or a cation of a salt, or both, with water. Salt hydrolysis usually

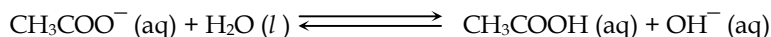
affects the pH of a solution. Depending upon the origin of salt and its nature (acidic, basic or neutral) in aqueous solution, salt hydrolysis can be categorized into following types.

1. Salts of strong bases and weak acids

The solution of a salt obtained from a strong base and a weak acid is basic. For example, the dissociation of sodium acetate (CH_3COONa) in water is given by

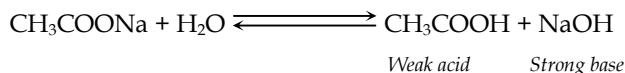


The hydrated Na^+ ion has no acidic or basic properties. The acetate ion CH_3COO^- , however, is the conjugate base of the weak acid CH_3COOH and therefore has an affinity for H^+ ions. The hydrolysis reaction is given by



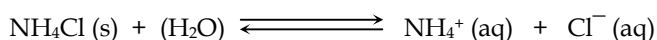
Because this reaction produces OH^- ions, the sodium acetate solution will be basic. In fact this is **anionic hydrolysis**.

For simplicity, this hydrolysis can be shown as:

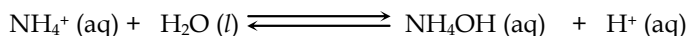


2. Salts of weak bases and strong acids

When a salt obtained from a strong acid such as HCl and a weak base such as NH_3 dissolves in water, the solution becomes acidic. For example, consider the process

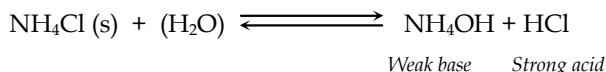


The Cl^- ion, being the conjugate base of a strong acid, has no affinity for H^+ and no tendency to hydrolyze. The ammonium ion NH_4^+ is the conjugate acid of weak base NH_3 and ionizes as follows:



Since H^+ ions are produced, the pH of the solution decreases. In fact this is cationic hydrolysis

For simplicity this hydrolysis can be shown as:



3. Salts of weak acids and weak bases

So far we have considered salts in which only one ion undergoes hydrolysis, for salts obtained from a weak acid and a weak base; both the cation and the anion hydrolyze.

Example: $\text{CH}_3\text{COONH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{NH}_4\text{OH}$

However, whether a solution containing such a salt is acidic, basic, or neutral depends on the relative strengths of the weak acid and the weak base.

- **$K_b > K_a$:** If K_b for the anion is greater than K_a for the cation, then the solution must be basic because the anion will hydrolyze to a greater extent than the cation. At equilibrium, there will be more OH^- ions than H^+ ions.
- **$K_b < K_a$:** Conversely, if K_b for the anion is smaller than K_a for the cation, the solution will be acidic because cation hydrolysis will be more extensive than anion hydrolysis.
- **$K_a \sim K_b$:** If K_a is approximately equal to K_b , the solution will be nearly neutral.

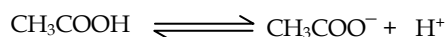
4. Salts of strong acids and strong bases

This type of salt in water gives cation and anion, and both of these ions do not hydrolyze. Therefore, the aqueous solution of such type of salt is neutral. For example, potassium chloride (KCl) when dissolved in water dissociates to give K^+ and Cl^- ions. Neither of these ions hydrolyze and therefore resulting solution will be neutral.

3.12 Common Ion Effect

It is previously discussed that weak electrolytes ionizes feebly while strong electrolytes ionize almost completely. What will happen to the ionization of weak electrolyte if strong electrolyte having one common ion is added?

Weak electrolytes are partially ionized and these are in equilibrium with their ions. If we consider a weak electrolyte acetic acid, then equilibrium of acetic acid with its ions is represented as,



Now applying the law of mass action,

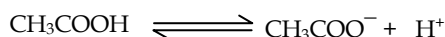
$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

In such equilibrium condition, if strong electrolyte having a common ion (say CH_3COONa) is added then some sort of change takes place.

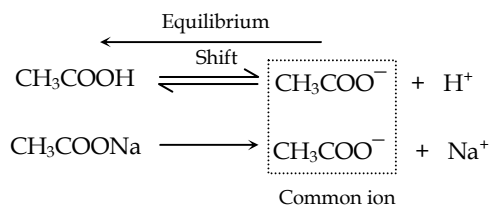
The concentration of CH_3COO^- increases. To keep the value of K_a constant either the concentration of H^+ should decrease or the concentration of CH_3COOH should increase or both take place according to Le-Chatelier's principle. This happens with H^+ ions of solution combine with CH_3COO^- ions to furnish undissociated CH_3COOH . This type of phenomenon takes place only when there is ion common to that furnished by weak electrolyte. The addition of common ion acts as stress in the equilibrium and the equilibrium shifts to LHS. This type of phenomenon is often called **common ion effect** which is defined as the suppression of ionization of weak electrolyte by the addition of strong electrolyte having a common ion.

Let us illustrate this by following example.

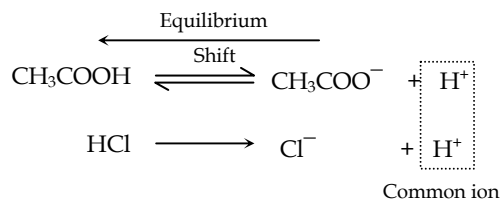
Acetic acid ionizes as follows.



When sodium acetate is added to the equilibrium,



The concentration of acetate ion increases which consumes more H^+ ions of solution to give more undissociated acetic acid. In this way ionization of acetic acid is suppressed. In this above case, if HCl is added instead of sodium acetate, the same type of phenomenon takes place due to common H^+ ions.

**Self Test**

3.7 What happens when (a) NH_4Cl (b) NaOH is added into the following equilibrium reaction?

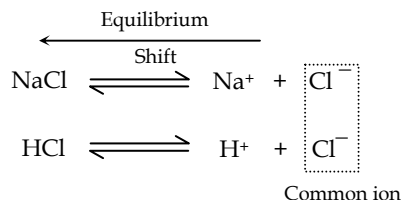
Effect of common ion on solubility

When a highly soluble salt is added to a sparingly soluble salt having at least one ion common then the solubility of sparingly soluble salt is decreased. This forms the fundamental principle in qualitative salt analysis. For example, when small amount of NaCl is added to AgCl solution then more precipitation of AgCl occurs. This is due to common chloride ion. In other words, the degree of dissociation of AgCl is suppressed due to common ion, which is also the decrease in the solubility of AgCl .

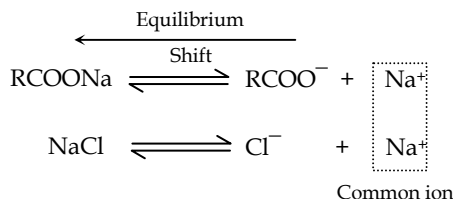
Application of common ion effect

The applications of common ion effect are as follows:

1. **Salting out of sodium chloride:** When dry HCl gas is passed into the saturated sodium chloride solution then chloride ions act as common ion and equilibrium shifts towards the back side. Due to this precipitation of NaCl takes place.



2. **Salting out of soap:** Soap is sodium salt of higher fatty acid. Sodium chloride is added into saturated soap solution to precipitate the soap. Here, Na^+ ion acts as common ion.



3. **In qualitative salt analysis:** Qualitative salt analysis is based on common ion effect and solubility product principle. This is discussed in solubility product principle.

3.13 Solubility Product and Solubility Product Principle

Solubility

Solubility of a substance is defined as the amount of solute (in gram) dissolved per hundred gram of solvent at a particular temperature to make it a saturated solution. At 25°C temperature 36 gram of sodium chloride is dissolved in 100 gram water to make it a saturated solution of sodium chloride. Here, solubility of NaCl at 25°C is 36.

$$\text{Solubility (s)} = \frac{\text{Weight of solute in gram}}{\text{Weight of solvent in gram}} \times 100$$

Solubility of a substance depends upon the property of solvent, nature of solute, temperature and presence of common ion. Chemists work so much with aqueous solution and hence need the knowledge of solubility of salt. In fact no salt is completely insoluble in water. Hence it is difficult to define a salt as soluble salt or insoluble salt. However, a salt can be said to be soluble if it can make a solution more concentrated than 0.01M (centimolar) near room temperature. On the same basis, a salt is said to be insoluble if its solubility is much less than 0.01 M.

Since no salt is completely insoluble in water, it is advisable to call it a sparingly soluble salt rather than insoluble salt.

Ionic product and solubility product

When a sparingly soluble salt such as AgCl is added into water, very small amount of it should be dissolved and that dissolved part of salt is almost completely dissociated in solution. The solubility of salt is often expressed in terms of mole per liter. Let us take an example of solubility of AgCl.

For the equilibrium, $\text{AgCl (solid)} \rightleftharpoons \text{Ag}^+ (\text{aq}) + \text{Cl}^- (\text{aq})$

We can apply law of mass action ,

$$\text{Equilibrium constant (K)} = \frac{[\text{Ag}^+][\text{Cl}^-]}{[\text{AgCl}]} \quad \dots \text{(i)}$$

Since concentration of solid silver chloride is almost constant,

$$K \cdot \text{AgCl} = [\text{Ag}^+][\text{Cl}^-]$$

$$\text{or } K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-] \quad \dots \text{(ii)}$$

$$\text{For simplicity, we write, } K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-] \quad \dots \text{(iv)}$$

If S mole lit^{-1} be the solubility of AgCl, equilibrium concentration of both Ag^+ and Cl^- ions will be S mol lit^{-1} , then $K_{\text{sp}} = \text{S mol. lit}^{-1} \times \text{S mol. lit}^{-1} = \text{S}^2 \text{ mol}^2 \cdot \text{lit}^{-2}$

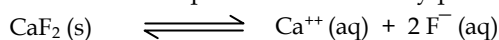
Here, K_{sp} is solubility product of AgCl. Since solubility of a salt is constant at a particular temperature, so does the solubility product.

The quantity $[\text{Ag}^+][\text{Cl}^-]$ is ionic product. The **ionic product** is defined as the product of molar concentration of the constituent ions, each concentration term raised to the power equals to the stoichiometric coefficient present in balanced ionic equilibrium.

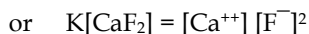
Note:

At equilibrium (in saturated solution of sparingly soluble salt), solubility product is equals to the ionic product.

Let us take another example to write solubility product expression.



$$K = \frac{[\text{Ca}^{++}][\text{F}^-]^2}{[\text{CaF}_2]}$$



Since concentration of CaF_2 is almost constant, we write K_{sp} for $K[\text{CaF}_2]$. So,

$$K_{\text{sp}} = [\text{Ca}^{2+}] [\text{F}^-]^2 \quad \dots \text{ (v)}$$

The quantity $[\text{Ca}^{2+}] [\text{F}^-]^2$ is ionic product of CaF_2 .

If molar solubility of CaF_2 is S mol/L, at equilibrium $[\text{Ca}^{2+}] = S$ mol/L and $[\text{F}^-] = 2S$ mol/L. Putting these values in equation (v), we get

$$K_{\text{sp}} = S \times (2S)^2 = 4S^3$$

For a general type of salt B_xA_y

The solubility product is, $K_{\text{sp}} = [\text{B}^{y+}]^x \cdot [\text{A}^{x-}]^y$

Hence, solubility product of a sparingly soluble salt is defined as the product of molar concentration of its constituent ions in its saturated solution, each concentration term being raised to power equal to stoichiometric coefficient present in balanced ionic equilibrium. Simply speaking, ionic product of saturated solution of sparingly soluble salt is **solubility product** of the salt.

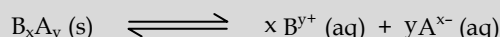
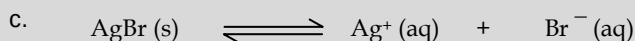
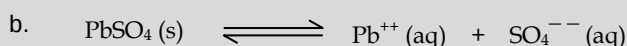
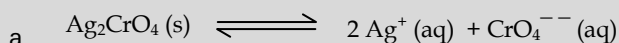
In the above illustration, we considered the case of saturated solution. However, if we consider the case of unsaturated or supersaturated solution, the case will be different.

In case of unsaturated solution, molar concentration of each ion is less than that in case of saturated solution. In such cases the ionic product is obviously less than in the case of saturated solution.

In case of supersaturated solution, the molar concentration of each ion is greater than that on saturated solution and hence ionic product is obviously greater than that the saturated solution.

Self Test

3.8 Write solubility product expression for

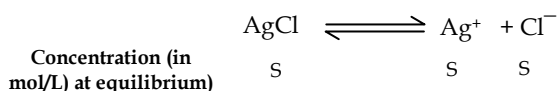


Relation between solubility and solubility product

Relation between solubility and solubility product of some different types of salt are given below:

- For AB types of salt:** Some examples of this type of salts are AgCl , AgBr , AgI , PbSO_4 , BaSO_4 , etc.

Let us take an example of AgCl . where S mol L^{-1} be the solubility of AgCl . It furnishes S mol L^{-1} which will be the solubility of AgCl . Then it furnishes S mol L^{-1} of Ag^+ and Cl^- ions in solution.



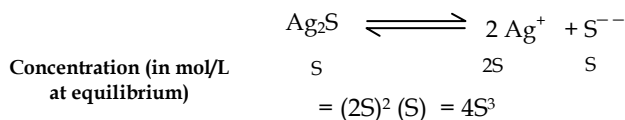
Now, Solubility product, $K_{\text{sp}} = [\text{Ag}^+] [\text{Cl}^-] = S \text{ mol L}^{-1} \times S \text{ mol L}^{-1} = S^2 \text{ mol}^2 \text{ L}^{-2}$

$$\text{Solubility } (S) = \sqrt{K_{\text{sp}}}$$

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2. **For A₂B or AB₂ type of salt:** Some examples of this type are Ag₂S, Ag₂CrO₄, CaF₂, BaF₂, Cu(OH)₂, etc. Let us take an example of Ag₂S, where solubility of Ag₂S be S mol L⁻¹. One mole of Ag₂S furnishes 2 mole of Ag⁺ ions and one mole of S²⁻ ions.

Now, Solubility product, $K_{sp} = [Ag^+]^2 [S^{2-}]$



$$\text{Solubility (S)} = \sqrt[3]{\frac{K_{sp}}{4}}$$

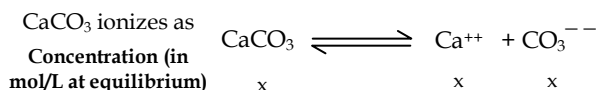
3. **In general: for A_x B_y salt:**

$$K_{sp} = x^x \cdot y^y \cdot s^{x+y}$$

Example, for Bi₂S₃: $K_{sp} = 2^2 \cdot 3^3 \cdot s^{2+3}$ or $K_{sp} = 108 S^5$ or $S = \sqrt[5]{\frac{K_{sp}}{108}}$

Example 3.5 The solubility product of CaCO₃ is 9.3×10^{-8} . Calculate its solubility in gram per litre.

Solution



Where x is the solubility of CaCO₃

$$\text{Solubility product } K_{sp} = [Ca^{++}] [CO_3^{--}]$$

$$9.3 \times 10^{-8} = x \cdot x$$

$$\therefore x = 3.049 \times 10^{-4} \text{ mol/L}$$

Note:

Molar concentration of Ca²⁺ in equilibrium = molar solubility of CaCO₃ = x mol/L

Mol. wt. of CaCO₃ = 100

Solubility in g/L = molarity $\times$ Mol.wt. = $3.09 \times 10^{-4} \times 100 = 3.049 \times 10^{-2} \text{ g/L}$

Example 3.6 The solubility of AgCl in water at 298 K is $1.43 \times 10^{-3} \text{ gL}^{-1}$, calculate its solubility in 0.5 M KCl solution.

Solution

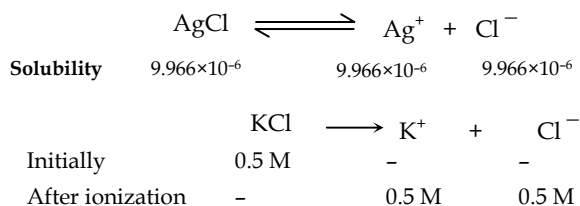
Method 1:

$$\text{Solubility of AgCl} = 1.43 \times 10^{-3} \text{ g/L}$$

$$\text{Mol. wt. of AgCl} = 108 + 35.5 = 143.5$$

$$\therefore \text{Solubility of AgCl in mole/L} = \frac{\text{g/L}}{\text{Mol. wt}} = \frac{1.43 \times 10^{-3}}{143.5} = 9.965 \times 10^{-6} \text{ mol/L}$$

$$\text{Solubility product of AgCl (K}_{sp}) = [Ag^+] [Cl^-] = (9.9654 \times 10^{-6}) \times (9.965 \times 10^{-6}) = 9.930 \times 10^{-11}$$



Total concentration of $\text{Cl}^- = (9.965 \times 10^{-6} + 0.5)$

Now, solubility product of AgCl (K_{sp}) = $[\text{Ag}^+][\text{Cl}^-]$

$$(9.965 \times 10^{-6})^2 = (9.965 \times 10^{-6} + 0.5) [\text{Ag}^+]$$

$$\therefore [\text{Ag}^+] = \frac{(9.965 \times 10^{-6})^2}{(9.965 \times 10^{-6} + 0.5)} = 1.985 \times 10^{-10} \text{ mol/L}$$

Method 2:

Since molar solubility of AgCl in pure water is $9.965 \times 10^{-6} \text{ mol/L}$

$$\begin{aligned} \text{its } K_{\text{sp}} &= [\text{Ag}^+][\text{Cl}^-] \\ &= 9.965 \times 10^{-6} \times 9.965 \times 10^{-6} \\ &= 9.930 \times 10^{-11} \end{aligned}$$

Lets suppose the molar solubility of AgCl in 0.5 M KCl is $S \text{ mol/L}$,

Equilibrium concentrations of different ions will be

$$[\text{Ag}^+] = S \text{ mol/L}$$

$$[\text{Cl}^-] = S + 0.5 \text{ M} \quad (S \text{ mol/L from AgCl are } 0.5 \text{ mol/L from KCl})$$

$\approx 0.5 \text{ M}$ (as S is very very small and can be neglected)

$$\text{So, } K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-]$$

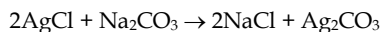
$$\text{or, } 9.930 \times 10^{-11} = S \times 0.5$$

$$\text{or, } S = \frac{9.930 \times 10^{-11}}{0.5} = 1.98 \times 10^{-10} \text{ mol/L}$$

Example 3.7 A sample of AgCl is treated with 5 mL of $2 \text{ M Na}_2\text{CO}_3$ solution to produce Ag_2CO_3 . The remaining solution contained 0.003 gram Cl^- per litre. Calculate the solubility product of AgCl . (K_{sp} of $\text{Ag}_2\text{CO}_3 = 8.2 \times 10^{-12}$)

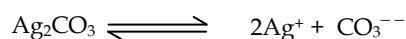
Solution

The reaction takes place as follows:



$$\text{Concentration of } \text{Cl}^- = \frac{\text{Gram lit}^{-1}}{\text{Gram ionic weight.}} = \frac{0.003}{35.5} = 8.45 \times 10^{-5} \text{ mol lit}^{-1}$$

For Ag_2CO_3



$$K_{\text{sp}} = [\text{Ag}^+]^2 [\text{CO}_3^{--}]$$

$$[\text{Ag}^+] = \sqrt{\frac{K_{\text{sp}}}{[\text{CO}_3^{--}]}} = \sqrt{\frac{8.2 \times 10^{-12}}{2}} = 2.024 \times 10^{-6} \text{ mol lit}^{-1}$$

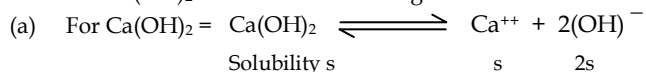
For AgCl

$$\begin{aligned} K_{\text{sp}} &= [\text{Ag}^+][\text{Cl}^-] \\ &= [2.024 \times 10^{-6}][8.45 \times 10^{-5}] \text{ mol}^2 \text{ lit}^{-2} \\ &= 1.71 \times 10^{-10} \text{ mol}^2 \text{ lit}^{-2} \end{aligned}$$

Example 3.8 A 500 mL of saturated solution of Ca(OH)_2 is mixed with equal volume of 0.4 M NaOH . What mass of Ca(OH)_2 is precipitated? (K_{sp} of $\text{Ca(OH)}_2 = 4.42 \times 10^{-5}$ at 25°C)

Solution

Strategy: Number of mol of Ca(OH)_2 precipitated = number of mol Ca(OH)_2 dissolved initially - number of mol of Ca(OH)_2 dissolved after adding 0.4 M NaOH .



$$\begin{aligned} K_{\text{sp}} &= [\text{Ca}^{++}][\text{OH}^-]^2 \\ &= s \cdot (2s)^2 = 4s^3 \end{aligned}$$

$$\text{Solubility (s)} = \sqrt[3]{\frac{K_{sp}}{4}} \quad \dots(i)$$

$$(b) \text{ For 500 mL of saturated } \text{Ca(OH)}_2 = \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{4.42 \times 10^{-5}}{4}} = 0.022273 \text{ mol lit}^{-1}$$

When NaOH is added to Ca(OH)_2 solution then some Ca^{++} or Ca(OH)_2 get precipitated

$$\text{After dilution concentration of solution} = \text{Initial concentration} \times \frac{\text{Initial volume}}{\text{Final volume}}$$

$$\text{Concentration of } \text{Ca}^{++} \text{ after mixing (M}_2) = \frac{0.022273 \times 500}{1,000} = 0.011136 = 1.1136 \times 10^{-2} \text{ mol/lit}$$

One mole of Ca(OH)_2 gives 2 mole of OH^- ions

$$\text{Total } [\text{OH}^-] = [\text{OH}^-] \text{ from } \text{Ca(OH)}_2 + [\text{OH}^-] \text{ from NaOH}$$

$$\begin{aligned} &= (2 \times 0.022273) \times \frac{500}{1,000} + 0.4 \times \frac{500}{1,000} \\ &= 0.2223 \text{ M} \end{aligned}$$

But due to addition of NaOH, Ca(OH)_2 precipitates due to common ion effect and concentration of Ca^{++} is suppressed. Then concentration of Ca^{++} in solution is calculated as;

Even after the precipitation, the K_{sp} value remains same while $[\text{Ca}^{++}]$ is solution change.

$$K_{sp} = [\text{Ca}^{++}_{\text{dissolved}}] [\text{OH}^-]^2$$

$$[\text{Ca}^{++}_{\text{dissolved}}] = \frac{K_{sp}}{[\text{OH}^-]^2} = \frac{4.42 \times 10^{-5}}{(0.2223)^2} = 8.84 \times 10^{-4} \text{ mol/lit}$$

$$\begin{aligned} \text{No. of mol. of } \text{Ca(OH)}_2 \text{ precipitated} &= \text{No. of mol. of } \text{Ca}^{++} \text{ precipitated} \\ &= \text{No. of mol. of } \text{Ca}^{++} \text{ before precipitation} - \text{No. of mol. of } \text{Ca}^{++} \text{ after precipitation} \\ &= 1.1136 \times 10^{-2} - 8.84 \times 10^{-4} \\ &= 0.010252 \times 10^{-4} \text{ mol} \\ &= 0.010252 \times 10^{-4} \times 74 \text{ g [weight = No. of mol.} \times \text{mol. wt.]} \\ &= 0.758 \text{ gram} \end{aligned}$$

Solubility Product Principle

An unsaturated solution has a capacity to dissolve more amount of solute at the same temperature. A saturated solution neither dissolves more amount of solute nor gives precipitate. A supersaturated solution gives precipitate of a salt. The same case can be studied in terms of ionic product and solubility product (K_{sp}).

- ◆ When ionic product is less than solubility product, the solution is unsaturated and can dissolve more amount of solute at the same temperature. Dissolution takes place here.
- ◆ When ionic product is equal to the solubility product, the solution is saturated and the solution neither dissolves more solute nor gives precipitate.
- ◆ When ionic product is greater than the solubility product the solution is supersaturated and precipitation takes place. This principle is called solubility product principle.

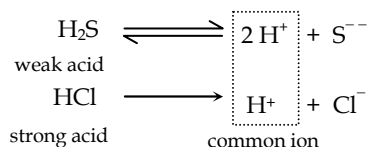
When two solutions are mixed, precipitate of a salt is formed if ionic product is greater than solubility product (K_{sp}).

Application of solubility product principle in salt analysis (in identification of basic radicals)

Solubility product principle helps to predict whether a solution gives or does not give precipitate in qualitative salt analysis. In presence of a common ion, the ionic product in solution is changed. When ionic product is greater than solubility product, precipitation of salt takes place.

Common ion effect provides the most important method to monitor the concentration of precipitant for the selective precipitation of metallic ions in qualitative analysis.

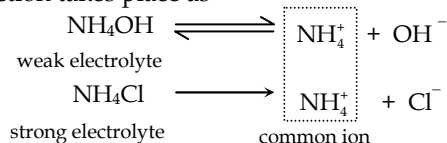
- 1. Precipitation of sulphides of group II metal ions:** Group II metal ions are Cd^{++} , Hg^{++} , Cu^{++} , Pb^{++} , Sn^{++} , As^{+++} , Sb^{+++} , Bi^{+++} . These metal ions are precipitated in the form of their sulphide (their sulphides have low solubility product). Therefore, H_2S gas is passed in presence of HCl . In absence of HCl , not only of group II but also of group IIIB metal ions are precipitate in the form of their metal sulphides due to high concentration of S^{--} ion. Precipitation of metal salt is based on the fact, if ionic product is greater than solubility product, precipitation occurs. To precipitate only group II metal ions the concentration of S^{--} should be decreased in such an extent that the ionic product would be greater than solubility product for sulphides of group II metal ions, while ionic product would be less than solubility product for sulphides of group IIIB metal ions. To suppress the concentration of S^{--} from H_2S , a strong electrolyte HCl is used.



Therefore, presence of HCl suppresses the ionization of H_2S and concentration of S^{--} would be decreased which is still sufficient to precipitate group II metal ions and insufficient to precipitate group IIIB metal ions (K_{sp} of group II metal sulphide is less than that of group III B metal sulphide) as their sulphides.

- 2. Precipitation of group III A metal ions:** Group III A includes Fe^{++} , Fe^{+++} , Cr^{+++} , Al^{+++} as basic radicals. In group IIIA, NH_4Cl is added along with NH_4OH to the solution. Here NH_4OH is precipitating agent. If only NH_4OH is used without NH_4Cl , not only of group IIIA metal ions are precipitated but also metal ions of subsequent group (eg. Mn^{++} , Mg^{++} etc.) are also precipitated out. The K_{sp} of hydroxides of group IIIA metal ion is relatively low. Therefore, suppressed concentration of OH^- ions still maintain $\text{IP} > K_{sp}$, hence they get precipitated. For $\text{Mg}(\text{OH})_2$, $\text{Mn}(\text{OH})_2$ etc. it becomes $\text{IP} < K_{sp}$ as they have high K_{sp} values.

Therefore, the amount of OH^- ion in the solution is sufficient to precipitate group IIIA metal ions but not others. Presence of NH_4Cl suppresses the ionization of NH_4OH due to common ion effect. The reaction takes place as



As the OH^- concentration is decreased, ionic product value of system is also decreases.

- 3. Precipitation of group IV metal ions:** Group IV metals ions are Ba^{++} , Sr^{++} , Ca^{++} . These basic radicals are precipitated in the form of their carbonate salt. $(\text{NH}_4)_2\text{CO}_3$ is the precipitating reagent. The reaction takes place in alkaline medium (NH_4OH is required). NH_4Cl suppresses the degree of ionization of $(\text{NH}_4)_2\text{CO}_3$ by common ion effect which decrease the concentration of CO_3^{--} ions which is sufficient to precipitate Ba^{++} , Sr^{++} , Ca^{++} (K_{sp} of group IV metal carbonate is fairly low). This concentration of CO_3^{--} ion is not sufficient to precipitate MgCO_3 as it has high solubility product.

■ Solved Questions

1. What is degree of ionization?

- Degree of ionization or dissociation (α) of an electrolyte is defined as the ratio of number of molecules which undergoes ionization or dissociation to the total number of molecules taken.

$$\alpha = \frac{\text{No. of molecules which undergo ionization or dissociation}}{\text{Total no. of molecules present}}$$

2. What is pH? How pH and pOH are related at 25°C?

- Negative common logarithm of molar concentration of hydrogen ion is called pH. The term pH is originated from the word *potenz de hydrogen*.

pH is defined as $\text{pH} = -\log [\text{H}^+]$

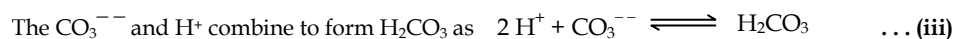
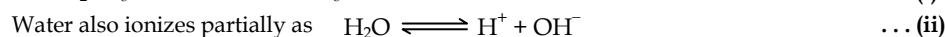
or, $\text{pH} = +\log \frac{1}{[\text{H}^+]}$

at 25°C $\text{pH} + \text{pOH} = 14$

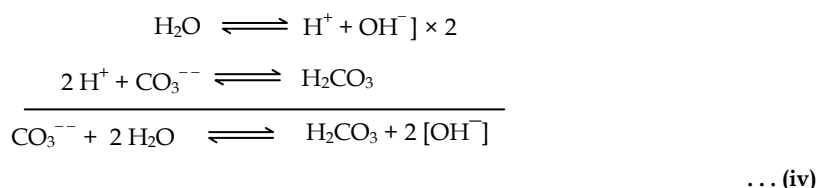
3. Why is aq. Na_2CO_3 basic?

- Aqueous Na_2CO_3 is basic due to anionic hydrolysis which can be explained as follows.

Na_2CO_3 dissociates as



The hydrogen ions necessary for equation (iii) is provided by further ionization of water in equation (2). Now adding equation (ii) and (iii) we get



Due to further ionization of water, thus produced H^+ are consumed by CO_3^{--} to form H_2CO_3 and solution contains excess OH^- . Therefore, the solution is basic. From equation (iv) it is seen that the alkaline nature of Na_2CO_3 is mainly due to hydrolysis of anion.

4. What is solubility product principle?

- The solubility product principle states that, "when two solutions containing ions are mixed together if ionic product is greater than solubility product, the precipitation takes place, otherwise not." It can be expressed as
- If $\text{I.P.} < K_{\text{sp}}$, the solution is unsaturated and can dissolve further salt at the same temperature.
 - If $\text{I.P.} = K_{\text{sp}}$, the solution is saturated and can't dissolve further salt at the same temperature.
 - If $\text{I.P.} > K_{\text{sp}}$, the solution is supersaturated solution which gives precipitation on slight disturbance.

■ Questions

■ Very Short Answer Questions:

- What is Ostwald's dilution law? How is degree of ionization related to concentration of solution?
- Why is Ostwald's dilution law not applicable for strong electrolyte?
- What is auto ionization of water?
- What are pH and pOH? How are they related?
- Define Bronsted-Lowry concept of acid and base. Give examples.
- Define Lewis concept of acid and base with examples.

- What is conjugate acid base pair? Give an example.
- What is buffer solution? Define buffer action and buffer range?
- What are solubility, solubility product, ionic product?
- What will happen when HCl gas is passed over a saturated solution of NaCl? Also explain the principle involved.
- Write suitable example to show that water acts as Bronsted-Lowry acid and base.
- Write the conjugate acid and conjugate base of NH_3 .
- Why is aqueous solution of NaCl neutral whereas that of Na_2CO_3 is basic?
- How do the following vary with temperature (a) Degree of ionization (b) ionisation constant (c) pH (d) Ionic product of water (K_w)?
- Why is NH_4Cl added before adding NH_4OH in the salt analysis of group IIIB metal ions?
- Why is HCl added before adding H_2S in the salt analysis of group II metal ions?

■ Short /Long Answer Questions:

- Explain common ion effect? Write its applications.
- What is solubility product principle? How is this principle helpful in the qualitative salt analysis?
- Write notes on:

a. Degree of ionization	b. Salt hydrolysis	c. pH and pH scale
d. Ionic product of water	e. Solubility product principle	
f. Common ion effect	g. Ostwald Dilution Law	
- Whether aqueous solution of the following compounds are acidic, basic or neutral. Give reason.

a. CuSO_4	b. FeCl_3	c. Na_2CO_3
d. NH_4Cl	e. CaCl_2	f. KCN
- State and explain Ostwald's dilution Law. Why is it not applicable for the strong electrolytes?

■ Numerical Problems

- What will be the H^+ ion concentration of a solution having pH 5.5?
Ans: $3.16 \times 10^{-6} \text{ mole L}^{-1}$
- 10^{-2} mol of KOH is dissolved in 10 litres of water. What will be the pH of the solution?
Ans: pH = 11
- Two litres of 1M HCl is mixed with one litre of 1M NaOH solution. Calculate the strength of the salt formed and the pH of the resulting solution.
Ans: Strength = 0.33M, pH = 0.48
- The solubility of CaF_2 in water at 18°C is $2.05 \times 10^{-4} \text{ mol L}^{-1}$. Calculate its solubility product.
Ans: $3.446 \times 10^{-11} \text{ mol/L}$
- The solubility product of BaSO_4 in water at 25°C is $1 \times 10^{-10} \text{ mol L}^{-1}$. Calculate the solubility of BaSO_4 in g/L. [Ba = 137]
Ans: Solubility of BaSO_4 = $2.33 \times 10^{-3} \text{ g/L}$
- A 500 mL of a saturated solution of $\text{Ca}(\text{OH})_2$ is mixed with equal volume of 0.4 M NaOH. What mass of $\text{Ca}(\text{OH})_2$ is precipitated. (K_{sp} of $\text{Ca}(\text{OH})_2$ is 4.42×10^{-5} .)
Ans: 0.754g
- 4.9 g of H_2SO_4 is present in 1000 ml of its solution. What is the pH of the solution?
Ans: 1
- Equal volumes of the solutions with pH = 2 and pH = 4 are mixed together. Calculate the pH of the resulting solution.
Ans: 2.3
- Equal volumes of the solutions with pH = 8 and pH = 10 are mixed together. Calculate the pH of the resulting solution.
Ans: 9.7

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10. What mass of NaOH should be dissolved in 1L of solution to prepare solution having pH = 12 at 25°C?
Ans: 0.4 g
11. Calculate the pH of 1 g/L NaOH solution
Ans: 12.4
12. A sample of AgCl is treated with 5 mL of 2 M Na₂CO₃ solution to produce Ag₂CO₃. The remaining solution contained 0.003 g of Cl⁻ per litre. Calculate the solubility product of AgCl (K_{sp} of Ag₂CO₃ = 8.2 × 10⁻¹²)
Ans: 1.20 × 10⁻¹⁰
13. The pH of 0.1 M HCN solution is 5.2. What is the value of ionization constant for the acid?
Ans: 3.98 × 10⁻¹⁰
14. Calculate the hydroxide ion concentration having pH = 10.5.
Ans: 3.16 × 10⁻⁴ mol/L
15. A solution having pH 6 is diluted 100 times. Calculate the pH of the resulting solution.
Ans: 6.96

Multiple Choice Questions

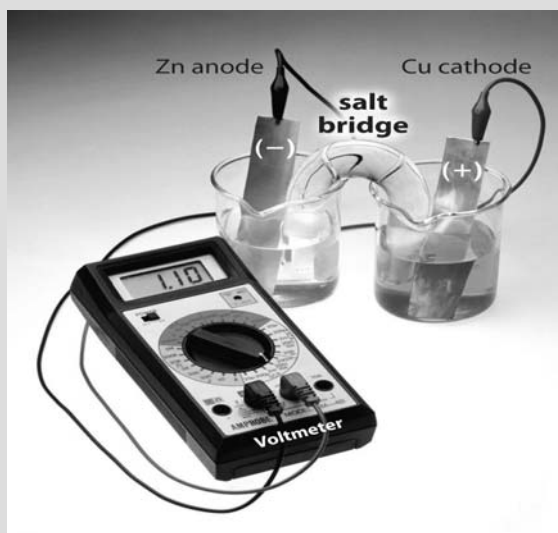
- The degree of dissociation of an electrolyte,
 - decreases with dilution
 - increases with dilution
 - may increase or decrease with dilution
 - is not affected by dilution
- In the reaction $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$, which of the following constitutes conjugate acid base pair?
 - NH₃ and H₂O
 - NH₄⁺ and OH⁻
 - H₂O and OH⁻
 - NH₃ and OH⁻
- The pH of 10⁻¹² M HCl is
 - 12.0
 - <7
 - 7.0
 - 12
- Which of the following can act both as Lowry - Bronsted acid and base?
 - CO₃²⁻
 - HSO₄⁻
 - BF₃
 - H₃O⁺
- The p^H of 0.015 N NaOH is
 - 1.82
 - 8.23
 - 12.18
 - 10.57
- The sum of pH and pOH at 25°C,
 - 7
 - 0
 - 7
 - 14
- The solution is 1% ionized having concentration 1 M. Find the dissociation constant:
 - 1 × 10⁻²
 - 1 × 10⁻⁴
 - 1 × 10⁻³
 - 1 × 10⁻⁵
- The solubility product of a sparingly soluble salt A₂B is 4.0 × 10⁻¹² at 298 K. Its molar solubility will be,
 - 4.0 × 10⁻³
 - 4.0 × 10⁻⁶
 - 1.0 × 10⁻⁶
 - 1.0 × 10⁻⁴
- Which of the following can act as a base according to the Lewis concept?
 - Cl⁻
 - Na⁺
 - H
 - K
- The degree of ionization of an electrolyte does not depend upon on,
 - temperature
 - nature of the solute
 - size of the solvent molecules
 - concentration of the solution
- pH of 0.001 M NaCl solution is
 - 7
 - <7
 - >7
 - can't be decided
- 10mL of 0.1N NH₄OH and 20 mL of 0.05 N HCl are mixed together. The pH of resulting solution is:
 - 7
 - <7
 - >7
 - can't be decided
- Solubility of AgCl is minimum in
 - water
 - 0.1 M NaCl
 - 0.1 M KCl
 - 0.1 M BaCl₂

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13
b	c	b	b	c	d	b	d	a	c	a	b	D

Electrochemistry

4



Galvanic cell is a device which converts chemical energy into the useful work (electrical energy).

4.1 Introduction

If a spoon is to be copper plated, assemble of electrodes and electrolyte is made in such a way that copper is made anode, spoon is made cathode and one of the soluble salt of copper is made electrolytic solution. Here copper is connected to the positive terminal and spoon is connected to the negative terminal of battery. Then electrolysis process is allowed to take place for hours. In this example, electricity produced from battery is used to cause chemical change. This type of cell is called electrolytic cell.

On the other hand, if a dry cell is considered, electricity is produced in the cost of electrochemical reaction within it. Dry cell is electricity generator. This type of cell is called galvanic cell or voltaic cell.

Both electrolytic cell and Galvanic cell is the domain of electrochemistry. Electrochemistry is that discipline which deals on the relationship between electrical energy and chemical change (redox reactions). The conduction in electrochemical cell is carried out either by electronic conductor or by electrolytic solution. Electrochemistry is important from theoretical as well as practical aspects. Purification of metals, electroplating of articles, commercial production of a number of chemicals such as Na, NaOH, Cl_2 etc. involve the principle of electrochemistry.

Galvanic (voltaic) cells have been used in torches, cell phones, digital cameras, hearing aids, digital watches, etc. Fuel cells, which utilize the redox reactions between the supplied gases to produce electricity have gained enormous amount of interest nowadays owing their high fuel efficiency, environment friendly operation and ease of maintenance.

4.2 Electronic and Electrolytic Conductors

The substances which allow the passage of electricity through them are called conductors. Metals, some alloys, graphite, acids, bases and salts can conduct electricity though in different extent. On the basis of mechanism of conduction the conductors can be classified into electronic (metallic) conductors and electrolytic (ionic) conductors.

Differences between electronic and electrolytic conductors

Electronic Conductors	Electrolytic Conductors
1. Those conductors which conduct electricity by the flow of electrons are called electronic conductors.	1. Those conductors which conduct electricity by the flow of cations or anions are called electrolytic conductors.
2. Metals, some alloys and graphite are examples of electronic conductors.	2. Acid, bases and salt solutions are example of electrolytic conductors.
3. Electronic conduction is faster than electrolytic conduction.	3. Electrolytic conduction is slower than electronic conduction.
4. During electronic conduction no chemical change takes place.	4. During electrolytic conduction chemical change takes place.
5. Transport of matter does not take place except the migration of electrons.	5. Transport of matter (ions) takes place.
6. On increasing the temperature, the electronic conduction decreases due to increase in hindrance in the flow of electrons.	6. On increasing the temperature, the electrolytic conduction increases usually due to decrease in viscosity of medium and increase in the velocity of ions.

4.3 Electrolytes: Strong and Weak Electrolytes

Refer Unit 3.2 (Page no. 68)

4.4 Arrhenius Theory of Ionization

Refer Unit 3.3 (Page no. 69)

4.5 Electrolytic Cells and Voltaic Cells

An electrochemical cell is an assemble of two electrodes dipped in same or different electrolyte solutions in which either electrical energy is produced as a result of chemical reaction or chemical reaction is driven by external supply of electrical energy. Simply speaking, the devices in which the electrochemical phenomena take place are called electrochemical cells. There are two types of electrochemical cells.

1. **Electrolytic cell:** The cell in which chemical reaction is performed by the external supply of electric current is called electrolytic cell. e.g. cell used in purification of metal and in electrolysis.
2. **Galvanic or voltaic cell:** The cell in which electrical energy is produced due to spontaneous chemical reaction (redox reaction) is called galvanic cell or voltaic cell. Panel cell, dry cell etc. are the examples of galvanic cell.

Differences between electrolytic cell and galvanic (voltaic) cell

Electrolytic cell	Galvanic cell
1. The cell in which chemical change is brought by the application of electrical energy is called electrolytic cell.	1. The cell in which electrical energy is produced as a result of chemical reaction is called galvanic cell.
2. The redox reaction is non-spontaneous.	2. The redox reaction is spontaneous.
3. Two electrodes are dipped in the same solution with in a same container.	3. Two electrodes with their respective electrolytes are usually set up in two container or these are separated by porous membrane.
4. Anode is positive and cathode is negative electrode.	4. Anode is negative and cathode is positive electrode.
5. Only one electrolytic solution is used.	5. Usually two different electrolytic solutions are used
6. No salt bridge or porous plug is used.	6. Salt bridge or porous plug is used.

4.6 Electrolysis and Criteria of Product Formation

(*Electro* = electricity, *lysis* = break down)

Electrolysis is the process of decomposition of electrolytes by the passage of electricity through its aqueous solution or molten form.

The apparatus used to carry out electrolysis is called electrolytic cell or voltameter.

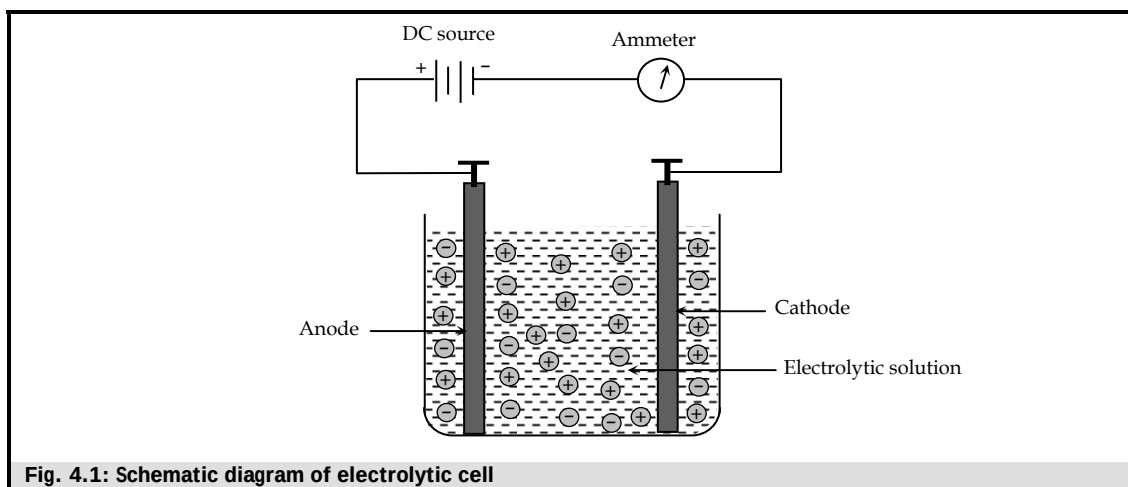


Fig. 4.1: Schematic diagram of electrolytic cell

Electrolytic cell consists two metallic rods or metallic plates dipped into electrolyte solution. The metallic rods or metallic plates are called **electrodes**. These electrodes are connected to the source of direct current. The electrode which is connected to the positive terminal of cell is called **anode** and the electrode which is connected to the negative terminal of cell is called **cathode**. The electrolytic solution may be aqueous solution or molten or fused state of electrolytes. The process of electrolysis can be explained on the basis of Arrhenius theory of ionization. When electric current is passed, the ions migrate towards the oppositely charged electrodes. It means, the anions (negatively charged ions) migrate towards anode (positive electrode) and cations (positively charged ions) migrate towards the cathode (negative electrode). Then these ions get discharged and become neutral chemical species.

Anions reach to anode and get oxidized by the loss of electrons. Oxidation takes place at anode. On the other hand cations reach to cathode and get reduced by the gain of electrons. Hence reduction takes place at cathode. The conversion of ions into corresponding neutral chemical species in corresponding electrode is called **primary change**. Thus obtained products may remain as such or they may undergo change to give final products. If products formed from primary change in electrolysis process further undergo change to give final products, it is called **secondary change**.

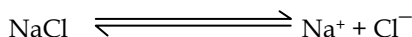
It is discussed that the products obtained in electrolysis are the substances obtained by the discharge of ions in corresponding electrodes. If molten or fused electrolyte is taken then the ions are cations and anions of electrolyte. If aqueous solution of electrolyte is taken then the ions present are some H^+ and OH^- ions obtained by ionization of water along with the cations and anions of electrolyte. When electric current is passed then cations of electrolyte and H^+ ions from water move towards cathode. Out of these two cations only one cation is discharged. Similarly anions of electrolyte and OH^- ions from water move towards anode at that condition. Here out of two anions, only one anion is discharged. Out of the different ions which one gets discharged depends upon relative discharge potentials and nature of electrode used.

In summary, the products obtained during electrolysis depend upon the following factors.

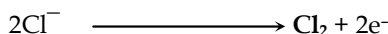
- i. Nature of electrolytes (Types of electrolytes)
- ii. State of electrolytes (Molten or aqueous solution)
- iii. Concentration of electrolyte solution
- iv. Nature of electrodes (Attackable or non-attackable)
- v. Charge density flown during electrolysis.

Examples:**i. Electrolysis of molten NaCl in presence of graphite anode and iron cathode (or Pt electrodes):**

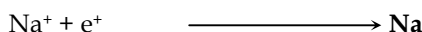
The platinum electrodes don't take part in oxidation reduction reaction with electrolyte solution so they are not-attackable electrode. When electricity is passed through molten NaCl then the ions are discharged at the corresponding electrodes.



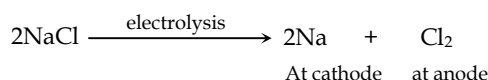
Half-reaction at anode (Oxidation)



Half-reaction at cathode (Reduction)

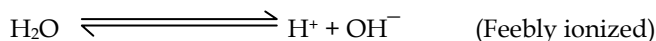
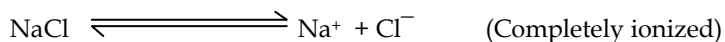


Cell Reaction,



As a result of electrolysis of molten NaCl, chlorine is obtained at anode and sodium is obtained at cathode. This forms the basis of **Down's process** in the extraction of sodium.

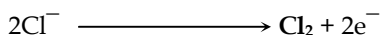
ii. Electrolysis of aqueous NaCl solution in presence of platinum electrodes: Aqueous NaCl solution consists Na^+ ions and Cl^- ions in addition with some H^+ ions and OH^- ions obtained by the ionization of H_2O



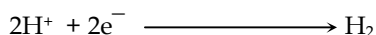
When electricity is passed through aqueous NaCl solution both cations (Na^+ and H^+) ions move towards the cathode. Out of these two, H^+ ion has lower discharge potential (The potential at which ions are converted into neutral atom is called discharge potential) than Na^+ ion (If the electrode is inert). So H^+ ions are discharged in preference to Na^+ ions.

Similarly both Cl^- and OH^- ions move towards anode. Here Cl^- has lower discharge potential and is discharged in preference to OH^- ions. So, following discharge reaction takes place.

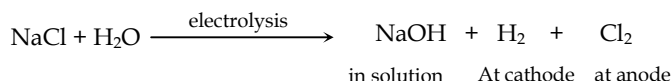
Half-reaction at anode (Oxidation)



Half reaction at cathode (Reduction)

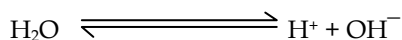
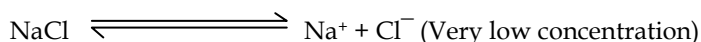


Cell Reaction:



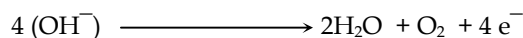
Therefore, when aqueous solution of NaCl is electrolysed, Cl_2 is obtained at anode and H_2 is obtained at cathode.

If very dilute solution of aqueous NaCl is electrolysed, the concentration of Na^+ and Cl^- is very low. So this type of electrolysis is almost equivalent to electrolysis of water. Here OH^- gets preference over Cl^- for discharge.

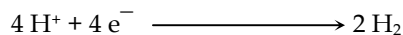


100 Modern Approach to Chemistry-II

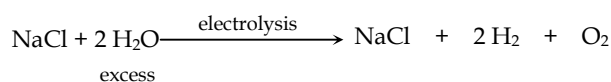
Half reaction at anode (Oxidation)



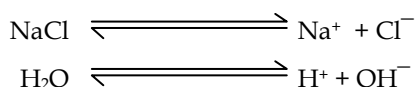
Half reaction at cathode (Reduction)



Cell Reaction

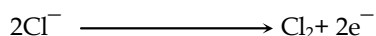


- iii. **Electrolysis of aqueous NaCl solution using graphite anode and mercury cathode:** The aqueous NaCl solution consists Na^+ and Cl^- ions along with some H^+ and OH^- ions.



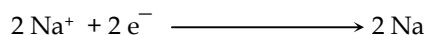
Half reaction at anode (Oxidation)

Out of Cl^- and OH^- ions, the Cl^- ions have lower discharge potential and are discharged in preference to OH^- ions.



Half reaction at cathode (Reduction)

The discharge potential of H^+ ions is lower than that of Na^+ ions under ordinary condition. If mercury is used as cathode, the discharge potential of H^+ ion will be high and it is discharged only at high voltage. This is called over voltage of hydrogen. In this condition Na^+ will be discharged in preference to H^+ ions. For Hg electrode, ease of discharge of cations: $\text{Na}^+ > \text{H}^+$



Thus obtained Na reacts with Hg of cathode and forms amalgam.



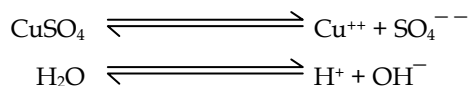
The sodium amalgam if reach to another chamber having mercury anode reacts with water to produce NaOH and H_2 .



*This forms the basic principle of manufacture of NaOH in **Castner Kellner Cell**.*

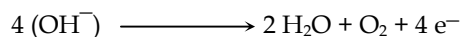
- iv. **Electrolysis of CuSO_4 solution using non attackable electrodes (Pt electrodes)**

Reaction:



Half reaction at anode

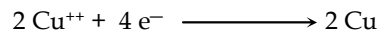
SO_4^{--} and OH^- ions migrate to anode. Discharge potential of OH^- is less than that of SO_4^{--} . So OH^- ions are discharged in preference to SO_4^{--} .



∴ Oxygen is released at anode or: Water is oxidized at anode.

Half reaction at cathode

Cu^{++} and H^+ migrate to cathode. Discharge potential of Cu^{++} is less than that of H^+ . So Cu^{++} is discharged in preference of H^+ .



∴ Copper is deposited at cathode.

v. Electrolysis of CuSO_4 solution using attackable electrodes

If copper electrodes are used then it act as attackable electrode.

Reaction:

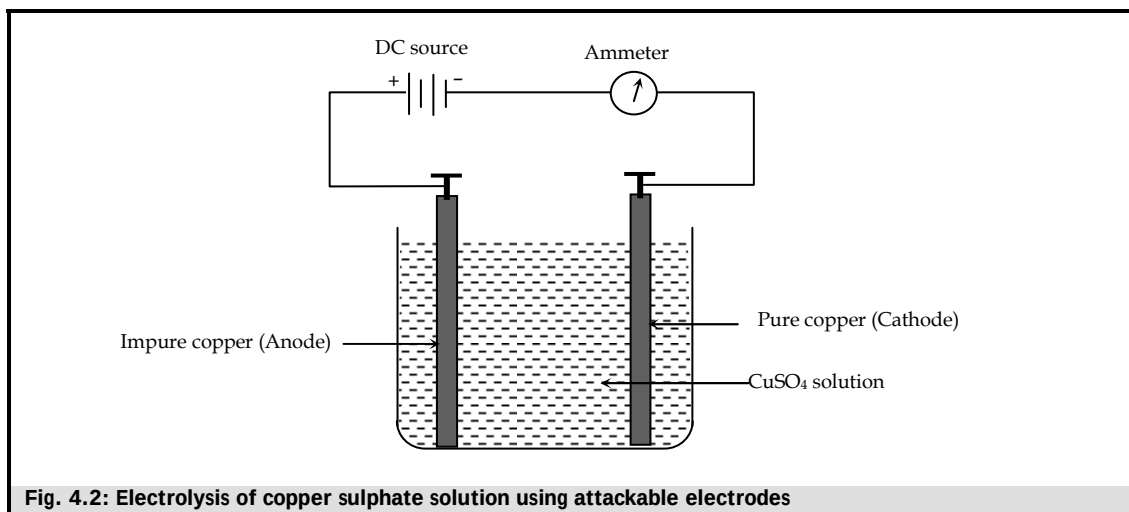
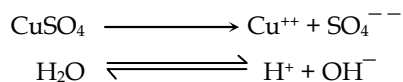
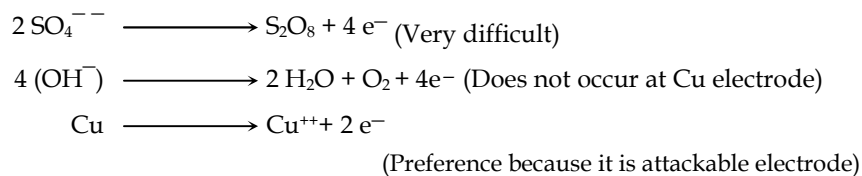


Fig. 4.2: Electrolysis of copper sulphate solution using attackable electrodes

Half reaction at anode (Oxidation)

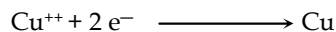
Reaction:



So, Cu anode undergoes dissolution.

Half reaction at cathode (Reduction)

Cu^{++} and H^{+} go to cathode. Discharge potential of Cu^{++} is lower than of H^{+} ions. So Cu^{++} ions are discharged in preference to H^{+} .



So, copper is deposited at cathode.

Note:

If the electrode is inert (non-attackable)

Discharge potential: $\text{Na}^{+} > \text{K}^{+} > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{Zn}^{2+} > \text{Fe}^{2+} > \text{H}^{+} > \text{Cu}^{2+} > \text{Ag}^{+} > \text{Au}^{3+}$

Ease of discharge at cathode: $\text{Na}^{+} < \text{K}^{+} < \text{Ca}^{2+} < \text{Mg}^{2+} < \text{Zn}^{2+} < \text{Fe}^{2+} < \text{H}^{+} < \text{Cu}^{2+} < \text{Ag}^{+} < \text{Al}^{3+}$

Discharge potential: $\text{SO}_4^{2-} > \text{NO}_3^{-} > \text{OH}^{-} > \text{X}^{-}$

Ease of discharge at anode: $\text{SO}_4^{2-} < \text{NO}_3^{-} < \text{OH}^{-} < \text{X}^{-}$

4.7 Quantitative Aspect of Electrolysis

The quantitative relationship between the amount of electricity passed and the mass of ion discharged in the voltameter was first investigated by *Michael Faraday* and he enunciated the following two laws in 1833. These laws are known as Faraday's law of electrolysis.

4.7.1 Faraday's First Law of Electrolysis

Faradays first law of electrolysis states "**The amount of substance discharged or liberated at the electrode during electrolysis is directly proportional to the quantity of electricity (charge) passed through the electrolyte solution.**"

Amount of substance discharged $\propto$ charge supplied

$$W \propto Q.$$

$$\text{or, } W = ZQ \quad \dots \text{(i)}$$

$$\text{or, } W = ZIt \text{ [Charge (Q) = current (I) } \times \text{ time (t)]} \quad \dots \text{(ii)}$$

where Z = proportionality constant and known as electrochemical equivalent. (ECE)

W = Amount of substance discharged,

Definition of Electrochemical Equivalent

We have, $W = ZIt$

If 1 ampere current is passed through the electrolyte solution for 1 sec (i.e. one *coulomb* charge) then, $W = Z$

Hence, **electrochemical equivalent of a substance is defined as the mass of substance (in gram) deposited or liberated by the passage of one coulomb charge.**

1 Faraday (96,500 *coulomb*) charge deposits 1 gram equivalent of any substance.

Let E be the equivalent weight of a substance

96,500 *coulomb* deposits E g of substance

1 *coulomb* deposits $\frac{E}{96500}$ g of substance.

$$\text{By definition, } Z = \frac{E}{96,500}$$

Unit of Z is *gram coulomb*⁻¹ or *gram ampere*⁻¹ *sec*⁻¹

$\therefore$ Faraday's first law can be mathematically expressed as,

$$W = ZIt.$$

$$W = \frac{EIt}{96,500}$$

$$\therefore \text{ Electrochemical equivalent of H} = \frac{1.008}{96500} = 0.0000104 \text{ g/coulomb}$$

$$\text{Electrochemical equivalent of Cu} = \frac{31.75}{96500} = 0.0003290 \text{ g/coulomb}$$

$$\text{Electrochemical equivalent of Ag} = \frac{107.7}{96500} = 0.0011160 \text{ g/coulomb}$$

$$\text{Electrochemical equivalent of Zn} = \frac{32.6}{96500} = 0.0003378 \text{ g/coulomb}$$

4.7.2 Faraday's Second Law of Electrolysis

It is dealt that when electric current is passed into the electrolytic solution then certain amount of substances are discharged or liberated at the electrode.

If such voltmeters are connected in series, the amount of substance deposited or liberated at the respective electrodes depends upon the equivalent weight or chemical equivalent of the respective substance. This is the Faraday's second law of electrolysis.

It states, **When same quantity of electricity is passed through different electrolytic solution connected in series, the mass of substance deposited or liberated at the respective electrodes is directly proportional to its chemical equivalent or equivalent weight.**

Mass of substance discharged (W) $\propto$ chemical equivalent (E)

$$W \propto E$$

or, $W = kE$ (k is proportionality constant)

$$\text{or, } \frac{W}{E} = k.$$

i.e. $\frac{\text{Mass of substance deposited}}{\text{Chemical equivalent}} = \text{Constant.}$

$$\text{Accordingly, } \frac{W_1}{E_1} = \frac{W_2}{E_2} = \frac{W_3}{E_3}$$

Let the same quantity of electricity be passed through the different voltmeters connected in series containing acidulated water, CuSO_4 and AgNO_3 as shown in the figure.

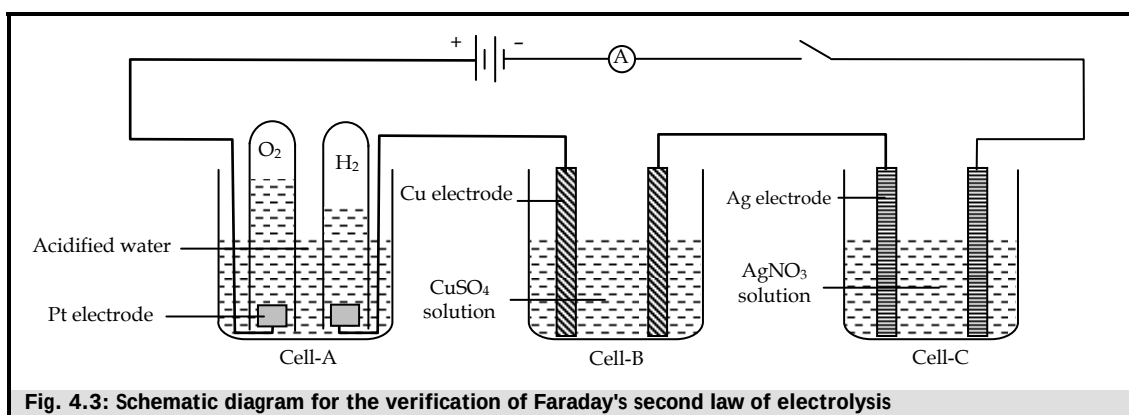


Fig. 4.3: Schematic diagram for the verification of Faraday's second law of electrolysis

Since the connection is in series, the same quantity of electricity is passed to all the voltmeters.

So, according to Faraday's second law of electrolysis,

For cell A: Mass of hydrogen displaced $\propto$ Chemical equivalent (or Equivalent weight) of hydrogen.

$$\therefore \frac{\text{Mass of hydrogen displaced}}{\text{Equivalent weight of hydrogen}} = \text{Constant} \quad \dots \text{ (i)}$$

For cell B: Mass of copper deposited $\propto$ Equivalent weight of Cu

$$\therefore \frac{\text{Mass of Cu deposited}}{\text{Equivalent weight of Cu}} = \text{Constant} \quad \dots \text{ (ii)}$$

For cell C: Mass of silver deposited $\propto$ Equivalent weight of Ag

$$\therefore \frac{\text{Mass of Ag deposited}}{\text{Equivalent weight of Ag}} = \text{Constant} \quad \dots \text{ (iii)}$$

From equation (i), (ii) and (iii)

$$\frac{\text{Mass of hydrogen displaced}}{\text{Equivalent weight of hydrogen}} = \frac{\text{Mass of Cu deposited}}{\text{Equivalent weight of Cu}} = \frac{\text{Mass of Ag deposited}}{\text{Equivalent weight of Ag}}$$

Relation between electrochemical equivalent (Z) and chemical equivalent (E)

From Faraday's first law of electrolysis, $W = ZIt$.

When $Q = 1 \text{ coulomb}$ then $W = Z$. Hence electrochemical equivalent can be defined as the mass of substance deposited by passing one *coulomb* of charge in voltameter. **The amount of substance in gram liberated or deposited by passing 96,500 coulomb charges is chemical equivalent or equivalent weight.**

Also we know,

One Faraday or 96500 *coulomb* charge deposits one gram equivalent of any substance (E) g of substance.

or, 1 *coulomb* charge deposits $= \frac{E}{96,500} \text{ g}$ of the substance

By the definition of electrochemical equivalent, $Z = \frac{E}{96,500}$

Also, $Z = \frac{1}{96,500} \times E$.

$Z \propto E$ where $\frac{1}{96,500}$ is constant.

For two substance 1 and 2.

$Z_1 \propto E_1$ and $Z_2 \propto E_2$

$$\therefore \frac{Z_1}{E_1} = \frac{Z_2}{E_2}$$

$\therefore$ Electrochemical equivalent of element is directly proportional to its equivalent weight.

Alternatively,

From Faraday's first law of electrolysis,

For two different substances in the voltameter if same quantity of charge is passed,

$$W_1 = Z_1 Q \quad \dots (i)$$

$$W_2 = Z_2 Q \quad \dots (ii)$$

Now equation (i) is divided by equation (ii), we get

$$\frac{W_1}{W_2} = \frac{Z_1}{Z_2} \quad \dots (iii)$$

From Faraday's second law of electrolysis,

$$\frac{W_1}{W_2} = \frac{E_1}{E_2} \quad \dots (iv)$$

From equation (iii) and (iv)

$$\frac{Z_1}{Z_2} = \frac{E_1}{E_2}$$

$$\text{or } \frac{Z}{E} = \text{constant}$$

$$\text{or } Z = \text{constant} \times E$$

$$\text{or } Z \propto E$$

Note:

96500 C (1F) charge can discharge or deposit or liberate one gram equivalent of any substance.

■ Solved Numerical Problems

Example 4.1 Show that 1 Faraday charge is approximately equal to 96,500 coulombs.

Solution

One Faraday is the charge of one mole of electrons

1F = Charge of one mole of electrons

= 1 mole of electron $\times$ charge of one electron.

= $6.023 \times 10^{23} \times 1.60 \times 10^{-19}$ C.

= 96488.46 *Coulomb*.

$\approx$ 96500 *Coulomb*.

Example 4.2 5 Ampere current is passed into a voltmeter containing CuSO_4 solution for 15 minutes (having Cu-electrode): (a) Calculate the mass of copper metal deposited. (b) If mass of each electrode is 5 g initially what will be the masses of anode and cathode after electrolysis? (c) What will be the mass of Cu metal deposited if current efficiency is only 75%?

Solution

(a) **Method 1:**

Total quantity of electricity (Q) = Current $\times$ time in second

= $5 \text{ A} \times (15 \times 60 \text{ sec.})$

= 4500 *coulomb*.

Atomic mass of Cu = 63.5

Equivalent mass of Cu = $\frac{63.5}{2} = 31.75$

We know,

96500 *coulomb* charge deposits 31.75g of Cu

4500 *coulomb* charge deposits $\frac{31.75}{96500} \times 4500 = 1.480 \text{ g}$

Method 2:

$$W = \frac{EIt}{96500} = \frac{31.75 \times 5 \text{ Ampere} \times (15 \times 60 \text{ sec})}{96500} = 1.480 \text{ g}$$

(b) After electrolysis mass of anode is decreased and mass of cathode is increased

$\therefore$ Mass of anode = $5 - 1.48 \text{ g} = 3.52 \text{ g}$

$\therefore$ Mass of cathode = $5 + 1.48 \text{ g} = 6.48 \text{ g}$

(c) If current efficiency is 75%

The quantity of electricity (Q) = Current $\times$ time in second $\times$ current efficiency

$$Q = 5 \text{ A} \times (15 \times 60 \text{ sec}) \times \frac{75}{100}$$

= 3375 *coulomb*

96500 *coulomb* charge deposits = 31.75 g of Cu.

3375 *coulomb* charge deposits = $\frac{31.75}{96500} \times 3375 \text{ g}$

= 1.110 g

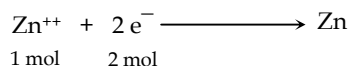
Example 4.3 Calculate the number of *coulomb* of electricity required:

a. to discharge 0.3 mol of Zn^{2+}

b. to deposit 4g of Mg from MgSO_4 .

Solution

(a) **Method 1:**



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1 mole of Zn^{++} is discharged by 2 moles of electrons.

0.3 mole of Zn^{++} is discharged by 2×0.3 mol of electrons.

$$= 0.6 \text{ mol of electron} = 0.6 \text{ Faraday} = 0.6 \times 96500 \text{ coulomb}$$

$$= 57900 \text{ coulomb}$$

[Charge of 1 mol of electron is 1 F or 96500 coulomb]

Method 2:

1 mole of Zn^{++} = 65.3 g, Eq. wt. of Zn = 32.65

$\therefore$ 0.3 mole of Zn^{++} = 0.3×65.3 g

$$W = \frac{EQ}{96500}$$

$$Q = \frac{W \times 96500}{E} = \frac{0.3 \times 65.3 \times 96500}{32.65} = 57900 \text{ coulombs.}$$

(b) 1 gram equivalent of Mg = 12 g

Since, 1 g equivalent of metal is deposited by 96500 coulomb of charge,

12 g of Mg is deposited by 96500 coulomb.

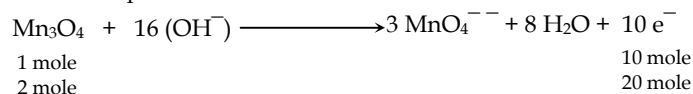
$$4 \text{ g of Mg is deposited by } \frac{96500}{12} \times 4 = 32166.66 \text{ coulomb}$$

[Do yourself by using second method]

Example 4.4 Calculate the charge involved to oxidize 2 moles of Mn_3O_4 to MnO_4^- in alkaline medium.

Solution

The balanced chemical equation is



From the above balance chemical equation,

Oxidation of 1 mole of Mn_3O_4 involves 10 mole of e^-

Oxidation of 2 mole of Mn_3O_4 involves 20 moles of e^- = 20 Faraday = 20×96500 coulomb

$$= 1930000 \text{ coulombs}$$

Example 4.5 75 milliampere current was passed through acidified water and metal nitrate solution for same time. 60 cc of H_2 at NTP and 0.57 g of metal deposited were measured. Calculate the equivalent weight of metal.

Solution

From the Faraday's second law of electrolysis, mass of hydrogen evolved = volume of H_2 in cc $\times 0.000089$ g/cc

$$[\because \text{Density of } \text{H}_2 = \frac{2}{22400} \text{ g/cc} = 0.000089 \text{ g/cc}]$$

$$= 60 \times 0.000089 \text{ g} = 0.00534 \text{ g}$$

$$\frac{\text{Mass of hydrogen evolved}}{\text{Eq. wt. of hydrogen}} = \frac{\text{Mass of metal deposited}}{\text{Equivalent weight of metal}}$$

$$\text{or, } \frac{0.00534}{1.008} = \frac{0.57}{\text{Equivalent weight of metal}}$$

$$\therefore \text{Equivalent weight of metal} = \frac{0.57 \times 1.008}{0.00534} = 107.59$$

Example 4.6 Same quantity of electricity is passed through acidified water and a metal sulphate solution. The volume of H_2 gas produced is 55.30 cc at 730 mm Hg and 17°C . The mass

of metal deposited is 0.0397 g. The specific heat of the metal is 0.236. Find the formula of the metal sulphate.

Solution

For H₂ gas

At given condition

$$P_1 = 730 \text{ mm Hg}$$

$$V_1 = 55.30 \text{ cc}$$

$$T_1 = 17 + 273$$

At NTP

$$P_2 = 760 \text{ mm Hg}$$

$$V_2 = ?$$

$$T_2 = 273 \text{ K}$$

$$= 290 \text{ K}$$

From combined gas equation, we have

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\text{or } V_2 = \frac{P_1 V_1}{T_1} \times \frac{T_2}{P_2} = \frac{730 \times 55.30}{290} \times \frac{273}{760} = 50.00 \text{ cc}$$

$$\text{Mass of H}_2 = \text{Vol. of H}_2 \text{ at NTP in cc} \times \frac{2}{22400} \text{ g/cc} = 50 \times 0.000089 \text{ g} = 0.00445 \text{ g}$$

From Faraday second law of electrolysis,

$$\frac{\text{Mass of H}_2 \text{ displaced}}{\text{Equivalent weight of hydrogen}} = \frac{\text{Mass of metal deposited}}{\text{Equivalent weight of metal}}$$

$$\text{or } \frac{0.00445}{1.008} = \frac{0.0397}{\text{Equivalent weight of metal}}$$

$$\therefore \text{Equivalent weight of metal} = \frac{0.0397 \times 1.008}{0.00445} = 8.99$$

From Dulong and Petit's law, we have

$$\text{Approximate atomic weight} \times \text{specific heat} \approx 6.4$$

$$\therefore \text{Approximate atomic weight} = \frac{6.4}{\text{sp heat}} = \frac{6.4}{0.236} = 27.11$$

$$\text{Valency} = \frac{\text{Approximate atomic weight}}{\text{Equivalent weight}} = \frac{27.11}{8.99} = 3.016$$

Valency of metal can't be fractional. Therefore, valency of metal = 3

$$\therefore \text{Formula of metal sulphate } M_2(SO_4)_3.$$

Example 4.7 A metallic spoon is coated with silver by passing a current of 5 ampere through AgNO₃ solution for 5 hrs. What is the thickness of silver plating if the area of the spoon is 12 cm² (density of silver is 10.5 g cm⁻³).

Solution

If ρ is the density of metal, θ is thickness of plating and A is surface area of the tray

$$W = \frac{EIt}{F}$$

$$\text{or, } \rho \times V = \frac{EIt}{F}$$

$$\text{or, } \rho A \theta = \frac{EIt}{F}$$

$$\theta = \frac{EIt}{\rho AF} = \frac{108 \times 5 \times (5 \times 60 \times 60)}{10.5 \times 12 \times 96500} = 0.799 \text{ cm}$$

Solved Questions

1. What is E.C.E. (Electro-chemical Equivalent)?

- Electrochemical equivalent of a substance is defined as the mass of substance deposited or liberated by the passage of 1 ampere current for one second. Example: e.c.e of $H_2 = \frac{1.008}{96500}$ g/coulomb

OR

Electrochemical equivalent of a substance is defined as the mass of substance deposited or liberated by the passage of 1 coulomb charge.

2. State Faraday's first law of electrolysis

- The amount of substance discharged or liberated at the electrode during electrolysis is directly proportional to the quantity of charge or electricity passed through the electrolyte.

Mathematically,

Amount of substance discharged (W) $\propto$ quantity of charged passed (Q)

$$W \propto Q.$$

$$W = ZQ$$

$$W = ZIt \quad [\text{charge (Q) = current (I) } \times \text{ time (t)}]$$

where z = proportionality constant and known as electrochemical equivalent. (ECE)

3. State Faraday's second law of electrolysis.

- It states that "when the same quantity of electricity is passed through the different electrolytes connected in a series, the amount of substance deposited or liberated at the respective electrodes are proportional to their chemical equivalent."

Mathematically,

Amount of substance discharged (W) $\propto$ chemical equivalent (E)

$$W \propto E$$

$$W = kE$$

$$\frac{W}{E} = k.$$

$$\text{i.e. } \frac{\text{Mass of substance deposited}}{\text{Chemical equivalent}} = \text{Constant.}$$

Questions

Very Short Answer Questions:

- Define the following terms:
 - Electro-chemical equivalent
 - Chemical equivalent
 - One Faraday charge
 - One ampere current
 - Electrochemical cells
- Write the differences between electronic conduction and electrolytic conduction.
- Justify the fact that value of 1 F is 96500 C
- Define the term electrochemical equivalent (ECE). Derive its relation with chemical equivalent.
- Calculate the charge on:
 - 1 mol of $Cr_2O_7^{2-}$
 - 4 g equivalent of Na^+
 - 3×10^{23} electrons

Short /Long Answer Questions:

- State and explain Faraday's first law of electrolysis.
- State and explain Faraday's second law of electrolysis.
- Distinguish between (a) Electrolytic and voltaic cell (b) Electrochemical equivalent and chemical equivalent.

4. Write short notes on: (a) Faraday's laws of electrolysis (b) Arrhenious theory of ionization (c) Electronic and electrolytic conductors.

■ Numerical Problems

- How many coulombs of electricity are required to discharge 0.1 mole of Na^+ ?
[1 Faraday = 96500 coulomb]
Ans: 9650 coulomb
- How long a current of 3 ampere has to be passed through a solution of AgNO_3 to coat a metal surface of 80 cm^2 with 0.005 mm thickness? (density of Ag = 10.5 g/cc).
Ans: 125.09 seconds
- What current is required to deposit whole copper from 1 litre of 1M CuSO_4 solution by passing electricity through it for 10 minute.
Ans: 321.66 A
- A current of 2.5 ampere is passed through the solution of a metal sulphate for 30 minutes and deposits 1.52 g of metal at cathode. Find the equivalent weight of the metal.
Ans: 32.59
- How many grams of silver could be plated out on a serving tray by passing electricity through a solution of Ag (I) salt for 8 hours at a current of 9 ampere? What is the area of the tray, if thickness of the silver plating is 0.002 cm ? Density of silver is 10 g/cm^3 . (atomic mass of Ag = 107.8).
Ans: Weight of Ag = 289.5 g; Area of tray = 14475 cm^2
- 1.52 g of a trivalent metal M was deposited at cathode by passing a current of 2.5 ampere through its salt solution (metal sulphate) for 30 minutes. What is the atomic mass of M?
Ans: 97.78 amu
- How many number of coulombs are required to deposit 81 g of aluminium when the electrode reaction is: $\text{Al}^{+++} + 3\text{e}^- \rightarrow \text{Al}$?
Ans: 868,500 coulombs
- Equal amount of current was passed through an aqueous solution of trivalent metallic salt and dil. H_2SO_4 . The volume of H_2 liberated was 96.5 mL at 27°C and 765 mm Hg pressure and mass of metal deposited was 0.74 g. Calculate atomic mass of the metal.
Ans: 284.45 amu

4.8 Electrolytic Conduction

In the previous heading we have discussed about conductors. The conductor may be electronic or electrolytic. When current is passed the electrolytic conductors undergo decomposition due to the discharge of ions. Some electrolytes ionize almost completely and furnish more number of ions which are called **strong electrolytes**. The example includes strong acid (HCl, H₂SO₄, HNO₃, etc.), strong base (NaOH, KOH, etc.), most of the inorganic salts (NaCl, KCl, etc.). Some electrolytes undergo feeble ionization and furnish low number of ions in solution and offer poor conduction. These are called **weak electrolytes**. These include weak acid (CH₃COOH, HCN, H₂CO₃), weak base (Cu(OH)₂, Zn(OH)₂, Al(OH)₃, NH₄OH etc.) and few inorganic sparingly soluble salts.

Electrolytic conduction not only depends upon the nature of electrolytes but also to the nature of solvents. When electrolytes are added into solvents, some sort of interaction take place between ions and solvent. Electrolytes ionize more in polar solvent than in non-polar solvent. The viscosity of solvent is another contributing factor. Inter-ionic attraction, solvation of ions and viscosity of solvents are influenced by temperature. Therefore, effect of temperature can't be ruled out.

Another most important factor that affects the electrolytic conduction is concentration of solution. Before discussion about this we should be familiar with Ohm's law and related terms which are given below.

Self Test

4.1 Write factors that affect electrolytic conduction.

Ohm's Law: Every substance offers resistance to the flow of electricity to some extent. The hindrance offered by a substance to the flow of electric charge through the solution is called resistance. It is denoted by R . The relationship among voltage or emf (E), resistance (R) and current flowing (I) is shown by Ohm's law. **Ohm's law states, "physical conditions (temperature, mechanical strain, etc.) remaining constant, the current flowing through a conductor is directly proportional to the potential difference (V) of the two points."**

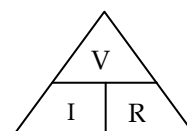
$$I \propto V$$

$$\text{or } I = \frac{1}{R} \cdot V$$

Where R is proportionality constant and is called resistance of the conductor

$$\text{or } V = IR$$

The resistance (R) of a conductor is



- i. Directly proportional to the length of the conductor (l) i.e. $R \propto l$

... (i)

- ii. Inversely proportional to the area of cross-section of the conductor (A). i.e., $R \propto \frac{1}{A}$

... (ii)

From equation (i) and (ii)

$$R \propto \frac{l}{A}$$

$$\text{or } R = \rho \frac{l}{A}$$

... (iii)

Where ρ is proportionality constant and is called **resistivity or specific resistance**.

Definition of resistivity

When $l = 1 \text{ cm}$ and $A = 1 \text{ cm}^2$ then from relation (iii), we get $R = \rho$.

Hence, **specific resistance or resistivity of a wire is defined as the resistance offered by the wire of unit length and unit area of cross section.**

From above relation,

$$\rho = \frac{RA}{l}$$

The S.I. unit of resistance is *Ohm* (Ω), area of cross-section is m^2 and length of wire is m , therefore, the unit of resistivity is *Ohm meter*.

$$\rho = \frac{\text{Ohm} \times \text{meter}^2}{\text{meter}} = \text{Ohm meter} = \Omega \text{ m}$$

Graphical representation of Ohm's law

The equation $I = \frac{1}{R} V$ is in the form of $y = mx$.

So, if a graph of current (I) versus voltage (V) is plotted, a straight line passing through the origin is obtained. This justifies the validity of Ohm's law. The metallic conductor is called Ohmic conductor.

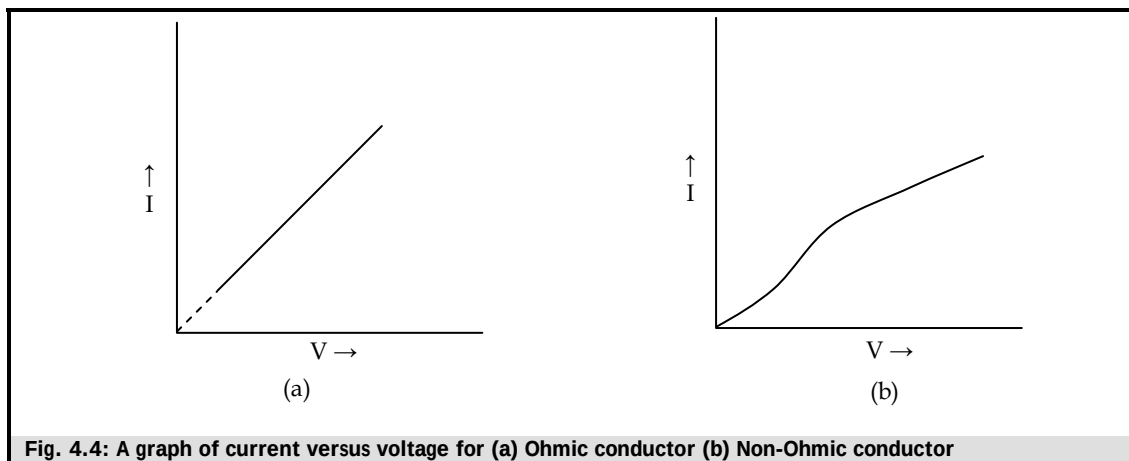


Fig. 4.4: A graph of current versus voltage for (a) Ohmic conductor (b) Non-Ohmic conductor

In case of electrolytic solution (e.g. CuSO_4 solution, NaCl solution etc.) the graph of current versus voltage is not straight line. This type of conductor is called non-Ohmic conductor.

Resistance in electrolytic solution: Like ohmic conductor, electrolyte solution also obeys Ohm's law. With respect to electrolyte solution, the Ohm's law can be stated as "the current flowing through an electrolyte solution is directly proportional to the emf or potential difference applied".

Specific Resistance (ρ)

It is already mentioned that like Ohmic conductor electrolyte solutions also obey the Ohm's law.

If electrolyte solution is kept in a tube having length $l \text{ cm}$ and area of cross-section $A \text{ cm}^2$ then resistance (R) offered by electrolyte solution is directly proportional to the length (l) of tube and inversely proportional to the area of cross-section (A) of the tube.

When $l = 1 \text{ cm}$ and $A = 1 \text{ cm}^2$ then $R = \rho$

Hence, **specific resistance or resistivity (ρ) is defined as the resistance offered by electrolyte solution kept in a tube having unit length and unit area of cross-section.**

A tube having unit length and unit area of cross-section will have unit volume. Hence, alternatively, **specific resistance is defined as the resistance offered by electrolyte solution kept in a cube having unit volume.**

In short, the resistance offered by 1 cc electrolyte solution is specific resistance.

Conductance (G)

The reciprocal of resistance is called conductance. It may be represented by G.

$$\text{Conductance (G)} = \frac{1}{\text{Resistance (R)}}$$

In case of electrolyte solution, the term conductance is frequently used rather than resistance.

Unit of conductance

The unit of conductance is Ohm^{-1} or Ω^{-1} or *mho* or *siemens*.

Types of conductance

Electrolytic conduction depends upon number of ions and their mobility in the solution. For pure electronic conductor the number of atoms per unit volume is fixed. However, electrolyte solution is prepared by dissolving electrolyte into solvent. The degree of ionization is also different for different electrolyte. Therefore, it is convenience to study conductance in terms of,

1. Specific conductance or conductivity (κ)
2. Equivalent conductance (λ)
3. Molar or molecular conductance (μ)

1. Specific conductance or conductivity (κ)

The reciprocal of specific resistance is called specific conductance and it is denoted by κ (kappa). From the above relation,

$$R = \rho \frac{\ell}{A}$$

$$\frac{1}{\rho} = \frac{1}{R} \cdot \left(\frac{\ell}{A} \right)$$

$$\kappa = G \frac{\ell}{A} \quad \left[\text{Where } \kappa = \frac{1}{\rho} \text{ and } G = \frac{1}{R} \right]$$

When length between two parallel electrodes (ℓ) = 1 cm and area of cross-section of electrode (A) = 1 cm² then $\kappa = G$.

Hence, **specific conductance may be defined as the conductance offered by a solution kept between two parallel electrodes each of cross-sectional area 1 cm² and separated by 1 cm length.**

or

Specific conductance of a solution may be defined as the conductance offered by 1 cc (centimeter cube) electrolyte solution.

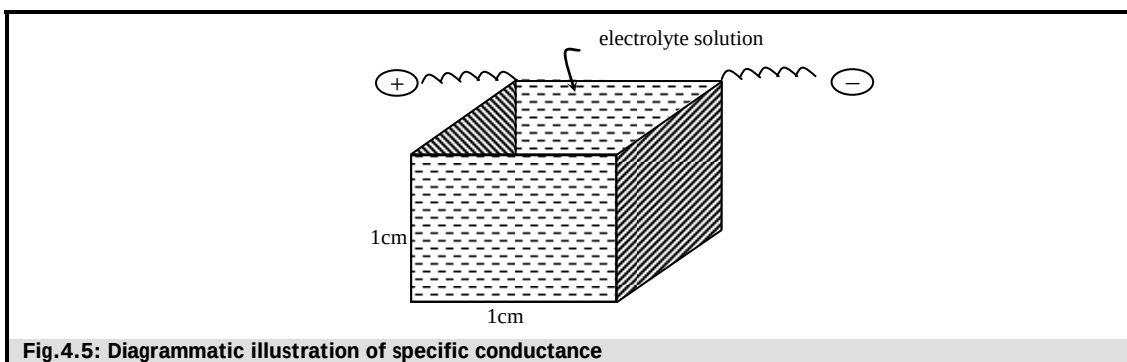


Fig.4.5: Diagrammatic illustration of specific conductance

Conductance of 1 cc electrolytic solution = Specific conductance.

Unit of Specific conductance

$$R = \rho \frac{\ell}{A}$$

$$\frac{1}{\rho} = \frac{1}{R} \frac{\ell}{A}$$

$$\kappa = \frac{1}{R} \cdot \frac{\ell}{A} = \frac{1}{\text{Ohm}} \cdot \frac{\text{Cm}}{\text{Cm}^2}$$

∴ Unit of κ = $\text{Ohm}^{-1} \text{cm}^{-1}$ or mho cm^{-1} or *simens* cm^{-1} .

2. Equivalent conductance (λ)

The conductance offered by 1 cc solution is not much informative because 1cc solution is not well defined in terms of the amount of electrolyte present in that volume. Therefore, it is a common practice to express of the solution in terms of its concentration. If electrolyte solution is prepared by dissolving certain number of gram equivalent of electrolyte in particular volume of solution, conductance is measured in terms of equivalent conductance. Equivalent conductance is the measure of conducting power of all ions produced by one gram equivalent of electrolyte in solution.

Hence, it is defined as the conductance offered by solution prepared by dissolving 1 gram equivalent of solute when the whole solution is placed between two parallel electrodes at 1 cm apart with area of cross-section 1 cm^2 of each electrode.

Equivalent conductance (λ) is calculated from the specific conductance and concentration of the solution.

Let 1 gram equivalent electrolyte is dissolved to prepare V cc of solution and κ be the specific conductance.

Now, by definition of specific conductance

conductance of 1 cc solution = κ .

conductance of V cc solution = $\kappa \times V$.

Equivalent conductance $\lambda = \kappa \times V$

... (i)

Concentration in normality (N) = $\frac{\text{Number of g eqvt. (n)}}{\text{Vol. of solution in litre (V)}}$

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If 1 gram equivalent of electrolyte is dissolved to prepare V cc (or $\frac{V}{1000}$ liter) solution,

$$N = \frac{1}{V/1000}$$

$$\text{or } N = \frac{1000}{V}$$

$$\text{or } V = \frac{1000}{N}$$

Putting the value of V in equation (i), we get

$$\lambda = \frac{\kappa \times 1000}{N} \quad \dots (ii)$$

$$\text{Also, } \kappa = \frac{l}{R.A}$$

Now putting the value of κ in equation (ii)

$$\therefore \lambda = \frac{l \times 1000}{R.A.N} \quad \dots (iii)$$

[Where, l = length, R = resistance, N = normality of solution, A = area of cross section]

Unit of Equivalent conductance

$$\lambda = \frac{\text{Simens } cm^{-1} \cdot cm^3}{equiv} = \text{Simens } cm^2 equiv^{-1}$$

3. Molecular conductance or molar conductance (μ)

Molecular conductance is the measure of conducting power of all ions produced by one mole of electrolyte in solution.

Hence, **it is defined as the conductance offered by solution prepared by dissolving 1 mole of solute when the whole solution is placed between two parallel electrodes at 1 cm apart with area of cross-section of each electrode 1 cm².**

Molar conductance is calculated from specific conductance and concentration of the solution.

Let one mole of solute be dissolved to prepare V cc of solution and κ be the specific conductance.

Conductance of 1 cc solution = κ

Conductance of V cc solution = κV

By definition $\mu = \kappa V \quad \dots (i)$

$$\text{Concentration in Molarity (M)} = \frac{\text{Number of moles (n)}}{\text{Volume of solution in litre (V)}}$$

If 1 mole of electrolyte is dissolved to prepare V cc (or $\frac{V}{1000}$ liter) solution then,

$$M = \frac{1}{V/1000}$$

$$\text{or } M = \frac{1000}{V}$$

$$\text{or } V = \frac{1000}{M}$$

Putting the value of V in equation (i), we get

$$\mu = \frac{\kappa \times 1000}{M} \quad \dots \text{(ii)}$$

$$\text{Also, } \kappa = \frac{l}{R.A}$$

Now putting the value of κ in equation (ii)

$$\therefore \mu = \frac{l \times 1000}{R.A.M} \quad [\text{Where, } l = \text{length, } R = \text{resistance, } M = \text{molarity of solution}] \quad \dots \text{(iii)}$$

Unit of molar conductance:

$$\mu = \frac{\text{Simens } \text{cm}^{-1} \cdot \text{cm}^3}{\text{mol}} = \text{Simens } \text{cm}^2 \text{ mol}^{-1}$$

Relationship between equivalent conductance (λ) and molar conductance (μ)

From the above relation, we have

$$\text{Equivalent conductance } \lambda = \frac{l \times 1000}{RAN} \quad \dots \text{(i)}$$

$$\text{Molar conductance } \mu = \frac{l \times 1000}{RAM} \quad \dots \text{(ii)}$$

Now equation (i) is divided by equation (ii), we get

$$\frac{\lambda}{\mu} = \frac{M}{N}$$

$$\text{or } \lambda \times N = \mu \times M$$

$$\text{or } \text{Eqvt. conductance} \times \text{Normality} = \text{Molar conductance} \times \text{Molarity}$$

Conductivity Cell and Cell Constant

We know,

$$R = \rho \frac{\ell}{A}$$

Where R = resistance

ρ = specific resistance

ℓ = distance between electrodes

A = cross-sectional area of electrode.

For a given cell, the cross-sectional area of electrode and length between two electrode is fixed.

Therefore, $\frac{\ell}{A}$ remains constant which is called cell constant.

$$\therefore \text{Cell constant} = \frac{\ell}{A}$$

Hence, **cell constant is defined as the ratio of the distance between the plates of two electrodes to the area of cross-section of electrode plate.**

Here, $R = \rho \frac{\ell}{A}$

$$\frac{1}{\rho} = \frac{1}{R} \left(\frac{\ell}{A} \right)$$

$$\kappa = G \times \text{cell constant}$$

$$\therefore \text{Cell constant} = \frac{\kappa}{G}$$

$$\therefore \text{Cell constant} = \frac{\text{Specific conductance } (\kappa)}{\text{Observed conductance } (G)}$$

Alternatively, cell constant is the ratio of specific conductance to the observed conductance.

Unit of cell constant

$$\text{Cell constant} = \frac{l}{A} = \frac{cm}{cm^2} = cm^{-1}$$

In fact cell constant is determined by measuring specific conductance and observed conductance rather than by measuring the distance between two parallel plates and area of cross-section of electrodes. A solution of known specific conductance is taken and observed conductance is determined. KCl solution is mostly used for this purpose. Its conductance has been measured with sufficient accuracy (calibration of the conductivity cell).

The conductance of electrolyte solution is measured by using **conductivity cell**. In market, different types of conductivity cells are found. Some models of conductivity cells are shown in figure 4.6.

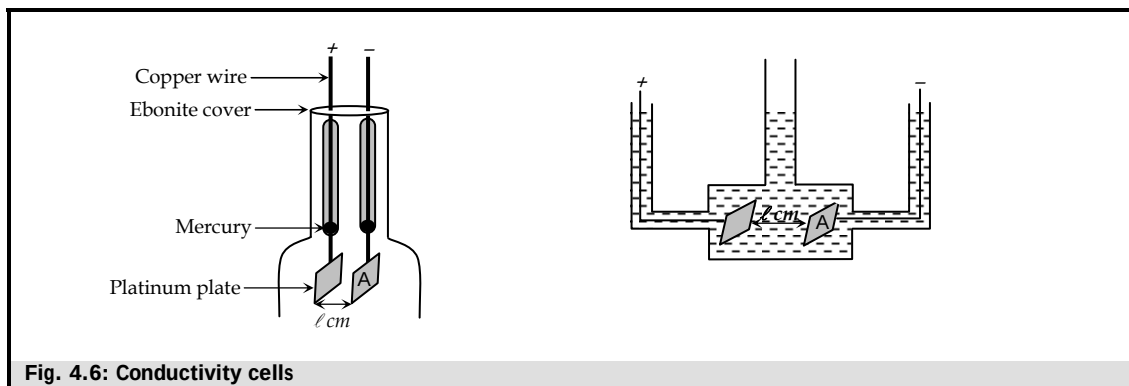


Fig. 4.6: Conductivity cells

Conductivity Water

For the measurement of the conductance of electrolyte solution pure water should be used to prepare solution. Distilled water has also certain level of conductance. Conductivity water is prepared by double distillation of distilled water containing small amount of $KMnO_4$ and KOH so that volatile impurities are oxidised. The specific conductance of the conductivity water is $8 \times 10^{-5} \text{ Ohm}^{-1}m^{-1}$ at 298 K. This conductivity is mainly due to dissolved CO_2 from air. The conductivity water is also called **equilibrium water**. Purest water free from air is called **ultra pure water**. The specific conductance of ultra pure water is $5.8 \times 10^{-6} \text{ Ohm}^{-1}m^{-1}$ at 298 K. For precise experimental work ultra pure water is used.

4.9 Variation of Conductivities of Electrolyte Solution with Dilution

Conductance of electrolyte solution mainly depends upon the following two factors.

1. Number of ions present in the solution
2. Mobility of ions

Other factors such as temperature, viscosity of solvent, hydration of ions, concentration or dilution of solution ultimately affect the above factors. The effects of dilution on different types of conductivities of solution are discussed in different headings.

1. Effect of dilution on specific conductance

Specific conductance of a solution is the conductance offered due to the conduction of ions in 1cm^3 of the solution. When the solution is diluted the following changes occur.

- a. Degree of ionization increases in case of weak electrolyte. Therefore, number of ions in solution increase.
 - b. The number of ions per centimeter cube decreases.
 - c. As the number of ions per centimeters cube decreases they become far apart from one another.
- But the effects of increase in number of ions due to ionization and increase in mobility due to decrease in inter-ionic interaction are far less than the effect caused by decrease in number of ions per unit volume due to dilution.

Overall effect is decrease in specific conductance with increase in dilution.

In brief, specific conductance decreases on dilution due to decrease in number of ions per unit volume of the solution.

2. Effect of dilution on equivalent conductance or molar conductance

On dilution, equivalent and molar conductance increase whether the solution is of strong electrolyte or weak electrolyte. But the reasons for increase in conductance are different.

- a. **Increase in equivalent or molar conductance on dilution for weak electrolyte:** Molar or equivalent conductance is the total conductance of all ions present in a solution containing 1 mole (for molar conductance) or 1 g equivalent (for equivalent conductance) electrolyte, assuming that their ions are present in unit volume of solution. Therefore, the total volume of solution has no effect on their values. But when the solution of weak electrolyte is diluted, the degree of ionization is increased and number of ions in the solution is increased which increases the equivalent or molar conductance.

The variation of equivalent conductance (or molar conductance) with dilution for weak electrolyte can be shown graphically as in figure 4.7.

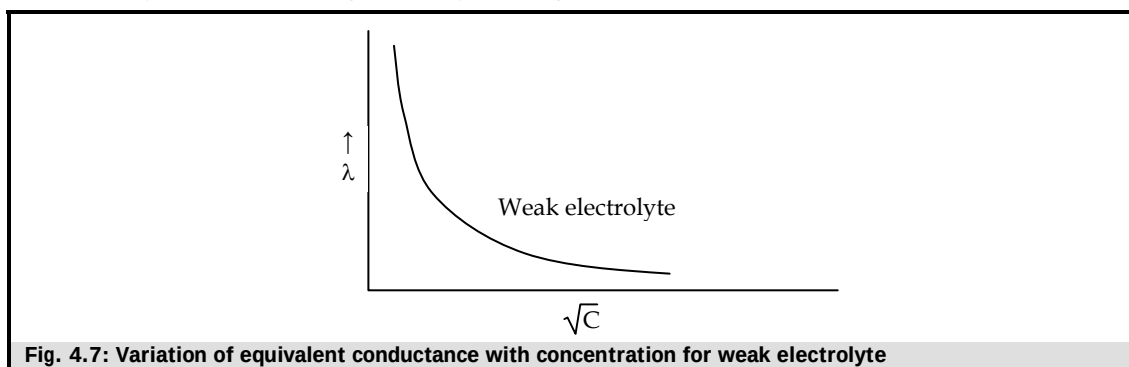


Fig. 4.7: Variation of equivalent conductance with concentration for weak electrolyte

- b. **Increase in equivalent or molar conductance with dilution for strong electrolyte:** Strong electrolytes are ionized almost completely at all dilution. Therefore, increase in equivalent or molar conductance with dilution for strong electrolytes is not due to the increase in degree of ionization but due to some other factors.

At concentrated solution, the ions are sufficiently near to one another and inter-ionic forces are predominant. It tends to decrease the mobility of ions. When strong electrolyte solution is diluted, the ions are dispersed in large volume and inter-ionic distance increases. This decreases the electrostatic forces of attraction.

$$F \propto \frac{q_1 q_2}{r^2}$$

Due to decrease in inter-ionic forces of attraction, the mobility of ions towards the corresponding electrode increases and equivalent or molar conductivity increases.

The variation of equivalent conductance with dilution for strong electrolyte solution can be shown graphically as in figure 4.8.

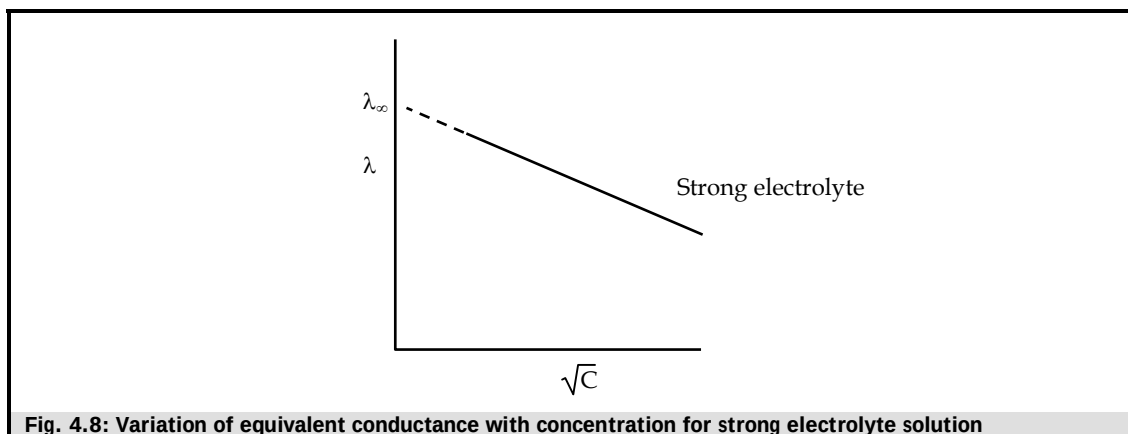


Fig. 4.8: Variation of equivalent conductance with concentration for strong electrolyte solution

The molar or equivalent conductivity of strong electrolyte solution varies with concentration according to equation.

$$\lambda_c = -b\sqrt{C} + \lambda_\infty$$

Where, λ_c = Equivalent conductance of electrolyte at concentration C .

λ_∞ = Equivalent conductance of electrolyte at infinite dilution.

C = Equivalent concentration

b = Constant, the value of b depends upon the type of electrolyte used, nature of solvent and temperature.

In fact the above equation is simplified form of Debye – Huckel – Onsager equation.

The increase in equivalent or molar conductance of strong electrolyte solution on dilution is due to decrease in asymmetric effect or relaxation effect and electrophoretic effect. So, mobility of ions increases. These terms are not described here. This is beyond the scope of this text.

4.10 Effect of Temperature on Conductance

There are mainly two types of conductances. These are

- Electronic conductance or metallic conductance
- Electrolytic conductance

The electronic conductance decreases with the increase of temperature. This can be explained as follows. Electronic conductance is due to flow of electrons. When temperature is increased, the drift velocity of electron is increased and these electrons tend to move randomly in several directions

rather than in particular direction with particular orientation. On the other hand, the electrolytic conductance increases with increasing temperature. The reason is explained as: The extent of increase in conductance of electrolyte solution is due to increase in temperature vary from one condition to another condition. There is in average, 2-3% increase in conductance of electrolyte solution for each degree rise in temperature. The conductance of an electrolyte solution depends on two factors.

- i. The number of ions present in unit volume of solution.
- ii. The speed of ions towards the respective electrode.

For strong electrolytes, due to increase in temperature there is hardly any increase in number of ions in unit volume of solution but the speed of ions increases due to decrease in viscosity of solvent.

For weak electrolyte, the effect of temperature on the conductance depends upon enthalpy change accompanying the process of ionization.

If the process of ionization is exothermic, the degree of ionization is decreased due to increase in temperature and conductance is decreased. If the process of ionization is endothermic, the ionization is increased due to increase in temperature and conductance is increased.

4.11 Kohlrausch's Law of Independent Migration of Ions

In 1875, Kohlrausch studied the equivalent conductance of different electrolyte pairs with common cation and common anion. On the basis of these observations, he gave a law which is called Kohlrausch's law of independent migration of ions.

It is defined as "At infinite dilution, where dissociation of all electrolytes is complete and inter-ionic effects fade out, each cation or anion migrates independently of its counter ion and contributes a definite share to the total equivalent conductance."

OR,

Kohlrausch's law states "at infinite dilution, the equivalent conductance of an electrolyte is equal to the sum of equivalent conductance of its cation and anion."

Mathematically,

$$\lambda^0 = \lambda_{+}^0 + \lambda_{-}^0$$

λ_0 = Equivalent conductance at infinite dilution.

λ_{+}^0 = Equivalent conductance of cation at infinite dilution

λ_{-}^0 = Equivalent conductance of anion at infinite dilution

For example,

At infinite dilution, the equivalent ionic conductance of Na^{+} is $50.0 \text{ Sm}^2 \text{ eqvt}^{-1}$ and equivalent ionic conductance of Cl^{-} ion is $76 \text{ Sm}^2 \text{ eqvt}^{-1}$. The equivalent conductance of NaCl at infinite dilution is $126 \text{ Sm}^2 \text{ eqvt}^{-1}$.

$$\lambda_{\text{NaCl}}^0 = \lambda_{\text{Na}^{+}}^0 + \lambda_{\text{Cl}^{-}}^0$$

$$126 = 50 + 76$$

Solved Questions

1. Define specific conductance.

- ↪ Specific conductance of a solution may be defined as the conductance offered by 1 cc (centimeter cube) electrolyte solution. Mathematically, $\kappa = G \times \frac{l}{A}$

2. Define equivalent conductance.

- ↪ It is defined as the conductance offered by solution prepared by dissolving 1 gram equivalent of solute when the whole solution is placed between two parallel electrodes at 1 cm apart with area of cross-section of each electrode 1 cm². Mathematically, $\lambda = \kappa \times V$ or, $\lambda = \frac{l \times 1000}{\text{RAN}}$

3. Define molecular conductance.

- ↪ It is defined as the conductance offered by solution prepared by dissolving 1 mole of solute when the whole solution is placed between two parallel electrodes at 1 cm apart with area of cross-section of each electrode 1 cm². Mathematically, $\mu = \kappa \times V$ or, $\mu = \frac{l \times 1000}{\text{RAM}}$

Questions

Very Short Answer Questions:

- Define the following terms:
 - Conductance
 - Resistivity
 - Specific conductance
 - Equivalent conductance
 - Molecular conductance
 - Cell constant
 - Electrolytic conductance
- Why do specific conductance decrease and equivalent conductance increase with dilution?
- Write graphical representation showing variations of conductance with concentration.

Short /Long Answer Questions:

- Write short notes on variation of electrolytic conductance with concentration.
- Write short notes: (a) conductance of solutions (b) molar and equivalent conductance.

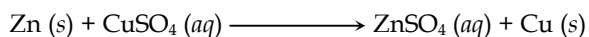
Numerical Questions

- Find the molar conductivity of 0.01 M acetic acid having specific conductivity $1.46 \times 10^{-4} \text{ Ohm}^{-1} \text{ cm}^{-1}$.
Ans: $14.6 \text{ S cm}^2 \text{ mol}^{-1}$
- Calculate the equivalent conductivity of 0.12 (N) solution of an electrolyte, whose conductivity is 0.024 S cm^{-1} .
Ans: $200 \text{ m}^{-1} \text{ cm}^2 \text{ eqv}^{-1}$ ($200 \text{ S cm}^2 \text{ eqv}^{-1}$)
- Calculate the equivalent conductance of 0.1N KCl solution having specific resistance 83.3 Ohm cm^{-1} .
Ans: $120 \text{ ohm}^{-1} \text{ cm}^2 \text{ eqvt}^{-1}$
- The conductivity of electrolytic solution is 0.3704 S cm^{-1} at 298K. A conductivity cell when filled with this solution gave a resistance of 1.5 ohm, find out the cell constant of this cell.
Ans: 0.556 cm^{-1}
- 0.028 N KCl solution has specific conductance of $0.00277 \text{ S cm}^{-1}$ having resistance of 520 ohm. 0.04N solution of unknown salt kept in the same cell showed the resistance of 300 Ohm at same condition. Calculate (i) cell constant of the cell and (ii) equivalent conductance of unknown salt.
Ans: (i) 1.44 cm^{-1} (ii) $119.5 \text{ S cm}^2 \text{ eq}^{-1}$

4.12 Galvanic Cell

4.12.1 Introduction

If a zinc rod is immersed into CuSO_4 solution, the solution is found to be warm due to following reaction.



The above reaction is redox reaction where metallic zinc is oxidized to Zn^{++} (or ZnSO_4) and Cu^{++} (or CuSO_4) is reduced to metallic copper. The reaction is exothermic which makes solution warmer.

But the reaction does not provide a useful work done. If the reducing agent (Zn) and oxidising agent (Cu^{2+}) are taken in separate containers connected by external metallic conductor (as shown in Fig. 4.9), the electron flow takes place through Zn electrode to Cu electrode. Now, the same reaction can be utilized for useful work and such cell is called galvanic cell.

An electrochemical device which converts chemical energy into electrical energy is called **galvanic cells** or **voltaic cell**. Galvanic cell is named in the honour of **Luigi Galvani (1780)** and voltaic cell is named in the honour of **Alessandro Volta (1800)** who were the pioneer to perform experiments on the conversion of chemical energy into electrical energy.

4.12.2 Construction of Galvanic Cell

Let us take an example where zinc rod is immersed into aqueous solution of zinc sulphate and copper rod is immersed into aqueous solution of copper sulphate as shown in figure 4.9. These two half cells are externally connected to emf measuring device such as voltmeter or potentiometer and internally connected by salt bridge as shown in figure 4.9.

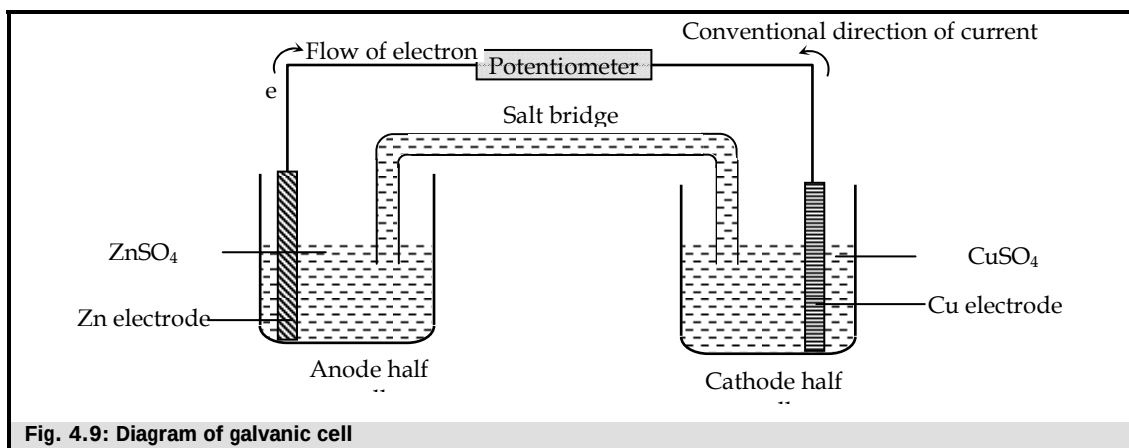


Fig. 4.9: Diagram of galvanic cell

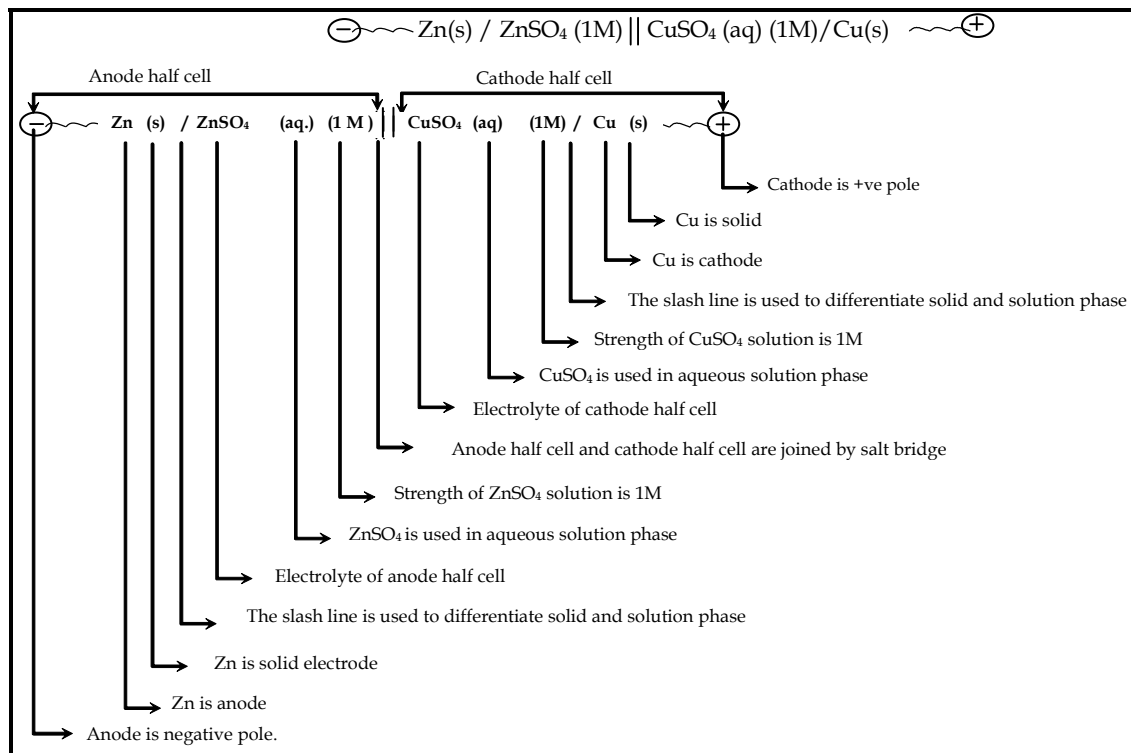
Cell notation

It is inconvenient to draw the diagram of galvanic cell. Therefore, this cell can be represented in symbolic form which is called cell notation. While writing cell notation, the following pattern is adopted (According to IUPAC recommendation).

Anode/ Electrolyte solution of anode // Electrolyte solution of cathode/ Cathode

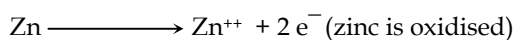
In the above format, the state of anode, cathode and electrolyte solution should be mentioned whether these are in solid, liquid or gaseous state. In case of electrolyte solution it is customary to mention the concentration of the solution too. If gas electrode is used then pressure of gas is written

in bracket. In the above cell notation a single vertical slash line is used to distinguish different phases such as solid-liquid or liquid-gas. The double vertical slash lines indicate salt bridge. In the galvanic cell of Zn and Cu (Daniell cell), the cell notation is given below.

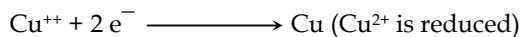


Electrode reactions

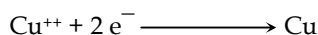
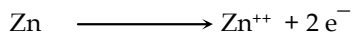
An anode (Oxidation-Half reaction): Electrode having lower reduction potential acts as anode. At anode half cell, oxidation reaction takes place. The electrode reaction taking place at anode half cell of above cell is given below.



At cathode (Reduction-Half reaction): Electrode having higher reduction potential acts as cathode. At cathode half cell, reduction reaction takes place. The electrode reaction taking place at cathode half cell is given below.

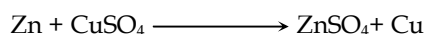


The complete cell reaction is obtained by adding anode half reaction and cathode half reaction.



Complete cell Reactions: $\text{Zn} + \text{Cu}^{++} \longrightarrow \text{Zn}^{++} + \text{Cu}$

This is the ionic reaction that takes place in galvanic cell. The molecular reaction is,



The above reaction informs that when zinc rod is immersed into ZnSO₄ solution and copper rod is immersed into CuSO₄ solution followed by joining these two half cells by salt bridge, the zinc rod gradually becomes thinner while copper rod gradually gets thickened. Zinc atom from zinc metal

goes into the solution in the form of zinc ions while copper ions accept electron and get reduced to form metallic copper. As a result of chemical reaction following changes take place.

- Mass of metal at anode decreases.
- Concentration of metal ion at anode increases.
- Concentration of metal ion at cathode decreases.
- Mass of metal at cathode increases.
- As the metallic atoms of anode pass into the solution in the form of ions leaving behind the electrons at anode, the anode is rich in electron and acts as negative electrode. Consequently, the cathode acts as positive electrode.
- The electrons flow from anode to cathode but the conventional direction of flow of current is from cathode to anode.

If the cell reactions are carried out without using salt bridge, the production of current is stopped after sometimes. The role of salt bridge in electrochemical cell is discussed below.

4.12.3 Salt Bridge

Salt bridge is 'U' shaped glass tube containing inert electrolyte solution such as KCl, KNO₃ or NH₄NO₃ in agar-agar or jelly medium. To understand its importance in a galvanic cell, let us take an example of Daniell cell where Zn strip is dipped in ZnSO₄ solution and Cu strip is dipped into CuSO₄ solution.

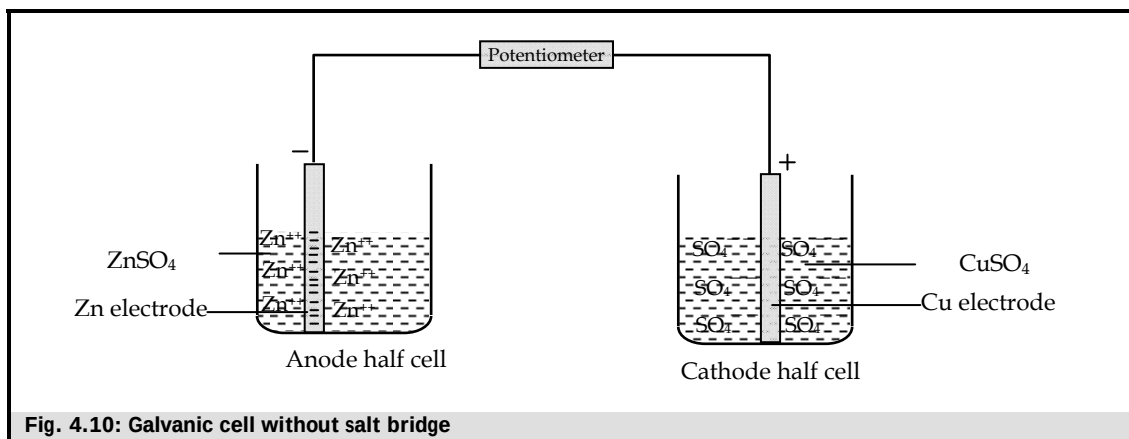


Fig. 4.10: Galvanic cell without salt bridge

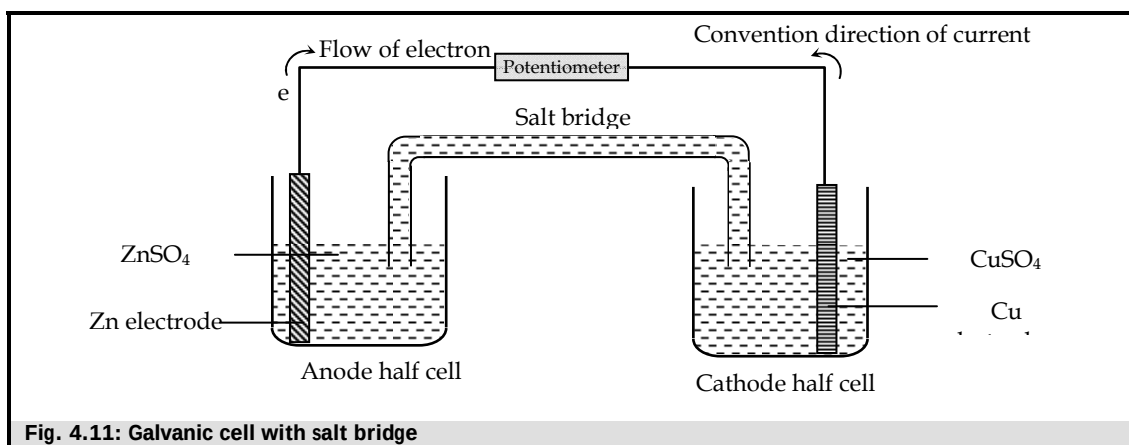


Fig. 4.11: Galvanic cell with salt bridge

Initially, here we discuss what will happen if salt bridge is not used in cell. Then, we will discuss how the salt bridge avoids the undesirable circumstances in the galvanic cell. Out of Zn and Cu strip, Zn has more tendency to get oxidized while Cu^{2+} has more tendency to get reduced. Therefore, zinc atoms lose electrons and zinc ions pass into solution. This makes zinc electrode rich in electrons and it acts as negative electrode. The electrolyte solution of anode half cell will be rich in Zn^{++} ions. The solution of anode half cell consists more cations than anions in average. Therefore, the electrolyte solution can be considered to be positively charged in net. On the other hand, electron driven by anode will be consumed by copper ions of cathode half cell to form metallic copper. As copper ions get reduced the cathode solution will be rich in sulphate ions in average.

Maintenance of this type of charge reparation requires high amount of energy and therefore, the flow of electrons stops. Therefore, salt bridge is used to balance this unbalanced charge between these two compartments. The anion present in salt of salt brings (e.g. Cl^-) migrates towards anode half-cell to balance excess positive charge there. On doing so, heavily crowded of positively charged particles are counter-balanced. Similarly, anion present in the salt of salt bridge (e.g. K^+) migrates towards cathode half-cell to balance excess negative charge there. On doing so, heavily crowded negatively charged particles are counter-balanced.

It is believed that the salt bridge balances unbalanced charges solutions without extensive intermixing of the ions. Therefore, the external circuit is completed by metallic conduction where as the internal circuit is completed by salt bridge.

The function of salt bridge can be summarized as,

- i. Salt bridge avoids or minimizes the liquid-liquid junction potential. Due to different mobility of ions in electrolyte solution, different ions migrate at different speeds. Due to this, there is separation of charge and formation of two layers of liquid. A potential is developed at the junction of two liquid layers which is called liquid-liquid junction potential or simply **junction potential**.
- ii. Salt bridge helps to complete the internal circuit of galvanic cell.
- iii. Salt bridge maintains electrical neutrality without direct mixing of electrolyte solution.
- iv. Salt bridge avoids the crowding of ions around electrodes.

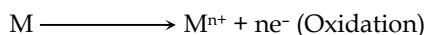
The criteria of electrolytes used in salt bridge are given below.

- i. The electrolyte solution should be inert towards electrode reaction i.e., the electrolyte of salt bridge should not take part in reaction. For example: KCl cannot be used in salt bridge, if electrolyte solution contains Ag^+ ions as there is the formation of insoluble AgCl.
- ii. The transport number or mobility of cations and anions should be same or nearly same. In salt bridge, KCl is preferred because the transport number of K^+ ions is nearly equal that of Cl^- ions. The fraction of current carried by an ion is called **its transport number**.
- iii. The concentration of electrolyte should be preferably high.

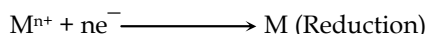
4.13 Electrode Potential and Standard Electrode Potential

A metal dipped in the solution of its own ion constitutes an electrode. When a metal is immersed into its soluble salt solution, three cases may arise as shown in fig. 4.12.

- i. The metal atoms of electrode pass into the solution in the form of positive ion leaving behind electron on the electrode and the electrode becomes negative.



- ii. The metal ions of electrolyte solution accept electrons from electrode and get reduced. This makes electrode electron deficient and the electrode becomes positive.



- iii. Neither the metal atoms loses electron nor the metal ions gain electron.

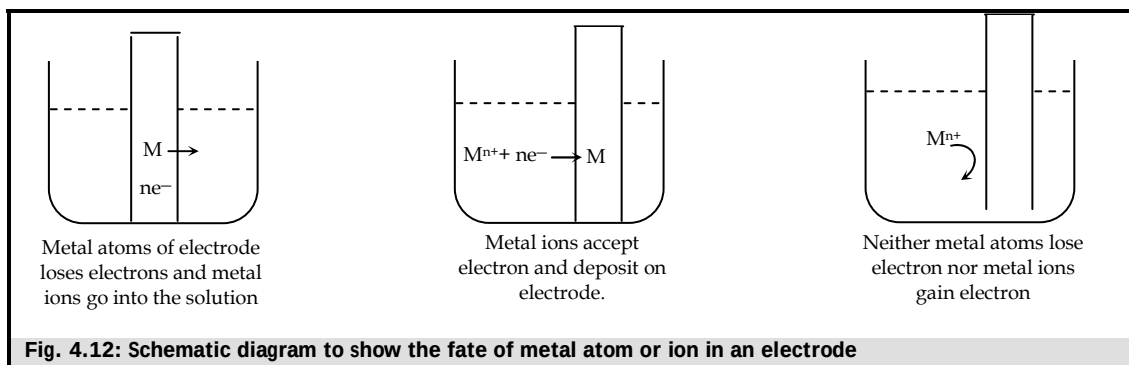


Fig. 4.12: Schematic diagram to show the fate of metal atom or ion in an electrode

When an electrode is immersed into electrolyte solution, the fate of atom of electrode or ions of solution is discussed above. This will be more clear from the Nernst theory of electrode potential. It will be discussed in the next heading.

When an electrode is dipped into electrolyte solution of its soluble salt, either the metal atom loses electrons and positive ions dive into the electrolyte solution or the ions of electrolyte accept electrons and deposit on the electrode. During this process, there is separation of charge on the electrode and electrolyte interface. Potential difference exists across the double electrical layer of electrode system.

This potential developed in electrode electrolyte interface due to the formation of an electrical double layer is called single electrode potential or simply **electrode potential**.

The adjoining figure portrays how single electrode potential is developed. To explain this, let us take an example. When zinc rod is immersed into zinc sulphate solution, zinc atoms lose electrons and zinc ions are formed. These ions are pulled towards the solution by the attraction of SO_4^{2-} ions. Then zinc ions dive into the solution leaving behind electrons on the electrode.

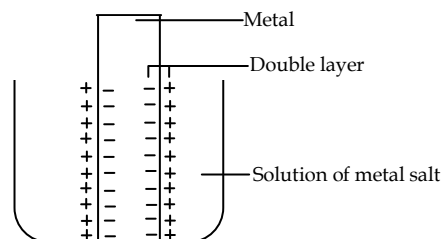


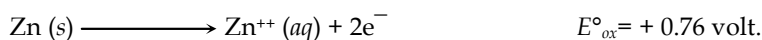
Fig. 4.13: Diagram of show the origin of single electrode potential.

When the process continues, electrode acquires negative charge while electrolyte solution acquires net positive charge. Later on, the positively charged ions of electrolyte solutions are attracted towards the negative electrode and a double layer of charge is developed at the solid liquid interface. This double layer of charge develops potential. In this way single electrode potential is originated.

On contrary, there is probability of deposition of zinc ions on electrode from electrolyte solution by accepting electrons. In such situation the electrode becomes electron deficient and it is regarded as positive electrode. In this case electrolyte interface develops net negative charge. To know, whether zinc atoms lose electrons and get oxidized or zinc ions gain electrons and get reduced, we should know about another half cell in the circuit. It will be more clear after studying electrochemical series.

Electrode potential is defined as the potential developed at the electrode electrolyte interface when an electrode is dipped into electrolyte solution having its ions. It is denoted by E . Electrode potential is the measure of the tendency of an electrode to lose or gain electrons to or from electrolyte solution of its salt. The electrode potential varies with the nature of electrode, concentration of electrolyte solution, pressure (in case of gas electrode) and temperature.

If electrode potential is measured for an electrode at its 1 molar concentration of electrolyte solution at 25°C, it is called standard electrode potential which is denoted by E° . If electrode potential is developed due to loss of electron (oxidation) by electrode, it is called **oxidation potential**. Or, oxidation potential shows a capacity of the electrode material to be oxidised (or to lose e^-). If oxidation potential is measured at standard condition, it is called **standard oxidation potential**. It is denoted by E_{ox}° . The standard oxidation, potential of zinc electrode is + 0.76 volt ($E_{Zn/Zn^{++}}^\circ = 0.76$ V).



If electrode potential is developed due to reduction, it is called **reduction potential**. Reduction potential at standard condition is called **standard reduction potential**. Reduction potential shows a capacity of the electrode material to be reduced or to gain electron. It is denoted by E_{red}° . The standard reduction potential value of zinc electrode is - 0.76 V. ($E_{Zn^{++}/Zn}^\circ = -0.76$ V).



Since oxidation is the reverse process of reduction the oxidation potential for an electrode will have same magnitude to that of reduction potential but with opposite sign.

According to IUPAC conventions, reduction potential value is used in electrochemical calculation.

Nernst theory of electrode potential

In Galvanic cell, anode is negative electrode and cathode is positive electrode. A much deeper insight about the polarity can be realized from Nernst theory of electrode potential. According to this theory, all metals and non-metals have definite tendency to pass into ionic state by the loss of electrons when it is dipped into electrolyte solution.

The property of atoms of electrode to pass into solution in the form of positive ion is called solution pressure. When a positive ion tends to enter into solution, it is opposed by similar ions of electrolyte solution. Therefore, cations develop a tendency to get deposited on the electrode. The property of ions of electrolyte which creates pressure to get deposited at the electrode by accepting electrons is called **osmotic pressure**. When an electrode is immersed into electrolyte solution, the solution pressure tends to develop positive ion by the loss of electrons while osmotic pressure opposes this phenomenon. The electrode potential of an element or electrode is equal to the difference between the solution pressure and the osmotic pressure of ions. Depending upon the magnitude, three cases may arise here.

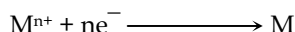
- i. **Solution pressure > Osmotic pressure:** When solution pressure is greater than osmotic pressure, atoms of electrode become positive ions and pass into the solution leaving behind electrons at the electrode until the charge in the electrolyte solution oppose further action. In such situation the electrode acquires negative charge while electrolyte solution acquires net

positive charge. Oxidation takes place here.



The potential developed in this condition is called oxidation potential.

- ii. **Solution pressure < Osmotic pressure:** When osmotic pressure is greater than solution pressure, the electrode fails to send the positive ions in solution. Further the positive ions of solution acquire electrons and get deposited. Due to this, electrode acquires positive charge while electrolyte solution acquires net negative charge. Oxidation takes place here.



The potential developed in this condition is called reduction potential.

- iii. **Solution pressure = Osmotic pressure:** When solution pressure is equal to osmotic pressure, no charge separation occurs at electrode-electrolyte interface without developing any potential difference. This type of electrode is often called null electrode.

Measurement of Electrode Potential

Potentiometer or voltmeter is used to measure the potential difference. Therefore, the electrode for which the electrode potential is measured should be connected to the reference electrode (electrode whose electrode potential is known) by means of potentiometer or voltmeter and solutions are connected by a salt bridge. Standard hydrogen electrode (SHE) is the **primary reference** electrode, whose standard electrode potential is assigned to be 0.00V.

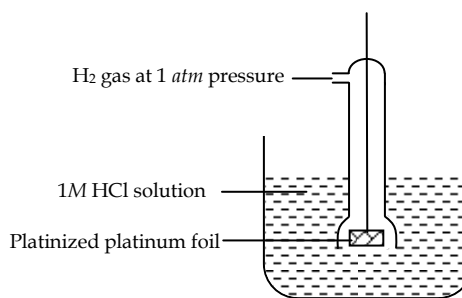
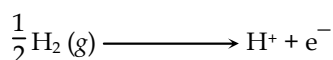


Fig. 4.14: Schematic diagram of hydrogen electrode

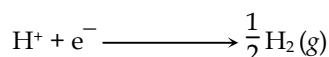
Standard hydrogen electrode: Hydrogen electrode is a gas electrode. It is reversible to H^+ ions of solution. A platinum foil is connected to a conducting wire and whole of this part is covered by glass jacket having an inlet. The platinized platinum foil is immersed in 1M HCl solution. Pure and dry H_2 gas is admitted through the narrow inlet maintaining 1 atm pressure (more correctly 1 bar pressure).

The hydrogen gas is preferentially adsorbed by platinized platinum foil and desorbed in the form of tiny bubbles as if the platinized foil were the sources of hydrogen gas. This hydrogen gas is reversible to H^+ ions of solution. Depending upon the counter electrode, hydrogen electrode acts as anode half cell or cathode half cell. If hydrogen gas is maintained at 1 atm pressure, 25°C temperature and platinized platinum foil is immersed into 1M HCl solution, hydrogen electrode is called **standard hydrogen electrode (SHE)** or **normal hydrogen electrode (NHE)**.

If hydrogen electrode acts as anode half cell, the half cell notation is, $Pt, H_2(g, 1 atm)/H^+ (1M)$ and oxidation half reaction is



If hydrogen electrode acts as cathode half cell, the half cell notation is, $H^+ (1 M)/H_2 (g, 1 atm), Pt$ and reduction half reaction is



Electrode potential of hydrogen electrode at 25°C, 1 atm pressure and 1M concentration of HCl solution is arbitrarily taken as zero. In fact, electrode potential of hydrogen electrode is neither

zero nor independent of temperature. All other electrodes are calibrated with respect to standard hydrogen electrode.

Hydrogen electrode is some what difficult to handle and can't be used in presence of oxidising or reducing agent. The platinum black is easily poisoned by substance such as mercury and hydrogen sulphide. Therefore, other reference electrodes such as calomel electrode and silver-silver chloride electrode are used. Such electrodes are called **secondary reference** electrodes.

Measurement of standard electrode potential: Standard electrode potential of an electrode is measured by coupling it with standard hydrogen electrode. The emf of cell is measured by potentiometer or voltmeter. The electrode whose emf is to be measured is immersed into its electrolyte solution at 1M concentration. A salt bridge is used to complete internal circuit between the electrode and standard hydrogen electrode. The circuit is complete and emf of cell is read as shown figure 4.15.

Here, measurement of electrode potential of Zn and copper electrodes is illustrated.

Measurement of electrode potential of Zn electrode: A pure zinc electrode is immersed into 1M ZnSO_4 solution at 25°C . Then it is coupled with standard hydrogen electrode as shown in figure. Out of zinc and hydrogen, zinc has more tendency to lose electron and get oxidised. Therefore, zinc electrode acts as anode half cell while standard hydrogen electrode acts as cathode half cell.

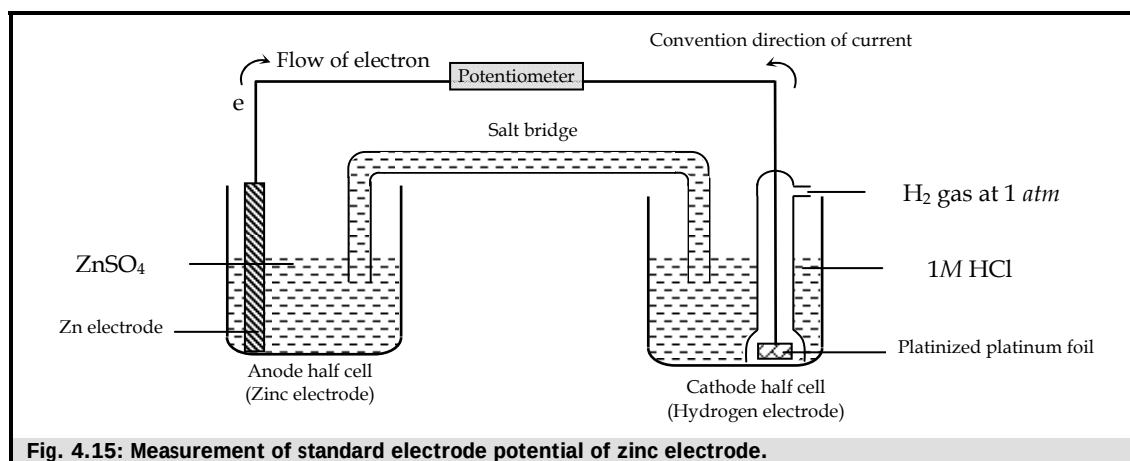


Fig. 4.15: Measurement of standard electrode potential of zinc electrode.

Cell notation of this combination is



The driving force for electron to move from one electrode to the another one is called emf of the cell or cell potential.

The cell potential of this combination is measured 0.76 V.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$0.76 \text{ V} = 0 - E^\circ_{\text{anode}}$$

$$\therefore E^\circ_{\text{anode}} = -0.76 \text{ volt (Reduction potential)}$$

Measurement of electrode potential of Cu electrode: A pure copper electrode is immersed into 1M CuSO_4 solution at 25°C . Then it is coupled with standard hydrogen electrode as shown in Fig. 4.16. Out of copper and hydrogen, hydrogen has more tendency to loss electron. Therefore hydrogen electrode acts as anode while copper electrode acts as cathode.

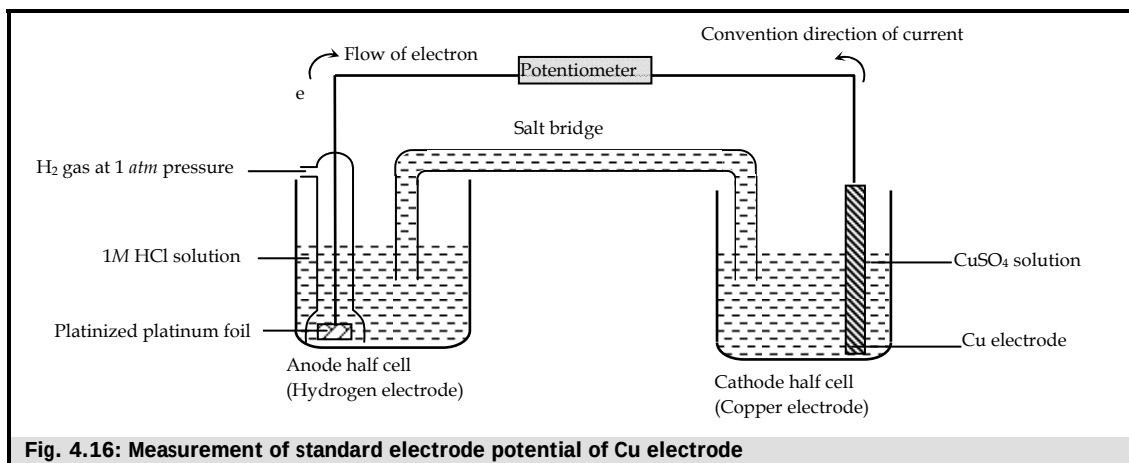


Fig. 4.16: Measurement of standard electrode potential of Cu electrode

Cell notation: $\text{Pt}, \text{H}_2 (\text{g}, 1 \text{ atm}) / \text{H}^+ (\text{aq}, 1 \text{ M}) // \text{CuSO}_4 (\text{aq}, 1 \text{ M}) / \text{Cu}(\text{s})$

The cell potential is found to be 0.34V

$$\begin{aligned} E^\circ_{\text{Cell}} &= E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} \\ &= +0.34 = E^\circ_{\text{cathode}} - 0 \end{aligned}$$

$\therefore E^\circ_{\text{cathode}} = +0.34 \text{ volt (Reduction potential)}$

4.14 Electrochemical Series (ECS)

A series of electrode materials which is arranged on the basis of increasing reduction potential or decreasing oxidation potential value is called **electrochemical series**. It is also called activity series. The electrochemical series is given below.

Table 4.1: Electrochemical series

Electrode reaction	E°_{red}	E°_{oxd}
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$	- 3.05	+ 3.05
$\text{K}^+ + \text{e}^- \rightarrow \text{K}$	- 2.92	+ 2.92
$\text{Ba}^{++} + 2\text{e}^- \rightarrow \text{Ba}$	- 2.90	+ 2.90
$\text{Sr}^{++} + 2\text{e}^- \rightarrow \text{Sr}$	- 2.89	+ 2.89
$\text{Ca}^{++} + 2\text{e}^- \rightarrow \text{Ca}$	- 2.87	+ 2.87
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	- 2.71	+ 2.71
$\text{Mg}^{++} + 2\text{e}^- \rightarrow \text{Mg}$	- 2.37	+ 2.37
$\text{Al}^{+++} + 3\text{e}^- \rightarrow \text{Al}$	- 1.66	+ 1.66
$\text{Mn}^{++} + 2\text{e}^- \rightarrow \text{Mn}$	- 1.18	+ 1.18
$\text{Zn}^{++} + 2\text{e}^- \rightarrow \text{Zn}$	- 0.76	+ 0.76
$\text{Cr}^{+++} + 3\text{e}^- \rightarrow \text{Cr}$	- 0.76	+ 0.76
$\text{Fe}^{++} + 2\text{e}^- \rightarrow \text{Fe}$	- 0.44	+ 0.44
$\text{Cd}^{++} + 2\text{e}^- \rightarrow \text{Cd}$	- 0.40	+ 0.40
$\text{Co}^{++} + 2\text{e}^- \rightarrow \text{Co}$	- 0.27	+ 0.27
$\text{Ni}^{++} + 2\text{e}^- \rightarrow \text{Ni}$	- 0.25	+ 0.25
$\text{Sn}^{++} + 2\text{e}^- \rightarrow \text{Sn}$	- 0.13	+ 0.13
$\text{Pb}^{++} + 2\text{e}^- \rightarrow \text{Pb}$	- 0.12	+ 0.12

$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00	0.00
$\text{Cu}^{++} + 2\text{e}^- \rightarrow \text{Cu}$	+ 0.34	- 0.34
$\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$	+ 0.54	- 0.54
$\text{Hg}^{++} + 2\text{e}^- \rightarrow \text{Hg}$	+ 0.78	- 0.78
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	+ 0.80	- 0.80
$\text{Pd} + 2\text{e}^- \rightarrow \text{Pd}$	+ 0.98	- 0.98
$\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$	+ 1.07	- 1.07
$\text{Pt}^{++} + 2\text{e}^- \rightarrow \text{Pt}$	+ 1.20	- 1.20
$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	+ 1.36	- 1.36
$\text{Au}^{+++} + 3\text{e}^- \rightarrow \text{Au}$	+ 1.50	- 1.50
$\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$	+ 2.87	- 2.87

Applications of electrochemical series

- To compare the strength of oxidising and reducing agents.** Lower the reduction potential of an electrode material, greater is the reducing strength. For example, the standard reduction potential of lithium is - 3.05 volt. Therefore, the reduced form Li has more reducing strength.

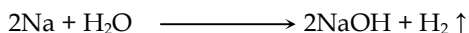
Similarly, the standard reduction potential of fluorine is highest. Therefore, it has lower reducing or most oxidising strength.

Similarly, Li is better reducing agent than K and K^+ is better oxidising agent than Na^+ .

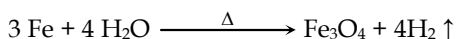
In ECS, the reducing character goes on decreasing while oxidising character goes on increasing on moving from top to bottom. From the above table, it is concluded that Li is best reducing agent (in aqueous medium) and F_2 is best oxidising agent.

- To predict whether a metal can or cannot liberate H_2 gas from water or acid.**

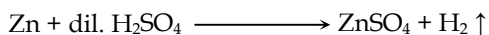
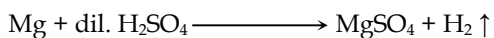
- Elements upto Na in electro chemical series (ECS) react with cold water to produce H_2 gas.



- Elements upto iron react with steam on heating to liberate hydrogen gas.

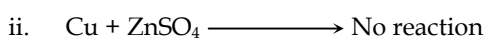
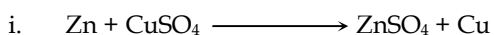


- Elements lying above hydrogen in electrochemical series can displace H_2 gas from dilute, mineral and non-oxidising acid.

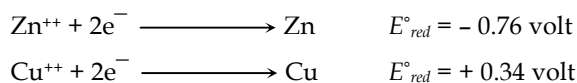


Points (a), (b) and (c) show that reactivity of elements go on decreasing from top to bottom in the electrochemical series which is applicable for metal only.

- To predict whether a metal can or cannot displace other metals from their salt.** Metals having lower reduction potential can displace those metals having higher reduction potential from their salt. It means, metal lying above in electrochemical series can displace only those metals from their salts which lie below in ECS. For example,



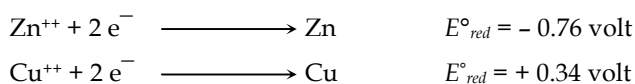
This can be understood as,



Zn⁺⁺ has low tendency to get reduced or Zn has more tendency to get oxidised. In reaction (i) Zn displace Cu from Cu⁺⁺ or CuSO₄ because Zn has more tendency to get oxidised. In reaction (ii) Zn is already in oxidised state and Cu can't displace it.

4. **To determine anode and cathode in galvanic cell.** Electrode material having lower reduction potential gets easily oxidised and becomes anode while metal having higher reduction potential gets easily reduced and becomes cathode. In other words, element lying above in ECS acts as anode and electrode lying below in series acts as cathode during the construction of galvanic cell.

Example,



Zn has lower reduction potential while Cu has higher reduction potential. If a galvanic cell of Zn and Cu is to be constructed, Zn becomes anode while Cu becomes cathode. Anode is the electrode where oxidation takes place and cathode is the electrode where reduction takes place. Therefore, the element which has lower reduction potential (higher oxidation potential) get oxidised in preference and acts as anode while metal having higher reduction potential (lower oxidation potential) get reduced in preference and acts as cathode.

5. **To calculate cell of emf.** Standard emf of cell is calculated from the following relation.

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

In this relation, all electrode potentials will be in terms of reduction.

In the galvanic cell of Zn and Cu,

$$\begin{aligned} E_{\text{cell}}^{\circ} &= E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = E_{\text{Cu}}^{\circ} - E_{\text{Zn}}^{\circ} \\ &= +0.34\text{V} - (-0.76) \text{ V [Taking both reduction potential value]} \\ &= (0.34 + 0.76) \text{ V} = 1.10 \text{ Volt.} \end{aligned}$$

6. **To Predict whether a cell reaction is spontaneous or not.** If emf of cell reaction is positive, the cell reaction is feasible in the given direction. If emf of cell reaction is negative, cell reaction is not feasible or non-spontaneous in the given direction.

In above cell, the emf of cell is positive, therefore, the reaction is spontaneous. In some cases, if emf of cell is found to be negative it shows that the redox reaction is taking place non-spontaneously i.e. the cell is under charging.

7. **Calculation of Gibbs free energy (ΔG).** The Gibbs free energy change is related with emf of cell as $\Delta G = -nFE$

At standard condition, $\Delta G^{\circ} = -nFE^{\circ}$

Where, ΔG = change in Gibbs free energy

n = number of moles of electrons lost or gained

F = Faraday constant, (96500 C)

E = Emf of cell

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In the galvanic cell of Zn and Cu, two moles of electrons are lost by Zn and that are gained by Cu. In this case,

$$n = 2 \quad F = 96,500$$

$$E^\circ = 1.10 \text{ volt}$$

$$\begin{aligned} \therefore \Delta G &= -2 \times 96,500 \text{ coulomb} \times 1.10 \text{ volt} \\ &= -212300 \text{ Joule } (\because \text{coulomb} \times \text{volt} = \text{Joule}) \\ &= -212.3 \text{ kJ} \end{aligned}$$

8. To calculate equilibrium constant (K_{eq}).

We have the relation, $\Delta G = -RT \ln K_{eq}$.

From the relation the value of K_{eq} for a reaction can be calculated.

Work done by Galvanic Cell

In a galvanic cell emf is produced as a result of redox reaction. This emf can be utilized for useful work. If nF coulombs of charge is produced against applied potential difference, the work done is given by,

$$\begin{aligned} \text{Work (W)} &= \text{Number of coulomb of charge} \times \text{Potential difference} \\ &= nF \text{ coulombs} \times E \text{ volts} \end{aligned}$$

$$W = nFE \text{ joule } [\because \text{Joule} = \text{volt} \times \text{coulomb}] \quad \dots \text{ (i)}$$

At standard condition, $W = nFE^\circ$

According to thermodynamics, the net work done is related with change in Gibbs free energy as,

$$W = -\Delta G \quad \dots \text{ (ii)}$$

From relation (i) and (ii)

$$\Delta G = -nFE \quad \dots \text{ (iii)}$$

If the redox reaction is carried out under standard condition, equation (iii) can be expressed as

$$\Delta G^\circ = -nFE^\circ \quad \dots \text{ (iv)}$$

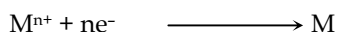
4.15 Nernst Equation

The emf of cell depends upon electrode potential value. The electrode potential value of an electrode depends upon the following factors.

- Nature of electrode
- Concentration of electrolyte solution
- Temperature

Nernst derived an expression to calculate emf of cell in the conditions other than standard one. A mathematical expression which shows the relationship between emf of a cell under ordinary condition to emf at standard condition, temperature and concentration is called Nernst equation.

For a general reduction half reaction,



According to Nernst equation, the reduction potential is given by

$$E = E^\circ - \frac{RT}{nF} \ln \frac{[M]}{[M^{n+}]} \quad \dots (i)$$

Where, E = Emf of cell at non standard condition.

E° = Standard emf of cell

R = Universal gas constant, $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$

n = Number of mole of electrons transferred

F = Faraday constant, $F = 96,500 \text{ C}$

$[M]$ = Molar concentration of reduced species

$[M^{n+}]$ = Molar concentration of oxidized species.

In terms of common logarithm, equation (i) can be expressed as,

$$E = E^\circ - \frac{2.303 RT}{nF} \log \frac{[M]}{[M^{n+}]} \quad [\because \ln x = 2.3031 \log x] \quad \dots (ii)$$

At 25°C , the value of $\frac{2.303RT}{F}$ is 0.0591

Now, equation (ii) becomes

$$E = E^\circ - \frac{0.0591}{n} \log \frac{[M]}{[M^{n+}]} \quad \dots (iii)$$

In the case of metal-metal ion electrode, the activity or molar concentration of pure metal is taken as one.

Example 4.8 Calculate the emf of given cell.



Given, $E^\circ_{\text{Fe}^{++}/\text{Fe}} = -0.44\text{V}$

$E^\circ_{\text{Cu}^{++}/\text{Cu}} = +0.34 \text{ V}$

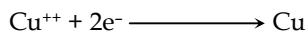
Solution

$$\begin{aligned} E^\circ_{\text{Cell}} &= E^\circ_{\text{C}} - E^\circ_{\text{A}} \\ &= +0.34 - (-0.44) \text{ V} \\ &= +0.78 \text{ V} \end{aligned}$$

Example 4.9 What will be the electrode potential of Cu electrode when it is immersed in decimolar concentration of CuSO_4 solution at 20°C ? Given standard electrode potential of Cu = + 0.34 V.

Solution

For a reaction,



Temperature (T) = $(20 + 273) \text{ K} = 293 \text{ K}$

Concentration of $\text{Cu}^{++} = 0.1 \text{ M}$

Activity or molar concentration of pure metal is arbitrarily taken as 1.

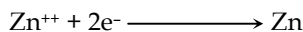
Now applying the Nernst equation,

$$\begin{aligned} E_{\text{Cu}} &= E^\circ_{\text{Cu}} - \frac{RT}{nF} \ln \frac{[\text{Cu}]}{[\text{Cu}^{++}]} = 0.34 - \frac{8.314 \times 293}{2 \times 96500} \ln \frac{1}{(0.1)} \\ &= 0.34 - 0.01262 \times \ln 10 = 0.34 - 0.01262 \times 2.302 = 0.34 - 0.029 = 0.31 \text{ volt.} \end{aligned}$$

Example 4.10 What will be the electrode potential of Zn electrode when it is immersed in decimolar concentration of ZnSO_4 solution at 25°C ? (Standard reduction potential of Zn is -0.76 volt).

Solution

Since standard reduction potential is given. Here we write reduction reaction.



Applying the Nernst equation,

$$E = E^\circ - \frac{RT}{nF} \ln \frac{[\text{Zn}]}{[\text{Zn}^{++}]}$$

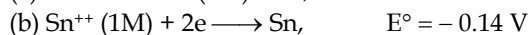
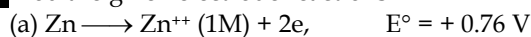
Also 25°C ,

$$E = E^\circ - \frac{0.0591}{n} \log \frac{[\text{Zn}]}{[\text{Zn}^{++}]}$$

For pure metal, the activity or molar concentration is arbitrarily taken as one

$$\begin{aligned} E &= E^\circ - \frac{0.0591}{n} \log \frac{1}{[\text{Zn}^{++}]} \\ &= -0.76 - \frac{0.0591}{2} \log \frac{1}{(0.1)} = -0.76 - 0.02955 \log 10 = -0.76 - 0.02955 = -0.789 \text{ volt.} \end{aligned}$$

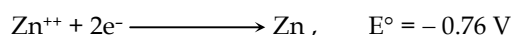
Example 4.11 You are given electrode reactions



Calculate the emf of this combination and write down the cell notation.

Solution

Before using the formula, $E_{\text{Cell}}^\circ = E_{\text{C}}^\circ - E_{\text{A}}^\circ$, we have to find out which one is anode and which one is cathode. For this, single electrode potential values should be in terms of reduction. Here, electrode potential of electrode (b) is in terms of reduction, however, that of (a) is in terms of oxidation. Therefore, we have to change electrode potential of (a) in terms of reduction as,



Now, it is clear that electrode (a) has low value of standard reduction electrode potential and it becomes anode whereas electrode (b) has high value of standard reduction electrode potential and it becomes cathode. Therefore, cell notation is



$$\begin{aligned} \text{emf of cell is, } E_{\text{Cell}}^\circ &= E_{\text{C}}^\circ - E_{\text{A}}^\circ \\ &= -0.14 - (-0.76) \text{ V} \\ &= 0.62 \text{ V} \end{aligned}$$

Self Test

4.2 Calculate the emf of cell having Zn and Cu when both electrodes are immersed in their corresponding metal sulphate electrolyte at 0.1 M concentration at 25°C .

4.16 Some Commercial Cells

In the previous units, we have discussed that every redox reaction can be used to produce electrical energy. However, every such redox reaction can not be suitable to develop commercial cells. This is

because electrochemical cells having salt bridge have high internal resistances. If large quantity of current is to be produced, voltage drops sharply. Besides this such types of cells may not be portable.

Commercial cells should be easy to carry and drop in voltage should be negligible during its use. When a number of cells are connected in series, it is termed as battery.

The commercial cells are classified as follows.

1. Primary cells
2. Secondary cells
3. Fuel cells

1. **Primary cells:** Primary cells are those electrochemical devices where spontaneous cell reaction occur in only one direction and become dead after reaching to equilibrium state. The electrochemical cell reactions can not be reversed. It means the primary cells can't be recharged. The examples of primary cells are dry cells and button cells. Here, working principle of dry cell is described.

Dry cell was invented by a French engineer George Leclanche in 1866. This type of cell is much familiar and is used in torches, toys, tape recorders and many other devices. It is one of the most widely used primary cell.

Construction of cell: Dry cell consists of zinc cylinder which acts as anode. A graphite rod is kept at the centre which acts as cathode. A paste of NH_4Cl and ZnCl_2 is filled (in the form of paste in water) near to the zinc electrode.

The graphite rod is surrounded by black paste of MnO_2 and carbon powder. A porous paper separates the electrolyte and zinc container. The zinc cylinder may have an outer insulation.

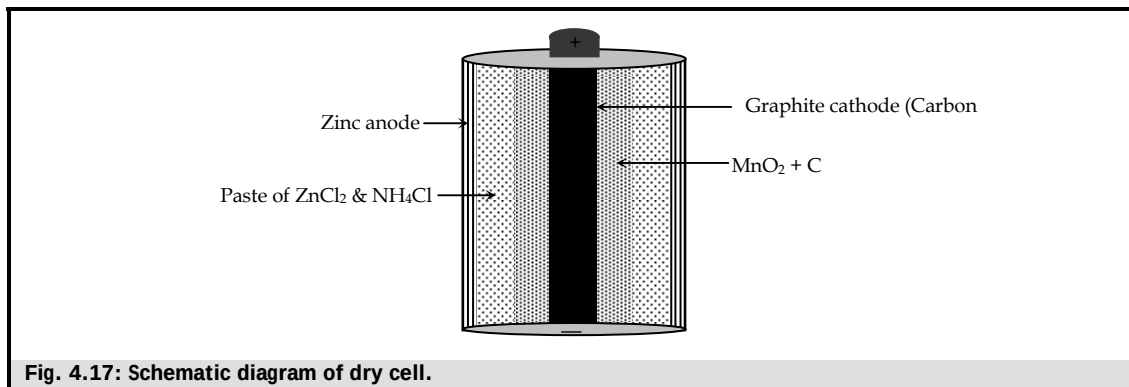
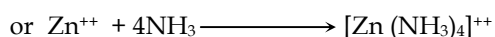
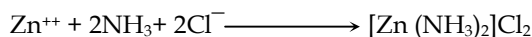
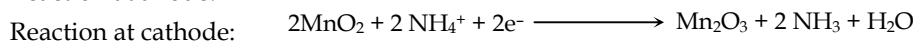


Fig. 4.17: Schematic diagram of dry cell.

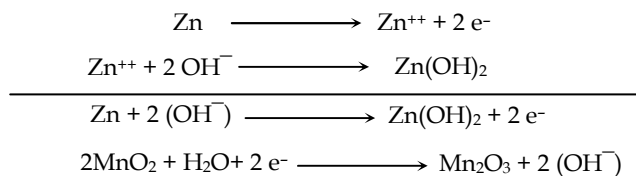
Electrode reactions



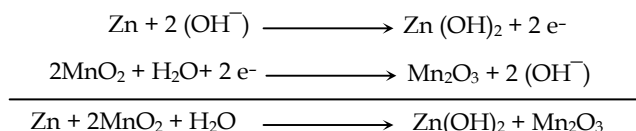
When the cell is in operation, zinc loses electrons to form Zn^{++} ions which dissolve in electrolyte solution. The released electrons pass to the external circuit and are taken by MnO_2 . At the cathode manganese is reduced from oxidation state of +4 to +3. Besides this, NH_4^+ ions are discharged at the cathode to form NH_3 . However, ammonia is not liberated as gas but it

combines with Zn^{++} to form diammine zinc (II) complex (poisonous).

- ☛ The cell produces a potential of 1.5 volt.
- ☛ The dry cell has no indefinite life because, NH_4Cl being acidic continuously corrodes the zinc container even when the cell is not in operation.
- ☛ In alkaline version of dry cell, NH_4Cl is replaced by NaOH or KOH . The alkaline cells have longer life. The electrode reaction taking place can be interpreted as



Complete cell reaction is obtained by adding reaction at anode and at cathode.

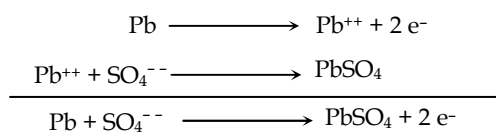
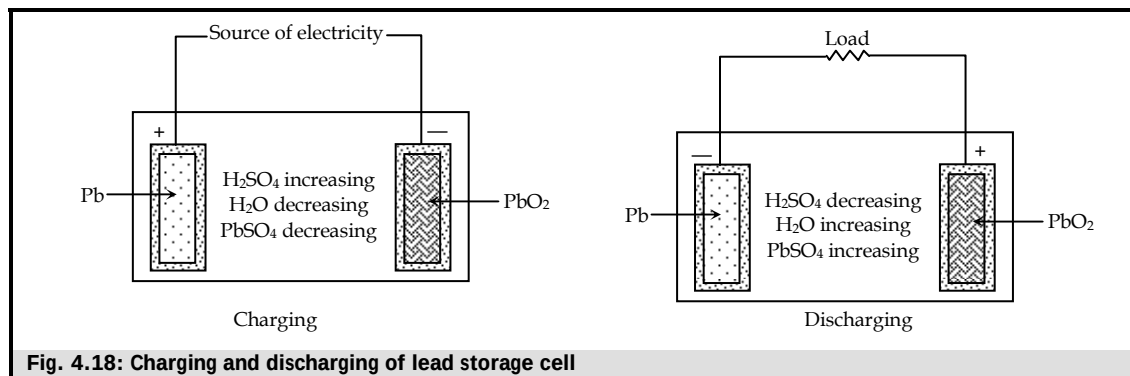


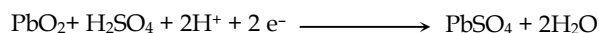
2. **Secondary cells:** Those electrochemical cells whose electrode reaction can be reversed by the application of external electric energy and can be recharged are called secondary cells. The example includes lead storage cell, nickel cadmium cell etc.

Lead storage cell acts as galvanic cell during discharge and electrolytic cell during charging. During the discharge process, the products are insoluble so that they adhere to the electrode and don't fall into the electrolyte solution.

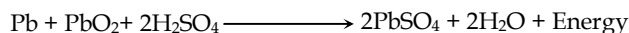
Construction: A lead storage cell or lead acid accumulator contains a number of voltaic cell in series to form a battery. Usually, 3 to 6 voltaic cells are connected in series to get 6 to 12 volt battery.

Spongy lead acts as anode and a grid of lead packed with PbO_2 acts as cathode. An aqueous solution of H_2SO_4 having specific gravity 2.15 is used as electrolyte.





The complete cell reaction is



During this process PbSO_4 precipitates at both anode and cathode and the potential of each cell is about 2V.

☞ In the above reaction, it found that

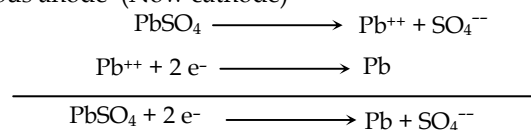
- The weight of anode (Pb) and cathode (PbO_2) go on decreasing.
- Concentration of H_2SO_4 goes on decreasing.
- The amount of product PbSO_4 and H_2O go on increasing.

Here, the cells deliver electrical energy. Hence, it acts as voltaic cell during discharging (or working). When both anode and cathode are covered by PbSO_4 , the cell reaction stops and no more energy is obtained.

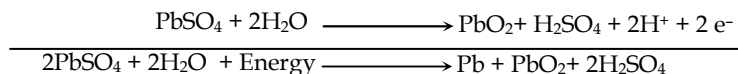
Recharging of battery: To recharge the battery, it is connected to external source of direct current. The positive pole of battery is connected to the positive pole of source of current and negative pole of battery is connected to the negative pole of source of current. In this stage, the cell gain the electric current and acts as electrolytic cell.

The following reaction takes place:

To the previous anode (Now cathode)



To the previous cathode (Now anode)



During the recharging process,

- The electrodes of the cell are restored to their original conditions.
- Concentration of H_2SO_4 is increased.

3. Fuel Cells

Fuel cell is a device in which chemical energy of fuels are directly converted into electricity.



In ordinary combustion reaction, oxidation of fuels takes place and heat energy is produced which is converted into other forms of energy. In case of fuel cell, electric energy is directly produced by the oxidation of fuel. Examples of fuel cells are as,

- Hydrogen-oxygen fuel cell
- Propane-oxygen fuel cell
- Methyl alcohol-oxygen fuel cell
- Phosphoric acid fuel cell
- Biochemical fuel cell.

Here, hydrogen-oxygen fuel cell is described. It is simplest form of fuel cell. The cell consist of two porous electrodes. The electrodes are made of graphite impregnated with finely divided platinum or alloy of palladium and silver. Nickel electrodes can also be used. Whatever may be the electrodes used, these should fulfill the following requirements.

- Electrodes should be good electron sources or sinks.
- Electrode should not be affected by electrode reactions.
- The electrodes should behave as catalyst for the reactions which take place on the surface of electrodes.

The electrolytes used may be KOH or H_2SO_4 or ion exchange resins saturated with water. Pure H_2 and O_2 gases are bubbled through anode and cathode compartment, respectively under a pressure of 50 atm at 250°C . The gases and electrolyte diffuse through the electrode pores and chemical reaction takes place. Electric current is obtained from anode and cathode source.

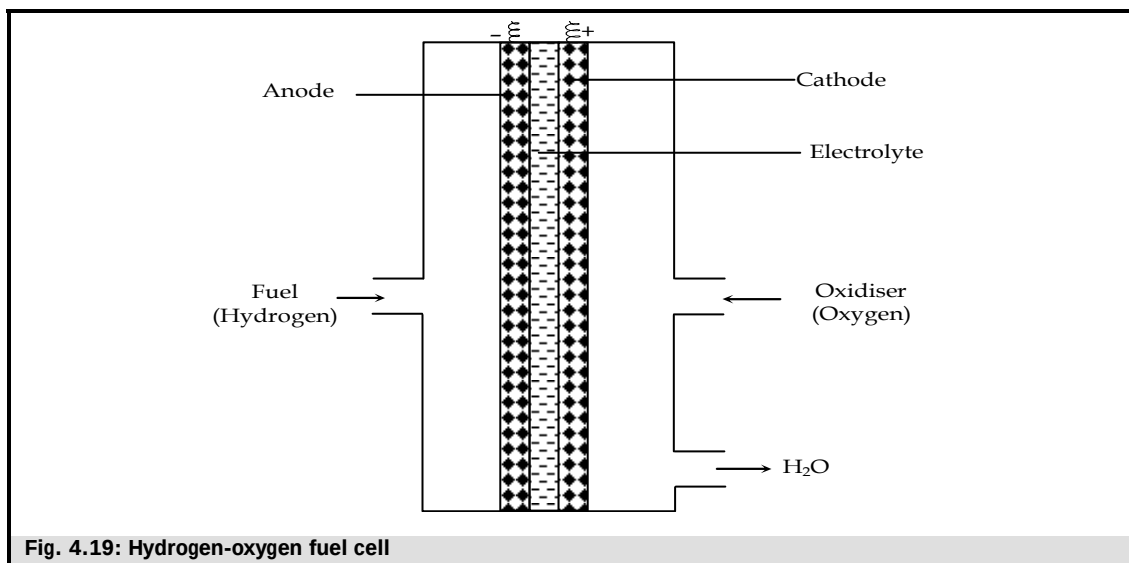
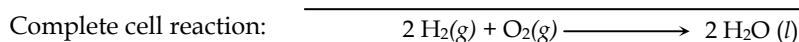
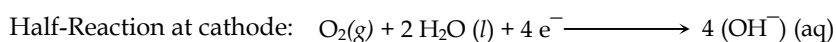
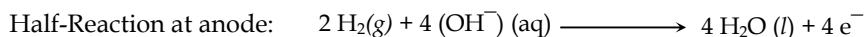


Fig. 4.19: Hydrogen-oxygen fuel cell

The electrode reaction takes place as follows.



Standard emf of cell $E^\circ = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ (Both Reduction Potential)

$$= 0.40 - (-0.83) = 0.40 + 0.83 = 1.23 \text{ volt.}$$

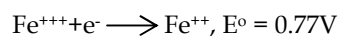
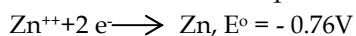
In practice emf of cell obtained is 0.80 to 1.0 volt.

Application

- It is auxiliary source of energy in spacecraft, submarine or in military vehicles.
- Fuel cell has higher efficiency than that in thermo-chemical reaction.
- The weight of fuel battery is relatively low in comparison to engine-generator set.
- In $\text{H}_2\text{-O}_2$ fuel cell, the product is water which can be used for drinking purpose. So it is environmentally friendly.

Solved Numerical Problems

1. The standard electrode potential for the following electrode are;



- i. Represent a suitable galvanic cell and point out which one will be cathode?
- ii. Write cell reaction and cell notation.

- iii. With 1 M solutions of the ions what will be emf?
 iv. Will the reaction $\text{Zn}^{2+} + 2\text{Fe}^{3+} \longrightarrow \text{Zn} + 2\text{Fe}^{2+}$ occur? Give reason.

Solution

- (i) Here Zn^{2+}/Zn has lower reducing potential value than $\text{Fe}^{3+}/\text{Fe}^{2+}$ electrode. So, Zn^{2+}/Zn electrode is an anode and $\text{Fe}^{3+}/\text{Fe}^{2+}$ electrode is a cathode.
 (ii) Electrode and cell reactions:
 At anode: $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$ (Oxidation)
 At cathode: $\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$ $\times 2$ (Reduction)
 Cell reaction: $\text{Zn} + 2\text{Fe}^{3+} \longrightarrow \text{Zn}^{2+} + 2\text{Fe}^{2+}$ (Redox reaction)
 The cell notation is
 $\text{Zn}/\text{Zn}^{2+} (1\text{M}) // \text{Fe}^{3+}(1\text{M}) / \text{Fe}^{2+}, \text{Pt}$
 (iii) E°_{cell} (Emf of cell) = $E^\circ_{\text{C}} - E^\circ_{\text{A}} = 0.77 - (-0.76) = 1.53 \text{ V}$
 (iv) For $\text{Zn}^{2+} + 2\text{Fe}^{2+} \longrightarrow \text{Zn} + 2\text{Fe}^{3+}$, oxidation should take place on $\text{Fe}^{3+}/\text{Fe}^{2+}$ electrode and reduction must take in Zn^{2+}/Zn electrode. In such case,
 $E^\circ_{\text{cell}} = E^\circ_{\text{C}} - E^\circ_{\text{A}} = -0.76 - 0.77 = -1.53 \text{ V}$
 Since, Emf is negative, the above reaction is non-spontaneous. The reaction will not be feasible.

■ Solved Questions

1. What is electrode potential?

- ↪ Electrode potential is defined as the potential developed at the solid liquid interface when a metal rod is dipped into its electrolyte solution. This arises due to the formation of electrical double layer between electrolyte and electrode interface. Example: Standard electrode potential of Zn^{2+}/Zn electrode is -0.76 V .

2. How is single electrode potential originated?

- ↪ When an electrode is dipped into electrolyte solution of its soluble salt then either the metal atom loses electrons and positive ions dive into the electrolyte solution or the ions of electrolyte accept electrons and deposit on the electrode. During this process there is separation of charge on the electrode and electrolyte interface and single electrode potential is originated.

3. What is electrochemical series?

- ↪ A series of electrode materials which is arranged on the basis of increasing reduction electrode potential or decreasing oxidation electrode potential value is called electrochemical series. It is also called activity series. In this series active metals lie above the series and less active metals lie below in this series.

4. What is SHE?

- ↪ If hydrogen gas is passed through 1M HCl solution at 1 atm pressure and 25°C by using platinized platinum foil immersed into 1M HCl solution, it is called standard hydrogen electrode (SHE) or normal hydrogen electrode (NHE). Its standard electrode potential is 0 volt. Therefore, it is used as reference electrode to measure the standard single electrode potential of different electrode.

■ Questions

■ Very Short Answer Questions:

- Define the following terms:

a. Electrode	b. Electrode potential
c. Standard electrode potential	d. Standard oxidation potential
e. Standard reduction potential	f. Salt bridge
g. Primary cell	h. Secondary cell
i. Fuel cell	

- Construct a galvanic cell in which the cell reaction is:
 $\text{Fe(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{H}_2(\text{g})$
- Standard hydrogen electrode can act as both anode or cathode. Explain.
- Mention one important application of standard hydrogen electrode giving example.
- What is meant by single electrode potential? Name any two factors which affect its magnitude.
- What is the important application of standard hydrogen electrode (SHE)
- Discuss the origination of single electrode potential.
- Discuss the meaning of standard oxidation potential of Cu/Cu^{2+} is -0.34V .
- Why can't the CuSO_4 solution be stored in Zn vessel? ($E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76\text{V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34\text{V}$)

■ Short /Long Answer Questions:

- Write short notes on (a) Standard electrode potential (b) electrochemical series.
- What is electrochemical series? Discuss its applications.

■ Numerical Problems

- Can a solution of 1M CuSO_4 be stored in a vessel made of nickel metal? If not, why?
 Given: $E^\circ_{\text{Ni}^{++}/\text{Ni}} = -0.25\text{V}$; $E^\circ_{\text{Cu}^{++}/\text{Cu}} = 0.34\text{V}$
Ans: No, Ni dissolves spontaneously
- Will the reaction occur: $\text{Zn}^{++} + 2\text{Fe}^{++} \rightarrow \text{Zn} + 2\text{Fe}^{+++}$
 Given standard reduction potentials are:
 $E^\circ_{\text{Zn}^{++}/\text{Zn}} = -0.76\text{V}$, $E^\circ_{\text{Fe}^{2+}/\text{Fe}^{3+}} = +0.77\text{V}$
 Give reason.
Ans: No. E° is negative
- Can a solution of 1M ZnSO_4 be stored in a vessel made up of Nickel?
 Given
 $E^\circ_{\text{Ni}^{++}/\text{Ni}} = -0.25\text{V}$ $E^\circ_{\text{Zn}^{++}/\text{Zn}} = -0.76\text{V}$
Ans: Yes. Ni does not dissolve
- Predict which one of the following reactions occur spontaneously?
 i. $2\text{Fe}^{2+} + \text{Sn}^{4+} \longrightarrow 2\text{Fe}^{3+} + \text{Sn}^{2+}$
 ii. $2\text{Fe}^{3+} + \text{Sn}^{2+} \longrightarrow 2\text{Fe}^{2+} + \text{Sn}^{4+}$
 Given standard reduction potentials of
 $\text{Fe}^{3+}/\text{Fe}^{2+}$ and $\text{Sn}^{4+}/\text{Sn}^{2+}$ are $+0.77\text{V}$ and $+0.15\text{V}$, respectively.
Ans: ii, E° = Positive
- You are given zinc rod, copper rod, zinc sulphate and copper sulphate solutions and standard electrode potential of zinc and copper are -0.76V and 0.34V respectively.
 a. Represent an electrochemical cell indicating anode and cathode
 b. Write net cell reactions
 c. What will be the emf of the cell?
Ans: c 1.10 V
- You are given standard reduction potential of Cu^{2+}/Cu and Fe^{2+}/Fe as $+0.34\text{V}$ and -0.44V respectively.
 a. Construct a Galvanic cell indicating anode and cathode
 b. Write the cell reaction and calculate the standard emf of the cell.
Ans: Standard EMF of the cell = 0.78V
- Construct a Galvanic cell in which the cell reaction is
 $\text{Fe(s)} + \text{H}_2\text{SO}_4(\text{aq}) \longrightarrow \text{FeSO}_4(\text{aq}) + \text{H}_2$ ($E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44\text{V}$)
- Design a Galvanic cell in which the cell reaction is
 $\text{Zn(s)} + 2\text{Ag}^+(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{Ag(s)}$.
 Predict
 a. Which of the electrode is negatively charged?

- b. Electrode reactions.
9. Consider the following standard electrode potential values:
- $$\text{Al}^{3+} + 3\text{e}^- \longrightarrow \text{Al} \quad E^\circ = -1.66 \text{ V}$$
- $$\text{Zn}^{2+} + 2\text{e}^- \longrightarrow \text{Zn} \quad E^\circ = -0.76 \text{ V}$$
- Answer the following questions:
- Construct the Galvanic cell
 - Identify anode and cathode of the cell
 - Write electrode reactions and cell reaction
 - Calculate emf of the cell
 - Write the reaction involved during charging of the cell.

Multiple Choice Questions

- The number of coulombs of electricity required to deposit 0.3 mole of Cu^{++} is
 - 9650
 - 2×9650
 - 4×9650
 - 6×9650
- By passing 0.50 A current to an aqueous solution half gram of an element is liberated. The time of passing the current in seconds is (Eq., Wt. = 96.5)
 - 1000 s
 - 500 s
 - 1000 s
 - 2000 s
- Electricity can be passed through molten lead (II) bromide because of the presence of
 - free electron
 - moveable ions
 - moveable atoms
 - lead in molten state
- In Faraday's law, the relation between electrochemical equivalent (ECE) and chemical equivalent (C.E) is
 - $\text{C.E.} = F \times \text{ECE}$
 - $\text{ECE} = F \times \text{CE}$
 - $\text{ECE} = \frac{F}{\text{CE}}$
 - $\text{CE} = \frac{F}{\text{ECE}}$
- The electrode reaction on the left hand side electrode of an electro-chemical cell with positive cell potential is
 - displacement reaction
 - oxidation reaction
 - reduction reaction
 - decomposition reaction
- What weight of copper will be deposited by passing two faradays of electricity through a solution of cupric salt? (At. wt. of Cu = 63.5)
 - 31.50 g
 - 2.00 g
 - 4.00 g
 - 63.5 g
- Given that,

$$\text{Zn}^{++} + 2\text{e}^- \longrightarrow \text{Zn} \quad [E^\circ = -0.763]$$

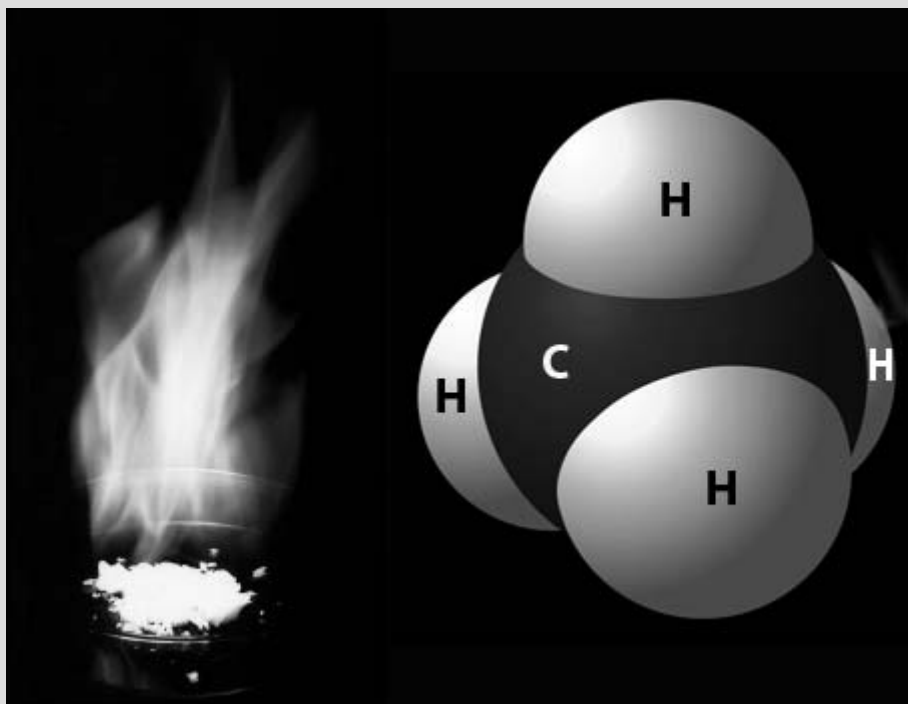
$$\text{Mg}^{++} + 2\text{e}^- \longrightarrow \text{Mg} \quad [E^\circ = -2.363]$$
 When Zn dust is added to MgCl_2 solution then
 - Zn is dissolved
 - ZnCl_2 is formed
 - Mg Crystallize out
 - No reaction
- One Faraday of electricity is equivalent to:
 - 1602×10^{-12} coulombs
 - 96500 coulombs
 - 95.5 coulombs
 - 22400 coulombs
- In a cell, oxidation takes place in.....
 - anode
 - cathode
 - electrolyte
 - none of these
- CuSO_4 can not be safely stored in
 - Al vessels
 - Ag vessels
 - Au vessels
 - Pt vessels
- In non-spontaneous reaction, E_{cell} is
 - Zero
 - positive
 - negative
 - can't be decided
- On electrolysis of CuSO_4 solution using copper anode, mass of the anode
 - increases
 - decreases
 - remains constant
 - can't be decided

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13
d	c	b	a	a	b	d	d	b	a	a	b	b

Energetics of Chemical Reaction

5



Combustion of hydrocarbons like methane is an exothermic reaction. So, this reaction can be used to produce heat.

5.1 Introduction

Any physical or chemical transformation is accompanied by change in energy. Flow of energy takes place from system to the surrounding and vice-versa. Measurement of change in temperature, pressure, concentration, volume etc of a system from initial state to final state can provide the information about change in energy and its related quantities such as heat and work. Different forms of energy are convertible from one to another and these are related quantitatively.

The branch of science which deals with the study of different forms of energy and their quantitative aspect is called thermodynamics. When study of different forms of energy is limited to chemical change, it is termed as chemical thermodynamics. Initially the term thermodynamics was introduced by mechanical engineers who were concerned mainly in the conversion of heat into mechanical work. In this sense *thermo* means heat and *dynamics* means motion which results into mechanical work.

Now, the study of thermodynamics is broad and it is based on different generalizations. Three laws of thermodynamics are first, second and the third law. In this Unit, we will discuss about the first law of thermodynamics. In the next (Unit 6), we shall discuss about the second law of thermodynamics with a brief account of the third law.

5.2 Some Thermodynamic Terms

Every discipline has its own terminology. Some basic terms involved in thermodynamics are briefly discussed below.

1. **System:** System is a part of the universe which is under consideration for the purpose of theoretical or experimental study.
2. **Surrounding:** Surrounding is a part of the universe with which a system can interact. A space of immediate vicinity of the system under study is surrounding.
3. **Boundary:** A real or imaginary interface which separates the system from surrounding is called boundary.

Let us consider a gas is taken in a cylinder fitted with piston. Let the cylinder is kept in a water bath maintained at 25°C. In this case the gas within cylinder is system. The wall of cylinder is boundary and the water bath is surrounding. The boundary may be permeable or impermeable or rigid.

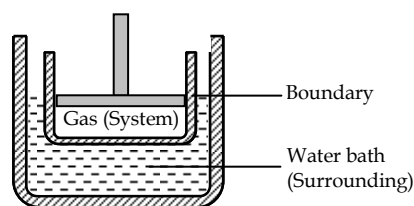


Fig. 5.1: Schematic diagram to illustrate system, surrounding and boundary

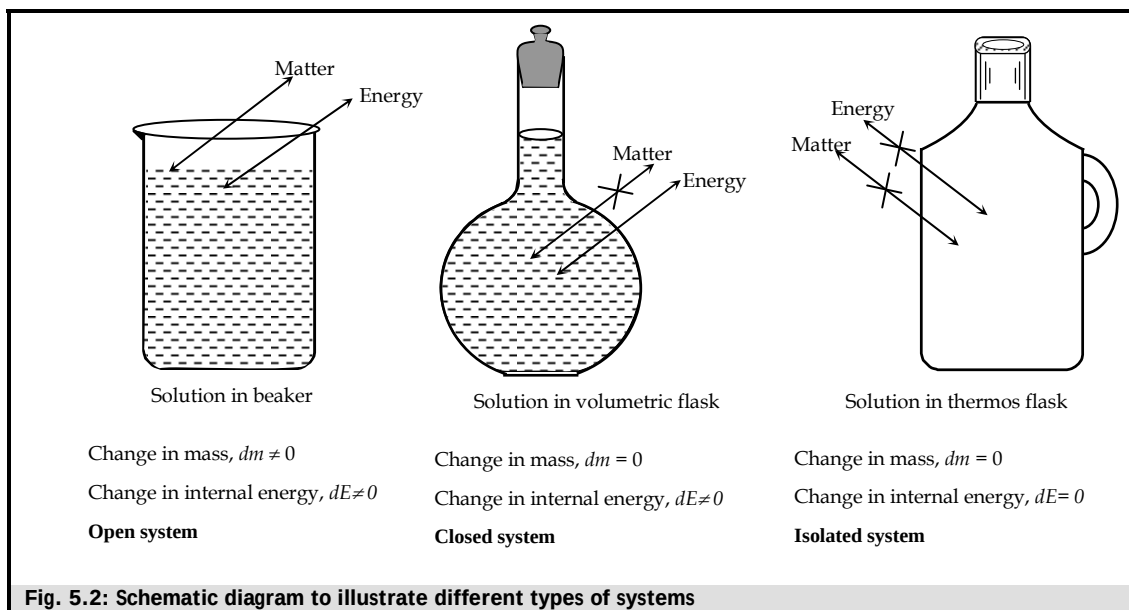
A permeable wall allows the passage of matter and energy. An impermeable wall prevents the flow of mass.

Classification of System on the Basis of Exchange of Energy between System and Surrounding

Depending upon the mode of exchange of heat and energy, a system can be classified into following types.

- a. **Open system:** A system which can exchange matter as well as energy with the surrounding across the boundary is called open system e.g, solution in an open beaker.

- b. **Closed system:** A system which can exchange energy but cannot exchange matter with the surrounding across the boundary is called closed system. e.g. hot solution in stoppered volumetric flask.
- c. **Isolated system:** A system which can neither exchange energy nor exchange matter with the surrounding across the boundary is called isolated system. e.g. hot water kept in thermos flask.



Classification of System Based Upon the Composition

On the basis of composition of system, it can be classified into homogeneous and heterogeneous system.

- Homogeneous system:** The system which is uniform throughout the given mass is called homogeneous system. e.g., pure solid, pure liquid, pure gas, alloys, true solution of solute in solvent, mixture of gases etc. A homogeneous system consists only one phase. A phase is a homogeneous, physically distinct and mechanically separable portion of a system.
 - Heterogeneous system:** The system which is not same throughout the given mass is called heterogeneous system. A heterogeneous system consists of two or more phases. Ice in contact with water has two phases; one phase is ice and another phase is water. Therefore, it is heterogeneous system. Similarly, water in contact with water vapour is also heterogeneous system.
- Universe:** The whole part of system, surrounding and boundary is called universe.
System + surrounding = universe
 - State functions (or state variables):** If we say water is in a beaker, it is not well defined term because different questions arise to define the system completely. These questions may be,
What is the volume of water?
What is the temperature of the water?
At what pressure, the system (water) is measured?

If all these questions are answered, the water in the beaker is well defined. A system is defined by measurable properties such as temperature (T), pressure (P), volume (V) and amount of substance or number of moles (n) which are called **state variables**. These properties are called state variables because these define the state of a system.

A thermodynamic system is said to be in a particular state when all of its properties are fixed. The state of a system is particularly defined in terms of temperature (T), pressure (P), volume (V) and composition. Change in the magnitude of these properties alters the state of a system. Let us consider an example of 1 kg ice at 0°C and 1 atm pressure. If temperature is increased, it undergoes into another state i.e. water. If temperature is further increased upto 100°C , the liquid form changes to vapour form. Besides temperature, when pressure is changed, the normal boiling point is also changed. In this way change in magnitude of state variables changes a system from its one initial state to another state. Hence, these thermodynamic variables are also called **state variables** or **state functions**. It can also be justified that when a system is brought from initial state to final state, it is accompanied by change in state variables. It does not matter how is the changes brought about.

State functions can also be defined as those thermodynamic properties which depend upon only initial and final states of a system but not upon the path of the change. Therefore, state function is also called path independent function. For example when a sample of water is heated from 10°C to 25°C , the change in temperature is equal to the difference between final temperature and initial temperature.

$$\Delta T = T_{\text{final}} - T_{\text{initial}} = 25^{\circ}\text{C} - 10^{\circ}\text{C} = 15^{\circ}\text{C}$$

It does not matter how the change in temperature is brought about.

For pure substance, the composition is always hundred percent. The remaining state variables are T , P and V . For ideal gas, these are related by equation of state as, $PV = nRT$. Here R is universal gas constant and n is number of mole of gas. Out of three variables, P , V and T , if two variables are known then the remaining variables are determined by the equation of state. The variables, P and T must be specified to define a system. These are called independent state variables. Then the remaining variable, volume, is determined by using the equation $PV = nRT$. Volume depends upon value of pressure and temperature. Here volume is dependent function of temperature and pressure, i.e. $V = f(T, P)$. State variables can be integrated between the

approximate limits e.g. $\int_{E_1}^{E_2} = E_2 - E_1$.

Table 5.1: Some state functions

Name of state function	Symbol
Temperature	T
Pressure	P
Volume	V
Number of mole	n
Internal energy	E
Enthalpy	H
Entropy	S
Gibbs free energy	G
Helmholtz free energy	A
Chemical potential	μ
Heat capacity	C_p and C_v

6. **Path functions (Path variables):** Heat (q) and work (w) are thermodynamic variables but not state function because their values depend upon the path via which the change is brought about. Work done carried out reversibly is greater than the work done carried out irreversibly. Similarly, reversibly exchanged heat is greater than the irreversibly exchanged heat. These facts can be proved mathematically. However, this is beyond the scope of the text in this book.

An analogy of state function can be exemplified in mountain climbing. Suppose we have to climb Mount Everest from its base camp. There may be various paths to climb up the mountain but the vertical distance from the base camp to the peak of the Mount Everest is same. The height of the mountain peak cannot be changed by changing the climbing path. The amount of work done and extent of fatigue depends upon the path chosen. In this example, height of the mountain peak from its base camp is analogous to state function while work done to climb up the mountain is not analogous to the state function. Path variables cannot be integrated between

the approximate limits. e.g. $\int_{w_1}^{w_2} \neq w_2 - w_1$.

7. **Extensive and intensive properties:** The macroscopic properties of a system can be classified into two types;
- Extensive properties
 - Intensive properties
- Extensive properties:** Those properties which depend upon the amount of substance (or number of moles) present and are additive in nature are called extensive properties. For example, mass of a system is extensive property because sum of masses of each subsystem is equal to the total mass of the system. If 40 g of NaOH is divided into three parts each having 20 g, 15 g and 5 g then total mass (20 + 15 + 5 = 40 g) remains same.
 - Intensive properties:** Those properties which do not depend upon the amount of substance (or number of moles) present and are non-additive in nature are called intensive properties. When a system is divided into n parts or subsystem and each part has the same value of property equal to that of the system, then these properties are intensive properties. For example, if 500 mL of water at 100°C is divided into five vessels at constant temperature then temperature of water in each vessel is also 100°C. Here each drop of water has 100°C temperature. Therefore, temperature is intensive property. If extensive properties are expressed in terms of per mole or per gram, etc., these become intensive properties.

Table 5.2: Some examples of some extensive and intensive properties

Extensive property	Intensive property
Volume	Molar volume
Number of moles	Temperature
Mass	Density
Energy	Specific gravity
Internal energy	Boiling point, freezing point
Enthalpy	Molar enthalpy
Entropy	Molar entropy
Heat capacity	Specific heat capacity
Free energy	Refractive index
	Chemical potential
	Free energy per mole
	Pressure

8. **Thermodynamic processes:** It is obvious that when liquid is heated it is changed into vapour. Here, state is changed from liquid to vapour. This is one of the examples of thermodynamic process. Any process which brings about the change in the state of a system is called thermodynamics operation or process. Various operations such as contraction, expansion, freezing, sublimation, vaporization, etc. are thermodynamic processes.

The thermodynamic processes can be classified on the following basis.

- a. **Classification of thermodynamic processes on the basis of constant variable:** During thermodynamic process, some variables can be kept constant. Accordingly thermodynamic processes are classified as follows.

- i. **Isothermal process:** If temperature is kept constant during various operations, the process is called isothermal process. To keep the temperature constant, heat is either removed or supplied. For isothermal process $dT = 0$ but $dq \neq 0$.
- ii. **Adiabatic process:** If no heat is exchanged (supplied or lost) between system and surrounding during various operations, the process is called adiabatic process. To keep the heat constant, temperature must be changed. Hence, adiabatic process is always accompanied by change in temperature. In adiabatic process $dq = 0$ but $dT \neq 0$. Joule-Thomson expansion is an example of adiabatic process.
- iii. **Isobaric process:** If pressure is kept constant during various operations then the process is called isobaric process. In isobaric process $dp = 0$.

Conversion of 1 mole of water at 20°C and 1 atm pressure into 100°C and 1 atm pressure is isobaric process. All the reaction carried out in open vessels are said to be carried out in isobaric condition as the pressure (atmospheric pressure) is constant.

- iv. **Isochoric process:** If volume is kept constant during various operations, the process is called isochoric process. In isochoric process $dV = 0$.

Combustion of chemical substance in a bomb calorimeter is an example of isochoric process.

- v. **Cyclic process:** If a system undergoes various changes and ultimately returns into initial state, the process is called cyclic process.

For cyclic process $dE = 0$, $dH = 0$. Here, dE and dH are infinitesimal change in internal energy and enthalpy respectively.

These thermodynamic processes are shown diagrammatically as follows.

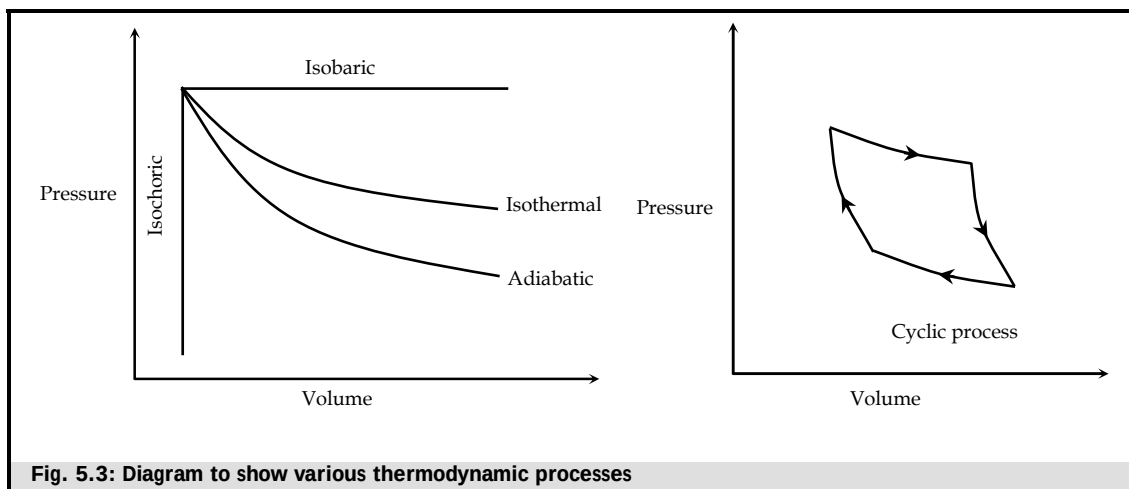


Fig. 5.3: Diagram to show various thermodynamic processes

- b. **Classification of thermodynamic processes on the basis of way of achieving the final state:** When a thermodynamic process is carried out, the final state of the system can be achieved either in single step or in multiple steps. On the basis of this, there are two types of thermodynamic processes. These are;

- i. Reversible process.
 - ii. Irreversible process.
- i. **Reversible process:** The process which takes place infinitesimally slowly such that its direction at any point can be reversed by infinitesimal change in the state of the system is called reversible process. In actual practice, it is difficult to perform experiments under reversible condition. However, the concept of reversibility serves as a means to study many thermodynamic processes which are of immense importance to chemists.

Evaporation of liquid within a closed vessel at constant temperature can be considered as reversible process. Chemical reactions taking place in dry cell, batteries, etc. are very near to the reversible process. Reversible processes are carried out slowly so they can be considered to be carried out in infinitesimal number of steps. At each step, the system is at equilibrium because the infinitesimal change in the state of the system has negligible effect.

To illustrate the concept of reversible process, let us take an example of compression and expansion of gas.

Let us take a sample of gas in a cylinder fitted with weightless and frictionless piston.

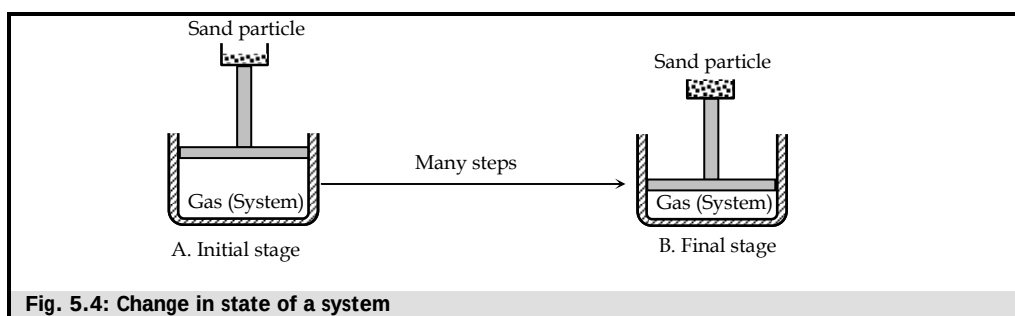


Fig. 5.4: Change in state of a system

Let the gas be at initial state as shown in the figure 5.4 (A). The amount of sand particles is gradually increased into the piston at different time interval. At each time, the weight of particle pushes the piston downward as equivalent to $W = mg$. Due to this, the pressure of gas within the cylinder increases but infinitesimally at each step. When the process is repeated till large amount of sand (say 500 gram) particles are added then the piston reaches at final step as shown in figure B. In this example, the final state is reached reversibly from initial state in infinitesimally large number of steps. If each particle of sand is removed one at each time, previous stage can be achieved. This illustrates an example of reversible process.

- ii. **Irreversible process:** When a process is carried out rapidly such that the system does not get a chance to attain equilibrium between initial and final state, it is called irreversible process. Such types of processes are carried out in single step or in few steps. Equilibrium stage can be achieved only in the initial and final states. All natural processes are irreversible processes. In the above example of reversible expansion of gas, if 500 g sand is added in single step, the piston goes down and reaches to stage B. When the sand is removed suddenly, the piston goes up accordingly. This illustrates the irreversible expansion of gas. Expansion of gas against vacuum is an example of irreversible expansion.

Table 5.3: The differences between reversible process and irreversible process

Reversible process	Irreversible process
1. When a process takes place infinitesimally slowly through a succession of equilibrium steps, it is called reversible process.	1. When a process takes place rapidly from initial to final step, it is called irreversible process.
2. It is slow process. It takes place in infinite number of steps.	2. It is fast process. It takes place in single step or in few steps.
3. All steps in the operation are at equilibrium.	3. Only initial and final steps are at equilibrium.
4. Work done in reversible process is maximum	4. Work done in irreversible process is lower than that in reversible process.
5. Heat exchanged reversibly is maximum.	5. Heat exchanged irreversibly is lower than that in reversible process.

5.3 State of Equilibrium

A state of a reversible process in which a macroscopic property (eg, temperature, pressure, concentration, density, etc.) does not change with time is called state of equilibrium. A system undergoes changes until it achieves the state of equilibrium. In the light of thermodynamics, an equilibrium condition must satisfy the following two conditions.

- The macroscopic properties should not change with time. The properties of a system due to bulk of mass rather than atomic or molecular nature are called macroscopic properties. Properties such as temperature, pressure, concentration, density etc are macroscopic properties. However, the submicroscopic properties may change continuously. For example, in equilibrium of colloidal system, the Brownian motion is not stopped.
- The system should be at equilibrium without any external assistance.

For a system to be at equilibrium both of the criteria mentioned above should be fulfilled. To illustrate this let us take an example of a water tank through which 10 mL water is lost in one hour by leakage. The same volume of water is obtained by leakage from tap into this tank in the same rate. Here, the level of water remains same. Its macroscopic properties do not change with time. The first condition is satisfied. But second condition is not satisfied because the level of water is maintained from external supply (leakage of water from tap into the tank) rather than the system itself. Hence, the above example signifies a **steady state** or **stationary state**, equilibrium state and steady state. Both are seen dynamic in nature. An insight view reveals that an equilibrium state is static on macroscopic level and dynamic on microscopic level while the steady state is dynamic on macroscopic as well as microscopic level.

In thermodynamics, the equilibrium state is attained to balance temperature, pressure and concentration. Accordingly, the thermodynamic equilibrium involves the following equilibria.

- Thermal equilibrium
 - Mechanical equilibrium
 - Chemical equilibrium
- i. **Thermal equilibrium:** When equilibrium condition is obtained due to difference in temperature, it is called thermal equilibrium. In thermal equilibrium each part of a system is at same temperature and this temperature is equal to the temperature of surrounding. When hot water and cold water are mixed, hot water loses temperature while cold water gains

temperature until equilibrium temperature is obtained. Ultimately, the temperature of mixture should be equilibrium with the temperature of surrounding.

- ii. **Mechanical equilibrium:** When equilibrium condition is obtained due to difference in pressure, it is called mechanical equilibrium. In mechanical equilibrium, there is no unbalanced force in the interior part of the system. Besides this, there is no unbalanced force between system and surrounding. If there exist unbalanced forces in the interior part of the system, the state of the system changes on passing the time. If unbalanced forces exist between system and surrounding, system as well as surrounding undergoes a change.

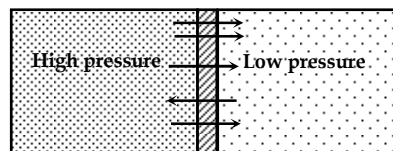


Fig. 5.5: Schematic diagram to illustrate mechanical equilibrium

When two gases at different pressure are separated followed by its removal, gases intermix till equilibrium.

- iii. **Chemical equilibrium:** When equilibrium condition is obtained due to difference in chemical composition, it is called chemical equilibrium. After attaining chemical equilibrium, the chemical composition of the system becomes uniform. Thermodynamically, a system is said to be in chemical equilibrium when chemical potential (μ) of each part of the system remains same.

For example, when a concentrated solution and dilute solution are mixed then their concentration undergo change until each part of the mixture has same concentration.

Types of equilibrium based upon stability

Equilibrium condition can be classified into following types.

- i. Stable equilibrium ii. Unstable equilibrium iii. Metastable equilibrium

- i. **Stable equilibrium:** A system is said to be in a state of stable equilibrium if it returns to its original state on removing the constraint causing to displace it. The adjoining figure shows a ball resting in a shallow pit. This illustrates stable equilibrium. If the ball is attempted to displace in either direction, ultimately it comes to its original position.

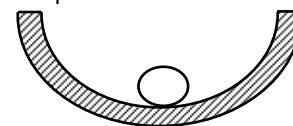


Fig. 5.6: Stable equilibrium

- ii. **Unstable equilibrium:** A system is said to be in a state of unstable equilibrium if it does not return to its original state on removing the constraint causing to displace it. In fact, this type of equilibrium is not real equilibrium. A ball resting on a hillock in the adjoining figure illustrates unstable equilibrium.

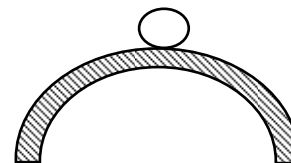


Fig. 5.7: Unstable equilibrium

- iii. **Metastable equilibrium:** A system is said to be in a state of metastable equilibrium if the system lies between the stages of stable and unstable equilibria. For a small disturbance the system remains at equilibrium. If the disturbance is extensive, the system is at unstable equilibrium. Supersaturated solution is an example of metastable equilibrium. For a very small disturbance the supersaturated solution remains intact. If the

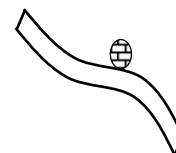


Fig. 5.8: Metastable equilibrium

level of disturbance is increased, precipitation takes place instantly. Similarly, a super cooled liquid is also an example of metastable equilibrium. The adjoining figure depicts an example of metastable equilibrium.

5.4 Heat and Work (Modes of Energy Transfer)

Change in state of a system is always accompanied by transfer of energy. The energy is transferred either in the form of heat or in the form of work.

Unit of Energy: SI unit of energy is joule (J). It can also be expressed in terms of kJ, calorie (cal), kcal etc.

Transfer of energy in the form of heat

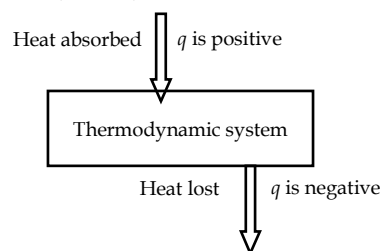
When a system and surrounding are at **two different temperatures**, the energy is transferred in the form of heat. If system is at higher temperature, the surrounding, heat is lost from the system. If surrounding is at higher temperature than the system, heat is gained by the system.

Sign convention of heat exchanged

If heat is gained by a system, it is taken as positive.

If heat is lost by a system, it is taken as negative.

Unit of heat The S.I. unit of heat is joule. It is also expressed in calorie 1 calorie = 4.184 Joule



Transfer of energy in the form of work

If system and surrounding are at two different pressures, energy is exchanged in the form of mechanical work. Let us consider a certain volume of gas enclosed in a container having a weightless and frictionless piston. If the pressure of gas is more than the external pressure, the piston is pushed outward. Here, work is done by the system. On the other hand if external pressure is greater than the pressure of gas inside the vessel, the piston is pushed inward. Here, work is done on the system. In both cases, energy is transferred in the form of work.

$$\text{Work (} W \text{)} = \text{Force (} F \text{)} \times \text{displacement (} d \text{)} \quad \dots (i)$$

$$\text{We know, Pressure (} P \text{)} = \frac{\text{Force (} F \text{)}}{\text{Area (} A \text{)}}$$

$$\text{or, Force (} F \text{)} = \text{Pressure (} P \text{)} \times \text{Area (} A \text{)}$$

$$\text{Now, Relation (i) becomes Work (} W \text{)} = \text{Pressure (} P \text{)} \times \text{Area (} A \text{)} \times \text{Displacement (} d \text{)}$$

$$\text{or, Work (} W \text{)} = \text{Pressure (} P \text{)} \times \text{change in volume (} \Delta V \text{)}$$

$$\text{or, } W = P \cdot \Delta V.$$

$$\text{Here, change in volume (} \Delta V \text{)} = \text{Area (} A \text{)} \times \text{Displacement (} d \text{)}$$

Unit of work: In CGS system, the unit of work is erg. The S.I. unit of work is joule (J). The units erg and joule are related as, 1 Joule = 10^7 erg

Sign convention of work done

- In terms of energy:** When work is done on the system by surrounding, the internal energy of the system is increased and work done on the system is taken as positive. If work is done by the system to the surrounding, the energy of the system is decreased and work done by the system is taken as negative. IUPAC recommends this system.
- In terms of volume:** When work is done on the system by surrounding, the volume of the system is decreased. Let V_1 be the initial volume and V_2 be the final volume of gas enclosed in cylinder. If work is done on the system V_2 becomes smaller than V_1 then $V_2 - V_1$ becomes negative. Therefore, $W = P \cdot \Delta V = \text{negative value}$

If work is done by the system, then volume of system is increased. It means V_2 becomes greater than V_1 . In this case $V_2 - V_1$ becomes positive. Therefore, $W = P \cdot \Delta V = \text{positive value}$

Sign convention: Heat gained by the system = Positive

Heat lost by the system = Negative

Work	In terms of energy change (IUPAC convention)	In terms of volume change
Work done on the system	+ ve	- ve
Work done by the system	- ve	+ ve

5.5 Internal Energy (E)

Every substance is associated with energy. Each atom, ion or molecule of a chemical substance has a definite quantity of energy within it. Molecule of a chemical substance possesses translational energy (E_t), rotational energy (E_r), vibrational energy (E_v), electronic energy (E_e), bond energy (E_b), intermolecular interaction energy (E_i). Hence, total energy of molecule is given by

$$\text{Total energy (E)} = E_t + E_r + E_v + E_e + E_b + E_i$$

Energy of a system includes both kinetic energy and potential energy.

The sum of all kinds of energy including kinetic energy and potential energy associated with a system is called internal energy or intrinsic energy. It is denoted by E or U (IUPAC recommends U). In thermodynamics, internal energy is dealt rather than energy of individual chemical species.

Some points about internal energy are given below,

- Internal energy is a state function. It does not depend upon the path. It only depends upon the state variables such as temperature (T), pressure (P) and volume (V). Out of these three variables, when two variables are known, the third variable can be found by using equation of state ($PV = RT$). Therefore, internal energy is a function of any two variables out of T , P and V .
- Internal energy is extensive property. Its value depends upon the amount of substance taken. For example, if amount of substance is increased, the energy of the system is also increased.
- The absolute value of internal energy cannot be determined. However, change in internal energy (ΔE), can be determined. If E_1 be the initial internal energy and E_2 be the final internal energy, change in internal energy is given by, $\Delta E = E_2 - E_1$

5.6 The First Law of Thermodynamics

In any physical or chemical transformation, there is change in energy. In some cases, energy is lost by a system to the surrounding while in some other cases energy is gained by a system from the surrounding. During these transformations, one form of energy may change into another form of energy. But the total energy of system and surrounding does not change. It means total energy of the universe remains conserved. This forms the basis of the first law of thermodynamics. The first law of thermodynamics can be defined in different ways as:

- ♦ The total energy of a system and surrounding remains conserved.
- ♦ Energy can neither be created nor be destroyed but one form of energy can be converted into equivalent amount of other form of energy.
- ♦ Whenever a particular type of energy is disappeared, an equivalent amount of another form of energy must be reappeared.

- ♦ It is impossible to construct a perpetual motion machine which can produce work without spending energy on it.

Formulation of the first law of thermodynamics

The first law of thermodynamics is nothing but it is an application of law of conservation of energy to the thermodynamics system. Suppose a thermodynamic system possesses internal energy E_1 initially. Let q quantity of heat is supplied to it. Thus supplied heat is used to increase the internal energy of the system or does some work or both. Let E_2 be the internal energy in the final state of the system.

$$\text{So, increase in internal energy} = E_2 - E_1 \quad \dots \text{(i)}$$

Heat absorbed = Increase in internal energy + work done by the system.

$$\text{or } q = (E_2 - E_1) + W$$

$$\text{or } q = \Delta E + W \quad \dots \text{(ii)}$$

Here, work done by the system is taken as positive and heat gained by the system is also taken as positive.

W includes all types of works which may be mechanical work, electrical work, gravitational work and work to increase the surface area or work in the magnetic field. If mechanical work is considered, $W = P \cdot \Delta V$ holds true.

$$\text{Now, equation (ii) becomes, } q = \Delta E + P \cdot \Delta V \quad \dots \text{(iii)}$$

$$\text{For infinitesimal change equation (iii) can be expressed as, } dq = dE + P \cdot dV \quad \dots \text{(iv)}$$

Where, dq = infinitesimal increase in heat.

dE = infinitesimal increase in internal energy.

dV = infinitesimal change in volume.

Different forms of the first law of thermodynamics

Mathematically, the first law of thermodynamics is defined as $q = \Delta E + P \cdot \Delta V$

Case I: For isobaric process, there is no change in pressure. Hence, $q_p = \Delta E + P \Delta V$

Case II: For isothermal process, $\Delta E = 0$, $\therefore q = W$

Case III: For isochoric process $dV = 0$, $\therefore q_v = \Delta E$

In this case the supplied heat is converted to increase the internal energy of the system. In isochoric process there is no work of expansion.

Case IV: For adiabatic process, $q = 0$

In adiabatic process, there is no heat gained or lost. Therefore, the above equation change to

$$q = \Delta E + W$$

$$0 = \Delta E + W$$

$$\text{or } W = -\Delta E$$

In adiabatic process work is done by the compensation of internal energy.

5.7 Enthalpy

When we consider the first law of thermodynamics at constant volume and at constant pressure we get the following relations.

At constant volume, $q_v = \Delta E$

At constant pressure, $q_p = \Delta E + P \cdot \Delta V$

From these relations, it is revealed that besides internal energy, molecules also possess some additional energy in the form of pressure-volume work done. Here, it is to be remembered that work done is also a form of energy. Therefore, total heat content of a system at constant pressure is equal to sum of internal energy and pressure-volume work. The total heat content of a system is called enthalpy.

Mathematically, enthalpy is defined as, $H = E + PV$

Where,

H = Enthalpy, E = Internal energy, P = Pressure, V = Volume

Chemical reactions are usually carried out at constant pressure. Therefore, it is better to express energy change in terms of enthalpy rather than internal energy.

Some points about enthalpy

- Enthalpy is a state function. It depends upon the state variables of a system. It does not depend upon the path.
- Enthalpy is extensive property. It depends upon the amount of substance taken.
- The absolute value of enthalpy cannot be determined but change in enthalpy for a reaction can be determined.

At constant pressure, let enthalpy, internal energy and volume of the system change from H_1 , E_1 and V_1 , to H_2 , E_2 and V_2 respectively, then

$$\text{Initially, } H_1 = E_1 + PV_1 \quad \dots \text{ (i)}$$

$$\text{Finally, } H_2 = E_2 + PV_2 \quad \dots \text{ (ii)}$$

Subtracting relation (i) from relation (ii), we get

$$\begin{array}{r} H_2 = E_2 + PV_2 \\ \underline{H_1 = E_1 + PV_1} \\ \hline (H_2 - H_1) = (E_2 - E_1) + P(V_2 - V_1) \end{array}$$

$$\Delta H = \Delta E + P \cdot \Delta V \quad \dots \text{ (iii)}$$

$$\text{For infinitesimal change, } dH = dE + P \cdot dV \quad \dots \text{ (iv)}$$

- For a chemical reaction, $\Delta H = \sum H_{\text{product}} - \sum H_{\text{reactant}}$

For exothermic reaction, $\sum H_{\text{product}} < \sum H_{\text{reactant}}$, ΔH becomes negative

For endothermic reaction, $\sum H_{\text{product}} > \sum H_{\text{reactant}}$, ΔH becomes positive.

- Enthalpy change at constant pressure and volume

$$\text{From the first law of thermodynamics, } q = \Delta E + P \cdot \Delta V \quad \dots \text{ (i)}$$

At constant volume, $\Delta V = 0$ and q is replaced by q_v . Now expression (i) becomes $q_v = \Delta E \quad \dots \text{ (ii)}$

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At constant pressure, the equation (i) remains same. q is replaced by q_p .

$$\begin{aligned} q_p &= \Delta E + P \Delta V \\ &= (E_2 - E_1) + P(V_2 - V_1) & [\Delta E = E_2 - E_1 \text{ and } \Delta V = V_2 - V_1] \\ &= E_2 - E_1 + PV_2 - PV_1 \\ &= (E_2 + PV_2) - (E_1 + PV_1) \\ &= H_2 - H_1 & [\because H = E + PV] \end{aligned}$$

$$\therefore q_p = \Delta H \quad \dots \text{(iii)}$$

If H_2 is the enthalpy of products and H_1 is the enthalpy of reactants, ΔH is enthalpy change of reaction or heat of reaction. Hence heat of reaction is the measure of heat of a process at constant pressure.

Self Test

5.1 How are ΔH and ΔE related?

5.2 What are q_p and q_v ? How are they related?

Relation between enthalpy change and internal energy change:

Let H_1 , E_1 and V_1 be the enthalpy, internal energy and volume of a reactant system. At constant pressure, when reactants are changed into products then enthalpy, internal energy and volume becomes H_2 , E_2 and V_2 , respectively.

Here,

$$H_1 = E_1 + PV_1 \quad \dots \text{(i)}$$

$$H_2 = E_2 + PV_2 \quad \dots \text{(ii)}$$

The enthalpy change for the reaction can be obtained by subtracting the enthalpy of reactants from the enthalpy of products.

Now subtracting equation (i) from equation (ii)

$$\begin{aligned} H_2 &= E_2 + PV_2 \\ \underline{H_1 = E_1 + PV_1} \\ (H_2 - H_1) &= (E_2 - E_1) + P(V_2 - V_1) \\ \Delta H &= \Delta E + P \Delta V \quad \dots \text{(iii)} \end{aligned}$$

The equation (iii) holds good at constant pressure.

- ♦ At constant pressure: Enthalpy change of a reaction is equal to the sum of internal energy change and pressure volume work.
- ♦ At constant volume: $\Delta V = 0$. Now equation (iii) becomes $\Delta H = \Delta E \quad \dots \text{(iv)}$
At constant volume, enthalpy change is equal to internal energy change
- ♦ At constant temperature: For ideal gas $PV = nRT$

Now enthalpy of a system can be expressed as, $H = E + nRT$

Let a system changes from initial condition to final condition at constant temperature. Here, H_1 , E_1 , and n_1 are enthalpy, internal energy and number of moles of the system initially whereas H_2 , E_2 and n_2 are enthalpy internal energy and number of mole of the system finally.

$$H_1 = E_1 + n_1 RT \quad \dots \text{(v)}$$

$$H_2 = E_2 + n_2RT$$

... (vi)

Subtracting equation (v) from equation (vi)

$$\begin{aligned} H_2 &= E_2 + n_2RT \\ \underline{H_1} &= \underline{E_1 + n_1RT} \\ (H_2 - H_1) &= (E_2 - E_1) + (n_2 - n_1)RT \end{aligned}$$

$$\Delta H = \Delta E + \Delta nRT$$

... (vii)

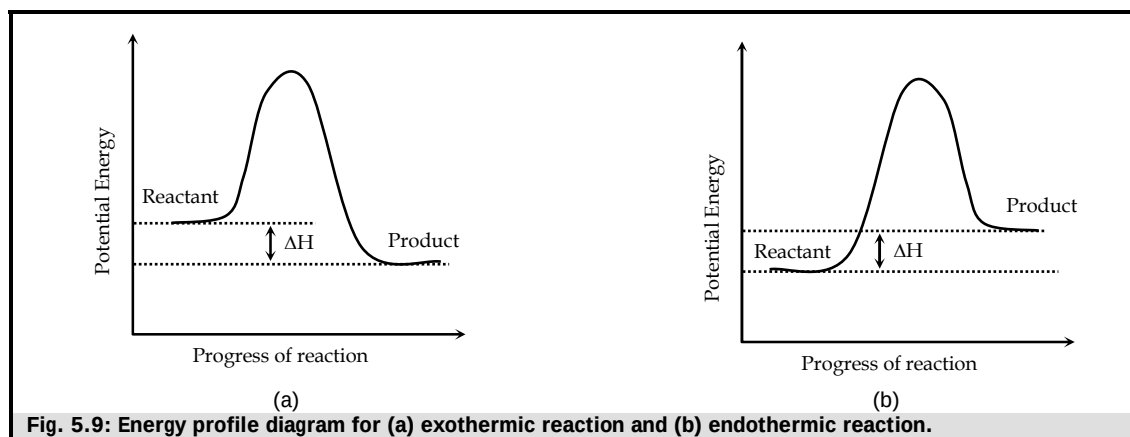
Endothermic and Exothermic Process

Physical change involves the change in state of matter although the constituents of matter are unaltered. The example includes melting of ice into water, vaporization of water, condensation of water vapour etc. Although the phase of the system changes, the chemical constituent H_2O remains same.

Chemical change involves breaking of some or all pre-existing bonds with the formation of new bonds. In chemical change totally new constituent of matter is formed.

Both physical and chemical changes are accompanied by change in energy. These involve breaking of all or some pre-existing bonds with the formation of new bonds. Breaking of bonds requires energy while formation of bond releases the energy. **If energy absorbed is greater than the energy released, net energy is absorbed in the system and this type of reaction is called endothermic reaction. On the other hand if energy absorbed is less than the energy released, net energy is released from the system and this type of reaction is called exothermic reaction.**

The endothermic and exothermic reactions are shown by energy profile diagram (fig. 5.9).

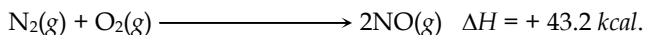


In chemical transformation, reactants don't directly change into products. The reactants have to surmount the higher energy path and ultimately change into products. Whatever may be the change, reactants initially absorb energy (endothermic process) and reach to high energy content, often called activated complex, then the energy rich complex changes into product with the release of energy (exothermic process). The net change in the energy is the key factor which determines whether the reaction is exothermic or endothermic.

The process which involves the gain of energy is called endothermic process. In endothermic process the energy of product (H_P) is higher than the energy of reactant (H_R).

$$\text{Enthalpy change of reaction } (\Delta H) = H_P - H_R = +ve \quad [\because H_P > H_R]$$

Example:

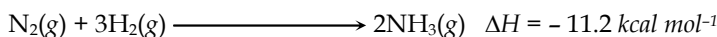


Since heat is absorbed, the reaction is endothermic. Melting of ice is also endothermic process.

The process which involves the loss of energy is called exothermic process. In exothermic process, the energy of product is lower than the energy of reactant.

The enthalpy change of reaction (ΔH) = $H_P - H_R$

For example,

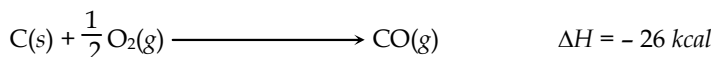


Since heat is liberated the reaction is exothermic.

5.8 Enthalpy Change

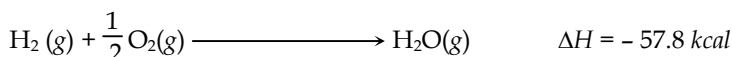
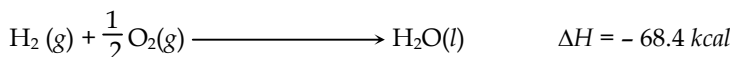
When a chemical reaction takes place heat is either evolved or absorbed. **The quantity of heat change involved when specified numbers (number of moles of reactants in a balanced chemical equation) of reactants are completely reacted under the given set of condition is called heat of reaction or enthalpy of reaction.** The following points are to be noted while studying the enthalpy of reaction.

- i. To express the enthalpy of reaction, the chemical equation must be balanced.
- ii. While balancing the chemical equation, the coefficients of chemical species need not always to be whole number. For example, for the formation of CO from carbon and oxygen the chemical equation is:



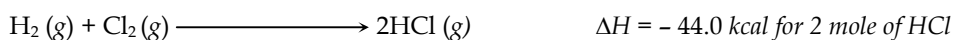
- iii. In a chemical equation, the physical state of all the chemical constituents must be written because heat change will be different for different states of matter.

Example,



When one mole of hydrogen gas reacts with half mole of oxygen gas to give one mole of liquid water, energy released is 68.4 kcal at 25°C. But, when one mole of hydrogen gas reacts with half mole of oxygen to give one mole of water vapour, energy released is 57.8 kcal. This suggests conversion of 1 mole of liquid water into 1 mol of water vapour is endothermic process and it demands 10.6 kcal energy at 25°C.

- iv. For exothermic process, the enthalpy change of reaction is negative and for endothermic process, the enthalpy change of reaction is positive.
- v. The value of ΔH should correspond to the balance equation. Enthalpy change is usually expressed in per mole.



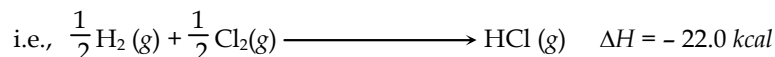
Since two moles of HCl (g) are formed, the ΔH value corresponds for the formation of 2 moles of HCl (g).

The enthalpy of formation of 2 mol of HCl = - 44 kcal

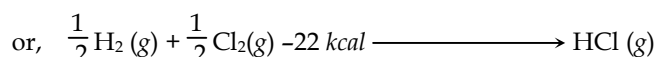
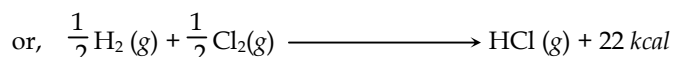
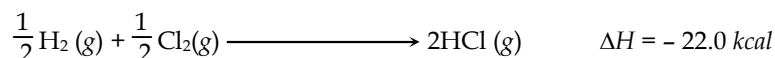
The enthalpy of formation of 1 mol of HCl = - 22 kcal

Therefore, enthalpy of formation of HCl is - 22 kcal per mol.

In practice, the term per mole is omitted. Enthalpy of formation of HCl is -22 kcal means -22 kcal energy is released when one mole of HCl is formed from half mole of hydrogen and half mole of chlorine.



- vi. The same chemical equation can be expressed in different ways. For example,



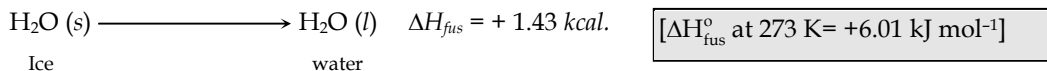
- vii. Enthalpy of reaction will be different at different temperature and pressure. To compare, the enthalpy of reactions at 25°C temperature, 1 atmospheric pressure are taken as standard reference state and the chemical substance should be in pure and standard state. Enthalpy of one mole of pure substance at 25°C and 1 bar pressure is called **standard enthalpy**. It is denoted by H° .

The quantity of heat change involved when the number of mole of reactants of balanced chemical equation are completely reacted at 25°C and 1 bar pressure is called **standard heat of reaction** or **standard enthalpy of reaction**. It is denoted by ΔH° .

A. Enthalpy Change in Physical Transformation

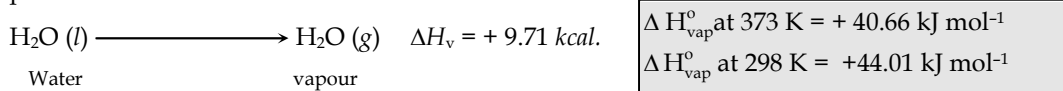
The arrangement of constituent particles and mode of cohesion determines the state of matter. Accordingly, the three states of matter possess different magnitude of interparticle forces. When heat energy is supplied solid changes to liquid and liquid changes to vapour. Thus supplied energy is used to break the interparticle forces. Hence, change in state of matter involves enthalpy change. The enthalpy changes in different phase transformations are given below.

- a. **Enthalpy of fusion (ΔH_{fus}) or heat of fusion.** It is defined as the quantity of heat change involved when 1 mole of pure solid is converted into liquid state at its melting point.



Higher value of heat of fusion indicates the higher magnitudes of intermolecular forces of attraction.

- b. **Enthalpy of vaporization (ΔH_{v}) or heat of vaporization.** It is defined as the quantity of heat change involved when one mole of pure liquid is converted into vapour state at its boiling point.

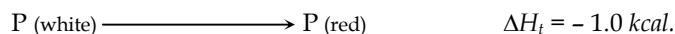
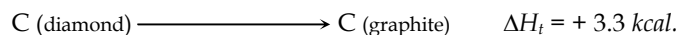


Higher value of heat of vaporization indicates the higher magnitude of intermolecular forces of attraction.

- c. **Enthalpy of sublimation (ΔH_s) or heat of sublimation.** It is defined as the quantity of heat change involved when 1 mol of pure solid is directly converted into vapour state below its melting point.



- d. **Enthalpy of transition (ΔH_t) or heat of transition.** It is defined as the quantity of heat change involved when 1 mol of an element changes from its one allotropic form to another allotropic form at its transition temperature.



B. Enthalpy Change in Chemical Reaction

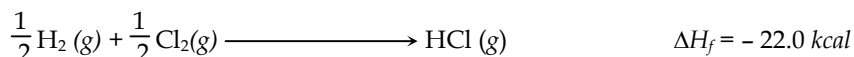
Chemical reactions are accompanied by change in energy. The enthalpy changes in chemical reactions are given below.

- Heat of formation or enthalpy of formation.
- Heat of combustion or enthalpy of combustion
- Heat of neutralization or enthalpy of neutralization.
- Heat of solution or enthalpy of solution
- Heat of hydration or enthalpy of hydration.

Some of these are discussed below

- a. **Enthalpy of formation: (Heat of formation) ΔH_{form}** : It is defined as the quantity of heat change involved when one mole of chemical substance is formed from its constituent elements.

Example:



- ◆ Enthalpy of pure compound is taken as enthalpy of formation from its constituent elements at standard state.
- ◆ By convention, the standard heat of formation of all elements in their most stable form is taken as zero. If enthalpy of formation of a substance is positive, the compound is energetically less stable than its elements at its standard state.
- ◆ Standard heat of reaction can be calculated from the standard heat of formation as
 $\Delta H^\circ = \text{Total } \Delta H_f^\circ \text{ of product} - \text{Total } \Delta H_f^\circ \text{ of reactant.}$

or $\Delta H^\circ = \sum (\Delta H_f^\circ)_P - \sum (\Delta H_f^\circ)_R$

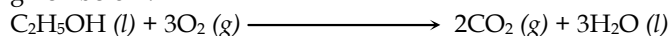
The quantity of heat change involved when one mole of chemical substance is formed from its constituent elements under standard conditions of all the reactants and products is called standard heat of formation. It is denoted by ΔH_f° or $\Delta_f H^\circ$.

Table 5.4: Standard heat of formation of some compounds at 25 °C and 1 atm:

Chemical substance	ΔH_f° in kJ mol^{-1}
$\text{H}_2 (\text{g})$	0.0
$\text{H}_2\text{O} (\text{g})$	- 241.8
$\text{H}_2\text{O} (\text{l})$	-285.8
$\text{HCl} (\text{g})$	-92.3
$\text{HBr} (\text{g})$	- 36.23
$\text{HI} (\text{g})$	+ 26.48
$\text{NH}_3 (\text{g})$	- 46.2

CO (g)	- 110.5
CO ₂ (g)	- 393.5
NH ₄ NO ₃ (s)	-368.0
NaCl (g)	-411.0
CH ₄ (g)	-74.85
C ₂ H ₆ (g)	-83.85
C ₆ H ₆ (g)	+ 82.93
C ₆ H ₆ (l)	+ 49.03
C ₆ H ₆ (g)	+ 82.93
C ₂ H ₅ OH (l)	- 277.6

Example 5.1 The standard heat of formation of C₂H₅OH (l), CO₂(g) and H₂O(l) are - 277.6, - 393.5 and - 285.8 kJ, respectively. Calculate the standard heat of reaction for the reaction given below:

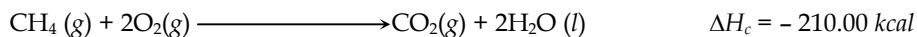


Solution

The ΔH_f° for O₂ is arbitrarily taken as zero.

$$\begin{aligned}\Delta H^\circ &= \sum (\Delta H_f^\circ)_{\text{Product}} - \sum (\Delta H_f^\circ)_{\text{Reactant}} \\ &= [2 \times (\Delta H_f^\circ \text{ for CO}_2) + 3 \times (\Delta H_f^\circ \text{ for H}_2\text{O})] - [\Delta H_f^\circ \text{ for C}_2\text{H}_5\text{OH}] \\ &= [2 \times (-393.5) + 3 \times (-285.8)] - [-277.6] \\ &= (-787 - 857.4) + 277.6 \\ &= -1366.8 \text{ kJ}\end{aligned}$$

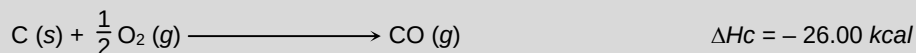
- b. **Enthalpy of combustion (Heat of combustion):** Heat of combustion is defined as the quantity of heat change involved when one mole of chemical substance is burnt completely with oxygen. It is denoted by ΔH_c . For example,



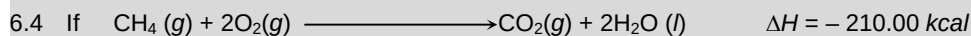
- ♦ Heat of combustion is always negative because combustion is exothermic reaction.

Self Test

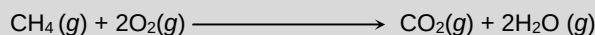
5.3 Study the following reaction.



What is the heat of combustion of carbon? Why?

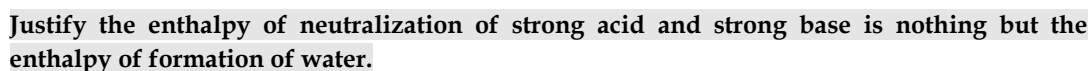


What is the enthalpy change for



Ans $\Delta H = -190.58 \text{ kcal}$

Calorific value of food and fuel: Food acts as biofuel. Food and fuel both are used for the production of energy. The energy released in the oxidation of food and fuel is usually measured in terms of energy per gram rather than per mole. This is because, food and fuel are mostly mixture of different composition and it is difficult to calculate molecular mass of each kind of molecule present in food. For example, if one takes rice, it does not consist only glucose but a mixture of varieties of constituents.



Bond Energy or Bond Enthalpy

When bond is formed between two atoms, energy is released because these two atoms have to come near to each other and work is done. On the contrary when bond is broken, energy must be supplied. Hence, formation of bond is exothermic process while breaking of bond is endothermic process.

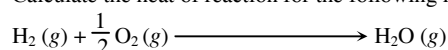
The bond energy is defined as the average amount of energy required to break all bonds in one mole of particular substance. It is measured in kcal mol^{-1} or kJ mol^{-1}

$$\text{Bond energy } (\Delta H) = \text{Bond energy of reactant} - \text{Bond energy of product}$$

Table 5.5: Bond energies of some common bonds:

Bond	Bond energy
Cl—Cl	243 kJ mol^{-1}
O = O	499 kJ mol^{-1}
C—H	414 kJ mol^{-1}
O—H	460 kJ mol^{-1}
H—H	$24.85 \text{ kJ mol}^{-1}$

Example 5.2 Calculate the heat of reaction for the following reaction:



Given bond energy:

$$\text{H—H} = 104 \text{ kcal mol}^{-1} \quad \text{O=O} = 118 \text{ kcal mol}^{-1} \quad \text{O—H} = 109 \text{ kcal mol}^{-1}$$

Solution

Heat of reaction (ΔH) = Bond enthalpy of reactants - Bond enthalpy of products

$$\text{or } \Delta H = (\Delta H \text{ of H—H}) + \frac{1}{2} (\Delta H \text{ of O = O}) - 2(\Delta H \text{ of O—H}).$$

$$= 104 + \frac{1}{2} \times 118 - 2 \times 109 = 104 + 59 - 218 = -55 \text{ kcal mol}^{-1}$$

$\therefore$ The heat of reaction is $-55 \text{ kcal mol}^{-1}$.

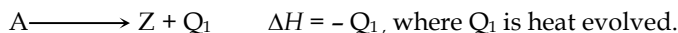
5.9 Hess's Law of Constant Heat Summation

Enthalpy change (ΔH) depends upon only the initial state and final state of the system and it does not depend upon the ways through which it is brought. Enthalpy is state function or it is path independent function. This fact was first generalized by G.H. Hess in 1840 and gave a law which is called Hess's law of constant heat summation. It states that at constant pressure **the amount of heat evolved or absorbed in a chemical reaction is always the same, regardless of the path by which the change occurs.** This law is concerned with conservation of energy.

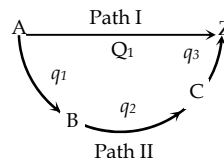
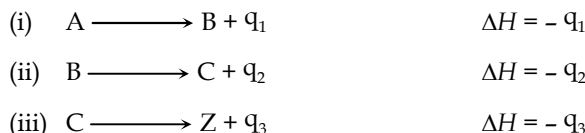
To reach upto the conclusion of Hess's law, we can add or subtract the chemical reactions algebraically whose enthalpy values are known in order to obtain the desired reaction. To balance the chemical equation, we can multiply by some number to a reaction so as to cancel out the intermediate product. Accordingly, we calculate the values of ΔH of these reactions to obtain the enthalpy of the net reaction.

Let us take a general reaction where A changes to Z.

When reactant A changes directly to Z



When reactant A changes to product Z by many steps



Total heat evolved in this case is $Q_2 = q_1 + q_2 + q_3$.

According to Hess's law $Q_1 = Q_2$.

Hess's law is the consequence of the first law of thermodynamics. This fact can also be justified as follows.

When, $Q_1 > Q_2$, (where $Q_2 = q_1 + q_2 + q_3$)

If reaction change is carried out by converting reactant (A) through high energy releasing path (i.e. path I) and converting the product back through a low energy releasing path (i.e. path II), in each complete cycle there is some release of energy i.e. energy released in one cycle = $Q_1 - Q_2$.

If the process is repeated many times, tremendous amount of heat is released which is against the first law of thermodynamics. Hence even according to the first law of thermodynamics Q_1 must be equal to Q_2 .

Illustration of Hess's law of Constant Heat Summation

Let us take the example of burning of carbon to CO_2 . Carbon can be burnt to give CO_2 directly or it may be changed first to carbonmonoxide then only to carbondioxide as shown in figure 5.10.

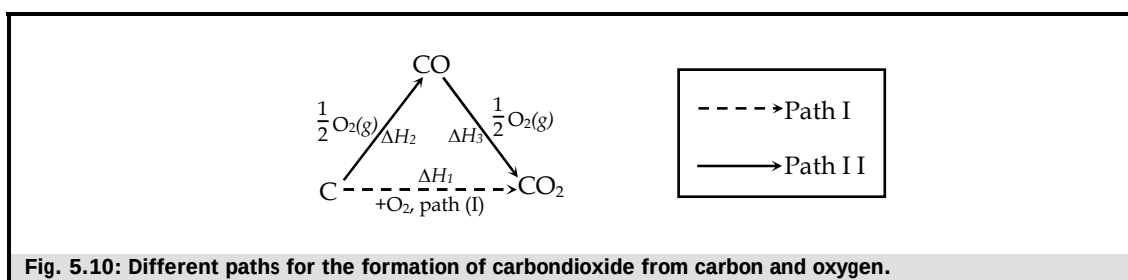
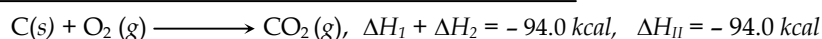
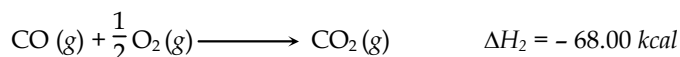
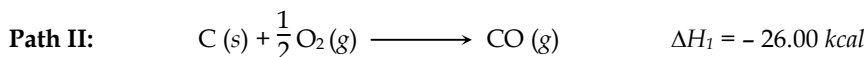
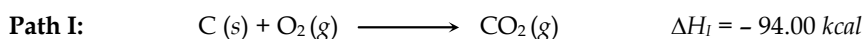


Fig. 5.10: Different paths for the formation of carbondioxide from carbon and oxygen.



Where, $\Delta H_{II} = \Delta H_1 + \Delta H_2$

From the above examples, it is found that total heat evolved is same whatever may be the path. This example justifies the Hess's law of constant heat summation.

Self Test

5.6 Explain the Hess's law of constant heat summation in the light of 1st law of thermodynamic.

Applications of Hess's Law

Hess's law is very useful in the field of chemistry for the determination of heat of a reaction. The applications are as follows.

- a. **Determination of heat of formation of substances which otherwise can't be measured experimentally:** Substance like carbon monoxide, methane, benzene can't be prepared by combining their constituent elements. Therefore, it is not possible to determine the heat of formation directly. Instead the heat of formation is calculated by applying Hess's law.
- b. **Determination of heat of transition:** The heat of transition of one allotropic form of element to another allotropic form can be calculated by applying Hess's law.
- c. **Determination of heats of various reactions:** By using Hess's law, enthalpies of many reactions can be calculated which otherwise can't be found out.

■ Solved Questions**1. Entropy is extensive property but temperature is intensive property. Explain.**

- Entropy is extensive property because it depends upon the amount of substance present. Larger the mass, larger will be the entropy. Entropy is the measure of randomness or disorder of a system. It is additive property.

Temperature is intensive property. It does not depend upon the quantity of substance. It is non-additive property. For example in a water system at 50°C every water drop is at 50° C.

2. State first law of thermodynamics.

- First law of thermodynamics states "Whenever energy of a particular type is disappeared, equivalent amount of another type of energy must be produced". Or "Energy can neither be created nor be destroyed"

If a quantity of heat (q) is supplied to a system, part of it goes to increase the internal energy (ΔE) of the system and part of it will be spent for the work done (W) by the system on the surrounding. Accordingly,

$$q = \Delta E + W \quad \dots (i)$$

Where,

q = heat supplied to a system

ΔE = change in internal energy of a system = E₂ - E₁

W = work done by the system

3. State Hess law of constant heat summation.

- Hess law of constant heat summation states "The amount of heat evolved or absorbed in a chemical reaction is always the same, regardless of whether the process takes place in a single step or in a series of steps".

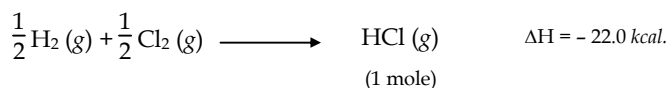
Example: write a suitable example.

4. What is meant by enthalpy of formation?

- The amount of heat involved when one mole of compound is formed from its constituent elements is called heat of formation.

Heat of formation of a compound is equal to its enthalpy of the compound.

Example



∴ Heat of formation of hydrogenchloride gas is - 22.0 kcal mol⁻¹.

Questions

Very Short Answer Questions:

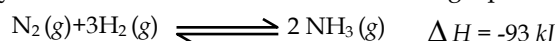
- Define the terms:
 - Extensive properties
 - Intensive properties
 - State function
 - Internal energy
 - Enthalpy
 - Isothermal process
- State the first law of thermodynamics.
- How are enthalpy change and internal energy change related?
- Neither heat nor work is a state function but their difference is state function. Comment it.
- Draw energy profile diagrams for exothermic and endothermic reactions.
- Define bond enthalpy.
- Define the terms:
 - Enthalpy of combustion
 - Enthalpy of formation
 - Enthalpy of neutralization
- State whether the following properties are extensive or intensive:
 - Temperature
 - Molar volume
 - Internal energy
 - Entropy
 - Enthalpy
- Deduce the relation $\Delta H = \Delta E + P\Delta V$, where the symbols have their usual meanings.

Short /Long Answer Questions:

- State and explain first law of thermodynamics and hence deduce $H = E + PV$, where all the symbols have their usual meanings.
- State and explain Hess's Law of constant heat summation. Also mention the applications of Hess's law of constant heat summation.
- Distinguish between:
 - Internal energy and enthalpy
 - Exothermic and endothermic reaction.
 - Reversible process and irreversible process
- State the first law of thermodynamics. What is its main limitation?

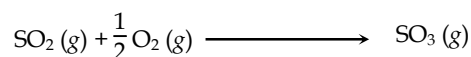
Numerical Problems

- Calculate the enthalpy of formation of NH_3 from the following equation.



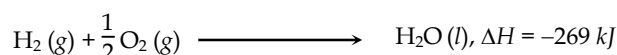
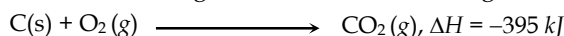
Ans: $\Delta H_f = -46.81 \text{ kJ}$

- The standard heat of formation of $\text{SO}_2(\text{g})$ and $\text{SO}_3(\text{g})$ are -296.6 kJ and -396 kJ respectively. Calculate ΔH for the reaction:



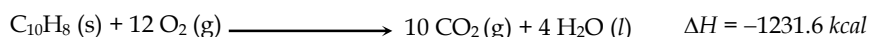
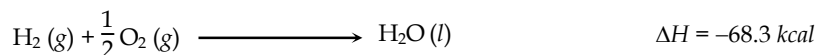
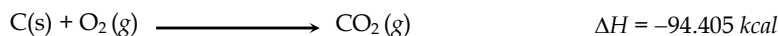
Ans $\Delta H = -99.4 \text{ kJ}$

- Calculate the heat of combustion of glucose from the following data:



Ans: -2815 kJ

4. Calculate the heat of formation of naphthalene ($C_{10}H_8$) from the following data:



Ans: Heat of formation of naphthalene = 14.35 kcal

5. Estimate the enthalpy change for the reaction



Given:

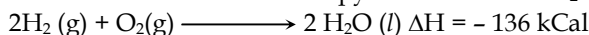
bond energy of H-H. = 435 kJ/mol,

bond energy of Cl-Cl = 243 kJ/mol,

bond energy of H - Cl = 430 kJ/mol

Ans: -182 kJ

6. Calculate the standard enthalpy of formation of $H_2O(l)$ in the following reaction:



Ans: -68 kcal mol⁻¹

7. Enthalpy of formation of benzene, water and carbondioxide are 55 kJmol⁻¹, -395 kJ mol⁻¹ and -285 kJ mol⁻¹, respectively. Calculate the enthalpy of combustion of benzene.

Ans: -2950 kJ mol⁻¹

8. Calculate the enthalpy of formation of ethane at 298 K, if the enthalpies of combustion of carbon, hydrogen and ethane are -94.14, -68.47 and -373.3 kCal mol⁻¹ respectively.

Ans: -20.39 kCal mol⁻¹

Multiple Choice Questions

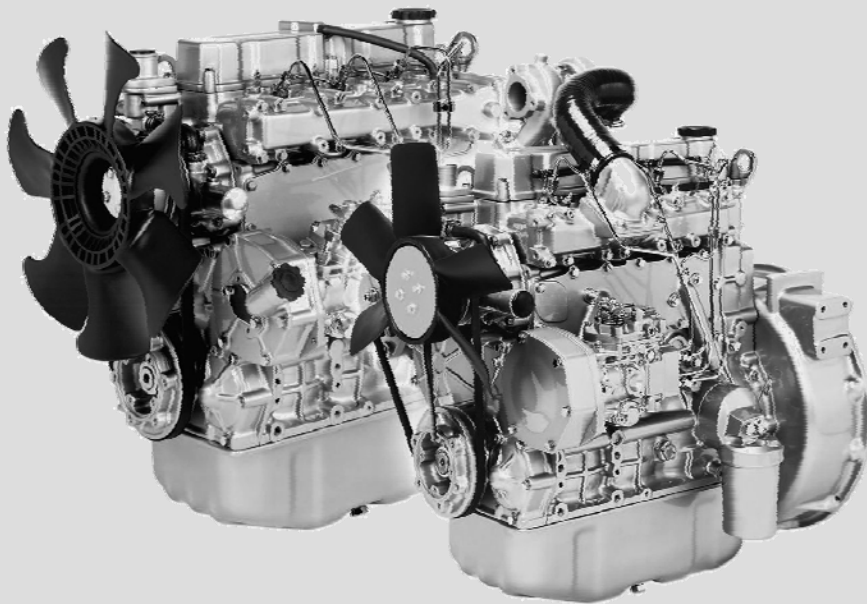
- In an adiabatic process
 - pressure is constant
 - the gas is expanded isothermally
 - there is perfect heat insulation
 - system exchanges heat with surrounding
- Hess's law deals with:
 - change in heat of reaction
 - rate of reaction
 - influence of pressure on volume of a gas
 - emf of redox reaction.
- According to the first law of thermodynamics.
 - heat can be flown from colder body to hotter body
 - energy can neither be created nor be destroyed
 - the efficiency of a heat engine can never be 100%
 - entropy of universe is increasing day by day
- H , E , P & V are related as,
 - $H = E + P - V$
 - $H = E + PV$
 - $H = E - PV$
 - $H = P - EV$
- In isothermal process
 - $dq = 0$
 - $dq \neq 0$
 - $dT \neq 0$
 - $dV = 0$
- Which of the following is a path function?
 - Enthalpy
 - Entropy
 - Heat
 - Temperature
- ΔH is
 - q_p
 - q_v
 - ΔE
 - Δn
- Enthalpy change of combustion is
 - +ve
 - ve
 - 0
 - +ve or -ve
- Direction of the spontaneous process is predicted by
 - First law
 - Hess law
 - Second law
 - Third law

Answer Key

1	2	3	4	5	6	7	8	9
c	a	b	b	b	c	a	b	c

Chemical Thermodynamics

6



An Internal Combustion Engine: Knowledge of thermodynamics is necessary to explain the working of internal combustion engine.

6.1 Introduction

The first law of thermodynamics is concerned with the law of conservation of energy as it explains about the exact equivalence of different forms of energy. If a hot iron ball at 100°C is connected to cold iron ball at 10°C , heat flows from hot ball to cold ball. The first law of thermodynamics explains that heat lost by hot ball is equal to heat gained by cold ball. But it does not tell us about the direction of flow of heat. There are so many natural phenomena which occur along a particular direction only. In one direction, these processes are spontaneous but not in opposite direction. To answer the feasibility of a process along a particular direction, second law of thermodynamics is required. Before discussing about the second law of thermodynamics, we shall discuss about spontaneous process and criteria for spontaneity.

6.2 Spontaneous and Non-spontaneous Process

A process which takes place itself (or by initiation at once) is called spontaneous process. **All processes occurring in nature are spontaneous processes or it can be said that all the irreversible processes are spontaneous processes.** Here, it is to be noted that the term spontaneous does not mean the process occurs at any time but the process occurs only under the given set of conditions. The time factors and mechanism of reaction are the domain of chemical kinetics but not the domain of thermodynamics.

Dissolution of glucose into water and burning of petrol in air both are spontaneous process. When glucose is kept in water it undergoes dissolution. On the other hand, petrol does not burn itself below its ignition temperature. It demands initiation. When a burning flame is exposed to the petrol, then only combustion starts. The reason behind this is inflammable substances such as diesel, petrol and kerosene are energy rich compounds and these are thermodynamically less stable. These require relatively higher activation energy to burn. When heat is supplied initially, the fuel starts to burn with the release of energy and that energy sustains activation energy required for further steps. It means fuels are thermodynamically less stable because these have high potential energy but these are kinetically stable because these require more activation energy.

Some examples of spontaneous process are given below.

- i. Neutralization of acids and bases
- ii. Rusting of iron
- iii. Evaporation of water
- iv. Dissolution of ammonium nitrate in water.
- v. Condensation of water vapour into water.
- vi. Freezing of water into ice.
- vii. Mixing of gases

Those processes which neither take place themselves nor take place by initiation are called non-spontaneous processes. Some examples of non-spontaneous processes are given below.

- i. Flow of water from lower level to upper level.
- ii. Flow of heat from cold body to hot body.
- iii. Flow of gases from low pressure region to high pressure region etc.

6.3 Driving Forces for Spontaneity

In the previous unit we have discussed about spontaneous and non-spontaneous processes. It arises a question to us that why some processes are spontaneous while some are not. The answer is, there must be some driving forces which are responsible for a process to be spontaneous. These driving forces for spontaneity are discussed below.

- 1. Decrease in enthalpy and spontaneity:** When a system loses energy, it will have minimum energy and becomes more stable. It means, to be stable, a system loses energy and the process is more likely to be spontaneous. This is understood by the following examples.

- i. Rolling down of a stone from top of hill is spontaneous process because while doing so the potential energy of the stone decreases.

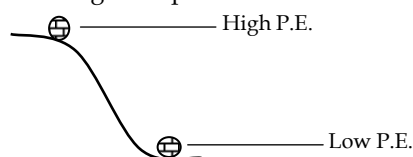
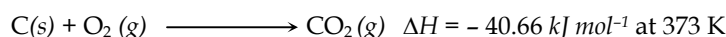
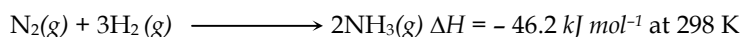
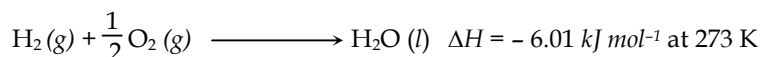


Fig. 6.1: Showing P.E. of a stone at higher and lower level (P.E. = mgh)

- ii. Water flows down spontaneously from a height because the potential energy of water at lower height is minimum.
- iii. Heat flows spontaneously from hot body to cold body because heat content of hot body decreases.
- iv. The following reactions are some examples of exothermic reactions. The heat content of the products are less than that of reactants. Therefore, exothermic reactions are more likely to be spontaneous.



From these examples, it can be concluded that decrease in enthalpy might be responsible for a process to be spontaneous or feasible. But it suffers from following limitations.

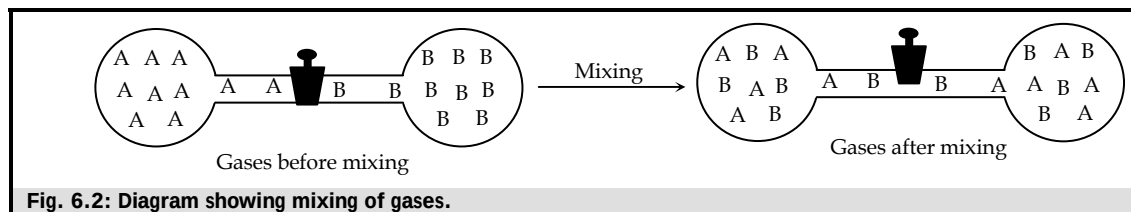
Limitations

- i. There are some endothermic reactions which are spontaneous. These include,
- Melting of ice, $\text{H}_2\text{O}(\text{s}) \longrightarrow \text{H}_2\text{O}(\text{l}) \quad \Delta H = +5.86 \text{ kJ mol}^{-1}$
- Evaporation of water $\text{H}_2\text{O}(\text{l}) \longrightarrow \text{H}_2\text{O}(\text{g}) \quad \Delta H = +44.0 \text{ kJ mol}^{-1}$
- Dissolution of some salt such as $\text{NH}_4\text{Cl} + \text{water} \longrightarrow \text{NH}_4\text{Cl}(\text{aq}) \quad \Delta H = +15.1 \text{ kJ mol}^{-1}$
- ii. All exothermic reactions may not proceed to completion.
- iii. There are some spontaneous processes which do not involve remarkable enthalpy change. Examples: mixing of water and milk, mixing of gases, etc.

Conclusion

Decrease in enthalpy is not the sole criterion for predicting the spontaneity of a process. There must be some other factors for a process to be spontaneous.

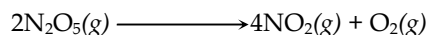
2. **Randomness and spontaneity:** There are some spontaneous processes which do not involve remarkable enthalpy change. Therefore, enthalpy change only cannot be the sole criterion for the spontaneity of reaction. If such processes are visualized, it is found that there occurs increase in the randomness or disorder of the system. For example,
- Mixing of gases involve the increase in randomness or disorders of the system.



Before mixing two types of gases, A and B are at their corresponding bulb. When the stop cock is opened both gases diffuse and disperse randomly.

- Many endothermic spontaneous processes are accompanied by increase in randomness or disorder. Some examples are given below.
 - In melting of ice, molecules experience more randomness in liquid state than in solid state.

$$\text{H}_2\text{O (s)} + \text{heat} \longrightarrow \text{H}_2\text{O (l)} \quad \Delta H^\circ = 6.01 \text{ kJmol}^{-1}$$
 - Evaporation of water, $\text{H}_2\text{O (l)} + \text{heat} \longrightarrow \text{H}_2\text{O (g)} \quad \Delta H^\circ = 44.01 \text{ kJmol}^{-1}$
 - Dissolution of glucose into water is spontaneous process because in solid state the glucose molecules are fixed but in aqueous solution each molecule achieves more randomness.
 - In the decomposition of N_2O_5 , two molecules of reactant yield five molecules of products. So the process is accompanied by increase in randomness.



From these discussions, it is revealed that increase in randomness might be responsible for a process to be spontaneous. But it suffers from following limitations.

Limitations

If the randomness factor were the only criterion for spontaneity, the process like liquefaction of gases or condensation of vapour and solidification of liquid or freezing of liquid would not have been spontaneous processes. In liquefaction of gases and solidification of liquid randomness decreases, still these are spontaneous processes.

Conclusion

Just as decrease in enthalpy cannot be the sole criterion for predicting spontaneity as well as increase in randomness cannot be the sole criterion for a process to be spontaneous.

- Decrease in enthalpy, increase in randomness and spontaneity:** From the above study, it is found that exothermic as well as some endothermic reactions are spontaneous. On the other hand, some processes accompanied by increase in randomness as well as some processes accompanied by decrease in randomness are spontaneous. Hence neither the decrease in enthalpy nor the increase in randomness alone is the criterion for spontaneity but the resultant of the two factors may be the criterion for a process to be spontaneous.

Hence, overall tendency for a process to be spontaneous is the resultant of tendency to have **minimum energy** and **maximum randomness (or disorder)** of the system.

6.4 Measure of Randomness and Concept of Entropy

To measure the degree of randomness (or disorder) of a system, a new term was introduced by Clausius which is called entropy. It is denoted by the symbol S .

Definitions of entropy

1. **Entropy is the measurement of randomness (or disorder) of a system.** Higher the randomness or disorder of a system, greater will be the entropy. In solid, the atoms, ions or molecules are compactly close to each other and solid state has lower entropy. When solid is converted into liquid, randomness of atoms, ions or molecules is increased and increase in entropy. On the same background gaseous state have highest entropy.

$$S_{\text{solid}} < S_{\text{liquid}} < S_{\text{gas}}$$

2. **Entropy is the measurement of unavailable heat energy.** Let us take an example of heating of water. When water is heated, the heat absorbed is utilized to increase the temperature. This happens upto boiling temperature. As soon as the boiling point reaches, thus supplied heat is not utilized to increase the temperature rather it changes the phase of the system from liquid to gas. It can be said that some amount of heat are not available to increase the temperature of the system but are utilized to increase the randomness of a system.
3. **In terms of probability, entropy is mathematically defined as, $S = k \ln W$**
where, S = entropy; k = Boltzmann constant; W = Probability (number of ways of change that can be carried out)
4. **Mathematical definition of entropy.**

Mathematically, entropy is defined as, $S = \frac{q_{\text{rev}}}{T}$

Unit of entropy

We know, $S = \frac{q_{\text{rev}}}{T}$

The unit of entropy in CGS system is calorie per Kelvin (cal K^{-1}). It is also called *entropy unit (eu)*. In S.I. system, the unit of entropy is Joule per Kelvin (J.K^{-1}).

Some points about entropy

- Like internal energy (E) and enthalpy (H), the entropy is also a state function.
- Entropy is an extensive property
- Unlike internal energy (E) and enthalpy, the absolute value of entropy of a substance can be calculated. It is dealt in the third law of thermodynamics.

If S_1 and S_2 are entropies of a substance at initial and final condition, the change in entropy is given by, $\Delta S = S_2 - S_1$

Also, $\Delta S = \frac{q_{\text{rev}}}{T}$

In terms of infinitesimal change $dS = \frac{dq_{\text{rev}}}{T}$.

Entropy change during a process is defined as the quantity of heat changed reversibly divided by absolute temperature at which the heat change occurs.

Self Test

6.1 Write the physical significance of the term entropy.

6.5 Probability and Entropy

It is discussed that entropy is the measurement of randomness of a system. Increase in randomness is one of the driving forces to cause spontaneity and randomness is directly related to probability. A highly ordered state is less probable while a highly disordered state is more probable. For a system to be highly ordered, there is only one way (or few ways) while for the system to be highly disordered, there are plenty of ways. In the nature, that event takes place which has high probability to occur. An ordered state has low probability and small entropy while a disordered state has high probability and high entropy. This fact can be understood from the following examples.

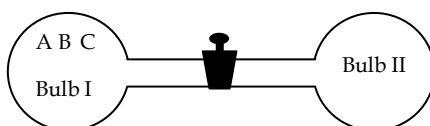
Example 1

Let us consider one colour playing cards are arranged in sequence and tossed. These are collected and again reshuffled. Will we get the state of previous sequence? Obviously no. Is it impossible to attain the previous state? The answer is, possible but highly improbable. Mathematically, it can be understood as, the 13 cards can be arranged by $13!$ ways. Out of these ways, to achieve the previous sequence there is only one way. So there is only one way to achieve previous state while there are $(13! - 1)$ ways not to achieve the previous state. Disordered state is highly probable state.

Example 2

Let us consider two glass bulbs connected by a glass tube fitted with stop cock. When two different types of gases are enclosed in two bulbs and stop cock is opened, these gas molecules undergo intermixing and randomness of molecules is increased. Will all the gas molecules collect only at bulb A? It is not impossible but highly improbable. This can be understood by taking an example of three gas molecules.

If we consider 3 gas molecules A, B, C these can be arranged by 2^3 ways i.e 8 ways as given below.



Way of arrangement	Molecules in Bulb	
	I	II
1	A, B, C	-
2	A, B	C
3	A, C	B
4	B, C	A
5	..	A, B, C
6	A	B, C
7	B	A, C
8	C	A, B

In case of 3 gas molecules out of 9 chances, there is only one chance for all the gas molecules to be occupied at bulb I. When number of gas molecules increases tremendously the improbable chance also increases accordingly. If we consider 1000 gas molecules, the probability of finding all gas molecules in bulb II even when the stop cock is opened is one out of 2^{1000} chances.

From the above examples, it is clear that highly ordered state is less probable and highly disordered state is more probable. Mathematically, entropy is related to probability as,

$$S = k \ln W$$

(Where, S = entropy, k = Boltzmann constant, W = number ways in which change can occur or probability)

Self Test

6.2 Find out whether the following processes involve increase or decrease in entropy.

- A. i. Melting of ice
iii. Condensation of vapor
v. Sublimation of gas
- ii. Evaporation of water
iv. Liquefaction of gas
vi. Sublimation of camphor
- B. i. $2\text{KClO}_3(\text{s}) \longrightarrow 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$
ii. $\text{NH}_4\text{Cl}(\text{s}) + \text{water} \longrightarrow \text{NH}_4^+(\text{aq}) + \text{Cl}^-(\text{aq})$
iii. $\text{CaCO}_3 \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
iv. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3$
- C. i. Gathering of students in canteen from classroom after ringing the bell.
ii. Sitting of students in their seats in classroom after returning from playing ground.
iii. Running of football players to goal football from their initial standing position.

6.6 Entropy Change

Physical or chemical transformations are accompanied by change in entropy. Here entropy change in phase transformation and chemical transformation are discussed in separate headings.

A. Entropy change in phase transformation

Melting of solid, evaporation of liquid, condensation of vapour, freezing of liquid, sublimation of solid substance, allotropic modification, etc. are the examples of physical changes. Such changes take place at constant temperature. For example, when ice is heated, it starts to melt. During melting, temperature does not rise and heat is absorbed reversibly. If q_{rev} heat change takes place at temperature T in a process, the entropy is given by

$$S = \frac{q_{rev}}{T} \quad \dots(\mathbf{i})$$

Phase transformation takes place at constant temperature and pressure. At constant pressure $q_{rev} = \Delta H$ holds true according to the first law of thermodynamics. Now equation (i) becomes,

$$S = \frac{\Delta H}{T} \quad \dots(\text{ii})$$

The different types of phase transformations are given below.

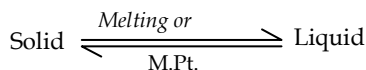
1. **Entropy of fusion: (ΔS_{fusion}):** When a solid melts, there exists equilibrium between solid and its liquid phase at melting point. **The entropy change involved when one mole of solid changes into its liquid state at melting point is called entropy of fusion. It is denoted by ΔS_{fusion} .**

At melting point, the heat absorbed (q_{rev}) is equals to latent heat of fusion (ΔH_f).

$$\Delta S_{\text{fusion}} = \frac{\Delta H_f}{T_m} \quad \dots(\text{iii. a})$$

Where, T_m = melting temperature in Kelvin scale

For a change,



or $\Delta S_{\text{fusion}} = S_{\text{liquid}} - S_{\text{solid}} \quad \dots(\text{iii. b})$

Where, ΔS_{fusion} = entropy of fusion, S_{liquid} = molar entropy of liquid and S_{solid} is molar entropy of solid.

Self Test

- 6.3 Calculate the entropy change for the conversion of 1 mole of ice into water at its melting point. Melting point of ice is 0°C . The enthalpy of fusion of ice is $6.025 \text{ kJ mol}^{-1}$

Ans: $22.06 \text{ JK}^{-1} \text{ mol}^{-1}$

- 6.4 In the above case, calculate the entropy change for 2 gram of ice.

Ans: 2.45 JK^{-1}

- 6.5 30.39 kJ energy is required to melt one mole of NaCl. The entropy of fusion is $28.4 \text{ J mol}^{-1} \text{ K}^{-1}$. Calculate the melting point of sodium chloride.

Ans: 797°C

2. **Entropy of vaporization ($\Delta S_{\text{vaporisation}}$):** When liquid is heated, it boils at boiling point. At that temperature liquid and vapour phase attain equilibrium condition.

The entropy change involved when one mole of liquid is changed into its vapour state at boiling point is called entropy of vaporization. It is denoted by $\Delta S_{\text{vaporisation}}$.

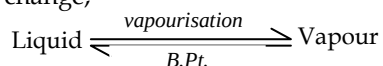
At boiling point, the heat absorbed reversibly (q_{rev}) is equal to the latent heat of vaporization or enthalpy of vaporization ($\Delta H_{\text{vaporisation}}$)

$$\Delta S_{\text{vaporisation}} = \frac{\Delta H_v}{T_b} \quad \dots(\text{iv. a})$$

Where,

T_b is boiling point of liquid.

For a change,



or, $\Delta S_v = S_{\text{vapour}} - S_{\text{liquid}} \quad \dots(\text{iv. b})$

where,

ΔS_v = entropy of vaporization

S_{vapour} = molar entropy of vapour phase

S_{liquid} = molar entropy of liquid phase.

Self Test

- 6.6 The boiling point of ethanol is 78.5°C . If its enthalpy of vaporization is $38.59 \text{ kJ mol}^{-1}$ what will be the molar entropy change?

Ans: $109.78 \text{ JK}^{-1} \text{ mol}^{-1}$

- 6.7 Calculate the entropy change for 0.25 mole of ethanol.

Ans: 27.445 JK^{-1}

- 7.8 If entropy of vaporization of water is $109 \text{ JK}^{-1} \text{ mol}^{-1}$ what will be the latent heat of vaporization?

Ans: $40.657 \text{ kJ mol}^{-1}$

3. **Entropy of sublimation ($\Delta S_{\text{sublimation}}$):** Entropy of sublimation is defined as the entropy change involved when one mole of solid substance is directly converted into vapour state below the melting point of solid. It is denoted by $\Delta S_{\text{sublimation}}$.

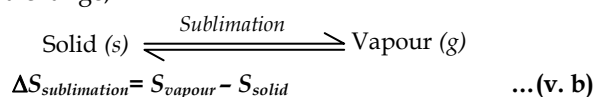
$$\Delta S_{\text{sublimation}} = \frac{\Delta H_{\text{sublimation}}}{T_s} \quad \dots(\text{v. a})$$

Where,

$\Delta H_{\text{sublimation}}$ = enthalpy of sublimation

T_s = sublimation temperature

For a change,



Where,

$\Delta S_{\text{sublimation}}$ = entropy of sublimation

S_{vapour} = molar entropy of vapour phase

S_{solid} = molar entropy of solid phase

4. **Entropy of transition ($\Delta S_{\text{transition}}$):** At a particular temperature, two allotropes of an element exist at equilibrium and this temperature is called transition temperature (T_t).

Entropy of transition is defined as the entropy change involved when one mole of one allotropic form of an element is changed into another allotropic form at transition temperature.

$$\Delta S_{\text{transition}} = \frac{\Delta H_{\text{transition}}}{T_t} \quad \dots(\text{vi. a})$$

Where, $\Delta S_{\text{transition}}$ = entropy of transition, $\Delta H_{\text{transition}}$ = enthalpy of transition, T_t = transition temperature

For a change, Diamond $\rightleftharpoons$ Graphite

$$\Delta S_t = S_{\text{graphite}} - S_{\text{diamond}}$$

Where, ΔS_t = entropy of transition, S_{graphite} = molar entropy of graphite, S_{diamond} = molar entropy of diamond

B. Entropy change in chemical reaction

In a chemical reaction, one type of chemical species are entirely changed into another type. This involves change in entropy in a chemical reaction. Entropy change of a chemical reaction is calculated by subtracting the sum of entropy of all reactants from the sum of entropy of all products.

For a general reaction, $a A + b B \longrightarrow c C + d D$

Let S_A , S_B , S_C and S_D are entropies of chemical species A, B, C and D respectively. Here, a , b , c and d are stoichiometric coefficients for these chemical species. The entropy change for the reaction is given by

$$\Delta S = \sum S_{\text{products}} - \sum S_{\text{reactants}}$$

$$\text{or, } \Delta S = (c S_C + d S_D) - [a S_A + b S_B]$$

If standard entropies are taken for each chemical species, entropy change in a reaction is called standard entropy change. It is denoted by ΔS° .

$$\Delta S^\circ = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}}$$

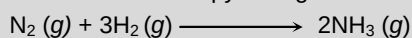
Entropy of 1 mole of pure substance at 25°C and 1 atmosphere pressure is called standard entropy.

Since, absolute value of entropy of a pure solid substance at 0K is zero, the standard entropies of all substances above 0K are always greater than zero. The standard entropies of some elements and compounds are given in table 6.1.

Chemical Substance	S° in $\text{cal K}^{-1}\text{mol}^{-1}$	S° in $\text{JK}^{-1}\text{mol}^{-1}$
$\text{H}_2(\text{g})$	32.21	130.6
$\text{H}_2\text{O}(\text{g})$	45.11	188.7
$\text{H}_2\text{O}(\text{l})$	16.72	69.96
$\text{HCl}(\text{g})$	44.62	186.7
$\text{HNO}_3(\text{l})$	37.19	155.6
$\text{H}_2\text{SO}_4(\text{l})$	37.50	157.0
$\text{N}_2(\text{g})$	45.77	191.5
$\text{NH}_3(\text{g})$	46.01	192.5
$\text{Na}(\text{s})$	36.72	153.6
$\text{NaCl}(\text{s})$	17.30	72.88
$\text{O}_2(\text{g})$	49.0	205.0
$\text{S}(\text{s})$	7.62	31.9
$\text{SO}_2(\text{g})$	59.40	248.5
$\text{Cl}_2(\text{g})$	53.29	223.0
$\text{CO}_2(\text{g})$	51.06	213.6

Self Test

6.9 Calculate the entropy change for the reaction



Given S° of N_2 , H_2 and NH_3 are 191.5, 130.6, 192.5 $\text{JK}^{-1}\text{mol}^{-1}$, respectively.

[Ans -198.3 $\text{JK}^{-1}\text{mol}^{-1}$]

6.10 Calculate the standard entropy for the reaction,



Consult data from Table 7.1.

C. Entropy change of system, surrounding and universe

System interacts with surrounding. In such interactions, heat or energy is exchanged. This causes the change in entropy of system and surrounding. System and surrounding comprise universe. So interaction of system and surrounding causes entropy change of universe too. Let us consider some examples to understand entropy change of system and surrounding.

- Spreading of red dyes, in water within beaker does not involve any exchange of matter and energy with surrounding. Here the mixture of red dyes and water forms a system. Since no heat or energy is exchanged to the surrounding, it is regarded as **isolated system**. Such processes always involve increase in entropy. For spontaneous processes in an isolated system, the entropy change is positive.
- Let hot water is taken in a breaker and it is regarded as system. The outer part (i.e. with which the hot water interacts) forms surrounding. In this case, system loses energy as well as matter and its entropy is decreased. In this case energy is gained by surrounding and its entropy is increased. In all spontaneous processes, entropy keeps on increasing till equilibrium is achieved. At equilibrium the entropy is maximum and change in entropy is zero.
- Depending upon the nature, entropy change of system or surrounding may be positive or negative. But for a process to be spontaneous, the sum of entropy change of system and surrounding must be positive.

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$$

a. Study the following data for the process, $\text{H}_2\text{O}(\text{l}) \longrightarrow \text{H}_2\text{O}(\text{s})$ at different temperature at 1 atm pressure.

Temperature	Entropy change in $\text{JK}^{-1} \text{mol}^{-1}$		
	ΔS_{sys}	ΔS_{surr}	ΔS_{univ}
-1°C	-21.85	+ 21.93	
0°C	-21.99	+ 21.99	
$+1^\circ\text{C}$	-22.13	+ 22.05	

- Calculate ΔS_{univ} at the mentioned temperature above.
- At what temperature, freezing of water is spontaneous?
- At what temperature freezing of water is non-spontaneous?

6.7 Criteria of Spontaneity in Terms of Entropy Change (Entropy and Spontaneous Process)

For a process to be spontaneous in isolated system, the entropy change should be positive. In case of an open system, spontaneity of the process depends upon the entropy change of system and entropy change of surrounding. Entropy change of system, surrounding and universe is related as

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$$

For a process to be spontaneous, the total entropy change of system and surrounding should be positive or the entropy change of universe should be positive. A process may be spontaneous even if the entropy change of system is negative as exemplified by condensation of vapour and freezing of liquid into solid. This is possible because in these cases, the entropy change of surrounding becomes positive and it outweighs the entropy change of system. This makes entropy change of universe positive. [See the following interplay table].

The various possibilities of entropy change of system and surrounding can be justified from the following interplay of ΔS_{sys} and ΔS_{surr} . Here, ΔS_{sys} is entropy change of system, ΔS_{surr} is entropy change of surrounding, ΔS_{univ} is entropy change of universe. $|\Delta S_{\text{sys}}|$ is modules value of ΔS_{sys} . Its magnitude value is considered.

S.No.	ΔS_{sys}	ΔS_{surr}	$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$	Fate of the process
1.	+ ve	+ ve	+ve	Spontaneous
2.	+ ve	- ve	i. +ve if $ \Delta S_{\text{sys}} > \Delta S_{\text{surr}} $ ii. 0 if $ \Delta S_{\text{sys}} = \Delta S_{\text{surr}} $ iii. - ve if $ \Delta S_{\text{sys}} < \Delta S_{\text{surr}} $	Spontaneous Equilibrium Non spontaneous
3.	-ve	- ve	- ve	Non spontaneous
4.	- ve	+ ve	i. + ve if $ \Delta S_{\text{sys}} < \Delta S_{\text{surr}} $ ii. 0 if $ \Delta S_{\text{sys}} = \Delta S_{\text{surr}} $ iii. - ve if $ \Delta S_{\text{sys}} > \Delta S_{\text{surr}} $	Spontaneous Equilibrium Non spontaneous

From the above table following points are concluded

- If entropy changes of system as well as that of surrounding are positive, the entropy change of universe becomes positive. The process is obviously spontaneous.
- If entropy change of system is positive and entropy change of surrounding is negative, the process is spontaneous only when system suffers larger entropy change than do the surrounding.
If entropy change of system and surrounding have same magnitude then entropy change of universe becomes zero. The process attains thermodynamic equilibrium. If entropy change of surrounding is greater than entropy change of system, the entropy change of universe becomes negative and the process is non-spontaneous in the given direction. However, the process is spontaneous or feasible in opposite direction.
- If entropy change of system and surrounding both are negative, the entropy change of universe is negative. The process is obviously non-spontaneous in the given direction.
- If entropy change of system is negative and entropy change of surrounding is positive, three cases arise (explain as in (ii) points).

Melting of ice at 1 atm and above 0°C is spontaneous process. Justify the statement.

Solution

When surrounding is at higher temperature than system, heat is absorbed by ice from surrounding to melt.

Let, T_{sys} is temperature of system. T_{surr} is temperature of surrounding. ΔH is latent heat of fusion or enthalpy of fusion.

The entropy change of system is

$$\Delta S_{\text{sys}} = + \frac{\Delta H}{T_{\text{sys}}} \quad (\text{Positive sign indicates that heat is gained by the system}) \quad \dots(i)$$

Entropy change of surrounding is

$$\Delta S_{\text{surr}} = - \frac{\Delta H}{T_{\text{surr}}} \quad (\text{Negative sign indicates that heat is lost by the surrounding}) \quad \dots(ii)$$

$$\text{Now, } \Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = \frac{\Delta H}{T_{\text{sys}}} + \left(-\frac{\Delta H}{T_{\text{surr}}} \right)$$

$$\begin{aligned} \Delta S_{\text{univ}} &= \frac{\Delta H}{T_{\text{sys}}} - \frac{\Delta H}{T_{\text{surr}}} \quad \dots(iii) \\ &= +ve \quad [\because T_{\text{surr}} > T_{\text{sys}}] \end{aligned}$$

Since entropy change of universe is found to be positive, the melting of ice is spontaneous at 1 atm pressure and above 0°C.

Self Test

- 6.6 Show that melting of ice and freezing of water is in equilibrium at 0°C and 1 atm pressure.
 6.7 Freezing of water at 1 atm pressure and below 0°C is spontaneous
 6.8 Freezing of water at 1 atm pressure and above 0°C is non-spontaneous.
 6.9 Melting of ice at 1 atm pressure and below 0°C is non-spontaneous

6.8 Limitation of the First Law of Thermodynamics and Dawn of Second Law of Thermodynamics

The first law of thermodynamics suffers from the following limitations.

- It does not answer, why is natural process unidirectional?
- It does not explain anything about the source of heat and direction of flow of heat.
- It does not explain why some chemical reactions do not proceed to completion.
- The first law states that one form of energy changes into another form but it does not define the extent of convertibility.

To solve the limitations of the first law of thermodynamics, the necessity of the second law of thermodynamics was felt. The most important aspect of the second law of thermodynamics is that it forms a basis to predict whether a particular process occurs or not under the given set of conditions and if they do occur, to what extent. The second law of thermodynamics is defined in different ways. In terms of entropy the second law of thermodynamics is defined as **entropy of the universe is increasing**. In thermodynamics, the term universe means system and surrounding. Hence, entropy of universe is defined as,

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

According to the second law of thermodynamics, a process will occur spontaneously if entropy of the universe is increased.

Some other statements of the second law of thermodynamics

- Clausius and Kelvin statement:** "It is impossible to construct a heat engine which will continuously abstract heat from a body and convert the whole heat energy into work without leaving changes in the working system"
- Planck statement:** "It is impossible to construct a machine operating in cycles that will convert heat into work without producing any other changes in the surroundings."
- "Heat cannot spontaneously pass from a colder to a warmer body" (R.J.E. Clausius)
- "Every system which is left to itself will, on the average, change toward a condition of maximum probability" (G.N. Lewis)
- "The state of maximum entropy is the most stable state for an isolated system" (Enrico Fermi)
- "There is a general tendency in nature for energy to pass from more available to less available form" (J.A.V. Butter)

If an isolated process is considered (e.g. mixing of a drop of ink into water in a beaker), the process proceeds in the forward direction so long as the entropy goes on increasing. The entropy of the system become maximum then entropy change becomes zero. This is the state where equilibrium condition is achieved.

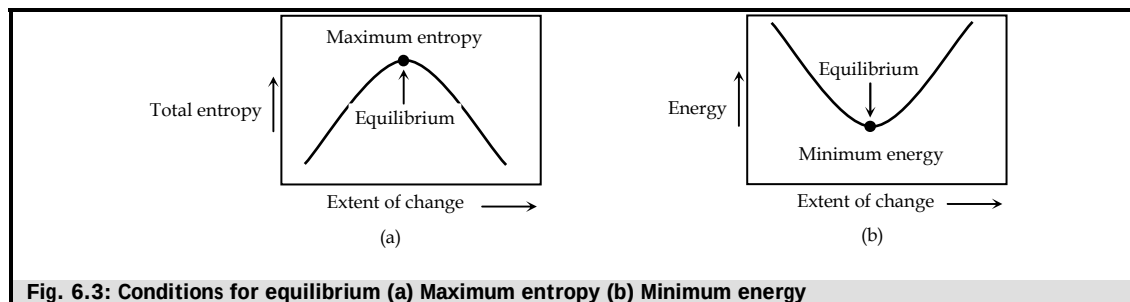


Fig. 6.3: Conditions for equilibrium (a) Maximum entropy (b) Minimum energy

If entropy change of universe is negative the process is obviously non-spontaneous. It is summarized as follows.

Entropy change of universe	Sign	Fate of the process
ΔS_{univ}	+ ve	Spontaneous
ΔS_{univ}	0	Equilibrium
ΔS_{univ}	- ve	Non-spontaneous

Application of the second law of thermodynamics

On the basis of validity of the second law of the thermodynamics Carnot theorem is deduced. The possibility of obtaining mechanical work from heat energy led to the development of that engines which may be external combustion engine or internal combustion engines. Carnot cycle is used to demonstrate the maximum convertibility of heat into work. The consequences of Carnot theorem led to calculate the efficiency of heat into work transformation. It introduced the concept of entropy and Kelvin scale of temperature.

6.9 The Third Law of Thermodynamics

We have discussed in the earlier topics that on increasing the temperature, the molecular motions and randomness increase. On the contrary, if temperature of a substance is decreased, the randomness is decreased and entropy of substance at 0 Kelvin is the minimum value. This forms the basis of the third law of thermodynamics. The third law of thermodynamics states **"the entropy of perfectly crystalline substance at absolute zero temperature is zero."**

$$\lim_{T \rightarrow 0} S = 0$$

This statement implies that every substance has finite positive entropy above absolute zero temperature. At absolute zero all perfect crystals¹ have same entropy.

6.10 Concept of Free Energy

1. Gibbs Free Energy (G)

The concept of entropy (S) is the fundamental consequence of the second law of thermodynamics. There is another thermodynamic function which makes use of the entropy in its derivation. This thermodynamic function is **'free energy function (G)'**.

¹ A perfect crystal is that where each point of the crystal must be at a crystal lattice point and it should have lowest energy. On the other hand perfect order means zero entropy. A perfect crystal has no disorder and has zero entropy at absolute zero. Glassy solids or liquids, ice etc. are not perfect crystals because there exist some randomness in the arrangement of atoms or molecules. Entropy of such type of substance can't be zero at absolute zero temperature.

We have dealt that energy can be converted into work. But it does not mean that all the energy can be converted into work. The energy which can be converted into useful work is called available energy. The available energy performs the operation on engine or motor. The energy which is not converted into useful work but consumed against friction, operation of piston etc is called unavailable energy. The measure of unavailable energy is entropy while the measure of available energy present in a system is free energy.

Total energy = isothermally available energy + isothermally unavailable energy

The isothermally available energy present in a system is called free energy or Gibb's free energy. It is denoted by the symbol G . Mathematically Gibbs free energy is defined as,

$$G = H - TS \quad \dots(\text{i})$$

Where,

G = Gibbs free energy; H = enthalpy; T = Temperature; S = Entropy

Gibbs free energy is also extensive property. It is state function too. It means the value of Gibbs free energy depends upon initial and final condition of a system but not upon the path.

Let G_1 , H_1 and S_1 are Gibb's free energy, enthalpy and entropy of a system initially. At constant temperature, when enthalpy and entropy are changed from H_1 and S_1 to H_2 and S_2 respectively then Gibbs free energy changes from G_1 to G_2 .

$$\text{Initially, } G_1 = H_1 - TS_1 \quad \dots(\text{ii})$$

$$\text{Finally, } G_2 = H_2 - TS_2 \quad \dots(\text{iii})$$

Now equation (ii) is subtracted from equation (iii), we get,

$$G_2 - G_1 = (H_2 - H_1) - T(S_2 - S_1)$$

$$\text{or } \Delta G = \Delta H - T\Delta S \quad \dots(\text{iv})$$

Equation (iv) is also called **Gibb's-HelmHoltz equation**. It is very useful equation in chemical thermodynamics.

2. Standard Free Energy of Formation (ΔG°_f)

It is defined as the change in free energy when 1 mole of substance is formed from its constituent element provided that all reactants and products are at their standard state (i.e. one bar pressure and 25°C). It is denoted by (ΔG°_f)

The standard free energy of formation of an element in standard state is taken as zero. The standard free energy of formation of some common substances are given in table 6.2.

Table 6.2: Standard free energy formation (ΔG°_f).	
Chemical Substance	(ΔG°_f) in kJ mol^{-1}
C (graphite)	0
H_2 (g)	0
H_2O (g)	-228.6
H_2O (l)	-237.2
HCl (g)	-95.3
HBr (g)	-53.22
HI (g)	+1.30
NH_3 (g)	-16.63
B_2H_6 (S)	+82.1

P ₄ (g)	+24.35
CO ₂ (g)	-394.4
CO (g)	-137.3
CH ₄ (g)	-50.80
C ₂ H ₆ (g)	- 32.90
C ₂ H ₂ (g)	+ 209.20
C ₂ H ₄ (g)	+68.12
C ₆ H ₆ (g)	+129.60
C ₂ H ₅ OH (l)	-104.76

Free energy change can also be calculated by using the relation $\Delta G = \Delta H - T \cdot \Delta S$. Since ΔH_f° for element is taken as zero, we cannot calculate ΔG_f° for an element using the relation $\Delta G^\circ = \Delta H^\circ - T \cdot \Delta S^\circ$.

3. Standard Free Energy Change (ΔG°)

Free energy depends upon temperature and molar concentration of chemical species. The standard state chosen for the measurement of Gibbs free energy are 25°C temperature and 1 molar concentration of reactants as well as products. **Standard free energy change is defined as the free energy change involved when all the reactants in their standard states are changed into products in their standard states. It is denoted by ΔG° .** When standard free energy of all reactants and products are known, standard free energy of reactions can be calculated as,

Standard free energy of reaction (ΔG°) = Standard free energy of formation of products (G_f°)_P
- Standard free energy of formation of reactants (G_f°)_R.

$$\text{or } \Delta G^\circ = \Sigma(\Delta G_f^\circ)_{\text{product}} - \Sigma(\Delta G_f^\circ)_{\text{Reactant}}$$

4. Relation between Standard Free Energy Change (ΔG°) and Equilibrium Constant (K_{eq})

The Gibbs free energy change depends upon the concentration of reactants and products. If a reaction takes place in non-standard condition then the free energy change (ΔG) of reaction is different from the standard free energy change (ΔG°). ΔG and ΔG° are related as

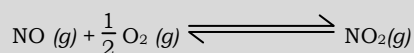
$$\Delta G = \Delta G^\circ + RT \ln K_{eq} \quad \dots (i)$$

At equilibrium change in Gibbs free energy (ΔG) is zero. So, relation (i) becomes

$$\Delta G^\circ = - RT \ln K_{eq} \quad \dots(ii)$$

This is very useful equation in the field of thermodynamics. It helps to calculate standard Gibbs free energy change from equilibrium constant and vice-versa.

An equilibrium reaction at 25 °C and 1 atm pressure is given below.



Given,

$$\Delta H_f^\circ \text{ of NO}_2 \text{ (g)} = 33.85 \text{ kJ mol}^{-1}, \text{ NO (g)} = 90.37 \text{ kJ mol}^{-1}$$

$$\Delta S^\circ \text{ at } 25^\circ\text{C for the reaction} = -70.8 \text{ JK}^{-1} \text{ mol}^{-1}$$

Calculate

- Standard heat of formation
- Standard Gibbs free energy
- Equilibrium constant

Solution

(a) Standard heat of formation of element (eg O₂) is zero.

$$\begin{aligned}\Delta H_f^\circ \text{ for reaction} &= \Sigma (\Delta H_f^\circ)_{\text{products}} - \Sigma (\Delta H_f^\circ)_{\text{reactants}} \\ &= (\Delta H_f^\circ)_{\text{NO}_2} - \left[\frac{1}{2} (\Delta H_f^\circ)_{\text{O}_2} + (\Delta H_f^\circ)_{\text{NO}} \right] \\ &= 33.85 - \left[\frac{1}{2} \times 0 + 90.37 \right] = 33.85 - 90.37 = -56.52 \text{ kJ} = -56520 \text{ Joule}\end{aligned}$$

(b) Gibbs Helmholtz equation is

$$\begin{aligned}\Delta G^\circ &= \Delta H^\circ - T \Delta S^\circ \\ &= (-56520) - 298 \times (-70.8) = -56520 + 210984 = -35421.6 \text{ J mol}^{-1}\end{aligned}$$

(c) Standard Gibbs free energy is related to equilibrium constant as,

$$\Delta G^\circ = -RT \ln K_{eq}$$

$$\ln K_{eq} = -\frac{\Delta G^\circ}{RT}$$

$$\text{or } \ln K_{eq} = \frac{-(-35421.6)}{8.314 \times 298} = 14.296$$

$$\text{Taking anti } \ln \text{ we get } K_{eq} = e^{14.296} = 1.618 \times 10^6$$

5. Physical Significance of Free Energy

Decrease in Gibbs free energy is the measurement of useful work done. This can be proved as follows.

$$\text{From the first law of thermodynamics, } q_{rev} = \Delta E + W_{rev} \quad \dots(i)$$

When a quantity of heat is supplied to a system, it increases the internal energy and performs some work. Work done reversibly is greater than work done irreversibly. W_{rev} is sum of maximum work of expansion ($P \cdot \Delta V$) and net useful work (e.g. electrical work, etc.) done. Now equation (i) becomes,

$$q_{rev} = \Delta E + W_{expansion} + W_{useful}$$

$$\text{or } q_{rev} = \Delta E + P \cdot \Delta V + W_{useful}$$

$$\text{or } q_{rev} = \Delta H + W_{useful} \quad [\because \Delta H = \Delta E + P \cdot \Delta V] \quad \dots(ii)$$

If the process is carried out reversibly then according to the second law of thermodynamics,

$$\Delta S = \frac{q_{rev}}{T}$$

$$\text{or } q_{rev} = T \cdot \Delta S \quad \dots(iii)$$

From equation (ii) and (iii), we get

$$T \cdot \Delta S = \Delta H + W_{useful}$$

$$\text{or } -\Delta H + T \cdot \Delta S = W_{useful}$$

$$\text{or } -(\Delta H - T \cdot \Delta S) = W_{useful}$$

$$\text{or } -(\Delta G)_{T,P} = W_{useful} \quad \dots(iv)$$

Equation (iv) reveals that decrease in Gibbs free energy is the measurement of net work or useful work done reversibly at constant temperature and pressure.

Note:

$$q_{rev} = \Delta E + w_{rev}$$

$$q_{rev} = \Delta E + w_{exp} + W_{useful}$$

$$q_{rev} = \Delta E + P \cdot \Delta V + W_{useful}$$

$$T \cdot \Delta S = \Delta H + W_{useful}$$

$$-(\Delta H - T \cdot \Delta S) = W_{useful}$$

$$-\Delta G = W_{useful}$$

6. Relation between Gibbs Free Energy and Electromotive Force

In galvanic cell, electrical energy is produced as a result of spontaneous redox reaction. The redox reaction occurs spontaneously with the decrease in free energy. The decrease in free energy is utilized to do useful work. The free energy change and electrical work are related as,

$$\Delta G = -nFE.$$

Where, n = number of moles of electrons involved in chemical reaction

F = Faraday constant = 96500 coulomb (approximately)

E = Emf of cell

At standard condition, $\Delta G^\circ = -nFE^\circ$

7. Criteria of Spontaneity in Terms of Gibbs Free Energy Change (ΔG) (Prediction for the feasibility of reaction)

One of the criteria of spontaneity of a process is that entropy change of universe should be positive which is discussed in the previous unit. To predict the spontaneity of a process in terms of entropy change of universe, one should know the entropy change of system and that of surrounding but chemists are more concerned with system rather than surrounding. This necessitates the introduction of another thermodynamic function so that the spontaneity of a process could be predicted by knowing the entropy change of system only. That thermodynamic function is Gibbs free energy (G).

$$\text{We have, } G = H - TS. \quad \dots(i)$$

Where, H is enthalpy, T is Kelvin temperature and S is entropy. The change in Gibbs free energy is given as, $\Delta G = \Delta H - T \cdot \Delta S$... (ii)

Derivation of criteria of spontaneity in terms of ΔG

To come up to the criteria of spontaneity in terms of Gibbs free energy change, we will start from the criteria of spontaneity in terms of entropy change of universe.

For a process to be spontaneous, ΔS_{univ} should be positive. The relation is,

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} \quad \dots(i)$$

Since it is difficult to measure ΔS_{surr} , it should be omitted from equation (i) by some measurable terms.

Suppose heat is lost by the surrounding and gained by the system.

$$\text{At constant temperature and pressure, } \Delta S_{surr} = -\frac{\Delta H}{T}$$

Now, Equation (i) becomes,

$$\Delta S_{univ} = \Delta S_{sys} - \frac{\Delta H}{T}$$

$$\text{or, } T\Delta S_{univ} = T\Delta S_{sys} - \Delta H$$

$$\text{or, } T\Delta S_{univ} = -(\Delta H - T\Delta S_{sys})$$

$$\text{or, } T\Delta S_{univ} = -\Delta G_{sys}$$

$$\text{For convenience the subscript 'sys' is dropped. Then } \Delta G = -T\Delta S_{univ} \quad \dots(ii)$$

For a process to be spontaneous ΔS_{univ} must be positive. It implies that for the process to be spontaneous ΔG must be negative.

For the process to be at equilibrium ΔS_{univ} should be Zero. It shows that when ΔG is zero, the process attains equilibrium condition. When ΔS_{univ} is negative, the process is non-spontaneous in the forward direction. It implies that when ΔG is positive, the process will be non-spontaneous in the forward direction. It can be understood from the following table.

Value of ΔS_{univ}	$\Delta G = -T \Delta S_{univ}$	Process
+ve	$\Delta G = -ve$	Spontaneous
0	$\Delta G = 0$	Equilibrium
-ve	$\Delta G = +ve$	Non-spontaneous

Criteria for Spontaneity (Relation among ΔG , ΔH , ΔS and Temperature):

Spontaneity of a process can be explained in terms of entropy change of universe (ΔS_{univ}) and Gibbs free energy change (ΔG). Out of these two, Gibbs free energy change (ΔG) is preferred to entropy change of universe (ΔS_{univ}) because prediction of spontaneity in terms of ΔG involves the entropy change of system only where as prediction of spontaneity in terms of ΔS_{univ} involves the entropy change of system and surrounding.

The Gibbs Helm-Holtz equation is $\Delta G = \Delta H - T\Delta S$. This equation includes both energy factor (ΔH) and entropy factor (ΔS). The interplay of energy factor and entropy factor for the prediction of spontaneity of a process is given below.

S.No.	ΔH	ΔS	$\Delta G = \Delta H - T \cdot \Delta S$	Fate of the process
1.	- ve	+ ve	-ve	Spontaneous
2.	- ve	- ve	i. if $ \Delta H > T \cdot \Delta S $, $\Delta G = -ve$ ii. if $ \Delta H < T \cdot \Delta S $, $\Delta G = +ve$ iii. if $ \Delta H = T \cdot \Delta S $, $\Delta G = 0$	Spontaneous Non spontaneous Equilibrium
3.	+ve	- ve	+ ve	Non spontaneous
4.	+ ve	+ ve	i. if $ \Delta H < T \cdot \Delta S $, $\Delta G = -ve$ ii. if $ \Delta H > T \cdot \Delta S $, $\Delta G = +ve$ iii. if $ \Delta H = T \cdot \Delta S $, $\Delta G = 0$	Spontaneous Non spontaneous Equilibrium

Here, $|\Delta H|$ means modulus value (magnitude only) of enthalpy change or value of enthalpy change ignoring sign.

From the above table it is concluded that

- If the reaction is exothermic ($\Delta H = -ve$) and entropy change of system is positive, the change in Gibbs free energy is always negative and reaction occurs spontaneously at all temperature.
- If the reaction is exothermic ($\Delta H = -ve$) but the entropy change of the system is negative then different cases arise.

Case I : Most likely at low temperature: The magnitude of $T \cdot \Delta S$ will be less than the magnitude of ΔH . So Gibbs free energy change becomes negative. Then the process is spontaneous.

Case II: Mostly likely at high temperature: The magnitude of $T \cdot \Delta S$ will be higher than the magnitude of ΔH . So Gibbs free energy change becomes positive. Then the process is non-spontaneous.

Case III: At a certain temperature, the magnitude of $T \Delta S$ becomes equal to the magnitude of ΔH . So Gibbs free energy change becomes zero. Then the process is at equilibrium.

- For endothermic reaction if the entropy change of reaction is negative the change in Gibbs free energy always becomes positive irrespective of temperature. Then the process is non-spontaneous.
- If reaction is endothermic and entropy change of system is positive, different case arises.
[Explain yourself as in (ii) above]

■ Solved Questions

1. What is the physical concept of entropy?

- Entropy is the measurement of randomness or disorder of a system. Higher the randomness or disorder of a system, greater the entropy of the substance. More ordered system has less entropy. Among the three states of matter the order of entropy is as follows.

Entropy of gas >> entropy liquid > entropy of solid.

2. Define standard free energy of a reaction.

- Standard free energy change is defined as the free energy change for a process at 25°C and 1 atmospheric pressure in which all the reactants in their standard state changes to product in their standard state. i.e.

$$\Delta G^\circ = \Delta H^\circ - T \cdot \Delta S^\circ.$$

The standard free energy of formation of an element in its standard state is taken to be zero.

3. What is meant by spontaneous process? Write an example of it.

- When a process takes place itself then the process is called spontaneous process. However the process may be initiated by some external agency at once. An example of spontaneous process is evaporation of water.

■ Questions

■ Very Short Answer Questions:

- Define the following terms:
 - Spontaneous process
 - Entropy
 - Entropy unit
- Write short definition
 - Entropy of fusion
 - Entropy of vaporization
 - Entropy of sublimation
 - Entropy of transition
- Define second law of thermodynamics and also define thermodynamic efficiency of heat engine.
- Define Gibbs free-energy change. Write the mathematical relation to predict the spontaneity.
- How would you predict the spontaneity using the relation?

$$T \Delta S_{\text{units}} = - \Delta G_{\text{sys.}}$$
- Define free energy. Derive an expression to relate Gibbs free energy change with work.
- Mention the proper conditions of a chemical reaction to become spontaneous, if its ΔH and ΔS are positive.
- Distinguish spontaneous and non-spontaneous processes giving one example of each.
- Give the physical meaning of entropy. Write its unit.
- Mention two criteria which must be met for a process to be spontaneous, regardless of the temperature.

■ Short /Long Answer Questions:

- Discuss the criteria of spontaneity, non spontaneity and equilibrium of exothermic and endothermic reactions on the basis of free energy and entropy change.
- "The decrease in enthalpy or increase in entropy." is not a sole criteria for the feasibility of a process." Explain.

■ Numerical Problems

- The latent heat of fusion of ice is 336 Jg⁻¹. Calculate the molar entropy of fusion of ice at its normal melting point.
Ans: 22.15 Jmol⁻¹ K⁻¹
- The standard enthalpy of vaporization of water is 40.75 kJ mol⁻¹ at its normal boiling point calculate the entropy of vaporization for the process.
Ans: 109.24 Jmol⁻¹

3. Enthalpy of fusion per mol of ice is 6008 J mol^{-1} . Calculate the entropy change in the conversion of 1 mole of ice at 0°C and 1 atm to liquid at 0°C and 1 atm.
Ans: $22 \text{ J mol}^{-1} \text{ K}^{-1}$
4. For a reaction $\Delta G = 38.5 \text{ kJ mol}^{-1}$. Calculate the equilibrium constant for the reaction.
Ans: 1.8×10^{-7}
5. Equilibrium constant for a reaction is 1.8×10^{-7} . Calculate ΔG° for the reaction.
Ans: 38.5 kJ mol^{-1}

Multiple Choice Questions

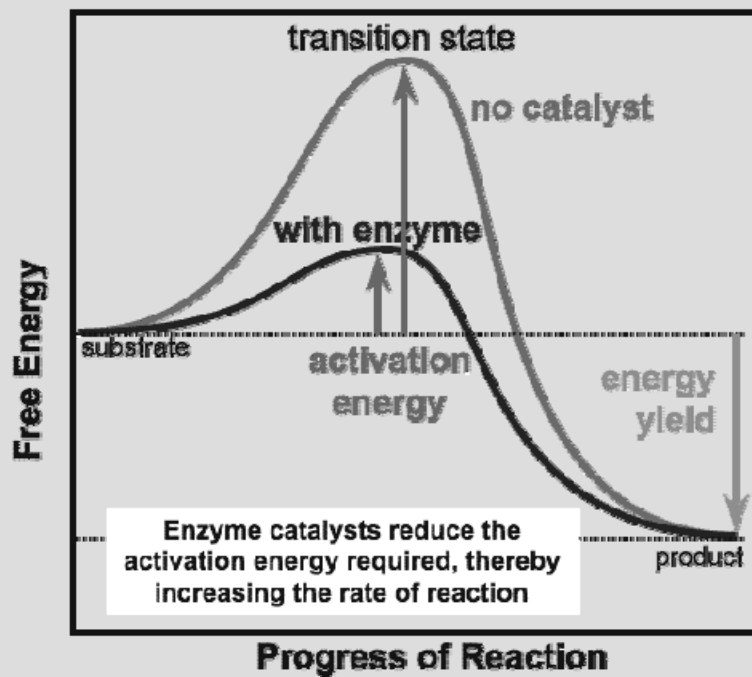
- A chemical reaction will be spontaneous if
 - activation energy for the reaction is low
 - entropy decreases during a reaction
 - free energy decreases
 - free energy increases
- In spontaneous reactions, in the free energy the system always,
 - increases
 - remain unaffected
 - decreases
 - first increases and then decreases
- Decrease in free energy of a reacting system indicates the reaction to be:
 - a reversible reaction
 - an endothermic reaction
 - a spontaneous reaction
 - a primary reaction
- A spontaneous reaction is impossible when
 - both ΔH and ΔS are negative
 - both ΔH and ΔS are positive
 - ΔH is negative and ΔS is positive
 - ΔS is negative and ΔH is positive
- For a reaction to occur at all possible temperature, which of the following condition is required?
 - $\Delta H > 0, \Delta S > 0$
 - $\Delta H < 0, \Delta S > 0$
 - $\Delta H = 0, \Delta S > 0$
 - $\Delta H < 0, \Delta S < 0$
- A reaction is not feasible when
 - $\Delta H = +ve, \Delta S = +ve$
 - $\Delta H = -ve, \Delta S = -ve$
 - $\Delta H = +ve, \Delta S = -ve$
 - $\Delta H = -ve, \Delta S = +ve$
- According to the equation $\Delta G = \Delta H - T \Delta S$. Spontaneous reaction will occur when the value of ΔG will be.
 - Positive
 - Negative
 - Zero
 - Infinitive
- In precipitation reactions entropy of the system
 - decreases
 - increases
 - remains same
 - zero
- Standard enthalpy of graphite is
 - zero
 - >0
 - <0
 - can't be generalized

Answer Key

1	2	3	4	5	6	7	8	9				
c	c	c	d	b	c	b	a	a				

Chemical Kinetics

7



Positive catalysts help to decrease energy of activation, leading to the increase in the rate of reaction.

7.1 Introduction

There are so many chemical reactions existing in the nature with their own identities and peculiarities. These reactions can be studied from different point of views. Some chemical reactions such as reaction of sodium chloride solution with silver nitrate solution are so fast that simple techniques of measurement for the rate of reaction cannot be applied. On the other hand, the rusting of iron is slow reaction as it takes some hours to some days for completion. In the light of velocities for the transformation of reactants into products, chemical reactions may be classified into fast reactions and slow reactions. The task of the chemists lies within two extremities; at one end reactions are to be studied on a femto second (10^{-15} second) or shorter time scale and at another end reactions are to be studied on geological time scale. This is the domain of chemical kinetics. Chemical kinetics is the branch of chemistry which deals about rate of chemical reaction, factors affecting it and mechanism of chemical reaction. The rate of reaction is determined by knowing a series of intermediate steps or mechanism of the reaction. Knowing the mechanism of a chemical reaction is the first step toward the understanding of reaction and hence it helps to monitor the chemical reaction. In this level, we will deal about the basic concept on rate of reaction, way of measurement of rate of those chemical reactions which are neither too fast nor too slow.

7.2 Rate of Chemical Reaction

Rate of reaction accounts how fast the reactants transform into products.

Let us consider a reaction, $A \rightarrow B$

During chemical reaction, reactant A is gradually converts into B . Concentration of A goes on decreasing while that of B goes on increasing. Disappearance of reactant A or appearance of product B is the indication of the chemical transformation. Change in concentration is the difference between final concentration and initial concentration. Therefore, the value of change in concentration of reactant comes to be negative and rate of reaction appears to be negative. But the rate of reaction is never negative. Due to this reason negative sign is introduced to express the rate of reaction in terms of reactant. Rate of reaction or velocity of reaction is defined as the change in concentration of reactants or products per unit time.

$$\text{Rate of reaction} = \frac{\text{Change in concentration}}{\text{Change in time}}$$

If rate of reaction is expressed in terms of reactants, negative sign is introduced and if rate of reaction is expressed in terms of products, positive sign is introduced.

For above reaction

$$\text{Rate of reaction} = - \frac{\Delta[A]}{\Delta t} \quad (\text{in terms of reactant})$$

$$\text{or} \quad \text{Rate of reaction} = \frac{\Delta[B]}{\Delta t} \quad (\text{in terms of product})$$

The rate of reaction or velocity of reaction can be classified into following types:

1. Average rate of reaction
2. Instantaneous rate of reaction

1. Average rate of reaction

Average rate of reaction may be defined as the change of concentration of reactants or products per unit time interval (where the time interval is measurable). It is denoted by $\frac{\Delta x}{\Delta t}$. Here, Δx and Δt are finite change in concentration and time, respectively.

$$\text{Average rate of reaction} = \frac{\text{Change in concentration}}{\text{Change in time}} = \frac{\Delta x}{\Delta t}$$

Here, changes in concentration = Final concentration – Initial concentration.

For reactants, the change in concentration is negative while that for product it is positive. For the above reaction, if rate of reaction is expressed in terms of reactant A, the rate of reaction = $-\frac{\Delta x}{\Delta t}$. If the rate of reaction is studied in terms of product, the change in concentration is found to be positive.

Therefore, Rate of reaction = $+\frac{\Delta x}{\Delta t}$

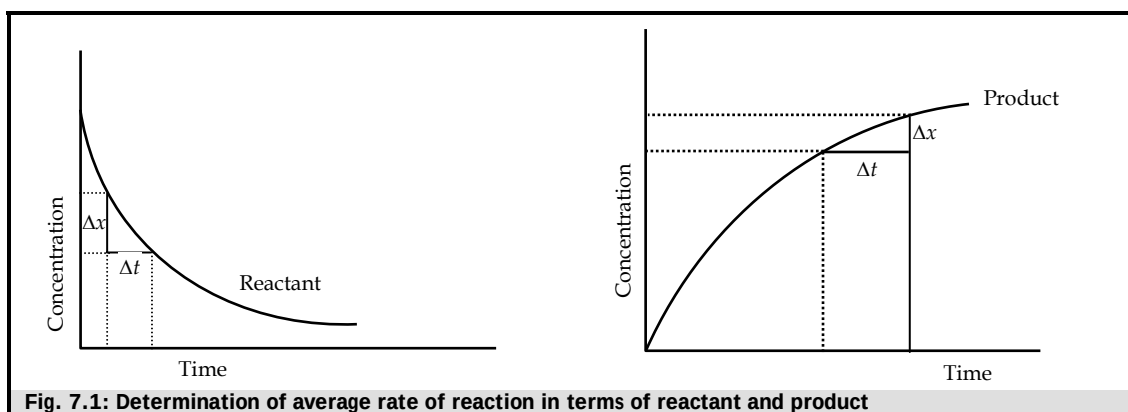


Fig. 7.1: Determination of average rate of reaction in terms of reactant and product

2. Instantaneous rate of reaction

One of the implications in the measurements of average velocity of chemical reaction is that the concentration of reactants decreases with time and so does the rate of reaction. But calculation shows constant rate of reaction for the particular interval of time. Besides this, the rate of reaction at a particular instant of time cannot be measured by average rate method. Therefore, to know the rate of reaction at a particular instant of time during the course of time, the term *instantaneous rate* of reaction is introduced. **Instantaneous rate of reaction may be defined as the rate of change of concentration of reactants or products at a particular moment of time.** It is denoted by $\frac{dx}{dt}$.

Where dx is infinitesimal change in concentration of reactants or products and dt is infinitesimal change in time.

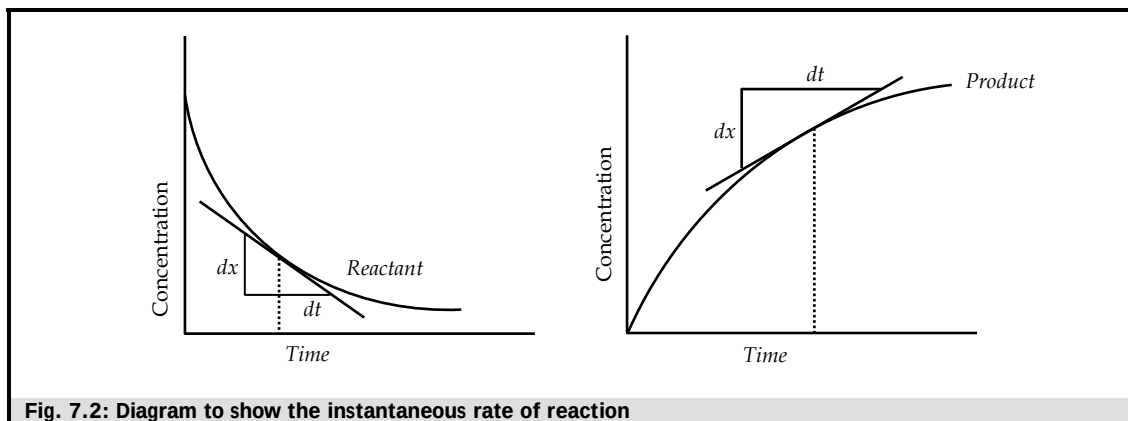
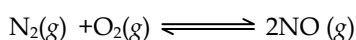


Fig. 7.2: Diagram to show the instantaneous rate of reaction

Instantaneous rate of reaction is determined by measuring the slope of a tangent plotted at a concentration-time graph corresponding to a particular instant of time.

As the time interval of average rate of reaction Δt decreases and tends to be zero then the average rate of reaction becomes instantaneous rate of reaction. Mathematically, average rate of reaction and instantaneous rate of reactions are related as $\lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$.

Let us take an examples to study the rate of chemical reaction.



In this reaction, one mole of nitrogen and one mole of oxygen disappear to produce two moles of nitric oxide. It means, for the appearance of two moles of NO, each mole of reactant is consumed.

$$\text{Rate of reaction in terms of N}_2 (r) = -\frac{d[\text{N}_2]}{dt}$$

$$\text{Rate of reaction in terms of O}_2 (r) = -\frac{d[\text{O}_2]}{dt}$$

$$\text{Rate of reaction in terms of NO (2r) = } \frac{d[\text{NO}]}{dt} \text{ or } r = \frac{d[\text{NO}]}{2 dt}$$

These expressions show that, the rate of reaction depends upon the chemical species chosen from the balanced chemical equation.

To express the rate of chemical reaction irrespective of the choice of chemical species, it is customary to express the rate of reaction for each mole of chemical species involved in the balanced chemical equation. For this purpose, rate expression of each chemical species is divided by stoichiometric coefficient of the balanced chemical equation. This is often called **equivalence rate of reaction**.

Hence equivalence rate of reaction is defined as **the change in concentration of each mole of reactants or products per unit time interval**. This type of chemical reaction is irrespective of the choice of chemical species involved to express the rate of reaction. If above reaction is to be expressed in terms of equivalence rate of reaction, it is written as,

$$-\frac{d[\text{N}_2]}{dt} = -\frac{d[\text{O}_2]}{dt} = +\frac{1}{2} \frac{d[\text{NO}]}{dt}$$

Self Test

- 7.1 What is chemical kinetics?
- 7.2 What are average rate of reaction and instantaneous rate of reaction? How are they related?
- 7.3 How is instantaneous rate of reaction superior to average rate of reaction in the determination of rate of chemical reaction?

7.3 Concept of Activation Energy and Threshold Energy

Reactants react to give products only under the given set of conditions. Whether the reaction is exothermic or endothermic, energy of reactant is insufficient to give product. Hence, energy must be added so that the reactants can surmount the energy barrier to give product. The concept of different types of energy for reaction system is clearly understood from figure 7.3 which is called energy profile diagram. In this diagram AB represents the energy associated with reactant.

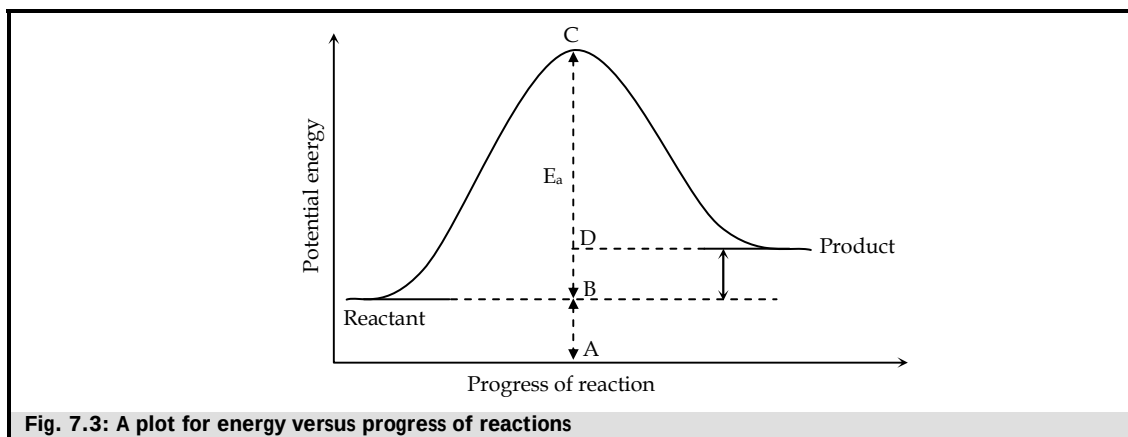


Fig. 7.3: A plot for energy versus progress of reactions

The minimum amount of energy **that must be supplied** to the reactants via collisions to cross the energy barrier for giving products is called **activation energy**. In the energy profile diagram, energy equivalent to BC represents the activation energy.

The minimum amount of energy **that must be associated** with the reactant molecules to cross the energy barrier for giving product is called **threshold energy**. In the above diagram energy equivalent to AC represents the threshold energy.

$$\text{Threshold energy} = \text{Energy of reactant} + \text{Activation energy}.$$

7.4 Theory of Chemical Reaction

There are different theories about the rate of reaction. Out of these, two important theories of rate of reaction are:

1. Collision theory of reaction
2. Activated complex formation theory

1. Collision theory of reaction

According to this theory, a chemical reaction can only take place when reactant molecules collide with each other. Chemical reaction results by breaking of pre-existing some or all bonds followed by formation of new bonds. The breaking of bonds takes place by the collision of molecules. The number of collisions that takes place per second in given volume of the reacting system is called

collision frequency. It is denoted by Z . Greater the collision frequency, greater the rate of reaction. If all the collisions would be effective chemical reactions would be completed in fractions of seconds. It shows that all the collisions are not effective. The impact of collision frequency on the rate of reaction can be explained on the basis of the following factors. These factors also affect the rate of chemical reaction.

- i. **Energy factors:** All the collisions are not effective. Only those collisions are effective which occur with sufficient amount of energy. For a reaction to take place, reactant molecules should have at least certain minimum amount of energy. The minimum quantity of energy that must be associated with reactant molecules to give products by crossing the energy barrier is called threshold energy. Those molecules which lack the threshold energy can't give products although these have good agreement of collision frequency.

According to kinetic theory of gas, the fraction of molecules having energy more than threshold energy at temperature T is, $f = e^{-E_a/RT}$

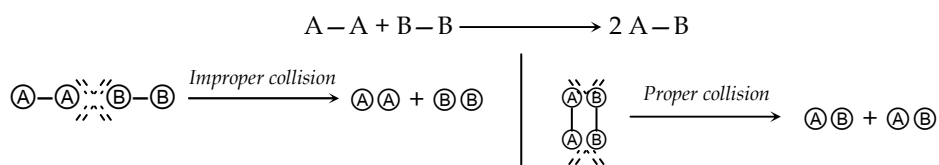
Where, f = fraction of molecules having sufficient quantity of energy.

E_a = activation energy

R = universal gas constant

T = temperature

- ii. **Orientation factors (p):** High degree of collision and sufficient quantity of energy possessed by reactant molecules still are not the sufficient criteria for the reactions to yield products because all the collisions may not be fruitful products. Some of the collisions, although possess sufficient quantity of energy, may become futile. The most important factor is orientation factor. When molecules collide with each other in proper alignment or with proper orientation then only reactants can give the products. All the collisions are not effective. Only those collisions which occur with proper orientation with sufficient quantity of energy can give the products. Such types of collisions are called **effective collisions**. Orientation factor is also called steric factor or probability factor because effective collision of molecules is the consequence of probability. When molecules are properly oriented, rearrangement of atoms in a chemical reaction takes place with minimum expenditure of time and energy. For a hypothetical reaction,



According to this theory,

Rate of reaction = Collision frequency (Z) $\times$ Fraction of molecules having sufficient energy (f)

Rate of reaction is directly proportional to rate constant (k).

Therefore, above relation can be expressed as,

$$k = Z \times f$$

$$\text{or, } k = Z \cdot e^{-E_a/RT} \quad \dots (i)$$

Here, E_a represents molecules achieving activation energy so that they have attained threshold energy. When orientation factor is introduced, the above relation becomes.

$$k = P \cdot Z \cdot e^{-E_a/RT} \quad \dots (ii)$$

Here P is orientation factor or steric factor.

Simply, equation (ii) can be expressed as, Rate of reaction = $P.Z.f$... (iii)

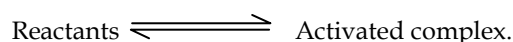
This theory is quite satisfactory for gaseous reactions. However, this theory does not tell us about the intermediate product formation. There are so many chemical reactions which proceed by the formation of intermediate product.

2. Activated complex formation theory

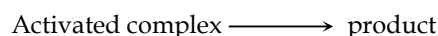
Collision theory is inadequate to explain the rate of reaction involving complex molecules. In 1935, Henry Eyring and Polyani gave a new theory to explain the dependence of reaction rate on concentration, temperature and other various factors in which activated complex or transition state is formed during transformation of reactant into product. This theory is called activated complex formation theory or transition state theory or theory of absolute reaction rate.

The main postulates of this theory are as follows:

- The reactant molecules are transformed into high energy containing intermediate complex before changing into product and such intermediate complex associated with high energy is called activated complex or transition state.
- The activated complex is in equilibrium with reactant and it has transient existence.



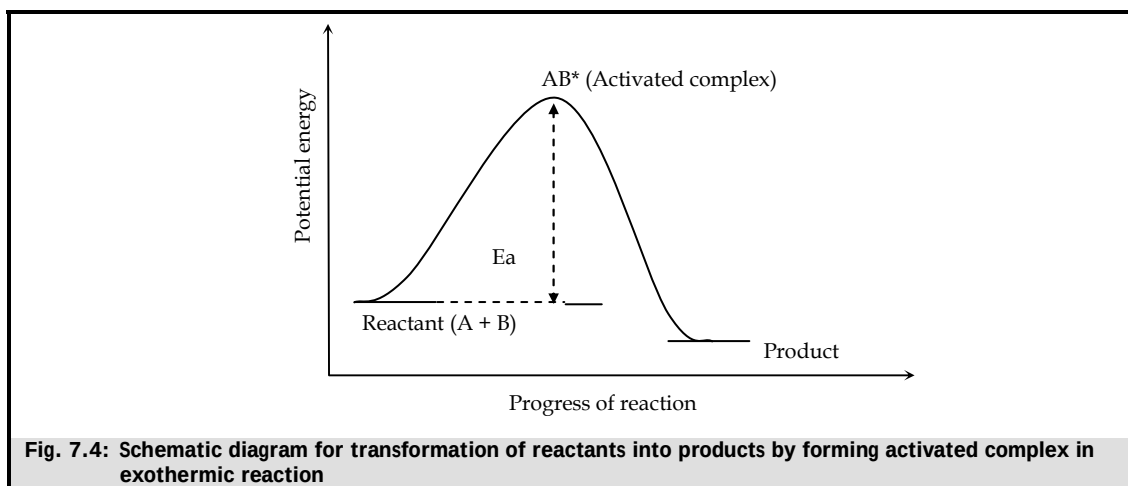
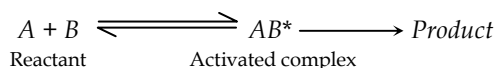
- The activated complex decomposes into products.



- The rate of reaction is given by the rate of decomposition of the activated complex to products.

$$\text{Rate of reaction } \frac{dx}{dt} = [\text{activated complex}] \times \text{rate of dissociation of activated complex}$$

Above points may be summarized by the following reaction and energy profile diagram (fig. 7.4).



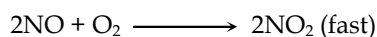
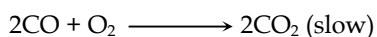
Self Test

7.4 Draw the energy profile diagram of an endothermic reaction.

7.5 Factors Affecting the Rate of Chemical Reaction

All the molecules present in reacting vessels may not yield product. There may be only some molecules which have sufficient energy to react. The rate of product formation is affected by the number of collisions of each reacting molecules at a specified time. All the collisions are not effective. Only those collisions are effective which collide in proper orientation with sufficient energy. There are following factors which affect the rate of reactions.

1. **Nature of reactant or product:** Ionic compounds are previously ionized so the rates of reaction are usually faster than that of covalent compounds. Ionic reactions occur in electrovalent compound while molecular reactions occur in covalent compounds. Oxidation of nitric oxide is faster than the oxidation of CO under similar conditions. This is because different amount of energy is required to break a bond and different amount of energy is released in the formation of a new bond.



2. **Temperature:** Rate of chemical reaction depends upon temperature. When temperature is increased kinetic energy of molecules is increased and it increases the degree of collision. Besides this, due to increase in kinetic energy, the fraction of molecules having sufficient quantity of energy increases which increases the rate of reaction. Increase in temperature increases the rate of reaction.

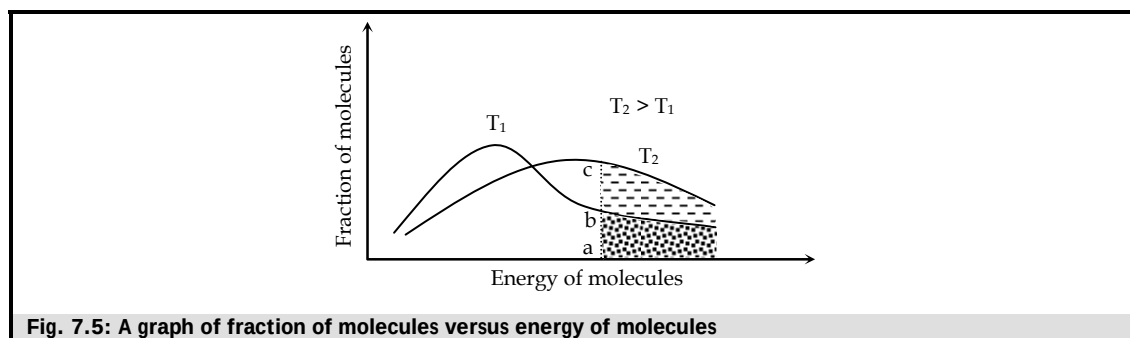


Fig. 7.5: A graph of fraction of molecules versus energy of molecules

In figure 7.5, molecules lying right side from line *ab* have sufficient quantity of energy at temperature T_1 . When temperature is increased to T_2 , molecules lying right side from line *ac* have sufficient quantity of energy. This shows that with increase in temperature, the fraction of molecules having sufficient energy increases. Due to increase in energy of chemical species degree of collisions increases and it increases the rate of chemical reaction.

For each 10° rise in temperature, the rate of chemical reaction is increased by 2 to 3 fold. This is often expressed in terms of **temperature coefficient (T_a)**. Temperature coefficient is defined as the ratio of rate constants for a chemical reaction at two different temperatures different by 10°C .

If K_t be the rate constant for a reaction at $t^\circ\text{C}$ and $K_{(t+10)}$ be the rate constant for a reaction at $(t+10)^\circ\text{C}$.

Then temperature coefficient (T_a) = $\frac{K_{t+10}}{K_t}$

Generally, temperature chosen are 25°C and 35°C,

$$T_{\alpha} = \frac{K_{35}}{K_{25}}$$

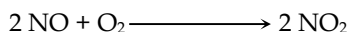
Where, K_{35} = Rate constant for a reaction at 35°C and K_{25} = Rate constant for a reaction at 25°C . The effect of temperature on the rate of reaction can be explained in terms of *Arrhenius equation*.

Arrhenius equation is $K = A e^{\frac{-E_a}{RT}}$

Here, K is rate constant, A is pre-exponential factor or Arrhenius factors, E_a is activation energy, R is universal gas constant and T is absolute temperature.

$$K = \frac{A}{e^{\frac{E_a}{RT}}}$$

Increase the value of T decreases the value of $e^{\frac{E_a}{RT}}$ and it increases the value of K . Grater value of K signifies the high rate of reaction. In the oxidation of NO to NO₂ rate of reaction decreases slightly on increasing the temperature.



In enzyme catalysed biochemical reactions, the rate of reaction is high at a certain temperature, called optimum temperature. Below and above this temperature the rate of reaction is low.

In explosive type of reaction, the rate of reaction increases abruptly on increasing the temperature. The natures of graph for different reactions are given below.

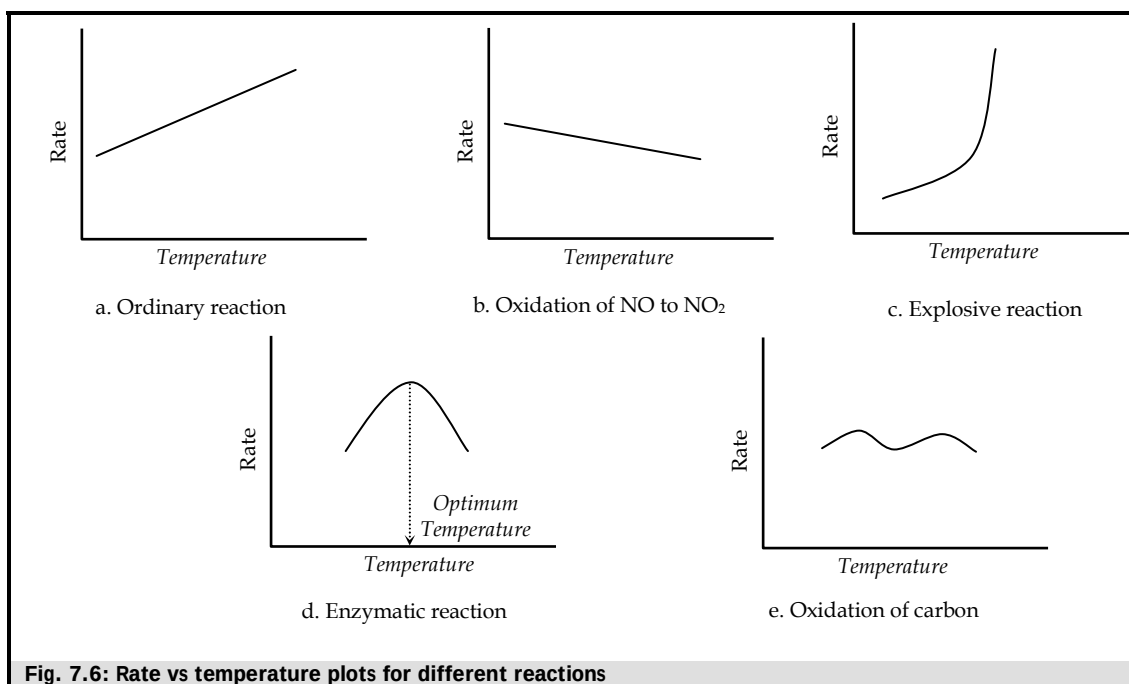


Fig. 7.6: Rate vs temperature plots for different reactions

3. **Concentration:** Rate of chemical reaction is appreciably affected by concentration. High concentration of reactants possesses more numbers of chemical species per unit volume. Due to increase in molecular density, the degree of collisions increases and thus the rate of reaction increases. Increase in concentration increases the rate of reaction. If a burning splinter is introduced into a gas jar filled with oxygen it glows more brightly. This is due to more concentration of oxygen. The effect of concentration on the rate of reaction was initially studied by **Guldberg** and **Waage** in 1867 and put forward a law which is called **law of mass action**. The law of mass action states that the rate at which a chemical substance reacts is directly proportional to its active mass or molar concentration and the rate of chemical reaction is directly proportional to the product of active masses of reactants, where each concentration term raised to power equal to stoichiometric coefficient present in balanced chemical equation.

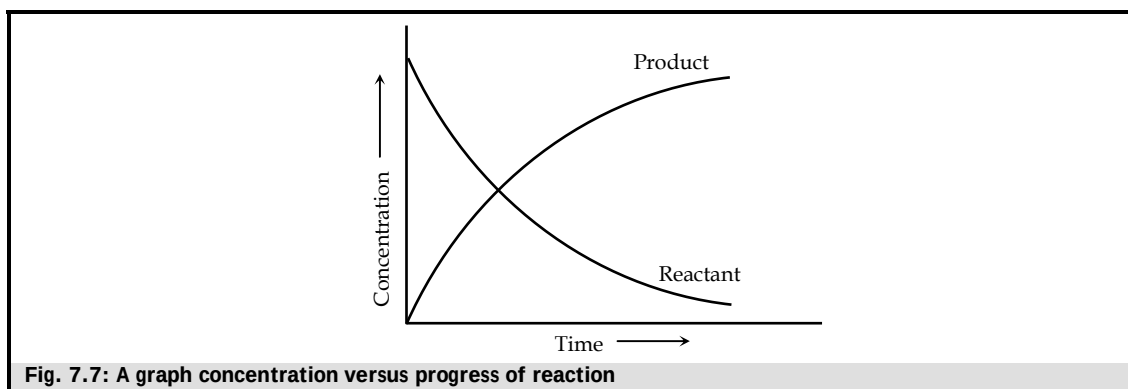
For a general reaction, $aA + bB \longrightarrow \text{products}$

According to the law of mass action,

$$\text{Rate} \propto [A]^a [B]^b$$

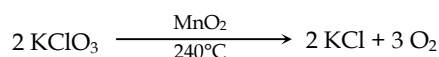
or $\text{Rate} = k[A]^a [B]^b$

The effect of concentration on the rate of reaction can be expressed graphically as in figure 7.7.

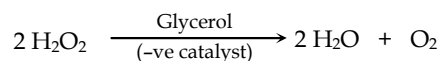


Initially, concentration of reactant is more and rate of reaction is also high. As the time passes, the concentration of reactant goes on decreasing and rate of reaction also goes on decreasing.

4. **Catalyst:** Catalysts are those chemical substances which increase or decrease the rate of chemical reaction by providing the alternative path for the reaction. Catalysts are specific in nature and catalyse a particular reaction. Catalyst takes part in chemical reaction without changing in its mass and composition. A positive catalyst increases the rate of reaction by decreasing the activation energy or threshold energy. Potassium chlorate should be heated upto 360°C in absence of catalyst to liberate oxygen gas. If MnO_2 is used as catalyst, potassium chlorate decomposes by the temperature of 240°C to 260°C . Here, MnO_2 acts as positive catalyst.



Glycerol acts as negative catalyst in the decomposition of H_2O_2 .



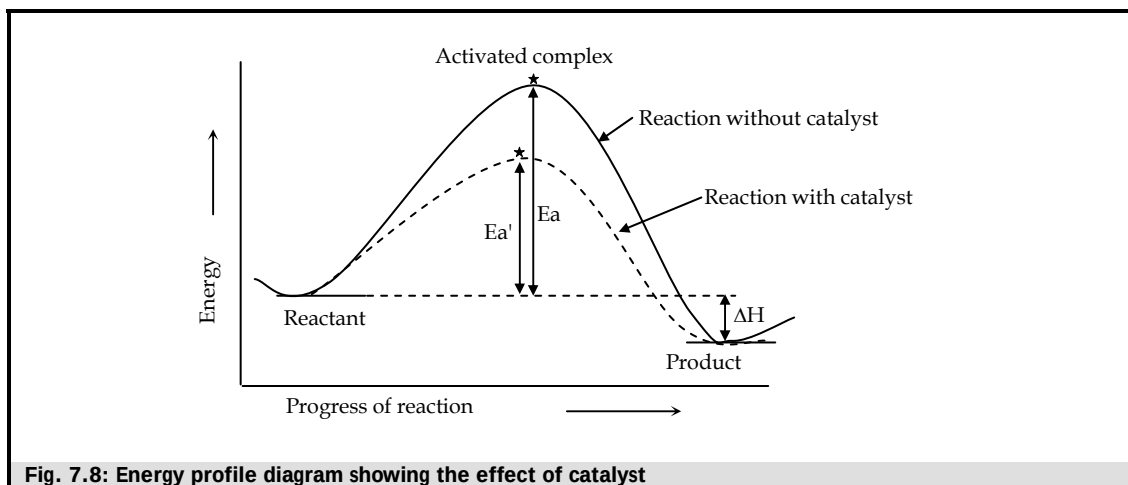


Fig. 7.8: Energy profile diagram showing the effect of catalyst

The effect of catalyst on the rate of reaction can also be understood with the help of Arrhenius equation $k = A e^{-E_a/RT}$.

$$\text{or, } k = \frac{A}{e^{E_a/RT}}$$

$$\text{Hence, rate of reaction} \propto \frac{1}{\text{activation energy}}$$

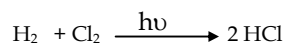
When activation energy is decreased, rate of reaction is increased. This is possible by the use of positive catalyst. A negative catalyst increases the activation energy or threshold energy for a reaction and it creates the alternative path having higher energy. This decreases the rate of chemical reaction.

Self Test

7.4 Draw the energy profile diagram to show the effect of catalyst for endothermic reaction.

- Radiation or light:** Radiations may initiate some chemical reactions which are otherwise not possible. Radiation consists of photon. Photon consists energy as shown by the relation $E = h\nu$. Some reactions are explosive in presence of light.

Photosynthesis occurs only in presence of sunlight. Chemistry of vision also involves light. H_2 and Cl_2 react on the surface of water in presence of sunlight to produce HCl as:



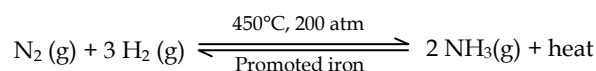
The chemical reaction which occurs by the absorption of light is called photochemical reaction. Unlike chemical reaction, photochemical reaction depends upon intensity of light. The rate of photochemical reaction depends upon the wavelength of light. Rates of photochemical reactions are not affected by temperature.

- Surface area:** Increase in surface area increases the number of molecules undergoing chemical reaction per unit time which increases the rate of chemical reaction. For example, the rate of production of H_2 gas is more when granulated zinc is reacted with *dil.* HCl instead of big lumps of zinc.

Self Test

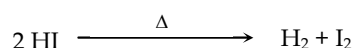
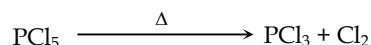
- 7.5 Calculate the surface area of cube of a zinc piece having edge length 1cm.
- 7.6 Zinc piece having edge length 1 cm in cubic structured is transformed into 1000 smaller cubes. Calculate
- Surface area of one small cube.
 - Surface area of all cubes.
 - Change in surface area when 1 cube is changed to 1000 smaller cubes.

7. **Pressure:** If the reactants are present in gaseous phase the rate of chemical reaction depends upon the pressure. Increase in pressure generally increases the number of collisions among the molecules, causing the increase in the rate of reaction.

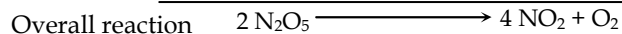
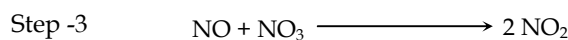
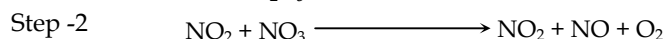
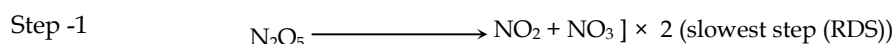
**7.6 Types of Reaction Based upon Mechanism**

There are different ways of classification of reactions. Depending upon the number of steps involved in the transformation of reactants into products, chemical reactions are classified into simple and complex reactions.

1. **Simple or elementary reaction:** Those chemical reactions which are supposed to take place in single step to form products are called simple or elementary reactions. Some simple or elementary reactions are given below.



2. **Complex reaction:** Those chemical reactions, which are supposed to take place in more than one elementary steps to give products are called complex chemical reactions. Complex chemical reactions are assumed to be composed of many elementary steps. For example decomposition of N_2O_5 is supposed to proceed in following steps.

**7.7 Concept of Rate Determining Step**

Elementary reaction completes in single step. Therefore the rate of reaction means, the rate of the elementary step. However, there are some complications to determine the rate, where complex reaction takes place.

Let us take an example of decomposition of N_2O_5 , mentioned in section 7.6

The decomposition of N_2O_5 takes place in three different steps where each step has its own velocity or rate. Which step would determine the rate of overall reaction? Obviously, the slowest step. It is because the rate of reaction can not be faster than the rate of slowest step. Hence, the slowest step is

the rate determining step (RDS). The slowest step or rate determining step is obtained experimentally. A chemical reaction can not proceed faster than its slowest step.

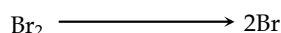
Self Test

7.7 A motor cycle company can manufacture 20 sets of wheels, 4 sets of engines, 10 sets of rems and 1 set of body part in a day. How many motor-cycles are manufactured by the company in one day?

7.8 Molecularity of Reaction

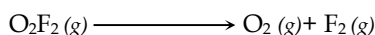
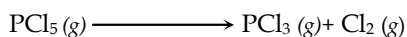
Each reaction has its own stoichiometry giving the number of molecules taking part in the chemical reaction. It is a common practice to express a chemical reaction in terms of molecularity. In the elementary reaction, molecularity of a reaction is defined as the number of reactant species (atoms, ions or molecules) reacting in an elementary step. In an elementary reaction, the total number of reactant molecules present in a balanced equation gives the molecularity of the reaction.

In the reaction,

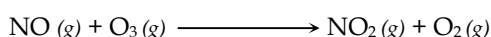
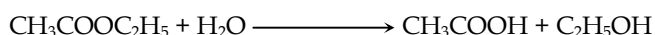
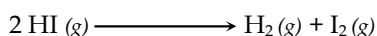


One bromine molecule undergoes decomposition to give two bromine atoms. Therefore, this is unimolecular reaction. In case of complex reaction, the number of reactant molecules of a slowest step is regarded as molecularity.

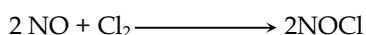
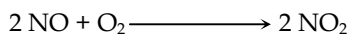
If only one reactant molecule undergoes decomposition or rearrangement in a chemical reaction, it is called unimolecular reaction. Some examples are given below.



If two reactant molecules or atoms or ions undergo simultaneous collisions to give the product, the reaction is called bimolecular reaction. Some examples are given below.



If three reactant molecules or atoms or ions undergo simultaneous collision to give product, the reaction is called termolecular or trimolecular reaction. Some examples are given below.



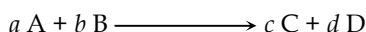
If we consider a complex chemical reaction, all atoms, ions or molecules in the balanced chemical equation do not undergo simultaneous collision. Thus total number of molecules present in the balanced chemical equation has no meaning to find out molecularity. In complex reaction, the complete reaction is considered to take place through different steps. Out of these different steps, one step is the slowest step. The number of reactant molecules which undergo simultaneous collision in the slowest step is the molecularity of given reaction.

Higher molecularity of reaction is rare. This can be explained on the basis of kinetic theory. For a chemical reaction to take place, reactant molecules must undergo simultaneous collision. However the chances of simultaneous collisions go on decreasing for higher molecularity. For example, the simultaneous collisions among three molecules are less probable than the simultaneous collision between two molecules.

7.9 Law of Mass Action and the Rate Law Equation

The effect of concentration on the rate of reaction was studied by **Guldberg** and **Waage** in 1867 and they gave a law which is called law of mass action. The law of mass action states, the rate of reaction is directly proportional to the product of molar concentration of reactants where each concentration term raised to power equal to stoichiometric coefficient present in the balanced chemical equation.

For a general reaction,



According to law of mass action, Rate of reaction (r_1) $\propto [A]^a [B]^b$... (i)

From equation (i) it is revealed that the rate of reaction depends upon ' a ' concentration term of A and ' b ' concentration term of B. The rate of overall reaction depends upon ($a+b$) concentration term.

If the reaction is studied experimentally, the rate of reaction may or may not depend upon the concentration terms present in balanced chemical equation. In the above example, the rate of reaction may or may not depend upon ' a ' concentration term of A and ' b ' concentration term of B. Let the rate of reaction depends upon ' p ' concentration term of A and ' q ' concentration term of B. Here, the values of p and q may or may not equal to a and b , respectively.

On the basis of this assumption, the rate of reaction is expressed as,

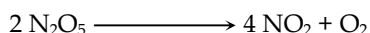
$$\text{Rate} \propto [A]^p [B]^q$$

$$\text{Rate} = k[A]^p [B]^q \quad \dots \text{(ii)}$$

Where k is proportionality constant and is called rate constant p and q are actual concentration terms on which rate of reaction depends. The value of p and q are determined experimentally. Their values may be zero, negative, positive or fractional. The equation (ii) is called **rate law equation**.

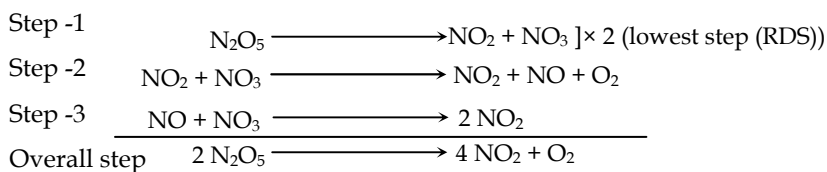
Rate law equation is a mathematical expression which shows the actual dependence of concentration term on the rate of reaction. Rate law equation is determined experimentally.

To make the concept clear between law of mass action and rate law equation let us take a specific example of decomposition of N_2O_5



According to law of mass action, Rate $\propto [N_2O_5]^2$... (iii)

The law of mass action, reveals that the rate of reaction depends upon two concentration terms of N_2O_5 . If the mechanism of the chemical reaction is studied, the following reaction pattern for the gaseous phase decomposition of energized N_2O_5 is obtained experimentally.



Out of these three different steps, the first step is the slowest step and it is rate determining step. The overall rate of reaction depends upon slowest step. Here, the slowest step consists only one reactant term.

According to the experimental finding,

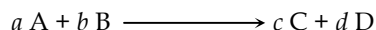
$$\text{Rate} \propto [N_2O_5]$$

$$\text{Rate} = k[N_2O_5] \quad \dots \text{(iv)}$$

The actual concentration term on which the rate of reaction depends is one. Therefore, equation (iv) is rate law equation.

Rate constant

Let us consider a general reaction



According to the law of mass action the rate of reaction is expressed as, $\text{Rate} \propto [A]^a [B]^b$... (i)

But, if the mechanism of the reaction is observed experimentally, the rate law equation may be same or different from what is expected from law of mass action. The mathematical expression showing the actual dependence of concentration term on the rate of reaction is rate law equation. For the above case, let the rate law equation is

$$\text{Rate} \propto [A]^p [B]^q$$

$$\text{Rate} = k[A]^p [B]^q \quad \dots \text{(ii)}$$

Here, p and q are actual concentration term on which the rate of reaction depends. Equation (ii) is rate law equation where k is proportionality constant and is called rate constant or velocity constant or specific rate of reaction.

From equation (ii), when molar concentration of A and B are unity then $\text{Rate} = k$.

Hence rate constant is defined as the rate of reaction when concentrations of all the reactant terms are unity. It is also called specific rate of reaction.

Characteristics of rate constant

Some important characteristics of rate constant are given below.

- i. Rate constant is the measure of rate of reaction. Larger the value of rate constant, greater will be the rate of reaction.
- ii. Rate constant is independent of the concentration of reactants.
- iii. Rate constant is the characteristics value of each reaction. Different reactions have different values of rate constants at a particular temperature.
- iv. Rate constant depends upon temperature. On changing the temperature, the value of rate constant is also changed.
- v. The unit of rate constant depends upon the order of reaction.

7.10 Order of Reaction

Order of reaction assigns the number of reactant molecules on which the rate of chemical reaction depends. Order of reaction with respect to one of the reactants is the power of concentration term of the reactant in the rate law equation. The over all order of reaction is the sum of the powers (exponents) of the concentration terms of reactants present in the rate law equation. As we have discussed earlier, molecularity of a reaction is a theoretical concept. Unlike molecularity of a reaction, the order of reaction is experimental value. Order of reaction is determined on the basis of rate law equation.

For a general reaction, $a A + b B \longrightarrow \text{Product}$

$$\text{Let, the rate law equation be, } \text{Rate} \left(\frac{dx}{dt} \right) = k [A]^p [B]^q$$

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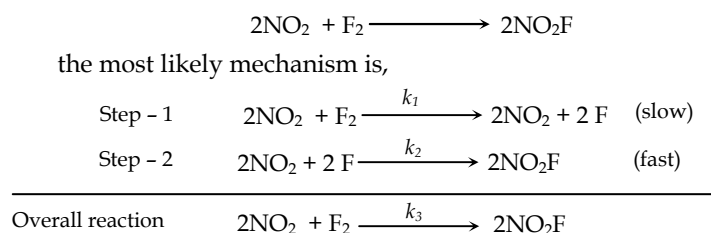
The order of reaction with respect to $A = p$

The order of reaction with respect to $B = q$

Overall order of reaction $= p + q$

From this, it is clear that order of a reaction is determined with respect to each kind of individual reactant as well as with respect to all the participating reactants. Overall order of reaction may also be defined as the sum of the powers of each concentration terms present in the rate law equation.

Since order of reaction is determined experimentally, it may be zero, positive, negative or fractional. If reaction environment for a chemical reaction is altered, the order of reaction is also changed. To understand the order of reaction let us take given example.



Apparently, both steps are trimolecular. Experiment shows that the first step is slowest step and it is rate determining step. The rate $\left(\frac{dx}{dt}\right)$ of the reaction can be expressed as,

$$\frac{dx}{dt} = k_1[\text{NO}_2]^2 [\text{F}_2]$$

This shows that, the rate of reaction depends upon the concentration of NO_2 and F_2 both.

The order of reaction with respect to $\text{NO}_2 = 2$

The order of reaction with respect to $\text{F}_2 = 1$

The overall order of reaction $= 2 + 1 = 3$.

Units of Rate Constant

Different the order of reactions, different will be the unit of rate constant. The units of rate constant are given below.

Let us consider a reaction, $\text{A} \longrightarrow \text{Product}$

Let the order of reaction be n . It means the rate of reaction depends upon ' n ' concentration term of A.

$$\text{Rate} \propto [\text{A}]^n$$

or, $\frac{dx}{dt} = k[\text{A}]^n$, where k is proportionality constant and is called rate constant

$$\text{or, } k = \frac{dx/dt}{[\text{A}]^n} = \frac{\text{mol lit}^{-1} \text{ time}^{-1}}{(\text{mol lit}^{-1})^n} = \text{mol}^{1-n} \text{ L}^{n-1} \text{ time}^{-1}$$

For zero order reaction, $n = 0$ unit of $k = \text{mol L}^{-1} \text{ time}^{-1}$

For first order reaction, $n = 1$ unit of $k = \text{time}^{-1}$

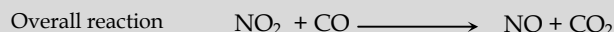
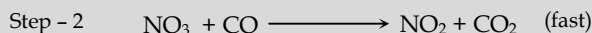
For second order reaction, $n = 2$ unit of $k = \text{mol}^{-1} \text{ L time}^{-1}$

For third order reaction, $n = 3$ unit of $k = \text{mol}^{-2} \text{ L}^2 \text{ time}^{-1}$

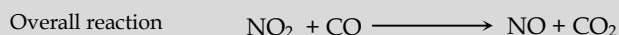
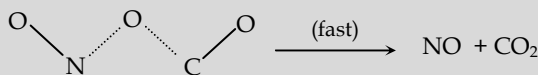
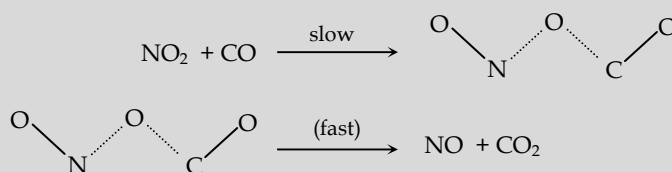
Self Test

7.8 The reaction between NO_2 and CO takes place as follows.

a. At lower temperature the reaction mechanism is



b. Above 500K temperature, the reaction mechanism is



7.8 Express the rate law equation for reaction of NO_2 and CO below approximate 500 K temperature. What is the order of reaction?

$$\text{Ans: } \frac{dx}{dt} = k[\text{NO}_2]^2 \quad (\text{second order reaction})$$

7.9 Express the rate law equation for the reaction of NO_2 and CO above approximate 500 K temperature. What is the order of reaction?

$$\text{Ans: } \frac{dx}{dt} = k[\text{NO}_2][\text{CO}] \quad (\text{second order reaction})$$

7.10 For a particular reaction, under different conditions, having same order, is it necessary to have same rate law equation?

Ans: No

Difference between molecularity and order of reaction

For some reactions, the molecularity and order are same, but it is not true for all the cases. The differences between molecularity and order of reaction are given below.

Molecularity	Order
1. It is the number of atoms or ions or molecules which involve in elementary step of reaction so as to give product in a chemical reaction.	1. It is the sum of the powers which are raised in each concentration term in a rate law equation.
2. It is always a natural number (whole number except zero).	2. It may be zero, negative, positive or fractional. For e.g., $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \xrightarrow{h\nu} 2\text{HCl}(\text{g}) \text{ is (Zero order)}$ $2\text{N}_2\text{O}_5(\text{g}) \longrightarrow 4\text{NO}_2(\text{g}) + \text{O}_2 \text{ is (1}^{\text{st}} \text{ order)}$ $\text{CH}_3\text{CHO}(\text{g}) \xrightarrow{\Delta} \text{CH}_4(\text{g}) + \text{CO}(\text{g}) \text{ (1.5 order)}$ $3\text{O}_3(\text{g}) \longrightarrow 3\text{O}_2(\text{g}) \text{ (-1 order w.r. to O}_2\text{)}$
3. It is determined theoretically.	3. It is always determined experimentally.

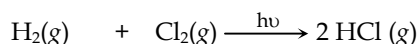
4. It is invariant. It means on changing the condition, the molecularity of reaction does not change.	4. It is variant. On changing the condition, the order of reaction may change. For example, hydrolysis of ethylacetate in acidic medium is of first order while hydrolysis in alkaline medium is of second order.
5. For simple reaction, the molecularity can be determined from balanced stoichiometric equation. It means, for elementary reaction, the molecularity is equal to the number of reactant molecules present e.g. $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \xrightarrow{h\nu} 2\text{HCl}(\text{g})$ The molecularity is 2.	5. Even for simple reaction, the order of reaction can not be determined from balanced stoichiometric elementary reaction. The order of reaction may not be equal to the number of reactant molecules present. e.g., For a reaction on the surface of water. $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \xrightarrow{h\nu} 2\text{HCl}(\text{g})$ The order of reaction is zero.

7.10.1 Zero-Order Reaction

The chemical reaction, in which rate of reaction does not depend upon the change in concentration of reactant is called zero-order reaction. In other words, zero-order reaction is independent to concentration of reactant. The rate of product formation depends upon the amount of reactant change.

Some examples of zero-order reaction are given below.

1. Formation of HCl gas from H₂ and Cl₂ gas in presence of sunlight on the surface of water.



At time = 0 *a mole* *b mole* 0

At time = t (*a - x*) mole (*b - x*) mole 2*x*

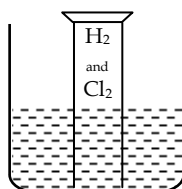


Fig.7.9: Reaction of hydrogen and chlorine in presence of radiation on the surface of water.

When light is passed, H₂ and Cl₂ combine to give HCl gas. Two volumes of reactants give two volumes of products. So there is no change in volume. As soon as HCl gas is formed it dissolves in water and water rushes up. Let *V*₁ be the volume of reactant and *V*₂ be the volume of product.

Number of mole of	Before reaction	After reaction	Result
a	$\frac{a}{V_1}$	$\frac{a-x}{V_2}$	$\frac{a}{V_1} = \frac{a-x}{V_2}$
b	$\frac{b}{V_1}$	$\frac{b-x}{V_2}$	$\frac{b}{V_1} = \frac{b-x}{V_2}$

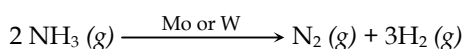
Due to chemical reaction, the amount of reactants change but the concentration of reactant does not change because the proportion of decreases in number of moles of reactant to volume in each step is same. In this case rate of reaction depends upon intensity of light. During chemical reaction, the amount of reactant changes but concentration of reactant does not change. The rate of reaction neither depends upon the concentration of H_2 nor the concentration of Cl_2 .

Order of reaction with respect to $H_2 = 0$

Order of reaction with respect to $Cl_2 = 0$

Overall order of reaction is zero.

2. **Catalytic decomposition of ammonia gas:** When decomposition of ammonia is carried out on the surface of molybdenum or tungsten,



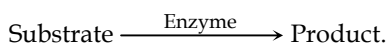
the rate of reaction does not depend upon the concentration of NH_3 . To carry out the chemical reaction the gas molecules should be desorbed on the catalyst surface. When catalyst surface is limited and concentration of NH_3 is high, rate of product formation depends upon the surface area of catalyst rather than the concentration of ammonia.

The rate law equation for the above reaction is

$$\frac{dx}{dt} = k[\text{NH}_3]^0$$

In this case, the rate of reaction is zero-order with respect to ammonia.

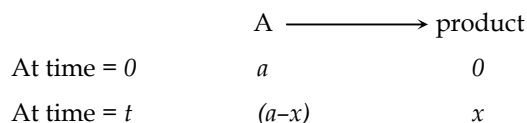
3. **Enzyme catalysed reaction:** In enzyme catalysed reaction, the rate of reaction depends upon the concentration of enzyme but not upon the concentration of substrate. So the reaction is first order reaction with respect to enzyme and zero-order reaction with respect to the substrate.



$$\text{Rate} = k [\text{Enzyme}]^1 [\text{Substrate}]^0$$

Kinetics of Zero-Order Reaction

Let us consider a reaction in which reactant A changes into product. Let 'a' be the initial concentration of reactant A. After certain time t, x mole of reactant changes into product.



For zero order reaction, rate of reaction depends upon zero concentration term of reactant.

$$\text{Rate} \propto [A]^0$$

or, $\frac{dx}{dt} = k_0(a-x)^0$, where k_0 is proportionality constant and is called rate constant or velocity constant for zero order reaction

$$\text{or, } \frac{dx}{dt} = k_0 \quad \dots (i)$$

Equation (i) is differential rate equation for zero order reaction.

$$\text{or } dx = k_0 dt$$

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Integrating equation (i), we get

$$\int dx = k_0 \int dt$$

$$\text{or } x = k_0 t + C \quad \dots \text{ (ii)}$$

where C is integration constant.

When $t = 0$, $x = 0$ then putting the value of t and x in equation (ii), we get $C = 0$. Now equation (ii) changes to

$$x = k_0 t$$

$$\text{or } k_0 = \frac{x}{t} \quad \dots \text{ (iii)}$$

Equation (iii) is **integrated rate equation** for zero order reaction.

When amount of reactant changes from $x = 0$ to $x = x$ with in time $t = 0$ to $t = t$, integrating the equation $dx = k_0 dt$, we get

$$\int_0^x dx = k_0 \int_0^t dt$$

$$\text{or, } [x]_0^x = k_0 [t]_0^t$$

$$\text{or, } (x - 0) = k_0(t - 0)$$

$$\text{or, } x = k_0 t$$

$$\text{or, } k_0 = \frac{x}{t}$$

Characteristics of zero-order reaction: Unit, Half life and Kinetic Plot

a. Unit of rate constant

$$k_0 = \frac{x}{t} = \frac{\text{concentration}}{\text{time}} = \frac{\text{mol. lit}^{-1}}{\text{time}} = \text{mol L}^{-1} \text{ time}^{-1}$$

- b. Half life period ($t_{1/2}$):** The time at which the concentration of reactant decreases to half of its initial value is called half life period. It is denoted by $t_{1/2}$. At half life period, time $t = t_{1/2}$ and $x = \frac{a}{2}$.

$$\text{Now the rate law equation becomes } k_0 = \frac{a/2}{t_{1/2}}$$

$$\text{or } t_{1/2} = \frac{a}{2k_0} \quad \dots \text{ (iv)}$$

$$\text{or } t_{1/2} \propto a.$$

For zero order reaction, half life period is directly proportional to the initial concentration of reactant.

c. Kinetic plots: For zero-order reaction,

$$k_0 = \frac{x}{t}$$

$$\text{or, } x = k_0 t$$

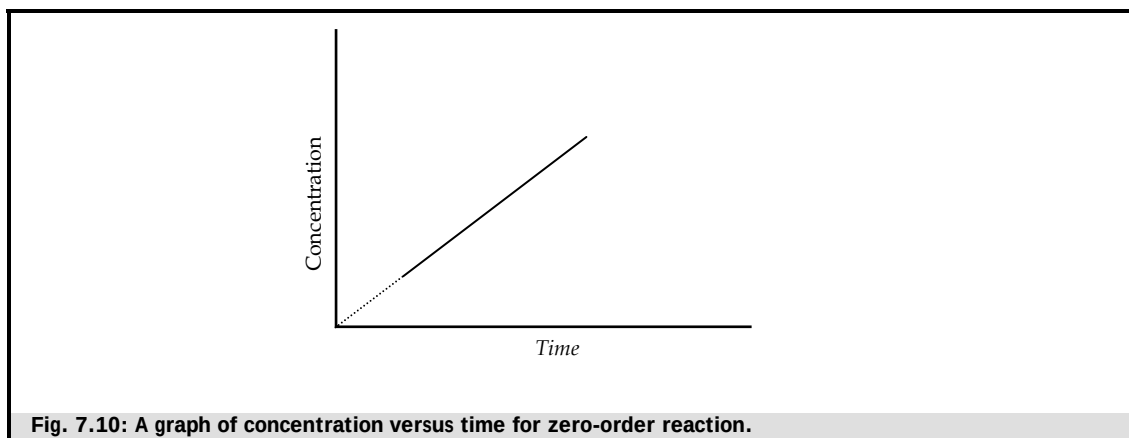


Fig. 7.10: A graph of concentration versus time for zero-order reaction.

This equation is in the form of $y = mx$ equation. Therefore, if a graph of concentration (x) versus time is plotted, a straight line passing through the origin is obtained (fig. 7.10). This verifies the validity of zero-order reaction.

Self Test

7.10 Plot a logarithmic graph for zero-order reaction.

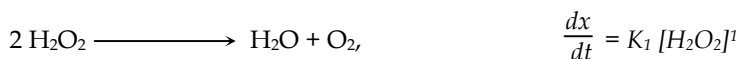
Hints, $x = k_0 t$, $\log x = \log k_0 + \log t$

7.10.2 First Order Reaction

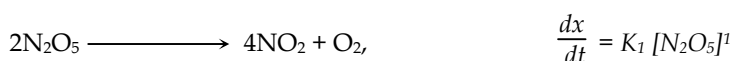
The chemical reaction in which rate of reaction depends upon only one concentration term is called first order reaction.

Some examples of first order reaction are given below.

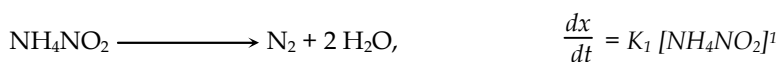
i. **Decomposition of H_2O_2**



ii. **Decomposition of N_2O_5**



iii. **Decomposition of NH_4NO_2**



iv. **Decomposition of SO_2Cl_2**



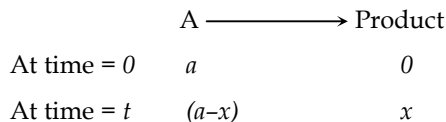
v. **Hydrolysis of ethyl acetate in acidic medium**



vi. **All radioactive disintegration reactions**

Kinetics of first order reaction: Differential and Integrated Rate Equation

Let us consider a general first order reaction in which reactant A changes to product. Let 'a' mol L⁻¹ be the initial concentration of reactant. After certain time *t*, *x* mol of reactant changes into product. Then concentration of reactant at time *t* is (a - *x*) mol L⁻¹.



For first order reaction, the rate of reaction depends upon only one concentration term.

$$\text{Rate} \propto [A]^1$$

$$\text{or, } \frac{dx}{dt} = (a - x)$$

$$\text{or, } \frac{dx}{dt} = k_1(a - x) \quad \dots \text{(i)}$$

Where, *k*₁ is proportionality constant and is called rate constant or velocity constant or specific rate of reaction. Equation (i) is the differential rate equation for the first order reaction.

$$dx = k_1(a - x)dt$$

$$\text{or, } \frac{dx}{(a - x)} = k_1 dt$$

Integrating the equation, we get

$$\int \frac{dx}{(a - x)} = \int k_1 dt$$

$$\text{or, } \frac{\ln(a - x)}{(-1)} = k_1 t + C \quad \left[\because \int \frac{dx}{(ax + b)} = \frac{\ln(ax + b)}{a} + C \right]$$

Where C is integration constant

$$\text{or, } -\ln(a - x) = k_1 t + C \quad \dots \text{(ii)}$$

when time *t* = 0, *x* = 0, then equation (ii) changes to $-\ln(a - 0) = k_1 \times 0 + C$

$$\text{or, } -\ln a = C$$

Putting the value of C in equation (ii), we get

$$\text{or, } -\ln(a - x) = k_1 t - \ln a$$

$$\text{or, } \ln a - \ln(a - x) = k_1 t$$

$$\text{or, } \ln \frac{a}{(a - x)} = k_1 t \quad \left[\because \ln a - \ln b = \ln \frac{a}{b} \right]$$

$$\text{or, } k_1 = \frac{1}{t} \ln \frac{a}{(a - x)} \quad \dots \text{(iii)}$$

$$\text{or, } k_1 = \frac{2.303}{t} \log \frac{a}{(a - x)} \quad \left[\because \ln a = 2.303 \log a \right] \quad \dots \text{(iv)}$$

Equation (iii) and (iv) are different forms of integrated rate equation for first order reaction.

Characteristics of first order reaction: Unit, Half-Life and the Kinetic Plot

a. Unit of rate constant

$$k_1 = \frac{1}{t} \ln \frac{a}{(a-x)} = \frac{1}{\text{time}} = \text{time}^{-1}$$

b. Half-life period ($t_{1/2}$): Half life period is defined as the time in which concentration of reactant decreases to half of its initial value. It is denoted by $t_{1/2}$.

At half life period, time $t = t_{1/2}$ and reactant changed into product $x = \frac{a}{2}$.

Now, the expression becomes,

$$k_1 = \frac{1}{t_{1/2}} \ln \frac{a}{\left(a - \frac{a}{2}\right)}$$

$$t_{1/2} = \frac{1}{k_1} \ln \frac{a}{a/2} = \frac{1}{k_1} \ln 2$$

$$\therefore t_{1/2} = \frac{0.693}{k_1} \quad \dots \text{(v)}$$

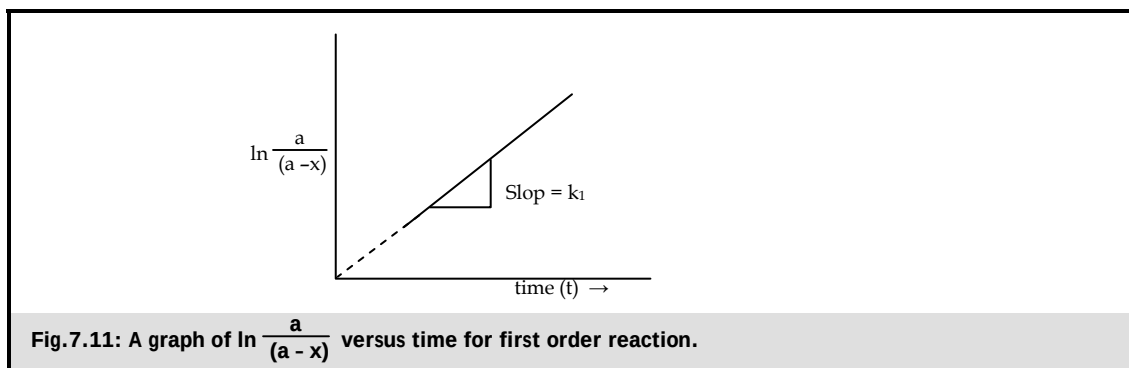
For first order reaction half life period is independent to initial concentration.

c. Kinetic plot: The rate constant for first order reaction is

$$k_1 = \frac{1}{t} \ln \frac{a}{a-x}$$

$$\text{or } \ln \frac{a}{a-x} = k_1 t \quad \dots \text{(vi)}$$

The equation (vi) is in the form of $y = mx$. If a graph of $\ln \frac{a}{a-x}$ versus time is plotted, a straight line passing through the origin is obtained.



Equation (vi) can be expressed as,

$$\ln a - \ln(a-x) = k_1 t$$

$$\text{or } \ln(a-x) = -k_1 t + \ln a \quad \dots \text{(vii)}$$

Equation (vii) is in the form of $y = -mx + c$. If a graph of $\ln(a-x)$ versus time is plotted a straight line with negative slop is obtained.

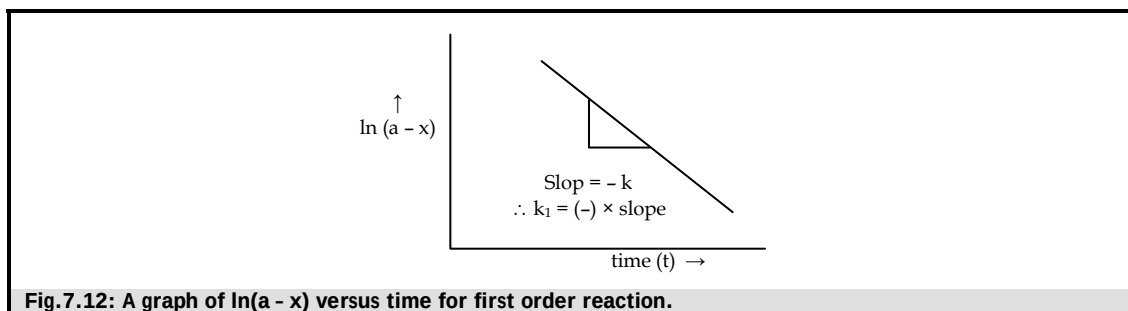


Fig.7.12: A graph of $\ln(a - x)$ versus time for first order reaction.

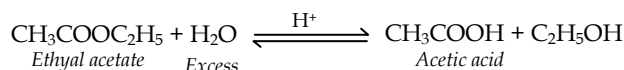
The natures of graphs mentioned above justify the validity of first order reaction.

7.10.3 Pseudo-Order Reaction

The chemical reaction which seems to be of higher order kinetics but actually follows the lower order kinetics when one of the reactants is present in large excess is called pseudo-order reaction or apparent order of reaction.

Let us take some examples of pseudo-order reaction.

i. Hydrolysis of ethyl acetate in acidic medium



The rate of chemical reaction depends upon concentration of both ethyl acetate and water. The rate law equation can be expressed as,

$$\text{Rate} \propto [\text{H}_2\text{O}] [\text{CH}_3\text{COOC}_2\text{H}_5]$$

$$\text{Rate} = k' [\text{H}_2\text{O}] [\text{CH}_3\text{COOC}_2\text{H}_5] \quad \dots \text{(i)}$$

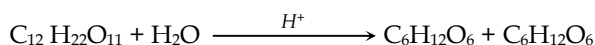
Where k' is proportionality constant and is called rate constant. Since water is present in large excess, its concentration remains constant. Therefore, equation (i) can be expressed as,

$$\text{Rate} = k [\text{CH}_3\text{COOC}_2\text{H}_5] \quad \dots \text{(ii)}$$

$$\text{Where, } k = k' [\text{H}_2\text{O}]$$

Although rate of reaction depends upon the concentration of water as well as ethyl acetate, the rate of reaction is practically determined by the concentration of ethyl acetate. Therefore, although the reaction seems to be second order, it actually follows the first order kinetics. This reaction is also called Pseudo-first order reaction.

ii. Hydrolysis of cane sugar



For the hydrolysis of cane sugar excess of water is used. During chemical reaction, the concentration of water is almost constant. So rate of chemical reaction practically depends upon the concentration of cane sugar. Hence this is an example of Pseudo first order reaction.

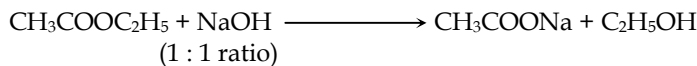
$$\text{Rate} = k [\text{C}_{12}\text{H}_{22}\text{O}_{11}]$$

7.10.4 Second Order Reaction

The chemical reaction in which rate of reaction depends upon two concentration terms of reactant is called second order reaction.

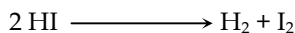
Some examples of second order reactions are given below.

i. **Hydrolysis of ethyl acetate in alkaline medium**



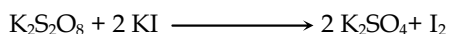
The rate law equation is, $\frac{dx}{dt} = k [\text{CH}_3\text{COOC}_2\text{H}_5] [\text{NaOH}]$

ii. **Dissociation of HI**



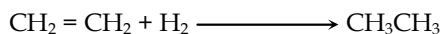
The rate law equation is, $\frac{dx}{dt} = k[\text{HI}]^2$

iii. **Reaction between potassium persulphate and potassium iodide**



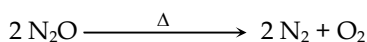
The rate law equation is, $\frac{dx}{dt} = k [\text{K}_2\text{S}_2\text{O}_8] [\text{KI}]$

iv. **Hydrogenation of ethylene**



The rate law equation is, $\frac{dx}{dt} = k[\text{C}_2\text{H}_4] [\text{H}_2]$

v. **Thermal decomposition of N_2O**



The rate law equation is, $\frac{dx}{dt} = k [\text{N}_2\text{O}]^2$

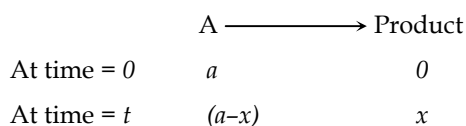
Kinetics of second order reaction

In second order reaction, the rate of reaction depends upon either two concentration terms of the same reactant or each concentration term of two different reactants. If rate of reaction depends upon each concentration term of different reactants with same initial concentration, the kinetics is similar to the rate involving two concentration terms of the same reactant.

Case I: Second order reaction in which rate of reaction depends upon two concentration terms of the same reactant.

Let us take a general example of second order reaction.

Let us consider a general second order reaction in which reactant *A* changes to product. Let *a mol L⁻¹* be the initial concentration of reactant. Then concentration of reactant at time *t* is *(a - x) mol L⁻¹*.



For second order reaction, the rate of reaction depends upon only one concentration term.

$$\text{Rate} \propto [\text{A}]^2$$

or, $\frac{dx}{dt} = (a - x)^2$

$$\text{or, } \frac{dx}{dt} = k_2(a-x)^2 \quad \dots \text{(i)}$$

Here, k_2 is proportionality constant and is called rate constant for second order reaction. Equation (i) is the differential rate law equation for the second order reaction. Now equation (i) can be expressed as

$$\text{or, } \frac{dx}{(a-x)^2} = k_2 dt$$

Integrating the equation, we get

$$\int \frac{dx}{(a-x)^2} = k_2 \int dt$$

$$\text{or, } \int (a-x)^{-2} dx = k_2 \int dt$$

$$\text{or, } \frac{(a-x)^{-2+1}}{(-2+1)(-1)} = k_2 t + C$$

$$\text{or, } \frac{1}{(a-x)} = k_2 t + C \quad \dots \text{(ii)}$$

When $t = 0$, $x = 0$. Now equation (ii) becomes

$$\text{or, } \frac{1}{(a-0)} = k_2 \times 0 + C$$

$$\text{or, } C = \frac{1}{a}$$

Now putting the value of c in equation (ii), we get

$$\text{or, } \frac{1}{(a-x)} - \frac{1}{a} = k_2 t$$

$$\text{or, } \frac{a-(a-x)}{a(a-x)} = k_2 t$$

$$\text{or, } \frac{a-a+x}{a(a-x)} = k_2 t$$

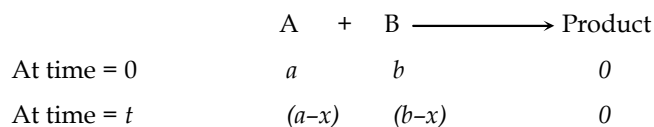
$$\text{or, } \frac{x}{a(a-x)} = k_2 t$$

$$\therefore k_2 = \frac{1}{t} \frac{x}{a(a-x)} \quad \dots \text{(iii)}$$

Equation (iii) is the integrated rate equation for second order reaction in which rate of reaction depends upon two concentration terms of the same reactant.

Case II: Second order reaction in which rate of reaction depends upon two different reactants with different initial concentration.

Let us consider a reaction.



Let, initial concentration of reactants A and B are $a \text{ mol L}^{-1}$ and $b \text{ mol L}^{-1}$ respectively. After certain time, x moles of reactants change into products. The concentrations of reactants A and B at time t are $(a-x) \text{ mol L}^{-1}$ and $(b-x) \text{ mol L}^{-1}$, respectively.

For second order reaction,

$$\text{Rate} \propto [A] [B]$$

$$\text{or, } \frac{dx}{dt} = (a-x)(b-x)$$

$$\text{or, } \frac{dx}{dt} = k_2(a-x)(b-x) \quad \dots \text{(i)}$$

Here, k_2 is proportionality constant and is called rate constant for second order reaction. Equation (i) is the differential rate law equation for the second order reaction in which rate of reaction depends upon concentration terms of two different reactants with different initial concentrations.

The integrated rate law equation for such reactions is,

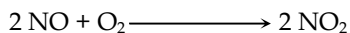
$$k_2 = \frac{1}{t(a-b)} \ln \frac{b(a-x)}{a(b-x)} \quad \dots \text{(ii)}$$

7.10.5 Third Order Reaction

The chemical reaction in which rate of reaction depends upon three concentration terms of reactant is called third order reaction.

Some examples of third order reactions are given below.

i. Combination of NO and O₂



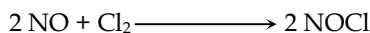
$$\text{The rate law equation is } \frac{dx}{dt} = k[\text{NO}]^2 [\text{O}_2]$$

ii. The combination of NO and Br₂



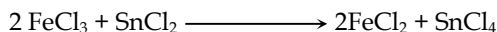
$$\text{The rate law equation is } \frac{dx}{dt} = k[\text{NO}]^2 [\text{Br}_2]$$

iii. Combination of NO and Cl₂



$$\text{The rate law equation is, } \frac{dx}{dt} = k[\text{NO}]^2 [\text{Cl}_2]$$

iv. Reaction between FeCl₃ and SnCl₂



$$\text{The rate law equation is, } \frac{dx}{dt} = k[\text{FeCl}_3]^2 [\text{SnCl}_2]$$

7.11 Methods of Determination of Order of Reaction

Order of reaction can be determined by any of the following methods.

1. Initial concentration measurement method
2. Graphical method
3. Half-life period measurement method
4. Integration method

1. **Initial concentration method:** When concentration of reactant is changed, it may change the rate of reaction. The extent of change in rate of reaction by changing the concentration of reactant is different for different order of reaction. This method is illustrated here.

For a reaction, $A \longrightarrow \text{Product}$

- i. **For zero order reaction,**

$$\text{Rate} = k_0 [A]^0$$

When concentration is increased or decreased, there is no change in rate of reaction.

- ii. **For first order reaction,**

$$\text{Initial Rate} = k_1 [A]$$

$$\text{New rate} = k_1 [nA] = n k_1 A$$

$$\text{New rate} = n \times \text{initial rate}$$

When concentration is increased or decreased by n times the rate of reaction is also increased or decreased by n times correspondingly

- iii. **For second order reaction,**

$$\text{Initial rate} = k_2 [A]^2$$

If the concentration of A is increased or decreased by n times

$$\text{New rate} = k_2 [nA]^2 = n^2 \times k_2 [A]^2 = n^2 \times \text{initial rate}$$

When concentration is increased or decreased by n times the rate of reaction is also increased or decreased by n^2 times

- iv. **For third order reaction,**

$$\text{Initial rate} = k_3 [A]^3$$

If the concentration is increased or decreased by n times

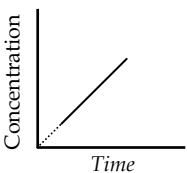
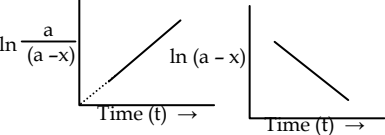
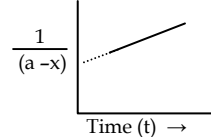
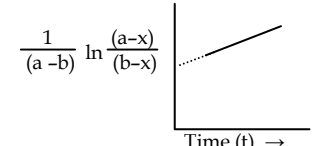
$$\text{New rate} = k_3 [nA]^3 = n^3 \times k_3 [A]^3 = n^3 \times \text{initial rate}$$

When the concentration is increased or decreased by n times the rate of reaction is also increased or decreased by n^3 times.

If concentration is increased by	Rate of reaction is increased by			
	Zero order reaction	1 st order reaction	2 nd order reaction	3 rd order reaction
2 times	$2^0 = 1$	$2^1 = 2$	$2^2 = 4$	$2^3 = 8$
3 times	$3^0 = 1$	$3^1 = 3$	$3^2 = 9$	$3^3 = 27$
4 times	$4^0 = 1$	$4^1 = 4$	$4^2 = 16$	$4^3 = 64$

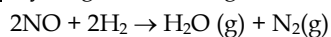
2. **Graphical method:** Different nature of graphs verify the order of reactions. These are described already.

3. **Half-life period method:** To determine the order of reaction, the co-relations between initial concentration of reactant and half life period are studied. For example, half-life period is directly proportional to initial concentration of reactant for zero-order reaction. Half-life period is independent to initial concentration of reactant for first order reaction. For n^{th} order of reaction, $t_{1/2} \propto a^{1-n}$. If two reactions are progress with different initial concentration a_1 and a_2 with respective half-lives $(t_{1/2})_1$ and $(t_{1/2})_2$ then $\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{a_1}{a_2}\right)^{1-n}$.
4. **Integration method:** In this method, the value of rate constant is calculated by using integrated rate equation. For different experimental data carried out under similar conditions, the value of K is found to be almost same. The rate constants for different order of reactions.

Order or reaction	Differential rate equation	Integrated rate equation	Half life period	Kinetic plot
0	$\frac{dx}{dt} = k_0$	$k_0 = \frac{x}{t}$	$t_{1/2} = \frac{a}{2k_0}$	
1	$\frac{dx}{dt} = k_1 (a - x)$	$K_1 = \frac{1}{t} \ln \frac{a}{(a-x)}$	$t_{1/2} = \frac{0.693}{k_1}$	
2	$\frac{dx}{dt} = k_2 (a - x)^2$	$K_2 = \frac{1}{t} \frac{x}{a(a-x)}$	$t_{1/2} = \frac{1}{ak_2}$	
	$\frac{dx}{dt} = k_2(a-x)(b-x)$	$K_2 = \frac{1}{t(a-b)} \ln \frac{b(a-x)}{a(b-x)}$	Different initial concentration corresponds to different half life period	

Some Solved Numerical Problems

Example 5.1: Hydrogen and nitrogen oxide react according to the following equation.



Experiments were performed at 800°C in order to determine the order of reaction and the following results were obtained.

	Initial concentration of nitrogen oxide (mole litre ⁻¹)	Initial concentration of hydrogen (mole litre)	Initial rate of production of nitrogen (mole litre ⁻¹ sec ⁻¹)
i.	6×10^{-3}	1×10^{-3}	3×10^{-3}
ii.	6×10^{-3}	2×10^{-3}	6×10^{-3}
iii.	6×10^{-3}	3×10^{-3}	9×10^{-3}
iv.	1×10^{-3}	6×10^{-3}	0.5×10^{-3}
v.	2×10^{-3}	6×10^{-3}	2.0×10^{-3}
vi.	3×10^{-3}	6×10^{-3}	4.5×10^{-3}

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- What is the order of this reaction with respect to (i) nitrogen oxide (ii) hydrogen?
- Write the rate equation for the reaction of nitrogen oxide with hydrogen.
- What is the unit of rate constant, K?
- Why are chemists interested in obtaining order of reaction and rate equations?

Solution

Let rate law equation be

$$\text{Rate} = K[\text{NO}]^x [\text{H}_2]^y \quad \dots (i)$$

- (a) (i) To find the order of reaction with respect to nitrogen oxide.

Let us take the case where concentration of hydrogen is constant.

$$\text{Rate} = K[\text{NO}]^x$$

From (iv)

$$0.5 \times 10^{-3} = K (1 \times 10^{-3})^x$$

$$\therefore K = \frac{0.5 \times 10^{-3}}{(1 \times 10^{-3})^x} \quad \dots (1)$$

From (v)

$$2 \times 10^{-3} = K (2 \times 10^{-3})^x$$

$$\therefore K = \frac{2 \times 10^{-3}}{(2 \times 10^{-3})^x} \quad \dots (2)$$

From equation (1) and (2),

$$\frac{0.5 \times 10^{-3}}{(1 \times 10^{-3})^x} = \frac{2 \times 10^{-3}}{(2 \times 10^{-3})^x}$$

$$\left(\frac{2 \times 10^{-3}}{(2 \times 10^{-3})^x} \right)^x = \frac{2 \times 10^{-3}}{(2 \times 10^{-3})^x}$$

$$2^x = 4$$

$$2^x = 2^2$$

$$x = 2$$

$\therefore$ Order of reaction with respect to NO is 2.

- (ii) To find the order of reaction with respect to hydrogen. Let us take the cases where concentration of NO is constant.

$$\text{Rate} = K[\text{H}_2]^y$$

From (i)

$$3 \times 10^{-3} = K (1 \times 10^{-3})^y$$

$$\therefore K = \frac{3 \times 10^{-3}}{(1 \times 10^{-3})^y} \quad \dots (1)$$

From (ii)

$$6 \times 10^{-3} = K (2 \times 10^{-3})^y$$

$$\therefore K = \frac{6 \times 10^{-3}}{(2 \times 10^{-3})^y} \quad \dots (2)$$

From equation (1) and (2),

$$\frac{3 \times 10^{-3}}{(1 \times 10^{-3})^y} = \frac{6 \times 10^{-3}}{(2 \times 10^{-3})^y}$$

$$\left(\frac{2 \times 10^{-3}}{1 \times 10^{-3}} \right)^y = \frac{6 \times 10^{-3}}{3 \times 10^{-3}}$$

$$2^y = 2$$

$$2^y = (2)^1$$

$\therefore y = 1$.

Hence, the reaction is 1st order with respect to H₂.

- (b) Rate equation from equation (i) is
 $\text{Rate} = K [\text{NO}]^x [\text{H}_2]^y$
 Putting the value of x and y
 $\text{Rate} = K [\text{NO}]^2 [\text{H}_2]$
- (c) Unit of rate constant K is $\text{mole}^{-2} \text{lit}^2 \text{sec}^{-1}$
- (d) Rate equation shows the actual dependence of concentration on the rate of equation. But actual dependence of concentration can't be found out by balanced chemical equation. The order of chemical reaction shows the actual dependence of concentration to the rate equations. The relation of initial concentration and half life also can be found out from order of reaction. Therefore, chemists are interested in obtaining order of reaction and rate equation.

Example 5.2: For a hypothetical reaction, $2\text{M} + \text{N} \rightarrow \text{product}$

The following data are given.

Expt. No.	Initial conc of M (mol L^{-1})	Initial conc of N (mol L^{-1})	Initial rate ($\text{mol L}^{-1} \text{sec}^{-1}$)
I	0.10	0.20	3×10^2
II	0.30	0.40	3.6×10^3
III	0.30	0.80	1.44×10^4
IV	0.30	1.60	A
V	0.60	0.80	B
VI	0.10	0.40	C

From the above data:

- Find the overall order of the reaction.
- Calculate the value of A, B and C.

Solution

Reaction, $2\text{M} + \text{N} \longrightarrow \text{Products}$

Let the rate law equation be $\frac{dx}{dt} = K[\text{M}]^x [\text{N}]^y$

From expt. no. I, $3 \times 10^2 = K (0.10)^x \cdot (0.20)^y$... (1)

From expt. no. II, $3.6 \times 10^3 = K (0.30)^x \cdot (0.40)^y$... (2)

From expt. no. III, $1.44 \times 10^4 = K (0.30)^x \cdot (0.80)^y$... (3)

Now, equation (3) is divided by equation (2)

$$\frac{1.44 \times 10^4}{3.6 \times 10^3} = \frac{K (0.30)^x \cdot (0.80)^y}{K (0.30)^x \cdot (0.40)^y}$$

$$\text{or } 4 = \left(\frac{0.80}{0.40}\right)^y$$

$$\text{or } 4 = 2^y$$

$$\text{or } 2^2 = 2^y$$

$$\therefore y = 2$$

Now equation (2) is divided by equation (1), we get

$$\frac{3.6 \times 10^3}{3 \times 10^2} = \frac{K(0.30)^x \cdot (0.40)^y}{K(0.10)^x \cdot (0.20)^y}$$

$$12 = \left(\frac{0.30}{0.10}\right)^x \left(\frac{0.40}{0.20}\right)^x$$

$$12 = 3^x \cdot 2^y$$

$$\text{or } 12 = 3^x \cdot 2^2$$

$$\text{or } 12 = 3^x \cdot 4$$

$$\text{or } 3^x = 3$$

$$\therefore x = 1$$

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Now, putting the value of x and y in rate law equation $\frac{dx}{dt} = K[M]^1 [N]^2$

- i. The order of reaction w.r to M = 1

The order of reaction w.r to N = 2

Overall order of reaction = 1 + 2 = 3

- ii. Calculation of value of A, B and C.

Calculating the value of K,

From experiment no (1)

$$3 \times 10^2 = K (0.10)^1 \cdot (0.20)^2$$

$$K = \frac{3 \times 10^2}{(0.10)(0.20)^2} = 75000 = 7.5 \times 10^4 \text{ mol}^{-2} \text{ lit}^2 \text{ sec}^{-1}$$

From experiment no (iv)

$$\frac{dx}{dt} = K(0.30)^1 (1.60)^2$$

$$= 7.5 \times 10^4 \times 0.30 \times (1.60)^2$$

$$A = 5.76 \times 10^4 \text{ mol lit}^{-1} \text{ sec}^{-1}$$

From experiment no (v)

$$\frac{dx}{dt} = K(0.60) (0.80)^2$$

$$= 7.5 \times 10^4 \times 0.60 \times (0.80)^2$$

$$B = 2.88 \times 10^4 \text{ mol lit}^{-1} \text{ sec}^{-1}$$

From experiment no. (vi)

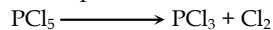
$$\frac{dx}{dt} = K(0.10)^1 (0.40)^2$$

$$C = 7.5 \times 10^4 \times 0.10 \times (0.40)^2 = 1.2 \times 10^3 \text{ mol. lit}^{-1} \text{ sec}^{-1}$$

■ Solved Questions

1. Give a chemical reaction which follows the first order kinetics.

- ↪ Decomposition of PCl_5 into PCl_3 and Cl_2 is an example of 1st order chemical reaction.



It is first order reaction because the rate of chemical reaction depends upon only one concentration of PCl_5 . It can be written as

$$\text{Rate} \propto [\text{PCl}_5]^1$$

$$\frac{dx}{dt} = K_1 [\text{PCl}_5]$$

2. What is activation energy?

- ↪ The minimum amount of energy **that must be supplied** to the reactant to cross the energy barrier so as to give the product is called activation energy.

3. If 75% of a first order reaction is completed in 32 minutes, when will be 50% of the reaction completed?

- ↪ Integrated rate equation for first order reaction is

$$t = \frac{2.303}{K} \log \frac{a}{a-x}$$

Using the equation for both situations, we get

$$32 = \frac{2.303}{K} \log \frac{a}{a - \frac{75}{100}a}$$

$$\text{or, } 32 = \frac{2.303}{K} \log 4$$

... (i)

$$\text{and } t = \frac{2.303}{K} \log \frac{a}{a - \frac{50a}{100}}$$

$$\text{or, } t = \frac{2.303}{K} \log 2 \quad \dots \text{ (ii)}$$

Dividing equation (i) by (ii), we get

$$\frac{32}{t} = \frac{\log 4}{\log 2}$$

$$\therefore t = 16 \text{ minutes}$$

4. A first order reaction has half-life period of 69.3 seconds. What will be the rate of reaction when the concentration of the reactant is 0.1 mol/L?

➤ For first order reaction:

$$K = \frac{0.693}{t_{1/2}}$$

$$\text{So, } K = \frac{0.693}{69.3} \text{ sec}^{-1} = 10^{-2} \text{ sec}^{-1}.$$

Putting the value of K in first order rate equation, Rate = K[A]

$$\text{Rate} = 10^{-2} \times 0.1 = 10^{-3} \text{ Mol lit}^{-1} \text{ sec}^{-1}.$$

5. The reaction $X + Y \longrightarrow$ products is a second order reaction. Write three different rate law expressions which may be true to the above reaction.

➤ Order of reaction is the sum of the powers in the rate law equation. To be, $x + y \longrightarrow$ product, a second order reaction the following expressions can be written which may be true.

i. Rate = $k_2 [x]^2 [Y]^0$ order of reaction = $2 + 0 = 2$

ii. Rate = $k_2 [x]^0 [Y]^2$ order of reaction = $0 + 2 = 2$

iii. Rate = $k_2 [x] [Y]$ order of reaction = $1 + 1 = 2$.

6. What is meant by order of a reaction? State the order of reaction having rate constant $2 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$.

➤ The sum of the power of concentration in the rate law equation is called order of chemical reaction.

$$\text{Given rate constant} = 2 \times 10^{-2} \text{ mol lit}^{-1} \text{ sec}^{-1}$$

$$\text{Unit of K for } n^{\text{th}} \text{ order reaction} = (\text{mol})^{1-n} \text{ lit}^{n-1} \text{ time}^{-1}.$$

$$\text{or } (\text{mol})^{1-n} \text{ lit}^{n-1} \text{ time}^{-1} = \text{mol lit}^{-1} \text{ sec}^{-1}$$

$$\text{or } 1 - n = 1.$$

$$\therefore n = 0.$$

Hence the given reaction is of zero order reaction.

7. Define the half life of a reaction.

➤ Half-life period is the time required for reactant to change into half of its initial amount. It is denoted by $t_{1/2}$



Questions

Very Short Answer Questions:

- Define the following terms:
 - Rate of reaction
 - Average rate of reaction
 - Instantaneous rate of reaction
 - Equivalence rate of reaction
 - Effective collision
 - Proper orientation
 - Rate law equation
 - Pseudo-order reaction
 - First order reaction
 - Half-life period
 - Activation energy
 - Threshold energy
- Draw the energy profile diagram to show the effect of catalyst on the rate of reaction. Justify.
- What is the order of reaction whose rate constant has same unit as the rate of reaction?

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- Define the half-life period of a reaction. The half-life periods of two reactions A and B are 3.21×10^2 min. and 569 min respectively. Which of these undergoes nuclear reaction faster? is a faster reaction?
- Write the difference between molecularity and order of reaction.
- Identify the order of reaction, if unit of rate constant are (a) min^{-1} (b) mol L^{-1} .
- If the rate of reaction is doubled when concentration of A is doubled but there is no effect in rate with change in concentration of B. (a) Write rate law equation (b) Find out the unit of rate constant.
- Write all possible rate law expression for the reaction:
 $M + N \longrightarrow \text{Products}$, if the order of the reaction is 3.
- What are the effects of collision frequency, energy of activation and orientation factor in the rate of reaction?
- Write the rate expression for the reaction:
 $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$
- Distinguish rate and rate constant of a reaction.

■ Short /Long Answer Questions:

- Explain the factors which affect the rate of reaction.
- Show your acquaintance with collision theory of reaction.
- What do you mean by rate of a reaction? How do the following factor affect rate of reactions:
(a) Surface area (b) temperature (c) catalyst.
- Write short notes on (a) Half life period (b) First order reaction (c) Collision theory of reactions

■ Numerical Problems

- The rate of reaction for, $A \rightarrow B$ increases by two times when its concentration is increased by 4 times. What is rate law? (Rate = $[A]^{1/2}$).
Ans: Rate = $[A]^{1/2}$
- The rate constant of a reaction is $2.4 \times 10^{-4} \text{ L mole}^{-1}\text{s}^{-1}$. What is the order of reaction?
Ans: 2nd
- Find the three-fourth life of a first order reaction with rate constant $7.4 \times 10^{-5} \text{ s}^{-1}$.
Ans: $1.87 \times 10^4 \text{ s}$
- Calculate the half-life period of a first order reaction when the rate constant is 5 year^{-1} .
Ans: 0.1386 years.
- Suppose that the rate law for the reaction $A \rightarrow B$ has been found to be of the form Rate = $k[A]^m$.
From the following data, determine the overall order of the reaction and the order with respect to A.

Initial Concentration of A (M)	Initial Rate (M/Sec)
0.05	3×10^{-5}
0.10	12×10^{-5}
0.20	48×10^{-5}

Ans: Order of reaction with respect to A is 2 and the over all order of the reaction is also 2

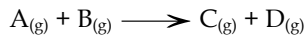
- For the reaction $\text{Cl}_2(\text{g}) + 2\text{NO}(\text{g}) = 2\text{NOCl}(\text{g})$, the obtained data was

Initial concentration of Cl_2 (mole litre ⁻¹)	Initial concentration of NO (mole litre ⁻¹)	Initial rate of production of NOCl (mole litre ⁻¹ sec ⁻¹)
0.02	0.01	2.4×10^{-4}
0.02	0.03	2.1×10^{-3}
0.04	0.03	2.32×10^{-3}

What is order of reaction with respect to (i) Cl_2 and (ii) NO , (iii) rate law and rate constant?

Ans: (i) 2 (ii) 1 (iii) 1.2×10^{-2}

7. For the gaseous reaction,



It is found that, rate = $K[\text{A}]^2 [\text{B}]^1$

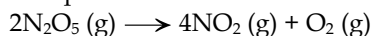
How many times does the rate of reaction increase or decrease if (a) the partial pressures of both A and B are doubled, (b) the partial pressure of A doubles but that of B remains constant. (c) the volume of reacting vessel is doubled (d) an inert gas is added which doubles the overall pressure while the partial pressure of A and B remains constant. (e) the temperature rises by 30°C .

Ans: (a) Rate increases by 8 folds (b) rate increases by 4 folds (c) rate decreases by 4 folds (d) no change in the rate of reaction (e) rate increases by 8 to 27 folds.

8. The half life period of first order reaction is 3 hours. Find the time required to complete 87.5% of the reaction.

Ans: $t = 9.002$ hours

9. Compare the rate of reaction of all the components of the following reaction:



Find the rate of each component in mol s^{-1} , when 2.24 litre of O_2 at NTP are produced in 30 minutes.

Ans: Rate of disappearance of $\text{N}_2\text{O}_5 = 1.11 \times 10^{-4} \text{ mol s}^{-1}$

Rate of formation of $\text{NO}_2 = 2.22 \times 10^{-4} \text{ mol s}^{-1}$ Rate of formation of $\text{O}_2 = 5.55 \times 10^{-5} \text{ mol s}^{-1}$

10. For a hypothetical reaction

$2\text{M} + \text{N} \rightarrow \text{product}$; The following data are given.

Expt. No.	Initial conc of M (mol L^{-1})	Initial conc of N (mol L^{-1})	Initial rate ($\text{mol L}^{-1} \text{sec}^{-1}$)
I	0.10	0.20	3×10^2
II	0.30	0.40	3.6×10^3
III	0.30	0.80	1.44×10^4
IV	0.30	1.60	A
V	0.60	0.80	B
VI	0.10	0.40	C

From the above data:

- Find the overall order of the reaction.
 - Calculate the value of A, B and C.
 - Determine the unit of rate constant
11. For a first order reaction, it takes 4 minute for initial concentration of 0.8 mol/L to become 0.2 mol/L. What time will it take to become the concentration to be 0.04 mol/L?
12. A first order reaction takes 100 minutes to complete 60% of the reactants into products. What time will take to complete 90% of the reactant into product?

Ans: 8.64 min

Ans: 251.31 min

Multiple Choice Questions

- A catalyst increases the rate of a chemical reaction by
 - decreasing the free energy
 - decreasing the heat of reaction
 - decreasing the activation energy
 - increasing the average speed of the reacting molecules.
- A catalyst speed up a chemical reaction by
 - increasing the activation energy
 - providing an alternative reaction path of lower activation energy
 - shifting the equilibrium towards the side the product (s).
 - increasing the reaction enthalpy
- In the reaction, $A + B \longrightarrow C + D$

D[A]	d[B]	Rate constant
0.1	0.1	1.5×10^{-3}
0.1	0.2	1.5×10^{-3}
0.2	0.2	3×10^{-3}

The order of reaction is

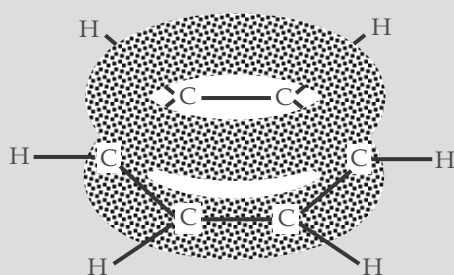
 - 1st order with respect to A
 - 2nd order with respect to A
 - 1st order with respect to B
 - 1st order with respect to A and B both
- The average minimum energy that must be associated with the reactant molecules in order to be capable of forming products:
 - activation energy
 - collision energy
 - threshold energy
 - transition energy
- The rate of a reaction is $5.4 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$ then order of reaction is
 - zero
 - one
 - two
 - Three
- The rate constant of a chemical reaction is changed if,
 - the reaction temperature is changed
 - the reaction particles size is changed
 - the initial concentration of reactant is changed
 - the product concentration is increased
- The order of the reaction is
 - Sum of stoichiometric coefficient of all reactants
 - Sum of stoichiometric coefficient of all reactants on which rate depends
 - Sum of power of concentration of the reactants on which rate depends
 - Sum of power of concentration of all the reactant
- The rate of reaction depends upon
 - initial concentration of reactants
 - temperature
 - pressure
 - volume
- The factor which does not influence the rate of reaction is
 - natural of reactants
 - concentration
 - temperature
 - molecularity
- 75% of a first order reaction was completed in 32 min, when was 50% of the reaction completed?
 - 24 min
 - 16 min
 - 8 min
 - 4 min

Answer Key

1	2	3	4	5	6	7	8	9	10
c	b	a	c	a	a	c	b	d	b

Aromatic Hydrocarbons

8



Benzene molecule: π electron clouds above and below the plane of the ring.

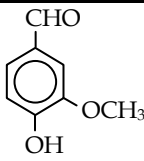
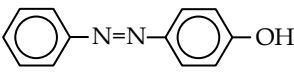
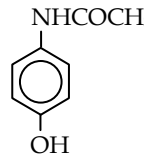
Benzene is the simplest and the important member of aromatic hydrocarbons (arenes). It is mainly used as a precursor to other industrially important compounds like polystyrene, nylon, resins, adhesives, rubbers, lubricants, dyes, detergents, drugs, explosives and pesticides.

8.1 Introduction

The term **aromatic** is derived from Greek word *aroma* meaning pleasant odour or fragrance. Originally, the term aromatic was used to represent a group of organic compounds obtained from natural sources which had pleasant odour and the term aliphatic is used to represent to distinguish them from aliphatic compounds usually lack such odour (Greek: *aliphos*= fat).

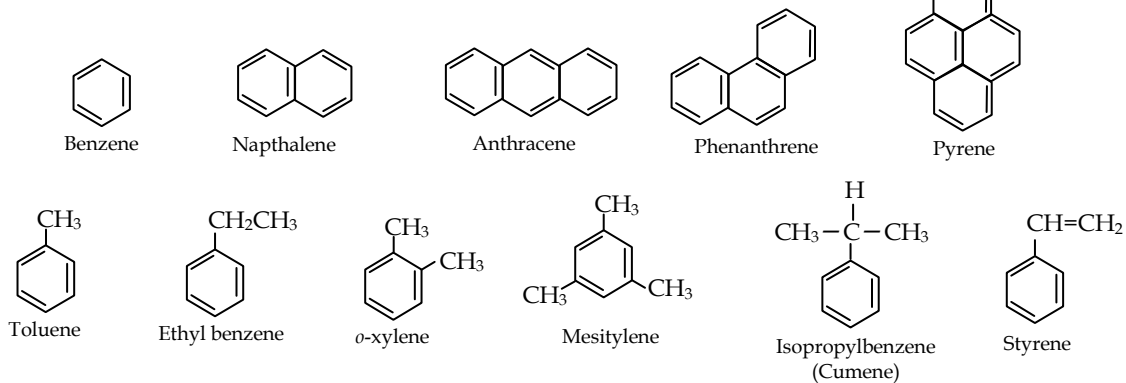
Later, it was found that almost such compounds contained benzene or derivatives of benzene. Hence **aromatic compounds are benzene and compounds that resemble benzene in chemical behaviour.**

Today's classification is, however, based on molecular structure and chemical behavior of compounds rather than in the sense of aroma. Aliphatic compounds represent all open chain compounds and those cyclic compounds which are similar to open chain compounds in chemical behavior. Aromatic compounds represent only cyclic organic compounds which are planar with cyclic cloud of pi electrons and quite stable towards addition reaction though unsaturated.

	 Vanillin	It is a popular flavoring agent in foods, beverages, and pharmaceuticals which is obtained naturally from seeds of vanilla (an orchid).
	 <i>p</i> -Hydroxyazobenzene (an azo dye)	It is used for coloring textiles, foodstuffs, etc.
	 Paracetamol	It is used to lower the body temperature in high fever (antipyretic).

Aromatic compounds play key roles in biochemistry of all living beings. They serve as basic building-blocks of proteins. All five bases in DNA and RNA are aromatic compounds. Aromatic compounds have high commercial value since they are indispensable part of our society providing drugs, dyes pesticides, polymers, etc.

Aromatic hydrocarbons are called **arenes**. Arenes have general formula C_nH_{2n-6y} , where, n = number of carbons and y = number of rings. Some examples are;



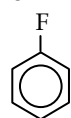
8.2 Characteristics of Aromatic Compounds

1. They are cyclic organic compounds having planar ring (usually having 5 or 6 carbon atoms per molecule) and cyclic cloud of pi electrons.
2. Although unsaturated, they do not give addition reactions easily. They show apparent unsaturation.
3. They give electrophilic substitution reactions regardless of functional group.
4. They have high proportion of carbon as compared to its aliphatic analog. Hence, they burn with sooty flame. Aliphatic compounds burn with non sooty flame.
5. They follow Huckel's $4n + 2$ rule.

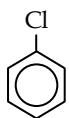
8.3 Nomenclature of Benzene Derivatives

Nomenclature of monosubstituted derivatives of benzene

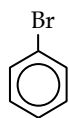
Many monosubstituted benzene are named by taking name of substituent as prefix of the word - benzene



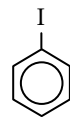
Fluorobenzene



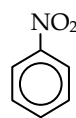
Chlorobenzene



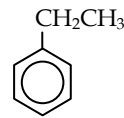
Bromobenzene



Iodobenzene

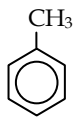
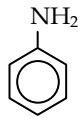
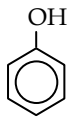


Nitrobenzene

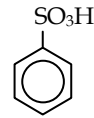


Ethylbenzene

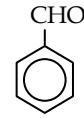
Some have special name without any significance of substituent in name.

Toluene
(Methylbenzene)Aniline
(Aminobenzene)Phenol
(Hydroxybenzene)

Benzoic acid



Benzenesulfonic acid

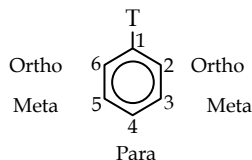


Benzaldehyde

Nomenclature of di-substituted derivatives of benzene: Concept of *ortho*, *meta* and *para*

If two substituents are present, three positional isomers are possible. The relative positions of substituents in disubstituted benzene are given by *ortho* (*o*-), *meta* (*m*-) and *para* (*p*-) or by numbers.

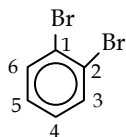
Taking the position of substituent as 1, other positions are given:



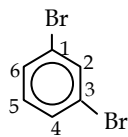
1. Positions 2 and 6 are equivalent and are called *ortho* or *o*- with respect to position 1.
2. Positions 3 and 5 are equivalent and are called *meta* or *m*- with respect to position 1.

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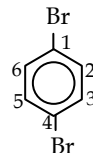
3. Position 4 is called *para* or *p*- with respect to position 1.



1,2-Dibromobenzene
(*o*-Dibromobenzene)

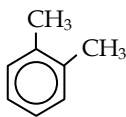


1,3-Dibromobenzene
(*m*-Dibromobenzene)

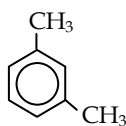


1,4-Dibromobenzene
(*p*-Dibromobenzene)

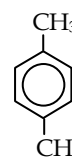
Dimethyl benzene has special name 'xylene'.



1, 2-Dimethylbenzene
(*o*-xylene)

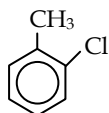


1, 3-Dimethylbenzene
(*m*-xylene)

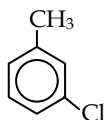


1, 4-Dimethylbenzene
(*p*-xylene)

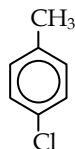
If two substituents are different and one of them gives special name to the molecule, the compound is named as derivative of that special name.



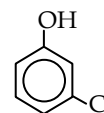
2-Chlorotoluene
(*o*-Chlorotoluene)



3-Chlorotoluene
(*m*-Chlorotoluene)

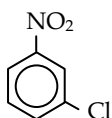


4-Chlorotoluene
(*p*-Chlorotoluene)

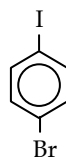


m-Chlorophenol

If two substituents are different and none of them gives special name to the molecule, then substituents are written in alphabetical order.

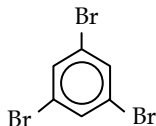


m-Chloronitrobenzene

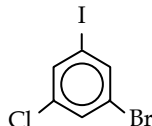


1-Bromo-4-iodobenzene
(*p*-Bromoiodobenzene)

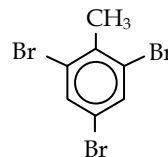
Nomenclature of polysubstituted derivatives of benzene



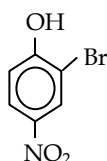
1,3,5-Tribromobenzene



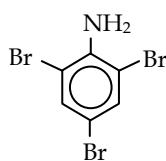
1-Bromo-3-chloro-5-iodobenzene



2,4,6-Tribromotoluene

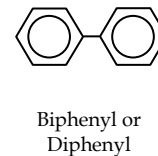
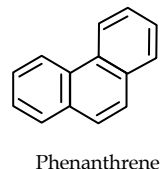
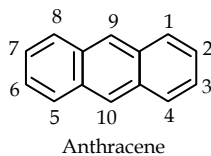
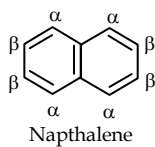
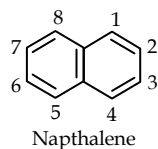


1-Bromo-4-nitrophenol

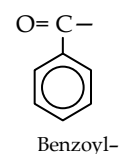
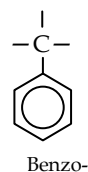
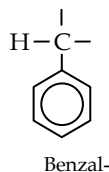
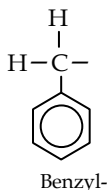
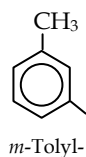
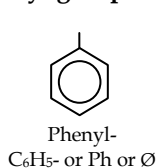


2,4,6-Tribromoaniline

Some polynuclear hydrocarbons



Some aryl groups



8.4 Structure of Benzene

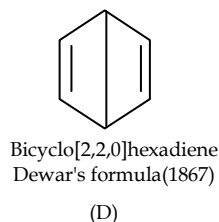
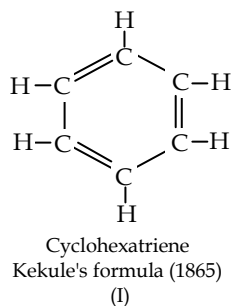
Since aromatic compounds resemble benzene, understanding of structure of benzene is important for the study of aromatic compounds.

Benzene (phene) was first isolated by Faraday in 1825 from cylinders of compressed illuminating gas obtained from pyrolysis of whale oil. Hofmann found benzene in coal tar in 1845.

1. **Elemental analysis and molecular weight determination indicate that molecular formula of benzene is C_6H_6 .** It has eight hydrogen atoms less than its saturated analog hexane, C_6H_{14} . This indicates that benzene is highly unsaturated. A number of structures can be written for this formula like:

- a. $\text{CH}_3-\text{C}\equiv\text{C}-\text{C}\equiv\text{C}-\text{CH}_3$
- b. $\text{CH}_2=\text{CH}-\text{C}\equiv\text{C}-\text{CH}=\text{CH}_2$
- c. $\text{CH}\equiv\text{C}-\text{CH}_2-\text{CH}_2-\text{C}\equiv\text{CH}$
- d. = CH_2

Kekule (1865) was the first to suggest a ring structure for benzene (I) but did not prove. Dewar proposed bicyclic structure (D).

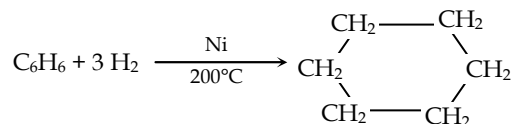


2. Evidences in support of Kekule's structure of benzene

- i. **Benzene gives only one monosubstituted product.** This indicates that all six hydrogen atoms are equivalent. This permits the structure (I).

- ii. **Benzene adds a maximum of six hydrogen atoms on heating in presence of catalyst to give hexahydrobenzene.**

Bayer proved that cyclohexane and hexahydrobenzene were identical hence establishing the cyclic structure of benzene.



- iii. Benzene adds a maximum of six chlorine atoms in presence of sunlight.

3. Objections to Kekule's structure of benzene

- i. **Benzene is quite stable towards addition reaction.** If structure I is correct, it should give addition reaction readily.

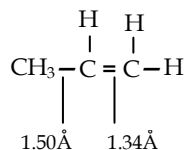
Benzene + Bayer's reagent = No reaction

Benzene + Br_2/CCl_4 = No reaction

Benzene + HX = No reaction

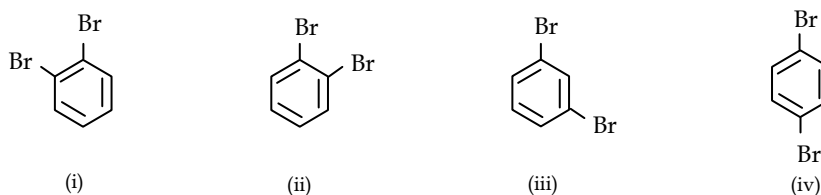
- ii. **All carbon-carbon bonds in benzene are equal and are intermediate in length between single and double bonds.**

Carbon -carbon single bond is longer ($1.50\text{\AA}$ in propene) than the carbon- carbon double bond ($1.34\text{\AA}$).

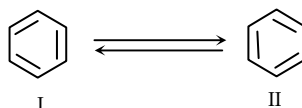


If Kekule's structure is correct, there should have three long bonds and three short bonds in benzene. X-ray diffraction studies show that all six carbon-carbon bonds in benzene are equal and have a length of $1.39\text{\AA}$ which is intermediate in length between single and double bonds.

- iii. **Benzene gives only three disubstituted products.** Theoretically, structure (I) should give four isomeric di-substituted derivatives:



Kekule, later (1872) proposed another structure (II) for benzene and suggested that there is rapid equilibrium between structure (I) and (II), each molecule spending half of its time in (I) and the other half in (II) hence, cannot be separated. Therefore, two 1, 2-disubstituted benzenes cannot be separated.

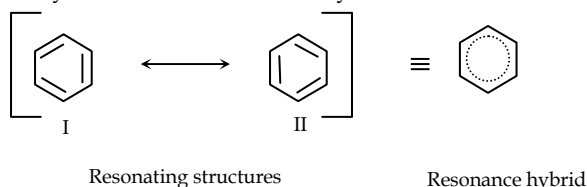


Hence, (I) and (II) cannot be distinguished.

- iv. Experimental heat of hydrogenation of benzene is found to be 36 kcal mol^{-1} less than calculated heat of hydrogenation of hypothetical cyclohexatriene ($85.5 \text{ kcal mol}^{-1}$). This indicates that benzene is 36 kcal mol^{-1} more stable than hypothetical cyclohexatriene. Such energy of stability of benzene by 36 kcal mol^{-1} is known as resonance energy.

4. Resonance theory

According to this theory, benzene is resonance hybrid of I and II.



The resonance hybrid explains the following features of benzene;

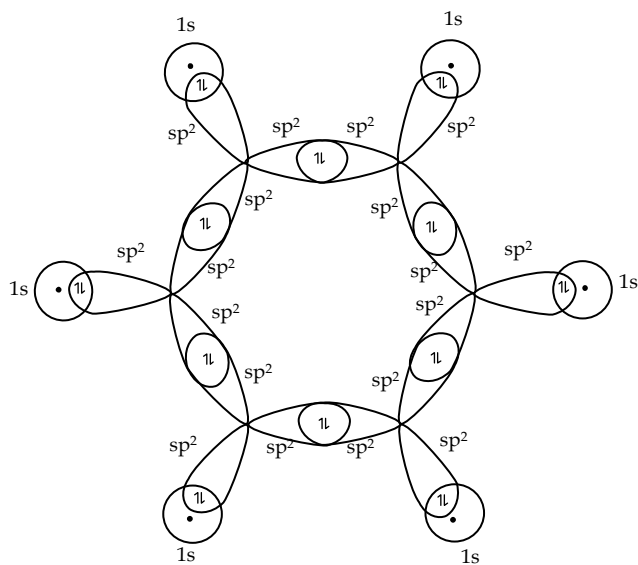
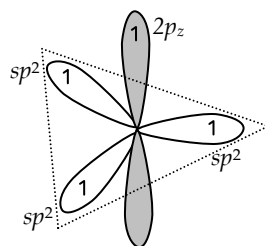
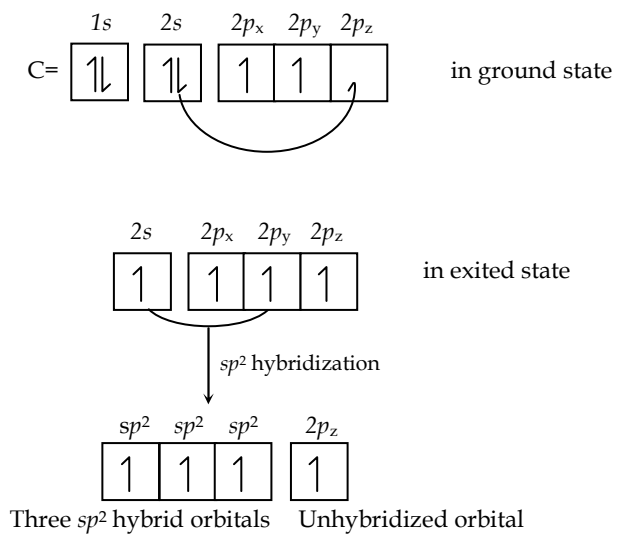
- i. A resonance hybrid is more stable than the resonating structures. Since, I and II are exactly equivalent, stabilization due to resonance is high. Hence, *benzene is quite stable towards addition reactions.*
- ii. Since I and II are exactly equivalent, each structure has equal contribution in resonance hybrid. Hence, in benzene, all carbon-carbon bonds are identical and are intermediate in length between single and double bonds.

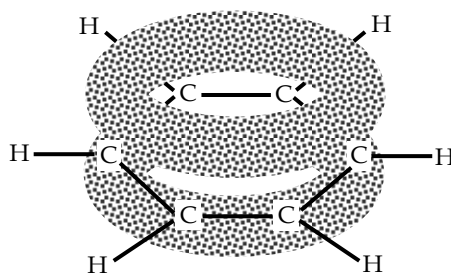
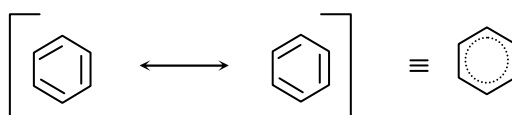
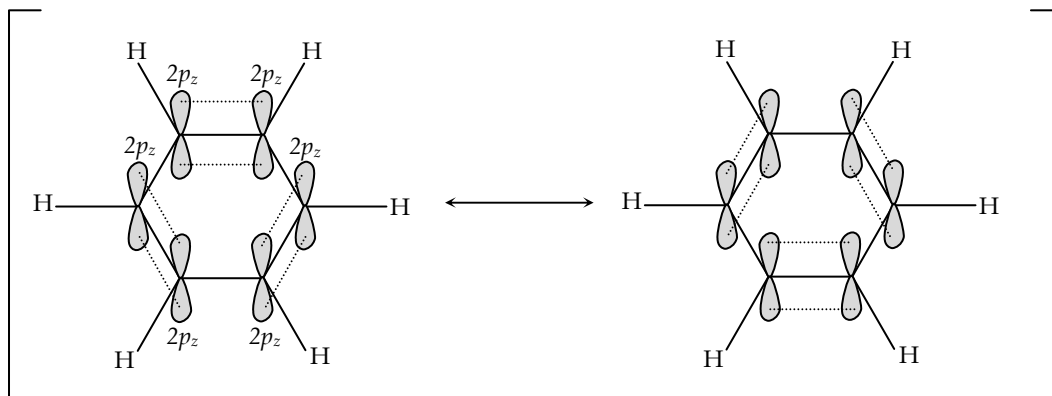
5. Orbital picture of benzene

Carbon has two half filled orbitals in ground state ($1s^2, 2s^2, 2p_x^1 2p_y^1, 2p_z^0$). It has four half filled orbitals in excited state since one electron is promoted from $2s$ to empty $2p$ orbital during chemical reaction ($1s^2, 2s^1, 2p_x^1 2p_y^1, 2p_z^1$).

Now, one $2s$ and two $2p$ orbitals undergo sp^2 hybridization to form three sp^2 hybrid orbitals. Three sp^2 orbitals are of equivalent energy and identical shape and size which take trigonal planar geometry with angle 120° . Unhybridized $2p_z$ half filled orbital remains perpendicular to the plane of hybrid orbitals.

Unhybridized half filled $2p_z$ orbitals of each carbon atom which are perpendicular to the plane of the ring, can overlap laterally on either side forming three pi bonds. But, the system is cyclic and planar, hence pi electrons are not localized but delocalized forming cyclic pi electron clouds above and below the plane of the molecule. This delocalization stabilizes the molecule. Hence **benzene is a flat molecule (all carbon and hydrogen atoms lie in a plane with bond angle 120°) with delocalized cyclic pi electrons above and below the plane of the molecule.**





Benzene molecule: π electron clouds above and below the plane of the ring.

8.5 Huckel's ($4n+2$) Rule for Aromaticity (Theoretical Criteria for Aromatic Character)

The term aromaticity is related with chemical stability due to resonance. All aromatic compounds show aromaticity. But some other classes of compounds like heterocyclic compounds also show aromatic behaviour (aromaticity).

Theoretically, to be aromatic an organic compound must fulfill the following conditions;

1. It must be cyclic and planar.
2. It must have cyclic cloud of delocalized pi electrons above and below the plane of the molecule.
3. π - electron cloud must contain a total of $(4n+2)$ π - electrons, where n is an integer ($n=0,1,2,3..$)

Hence aromatic compounds are cyclic planar molecules having cyclic cloud of 2 ($n=0$), 6 ($n=1$), 10 ($n=2$), 14 ($n=3$), 18 ($n=4$), π - electrons. This is called Huckel rule or $4n+2$ rule for aromaticity.

The Huckel number six is most common in aromatic system and hence the term **aromatic sextet** is commonly used for 6π electrons.

Example 8.1 Predict the aromaticity of the following species applying Huckel's rule.

- a. Benzene b. Cycloheptatriene c. Napthalene
d. Cyclopropenyl cation e. Cycloheptatrienyl cation

Solution

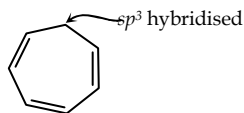
- a. Benzene



- The compound is cyclic and planar since all six ring carbon atoms are sp^2 hybridized.
- π electrons = $6: 4n + 2 = 6$, where $n = 1$

Planar structure permits the delocalization of pi electrons in a ring. Hence it is aromatic.

b. Cycloheptatriene

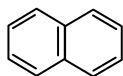


The compound is cyclic but nonplanar since one ring carbon atom is sp^3 hybridized.

π electrons = 6 : $4n + 2 = 6$, where $n = 1$

Nonplanar structure does not permit delocalization of π electrons in a ring. Hence it is not aromatic.

c. Napthalene



i. The compound is cyclic and planar since all ring carbon atoms are sp^2 hybridized.

ii. π electrons = 10 : $4n + 2 = 10$, where $n = 2$

Planar structure permits the delocalization of π electrons in a ring. Hence it is aromatic.

d. Cyclopropenyl cation

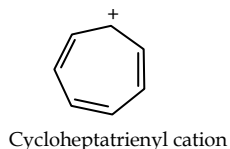


i. The compound is cyclic and planar since all ring carbon atoms are sp^2 hybridized.

ii. π electrons = 2 : $4n + 2 = 2$, where $n = 0$

Planar structure permits the delocalization of π electrons in a ring. Hence it is aromatic.

e. Cycloheptatrienyl cation



i. The compound is cyclic and planar since all ring carbon atoms are sp^2 hybridized.

ii. π electrons = 6 : $4n + 2 = 6$, where $n = 1$

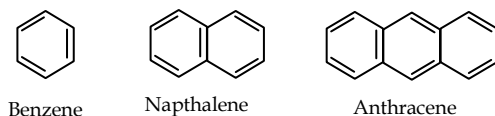
Planar structure permits the delocalization of π electrons in a ring. Hence it is aromatic.

Hence, **an aromatic compound may be defined as a cyclic planar molecule which has cyclic cloud of delocalized $4n+2$ π electrons and has unusual stability towards addition reaction due to delocalization of π electrons.**

Benzenoid and non-benzenoid aromatic compounds

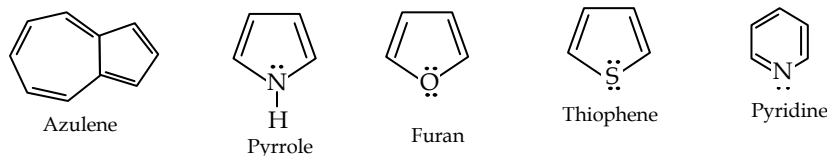
Aromatic compounds having benzene ring are called benzenoid aromatic compounds.

Examples:



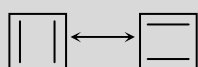
Aromatic compounds having no benzene ring are called non-benzenoid aromatic compounds.

Examples:



Concept of Antiaromaticity (Breslow, 1967)

There are some cyclic unsaturated organic compounds having $4n$ π electrons which are less stable than acyclic analog. Such organic compounds are called antiaromatic and characteristics shown by them are called antiaromaticity. For example, 1, 3- cyclobutadiene



Although, it has two resonating structures, it is less stable than acyclic compound, 1, 3-butadiene. Here, resonance instead of stabilization, destabilizes.

Some anti-aromatic species are:



Cyclopropenyl cation
 4π electrons



Cyclopentadienyl cation
 4π electrons



Cyclooctatetraene
 8π electrons

Aromatic, Non-aromatic and Anti-aromatic compounds

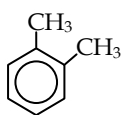
Aromatic compounds: Cyclic organic compound which is more stable than its open chain analog

Non-aromatic compounds: Cyclic organic compound which has equal stability to its open chain analog

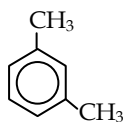
Anti-aromatic compounds: Cyclic organic compound which is less stable than its open chain analog

8.6 Isomerism in Benzene Derivatives

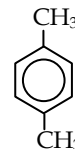
- Mono-substituted derivative of benzene:** All six hydrogen atoms in benzene are equivalent. Hence, replacement of any of six hydrogen atom gives only one mono-substituted benzene.
- Di-substituted derivatives of benzene:** Three positional isomers are possible.



1, 2-Dimethyl benzene
(*o*-xylene)

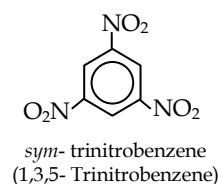
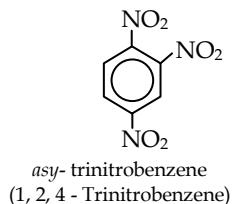
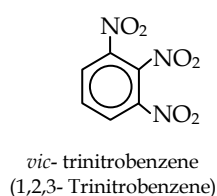


1, 3-Dimethyl benzene
(*m*-xylene)



1, 4-Dimethyl benzene
(*p*-xylene)

- Tri-substituted derivatives of benzene:** The number of isomers depends upon nature of substituents:
 - If the three substituents are identical, then *three* isomers are possible. (Concept of vicinal, asymmetrical and symmetrical)

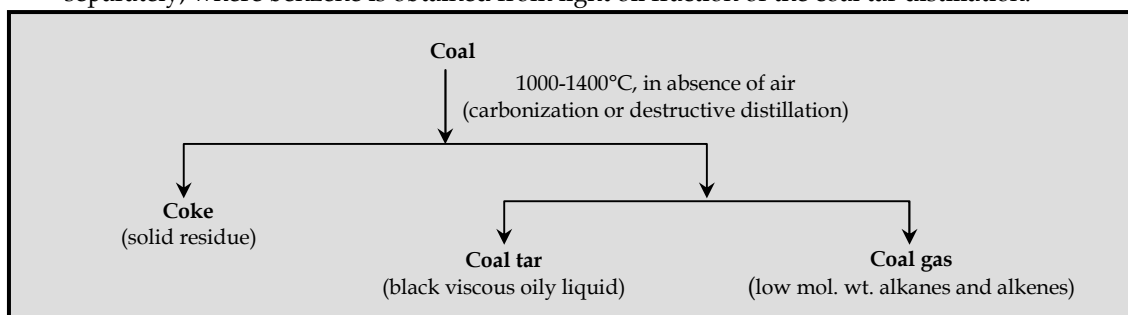


- ii. If two substituents are identical and the third is different, then *six* isomers are possible.
- iii. If all three substituent are different, then *ten* isomers are possible.

8.7 Commercial Source of Benzene

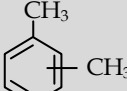
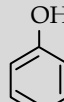
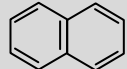
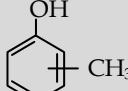
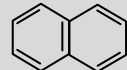
Before 1950s, benzene was mainly obtained from coal tar. It was obtained as a byproduct of coke production in steel industries. But, demand of benzene increased especially due to the growing polymer industry. Today, benzene is mostly produced from petroleum.

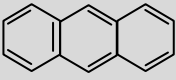
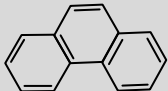
- From coal tar:** Coal after destructive distillation gives coal-tar as one of the fractions. The coal tar contains more than 70 different aromatic compounds like benzene, phenol, naphthalene, anthracene, etc. The coal tar is subjected to fractional distillation to get aromatic compounds, separately, where benzene is obtained from light oil fraction of the coal tar distillation.



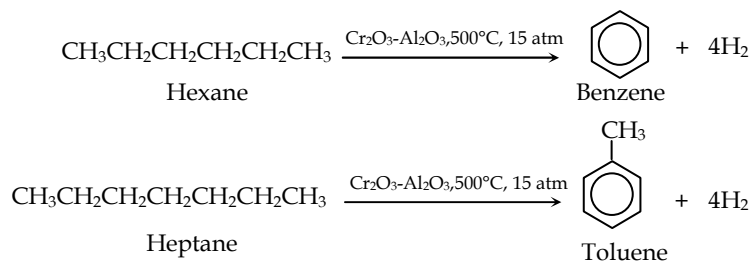
Coal tar is rich in aromatic compounds. It is subjected to fractional distillation to different fractions.

Main fractions of coal tar

SN	Fraction	Temperature range	%	Chief constituents
1	Light oil or Crude naptha	upto 170°C	7-8	 Benzene  Toluene  o-,m-,p-xylenes
2	Middle oil or Carbolic oil	170-230°C	8-10	 Phenol  Napthalene
3	Heavy oil or Creosote oil	230-270°C	8-10	 Cresols  Napthalene

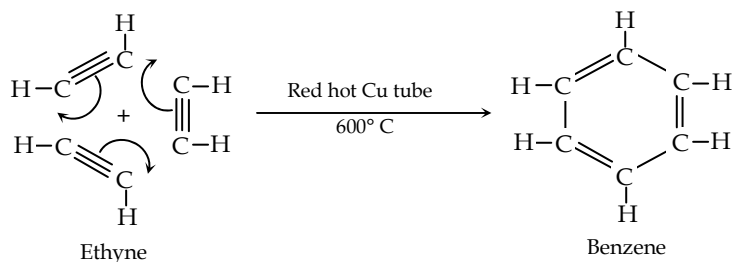
4	Green oil or Anthracene oil	270-360°C	16-20	 Anthracene  Phenanthrene
5	Pitch	residue	50-60	90-95% carbon

2. **From petroleum:** n-alkanes generally with C₆–C₈ on cyclization followed by aromatization give arenes.

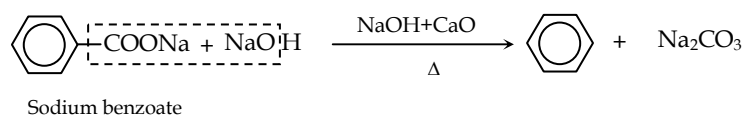


8.8 Preparation of Benzene

1. **From acetylene:** When acetylene is passed through a copper tube heated to about 600°C, benzene is formed. Benzene was prepared for the first time by Berthelot (1870) by this method. The process is called **trimerization** or **polymerization** or **aromatization**.

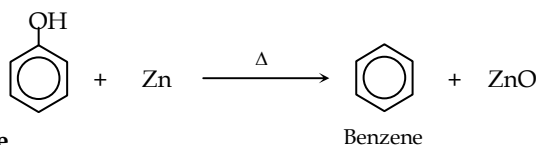


2. **From sodium benzoate:** When sodium benzoate is heated with sodalime, benzene is formed. The reaction is called sodalime **decarboxylation reaction**.



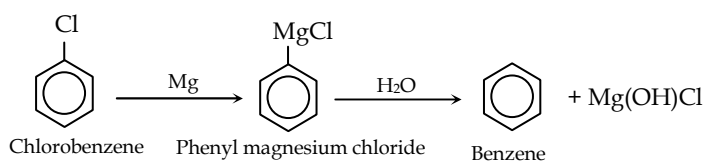
This reaction is used as **laboratory method** for the preparation of benzene.

3. **From phenol:** When phenol is distilled with zinc dust, benzene is formed.

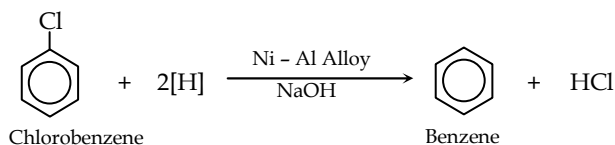


4. **From chlorobenzene**

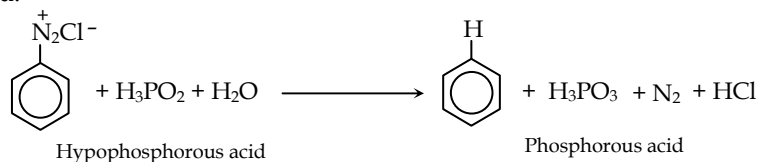
- i. When chlorobenzene is treated with Mg metal followed by hydrolysis, benzene is obtained.



- ii. When chlorobenzene is reduced with Ni-Al alloy in presence of NaOH, benzene is obtained.



5. **From diazonium salt:** When diazonium salt is mixed with aqueous hypophosphorous acid, benzene is formed.



Note:

Benzene diazonium salt is obtained by diazotization reaction of aniline (see in Aromatic Amine).

8.9 Physical Properties

1. Benzene is a colourless liquid with aromatic odour.
2. It is insoluble in water but soluble in organic solvents.
3. The boiling point of benzene is 80.4°C and melting point is 5.5°C.
4. The vapour of benzene is highly inflammable, toxic and burns with sooty flame.
5. It is a very good solvent for fats, resins, sulphur, iodine, etc.

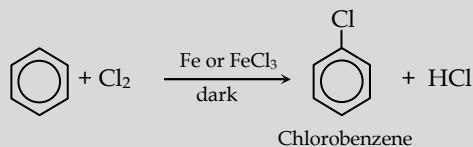
8.10 Chemical Properties

Benzene, a typical aromatic compound, undergoes mainly three categories of reactions:

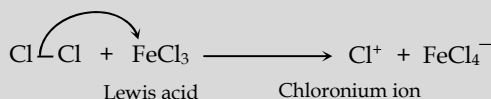
- A. Electrophilic aromatic substitution reactions
- B. Addition reactions
- C. Combustion

A. Electrophilic aromatic substitution reactions: Typical reaction of aromatic compounds

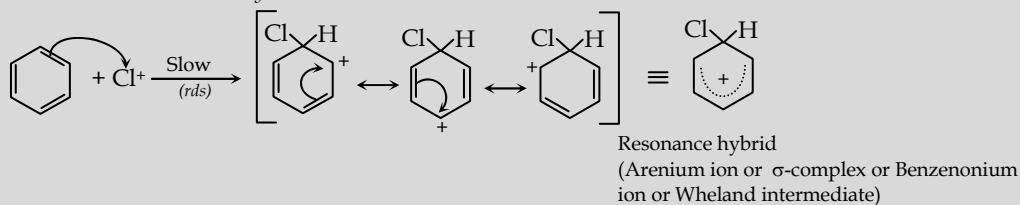
Benzene serves as a **source of electron** due to delocalized cyclic pi electron cloud above and below the plane of the molecule. So, electrophile attacks benzene ring to give substitution reactions. A substitution reaction initiated by an electrophile is called electrophilic substitution reaction. This is the typical reactions of benzene and all aromatic compounds.

Mechanism of electrophilic aromatic substitution reaction

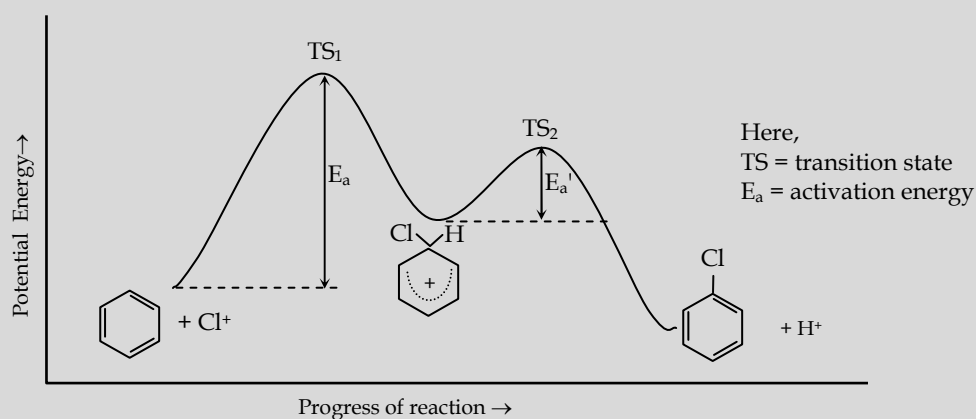
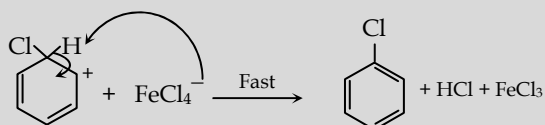
Generation of electrophile: Catalyst reacts with the reagent to generate electrophile.



Step I: Electrophile is attracted toward π electron cloud of benzene ring to form arenium ion as intermediate. This step is slow since aromatic character is destroyed.

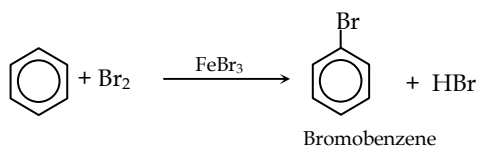
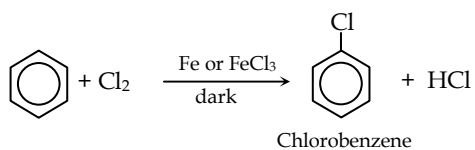


Step II: Abstraction of proton by nucleophile to give product. This step is fast since aromatic character is restored.

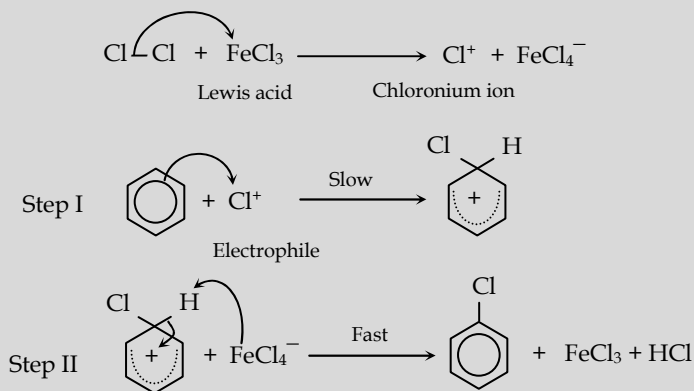


Energy profile diagram for electrophilic aromatic substitution

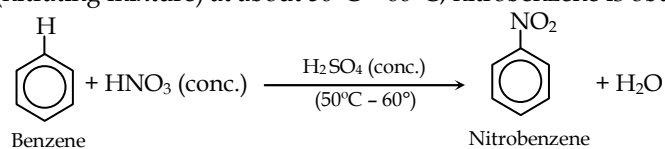
- Halogenation:** Benzene reacts with limited amount of Cl_2 or Br_2 at ordinary temperature (not at high temperature and in presence of sunlight) and in presence of Lewis acid like FeCl_3 , AlCl_3 , FeBr_3 to give corresponding halobenzene. Iron is most commonly used since it is converted to FeCl_3 or FeBr_3 .



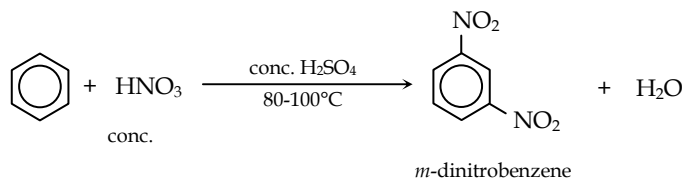
Simplified mechanism:



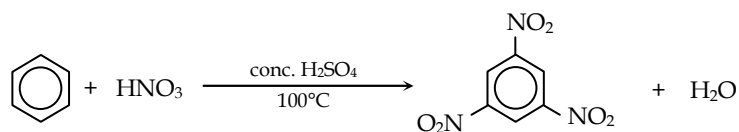
2. **Nitration:** When benzene is heated with a mixture of concentrated nitric acid and concentrated sulphuric acid (nitrating mixture) at about 50°C – 60°C, nitrobenzene is obtained.



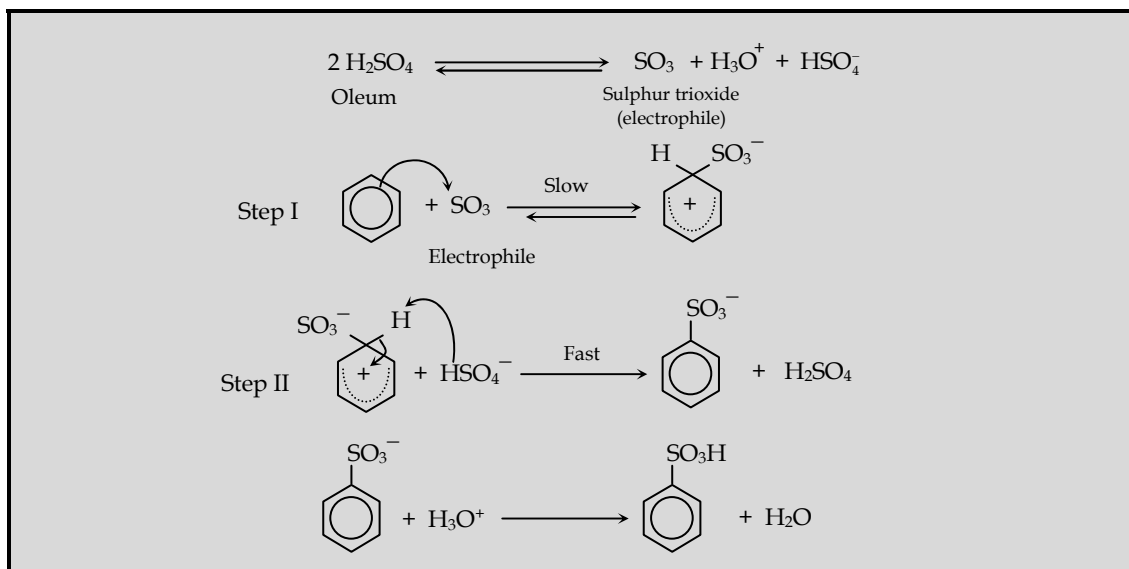
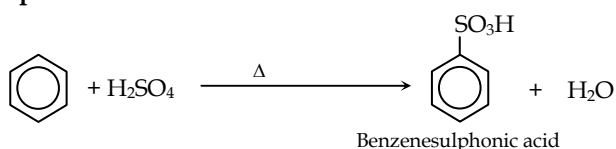
When benzene is heated with a mixture of conc. HNO₃ and conc. H₂SO₄ at about 80–100°C, m-dinitrobenzene is obtained.



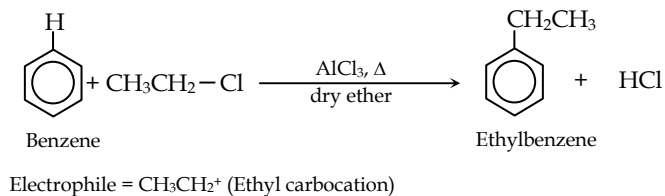
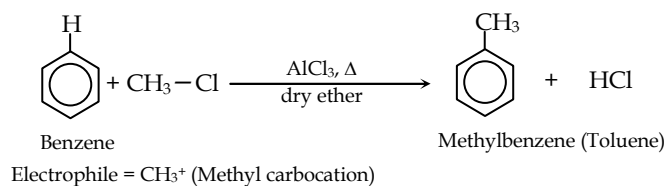
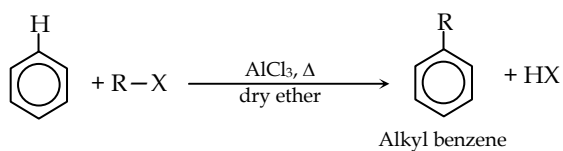
When benzene is heated with a mixture of fuming HNO₃ and conc. H₂SO₄, 1,3,5-trinitrobenzene is obtained.



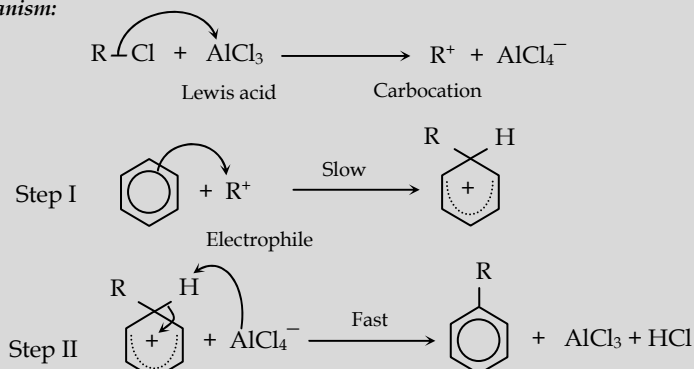
3. **Sulphonation:** When benzene is heated with concentrated or fuming sulphuric acid or oleum or pyrosulphuric acid (sulphuric acid containing SO_3), benzenesulphonic acid is obtained. The reaction is called **sulphonation reaction**.



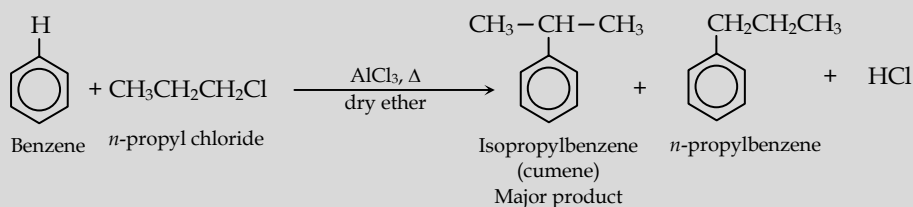
4. **Friedel-Crafts alkylation:** When benzene reacts with alkyl halide ($\text{R}-\text{X}$) in presence of anhydrous AlCl_3 , BF_3 or FeCl_3 etc., alkyl group replaces the hydrogen of benzene nucleus to give alkyl benzene. The reaction is called **Friedel-Crafts alkylation**.



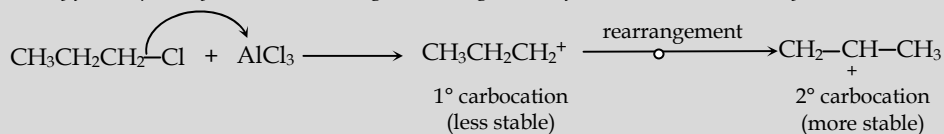
Simplified mechanism:



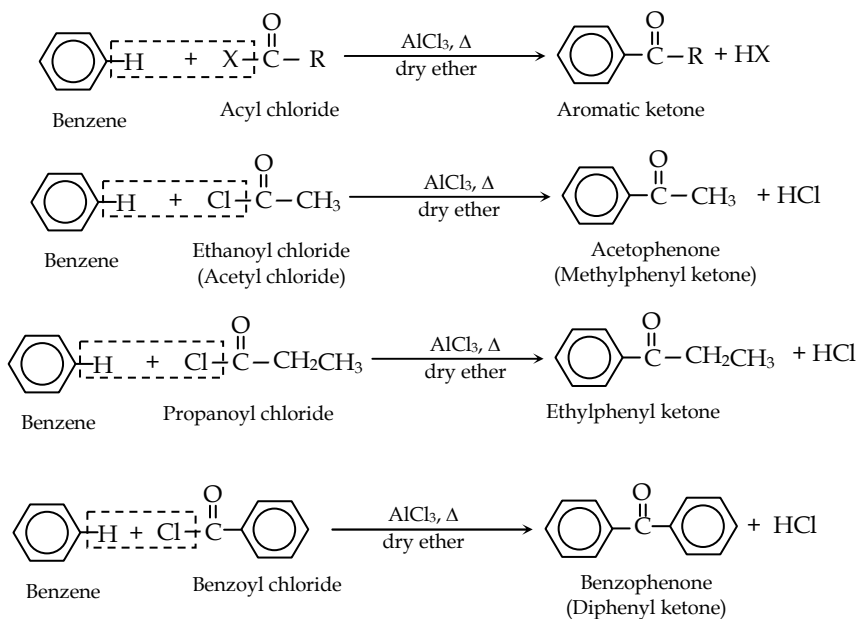
Benzene reacts with *n*-propyl chloride to give isopropylbenzene as major product rather than *n*-propylbenzene.

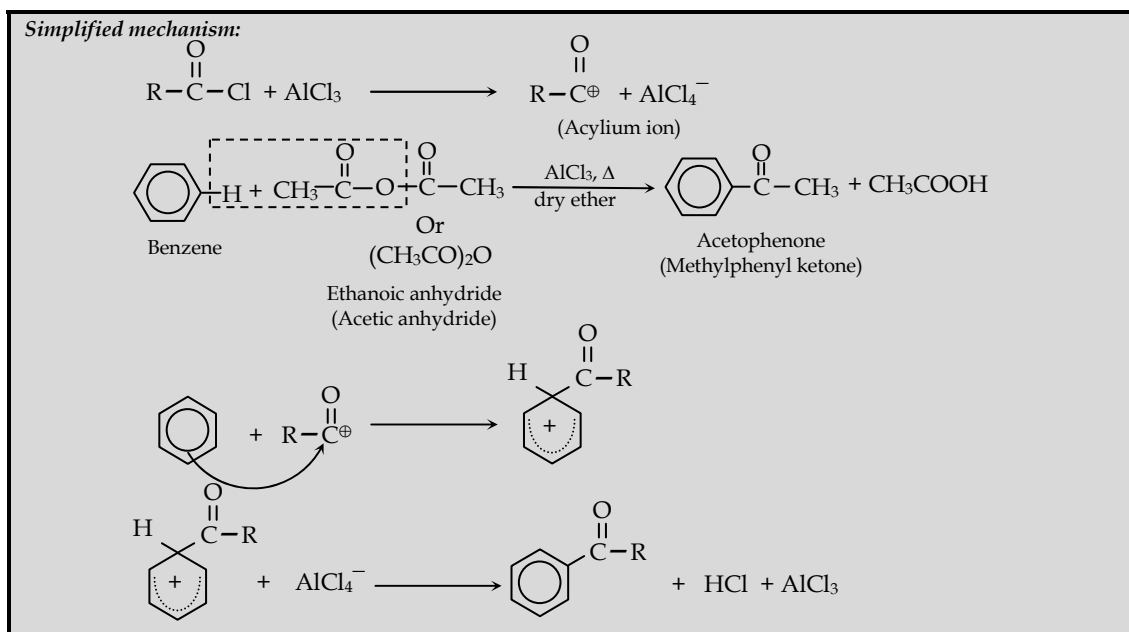


This is because initially formed primary carbocation undergoes rearrangement to form more stable secondary carbocation.



5. **Friedel-Crafts acylation:** When benzene, reacts with acyl halide RCOCl or anhydride in presence of anhydrous AlCl₃, BF₃, or FeCl₃ etc. Lewis acid, acyl group replaces the hydrogen of benzene nucleus to give aromatic ketone (acylbenzene). The reaction is called as **Friedel-Crafts acylation**.

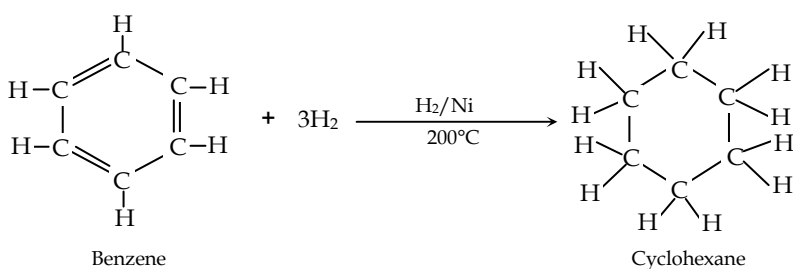




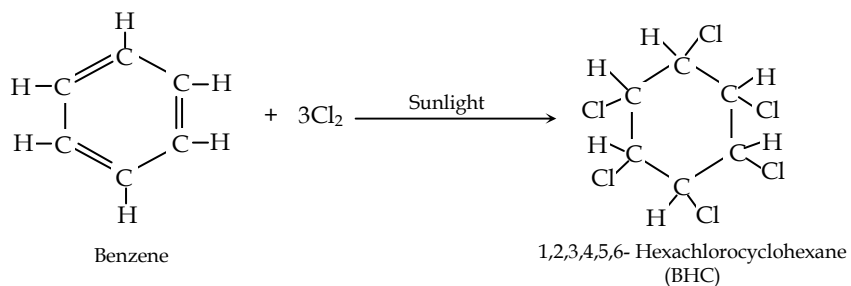
B. Addition reactions

Benzene and all aromatic compounds are stabilized due to delocalization of pi electrons. Addition reactions destroy their aromatic character. Hence all aromatic compounds are quite stable towards addition reactions. However, benzene gives addition reactions under drastic conditions.

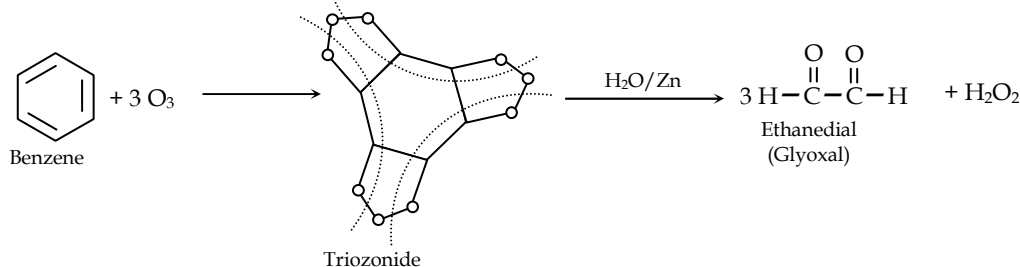
- 1. Addition of hydrogen (hydrogenation):** Benzene reacts with hydrogen in presence of nickel as catalyst on heating to give cyclohexane.



- 2. Addition of chlorine:** Benzene reacts with Cl₂ in presence of sunlight to give BHC which is an insecticide.

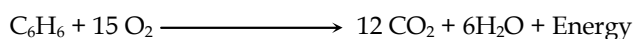


3. **Addition of ozone followed by hydrolysis (Ozonolysis):** Benzene reacts with ozone to form triozonide which on hydrolysis in presence of Zn dust gives glyoxal.

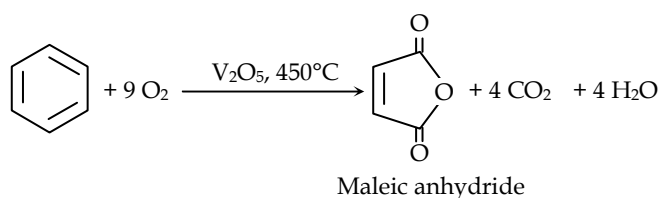


C. Combustion/Oxidation

1. On complete combustion, benzene gives CO₂ and water.



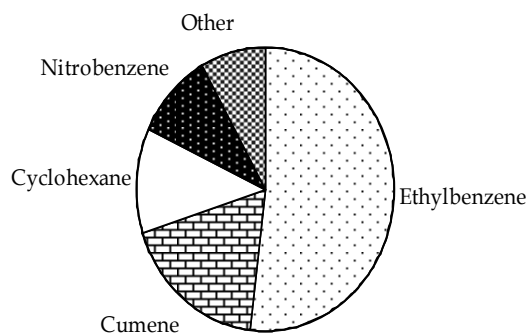
2. Benzene is partially oxidized in presence of V₂O₅ to give maleic anhydride.



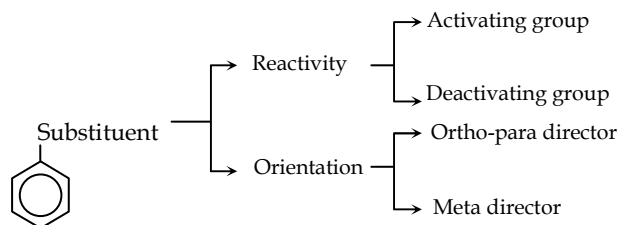
8.11 Uses of Benzene

Today, more than half of the benzene produced is converted to ethylbenzene which is converted then to styrene for polystyrene. The next largest use of benzene is in the preparation of cumene which is then converted to phenol for resins and adhesives. The next use is to prepare cyclohexane which is used in the manufacture of nylon. Other uses including, the preparation of aniline (for dyes) and dodecylbenzene (for detergents). Benzene is also used to make some types of rubbers, lubricants, detergents, drugs, explosives, and pesticides. Benzene was used as solvent in dry cleaning but in these days it was discouraged because of its carcinogenic effect.

World Consumption of Benzene in 2010



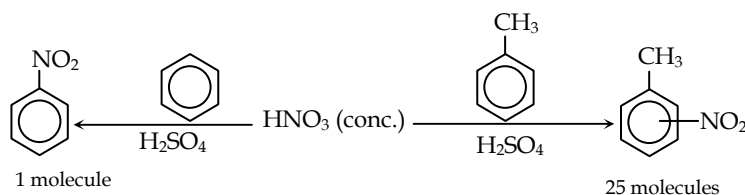
8.12 Electrophilic Aromatic Substitution Reactions in Substituted Benzene



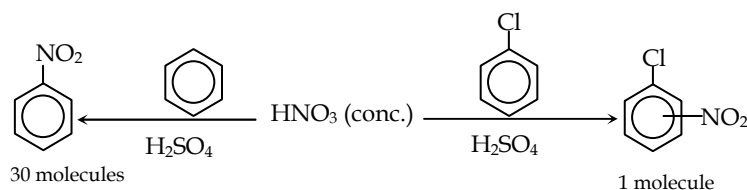
A substituent present in benzene governs the **reactivity and orientation** of the benzene ring.

Effect on reactivity

A substituent makes the benzene ring either more reactive or less reactive as compared to benzene in electrophilic aromatic substitution reaction. A substituent which makes the benzene ring more reactive than benzene in electrophilic aromatic substitution is called activating group. For example, methyl benzene undergoes nitration about 25 times faster than benzene. Hence $-\text{CH}_3$ is an activating group.



Similarly, a substituent which makes the benzene ring less reactive than benzene in electrophilic aromatic substitution is called deactivating group. For example, chlorobenzene undergoes nitration about 30 times slower than benzene. Hence $-\text{Cl}$ is deactivating group.

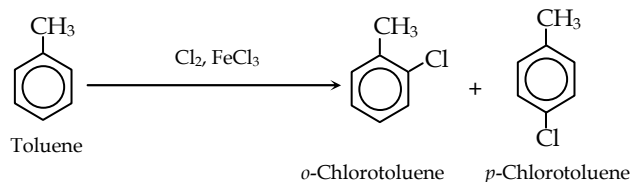


Effects of substituent on electrophilic aromatic substitution reactions is given below.

Activating group and ortho, para director	Deactivating groups
Strongly activating $-\text{NH}_2$ ($-\text{NHR}$, $-\text{NR}_2$) $-\text{OH}$	Meta directors $-\text{NO}_2$ $-\text{NH}_3^+$ $-\text{CN}$
Moderately activating $-\text{OCH}_3$ ($-\text{OC}_2\text{H}_5$ etc.) $-\text{NHCOCH}_3$	$-\text{COOH}$ ($-\text{COOR}$) $-\text{SO}_3\text{H}$ $-\text{CHO}$, $-\text{COR}$
Weakly activating $-\text{C}_6\text{H}_5$ $-\text{CH}_3$ ($-\text{C}_2\text{H}_5$, etc.)	Ortho and para directors $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$

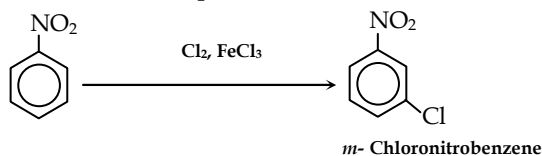
Effect on orientation

A substituent which directs the incoming group chiefly at ortho and para to it in electrophilic aromatic substitution reaction is called ortho- para director. For example,



Hence $-\text{CH}_3$ is called ortho- para director.

A substituent which directs the incoming group chiefly at meta to it in electrophilic aromatic substitution is called meta director. For example,



Hence $-\text{NO}_2$ is called meta director.

Note (for Orientation and reactivity of substituted benzene)

The group attached in the benzene ring can be +M group or -M group (M = mesomeric):

1. **+M groups:** Groups like $-\ddot{\text{Cl}}:$, $-\ddot{\text{Br}}:$, $-\ddot{\text{I}}:$, $-\ddot{\text{O}}\text{H}$, $-\ddot{\text{N}}\text{H}_2$, $-\ddot{\text{O}}\text{R}$, $-\ddot{\text{N}}\text{R}_2$, $-\text{CH}_3$, etc. make ortho and para position electron rich. (see resonance in chlorobenzene, phenol etc.) So, electrophile is likely to attack in these positions. Thus, these groups are ortho and para directors for electrophilic substitution reactions. As they create make the ring electron rich, they are activators of the ring (except halo group, as halo groups are strongly electron withdrawing (-I) groups. In halo groups their -I effect is more powerful than +M effect).

2. **-M groups:** Groups like $\text{O}=\text{N}-$, $\text{O}=\text{C}-$, $\text{O}=\text{C}-\text{H}$, $\text{O}=\text{C}-\text{OH}$ etc. make the ortho and para positions electron deficient (see resonance in nitrobenzene, benzaldehyde etc.). Thus, electrophile does not attack in these positions. Only the position left for electrophilic attack is meta. So, these groups are meta directors. As they make the ring electron deficient, they are the deactivator of ring.

Explanations in details are given in respective units.

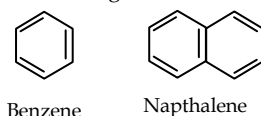
Solved Questions

1. Define aromatic compound. What are benzenoid aromatic compounds and nonbenzenoid aromatic compounds?

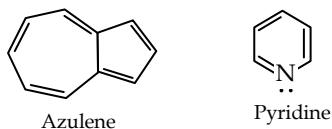
→ An aromatic compound may be defined as a cyclic planar molecule which has cyclic cloud of delocalized $4n+2$ pi electrons and has unusual stability towards addition reaction due to delocalization of pi electrons though unsaturated.

Aromatic compounds can be divided into benzenoid and nonbenzenoid aromatic compounds on the basis of benzene ring which is the most common structure in aromatic compounds.

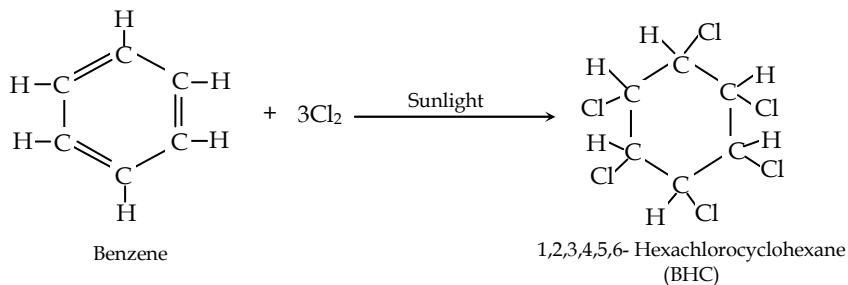
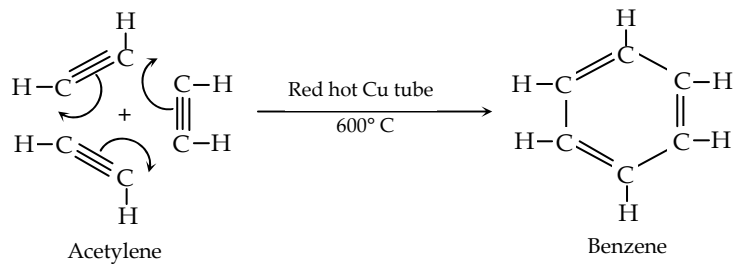
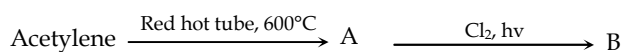
Aromatic compounds having benzene ring are called benzenoid aromatic compounds. For example,



Aromatic compounds having no benzene ring are called non-benzenoid aromatic compounds. For example,



2. Identify A and B giving complete reactions.



Questions

Very Short Answer Questions:

- How do aromatic compounds differ from aliphatic compounds?
- Define the following with examples:
 - Friedel-Crafts alkylation
 - Friedel-Crafts acylation
 - Nitration of benzene
 - Chlorination of benzene
- Why does benzene not decolorize pink color of Baeyer's reagent?
- How can you convert acetylene into toluene?

Short /Long Answer Questions:

- What is Huckel's rule for aromaticity? Predict the aromaticity of the following species;
 - Tropyllium cation
 - Cycloheptatriene
 - Cyclopentadienyl cation
 - Anthracene
 - Pyrrole
 - Pyridine
 - Azulene
 - Cyclopropenyl cation
 - Cyclooctatetraene
- Identify A and B
 - Sodium benzenesulphonate $\xrightarrow{\text{NaOH}+\text{CaO}, \Delta}$ A $\xrightarrow{\text{H}_2/\text{Ni}, 200^\circ\text{C}}$ B
 - Hexane $\xrightarrow{\text{Al}_2\text{O}_3-\text{Cr}_2\text{O}_3, 500^\circ\text{C}, 15 \text{ atm}}$ A $\xrightarrow{\text{Br}_2, \text{FeBr}_3, \text{dark}}$ B
 - Hexane $\xrightarrow{\text{Pt}, 500^\circ\text{C}, 10-20 \text{ atm}}$ A $\xrightarrow{\text{CH}_3\text{COCl}, \text{AlCl}_3, \Delta}$ B
 - Phenol $\xrightarrow{\text{Zn dust}, \Delta}$ A $\xrightarrow{\text{CH}_3\text{CH}_2\text{Cl}, \text{AlCl}_3, \Delta}$ B
 - Benzene $\xrightarrow{\text{O}_3}$ A $\xrightarrow{\text{H}_2\text{O}, \text{Zn dust}, \Delta}$ B
 - Benzene diazonium chloride $\xrightarrow{\text{aq. H}_3\text{PO}_2, \Delta}$ A $\xrightarrow{\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}, \text{AlCl}_3, \Delta}$ B
- What happens when:
 - Sodium benzoate is heated with sodalime?
 - Phenol is heated with zinc dust?
 - Benzene is heated with acetic anhydride in presence of anhydrous AlCl_3 ?
 - Benzene is reduced?
 - Benzene is ozonolysed?

Multiple Choice Questions

- Benzene was discovered by
 - Faraday
 - Dalton
 - Hofmann
 - Kekule
- Coal tar is the main source of
 - Aliphatic compounds
 - Aromatic compounds
 - Heterocyclic compounds
 - Organic compounds
- Which is impossible to obtain from the fractionation of coal tar?
 - Light oil
 - Heavy oil
 - Middle oil
 - Vegetable oil
- Which of the followings is the major source of benzene?
 - Light oil
 - Green oil
 - Carbolic oil
 - Coal tar
- Typical reaction of aromatic compounds is
 - Electrophilic addition
 - Nucleophilic addition
 - Free radical addition
 - Electrophilic substitution

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6. Benzene when reacts with hydrogen in presence of nickel at high temperature to give cyclohexane, the reaction is
 - a. addition
 - b. reduction
 - c. oxidation
 - d. substitution
7. Benzene when reacts with ozone followed by hydrolysis in presence of zinc dust to give
 - a. Ethanedial
 - b. Oxalic acid
 - c. Acetaldehyde
 - d. Glycol
8. The electrophile in sulphonation of benzene is
 - a. SO_3^+
 - b. SO_3
 - c. HSO_3
 - d. HSO_4^-
9. Benzene reacts with the *n*-propyl chloride in presence of anhydrous AlCl_3 to give ... as major product.
 - a. *n*-propyl benzene
 - b. Isopropyl benzene
 - c. Xylene
 - d. Mesitylene
10. Friedel-Crafts reaction is given by
 - a. Nitrobenzene
 - b. Aniline
 - c. Benzene
 - d. All the three
11. Friedel-Crafts reaction is given by
 - a. Aliphatic compounds
 - b. Aromatic compounds
 - c. Alicyclic compounds
 - d. Benzene only
12. In Friedel-Crafts acylation, electrophile is
 - a. Carbonium ion
 - b. Carbanion
 - c. Acyl chloride
 - d. Acylium ion
13. Nitration takes place faster in
 - a. Toluene
 - b. Phenol
 - c. Benzene
 - d. Nitrobenzene
14. Benzene is more stable than alkene due to
 - a. Electron losing tendency of ring
 - b. Electron gaining tendency of ring
 - c. Resonance of ring
 - d. Electromeric effect
15. Structure of benzene is
 - a. Planar
 - b. Chair form
 - c. Trigonal bipyramidal
 - d. Tetrahedral
16. If *n* is number of carbons and *y* is number of benzene ring then the general formula of arene is
 - a. $\text{C}_n\text{H}_{2n-2y}$
 - b. $\text{C}_n\text{H}_{2n-3y}$
 - c. $\text{C}_n\text{H}_{2n+2y}$
 - d. $\text{C}_n\text{H}_{2n-6y}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
a	b	d	a	d	a	a	b	b	c	b	d	b	c	a	d

Haloalkanes and Haloarenes

9



Ethyl chloride

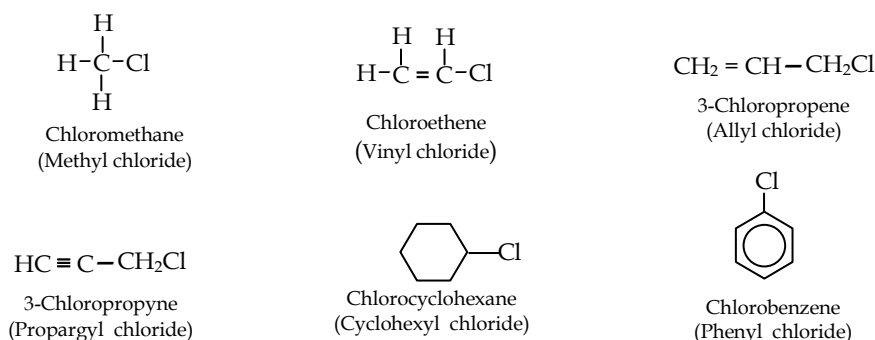
Ethyl chloride is a haloalkane. It acts as a topical anesthetic by its chilling effect when sprayed on skin, such as in minor surgical procedures, in minor sports injuries etc.

9.1 Haloalkanes

9.1.1 Introduction

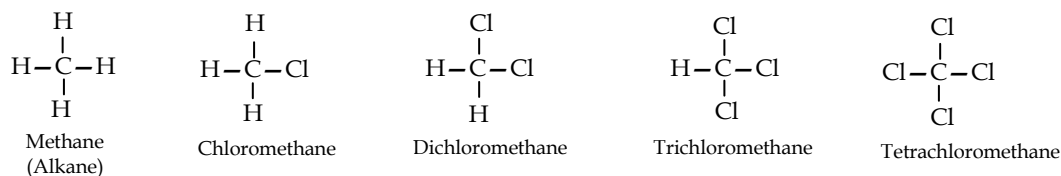
Organic compounds derived from hydrocarbons (alkanes, alkenes, alkynes, cycloalkanes and arenes) by replacement of one or more hydrogen atoms with equal number of halogen atoms are called **halogen derivatives of hydrocarbons**.

Examples:



Halogen derivatives of alkanes, alkenes and alkynes are called **haloalkanes**, **haloalkenes** and **haloalkynes**, respectively. Halo group ($-\text{Cl}$, $-\text{Br}$, $-\text{I}$) is the functional group of halogen derivatives of hydrocarbons since it determines their characteristic chemical properties as well as physical properties.

When one or more hydrogen atoms in alkanes are replaced by halogen atoms, the resulting organic compounds are called **halogen derivatives of alkanes** or **halogenoalkanes** or simply **haloalkanes**.



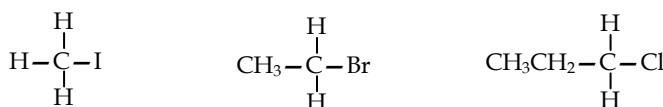
Depending upon number of halogen atoms per molecule, haloalkanes can be classified into monohaloalkanes, dihaloalkanes, trihaloalkanes and so on.

9.1.2 Monohaloalkanes (Alkyl halides)

9.1.2.1 Introduction

When one of the hydrogen atoms in alkanes is replaced by a halogen atom, the resulting organic compounds are called monohaloalkanes or alkyl halides.

Examples:



Alkyl halides have general molecular formula $C_nH_{2n+1}X$. Its general structural formula is $R-X$ where X represents Cl , Br or I .

9.1.2.2 Classification

Depending upon type of carbon to which halogen atom is bonded, monohaloalkanes are classified into three different classes; primary alkyl halides, secondary alkyl halides and tertiary alkyl halides.

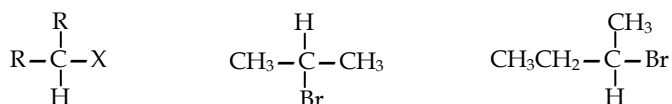
- Primary (1°) alkyl halides (RCH_2X):** If halogen atom is bonded to primary carbon, the alkyl halide is called primary alkyl halide or primary haloalkane.

Examples:



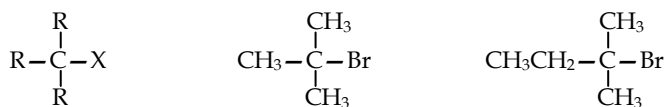
- Secondary (2°) alkyl halides (R_2CHX):** If halogen atom is bonded to secondary carbon, the alkyl halide is called secondary alkyl halide or secondary haloalkane.

Examples:



- Tertiary (3°) alkyl halides (R_3CX):** If halogen atom is bonded to tertiary carbon, the alkyl halide is called tertiary alkyl halide or tertiary haloalkane.

Examples:



9.1.2.3 Nomenclature

The functional group of haloalkanes is halo group because the properties of haloalkanes are governed by halogen atom. However, during writing IUPAC name, it is not taken as functional group and treated as substituent. Alkyl halides are named taking Halo- as substituent's name.

Formula	IUPAC name	Common name
CH_3Cl	Chloromethane	Methyl chloride
CH_3CH_2Cl	Chloroethane	Ethyl chloride
$CH_3CH_2CH_2Cl$ $CH_3-\underset{\substack{ \\ Cl}}{CH}-CH_3$	1- Chloropropane 2- Chloropropane	<i>n</i> -Propyl chloride Isopropyl chloride

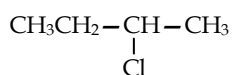
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$	1-Chlorobutane	<i>n</i> -Butyl chloride
$\begin{array}{c} \text{CH}_3\text{CH}_2-\text{CH}-\text{CH}_3 \\ \\ \text{Cl} \end{array}$	2-Chlorobutane	<i>sec</i> -Butyl chloride
$\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{Cl} \\ \\ \text{CH}_3 \end{array}$	1-Chloro-2-methylpropane	Isobutyl chloride
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{C}-\text{Cl} \\ \\ \text{CH}_3 \end{array}$	2-Chloro-2-methylpropane	<i>tert</i> -Butyl chloride

9.1.2.4 Isomerism

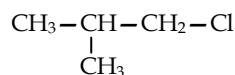
Alkyl halides can show following types of structural isomerism.

- Chain isomerism:** Isomers that differ only in chain length are called chain isomers.

Example:



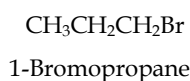
2-Chlorobutane



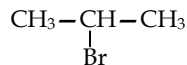
1-Chloro-2-methylpropane

- Positional isomerism:** Isomers having same chain length but differ in the position of functional group or substituents are called positional isomers.

Example:



1-Bromopropane



2-Bromopropane

Molecular formula	CH_3X	$\text{C}_2\text{H}_5\text{X}$	$\text{C}_3\text{H}_7\text{X}$	$\text{C}_4\text{H}_9\text{X}$	$\text{C}_5\text{H}_{11}\text{X}$
Number of structural isomers	-	-	2	4	8

Self Test

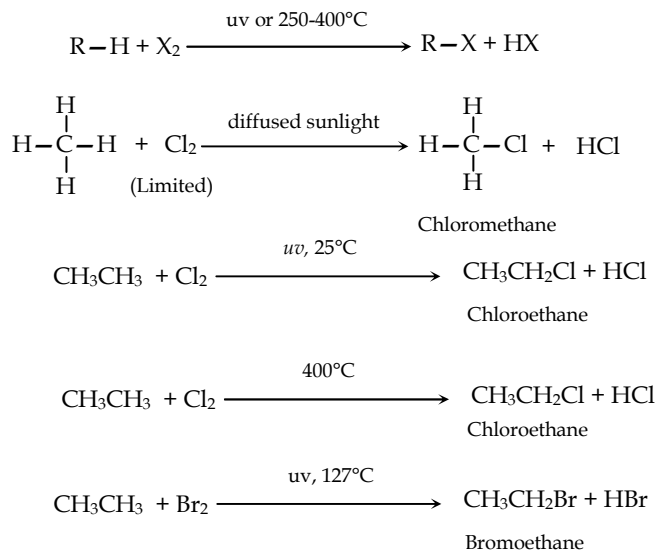
- What are haloalkanes? Write the isomers and give their IUPAC name.
 - $\text{C}_3\text{H}_7\text{Cl}$
 - $\text{C}_2\text{H}_4\text{Cl}_2$
 - $\text{C}_3\text{H}_6\text{Cl}_2$
 - $\text{C}_3\text{H}_5\text{Cl}_3$
- Write the structural isomers of $\text{C}_4\text{H}_9\text{Br}$ and give their IUPAC name, common name and classify them as primary, secondary and tertiary alkyl halide.
- Write the structural isomers of $\text{C}_5\text{H}_{11}\text{Cl}$, give their IUPAC name and indicate them as primary, secondary and tertiary alkyl halide. *Hint: 8 isomers*
- Draw the structure and give IUPAC name of;
 - n*-Pentyl chloride
 - neo* Pentyl chloride
 - tert*- Butyl chloride
 - tert*- Pentyl chloride
 - Isobutyl bromide
 - Isopentyl bromide
 - Isohexyl bromide
 - sec*-Butyl bromide

9.1.2.5 General Methods of Preparation

Alkyl halides are prepared as follows;

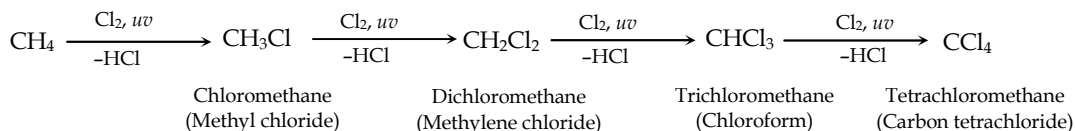
- From alkanes
- From alkenes
- From alcohols
- By halide exchange method
- From silver carboxylate

1. **From alkanes: Direct halogenation:** Alkane reacts with limited amount of halogen (Cl_2 or Br_2) in presence of sunlight or heat to form alkyl halide.

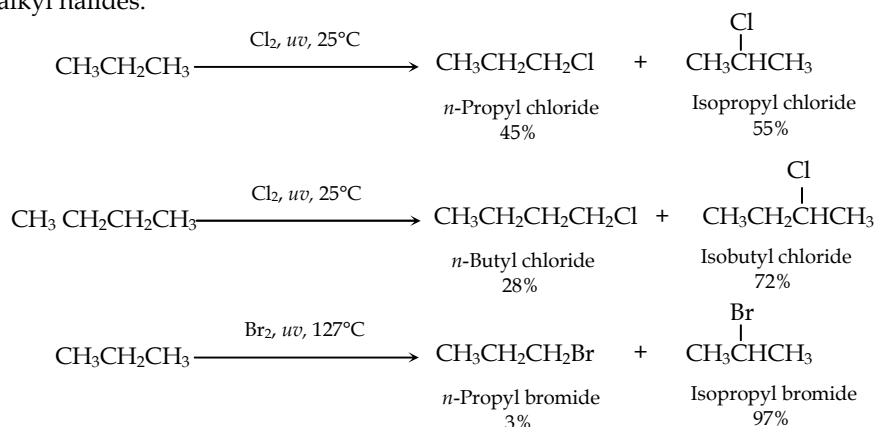


Generally, this is not a good method for the preparation of haloalkanes due to following reasons:

- i. Excess X_2 (Cl_2 or Br_2) leads to formation of a mixture of polyhaloalkanes.



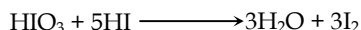
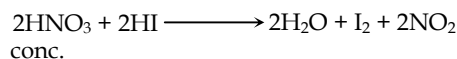
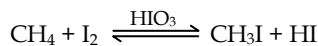
- ii. Higher alkanes give a mixture of all possible isomeric alkyl halides which are difficult to separate. Hence, halogenation of alkanes is not suitable method for the laboratory preparation of alkyl halides.



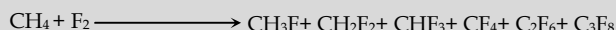
Reactivity of hydrogen atom towards halogenation is $3^\circ > 2^\circ > 1^\circ > \text{CH}_4$.

Reactivity of halogen towards halogenation is $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$

Iodination is reversible (as HI is powerful reducing agent), hence, it is carried out in presence of oxidizing agents like HIO_3 , HNO_3 , HgO etc which oxidizes HI and equilibrium is shifted towards forward direction.

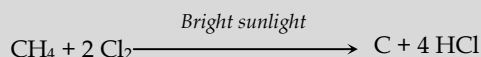


1. F_2 is the most reactive of all halogens. It reacts with alkanes explosively producing large amount of heat and a complex mixture. For example, methane reacts with F_2 as:



Monofluoroalkanes, however, can be obtained by diluting F_2 with inert gas like N_2 to make the reaction less vigorous and carrying out the reaction in an apparatus which carry out the heat produced.

2. Cl_2 reacts with alkanes in bright sunlight explosively. Hence, reaction is carried out in diffused sunlight.

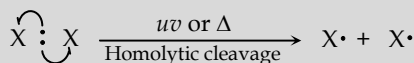


3. Br_2 reacts with alkanes in sunlight slowly. Hence heat is also needed in addition to sunlight.

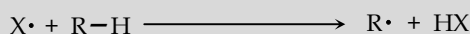
Mechanism of halogenation of alkanes: Free radical substitution

The halogenation of alkanes proceeds through free radical substitution. It involves the following steps.

i. Chain initiation step:



ii. Chain propagation step:



iii. Chain termination step:



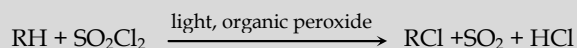
Or,



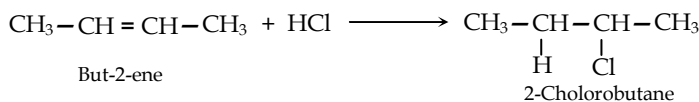
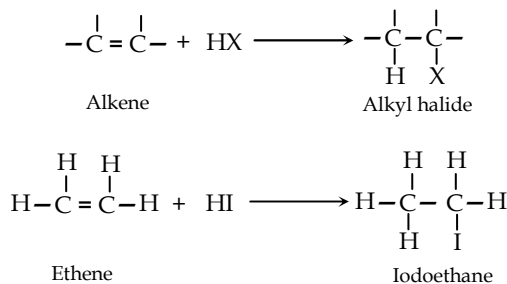
Or



Direct chlorination of alkanes can also be carried out using sulphuryl chloride (SO_2Cl_2) in presence of light and traces of an organic peroxide.

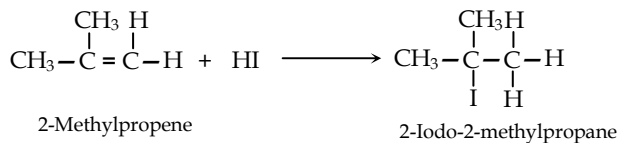
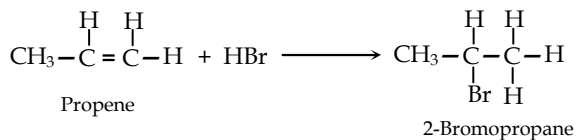


2. **From alkenes: Hydrohalogenation:** Alkene reacts with hydrogen halide ($\text{HX} = \text{HCl}, \text{HBr}, \text{HI}$) to give respective alkyl halide. The reaction is frequently carried out by passing the dry gaseous hydrogen halide directly into alkene. Aqueous solution of hydrogen halide is not used to avoid chance of hydrolysis of alkyl halide.



Addition of hydrogen halide onto unsymmetrical alkene: Markovnikov's addition (1869)

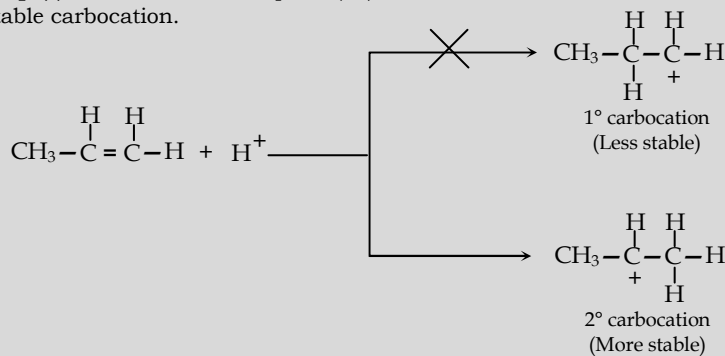
According to Markovnikov's rule, *when an unsymmetrical reagent is added to unsymmetrical unsaturated organic compound, negative part of the reagent adds to the unsaturated carbon having less number of hydrogen.*



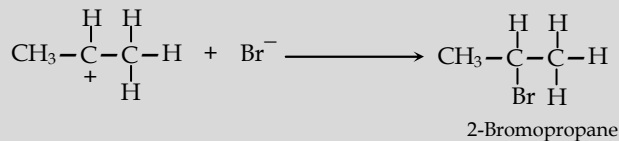
Mechanism of Markovnikov's addition: Electrophilic addition: Addition of hydrogen halide onto unsymmetrical alkene follows Markovnikov's rule and the reaction proceeds through electrophilic addition as follows.



Step (i): Addition of electrophile (H^+) onto carbon to carbon double bond of alkene to form more stable carbocation.

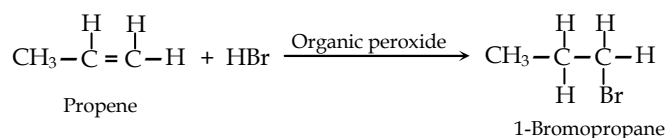


Step (ii): Addition of Br^- on carbocation to form Markovnikov's addition product.



Addition of hydrogen bromide onto unsymmetrical alkene in presence of organic peroxide: Anti-Markovnikov's addition/ peroxide effect/ Kharasch effect (1933)

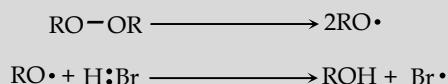
According to anti-Markovnikov's rule, when HBr is added to unsymmetrical unsaturated organic compound in presence of traces of organic peroxide, bromine adds to the unsaturated carbon having more number of hydrogen.



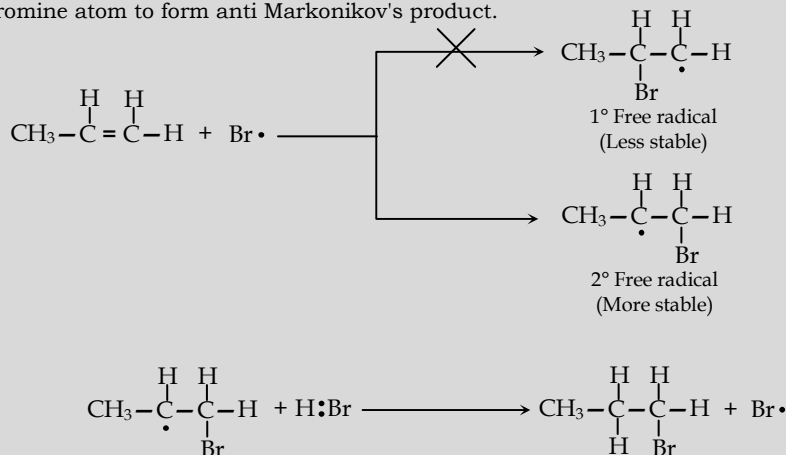
Mechanism of anti-Markovnikov's addition: Free radical addition

Addition of hydrogen bromide onto unsymmetrical alkene in presence of organic peroxide follows anti Markonikov's rule and the reaction proceeds through free radical addition as follow:

Step (i): Chain initiation steps: Generation of bromine free radical.



Step (ii): Chain propagation steps: Addition of bromine free radical onto carbon to carbon double bond of alkene to form more stable free radical followed by addition of stable free radical to new bromine atom to form anti Markonikov's product.



Step (iii): Chain termination steps: The reactive particles combine with one another or is captured by the wall of the reaction vessel, and a chain is terminated.

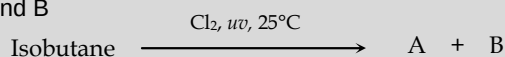
Only HBr in presence of traces of organic peroxide follows anti Markonikov's rule. Other reagents like HCl, HI do not follow this rule. This can be explained as follows.

H-Cl is highly polar hence does not undergo homolysis easily. HI undergo homolysis to give iodine free radicals which instantly combine to give I₂.

$H-X \longrightarrow H^\bullet + X^\bullet$	Homolytic bond dissociation energy ($kcal\ mol^{-1}$)
$H-Cl \longrightarrow H^\bullet + \text{Cl}^\bullet$	103 $kcal\ mol^{-1}$
$H-Br \longrightarrow H^\bullet + \text{Br}^\bullet$	88 $kcal\ mol^{-1}$
$H-I \longrightarrow H^\bullet + \text{I}^\bullet$	71 $kcal\ mol^{-1}$

Self Test

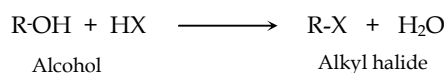
9.5 Identify A and B



9.6 Convert propene into

- a. 1-Chloropropane b. 2-Chloropropane

3. **From alcohol:** Alkyl halides are almost always prepared from corresponding alcohols. Alcohol reacts with halogen acid, phosphorous halides or thionyl chloride to give alkyl halide.
- a. **Reaction with halogen acid (HX):** Alcohol reacts with halogen acid (HCl, HBr, HI) to give corresponding alkyl halide.



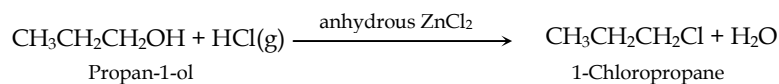
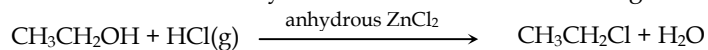
The reaction is carried out by passing the dry hydrogen halide gas into the alcohol or by heating alcohol with the concentrated aqueous acid. The conditions of reaction depend on both the nature of alcohol and the halogen acid.

Reactivity of alcohol: $3^\circ > 2^\circ > 1^\circ$

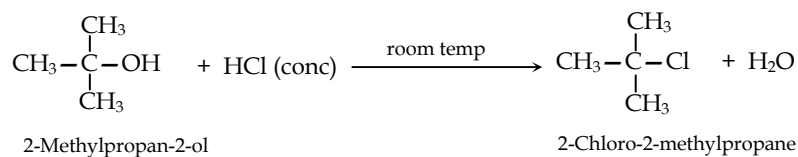
With HCl

Reactivity of HX: $HI > HBr > HCl$

HCl is the least reactive of halogen acids. Hence HCl requires anhydrous $ZnCl_2$ to react with primary and secondary alcohol to give chloroalkane. The reaction of primary and secondary alcohol with HCl in presence of anhydrous $ZnCl_2$ to give chloroalkane is called **Grooves' process**. A mixture of conc. HCl and anhydrous $ZnCl_2$ is called Lucas reagent.

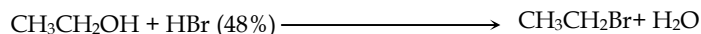


Tertiary alcohol being more reactive does not need any catalyst even at room temperature.

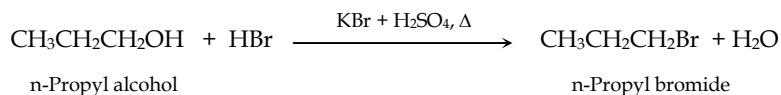


With HBr

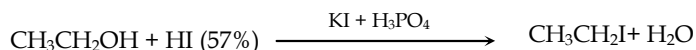
HBr is more reactive than HCl. Concentrated HBr is used to prepare alkyl bromides. Primary alcohol requires little concentrated H_2SO_4 but secondary and tertiary alcohol do not need any catalyst.



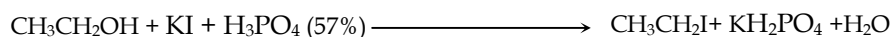
The reaction can be carried out by heating alcohol with NaBr or KBr in concentrated H_2SO_4 .



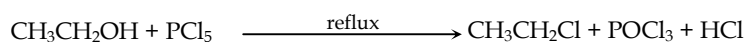
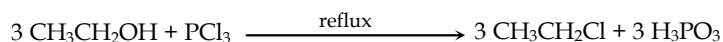
HI is the most reactive of all HX. Concentrated HI is used to prepare alkyl iodides. No catalyst is required.



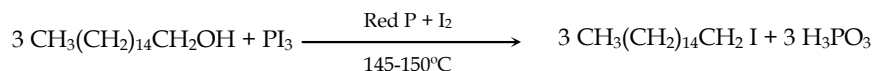
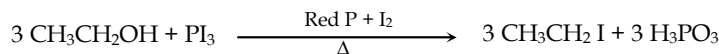
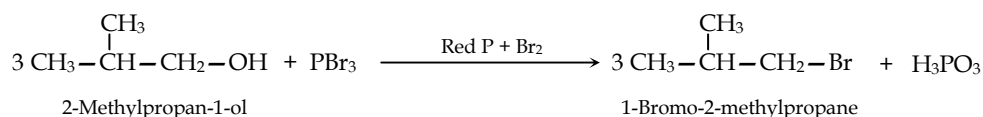
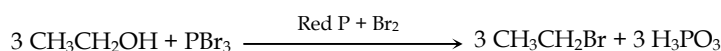
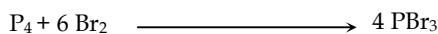
The reaction can be carried out by heating alcohol with NaI or KI in 95% H_3PO_4 .



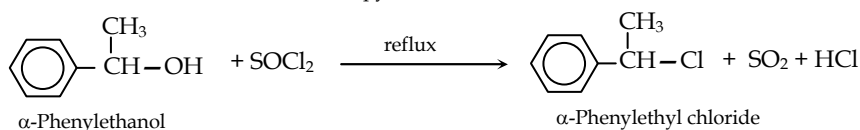
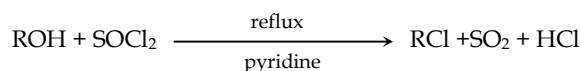
- b. Reaction with phosphorous halide (PX_3 or PCl_5):** When alcohol is refluxed with phosphorous trihalide or phosphorous pentahalide, corresponding alkyl halide is formed.



PBr_3 , PI_3 are unstable hence a mixture of red phosphorous and Br_2 or I_2 is used.

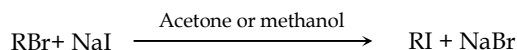
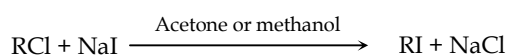
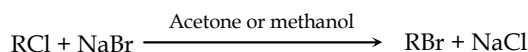


- c. Reaction with thionyl chloride (SOCl_2):** Alcohol reacts with thionyl chloride in presence or absence of pyridine to give alkyl chloride. This is called **Darzens procedure** (1911).



Advantages of this method

- i. The product is obtained in pure form since the side products are acidic gases, which can be removed by heating them or by absorbing them by pyridine.
 - ii. Any excess SOCl_2 can be removed by distillation easily due to its low boiling point (b.pt. 76°C).
4. **By halide exchange method:** Alkyl iodides can be prepared by treating corresponding alkyl chlorides or bromides with NaI in presence of acetone or methyl alcohol. This reaction is called **Finkelstein or Conant-Finkelstein reaction**.



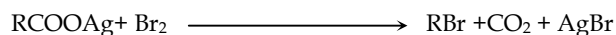
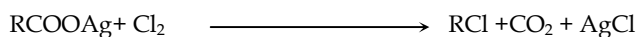
Fluoroalkanes can be obtained indirectly by using inorganic fluorides like AgF , Hg_2F_2 etc.



This method was used by Swarts (1898) and called **Swarts Reaction**.

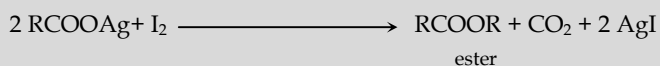
5. **From silver salt of carboxylic acid: Hunsdiecker or Borodine-Hunsdiecker reaction(1935)**

Silver salt of carboxylic acid reacts with chlorine or bromine in carbon tetrachloride as solvent to form corresponding alkyl halide. This is called Hunsdiecker or Borodine-Hunsdiecker reaction.



i. This reaction follows free radical mechanism

ii. Iodine, however, forms ester with silver salt of carboxylic acid. This is called **Birnbaum-Simonini reaction (1892)**.

**9.1.2.6 Physical Properties**

1. CH_3Cl , CH_3Br , and $\text{C}_2\text{H}_5\text{Cl}$ are colourless gases, some higher members (upto C_{18}) are sweet smelling liquids and higher members are odourless solids.
2. Alkyl halides are insoluble in water but soluble in organic solvents like benzene, ether, chloroform etc. Although alkyl halides are polar compounds, their polarity is not sufficient to form intermolecular H-bond with water.
3. They burn with green edged flame.
4. Alkyl bromides and iodides are heavier than water whereas alkyl chlorides are lighter than water. (Generally organic compounds are lighter than water. To be heavier than water, the organic compounds must have heavier atom like Br, I or more atoms like Cl).

5.

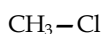
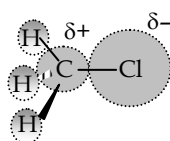
- a. For same halogen atom, the boiling points of alkyl halides increase with increase in carbon number.



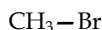
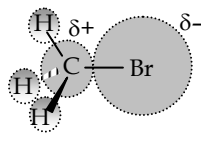
- b. For same alkyl group, the boiling points of alkyl halides



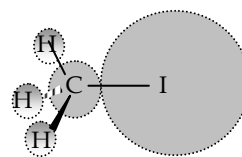
This is because of the fact that, with the increase in size and mass of the molecule, the magnitude of Vander Waals' forces of attraction increases. For example,



b.pt = -24°C



b.pt = 5°C



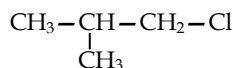
b.pt = 43°C

- c. In isomeric alkyl halides, boiling point decreases with increase in branching. This is because with branching, the shape of molecule tends to be spherical i.e. surface area of molecule decreases due to which intermolecular force of attraction (magnitude of Vander Waals' forces of attraction) decreases. For example,



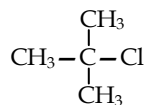
n-Butyl chloride

b. pt. 78°C



Isobutyl chloride

b. pt. 68°C



tert-Butyl chloride

b.pt. 51°C

Self Test

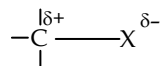
9.7 Explain

- 1-Chloropropane is not soluble in water
- n*-Propyl chloride is liquid (b. pt. = 47°C) but *n*-Butane is gas (b. pt. = 0°C)
- Ethyl chloride is gas but ethyl iodide is liquid
- tert*-Butyl chloride has lower boiling point than *n*-Butyl chloride

9.8 Write all possible monobromoderivatives of propane and arrange them in the decreasing order of their boiling point.

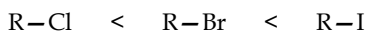
9.1.2.7 Chemical Properties

Alkyl halides are a highly reactive class of aliphatic compounds which undergo a wide range of chemical reactions forming a large variety of other organic compounds. Hence they are extremely important in organic synthesis. The high reactivity of alkyl halides is due to polar nature of carbon to halogen bond.

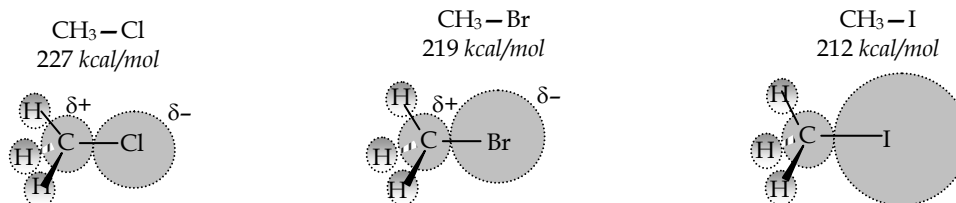


Element	Electronegativity (Pauling scale)
C	2.4
F	4.0
Cl	3.0
Br	2.8
I	2.5

In general, for a given alkyl group, the reactivity is in the order



This can be explained on the fact that C-X bond dissociation energy decreases in the order C-Cl > C-Br > C-I. For example, *heterolytic bond dissociation energy* for methyl halides are:

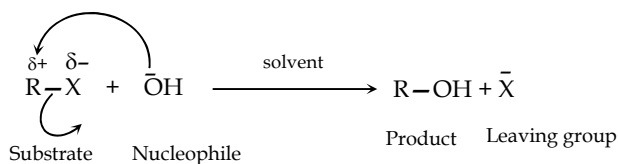


The chemical properties of alkyl halides can be categorized as follows;

- | | |
|---------------------------------------|--------------------------|
| i. Nucleophilic substitution reaction | ii. Elimination reaction |
| iii. Reaction with metals | iv. Reduction |
| v. Rearrangement reaction | |

A. Nucleophilic substitution reaction

In alkyl halides, carbon to halogen bond is polar since halogen is more electronegative than carbon. Therefore, nucleophile attacks the carbon bearing the halogen atom to give substitution reaction. A substitution reaction initiated by a nucleophile is called nucleophilic substitution reaction.



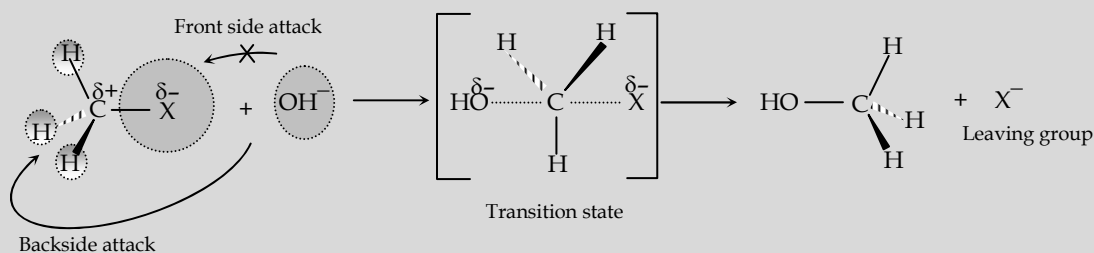
Depending upon nature of substrate, nature of nucleophile and nature of solvent, nucleophilic substitution reaction occurs by two different mechanisms;

S_N2 reaction and S_N1 reaction.

S_N2 reaction: S_N2 stands for bimolecular nucleophilic substitution reaction. Here, rate of reaction depends upon concentration of substrate and concentration of nucleophile.

Rate = k[RX] [nucleophile]

Generally, primary alkyl halides follow with this type of mechanism. Here, nucleophile attacks the substrate from backside (rear side, opposite side) of the leaving group. Nucleophilic attack from front side (from the side of the leaving group) is difficult since nucleophile and leaving group both are electron rich.



S_N2 reaction mechanism: Complete inversion of configuration from substrate to product.

As the nucleophile starts to form bond with the substrate, leaving group also starts to break the bond with the substrate and finally leaving group leaves to give the product. This means the reaction takes place in a single step without any intermediate. Due to backside attack of nucleophile, the product is obtained with complete inversion of configuration like inversion of an umbrella inside out in a strong gale. This is called *Walden inversion*.

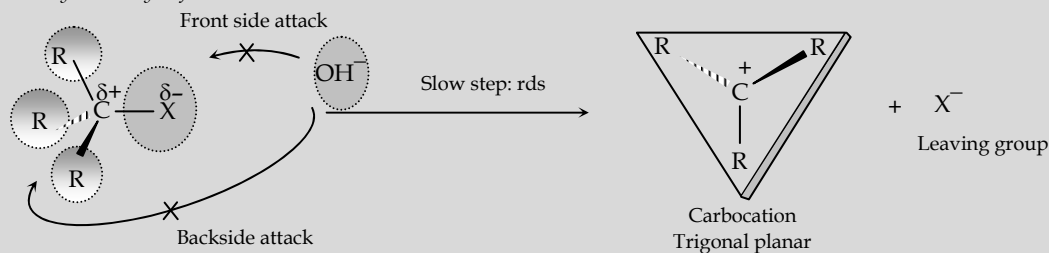
S_N1 reaction: S_N1 stands for unimolecular nucleophilic substitution reaction. Here, rate of reaction depends upon concentration of substrate only but not upon concentration of nucleophile.

$$\text{Rate} = k[\text{RX}]$$

Tertiary alkyl halides follow with this type of mechanism. The reaction occurs in two different steps;

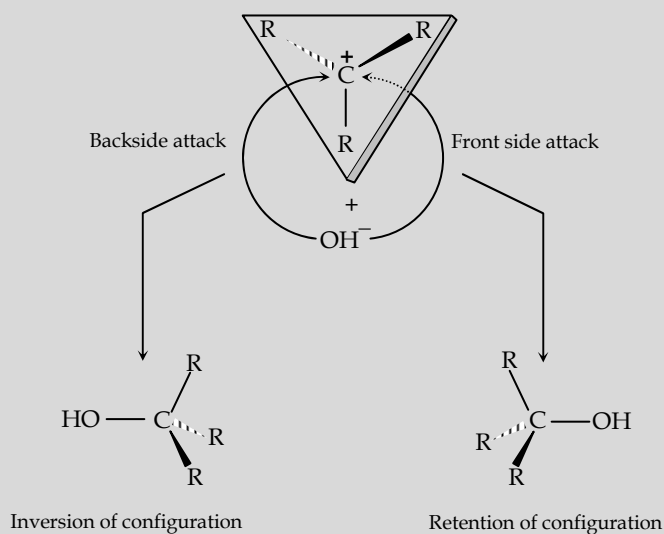
Step I: Substrate undergoes heterolysis slowly to form carbocation

Nucleophilic attack from front side (from the side of the leaving group) is difficult since nucleophile and leaving group both are electron rich. Nucleophilic attack from backside (rear side, opposite side) of the leaving group is also difficult due to bulky alkyl groups (steric hindrance). Hence, nucleophile cannot attack the substrate directly. Here, substrate undergoes heterolysis slowly to form carbocation.

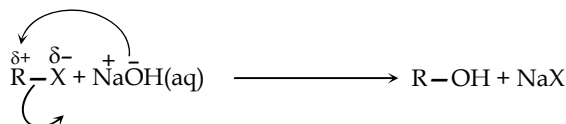


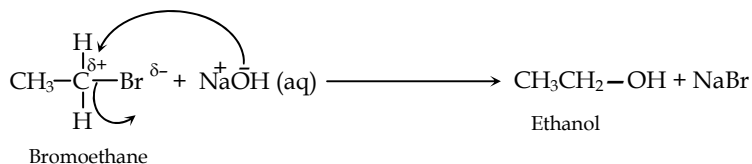
Step II: Nucleophile attacks the carbocation to give product.

Since carbocation is planar, nucleophile attacks from either side to give products with retention of configuration and inversion of configuration. However, just formed carbocation is not totally free. For certain time, it remains associated with the leaving group. For that time period, nucleophilic attack from the side of the leaving group is hindered. Hence in S_N1 reaction, the product is obtained with inversion of configuration and retention of configuration but the former always exceeds the latter.

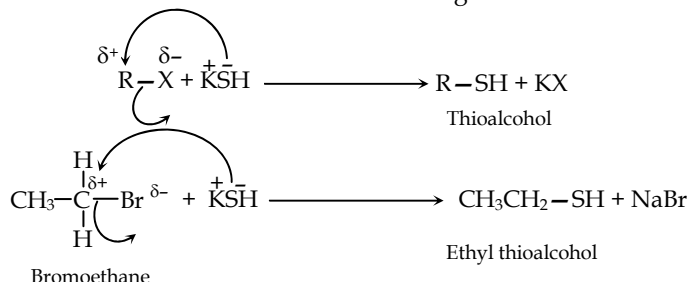


- 1. Action of aqueous caustic alkali (Substitution by OH^-): Formation of alcohol:** Alkyl halide reacts with aqueous solution of caustic alkali (NaOH or KOH) to give alcohol.



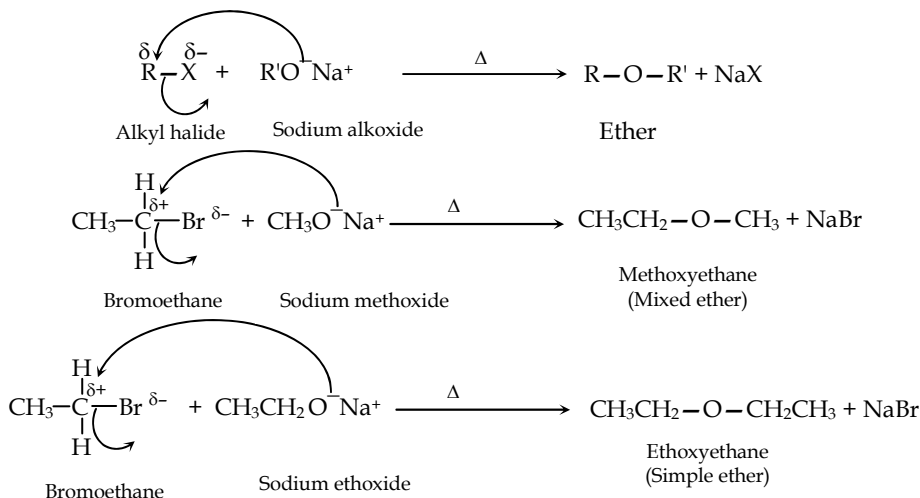


2. **Action of alcoholic potassium bisulphide (Substitution by SH^-): Formation of thioalcohol:** Alkyl halide reacts with alcoholic solution of KSH to give thioalcohol.



Water must be avoided as it causes hydrolysis of haloalkanes.

3. **Action of alcoholic sodium alkoxide (Substitution by RO^-): Formation of ether (Williamson's synthesis):** When alkyl halide is heated with sodium or potassium alkoxide, ether is obtained. This method is called Williamson's synthesis. This is a nucleophilic substitution reaction and alkoxide ion (RO^-) acts as a nucleophile.

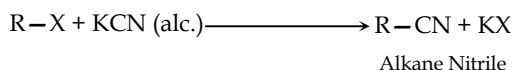


The importance of this method lies on its versatility; simple as well as mixed ethers can be obtained by this method.

Self Test

9.9 Convert ethyl alcohol into ethyl methyl ether.

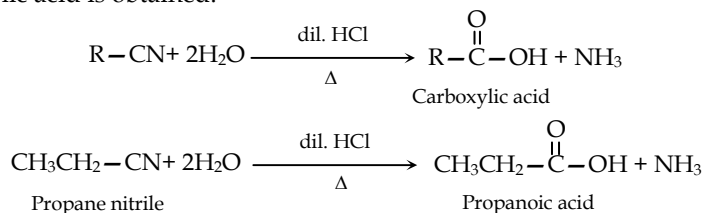
4. **Action of alcoholic potassium cyanide and alcoholic silver cyanide (Substitution by CN^-)**
- a. **Action of alcoholic NaCN or KCN: Formation of alkane nitrile:** When alkyl halide is heated with ethanolic potassium cyanide, alkane nitrile (cyanide) is obtained (little isocyanide is also obtained).



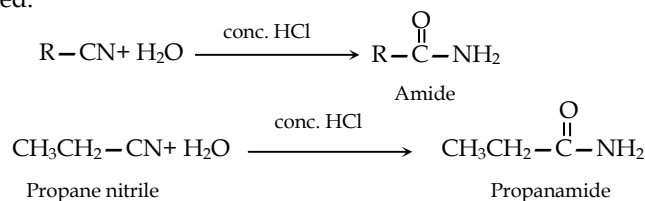


Nitrile gives different classes of organic compounds with different reagents.

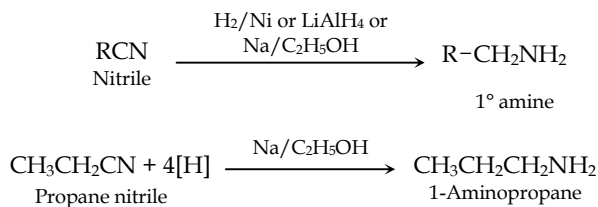
- i. **Formation of carboxylic acid (Complete hydrolysis):** When nitrile is treated with *dilute HCl*, carboxylic acid is obtained.



- ii. **Formation of amide (Partial hydrolysis):** When nitrile is treated with concentrated HCl, amide is obtained.

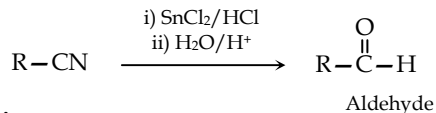


- iii. **Formation of primary amine (Reduction):** When nitrile is reduced with H_2/Ni , LiAlH_4 or $\text{C}_2\text{H}_5\text{OH}/\text{Na}$, primary amine is obtained.



Reduction of nitrile into primary amine with $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ is called **Mendius reaction**.

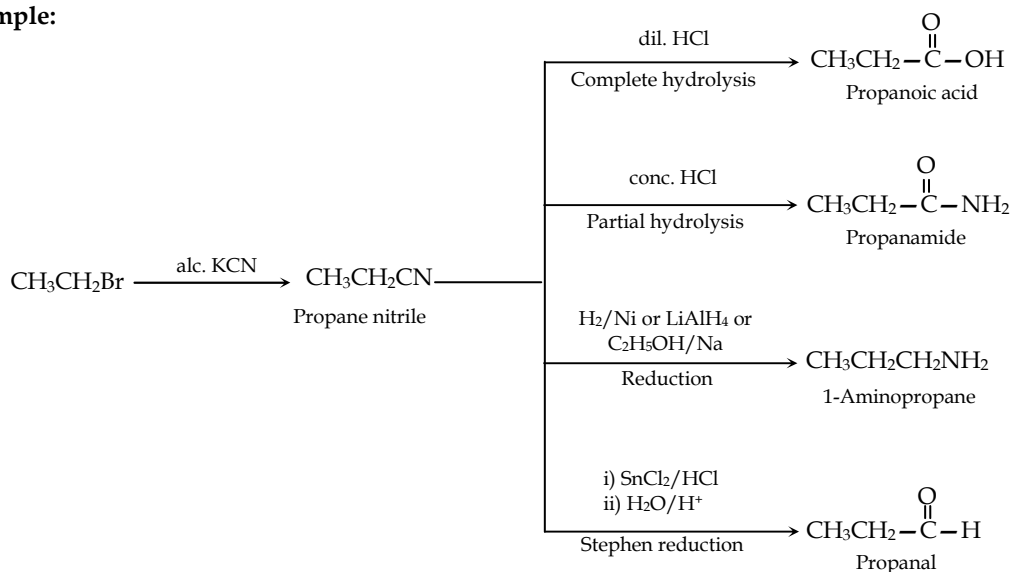
- iv. **Formation of aldehyde (Stephen's reduction):** When nitrile is treated with SnCl_2 and concentrated HCl followed by hydrolysis, aldehyde is obtained.



Advantage of nitrile synthesis

Nitrile synthesis has advantage of increasing carbon chain length by one carbon atom.

Example:

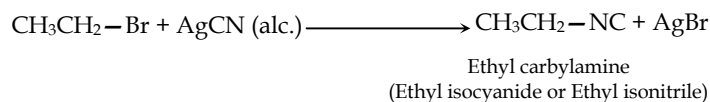
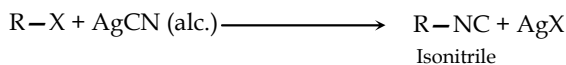


Self Test

9.10 Starting from ethanol, outline the formation of propanoic acid.

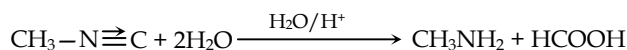
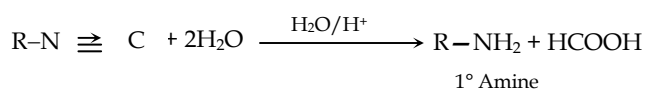
9.11 Convert ethyl alcohol into *n*-propyl amine.

- b. **Reaction with alcoholic AgCN: Formation of isonitrile:** When alkyl halide is heated with ethanolic silver cyanide, isonitrile (isocyanide) is obtained (little nitrile is also obtained).

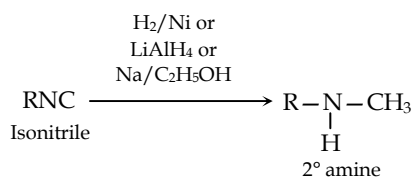


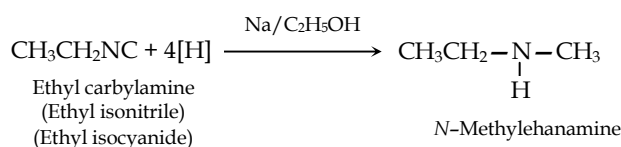
Isonitrile gives different classes of organic compounds with different reagents.

- i. **Formation of primary amine (hydrolysis):** Isonitrile is hydrolyzed to primary amine and formic acid.

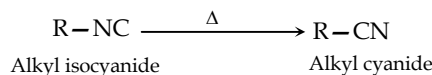


- ii. **Formation of secondary amine (Reduction):** Isonitrile is reduced to secondary amine by H_2/Ni or LiAlH_4 .

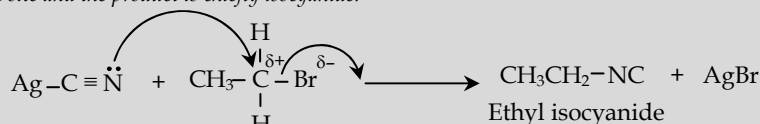




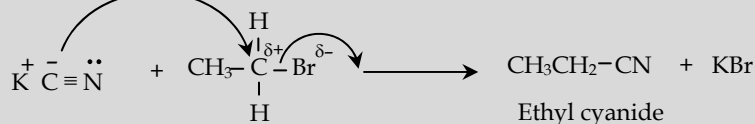
iii. **Formation of nitrile (rearrangement):** When isonitrile is heated for a long time, nitrile is obtained.



In AgCN , silver to carbon bond has covalent character, hence nitrogen atom with its lone pair of electron acts as nucleophilic site and the product is chiefly isocyanide.



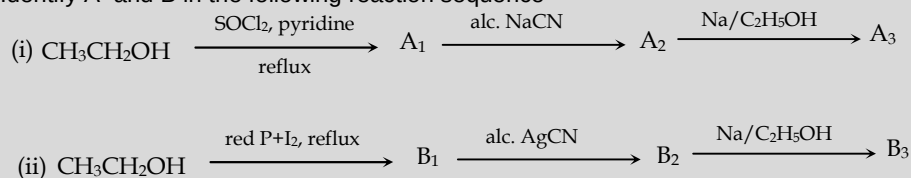
KCN is ionic compound. Here, carbon atom has negative charge and acts as nucleophilic site and the product is chiefly cyanide.



A nucleophile in which more than one atom can act as nucleophilic site is called ambident nucleophile. NO_2^- , CN^- etc. are ambident nucleophile.

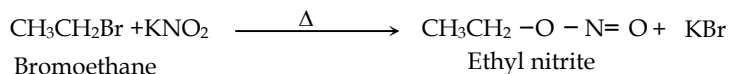
Self Test

9.12 Identify A and B in the following reaction sequence

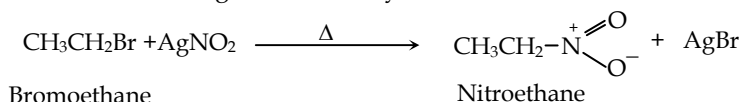


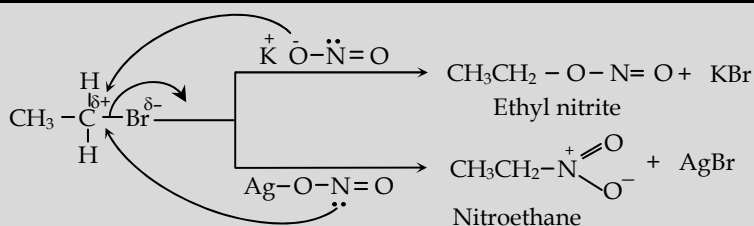
5. Action of KNO_2 or AgNO_2

- a. **Action of alcoholic KNO_2 : Formation of nitrite:** When a primary alkyl halide (especially bromides and iodides) is heated with ethanolic solution of potassium nitrite, alkyl nitrite is obtained along with little nitroalkane.



- b. **Action of alcoholic AgNO_2 : Formation of nitro compound:** When a primary alkyl halide (especially bromides and iodides) is heated with aqueous ethanolic solution of silver nitrite, nitroalkane is obtained along with little alkyl nitrite.



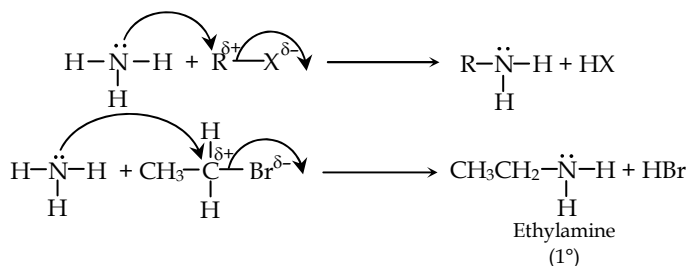


KNO_2 is ionic compound. Here, oxygen atom has negative charge and acts as nucleophilic site and the product is chiefly alkyl nitrite. In AgNO_2 , silver to oxygen bond has covalent character, hence nitrogen atom with its lone pair of electron acts as nucleophilic site and the product is chiefly nitroalkane. (an atom with negative charge is more nucleophilic than the atom having lone pair of electron).

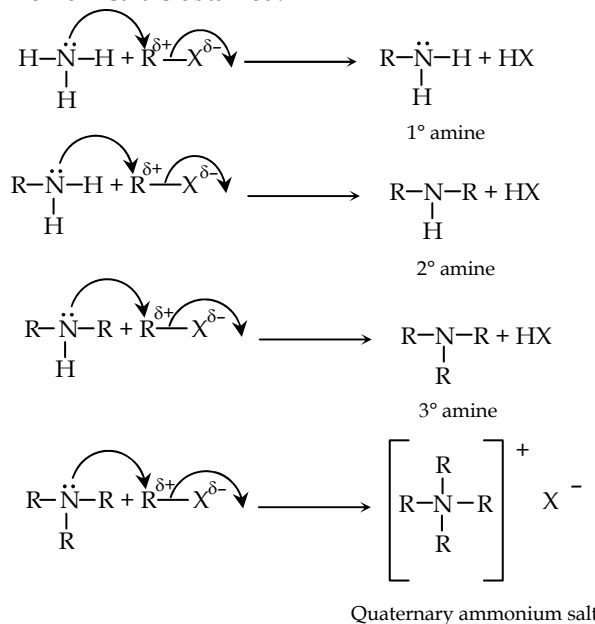
A nucleophile in which more than one atom can act as nucleophilic site is called ambident nucleophile. NO_2^- , CN^- etc. are ambident nucleophile.

6. **Action of Ammonia: Formation of amine (Hofmann's ammonolysis):** When an alkyl halide is heated with aqueous or alcoholic solution of ammonia at 100°C in a sealed tube or allowed to stand at room temperature, amine is formed. By varying the proportions of ammonia and alkyl halide, particular class of amine can be obtained in large proportion. The process of cleavage of the C-X bond by ammonia molecule is called *ammonolysis*.

- i. If ammonia is excess, primary amine is obtained.

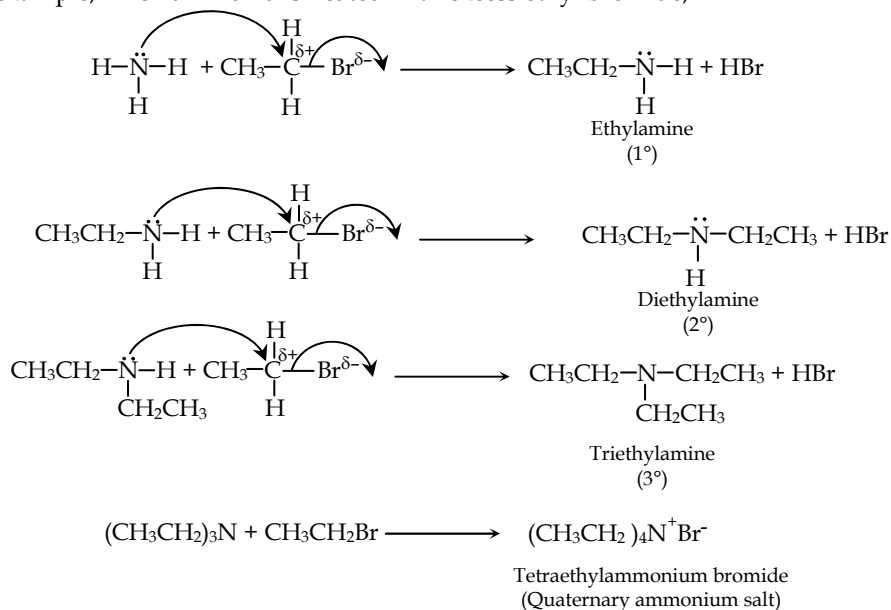


- ii. If alkyl halide is excess, a mixture of primary, secondary and tertiary amines and quaternary ammonium salt is obtained.

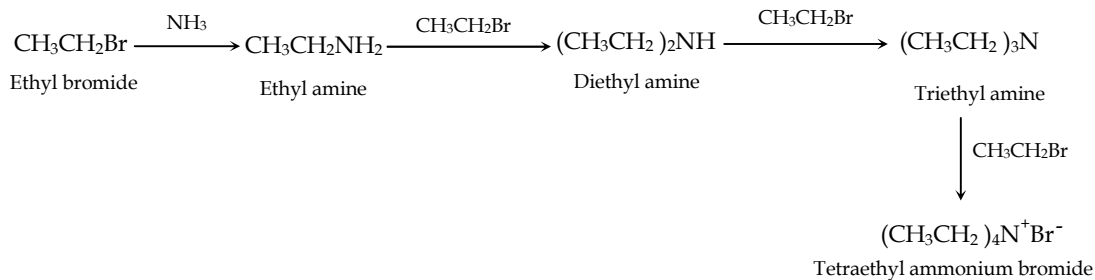


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For example, when ammonia is heated with excess ethyl bromide,

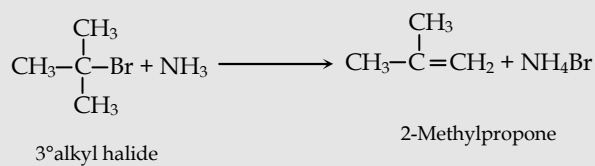


Or,



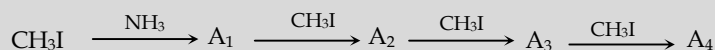
Note:

- The order of reactivity of halides with amines is $\text{RI} > \text{RBr} > \text{RCl}$
- This is nucleophilic substitution reaction. Here, ammonia, primary and secondary amine acts as nucleophile. Primary alkyl halide gives good yield but tertiary alkyl halide gives elimination reaction.

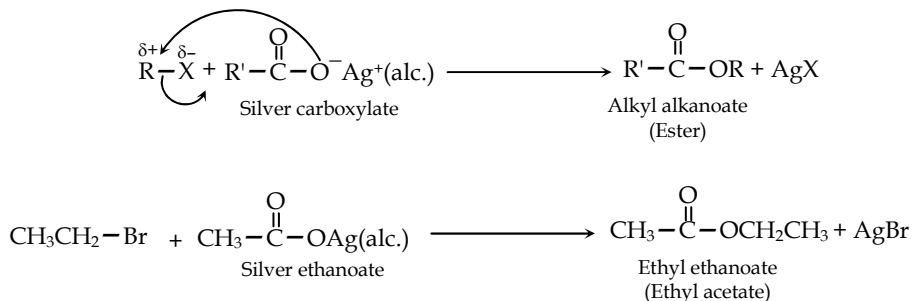


Self Test

9.13 Identify A in the following reaction sequence:

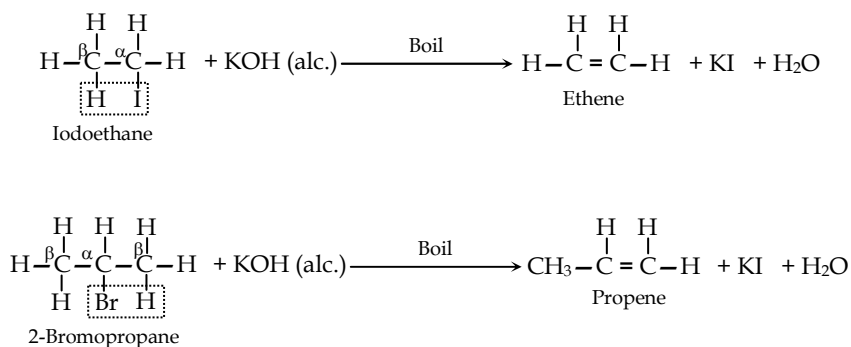


7. **Action of alcoholic silver salt of carboxylic acid: Formation of ester:** When alkyl halide is refluxed with alcoholic solution of silver salt of carboxylic acid, ester is formed.



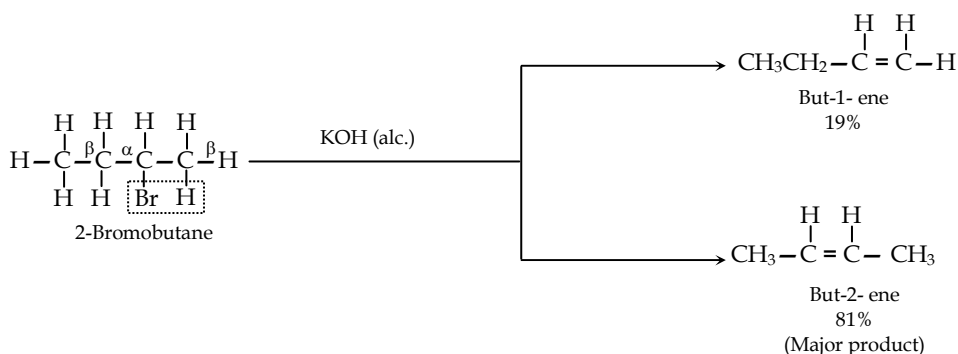
B. Elimination reaction: Dehydrohalogenation

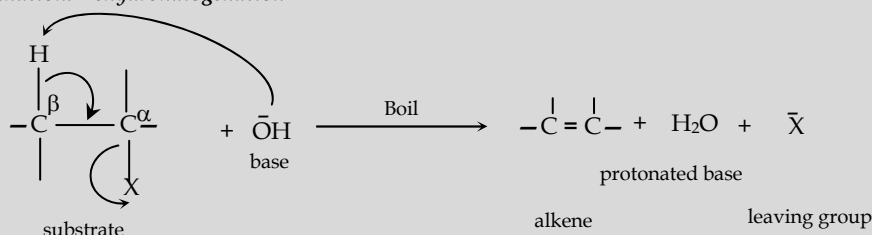
When alkyl halide is boiled with alcoholic solution of caustic alkali, elimination of elements of hydrogen halide takes place to give alkene. This is an elimination reaction and called dehydrohalogenation. This reaction belongs to 1,2 elimination since eliminating groups are removed from adjacent carbons and also called β -elimination since β -hydrogen is removed.



In dehydrohalogenation (also in dehydration), if there is possibility of forming two different alkenes, the more substituted alkene is formed as major product. This is called *Saytzeff rule*.

Example:



1,2-Elimination: Dehydrohalogenation

In dehydrohalogenation, OH^- (from alcoholic KOH) acts as a **base** (not nucleophile) and abstracts H^+ from β carbon.

Reactivity of alkyl halides towards dehydrohalogenation is $3^\circ > 2^\circ > 1^\circ$

Self Test

9.14 Write the structure of all alkenes obtained on heating following alkyl halides with alcoholic potassium hydroxide.

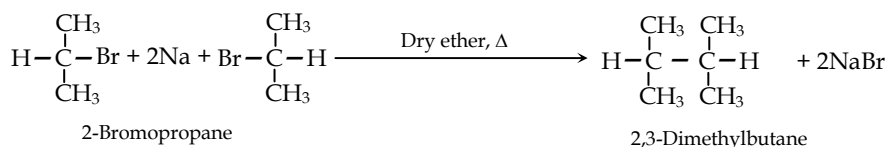
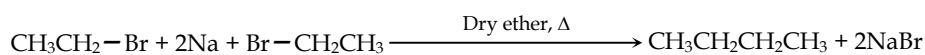
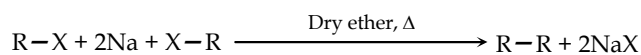
- a. 1-Chloropentane b. 2-Chloropentane
c. 3-Chloropentane d. 2-Chloro-2-methylbutane

9.15 Convert 1-Bromopropane into 2-Bromopropane and vice versa.

9.16 Convert 1-Chloropropane into 2-Chloropropane and vice versa.

C. Reaction with Metals

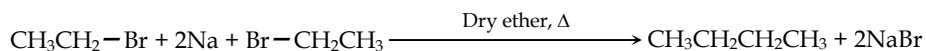
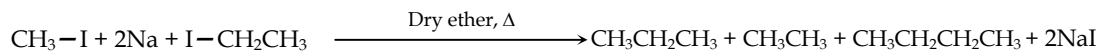
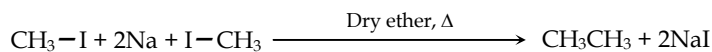
- a. **Reaction with sodium: Wurtz reaction:** When alkyl halide is heated with sodium metal in dry ether as solvent, two molecules of alkyl halide combine to give higher alkane. This reaction is called Wurtz reaction.



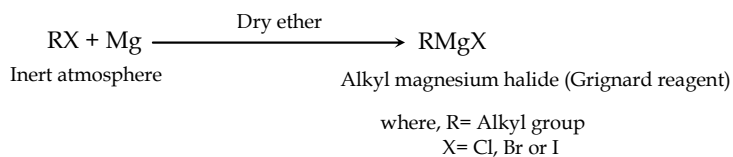
This method is suitable for the preparation of symmetrical alkanes (alkanes having even number of carbon atoms) of higher carbon number.

Limitations of Wurtz reaction

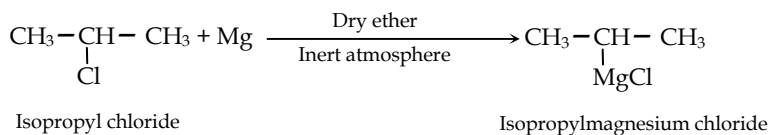
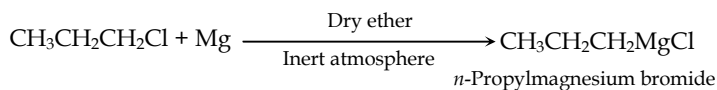
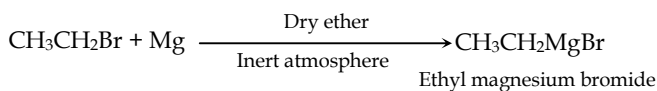
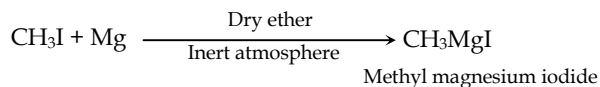
- i. This method is not suitable for the preparation of unsymmetrical alkanes and alkanes having odd number of carbon atoms. To prepare unsymmetrical alkanes, two different alkyl halides have to be used which gives a mixture of three different alkanes. For example, when a mixture of methyl iodide and ethyl iodide is heated with sodium metal in anhydrous ether, a mixture of ethane, propane and butane is formed.



- ii. Methane can not be prepared by this method.
- iii. Only primary and secondary alkyl halides give Wurtz reaction. 3° alkyl halide gives alkene.
- b. **Reaction with magnesium: Formation of Grignard reagent:** Alkyl halide reacts with magnesium metal in anhydrous ether to form **alkylmagnesium halide** which is commonly called Grignard reagent¹.



Examples,



1. Victor Grignard discovered and developed the use of Grignard reagent in 1900, so he was awarded Nobel prize for chemistry in 1912.

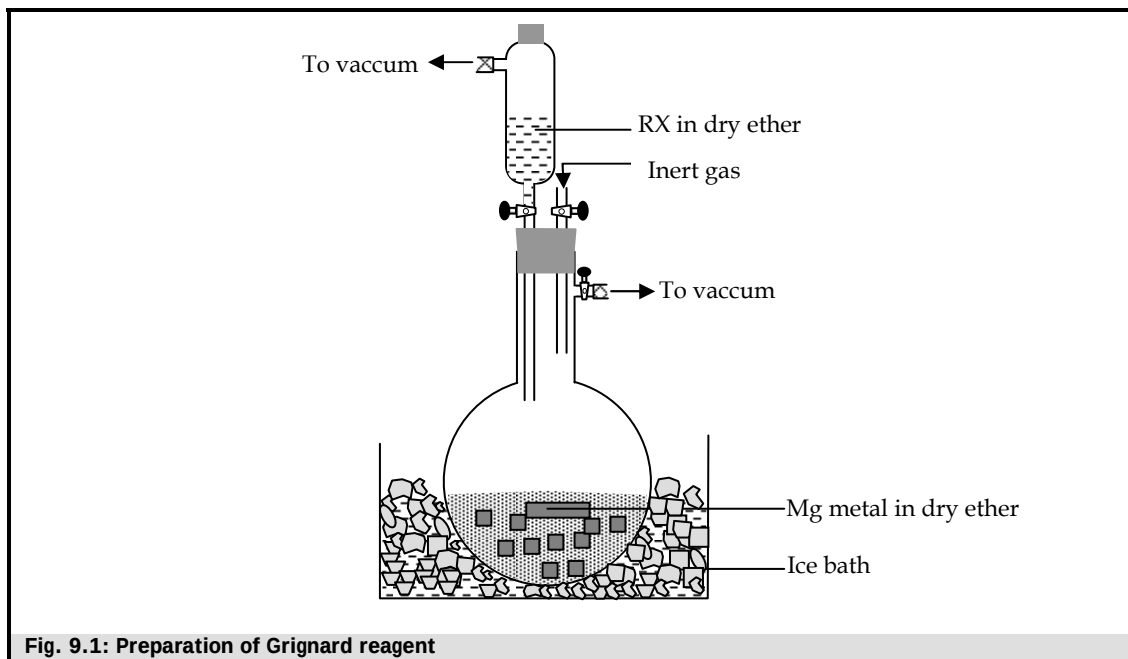
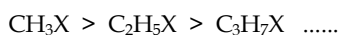
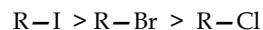


Fig. 9.1: Preparation of Grignard reagent

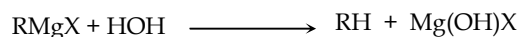
It is an **organometallic compound**². It is extremely valuable and versatile reagent from which various classes and varieties of organic compounds can be synthesized like alkanes, alcohols, carboxylic acids etc.

Rate of formation of Grignard reagent

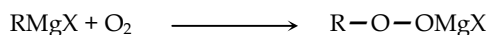


Precautions for the preparation of Grignard reagent

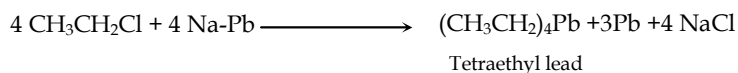
- i. The apparatus, solvent and the reactants must be dry. If water is present, it reacts with Grignard reagent to form alkane.



- ii. The reaction must be carried out in inert atmosphere. Otherwise oxygen of air reacts with Grignard reagent to form hydroperoxide.



- c. **Reaction with lead:** Ethyl chloride reacts with lead sodium alloy to form tetraethyl lead (TEL) which is used as anti-knocking additive to petrol.



² Organometallic compounds are those compounds in which metal atom is directly bonded to carbon.

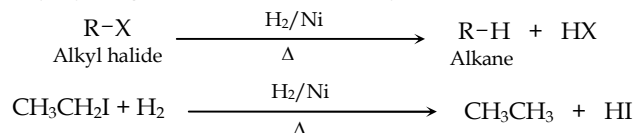
C-M, where M = Li, Mg, Al, Zn, Cd, Hg, Sn, Pb, Cu etc. For example,

RLi (Alkyl lithium), *Al(C₂H₅)₃* (Triethyl aluminium), *(C₂H₅)₄Pb* (Tetraethyl lead), *R₂CuLi* (Lithium dialkyl copper) etc.

D. Reduction

Alkyl halide can be reduced to alkane by different ways using different reagents.

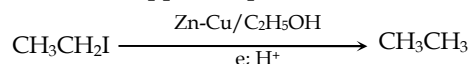
- a. **Catalytic hydrogenolysis:** Alkyl halides (alkyl iodides, bromides and chlorides) are reduced into alkanes by hydrogen in presence of catalyst like Ni, Pt or Pd.



- b. **Reduction with metal hydride**

- Reduction with LiAlH_4 (Lithium aluminium hydride): Primary and secondary alkyl halides (not tertiary alkyl halides) are reduced into alkanes by LiAlH_4 .
- Reduction with NaBH_4 (Sodium borohydride): Secondary and tertiary alkyl halides (not Primary) are reduced into alkanes by NaBH_4 .

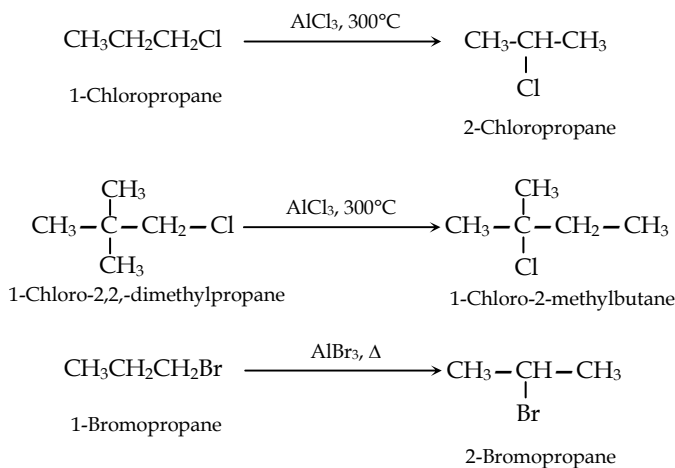
- c. **Reduction with dissolving metals:** Alkyl halides are reduced into alkanes by $\text{Zn}/\text{CH}_3\text{COOH}$, Zn/HCl , Zn/NaOH , zinc-copper couple and ethanol, etc.



The reaction occurs due to transfer of electron from metal to organic molecule being reduced followed by transfer of proton from solvent. Nascent hydrogen is not responsible.

E. Rearrangement reaction

Alkyl halides on heating with aluminium chloride undergo rearrangement reaction.

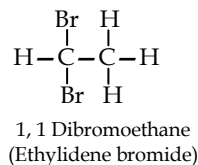


9.1.3 Dihaloalkanes

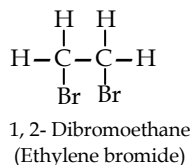
When any two of the hydrogen atoms in alkanes are replaced by halogen atoms, the resulting organic compounds are called dihaloalkanes. They have general formula $\text{C}_n\text{H}_{2n}\text{X}_2$.

There are following types of dihaloalkanes.

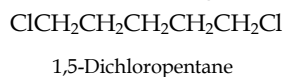
- i. **1, 1-Dihalides (geminal dehalide or gem-dihalide) or alkylidene dihalide:** two halogen atoms on the *same* carbon atom



- ii. **1, 2-Dihalides (vicinal dihalide or vic-dihalide) or alkylene halide:** two halogen atoms on the *adjacent* carbon atoms



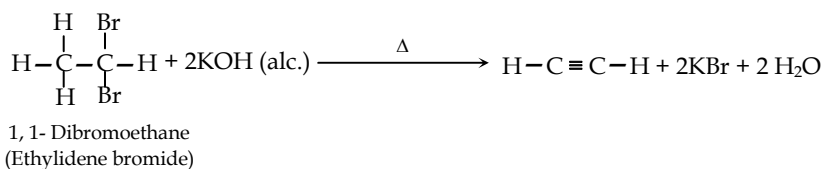
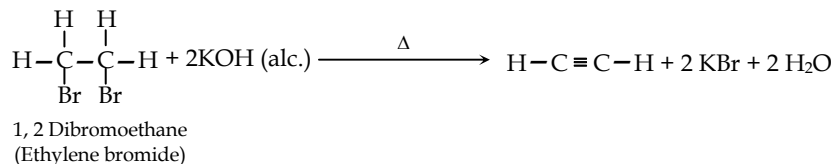
- iii. **Polymethylene dihalides (α,ω -dihalides)** two halogen atoms on the *terminal* carbon atoms of the chain



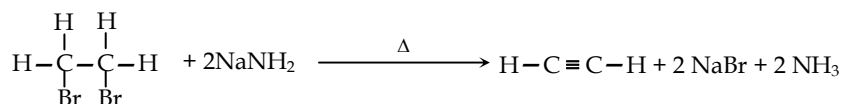
- iv. Dihalogen compounds having two halogen atoms other than mentioned above

Some Reactions

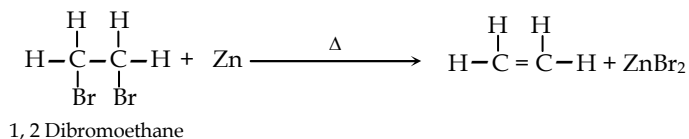
1. **Dehydrohalogenation:** When 1, 1-dihalide or 1,2-dihalide is boiled with alcoholic solution of caustic alkali, elimination of elements of hydrogen halide takes place to give alkyne. This is an elimination reaction and called dehydrohalogenation.



Alternatively, sodamide in liquid ammonia can be used for the better yield of alkyne.



2. **Dehalogenation:** When 1, 2-dihalide is heated with zinc dust, dehalogenation takes place to give alkene.



Self Test

9.17 Convert ethane into ethyne.

9.18 Write the structural isomers of $C_2H_4Br_2$ and give their IUPAC as well as common name. Convert one isomer into another and vice versa.

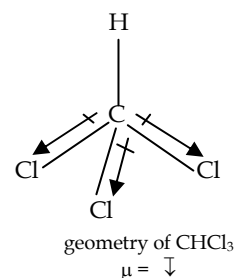
9.19 Write the structure and give IUPAC name of

- | | |
|-----------------------|--------------------------|
| i. Methylene chloride | ii. Ethylidene bromide |
| iii. Ethylene bromide | iv. Propylidene chloride |

9.1.4 Trihaloalkanes: Chloroform

Some important trihalogen derivatives of alkanes are as follows;

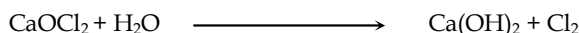
Formula	IUPAC name	Common name	Medicinal value
$CHCl_3$	Trichloromethane	Chloroform	Anaesthetic
$CHBr_3$	Tribromomethane	Bromoform	Medicine for whooping cough
CHI_3	Triiodomethane	Iodoform	Antiseptic

9.1.4.1 Laboratory Preparation of Chloroform**Theory**

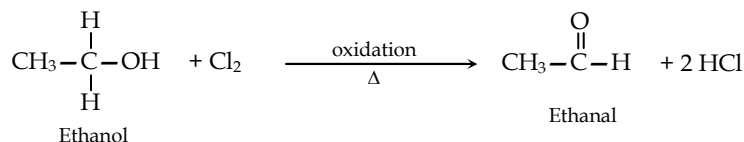
Chloroform is prepared in laboratory by distilling ethyl alcohol or acetone with aqueous paste of bleaching powder. In the reaction, aqueous bleaching powder acts as oxidizing agent, chlorinating agent as well as hydrolyzing agent.

Preparation of chloroform involves following steps of reactions

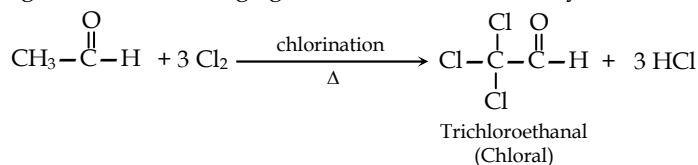
Bleaching powder reacts with water

**1. Preparation from Ethyl Alcohol**

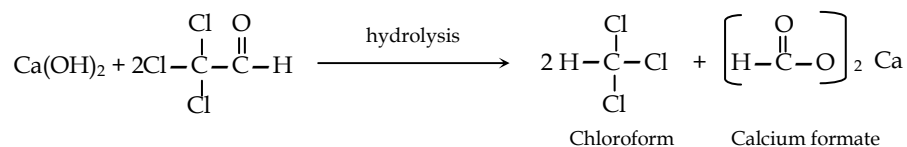
- a. **Oxidation of ethyl alcohol into acetaldehyde:** The liberated chlorine gas acting as an oxidizing agent, oxidizes ethyl alcohol to acetaldehyde.



- b. **Chlorination of acetaldehyde into trichloroacetaldehyde (Chloral):** The excess of liberated chlorine acting as the chlorinating agent, chlorinates acetaldehyde into chloral.

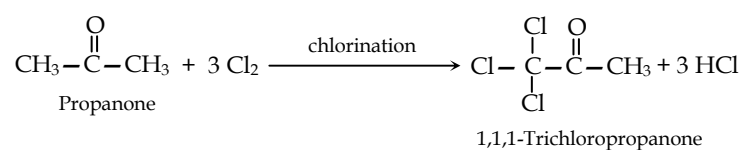


- c. **Hydrolysis of chloral into chloroform:** The calcium hydroxide obtained from hydrolysis of bleaching powder, hydrolyzes chloral to chloroform.



2. Preparation from acetone

- a. **Chlorination of acetone into trichloroacetone:** The liberated chlorine gas from bleaching powder acting as the chlorinating agent, which chlorinates acetone into trichloroacetone.



- b. **Hydrolysis of trichloroacetone into chloroform:** The calcium hydroxide obtained from hydrolysis of bleaching powder, hydrolyzes trichloroacetone to chloroform.

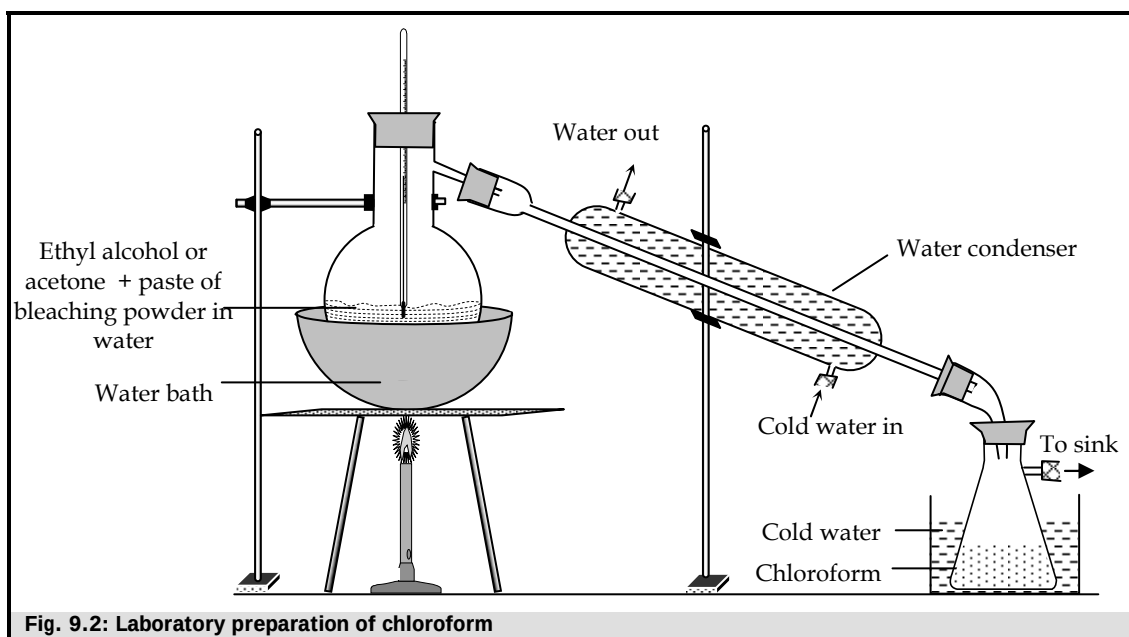
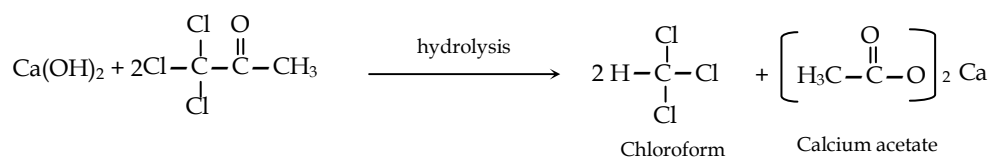


Fig. 9.2: Laboratory preparation of chloroform

Procedure

A paste of bleaching powder in water is prepared by mixing 100 g of bleaching powder in 200 ml water and transferred into one liter round bottom flask. 25 ml of ethyl alcohol or acetone is added into the flask. The flask is fitted with water condenser provided with receiver then heated gently on a water bath. Chloroform formed distills over which is collected in the water cooled receiver. It is transferred into a separating funnel, where chloroform forms the lower layer and such layer is separated.

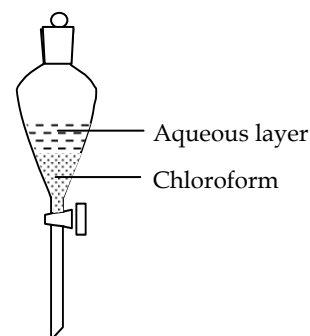


Fig. 9.3: Separating funnel, separation of chloroform from reaction mixture

Purification

The main impurities in chloroform are acids and water. To remove these impurities following steps are applied:

1. Chloroform is treated with dilute NaOH in a separating funnel to dissolve acidic impurities and aqueous layer is rejected.
2. Chloroform is washed with water 2-3 times to remove water soluble impurities and aqueous layer is again rejected.
3. Chloroform is dried with anhydrous CaCl_2 .
4. Finally, chloroform is redistilled at about 60 to 65°C to get pure and dry chloroform.

In this way, chloroform is prepared in laboratory in pure and dry state.

Industrial Preparation

i. Manufacture from carbon tetrachloride

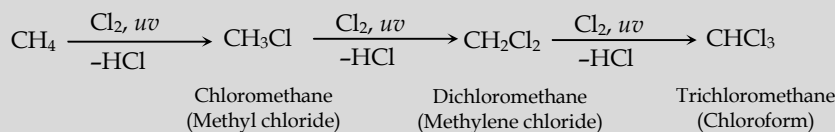
Chloroform can be prepared on a large scale by partial reduction of carbon tetrachloride using iron fillings and water.



Limitation of the method: Chloroform manufactured by this method is used as a solvent only as it is not enough pure for its use as an anesthetic agent.

ii. Manufacture from methane

Chloroform can also be prepared by the chlorination of methane with calculated quantity of chlorine in the presence of sunlight.



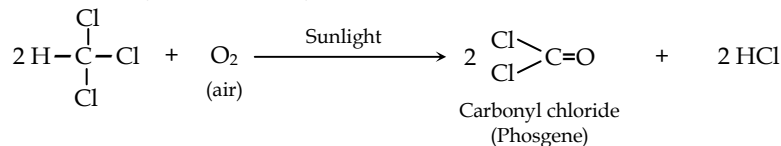
In the reaction, a mixture of methyl chloride, methylene chloride, chloroform and carbon tetrachloride is obtained, from which chloroform is separated by fractional distillation.

9.1.4.2 Physical Properties of Chloroform

1. It is colorless sweet smelling liquid.
2. It boils at 61°C and its freezing point is -63°C.
3. It is heavier than water (sp. gr. = 1.485 at 25°C)
4. The vapor of chloroform induces unconsciousness therefore it can be used as an anesthetic agent.
5. It acts as solvent, which dissolves substances like oils, fats, waxes etc.

9.1.4.3 Chemical properties of chloroform

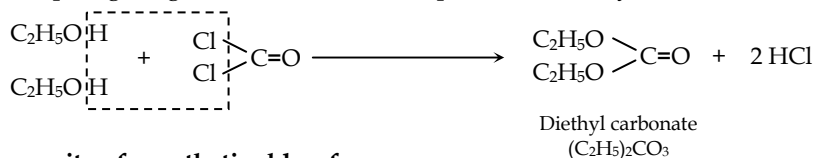
1. **Reaction with air:** Chloroform reacts with *air* in presence of *sunlight* slowly forming carbonyl chloride which is commonly called *phosgene*. Phosgene is a highly poisonous gas which attacks to the central nervous system and may cause death.



How to store chloroform?

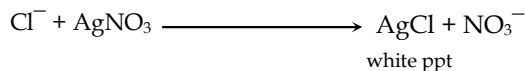
To prevent the formation of poisonous phosgene gas, chloroform is stored in the following ways for the given reasons:

- It is always kept in dark coloured bottles so that it could be protected from exposure to sunlight.
- It is filled well up to the stopper so that no air be left inside the bottles.
- A small amount of ethanol (1%) is added to the chloroform while filling the bottle. Ethanol converts phosgene gas, if formed, into non-poisonous diethyl carbonate.

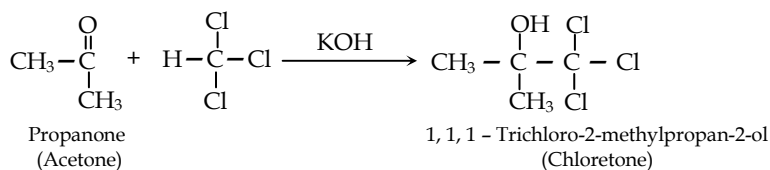


Test for the purity of anesthetic chloroform

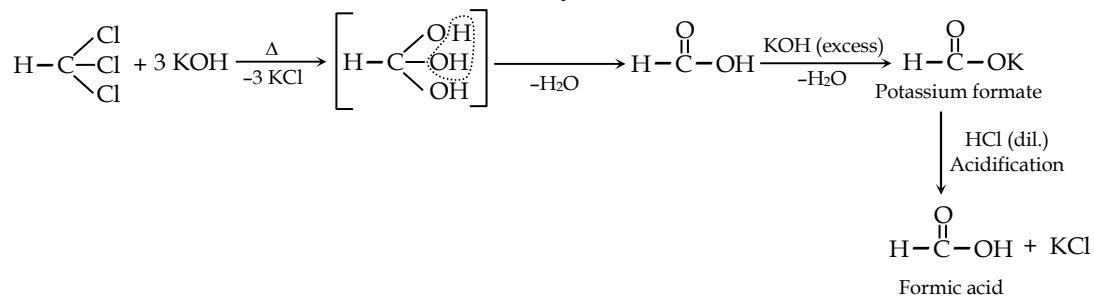
Impure chloroform containing phosgene (due to the oxidation of chloroform in air) produces white precipitate with aqueous AgNO₃ due to Cl⁻ formed (from HCl) during the phosgene formation.



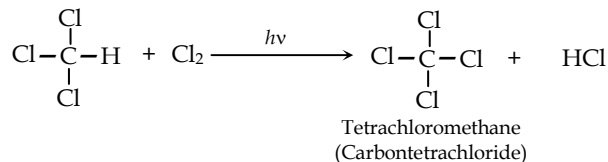
2. **Reaction with acetone:** Chloroform undergoes condensation with acetone in presence of caustic alkali to form chloretone (colorless crystalline solid, *m.pt.* 97°C). Chloretone is a hypnotic drug (used as a sleep-inducing agent).



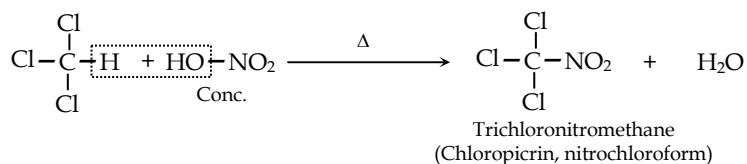
3. **Reaction with aqueous caustic alkali: Hydrolysis:** When chloroform is boiled with aqueous solution of alkali like NaOH or KOH followed by acidification, formic acid is obtained.



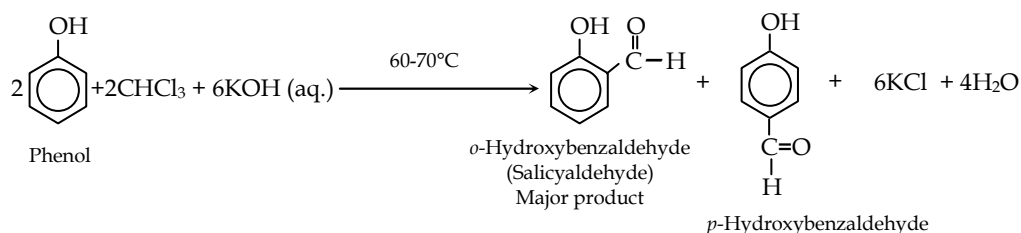
4. **Reaction with chlorine:** Chloroform reacts with chlorine in presence of sunlight to form carbontetrachloride.



5. **Reaction with HNO₃:** When chloroform is heated with concentrated nitric acid, nitrochloroform or chloropicrin is formed which is used as an insecticide and has been used as a war gas or tear gas.

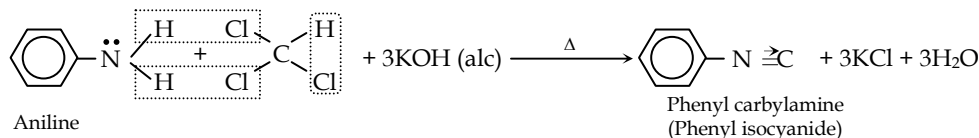
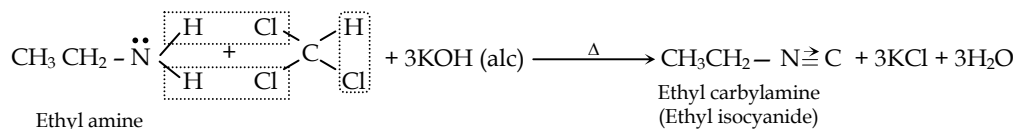
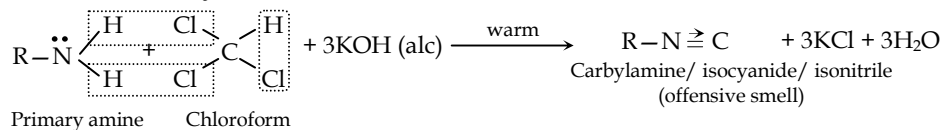


6. **Reaction with Phenol: Reimer-Tiemann Reaction:** When chloroform is refluxed with phenol in presence of aqueous solution of caustic alkali followed by acidification, salicylaldehyde is obtained as major product. The reaction is called Reimer-Tiemann reaction.



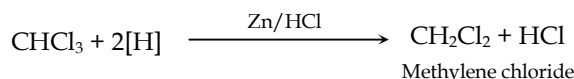
See phenol for mechanism

7. **Reaction with primary amine: Carbylamine Reaction:** When chloroform is warmed with a primary amine (aliphatic or aromatic amine) in presence of alcoholic KOH solution, isocyanide (carbylamine) is formed which gives an unpungent smell like that of rotten onion. This reaction is known as *carbylamine reaction*.

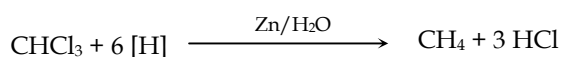


This reaction serves as the test for both chloroform and primary amines under the name **carbylamine test** or **isocyanide test**.

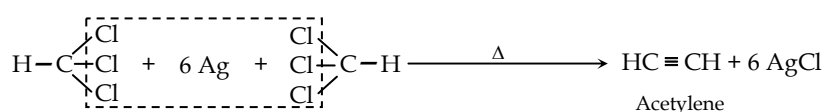
8. **Reduction:** Chloroform undergoes reduction with Zn/HCl (acidic reducing agent) in presence of ethyl alcohol to give methylene chloride.



Chloroform on reduction with Zn/H₂O (neutral reducing agent) gives methane.



9. **Reaction with silver:** When chloroform is heated with metallic silver powder, acetylene gas is evolved.



9.1.4.4 Uses of chloroform

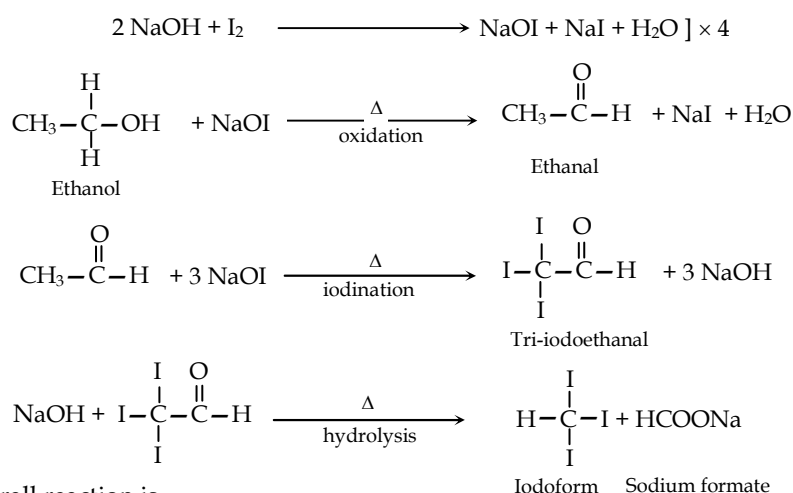
Chloroform is used

1. as a solvent in organic reactions and in industry for fats, waxes, resins, rubber, etc.
2. as laboratory reagent for testing primary amines (Carbylamine test)
3. for the preparation of chloretone and chloropicrin.
4. as an anesthetic: Chloroform was a popular anaesthesia from the mid 1800s to around 1900. Sir James Young Simpson, and Scottish Obstetrician, introduced the medical use of chloroform as an anesthetic in 1847 in Scotland specially for childbirth.

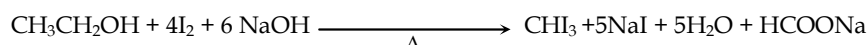
9.1.5.5 Trihaloalkane: Iodoform

Preparation

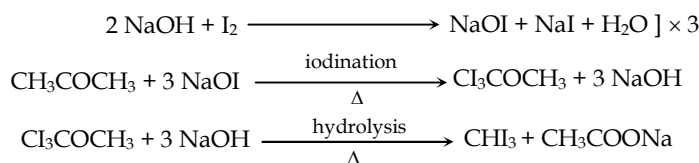
1. **From ethyl alcohol:** When ethyl alcohol is warmed with I₂ and NaOH or Na₂CO₃, pale yellow precipitate of iodoform is obtained.



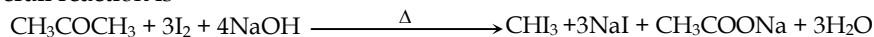
The overall reaction is



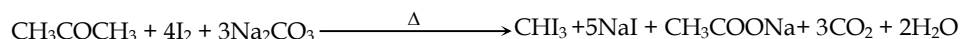
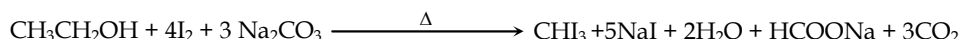
2. **From acetone:** When acetone is warmed with I₂ and NaOH or Na₂CO₃, pale yellow precipitate of iodoform is obtained.



The overall reaction is



When sodium carbonate is used instead of NaOH, the reactions are



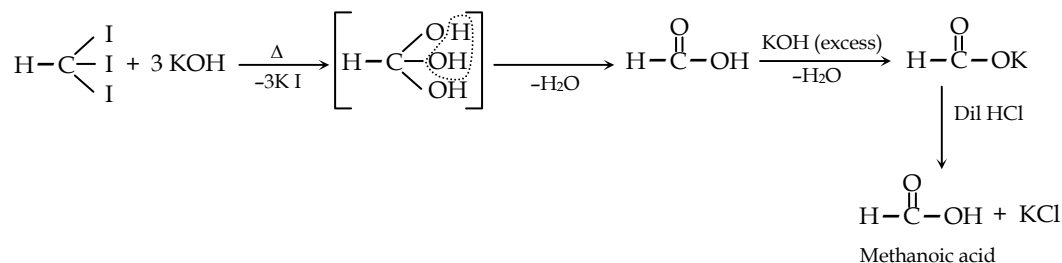
Physical Properties

1. Iodoform is pale yellow crystalline solid with characteristic odour.
2. Its melting point is 119°C.
3. It is insoluble in water but readily soluble in ethyl alcohol and ether.

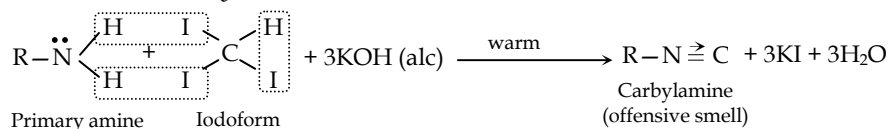
Chemical properties

Chemically, iodoform resembles chloroform.

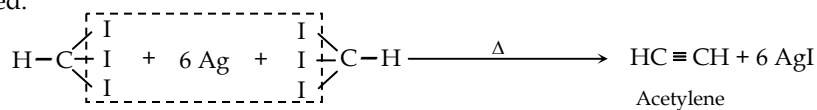
1. **Reaction with aqueous caustic alkali: Hydrolysis:** When iodoform is boiled with aqueous solution of alkali like NaOH or KOH followed by acidification, formic acid is obtained.



2. **Reaction with primary amine: Carbylamine Reaction:** When iodoform is warmed with a primary amine (aliphatic or aromatic amine) in presence of alcoholic KOH solution, isocyanide (carbylamine or isonitrile) is formed which gives a pungent smell like that of rotten onion. This reaction is known as *carbylamine reaction*.



3. **Reaction with silver:** When iodoform is heated with metallic silver powder, acetylene gas is evolved.



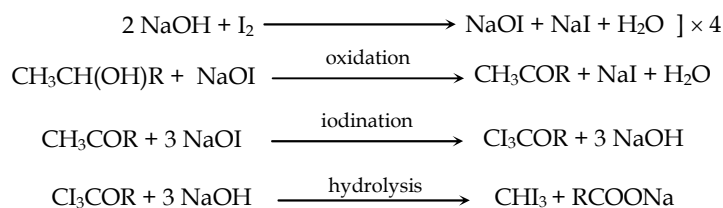
Uses of Iodoform

It is used as an antiseptic, its antiseptic property is not due to itself but due to the liberated I_2 .

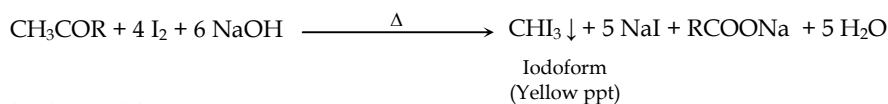
Iodoform Test

1. For alcohols

Alcohols having $\text{CH}_3-\overset{\text{OH}}{\underset{|}{\text{CH}}}-$ structure on heating with excess I_2 and aqueous alkali (NaOH) give yellow crystalline precipitate of iodoform. This test is called iodoform test.

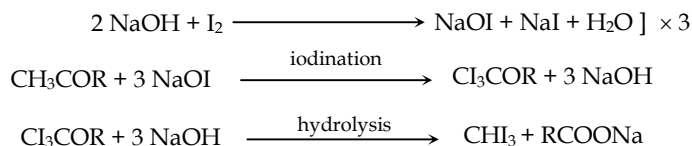


The overall reaction is

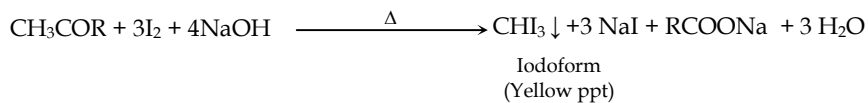


2. For aldehydes and ketones

Aldehydes or ketones having $\text{CH}_3-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-$ structure on heating with excess I_2 and aqueous alkali (NaOH) give yellow crystalline precipitate of iodoform. This test is called iodoform test.



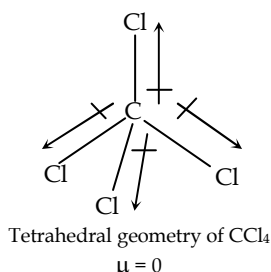
The overall reaction is



Carbonyl compounds like HCHO , $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$, $\text{C}_6\text{H}_5\text{CHO}$ (benzaldehyde) etc. do not contain CH_3- groups in α position. So they do not give iodoform test. The compounds like CH_3CHO , CH_3COCH_3 , $\text{CH}_3\text{COCH}_2\text{CH}_3$ etc. contain $\text{CH}_3\text{CO}-$ group. So they give positive iodoform test.

9.1.6 Tetrahaloalkane: Carbon tetrachloride

Tetrachloromethane (Carbon tetrachloride): It is a colorless liquid with boiling point 77°C. It is insoluble in water but soluble in organic solvent.

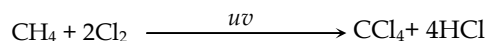


It is prepared industrially by different ways:

1. By the action of Cl₂ on CS₂

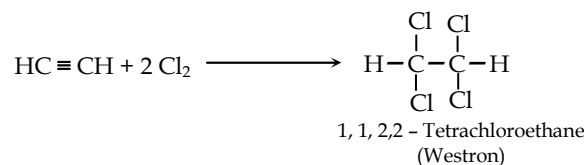


2. By the action of Cl₂ on CH₄



Carbon tetrachloride is nonpolar, aprotic solvent. It is used industrially as solvent for fats, oils, resins, etc. It is also used as a fire extinguisher under the name of *Pyrene*.

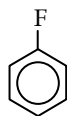
1,1,2,2-Tetrachloroethane (Westron): It is used noninflammable liquid with boiling point 146°C and used as solvent for fats, oils, paints, varnishes, rubber etc.



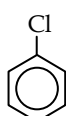
9.2 Haloarenes (Aryl halides)

9.2.1 Introduction

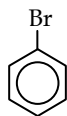
Halogen derivatives of arenes where halo group is directly bonded to aromatic ring are called haloarenes or aryl halides. It may be represented by Ar-X; where Ar is any aryl group and X is F, Cl, Br or I.



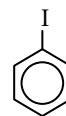
Fluorobenzene



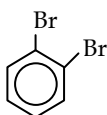
Chlorobenzene



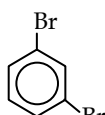
Bromobenzene



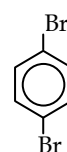
Iodobenzene



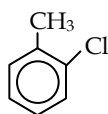
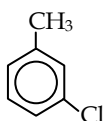
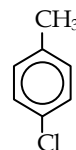
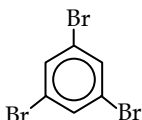
1,2-Dibromobenzene
(o-Dibromobenzene)



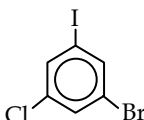
1,3-Dibromobenzene
(m-Dibromobenzene)



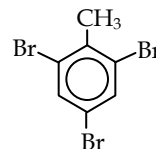
1,4-Dibromobenzene
(p-Dibromobenzene)

2-Chlorotoluene
(*o*-Chlorotoluene)3-Chlorotoluene
(*m*-Chlorotoluene)4-Chlorotoluene
(*p*-Chlorotoluene)

1,3,5-Tribromobenzene

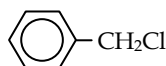


1-Bromo-3-chloro-5-iodobenzene



2,4,6-Tribromotoluene

If halogen atom is not directly bonded to aromatic ring e.g. ArCH_2X , they are called aryl substituted alkyl halide or **aryllalkyl halide** since they behave very much like alkyl halides. They are not aryl halides. For example, Benzyl Chloride



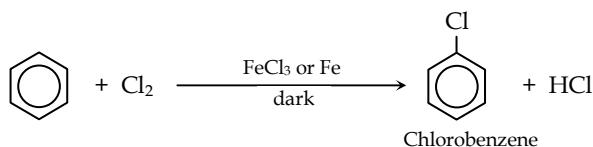
Benzyl Chloride

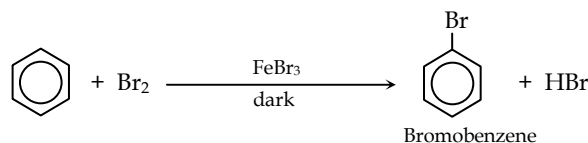
Triiodothyronine (T_3)	Thyroxine (T_4)	Thyroid gland
Organic compounds in which halogen atom is directly bonded to aromatic ring are also found in human beings. For example, thyroid hormones, triiodothyronine (T_3) and thyroxine (T_4). In both hormones, iodine atoms are directly bonded to aromatic ring. They are produced by the thyroid gland that are primarily responsible for regulation of metabolism. Iodine is necessary for the production of T_3 and T_4. A deficiency of iodine leads to decreased production of T_3 and T_4, enlarges the thyroid tissue and will cause the disease known as goitre. (i) List all the functional groups present in T_3 and T_4 and arrange them in the priority order .		

9.2.2 General Methods of Preparation

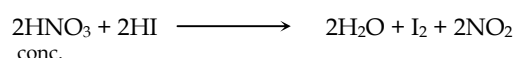
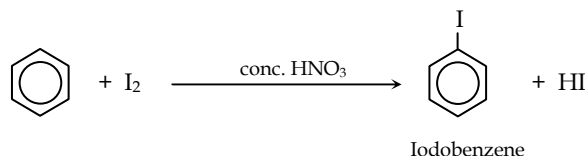
1. From aromatic hydrocarbons

Aromatic hydrocarbon reacts with limited amount of Cl_2 or Br_2 at ordinary temperature in dark and in presence of Lewis acid like FeCl_3 , AlCl_3 to give corresponding aryl halide. Iron is most commonly used since it is converted to FeCl_3 .





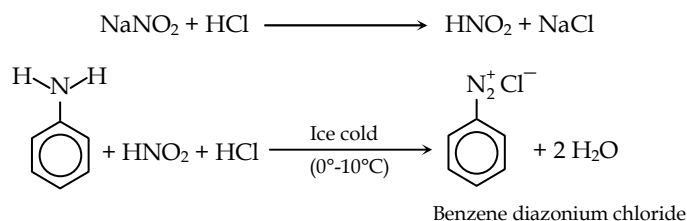
Direct iodination is carried out in presence of oxidizing agent like nitric acid, mercuric oxide etc. oxidizing agents are used to remove powerful reducing agent HI.



2. From diazonium salts

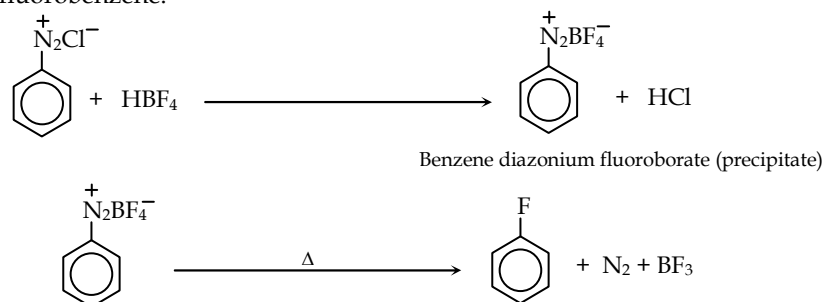
Primary aromatic amine reacts with nitrous acid in ice cold solution to form diazonium salt.

The reaction is called diazotization. (See also Amines)

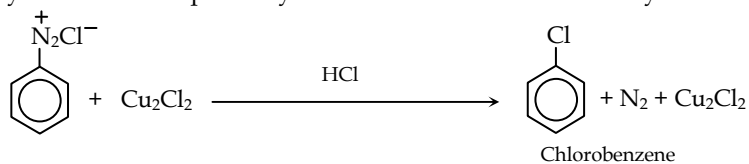


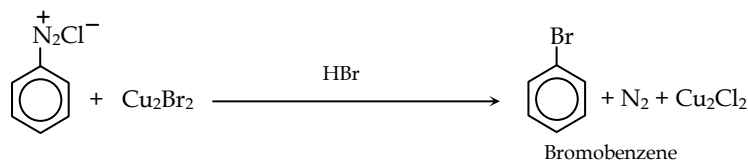
Diazonium salts are very important synthetic reagents for the preparation of a wide range of valuable aromatic compounds. Haloarenes are prepared as follows;

- i. **Preparation of fluorobenzene: Schiemann reaction (1927):** When fluoroboric acid is added to a diazonium salt solution, precipitate of diazonium fluoroborate is obtained which on gentle heating gives fluorobenzene.

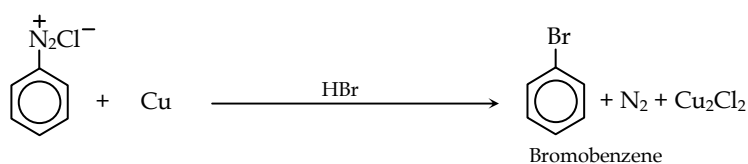
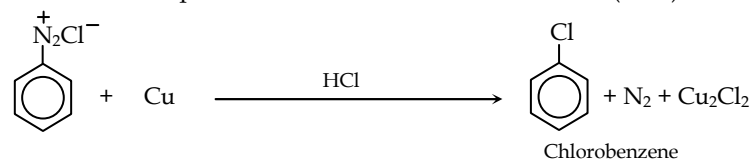


- ii. **Preparation of Chlorobenzene and Bromobenzene: Sandmeyer reaction (1884):** When diazonium salt solution is mixed with cuprous chloride or cuprous bromide dissolved in the corresponding halogen acid at room temperature or sometimes on warming, diazo group is replaced by -Cl and -Br respectively. This reaction is called Sandmeyer reaction.

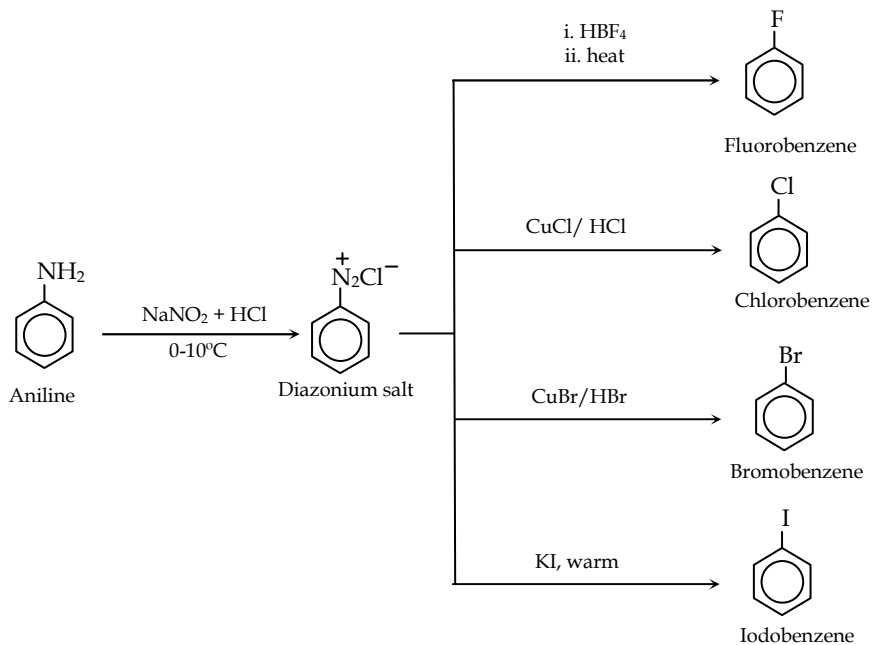
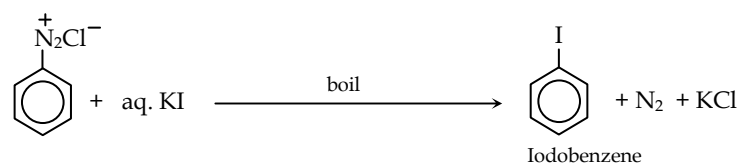




This reaction can be carried out by warming diazonium salt solution with copper powder in place of cuprous halide. The process is called **Gattermann reaction** (1890).



iii. **Preparation of Iodobenzene:** When diazonium salt solution is boiled with aqueous potassium iodide, diazo group is replaced by -I.



When a mixture of benzene vapour, hydrogen chloride and air is passed over heated copper chloride, chlorobenzene is formed. The process is called **Raschig process**. Chlorobenzene is commercially prepared by this method.

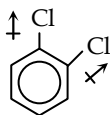


Self Test

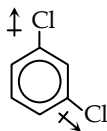
9.20 How do you prepare chlorobenzene from (i) benzene (ii) aniline

9.2.3 Physical Properties

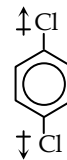
1. They are colourless oily liquids or crystalline solids.
2. Chlorobenzene is colourless liquid having aromatic odour.
3. They are insoluble in water but soluble in organic solvents.



o-Dichlorobenzene
m.pt. -17°C
b.pt 180°C



m-Dichlorobenzene
m. pt. -24°C
b. pt. 173°C



p-Dichlorobenzene
m. pt. 52°C
b. pt. 175°C

In isomeric dihalobenzenes, there is only slight variation in the boiling point and the order is: *ortho* > *para* > *meta*. However, there is considerable variation in their melting point. Para isomer has considerably high melting point. This is because para isomer is symmetrical to fit better in the crystal lattice.

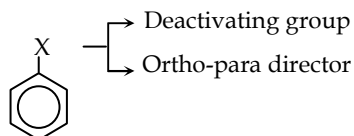
9.2.4 Chemical Properties

Chemical properties of chlorobenzene can be categorised as follows;

1. Electrophilic aromatic substitution reaction
2. Nucleophilic substitution reaction
3. Reduction
4. Reaction with metals
4. Other reactions

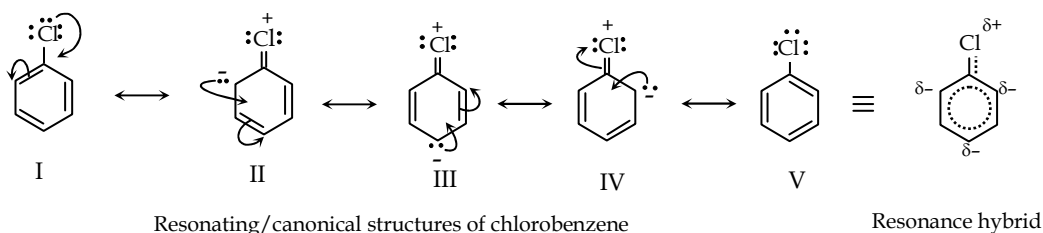
1. Electrophilic aromatic substitution reaction

Halogen atom (-X) present in benzene governs the reactivity and orientation in electrophilic aromatic substitution reaction.



Chlorine atom (-Cl) withdraws electron from benzene ring due to electron withdrawing inductive effect (-I effect) of Cl since chlorine is more electronegative than carbon.

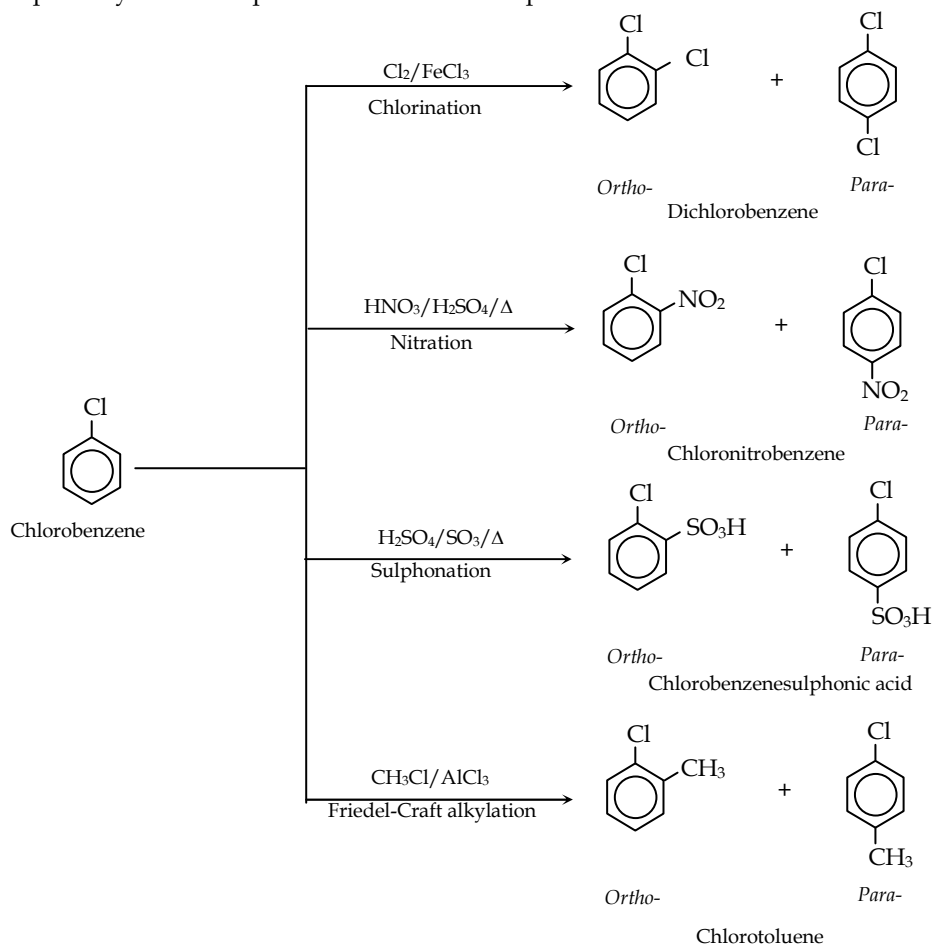
Chlorine atom releases electron to benzene ring due to resonance effect (+M effect).



Here, inductive effect is stronger than resonance effect and causes net electron withdrawal from benzene ring.

Hence, chlorine atom (Cl) decreases the electron density in benzene ring of chlorobenzene as compared to that in benzene. Hence chlorobenzene is less reactive than benzene in electrophilic aromatic substitution reaction (each position of chlorobenzene is less reactive than corresponding position in benzene). A substituent that makes the benzene ring less reactive than benzene in electrophilic aromatic substitution reaction is called deactivating group. Chloro group is deactivating group.

However, due to resonance, Cl releases electrons towards benzene ring not equally to all positions. It releases electron more to ortho and para positions than meta positions, so electron density is relatively higher in ortho and para positions than meta in positions. Hence, chlorobenzene gives ortho- para substituted products. A substituent that directs the incoming group chiefly ortho and para to it is called ortho-para director.



Self Test

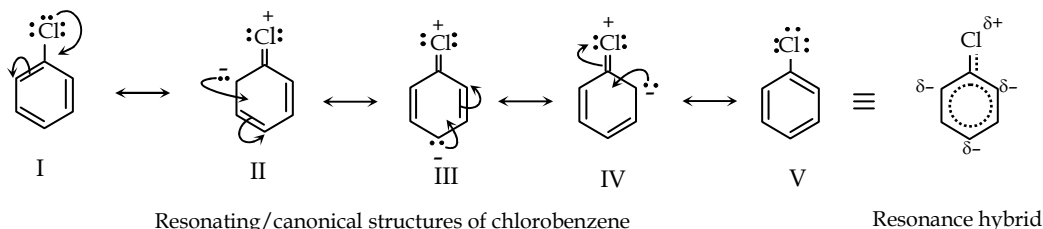
9.21 Chlorobenzene is less reactive than benzene in electrophilic aromatic substitution reaction, explain.

9.22 Chlorobenzene gives ortho para substituted products, explain.

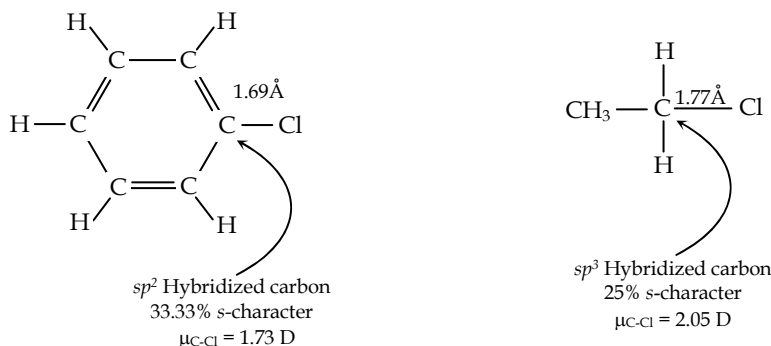
2. Nucleophilic substitution reaction

Haloalkanes undergo nucleophilic substitution reaction only with extreme difficulty. The low reactivity of haloarenes towards nucleophilic substitution reaction can be explained as follows;

- a. **Carbon to halogen bond in haloarenes are strong; Resonance effect:** Haloarene is resonance hybrid of structures I to V which indicates that carbon to halogen bond on haloarene has a partial double bond character. A double bond is shorter and stronger than the corresponding single bond. Hence it is difficult to break down the carbon to halogen bond.

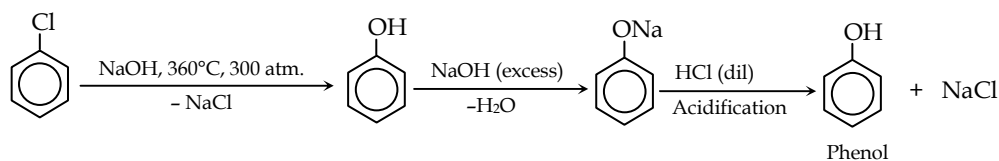


- b. **Carbon to halogen bond in haloarene is less polar; Hybridization effect:** In haloarene, halogen atom is bonded to sp^2 carbon whereas in haloalkane halogen atom is bonded to sp^3 carbon. An sp^2 carbon (33.33% s -character) is more electronegative than sp^3 carbon (25% s -character). Hence carbon to halogen bond in haloarene is less polar than that in haloalkanes.

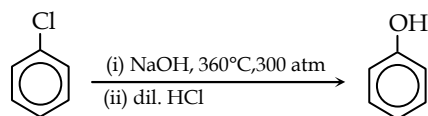


However, aryl halides undergo nucleophilic substitution reaction under vigorous conditions like high temperature, high pressure, catalyst, etc.

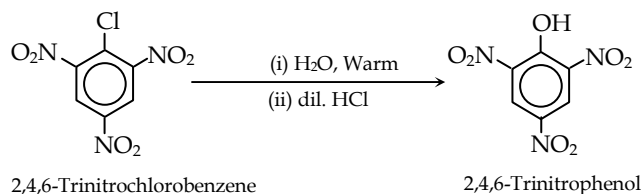
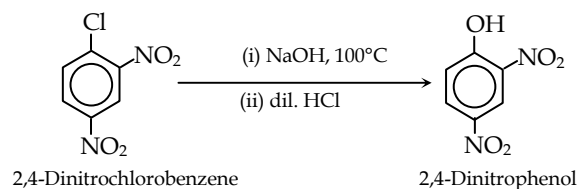
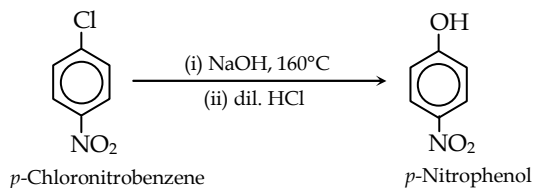
- i. **Action of NaOH (hydrolysis):** When chlorobenzene is heated with caustic soda at 360°C and 300 atmospheric pressure followed by acidification, phenol is formed. This method was used by Dow chemical company, USA for the commercial preparation of phenol. Hence this method is called **Dow process**.



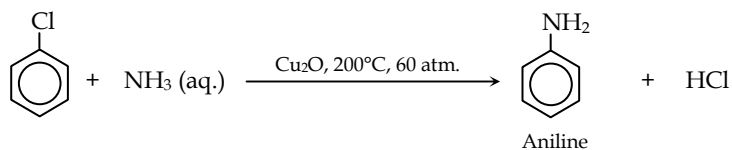
or



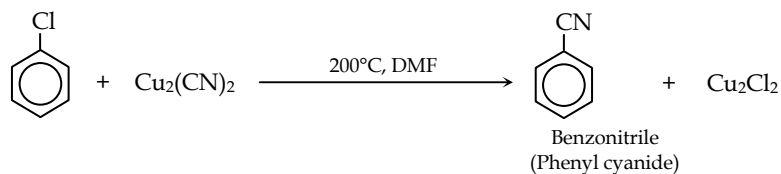
Powerful electron withdrawing group (eg. $-\text{NO}_2$, $-\text{CN}$, $-\text{COOH}$ etc.) present ortho and para to halogen atom increases the reactivity of benzene ring towards nucleophilic substitution reaction.



- ii. **Action of ammonia (Ammonolysis):** When chlorobenzene is heated with aqueous ammonia solution under pressure in presence of cuprous oxide, aniline is formed.

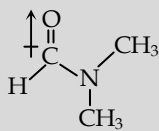
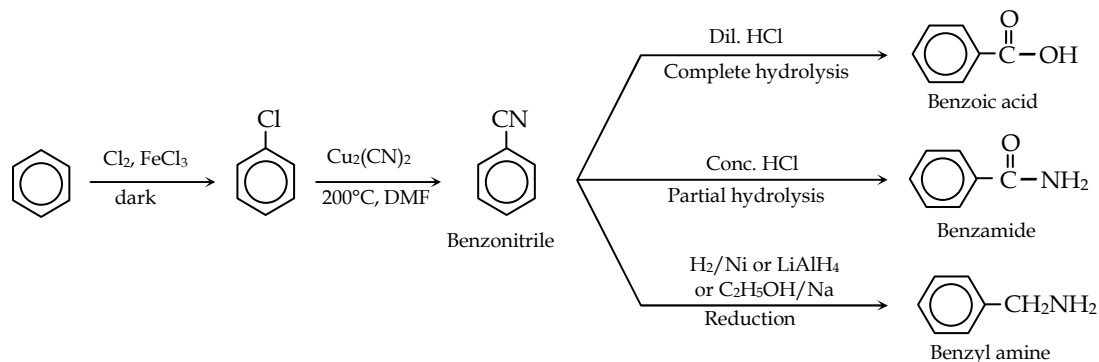


- iii. **Action of cuprous cyanide:** When chlorobenzene is heated with cuprous cyanide in DMF as solvent, phenyl cyanide is formed.



Application of nitrile synthesis

Nitrile synthesis has advantage of increasing carbon chain length by one carbon atom.

Example:

Dimethylformamide (DMF)

DMF stands for dimethyl formamide, which is a polar aprotic, non aqueous solvent.

Self Test

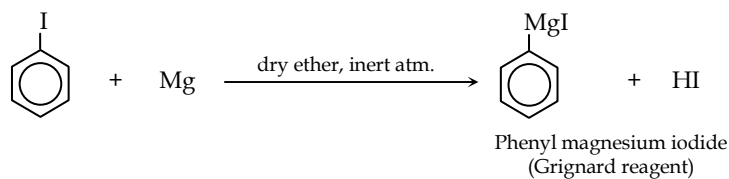
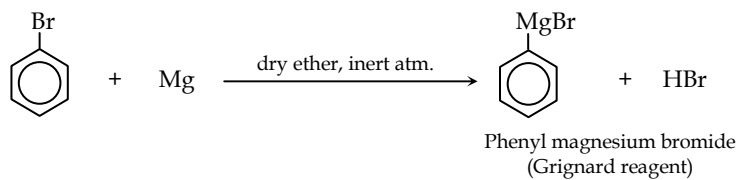
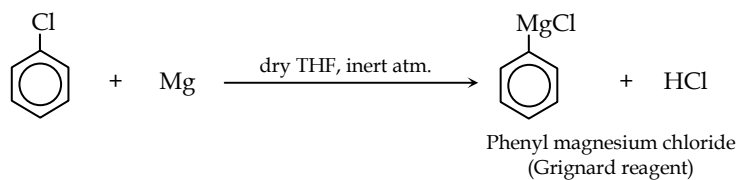
9.22 Convert benzene into (a) Benzoic acid (b) Benzamide (c) Benzyl amine

9.23 Nucleophilic substitution reaction is difficult in haloarene, explain.

9.24 Convert chlorobenzene into bromobenzene.

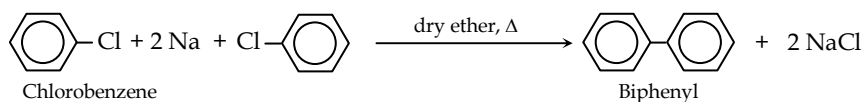
3. Reaction with metals

- a. **Reaction with magnesium: Formation of Grignard reagent:** Aryl halide reacts with magnesium metal in dry ether or THF to give Grignard reagent.

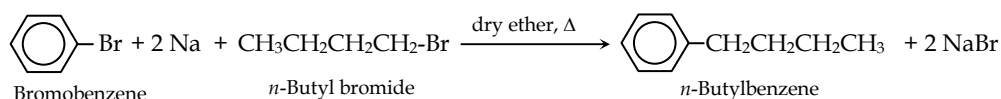
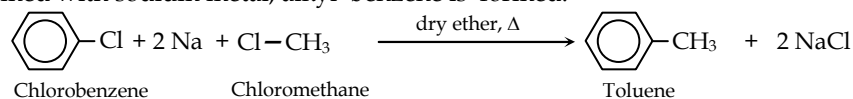


b. Reaction with sodium

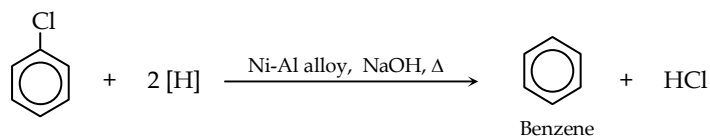
- i. **Fittig reaction (1863):** When ethereal solution of aryl halide is warmed with sodium metal, two molecules of aryl halides combine to give biphenyl compounds.



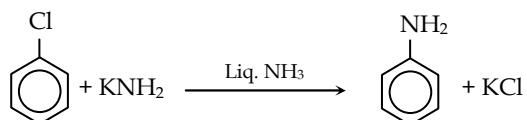
- ii. **Wurtz-Fittig reaction (1863):** When ethereal solution of aryl halide and alkyl halide is warmed with sodium metal, alkyl benzene is formed.

**4. Other reactions**

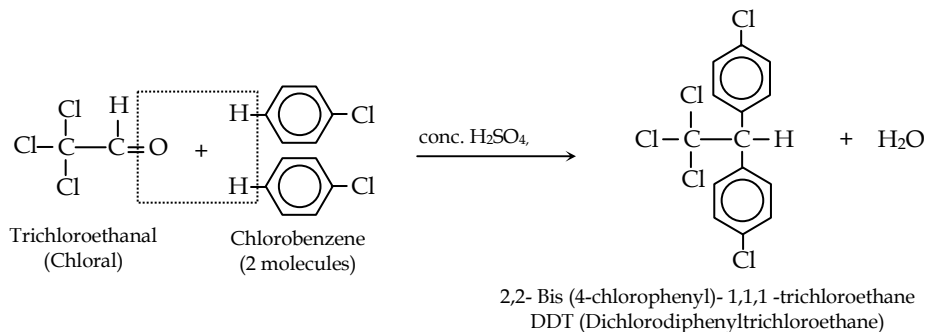
- a. **Reduction:** When chlorobenzene is heated with nickel aluminium alloy in presence of caustic alkali, it is reduced to benzene.



- b. **Reaction with NaNH₂:** When chlorobenzene is treated with NaNH₂ or KNH₂ in liquid ammonia, aniline is obtained. The reaction follows **elimination-addition mechanism** through **benzyne** intermediate.



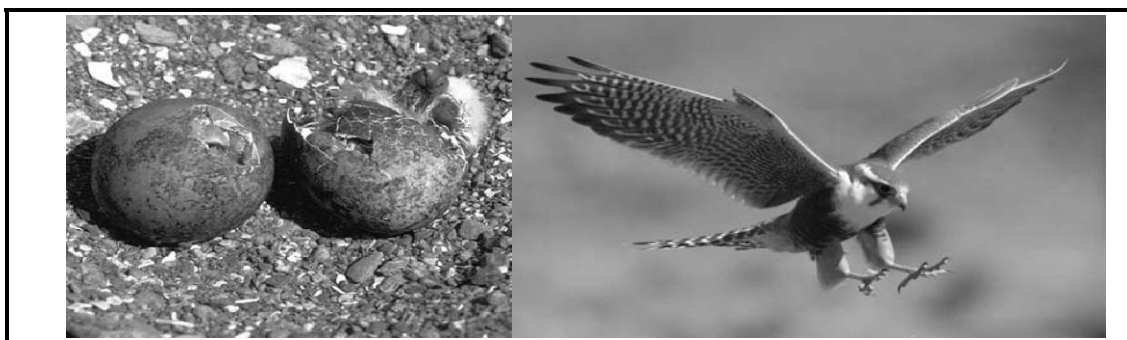
- c. **Reaction with Chloral:** When chlorobenzene is heated with chloral in presence of concentrated H₂SO₄, DDT is formed which is an insecticide. It is a colorless, tasteless crystalline compound.



Insecticidal property of DDT was discovered by the Swiss chemist **Paul Hermann Müller** in 1939. In the second half of World war II, it was used with great success to control malaria and typhus among civilians and troops. After the war, DDT was made available for use as an agricultural insecticide and extensively used in agriculture. Müller was awarded the **Nobel prize in physiology or medicine** for his discovery of the high efficiency of DDT as a contact poison against several arthropods in 1948.

DDT is toxic to a wide range of living organisms including human beings. It is a persistent organic pollutant and bioaccumulate concentrating more chemicals especially in predatory birds and apex consumers through the food chain. It causes egg shell thinning in birds of prey. Eggshell thinning lowers the reproductive rate of certain bird species by causing egg breakage and embryo deaths. It is considered a major reason for the decline of the bald eagle, peregrine falcon, brown pelican etc.

Because of its hazards in health and environment, the use of DDT as agricultural insecticide was outlawed in 2004 under Stockholm Convention and restricted to disease vector control.



9.2.5 Some Commercially Important Halogen Derivatives of Hydrocarbons

1. **Polyvinyl chloride (PVC):** It is prepared by polymerization of vinyl chloride. (Detail in chapter Polymers)
2. **Polytetrafluoroethane (Teflon):** It is prepared by polymerization of tetrafluoroethene. (Detail in chapter Polymers)
3. **Chlorofluorocarbons (CFCs):** Chlorofluorocarbons (CFCs) also known as *Freons* are compounds having only carbon, fluorine and chlorine and synthesized for their use in refrigeration, air conditioning and aerosol propellant. However, CFCs are to be phased out due to their role ozone layer depletion



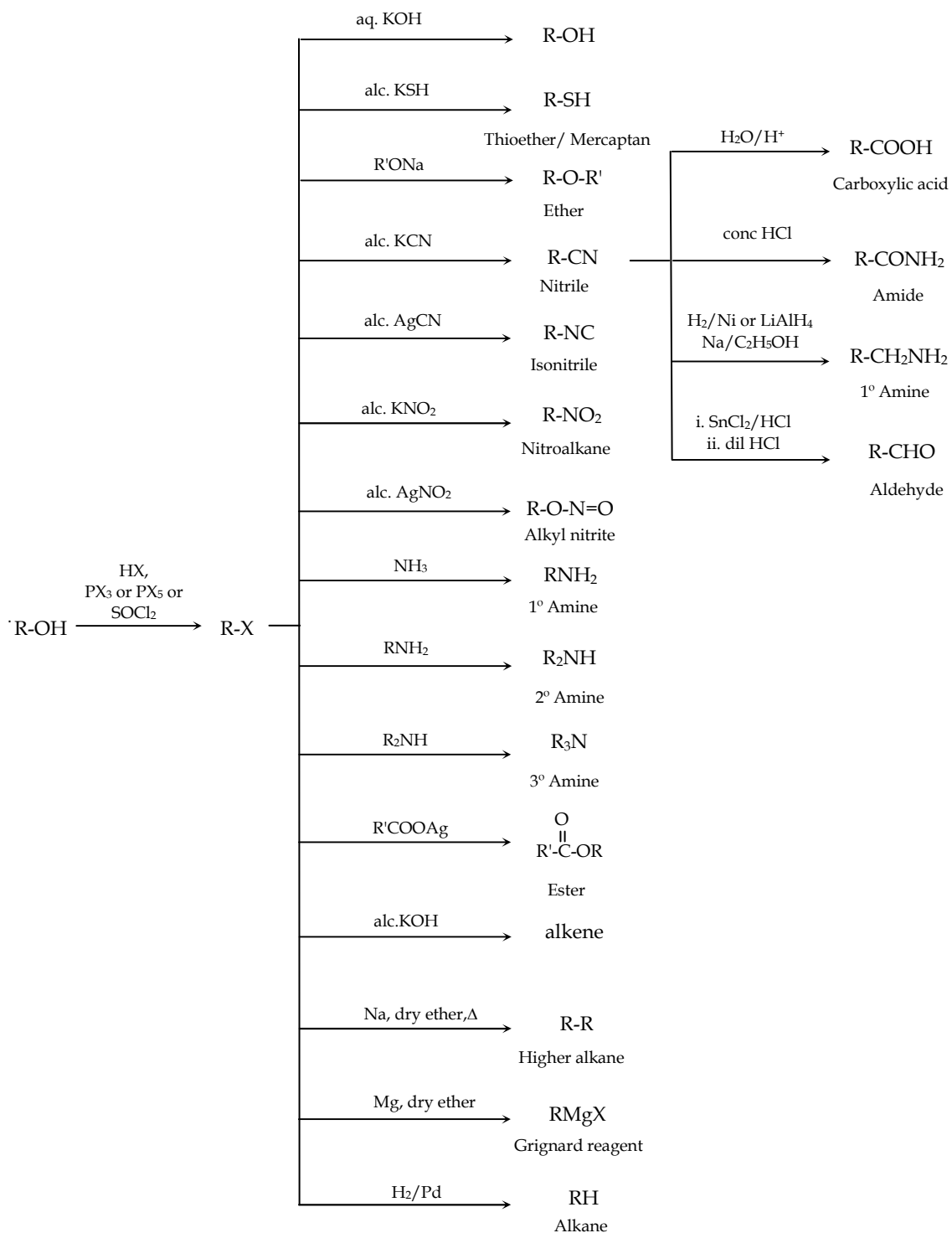
Trichlorofluoromethane
Freon 11 (CFC 11)



Dichlorodifluoromethane
Freon 12 (CFC 12)

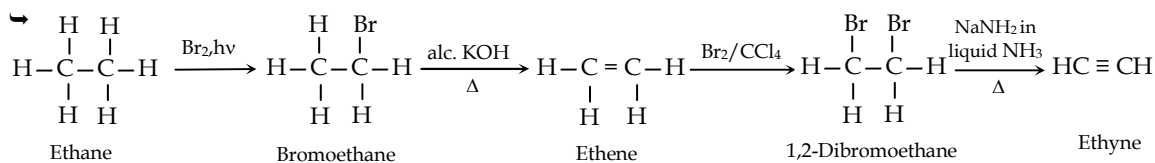
4. DDT and BHC are insecticides. Detail in chapter 16

Alkyl halides undergo a wide range of chemical reactions hence provide useful routes to form a large variety of other organic compounds. Hence they are extremely important in organic synthesis.

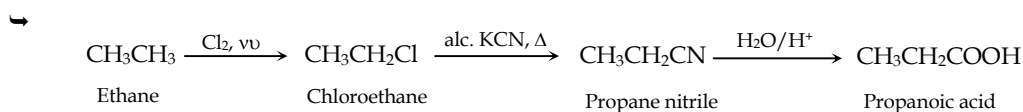


Solved Questions

1. Convert ethane into ethyne.

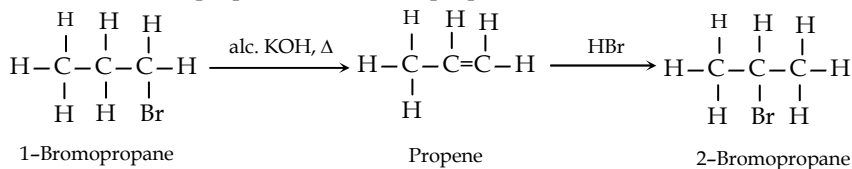


2. Convert ethane into propanoic acid.

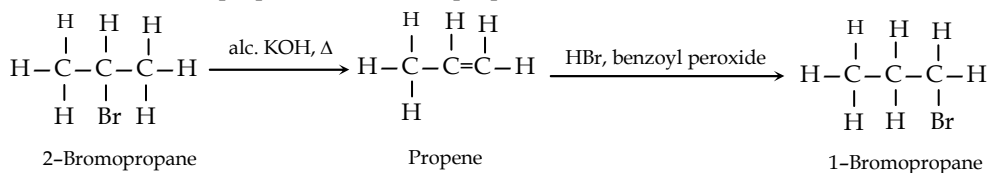


3. Convert 1-Bromopropane into 2-Bromopropane and vice-versa.

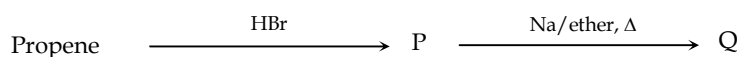
➤ Conversion 1-Bromopropane into 2-Bromopropane



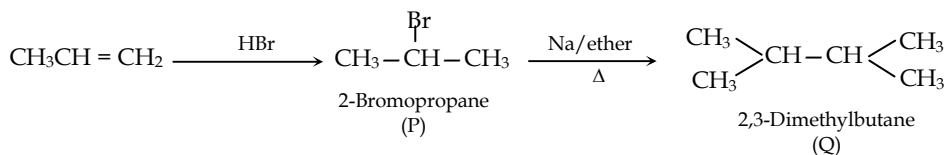
Conversion 2-Bromopropane into 1-Bromopropane



4. Write down the IUPAC name of major products P and Q in the following reaction sequence.



➤

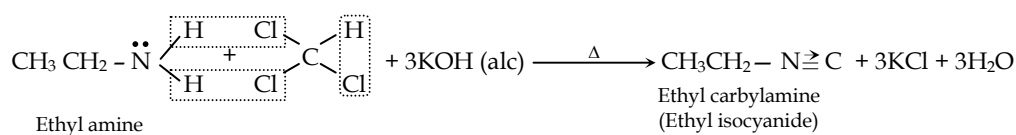


5. What would you do to distinguish chloroform from carbontetrachloride?

➤ Carbylamine test (isocyanide test) is carried out.

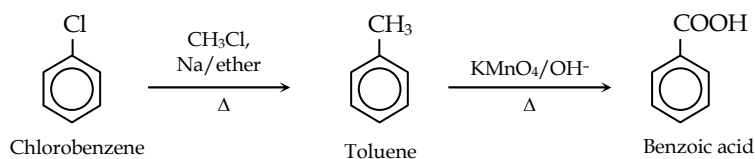
When chloroform is warmed with aliphatic or aromatic primary amine in presence of alcoholic KOH, offensive smell of isocyanide is obtained. CCl_4 does not give this test.

Example

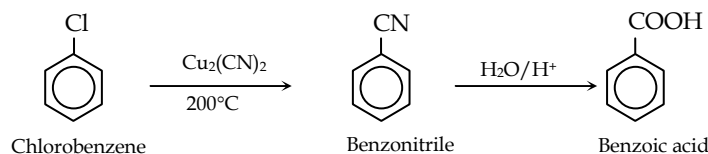


6. How do you get benzoic acid from chlorobenzene?

→



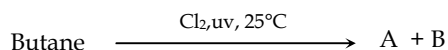
or,



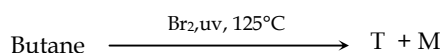
Questions

Very Short Answer Questions:

1. Identify A and B



2. Identify T and M

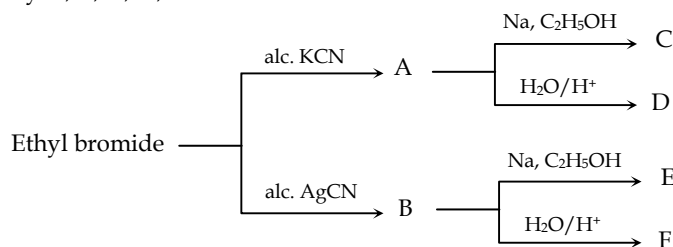


3. Identify T and M and indicate the major product giving reason.

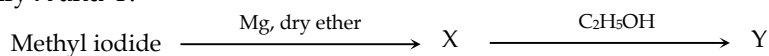


4. Write all steps when excess ethyl bromide reacts with alcoholic ammonia.

5. Identify A, B, C, D, E and F



6. Identify X and Y:



7. Why is chloroform stored in dark coloured bottles with little ethanol?

8. Give an example of

- | | | |
|----------------------------|-------------------------|----------------------|
| a. Carbylamine reaction | b. Dehydrohalogenation | c. Grooves process |
| d. Hofmann ammonolysis | e. Hydrogenolysis | f. Hydrohalogenation |
| g. Reimer-Tiemann reaction | h. Williamson synthesis | i. Wurtz reaction |

9. Give an example of

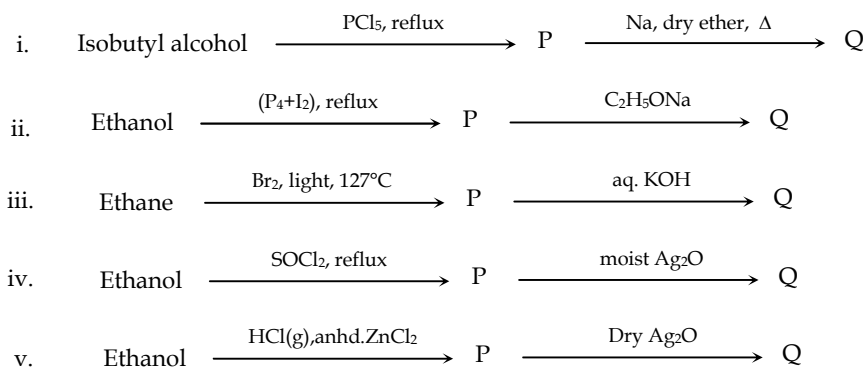
- | | | |
|-------------------------------|------------------------|--------------------------|
| a. Diazotization | b. Fittig reaction | c. Wurtz-Fittig reaction |
| d. Sandmayer reaction | e. Gattermann reaction | f. Schiemann reaction |
| g. Benzene into benzoic acid. | | |

10. A primary haloalkane (X) if allowed to react with KCN yields a compound (Y), which on acidic hydrolysis gives propanoic acid. Identify X and Y.

11. Why does chloroform not give white precipitate with aqueous silver nitrate?
12. How is Grignard reagent prepared? What precautions should be taken for its preparation?
13. Write the action of trichloromethane with monohydroxy benzene in presence of alcoholic NaOH.
14. Why does chlorobenzene give *ortho-para* substituted products?

■ Short /Long Answer Questions:

1. Identify P and Q



2. Starting from iodoethane, how do you get

a. Ethane	b. Ethene	c. Ethyne
d. Butane	e. Bromoethane	f. Methoxyethane
g. Ethoxyethane	h. Ethylethanoate	i. Ethyl thioalcohol
j. Nitroethane	k. Ethyl nitrite	l. Ethyl carbylamine
m. Propanoic acid	n. Propanamide	o. 1-Aminopropane
3. How is chloroform prepared in laboratory in pure and dry state? How does chloroform react when heated with
 - i. Aqueous KOH
 - ii. Phenol in presence of aqueous KOH
 - iii. Ethanamine in presence of alcoholic KOH
 - iv. Acetone in presence of alcoholic KOH
 - v. Conc. HNO_3
4. Give any two methods for the preparation of chlorobenzene. Why is nucleophilic substitution reaction is difficult in chlorobenzene as compared to chloroethane? Convert chlorobenzene into DDT.
5. How can you convert
 - a. 1- Bromopropane into 2- Bromo propane and vice-versa
 - b. Methane into ethane
 - c. Ethane into butane
 - d. Ethane into Ethene
 - e. Ethane into Ethanol
 - f. Chlorobenzene into Toluene
6. What happens when:
 - a. Chloroethane is hydrolized?
 - b. 2- Chlorobutane is heated with alcoholic KOH?
 - c. Chlorobenzene is treated with chloral in acidic medium?
 - d. Chlorobenzene is heated with Ni-Al in alkaline medium?
7. Write any three methods of preparation of iodoethene. What happens when iodoethane is heated with (i) sodium in presence of dry ether? (ii) alcoholic caustic soda.
8. Starting from trichloromethane, how would you prepare

a. Ethyne	b. Methane	c. Chloropicrin
d. Carbonyl chloride	e. Chlorethane	

1. The number of primary alkyl halide having molecular formula C_4H_9X is
 - a. 1
 - b. 2
 - c. 4
 - d. 3
2. For the reaction $C_2H_5OH + HX \rightarrow C_2H_5X + H_2O$, the order of reactivity is:
 - a. $HCl > HBr > HI$
 - b. $HI > HCl > HBr$
 - c. $HI > HBr > HCl$
 - d. $HBr > HI > HCl$
3. When propylene reacts with HCl in presence of traces of organic peroxide, it follows
 - a. Markonikov's rule
 - b. Anti-Markonikov's rule
 - c. Free radical addition
 - d. Chain reaction
4. When Chloroform is boiled with aqueous KOH and followed by acidification, it forms
 - a. Salicylaldehyde
 - b. Methanoic acid
 - c. Methoxymethane
 - d. Methanal
5. Alkyl halide reacts with alcoholic AgCN to give
 - a. Alkyl cyanide
 - b. Alkyl isocyanide
 - c. Alkyl isocyanate
 - d. Nitrile
6. Tear gas is
 - a. Chloreton
 - b. Ethyl carbonate
 - c. Chloropicrin
 - d. Methylene chloride
7. Which of the following is wrong statement?
 - a. $CHCl_3$ is used in surgery as an anesthetic
 - b. $CHCl_3$ is prepared from chlorination of chloral
 - c. $CHCl_3$ on exposure to sunlight and air gives phosgene
 - d. Chloropicrin is an insecticide prepared from nitration of chloroform
8. Which of the following is gem-halide?
 - a. $CH_3CH_2BrCH_2Br$
 - b. CH_3CHBr_2
 - c. $CH_3CHBrCH_2CH_2Br$
 - d. $BrCH_2CH_2Br$
9. The most common reaction that alkyl halide shows is
 - a. Electrophilic addition
 - b. Nucleophilic substitution
 - c. Free radical addition
 - d. Electrophilic substitution
10. The order of methyl halide to the formation of Grignard reagent is
 - a. $CH_3Br > CH_3Cl > CH_3I$
 - b. $CH_3I > CH_3Br > CH_3Cl$
 - c. $CH_3Cl > CH_3I > CH_3Br$
 - d. $CH_3Br > CH_3I > CH_3Cl$
11. Out of the following compound which is not an organometallic compound?
 - a. CH_3MgI
 - b. $(CH_3)_2Li$
 - c. C_2H_5ONa
 - d. TEL
12. Which out of the following is the best method to prepare chloroalkane?
 - a. Groove's method
 - b. Dorzen's method
 - c. Finkelstein method
 - d. Borodine Hunsdickner reaction
13. When But-1-ene reacts with HBr gives 1-bromobutane in presence of Peroxide, it follows
 - a. Markonikov's rule
 - b. Anti-Markonikov's rule
 - c. Saytzeff rule
 - d. None of above
14. Alkyl halide burns with
 - a. Green edged flame
 - b. Smoky flame
 - c. Yellow flame
 - d. Blue flame
15. Chloroform when reacts with Zn dust and water and Zn dust and HCl respectively forms
 - a. CH_4 and CH_2Cl_2
 - b. CH_2Cl_2 and CH_4
 - c. CCl_4 and CH_4
 - d. CH_4 and CCl_4
16. When Chloroform is boiled with KOH and followed by acidification, it forms
 - a. Methanol
 - b. Methanoic acid
 - c. Methoxy methane
 - d. Methanal
17. Isocyanide test can not be used to distinguish
 - a. $CHCl_3$ and CCl_4
 - b. 1° and 2° amine
 - c. Ethyl amine and trimethyl amine
 - d. Ethyl amine and aniline
18. Which of the following is used as refrigerant?
 - a. CH_2Cl_2
 - b. CF_2Cl_2
 - c. CCl_4
 - d. CF_4
19. Which of the following is used as a War gas?
 - a. CCl_3NO_2
 - b. CH_2Cl_2
 - c. $CHCl_3$
 - d. $COCl_2$
20. Acetone is warmed with iodine and dil. alkali gives
 - a. CHI_3
 - b. CH_2I_2
 - c. CH_3I
 - d. Cl_3COCH_3
21. $CHCl_3$ on oxidation with air gives
 - a. Phosgene
 - b. Formic acid
 - c. Chloropicrin
 - d. CCl_4
22. Chlorobenzene is prepared commercially by
 - a. Dow process
 - b. Down process
 - c. Darzens procedure
 - d. Groove's process
23. When benzene diazonium chloride is heated with the cuprous chloride in corresponding halogen acid,, chlorobenzene is formed. The reaction is
 - a. Sandmeyer reaction
 - b. Gattermann reaction
 - c. Scheimann reaction
 - d. Fittig reaction
24. Which of the following compound is non polar?
 - a. CH_3Cl
 - b. CH_2Cl_2
 - c. $CHCl_3$
 - d. CCl_4

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12
b	c	a	b	b	c	b	b	b	b	c	b
13	14	15	16	17	18	19	20	21	22	23	24
b	a	a	b	d	b	a	a	a	a	a	d

Alcohols and Phenols

10

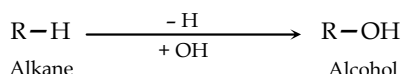


Power alcohol is being used nowadays as the source of fuel (which is used to save petrol).

10.1 Alcohols

10.1.1 Introduction

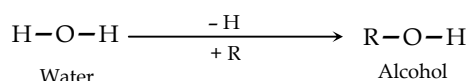
Aliphatic alcohols: Hydroxy derivatives of aliphatic hydrocarbons are aliphatic alcohols. Their general representation is ROH, where R— is alkyl group.



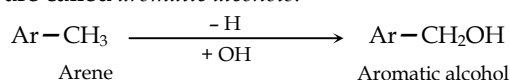
Examples:



Aliphatic alcohols are also regarded as alkyl derivatives of water.



Aromatic alcohols: Hydroxy derivatives of arenes where the —OH group is attached to the side chain of the aromatic ring are called *aromatic alcohols*.



Examples:



10.1.2 Classification of Aliphatic Alcohols

A. Classification based upon number of —OH groups per molecule

Aliphatic Alcohols are classified into different classes according to the number of —OH groups contained in their molecules.

- 1. Monohydric (Monohydroxy) alcohols:** Alcohols containing only one —OH group per molecule are called monohydric alcohols. The common name of monohydric alcohol is alkyl alcohol and their IUPAC name is *alkanol*. The general structural formula of it is R—OH and molecular formula is (C_nH_{2n+1}OH).

Examples: CH₃OH, CH₃CH₂OH, etc.

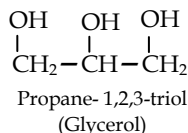
- 2. Dihydric (Dihydroxy) alcohols:** Alcohols containing two —OH groups per molecule are called *dihydric alcohols*. The general molecular formula is C_nH_{2n}(OH)₂.

Examples:



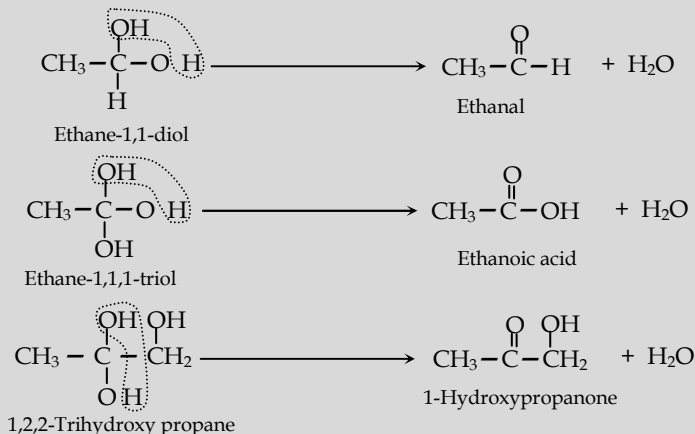
3. **Trihydric (Trihydroxy) alcohols:** Alcohols containing three $-OH$ groups per molecule are called *trihydric alcohols*. The general molecular formula is $C_nH_{2n-1}(OH)_3$.

Example:



Alcohol with two or more $-OH$ groups attached to the same carbon atom of the carbon chain of the molecule becomes unstable and hence it is decomposed to aldehyde, ketone or carboxylic acid.

Examples:



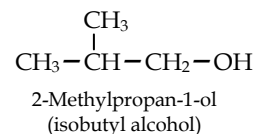
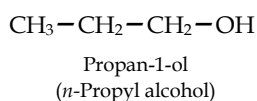
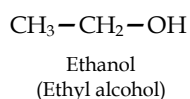
4. **Polyhydric alcohols:** Alcohols containing four or more $-OH$ groups per molecule are called *polyhydric alcohols*.

B. Classification of monohydric alcohols (based upon type of carbon, in which the hydroxy group is attached)

Monohydric alcohols are further classified as *primary* (1°), *secondary* (2°) and *tertiary* (3°) alcohols according to the type of carbon in which the hydroxyl ($-OH$) group is attached.

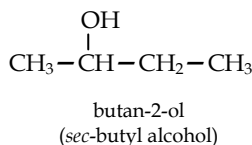
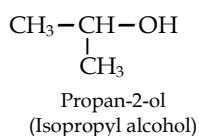
1. **Primary alcohols:** Alcohols in which the $-OH$ group is attached to a primary carbon atom are called primary alcohols. Primary alcohols are characterized by the presence of a structural unit of $-CH_2OH$ group, hence they are represented by general formula RCH_2OH .

Examples:



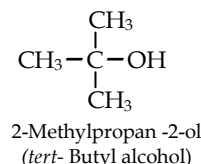
2. **Secondary alcohol:** Alcohols containing $-OH$ group attached to a secondary carbon atom of the molecule are called secondary alcohols. Secondary alcohols are characterized by the presence of structural unit of $>CHOH$ and hence they are represented by general formula R_2CHOH .

Examples:



3. **Tertiary alcohol:** Alcohols containing -OH group attached to a tertiary carbon atom are called tertiary alcohols. Tertiary alcohols are represented by general formula $\text{R}_3\text{C-OH}$, which are characterized by the presence of a characteristic structural unit of $\begin{array}{c} | \\ -\text{C}-\text{OH} \\ | \end{array}$

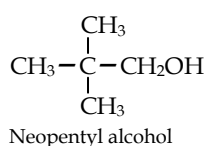
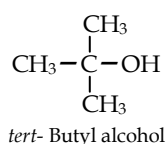
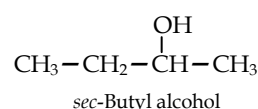
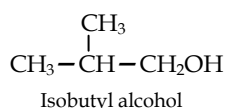
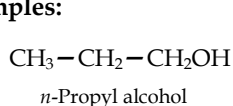
Example:



10.1.3 Nomenclature

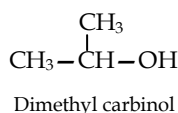
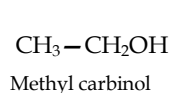
1. **The common naming system:** In the common naming system, the monohydric alcohols are named as *alkyl alcohols*. They are named by writing the name of alkyl group followed by the word 'alcohol'.

Examples:

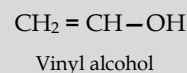
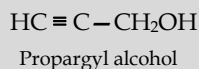
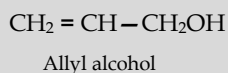


2. **Carbinol system of naming:** CH_3-OH is a first member of alcohol called carbinol and all other members of alcohol are considered as substituted products of carbinol.

Examples:

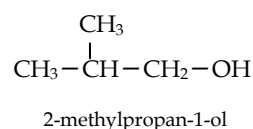
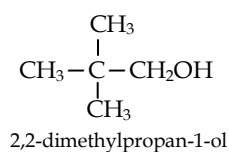
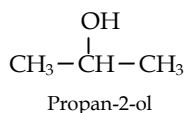
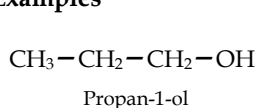


Other examples of alcohols which are famous for their common name are:



3. **The IUPAC system:** In the IUPAC system, the monohydric alcohols are named as *alkanols*. The numbering of parent is done so as to give the lowest number to the carbon atom carrying the -OH group.

Examples

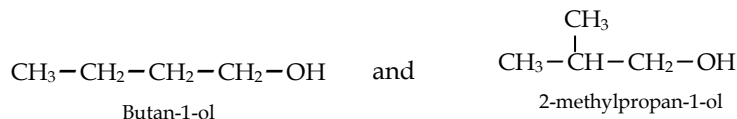


10.1.4 Isomerism in Monohydric Alcohols

Alcohols show the following types of structural isomerisms.

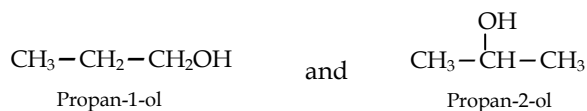
- Chain isomerism:** Alcohols having same molecular formula but different carbon skeleton gives rise chain isomerism in alcohols.

Example:



- Positional isomerism:** Different alcohols having same molecular formula but different structure due to the difference in the position of the -OH group in the identical carbon chain gives rise to positional isomerism.

Example:



- Functional isomerism:** Alcohols show functional isomerism with ethers. Alcohols and ethers are functional isomers each other as they have same general molecular formula of $\text{C}_n\text{H}_{2n+2}\text{O}$.

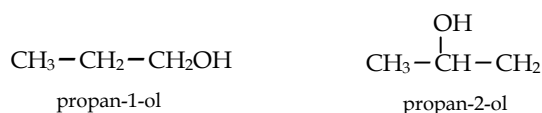
Examples:

- $\text{C}_2\text{H}_6\text{O}$ has two functional isomers.

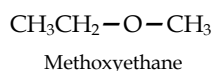


- $\text{C}_3\text{H}_8\text{O}$ has three isomers, out of them two compounds *viz.* 1-propanol or 2-propanol and methoxyethane are functional isomers.

Alcohols:



Ether:



Self Test

10. 1 Write the possible isomers of molecular formula $\text{C}_4\text{H}_{10}\text{O}$ with their IUPAC names.

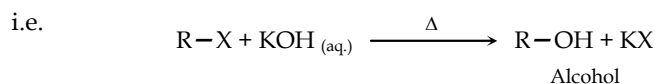
10.1.5 General Methods of Preparation of Monohydric Alcohols

Monohydric alcohols are generally prepared from:

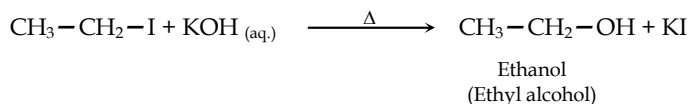
- Haloalkanes
- Aldehydes and ketones
- Esters
- Primary amines

1. Preparation of alcohols from haloalkanes: (Nucleophilic Substitution Reaction of Haloalkanes).

- a. **By hydrolysis of haloalkanes:** Haloalkanes undergo hydrolysis into alcohols by boiling with aqueous solution of alkali like NaOH or KOH.

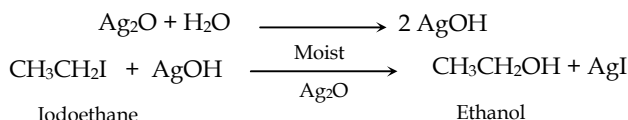


Examples:



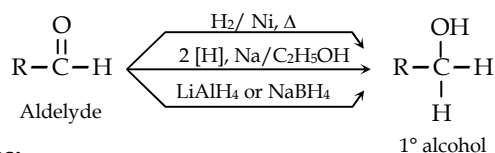
- i. Primary alkyl halides give good yield of alcohols. However, tertiary alkyl halides in this reaction give mainly alkene due to dehydrohalogenation.
- ii. Secondary alkyl halides give a mixture of alcohol and alkene.
- iii. This method is not very useful commercially for preparing aliphatic alcohols because alkyl halides are themselves obtained from alcohols. However, this method is utilized in the preparation of aromatic alcohols.

- b. **By action of moist silver oxide on haloalkanes:** Haloalkanes give corresponding alcohols upon heating with moist silver oxide. (Suspension of Ag_2O is moist ether).

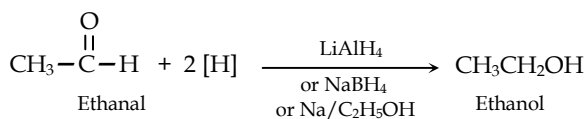
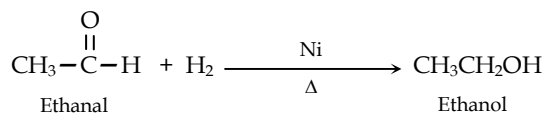
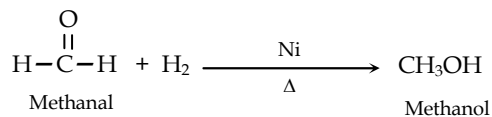
**2. Preparation of alcohols from aldehydes and ketones**

- a. **By reduction of aldehydes and ketones:** Aldehydes and ketones are easily reduced to the corresponding alcohols by the action of reducing agents like H_2/Ni or $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ or LiAlH_4 or NaBH_4 etc.

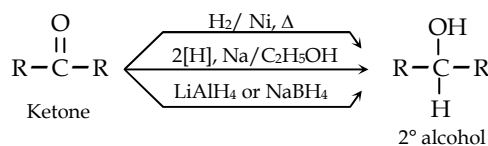
- i. Aldehydes are reduced to primary (1°) alcohols.



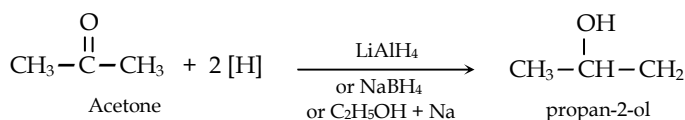
Examples:



ii. Ketones are reduced to secondary (2°) alcohols.

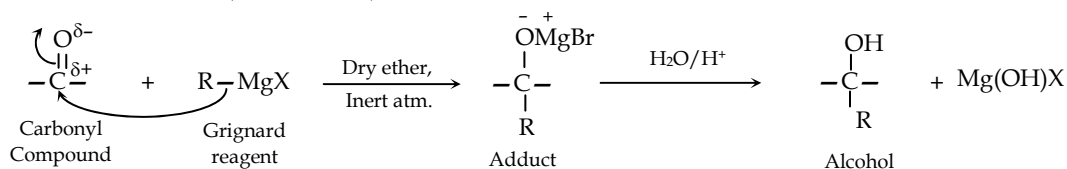


Example:



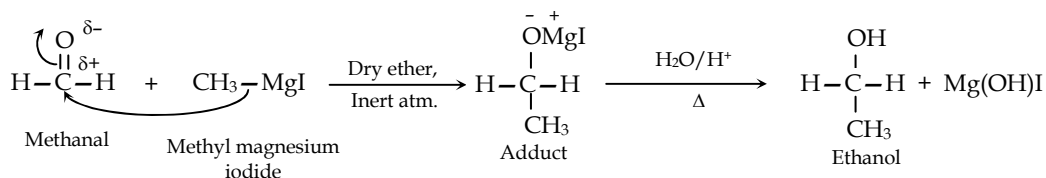
b. By Grignard synthesis of alcohols: Grignard reagent (RMgX) undergoes nucleophilic addition reaction to the carbonyl (>C=O) group of aldehydes or ketones form an intermediate addition product *adduct*, which on acidic hydrolysis with dilute HCl yields different types of alcohols.

According to the nature of aldehydes and ketones used in the Grignard synthesis, different classes of alcohols (1°, 2° and 3°) are obtained.

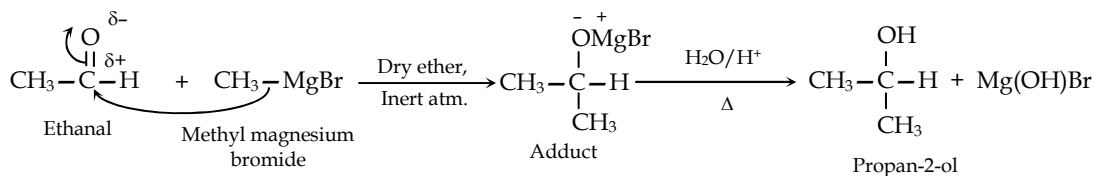


Examples:

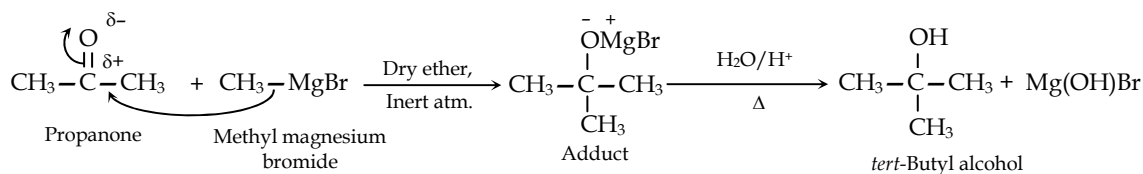
i. **Synthesis of primary alcohols:** Addition of Grignard reagent to formaldehyde followed by the hydrolysis of adduct gives primary alcohol.



ii. **Synthesis of secondary alcohols:** Addition of Grignard reagent to aldehydes other than formaldehyde, followed by hydrolysis of adduct yields secondary alcohols.

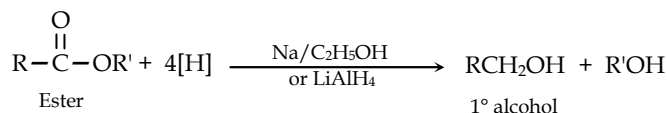


iii. **Synthesis of tertiary alcohols:** Addition of Grignard reagent to ketones followed by hydrolysis of adduct yields tertiary alcohols.

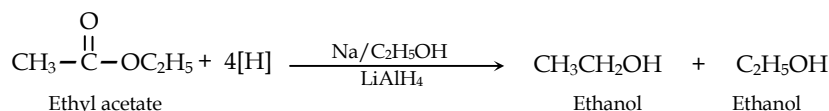


3. Preparation of alcohols from esters:

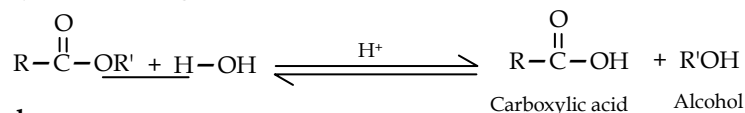
- a. **By reduction of esters:** Esters are easily reduced with different reducing agents like Na/C₂H₅OH or LiAlH₄ into alcohols.



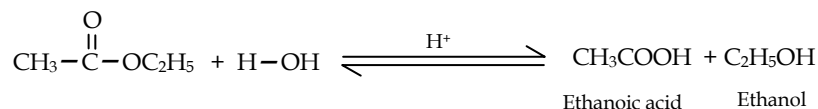
Example:



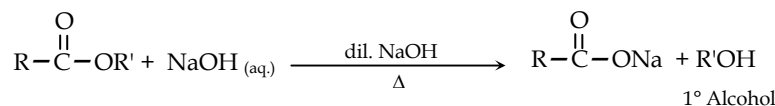
- b. **By acidic hydrolysis of esters:** When esters are subjected to acidic hydrolysis by boiling with dilute H₂SO₄, a mixture of alcohol and carboxylic acid is formed. Alcohol is obtained from the alcohol (R'O-) group of ester (RCOOR') while the carboxylic acid is obtained from acyl (R-CO-) group of ester (RCOOR').



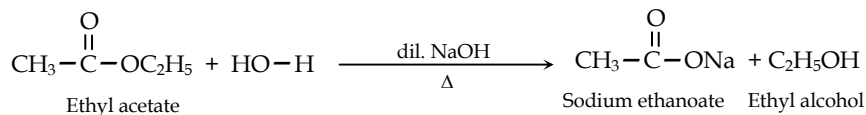
Example:



- c. **By alkaline hydrolysis of esters:** When esters are subjected to alkaline hydrolysis by boiling with dilute aqueous solution of alkali (NaOH or KOH), a mixture of alcohol and salt of carboxylic acid (carboxylate salt) is formed. Alcohol is obtained from the alkoxy (R'O-) group and the carboxylate salt is obtained from acyl (R-CO-) group of ester (RCOOR').

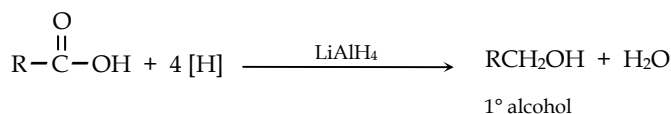


Example:

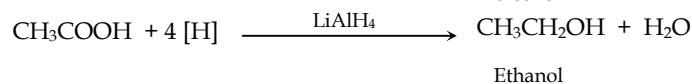


4. Preparation of alcohols from carboxylic acids:

Carboxylic acids can be reduced to primary alcohols by using the reducing agents like H₂/copper chromite, LiAlH₄ etc.

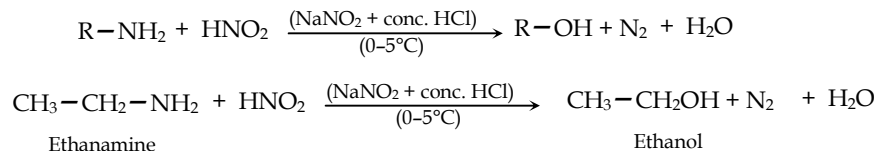


Example:

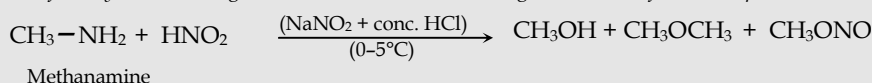


5. Preparation of alcohols from aliphatic primary amines:

Aliphatic primary amines react with cold solution of nitrous acid (mixture of cold and aqueous solution of NaNO_2 and conc. HCl) at about $0^\circ\text{--}5^\circ\text{C}$ to form an unstable alkyl diazonium salts (alkyl diazonium chloride, R-N=N-Cl). The salts at once decompose to liberate nitrogen gas and form primary alcohols.

**Note:**

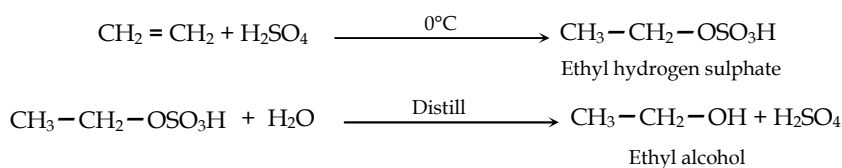
Preparation of methyl alcohol using this method is not suitable as it gives mixture of other compounds also

**10.1.6 Industrial Method of Preparation of Alcohols**

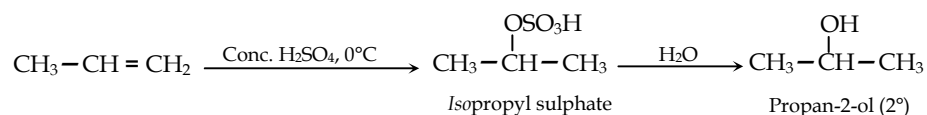
Alcohols are commercially prepared from:

- Alkenes
- Carbohydrates by fermentation
- From alkenes:** In commercial scale, alkenes are obtained from the thermal cracking or pyrolysis of petroleum products, and hence these alkenes are widely used for various synthetic purposes. Among them commercial preparation of alcohols is the most important.
 - By hydration of alkenes:** Hydration of alkene means addition of water molecule onto the $\text{C}=\text{C}$ double of the alkene molecule to yield alcohol.

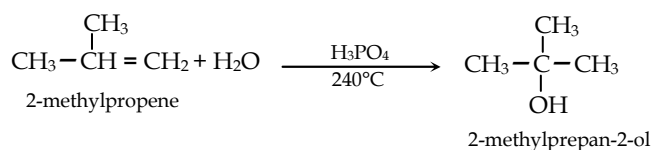
Hydration of alkene is carried out by passing the alkene gas through cold and concentrated H_2SO_4 or H_3PO_4 under pressure, where H_2SO_4 is added to the $\text{C}=\text{C}$ double bond of alkene to form an addition product of alkyl hydrogen sulphate. Alkyl hydrogen sulphate thus formed on subsequent hydrolysis by boiling it with water yields alcohol.



The addition of H_2SO_4 to unsymmetrical alkenes takes place according to Markownikov's rule and hence there is overall Markownikov's addition of water onto the $\text{C}=\text{C}$ double bond of unsymmetrical alkenes.

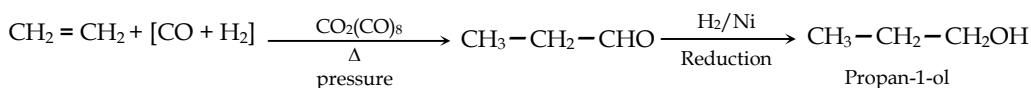


Direct hydration of alkene can also be carried out under high temperature and high pressure in presence of phosphoric acid (H_3PO_4) catalyst.



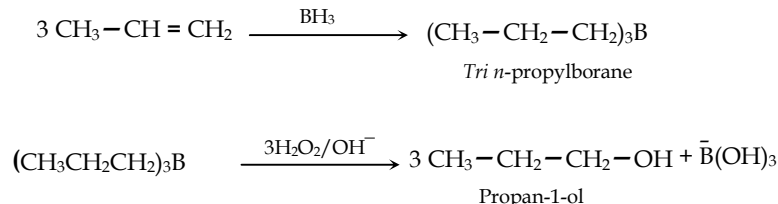
- b. **By Oxo-Process: Hydroformylation of alkene:** Alkenes react with carbon monoxide and hydrogen in the presence of *oxo*-catalyst (octacarbonyl/dicobalt), $\text{Co}_2(\text{CO})_8$, at high temperature and pressure to yield aldehydes. Catalytic hydrogenation of aldehydes with H_2/Ni gives primary alcohol having one more carbon atom than that of its parent alkene.

Oxo-process is hydroformylation reaction which means addition of hydrogen and formyl ($-\text{CHO}$) group onto the $\text{C}=\text{C}$ double of alkene molecule in presence of *oxo*-catalyst.



- c. **By Hydroboration-oxidation of alkene:** The indirect addition of water to alkenes can be performed through Hydroboration-oxidation reaction. In this method alkene is treated with diborane followed by treatment with alkaline H_2O_2 yields primary alcohol.

Hydroboration is a reduction process, which is carried out by treating alkene with diborane, $(\text{BH}_3)_2$ to give alkyl dialkyl trialkyl borane which upon oxidation with alkaline solution of H_2O_2 gives primary alcohol.

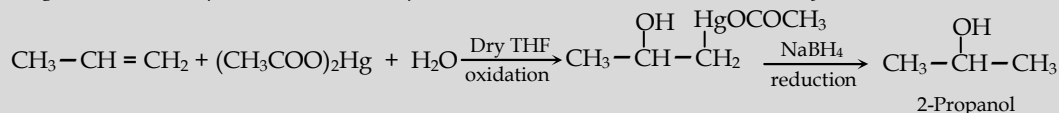


Each step the reaction here follows Markovnikov's rule. But the product formed here is anti-Markovnikov product.

By Oxymercuration-demercuration:

The indirect addition of water to alkenes can also be performed through oxymercuration-demercuration reaction. In this method alkene is treated with mercuric acetate in THF-water solution followed by treatment with NaBH_4 yields alcohol.

This is carried out by treating alkene with aqueous solution of mercuric acetate in dry THF, where alkene is oxidized organo-mercuric compound, which on subsequent reduction or demercuration with NaBH_4 yields alcohol.



Overall oxymercuration-demercuration is a hydration of alkene, in which addition of water occurs in Markovnikov's way in the $\text{C}=\text{C}$ double bond of alkene.

2. **From Carbohydrates by fermentation:** Many chemical reactions like petrification of meat, conversion of fruit juices and sugars into alcohol and vinegar are brought about by the enzymatic action of the living organisms. In all these reactions, the complex organic compounds are decomposed into simpler substances. Such process of slow decomposition of complex organic compound into simpler ones by the agency of microorganisms (Ferments) is known as

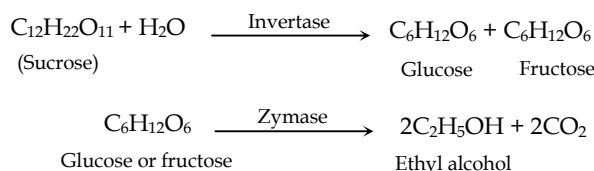
Fermentation (Latin, ferver = to boil). Fermentation is an **exothermic process**.

In the modern sense, fermentation is the process of slow decomposition of complex organic substances into its simpler form of organic compounds in the presence of complex nitrogenous substances secreted by the cells of microorganisms (e.g. yeast, bacteria) known as **Enzymes**. The enzymes act as a catalyst in all these reactions and are generally referred to as **biocatalysts**.

The Yeast containing various enzymes like invertase, zymase etc., which are used for the fermentation of carbohydrates like cane sugar juice, molasses, grapes, apples etc. into ethyl alcohol.

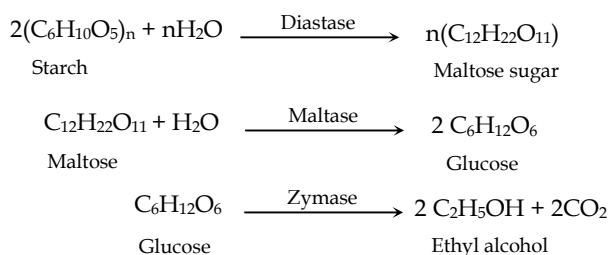
- i. **Manufacture of ethyl alcohol from sugar or molasses:** Molasses is a mother liquor left during the crystallization of sugar. It is a dark coloured viscous liquid which contains 30% sucrose and 32% invert sugar i.e. mixture of glucose and fructose.

Fermentation of sugar or molasses is carried out by adding quantitative amount of yeast to the solution of sugar or molasses at low concentration in presence of ammonium phosphate, $(\text{NH}_4)_3\text{PO}_4$ as a food for yeast. The mixture is left under warm condition at about 30°C for few days, when fermentation takes place with the slow effervescence of CO_2 gas with the boiling appearance.



- ii. **Manufacture of ethyl alcohol from starch (Weizmann method):** Fermentation of starch rich substances like rice, wheat, barley, potatoes, maize etc. into ethanol is carried out in usual manner as that has been used for fermentation of sugar or molasses. But in this process the starch containing materials are first cooked with water so as to release the granules of starch. The resulting mass is called "**Mash**". The Mash is then mixed with "**Malt**" and the mixture is left under warm condition at about 30°C for few days, when fermentation takes place.

Malt is especially germinated barley. Diastase enzyme from malt converts starch into maltose (a disaccharide). Maltase by enzymatic action of yeast gives ethanol.



The fermented liquor obtained from fermentation vessel is called "**Wash**", which is filtered and the filtrate on distillation gives a **distillate**. The distillate contains about 90% ethyl alcohol along with impurities of acetaldehyde and fusel fuel (fusel fuel mainly contains amyl alcohol, $\text{C}_5\text{H}_{11}\text{OH}$). The fractional distillation of distillate gives a major fraction of about **95% ethyl alcohol** and a minor fraction of acetaldehyde and fusel fuel. These two fractions are separately collected. The major fraction of ethyl alcohol is called **Rectified spirit**. The exact composition of rectified spirit is 95.87% ethyl alcohol and 4.13% water. The fermentation can be retarded by NH_4^+ salts.

10.1.7 Physical Properties of Alcohols

The -OH group of alcohol gives the characteristics physical and chemical properties whereas alkyl group (R-) depending upon its shape and size modifies their properties.

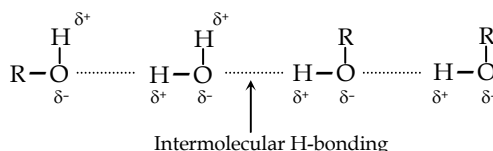
The -OH group of alcohol is quite polar and contains H-atom bonded to the highly electronegative oxygen.

1. State: The lower members are usually colorless liquids with the characteristic hospital odour, whereas higher members are usually colourless and odourless solids.

2. Solubility:

- i. The solubility of alcohols is higher than that of the solubility of corresponding aliphatic hydrocarbons (alkanes), alkyl halides (haloalkanes) and ethers of comparable molecular masses.

Reasons: The high solubility of alcohols in water is due to ability of forming hydrogen bonding between water and polar O-H group of alcohol molecules. Such hydrogen bonding is absent in alkanes and alkyl halides.



- ii. There is gradual decrease in the solubility of alcohols in its homologous series with increase in their molecular masses.

Reasons: The gradual increase in the molecular mass in the homologous series of alcohols is due to increase in the molecular mass of the non-polar lipophilic or hydrophobic alkyl group of the alcohols becomes predominant and masks the effect of polar -OH group and hence solubility goes on decreasing gradually from lower alcohols to higher alcohols.

- iii. The solubility of isomeric alcohols increases with branching of carbon chain in the molecules. It means that the solubility of isomeric alcohols increase in the order as $1^\circ < 2^\circ < 3^\circ$ alcohols.

Reason: The branching in the non-polar alkyl group of alcohol causes the decrease in available surface area of the molecule and hence enhancing the solubility. The argument is well supported by solubility data of isomeric alcohols with molecular formula $\text{C}_4\text{H}_9\text{OH}$:

Alcohols	Solubility: gm/100gm of water
<i>n</i> -Butyl alcohol	7.9
Isobutyl alcohol	10.0
<i>sec</i> -Butyl alcohol	12.5
<i>tert</i> -Butyl alcohol	> 12.5

3. Boiling point and melting point:

- i. The boiling points of alcohols are higher than that of the boiling points of corresponding aliphatic hydrocarbons (alkanes), alkyl halides (haloalkanes) and ethers of comparable molecular masses.

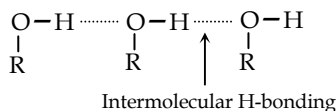
The given data illustrate the higher boiling point of alcohols with respect to other compounds.

Compounds	Molecular wt.	B. pt. (°C)
Propane (CH ₃ .CH ₂ .CH ₃)	44	- 65
Ethyl alcohol (CH ₃ .CH ₂ OH)	46	78
Dimethyl ether (CH ₃ .O.CH ₃)	46	- 25

Reasons: Alcohols molecules are closely and strongly associated together by strong intermolecular force of attraction of hydrogen bonding. Such hydrogen bonding is absent in alkanes, alkyl halides and ether. The presence of hydrogen bonding in alcohols requires an additional amount of energy to overcome the hydrogen bonding so the alcohols molecules boil at high temperature.

- ii. There is gradual increase in the boiling points of alcohols in its homologous series with the increase in their molecular masses.

Reason: The gradual increase in the molecular mass in the homologous series of alcohols also increases the magnitude of van der Waals' force, which causes the increase in the boiling points of alcohols.



- iii. The boiling points of isomeric alcohols decrease with branching of carbon chain in the molecules. It means that the boiling points of isomeric alcohols decrease in the order as 1° > 2° > 3° alcohols.

Reason: The gradual decreases in the boiling points of the isomeric alcohols are due to the regular decrease in the available surface area occupied by more branched alcohols molecules.

10.1.8 Important Terms Used in Alcohol Industries

- Alcoholate:** Lower alcohols especially methyl alcohol and ethyl alcohol interact with anhydrous CaCl₂ or MgCl₂ to form the solid crystalline derivatives known as *alcoholates*. Like water molecules are closely associated in the hydrated crystalline inorganic salts as 'water of crystallization', similarly, alcohol molecules are also closely associated in the crystalline alcoholates as '*alcohol of crystallization*'. It is the reason for that alcohol cannot be dried over anhydrous CaCl₂ or MgCl₂.

Example: CaCl₂.3C₂H₅OH MgCl₂.6C₂H₅OH

- Wood spirit (wood alcohol):** Methyl alcohol also called *carbinol* is commonly known as *wood spirit* because of one time it was obtained by *destructive distillation of wood*. It is highly poisonous that drinking of it causes blindness as well as nervous *unconscious*. A dose of 15 to 20 mL of CH₃OH is sufficient to cause death of a person.
- Rectified spirit and absolute alcohol:** Ethyl alcohol obtained by the fractional distillation of distillate from *wash* of fermentation as a major fraction containing concentration about 95% is called *Rectified spirit*, it includes 95.87% ethyl alcohol and 4.13% water is called *azeotrope or constant boiling mixture* which can not be purified by simple distillation.

The pure and anhydrous ethyl alcohol (100%) is called *absolute alcohol*. The absolute alcohol is obtained from rectified spirit by following methods:

- By distilling the rectified spirit over quicklime (CaO), followed by re-distillation over calculated amount of sodium metal in anhydrous condition.
- By *azeotropic distillation*: Rectified spirit is mixed with a very small amount of benzene make ternary azeotrope which distillation gives 100% alcohol.

4. **Power alcohol:** A mixture of absolute ethyl alcohol and gasoline (petrol) in the approximate proportion of about 20:80 in a co-solvent like benzene or ether is known as '*power alcohol*'. It is called power alcohol because it can be used to generate the power in internal combustion engines. The calorific value of alcohol is low but it has good antiknocking properties. Its octane number is 90. Alcohols thus help to increase octane number of fuel. It is also important to reduce dependency on petroleum import.
5. **Methylated spirit (Denatured alcohol):** Ethyl alcohol is made available only for industrial purpose by addition of some poisonous substances like methyl alcohol, acetone, pyridine or colouring matter, and hence it becomes unfit for drinking.
It has been found that some times urea is illegally added to ethyl alcohol to improve its alcoholic action, which is also unfit for drinking.
Such undrinkable ethyl alcohol containing poisonous substances especially methyl alcohol is called *denatured alcohol* or *methylated spirit*. The process of making methylated spirit is known as *denaturing*. Methylated alcohol is very important and excellent solvent in industries. It is much cheaper than that of ethyl alcohol, and hence it has been illegally sold for drinking and it causes serious physiological hazard to the human health including blindness. This alcohol is not taxable as it is to be used for industrial purposes only.
6. **Alcoholic beverages:** They are liquors for drinking purpose. They contain ethanol, water, colouring material and flavouring material. They may be distilled (e.g. Vodka, Gin, Whiskey, brandy etc.) or undistilled (e.g. Beer, Wine, etc.)

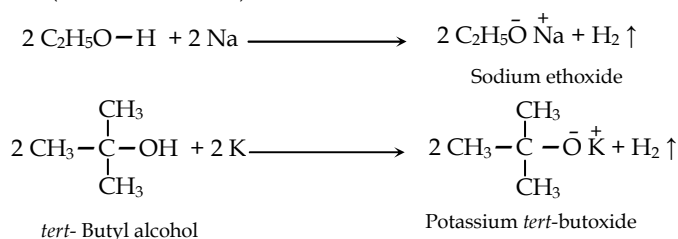
10.1.9 Chemical Properties of Alcohols

The chemical properties of alcohols can be studied under the following heads.

- A. Reactions involving the cleavage of O—H bond (Oxygen-Hydrogen bond) in alcohols: Reaction due to —H atom of alcohols:
- B. Reactions involving the cleavage of C—O bond (Carbon-Oxygen bond) in alcohols: Reaction due to —OH group of alcohols:

A. Reaction involving the cleavage of O—H (Oxygen-hydrogen) bond in alcohols: Reaction due to —H atom of alcohols:

1. **Reaction of alcohols with active metals: Acidic nature of alcohols:** When alcohols are treated with active metals like Na, K, Mg etc., these liberate hydrogen with the formation of metal alkoxides (alkali alcoholate).



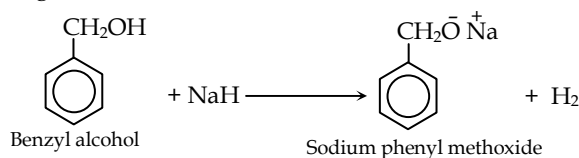
The active metals displace hydrogen from the —OH group of alcohol in the form of H₂ gas. This shows the acidic nature of alcohols.

- i. However, alcohols are very weak acids ($K_a=10^{-16}$ – 10^{-18}) even weaker than water ($K_a=10^{-14}$) that they neither turn litmus paper nor neutralize even strong alkali like NaOH. Alcohols do not react with aqueous alkali solution.

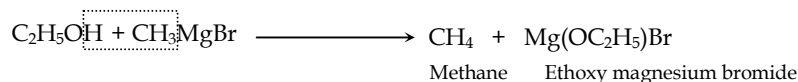
- ii. The weak acidic nature of alcohols than that of water is also explained on the basis of electron releasing inductive effect (+I effect) of alkyl (R-) group attached to -OH group in alcohols. The electron releasing tendency of alkyl group increases the electron density around the oxygen atom due to which the polarity of the O-H bond decreases and causes the breaking of O-H bond comparatively difficult in alcohols than that of water. In addition, the alkoxide (RO-) ion formed is also destabilized due to high concentration of negative charge on oxygen atom by electron releasing inductive effect of alkyl group. Thus, the release of H⁺ ion is difficult. This explains the weak acidic character of alcohols and hence makes alcohols weak acidic than water.

Order of acidity: H-OH > CH₃-OH > RCH₂-OH (1° alcohol) > R₂CH-OH (2° alcohol) > R₃C-OH (3° alcohol) > NH₃ > CH₄

2. **Reaction with metal hydrides:** Alcohols react with metal hydrides to form salts along with evolution of hydrogen gas.

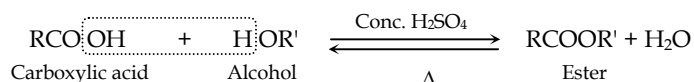


3. **Reaction with Grignard's reagent:** Alcohol contains active hydrogen atom so, it reacts with Grignard reagent to form alkane. In this reaction, the alkane is obtained from the alkyl part of the Grignard's reagent. This reaction is used for the estimation of alcoholic group under the name of *Zerewitnoff reaction*.



4. Formation of esters: Esterification reaction

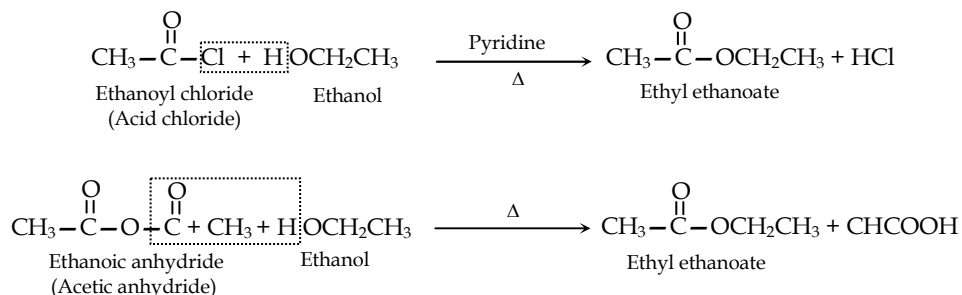
- a. **Reaction of alcohols with carboxylic acids: Basic nature of alcohol:** When a mixture of alcohol and carboxylic acid is heated in presence of calculated amount of conc. H₂SO₄, an ester is formed, which gives a characteristic pleasant fruity smell. This reaction is called *esterification test*.



The function of conc. H₂SO₄ is to act as a dehydrating agent since the reaction is reversible in nature and the equilibrium can be shift to the right by removing water as soon as it is formed.

This reaction has an important application for the detection of both alcoholic and carboxylic acid functional groups under the name of esterification test since the ester gives the characteristics fruity smell.

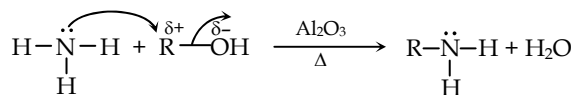
- b. **Reaction of alcohols with acid chlorides and acid anhydrides: Acylation:** When an alcohol is heated either with acid chloride or with acid anhydride in the presence of pyridine as catalyst, an ester is formed. Through this reaction, an acyl group (R-CO-) is introduced in the alcohol molecule by replacing hydrogen atom of the -OH group and hence the reaction is termed as '*acylation*' of alcohols.



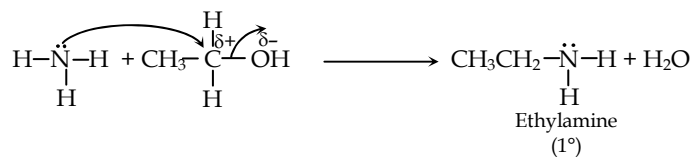
B. Reaction involving the cleavage of C–O (Carbon-oxygen) bond in alcohols:
Reaction due to –OH group of alcohols: Nucleophilic substitution reaction:

- Reaction of alcohols with ammonia: Ammonolysis of alcohols:** Like ammonolysis of alkyl halides, alcohols can also undergo ammonolysis when the vapor of alcohols mixed with ammonia is passed over heated alumina (Al_2O_3) or thoria (ThO_3) at about 400°C . The reaction results a mixture of amines (1° , 2° and 3°).

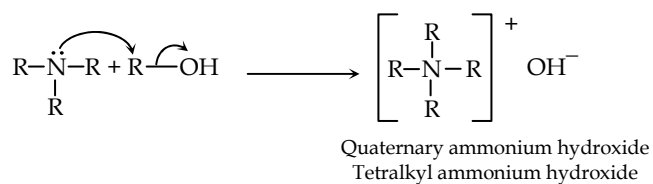
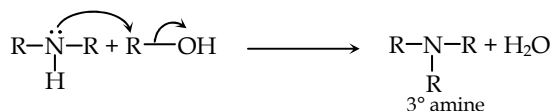
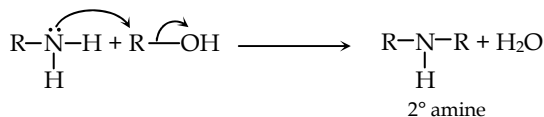
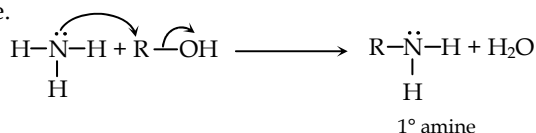
The 1° amine would be a major product, when large excess of ammonia gas is used as compared to alcohol.



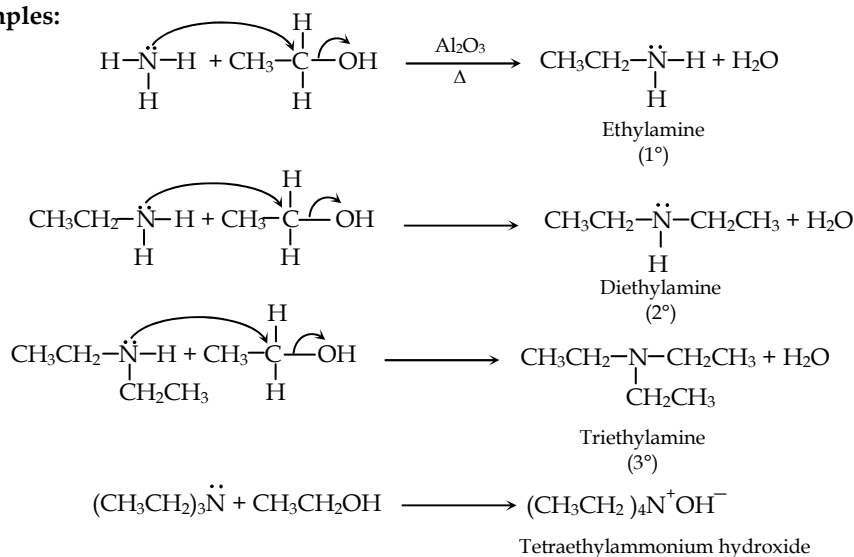
Example:



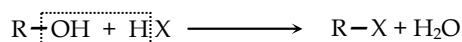
However, if the alcohol is in excess, the 1° amine formed at first stage of reaction as given above further reacts with excess of alcohol to form 2° amine then 3° amine and finally 4° quaternary ammonium hydroxide.



Examples:

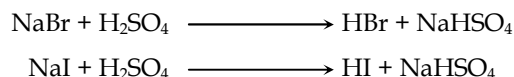


2. **Reaction of alcohols with hydrogen halides (HX):** Alcohols react readily with hydrogen halides to yield alkyl halides and water.



The reaction is carried out either by passing dry hydrogen halide gas into the heated alcohol or by heating the alcohol with the concentrated aqueous solution of hydrogen halide. In some reactions the use of catalyst may be needed.

Sometime hydrogen bromide is generated *in situ* by reaction between sulphuric acid and sodium bromide in presence of alcohol. Similarly hydrogen iodide is generated in the same manner by reaction between sulphuric acid and sodium iodide.

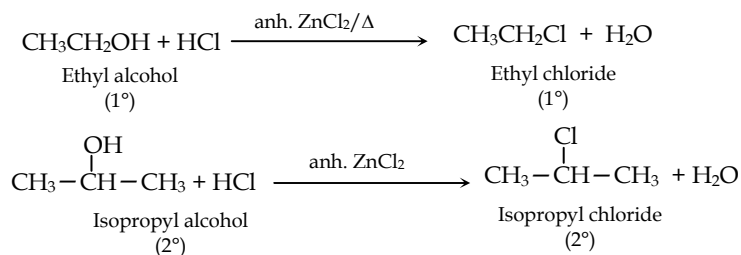


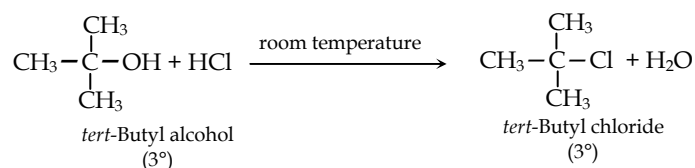
The order of reactivity of various alcohols for given hydrogen halide follows the sequence of: 3° alcohols > 2° alcohols > 1° alcohols > methyl alcohol

On the other hand, the order of reactivity of various hydrogen halides for a given alcohol follow the sequence of: HI > HBr > HCl

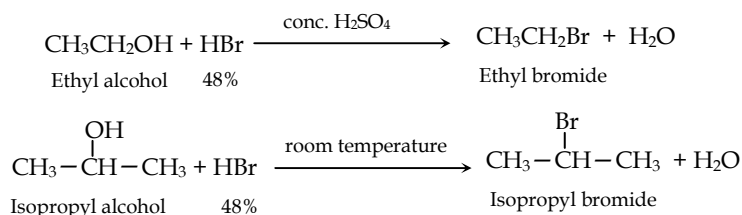
Examples:

- a. On this background, the least reactive hydrogen halides like hydrogen chloride, generally requires the presence of anhydrous ZnCl_2 for reaction with 1° and 2° alcohols. The mixture of hydrogen chloride and anhydrous ZnCl_2 is called **Lucas reagent**. On the other hand, the very reactive *tert*-butyl alcohol reacts with conc. HCl by simply shaken at room temperature.

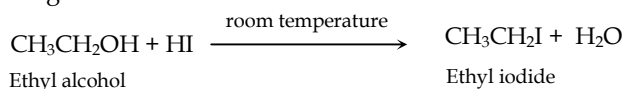




- b. Primary alcohol when reacts with hydrogen bromide in presence of conc. H_2SO_4 gives corresponding alkyl bromide. For secondary and tertiary alcohols reacts with hydrogen bromide even in absence of conc. H_2SO_4 .

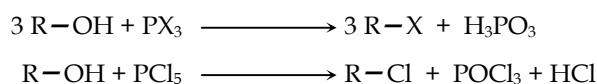


- c. Hydrogen iodide is highly reactive so it reacts with alcohols even in room temperature when shaken together.

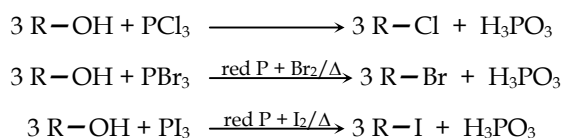


The order of reactivity of various alcohols for a hydrogen halide can be explained on the basis of electron releasing inductive (+I) effect of alkyl ($\text{R}-$) groups present in alcohols. The alkyl groups by their electron releasing effect tend to increase the electron displacement towards oxygen. In other words, the polarity of $\text{C}-\text{O}$ bond increases and this makes the breaking of the bond between carbon and oxygen easier. Therefore, the alcohols with greater number of alkyl groups attached to carbon will be more reactive. This justifies the above sequence of reactivity of alcohols in this type of reactions.

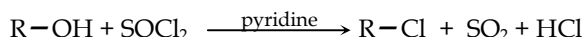
3. **Reaction of alcohols with phosphorus halides:** Alcohols are readily converted to the corresponding alkyl halides by reaction with phosphorus halides (PCl_5 or PX_3).



PBr_3 and PI_3 are generally prepared *in situ* from the reaction of red phosphorus with bromine and iodine respectively.



4. **Reaction of alcohols with thionyl chloride:** Alcohols react with thionyl chloride in presence of pyridine, to give chloroalkane.



The order of reactivity of various alcohols towards this reaction follows the same sequence of: 3° alcohols $>$ 2° alcohols $>$ 1° alcohols $>$ methyl alcohol

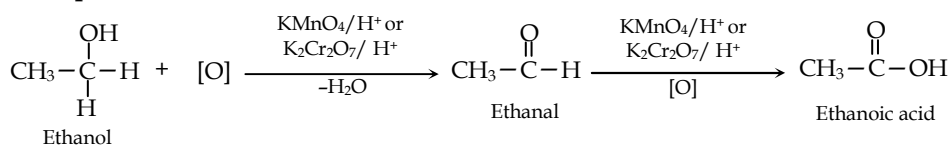
C. Other Reaction:

1. Oxidation of alcohols

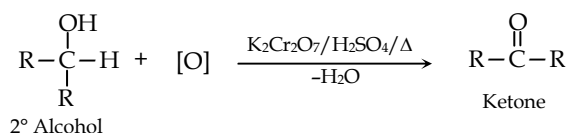
a. Oxidation using KMnO_4/H^+ or $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$

Oxidation of primary (1°) alcohols: Primary alcohols are oxidized first to aldehydes, which on further oxidation give carboxylic acids with the same number of carbon atoms as that of the parent alcohols.

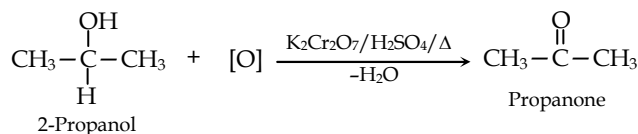
Examples:



Oxidation of secondary (2°) alcohols: Secondary alcohols are oxidized to ketones with the same number of carbon atoms as that of the parent alcohols.

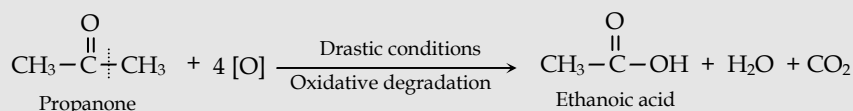


Examples:



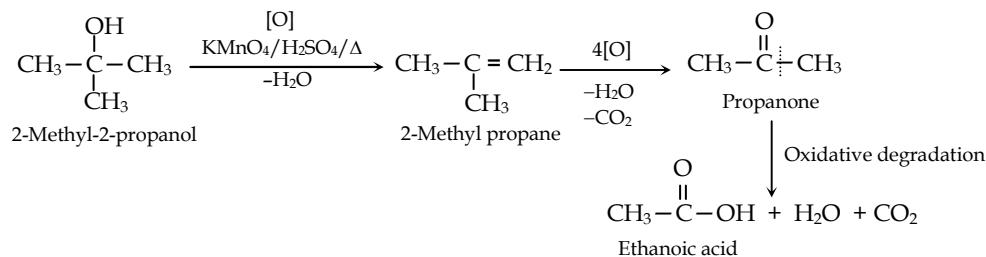
Note:

Unlike aldehydes, ketones are quite resistant to further oxidation under ordinary and mild conditions with these oxidizing agents. But upon prolonged treatment with strong oxidizing agent at high temperature, ketones are oxidized to carboxylic acids with lesser number of carbon atoms than that of the parent secondary alcohols. The reaction is called "**oxidative degradation**" of alcohols.



Oxidation of tertiary (3°) alcohols: Tertiary alcohols are quite resistant for oxidation with mild oxidizing agents in aqueous, alkaline or neutral conditions.

However, when oxidation is carried out under drastic acidic conditions with strong oxidizing agent like KMnO_4 conc. H_2SO_4 , tertiary alcohols first undergo dehydration into alkenes. The alkene formed is then oxidized to ketone, which is finally oxidized to carboxylic acid with lesser number of carbon atoms than that of the parent alcohol.



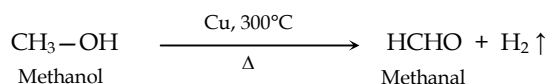
Oxidation using PCC

Pyridinium chlorochromate: PCC is obtained by treating CrO_3 with pyridine and HCl . This reagent is used in non-polar solvent like CH_2Cl_2 . It does not hydrate aldehyde. So, no further oxidation of aldehydes takes place here. PCC can oxidize 1° alcohol into aldehyde and 2° alcohol into ketones.

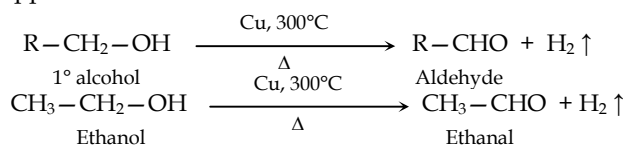
b. Catalytic dehydrogenation of alcohols: (Oxidation):

When vapour of alcohol is passed over heated copper at 300°C , different nature of products are formed depending upon the nature of alcohol used in the reaction.

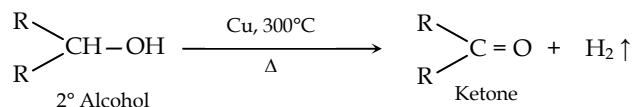
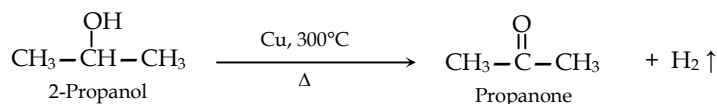
- i. Methyl alcohol forms formaldehyde upon catalytic dehydrogenation by passing its vapor over reduced copper at 300°C .



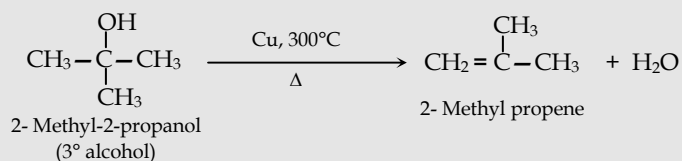
- ii. Primary alcohols form aldehydes upon catalytic dehydrogenation by passing their vapors over reduced copper at 300°C .



- iii. Secondary alcohols form ketones upon catalytic dehydrogenation by passing their vapors over reduced copper at 300°C .

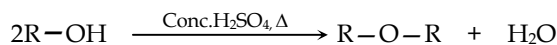
**Examples:****Note:**

Tertiary alcohols are resistant towards dehydrogenation due to absence of α -hydrogen. However, upon passing their vapors over reduced copper at 300°C , they get dehydrated into alkenes rather than dehydrogenation.

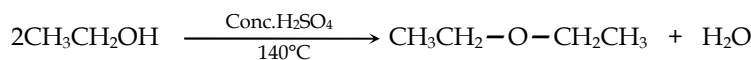


2. **Dehydration of alcohols:** Alcohols undergo dehydration reaction into different products depending upon the conditions of reaction and nature of dehydrating agents used in the reaction.

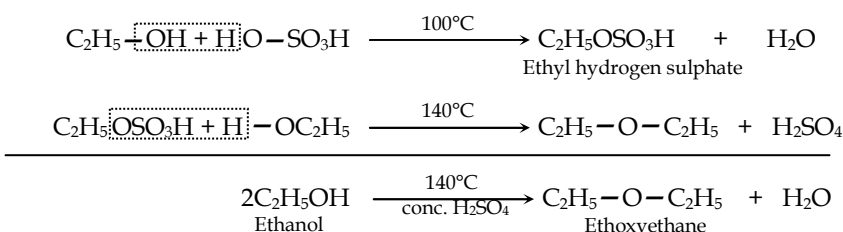
- a. **Dehydration of alcohols into ethers with catalytic amount of conc. H_2SO_4 at 140°C :**
Intermolecular dehydration of alcohols: Preparation of Ether: When excess of ethyl alcohol is heated with catalytic amount of conc. H_2SO_4 as a dehydrating agent at about 140°C , ethyl alcohol undergoes intermolecular dehydration to give diethyl ether.



Example:

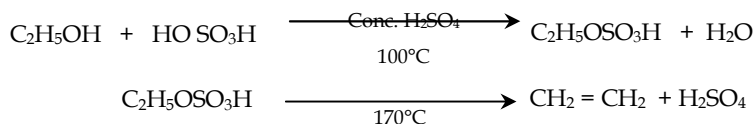


The reaction takes place in two steps: In first step, ethyl alcohol reacts with H_2SO_4 at $100^\circ C$ to form ethyl hydrogen sulphate. In second step, ethyl hydrogen sulphate further reacts with excess of ethyl alcohol at $140^\circ C$ to produce diethyl ether with the regeneration of H_2SO_4 . The regenerated H_2SO_4 can further reused to dehydrate addition amount of ethyl alcohol into diethyl ether and hence the method is known as '*Williamson's Continuous Etherification*'.



- b. Dehydration of alcohols into alkenes with excess of conc. H_2SO_4 at $170^\circ C$:**
Intramolecular dehydration of alcohols: Preparation of Alkenes: When calculated amount of ethyl alcohol is heated with excess of conc. H_2SO_4 as a dehydrating agent at about $170^\circ C$, ethyl alcohol undergoes intramolecular dehydration to ethene.

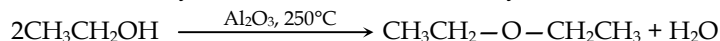
The reaction takes place in two steps: In first step, ethyl alcohol reacts with H_2SO_4 at $100^\circ C$ to form ethyl hydrogen sulphate. In second step, ethyl hydrogen sulphate decomposes at about $170^\circ C$ to produce ethene.



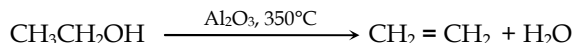
Dehydration of alcohols into alkenes can also be carried out by heating alcohol with phosphoric acid or boric acid at high temperature.

- c. Catalytic dehydration of alcohols with alumina (Al_2O_3):**

- i. Passing the vapor of ethyl alcohol over heated alumina to the temperature less than $300^\circ C$ can undergo intermolecular dehydration of alcohol into diethyl ether.



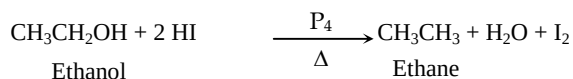
- ii. On the other hand, passing the vapor of ethyl alcohol over heated alumina as a catalyst to the temperature above than $300^\circ C$ can undergo intramolecular dehydration of alcohol into ethene.



3. Reduction of Alcohols

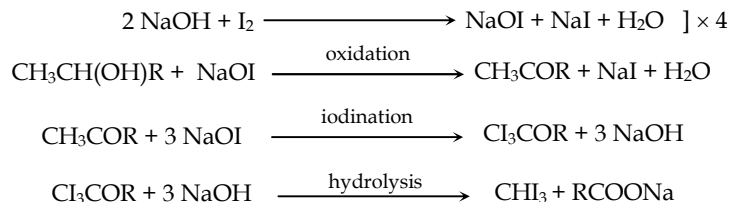
Alcohols can be reduced using reducing agents like HI/P_4 to get corresponding alkanes.

Example:

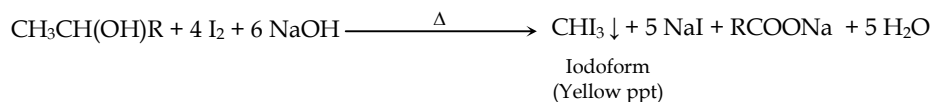


4. Iodoform Test

Alcohols having $\text{CH}_3-\overset{\text{OH}}{\underset{|}{\text{CH}}}-$ structure on heating with excess I_2 and aq. alkali (NaOH) give yellow crystalline precipitate of iodoform. This test is called iodoform test.



The overall reaction is



10.1.10 Distinction of Primary, Secondary and Tertiary Alcohols

There are several methods used to distinguish primary, secondary and tertiary alcohols. These methods are:

- Victor Meyer test:** Given alcohol is treated with PI_3 (red $\text{P}_4 + \text{I}_2$) to get corresponding iodoalkane. The iodoalkane thus obtained is treated with alc. AgNO_2 to get corresponding nitroalkane. The nitroalkane thus obtained is treated with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) and then with aqueous alkali

Colour of the solution observed:

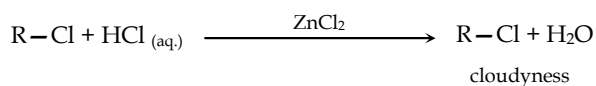
- ♦ If blood red colouration is obtained, the alcohol is primary (1°)
- ♦ If blue colouration obtained the alcohol is secondary (2°) alcohol.
- ♦ Colourless solution indicates presence of tertiary (3°) alcohol.

RCH_2OH (1° alcohol)	R_2CHOH (2° alcohol)	R_3COH (3° alcohol)
$\Delta \downarrow \text{Red P}_4 + \text{I}_2$ $\text{R}-\text{CH}_2-\text{I}$ 1° Iodoalkane $\text{alc.} \downarrow \text{AgNO}_2$ $\text{R}-\text{CH}_2-\text{NO}_2$ 1° Nitroalkane $\text{O}=\text{N}-\text{OH}$ $\text{R}-\text{C}-\text{NO}_2$ $\text{N}-\text{OH}$ Nitrolic acid $\text{aq. NaOH} \downarrow$ $\text{R}-\text{C}-\text{NO}_2$ $\text{N}=\text{O}^-\text{Na}^+$ Sodium nitroate (Blood red colouration)	$\Delta \downarrow \text{Red P}_4 + \text{I}_2$ $\text{R}-\text{CH}(\text{R})-\text{I}$ 2° Iodoalkane $\text{alc.} \downarrow \text{AgNO}_2$ $\text{R}-\text{CH}(\text{R})-\text{NO}_2$ H 2° Nitroalkane $\text{OH}-\text{N}=\text{O}$ $\text{R}-\text{C}-\text{NO}_2$ $\text{N}=\text{O}$ Pseudonitrol $\text{aq. NaOH} \downarrow$ Blue colouration	$\Delta \downarrow \text{Red P}_4 + \text{I}_2$ $\text{R}-\text{C}(\text{R})_2-\text{I}$ 3° Iodoalkane $\text{alc.} \downarrow \text{AgNO}_2$ $\text{R}-\text{C}(\text{R})_2-\text{NO}_2$ 3° Nitroalkane $\text{HNO}_2 \downarrow$ No reaction $\text{aq. NaOH} \downarrow$ Colourless solution

Self Test

10. 2 How would you distinguish propan-1-ol, propan-2-ol and 2-methylpropan-2-ol?

2. **Lucas test:** In this method, the given alcohol is treated with Lucas reagent (mixture of anhydrous ZnCl_2 dissolved in conc. HCl) at room temperature, where an appearance of cloudiness in the reaction mixture shows the formation of alkyl chloride.



Since the order of reactivity of alcohols is $3^\circ > 2^\circ > 1^\circ$:

- A 3° alcohol reacts immediately with Lucas reagent and shows cloudiness immediately.
- A 2° alcohol reacts steadily with Lucas reagent and shows cloudiness after few minutes.
- A 1° alcohol does not react appreciably with Lucas reagent at ordinary condition of temperature and does not show cloudiness. However the cloudyness appears only after heating the solution.

Note:

Iodoform test is given by alcohols with $\text{CH}_3-\text{CH}(\text{OH})-$ structure and aldehyde or ketone with $\text{CH}_3-\text{CO}-$ structure.

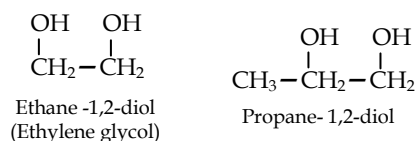
10.1.11 Laboratory Test for Ethanol (Already discussed)

1. Esterification test (Refer section 10.1.9A)
2. Iodoform test (for the compounds having $\text{CH}_3-\text{CH}(\text{OH})$ -structure. Refer section 10.1.9C)
3. Dichromate test and permanganate test (Refer section 10.1.9C)

10.1.12 Dihydric Alcohols

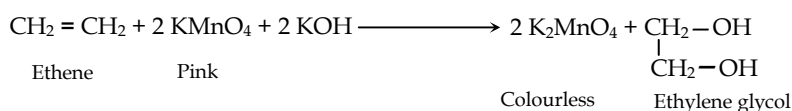
Alcohols containing two $-\text{OH}$ groups attached per molecule are called Dihydric alcohols. The common name of it is Alkyleneglycol and its IUPAC name is alkanediol. The general molecular formula of it is $\text{C}_n\text{H}_{2n}(\text{OH})_2$.

Examples:

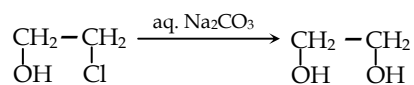
**Preparation of Dihydric alcohols:**

1. **By hydroxylation of alkenes:** Hydroxylation of alkenes means addition of $-\text{OH}$ (hydroxy) groups onto the $\text{C}=\text{C}$ double bond of alkene into diols. Hydroxylation of alkenes into diols is a redox reaction, which is carried out by the action of cold and alkaline solution of KMnO_4 known as *Baeyer's reagent*. Baeyer's reagent acts as an oxidizing agent, where it oxidizes alkenes into diols and itself get reduced to colourless product of MnO_2 . Hence this reaction has great importance in organic analysis to detect the presence of unsaturation center (multiple bonds: double or triple bonds) in organic compounds.

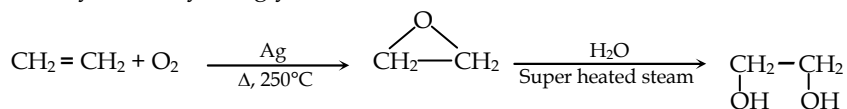
Ethylene glycol is prepared in the laboratory by hydroxylation of ethylene, when the ethylene gas is bubbled through a cold dilute alkaline solution of KMnO_4 (Baeyer's reagent). Ethylene is oxidized to ethylene glycol and the purple color of Baeyer's reagent is discharged.



2. **By hydrolysis of ethylene chlorohydrin:** Ethylene glycol can also be obtained by the hydrolysis of ethylene chlorohydrin.



3. **By hydrolysis of Epoxy-ethane: Industrial preparation of Ethylene glycol:** Ethylene glycol is commercially prepared by oxidation of ethylene. Ethylene on heating with air in silver as a catalyst at about 250°C forms epoxyethane (ethylene epoxide), which upon hydrolysis with super heated steam yields ethylene glycol.

**Physical Properties of ethylene glycol:**

- i. It is a colorless and viscous liquid with sweet taste.
- ii. It boils at 197°C .

- iii. It is highly soluble in both water and ethyl alcohol.

Its viscous nature and high boiling point is due to the intermolecular hydrogen bonding. Its high solubility in water and alcohols is also due to the formation of hydrogen bonding between molecules of water and alcohol.

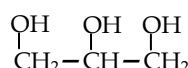
Uses of Ethylene Glycol

- It is used as a solvent.
- It is used as a coolant for aeroplane engines.
- It is used as an anti-freeze agent in the radiators of automobiles.
- It is used in the preparation of explosive.
- It is also used in the preparation of solvents like dioxane, polymethylene glycol, etc.
- It is also used in the preparation of polymer Dacron (Terylene).

10.1.13 Trihydric Alcohols

Alcohols containing three -OH groups attached per molecule are called **trihydric alcohols**. The common name of it is **glycerols** and its IUPAC name is **alkanetriol**. The general molecular formula of it is $C_nH_{2n-1}(OH)_3$. It can be purified by vacuum distillation as it decompose at its normal boiling point.

Example:



Propane-1,2,3-triol

Glycerol

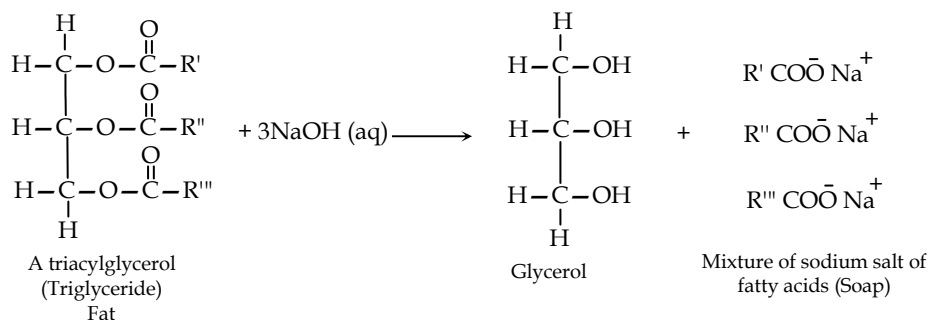
Oils and fats are triesters of glycerol with long chain of higher fatty acids. These higher fatty acids may be palmitic acid ($C_{15}H_{31}COOH$), stearic acid ($C_{17}H_{35}COOH$), oleic acid ($C_{17}H_{33}COOH$) etc. The triesters of glycerol are also known as Triglycerides. A glyceride may be represented by general formula as given below, where R, R' and R'' may be same or different alkyl groups.

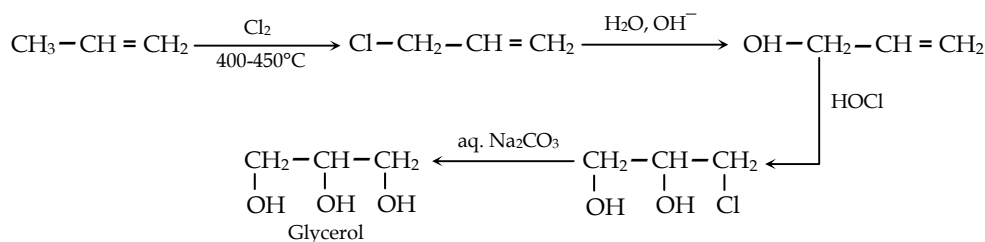
Preparation

1) From oils and fats: By saponification:

Glycerol is prepared in soap industries during the manufacture of soap by heating oils or fats (triglycerides) with caustic soda solution. Thus, the alkaline hydrolysis of triglycerides with NaOH solution into soap and glycerol is called **saponification**. Glycerol is the byproduct of saponification and the major product is soap. Soap is the sodium salt of higher fatty acid.

Reaction:





Uses of Glycerol

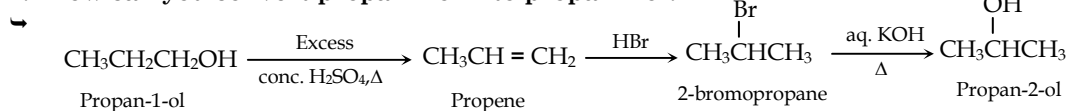
- i. It is used as lubricant.
- ii. It is used to prepare nitroglycerine, which is used in making dynamite.
- iii. It is used as soothing agent.
- iv. It is used in the manufacture of cosmetics and transparent soaps.
- v. It is used as moistening agent for skin in winter in the name of glycerine.
- vi. It is also used as a sweetening agent for sugar free sweet, confectionaries and beverages.
- vii. It is used as a humectant in non-drying printing inks, stamp pad inks etc.
- viii. It is used as an anti-freeze agent in the radiators of automobiles.
- ix. Glyceryltrinitrate is used as medicine to low the blood pressure.

Self Test

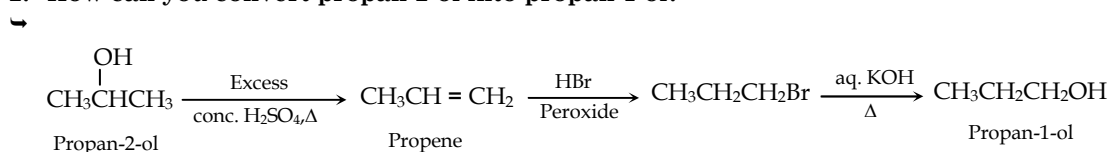
10.3 What are primary, secondary and tertiary alcohols? Starting from Grignard reagent, how would you obtain each of them? State what happens when the three types of alcohols are oxidized?

■ Some Solved Questions

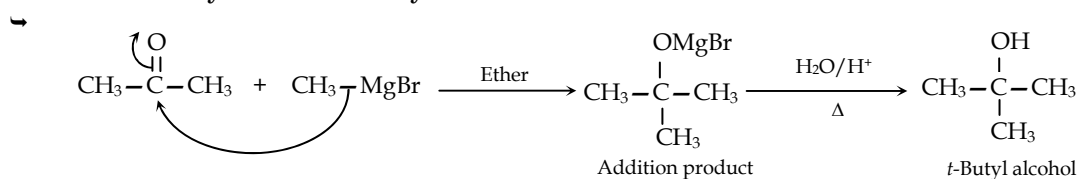
1. How can you convert propan-1-ol into propan-2-ol?



2. How can you convert propan-2-ol into propan-1-ol?

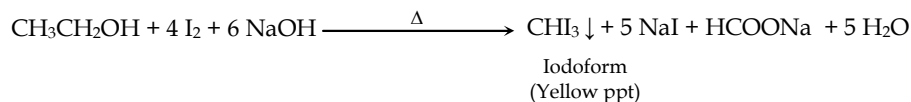


3. Outline the synthesis of t-butyl alcohol from acetone.



4. How can you distinguish methanol and ethanol?

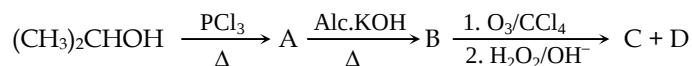
- Methanol (CH_3OH) does not have $\text{CH}_3\text{CH}(\text{OH})-$ structure. So it does not give iodoform test. Whereas ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) has $\text{CH}_3\text{CH}(\text{OH})-$ structure. So, it gives iodoform test.



Questions

Very Short Answer Questions:

- Describe Victor Meyer method of distinguishing 1° , 2° and 3° alcohols.
- Alcohols are not dehydrated by CaCl_2 . Why?
- Why are higher alcohols practically insoluble in water?
- What happens when an alcohol is dehydrated?
- Write structure of the tertiary alcohol having a molecular formula $\text{C}_4\text{H}_{10}\text{O}$ and write its IUPAC and common name.
- Write any two laboratory tests of ethanol.
- What is meant by esterification? Give an example.
- How can you distinguish propan-1-ol and propan-2-ol?
- Why is phenol more acidic than ethanol?
- What happens when propan-2-ol is treated with I_2 in presence of aqueous NaOH ?
- Identify A, B, C and D in the following reaction writing the related reactions.



- What is meant by fermentation? Write an example.

Short /Long Answer Questions:

- How can you distinguish:
 - Ethanol and methanol
 - Propan-1-ol and propan-2-ol by Victor Meyer method
 - t-butyl alcohol and sec-butyl alcohol
 - chloromethane and ethanol
- Outline the synthesis of the following alcohols using Grignard synthesis
 - Propan-2-ol
 - 2-methylpropan-2-ol
 - Isobutyl alcohol
 - n-propyl alcohol
- Write short notes on:
 - Lucas test
 - Victor Meyer test
 - Fermentation
 - Uses of ethanol
 - Oxo process
- A dihydric alcohol $\text{C}_2\text{H}_6\text{O}_2$ (A) undergoes stepwise oxidation with $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ to give a dicarboxylic acid $\text{C}_2\text{H}_2\text{O}_4$ (B), as final product. Identify A and B writing related reaction with their IUPAC names.
- How can you convert:
 - Ethanol into ethanoic acid
 - Ethanol into propanoic acid
 - Ethanol into acetylene
 - Ethanol into benzene
 - Methane into ethanoic acid
 - Ethanol into methanol
 - propan-1-ol into propan-2-ol
- What happens when:
 - Ethanol is dehydrated
 - Propan-2-ol is reacted with PCl_5
 - Ethanol is treated with acetic acid in presence of conc. H_2SO_4
 - Sucrose is fermented.
 - Propan-2-ol is treated with sodium.

7. Write down the isomers of C_3H_8O with their IUPAC and common names which one of them gives iodoform test. Why? Write the reactions involved.
8. An organic compound A reacts with PBr_3 to give B. Compound B produces C when heated with alcoholic KOH. The compound C undergoes ozonolysis to give ethanal and methanal. The compound A responds iodoform test. Identify A, B and C writing related reactions.

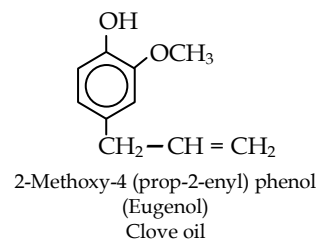
Multiple Choice Questions

1. Formation of alcohol and carboxylic acid for ester is an example of
 - a. Esterification
 - b. Trans-esterification
 - c. Ester-hydrolysis
 - d. Oxidation
2. Grignard reagent reacts with alcohols to form
 - a. Alkanes
 - b. Alkenes
 - c. Alkynes
 - d. All
3. The reaction of alcohol and carboxylic acid is
 - a. Esterification
 - b. Hydrolysis
 - c. Saponification
 - d. Hydrogenation
4. Which of the following compound does not give Iodoform test?
 - a. C_2H_5OH
 - b. CH_3CHO
 - c. CH_3COCH_3
 - d. $HCHO$
5. Which has the highest boiling point?
 - a. Propan-1-ol
 - b. Propane
 - c. Chloropropane
 - d. Diethyl ether
6. The boiling point order of alcohols is
 - a. $1^\circ > 2^\circ > 3^\circ$
 - b. $1^\circ > 3^\circ > 2^\circ$
 - c. $3^\circ > 2^\circ > 1^\circ$
 - d. $2^\circ > 3^\circ > 1^\circ$
7. The most suitable reagent to convert Alcohol into aldehyde is
 - a. $KMnO_4$
 - b. $K_2Cr_2O_7$
 - c. PCC (pyridine chlorochromate)
 - d. CrO_3
8. Drinking of which alcohol may cause blindness
 - a. Wood spirit
 - b. Ethanol
 - c. absolute alcohol
 - d. Rectified spirit
9. The grain alcohol refers to
 - a. Amyl alcohol
 - b. Methyl alcohol
 - c. Ethyl alcohol
 - d. Isopropyl alcohol
10. The amine is changed into corresponding alcohol by
 - a. acidified $KMnO_4$
 - b. HNO_2
 - c. Br_2/KOH
 - d. HNO_3

Answer Key

1	2	3	4	5	6	7	8	9	10
c	a	a	d	a	a	d	a	c	b

10.2 Phenols

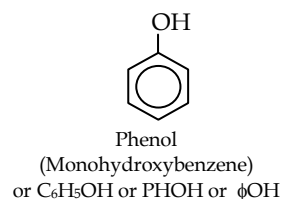


Phenols occur widely in nature. Eugenol (b.pt. 254°C) is oily liquid found in many essential oils and it is also a main constituent of clove oil, which is responsible for aroma of cloves. It is used in perfumeries, flavorings, essential oils and in medicines as dental antiseptic and anesthetic

10.1.1 Introduction

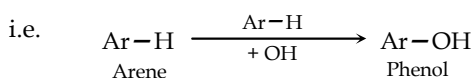
Phenols are a class of organic compounds of the general formula $\text{Ar}-\text{OH}$, where Ar is phenyl (C_6H_5-) or substituted phenyl ($\text{R}-\text{C}_6\text{H}_4-$), or some other aryl group like naphthyl (C_{10}H_7-), etc. In phenols the hydroxyl group ($-\text{OH}$) is directly attached to the aromatic ring. The most important member of phenols is phenol, where aromatic ring is benzene.

Both phenols (ArOH) and alcohols (ROH), contain the $-\text{OH}$ group, and as a result these two families resemble each other to a limited extent. In most of their properties and in their preparations, phenols and alcohols differ so greatly that they are classified as different families. Phenols differ from alcohols in having the $-\text{OH}$ group attached directly to an aromatic ring whereas in alcohols the $-\text{OH}$ group is attached to an alkyl group.



Definition

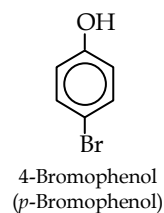
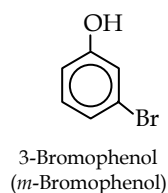
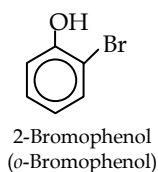
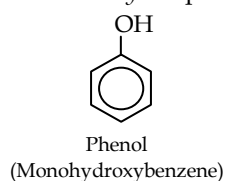
Hydroxy derivatives of arenes, where the $-\text{OH}$ group is directly attached to the aromatic ring are called phenols.



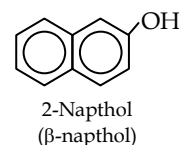
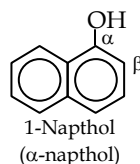
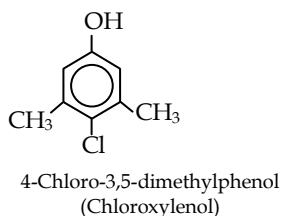
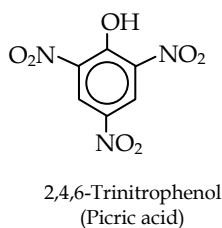
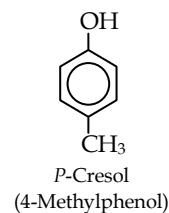
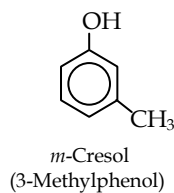
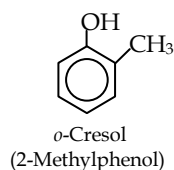
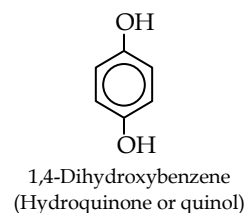
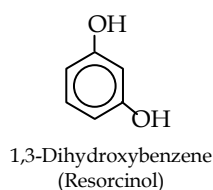
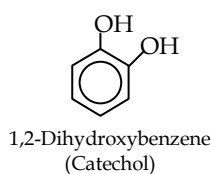
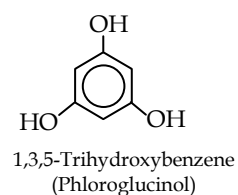
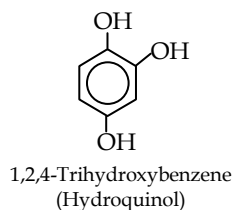
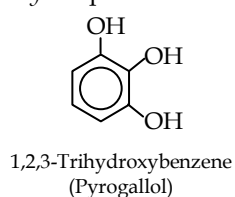
Phenol is colorless crystalline solid with melting point 43°C and boiling point 182°C . It is moderately soluble in water (solubility 9.3) but readily soluble in organic solvents. Phenol with 5% water is commonly called carbolic acid.

10.2.2 Classification

Like alcohols, phenols are classified as *mono*-, *di*- and *tri*-hydric phenols according to the number of $-\text{OH}$ groups contained in their molecules. When four or more $-\text{OH}$ groups are present, they are called *polyhydric phenols*. Some examples of *mono*-, *di*- and *tri*-hydric phenols are as follows:

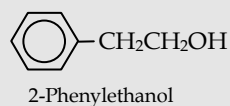
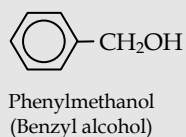
a. *Monohydric* phenols:

Methyl phenols are called cresols.

b. *Dihydric* phenols:c. *Trihydric* phenols:**Note:**

It may be noted that the aromatic compounds in which $-OH$ group is not directly attached to benzene ring are not phenols but they are called aromatic alcohols. These may be regarded as aryl alkyl alcohols since they behave very much like alcohols.

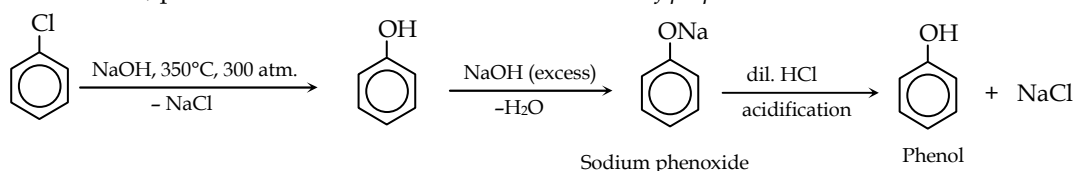
Examples:

**Self Test**

10.4 What are phenols? Draw the possible isomers of molecular formula C_7H_8O having benzene ring and indicate the phenols.

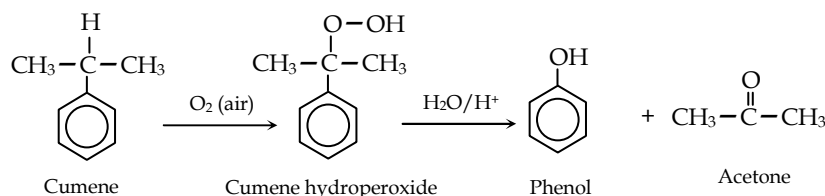
10.2.3 Methods of Preparation

1. **From coal tar:** Middle oil fraction of coal tar contains phenols and cresols.
2. **From chlorobenzene (Dow Process):** When chlorobenzene is heated with aqueous NaOH at high temperature (350°C) and pressure (300 atm.) in presence of copper catalyst followed by acidification, phenol is obtained. *It is an industrial method of preparation.*



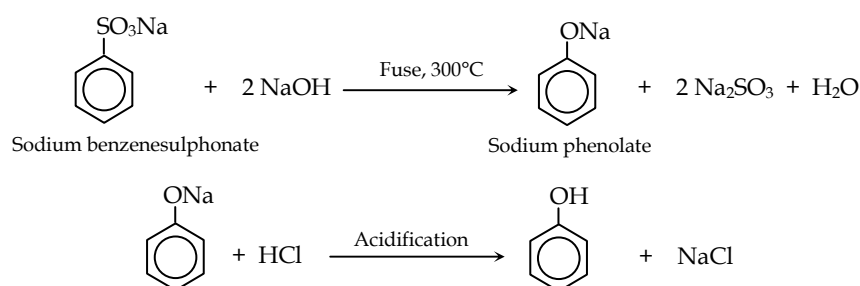
This method was first used by 'Dow chemical company' of USA in 1928 for the manufacture of phenol. Hence, this process is known as **Dows process**.

3. **From cumene (Isopropyl benzene):** Cumene reacts with air to form cumene hydroperoxide which on treatment with dilute acid gives phenol along with acetone.



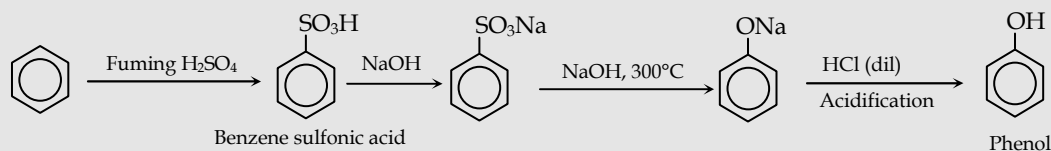
This is the most important commercial method because the raw materials are easily available from petroleum and the byproduct of the reaction is acetone, which is a valuable chemical.

4. **From sodium aryl sulphonate (alkali fusion of sulphonate):** When sodium benzene sulphonate is fused with NaOH at high temperature (300°C) followed by acidification, phenol is formed.

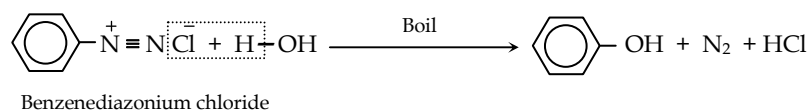


Note:

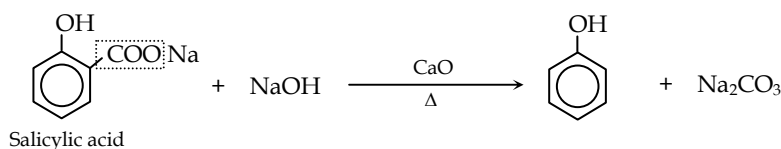
- i. Sodium benzene sulphonate is obtained from benzene by sulphonation with fuming H_2SO_4 followed by action of aq. NaOH on benzene sulphonic acid.
- ii. Conversion of benzene into phenol



5. **From benzene diazonium salt:** When aqueous solution of benzene diazonium salt is boiled in presence of little dilute sulphuric acid, where diazo group ($\text{N}\equiv\text{N}^+-$) is replaced by $-\text{OH}$ group, phenol is formed. This is laboratory method of preparation.



6. **By decarboxylation of salt of salicylic acid:** Phenol can also be obtained by the decarboxylation of salt of salicylic acid with soda lime.



Self Test

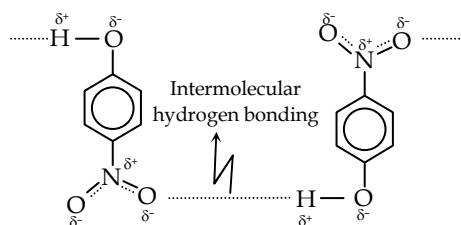
10.5 Outline the preparation of phenol from

- a. Aniline b. Benzene sulfonic acid c. Chlorobenzene d. Cumene

10.2.4 Physical Properties

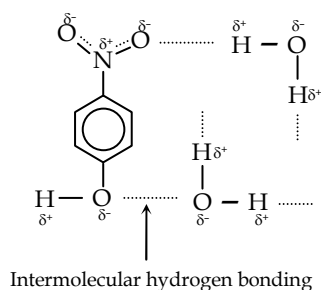
- The simplest phenols are colourless liquids or low melting solids.
- They have characteristic phenolic odour.
- They have quite high boiling point due to intermolecular hydrogen bonding in them.
- Phenol itself is somewhat soluble in water (9 g/100 g water) because of hydrogen bonding with the water while most other phenols are insoluble in water but soluble in organic solvents.
- The physical properties of isomeric nitrophenols:** *ortho*-nitrophenol has a much lower boiling point and much lower solubility in water than its other isomers (*para*-nitrophenol and *meta*-nitrophenols). Therefore, *o*-nitrophenol is steam-distillable.

In *p*-nitrophenol the relative positions of $-\text{OH}$ and $-\text{NO}_2$ groups permit them to form intermolecular hydrogen bonding between their molecules and hence has a high melting point (114°C) and boiling point than that of its other isomers.

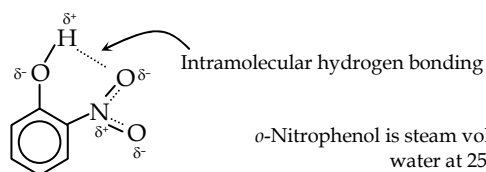


p-Nitrophenol is non volatile in steam distillation, its solubility is 1.69 g /100 g water at 25°C and its m.pt. 114°C

The high solubility of *p*-nitrophenol in water (1.69 g/100 g H_2O at 25°C) than its other isomers is because of hydrogen bonding with water molecules.



In *o*-nitrophenol the relative positions of -OH and -NO_2 groups permit them to form intramolecular hydrogen bonding within a single molecule, therefore *o*-nitrophenol has low melting point (45°C) and boiling point than that of its other isomers.



o-Nitrophenol is steam volatile and its solubility is 0.2 g /100 g water at 25°C and its m.pt. 45°C

The low solubility of *o*-nitrophenol in water (0.2 g/100 g H_2O at 25°C) than its other isomers is because of lack of hydrogen bonding with water molecules.

Note:

o-Nitrophenol can be separated from *p*-Nitrophenol by steam distillation.

Self Test

- 10.6 How do you account for the fact that *o*-Nitrophenol has lower boiling point and lower solubility in water as compared to its *p*-isomer?
- 10.7 Both phenol and ethanol have -OH functional group. Why is ethanol completely soluble in water, while phenol is slightly soluble?

10.2.5 Chemical Properties

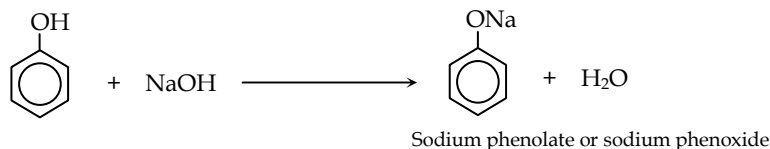
1. Acidic Nature: Salt formation: (Reaction due to H- atom of phenolic -OH group)

Phenols are fairly acidic compounds which can be shown as follows;

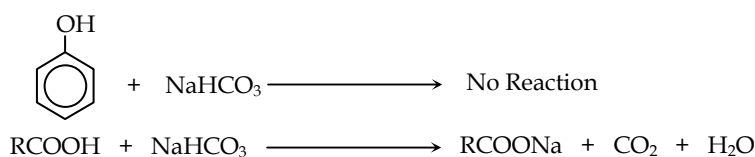
- i. **Reaction with active metals:** Phenol reacts with active metals such as sodium or potassium to form phenolate salt and hydrogen gas is liberated.



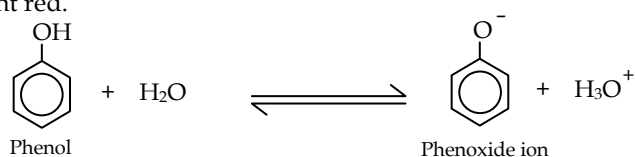
- ii. **Reaction with caustic alkalies (NaOH, KOH):** Phenol is soluble in solution of caustic alkali to form phenolate salt and water.



- iii. **Reaction with sodium carbonate and bicarbonate:** Phenol is weak acid hence it does not react with weak bases like sodium carbonate and bicarbonates. This reaction can be used to distinguish phenols from carboxylic acids since, acids react to give them with effervescence of CO_2 .



- iv. **Litmus test:** Phenol behaves as acid because of the presence of polar O–H group, which ionizes in aqueous solution to give H_3O^+ ions and that turns moist blue litmus into faint red.

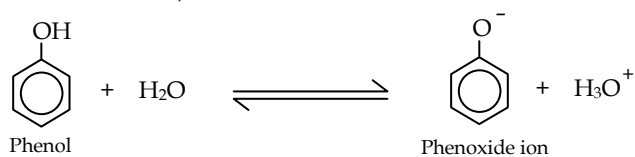


Self Test

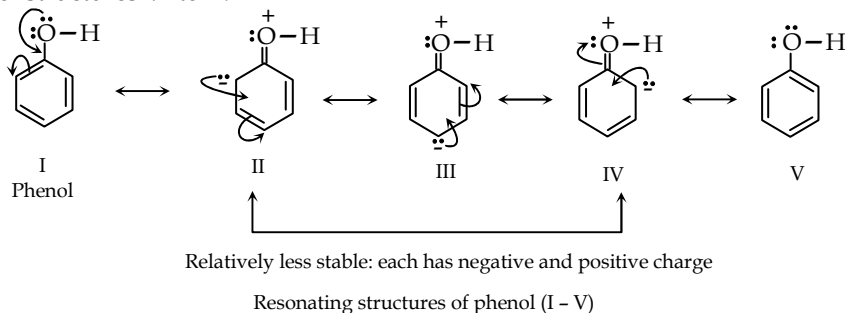
10.8 Phenol and formic acid are given to you separately, how would you chemically distinguish them?

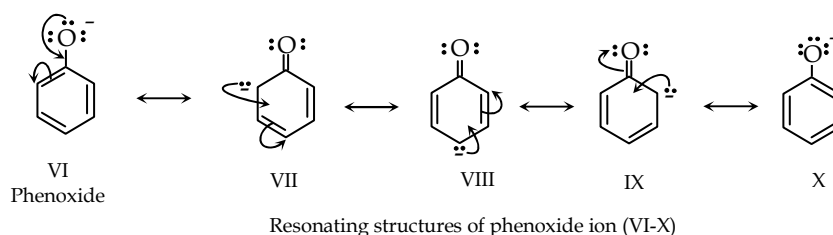
Explanation for acidic nature of phenol

Phenol in aqueous solution ionizes as;

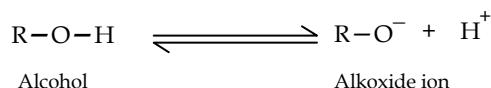


Phenol is resonance hybrid of structures I to V, similarly, phenoxide ion is also resonance hybrid of structures VI to X.





Resonance in phenoxide ion is more important than the resonance of phenol. The phenoxide ion (which is the conjugate base of phenol) is stabilized by resonance. Therefore, its corresponding acid (phenol) is fairly acidic in nature. Note that, more the stability of a conjugate base, stronger will be the corresponding acid.



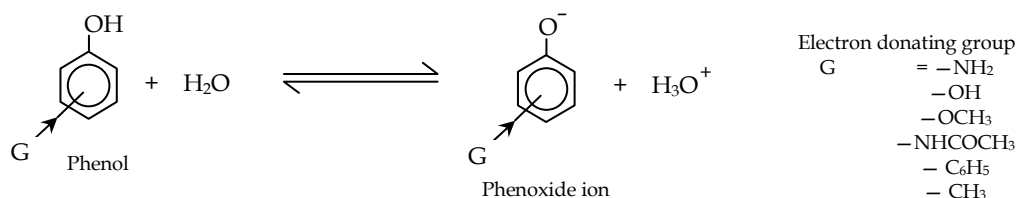
Alcohol and alkoxide ion have no resonating structures (highly unstable ion). Hence equilibrium does not shift towards ionization as in alcohol. Hence phenol is more acidic than alcohol.

Self Test

10.9 Why is phenol more acidic than alcohol though both involve cleavage of O-H bond?

Effect of substituent(s) on acidity of phenol

Any factor that stabilizes phenoxide ion increases the acidity of phenol. Electron donating substituent intensifies the negative charge in phenoxide ion and hence destabilizes the phenoxide ion to decrease acidity of phenol.

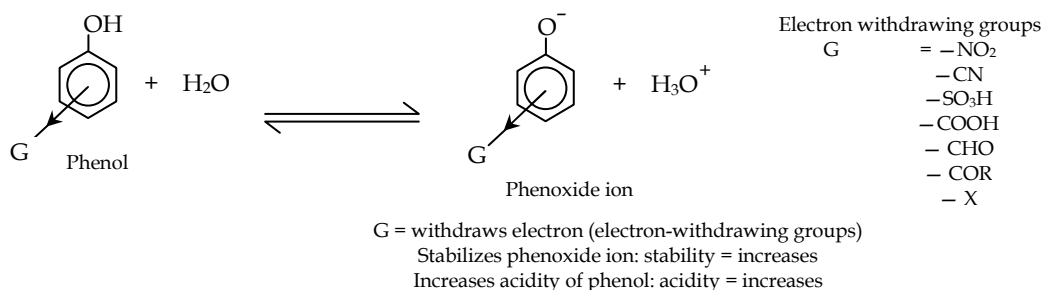


G = donates electron (electron-donating group)

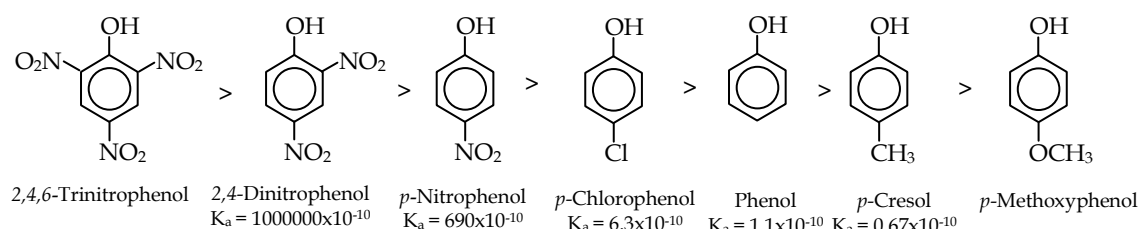
destabilizes phenoxide ion: stability decreases

decreases acidity of phenol : acidity decreases

Electron withdrawing substituent acts in opposite way.



Sequential order of acidity of phenols

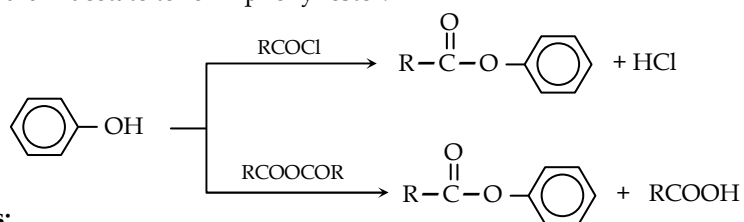


Self Test

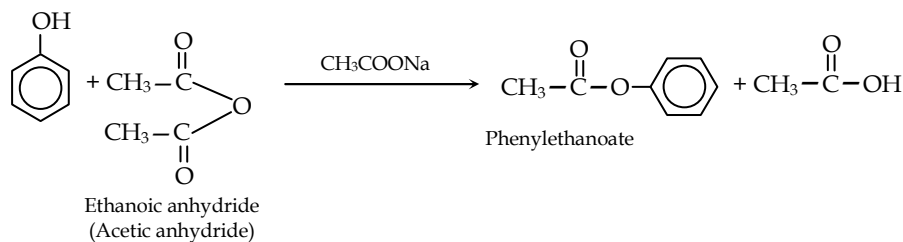
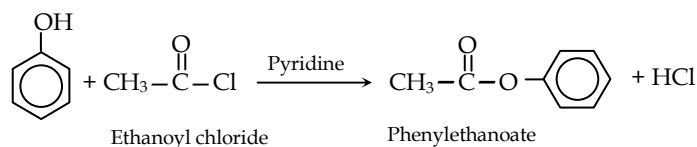
10.10 How do you account for the fact that 2,4,6-trinitrophenol reacts with aqueous sodium bicarbonate but phenol does not?

2. Reaction with acid chlorides and anhydrides: Acylation: Reaction due to H-atom of phenolic $-\text{OH}$ group: Formation of phenyl ester

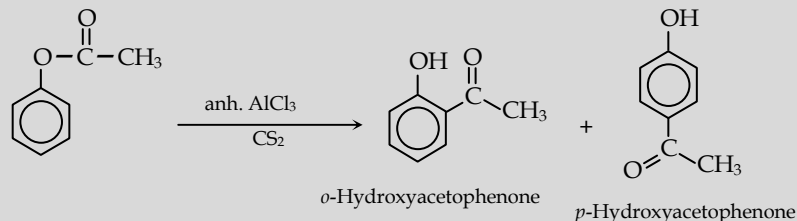
Phenol reacts with acid chlorides in presence of pyridine or with acid anhydrides in presence of base or sodium acetate to form phenyl ester.



Examples:

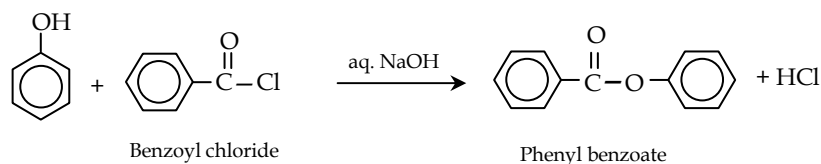


Fries rearrangement: When ester of phenol is heated with anhydrous aluminium chloride in presence of CS_2 as solvent, acyl group ($\text{CH}_3\text{CO}-$) migrates from phenolic oxygen to ortho- and para- position of the ring, thus forming hydroxyketone. This reaction is called **Fries rearrangement**.



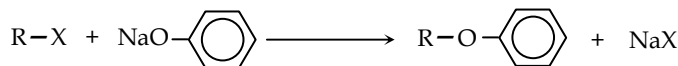
3. Benzoylation: Schotten-Baumann reaction

Phenol reacts with benzoyl chloride in presence of aqueous NaOH to form phenyl benzoate. This reaction is known as Schotten-Baumann reaction.

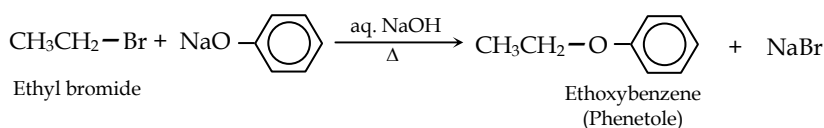
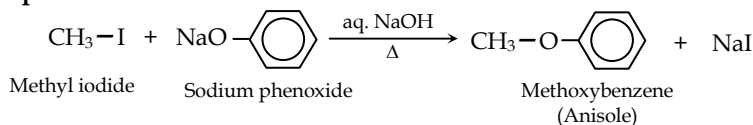


4. Reaction with alkyl halides: Alkylation: Formation of aryl ether: Williamson's synthesis

When sodium phenoxide is heated with alkyl halide in presence of aqueous NaOH , aryl ether is formed. This is an example of **Williamson's etherification**.



Examples:

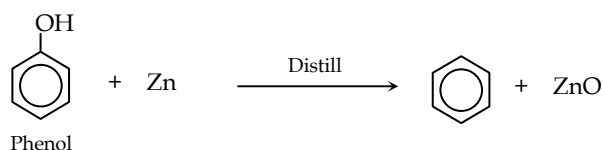


Self Test

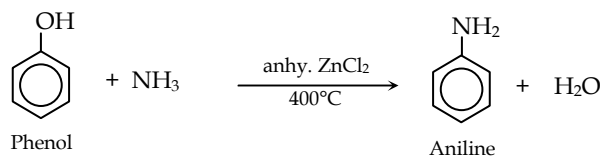
10.11 Convert phenol into (a) Methoxybenzene (b) Ethoxybenzene

5. Reaction due to phenolic -OH group

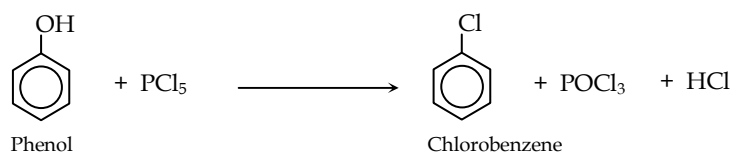
a. **Reaction with zinc dust:** When phenol is distilled with zinc dust, it gets reduced to benzene.



- b. **Reaction with ammonia:** Phenol reacts with ammonia at high temperature in presence of anhydrous ZnCl_2 to give aniline.



- c. **Reaction with PCl_5 :** Phenol reacts with phosphorous pentachloride to give only a small amount of chlorobenzene, along with the major product of triphenyl phosphate.

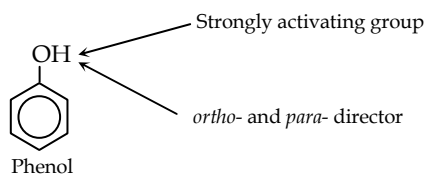
**Self Test**

10.12 Convert phenol into:

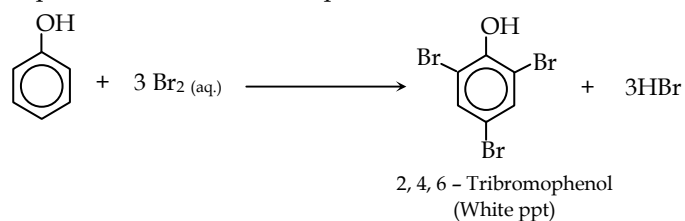
- a. Benzene b. Aniline c. Chlorobenzene

6. **Reaction due to benzene ring: Ring substitution: *Electrophilic aromatic substitution reaction***

In phenol, $-\text{OH}$ group is a strongly activating group, which activates the benzene ring towards electrophilic substitution reactions and direct the substitution at *ortho*- and *para*- positions.



- a. **Halogenation: Bromination: *Test of phenol*:** Phenol reacts with aqueous solution of bromine immediately to form a white precipitate of 2,4,6-tribromophenol. This reaction can be used to test phenol as well as quantitative estimation of phenol.



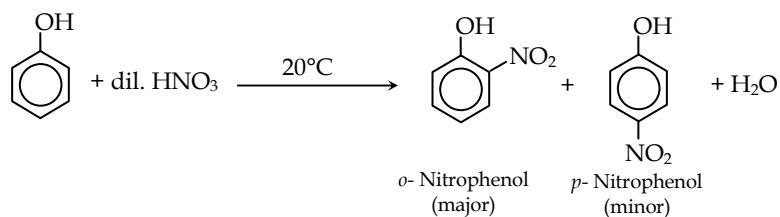
Due to strong activating effect of $-\text{OH}$, reactivity of benzene ring in phenol is very high, so polyhalogenation of phenol occurs even in absence of catalyst.

Monobromophenol is obtained by reacting phenol with bromine in presence of less polar solvents like CS_2 or CCl_4 at low temperature.



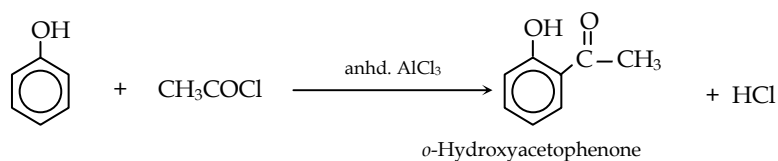
- Oc1ccccc1.O=[N+]([O-])O>>Oc1c([N+](=O)[O-])cc([N+](=O)[O-])cc1[N+](=O)[O-].O
- 2, 4, 6 - Trinitrophenol
(Picric acid)

To obtain mononitrophenol, it is necessary to carry out the reaction at low temperature with dilute nitric acid.

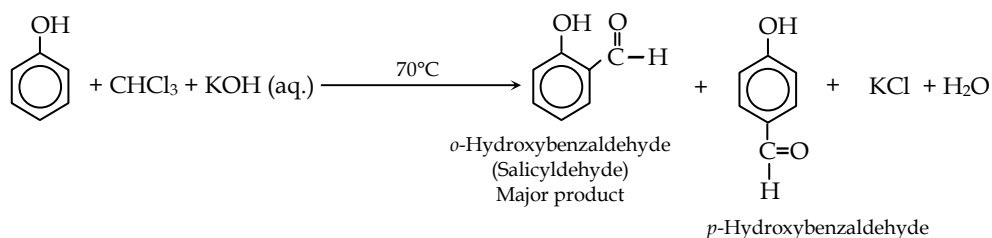


-
- Oc1ccccc1 (Phenol) + H2SO4 (conc.)
- At 20°C , $-\text{H}_2\text{O}$ → Oc1ccccc1S(=O)(=O)O (*o*-hydroxybenzenesulphonic acid)
- At 100°C , $-\text{H}_2\text{O}$ → Oc1ccc(S(=O)(=O)O)cc1 (*p*-hydroxybenzenesulphonic acid)
- The *p*-isomer at 100°C → Oc1ccccc1S(=O)(=O)O (*o*-hydroxybenzenesulphonic acid)

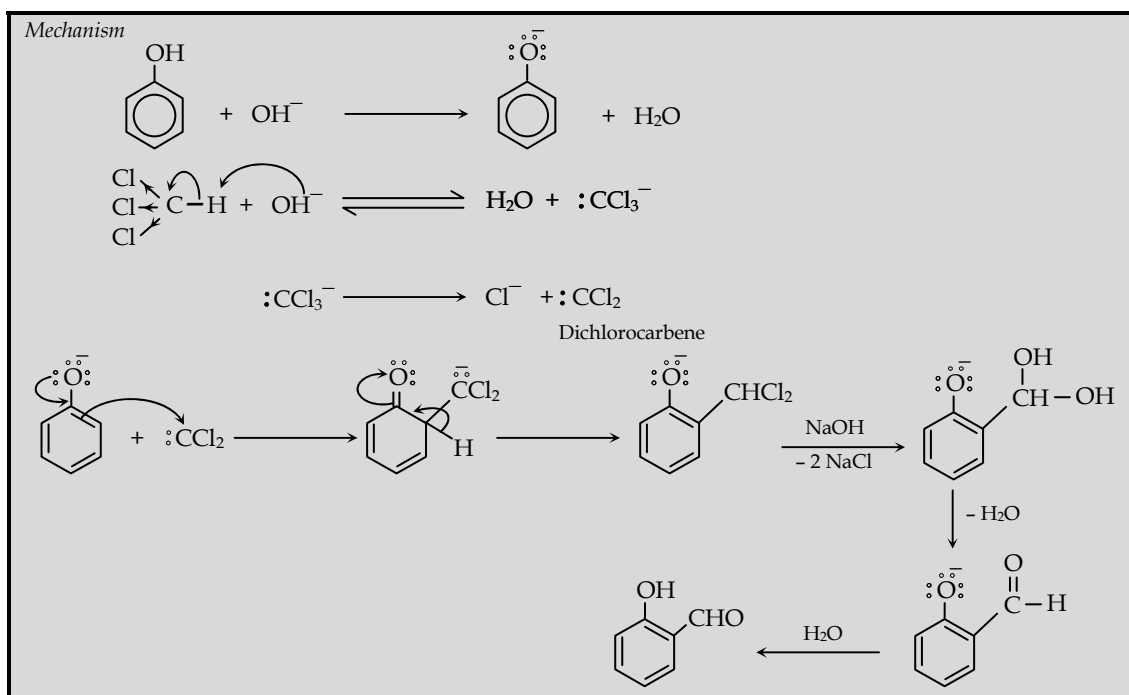
- Oc1ccccc1.ClC>>Oc1ccccc1C.Oc1ccc(C)cc1.Cl
- Phenol + CH₃Cl $\xrightarrow{\text{anhd. AlCl}_3}$ *o*-Cresol (minor) + *p*-cresol (major) + HCl



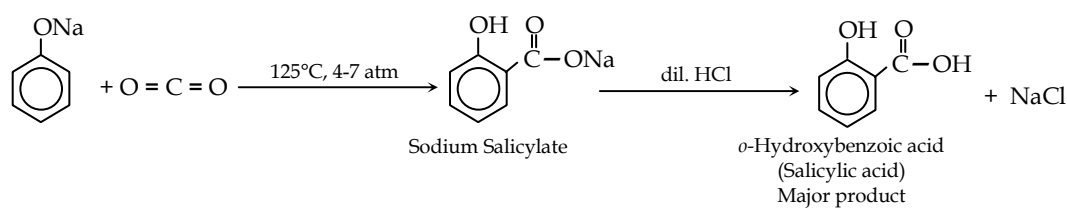
- e. **Reimer-Tiemann reaction: Formation of hydroxy aldehyde:** When phenol is refluxed with chloroform in presence of aqueous solution of caustic alkali, salicylaldehyde is obtained as major product. The reaction is called Reimer-Tiemann reaction.



If CCl_4 is used instead of CHCl_3 , salicylic acid is obtained.

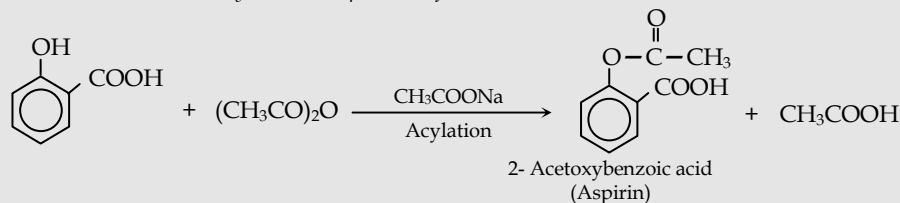


- f. **Kolbe's reaction: Carbonation:** When sodium or potassium salt of phenol is heated with carbon dioxide under pressure, salicylate salt is obtained, which upon acidification gives *salicylic acid* as major product. This reaction is called Kolbe's carbonation reaction.



Note:

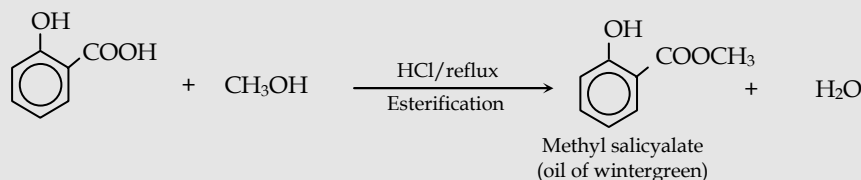
- i. Salicylic acid is a starting materials for the manufacture of a well known analgesic (pain killer) "aspirin". Aspirin is obtained when salicylic acid reacts with acetic anhydride in the presence of sodium acetate.



- ii. Salicylic acid can be used to prepared salol an intestinal antiseptic by heating it with phenol in presence of POCl_3 .

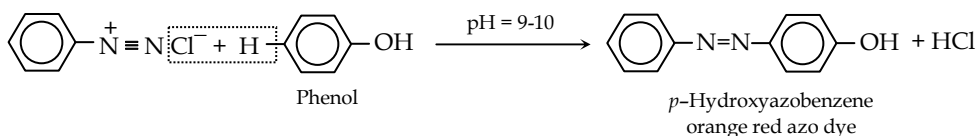


- iii. Methyl salicylate is prepared from salicylic acid, is used as a pleasant smelling liquid also called "oil of wintergreen" in perfumery.

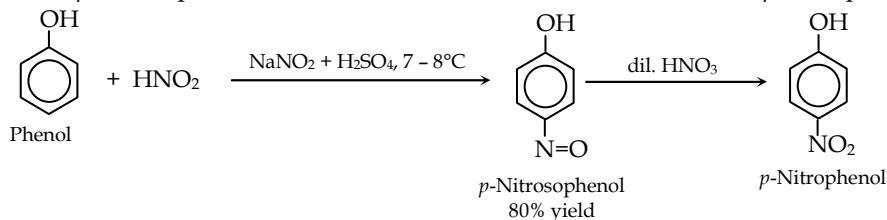
**Self Test**

10.13 Convert phenol into aspirin.

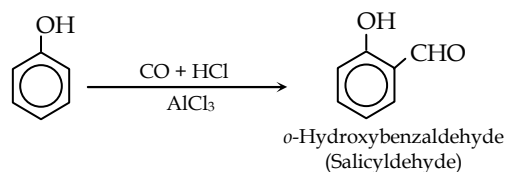
- g. **Reaction with benzenediazonium chloride (coupling reaction):** Phenol reacts with diazonium salt solution in slightly alkaline medium at low temperature to form *p*-Hydroxyazobenzene which appears as orange red compound called *azo dye*.



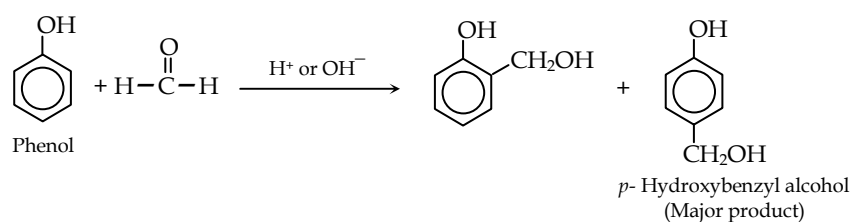
- h. **Nitrosation:** Phenol reacts with ice cold solution of nitrous acid to form *p*- nitrosophenol. Though, ^+NO (nitrosonium ion) is weak electrophile, the reaction occurs due to high reactivity of phenol. The *p*-nitrosophenol can be further oxidized with dilute HNO_3 to *p*- nitrophenol.



- i. **Gattermann -koch aldehyde synthesis:** When phenol is treated with a mixture of CO and HCl in presence of AlCl_3 followed by hydrolysis, salicylaldehyde is obtained.



- j. **Reaction with formaldehyde: Lederer-Manasse reaction:** Phenol condenses with formaldehyde in presence of dilute acid or alkali especially at low temperature to form *p*-hydroxybenzyl alcohol with little *o*-isomer. This is known as the **Lederer-Manasse reaction** (1894).



When the reaction is carried out at elevated temperature, the condensation occurs to form phenol formaldehyde resin (bakelite) which is thermosetting polymer.

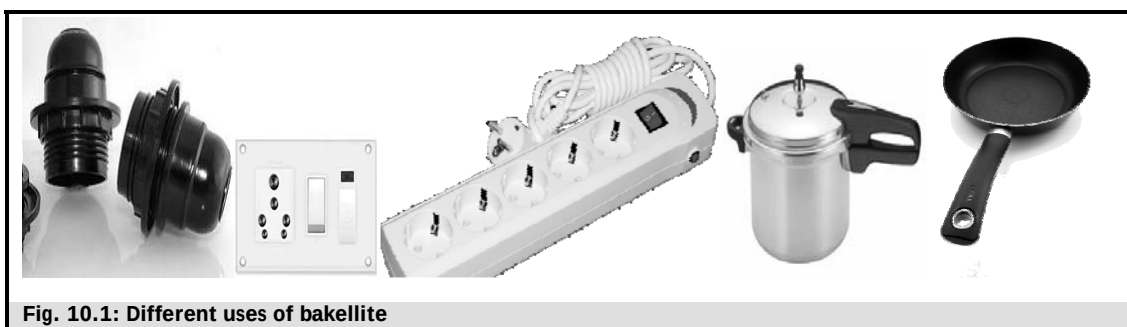
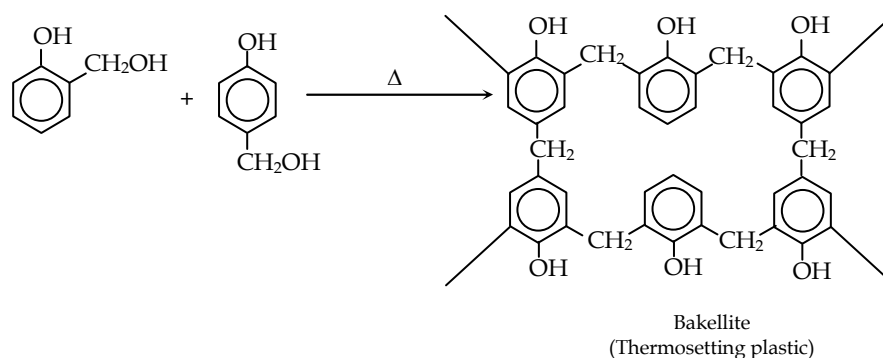


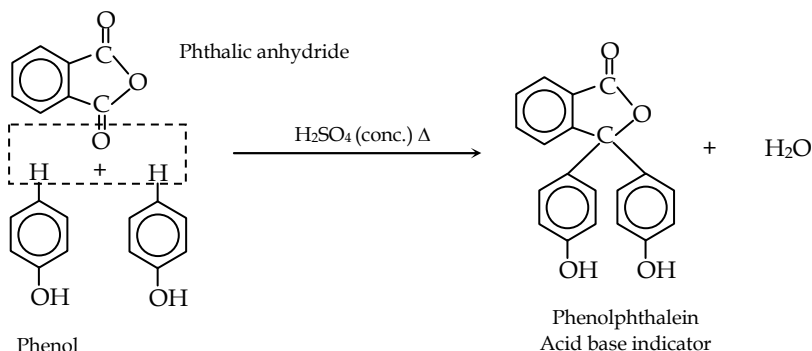
Fig. 10.1: Different uses of bakelite

Self Test

10.14 Phenol ranks near the top of the list of synthetic aromatic compounds. Its principal use is in the manufacture of bakelite, which is a thermo-setting polymer. Outline the reaction involved in the formation of bakelite from phenol and formaldehyde.

7. Condensation reaction: Reaction with phthalic anhydride

When phenol is heated with phthalic anhydride in presence of concentrated sulphuric acid, phenolphthalein is formed.



Self Test

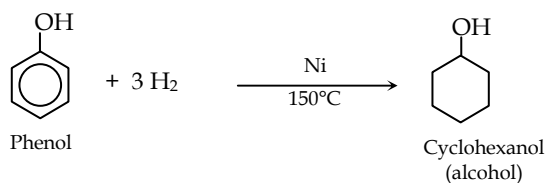
10.15 Convert benzene into picric acid.

10.16 Convert phenol into 1, 3, 5-tribromobenzene.

10.17 What happens when phenol is treated with phthalic anhydride in the presence of conc. H_2SO_4 ?

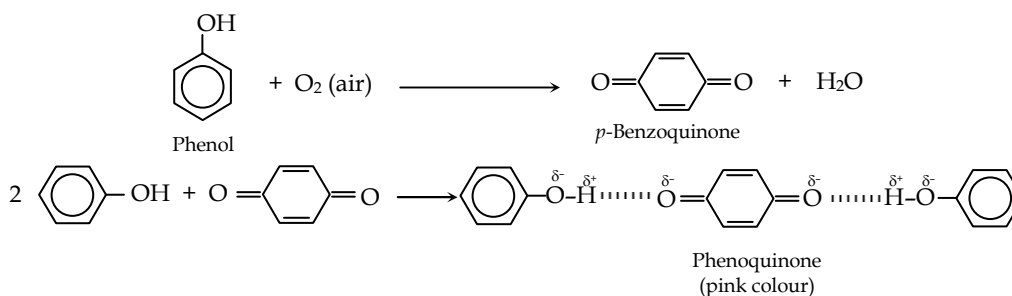
8. Catalytic hydrogenation

Phenol on catalytic hydrogenation at 150°C in presence of nickel catalyst gives cyclohexanol.



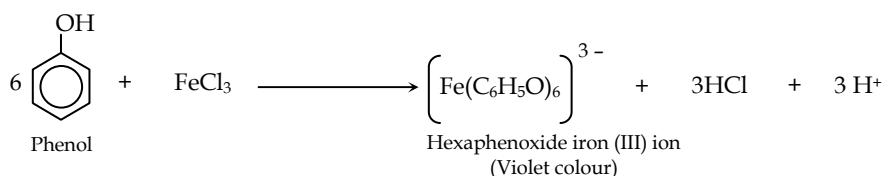
9. Air oxidation

Phenol gets slowly oxidized on exposure to air and turns pink in colour. The oxidation product is quinone, which forms an addition product with phenol having a pink colour.

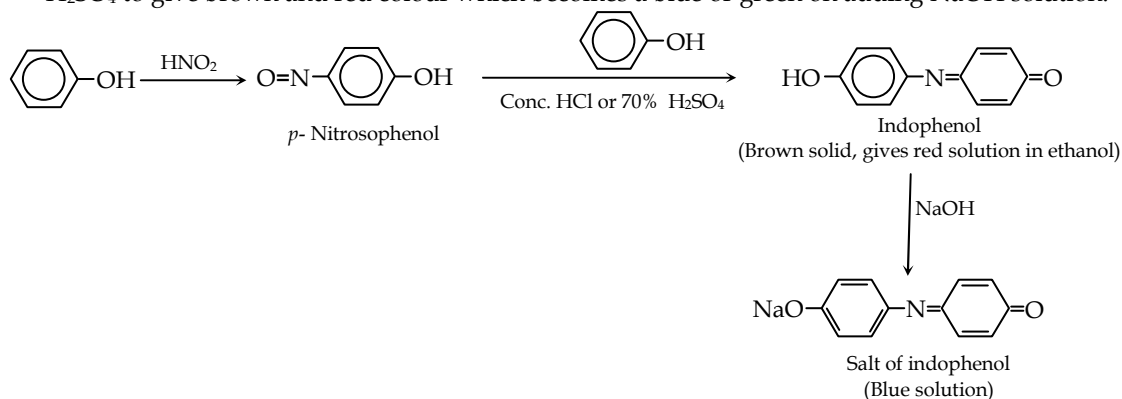


10.3 Tests for Phenols

- Solubility test:** A water insoluble compound soluble in aqueous NaOH but insoluble in NaHCO_3 is most likely a phenol.
- Ferric chloride test: Reaction with ferric chloride:** Phenol reacts with FeCl_3 solution to give green, blue, violet or red coloured water soluble complex. It is a coordination compound in which iron is hexa covalent.



- Libermann's test:** Many phenols react with nitrous acid ($\text{NaNO}_2 + \text{H}_2\text{SO}_4$) in presence of conc. H_2SO_4 to give brown and red colour which becomes a blue or green on adding NaOH solution.



Self Test

10.18 Suggest a suitable chemical test to distinguish phenol from: (a) Ethanol (b) Formaldehyde

10.4 Uses of Phenol

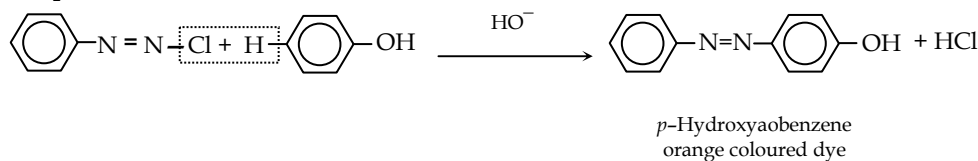
- Large quantities of phenol are used for the preparation of bakelite.
- For the manufacture of dyes, drugs, explosives.
- For the preparation of phenolphthalein.
- As disinfectant and germicides.

Solved Questions

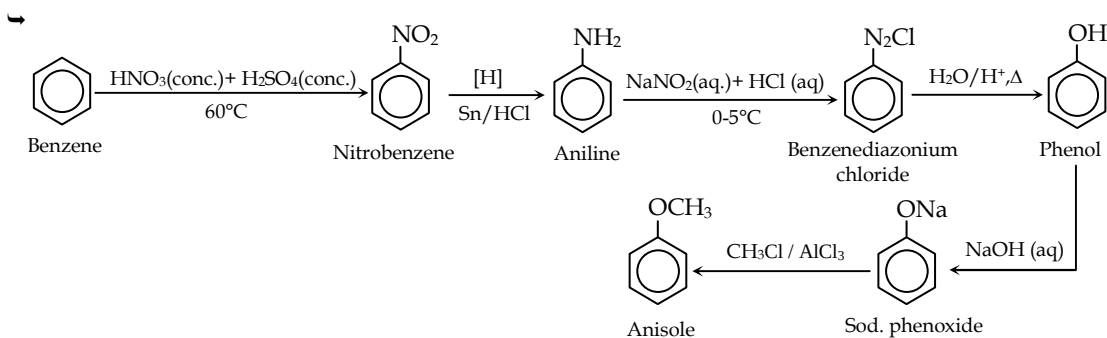
- Explain Diazo-coupling test with example.**

→ When benzene diazonium chloride is treated with aromatic compound like phenol, aniline, etc. azo dyes are obtained. This test is called diazo-coupling test.

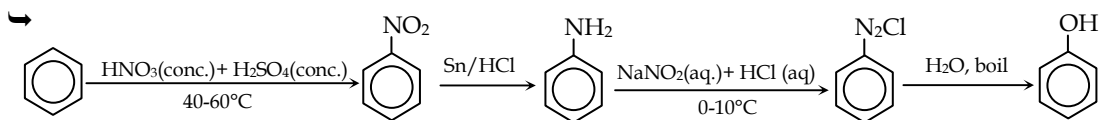
Example:



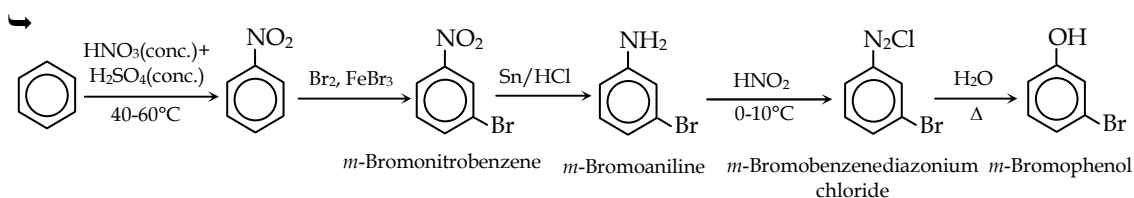
2. How can convert benzene into anisole?



3. How can you convert benzene into phenol?



4. How can you convert benzene into *m*-Bromophenol?



Questions

Very Short Answer Questions:

1. What happens when diazotized solution of aniline is hydrolysed?
2. The colour of phenol changes on standing in air, why?
3. What is Reimer-Tiemann reaction?
4. What happens when diazotized solution of aniline is treated with phenol?
5. Why is $-\text{OH}$ group in phenol *ortho*- and *para*- directing towards electrophilic substitution reaction?

Short /Long Answer Questions:

1. What happens when:
 - a. Phenol is treated with benzenediazonium chloride.
 - b. Phenol is carboxylated.
 - c. Phenol is treated with methanol in acidic medium.
 - d. Phenol is nitrated.

- How can you convert:
 - Benzene into phenol?
 - Phenol into toluene
 - Benzene into m-chlorophenol
 - Phenol into m-nitrobenzoic acid
 - Benzene into m-bromophenol
- Show your acquaintance with:
 - Reimer - Tiemann reaction
 - Kolbe reaction
 - Acidic nature of phenol
 - Diazo-coupling test
 - Nitration of phenol
- An aromatic compound A on reduction gives parent hydrocarbon B. B on nitration gives C. C on reduction in acidic solution gives D. On coupling with diazonium salt D gives diazoamino-benzene. Identify A, B, C and D writing related reactions.
- An organic compound A with a molecular formula C_6H_6O gives a parent hydrocarbon B. The compound A responds to $FeCl_3$ test. The compound B can also be obtained by the polymerisation of acetylene. Identify A and B writing related reactions. Why is compound A slightly acidic?

Multiple Choice Questions

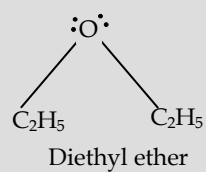
- Phenol and benzoic acid can be distinguished by
 - $AgNO_3$
 - Baeyer's reagent
 - Tollen's reagent
 - $NaHCO_3$
- $FeCl_3$ gives violet colour with
 - Alcohols
 - Amines
 - Phenol
 - Aldehydes
- Phenol on heating with Zn gives
 - Benzene
 - Aniline
 - Benzaldehyde
 - Cyclohexane
- Picric acid is an example of
 - Azo dye
 - Phthalein dye
 - Nitro dye
 - Anthracene dye
- O-acetoxibenzoic acid is
 - Paracetamol
 - Aspirin
 - Salol
 - Salicylaldehyde
- Oil of wintergreen is
 - Phenyl salicylate
 - Phenyl benzoate
 - Benzoic acid
 - Methyl salicylate
- Reimer-Tiemann reaction yields
 - Salicylaldehyde
 - Benzaldehyde
 - Benzoic acid
 - Aspirin
- Carbolic acid is
 - Dil. solution of formaldehyde
 - Dil. solution of ethanol
 - Dil. solution of phenol
 - Dil. solution of methanol
- Phenol formaldehyde resin is
 - Bakelite
 - Thermoplastic
 - Malamine
 - PMMA
- Which of the following is most acidic?
 - Phenol
 - o-nitrophenol
 - Picric acid
 - Dnisoole

Answer Key

1	2	3	4	5	6	7	8	9	10
d	c	a	c	b	d	a	c	a	c

Ethers

11

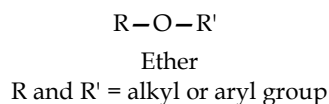


Diethyl ether, simply known as ether, is a highly volatile and inflammable colourless liquid commonly used as organic solvent. It was once used as general anesthesia in surgery.

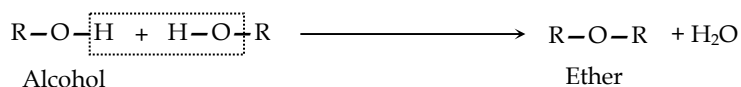
11.1 Aliphatic Ethers

11.1.1 Introduction

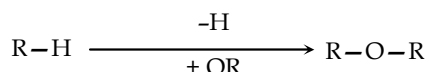
Ethers are a class of organic compounds having general structure R-O-R, where R may be alkyl or aryl group.



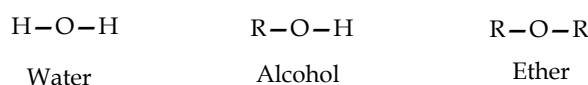
Ethers are regarded as anhydride of alcohols.



They are also referred as alkoxy derivative of alkane.

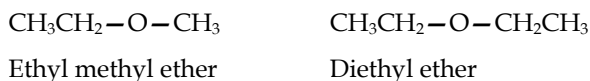


Ethers may be considered as derivative of water in which both hydrogen atoms have been replaced by alkyl or aryl groups.



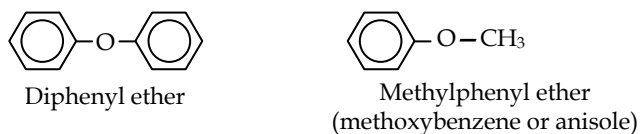
If R and R' are alkyl groups, ethers are called *aliphatic ethers*.

Examples:

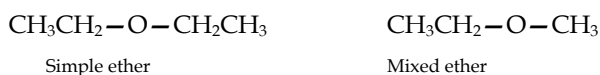


Diaryl or aryl alkyl derivatives of water are called *aromatic ethers*.

Example:



If R=R', ethers are called symmetrical ethers or simple ethers and if R≠R', ethers are called unsymmetrical ethers or mixed ethers.



Petroleum ether is not ether chemically. It is a fraction of petroleum as a mixture of pentane and hexane, which is used as solvent in dry cleaning.

11.1.2 Nomenclature

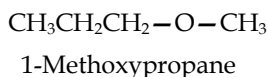
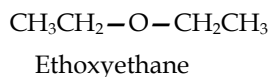
Formula	IUPAC name	Common name
$\text{CH}_3-\text{O}-\text{CH}_3$	Methoxymethane	Dimethyl ether
$\text{CH}_3\text{CH}_2-\text{O}-\text{CH}_3$	Methoxyethane	Ethylmethyl ether
$\text{CH}_3\text{CH}_2-\text{O}-\text{CH}_2\text{CH}_3$	Ethoxyethane	Diethyl ether
$\text{CH}_3\text{CH}_2\text{CH}_2-\text{O}-\text{CH}_3$	1-Methoxypropane	Methyl n-propyl ether
$\begin{array}{c} \text{CH}_3-\text{CH}-\text{O}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	2-Methoxypropane	Isopropylmethyl ether

11.1.3 Isomerism

Ethers can show following types of structural isomerism.

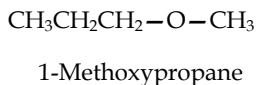
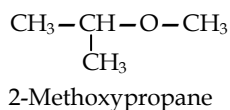
- Chain isomerism:** Isomers that differ only in chain length are called chain isomers.

Examples:



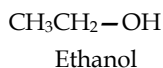
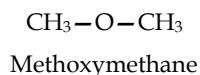
- Positional isomerism:** Isomers having same chain length but differ in the position of functional group or substituent is called positional isomers.

Examples:



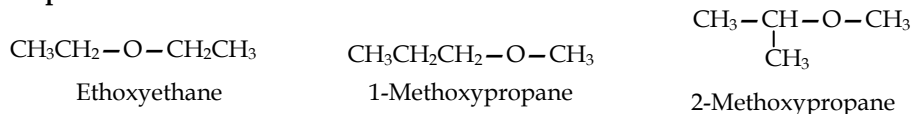
- Functional isomerism:** Isomers that differ in their functional group are functional isomers. Ether shows functional isomerism with alcohol.

Examples:



- Metamerism:** Like ketones and secondary amines, ethers can show metamerism. Isomers differing in alkyl group present on either side of the functional group are called metamers.

Examples:



Molecular formula	Total isomers	Alcohols	Ethers
$\text{C}_2\text{H}_6\text{O}$	2	1	1
$\text{C}_3\text{H}_8\text{O}$	3	2	1
$\text{C}_4\text{H}_{10}\text{O}$	7	4	3

Self Test

11.1 What are ethers? Give two examples of aliphatic as well as aromatic ethers.

11.2 Write the structure of ethers of molecular formula $\text{C}_4\text{H}_{10}\text{O}$ and point out the simple and mixed ethers.

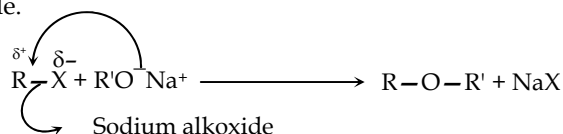
11.1.4 General Methods of Preparation

Ethers are generally prepared

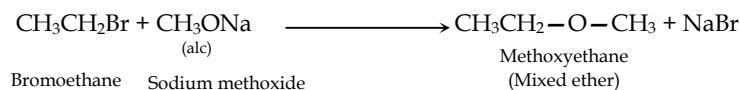
1. From alkyl halides
2. From alcohols

1. From alkyl halide

- a. **By Williamson's synthesis:** When alkyl halide is heated with sodium or potassium alkoxide, ether is obtained. This method is called **Williamson's synthesis**. This is nucleophilic substitution reaction when alkoxide ion (OR^-) acts as nucleophile substitutes the halogen atom from alkyl halide.

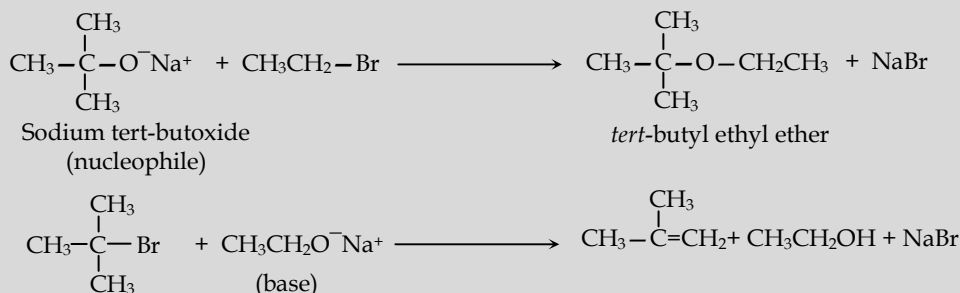


Examples:

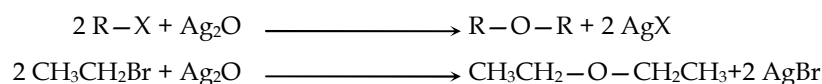


The importance of this method lies on its versatility that simple as well as mixed ethers can be obtained by this method.

Secondary and tertiary alkyl halides are not used in Williamson's synthesis: Substitutions vs. Elimination
 In Williamson's synthesis, for the preparation of unsymmetrical ether, primary alkyl halide and alkoxide of secondary and tertiary alcohol are used. Secondary and tertiary alkyl halides are not preferred because they undergo elimination reaction to give alkene in presence of alkoxide ion since alkoxide ions are not only acts as nucleophiles but also acts as strong bases.



- b. **By the action of dry silver oxide:** When alkyl halide is heated with dry silver oxide, ether is formed.



This method is only suitable for the preparation of simple ethers.

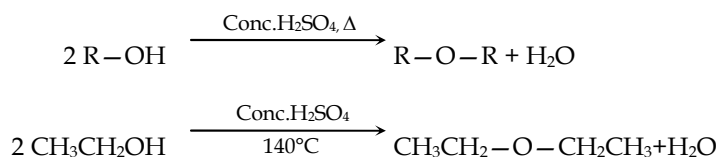
Self Test

11.3 Outline the preparation of ethyl methyl ether and diethyl ether by Williamson's method.

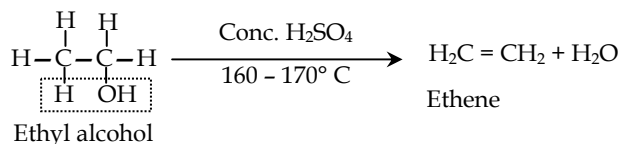
2. From alcohol

Dehydration of alcohol with concentrated H_2SO_4 , alumina or aluminum phosphate under proper reaction conditions gives ether.

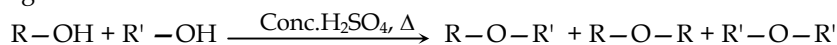
- a. **By using conc. H_2SO_4 :** When excess alcohol is heated with concentrated H_2SO_4 to proper temperature, ether is obtained by intermolecular dehydration:



If alcohol is not excess and temperature is high, elimination reaction takes place to give alkene by intra-molecular dehydration.

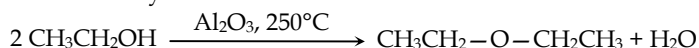


This method is not useful for preparing unsymmetrical ethers because a combination of two different alcohols give a mixture of three ethers.

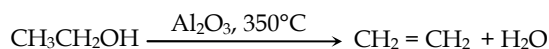
**Self Test**

11.4 A mixture of ethyl alcohol and n-propyl alcohol when heated with concentrated H_2SO_4 , gives a mixture of three different ethers. Identify them.

- b. **By using alumina:** When alcohol vapour is passed over heated alumina at 250°C , ether is obtained by intermolecular dehydration.



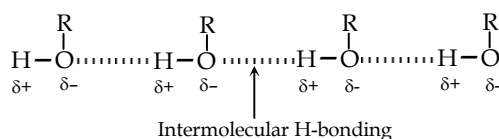
If temperature is higher, alkene is formed by elimination.



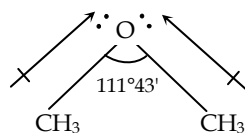
Dimethyl ether (b.pt. -23°C) is prepared industrially by passing methyl alcohol vapour over aluminum phosphate as catalyst at 350°C to 400°C under pressure of 15 atmosphere.

11.1.5 Physical Properties

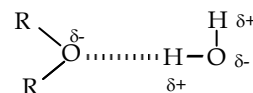
1. Dimethyl ether and ethylmethyl ether are gases. Other members are colorless volatile liquids with pleasant odour.
2. In alcohol, there exists intermolecular hydrogen bond due to polar nature of O-H bond. But in ethers, there does not exist intermolecular hydrogen bond since there is no polar O-H bond. Hence, ethers have much lower boiling point than alcohol of comparable molecular weight.



3. Like water, ether molecule is angular and has a net dipole moment. For example, in methoxymethane, bond angle is $111^{\circ}43'$ and dipole moment is 1.3D. Hence, ether molecule can form intermolecular hydrogen bond with water. Hence lower member of ethers are partially soluble in water.



Methoxymethane
($\mu = 1.3$ D)



Intermolecular H-bonding

Self Test

11.5 Write the isomers of C_2H_6O and compare their boiling point.

11.6 How do you account for the fact that, although diethyl ether has a much lower boiling point than n-butyl alcohol, it has the same solubility (8g per 100g) in water?

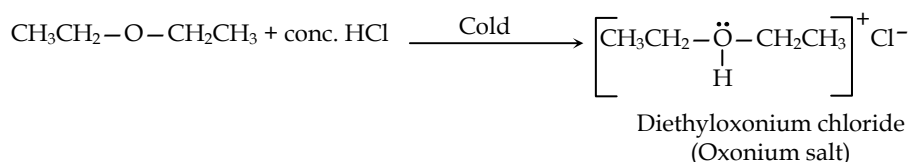
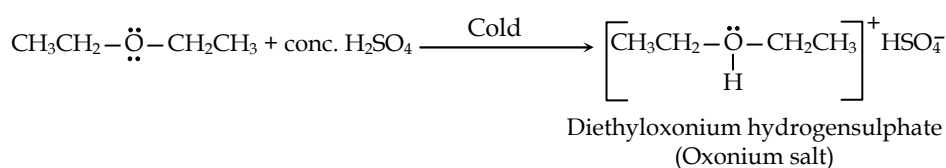
11.1.6 Chemical Properties

Ethers are relatively less reactive class of organic compounds and comparable to hydrocarbons in their chemical reactivity. They do not react with oxidizing agents, reducing agents, highly reactive alkali metals, caustic alkalis. Due their low reactivity, ethers are used as solvents in laboratories as well as in industries. However, ethers react with acids due of presence of lone pair of electron on oxygen atom which can accept proton from acid. The reactions of ethers can be categorized as follows;

- A. Reactions involving ethereal oxygen
- B. Reactions involving alkyl group
- C. Reactions involving cleavage of C-O bond

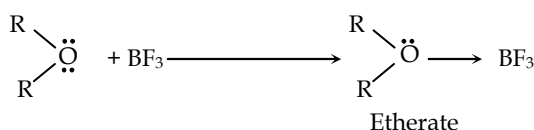
A. Reactions involving ethereal oxygen

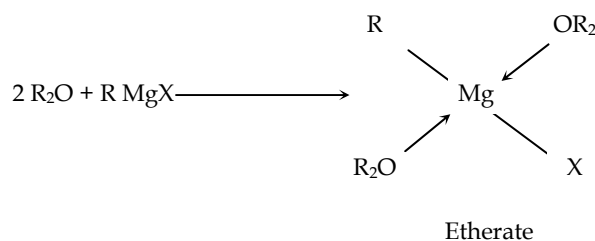
1. **Formation of oxonium salt:** Due to presence of lone pair of electron on oxygen atom, ether behaves as Bronsted-Lowry base and accepts H^+ from acids. Hence ethers dissolve in cold and concentrated mineral acids (HCl , H_2SO_4) to form oxonium salts.



Oxonium salts are unstable at high temperature and in dilute solution.

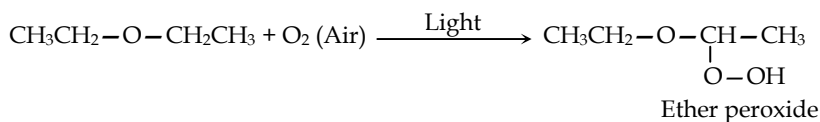
Ethers also form coordination complexes commonly known as etherate with Lewis acids like BF_3 , $RMgX$ etc.





Ethers can be distinguished from hydrocarbons from their solubility; ethers are soluble in cold concentrated sulphuric acid due to formation of oxonium salts but hydrocarbons do not.

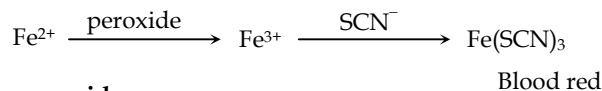
2. **Reaction with air:** Aliphatic ethers react with **air** slowly in presence of **light** to form ether peroxides.



Ether peroxide is explosive and thermally unstable. Hence care should be taken during distillation of old sample of ethers otherwise violent explosion may take place. Addition of small amount of cuprous compound like Cu_2O has been recommended to prevent formation of ether peroxide.

Detection of ether peroxide

To detect peroxide in ether, ether sample is shaken with little ferrous ammonium sulphate and potassium thiocyanate. Formation of blood red color indicates presence of peroxide in ether.



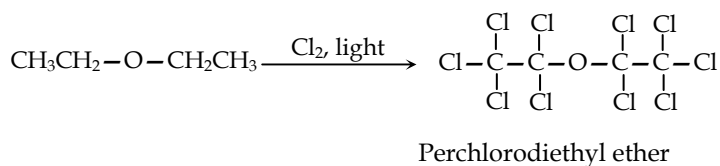
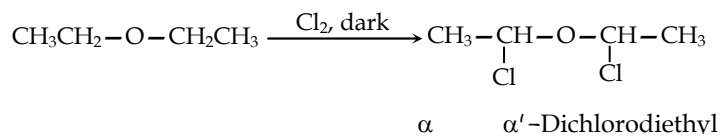
Removal of ether peroxide

To remove ether peroxide

- i. Ether is distilled with concentrated sulphuric acid, which oxidizes peroxide.
- ii. Ether is washed with ferrous sulphate solution, which reduces peroxide.

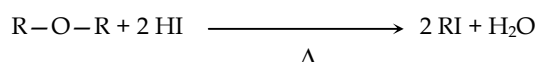
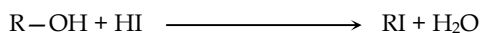
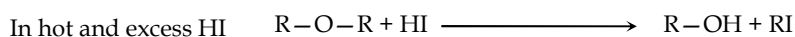
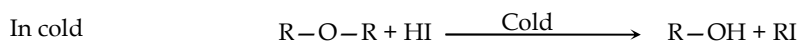
B. Reactions involving alkyl group

Halogenation: Ether reacts with Cl_2 or Br_2 to form halogen substituted ethers. The nature of substitution products depend upon the conditions of reaction.

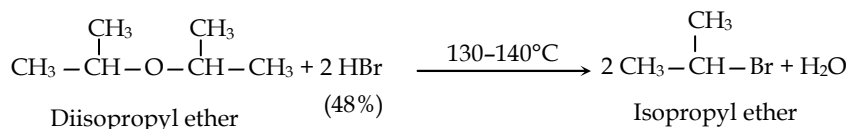
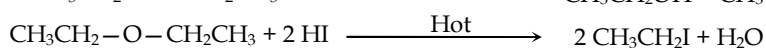
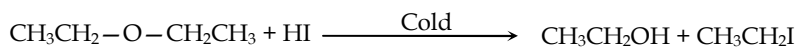


C. Reactions involving cleavage of C-O bond

1. **Reaction with HBr and HI:** Ethers readily react with **concentrated** HBr and HI in different conditions.



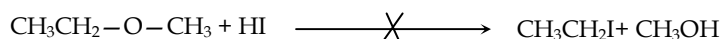
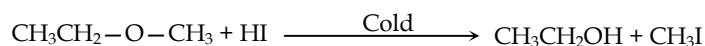
Examples:



Cleavage of unsymmetrical ether:

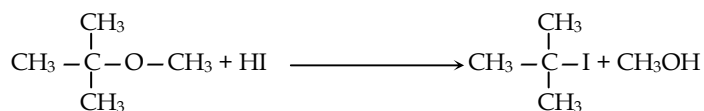
When unsymmetrical ether is treated with conc. HI, the cleavage in ether takes place in such a way that oxygen atom remains attached to the larger alkyl group. This is because alkoxide having the larger alkyl group is bulkier and better leaving group.

Example:

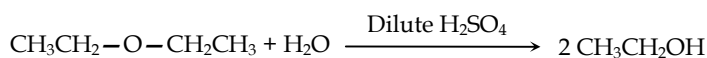


But, if 3° alkyl group is present, the bond breakage takes place in such a way that oxygen remains attached to the smaller alkyl group. This is because, formation of 3° haloalkane is more feasible due to the high stability of 3° carbocation.

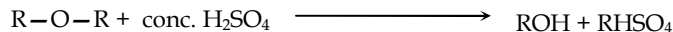
Example:



2. **Reaction with dilute sulphuric acid:** When ether is heated with dilute sulphuric acid under pressure, corresponding alcohol is obtained by its hydrolysis.

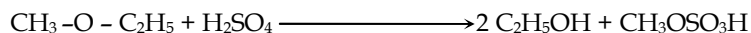


When ether is **warmed** with **concentrated sulphuric acid**, alcohol and alkyl hydrogen sulphate are formed.

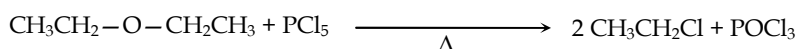
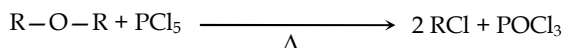


When unsymmetrical ether is heated with conc. H_2SO_4 , the bond breakage generally takes place in such a way that oxygen remains attached to the larger alkyl group.

Example:



3. **Reaction with phosphorous pentachloride:** When ether is heated with PCl_5 , alkyl chloride is formed.



Self Test

11.7 Give the reaction of diethyl ether with

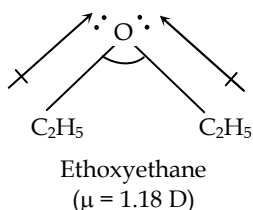
- a. conc. HCl in cold b. conc. HI in cold c. conc. H_2SO_4 in cold
d. conc H_2SO_4 in hot e. dil. H_2SO_4 in hot

11.8 Convert ethoxyethane into methoxyethane and vice versa.

11.1.7 Diethyl Ether

The use of ether as anesthetic was first publicly demonstrated on 16th October, 1846 at Massachusetts General Hospital, Boston, USA to remove a tumor from neck of a young man. The patient felt no pain during the surgery, which seemed like a miracle at the time. Citizen of Boston raised funds to construct a monument to commemorate humankind's triumph over pain.

Before anesthetics, surgery was a very painful and horrific practice. Surgeons had few options for dulling the intense pain. The most common techniques were plying the patient with opium or alcohol and then employing several strong men to hold the patient down to carry out a quick and brutal surgery. Patients would scream with pain throughout the entire procedure. It wasn't uncommon for people to commit suicide rather than go through the nightmare of surgery. Ether obviously made surgery a more comfortable experience for patients, and allowed surgeons to take more time in which they can work more effectively and develop more and more complex and beneficial surgical techniques. The use of ether as an anesthetic in the mid 19th century opened a new advances in surgical technology and medical history although it has been replaced by safer anesthetic.



Ethoxyethane is the most important member of ether. It is commonly called diethyl ether or sulphuric ether or simply ether.

11.1.8 Laboratory Preparation of Diethyl Ether

Principle: When an excess of ethyl alcohol is heated with concentrated sulphuric acid at 140°C, diethyl ether is formed. The reaction occurs in two steps.

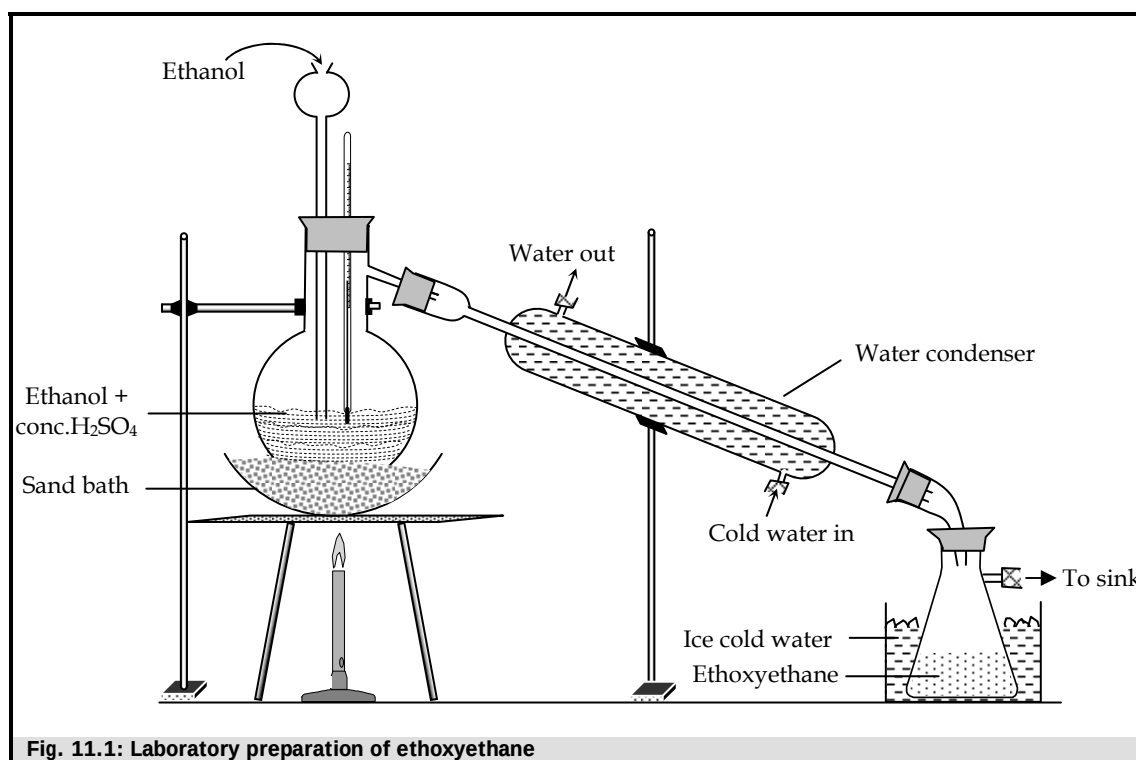
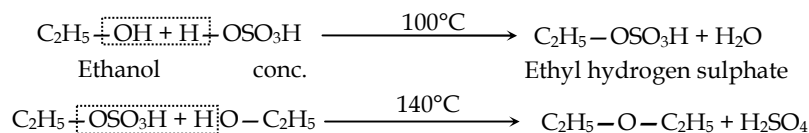


Fig. 11.1: Laboratory preparation of ethoxyethane

Procedure

About 100 ml of ethyl alcohol is taken in a distillation flask in which dropping funnel, thermometer, water condenser and receiver are fitted as shown in the figure. The rubber tube is attached to the side tube of receiver to carry away any uncondensed vapour of ether. The whole apparatus is made air tight because ether is highly inflammable. About 100 ml of concentrated sulphuric acid is added from dropping funnel and heated to about 140°C on sand bath. Ether so formed distills off and is collected in ice cooled receiver. As ether starts distilling, a continuous stream of fresh alcohol is added from the dropping funnel at rate of the formation of ether. In this process, ether can be obtained continuously, hence the process is called **Williamson's continuous etherification**.

Purification

The main impurities in diethyl ether are alcohol, water, SO₂ and CO₂. To remove these impurities following steps are applied.

- Ether is treated with dilute NaOH in a separating funnel, which dissolves CO₂, SO₂ and other acidic impurities and the aqueous layer is rejected.

- ii. Ether is washed with water 2-3 times to remove water soluble impurities and the aqueous layer is again rejected.
- iii. Ether is treated with 50% CaCl_2 solution which removes alcohol forming alcoholates.
- iv. The ether is dried over anhydrous CaCl_2 .
- v. Finally, ether is redistilled at about 34 to 40°C to get it in pure and dry state.

In this way, diethyl ether is prepared in the laboratory in pure and dry state.

11.1.9 Physical Properties

1. Diethyl ether is highly volatile and highly inflammable colourless liquid.
2. Its boiling point is 34.6°C and freezing point is -116°C.
3. It is sparingly soluble in water (8 g per 100 g water) but soluble in organic solvents.

11.1.10 Chemical Properties

Diethyl ether shows all chemical properties of aliphatic ethers.

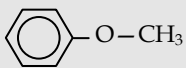
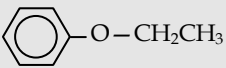
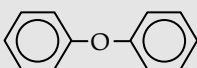
11.1.11 Uses

- i. The major use of diethyl ether is as solvent. It is used as solvent in industries for fats, oils, waxes, resins, etc. In laboratories, it is used as solvent in many organic reactions like Wurtz reaction, Fittig reaction, for the preparation of Grignard reagent and carrying out its reactions, in solvent extraction for the separation and purification of organic compounds, etc.
- ii. It is also used as refrigerant.
- iii. It can also be used as general anesthetic during surgery. However, it has been replaced by more safe anesthetic.
- iv. It is used as volatile starting fluid for diesel engines and gasoline engines in cold weather.

11.2 Aromatic Ethers

11.2.1 Nomenclature

Some common examples of aromatic ethers are:

Formula	IUPAC name	Common name
	Methoxybenzene	Methyl phenyl ether or Anisole
	Ethoxybenzene	Ethyl phenyl ether or phenetole
	Phenoxybenzene	Diphenyl ether

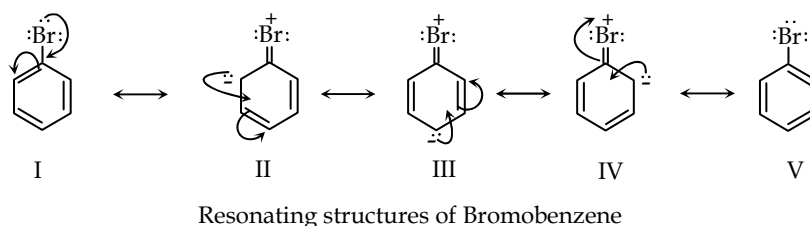
CH3Br + [Na+].[O-]c1ccccc1 $\longrightarrow$ COc1ccccc1 + [Na+].[Br-]

Methyl bromide Sodium phenoxide Methoxybenzene (Anisole)

c1ccccc1Br.[Na+].[O-]C>>c1ccccc1OC.[Na+].[Br-]

Methoxybenzene
(Anisole)

- carbon to halogen bond has partial double bond character due to resonance effect.
- carbon to halogen bond is less polar due to hybridization effect.



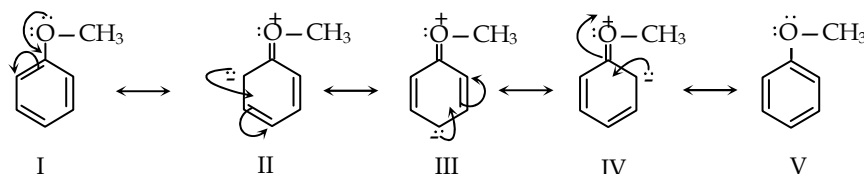
11.2.3 Chemical Properties

The chemical reaction of methoxybenzene can be divided into two categories.

A. Reaction due to cleavage of C-O bond

This can be explained as follows:

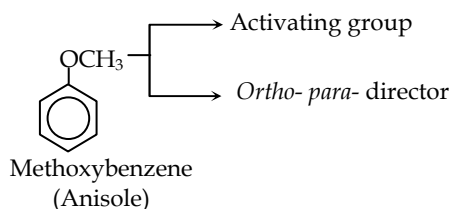
Oxygen to carbon of benzene ring bond (bond b) is stronger: Oxygen to carbon of benzene ring (bond b) has partial double bond character since anisole is resonance hybrid of structures I-V. Hence it is stronger than oxygen to carbon of alkyl group (bond a).



Resonating structures of anisole

B. Reaction due to benzene ring: Electrophilic substitution reaction

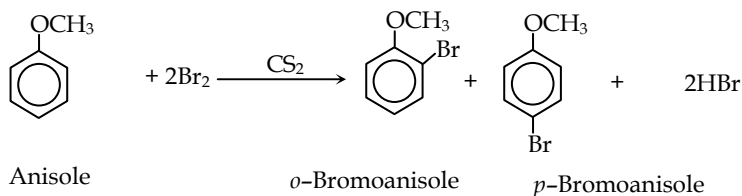
Like other aromatic compounds, anisole undergoes electrophilic substitution reaction. The $-OCH_3$ (methoxy group) is weakly activating group, which is *ortho*- and *para*- director towards electrophilic substitution reaction.



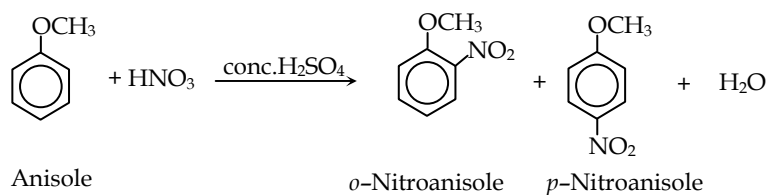
Self Test

11.9 Anisole gives *ortho*- and *para*- substituted products, explain.

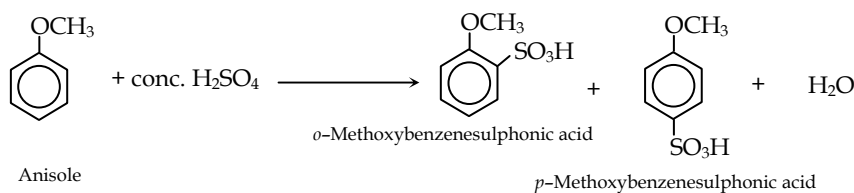
1. **Halogenation:** When anisole reacts with Br_2 in CS_2 , a mixture of *p*-Bromoanisole and *o*-Bromoanisole is formed.



2. **Nitration:** When anisole is heated with a mixture of conc. HNO_3 and conc. H_2SO_4 (nitrating mixture), a mixture of *o*-nitroanisole and *p*-nitroanisole is formed.



3. **Sulphonation:** When anisole is heated with conc. H_2SO_4 , a mixture of *o*-Methoxybenzenesulphonic acid and *p*-Methoxybenzenesulphonic acid is formed.





$$\text{R}-\text{O}-\text{R}$$

2-Methoxypropane

$$\text{CH}_3-\overset{\text{H}}{\underset{\text{H}}{\overset{\delta+}{\text{C}}}}-\text{I}^{\delta-} + \text{CH}_3\text{O}^-\text{Na}^+ \longrightarrow \text{CH}_3\text{CH}_2-\text{O}-\text{CH}_3 + \text{NaI}$$

Sodium methoxide Methoxyethane

$$\text{CH}_3\text{CH}_2-\text{O}-\text{CH}_2\text{CH}_3 + \text{O}_2 \xrightarrow[\text{(air)}]{\text{Light}} \text{CH}_3\text{CH}_2-\text{O}-\underset{\text{O}-\text{OH}}{\text{CH}}-\text{CH}_3$$

Ether peroxide

Ether peroxide is explosive and thermally unstable. Hence care should be taken during distillation of old sample of ethers otherwise violent explosion may take place.

Questions

Very Short Answer Questions:

- Write the possible isomers of C_3H_8O and indicate the isomer which does not decolorize acidified potassium permanganate solution.
- Write the alcohols having molecular formula $C_4H_{10}O$ with their IUPAC name and indicate the isomers which do not decolorize acidified potassium permanganate solution.
- How can you detect presence of organic peroxides in ether? Give any method to remove peroxides.
- Outline the preparation of tert-Butyl ethyl ether by Williamson's synthesis.
- Anisole is prepared by using sodium phenolate and methyl iodide but not sodium methoxide and chlorobenzene, why?
- What is unsymmetrical ether? Write an example with their IUPAC name.
- Name the isomer of C_2H_6O which reacts with excess HI to give iodomethane as major product. Give the reaction involved.
- Write the chemical equation for the reaction taking place when:
 - Sodium phenoxide reacts with iodomethane.
 - Ethoxyethane is exposed to light and air.
- An organic compound with a molecular formula $C_4H_{10}O$ does not react with Na and gives only single haloalkane when heated with excess conc. HI. Identify the organic compound.

Short /Long Answer Questions:

- Identify A and B
 - Monohydroxybenzene $\xrightarrow{NaOH}$ A $\xrightarrow{CH_3I}$ B
 - Phenol $\xrightarrow{(i) Na, (ii) CH_3I}$ A $\xrightarrow[(130^\circ C)]{(i) 57\% HI, 120-}$ B+C
 - Ethanol $\xrightarrow{(i) Na, (ii) CH_3I}$ A $\xrightarrow{HI, \Delta}$ B+C
 - Ethanol $\xrightarrow[140^\circ C]{Conc. H_2SO_4}$ A $\xrightarrow{0^\circ C}{Conc. HCl}$ B+C
 - Ethyl chloride $\xrightarrow{Ag_2O, \Delta}$ A $\xrightarrow{PCl_5, \Delta}$ B
- How is ethoxyethane prepared in laboratory in pure and dry state?
- An organic liquid 'A' has molecular formula $C_4H_{10}O$ which is soluble in cold and conc. sulphuric acid but doesn't decolorize pink color of acidified potassium permanganate. 'A' when refluxed with phosphorous pentachloride gives C_2H_5Cl . Identify 'A' giving necessary reactions. How does 'A' react with:
 - Cold HI
 - Hot excess HI
- Write all possible isomers of ethers of molecular formula $C_5H_{12}O$ and give their IUPAC names.
 - Convert ethoxyethane into methoxyethane
- What is Williamson synthesis? What is the advantage of this method over dehydration of alcohol?
 - Write the reaction when anisole is heated with conc. HI.

Multiple Choice Questions

- Diethyl ether is
 - Mixed ether
 - Petroleum ether
 - Sulphuric ether
 - Phenetole
- When ethyl halide reacts with sodium ethoxide, it gives diethyl ether. The reaction is
 - Williamson's synthesis
 - Wurtz reaction
 - Kolbe's reaction
 - Hoffmann's reaction
- When tert-butyl bromide is reacted with sodium ethoxide, the major product is
 - tert-Butyl methyl ether
 - Diethyl ether
 - Ethane
 - 2-methyl-1-propene
- When vapors of ethanol is passed over heated alumina at 523K, it gives
 - Ethoxyethane
 - Ethanal
 - Ethene
 - Ethylhydrogen sulphate
- The number of metamers that is represented by formula $C_4H_{10}O$ is
 - 2
 - 3
 - 4
 - 5
- The explosive substance formed when ether is exposed to air for sometime is
 - TNT
 - Super oxide
 - Oxide
 - Peroxide
- When ethyl hydrogen sulphate is heated to 140°C , it gives
 - Ethoxyethane
 - Ethanal
 - Ethene
 - 1,2-Ethandiol
- With cold and conc. HCl, Diethyl ether forms
 - Ethyl alcohol
 - Ethyl chloride
 - Oxonium salt
 - Ether peroxide
- Which of the following does not form dative bond with ether?
 - $RMgX$
 - BF_3
 - $AlCl_3$
 - $NaBH_4$
- When anisole reacts with HI, it gives,
 - Methanol
 - Phenol
 - Iodobenzene
 - Iodophenol

Answer Key

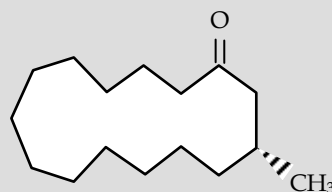
1	2	3	4	5	6	7	8	9	10
c	a	d	c	b	d	a	c	d	b

Aldehydes and Ketones

12



Musk deer (*Kasturi mriga*)
(कस्तुरी मृग)



3-Methylcyclopentadecanone
(Muscone)

Aldehydes and ketones are widespread in nature. Muscone is a ketone that is the primary contributor to the odour of musk. Natural muscone is obtained from musk, a glandular secretion of the musk deer which has been used in perfumery and medicine for thousands of years. Since obtaining natural musk requires killing the endangered animal, nearly all muscone used in perfumery today is synthetic.

12.1 Aliphatic Aldehydes and Ketones

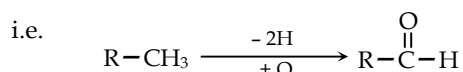
12.1.1 Introduction

The organic compounds containing the carbonyl ($>\text{C}=\text{O}$) group in their molecules are called carbonyl compounds. The carbonyl ($>\text{C}=\text{O}$) group may exist in two forms; if it exists in the form of $-\text{CHO}$ group, it is an **aldehydic functional group (formyl group)** and if it exists as such in the form of $-\text{CO}-$ group, it is a **ketonic functional group (oxo or keto group)**.

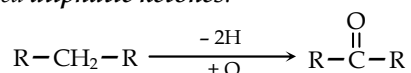
The aldehydic (formyl) functional group containing organic compounds are aldehydes, in which the carbonyl group is linked to an alkyl (R-) or aryl (Ar-) group and a hydrogen atom in the form of $\text{R}-\text{CHO}$ or $\text{Ar}-\text{CHO}$. On the other hand, ketonic functional group containing compounds are ketones, in which the carbonyl group is linked to two alkyl or aryl groups in the form of $\text{R}-\text{CO}-\text{R}$, $\text{R}-\text{CO}-\text{Ar}$ or $\text{Ar}-\text{CO}-\text{Ar}$. General molecular formula of aliphatic aldehydes and ketones is $\text{C}_n\text{H}_{2n}\text{O}$.

Definitions

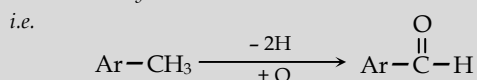
- Derivatives of alkanes, where two terminal hydrogen atoms are replaced by divalent oxygen atom are called **aliphatic aldehydes**.



- Derivatives of alkanes, where two non-terminal hydrogen atoms are replaced by divalent oxygen atom are called **aliphatic ketones**.



Derivatives of arenes, where two hydrogen atoms from side chain $-\text{CH}_3$ are replaced by divalent oxygen atom are called **aromatic aldehydes**.



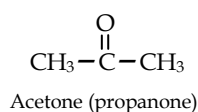
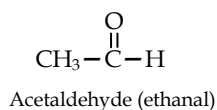
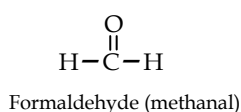
Derivatives of arenes, where two hydrogen atoms from side chain $-\text{CH}_2-\text{R}$ or $-\text{CH}_2-\text{Ar}$ are replaced by divalent oxygen atom are called **aromatic ketones**.

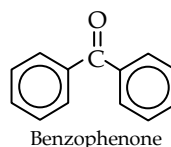
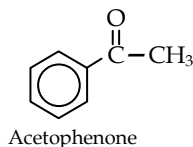
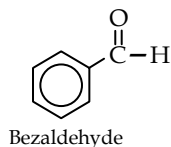


In other words, both aldehydes and ketones contain the carbonyl ($>\text{C}=\text{O}$) group and are often referred to collectively called as carbonyl compounds. It is the carbonyl group that largely determines the chemistry of aldehydes and ketones i.e. the presence of the carbonyl group in both aldehydes and ketones is responsible for their identical chemical properties. However, the presence of H-atom attached with the $-\text{CO}-$ group in aldehydes is more reactive than alkyl group (R-) attached with the $-\text{CO}-$ group in ketones and hence they differ from each other in certain aspects.

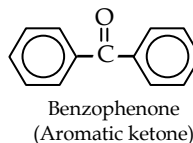
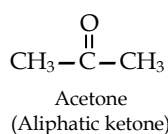
Examples:

- a. **Aliphatic aldehydes and ketones:**

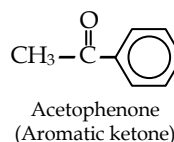
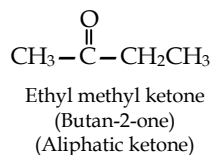


b. Aromatic aldehydes and ketones:

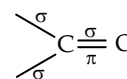
Simple or symmetrical ketones: Those ketones in which the -CO- group is attached with two identical alkyl or aryl groups.



Mixed or unsymmetrical ketones: Those ketones in which the -CO- group is attached with two different alkyl and aryl groups.

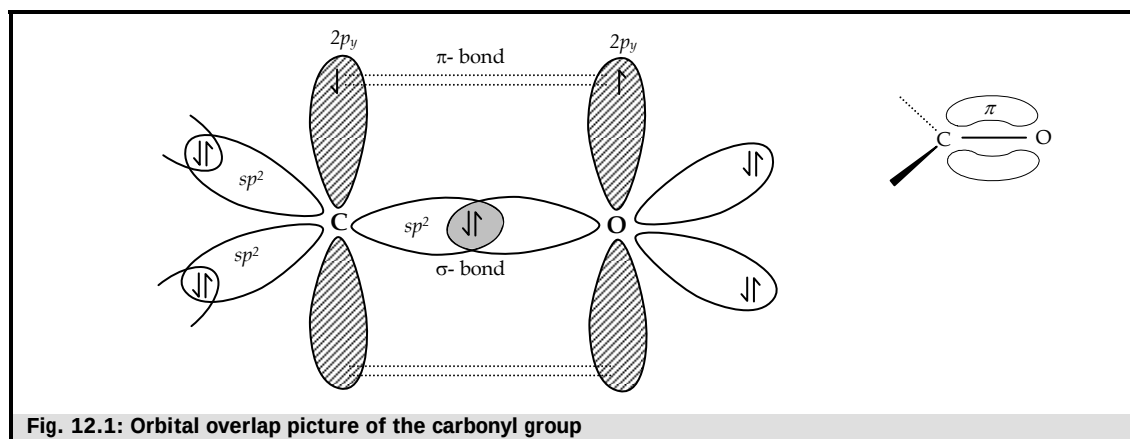
**12.1.2 Structures of Carbonyl Group**

Both aldehydes and ketones have common carbonyl group. Carbon of the carbonyl group is sp^2 hybridized. So, the geometry of the molecule across the carbonyl carbon is trigonal planar with the bond angle of 120° .



Carbon and oxygen atoms are linked together by a double bond, one of this bond is σ -bond and the another is π bond. The π bond is weak and polarisable, so the carbonyl compounds are likely to give addition reactions.

Since the carbonyl carbon atom of carbonyl group is sp^2 hybridized which contains singly filled unhybridized p -orbital that undergoes side ways on overlaps with singly filled p -orbital of oxygen atoms giving π bonds.

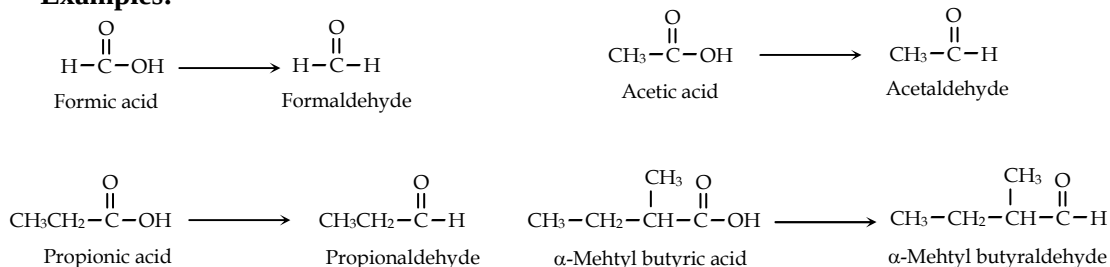


12.1.3 Nomenclature of Carbonyl Compounds

Nomenclature of Aldehydes

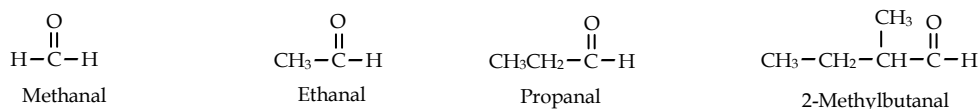
- a. **Common names:** The common names of aldehydes are derived from the names of the corresponding carboxylic acid, where *-ic acid* of parent carboxylic acid is replaced by *-aldehyde*.

Examples:



- b. **IUPAC names:** Aldehydes are named by replacing *-e* of alkane by *-al* as secondary suffix.

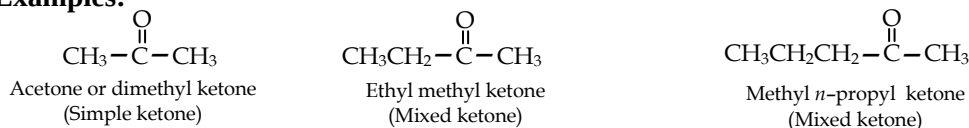
Examples:



Nomenclature of ketones

- a. **Common names:** Ketones are named by writing the names of alkyl groups attached to the carbonyl group followed by the word "ketone". First member of ketone (CH_3COCH_3) is known as acetone which has been in use for a long time as a solvent, nail-polish remover, etc.

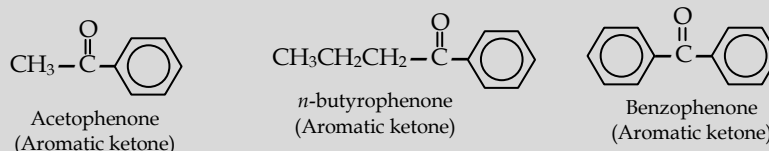
Examples:



Aromatic ketones

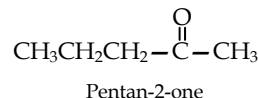
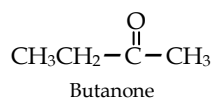
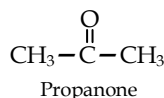
A ketone in which the carbonyl group attached to a benzene ring is named as a *-phenone*.

Examples:



- b. **IUPAC names:** In longest carbon chain carrying $>\text{C}=\text{O}$ group is considered as parent chain, and is named by replacing *-e* of the corresponding alkane with *-one*.

The position of various substituents is indicated by numbers, the carbonyl carbon of $-\text{CO}-$ group being given the lowest possible number.

Examples:**Self Test**

12.1 Give the structures and IUPAC names of the following:

- | | |
|------------------------------------|-------------------------|
| i. ethyl <i>neo</i> -pentyl ketone | ii. phenyl acetaldehyde |
| iii. vinyl acetone | iv. salicylaldehyde |
| v. acetyl acetone | |

12.2 Give structures and IUPAC names of carbonyl compounds of molecular formula $\text{C}_4\text{H}_8\text{O}$.

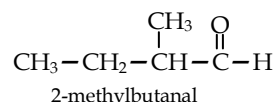
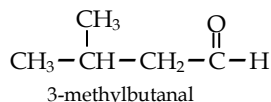
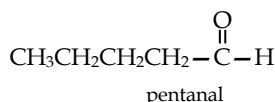
12.1.4 Structural Isomerism in Aliphatic Carbonyl Compounds

Aliphatic aldehydes and ketones exhibit the following types of structural isomerisms.

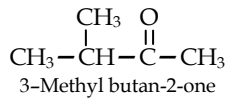
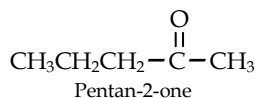
1. **Chain isomerism:** Aldehydes and ketones exhibit chain isomerism.

Examples:

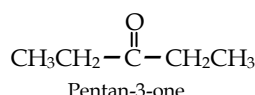
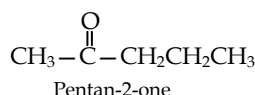
- i. Pentanal, 3-methylbutanal and 2-methylbutanal are chain isomers.



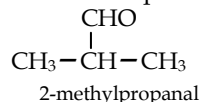
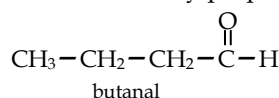
- ii. Pentan-2-one and 3-methylbutan-2-one are chain isomers.



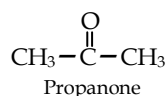
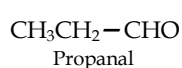
2. **Positional isomerism:** Ketones show positional isomerism when position of $>\text{C}=\text{O}$ group is different in the longest carbon chain of compound of same molecular formula. For example, Pentan-2-one and pentan-3-one are positional isomers and they are also *metamers* to each other.



But aliphatic aldehyde does not exhibit positional isomerism as the $-\text{CHO}$ group remains in the terminal position and the difference of position of $-\text{CHO}$ group gives rise to chain isomerism. For example, butanal and 2-methylpropanal are chain isomers not the positional isomers.

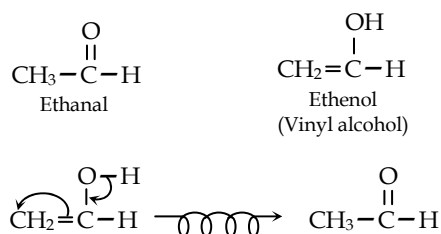


3. **Functional isomerism:** Aldehydes and ketones can be functional isomers of each other because both of them have the same molecular formula of $\text{C}_n\text{H}_{2n}\text{O}$. For example, propanal and propanone are functional isomers.

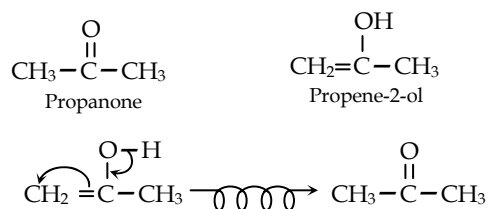


4. Tautomerism

- i. Ethanal and ethenol are tautomers, and they undergo keto-enol tautomerism.



- ii. Propanone and propen-2-ol are tautomers.

**Note**

Tautomers are interconvertible functional isomers.

Self Test

12.3 How many isomeric ketones are possible for 2,2-dimethylpropanal ?

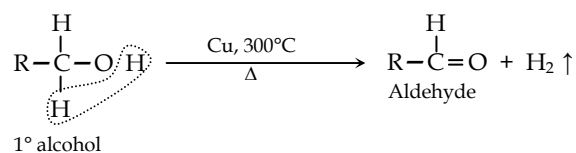
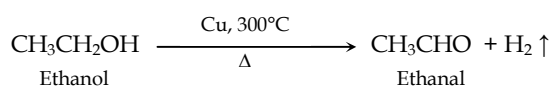
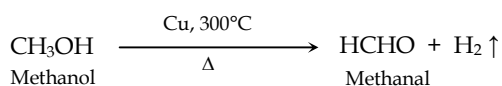
12.1.5 General Methods of Preparation of Aliphatic Aldehydes and Ketones

Some general method for the preparation of aliphatic aldehydes and ketones are discussed below:

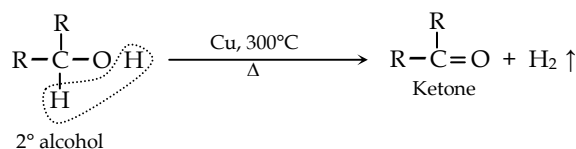
1. From Alcohols

a By catalytic dehydrogenation of alcohols

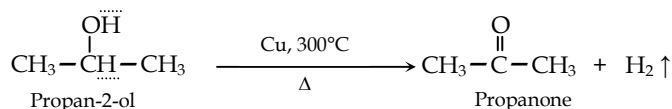
- i. When vapours of primary (1°) alcohol is heated over Cu or MnO at 300°C , an aldehyde is formed where hydrogen atoms are removed each from -OH group and α -carbon. The reaction is called as *catalytic dehydrogenation*.

**Examples:**

- ii. When vapour of secondary (2°) alcohol is heated over Cu at 300°C , ketone is obtained.

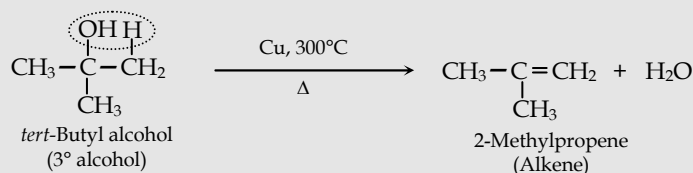


Example:



Note:

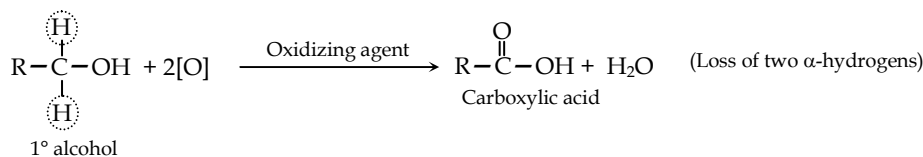
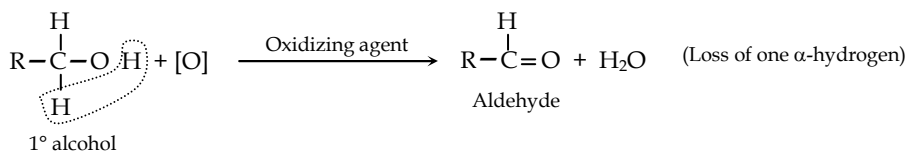
When vapour of tertiary (3°), alcohol is passed over heated Cu at 300°C , alcohol undergoes dehydration into respective alkene instead of dehydrogenation into aldehyde or ketone.



b. By oxidation of alcohols

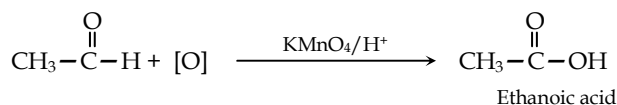
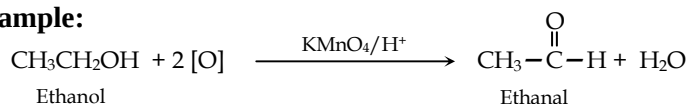
The oxidation of an alcohol involve the loss of one or more α -hydrogens from the carbon bearing the $-\text{OH}$ group. The nature of product formed depends upon the type of alcohol.

- i. **Oxidation of primary alcohols:** A primary alcohol contains two α -hydrogens, and can either lose one of them to form an aldehyde or both of them to form a carboxylic acid.



Oxidation of 1° alcohols to carboxylic acids is usually accomplished by use of acidified aqueous KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, CrO_3 in glacial acetic acid, etc. Oxidation of 1° alcohols yields aldehydes, and aldehydes are susceptible so they are further oxidized to carboxylic acid.

Example:

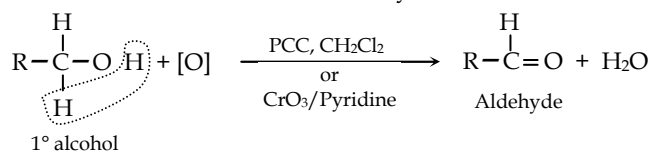


372 Modern Approach to Chemistry-II

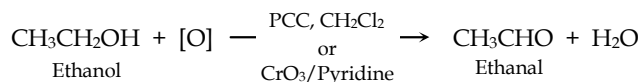
Therefore, the conversion of 1° alcohols to aldehydes requires special methods. One of the best and most convenient reagents for this conversion is pyridinium chlorochromate, PCC ($\text{C}_5\text{H}_5\text{NH}^+\text{CrO}_3\text{Cl}^-$) or collin's reagent (chromium trioxide pyridine complex) used in non-aqueous medium like CH_2Cl_2 .

Thus, 1° alcohols are oxidized to aldehydes with PCC or Collin's reagent in CH_2Cl_2 as a solvent. This reagent prevents the further oxidation of aldehydes to carboxylic acids.

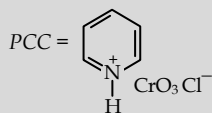
When primary alcohol is oxidized with pyridinium chlorochromate (PCC) or Collin's reagent in dichloromethane as a solvent, aldehyde is obtained.



Example:

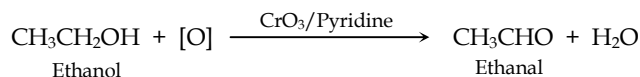


The pyridinium chlorochromate, $\text{C}_5\text{H}_5\text{NH}^+\text{CrO}_3\text{Cl}^-$ (PCC) is prepared by heating chromic acid with pyridinium chloride.



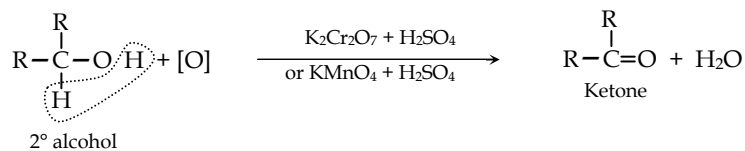
Primary alcohols can also be selectively oxidized up to aldehyde using CrO_3 /pyridine

Example:

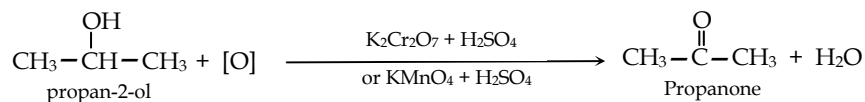


- ii. **Oxidation of secondary alcohols:** Oxidation of 2° alcohols can lose its only one α -hydrogen to form a ketone is generally straight forward.

When secondary alcohol is oxidized by using oxidizing agents like acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 , ketone is obtained.

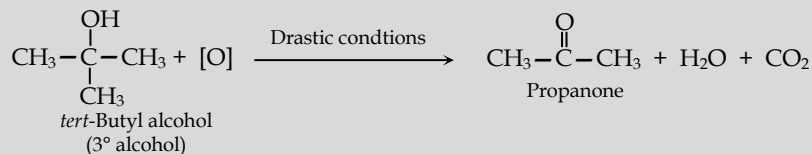


Example:



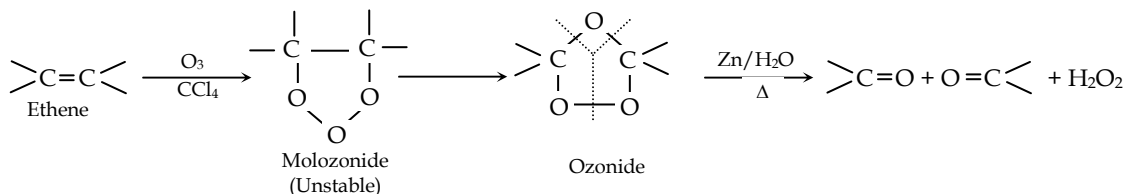
- Chromic anhydride in pyridine is called Sarett's reagent.
- Aqueous solution of CrO_3 , acetone and conc. sulphuric acid is called Jone's reagent and oxidation using Jone's reagent is called Jone's oxidation.
- Tertiary alcohols do not contain α -H. So they are not oxidized under ordinary condition. However, in drastic conditions they are oxidized to ketones having decrease no. of C atoms.

Example:

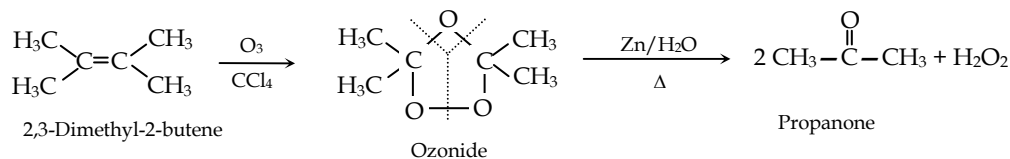
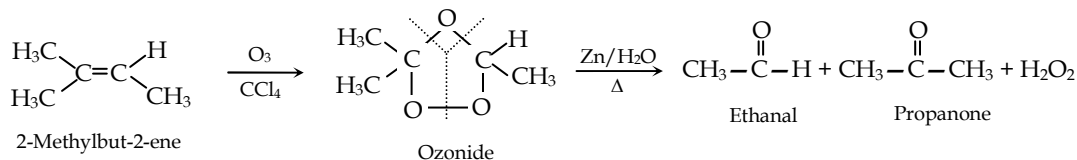
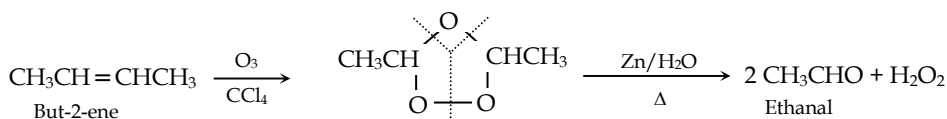
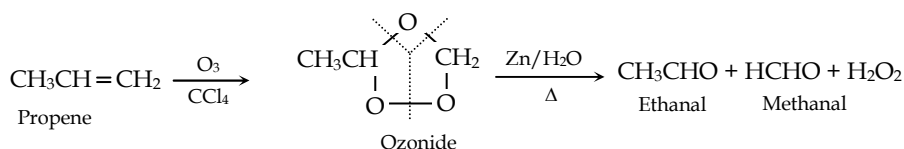
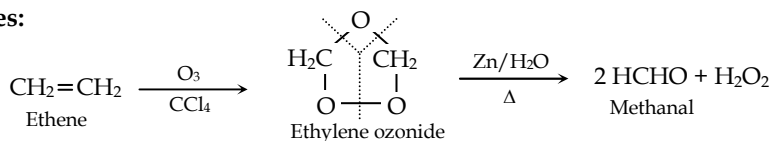


2 By Ozonolysis of Alkenes

When an alkene is treated with ozone in presence of CCl_4 as a solvent, ozonide is obtained. The ozonide when distilled with water in presence of zinc, gives carbonyl compounds (aldehyde and/or ketone). The nature of aldehyde and ketone so formed may depend upon the structure of alkene. Hence, this reaction is used to identify the structure of alkene.



Examples:



Note:

- Ozonides are generally explosive at high temperature so, they are not isolated in dry condition.
- Zinc absorbs H_2O_2 obtained as a side product. If H_2O_2 is not absorbed, it oxidizes the aldehyde so formed into carboxylic acid.
- Ozonolysis occurs in two steps.
 - a. Formation of ozonide (ozonation)
 - b. Decomposition of ozonide thus obtained by water (hydrolysis).
- Unbranched alkene is likely to give aldehydes only.
- Ozonolysis is an important reaction to determine the structure of alkene.
- Zn is used to remove H_2O_2 as: $\text{Zn} + \text{H}_2\text{O}_2 \longrightarrow \text{Zn}(\text{OH})_2$, otherwise, aldehydes get oxidised to carboxylic acids.

Self Test

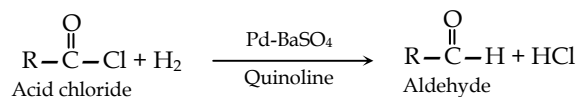
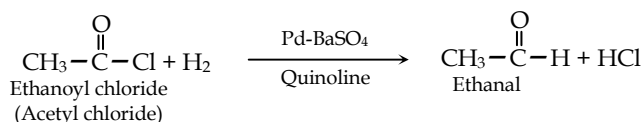
12.4 Which aldehydes are formed by reductive ozonolysis of 1,3-pentadiene?

- a. HCHO b. $\text{CHO}-\text{CHO}$ c. CH_3CHO d. CH_3COCH_3

3 From Acid Chloride

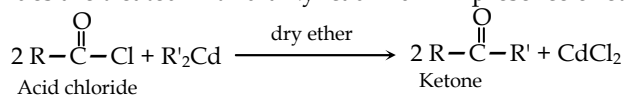
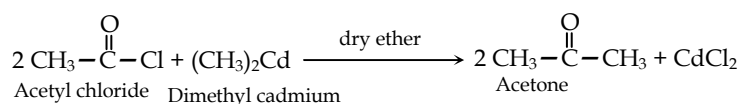
When acid chlorides are heated with H_2 in presence of $\text{Pd}-\text{BaSO}_4$ poisoned by quinoline, aldehydes are obtained. This reaction is called **Rosenmund Reduction**.

i.e.

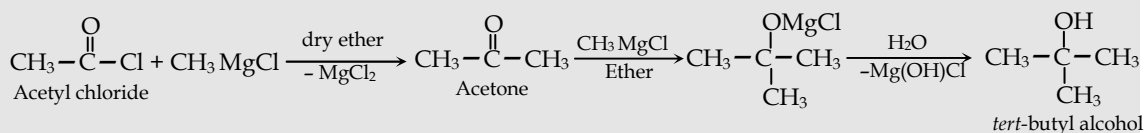
**Example:****Note:**

LiAlH_4 and H_2/Ni cannot be used for this reduction as they reduce the aldehyde further into 1° alcohols.

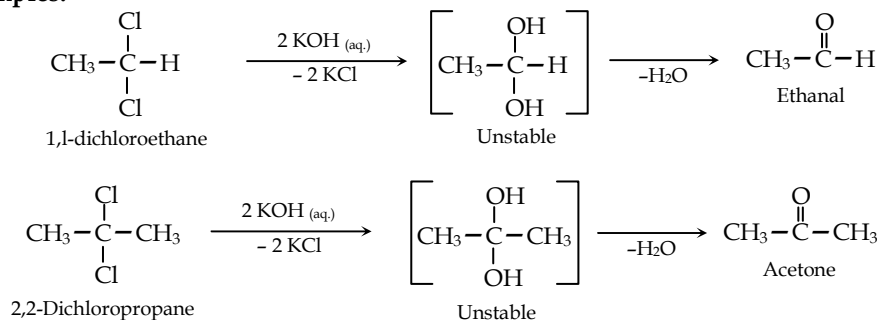
When acid chlorides are treated with dialkyl cadmium in presence of ether, ketones are obtained.

**Example:****Note:**

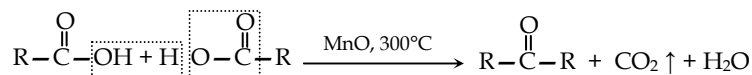
Ketones cannot be obtained by the reaction of acid chloride with Grignard reagent. It is because the ketone obtained would further react with Grignard reagent to form tertiary alcohol.

**4. From geminal dihalides**

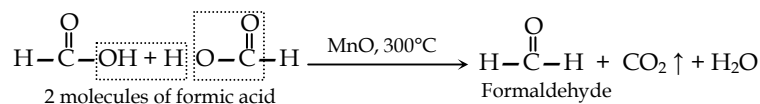
Geminal dihalides on base promoted hydrolysis (on heating with aqueous KOH) give aldehydes or ketones.

Examples:**5. By the dry distillation of fatty acids**

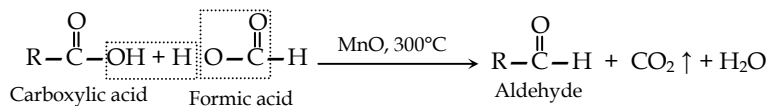
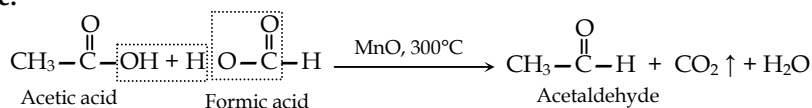
Fatty acids on heating with MnO at 300°C give aldehydes or ketones. General equation:



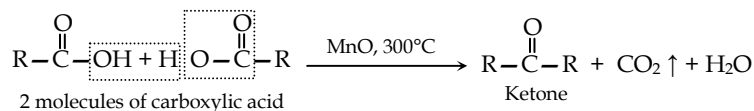
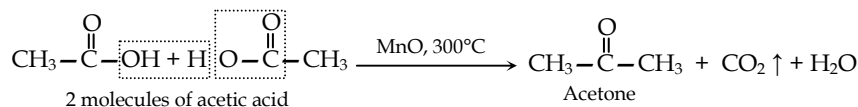
- a. Formaldehyde is obtained by passing the vapours of formic acid over heated MnO catalyst at 300°C.



- b. Aldehydes other than formaldehyde are obtained by passing the vapours of a mixture of a carboxylic acid and formic acid over heated MnO catalyst at 300°C.

**Example:**

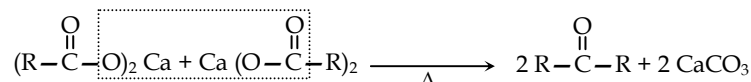
- c. Simple ketones are obtained by passing the vapours of a carboxylic acid (other than formic acid) over heated MnO catalyst at 300°C.

**Example:**

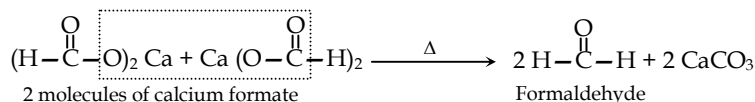
6. By the distillation of calcium salts of fatty acids

Calcium salts of fatty acid on heating (distillation) give carbonyl compounds.

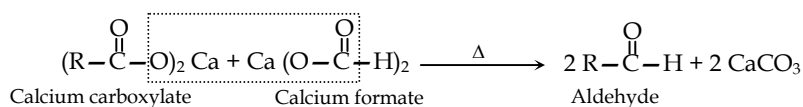
The general equation:



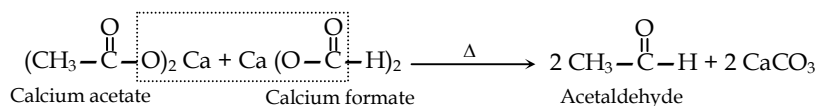
- a. Dry distillation of calcium formate (calcium salt of formic acid) alone gives formaldehyde.



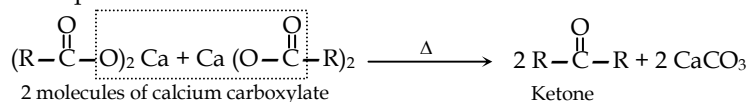
- b. Dry distillation of a mixture of calcium formate and calcium salt of other acids give aldehydes other than formaldehyde.



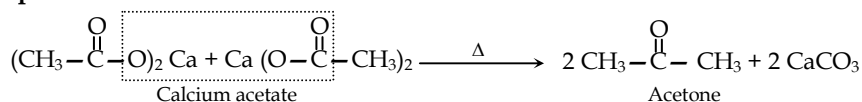
Example:



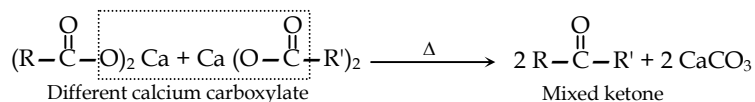
- c. Dry distillation of calcium carboxylate (calcium salts of carboxylic acid other than formic acid) alone gives simple ketone.



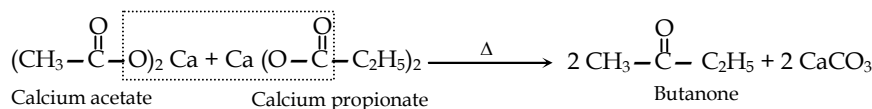
Example:



- d. Dry distillation of a mixture of different calcium carboxylates gives mixed ketone. But this method gives a mixture of ketone.



Example:



When calcium salt of dicarboxylic acid is heated, cyclic ketone is obtained.

Self Test

12.5 What happens when calcium salt of dicarboxylic acid is heated?

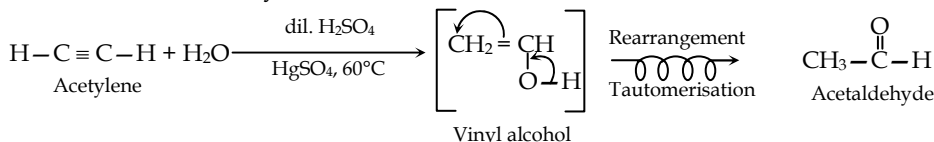
12.6 Calcium salt of an acid X on dry distillation with calcium formate produces *iso*-butyraldehyde as the product. What is X?

7. Catalytic hydration of alkynes

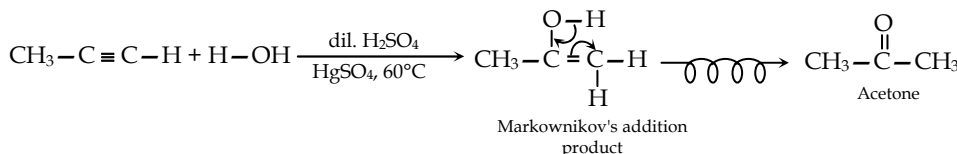
Hydration of alkynes with water in presence of catalyst like dilute solution of H_2SO_4 and HgSO_4 at 60°C give corresponding aldehydes or ketones.

Examples:

- i. Acetaldehyde is prepared by catalytic hydration of acetylene. When acetylene gas is heated with dil. H_2SO_4 in presence of dil. HgSO_4 solution as a catalyst, acetylene adds a molecule of H_2O onto its $\text{C}\equiv\text{C}$ triple bond, vinyl alcohol is formed. Vinyl alcohol undergoes tautomerization into more stable acetaldehyde.



- ii. Under similar condition higher alkynes other than acetylene give ketone.

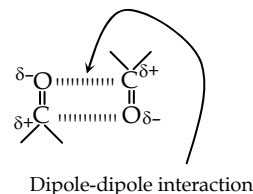


Note:

Formaldehyde cannot be prepared by this method.

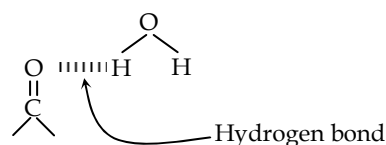
12.1.6 Physical Properties

- a. **State:** Formaldehyde is a gas at room temperature. Aldehydes and ketones upto C_{11} are colourless liquids. Higher members are colourless solids.
- b. **Odour:** Formaldehyde has irritating pungent odour while other lower aldehydes and ketones have pleasant odour. Many ketones have perfumery value.
- c. **Boiling points:** Aldehydes and ketones have fairly polar $C=O$ bond. The molecules have a very high dipole moment. So, the intermolecular force of attraction in the carbonyl compounds is relatively stronger (dipole-dipole interaction). However this force is weaker than that in alcohols (where the intermolecular force is hydrogen-bonding). So, aldehydes and ketones have boiling points greater than of alkanes and smaller than those of alcohols and carboxylic acids of nearly same molecular mass.
- 
- Dipole-dipole interaction



Compound	Name	Mol. wt.	Boiling point
$\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—CH}_3$	Pentane	72	36°C
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$	Butanal	72	76°C
$\text{CH}_3\text{—CH}_2\text{—}\overset{\text{O}}{\underset{\text{ }}{\text{C}}}\text{—CH}_3$	Butanone	72	80°C
$\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—OH}$	Butan-1-ol	74	118°C

- d. **Solubility:** Lower aldehydes and ketones are fairly soluble in water. This is because, the molecules of aldehydes and ketones can form hydrogen bonding with water. Moreover aldehydes and ketones are polar compounds and water is also a polar solvent.



But higher aldehydes and ketones are practically insoluble in water. This is because, higher aldehydes and ketones contain longer hydrophobic part in their molecule which make them water insoluble.

However, they are soluble in nonpolar solvents (organic solvents).

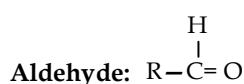
Self Test

12.7 Arrange the following in order of increasing boiling points:

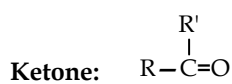
- a. Propanone b. Propanal c. Propane d. Propan-1-ol

12.1.7 Chemical Properties

Aldehydes and ketones resemble with each other closely in most of their properties due to a common structural features of a carbonyl ($>\text{C}=\text{O}$) group. However, there is a H-atom attached to the carbonyl group of aldehydes, and there are two alkyl groups attached to the carbonyl group of ketones.



(H-atom attached to $>\text{C}=\text{O}$)

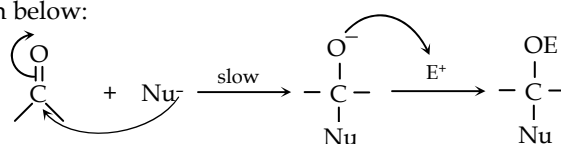


(Two alkyl groups R and R' attached to $>\text{C}=\text{O}$)

This difference in structure affects their properties in two ways:

- Aldehydes are quite easily oxidized, whereas ketones are oxidized only with difficulty.
- Aldehydes are usually more reactive than ketones towards nucleophilic addition.

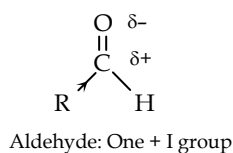
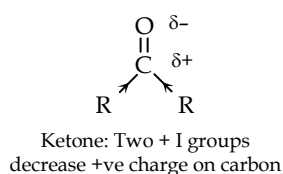
Main type of reactions given by carbonyl compounds are nucleophilic addition reactions. In this type of addition reactions, nucleophile is added in the first step (rate determining step). The mechanism of the reaction is shown below:



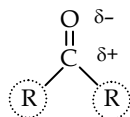
For these types of reactions aldehydes are more reactive than ketones.

Reasons:

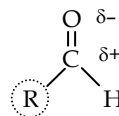
- a. **Electronic factor:** In ketones, carbonyl carbon is attached to two alkyl groups (e^- releasing groups) which make the carbonyl less electropositive and make the carbonyl group less reactive for nucleophilic attack.



- b. **Steric factor:** In aldehydes carbonyl carbon is attached to one bulky alkyl groups, whereas in ketones, the carbonyl carbon is attached with two bulky alkyl groups, which cause much steric hinderance for attack by nucleophile. Thus reactivity of ketone gets decreased.



Ketone: Two bulky alkyl groups



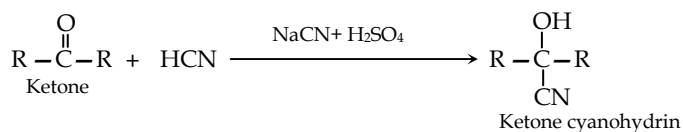
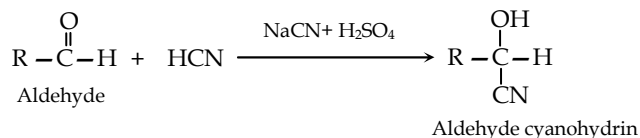
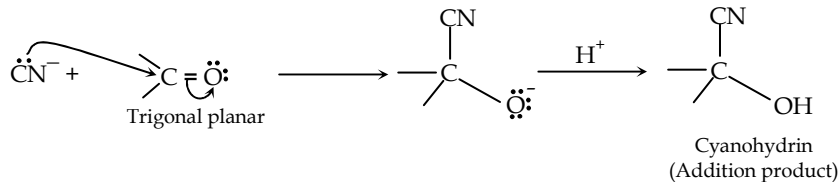
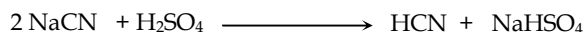
Aldehyde: One bulky alkyl group

Reactions of aliphatic aldehydes and ketones will be discussed under the following heads.

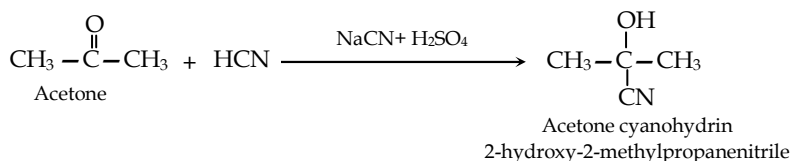
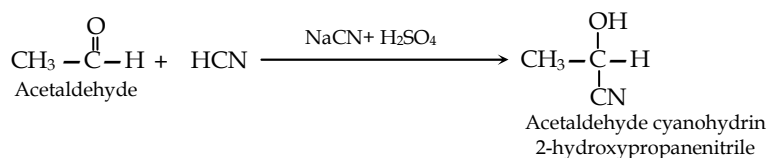
- A. Nucleophilic addition reactions
- B. Oxidation of aldehydes and ketones
- C. Reduction of aldehydes and ketones
- D. Other reactions of aldehydes and ketones.

A. Nucleophilic addition reactions of aldehydes and ketones

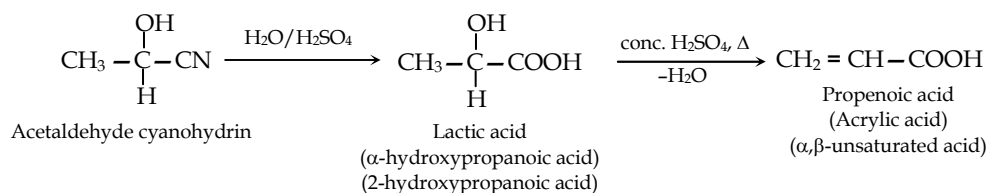
1. **Addition of HCN (hydrogen cyanide):** When HCN (mixture of NaCN + H₂SO₄) is added to aldehydes and ketones, cyanohydrins are formed. Aldehydes form aldehyde cyanohydrins and similarly ketones form ketone cyanohydrins. The reaction is often carried out by adding mineral acid (H₂SO₄) to a mixture of carbonyl compound and aqueous NaCN, when HCN is prepared *in situ* from NaCN and mineral acid (H₂SO₄).



Examples:



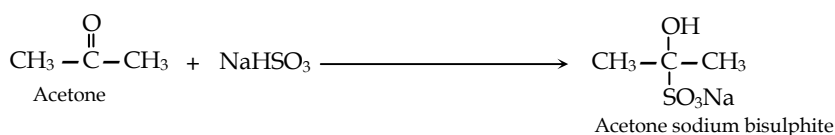
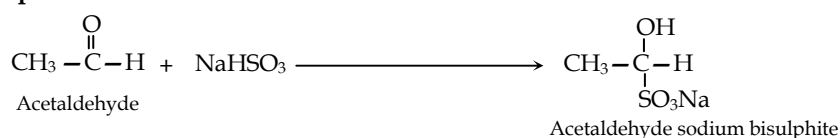
Application of cyanohydrins: Cyanohydrins can be hydrolyzed to α -hydroxy carboxylic acid. The hydroxy acid upon strong heating, is dehydrated to α,β -unsaturated carboxylic acid.



2. **Addition of NaHSO_3 (sodium bisulphite):** When aliphatic aldehydes or ketones are treated with saturated solution of NaHSO_3 in presence of ethanol, the crystalline precipitate of aldehyde or ketone sodium bisulphite is obtained (*Test of aldehydes and ketones*)

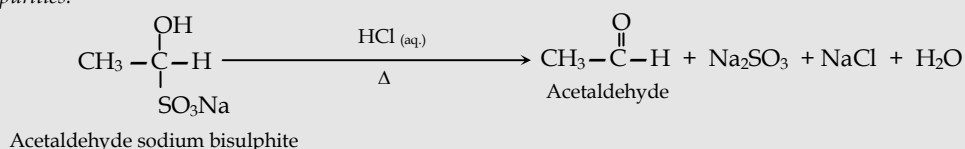


Examples:



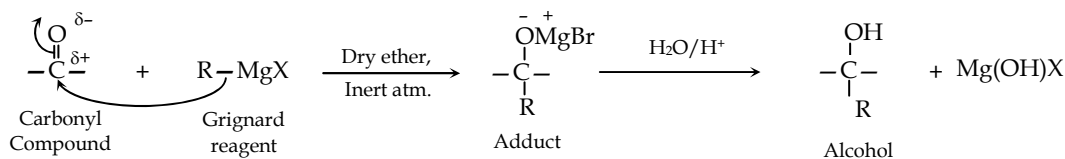
Note:

This crystalline precipitate formed is filtered from ethanol solution and can be decomposed by the action of acid or base back to its parent aldehyde or ketone. Therefore, this reaction is used for separation of carbonyl compounds from a mixture and other impurities.



3. **Addition of Grignard reagents: Grignard synthesis of alcohols: Nucleophilic addition reaction of Grignard reagent on aldehydes and ketones.**

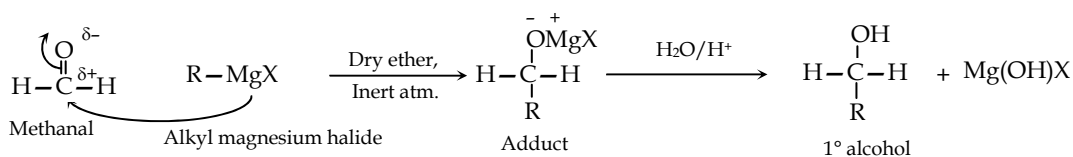
Grignard reagent (RMgX) undergoes nucleophilic addition reaction to the carbonyl ($>\text{C}=\text{O}$) functional group of aldehydes or ketones to form an addition product called **adduct**, which on acidic hydrolysis with dilute HCl yields different types of alcohols.



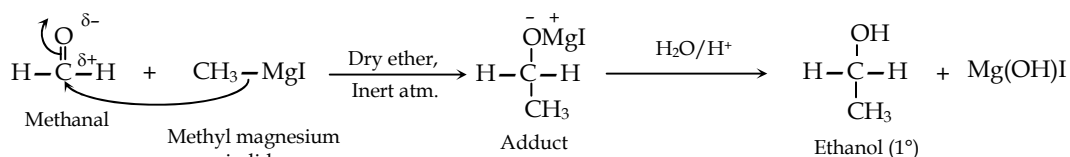
According to the structure of aldehydes and ketones used in the Grignard synthesis, different classes of alcohols (1° , 2° and 3°) are obtained.

Reactions

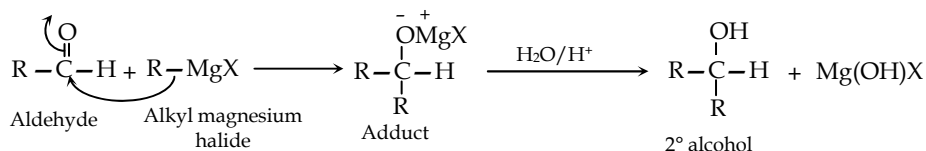
- i. Formaldehyde reacts with Grignard reagent followed by acid hydrolysis always gives primary (1°) alcohol.



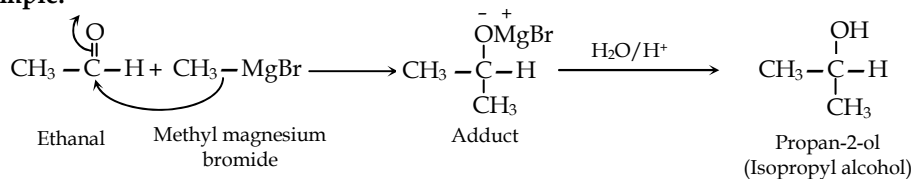
Example:



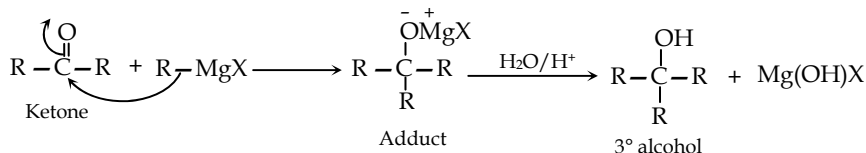
- ii. Higher aldehydes (other than formaldehyde) react with Grignard reagent followed by acid hydrolysis give secondary (2°) alcohols.



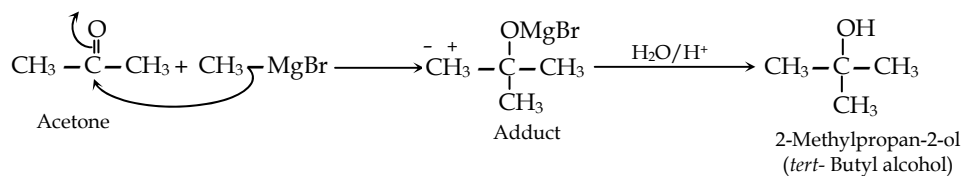
Example:



- iii. Ketones react with Grignard reagent followed by acid hydrolysis gives tertiary (3°) alcohols.



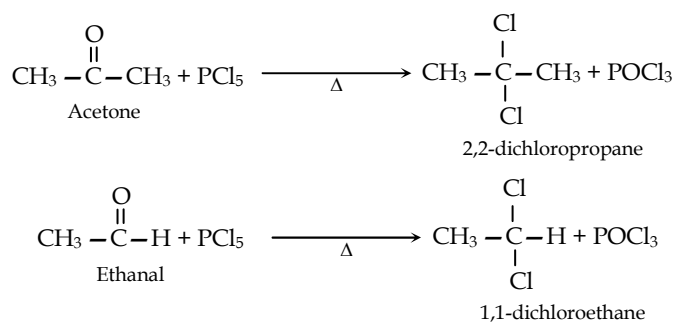
Example:



4. Reaction of phosphorus pentachloride (PCl₅): Formation of geminal dehalides:

Aldehydes and ketones react with PCl₅ to give gem-dihalides.

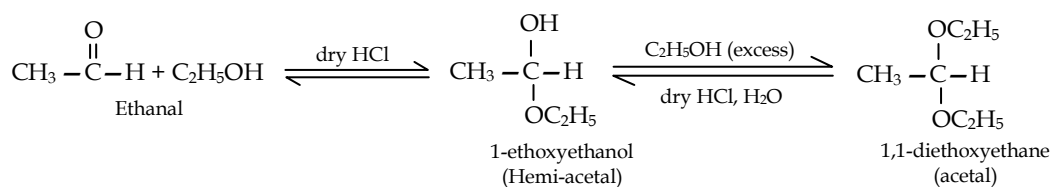
Examples:



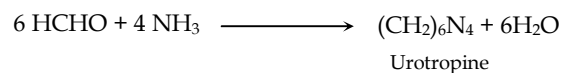
5. Addition of alcohols: Formation of acetals:

Aldehyde reacts with alcohols in the presence of dry HCl giving hemiacetal. If alcohol is in excess, acetal is obtained. Aliphatic ketones do not form acetals. Dry HCl shifts the equilibrium towards the right-hand side.

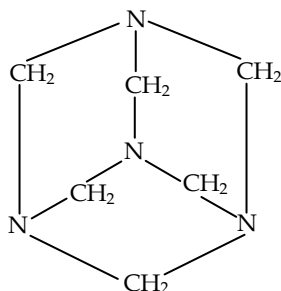
Example:



6. Addition of ammonia: Formaldehyde reacts with ammonia giving hexamethylene tetraamine (urotropine).



Structure of Urotropine:

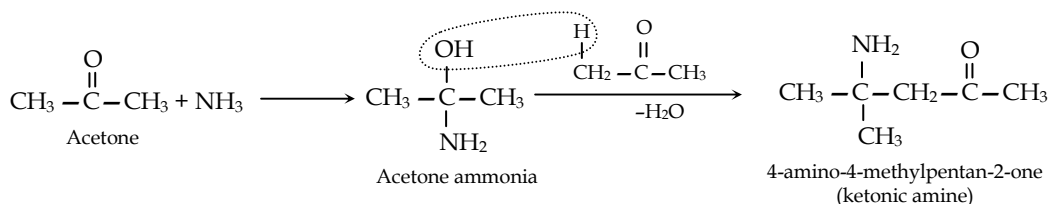


Urotropine is used as urinary antiseptic drug.

Acetaldehyde reacts with ammonia giving acetaldehyde ammonias.



Acetone reacts with ammonia giving acetone ammonias, which further reacts with excess of acetone to form ketonic amine.



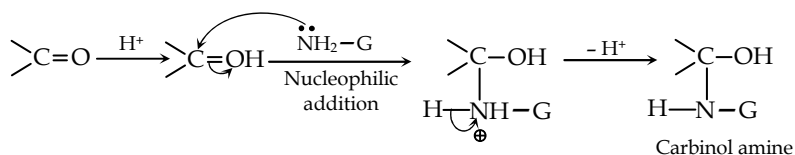
7. Addition of ammonia derivatives followed by loss of water: Nucleophilic addition followed by elimination of water

Ammonia derivatives are compounds that can be made by replacing one H- atom from NH_3 by a group G. It can be generally represented as $\text{NH}_2\text{-G}$.

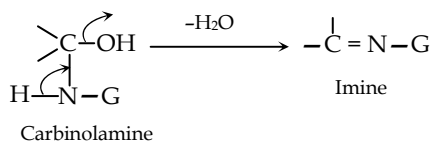
When aldehyde or ketone is treated with ammonia derivative, an organic compound containing carbon nitrogen ($\text{C}=\text{N}$) double bond known as *imine* is formed.

This reaction is completed in two steps and is represented as:

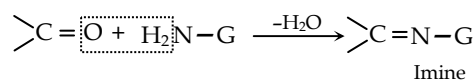
Step (i) Acid catalyzed nucleophilic addition of ammonia derivative ($\text{NH}_2\text{-G}$) onto carbonyl carbon of aldehyde or ketone to give carbinol amine.

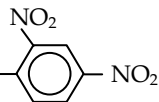
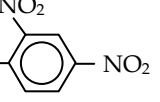
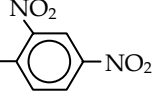


Step (ii) Elimination of water molecule gives *imine*.



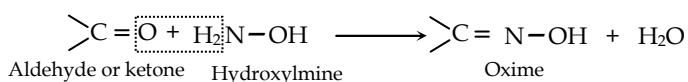
Both steps are summarized as:



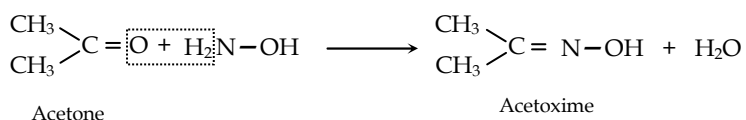
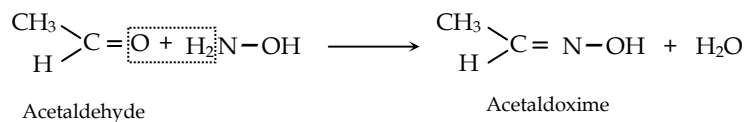
G	Ammonia derivatives (NH ₂ -G)	The product
—OH	NH ₂ —OH (Hydroxylamine)	>C=N—OH (Oxime)
—NH ₂	NH ₂ —NH ₂ (Hydrazine)	>C=N—NH ₂ (Hydrazone)
—NH— 	NH ₂ —NH—  (Phenyl hydrazine)	>C=N—NH—  (Phenylhydrazone)
—NH— 	NH ₂ —NH—  2,4-Dinitrophenyl hydrazine (2,4-DNP)	>C=N—NH—  (2,4-Dinitrophenyl hydrazone) yellow ppt
—NHCONH ₂	NH ₂ —NHCONH ₂ (Semicarbazide)	>C=N—NHCONH ₂ (Semicarbazone) crystalline ppt

i. Reaction with Hydroxylamine: Formation of Oximes

Aldehydes and ketones react with hydroxylamine giving oximes. Aldehydes form aldoximes and ketones form ketoximes.

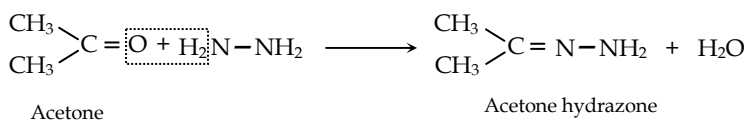
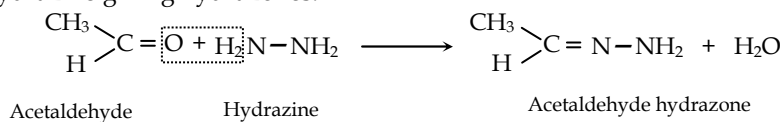


Examples:

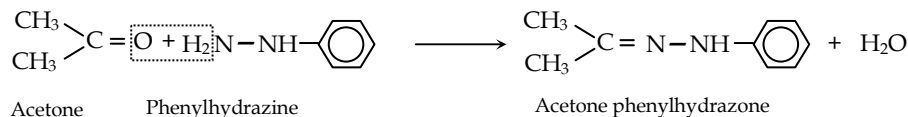
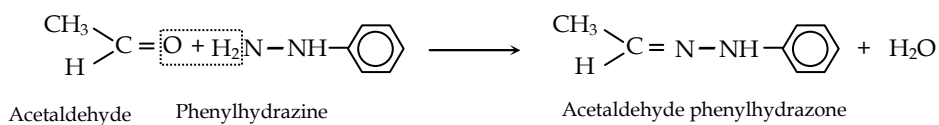


The Beckmann rearrangement: The oximes obtained above can be transformed to amides on reaction with strong acids like H₂SO₄, SOCl₂, etc.

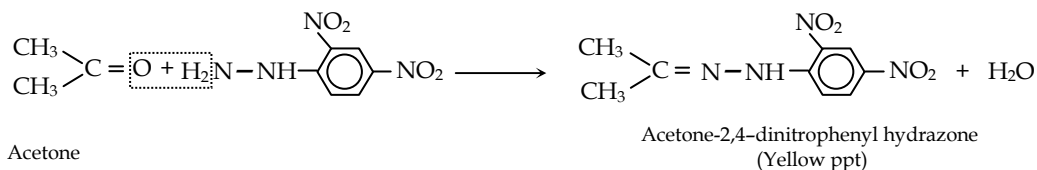
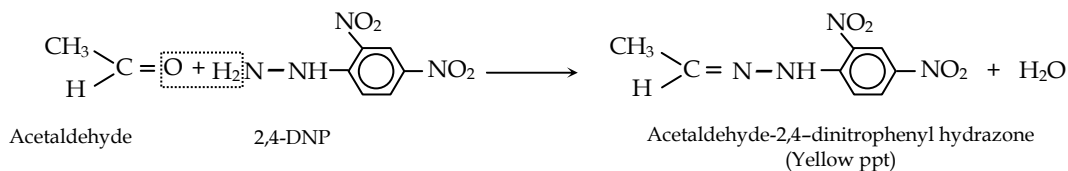
ii. Reaction with Hydrazine: Formation of Hydrazone: Aldehydes or ketones react with hydrazine giving hydrazones.



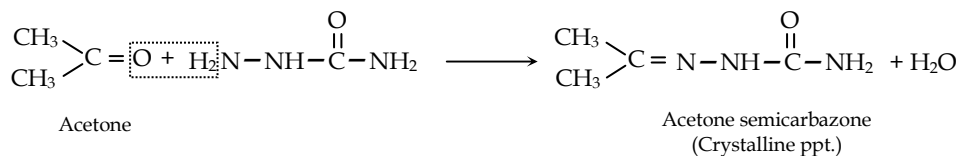
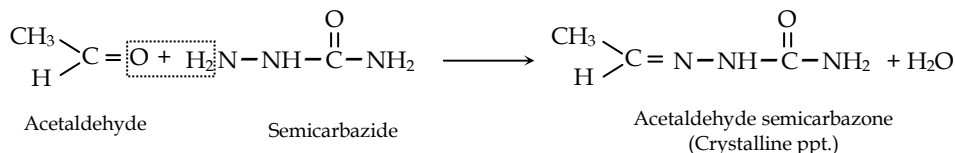
iii. Reaction with phenylhydrazine: Formation of phenylhydrazone: Aldehydes and ketones react with phenylhydrazine to give phenylhydrazone.



- iv. **Reaction with 2,4-dinitrophenyl hydrazine (2,4-DNP): Formation of 2,4-dinitrophenyl hydrazone:** *2,4-DNP test:* When aldehydes and ketones are treated with 2,4-dinitrophenyl hydrazine (2,4-DNP), yellow or red precipitate of aldehyde/ketone 2,4-dinitrophenyl hydrazone obtained. It is characteristic reaction of carbonyl compounds and is used to distinguish carbonyl compounds from other classes of organic compounds. Thus reaction serves as a test for carbonyl compounds.



- v. **Reaction with semicarbazide: Formation of semicarbazone:** Aldehydes and ketones react with semicarbazide to give crystalline precipitate of semicarbazone.

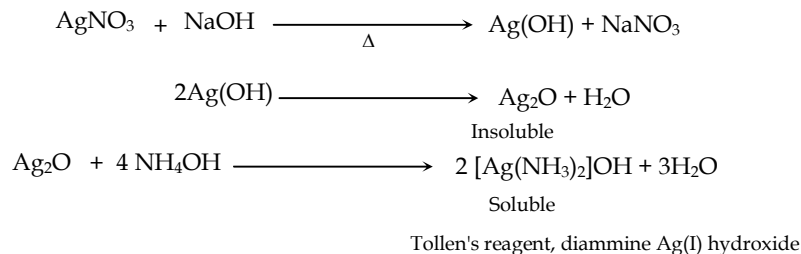


B. Oxidation

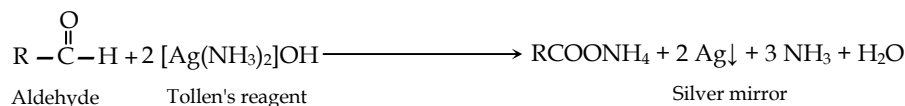
Aldehydes are not only oxidized by strong oxidizing agents like KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, etc. but they are also oxidized by mild oxidizing agents like Tollen's reagent, Fehling's solution, etc. and they show reducing properties. *Ketones are not oxidized under ordinary conditions.*

1. **Oxidation with Tollen's reagent (Silver mirror test):** The ammoniacal silver nitrate solution, $[\text{Ag}(\text{NH}_3)_2]\text{OH}$, is called Tollen's reagent. Tollen's reagent contains $\text{Ag}(\text{NH}_3)_2^+$ ion in water soluble form.

Reaction involved in the preparation of Tollen's reagent:

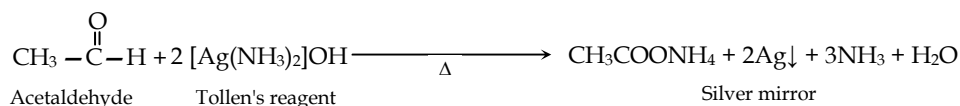
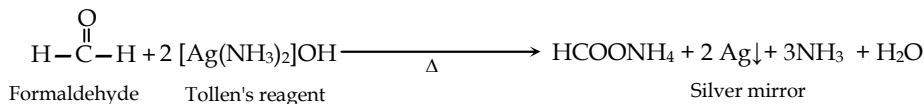


When an aldehyde is heated with Tollen's reagent, aldehyde reduces the Tollen's reagent into free metallic silver and aldehyde itself gets oxidized to carboxylic acid. The metallic silver appears in the form of mirror deposited around the inner wall of the reaction test tube. Therefore, this reaction is widely known as **silver mirror test of aldehyde**. Tollen's reagent oxidizes both aliphatic and aromatic aldehydes.



Examples:

Oxidation of formaldehyde, acetaldehyde, etc. with Tollen's reagent:

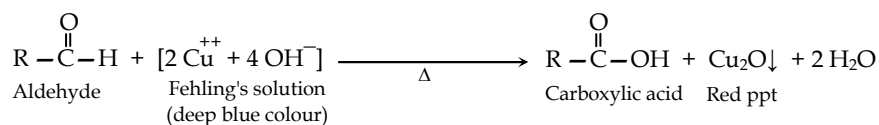


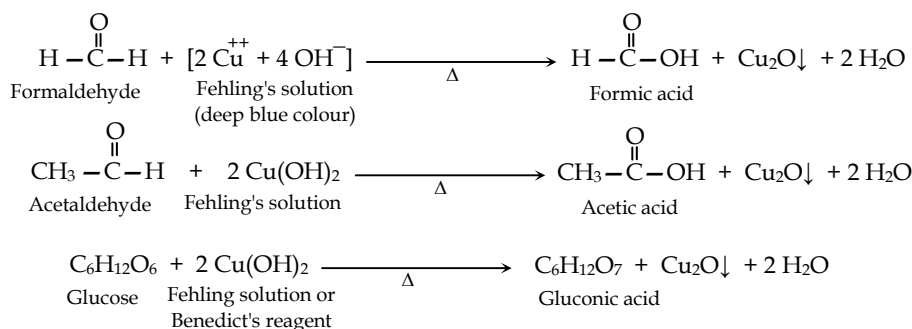
Ketones do not respond this reaction; therefore, aldehydes can be distinguished from ketones by silver mirror test in laboratory.

2. **Oxidation with Fehling's solution and Benedict's solution:** *Fehling's and Benedict's tests*

The alkaline solution of copper sulphate complexed by sodium potassium tartarate (*Rochelle salt*) is called Fehling's solution, whereas alkaline solution of copper sulphate complexed by sodium citrate is called Benedict's reagent. In Fehling's solution, the complexing agent is tartarate ion but in Benedict's reagent, the complexing agent is citrate ion.

When an aldehyde is heated with Fehling's solution, red precipitate of cuprous oxide (Cu_2O) is obtained.



Examples:**Note:**

Aliphatic aldehydes give Fehling's solution and Benedict's tests, whereas aromatic aldehydes like benzaldehyde does not response to this reaction; therefore, aliphatic aldehydes can be distinguished from aromatic aldehydes by performing the Fehling's solution and Benedict's solution tests in the laboratory. Similarly the Fehling's solution or Benedict's test are also not given by ketones.

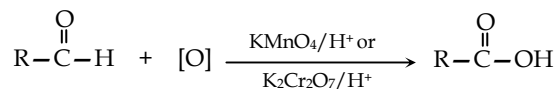
Self Test

12.8 How can you distinguished between:

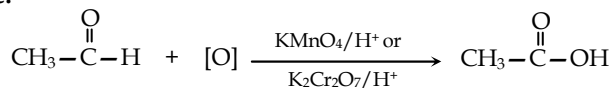
- Acetaldehyde and benzaldehyde
- Acetaldehyde and acetone

3. **Oxidation using acidified KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$:** These oxidizing agents can oxidize aldehydes into carboxylic acids. In these reactions, pink colour of KMnO_4 discharged or orange colour $\text{K}_2\text{Cr}_2\text{O}_7$ is changed to green. These tests are called **permanganate test** and **dichromate test**, respectively.

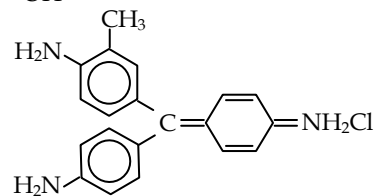
i.e



Example:



4. **Action of Schiff's Reagent:** Aldehydes give pink colour with Schiff's reagent (Rosaniline hydrochloride, pink colour of which is discharged by SO_2)

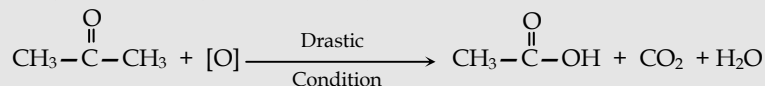


Rosaniline hydrochloride

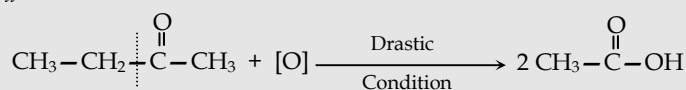
Note

Oxidation of ketones under drastic conditions: Ketones are not oxidized under ordinary conditions. However in drastic conditions, they are oxidized to give smaller carboxylic acids.

eg.



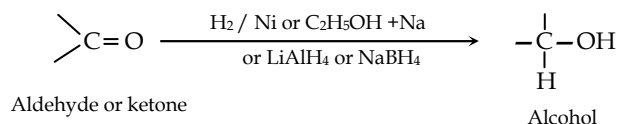
In case of unsymmetrical ketones cleavage takes place in such a way that smaller alkyl group remains attached to the carbonyl group (Popoff's rule").



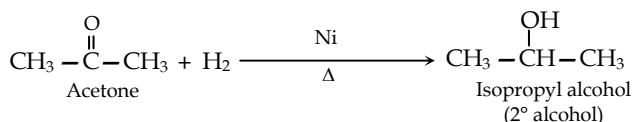
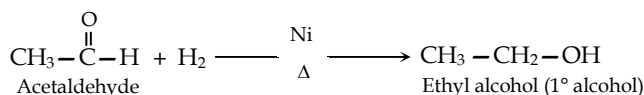
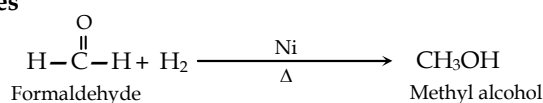
C. Reduction of aldehydes and ketones

1. Reduction to alcohols

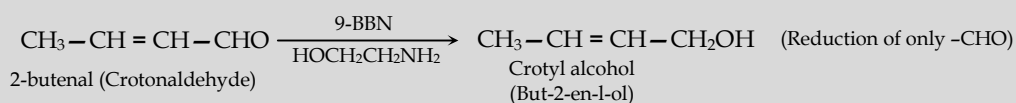
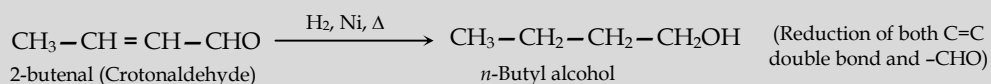
Aldehydes are reduced to primary (1°) alcohols, whereas ketones are reduced to secondary (2°) alcohols either by catalytic hydrogenation (H_2/Ni , Δ) or by use of chemical reducing agents like $\text{Na}/\text{C}_2\text{H}_5\text{OH}$, lithium aluminum hydride (LiAlH_4), sodium borohydride (NaBH_4) etc.



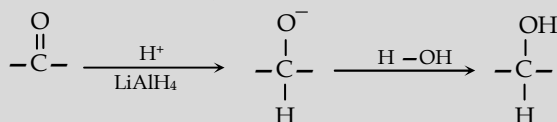
Examples



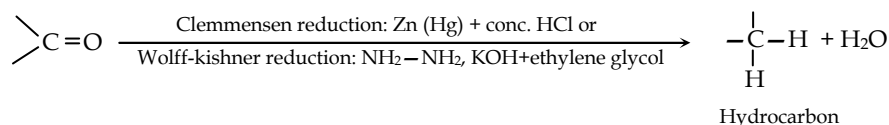
To reduce a carbonyl group that is conjugated with carbon-carbon double ($\text{C}=\text{C}$) bond without reducing the carbon and carbon double bond requires a chemoselective reducing agent like 9-BBN, (9-borabicyclo[3,3,1]nonane).



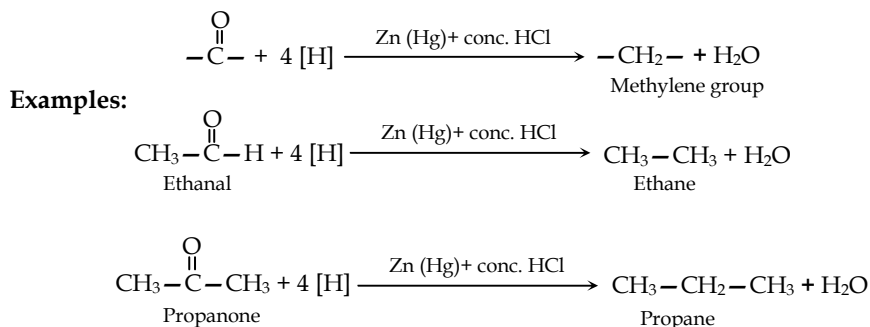
In fact, the reduction by LiAlH_4 takes place in two steps (a) Addition by LiAlH_4 to get alkoxide (b) Hydrolysis of the alkoxide. This concept is beyond the scope of this text.



2. **Reduction to alkanes (hydrocarbons): Wolff-Kishner reduction and Clemmensen reduction:** The conversion of the acyl (R—CO—) group into an alkyl ($\text{R—CH}_2\text{—}$) group can be accomplished by the Clemmensen reduction or the Wolff-kishner reduction. In both of these reduction reactions, the carbonyl ($>\text{C}=\text{O}$) group of aldehydes and ketones are reduced to the methylene ($-\text{CH}_2-$) group of hydrocarbons (alkanes).

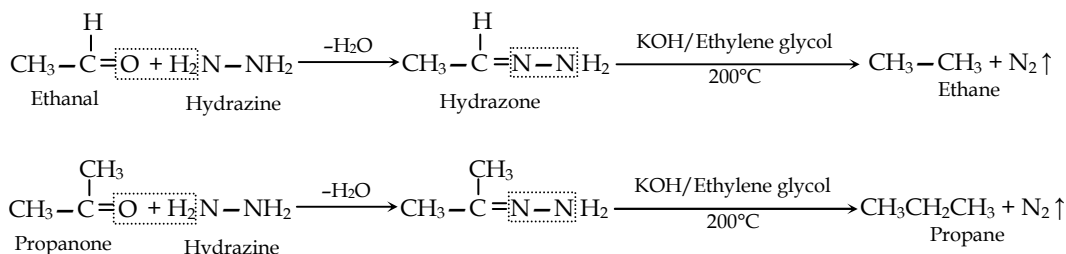


- i. **Clemmensen reduction:** Aldehydes or ketones are reduced to alkanes by amalgamated zinc in presence of conc. HCl. The reaction is known as **Clemmensen reduction**.



Clemmensen reduction is useful method for the reduction of the organic compounds which are base sensitive.

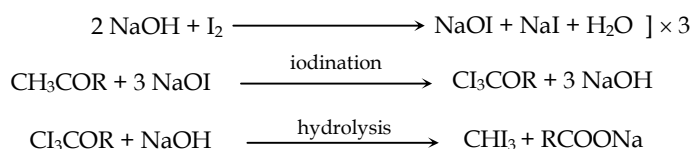
- ii. **Wolff-Kishner reduction:** Aldehydes and ketones are reduced into respective alkanes with a mixture of hydrazine (NH_2-NH_2) and alkali (KOH) in ethylene glycol is known as **Wolff-Kishner reduction**.



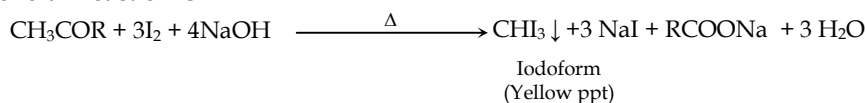
Wolff-Kishner reduction is useful method for the reduction of the organic compound which are acid sensitive.

D. Other reactions

1. **Iodoform test:** Aldehydes or ketones having $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}- \end{array}$ structure on heating with excess I_2 and aq. alkali (NaOH) give yellow crystalline precipitate of iodoform. This test is called iodoform test.



The overall reaction is



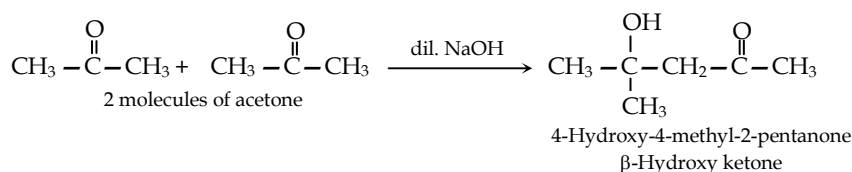
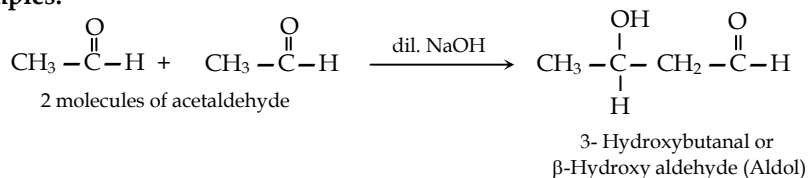
Carbonyl compounds like HCHO , $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$, $\text{C}_6\text{H}_5\text{CHO}$ (benzaldehyde) etc. do not contain CH_3- groups in α position do not give iodoform test. Whereas compounds like CH_3CHO , CH_3COCH_3 , $\text{CH}_3\text{COCH}_2\text{CH}_3$ etc. contain $\text{CH}_3\text{CO}-$ group, give positive iodoform test.

Self Test

12.9 How can you distinguish formaldehyde and acetaldehyde?

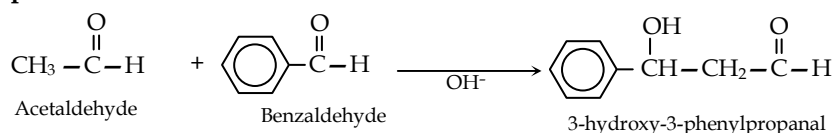
2. **Aldol condensation:** Aldehydes or ketones having α -hydrogen on treatment with dilute bases like NaOH or K_2CO_3 , undergo condensation reaction to give β -hydroxy aldehyde or β -hydroxy ketone. This reaction is called **aldol condensation**. β -hydroxy aldehyde or β -hydroxy ketone is also known as aldol since it is a compound containing both aldehyde and hydroxy functional groups.

Examples:



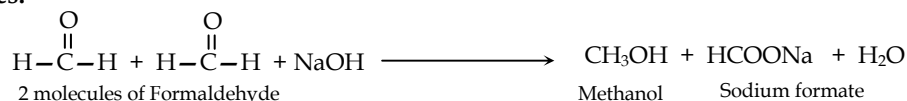
Aldol condensation between the molecules of different compounds is called ***crossed-aldol condensation***.

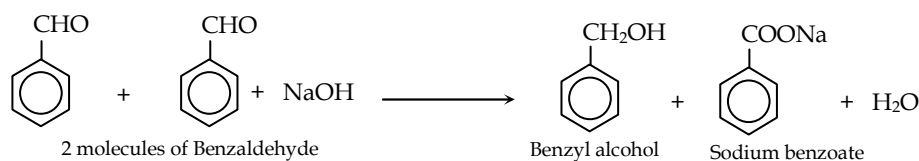
Example:



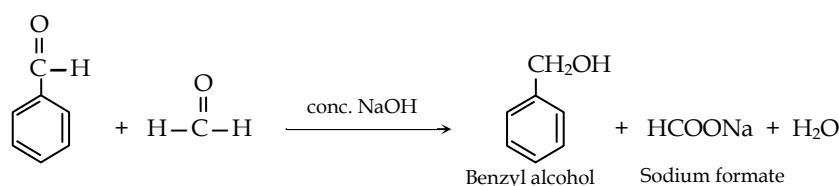
3. **Cannizzaro reaction:** Aldehydes that do not have α -hydrogen atoms (like formaldehyde, benzaldehyde etc.), when treated with concentrated aqueous solution of alkali (for example 50% aqueous NaOH) at room temperature and left over the reaction over night, self-oxidation-and-reduction reaction (also known as *disproportionation reaction*), takes place giving a mixture of alcohol and a salt of carboxylic acid. This reaction is known as ***cannizzaro reaction***. In the reaction, one molecule of aldehyde is reduced to corresponding alcohol and another molecule is oxidized to corresponding carboxylic acid, which gives its sodium carboxylate salt.

Examples:



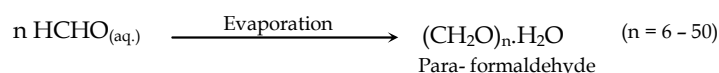


Cannizzaro reaction between molecules of different aldehydes is termed as ***crossed Cannizzaro reaction***.

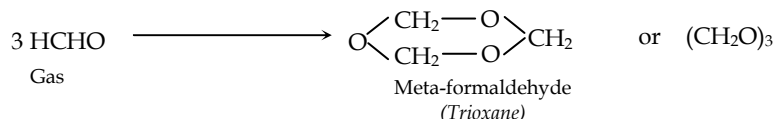


4. Polymerization of formaldehyde: Formaldehyde undergoes polymerization in various conditions giving various products.

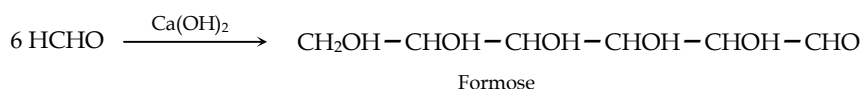
- i. Aqueous formaldehyde when allowed to evaporate upto dryness, white crystalline solid of paraformaldehyde is obtained.



- ii. Gaseous formaldehyde when allowed to stand for long time, metaformaldehyde (trioxane) is formed.



- iii. Formaldehyde upon treatment with weak base like $\text{Ca}(\text{OH})_2$ gives formose (hexose sugar).



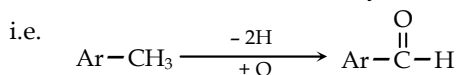
12.1.8 Uses of Aldehydes and Ketones

- Formaldehyde is used in preparation of urotropine and polymers (eg. bakelite).
- Formaldehyde is used as biological preservative (40% aqueous solution of formaldehyde in the form of **formalin**).
- Aldehydes are used as reducing agent in laboratory.
- Many ketones are used as solvents. For example: acetone is used as nail polish remover and for removing stains.

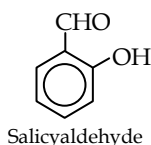
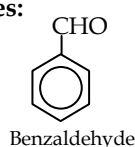
12.2 Aromatic Aldehydes and Ketones

12.2.1 Introduction

Derivatives of arenes, where two hydrogen atoms from side chain $-\text{CH}_3$ are replaced by divalent oxygen atom are called aromatic aldehydes.



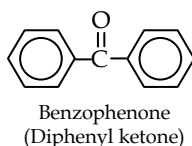
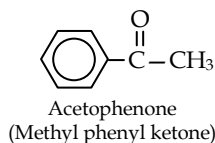
Examples:



Derivatives of arenes, where two hydrogen atoms from side chain $-\text{CH}_2-\text{R}$ or $-\text{CH}_2-\text{Ar}$ are replaced by divalent oxygen atom are called aromatic ketones.



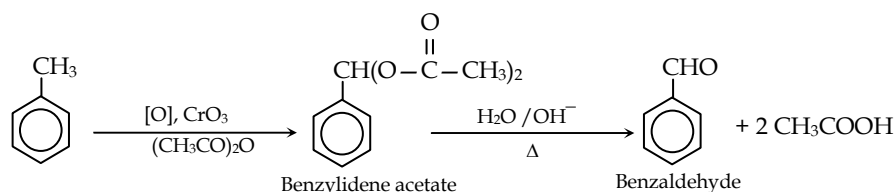
Examples:



12.2.2 Preparation of Aromatic Aldehyde (Benzaldehyde)

1. **By the oxidation of toluene:** Toluene can be oxidized to benzaldehyde using the following methods:

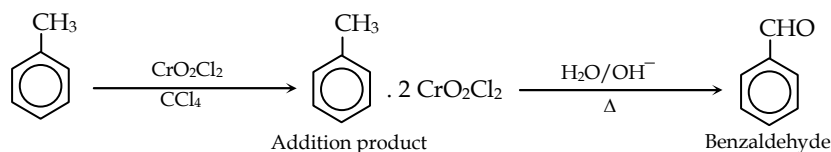
- a. When toluene is treated with chromic anhydride and acetic anhydride, benzylidene acetate is obtained, which on hydrolysis gives benzaldehyde.



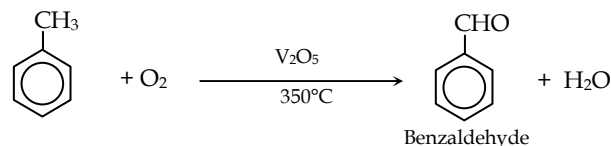
Note:

Formation of benzylidene acetate prevents further oxidation of benzaldehyde into benzoic acid.

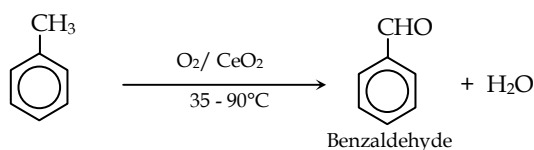
- b. Benzaldehyde on treatment with chromyl chloride in presence of CCl_4 gives brown coloured complex, which on decomposition by water gives benzaldehyde.



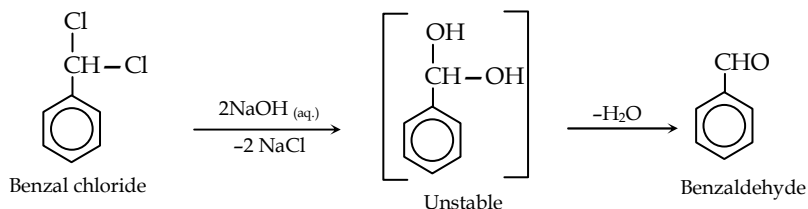
- c. Oxidation of toluene in air with vanadium pentoxide at high temperature gives benzaldehyde.



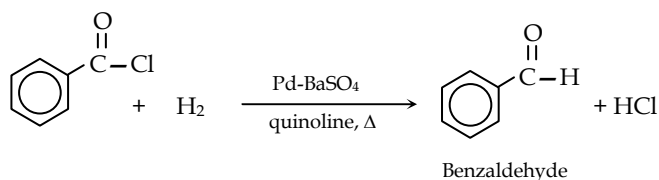
- d. Selective oxidation of toluene into benzaldehyde can be achieved by using active surface exposed cerium oxide (CeO_2) heterogeneous catalyst.



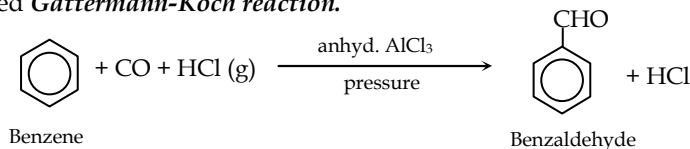
2. **By the hydrolysis of benzal chloride:** Benzal chloride on heating with aqueous alkali gives benzaldehyde.



3. **From acid chloride: By Rosenmund's reduction:** When benzoyl chloride is heated with H_2 in presence of palladium catalyst deposited in BaSO_4 and poisoned by quinoline, benzaldehyde is obtained. This reaction is called as **Rosenmund's reduction**.

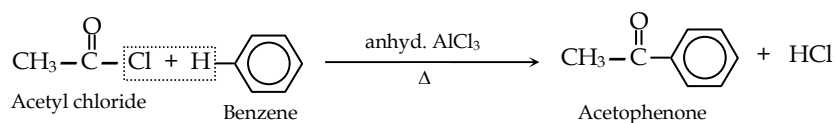


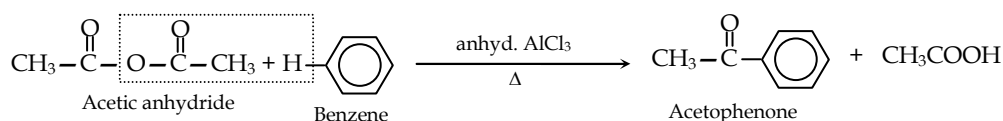
4. **By Gattermann-Koch synthesis: Formylation:** When benzene is treated with a mixture of CO and dry HCl gas in presence of anhydrous AlCl_3 under pressure, benzaldehyde is formed. The reaction is called **Gattermann-Koch reaction**.



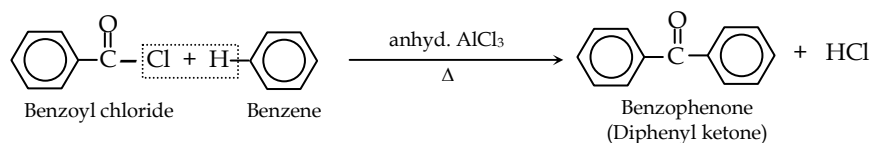
12.2.3 Preparation of Aromatic Ketones (Acetophenone)

When benzene is treated with acetyl chloride (CH_3COCl) or acetic anhydride ($\text{CH}_3\text{COOCOCH}_3$) in presence of anhydrous AlCl_3 as a lewis acid, acetophenone is formed. This reaction is **Friedel-Craft acylation**.





Similarly, benzophenone is prepared by heating benzene with benzoyl chloride in presence of anhydrous AlCl_3 .



12.2.4 Physical Properties of Benzaldehyde and Acetophenone

Benzaldehyde is a colourless liquid. Its boiling point is 179°C . It has an odour of bitter almonds. It is sparingly soluble in water.

Acetophenone is colourless liquid. Its boiling point is 202°C . It is sparingly soluble in water.

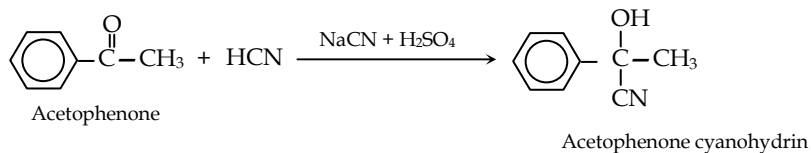
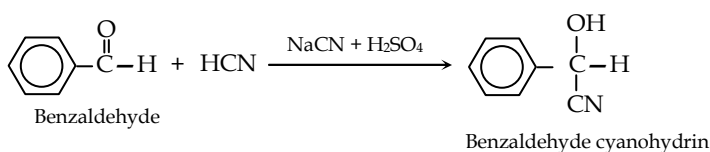
12.2.5 Chemical Properties of Benzaldehyde and Acetophenone

Most of the properties of aromatic aldehydes and ketones are similar to aliphatic aldehydes and ketones. The chemical reactions in benzaldehyde and acetophenone will be discussed under the following heads.

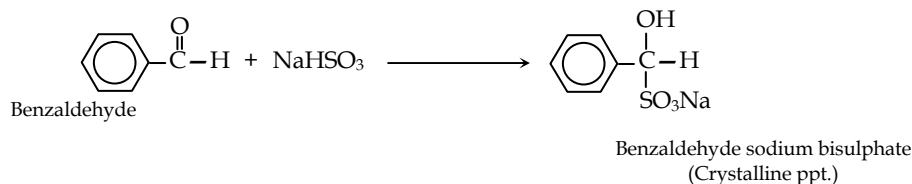
- A. Nucleophilic addition reactions
- B. Oxidation
- C. Reduction
- D. Electrophilic substitution reactions
- E. Other reactions.

A. Nucleophilic addition reaction:

1. **Reaction with HCN:** Benzaldehyde and acetophenone react with HCN (a mixture of $\text{NaCN} + \text{H}_2\text{SO}_4$) giving cyanohydrins.

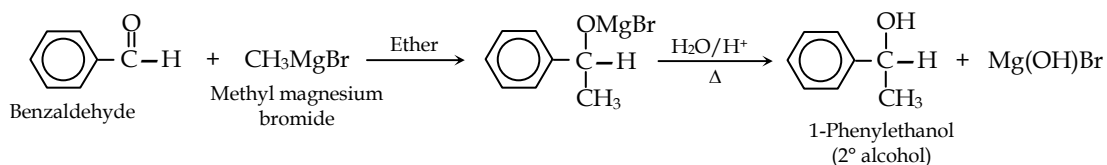


2. **Addition of NaHSO_3 :** Benzaldehyde reacts with NaHSO_3 solution to give crystalline precipitate of benzaldehyde sodium bisulphite.



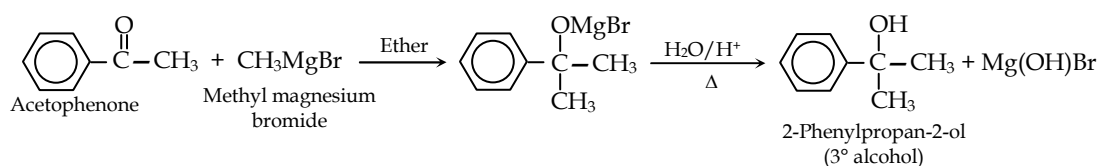
3. **Addition of Grignard reagent:** Nucleophilic addition of Grignard reagent on aromatic aldehyde followed by hydrolysis gives secondary (2°) alcohol.

Example:

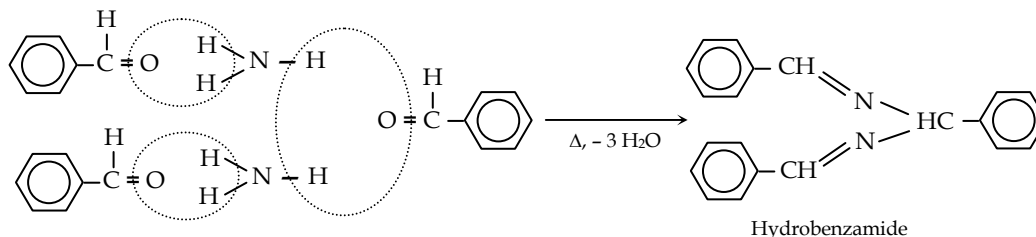


Nucleophilic addition of Grignard reagent on aromatic ketone followed by hydrolysis gives tertiary (3°) alcohol.

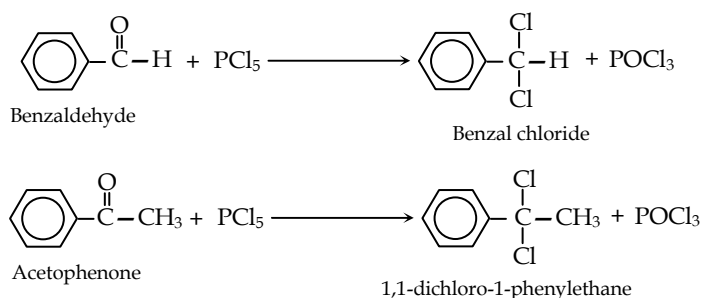
Example:



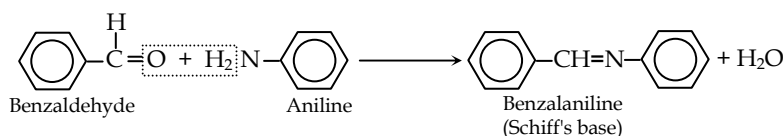
4. **Addition of ammonia:** Benzaldehyde reacts with ammonia giving hydrobenzamide.



5. **Reaction with PCl_5 :** When aldehyde or ketone is treated with PCl_5 , the carbonyl oxygen is replaced by two chlorine atoms and geminal-dichloride is formed.

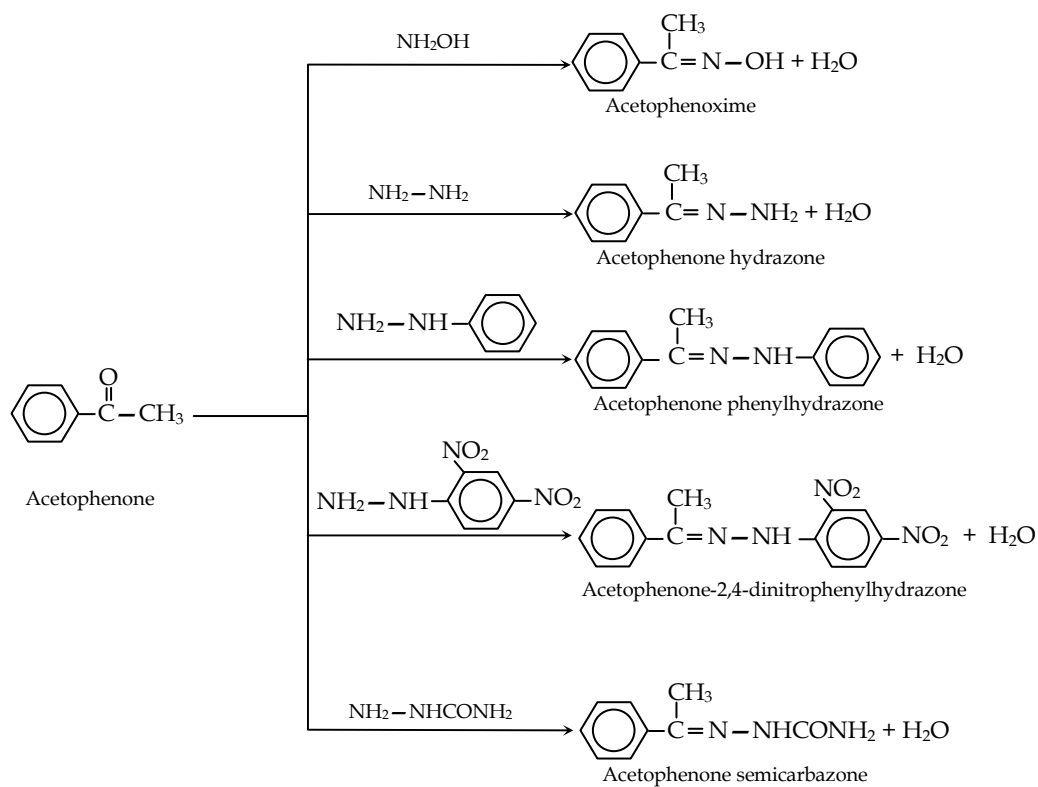
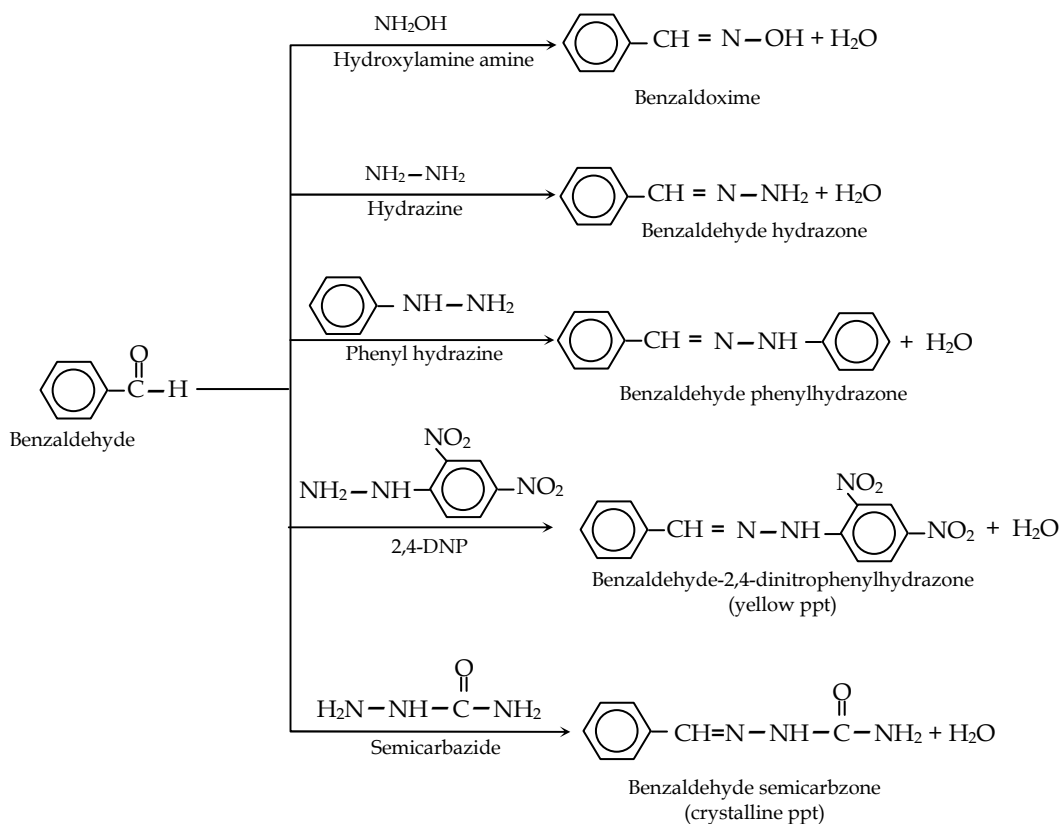


6. **Reaction of benzaldehyde with amine:** When benzaldehyde is treated with primary aliphatic or aromatic amine (aniline), Schiff's base is formed.



7. **Reaction with ammonia derivatives:** Nucleophilic addition followed by loss of water: Similar to aliphatic aldehydes and ketones, aromatic aldehydes and ketones also give

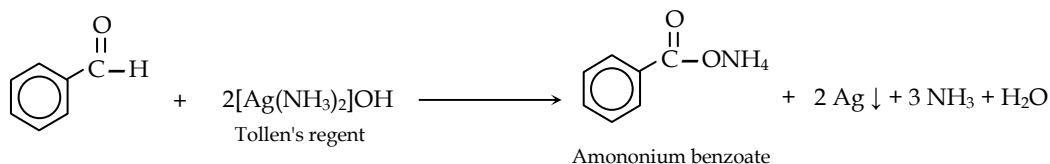
reaction with ammonia derivatives.



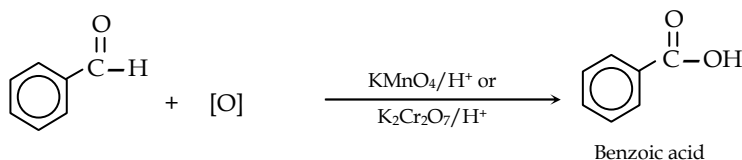
B. Oxidation

Aromatic aldehydes are also oxidized to the respective acids as in aliphatic aldehydes.

1. **Oxidation with Tollen's reagent: Silver mirror test:** Like aliphatic aldehydes, benzaldehyde also gives silver mirror test.



2. **Oxidation using KMnO_4/H^+ or $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$:** Benzaldehyde is oxidized to benzoic acid.

**Note:**

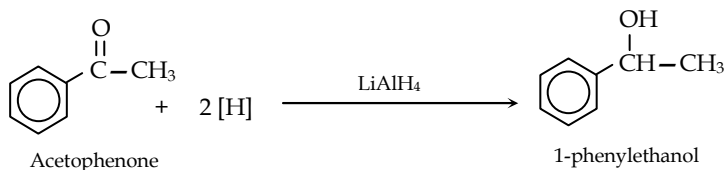
Benzaldehyde does not reduce Fehling's solution though it is an aldehyde. Aliphatic aldehydes give Fehling's solution test, whereas aromatic aldehydes like benzaldehyde do not respond this reaction. Therefore, aliphatic aldehydes can be distinguished from aromatic aldehydes by Fehling's solution test in laboratory.

C. Reduction

Aromatic aldehydes and ketones are reduced to aromatic alcohols or arenes.

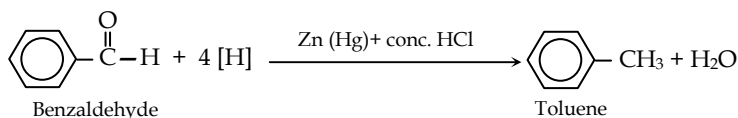
1. **Reduction to aromatic alcohols:** Aromatic aldehydes or ketones are reduced to aromatic alcohols using LiAlH_4 .

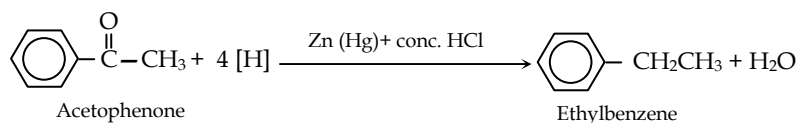
Examples:



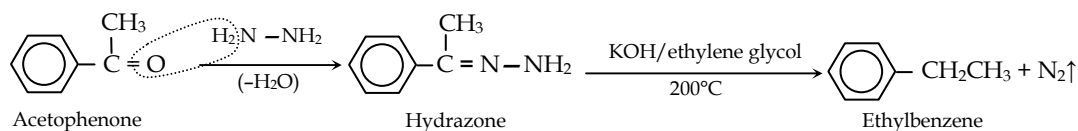
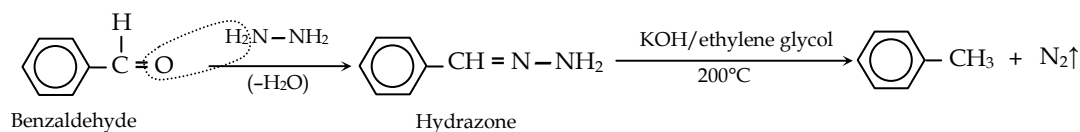
2. **Reduction to aromatic hydrocarbons (arenes):**

- i. **Clemmensen reduction:** Aromatic aldehydes or ketones are reduced by zinc-amalgam and conc. HCl, to give corresponding arenes (aromatic hydrocarbons).





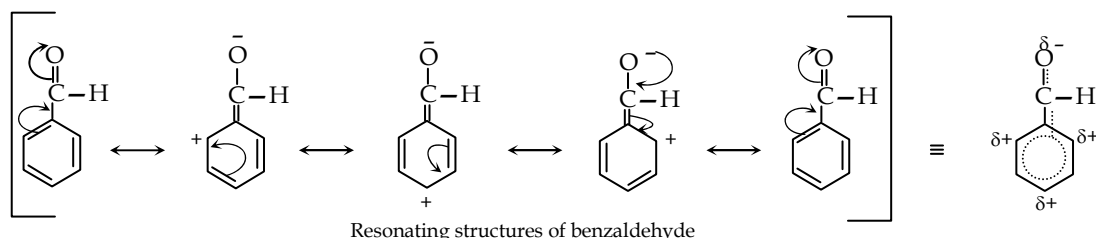
- ii. **Wolff-Kishner reduction:** Like aliphatic aldehydes and ketones, aromatic aldehydes and ketones are also reduced by alkaline solution of hydrazine to give corresponding by arenes. Reduction of aldehyde or ketone into respective hydrocarbon with basic solution of hydrazine in presence of ethylene glycol at 200°C is known as **Wolff-Kishner reduction**.

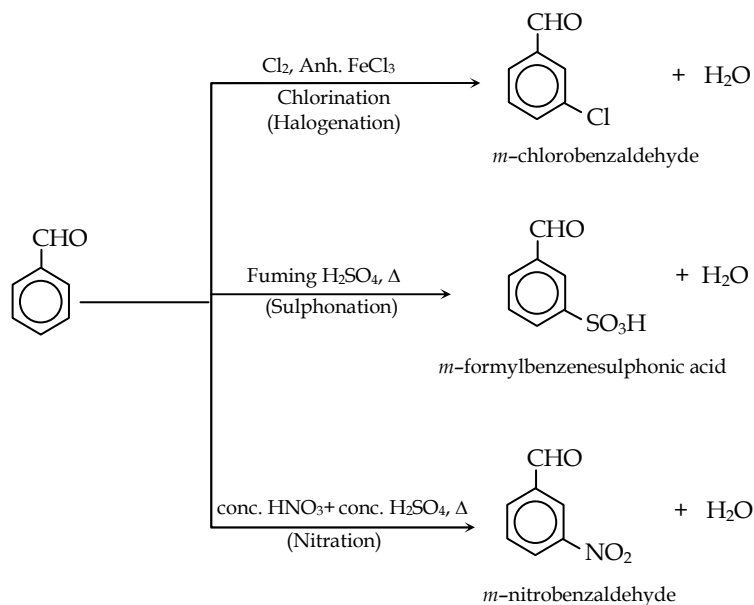


D. Reaction given by aromatic benzene ring: Electrophilic aromatic substitution reaction

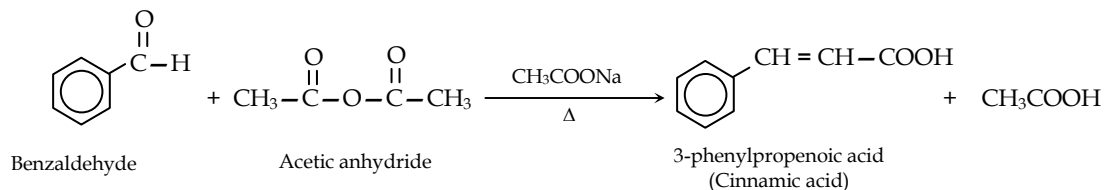
Benzaldehyde is less reactive than aliphatic aldehydes towards nucleophilic addition reaction. It can be explained on the basis of electron-releasing resonance effect of benzene ring. The carbonyl ($>\text{C}=\text{O}$) group is electron-withdrawing group, which pull the electron density from aromatic benzene ring towards the carbonyl oxygen of the $>\text{C}=\text{O}$ group. As a result of which, negative charge is developed on carbonyl oxygen and positive charge is developed in the aromatic benzene ring. Therefore, the electron-withdrawing carbonyl ($>\text{C}=\text{O}$) group always deactivates the benzene ring towards electrophilic substitution reaction.

The positive charge so developed on benzene ring is delocalized in the aromatic benzene ring to give the resonance hybrid structures. In these resonance structures, the positive charge (δ^+) is appeared in *ortho*- and *para*- positions rather than in *meta*- positions, which makes *ortho*-, *para*- positions less reactive than *meta*- positions towards electrophilic substitution reaction. Therefore, aromatic aldehydes and ketones are *meta*- directing towards electrophilic substitution reaction and hence the reaction takes place slowly at *meta*- positions to give *meta*-substituted products.

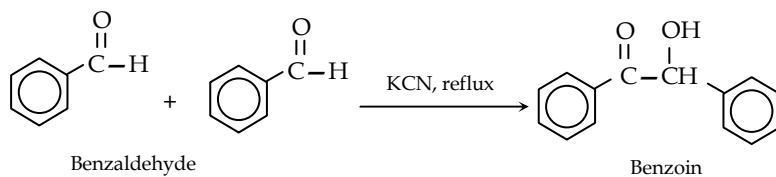


Examples:**E. Other reactions:**

- Perkin reaction:** When aromatic aldehyde (e.g. benzaldehyde) is heated with acid anhydride (e.g. acetic anhydride) in presence of salt of same carboxylic acid as a base, α, β -unsaturated carboxylic acid is obtained. This reaction is called *Perkin reaction*.



- Benzoin condensation:** When benzaldehyde is heated with alcoholic solution of KCN, condensation takes place giving **benzoin**. This reaction is called as *benzoin condensation*.

**12.2.6 Uses of Aromatic Aldehydes and Ketones****Uses of benzaldehyde**

- It is used as flavouring agent in perfumes.
- It is used to prepare various aromatic compounds like benzoin, hydrobenzamide, α, β -unsaturated acid etc.
- It is used for the preparation of dyes like malachite green.

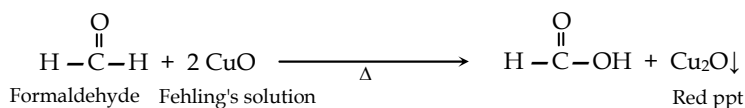
Uses of acetophenone

1. It is used as a precursor for the preparation of some resins and fragrances.
2. It is used as a raw material for the synthesis of some drugs like phenylpropanolamine.

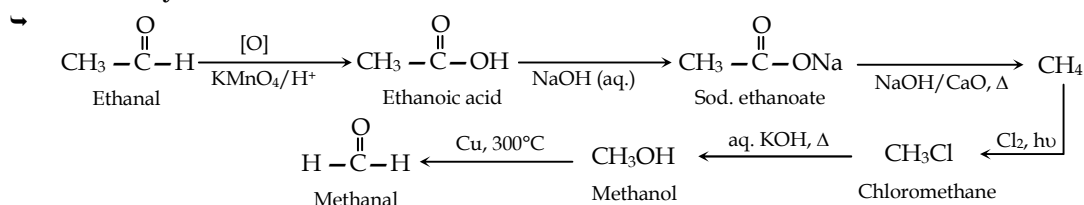
Solved Questions

1. How can you distinguish formaldehyde and benzaldehyde?

- Benzaldehyde does not reduce Fehling solution, whereas formaldehyde reduces it. So, when formaldehyde is heated with Fehling solution, red precipitate of CH_2O is obtained, whereas benzaldehyde does not



2. How can you convert ethanal into methanal?



Questions

Very Short Answer Questions:

1. How can you distinguish acetaldehyde and acetone?
2. For nucleophilic addition reactions, aldehydes are more reactive than ketones, why?
3. Write any two electrophilic addition reactions for benzaldehyde.
4. How can 2, 4-DNP test be utilized to distinguish carbonyl compounds from other compounds?
5. Give an example of:
 - (a) DNP test
 - (b) Aldol condensation
6. How can you obtain 3-hydroxybutanal from ethanal?
7. Write the structure and use of the main product obtained by the reaction of formaldehyde with ammonia.
8. Identify A, B and C in the following reaction writing related reactions.

$$\text{A} \xrightarrow[\Delta]{\text{I}_2/\text{NaOH}} \text{B} \xrightarrow[\Delta]{\text{Ag}} \text{C} \xrightarrow{\text{ozonolysis}} \text{H}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{H}$$
9. Why are aromatic carbonyl compounds less reactive than those of aliphatic ones?
10. Identify an aldehyde, which responds to iodoform test writing justification and related reactions.
11. Write isomers of $\text{C}_4\text{H}_8\text{O}$ with their IUPAC names. Which one of them gives iodoform test? Why?
12. What happens when acetophenone is:
 - (a) treated with hydroxylamine.
 - (b) reduced using $\text{Zn, Hg} + \text{HCl}$.

■ Short /Long Answer Questions:

- Write any five methods for the preparation of aldehydes.
- Write any five differences between aldehydes and ketones.
- How can you distinguish?
 - Methanal and Ethanal
 - Ethanal and Benzaldehyde
 - Pentan-2-one and Pentan-3-one
 - Benzaldehyde and Phenylethanl
 - Propanal and Propanone
- How can you convert:
 - Methanal into Ethanal
 - Ethanal into Methanal
 - Ethanal into Propanone
 - Ethanal into propanal
 - Phenol into benzaldehyde
 - Nitrobenzene into Acetophenone
 - Propane into 2-hydroxy-2-methyl propanoic acid
- What happen when:
 - Methanal reacts with ammonia?
 - Methanol reacts with Conc. NaOH?
 - Toluene is treated with chromic anhydride and acetic anhydride and the product is hydrolyzed?
 - Propanone is treated with semicarbazide?
 - Benzaldehyde is treated with alc. KCN?
- Show your acquaintance on/Write short notes on:
 - Iodoform test for carbonyl compound
 - Canizzaro Reaction
 - Oxidation of aldehydes
 - Relative reactivities of aldehydes and ketones
- A carbonyls compound, A, having molecular formula C_3H_6O is found to decolorise acidified $KMnO_4$ and responds to 2, 4-DNP test. The compound does not give iodoform test. Identify A, writing the related reactions. How can the compound A be converted into Ethane?
- An organic compound 'P' is found to reduce Fehling solutions. 'P' on oxidation using $KMnO_4/H^+$ gives compound 'Q' having same number of C as 'P'. The compound 'Q' responds to $NaHCO_3$ test and gives as ester 'R' with molecular formula $C_4H_8O_2$ when reacted with ethanol in presence of concentrated H_2SO_4 . Identify P, Q and R writing related reactions and write their IUPAC names.
- An alkene 'A' on ozonolysis gives two aldehydes 'B' and 'C'. Compound 'B' on clemensen reduction gives propane while the compound 'C' on treatment with HCN followed by acid hydrolysis gives 2-hydroxypropanoic acid. Identify A, B and C writing related reactions.

Ans: A = Pent-2-ene, B = Propanal, C = Ethanal)

- An organic compound 'A' on oxidation using $CrO_3/Pyridine$ gives compound 'B'. Both the compound 'A' and 'B' respond to iodoform test. The compound 'A' is a primary alcohol. Identify 'A' and 'B' writing related reactions. How can the compound A be converted into benzene?

(Ans: A = Ethanol, B = Ethanal)

Multiple Choice Questions

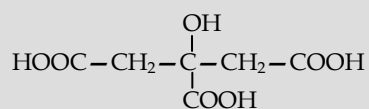
- Fehling solution test is not given by
 - Acetaldehyde
 - Benzaldehyde
 - Formaldehyde
 - Propanal
- The reagent that is used to detect sugar in urine is:
 - Bayer's reagent
 - Benedict's solution
 - AgNO_3
 - Tollen's reagent
- The compound having least solubility in water is:
 - Ethanal
 - Propanal
 - Acetophenone
 - Methanal
- Which of the following reagent reacts with both aldehyde and ketone?
 - Fehling's solution
 - Grignard's reagent
 - Schiff's reagent
 - Tollen's reagent
- Clemmenson reduction reaction is carried out by:
 - H_2/Pd
 - $\text{NH}_2\text{NH}_2 + \text{Glycol}$
 - $\text{LiAlH}_4/\text{ether}$
 - $\text{Zn-Hg}/\text{HCl}$
- Calcium acetate on distillation gives:
 - Ethane
 - Acetone
 - Acetaldehyde
 - Acetic acid
- Methanal and ethanal can be distinguished by
 - KMnO_4
 - $\text{K}_2\text{Cr}_2\text{O}_7$
 - AgNO_3
 - iodoform test
- When 2-propanol is heated with Cu at 300 C, gives:
 - Acetone
 - Acetaldehyde
 - Propene
 - Propyne
- Most reactive carbonyl compound is
 - HCHO
 - CH_3COCH_3
 - $\text{C}_6\text{H}_5\text{CHO}$
 - $\text{C}_6\text{H}_5\text{COCH}_3$
- A Carbonyl compound is treated with HCN and the product is hydrolysed. The final product is
 - Cyanohydrin
 - carboxylic acid
 - β -hydroxyaldehyde
 - α -hydroxy acid
- Active part of benedict reagent is
 - CuO
 - Cu
 - Ag_2O
 - Ag
- Methanal and ethanal can be distinguished by
 - Fehling's solution
 - Tollen's reagent
 - $\text{I}_2 + \text{NaOH}$
 - All of the above
- The alkene which gives ethanal only on ozonolysis is
 - 2-methylpropane
 - Ethene
 - But-1-ene
 - But-2-ene
- Which of the following can distinguish formaldehyde and benzaldehyde?
 - Fehling's solution
 - Tollen's reagent
 - $\text{I}_2 + \text{NaOH}$
 - NH_3

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14
b	b	c	b	d	b	d	a	a	d	a	c	d	a

Carboxylic Acids and Their Derivatives

13



Citric acid

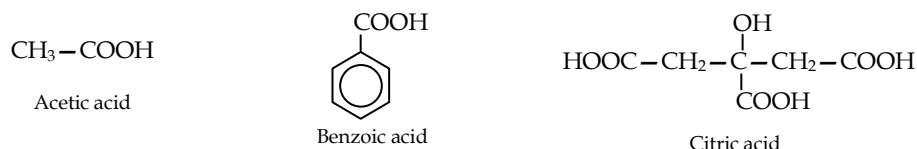
Citric acid is a carboxylic acid. Naturally, it occurs mainly in citrus fruits like lime, lemon etc. It is a commodity chemical, and produced in large scale every year by fermentation. It is used as a flavoring and preservative in food and beverages, especially soft drinks.

13.1 Aliphatic Carboxylic Acids

13.1.1 Introduction

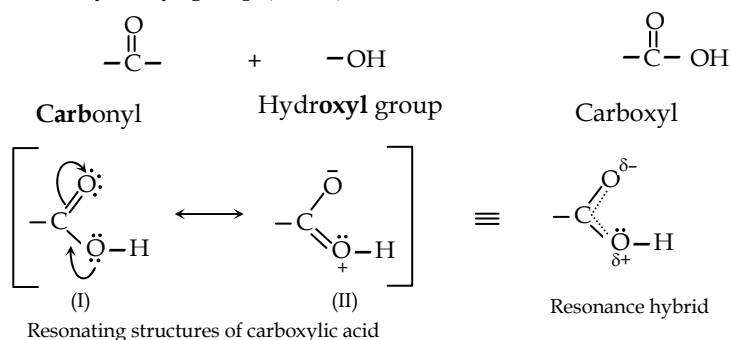
Organic compounds having **carboxyl** ($-\text{COOH}$) functional group with acidic property are carboxylic acids.

Example:



Vinegar is very dilute aqueous solution of acetic acid. Benzoic acid serves as food preservative and flavouring agent. Citric acid is responsible for sour taste of lemons.

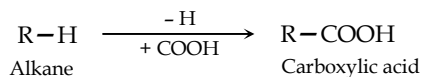
The term **carboxyl** is a composite word of the term '**carb**' derived from carbonyl group ($>\text{C}=\text{O}$) and '**oxyl**' derived from hydroxyl group ($-\text{OH}$) as:



Carboxyl group is resonance hybrid of (I) and (II). Hence, carboxylic acids have their own distinctive properties and do not show all the properties of carbonyl compounds and alcohols through it seems to have both carbonyl and hydroxyl part in them.

Carboxylic acids are broadly classified as (i) Aliphatic carboxylic acids and (ii) Aromatic carboxylic acids, depending upon whether $-\text{COOH}$ group is attached to alkyl (R-) chain and aromatic aryl (Ar-) group respectively.

The **aliphatic carboxylic acid** may be regarded as derivatives of alkanes in which one or more H-atoms have been substituted by $-\text{COOH}$ group as shown as below:



The general formula of aliphatic saturated monocarboxylic acids (homologous series) is $\text{C}_n\text{H}_{2n+1}\text{COOH}$ (*exception: Formic acid, HCOOH*).

The carboxylic acids, aliphatic as well as aromatic, are further classified as *mono-*, *di-*, *tri-* or *poly-* carboxylic acids according to the number of $-\text{COOH}$ groups present in a molecule of carboxylic acid.

1. **Aliphatic monocarboxylic acids:** The carboxylic acids containing only one —COOH group per molecule are called monocarboxylic acids.

Example:



The open chain aliphatic monocarboxylic acids are also called *fatty acids* because many higher members with even number of carbon occur in natural fats and oils as esters in substantial amount and they can be obtained by their hydrolysis.

The percentages of carboxylic acids in butter.

Saturated carboxylic acids

Myristic acid (8-13%)

Palmitic acid (25-32%)

Stearic acid (8-13%)

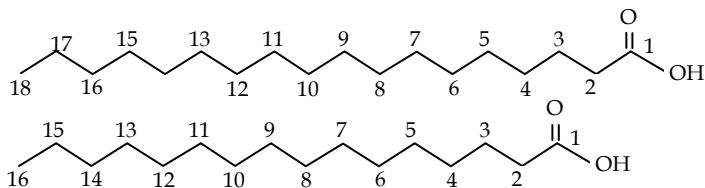
Unsaturated carboxylic acids

Oleic acid (22-29%)

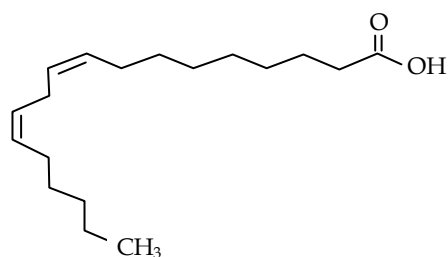
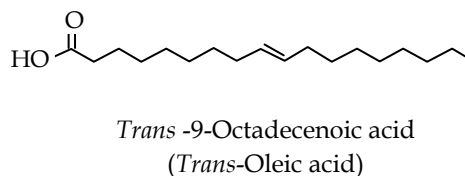
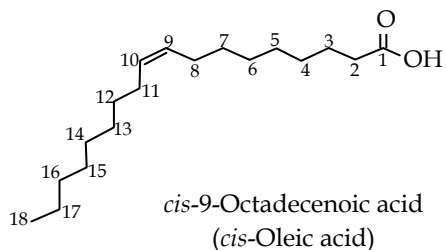
Linoleic acid (3%)

Linolenic acid (2-10%)

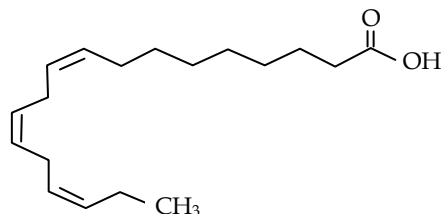
Octadecanoic acid (Stearic acid, $\text{C}_{17}\text{H}_{35}\text{COOH}$), melting point 70°C



Hexadecanoic acid (Palmitic acid, $\text{C}_{15}\text{H}_{31}\text{COOH}$), melting point 63°C



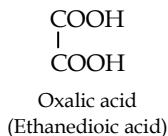
cis,cis-9,12-Octadecadienoic acid
(Linoleic acid), melting point -5°C



cis,cis,cis-9,12,15-Octadecatrienoic acid
(Linolenic acid), melting point -11°C

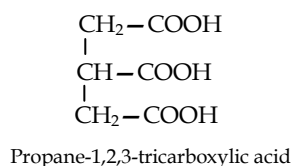
2. **Aliphatic dicarboxylic acids:** The carboxylic acids having two $-\text{COOH}$ groups per molecule are called dicarboxylic acids.

Example:



3. **Aliphatic tricarboxylic acids:** The carboxylic acids having three $-\text{COOH}$ groups per molecule are called tricarboxylic acids.

Example:



13.1.2 Nomenclature

Carboxylic acids are named according to following systems:

1. **Common System:** Common name of carboxylic acids are given on the basis of the source from which they are obtained. For example, the name *formic acid* was given since it was first obtained by the distillation of red ants (Latin: *Formica* = ants). Similarly, the name valeric acid is given since it is naturally found in valerian plant, which is a perennial flowering plant and so on.

Formula	IUPAC name	Common name	Source
HCOOH	Methanoic acid	Formic acid	Ant (Latin: <i>Formica</i> = Red ant)
CH_3COOH	Ethanoic acid	Acetic acid	Vinegar (Latin: <i>Acetum</i> = Vinegar)
$\text{CH}_3\text{CH}_2\text{COOH}$	Propanoic acid	Propionic acid	Milk (Greek: <i>Protos pion</i> = First fat)
$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	Butanoic acid	Butyric acid	Butter (Latin: <i>Butyrum</i> = Butter)
$\text{CH}_3(\text{CH}_2)_3\text{COOH}$	Pentanoic acid	Valeric acid	Valerian plant
$\text{CH}_3(\text{CH}_2)_4\text{COOH}$	Hexanoic acid	Caproic acid	Goat fat (Latin: <i>Caper</i> = Goat)

The positions of the substituents are indicated by the Greek letters- α , β , γ , δ , etc. The α -carbon is the one bearing the $-\text{COOH}$ group, the next one as β and so on as:

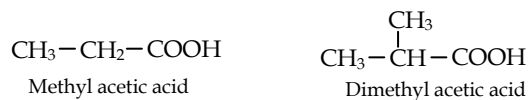


Example:



2. **Derived system:** According to this system, some monocarboxylic acids are named as alkyl derivatives of acetic acid.

Example:



3. IUPAC system

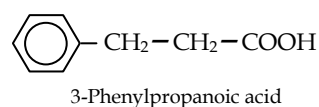
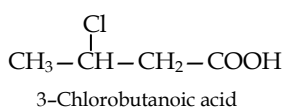
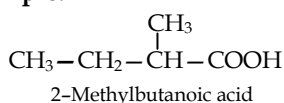
- a. **Monocarboxylic acids:** The longest chain of carbon atoms including $-\text{COOH}$ in monocarboxylic acid is considered the parent chain, and is named by replacing the $-e$ of the corresponding parent alkane with $-oic\ acid$ and hence IUPAC name of monocarboxylic acid is *Alkanoic acid*.

Carbon is always given as C-1, in numbering the carbon atom in the parent chain as:



The positions of the substituents are indicated by the number before the names of substituents, where the positions and names are separated by hyphen.

Example:

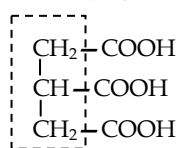


- b. **Dicarboxylic acids:** IUPAC name of dicarboxylic acid is *Alkanedioic acid*. The suffix $-dioic\ acid$ is added to the name of parent alkane containing the same number of carbon atoms.

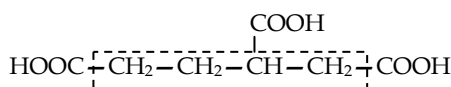
Formula	IUPAC name	Common name
$\text{HOOC}-\text{COOH}$	Ethanedioic acid	Oxalic acid
$\text{HOOC}-\text{CH}_2-\text{COOH}$	Propanedioic acid	Malonic acid
$\text{HOOC}-(\text{CH}_2)_2-\text{COOH}$	Butanedioic acid	Succinic acid
$\text{HOOC}-(\text{CH}_2)_3-\text{COOH}$	Pentanedioic acid	Glutaric acid
$\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$	Hexanedioic acid	Adipic acid
$\text{HOOC}-(\text{CH}_2)_5-\text{COOH}$	Heptanedioic acid	Pimelic acid
$\text{HOOC}-(\text{CH}_2)_6-\text{COOH}$	Octanedioic acid	Suberic acid
$\text{HOOC}-(\text{CH}_2)_7-\text{COOH}$	Nonanedioic acid	Azelaic acid

c. Tricarboxylic acids:

- i. If more than two carboxyl groups are bonded to unbranched longest carbon chain, carbon atoms of the carboxyl groups are not included in the parent chain.

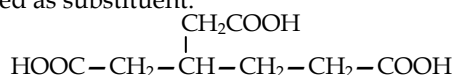


Propane-1,2,3-tricarboxylic acid



Butane-1,2,4-tricarboxylic acid

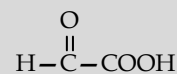
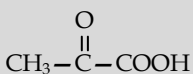
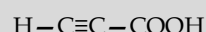
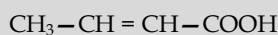
- ii. If all carboxyl groups are not directly bonded to unbranched longest carbon chain, the longest chain of carbon atoms between two carboxyl groups is selected as parents chain. Carbon atoms of the two carboxyl groups are included in the parent chain and the remaining is treated as substituent.



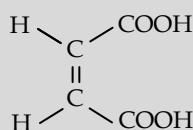
3-(Carboxymethyl)Hexane-1,6-dioic acid

Self Test

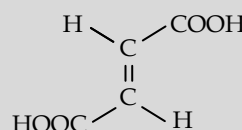
- 13.1 What are carboxyl acids? Why –COOH is called carboxyl group? Give examples of mono-, di- and tricarboxylic acids.
- 13.2 Write the isomers of the compound with formula $C_2H_4O_2$ and give their IUPAC name.
- 13.3 Write the isomers of the compound with formula $C_3H_6O_2$ and give their IUPAC name.

Some Other Examples of Carboxylic acidOxoethanoic acid
(Glyoxalic acid)2-Oxopropanoic acid
(Pyruvic acid/acetyl formic acid)Propenoic acid
(Acrylic acid)Propynoic acid
(Propiolic/ propargylic acid)

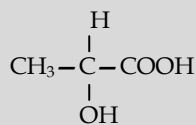
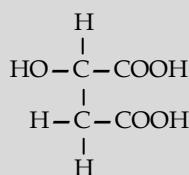
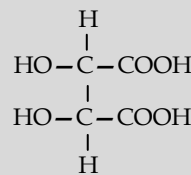
2-Butenoic acid

*cis*-Butenedioic acid (maleic acid)

m pt. = 130°C

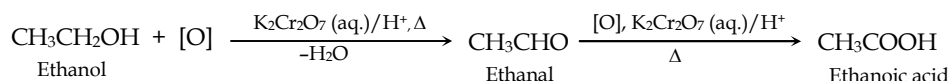
*trans*-Butenedioic acid (Fumaric acid)

m pt. = 302°C

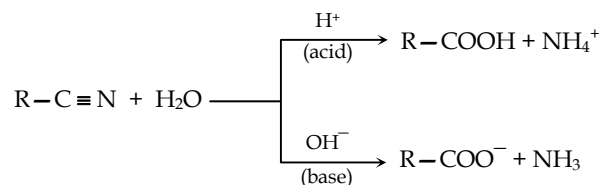
2-Hydroxypropanoic acid
(Lactic acid)2-Hydroxybutanedioic acid
(Malic acid)2,3- Dihydroxybutanedioic acid
(Tartaric acid)**13.1.3 General Methods of Preparation of Monocarboxylic Acids**

The straight carbon chain aliphatic carboxylic acid upto C_6 and those of even carbon number upto C_{18} are commercially available. Other carboxylic acids can be prepared by the methods outlined below.

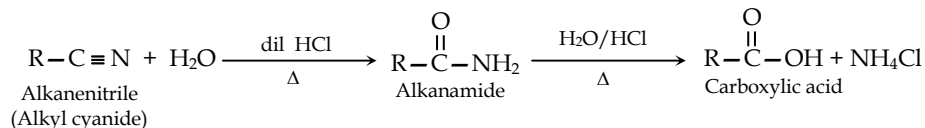
- From primary alcohols and aldehydes (By oxidation):** Oxidation of primary alcohol or aldehyde by using oxidizing agents like $\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$, $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$, $\text{KMnO}_4/\text{H}_2\text{SO}_4$ gives carboxylic acid of same number of carbon.

**Example:**

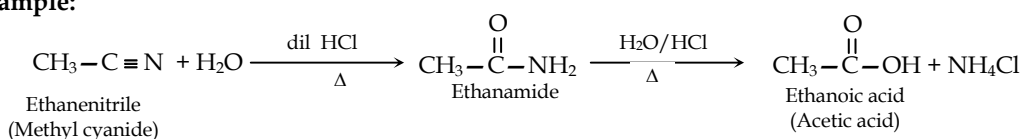
- From nitriles (By hydrolysis): Nitrile synthesis of carboxylic acid:** Hydrolysis of nitrile by boiling with aqueous solution of mineral acid or alkali gives carboxylic acid. The reaction is popularly known as **nitrile synthesis** of carboxylic acid.



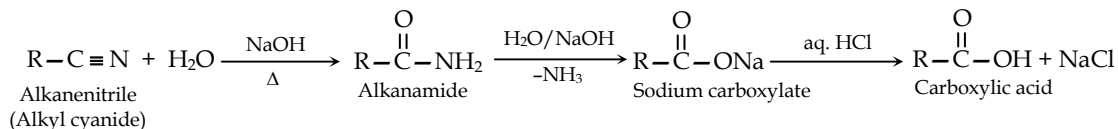
(i) *Acidic hydrolysis of nitrile:*



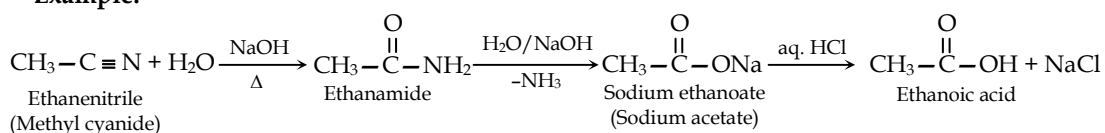
Example:



(ii) *Alkaline hydrolysis of nitrile:* If aqueous NaOH is used for the hydrolysis of alkyl nitrile, the sodium salt of carboxylic acid is formed at first, which upon acidification gives the corresponding carboxylic acid.

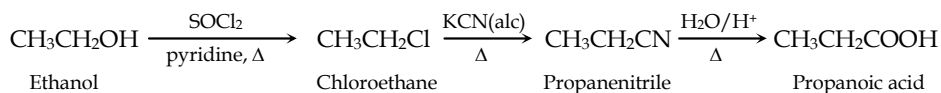


Example:

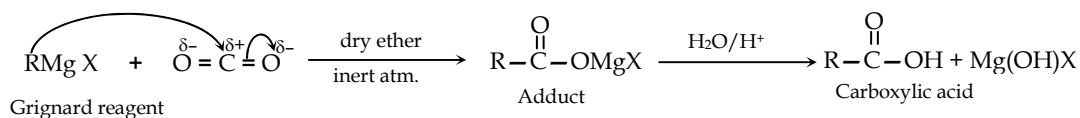
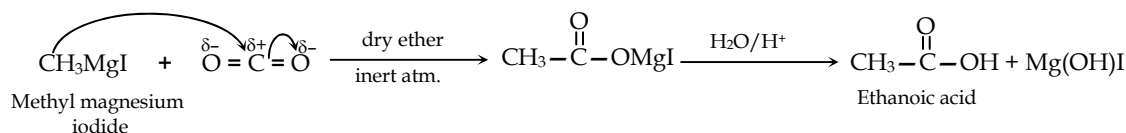


The nitrile synthesis of carboxylic acid is a good synthetic method for preparing carboxylic acids and has an advantage of increasing the length of a carbon chain than that of the parent alkyl halide as:

Example:



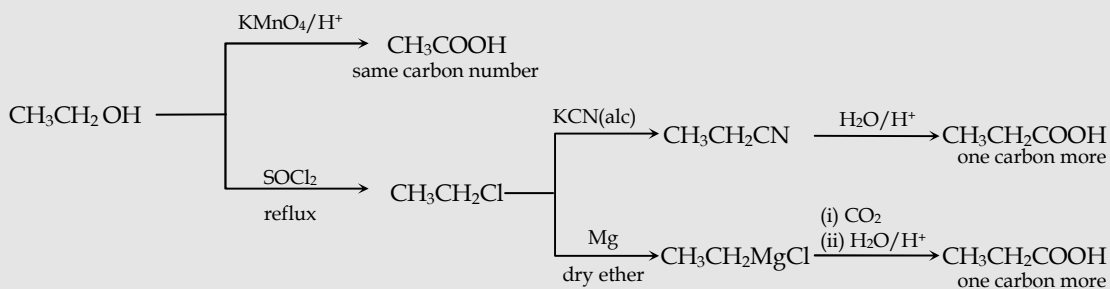
3. **From Grignard reagent: Grignard synthesis of carboxylic acid (Carbonation):** When carbon dioxide gas is bubbled through ether solution of the Grignard reagent followed by hydrolysis, carboxylic acid is obtained. This reaction is also carried out when ether solution of Grignard reagent is poured over crushed *dry ice* followed by hydrolysis. Dry ice serves not only as a source of CO₂ but also serves as cooling agent.

**Example:**

Carboxylic acid thus obtained always has one carbon atom more than that of the alkyl group of parent Grignard reagent and hence has advantage of increasing the length of the carbon chain.

Note:

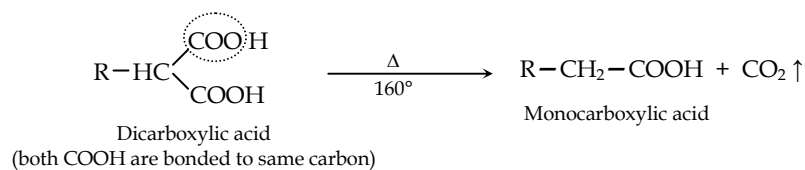
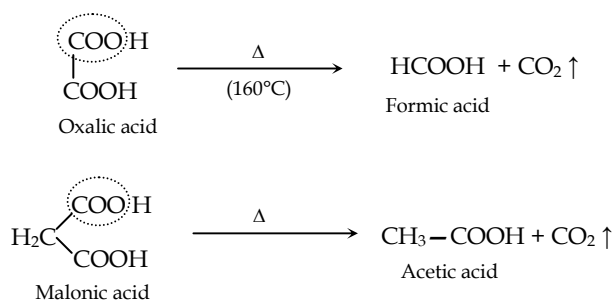
Preparation of carboxylic acid

**Self Test**

13.4 Convert ethanol into:

- i. Ethanoic acid ii. Propanoic acid

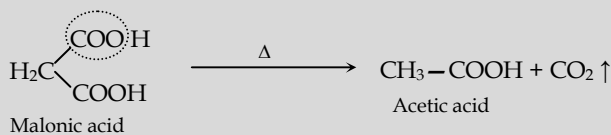
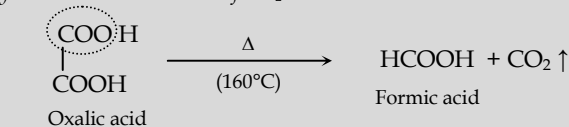
4. **From dicarboxylic acids (Decarboxylation):** When a dicarboxylic acid having two carboxyl groups attached to the same carbon atom is heated strongly, decarboxylation takes place and monocarboxylic acid is obtained.

**Examples:**

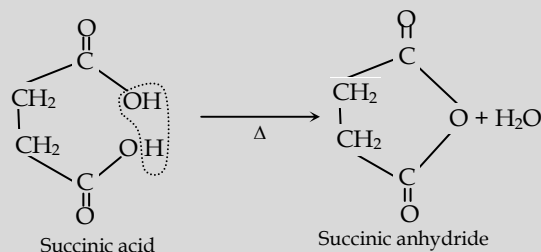
Effect of heat on dicarboxylic acids:

When dicarboxylic acids are heated, different products are obtained. The nature of products formed depends on the number of carbon atoms intervening two carboxyl groups.

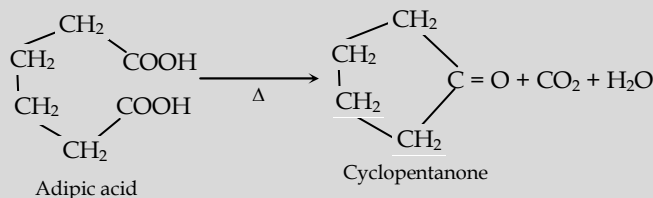
- i. Dicarboxylic acids with one or without intervening carbon atom between two -COOH groups on simple heating produce monocarboxylic acid with elimination of CO_2 .



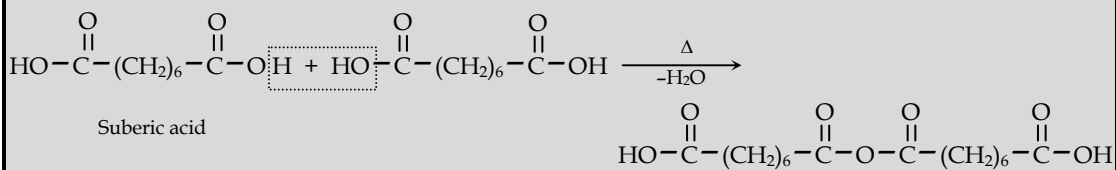
- ii. Dicarboxylic acids with two or three intervening carbon atoms between two -COOH groups on heating undergo cyclization with elimination of a molecule water to yield acid anhydrides with five or six membered ring respectively.



- iii. Dicarboxylic acids with four or five intervening carbon atoms between two -COOH groups on heating loses a molecule water and CO_2 to yield cyclic ketones with five or six membered ring respectively.



- iv. Dicarboxylic acids with more than five intervening carbon atoms between two -COOH groups undergo intermolecular dehydration to give linear polymer.

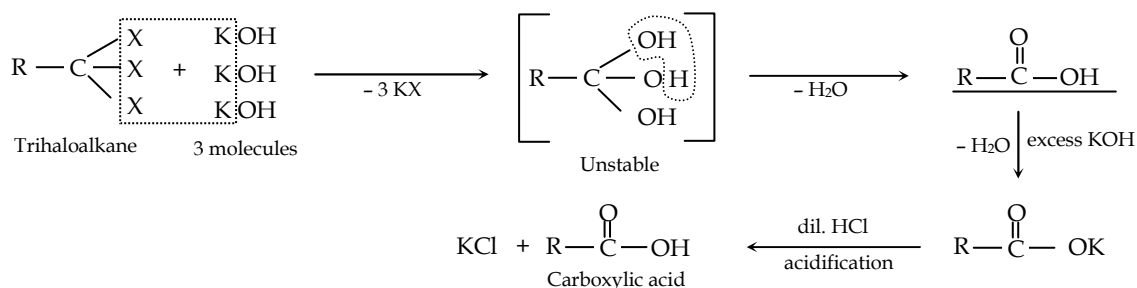
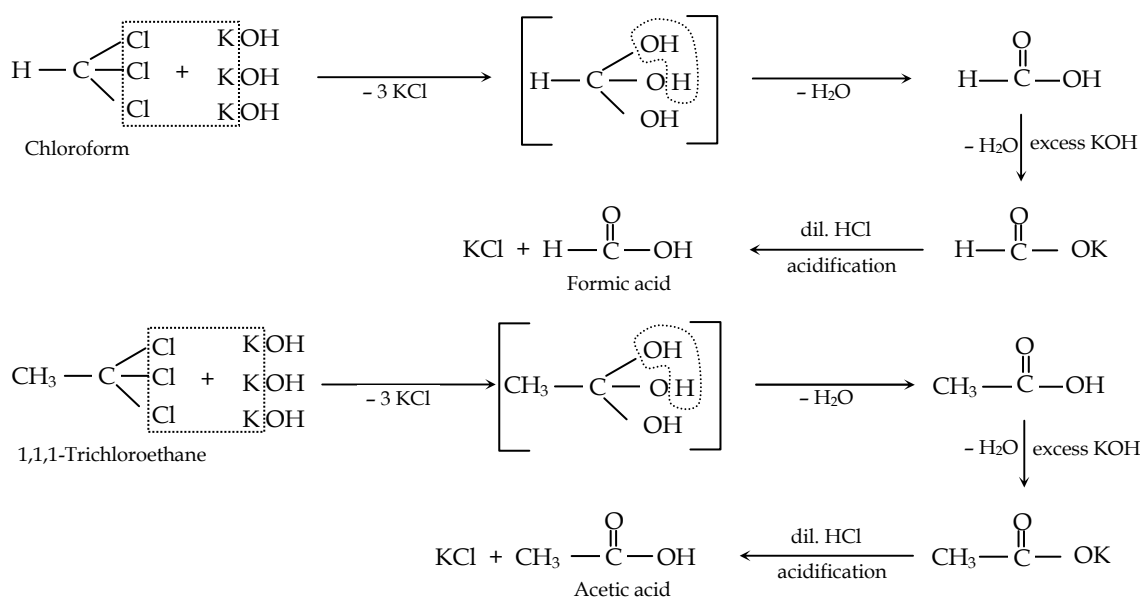


5. **From sodium alkoxide (alkali alcoholates):** When pure and dry carbon monoxide is passed through sodium alkoxide (alkali alcoholate) under pressure, sodium carboxylate is obtained, which on acidification with mineral acid gives monocarboxylic acid.



Example:

6. **From trihaloalkanes:** Hydrolysis of trihaloalkane containing three halogen atoms at the same carbon atom with hot and aqueous solution of caustic alkali (NaOH or KOH) followed by acidulation gives corresponding carboxylic acid.

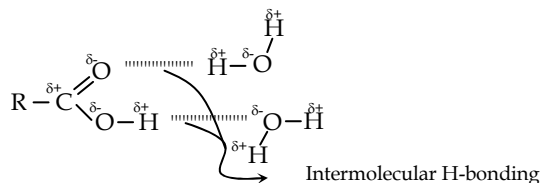
**Example:**

Hydrolysis of fat gives a mixture of monocarboxylic acids which can be fractionally distilled to get in a purity of over 90% straight chain monocarboxylic acid with 6 – 18 carbon atoms of even number of carbon atoms.
This is the most important commercial source of aliphatic monocarboxylic acid.

13.1.4 Physical Properties of Monocarboxylic Acids

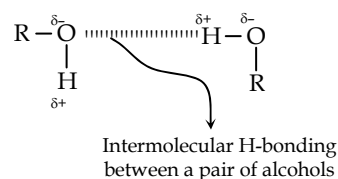
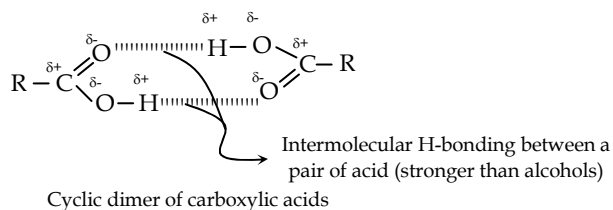
1. **Physical state:** The first three members (C_1 , C_2 , C_3) of aliphatic monocarboxylic acids are colourless liquids with pungent smell. The next three members (C_4 , C_5 , C_6) are oily liquids with unpleasant odour of rancid butter and higher members are colourless waxy solids and do not have distinct odour because of their low volatility.

2. **Solubility:** Carboxylic acids are polar and form intermolecular hydrogen bonds with water. The first four members of carboxylic acids (C_1 to C_4) are miscible with water due to formation of effective hydrogen bonding between the carboxylic acid and water.



The carboxylic acid up to five carbon atoms are partially soluble, and the higher acids are insoluble in water. Solubility of the carboxylic acid decreases rapidly with increase in the size of water insoluble hydrophobic alkyl part of the acid.

3. **Boiling Point:** The boiling points of lower member of carboxylic acids are higher than that of alcohols of comparable molecular masses. This is due to the fact that a pair of carboxylic acid molecules are held together by two intermolecular hydrogen bonds and, it is stronger (due to formation of special two point hydrogen bond) than those in alcohols. These two carboxylic acids exist as cyclic dimers. Also, O-H bond of carboxylic acid is more polar than O-H bond of alcohol due to *electron withdrawing effect* of carbonyl group ($>C=O$) in carboxylic acid.



Self Test

- 13.5 Saturated monocarboxylic acid have higher melting point as compare to their unsaturated analog. Explain.
- 13.6 Account for the fact that propanoic acid has higher boiling point (b. pt. 141°C) than that of the alcohol of comparable molecular masses, n-butyl alcohol (b. pt. 118°C).
- 13.7 The boiling point of methanoic acid is higher than ethanol though they have same molecular mass. Explain.

13.1.5 Chemical Properties of Monocarboxylic Acids

The chemical properties of monocarboxylic acids can be discussed under following heading:

A. Reaction due to H- atom of the $-\text{COOH}$ group: Acidic Nature:

Carboxylic acids are the most acidic of all organic compounds. Carboxylic acids undergo ionization when dissolved in water and produce H_3O^+ ions, which is responsible for their acidic behaviour. In aqueous solution, there exists a measurable equilibrium between acid and H_3O^+ .



According to law of mass action,

$$K_{\text{eq}} = \frac{[\text{RCOO}^-][\text{H}_3\text{O}^+]}{[\text{RCOOH}][\text{H}_2\text{O}]}$$

4.14 Modern Approach to Chemistry-II

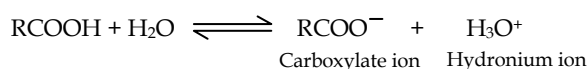
In very dilute solution,

$$K_{\text{eq}} [\text{H}_2\text{O}] = K_a = \frac{[\text{RCOO}^-] [\text{H}_3\text{O}^+]}{[\text{RCOOH}]}$$

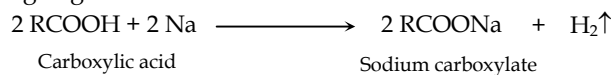
Here, K_a is ionization constant of the acid. Greater the value of K_a , more acidic is the acid. Hence, acidic strength can be expressed in terms of K_a or pK_a of the acid, $pK_a = -\log K_a$

The following reactions prove the acidic nature of carboxylic acids.

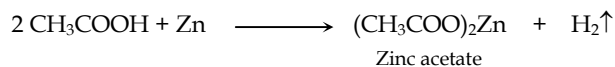
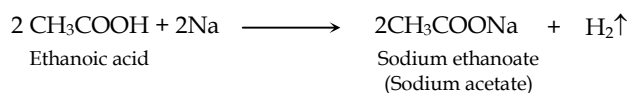
1. **Litmus test:** Aqueous carboxylic acid turns blue litmus red. This is due to the presence of H_3O^+ ion in aqueous solution.



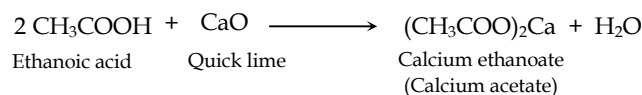
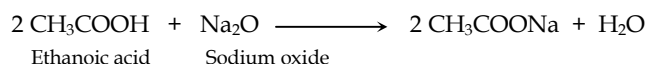
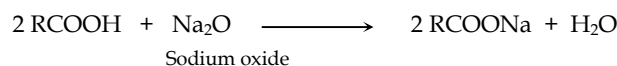
2. **Reaction with active metals:** Carboxylic acid reacts with active metals like Na, K, Zn, Mg, etc. to liberate hydrogen gas.



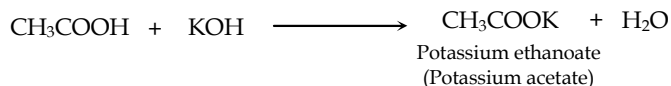
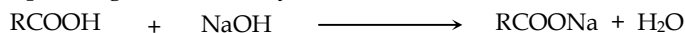
Example:



3. **Reaction with metal oxides:** Carboxylic acid reacts with metal oxides to give corresponding salts of carboxylic acid (carboxylate) and water.



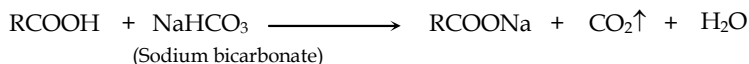
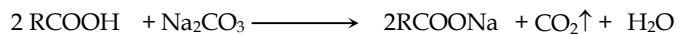
4. **Reaction with caustic alkali:** Carboxylic acid reacts with caustic alkali like NaOH or KOH, to give corresponding salt of carboxylic acid and water.



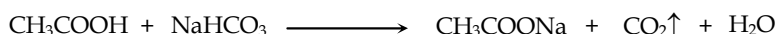
Note:

The concentration of ethanoic acid (acetic acid) in vinegar may be determined by titrating the vinegar solution with sodium hydroxide solution using phenolphthalein indicator.

5. **Reaction with metal carbonates and bicarbonates: Test of carboxylic acids:** Carboxylic acid reacts with sodium carbonate and sodium bicarbonate giving brisk effervescence of carbon dioxide. This reaction serves as the test of the carboxylic acid.



Example:



Note:

The two important classes of organic compound with acidic nature which change blue litmus red are carboxylic acids and phenols. However, phenols are weak acid and unable to decompose sodium carbonate and bicarbonate. Hence, reaction of carboxylic acid with sodium carbonate and bicarbonate can be used to distinguish carboxylic acids from phenols and other organic acids.

6. **Reaction with ammonia:** Carboxylic acids react with ammonia to form ammonium salts (ammonium carboxylate), which upon heating give amides.



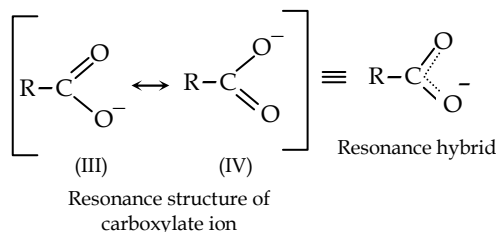
Self Test

13.8 Write the suitable chemical test to distinguish between the following pairs of compounds:

- | | |
|-----------------------------------|------------------------------|
| i. Ethyl alcohol and acetic acid. | ii. Acetone and acetic acid. |
| iii. Phenol and acetic acid. | iv. Phenol and benzoic acid |

Explanation for acidic nature of carboxylic acid:

In aqueous solution, there exists a measurable equilibrium between acid and H_3O^+ .

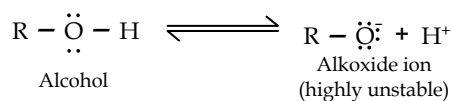


Since the carboxylate ion (conjugate base of the carboxylic acid) is stabilized by resonance, its corresponding acid (carboxylic acid) is fairly acidic.

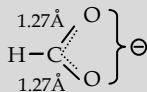
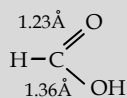
Note:

Conjugate base of the strong acid is relatively stable.

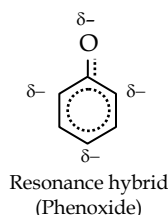
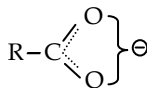
Though carboxylic acid and alcohol both contain $-\text{OH}$ group, alcohol is far less acidic than carboxylic acid due to the absence of resonance in alkoxide ion.



X-ray and electron-diffraction measurements show that the two carbon to oxygen bonds in carboxylate ion are of equal length. This fact supports that carboxylate ion is a resonance hybrid and each resonating structure has equal contribution.



Carboxylic acids are stronger acids than phenols. In resonance structures of carboxylate ion (Structures III and IV), the negative charge is identically distributed over two oxygen atoms while in resonance structure of phenoxide ion, it is present in oxygen and also in carbon (electro positive element). Hence carboxylate ion is more stabilized than phenoxide ion. Therefore, carboxylic acids ionize to the greater extent than phenols and behave as strong acids than phenols.



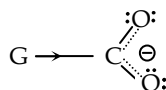
Distinction between alcohols, phenols and carboxylic acids:

Property	Alcohols	Phenols	Carboxylic acids
1. Action of litmus	No action	Turn blue litmus red	Turn blue litmus red
2. Action of NaHCO ₃	No action	No action	Effervescence of CO ₂
3. Action of NaOH	No action	Form sodium phenoxide	Form sodium carboxylate
4. Action of FeCl ₃	No action	Form violet blue, red etc.	No action

Effect of Substituent(s) on Acidity of Carboxylic Acids

a. Effect of electron donating substituents

Any factor that destabilises the carboxylate ion decreases the acidity of the acid. The electron-donating (releasing) substituent donates electron, intensify the negative charge on carboxylate ion and destabilizes the carboxylate ion and thus decreases the acidic nature of the corresponding carboxylic acid.



G = **d**onates electron

destabilises the anion by intensifying negative charge

decreases the acidic nature of the acid

Example:

Alkyl groups behave as electron releasing groups. The sequential order of electron donating effect is: H- < CH₃- < CH₃CH₂- < CH₃CH₂CH₂-.

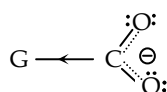
Hence formic acid is more acidic than acetic acid and acetic acid is more acidic than propionic acid and so on.

Acidic order: $\text{HCOOH} > \text{CH}_3\text{COOH} > \text{CH}_3\text{CH}_2\text{COOH}$.

Compound	K_a	p K_a
HCOOH	1.77×10^{-4}	3.75
CH_3COOH	1.75×10^{-5}	4.74
$\text{CH}_3\text{CH}_2\text{COOH}$	1.3×10^{-5}	4.87

b. Effect of electron withdrawing substituents

Any factor that stabilises the carboxylate ion increases the acidity of the acid. The electron-withdrawing substituent withdraws electron, disperse the negative charge on carboxylate ion and stabilizes the carboxylate ion and thus increases the acidic nature of carboxylic acid.



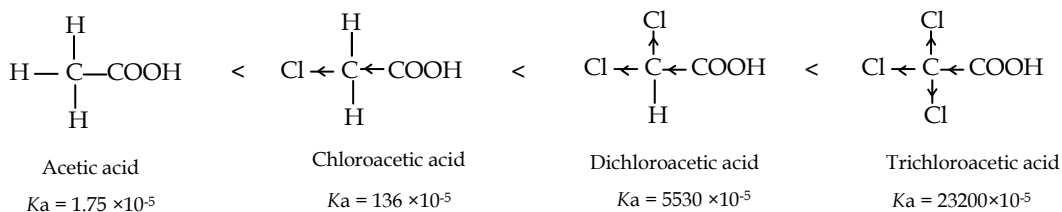
G = withdraws electron

stabilises the anion by dispersing negative charge

The following factors also govern the acidity of carboxylic acids due to the electron-withdrawing substituents:

Case I: Number of substituents:

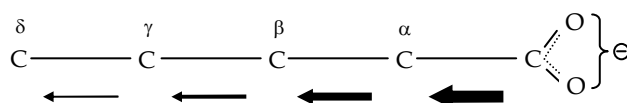
Chloride ($\text{Cl}-$) withdraws electron due to -I effect. Greater the number of electron withdrawing substituents, higher is the acidic nature of acids. Hence chloroacetic acid is more acidic than acetic acid. Dichloroacetic acid is even more acidic than chloroacetic acid and so on.



Case II: Nature of substituents:

The electron withdrawing inductive effect (-I effect) of halogen is $\text{F} > \text{Cl} > \text{Br} > \text{I}$. Hence fluoroacetic acid is more acidic than chloroacetic acid, chloroacetic acid is more acidic than bromoacetic acid and iodoacetic acid is the least acidic among the four haloacetic acids.

Compound	Common name	K_a	p K_a
$\text{F} \leftarrow \text{CH}_2\text{COOH}$	Fluoroacetic acid	260×10^{-5}	2.59
$\text{Cl} \leftarrow \text{CH}_2\text{COOH}$	Chloroacetic acid	136×10^{-5}	2.87
$\text{Br} \leftarrow \text{CH}_2\text{COOH}$	Bromoacetic acid	125×10^{-5}	2.90
$\text{I} \leftarrow \text{CH}_2\text{COOH}$	Iodoacetic acid	67×10^{-5}	3.16

Case III: Position or location of substituents:

Electron withdrawing inductive effect (-I effect) decreases rapidly with distance from the carboxyl group.

Compound	Common name	K_a
$\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{COOH}$	Butyric acid	1.52×10^{-5}
$\text{CH}_3-\text{CH}_2-\overset{\text{Cl}}{\underset{ }{\text{CH}}}-\text{COOH}$	α -Chlorobutyric acid	139×10^{-5}
$\text{CH}_3-\overset{\text{Cl}}{\underset{ }{\text{CH}}}-\text{CH}_2-\text{COOH}$	β -Chlorobutyric acid	8.9×10^{-5}
$\overset{\text{Cl}}{\underset{ }{\text{CH}_2}}-\text{CH}_2-\text{CH}_2-\text{COOH}$	γ -Chlorobutyric acid	2.96×10^{-5}

Electron withdrawing inductive effect (-I) effect decreases rapidly with distance or number of carbon atoms from the carboxyl group and becomes non effective after four atoms. Hence, β -Chlorobutyric acid is less acidic than α -Chlorobutyric acid, γ -Chlorobutyric acid is less acidic than β -Chlorobutyric acid and so on.

Self Test

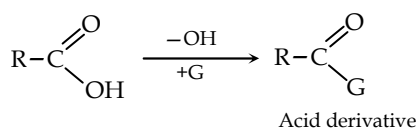
13.9 Why is methanoic acid stronger than ethanoic acid?

13.10 Which one of the following is a stronger acid in each of the following pairs?

- Propionic acid and acetic acid.
- Chloroacetic acid and acetic acid
- Chloroacetic acid and dichloroacetic acid.
- α -chloropropionic acid and β -chloropropionic acid.
- Acetic acid and benzoic acid.

B. Reaction due to replacement of -OH part of -COOH group:

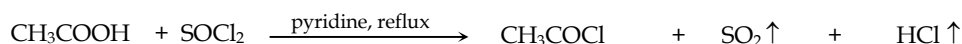
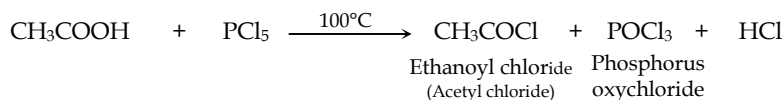
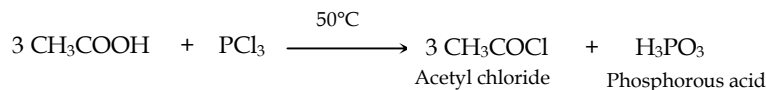
A category of reactions given by carboxylic acids where the -OH part of the -COOH group is replaced by electron rich species like -Cl, -OR', -NH₂ and -OCOR' groups to yield acid chloride (R-COCl), ester (R-COOR'), acid amide (R-CONH₂) and acid anhydride (R-COOCOR') respectively. These compounds are, therefore called **functional derivatives of carboxylic acids**.



G = Cl (acid chloride)
 = OR (ester)
 = NH₂ (acid amide)
 = OCOR (acid anhydride)

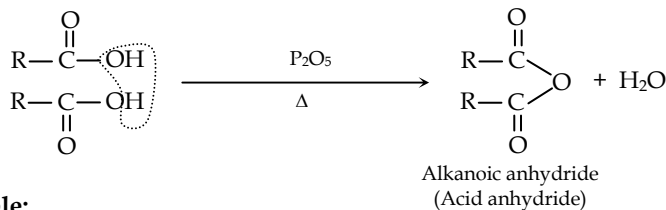
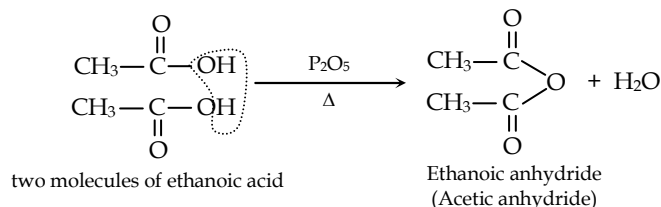
1. Formation of acid chlorides:

Carboxylic acids react with phosphorus trichloride (PCl_3), phosphorus pentachloride (PCl_5) or thionyl chloride (SOCl_2) to give corresponding acid chlorides.

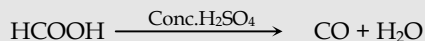
Examples

Thionyl chloride is more conveniently used than that of phosphorus chlorides to prepare acid chloride from carboxylic acid since the products formed besides the acid chloride are gases (HCl and SO_2), which can be easily separated from the acid chlorides. Any excess of thionyl chloride is easily removed from the reaction by distillation due to its low b.pt. (79°C).

- 2. Formation of acid anhydrides: Dehydration of carboxylic acid:** Monocarboxylic acid except formic acid on heating with strong dehydrating agent like P_2O_5 or conc. H_2SO_4 undergoes intermolecular dehydration (elimination of water molecule) into acid anhydrides.

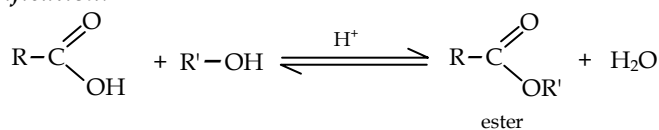
**Example:****Note:**

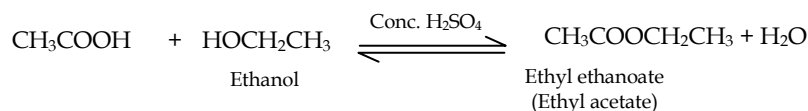
Formic anhydride is unknown because formic acid undergoes dehydration with conc. H_2SO_4 to give CO and water.

**3. Formation of ester:**

The carboxylic acids are converted into esters as follows:

- i. Direct method:** When a mixture of carboxylic acid and alcohol is heated in the presence of a little mineral acid like conc. H_2SO_4 or dry hydrogen chloride, ester is formed. This reaction is called *esterification*.



Example:**Note:**

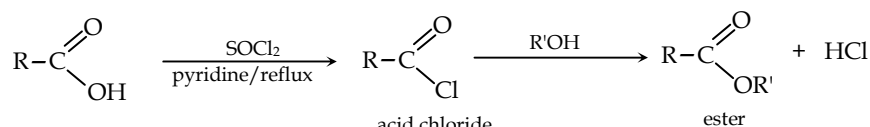
- i. This reaction can be used to test alcohol or carboxylic acid in laboratory because of distinct fruity smell of ester.
- ii. It should be noted that always -OH group from the acid and H-atom from the alcohol are eliminated as water in esterification reaction.

The reaction is reversible. A reversible reaction is not generally a good choice from synthetic point of view due to low yield of product. Therefore, the indirect method as given below is the method of choice for quantitative preparation of ester. However, in the esterification, the equilibrium can be shifted in the forward direction for quantitative yield of ester by using one of the reactants in excess and/or removing one of the products as soon as it is formed. Therefore, in order to shift the reaction in forward direction water is removed as soon as it is formed.

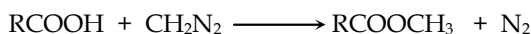
Reactivity of alcohol: $\text{CH}_3\text{OH} > 1^\circ > 2^\circ > 3^\circ$

Reactivity of acid: $\text{HCOOH} > \text{CH}_3\text{COOH} > \text{RCH}_2\text{COOH} > \text{R}_2\text{CHCOOH} > \text{R}_3\text{CCOOH}$

- ii. **Indirect method:** Carboxylic acids are frequently converted into their esters via the acid chlorides since it is irreversible and goes to completion.

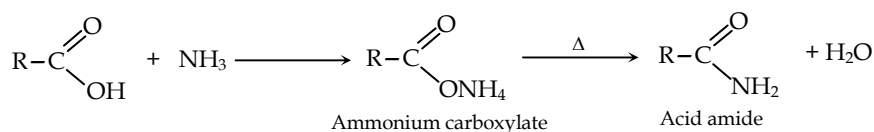
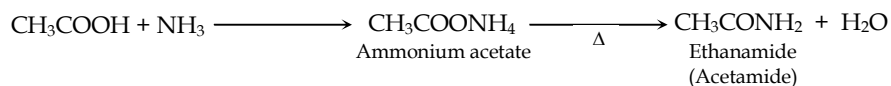


- iii. **Diazomethane method:** Carboxylic acids react with diazomethane to produce methyl esters.

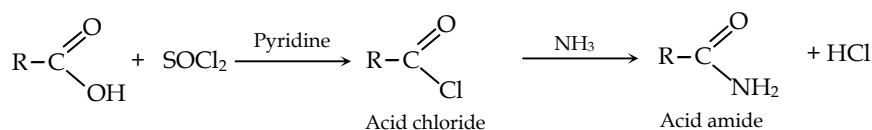


4. Formation of amide: Amides are prepared from carboxylic acids as follows:

- i. **Direct Method:** Carboxylic acid reacts with ammonia to form ammonium carboxylate (ammonium salt), which upon strong heating loses H_2O to give amide.

**Example:**

- ii. **Indirect Method:** Carboxylic acids are converted into amide via the formation of acid chloride, when acid chloride is treated with ammonia.



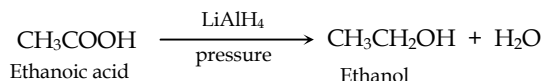
Example:



C. Reaction due to $>\text{C}=\text{O}$ part of $-\text{COOH}$ group:

Partial Reduction: Formation of alcohols: Carboxylic acid can be reduced to the corresponding primary alcohol with LiAlH_4 or with hydrogen in the presence of catalyst like copper chromite (CuCr_2O_4)

Example:

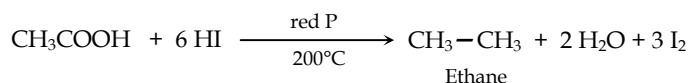
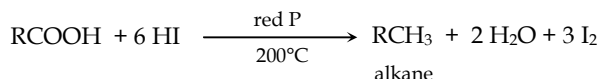


Note:

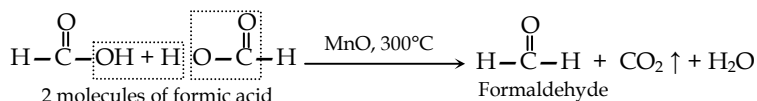
Reduction of carboxylic acids into alcohols does not take place with NaBH_4 .

D. Reaction due to complete $-\text{COOH}$ group:

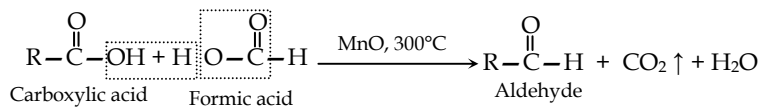
- Complete reduction: Formation of alkane:** On prolonged reduction of carboxylic acids with HI in red phosphorus at about 200°C in a sealed tube, corresponding alkanes are obtained, where $-\text{COOH}$ group is reduced to $-\text{CH}_3$ group.



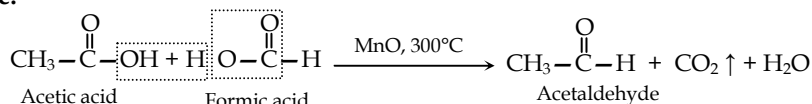
- Formation of carbonyl compounds:** When vapours of carboxylic acids are passed over heated alumina or manganous oxide at 300°C , carbonyl compounds are formed.
 - Formic acid gives formaldehyde.



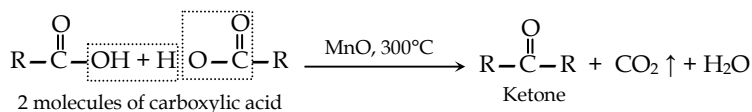
- A mixture of formic acid and other carboxylic acid gives aldehyde other than formaldehyde.



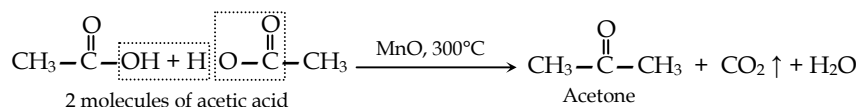
Example:



- All carboxylic acids other than formic acids give ketones.

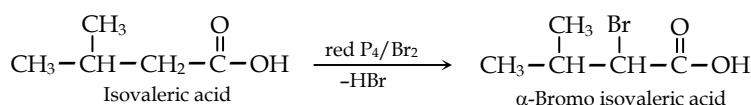


Example:

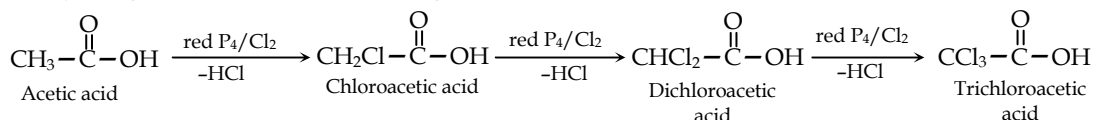


E. Reaction due to alkyl (R-) group of carboxylic acid:

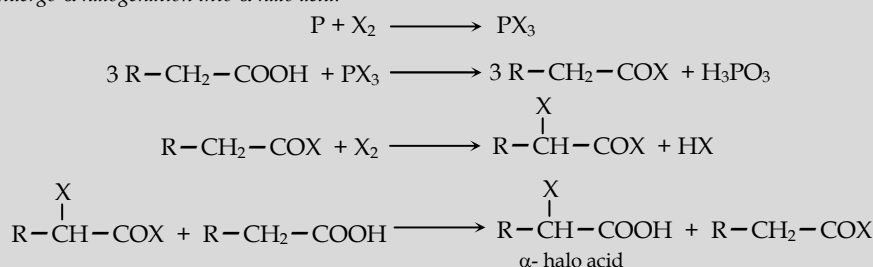
- α -Halogenation of aliphatic acids: Hell-Volhard-Zelinsky reaction: Reaction with halogens:** Carboxylic acid having α -hydrogen atom reacts with halogens (Cl_2 or Br_2) in presence of red phosphorus as a catalyst to yield α -halo carboxylic acid. This reaction is known as *Hell-Volhard-Zelinsky (HVZ) reaction*. For example,



Polyhalogenation is possible, if halogen is in excess. For example,



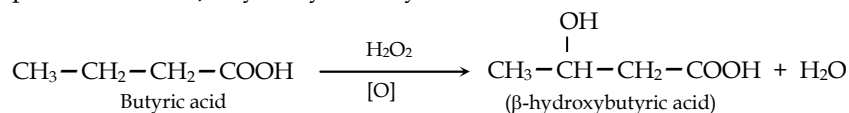
The function of the phosphorus is to convert a little of the acid into acid halide. The acid and acid halide sooner or later undergo α -halogenation into α -halo acid.



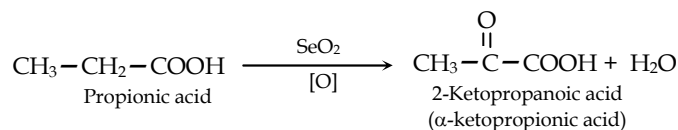
Formic acid does not undergo *Hell-Volhard-Zelinsky reaction* (α -halogenation) due to absence of α -hydrogen. This reaction is not applicable to fluorine and iodine.

2. Oxidation:

- When carboxylic acid is treated with H_2O_2 as a mild oxidizing agent, the alkyl group is oxidized at the β -position to form β -hydroxy carboxylic acid.



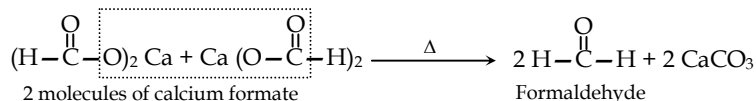
- Selective oxidation:** On the other hand, when carboxylic acid is heated with SeO_2 (selenium dioxide) as a selective oxidizing agent, the alkyl group is oxidized at the α -position to form α -keto acid.



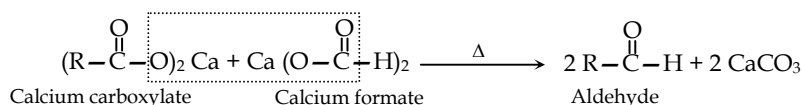
F. Reactions involving salt of carboxylic acids

1. **Reaction involving calcium salt of carboxylic acid:** When calcium salt of carboxylic acid is strongly heated (dry distillation), carbonyl compound is formed.

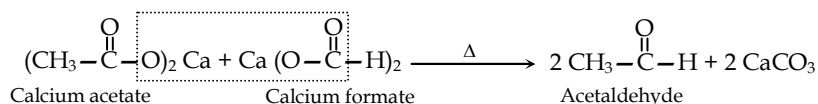
a. Dry distillation of calcium salt of formic acid gives formaldehyde.



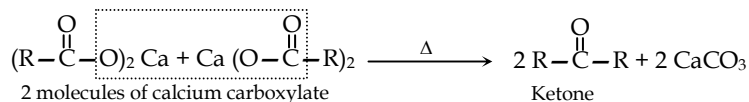
b. Dry distillation of a mixture of calcium salt of formic acid and salts of other acid give aldehydes other than formaldehyde.



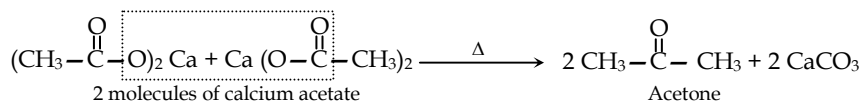
Example:



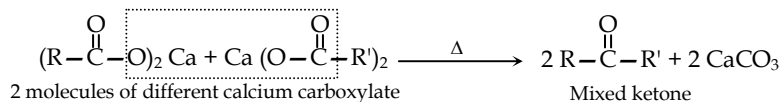
c. Dry distillation of calcium salts of carboxylic acids other than formic acid gives simple ketones.



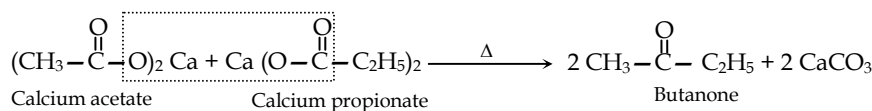
Example:



d. Dry distillation of a mixture of different calcium carboxylates gives mixed ketone. But this method gives a mixture of ketone.

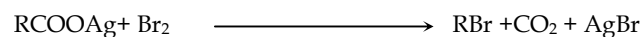
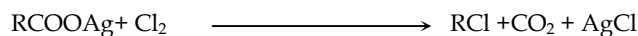


Example:



2. **Reactions involving silver salt of carboxylic acid:** Silver salt of carboxylic acid reacts with chlorine or bromine in carbon tetrachloride as solvent to form corresponding alkyl halide. This is called **Hunsdiecker or Borodine-Hunsdiecker reaction (1935)**.

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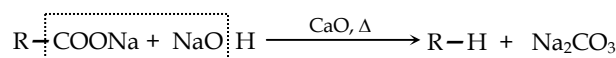


Note:

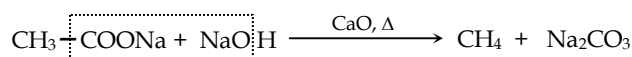
Iodine, however, forms ester with silver salt of carboxylic acid. This is called **Birnbbaum-Simonini reaction**(1892).



3. **Sodalime decarboxylation:** When sodium or potassium salt of carboxylic acid is heated with sodalime, alkane having one carbon less than the acid is formed. The reaction involves removal of carbon dioxide in presence of sodalime hence called sodalime decarboxylation.

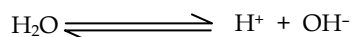
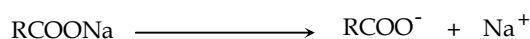


Example:

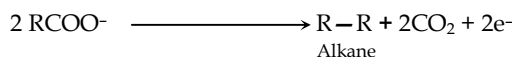


This reaction is utilized in decreasing the carbon chain in organic conversion.

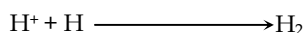
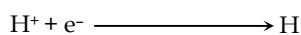
4. **Kolbe's electrolytic decarboxylation (1849):** Electrolysis of a concentrated aqueous solution of sodium or potassium salt of carboxylic acid gives higher alkane at anode.



At anode,



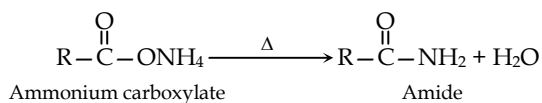
At cathode,



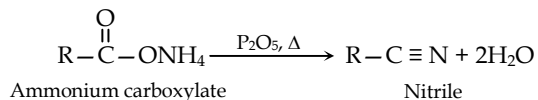
Since discharge potential of H^+ is lower than that of Na^+ , H^+ is preferably discharged.

5. **Reactions involving ammonium salt of carboxylic acid:**

- a. when ammonium salt of carboxylic acid is heated alone, amide is formed.



- b. when ammonium salt of carboxylic acid is heated with P_2O_5 , nitrile is formed.



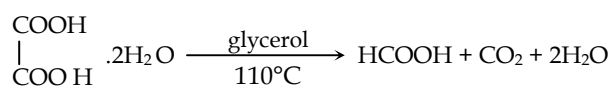
Formic Acid

Formic acid is the first member of aliphatic monocarboxylic acid. Its name is derived from Latin word "*Formica*" meaning "*ant*" since it was obtained from distillation of red ants for the first time. It occurs naturally in insects like bees, ants, wasps, etc. It is colourless liquid with pungent odour with boiling point 100.5°C . It is soluble in water, alcohol, ether, etc.

13.1.6 Laboratory Preparation

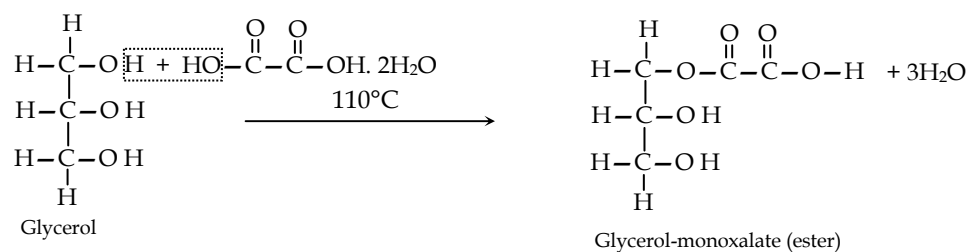
Principle

Formic acid is prepared in the laboratory by heating crystalline oxalic acid with anhydrous glycerol at about 110°C.

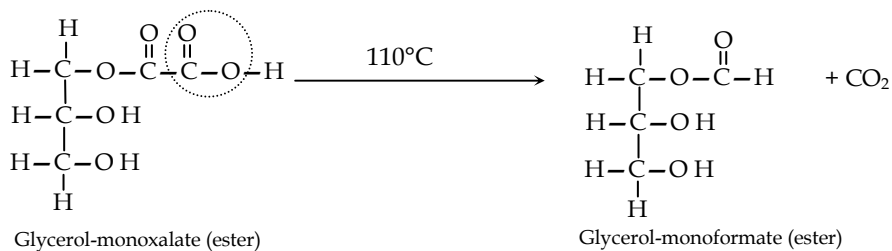


The reaction takes place as follows:

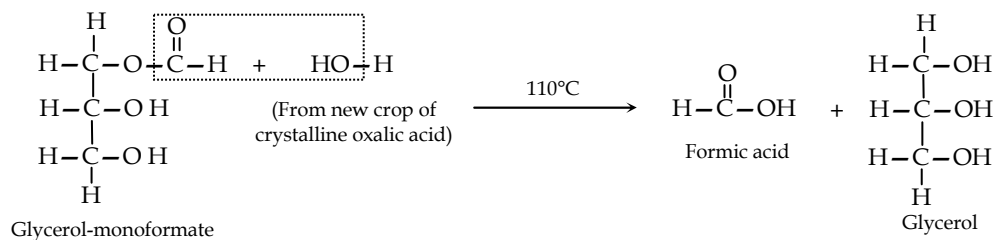
- i. Oxalic acid reacts with glycerol to form glycerol-monooxalate.



- ii. Glycerol-monooxalate is decomposed at 110°C into glycerol-monoformate.



- iii. When the effervescence of CO₂ ceases completely, further amount of crystalline oxalic acid [(COOH)₂·2H₂O] is added when glycerol-monoformate is hydrolyzed with water of crystallization of crystalline oxalic acid into formic acid.



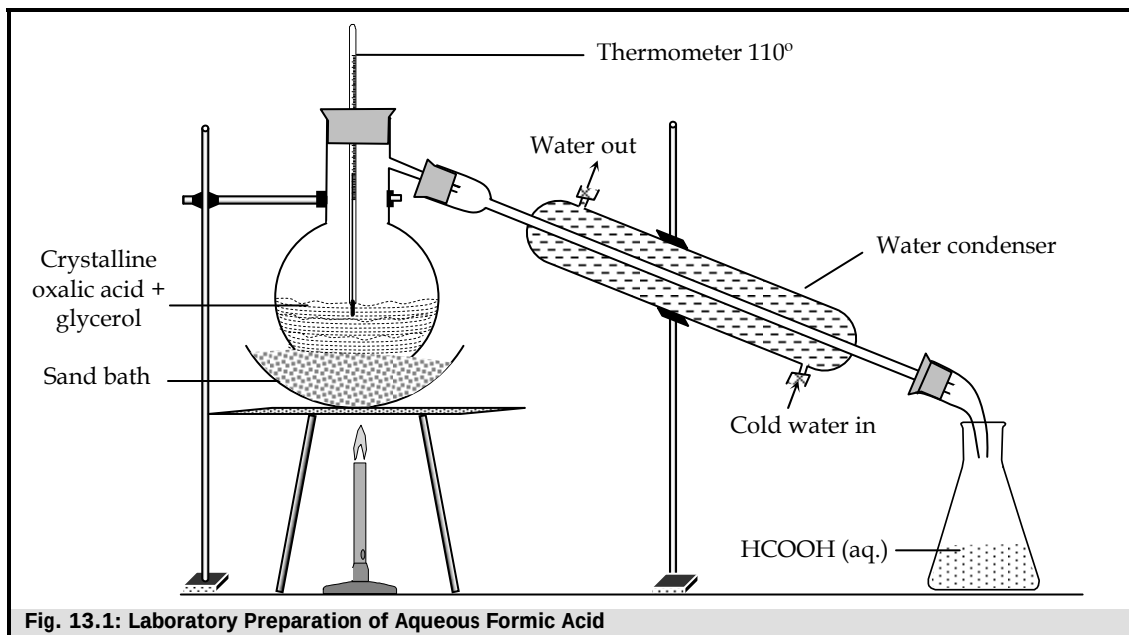


Fig. 13.1: Laboratory Preparation of Aqueous Formic Acid

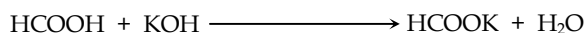
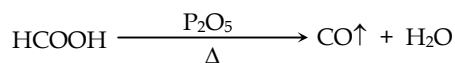
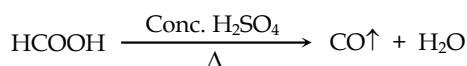
Procedure

In a distillation flask, a mixture of about 40 g crystalline oxalic acid and 50 g of anhydrous glycerol is taken. The apparatus is fitted as shown in the figure 13.1 and heated to 110°C. As the effervescence of CO₂ ceases, the flask is cooled then fresh batch of oxalic acid crystals (40 g) is again added, and the flask is heated to about 110°C. Formic acid formed distils off and is collected in the receiver as aqueous formic acid.

Preparation of anhydrous formic acid from aqueous formic acid

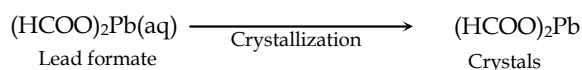
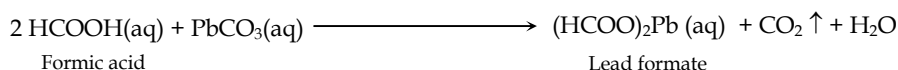
Aqueous formic acid cannot be subjected to fractional distillation to remove water since boiling point of formic acid (100.5°C) is very closer to that of the boiling point of water (100°C).

The dehydrating agents like anhydrous P₂O₅, concentrated H₂SO₄ and solid KOH can not be used for drying because formic acid itself reacts with them.



Hence, chemical method of separation is applied. This can be carried out by using water condenser as a separating device.

In this method, aqueous formic acid is neutralized with lead carbonate (PbCO₃) until the completion of effervescence of CO₂ to get aqueous saturated solution of lead formate. The solution is filtered then subjected to crystallization and recrystallization.



The crystals of lead formate are taken in the inner tube of a water condenser and the apparatus is fitted as in figure 13.2. Water condenser is heated by means of steam passing through the outer jacket of the condenser. H_2S gas is passed under pressure from upper end. The reaction takes place forming anhydrous formic acid which is collected in the receiver. The precipitate of PbS does not enter to the flask due to glass wool plug placed at the lower end of water condenser. The anhydrous formic acid may contain dissolved H_2S which is removed by redistilling anhydrous formic acid with fresh lead formate crystals. In this way, pure and dry anhydrous formic acid is prepared in the laboratory.

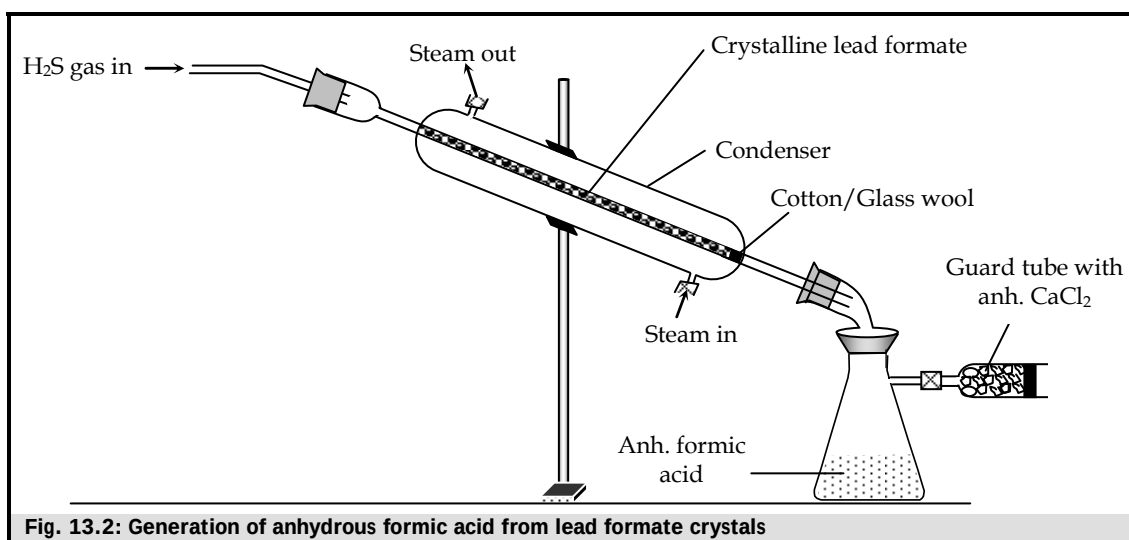
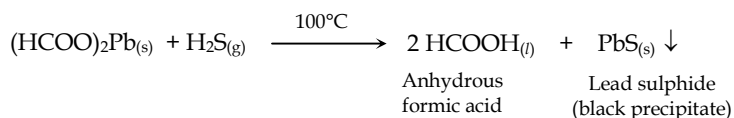


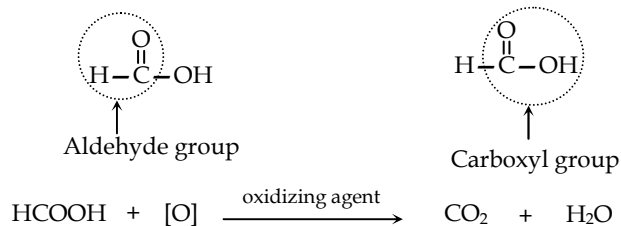
Fig. 13.2: Generation of anhydrous formic acid from lead formate crystals

Formic acid is commercially obtained by heating sodium hydroxide with carbon monoxide at 210°C under the pressure of 6-10 atmosphere followed by acidification.



13.1.7 Abnormal Behaviour of Methanoic Acid (Comparison of Properties of Formic Acid with Acetic Acid)

Formic acid differs from the other monocarboxylic acid. This is because besides the carboxyl group, it contains $-\text{CHO}$ group. So, formic acid behaves both as an acid as well as an aldehyde.



HCOOH	CH ₃ COOH
1. Tollens' test: When formic acid is warmed with Tollen's reagent in water bath, it is oxidized to CO₂ and Tollen's reagent is reduced to the silver. Hence silver mirror is deposited on the inner wall of the test tube. The only carboxylic acid that gives Tollen's or silver mirror test is formic acid. $\text{HCOOH} + 2 [\text{Ag}(\text{NH}_3)_2]\text{OH}$ Methanoic acid Tollen's reagent ↓ Warm $2 \text{Ag} \downarrow + \text{CO}_2 \uparrow + 2 \text{H}_2\text{O} + 4 \text{NH}_3$ Silver mirror	1. Tollens' test: It does not give Tollen's test.
2. Fehling's test: When formic acid is warmed with Fehling's solution in water bath, it is oxidized to CO₂ and Fehling's solution is reduced to Cu₂O, which gives red precipitate. The only carboxylic acid that gives Fehling's solution test is formic acid. $\text{HCOOH} + [2 \text{Cu}^{++} + 4 \text{OH}^-]$ Methanoic acid Fehling's solution ↓ Tartarate Warm $\text{Cu}_2\text{O} \downarrow + \text{CO}_2 \uparrow + 3 \text{H}_2\text{O}$	2. Fehling's test: It does not give Fehling test.
3. Reaction with acidified KMnO₄: When formic (permanganate test) acid is warmed with acidified solution of potassium permanganate, the pink colour of KMnO₄ is discharged. $5 \text{HCOOH} + 2 \text{KMnO}_4 + 3 \text{H}_2\text{SO}_4$ ↓ pink colour $\text{K}_2\text{SO}_4 + 2 \text{MnSO}_4 + 5 \text{CO}_2 + 8 \text{H}_2\text{O}$	3. Reaction with acidified KMnO₄: It does not react.
4. Reaction with mercuric chloride: When formic acid is treated with mercuric chloride, it gives white precipitate of mercurous chloride (Hg₂Cl₂) and then finally gives black mercury. $\text{HCOOH} + 2 \text{HgCl}_2 \longrightarrow \text{Hg}_2\text{Cl}_2 \downarrow + \text{CO}_2 + 2 \text{HCl}$ white ppt $\text{HCOOH} + 2 \text{Hg}_2\text{Cl}_2 \longrightarrow \text{CO}_2 + 2 \text{HCl} + 2 \text{Hg}$ $2 \text{HCOOH} + \text{HgCl}_2 \longrightarrow 2 \text{CO}_2 + 4 \text{HCl} + 2 \text{Hg} \downarrow$ black	4. Reaction with mercuric chloride: It does not react.

5. Action of conc. H₂SO₄: Dehydration: Formic acid is dihydrated with conc. H₂SO₄ into CO and H₂O. This reaction can be utilized for the laboratory preparation of carbon monoxide. $\text{HCOOH} \xrightarrow{\text{conc. H}_2\text{SO}_4} \text{H}_2\text{O} + \text{CO}$	5. Action of conc. H₂SO₄: No action
6. Action of Heat: When formic acid is heated to about 160°C, it decomposes into hydrogen and carbon dioxide. $\text{HCOOH} \xrightarrow[\text{pressure}]{160^\circ\text{C}} \text{H}_2 + \text{CO}_2$	6. Action of Heat: No action
7. Action with P₂O₅: When formic acid is heated with anhydrous P₂O₅, it gives CO and H₂O. $\text{HCOOH} \xrightarrow[\Delta]{\text{P}_2\text{O}_5} \text{CO} + \text{H}_2\text{O}$	7. Action with P₂O₅: It gives corresponding acid anhydride. $\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3 - \text{C} - \text{OH} \\ \text{CH}_3 - \text{C} - \text{OH} \\ \parallel \\ \text{O} \end{array} \xrightarrow{\text{P}_2\text{O}_5, \Delta} \begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3 - \text{C} \\ \parallel \\ \text{CH}_3 - \text{C} \\ \parallel \\ \text{O} \end{array} \text{O} + \text{H}_2\text{O}$ Ethanoic anhydride
8. Reaction with PCl₅: Formic acid reacts with PCl₅ to give methanoyl chloride (formyl chloride), which decomposes into carbon monoxide and HCl. $\begin{array}{c} \text{O} \\ \parallel \\ \text{H} - \text{C} - \text{OH} + \text{PCl}_5 \longrightarrow \text{H} - \text{C} - \text{Cl} + \text{POCl}_3 + \text{HCl} \end{array}$ $\text{HCOCI} \longrightarrow \text{CO} + \text{HCl}$ $\text{HCOOH} + \text{PCl}_5 \longrightarrow 2\text{HCl} + \text{CO} + \text{POCl}_3$	8. Reaction with PCl₅: Acetic acid reacts with PCl₃, PCl₅ or SOCl₂ to give acetyl chloride. $\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3 - \text{C} - \text{OH} + \text{PCl}_5 \longrightarrow \\ \text{CH}_3 - \text{C} - \text{Cl} + \text{POCl}_3 + \text{HCl} \end{array}$
9. Sodalime decarboxylation: Gives H₂ gas $\text{HCOOH} + \text{NaOH} \xrightarrow[\Delta]{\text{CaO}} \text{H}_2 + \text{Na}_2\text{CO}_3$	9. Sodalime decarboxylation: Gives methane $\text{CH}_3\text{COOH} + \text{NaOH} \xrightarrow[\Delta]{\text{CaO}} \text{CH}_4 + \text{Na}_2\text{CO}_3$

In above reaction number 1, 2, 3 and 4, formic acid behaves as a reducing agent and itself is oxidized to CO₂ whereas all other carboxylic acids do not show reducing behaviour.

Note:

Pure acetic acid is called *glacial acetic acid* since it solidifies forming ice like solid when cooled (melting point 16.6°C)

Self Test

13.11 Suggest a suitable chemical test to distinguish

- | | |
|--------------------------------|----------------------------------|
| a. Formic acid and acetic acid | b. Formic acid and ethyl alcohol |
| c. Formic acid and phenol | d. Formic acid and formaldehyde |

13.12 What happens when methanoic acid is:

- | | |
|---|-----------------------------------|
| a. Warmed with Tollen's reagent | b. Warmed with Fehling's solution |
| c. Added into acidified potassium permanganate solution | |

13.13 Why does methanoic acid give Tollen's test but ethanoic acid does not?

Note:

Formic acid does not give 2,4-DNP test of aldehydes.

13.1.8 Uses of Carboxylic Acids**1. Use of formic acid**

- It is used in leather tanning.
- It is used in textile industry for dyeing process.
- It is used in laboratory for preparation of carbon monoxide.
- It is used as medicine in the treatment of gout.

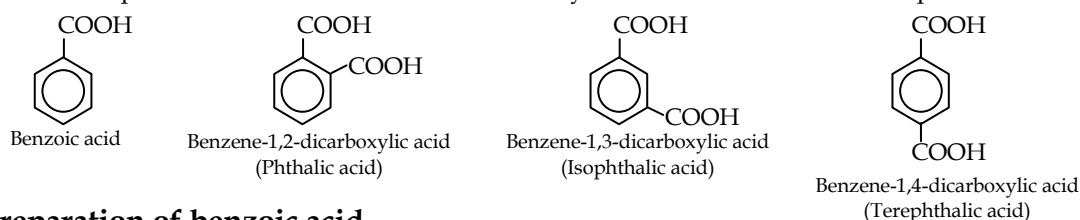
2. Use of acetic acid

- It is used in plastics, rubber industries.
- It is used as a coagulant in the manufacture of rubber from latex and caesin from milk.
- It is used as vinegar.
- It is used as solvent.

13.2 Aromatic Carboxylic Acids

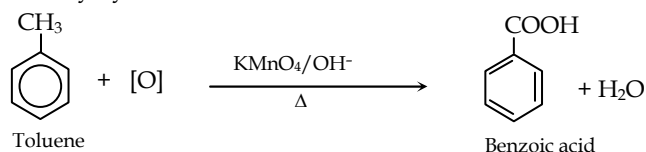
The organic compounds having carboxyl group ($-\text{COOH}$) directly bonded to aromatic ring are called **aromatic carboxylic acids**. They are regarded as the carboxyl derivatives of aromatic hydrocarbons when one or more ring hydrogen atoms have been substituted by carboxyl groups.

The most important members of the aromatic carboxylic acids are benzoic acid and phthalic acid.

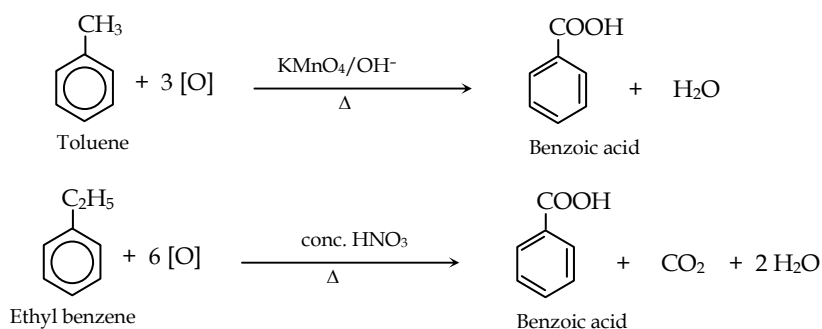
**Preparation of benzoic acid**

Benzoic acid is prepared by following methods:

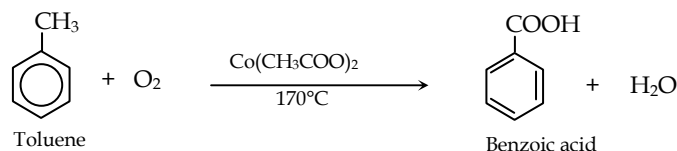
- By the oxidation of alkyl benzene:** Oxidation of alkyl benzene with acidic or alkaline KMnO_4 , acidified $\text{K}_2\text{Cr}_2\text{O}_7$, conc. HNO_3 or chromic anhydride gives benzoic acid. Many aromatic acids are obtained industrially by this method.



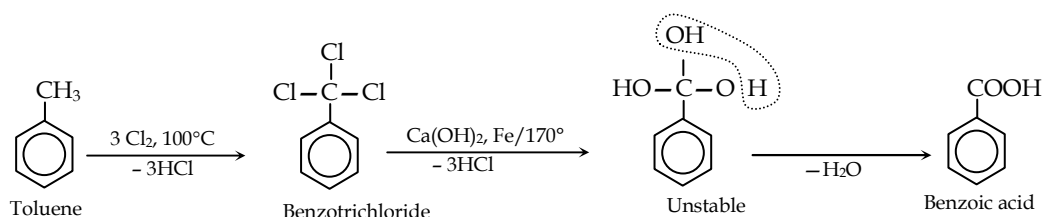
The alkyl side chain ($\text{R}-$) to the aromatic ring gets oxidized to $-\text{COOH}$ groups, which is irrespective of the size of the side chain.



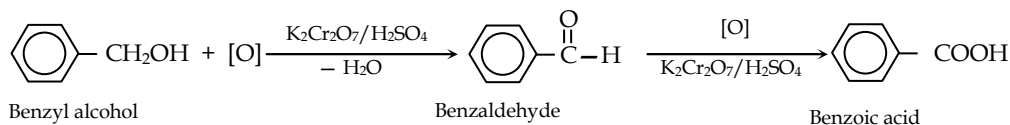
2. **By the oxidation of toluene:** Oxidation of toluene with air in presence of cobalt acetate or manganese acetate at 170°C, gives benzoic acid. (**Commercial method of preparation**).



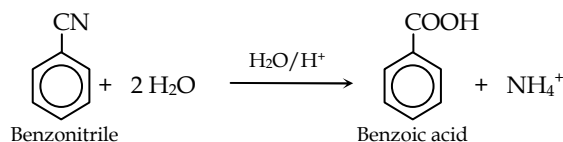
3. **By the hydrolysis of benzotrichloride:** When benzotrichloride is boiled with $\text{Ca}(\text{OH})_2$ (milk of lime) in presence of Fe, it gives benzoic acid. Benzochloride required for the reaction itself is prepared by passing chlorine through boiling toluene in presence of light at 110°C (**Commercial method of preparation**).



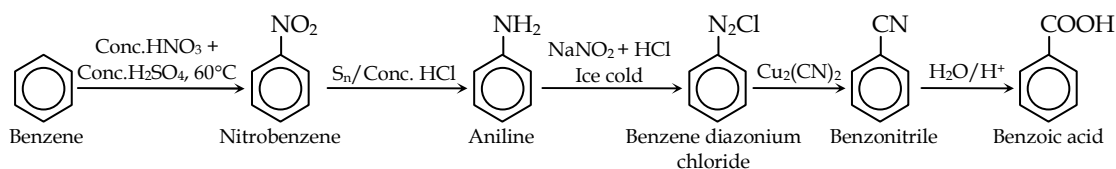
4. **By oxidation of benzyl alcohol:** Oxidation of benzyl alcohols with acidified KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$ gives benzoic acid



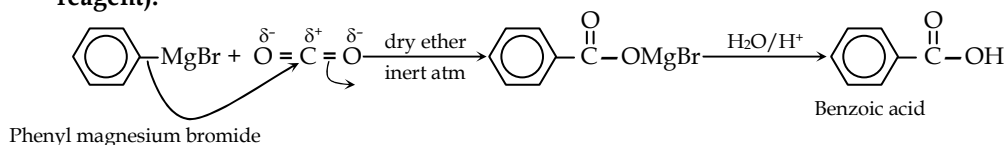
5. **From benzonitrile:** *Nitrile synthesis of carboxylic acid:* Hydrolysis of benzonitrile with dilute acid or dilute alkali gives benzoic acid.



The preparation can be performed from the following given compounds



6. **By Grignard synthesis:** By the reaction of CO_2 with phenyl magnesium bromide (Grignard reagent):



Self Test

13.14 Convert benzene into benzoic acid and vice versa.

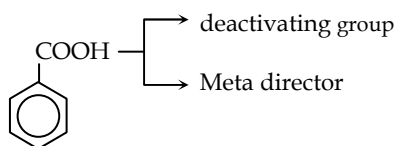
13.15 Which acid is formed when *p*-bromotoluene is oxidized?**Physical properties**

1. Benzoic acid is a white crystalline solid with melting point 121°C.
2. It is slightly soluble in cold water but highly soluble in hot water. It is also soluble in organic solvents.
3. It is steam volatile.

Chemical properties

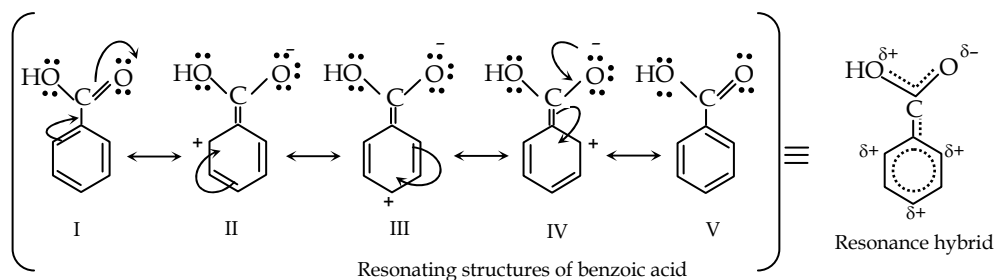
Reaction of benzoic acid can be categorized into:

- A. Reaction due to benzene ring
 - B. Reaction due to carboxyl group
- A. Reaction due to benzene ring: Electrophilic aromatic substitution reaction.**



Like other aromatic compounds, benzoic acid gives electrophilic aromatic substitution reaction.

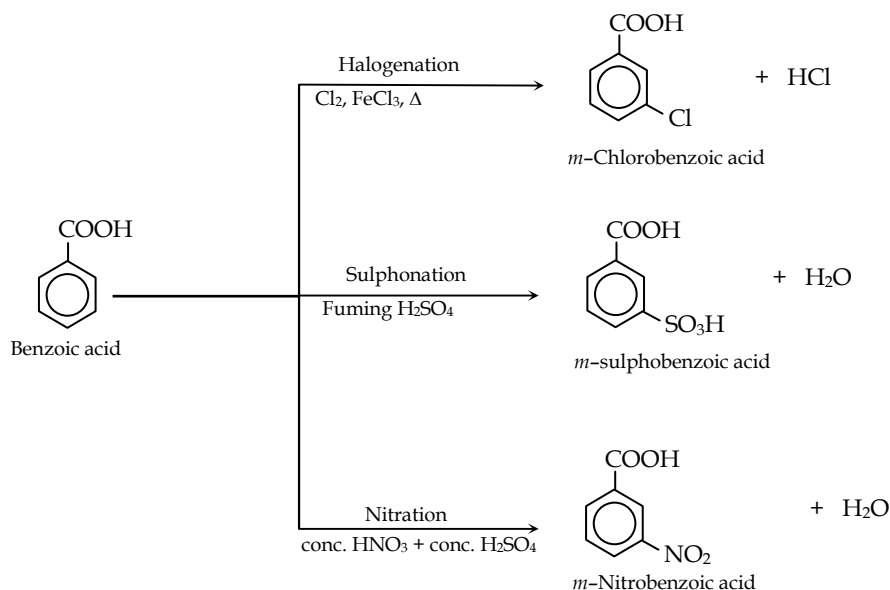
The carboxyl ($-\text{COOH}$) group is electron-withdrawing group which creates positive charge in the aromatic ring due to resonance effect. Therefore, $-\text{COOH}$ group deactivates the benzene ring towards electrophilic substitution reaction and hence makes the electrophilic attack difficult. The carboxyl ($-\text{COOH}$) group is *meta*-directing that directs the incoming electrophile towards *meta*-positions of the benzene ring to the $-\text{COOH}$ group, which can be described as follow:



In resonance hybrid of benzoic acid, the partial positive charge (δ^+) is developed at *ortho*- and *para*- positions. The electron density is low at *ortho*- and *para*- positions. So, electrophilic attack becomes difficult in these positions. Therefore, electrophile preferentially attacks at *meta*-positions to the $-\text{COOH}$ group to give *meta*-substituted products.

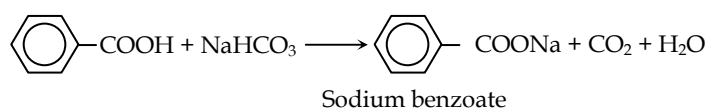
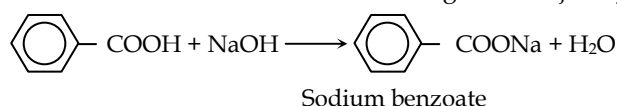
Hence, $-\text{COOH}$ group of benzoic acid is deactivating and *meta*-directing towards electrophilic substitution reaction. Deactivation is so strong that the Friedel-Crafts reaction does not take place.

Benzoic acid undergoes some common electrophilic reactions like nitration, sulphonation and halogenation vigorous at *meta*-positions.

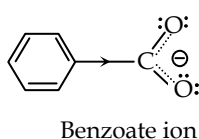
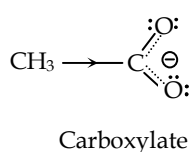


B. Reaction due to carboxyl group

1. **Acid nature:** Benzoic acid is acidic nature. The following reaction justify its acidic behaviour



Benzoic acid is less acidic than formic acid but more acidic than acetic acid. This is due to the fact that phenyl (C₆H₅-) group of benzoic acid is weaker electron releasing group than that of the methyl (CH₃-) group of acetic acid. The CH₃- group of acetate ion enhances the negative charge of the ion strongly, which makes it less stable than benzoate ion.

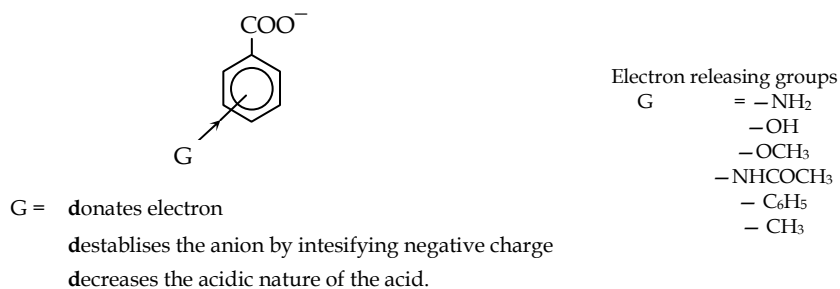


The order of acidic character is: HCOOH > C₆H₅COOH > CH₃COOH

13.2.1 Effect of substituents on acidity of carboxylic acids

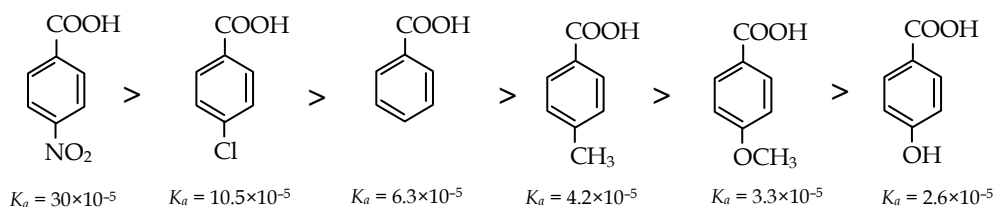
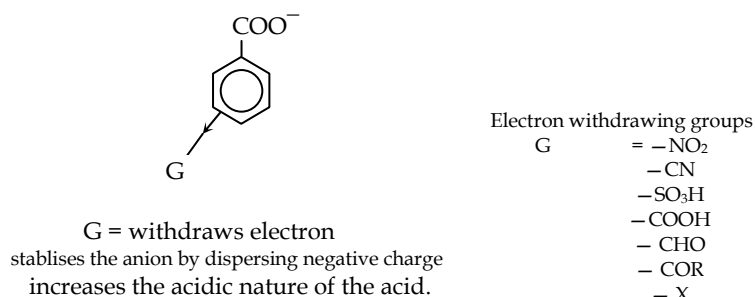
1. Effect of electron donating substituent

Any factor that destabilises the carboxylate ion decreases the acidity of the acid. The electron-releasing substituent donates electron, intensify the negative charge on carboxylate ion and destabilizes the carboxylate ion and thus decreases the acidic nature of carboxylic acid.

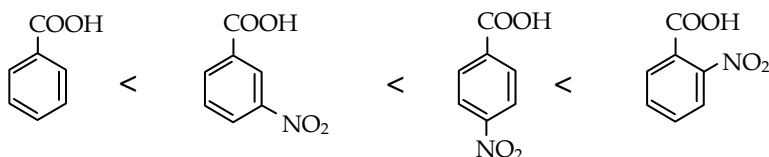


2. Effect of electron withdrawing substituent

Any factor that stabilises the carboxylate ion increases the acidity of the acid. The electron-withdrawing substituent withdraws electron, disperse the negative charge on carboxylate ion and stabilizes the carboxylate ion and thus increases the acidic nature of carboxylic acid.



3. Effect of the position of substitutions: The acidity is more pronounced at *ortho*-position as compared to that at *para*- and *meta*- positions. This is called ortho effect.

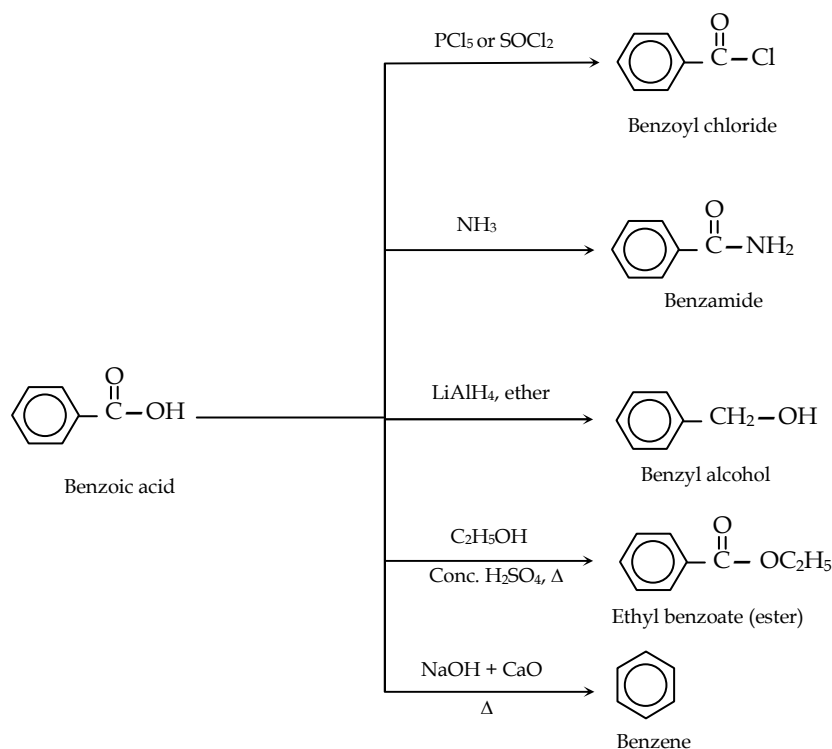


Self Test

13.16 Which one of the following is a stronger acid in each of the following pairs?

- m*-hydroxy benzoic acid and *p*-hydroxy benzoic acid.
- Ethanoic acid and benzoic acid.
- Methanoic acid and benzoic acid.

4. The other reactions of benzoic acid are summarized as:

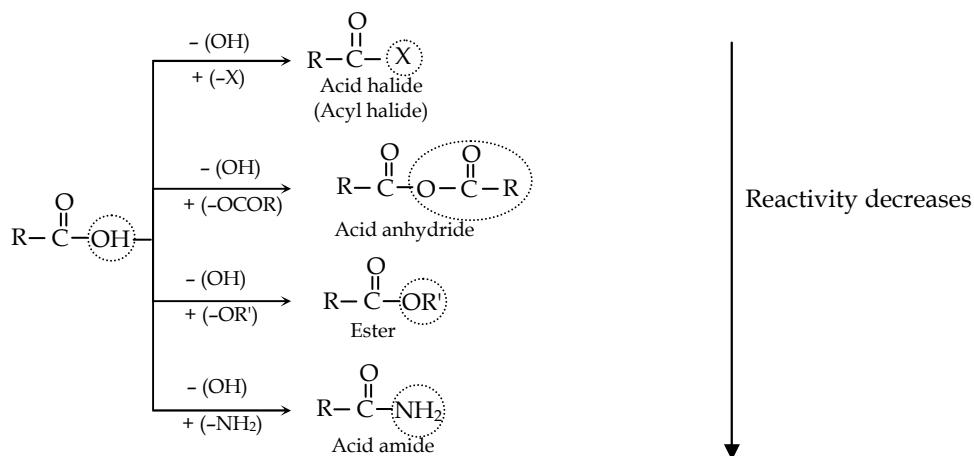


Uses of Benzoic acid

1. Salt of benzoic acid like sodium benzoate is used as food preservatives.
2. Esters of benzoic acid are used in perfumery.
3. It is used to make aniline blue dye.
4. It is used in synthesis of various organic compounds.

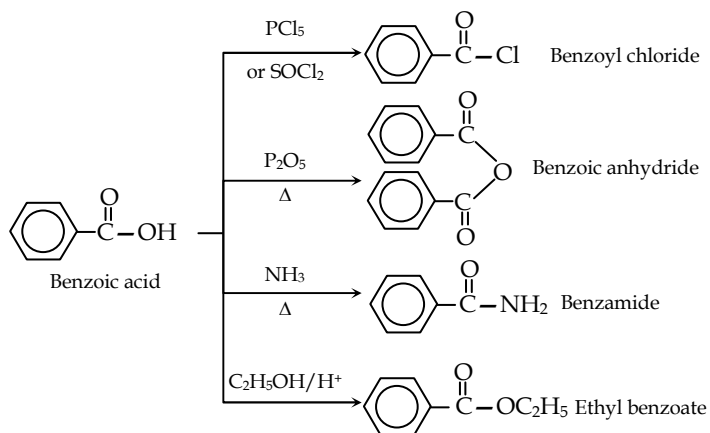
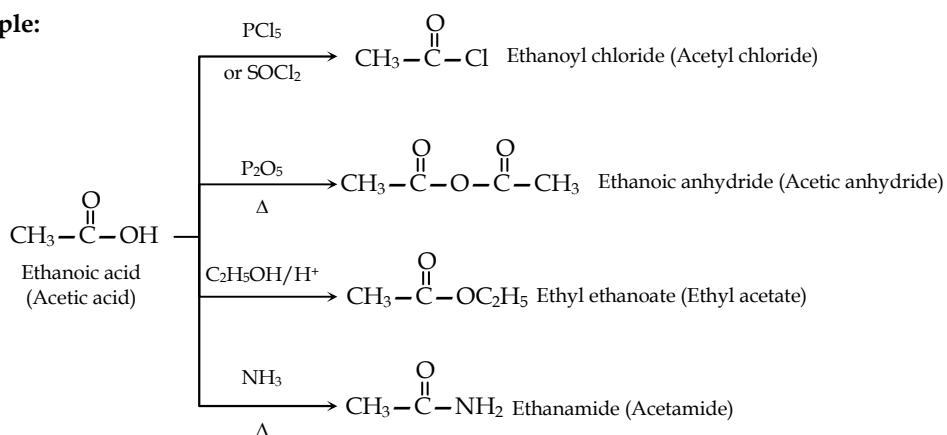
13.3 Functional Derivatives of Carboxylic Acids

Functional derivatives of carboxylic acids are closely related to those classes of organic compounds which can be structurally obtained by the replacement of $-\text{OH}$ part of the $-\text{COOH}$ group of the carboxylic acid by electron rich species. The electrons rich species includes $-\text{Cl}$, $-\text{OR}'$, $-\text{NH}_2$ and $-\text{OCOR}'$ groups which yield acid chloride ($\text{R}-\text{COCl}$), ester ($\text{R}-\text{COOR}'$), acid amide ($\text{R}-\text{CONH}_2$) and acid anhydride ($\text{R}-\text{COOCOR}'$) respectively. These compounds are, therefore called **functional derivatives of carboxylic acids**.



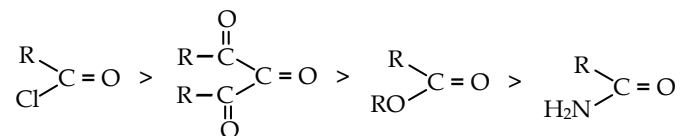
Acid derivatives where R and R' may be aliphatic or aromatic.

Example:



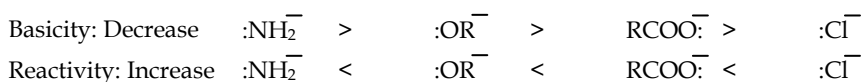
Relative reactivity of acid derivatives

The reactivity of the acid derivatives for nucleophilic substitution reaction in general, follows the order:

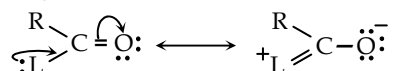


The above order of reactivity can be explained in terms of the: (i) Basicity of leaving group (ii) resonance effect.

- 1. Basicity of leaving groups:** Weaker the basic character of the leaving group (anion) is, more will be the ease with which the bond can be ruptured and hence more is the reactivity. Among the given compounds Cl^- ion is the weakest base and hence chlorides are most reactive. The correct order of basicity of the leaving group and their tendency to leave is:



- 2. Resonance effect:** In general, all the acid derivatives show resonance as follows:

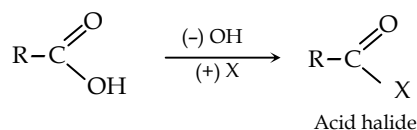


L = leaving group: $-\text{Cl}$, $-\text{OCOR}$, $-\text{OR}$, $-\text{NH}_2$

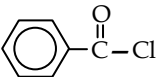
In the resonating structures, it is clear that the carbon to leaving group (L) bond acquires a double bond character due to which stabilization occurs. Now more is the stabilization, lesser is the reactivity and vice-versa. As the stabilization is the least in the case of acid chloride because the X-atom exerts a high magnitude of $-\text{I}$ electron withdrawing effect as compared to the other groups, therefore, its reactivity is the most.

13.3.1 Acid Halides (Acyl Halides)

Acid halides are halogen derivatives of carboxylic acid that are obtained by replacing $-\text{OH}$ group of carboxyl group by halogens ($-\text{Cl}$, $-\text{Br}$, $-\text{I}$).

**Nomenclature**

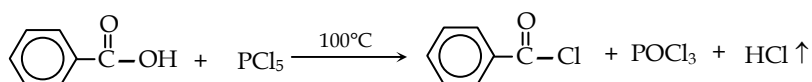
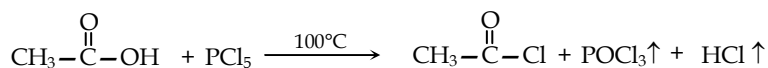
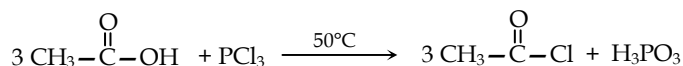
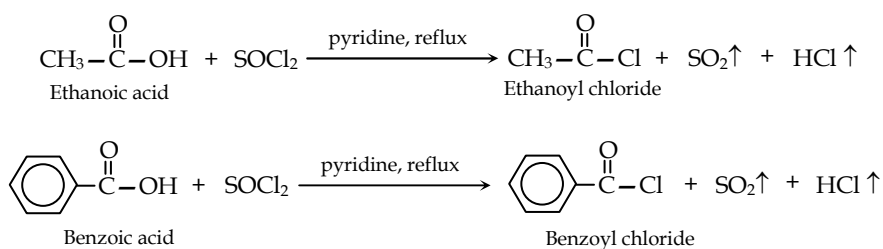
The names of acyl chlorides are derived from the common or IUPAC names of the parent acids by replacing the suffix *-ic acid* with *-yl chloride*.

Compounds	Common name	IUPAC name
$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{X}$	Acyl halide	Alkanoyl halide
$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$	Acetyl chloride	Ethanoyl chloride
$\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$	Propionyl chloride	Propanoyl chloride
	Benzoyl chloride	Benzoyl chloride

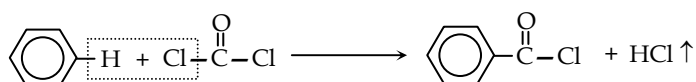
HCOCl is unstable.

Preparation

- From carboxylic acid:** When carboxylic acid is refluxed with thionyl chloride (SOCl_2), phosphorus trichloride (PCl_3) or phosphorus pentachloride (PCl_5), corresponding acid chloride is formed.



- Using phosgene:** Aromatic acid chloride can be prepared by treating arenes with phosgene.



Physical properties

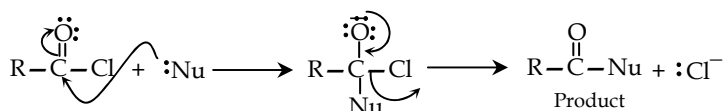
- Physical state:** Lower members of acid chlorides are colourless liquid having irritating odour and higher members are colourless solid.
- Boiling point:** The boiling point of acid chloride is less than that of parent acid due to absence of intermolecular H-bonding.
- Solubility:** Acid chlorides are insoluble in water due to lack of intermolecular H-bonding but soluble in organic solvents.

Self Test

13.17 Acyl chlorides have lower boiling points than corresponding acids, explain.

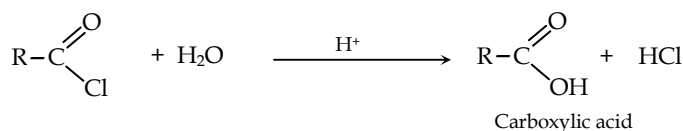
Chemical properties

The most common and typical reaction of acid chloride is nucleophilic substitution.

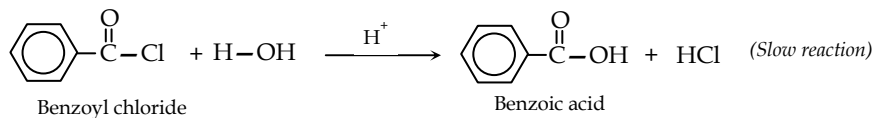
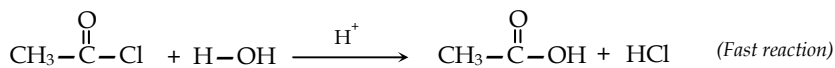


Acid chlorides are most reactive of the functional derivative of carboxylic acids. This is due to -I effect of -Cl and hence a partial positive charge on carbonyl carbon is intensified. Aromatic acid chlorides (ArCOCl) are considerably less reactive than the aliphatic acid chlorides (RCOCl).

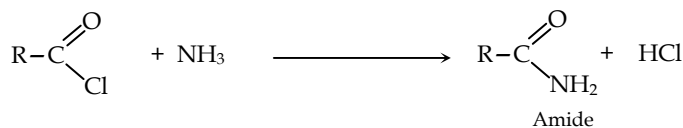
1. **Reaction with water: Hydrolysis:** Acid chloride reacts with water, to give parent carboxylic acid.



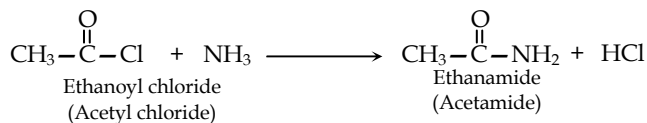
Examples:



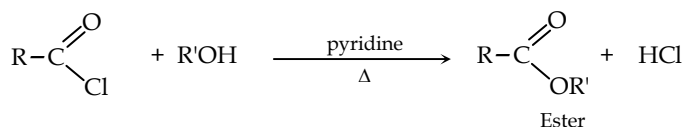
2. **Reaction with ammonia: Ammonolysis:** Acid chloride reacts with ammonia, to give amide.

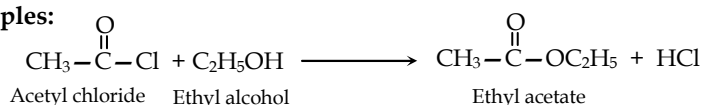


Examples:

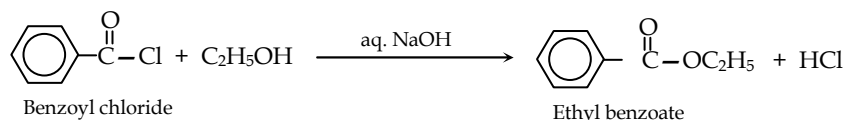
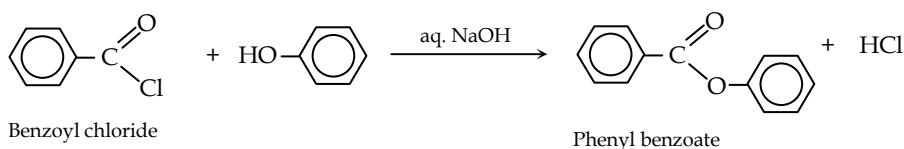


3. **Reaction with alcohol: Alcoholysis:** Acid chloride reacts rapidly with alcohols to give ester.



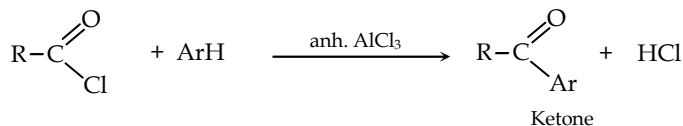
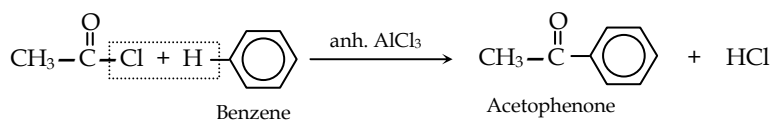
Examples:

4. **Schotten-Baumann reaction: Benzoylation:** The reaction of aromatic acid chlorides (Benzoyl chloride) with an alcohol or a phenol in presence of a base, usually aqueous sodium hydroxides or pyridine to give aromatic esters is called *Schotten-Baumann* reaction

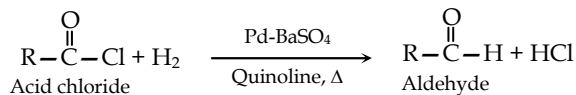
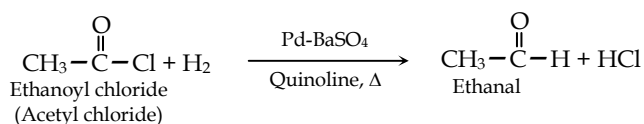
**Example:**

In this reaction, acid chloride is added in portions followed by vigorous shaking to a mixture of alcohol or phenol and a base. Base serves not only to neutralize the HCl that would be liberated, but also to catalyze the reaction.

5. **Reaction with benzene: Friedel-Crafts acylation:** When acid chloride is treated with benzene in presence of lewis acid (anh. AlCl_3) as a catalyst, aromatic ketone is formed. This reaction is an example of Friedel-Crafts acylation.

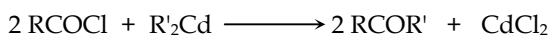
**Example:**

6. **Reduction into aldehyde: Rosenmund reaction:** When acid chlorides are heated with H_2 in presence of Pd-BaSO_4 partially poisoned by quinoline, aldehydes are obtained. This reaction is called Rosenmund reaction.

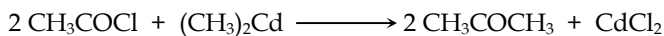
**Example:****Note**

If LiAlH_4 is used as a reducing agent here, aldehydes are further reduced to alcohols.

7. **Reduction with dialkyl cadmium:** Acid chlorides reacts with dialkyl cadmium to form ketones.

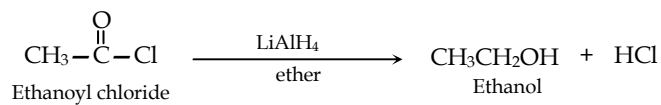


Example:

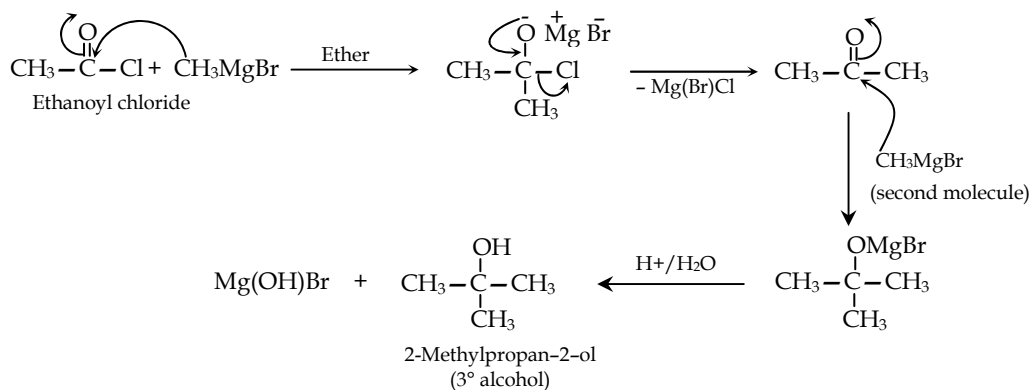


8. **Reduction into alcohol:** Acid chloride is reduced by LiAlH_4 to form primary alcohol.

Example:



9. **Reaction with Grignard reagent:** When acid chloride is treated with Grignard reagent (RMgX), an addition product of *adduct*, which upon hydrolysis gives tertiary (3°) alcohol.

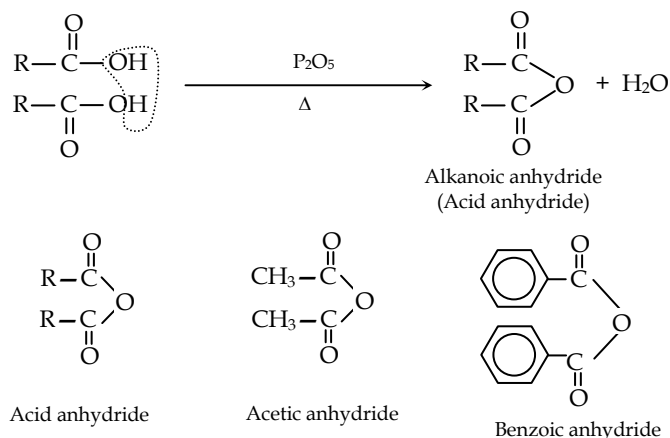


Self Test

13.18 Why does acetyl chloride fume on air?

13.3.2 Acid Anhydrides

Functional group of acid anhydride is $-\text{CO}-\text{O}-\text{CO}-$. The most important member of acid anhydride is acetic anhydride. The name is given since it is formed by intermolecular dehydration of the parent acid.

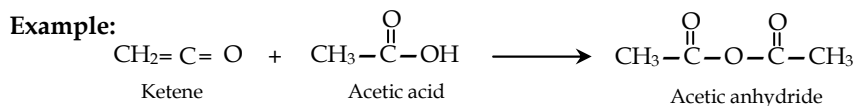


Nomenclature

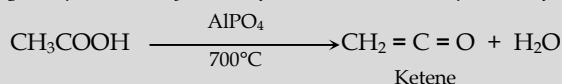
Compound	Common name	IUPAC name
$R-\overset{\overset{O}{\parallel}}{C}-O-\overset{\overset{O}{\parallel}}{C}-R$	Acid anhydride	Alkanoic anhydride
$CH_3-\overset{\overset{O}{\parallel}}{C}-O-\overset{\overset{O}{\parallel}}{C}-CH_3$ or $(CH_3CO)_2O$	Acetic anhydride	Ethanoic anhydride
	Benzoic anhydride	Benzoic anhydride

Preparation

1. **From ketene:** By the reaction of acetic acid with ketene, an acetic anhydride is formed.



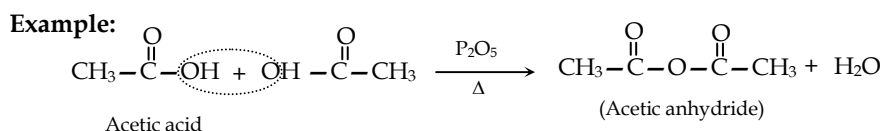
Ketene is prepared at high temperature dehydration of acetic acid to 700° in presence of aluminum phosphate (AlPO₄).



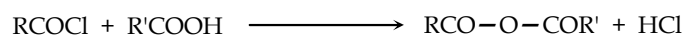
Ketene is an extremely reactive, it is also prepared in laboratory by pyrolysis of acetone, and ordinarily used as soon as it is prepared



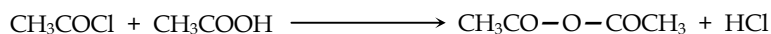
2. **From carboxylic acid:** When carboxylic acid is heated with strong dehydrating agent like P₂O₅, it gives acid anhydride.



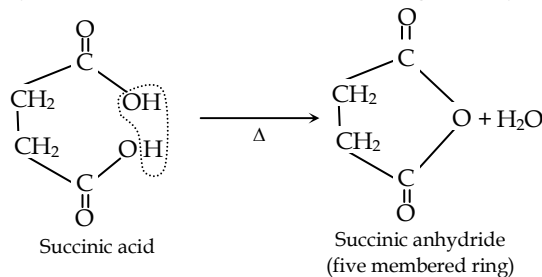
3. **From acid chloride:** Acid chloride reacts with carboxylic acid to form acid anhydride.



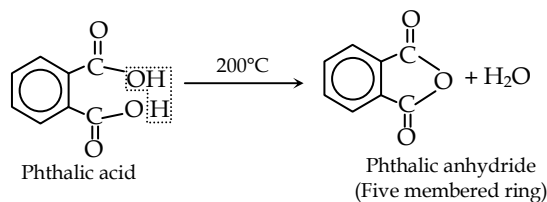
Example:



4. **By heating dicarboxylic acids:** Dehydration of certain dicarboxylic acid yield acid anhydride on simple heating only, where five- or six- membered ring of anhydride is produced.



Example:

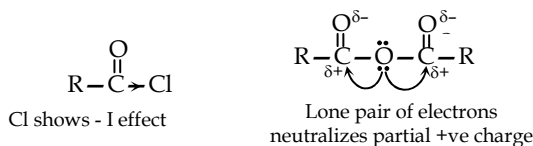


Physical properties

1. **Physical state:** Lower members of acid anhydrides are colourless liquids with sharp pungent smell and higher members are colourless solid.
2. **Solubility:** Lower members are soluble in water.
3. **Boiling point:** They have higher boiling point than their parent acid because of large molecular size.

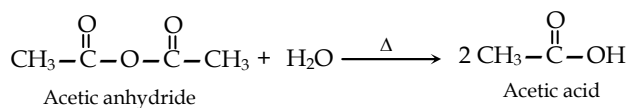
Chemical properties

Acid anhydrides give nucleophilic substitution reaction but less reactive than acid chlorides.



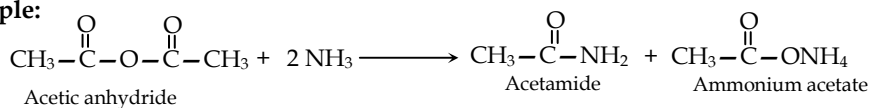
1. **Reaction with water: Hydrolysis:** Acid anhydride on hydrolysis yield parent carboxylic acid.

Example:



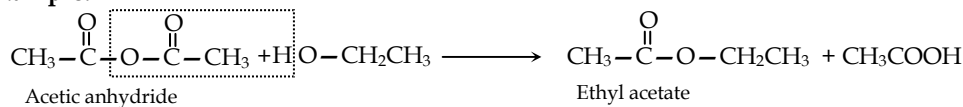
2. **Reaction with ammonia: Ammonolysis:** Acid anhydride reacts with ammonia to give amide.

Example:



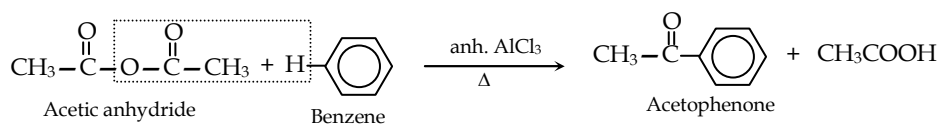
3. **Reaction with alcohol: Alcoholysis:** Acid anhydride reacts with alcohol to give ester and acid.

Example:



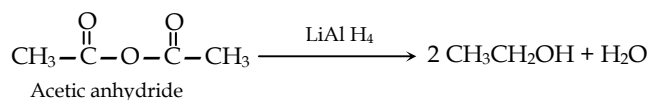
4. **Friedel-Crafts acylation: Formation of ketone:** Acid anhydride reacts with benzene in presence of lewis acid like anhydrous AlCl₃ as catalyst, aromatic ketone is formed. This reaction is called Friedel-Crafts acylation.

Example:



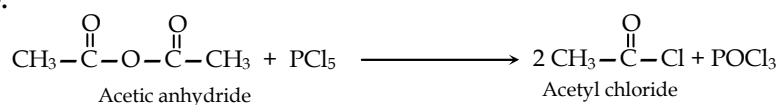
5. **Reduction into alcohol:** Acid anhydride is reduced by LiAlH_4 into primary (1°) alcohol.

Example:



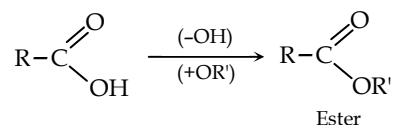
6. **Reaction with PCl_5 :** When acid anhydride is treated with PCl_5 , acid chloride is produced.

Example:



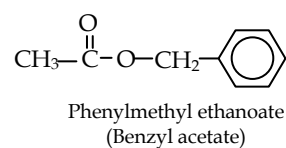
13.3.3 Esters

Ester is a functional derivative of carboxylic acid, which is obtained by replacing $-\text{OH}$ group of carboxylic acid by $-\text{OR}$ group.



Flowering jasmine plant

The natural flavours of many fruits and fragrance of flowers are blends of many organic compounds. Many of them are esters. For example, phenylmethyl ethanoate (an ester) is the major constituent of oil of jasmine. It is naturally obtained from jasmine plant. Now, it is produced by chemical synthesis for perfumes industry.



The flavours and fragrances of many esters are widely used in food flavourings and perfumes.

Ester	Flavour
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3\text{CH}_2-\text{CH}-\text{C}(=\text{O})-\text{OC}_2\text{H}_5 \end{array}$ Ethyl 2-methylbutanoate	Apple
$\begin{array}{c} \text{O} \quad \text{CH}_3 \\ \quad \\ \text{CH}_3-\text{C}-\text{O}-\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \end{array}$ 1-Methyl butylethanoate	Banana
$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CH}_2\text{CH}_2-\text{C}-\text{O}-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 \end{array}$ Butyl butanoate	Pineapple

Fats and oils are also esters of glycerol (trihydric alcohol) with fatty acids.

Esters are used as solvent in chemical industries. For example, ethylethanoate (ethyl acetate) is used as nail polish remover.

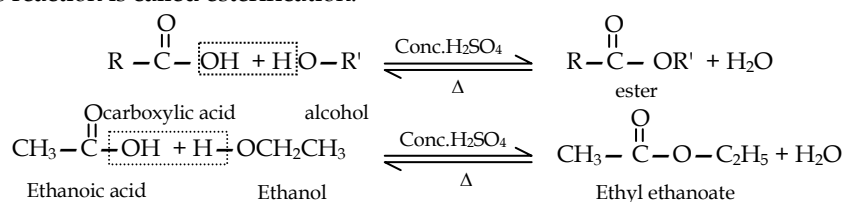
Nomenclature

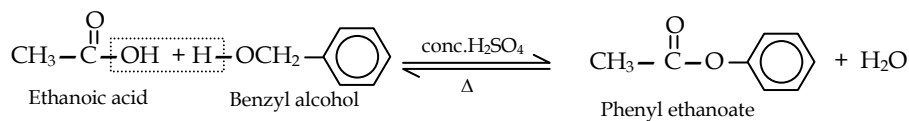
The ester are named by writing the alkyl or aryl group of $-\text{OR}$ part of the molecule before the names of the parent carboxylic acid and replacing the suffix $-ic$ of the common or IUPAC name of the acid by $-oate$.

Compound	Common name	IUPAC name
$\text{R}-\text{C}(=\text{O})-\text{OR}'$	Alkyl carboxylate	Alkyl alkanoate
$\text{H}-\text{C}(=\text{O})-\text{OCH}_3$	Methyl formate	Methyl methanoate
$\text{H}-\text{C}(=\text{O})-\text{OC}_2\text{H}_5$	Methyl acetate	Ethyl methanoate
$\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{C}_2\text{H}_5$	Ethyl acetate	Ethyl ethanoate
 $-\text{C}(=\text{O})-\text{O}-\text{CH}_3$	Methyl benzoate	Methyl benzoate

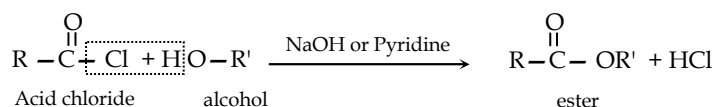
Preparation

- From carboxylic acids: Esterification:** When carboxylic acid is heated with an alcohol in presence of dehydrating agents like concentrated H_2SO_4 , fruity odour of an ester is obtained. This reaction is called esterification.

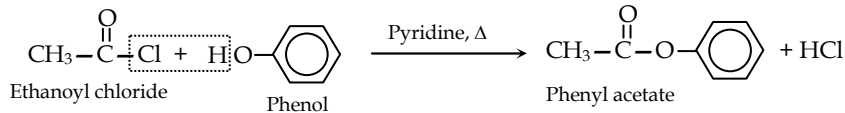
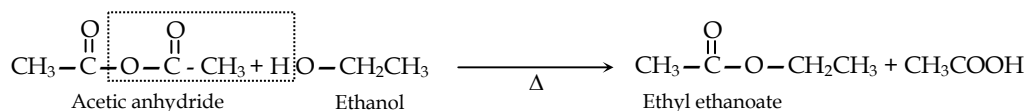
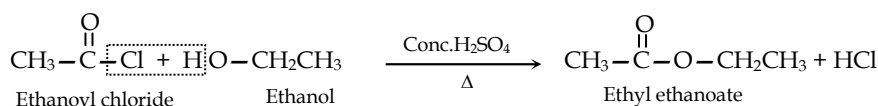
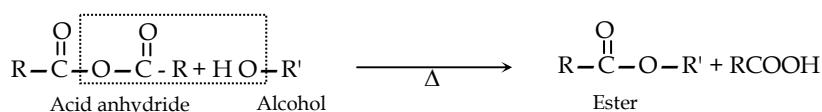




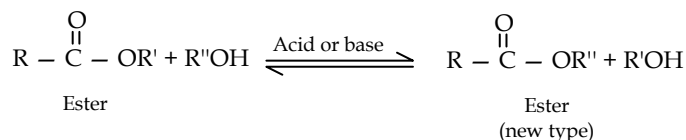
2. **Acylation: From acid chlorides or acid anhydrides:** When acid chloride or acid anhydride is heated with alcohol in presence of pyridine or conc. H_2SO_4 , ester is obtained.



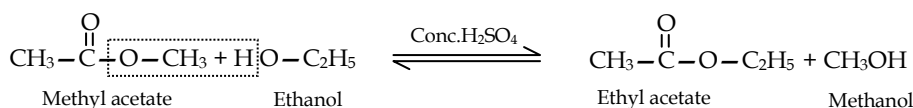
Examples:



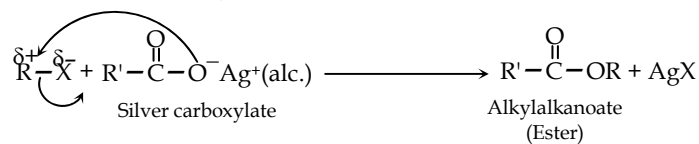
3. **By trans-esterification:** When an ester reacts with alcohol in presence of conc. H_2SO_4 , a new ester is formed. The reaction is called trans-esterification.



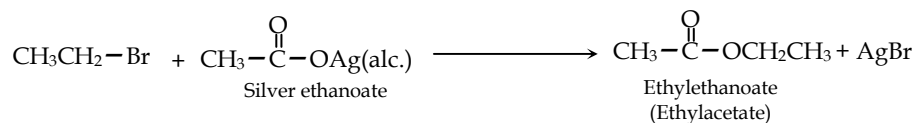
Examples:



4. **Action of alcoholic silver salt of carboxylic acid:** When alkyl halide is refluxed with alcoholic solution of silver salt of carboxylic acid, ester is formed.



Examples:

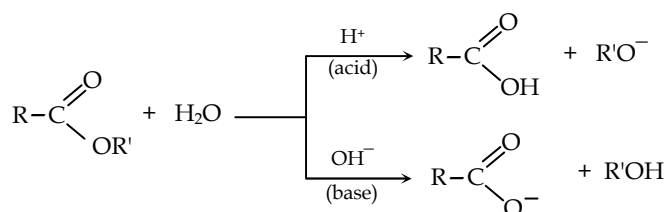


Chemical properties

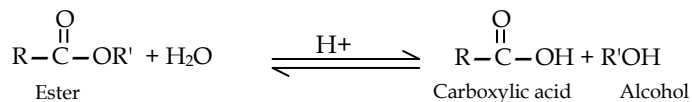
Esters are less reactive than acid chloride and acid anhydride but more reactive than amide towards nucleophilic substitution reaction.



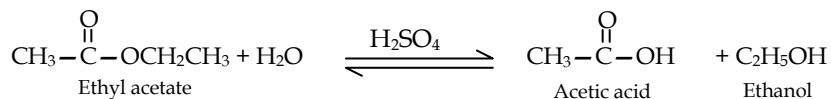
1. **Hydrolysis:** Ester can be hydrolyzed in acidic as well as in alkaline medium.



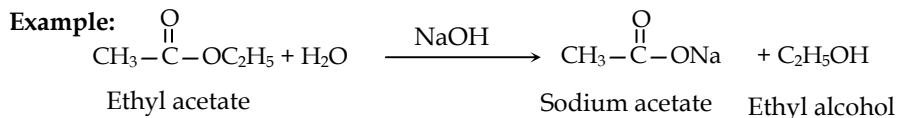
- a. **Conversion into acids: Acidic hydrolysis:** Hydrolysis of ester with dilute solution of mineral acid gives carboxylic acid and alcohol. The reaction is reversible.



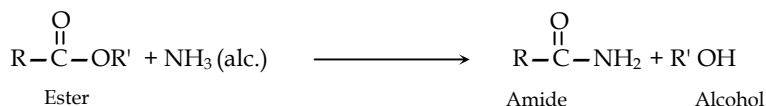
Examples:

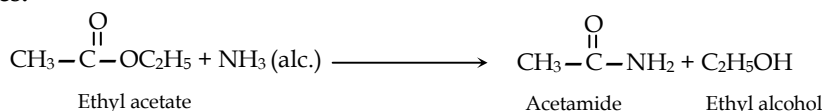


- b. **Alkaline hydrolysis:** Hydrolysis of ester with dilute solution of base gives carboxylate salt and alcohol.



2. **Conversion into amide: Reaction with NH₃: Ammonolysis:** On treatment of an ester with ammonia in presence of alcohol yields amide.

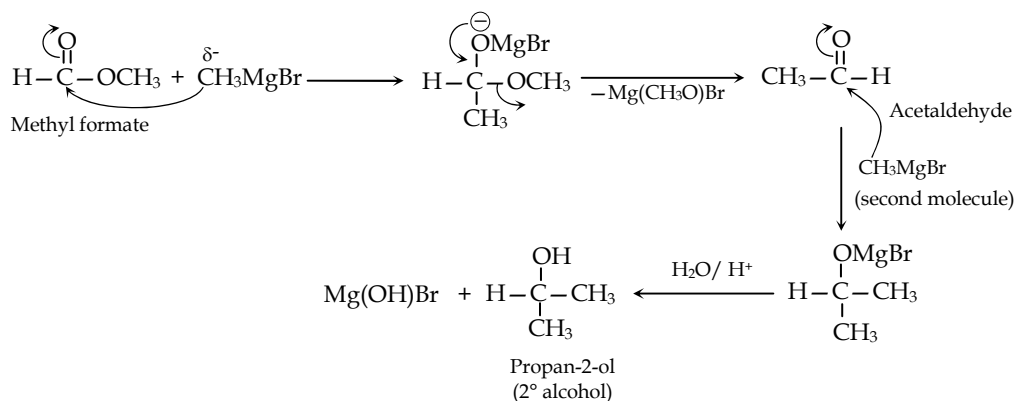




3. **Reaction with Grignard reagent:** An ester reacts with Grignard reagent, forms aldehyde or ketone at first, which further reacts with another molecule of Grignard reagent followed by hydrolysis, gives corresponding alcohol.

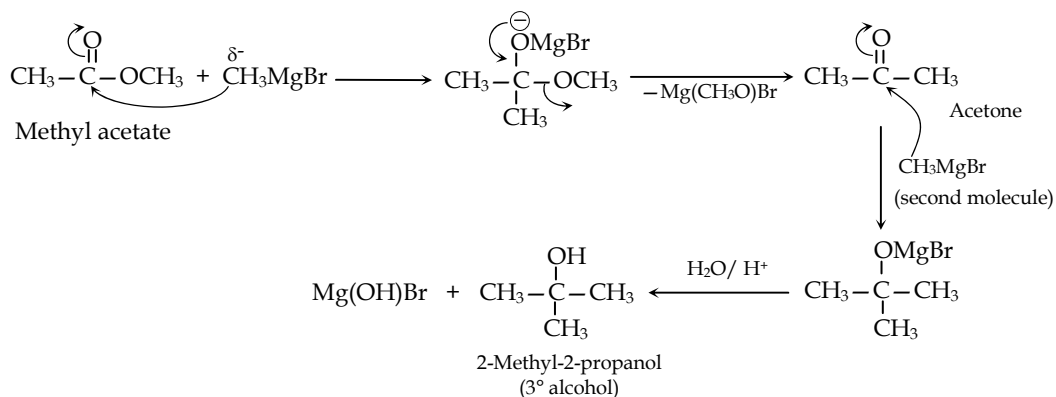
- i. Ester of formic acid reacts with Grignard reagent, then aldehyde is formed at first, which further reacts with another molecule of Grignard reagent followed by hydrolysis and gives secondary (2°) alcohol.

Example:



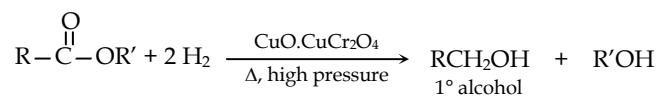
- ii. When an ester of carboxylic acid other than formic acid react with Grignard reagent, then ketone is formed at first, which further reacts with another molecule of Grignard reagent followed by hydrolysis and gives tertiary (3°) alcohol.

Example:

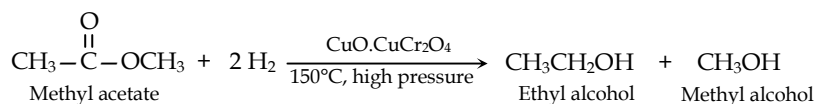


4. **Reduction into alcohol:** Esters are reduced to primary (1°) alcohol from acid part of the ester. The reduction can takes place in the following ways:

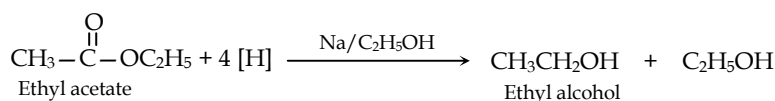
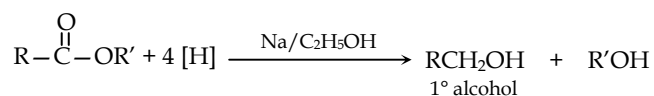
- i. **Catalytic hydrogenation: Hydrogenolysis:** Catalytic hydrogenation of ester with molecular hydrogen in presence of **Adkin catalyst** ($\text{CuO} \cdot \text{CuCr}_2\text{O}_4$) gives primary alcohol from acid part of ester.



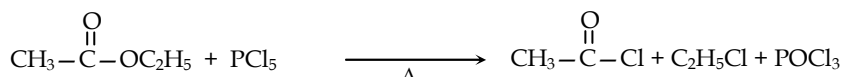
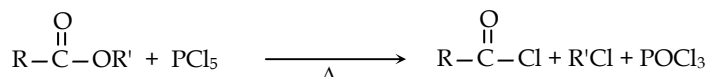
Example:



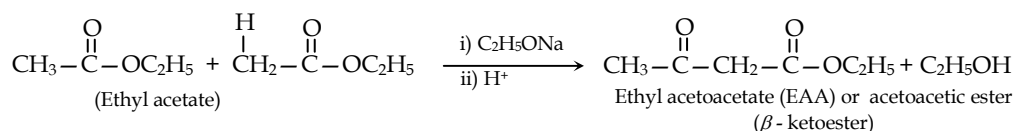
- ii. **Chemical reduction:** When an ester is reduced by reducing agent like Na/C₂H₅OH or LiAlH₄, primary alcohol is formed from acid part of ester.



5. **Conversion into acid chloride:** Action of phosphorous chloride on esters.



6. **Claisen condensation:** When an ester having at least one α-hydrogen atom undergoes self condensation reaction in presence of strong base like sodium ethoxide (C₂H₅ONa), ethyl acetoacetate or acetoacetic ester is formed. Acetoacetic ester is β-ketoester. The reaction is known as **Claisen condensation**.

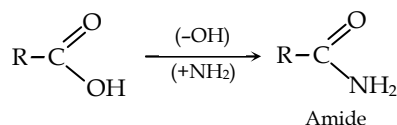


Self Test

13.19 Why is ethyl acetate more reactive than acetyl chloride?

13.3.4 Acid Amides

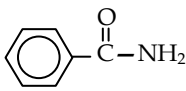
Acid amides are functional derivative of carboxylic acids in which -OH group of carboxylic acid is replaced by -NH₂ group. They are often simply called as amides.



The amide is aliphatic if R = alkyl group and aromatic if R = aryl group.

Nomenclature

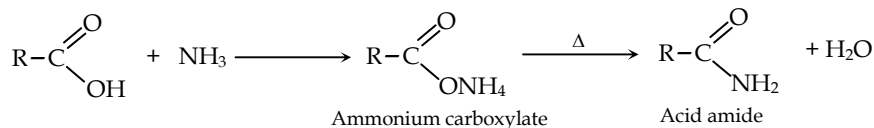
Acid amides are named by replacing the suffix *-ic acid* of common or IUPAC name of the parent carboxylic acids by *-amide*.

Compounds	Common name	IUPAC name
$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$	Acidamide	Alkanamide
$\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$	Formamide	Methanamide
$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$	Acetamide	Ethanamide
	Benzamide	Benzamide

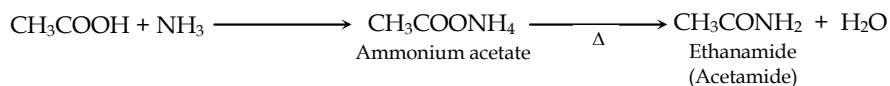
Preparation

1. Amides are prepared from carboxylic acids by two ways:

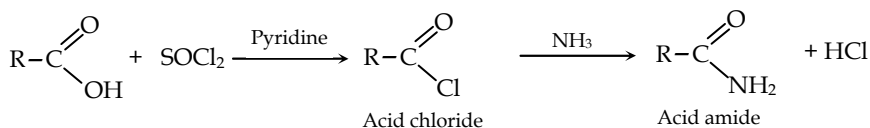
- i. Using ammonium salt: Carboxylic acid reacts with ammonia to form ammonium carboxylate (ammonium salt), which upon strong heating loses H_2O to give amide.



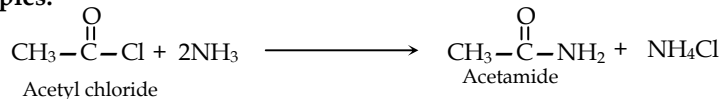
Examples:



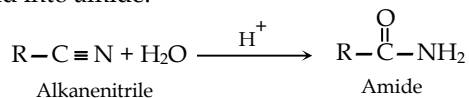
- ii. By using acid chlorides : Acid chloride reacts with ammonia to give amide.



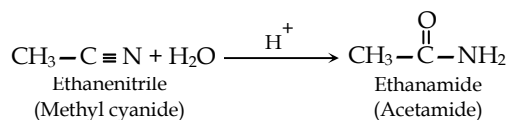
Examples:



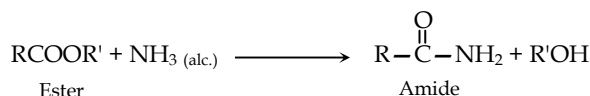
2. From alkane nitrile: By partial hydrolysis: Alkane nitrile ($\text{R}-\text{C}\equiv\text{N}$) is partially hydrolyzed with concentrated acid into amide.



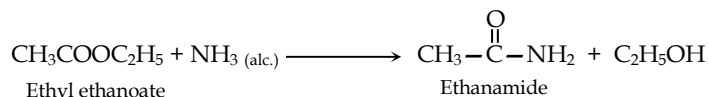
Examples:



- 3. From esters: Ammonolysis:** Action of alcoholic solution of ammonia on ester gives amide.

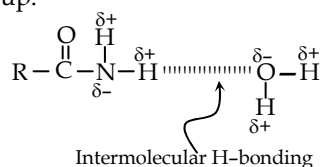


Examples:



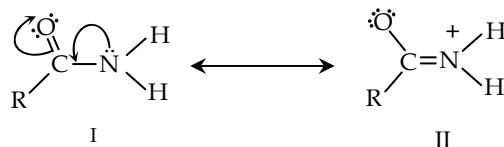
Physical properties

1. **Physical state:** The first member, formamide, is liquid. All the other amides are colourless crystalline solids.
2. **Solubility:** The amides with carbon number up to six are soluble in water due to the fact that they can form intermolecular hydrogen bonding with water. Solubility decreases with increase in size of alkyl group.



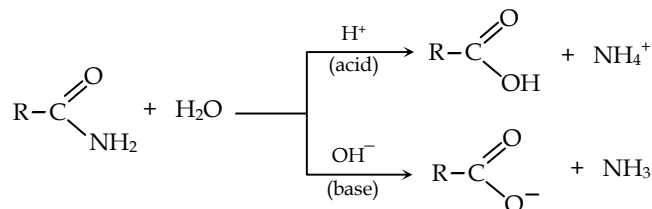
Chemical properties

Amides are the least reactive of functional derivatives of carboxylic acids towards nucleophilic substitution reaction. This is due to less polarity of C-N bond as compared to C-O bond and due to resonance.

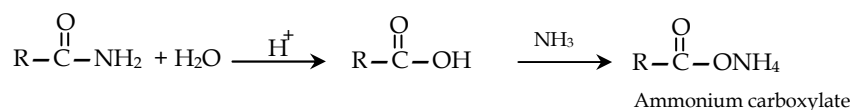


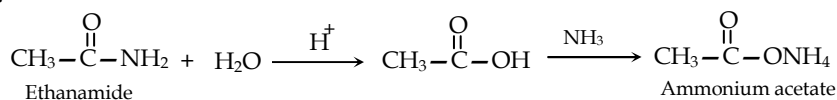
From the given resonating structures, it is clear that amide is amphoteric in nature. The presence of partial positive charge on nitrogen atom in resonance structure (II) can easily lose proton (H^+) to behave as an acid. On the other hand, presence of lone pair of electrons on nitrogen atom in resonance structures (I) can donate lone pair of electron to acid as a base. Thus, amides are weak base than amines.

- 1. Hydrolysis:** Hydrolysis of amide is nucleophilic substitution reaction, in which the $-\text{NH}_2$ group of amide is replaced by $-\text{OH}$ group.

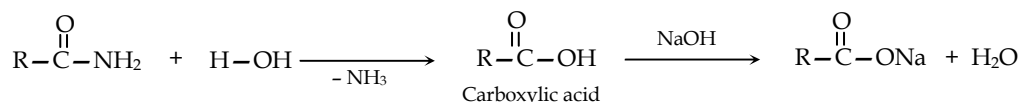
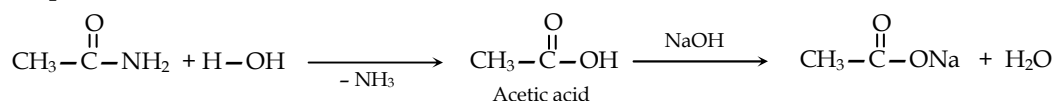


- a. **Acidic hydrolysis:** Hydrolysis of amide in acidic medium gives carboxylic acid, which in turn reacts with NH_3 to give ammonium carboxylate.

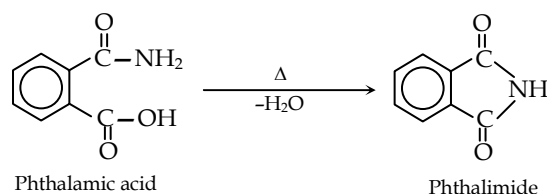


Example:

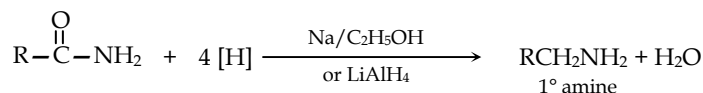
- b. Alkaline hydrolysis:** Hydrolysis of amide in basic medium gives carboxylic acid, which in turn reacts with alkali to give carboxylate salt and ammonia.

**Example:**

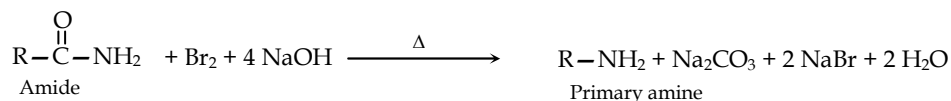
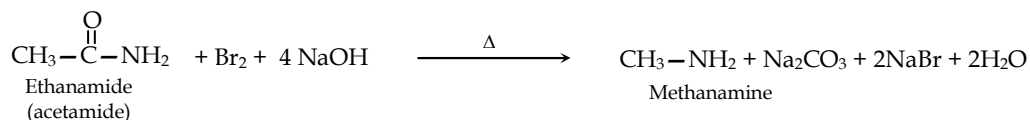
- 2. Conversion into imides:** An organic compound with both -CONH₂ and -COOH groups in its molecule on simple heating eliminates a molecule of H₂O and a cyclic ring compound is formed, the product thus, obtained is called imide.



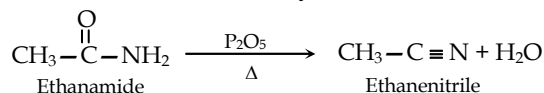
- 3. Reduction:** Amides are reduced by either LiAlH₄/ether or by Na/C₂H₅OH into corresponding primary amine.

**Example:**

- 4. Conversion of amides into primary amines: Hofmann bromamide reaction: Hoffman rearrangement:** (See amines unit 15 for details.)

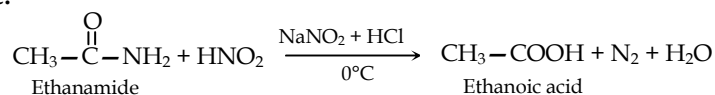
**Example:**

- 5. Reaction with P₂O₅:** When amide is heated with anhydrous P₂O₅, alkane nitrile is formed.



6. **Reaction with HNO_2 :** Amide reacts with nitrous acid to give carboxylic acid with the evolution of nitrogen gas.

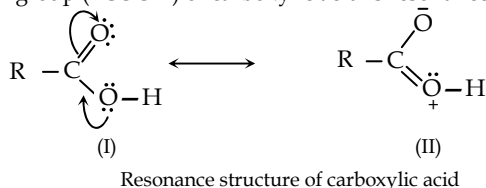
Example:



■ Solved Questions

1. **Why carboxylic acids do not give the characteristics reactions of carbonyl group?**

→ The functional carboxyl group ($-\text{COOH}$) of carboxylic acid is resonance hybrid of the following structures



In these structures, the carbonyl part ($>\text{C}=\text{O}$) of the carboxyl group does not have a double bond character but has a reduced double bond character. Hence, carboxylic acids do not give the characteristic reactions of carbonyl group.

■ Questions

■ Very Short Answer Questions:

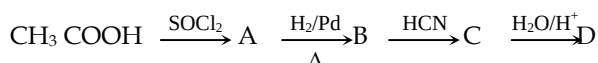
- Name the type of isomerism that can be shown by an acid with formula $\text{C}_4\text{H}_8\text{O}_2$.
- Write the functional isomer of ethanoic acid and give its IUPAC name.
- You are given two test tubes one containing methanoic acid and another containing ethanoic acid. Suggest a suitable chemical test to identify them.
- How would you prepare benzoic acid from (a) Benzene (b) Aniline
- Why are carboxylic acids stronger acids than phenols and alcohols?
- How is benzoic acid prepared by the carbonation of Grignard reagent?
- Why is formic acid not fractionated to get anhydrous formic acid?
- Why is ethanoic acid halogenated with halogens in red phosphorus but methanoic acid is not halogenated?
- Why is amide amphoteric in nature?
- In the esterification reaction of carboxylic acid with alcohol, the ester so formed is distilled as fast as it is formed, why?
- Why is benzamide less easily hydrolyzed than methyl benzoate?
- Suggest method to prepare benzoic acid from ethanol.
- How can the presence of carboxylic acid be detected?

■ Short /Long Answer Questions:

- How can you convert the following organic compounds:
 - Ethanoyl chloride to (i) methanol (ii) Acetaldehyde
 - Ethanoic acid to methanoic acid and vice versa.
 - Ethanoic acid to (i) Ethanoic anhydride (ii) Ethanamide (iii) Methanol (iv) Ethanol (v) Methane (vi) Methyl ethanoate (vii) Ethyl ethanoate (viii) Chloroform
 - Aniline to benzoic acid and vice versa.

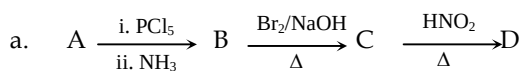
454 Modern Approach to Chemistry-II

- (e) Toluene to (i) benzoyl chloride (ii) *m*-nitrobenzoic acid
- (f) Benzyl alcohol to benzoic acid.
- (g) Benzene to phenyl acetic acid.
- (h) Benzamide to (i) Toluene (ii) Phenyl isocyanide
- (i) Benzoic acid to (i) benzophenone (ii) benzene
- (j) Benzoyl chloride to benzoic acid
2. What happen when:
 - (a) Ethanoyl chloride and ammonia is heated with Br₂ and aqueous KOH.
 - (b) Ethanoic acid is allowed to react with (i) NaOH (ii) NaOH/CaO (iii) PCl₅ (iii) Heated with P₂O₅ (iv) SOCl₂ (v) LiAlH₄ (vi) HI in presence of red phosphorus (vii) Hot MnO (viii) Warmed with ethanol in presence of conc. H₂SO₄. (ix) Ethanol/conc.H₂SO₄
 - (c) Methanoic acid act upon: (i) Conc. H₂SO₄ (ii) methanol / H⁺ (iii) Fehling's solution (iv) ammoniacal silver nitrate
 - (d) Benzoic acid is nitrated
 - (e) Ethyl acetate is treated with methyl magnesium iodide
 - (f) Acetyl chloride reacts with phenol in presence of anhydrous AlCl₃.
 - (g) benzamide is heated with bromine and aq. KOH.
3. How would you prepare ethanoic acid from:
 - (a) Ethane (b) Ethene (c) Ethyne (d) Iodomethane (e) Methanol (f) Methyl magnesium iodide (g) Tribromoethane (h) Ethanenitrile
4. Explain and give an example of each of: (a) Carboxylation reaction (b) decarbonylation reaction (c) Perkin condensation (d) Hell-Volhard-Zelinsky reaction (e) Claisen condensation (f) Kolbe's electrolytic decarboxylation (g) Friedel-Craft acylation
5. How is anhydrous formic acid prepared in the laboratory? Why does formic acid give silver mirror with tollen's reagent?
6. Why is P₂O₅ not used for the preparation of anhydrous formic acid? Suggest a suitable method for the preparation of anhydrous formic acid from impure farmic acid.
7. Identify each lettered in the following reaction sequence.

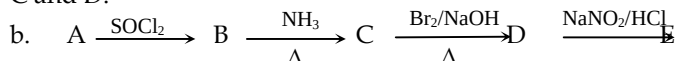


What major product would you obtain when B is treated with alkaline solution of hydrazine and ethylene glycol?

8. Consider the following reactions:



The compound A is a carboxylic acid. Calcium salt of A on heating gives acetone. Identify A, B, C and D.



The compound E is a primary alcohol which has positive iodoform test. Identify A, B, C, D and E.

Multiple Choice Questions

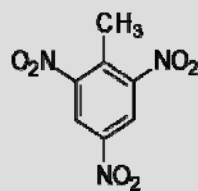
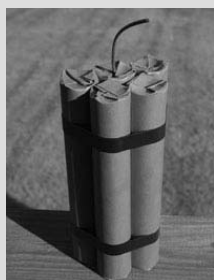
- Fatty acids include
 - All carboxylic acids
 - Aliphatic carboxylic acids
 - Aromatic carboxylic acids
 - Open chain monocarboxylic acids
- Which of the following can directly convert Carboxylic acid into alkane
 - LiAlH_4
 - NaBH_4
 - Na in ethanol
 - HI and Red P
- Which of the following has lowest boiling point?
 - Propan-1-ol
 - 1-Aminopropane
 - Ethanoic acid
 - Methoxymethane
- Which of the following order of acidity is correct?
 - $\text{HCOOH} > \text{CH}_3\text{COOH} > \text{C}_6\text{H}_5\text{COOH}$
 - $\text{HCOOH} > \text{C}_6\text{H}_5\text{COOH} > \text{CH}_3\text{COOH}$
 - $\text{C}_2\text{H}_5\text{COOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
 - $\text{C}_6\text{H}_5\text{COOH} > \text{HCOOH} > \text{CH}_3\text{COOH}$
- The reaction of formic acid with conc. H_2SO_4 gives
 - CO_2
 - CO
 - Oxalic acid
 - Acetic acid
- HCHO and HCOOH can be distinguished by
 - Tollen's reagent
 - Fehling's solution
 - Ferric chloride
 - Sodium carbonate
- Which of the following is a food preservative?
 - Sodium acetate
 - Sodium propionate
 - Sodium benzoate
 - Sodium oxalate
- Ethanamide is
 - Neutral
 - Basic
 - Acidic
 - Amphoteric
- The acid showing salt like behavior is
 - Acetic acid
 - Formic acid
 - Benzoic acid
 - Aminoacetic acid
- Salicylic acid and phenol can be distinguished by using.
 - NaHCO_3
 - Neutral FeCl_3
 - 5 % NaOH
 - Red litmus

Answer Key

1	2	3	4	5	6	7	8	9	10
d	d	d	b	b	d	c	d	d	a

Nitro Compounds

14



2,4,6-Trinitrotoluene (TNT)

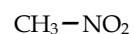
2,4,6-Trinitrotoluene (TNT) is a nitro compound. It is yellow solid and used as an explosive in quarrying, controlled demolition of buildings, in military warfare, etc.

14.1 Aliphatic Nitro Compounds (Nitroalkanes)

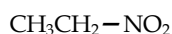
14.1.1 Introduction

Organic compounds having nitro group ($-\overset{+}{\text{N}}\begin{smallmatrix} \text{O} \\ \parallel \\ \text{O}^- \end{smallmatrix}$) as functional group are called nitro compounds.

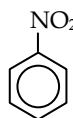
Examples:



Nitromethane



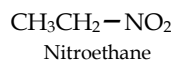
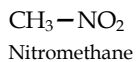
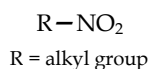
Nitroethane



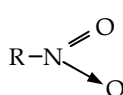
Nitrobenzene

When one of the hydrogen atom in alkane is replaced by nitro group ($-\text{NO}_2$), the resulting organic compound is called nitroalkane. Hence, nitroalkanes are nitro derivatives of alkanes.

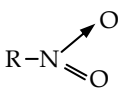
Example:



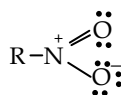
Nitro group is resonance hybrid of structures I and II or III and IV.



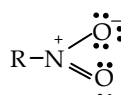
I



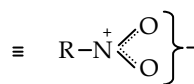
II



III



IV



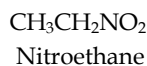
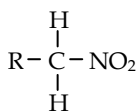
Resonance hybrid

14.1.2 Classification

Depending upon type of carbon to which nitro group is bonded, nitroalkanes are classified into three different classes; primary, secondary and tertiary nitroalkanes.

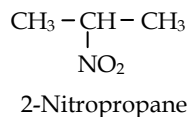
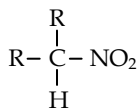
- Primary nitroalkanes:** If nitro group is bonded to primary carbon, the nitroalkane is called primary nitroalkane.

Examples:



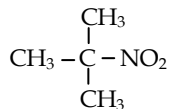
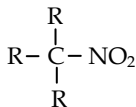
- Secondary nitroalkanes:** If nitro group is bonded to secondary carbon, the nitroalkane is called secondary nitroalkane.

Examples:



3. **Tertiary nitroalkanes:** If nitro group is bonded to tertiary carbon, the nitroalkane is called tertiary nitroalkane.

Example:



2-Methyl-2-nitropropane

14.1.3 Nomenclature

Nitro compounds are named taking Nitro- as prefix.

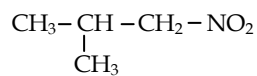
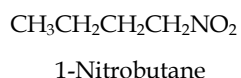
CH_3NO_2	Nitromethane
$\text{CH}_3\text{CH}_2\text{NO}_2$	Nitroethane
$\text{CH}_3\text{CH}_2\text{CH}_2\text{NO}_2$ $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_3 \\ \\ \text{NO}_2 \end{array}$	1-Nitropropane 2-Nitropropane
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NO}_2$ $\begin{array}{c} \text{CH}_3\text{CH}_2-\text{CH}-\text{CH}_3 \\ \\ \text{NO}_2 \end{array}$ $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{NO}_2 \\ \\ \text{CH}_3 \end{array}$ $\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{C}-\text{NO}_2 \\ \\ \text{CH}_3 \end{array}$	1-Nitrobutane 2-Nitrobutane 2-Methyl-1-nitropropane 2-Methyl-2-nitropropane

14.1.4 Isomerism

Nitroalkanes can show following types of structural isomerism.

1. **Chain isomerism:** Isomers that differ only in chain length are called chain isomers.

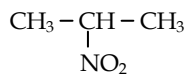
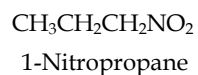
Examples:



2-Methyl-1-nitropropane

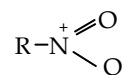
2. **Positional isomerism:** Isomers having same chain length but differ in the position of functional group or substituent are called positional isomers.

Examples:

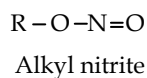


2-Nitropropane

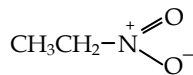
- c. **Functional isomerism:** Isomers that differ in their functional group are functional isomers. Nitroalkanes are isomeric with alkyl nitrites.

Example

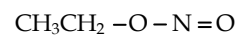
Nitroalkane



Alkyl nitrite

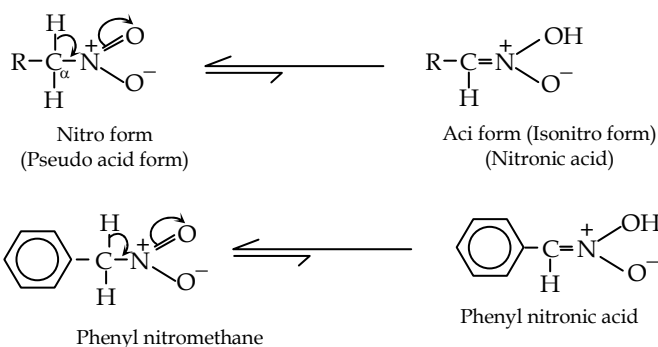


Nitroethane

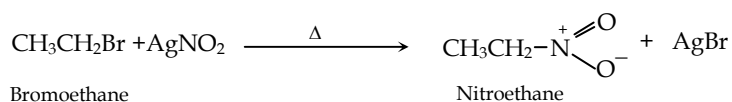


Ethyl nitrite

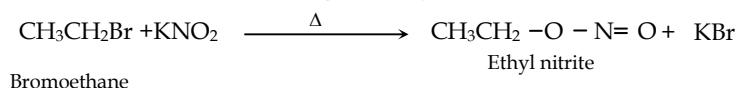
4. **Tautomerism:** Primary and secondary nitro compounds having α -hydrogen atoms show tautomerism, which is called *nitro-acinitro* tautomerism.

**14.1.5 General Methods of Preparation**

1. **From alkyl halides:** When a primary alkyl halide (especially bromides and iodides) is heated with aqueous ethanolic solution of silver nitrite, nitroalkane is obtained along with little alkyl nitrite.



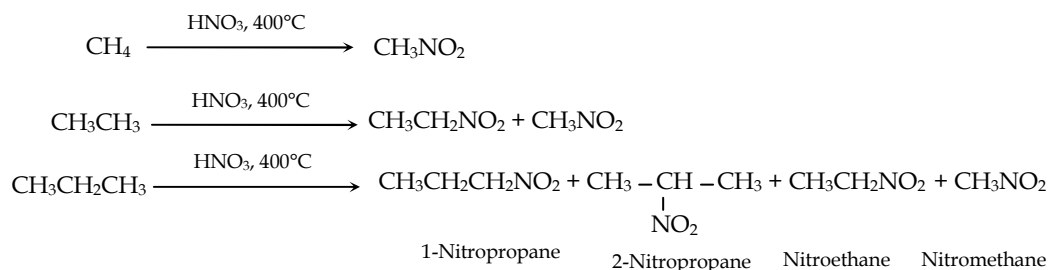
Potassium nitrite instead of silver nitrite gives alkyl nitrite.

**Self Test**

14.1 KNO_2 is not used instead of AgNO_2 to prepare nitroalkanes, give reason.

Hint: See in haloalkanes.

2. **From alkanes:** When alkane is heated with fuming nitric acid at about 400°C (*vapour phase nitration*), a mixture of all possible nitroalkanes are formed by carbon chain fission reaction of alkanes.

Examples:

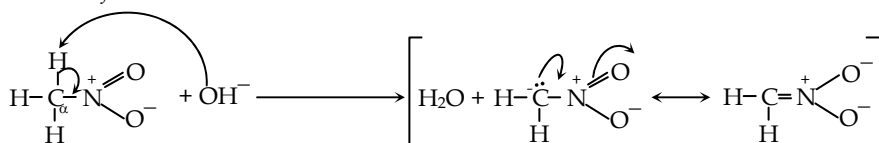
This is an example of substitution reaction where H-atom of alkane is substituted by NO_2 group and ease of abstraction of hydrogen atom is in the order $3^\circ > 2^\circ > 1^\circ$.

14.1.6 Physical Properties

1. Nitroalkanes are colorless liquids with pleasant odor.
2. They are sparingly soluble in water.

14.1.7 Chemical Properties

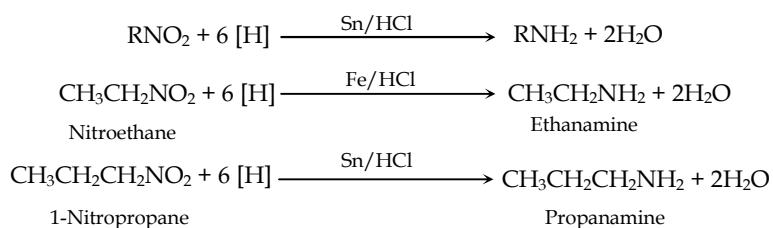
Primary and secondary nitro compounds having α -hydrogen atoms are acidic, hence, 1° and 2° nitro compounds react with aqueous NaOH to form salt.



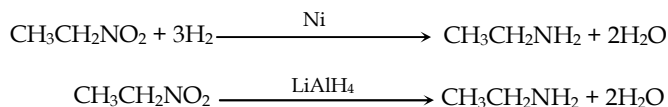
The acidic nature of α -hydrogen is due to i) -I effect of $-\text{NO}_2$ ii) resonance stabilization of anion formed.

1. Reduction

- Nitroalkane can be reduced in acidic solution using metal and acid usually Sn/HCl , Fe/HCl , Zn/HCl into primary amine.

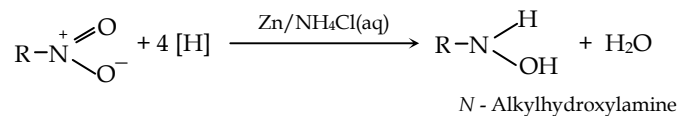


Reduction of nitro compounds with H_2 in presence of Ni or Pt catalyst or by LiAlH_4 , also give primary amines.



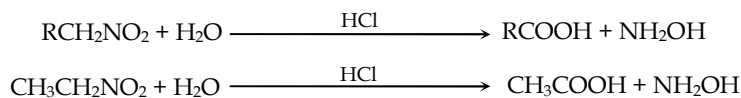
462 Modern Approach to Chemistry-II

- b. Nitroalkanes can be reduced in neutral solution using zinc dust in aqueous ammonium chloride into *N*-alkylhydroxylamine.

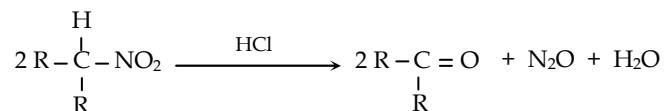


2. Hydrolysis

- a. Primary nitroalkanes are hydrolyzed by boiling with aq. HCl or 85% sulphuric acid to carboxylic acid and hydroxylamine.



- b. Secondary nitroalkanes are hydrolyzed by boiling with HCl to ketone and nitrous oxide.

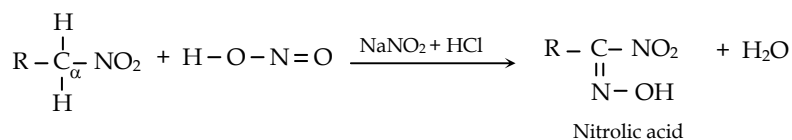


- c. Tertiary nitroalkanes do not undergo hydrolysis.

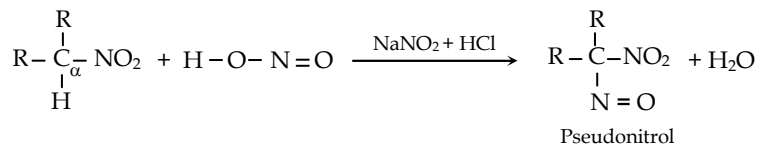
3. Reaction with nitrous acid

Different classes of nitroalkanes react with nitrous acid to give different products. Hence, this reaction can be used to distinguish 1°, 2° and 3° nitroalkanes as Hofmann's method.

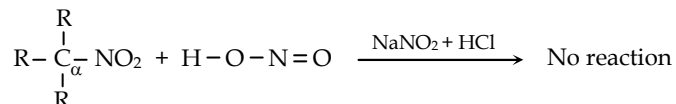
- a. Primary nitroalkanes form crystalline compound of nitrolic acids which is soluble in aqueous NaOH to give red solution.



- b. Secondary nitroalkanes form colorless crystalline compound pseudonitroles which is soluble in aqueous NaOH to give blue solution.

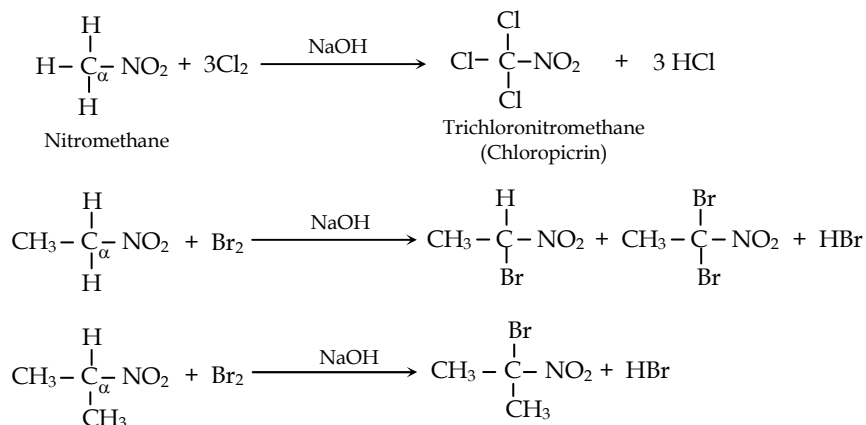


- c. Tertiary nitroalkanes do not react with nitrous acid since they do not have α-hydrogen atom.



4. Halogenation

Primary and secondary nitroalkanes when treated with Cl_2 or Br_2 in presence of NaOH , undergo α -halogenation.



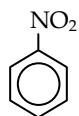
14.1.8 Differences between Nitroalkanes and Alkyl Nitrites

Nitroalkanes	Alkyl nitrites
1. Boiling point Nitroalkane has higher boiling point than that of isomeric alkyl nitrite since the former is polar compound. $\text{CH}_3\text{CH}_2-\text{N}^+=\text{O} \begin{array}{c} \diagup \text{O}^- \\ \diagdown \end{array}$ Nitroethane	1. Boiling point Alkyl nitrite has lower boiling point than that of isomeric nitroalkanes since the former is less polar compound. $\text{CH}_3\text{CH}_2-\text{O}-\text{N}=\text{O}$ Ethyl nitrite
2. Reduction Nitroalkane on reduction gives primary amine. $\text{CH}_3\text{CH}_2\text{NO}_2 + 6[\text{H}] \xrightarrow{\text{Fe/HCl}} \text{CH}_3\text{CH}_2\text{NH}_2 + 2\text{H}_2\text{O}$	2. Reduction Alkyl nitrite on reduction gives alcohol and ammonia or hydroxylamine. $\text{CH}_3\text{CH}_2-\text{O}-\text{N}=\text{O} \xrightarrow{\text{Fe/HCl}} \text{CH}_3\text{CH}_2\text{OH} + \text{NH}_3 + \text{H}_2\text{O}$
3. Hydrolysis Nitroalkane on hydrolysis with acid forms carboxylic acid. $\text{CH}_3\text{CH}_2\text{NO}_2 + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{CH}_3\text{COOH} + \text{NH}_2\text{OH}$	3. Hydrolysis Alkyl nitrite on hydrolysis with acid gives alcohol and nitrous acid. $\text{CH}_3\text{CH}_2-\text{O}-\text{N}=\text{O} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{CH}_3\text{CH}_2\text{OH} + \text{HNO}_2$
4. Reaction with nitrous acid Nitroalkanes react with nitrous acid to give different products, the nature of products formed dependent upon the classes of nitroalkanes (see section 14.1.7)	4. Reaction with nitrous acid Alkyl nitrite does not react with nitrous acid.

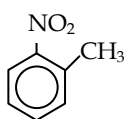
14.2 Aromatic Nitro Compounds

14.2.1 Introduction and Nomenclature

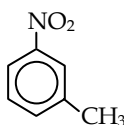
Nitro derivatives of arenes in which nitro group ($-\text{NO}_2$) is directly bonded to aromatic ring are called **aromatic nitrocompounds**.



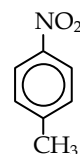
Nitrobenzene



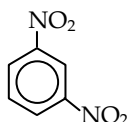
2-Nitrotoluene



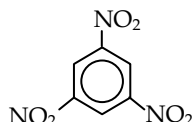
3-Nitrotoluene



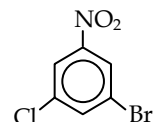
4-Nitrotoluene



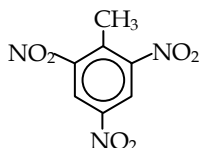
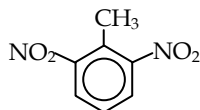
1,3-Dinitrobenzene



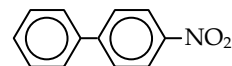
1,3,5-Trinitrobenzene



3-Bromo-5-chloronitrobenzene

2,4,6-Trinitotoluene
(TNT)

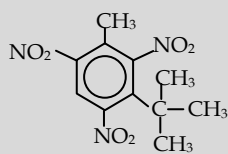
2,6-Dinitrotoluene



4-Nitrobiphenyl

Artificial musks

Polynitro derivatives of benzene having a tertiary butyl group possess musk like odour and hence, is used in perfumery as artificial musk.

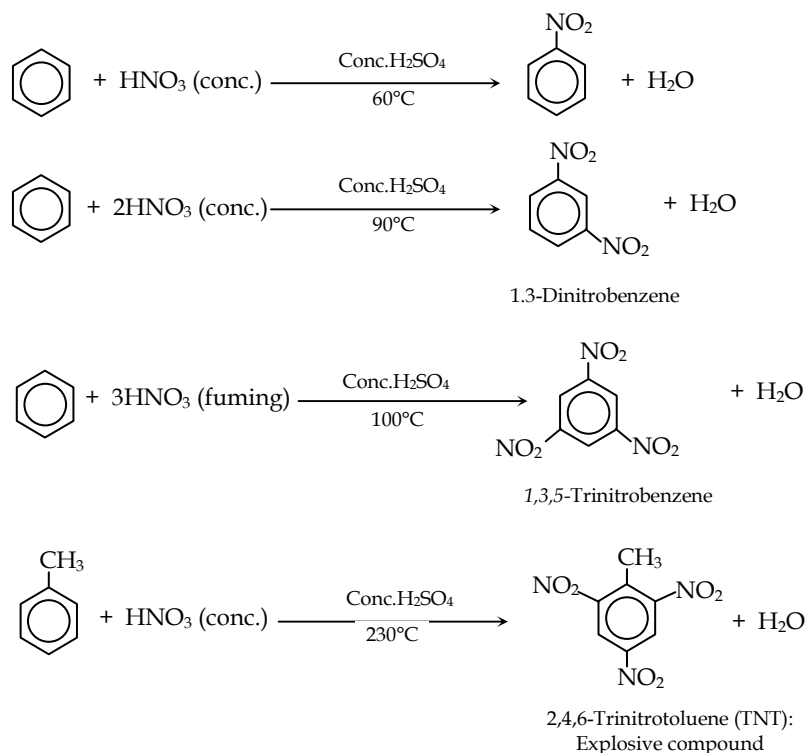


'Toluene-musk'

14.2.2 Preparation of Aromatic Nitro Compounds

By nitration of aromatic hydrocarbons (arenes):

Aromatic nitro compounds are prepared by direct nitration of aromatic hydrocarbons. The most commonly used nitrating reagent is a mixture of concentrated nitric acid and concentrated sulphuric acid.

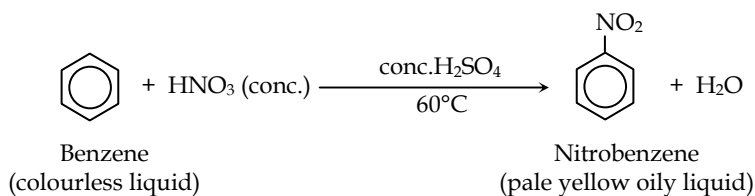


Other reagents used for nitration are (i) concentrated or fuming HNO_3 (ii) HNO_3 in organic solvents like acetic acid, nitromethane (iii) nitronium salt in organic solvents (iv) acyl nitrates (v) N_2O_5 in CCl_4 , N_2O_4 .

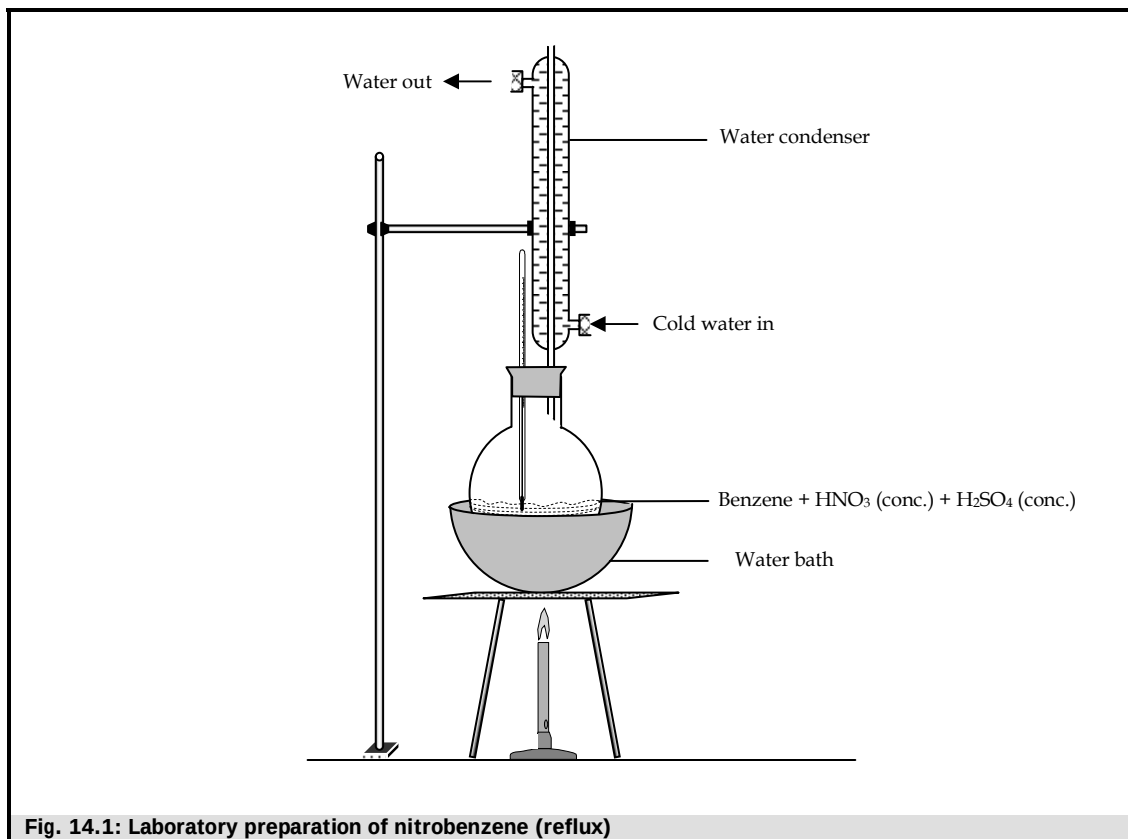
14.3 Nitrobenzene (Oil of Mirbane)

14.3.1 Laboratory Preparation of Nitrobenzene

Theory: Nitrobenzene is prepared in laboratory by refluxing benzene with a mixture of concentrated nitric acid and concentrated sulphuric acid to about $40\text{--}60^\circ\text{C}$.



This is an electrophilic aromatic substitution reaction, where the electrophile is nitronium ion (NO_2^+), which is generated from a mixture of conc. HNO_3 and conc. H_2SO_4 .



Procedure

50 ml of benzene is taken in a R.B. flask. A nitrating mixture is prepared by mixing 60 ml of concentrated HNO_3 with 60 ml of concentrated H_2SO_4 with shaking and cooling. The nitrating mixture is then added slowly into the R.B. flask a little at a time with shaking and cooling, after each addition. The temperature of mixture should be maintained below 60°C . When whole nitrating mixture is added, the mixture is refluxed on water bath for an hour at about 60°C (at higher temperature polynitration takes place). Appearance of pale yellow oily layer in the flask indicates the formation of nitrobenzene. It is allowed to cool and poured it in cold water (for the dilution of remaining acid). The layer of nitrobenzene is separated using separating funnel.

Purification

- Nitrobenzene thus obtained is washed with dilute Na_2CO_3 to remove acidic impurities.
- It is then washed with water several times to remove salts.
- It is dried with anhydrous CaCl_2 .
- Finally, it is distilled to about $210\text{--}215^\circ\text{C}$ using air condenser to get pure nitrobenzene.

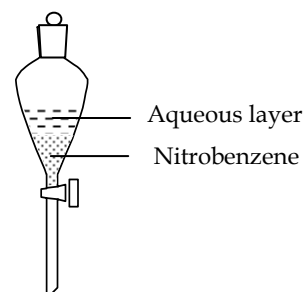
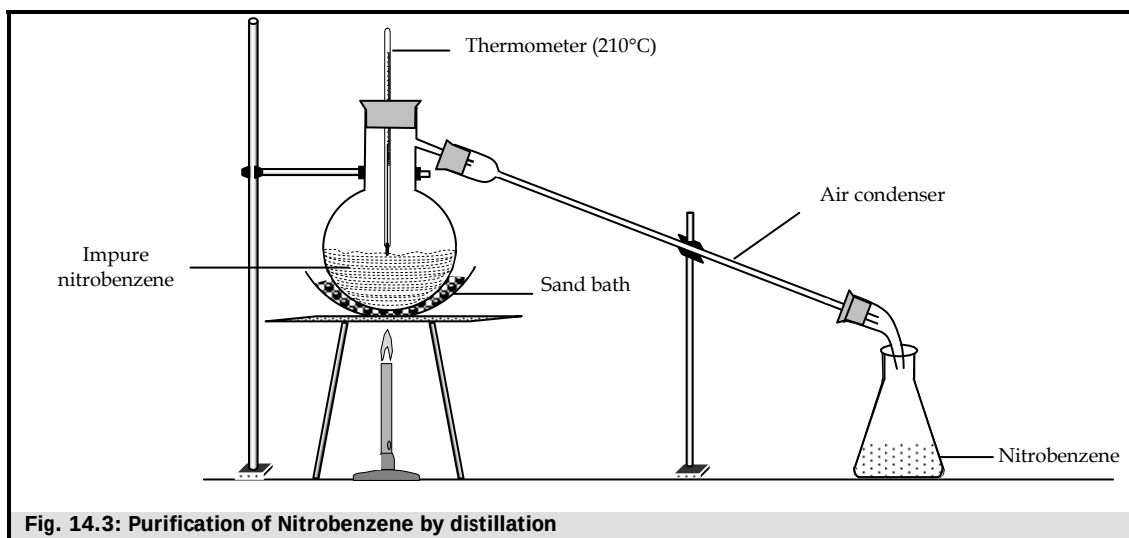
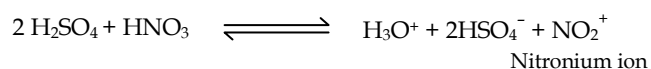


Fig. 14.2: Separating funnel, separation of nitrobenzene from reaction mixture

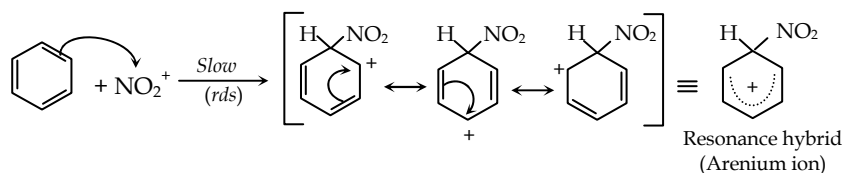


Mechanism of Nitration

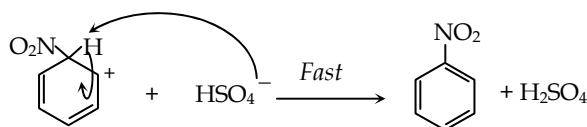
Generation of NO_2^+ Nitronium ion electrophile is generated from a mixture of conc. HNO_3 and conc. H_2SO_4 .



Step I: NO_2^+ (electrophile) is attracted toward π electron cloud of benzene ring to form arenium ion as an intermediate. This step is slow since aromatic character is destroyed and it determines the rate of reaction (rds).



Step II: Abstraction of proton(H^+) by HSO_4^- to give nitrobenzene. This step is fast since aromatic character is restored.



14.3.2 Physical Properties

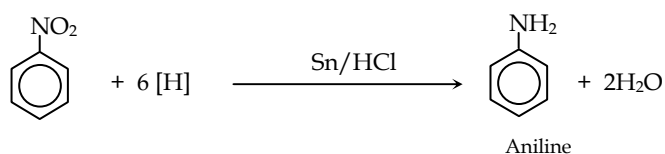
1. Nitrobenzene is pale yellow oily liquid with almond like odour.
2. It is insoluble in water but soluble in organic solvents.
3. All nitro compounds are more dense than water.
4. Its freezing point is 5.7°C (greenish-yellow crystal) and boiling point is 211°C .

14.3.3 Chemical Properties

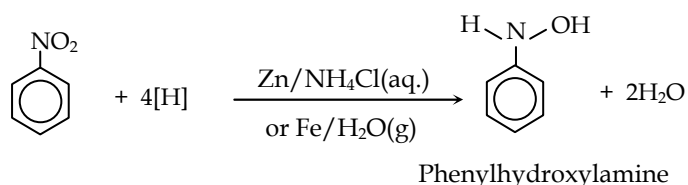
A. Reaction due to -NO₂: Reduction

Nitrobenzene can be reduced to different products depending upon conditions as well as the nature of reducing agents used in different medium.

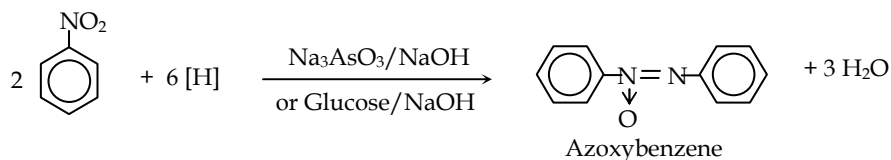
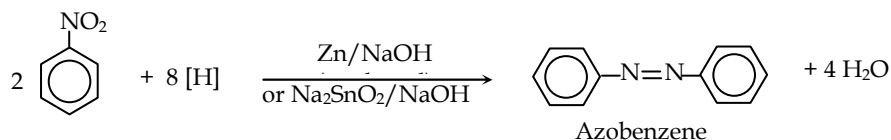
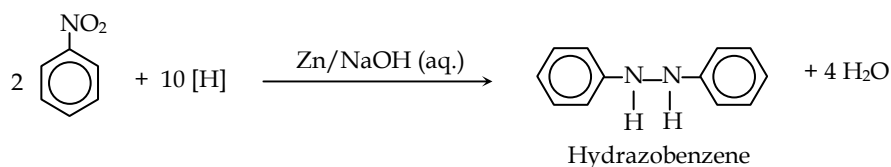
- 1. Reduction in acidic solution:** Nitrobenzene can be reduced into aniline in acidic solution using metal and acid usually Sn/HCl, Fe/HCl, Zn/HCl.



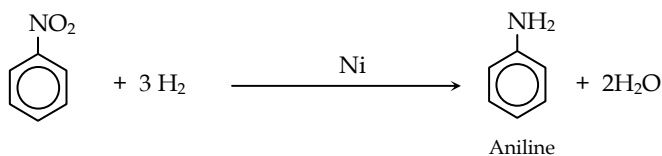
- 2. Reduction in neutral solution:** Nitrobenzene can be reduced into phenylhydroxylamine in neutral solution using zinc dust in aqueous ammonium chloride.



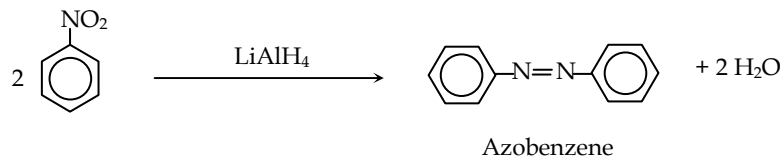
- 3. Reduction in alkaline solution:** Nitrobenzene can be reduced in different reduced products in alkaline solution using different reagents.



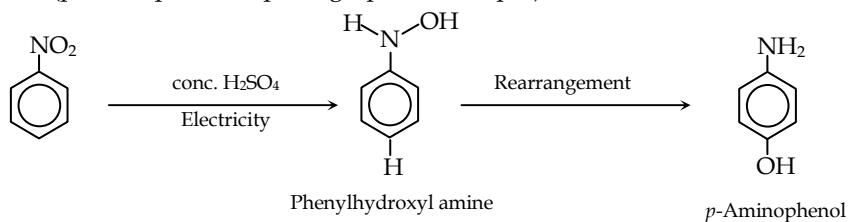
- 4. Catalytic reduction:** Nitrobenzene can be reduced catalytically into aniline with H₂ in presence of Ni, Pt or Pd catalyst.



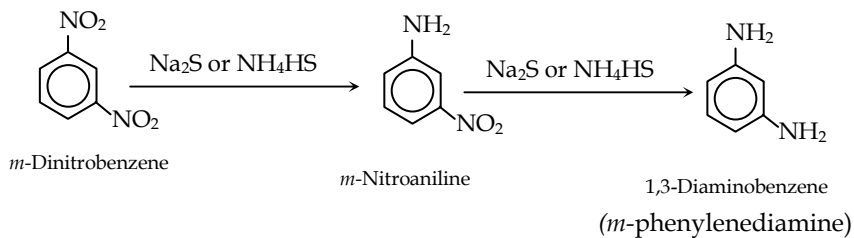
5. **Reduction with LiAlH_4 :** Nitrobenzene can be reduced into azobenzene with lithium aluminum hydride .



6. **Electrolytic reduction:** Nitrobenzene is reduced into p-Aminophenol by electrolysis in strongly acidic medium (p-aminophenol is photographic developer).

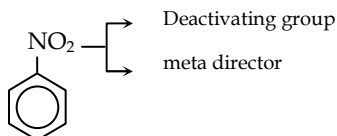


m-Dinitrobenzene can be reduced by sodium sulphide stepwise as;



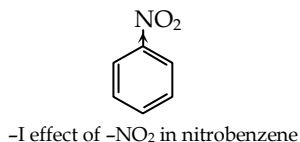
B. Reaction due to benzene ring: Electrophilic aromatic substitution reaction

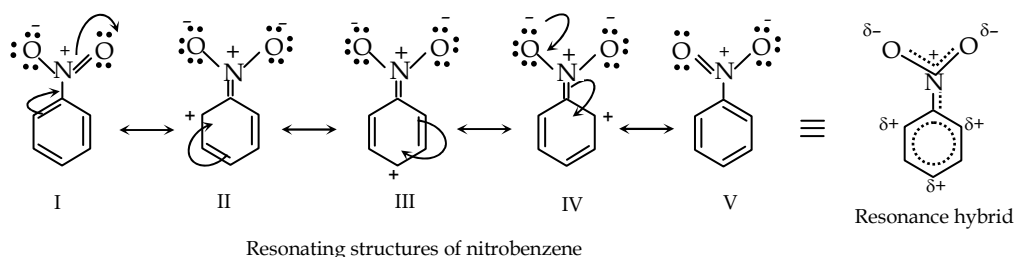
Nitro group ($-\text{NO}_2$) present in benzene governs the reactivity and orientation in electrophilic aromatic substitution reaction.



Nitrobenzene is less reactive than benzene in electrophilic aromatic substitution reaction:

Nitro group ($-\text{NO}_2$) withdraws electrons from benzene ring due to electron withdrawing inductive effect ($-I$ effect) of $-\text{NO}_2$ group since nitrogen is more electronegative than carbon. Similarly $-\text{NO}_2$ group also withdraws electron from benzene ring due to resonance effect to give the following resonance structures.

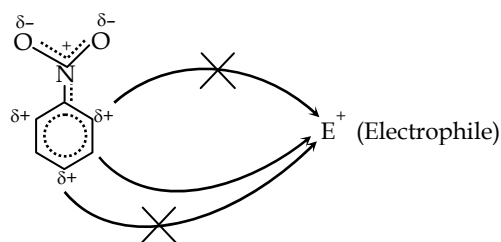




The electron-withdrawing $-\text{NO}_2$ group decreases the electron density in benzene ring of nitrobenzene as compared to that in benzene. A substituent which makes the benzene ring less reactive than benzene in electrophilic aromatic substitution is called **deactivating group**. Hence, $-\text{NO}_2$ is a deactivating group. Therefore, nitrobenzene is less reactive than benzene in electrophilic aromatic substitution reaction. Nitrobenzene is roughly one million times less reactive than benzene.

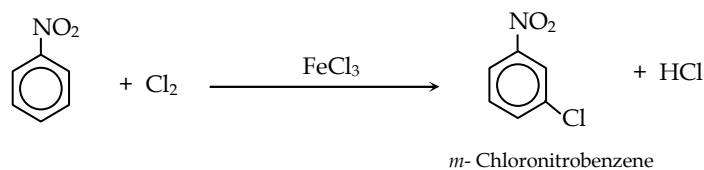
Nitrobenzene gives *meta* substituted products in electrophilic aromatic substitution reaction.

The electron-withdrawing deactivating $-\text{NO}_2$ group decreases the electron density in benzene ring of nitrobenzene as compared to benzene which makes each position of nitrobenzene less reactive than benzene towards electrophilic aromatic substitution reaction. However, due to resonance, the partial positive charge (δ^+) is developed at *ortho*- and *para*-positions. The electron density is low at *ortho*- and *para*-positions. So, electrophilic attack becomes difficult in these positions. Hence, in electrophilic aromatic substitution reaction, $-\text{NO}_2$ electrophile selectively attacks the *meta* position to give *meta* substituted products.

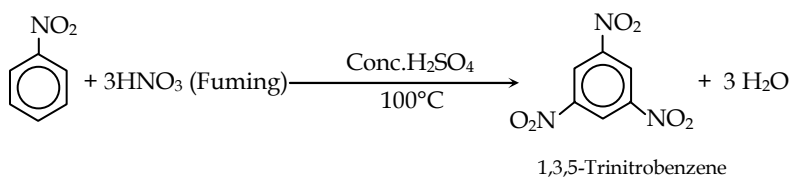
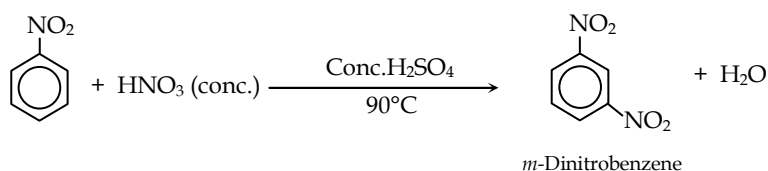


A substituent which directs the incoming electrophile chiefly at *meta* position to it in electrophilic aromatic substitution is called ***meta* director**. Hence, $-\text{NO}_2$ is *meta* director.

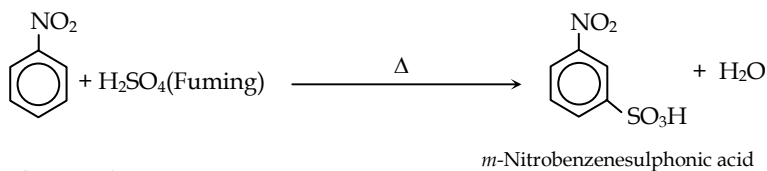
1. Halogenation



2. Nitration



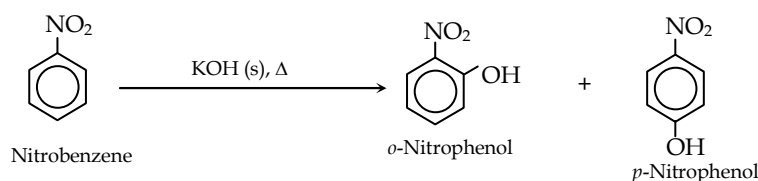
3. Sulphonation



4. Friedel-Crafts reaction

Nitrobenzene does not give Friedel-Crafts reaction. This is due to strongly deactivating effect of $-\text{NO}_2$.

Nucleophilic substitution: When nitrobenzene is fused with solid KOH, ortho and para nitrophenols are obtained. This is nucleophilic substitution reaction.

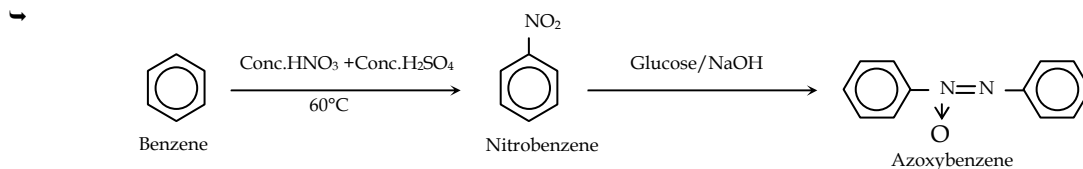
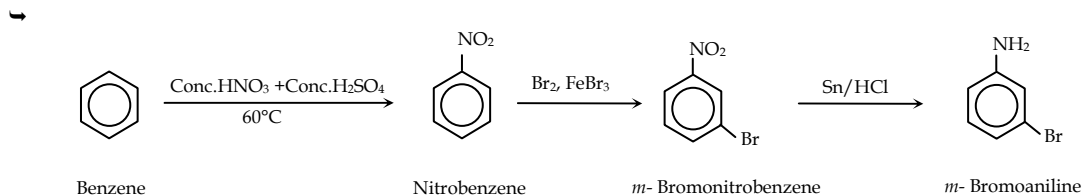
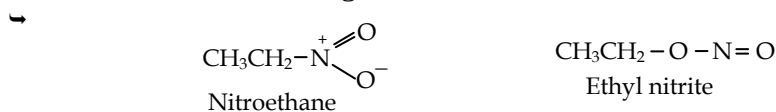


14.3.4 Uses of Nitrobenzene

- i. Nitrobenzene is used to prepare aniline.
- ii. It is also used in shoes polishes, paint solvent to mask unpleasant odour.
- iii. It is also used as inexpensive perfume for soaps.
- iv. It is also used as solvent.

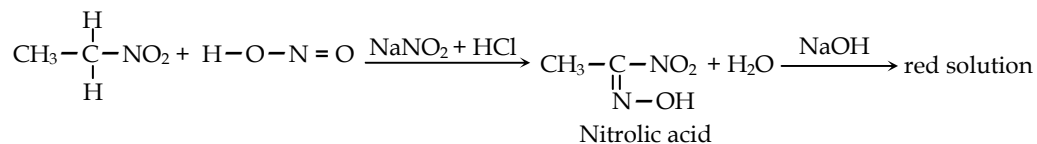
Solved Questions

1. Convert benzene into azoxybenzene.

2. How do you get *m*-Bromoaniline from benzene?3. Write the functional isomers of molecular formula $\text{C}_2\text{H}_5\text{NO}_2$ and suggest a suitable chemical test to distinguish them.

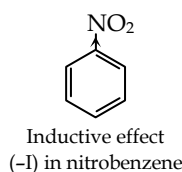
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Nitroethane gives red solution when treated with nitrous acid followed by dil. NaOH but ethyl nitrite does not.

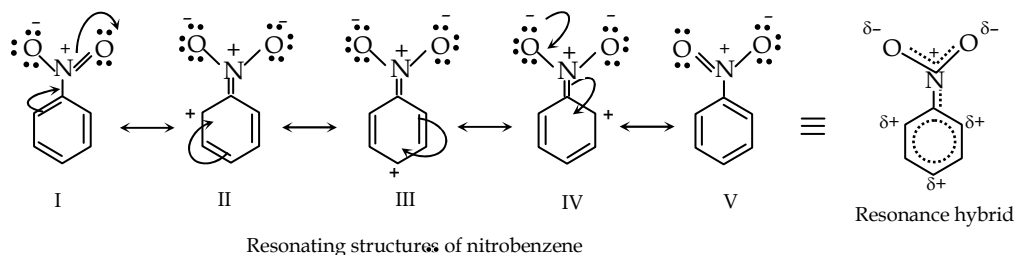


4. Nitro group (-NO₂) is *meta* director in electrophilic aromatic substitution reaction, explain.

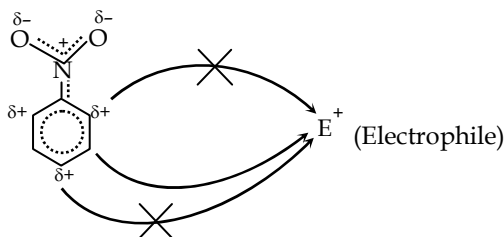
- Nitro group (-NO₂) withdraws electrons from benzene ring due to -I effect of -NO₂ since nitrogen is more electronegative than carbon.



Nitro group (-NO₂) withdraws electrons from benzene ring due to resonance effect too.



Hence, nitro group decreases the electron density in benzene ring of nitrobenzene as compared to benzene making each positions of nitrobenzene less reactive than corresponding positions in benzene towards electrophilic aromatic substitution reaction. However, due to resonance, the partial positive charge (δ^+) is developed at *ortho*- and *para*- positions. The electron density is low at *ortho*- and *para*- positions. So, electrophilic attack becomes difficult in these positions. Hence electrophile attacks the *meta* positions to give *meta* substituted products.

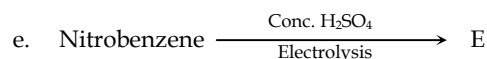
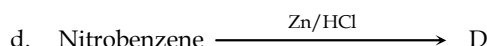
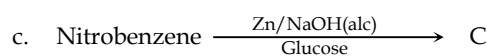
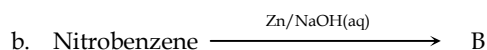
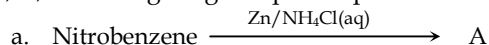


A substituent which directs the incoming group chiefly at *meta* to it in electrophilic aromatic substitution is called *meta* director.

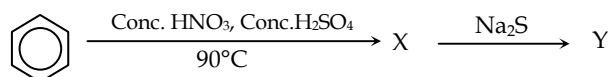
Questions

Very Short Answer Questions:

- Nitroethane boils at higher temperature than ethyl nitrite although both have the same molecular weight, explain.
- Suggest a suitable chemical test to distinguish nitroethane from ethyl nitrite.
- Identify A, B, C, D and E giving complete equations.



- Identify X and Y

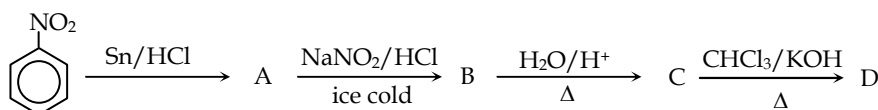


- Why is nitrobenzene less reactive than benzene in electrophilic aromatic substitution reaction?
- How can you distinguish
 - 1°, 2° and 3° nitroalkanes
 - Amines and nitroalkanes

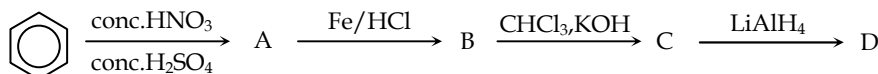
Short /Long Answer Questions:

- Identify A, B, C and D in the given reactions.

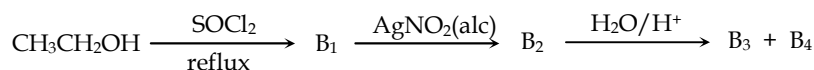
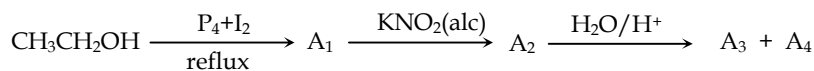
(a)



(b)



- How is nitrobenzene prepared in laboratory in pure and dry state?
 - Identify A and B giving complete reactions.



- How can you convert nitrobenzene into
 - Phenylhydroxylamine
 - Azobenzene
 - Azoxybenzene
 - p*-Aminophenol
 - m*-Nitroaniline
 - p*-Hydroxyazobenzene

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4. Write short notes on:
 - a. Reduction of nitrobenzene in acidic, neutral, and alkaline medium
 - b. Laboratory preparation of nitrobenzene
5. How can you convert
 - a. Ethane into nitroethane
 - b. Phenol into nitrobenzene
 - c. Nitroethane into ethene
 - d. Nitroethane into ethanol
6. What happens when:
 - a. 1-Nitropropane is hydrolysed
 - b. 2-Nitropropane is hydrolysed
 - c. Nitrobenzene is reduced using $\text{Zn} + \text{NaOH}$
 - d. Iodoethane is treated with alcoholic KNO_2
 - e. Nitroethane is reduced in neutral medium.

Multiple Choice Questions

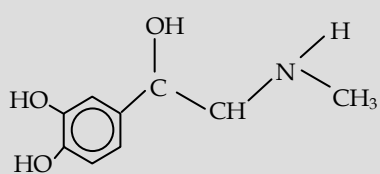
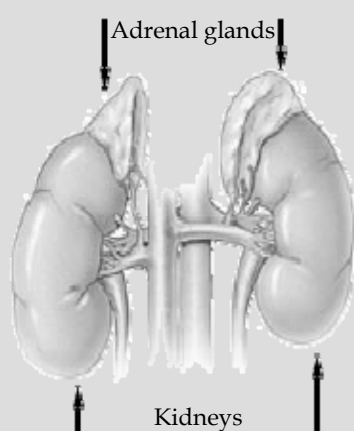
1. Nitro compounds do not exhibit
 - a. Functional isomerism
 - b. Tautomerism
 - c. Chain isomerism
 - d. Metamerism
2. Nitroethane and ethyl nitrites are
 - a. Tautomers
 - b. Metamers
 - c. Functional isomers
 - d. Isomers
3. Reduction of nitroalkane gives
 - a. Alcohol
 - b. Alkane
 - c. Alkyl amine
 - d. Ammonia
4. Hydrolysis of nitroethane gives
 - a. Acetic acid and hydroxyl amine
 - b. Alcohol and nitrous acid
 - c. Alcohol and ammonia
 - d. Ketone and Nitrous oxide
5. Hydrolysis of secondary nitroalkane gives
 - a. Carboxylic acid
 - b. Alcohol
 - c. Ketone
 - d. Hydroxyl amine
6. When treated with nitrous acid followed by caustic alkali, secondary nitroalkane gives
 - a. Red colour due to nitrolic acid
 - b. Blue colour due to pseudonitrol
 - c. Red colour due to pseudonitrol
 - d. Blue colour due to salt of nitrolic acid
7. Nitrobenzene is
 - a. Oil of wintergreen
 - b. Oil of vitriol
 - c. Paraffin oil
 - d. Oil of mirbane
8. Nitration of benzene with concentrated nitric acid in presence of sulphuric acid is
 - a. Electrophilic substitution
 - b. Nucleophilic substitution
 - c. Free radical substitution
 - d. Electrophilic addition
9. Which of the followings is the least reactive in electrophilic aromatic substitution?
 - a. Aniline
 - b. Toluene
 - c. Chlorobenzene
 - d. Nitrobenzene
10. Reduction of nitrobenzene with zinc and aqueous NaOH gives
 - a. Hydroxylamine
 - b. Azobenzene
 - c. Hydrazobenzene
 - d. Azoxybenzene
11. Reduction of nitrobenzene with zinc dust and aqueous ammonium chloride gives
 - a. Phenylhydroxylamine
 - b. Azobenzene
 - c. Hydrazobenzene
 - d. p-Aminophenol
12. When nitrobenzene is fused with caustic alkali, o- and p-nitrophenol are formed. The reaction is
 - a. Electrophilic substitution
 - b. Nucleophilic substitution
 - c. Free radical substitution
 - d. Electrophilic addition
13. Reduction of nitrobenzene with lithium aluminium hydride gives
 - a. Hydroxylamine
 - b. Azobenzene
 - c. Hydrazobenzene
 - d. p-Aminophenol
14. Electrolytic reduction of nitrobenzene in concentrated sulphuric acid gives
 - a. Hydroxylamine
 - b. Azobenzene
 - c. Hydrazobenzene
 - d. p-Aminophenol

Answer Key

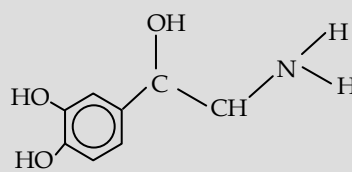
1	2	3	4	5	6	7	8	9	10	11	12	13	14
d	c	c	a	c	b	d	a	d	c	a	b	b	d

Amines

15



Adrenaline



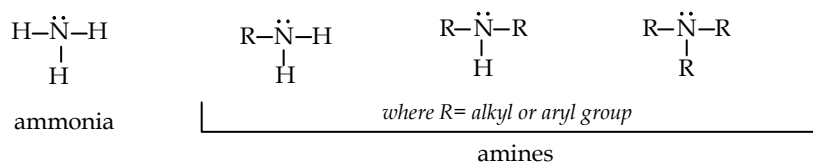
Noradrenaline

Adrenaline and noradrenaline are hormones and neurotransmitters produced by adrenal glands.

15.1 Introduction

When one, two or all three hydrogen atoms in ammonia molecule are replaced by alkyl or aryl groups, the resulting organic compounds are called amines.

Examples:



Amines are classified into two broad classes; aliphatic amines and aromatic amines.

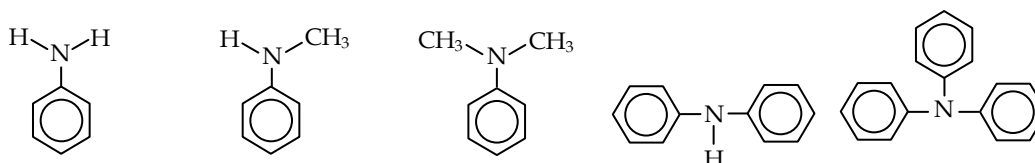
If R is alkyl group, the amines are called *aliphatic amines*.

Examples:



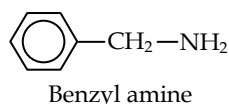
If R is aryl group (at least one), the amines are called *aromatic amines*.

Examples:



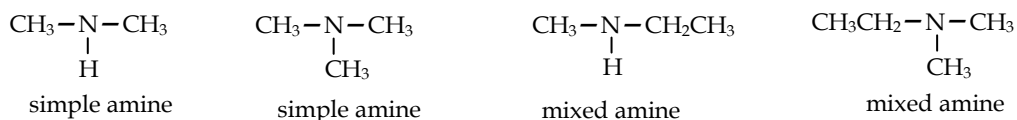
If aromatic ring is not directly bonded to amino group, e.g. ArCH_2NH_2 , the amines are called aryl substituted aliphatic amines aryl alkyl amine since they behave very much like aliphatic amines.

Example:



When all alkyl or aryl groups are same, the amines are called *simple amines* otherwise called *mixed amines*.

Examples:



In nature, amino functional group occurs in wide range of organic compounds like amino acids, dyes, drugs, vitamins, hormones etc.

15.2 Classification

Depending upon number of alkyl or aryl group present on nitrogen atom, amines are classified into three different classes; primary, secondary and tertiary amines.

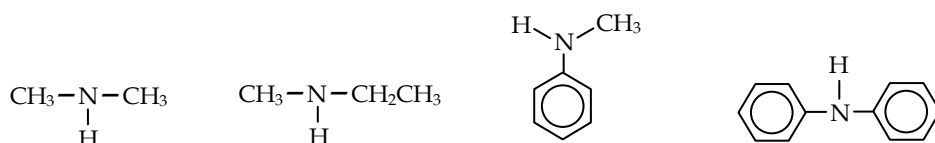
- Primary amines:** When only one hydrogen atom of ammonia molecule is replaced by alkyl or aryl group the amine is called primary or 1° amine. It is represented by RNH_2 . The functional group of primary amine is $-\text{NH}_2$ (amino group).

Examples:



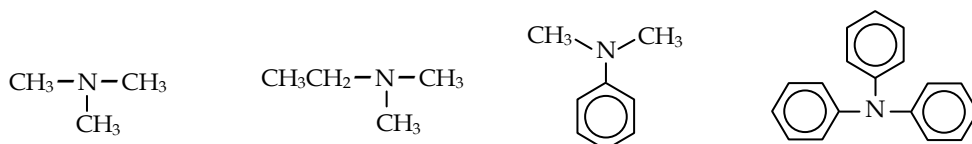
- Secondary amines:** When two hydrogen atoms of ammonia molecule are replaced by alkyl or aryl groups, the amine is called secondary or 2° amine. It is represented by R_2NH . The functional group of secondary amine is $-\text{NH}-$ (imino group).

Examples:



- Tertiary amines:** When all three hydrogen atoms of ammonia molecule are replaced by alkyl or aryl groups, the amine is called tertiary or 3° amine. It is represented by R_3N . The functional group of tertiary amine is $-\text{N}-$ (tertiary nitrogen atom).

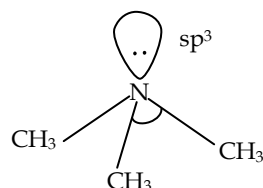
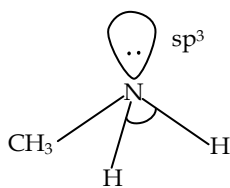
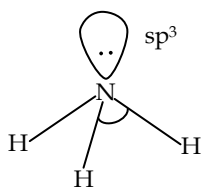
Examples:



Wurtz (1849) discovered amines by boiling alkyl isocyanates with alkali.

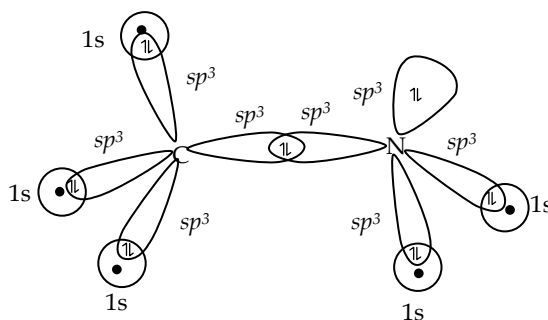


15.3 Structure of Amines



Like in ammonia molecule, nitrogen atom of an amine molecule is sp^3 hybridised to form four sp^3 orbitals. One sp^3 orbital is fulfilled and does not take part in bonding but remaining three sp^3 orbitals are half filled and participate in bonding.

In primary amine like methyl amine, one half filled sp^3 orbital of nitrogen atom overlaps with half filled sp^3 orbital of carbon atom forming N-C sigma bond. The remaining each half filled sp^3 orbital of nitrogen overlaps with half filled $1s$ orbital of hydrogen forming N-H sigma bonds.



Orbital picture of methyl amine

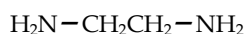
15.4 Nomenclature

In IUPAC system, aliphatic amines are named by taking **amino-** as prefix to the name of parent alkane and they are also named by taking the word- amine as suffix to the name of parent alkane. If alkyl group is present on nitrogen atom as substituent, it is indicated by writing *N* - before the substituent. In common system, amines are named by naming alkyl group followed by the ending - amine.

Structural formula	IUPAC name	Common name
CH_3NH_2	Aminomethane or Methanamine	Methyl amine
$\text{CH}_3\text{CH}_2\text{NH}_2$ $\begin{array}{c} \text{CH}_3-\text{N}-\text{CH}_3 \\ \\ \text{H} \end{array}$	Aminoethane or Ethanamine <i>N</i> -Methylaminomethane	Ethyl amine Dimethyl amine
$\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_3 \\ \\ \text{NH}_2 \end{array}$ $\begin{array}{c} \text{CH}_3\text{CH}_2-\text{N}-\text{CH}_3 \\ \\ \text{H} \end{array}$ $\begin{array}{c} \text{CH}_3-\text{N}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	1-Aminopropane 2-Aminopropane <i>N</i> -Methylaminoethane <i>N,N</i> -Dimethylaminomethane	<i>n</i> -Propyl amine Isopropyl amine Ethyl methyl amine Trimethyl amine

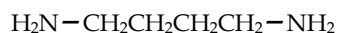
If two or more amino groups are present, then their position is indicated.

Example:



Ethane-1,2-diamine

Putrescine and *cadaverine* are amines which are largely responsible for the foul odour of decaying dead animal body (proteins).



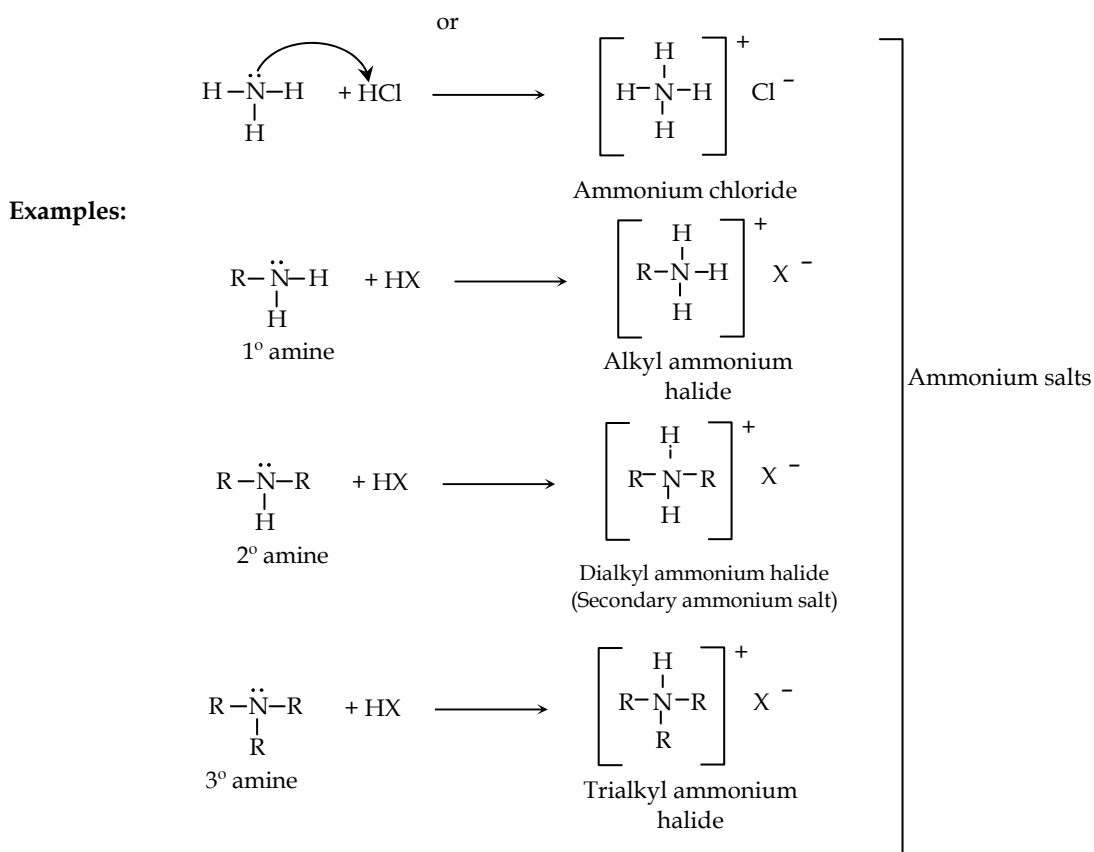
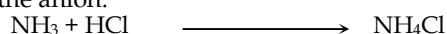
1,4-Diaminobutane
(Putrescine)



1,5-Diaminopentane
(Cadaverine)

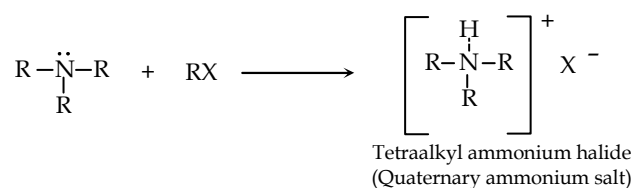
15.5 Salts of Amines

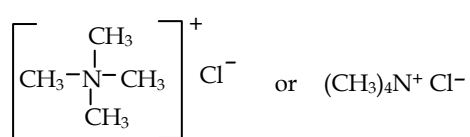
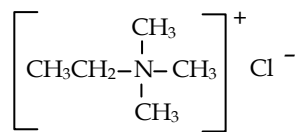
Like ammonia, all classes of amines are basic in nature hence react with acids to form ammonium salts. These salts are generally named by replacing *-amine* by *-ammonium* or *aniline* by *anilinium* and adding the name of the anion.



Tertiary amine reacts with alkyl halide to form a salt of type $\text{R}_4\text{N}^+\text{X}^-$ which is called **quaternary ammonium salt**. Here, nitrogen atom is quadricovalent unelectrovalent.

Examples:



Tetramethylammonium chloride
(Quaternary ammonium salt)Ethyltrimethylammonium chloride
(Quaternary ammonium salt)

Quaternary ammonium salts are white crystalline solids and soluble in water.

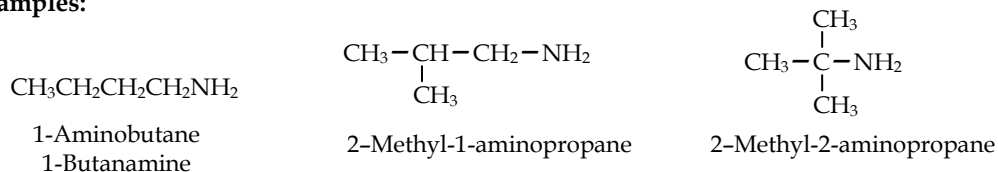
Quaternary ammonium salts having a long chain alkyl group have detergent properties. They are also known as *invert soaps* because their surface activity is due to positive ion (R_4N^+). Surface activity of detergents and soaps are due to negative ion (RCOO^-).

15.6 Isomerism

Amines can show following types of structural isomerism.

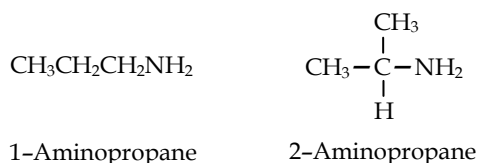
- Chain isomerism:** Isomers that differ only in chain length are called chain isomers.

Examples:



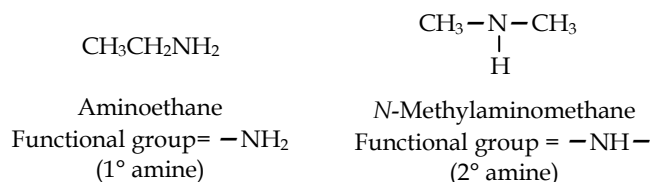
- Positional isomerism:** Isomers having same chain length but differ in the position of functional group or substituents are called positional isomers.

Examples:

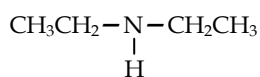
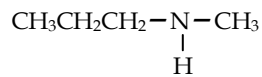
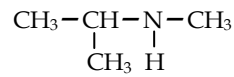


- Functional isomerism:** Isomers that differ in their functional group are functional isomers. 1°, 2° and 3° amines are functional isomers.

Examples:



- Metamerism:** Isomers that differ in alkyl group present on either side of the same functional group are called metamers. Secondary amines can show this type of isomerism. For example, $\text{C}_4\text{H}_{11}\text{N}$ has following three metamers.

Examples:*N*-Ethylaminoethane*N*-Methyl-1-aminopropane*N*-Methyl-2-aminopropane**Self Test**

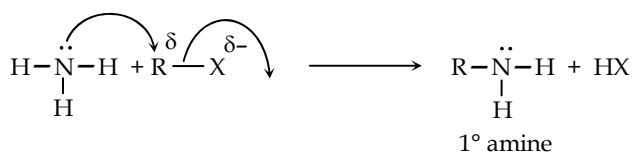
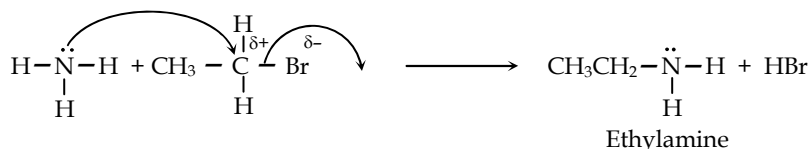
15.1 What are amines? Write structure of amines of molecular formula $\text{C}_3\text{H}_9\text{N}$, indicate their functional group, classify them as primary, secondary and tertiary amines and give their name. *Hint: 4 isomers*

15.2 Write structures of amines of molecular formula $\text{C}_4\text{H}_{11}\text{N}$, classify them as primary, secondary and tertiary amines and give their name. *Hint: 8 isomers*

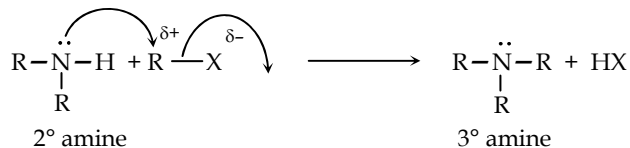
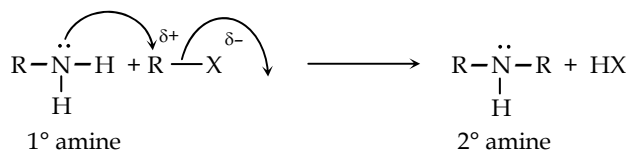
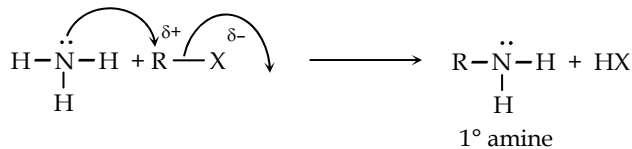
15.7 General Methods of Preparation**A. Methods Giving Mixture of Amines**

- From alkyl halides (Hofmann's ammonolysis of alkyl halides):** When an alkyl halide is heated with aqueous or alcoholic solution of ammonia at 100°C in a sealed tube or allowed to stand at room temperature, amine is formed. By varying the proportions of ammonia and alkyl halide, particular class of amine can be obtained in large proportion.

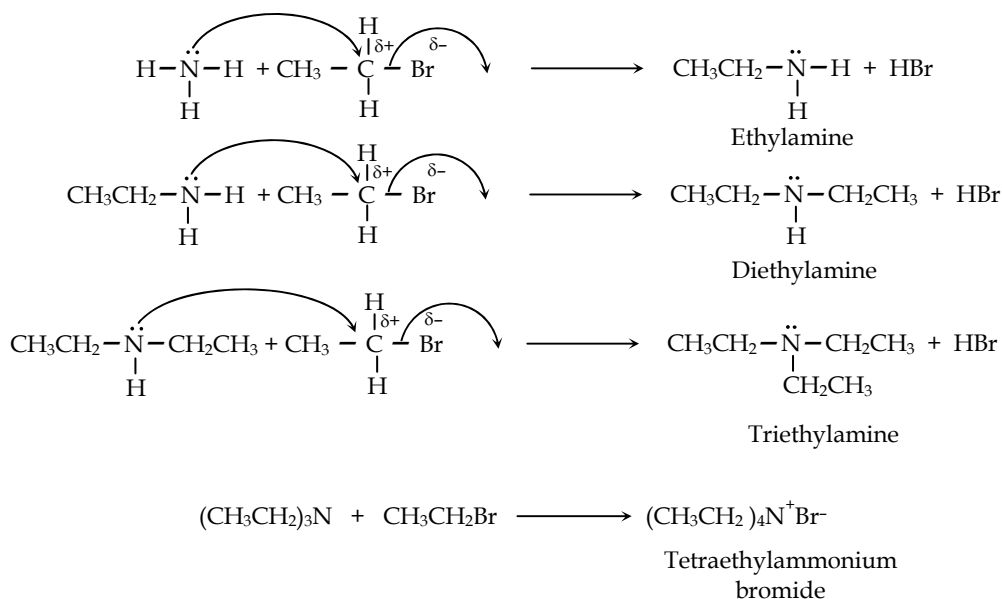
- If ammonia is in excess, primary amine is obtained.

**Example:**

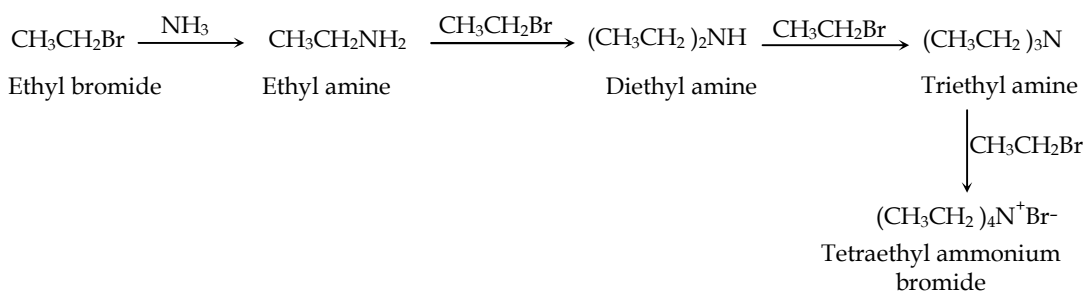
- If alkyl halide is in excess, a mixture of primary, secondary and tertiary amines and quaternary ammonium salt is obtained.



Example: when ammonia is heated with excess ethyl bromide,



or,



Note:

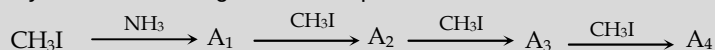
- This method has disadvantage of getting mixture of different classes of amines.
- The order of reactivity of alkylhalides with amines is $\text{RI} > \text{RBr} > \text{RCl}$.
- This is nucleophilic substitution reaction. Here, ammonia, primary and secondary amine acts as nucleophile. Primary alkyl halide gives good yield of amine by nucleophilic substitution but tertiary alkyl halide gives alkene by elimination reaction.



- This method is not suitable for the preparation of aryl amines since aryl halides are less reactive towards nucleophilic substitution reaction.

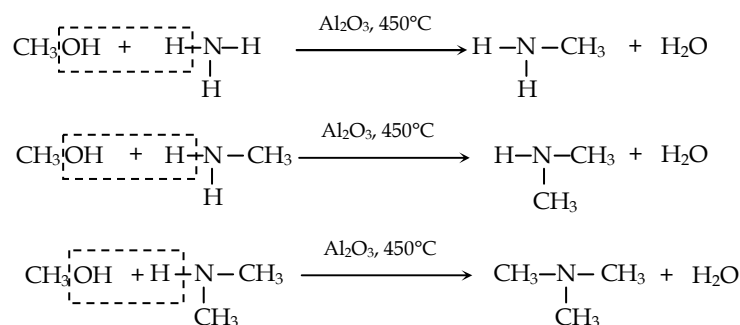
Self Test

15.3 Identify A in the following reaction sequence:



- From Alcohols:** When alcohol is heated with ammonia in presence of catalyst like alumina or copper chromite, a mixture of primary, secondary and tertiary amines is obtained.

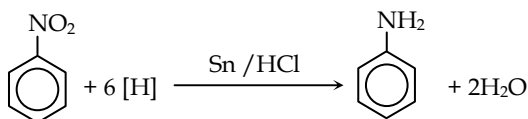
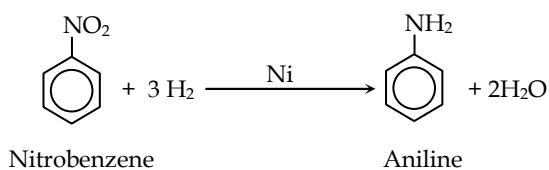
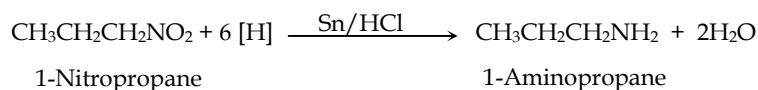
Example:



B. Preparation of Primary Amines

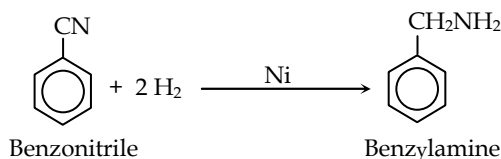
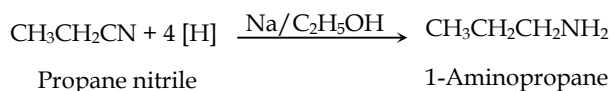
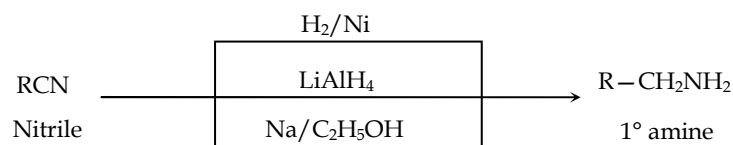
- From nitro compounds:** Catalytic reduction of nitro compounds with H_2 in presence of finely divided Ni or Pt catalyst or chemical reduction with metal and acid usually Sn/HCl, Fe/HCl give primary amines.

Examples:



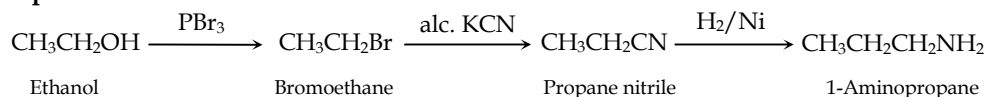
This is the most useful method of preparing aromatic amines. This is because starting material, aromatic nitro compound, is readily available (by direct nitration of aromatic hydrocarbons) and the product primary aromatic amine is the most important kind of amine.

- From nitriles:** Reduction of nitriles with H_2/Ni , $\text{Na}/\text{C}_2\text{H}_5\text{OH}$, LiAlH_4 give primary amines.



This method is useful for increasing carbon chain length by one carbon atom.

Examples:



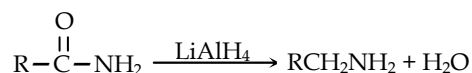
Reduction of nitrile into primary amine with Na/C₂H₅OH is called **Mendius reaction**.

Self Test

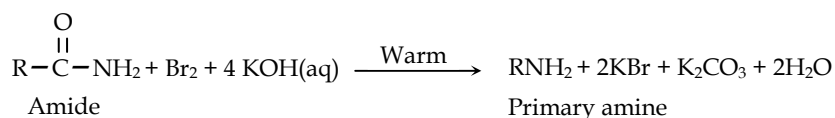
15.4 Convert ethyl amine into *n*-propyl amine.

3. **From amides:** From amides, amines can be obtained by two different ways:

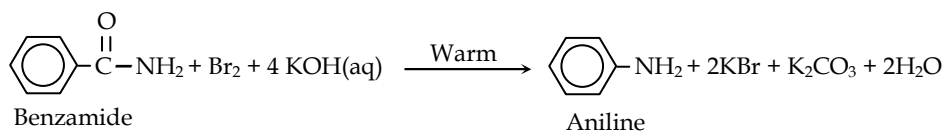
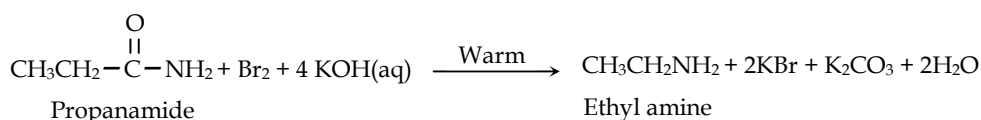
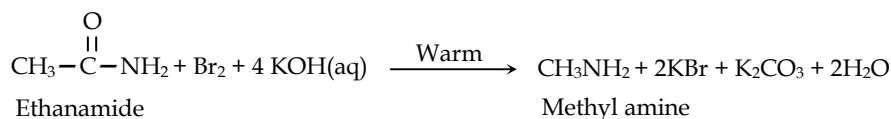
a. **By reduction:** Reduction of amide with LiAlH₄ or Na/C₂H₅OH gives primary amine.



b. **By Hofmann hypobromite rearrangement/Hofmann degradation (1881):** When acid amide is warmed with bromine (or chlorine) and aqueous or alcoholic caustic alkali, primary amine having one carbon less than the amide is formed. This reaction is commonly known as **Hofmann degradation**.

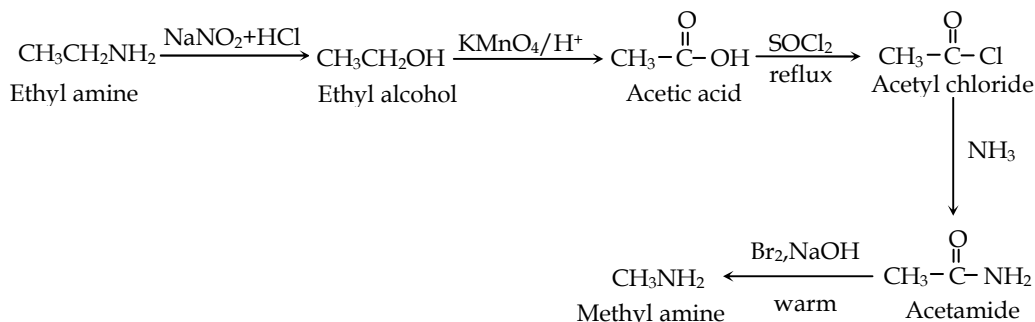


Examples:

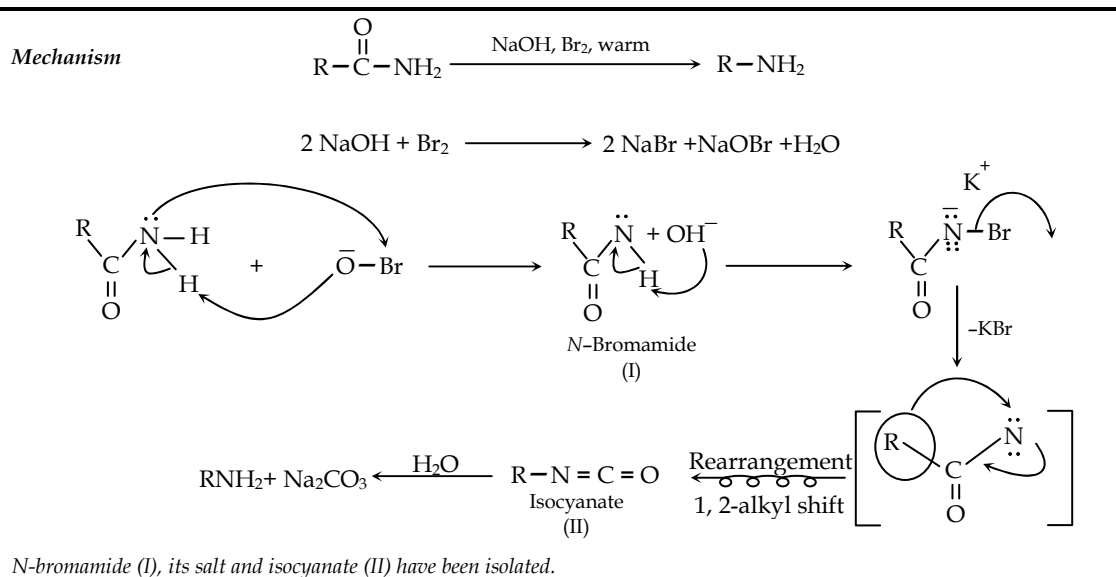


The overall result is removal of carbonyl group, $-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-$ from amide. Hence this reaction is **decarbonylation reaction**.

This method is utilized in descending the series by one carbon. This can be illustrated by the conversion of ethyl amine into methyl amine.

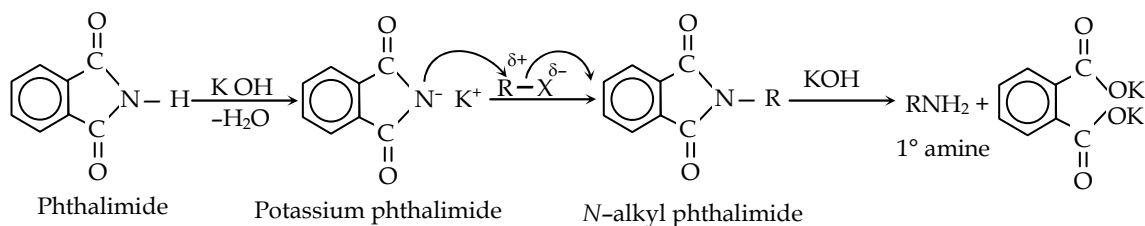
**Self Test**

15.5 What is Hofmann's degradation reaction? Illustrate its application by the conversion of ethyl amine into methyl amine.

**Note:**

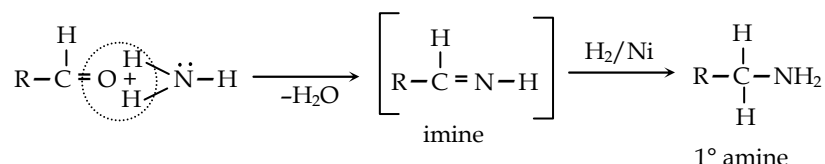
This reaction is also called Hofmann hypobromite reaction the actual reagent for this reaction is hypobromite ion, O^-Br . In Hofmann bromamide reaction, *N*-bromamide is formed as intermediate which can be isolated and Hofmann rearrangement takes place in a stage of this reaction.

4. **From alkyl halide (Gabriel phthalimide synthesis, 1887):** In this method, phthalimide is converted into its salt by treating with alcoholic caustic potash. The salt is then treated with primary alkyl halide to get *N*-alkylphthalimide, which is hydrolyzed by heating with 20% HCl or KOH to get primary amine and phthalic acid. If KOH is used potassium salt of phthalic acid is obtained.

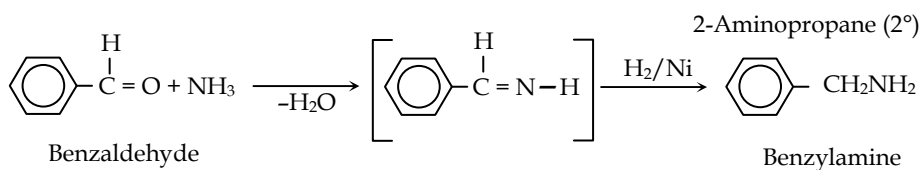
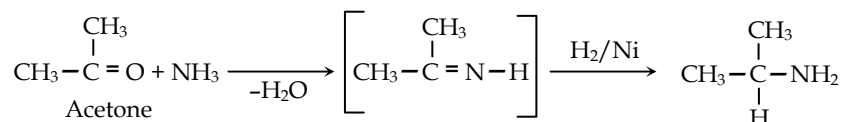
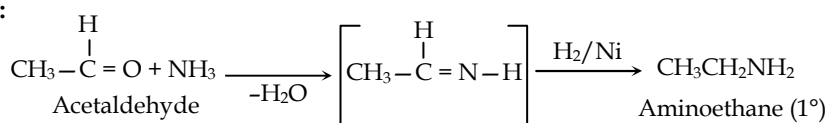


This method gives pure primary amine and phthalimide is regenerated from phthalic acid. Aromatic primary amines cannot be prepared by this method because aryl halides do not undergo nucleophilic substitution easily.

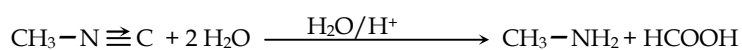
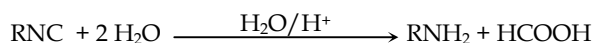
5. **From aldehydes and ketones (Reductive amination):** Aldehydes and ketones react with ammonia in presence of reducing agent like H_2/Ni or NaBH_3CN (sodium cyanohydridoborane) to give primary amine. This is called reductive amination of aldehydes and ketones.



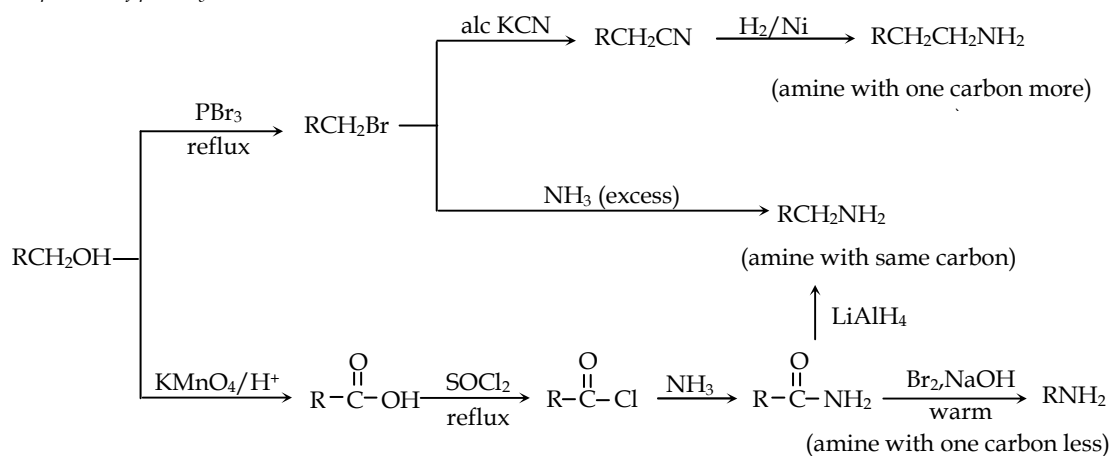
Example:



6. **From isocyanides:** Alkyl isocyanides are hydrolyzed to an primary amine and **formic acid** by dilute acids but not hydrolyzed by alkalis.



Preparation of primary amines: An overview

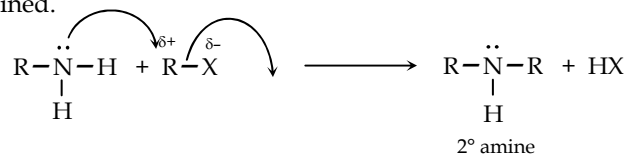


Self Test

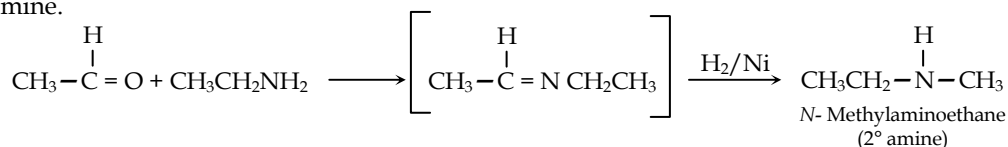
15.6 How do you prepare *n*-propyl amine from i) 1-Nitropropane ii) *n*-Propyl bromide
 iii) *n*-Propyl alcohol iv) *n*-Butyl alcohol v) Ethyl alcohol

C. Preparation of Secondary Amines

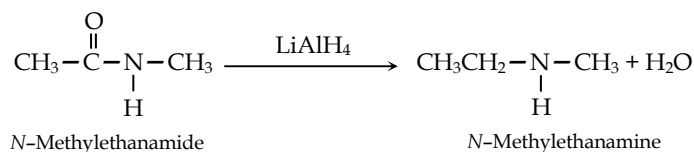
1. **From alkyl halides:** When primary amine treated with calculated amount of alkyl halide, secondary amine is obtained.



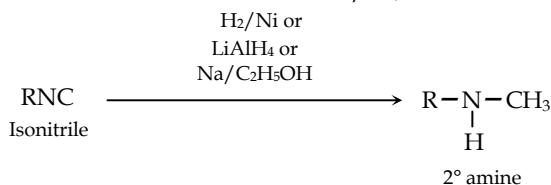
2. **From aldehydes and ketones:** Aldehydes and ketones react with primary amine in presence of reducing agent like H_2/Ni or NaBH_3CN (sodium cyanohydridoborane) to give secondary amine.



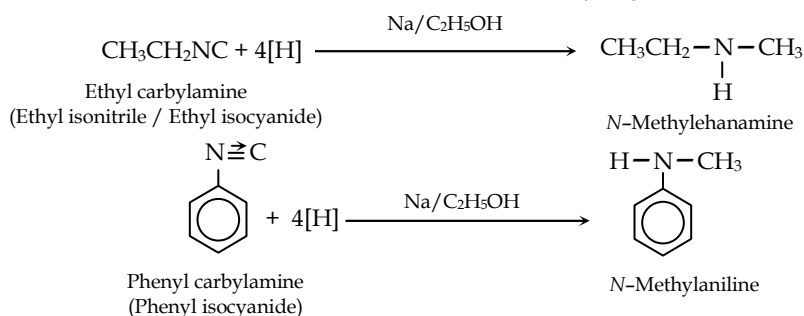
3. **From N-substituted amides:** Reduction of *N*-substituted amide with LiAlH_4 or $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ gives secondary amine.



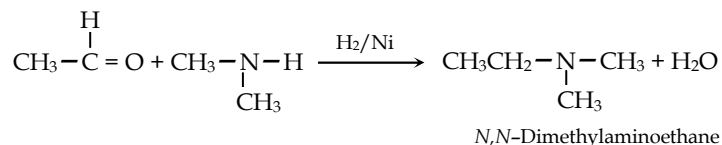
4. **From isonitriles:** Reduction of isonitrile with H_2/Ni , LiAlH_4 or $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ gives secondary amines.



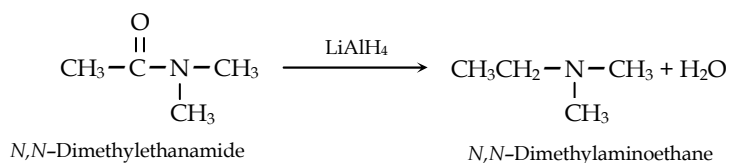
Examples:

**D. Preparation of Tertiary Amines**

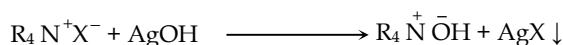
1. **From aldehydes and ketones:** Aldehydes and ketones react with secondary amine in presence of reducing agent like H_2/Ni or NaBH_3CN (sodium cyanohydridoborane) to give tertiary amine.



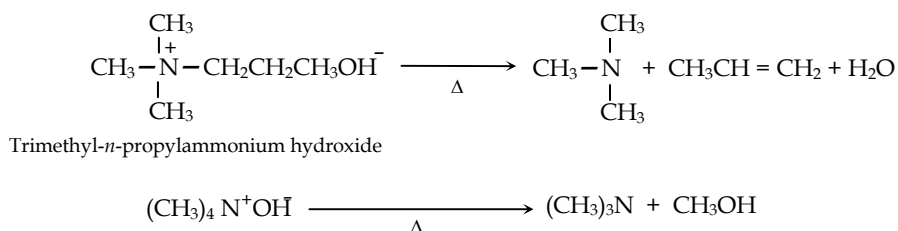
2. **From *N,N*-disubstituted amides:** Reduction of *N,N*-disubstituted amide with LiAlH_4 or $\text{Na/C}_2\text{H}_5\text{OH}$ give tertiary amine.



3. **From quaternary ammonium salt:** Quaternary ammonium salt reacts with moist silver oxide or methanolic KOH to give quaternary ammonium hydroxide.



When quaternary ammonium hydroxide is heated strongly (to 125°C or higher), it undergoes decomposition to give alkenes, tertiary amine and water.

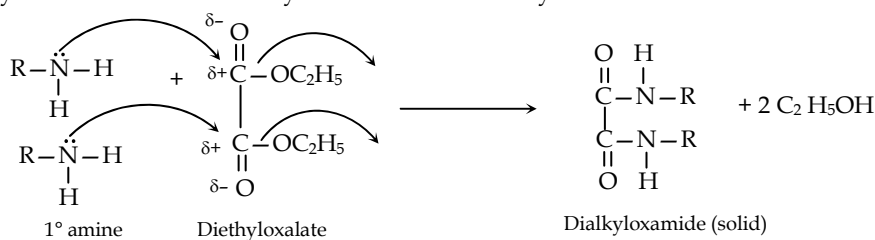


Note:

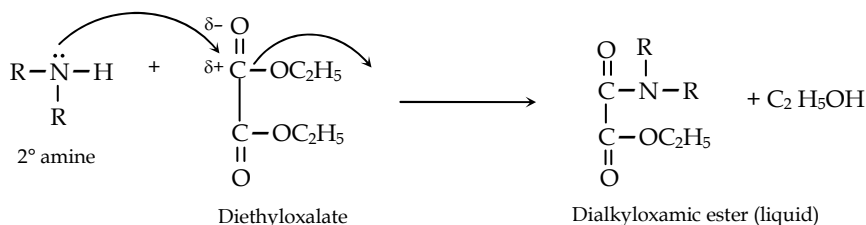
Quaternary ammonium hydroxides are white deliquescent crystalline solids. Their solution is strongly alkaline comparable to that of NaOH or KOH due to complete dissociation producing OH^- .

15.8 Separation of Amines

- Fractional distillation:** Primary, secondary and tertiary amines can be separated from their mixture by fractional distillation since they differ in their boiling point. The success of this method is due to high efficiency of fractionation apparatus.
- Hofmann method:** This method is a convenient method for the separation of different classes of amines from their mixture. In this method, the mixture of amines is treated with diethyl oxalate. Different classes of amines react differently.
 - Primary amine reacts with diethyl oxalate to form dialkyloxamide which is solid.



- ii. Secondary amine reacts with diethyloxalate to form dialkyloxamic ester which is generally liquid.

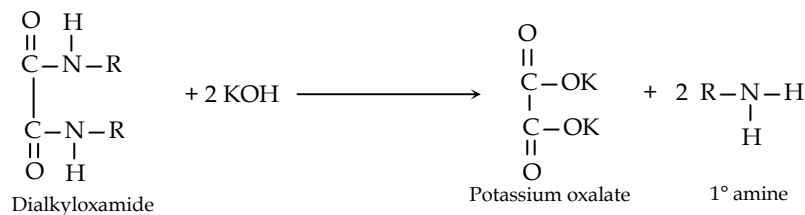


Dialkyl oxamic ester does not react further with another molecule of 2° amine due to steric hinderance.

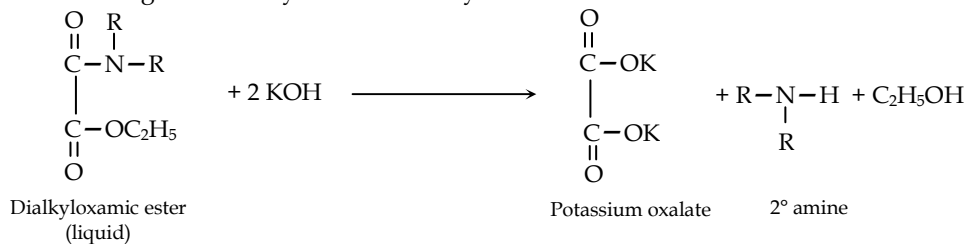
- iii. Tertiary amine does not react with diethyloxalate presumably due to lack of hydrogen with nitrogen.

The resulting mass (dialkyl oxamide, dialkyloxamic ester, 3° amine and ethyl alcohol) is subjected to fractional distillation. Here, 3° amine distills first followed by ethyl alcohol.

The remaining mass is filtered. The residue (dialkyloxamide) is treated with caustic alkali to recover primary amine.



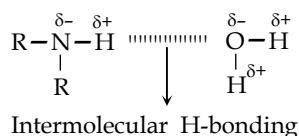
The filtrate (dialkyloxamic ester) is treated with caustic alkali followed by fractional distillation to get secondary amine and ethyl alcohol on different fractions.



In this way, different classes of amines are separated.

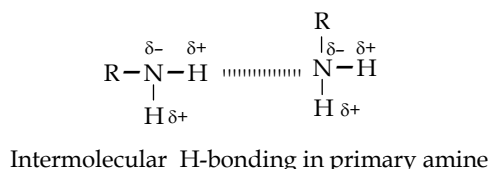
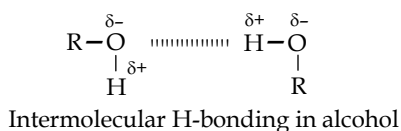
15.9 Physical Properties

- State:** Lower members of amines are gases and liquids, higher members are solid.
- Odour:** Methyl amines and ethyl amines have ammoniacal odour and higher alkyl amines have fishy odour.
- Solubility:** All classes of amines can form intermolecular H-bond with water. Hence, lower members are soluble in water. The solubility decreases with increase in size of water insoluble hydrocarbon part. Amines are soluble in organic solvents.



4. Boiling point

- i. There exists intermolecular H-bond in primary and secondary amines due to polar nature of N-H bond. Tertiary amines have no N-H bond and can not form intermolecular H-bond among their molecules. Hence, amines, except tertiary amines have higher boiling point than nonpolar compounds of comparable molecular weight. Order of boiling points of amines is primary > secondary > tertiary
- ii. Amines have lower boiling point than alcohols of comparable molecular weight.
This is because intermolecular H-bond in amine is weaker than that in alcohol since N-H bond is less polar than O-H bond.



Self Test

15.7 Ethyl amine has lower boiling point than that of ethyl alcohol, explain.

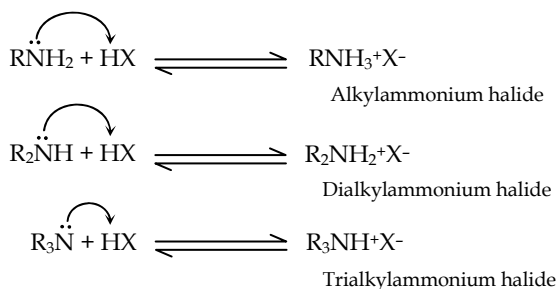
15.8 The boiling point of following isomeric amines is observed in the order: n-Propyl amine (B. Pt. 49°C) > Ethylmethyl amine (B. Pt. 36°C) > Trimethyl amine (3°C). How do you account for this order of their boiling point?

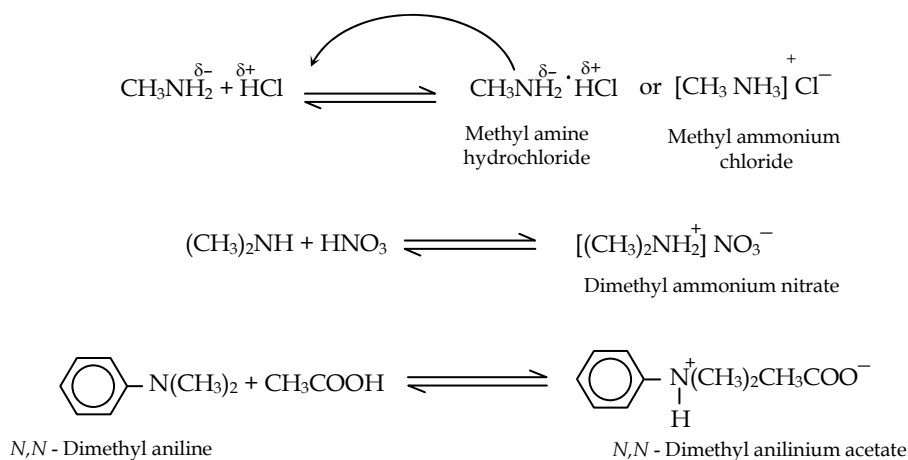
15.10 Chemical Properties

Amines have an unshared pair of electrons in nitrogen atom which governs the chemical behaviour of amines. Due to unshared pair of electrons, an amine can act as a nucleophile as well as a base. The number of alkyl or aryl groups bonded to nitrogen atom also decides the pattern of reaction and reactivity of amines. This is why 1°, 2° and 3° amines differ in many reactions.

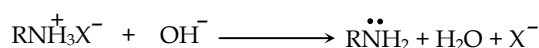
1. **Reaction with acids: Basic nature:** Like ammonia, all classes of aliphatic as well as aromatic amines are basic. Amines show appreciable basicity enough to turn red litmus into blue. They react with mineral acids and carboxylic acids to form salt.

Examples:





Amines are displaced by caustic alkalis form their salts.



Comparison of basic strength of amines

A convenient way to compare basic strength of amines is to measure the basicity constant (K_b), which measures the extent to which the amine accepts the H^+ .

Each amine has its characteristic K_b . Larger the value of K_b , or smaller the value of $\text{p}K_b$, the more basic the base.

Aliphatic amines have K_b of about 10^{-4} whereas aromatic amines have K_b of 10^{-10} . This indicates that aliphatic amines are somewhat more basic than ammonia and aromatic amines are considerably less basic than ammonia.

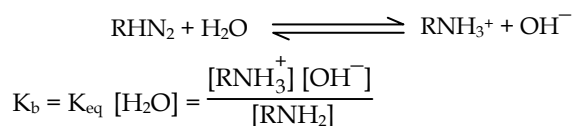
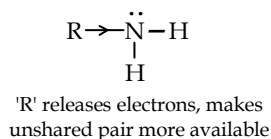
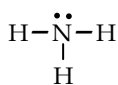


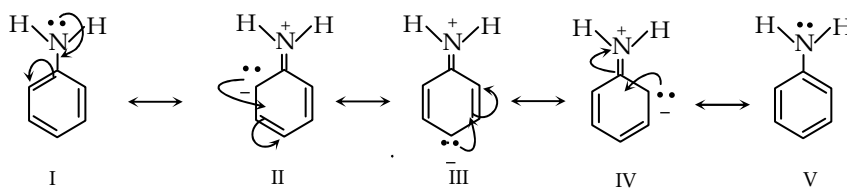
Table 15.1: Basicity constant (K_b) of amines (For ammonia, $K_b = 1.8 \times 10^{-5}$)

Aliphatic amines	$K_b = (10^{-4})$	Aromatic amines	$K_b = (10^{-10})$
Methylamine	4.5×10^{-4}	<i>p</i> -Phenylenediamine	140×10^{-10}
Dimethylamine	5.4×10^{-4}	<i>p</i> -Anisidine	20×10^{-10}
Trimethylamine	0.6×10^{-4}	<i>p</i> -Toluidine	12×10^{-10}
Ethylamine	5.1×10^{-4}	Aniline	4.2×10^{-10}
Diethylamine	10×10^{-4}	<i>p</i> -Chloroaniline	1×10^{-10}
Triethylamine	5.6×10^{-4}	<i>p</i> -Nitroaniline	0.001×10^{-10}

- a. **Aliphatic amines are more basic than ammonia:** Ammonia as well as amines are basic due to presence of a lone pair of electron present on nitrogen atom which is available for sharing with proton. Alkyl groups are electron releasing groups. Therefore, lone pair of electron is more available on nitrogen atom of aliphatic amines than that in ammonia. Hence, aliphatic amines are more basic than ammonia.



- b. **Aryl amines are less basic than ammonia:** Lone pair of electron present on nitrogen atom of aryl amines is involved in resonance with the ring hence less easily available for sharing with proton than that in ammonia. Hence aryl amines are less basic than ammonia.



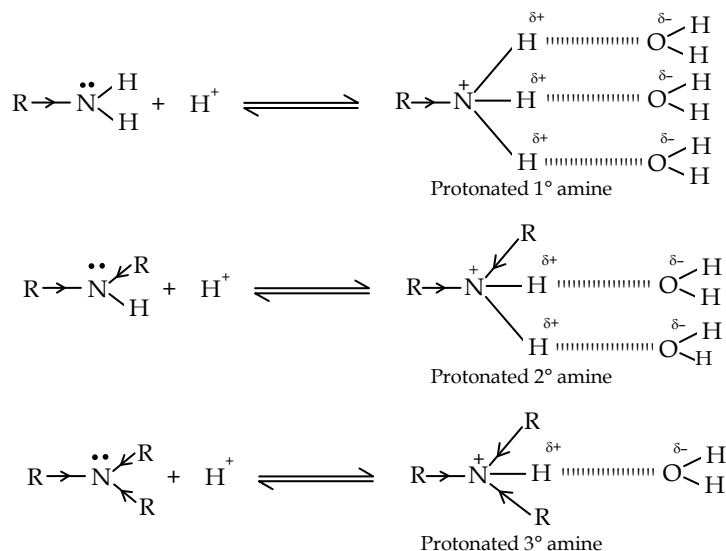
Resonating structures of aniline

- c. **Basic strength of 1°, 2° and 3° amines:** In gas phase, basic strength of different classes of aliphatic amines is found in the order $\text{R}_3\text{N} > \text{R}_2\text{NH} > \text{RNH}_2$ (This is found to be true for $\text{R} = -\text{CH}_3, -\text{C}_2\text{H}_5, -\text{C}_3\text{H}_7$). This can be explained on the basis of +I effect of alkyl group.

In aqueous solution, however, the basic strength is $\text{R}_2\text{NH} > \text{RNH}_2 > \text{R}_3\text{N}$ for $\text{R} = \text{CH}_3$. This can be explained as follows;

In aqueous solution, effect of water called hydration effect also comes into play in addition to inductive effect of alkyl group.

- i. **Hydration effect:** In aqueous solution, protonated amines are hydrated and hence stabilized due to formation of intermolecular hydrogen bond with water as shown.



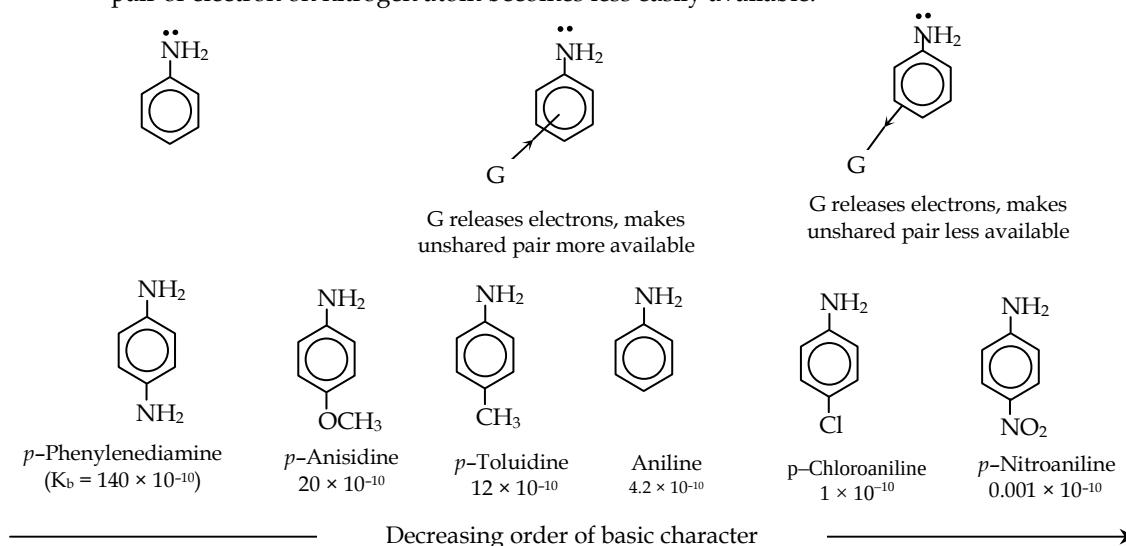
Protonated primary amine is the most hydrated and most stabilized whereas protonated tertiary amine is the least hydrated and least stabilized. Hence basic strength due to hydration effect is $1^\circ > 2^\circ > 3^\circ$.

- ii. **Inductive effect:** Alkyl group releases electrons towards nitrogen atom due to inductive effect and neutralizes the positive charge and stabilizes protonated amines. Protonated tertiary amine is stabilized the most and protonated primary amine is stabilized the least. Hence due to inductive effect of alkyl group, the basic strength is in the order $3^\circ > 2^\circ > 1^\circ$.

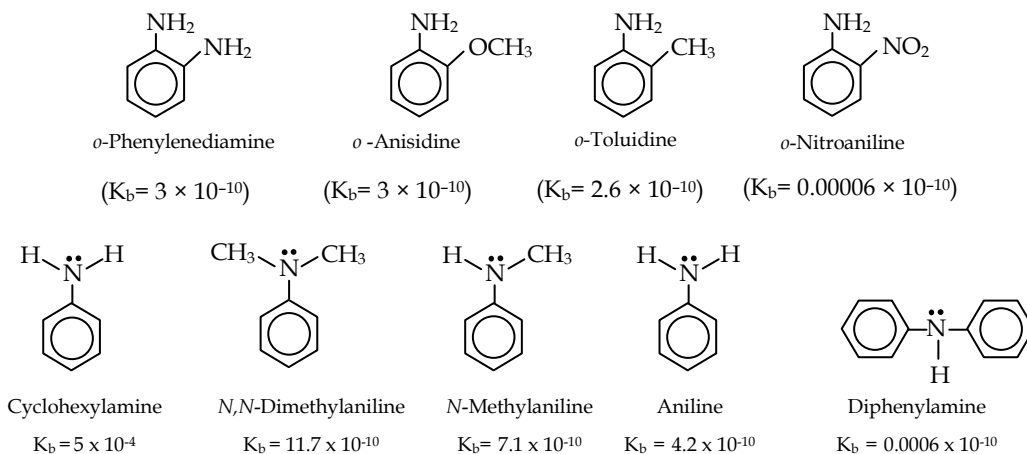
In aqueous solution, both effects come to play which are exactly opposite order. The net result of both effects is the basic strength of aliphatic amines in aqueous solution in the order $2^\circ > 1^\circ > 3^\circ$.

Steric Factor: In 3° amine, N is attached with three bulky alkyl groups around it, which cause large steric hinderance for attack by an acid. So, 3° amines are less basic than 1° and 2° amines (due to setric factor, the basic nature of amines is $1^\circ > 2^\circ > 3^\circ$).

- d. **Effect of substituent(s) on basicity of amines:** Electron releasing substituent increases the basicity of amines since lone pair of electron on nitrogen atom becomes more easily available. Electron withdrawing substituent decreases the basicity of amines since lone pair of electron on nitrogen atom becomes less easily available.



- e. **Ortho effect:** Any substituent, whether electron releasing or electron withdrawing, present on ortho position to amino group decreases the basicity. All the following compounds are less basic than aniline. (Compare K_b values).



Self Test

15.9 Explain

- i. Ethyl amine is more basic than ammonia ii. Aniline is less basic than ammonia
 iii. Aliphatic amine is more basic than aniline

15.10 Arrange aniline, *p*-nitroaniline and cyclohexyl amine in the decreasing order of their basic character. Give explanation for your arrangement.

Some multiple choice questions related to basicity of amines

1. In aqueous solution, the basicity of the following compounds is in the order;

- a. $(\text{CH}_3)_4\text{N}^+\text{OH}^-$ b. $\text{CH}_3\text{CH}_2\text{NH}_2$
 c. NH_3 d. $\text{C}_6\text{H}_5\text{NH}_2$

Answer: a) > b) > c) > d)

(Quaternary ammonium hydroxide > aliphatic amine > ammonia > aromatic amine)

2. Arrange the following compounds in the decreasing order of their basicity;

- a. Ammonia b. Methylamine
 c. Ethylamine d. Aniline

Answer: c) > b) > a) > d)

3. Arrange the following compounds in the decreasing order of their basicity in aqueous solution;

- a. Ammonia b. Methylamine
 c. Dimethylamine d. Trimethylamine

Answer: c) > b) > d) > a)

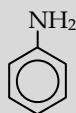
4. Which one is more basic?

- a. Aniline or *N*-Methylaniline b. *N*-Methylaniline or *N,N*-Dimethylaniline
 c. Aniline or Diphenylamine d. Aniline or Cyclohexylamine

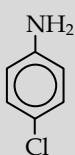
Answer:

- a. *N*-Methylaniline ($-\text{CH}_3$ releases electron on nitrogen)
 b. *N,N*-Dimethylaniline (Two $-\text{CH}_3$ release more electron)
 c. Aniline (Unshared pair of electron one delocalized in two rings)
 d. Cyclohexylamine (Aliphatic amine)

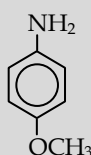
5. Arrange the following compounds in the decreasing order of their basicity;



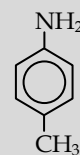
(a)



(b)



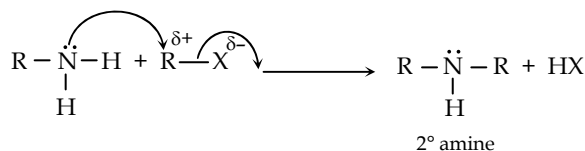
(c)

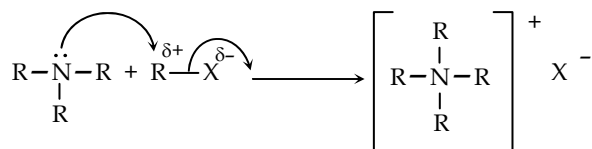
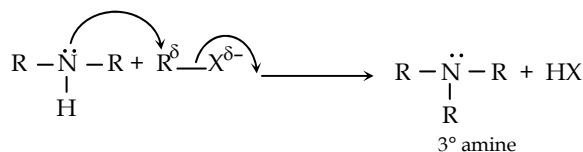


(d)

Answer: (c) > (d) > (a) > (b)

2. **Reaction with alkyl halides: Alkylation:** When an amine is treated with an alkyl halide, hydrogen of N atom are successively replaced by alkyl group to form amines of next higher class and finally forms quaternary ammonium salt.

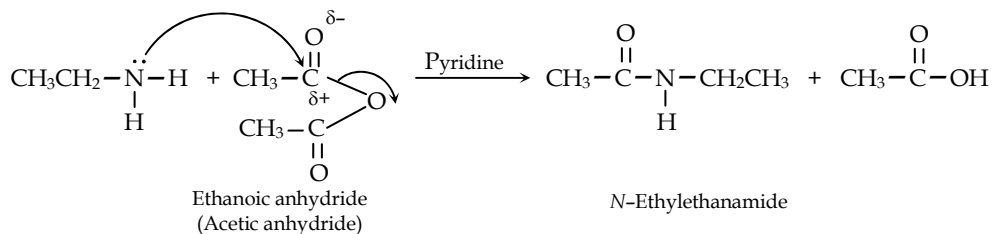
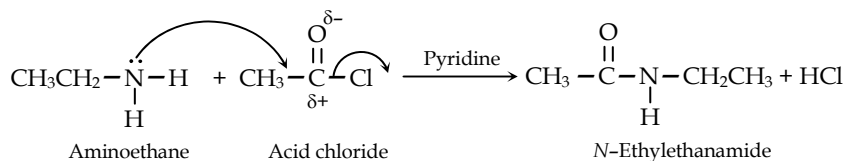
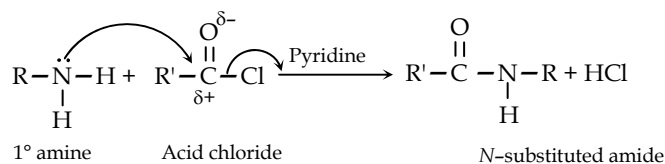




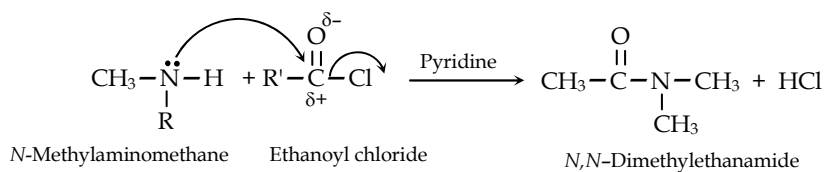
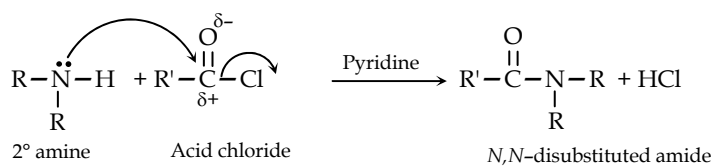
Quaternary ammonium salt

32. Reaction with acid chlorides and anhydrides: Acylation

- a. Primary amine reacts with acid chloride or acid anhydride to form *N*-substituted amide.



- b. Secondary amine reacts with acid chloride to form *N,N*-disubstituted amide.

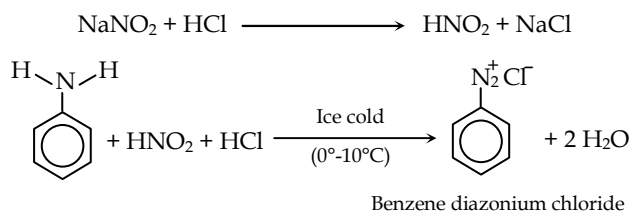


- c. Tertiary amines do not react since they have no hydrogen atom on nitrogen to stabilize the product.

4. **Reaction with nitrous acid:** Different classes of aliphatic and aromatic amines react with nitrous acid differently. Nitrous acid is unstable hence prepared *in situ* by mixing sodium nitrite and mineral acid usually HCl under cold condition.

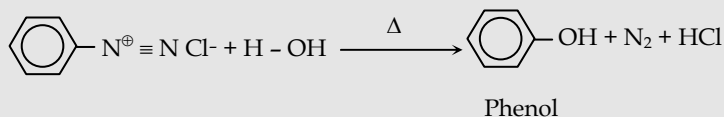
a. **Reactions of primary amines**

- i. **Primary aromatic amine** reacts with nitrous acid in ice cold solution to form diazonium salt. The reaction is called diazotization.

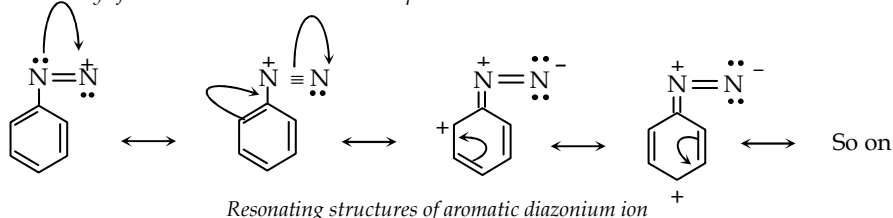


Note

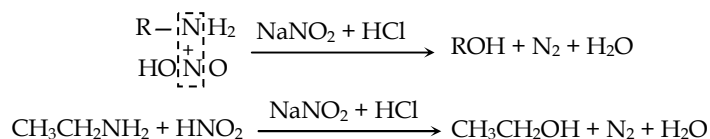
When the temperature of the reaction exceeds than 5°C , benzene diazonium chloride is hydrolysed into phenol.



The relative stability of aromatic diazonium salt over aliphatic diazonium salt is due to its resonance stabilization.

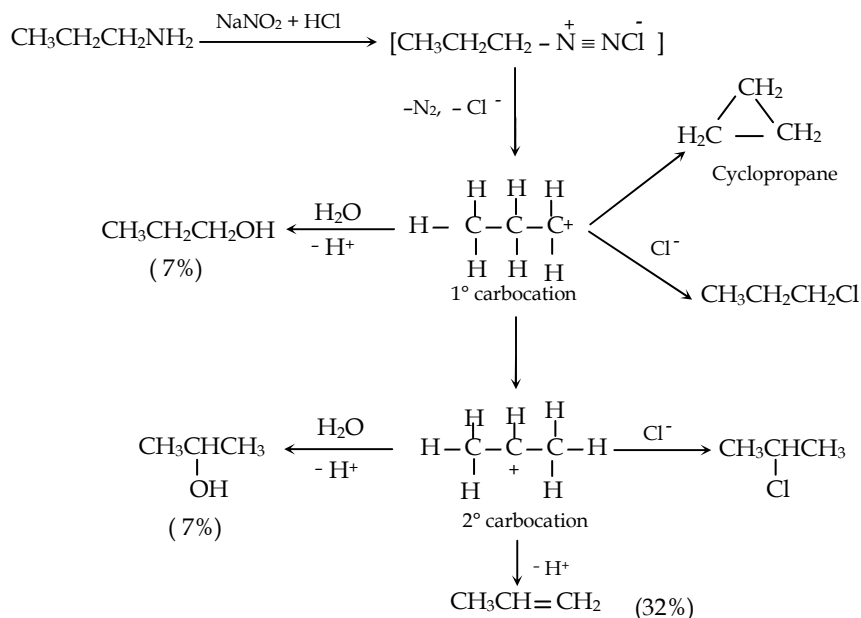


- ii. **Primary aliphatic amine** reacts with nitrous and in ice-cold solution to give alcohol. N_2 (as bubbles) is evolved quantitatively. Hence this reaction can be used as a test for primary aliphatic amine since none of other classes of amines evolve N_2 .

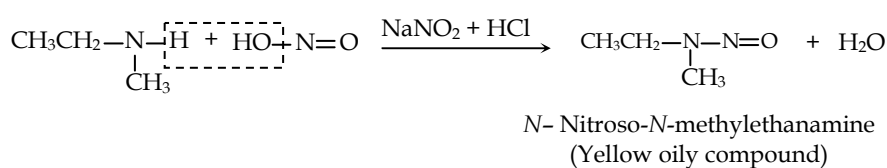
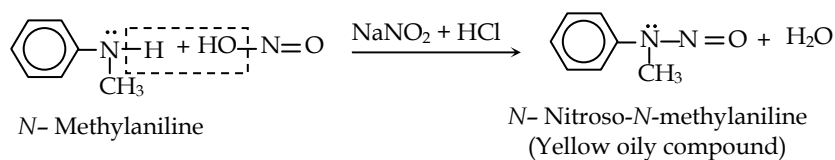


In fact, the products are isomeric alcohols, isomeric alkyl halides, alkene and very little cycloalkanes when structure permits. Hence, reaction of primary aliphatic amines with nitrous acid is not important from synthetic point of view.

Aliphatic diazonium salts are unstable even at low temperature hence decompose forming carbocation which then gives a mixture of compounds. For example, *n*-propylamine when treated with nitrous acid; the products are a mixture of propene, Propan-1-ol, Propan-2-ol, 1-Chloropropane, 2-Chloropropane etc. Very little cyclopropane is also observed.

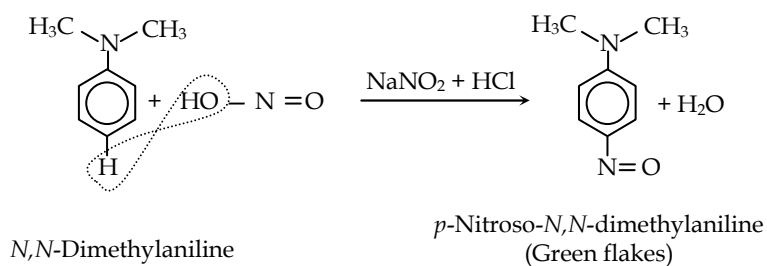


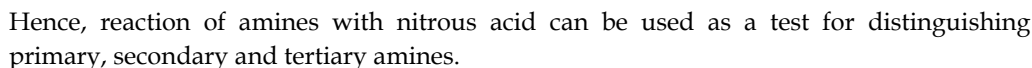
- b. **Secondary aliphatic as well as aromatic amine** reacts with nitrous acid to give *N*-nitroso amine which appears as insoluble yellow oily layer.



c. **Reactions of tertiary amines**

- i. **Tertiary aromatic amine** reacts with nitrous acid to form *p*-nitroso compound.

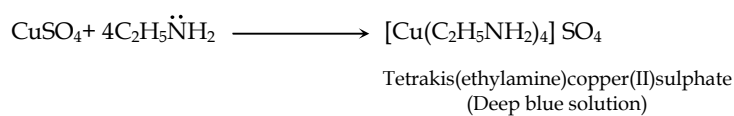
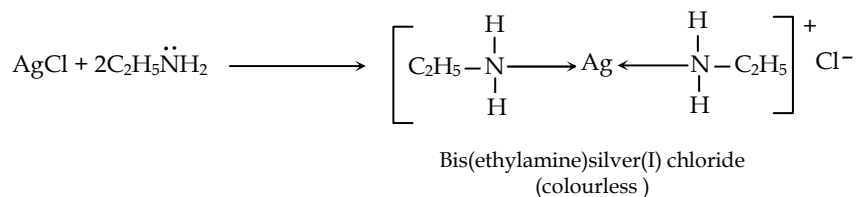
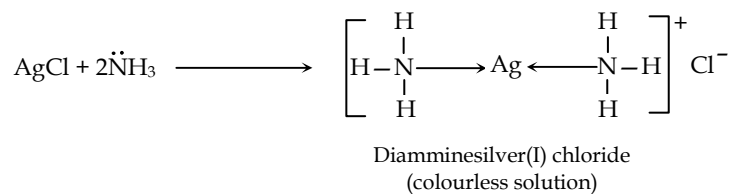




-
- Aniline + 3KOH (alc) $\xrightarrow{\Delta}$ Phenyl carbylamine (Phenyl isocyanide) + 3KCl + 3H₂O

7. **Reaction with metal ions:** Like ammonia, lower aliphatic amines like CH_3NH_2 , $\text{C}_2\text{H}_5\text{NH}_2$ etc. form complex with many transition metal ions. This is due to a lone pair of electron present on nitrogen atom which is accepted in vacant d-orbital of transition metal ions.

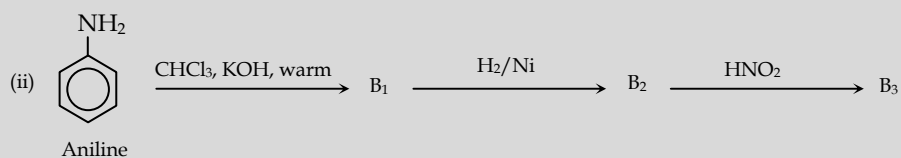
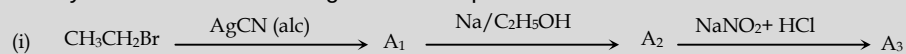
Examples:



Here, ammonia and amines are called ligands.

Self Test

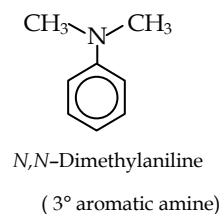
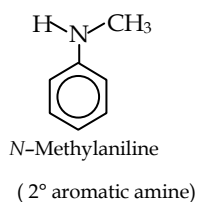
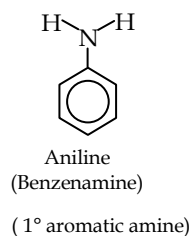
15.14 Identify A and B in the following reaction sequence:

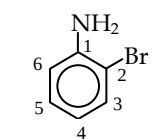
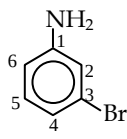
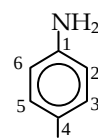
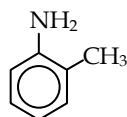
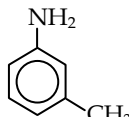
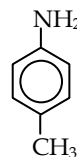
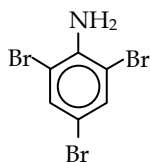


15.11 Aromatic Amine (Aniline)

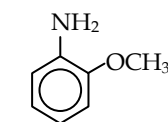
15.11.1 Nomenclature

Aromatic amines are named as derivatives of the simplest aromatic amine, aniline.



2- Bromoaniline
(o-Bromoaniline)3- Bromoaniline
(m-Bromoaniline)4- Bromoaniline
(p-Bromoaniline)2- Aminotoluene
(o-Toluidine)3- Amino toluene
(m-Toluidine)4- Bromoaniline
(p-Bromoaniline)

2, 4, 6-Tribromoaniline



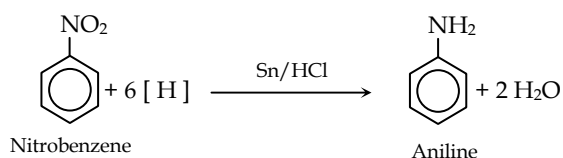
o-Anisidine

Self Test

15.15 Write isomeric amines of molecular formula $C_7H_{11}N$ having benzene ring, classify them as primary, secondary and tertiary amines and give their name. *Hint: 5 isomers*

15.11.2 Preparation of Aniline

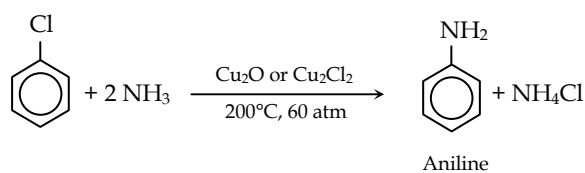
- From nitro compounds:** Reduction of nitro compounds with H_2 in presence of Ni or Pt catalyst (catalytic reduction) or with metal and acid usually Sn/HCl, Fe/HCl (chemical reduction) give primary amines.



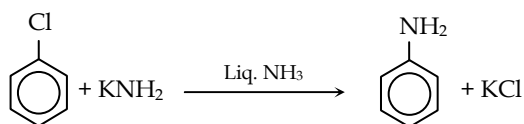
This is the most useful method of preparing aromatic amines. This is because starting material (aromatic nitro compounds) is readily available and the product is the most important kind of amine, primary aromatic amine.

2. From chlorobenzene

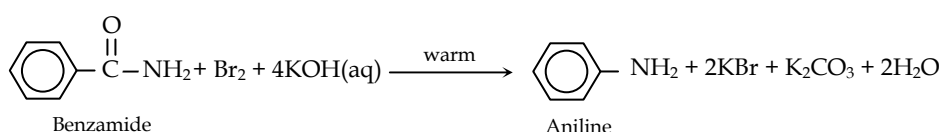
- When chlorobenzene is heated with aqueous ammonia solution under pressure in presence of cuprous oxide, aniline is formed.



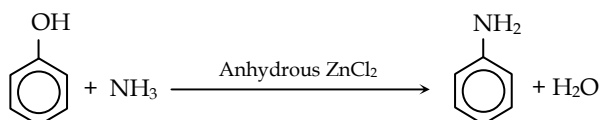
- ii. When chlorobenzene is treated with NaNH_2 or KNH_2 in liquid ammonia, aniline is obtained. The reaction follows **elimination-addition mechanism** through **benzyne** intermediate.



3. **From benzamide (Hofmann's degradation reaction):** When benzamide is warmed with bromine (or chlorine) and aqueous or alcoholic caustic alkali, aniline is obtained.



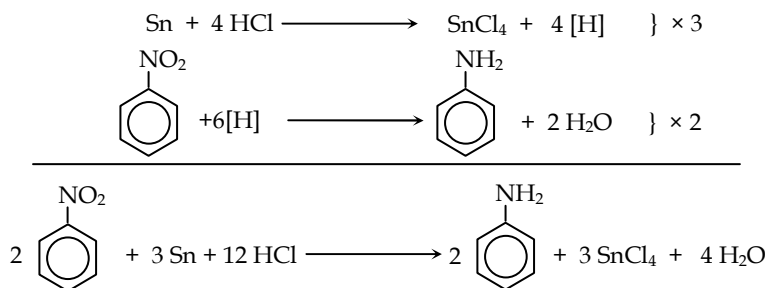
4. **From phenol:** When phenol is heated with NH_3 in presence of dehydrating agent anhydrous ZnCl_2 , it gives aniline.



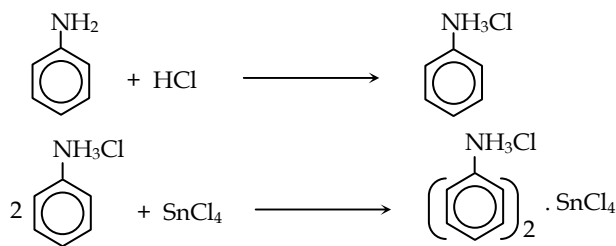
15.11.3 Laboratory Preparation of Aniline

Theory

Aniline is prepared in the laboratory by the reduction of nitrobenzene with granulated tin and concentrated hydrochloric acid.

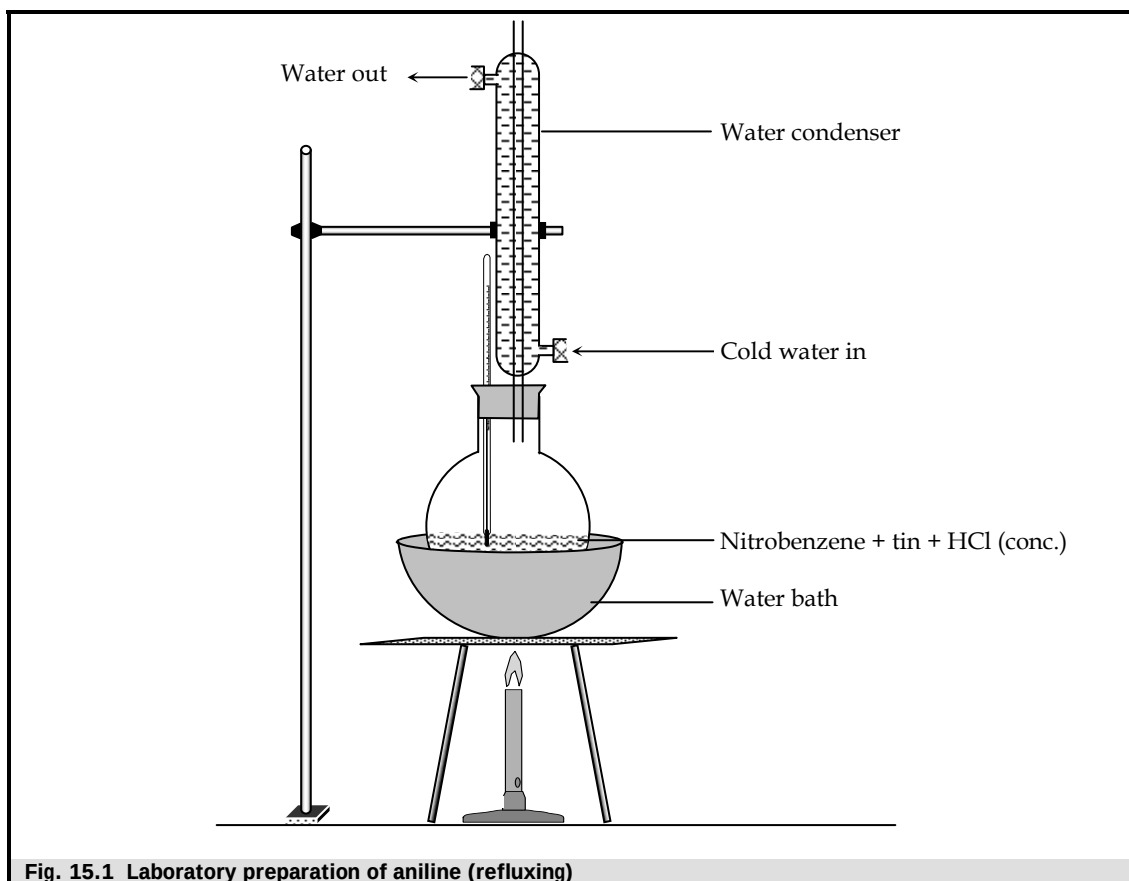
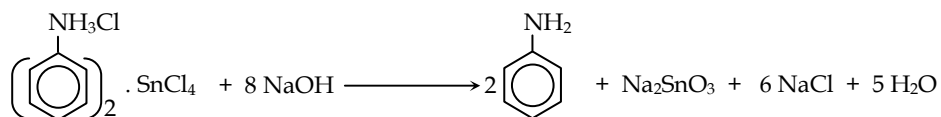


Aniline produced reacts with hydrochloric acid and stannic chloride to form addition compound.



aniline stannichloride (solid)

Aniline is recovered from addition compound by the action of caustic soda solution followed by **steam distillation**.



Procedure

About 25 g of nitrobenzene and 50 g of granulated tin are taken in a R B flask and 100 ml of concentrated HCl is added into the flask, about 20 ml at a time with shaking and cooling in cold water.

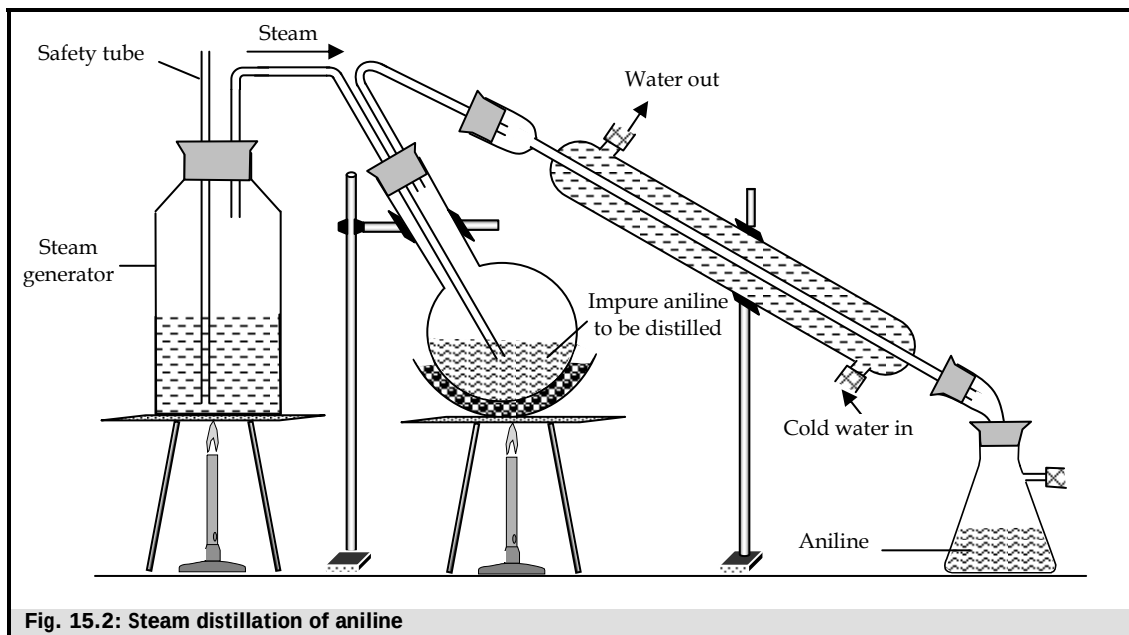


Fig. 15.2: Steam distillation of aniline

When whole acid is added, it is refluxed on water bath for an hour until the smell of nitrobenzene disappears. On cooling, the solid mass appears due to addition compound.

Concentrated NaOH is added to the solid mass to get clear and alkaline solution. Here, the dark brown layer of aniline separates out. The whole mass is subjected to *steam distillation*¹ and aniline is obtained as distillate.

Purification

Aniline obtained as distillate contains water. Aniline is slightly soluble in water (solubility is 3.7), hence, it is difficult to separate using separating funnel directly.

- Solid sodium chloride is added to aniline till saturation. Sodium chloride is more soluble in water than aniline hence layer of aniline separates out and separated.
- Aniline is then **extracted** with ether in small lots. Ether is evaporated using hot water.
- Aniline is dried over solid NaOH or KOH.
- Finally, aniline is redistilled to about 182-190°C to get pure aniline.

Self Test

15.16 What would you do to distinguish aniline from nitrobenzene chemically?

1 Steam distillation (Co-distillation with water): This method is applied for the separation and purification of organic compounds (solids and liquids) which are insoluble in water, volatile in steam and have high vapour pressure at boiling point of water. Impurities must be non-volatile. Here, steam lowers the boiling point of organic compound.

A liquid boils when its vapour pressure is equal to the atmospheric pressure. In steam distillation, two immiscible compounds, water and organic compound, exert their own vapour pressure independently and the mixture boils when sum of vapour pressure of organic compound (P_1) and vapour pressure of water (P_2) becomes equal to the atmospheric pressure (P).

$P_1 + P_2 = P$ or $P_1 = P - P_2$

This means organic compound (as well as water) boils below its normal boiling point. For example, aniline (b pt. 184°C) boils at 98.5°C in presence of steam.

15.11.4 Physical Properties of Aniline

- It is colorless liquid with unpleasant odour and very sensitive to oxidation, so it rapidly becomes brown when exposed to air.
- Its Boiling point is 184°C and solidifies at - 6.2°C.
- It is sparingly soluble in water (3.7 g/100ml water) but soluble in organic solvents.
- It is steam volatile and poisonous.

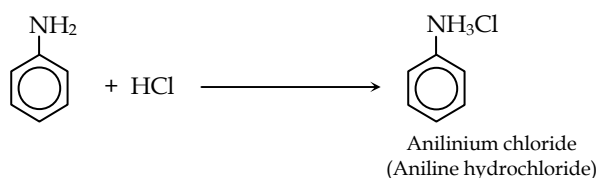
15.11.5 Chemical Properties of Aniline

Chemical properties of aniline can be divided into two groups;

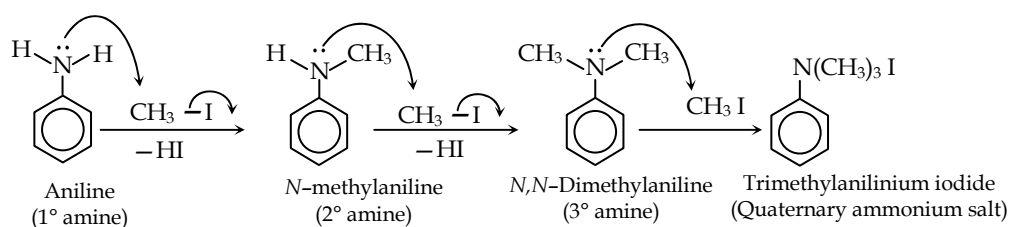
- Reaction due to -NH₂
- Reaction due to benzene ring

A. Reaction due to -NH₂ group

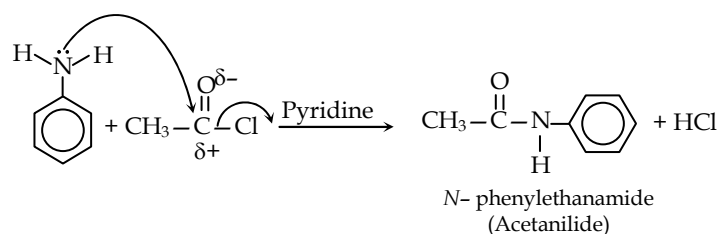
- Salt formation:** Aniline reacts with mineral acids to form salt. The basic nature of aniline is due to a lone pair of electron present on nitrogen atom which is available for sharing with a proton.

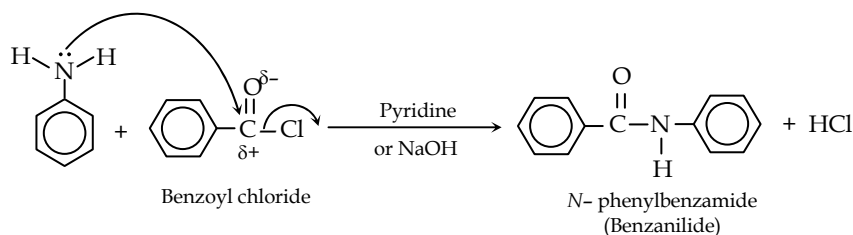
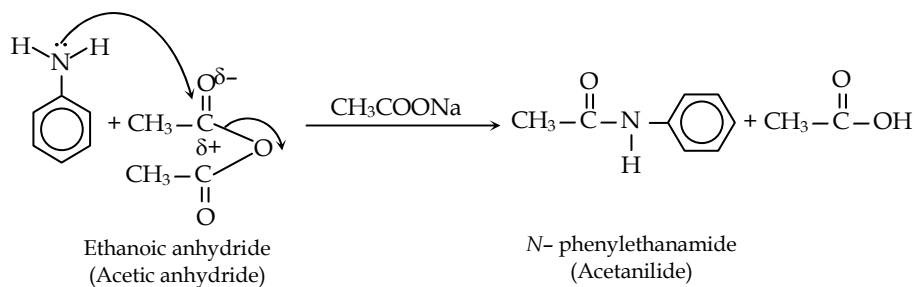


- Alkylation:** When aniline is treated with alkyl halide, alkylation occurs giving secondary amine, tertiary amine and finally quaternary ammonium salt.



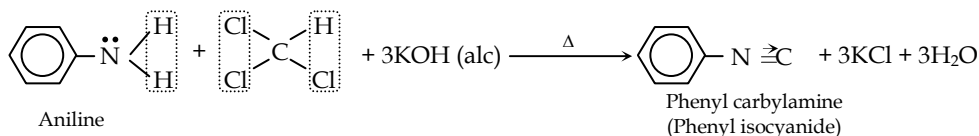
- Acylation:** When aniline is treated with acid chloride or anhydride, acylation occurs giving N-phenylamide called **anilide**.



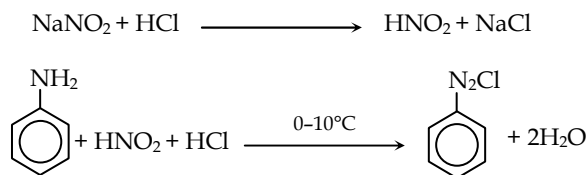


Reaction of aromatic acid chloride (like benzoyl chloride) with amine or alcohol or phenol in presence of base like aqueous sodium hydroxide or pyridine is called **Schotten-Baumann reaction**.

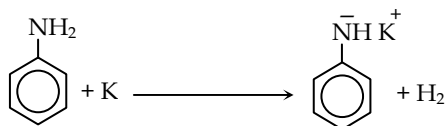
4. **Carbylamine reaction:** When aniline is warmed with chloroform and alcoholic caustic alkali, phenyl carbylamine (phenyl isocyanide) is formed which gives offensive smell (unpleasant odour).



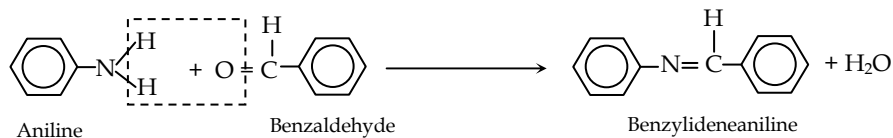
5. **Reaction with nitrous acid: Diazotization:** Aniline reacts with nitrous acid in ice cold solution to form diazonium salt. This process is called diazotization. Nitrous acid is prepared in situ taking sodium nitrite and a mineral acid.



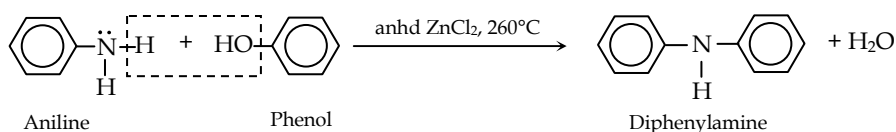
6. **Reaction with Potassium:** When aniline is heated with sodium or potassium, hydrogen gas is evolved.



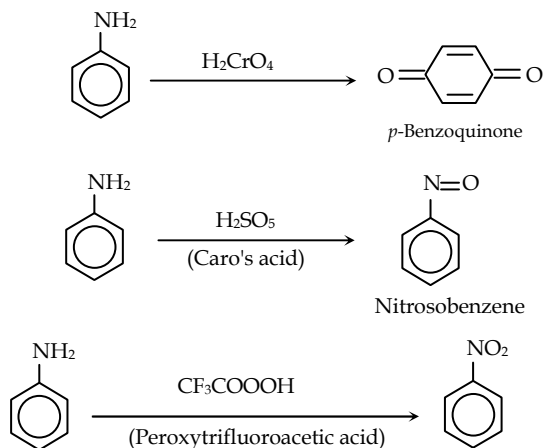
7. **Reaction with benzaldehyde:** Aniline undergoes condensation with aromatic aldehydes to form anils (Schiff bases).



8. **Reaction with phenol:** When aniline is heated with phenol in presence of zinc chloride, diphenyl amine is formed.



9. **Oxidation:** Aniline is easily oxidized by different oxidizing agent into different oxidation products.

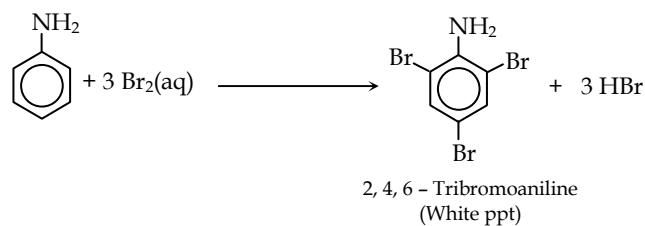


With potassium dichromate, a black dye (aniline black) is obtained. With bleaching powder, *p*-aminophenol (a purple colour) is obtained.

B. Electrophilic aromatic substitution reaction

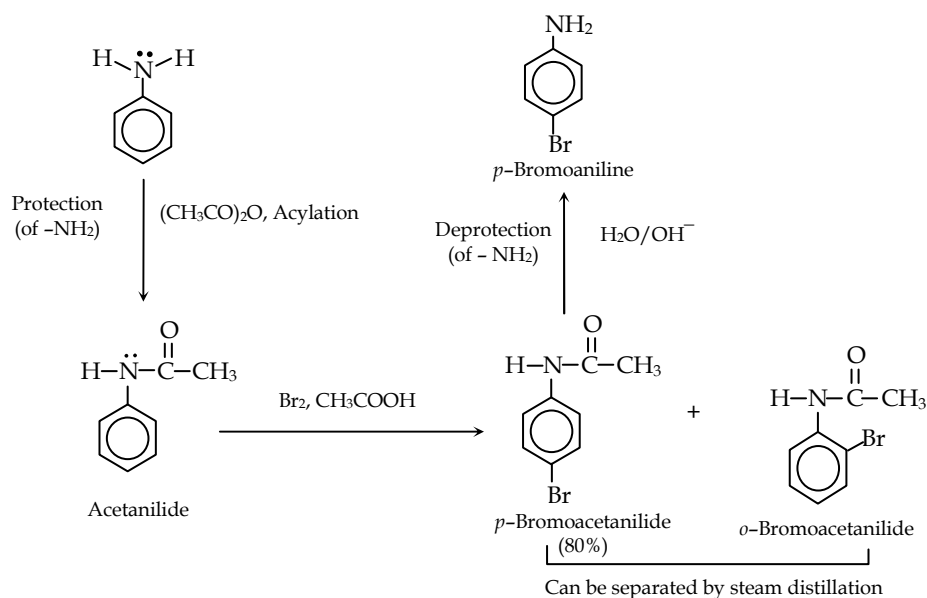
Amino group ($-\text{NH}_2$) in aniline determines the reactivity and orientation in electrophilic aromatic substitution reaction. $-\text{NH}_2$ is a strongly activating group and ortho para director. Aniline is roughly one million times as reactive as benzene in electrophilic aromatic substitution reaction.

1. **Halogenation:** Aniline reacts with aqueous chlorine or bromine immediately to form precipitate of 2,4,6- Trihaloaniline.



The yield is quantitative, hence, this reaction is used for the estimation of aniline. Due to strong activating effect of -NH_2 , reactivity of benzene ring in aniline is very high, so polyhalogenation occurs even in absence of catalyst.

To get *p*-bromoaniline, strongly activating effect of -NH_2 is lowered by acylation. This is called protection or masking of -NH_2 . -NHCOCH_3 (acetamido group) is ortho para director (predominantly *p*- director) and moderately activating group hence no polyhalogenation occurs. *p*-bromoacetanilide is separated then hydrolyzed to get *p*-bromoaniline. Here, -NH_2 is regenerated. This is called deprotection or demasking of -NH_2 .



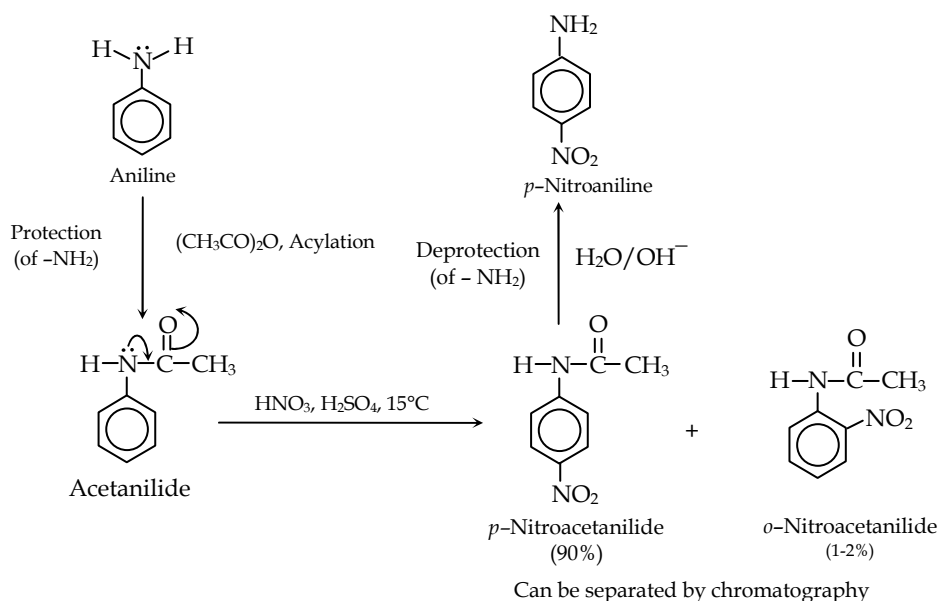
2. **Nitration:** Aniline is not directly nitrated using nitrating mixture to get *p*-nitroaniline. The main problem arises due to very high reactivity of aniline.

In strongly acidic nitrating medium, -NH_2 is converted into -NH_3^+ . Substitution is thus controlled by -NH_3^+ which is a meta director. (Nitration of aniline yields a mixture of about 66% meta and 33% ortho product).

-NH_2 (amino group) is strongly activating group. It increases the reactivity of benzene ring to a very large degree. Therefore, polynitration takes place giving a mixture of mono-, di- and tri-nitro compounds.

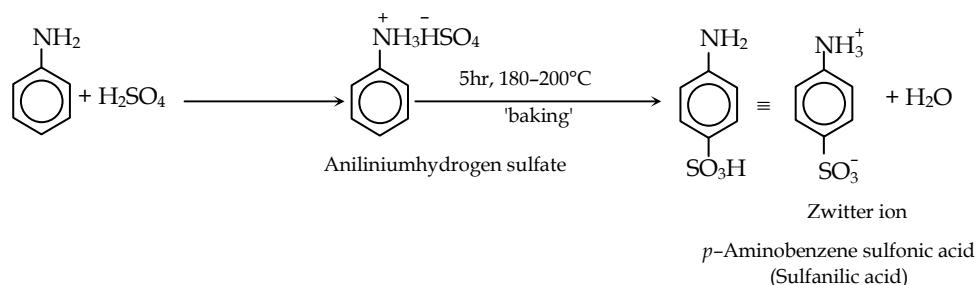
Concentrated HNO_3 is an excellent oxidizing agent. Therefore, HNO_3 not only nitrates but also oxidizes aniline into oxidation products. The aromatic ring structure gets destroyed.

To get *p*-nitroaniline, first of all, -NH_2 of aniline is protected by acylation. -NHCOCH_3 (acetamido group) is ortho para director (predominantly *p*- director) and only moderately activating group (-NH_2 is strongly activating group), hence no such problems occur on nitration. This is called **protection** or **masking** of -NH_2 .

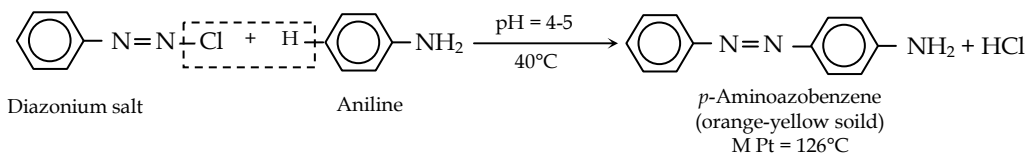


p-nitroacetanilide is separated then hydrolyzed to get *p*-nitroaniline. Here, $-\text{NH}_2$ is regenerated. This is called **deprotection** or **demasking** of $-\text{NH}_2$.

3. **Sulfonation:** Aniline reacts with concentrated sulphuric acid to form anilinium hydrogen sulfate which on heating at $180 - 200^\circ\text{C}$ produces *p*-aminobenzenesulphonic acid commonly called sulfanilic acid as major product which exists as *zwitter ion* (German, Zwitter = hermaphrodite).



4. **Friedel-Crafts reaction:** Aniline does not undergo Friedel-Crafts reaction because catalyst used (Lewis acid) coordinates with basic $-\text{NH}_2$ group. Due to this, N atom of aniline acquires positive charge and hence acts as a strong deactivating group for further reaction.
5. **Reaction with diazonium salt: Coupling reaction:** Aniline reacts with diazonium salt solution in slightly acidic medium to form *p*-aminoazobenzene which appears as orange-yellow compound. Electrophile of this reaction is diazonium ion.



Self Test

- | | | | |
|-------|--|------|----------------------------|
| 15.17 | What happens when | | |
| i. | Aniline is warmed with chloroform in presence of alcoholic caustic alkali ? | | |
| ii. | Aniline is added to diazotized solution of aniline? | | |
| iii. | Aniline is treated with bromine water? | | |
| 15.18 | Write the reaction involved when aniline is treated with excess methyl iodide. | | |
| 15.19 | Convert aniline into | | |
| i. | <i>N</i> -Methylaniline | ii. | <i>N</i> -Phenylethanamide |
| | | iii. | Diphenyl amine |
| 15.20 | Convert aniline into | | |
| i. | <i>o</i> -Nitroaniline | ii. | <i>m</i> -Nitroaniline |
| | | iii. | <i>p</i> -Nitroaniline |
| 15.21 | Convert aniline into | | |
| i. | <i>o</i> -Bromoaniline | ii. | <i>m</i> -Bromoaniline |
| | | iii. | <i>p</i> -Bromoaniline |

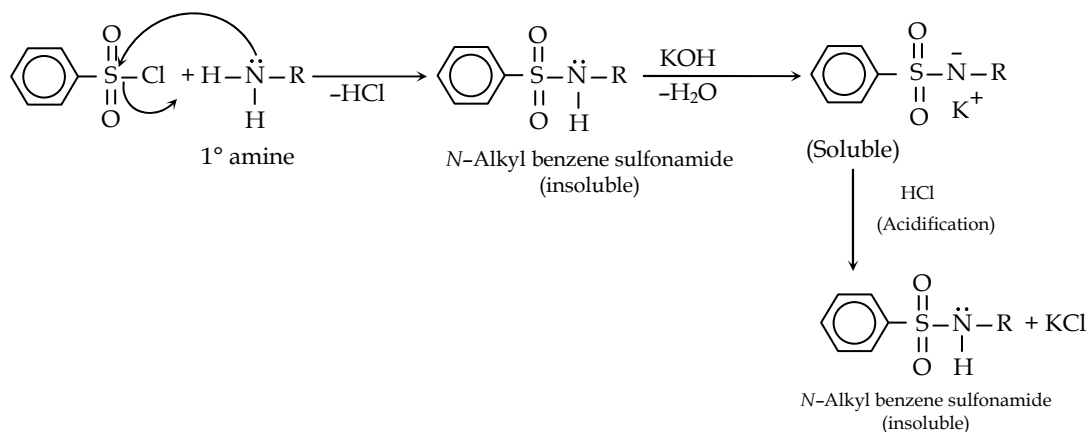
15.11.6 Uses of Aniline

- i. For the preparation of dyes, drugs
- ii. For the preparation of antioxidants in rubber industry

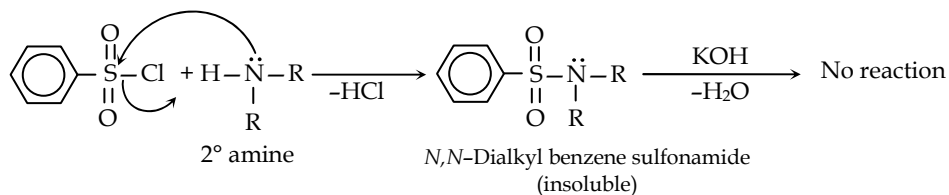
15.12 Test for Amines

1. **Hinsberg test:** Primary, secondary and tertiary amines can be distinguished best by Hinsberg test. In this method, amine is shaken with benzenesulfonyl chloride and aqueous caustic alkali. Different classes of amines react differently.

- i. Primary amine gives a clear solution which forms insoluble compound on acidification.



- ii. Secondary amine gives insoluble compound which is unaffected on acidification.



- iii. Tertiary amine is unaffected and remains as insoluble compound.

Note

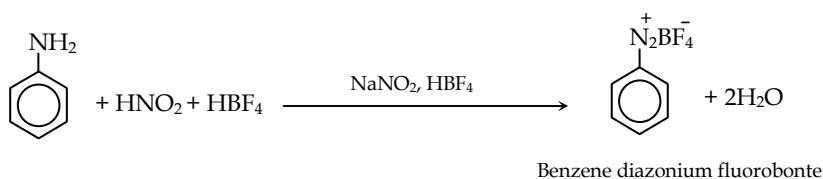
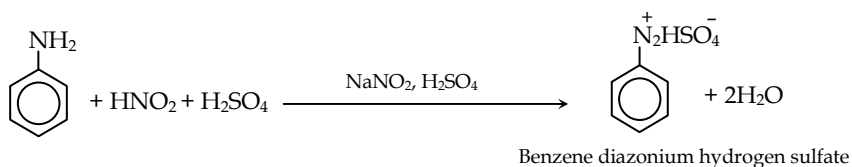
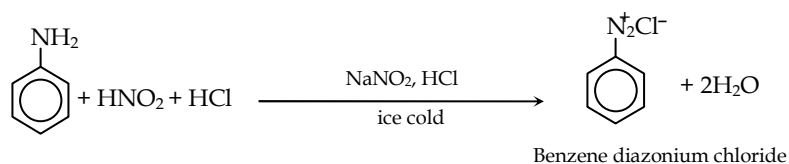
Now, *p*-toluenesulphonyl chloride (tosyl chloride) is used instead benzenesulphonyl chloride.

2. Nitrous acid test (discussed earlier)

3. Carbylamine test (discussed earlier)

15.13 Diazonium Salts and their Reactions

Primary aromatic amine reacts with nitrous acid in ice cold solution to form diazonium salt. This process is called diazotization. Nitrous acid is unstable hence prepared in situ taking sodium nitrite and a mineral acid. This reaction was discovered by **Griess** in 1858.



In the preparation of diazonium salt, usually, primary amine is dissolved or suspended in excess dilute mineral acid, cooled in ice bath and a cooled aqueous solution of sodium nitrite is added slowly with stirring.

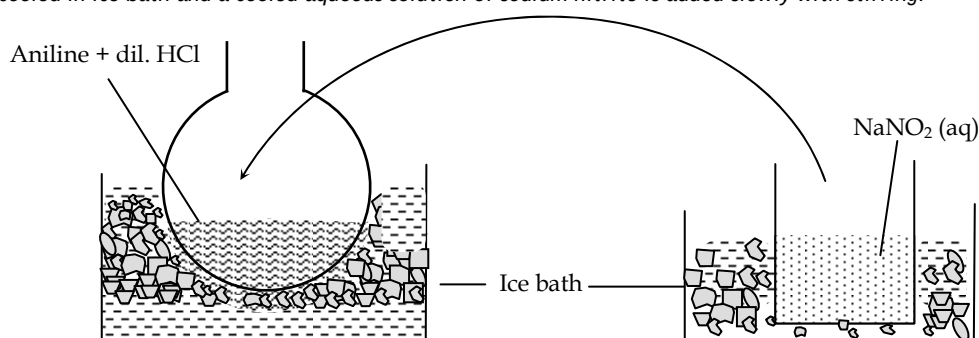


Fig. 15.3: Preparation of diazonium salt

Diazonium salts are colorless solids, highly soluble in water, explode especially nitrates in solids and decompose slowly even at ice bath. Hence they are used immediately after preparation.

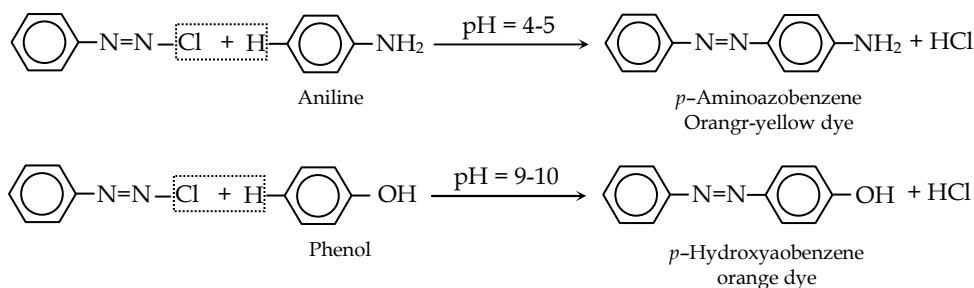
Reactions of Diazonium Salts

Diazonium salts are very important synthetic reagents for the preparation of wide range of valuable aromatic compounds. A large number reactions of diazonium salt may be divided into two groups;

1. Coupling reactions
2. Replacement reactions

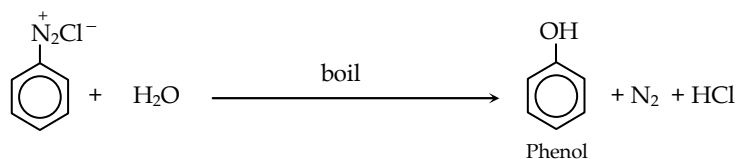
1. **Coupling reactions:** In this type of reaction, diazo-group is retained in the product. Diazonium salt reacts with aromatic compounds having strongly activating group ($-\text{NH}_2$, $-\text{NHR}$, $-\text{NR}_2$, $-\text{OH}$) to form azo compounds ($\text{Ar}-\text{N}=\text{N}-\text{Ar}$). The reaction is called coupling reaction. Azo compounds are colored hence they are of great importance to dye industry.

Examples:

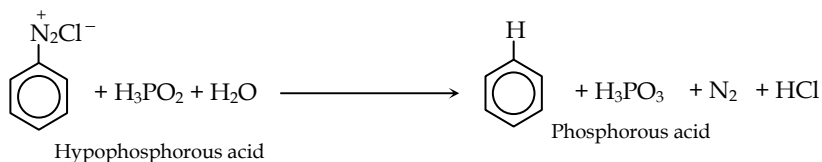


2. **Substitution reactions:** In this type of reactions, diazo-group is replaced by some other atom or group. Many atoms or groups can be attached to the aromatic ring by replacing diazo group.

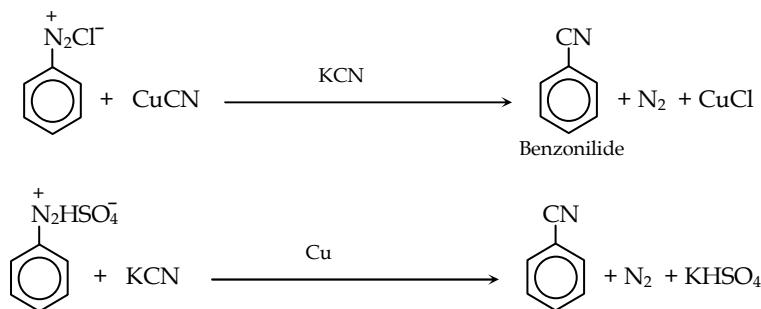
- Replacement by $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, and $-\text{I}$: Discussed in Haloarenes
- Replacement by $-\text{OH}$: When diazonium salt solution is boiled, steam distilled or added to boiling dilute sulphuric acid, diazo group is replaced by $-\text{OH}$ to form phenol.



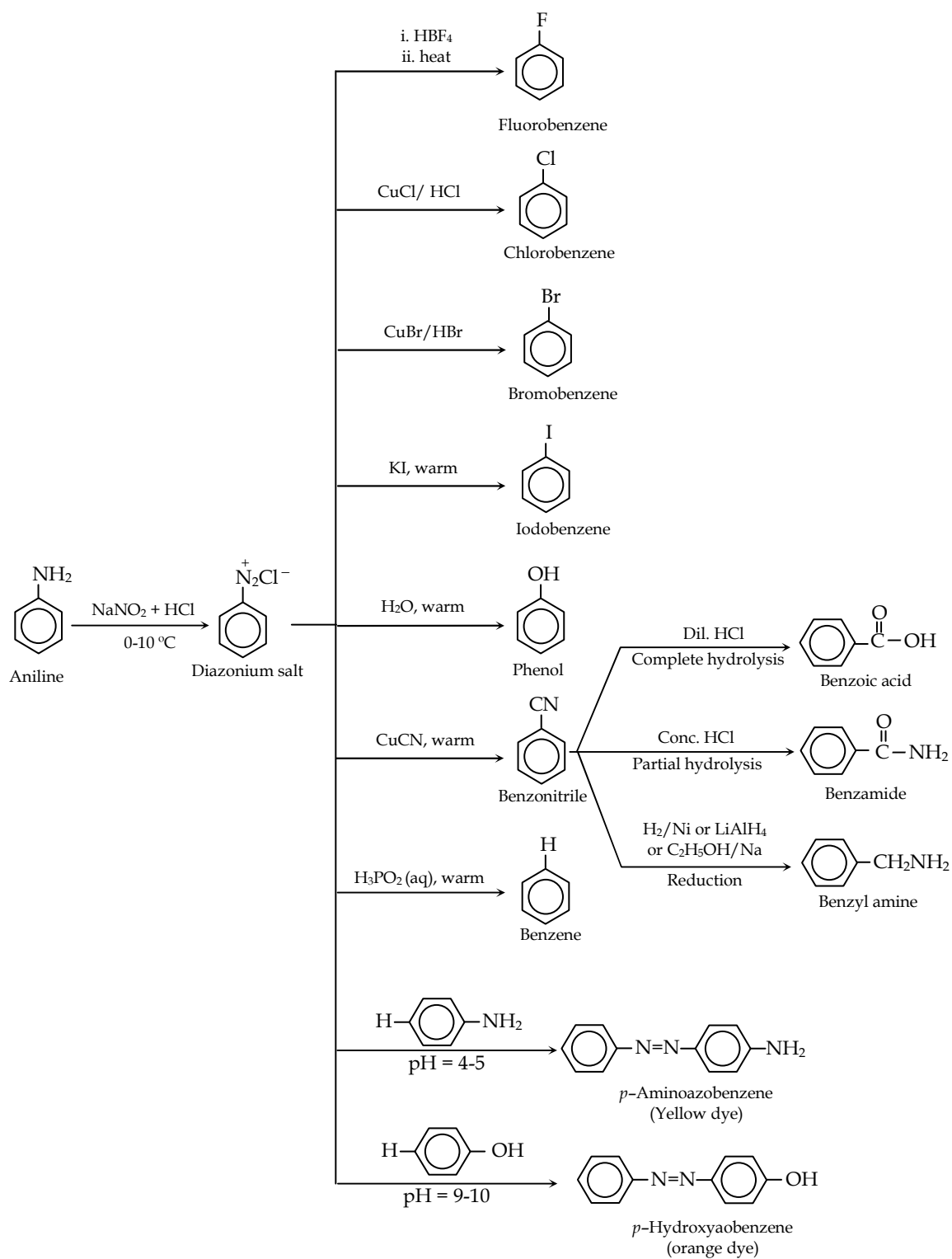
- Replacement by $-\text{H}$:** When diazonium salt solution is mixed with hypophosphorous acid (H_3PO_2), diazo group is replaced by $-\text{H}$.



- Replacement by $-\text{CN}$:** When cyanide salt solution is mixed with cuprous cyanide dissolved in aqueous potassium (a special case of Sandmeyer reaction) or aqueous KCN and copper powder (a special case of Gattermann reaction), diazo group is replaced by $-\text{CN}$.



Reactions of diazonium salt: A summary



Self Test

15.22 Starting from benzene outline the synthesis of following organic compounds;

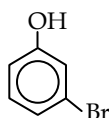
- | | | |
|-------------------------------|-----------------------------------|------------------|
| i. Sulfanilic acid | ii. Carboic acid | iii. Picric acid |
| iv. Salicylic acid | v. Benzoic acid | vi. Benzamide |
| vii. Benzyl amine | viii. <i>p</i> -Hydroxyazobenzene | |
| ix. <i>p</i> -Aminoazobenzene | | |

15.23 Convert nitrobenzene into

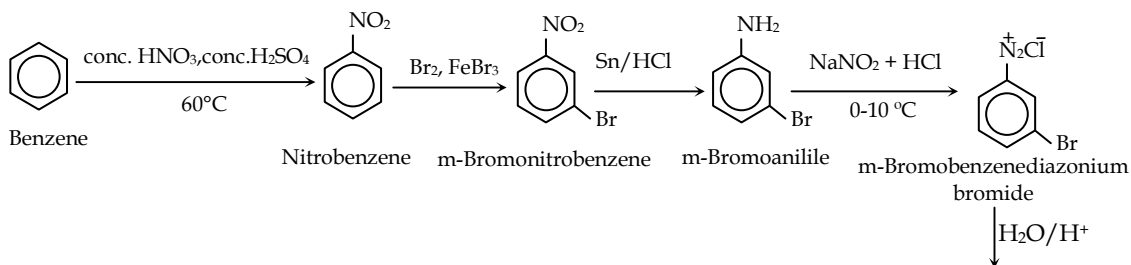
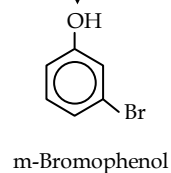
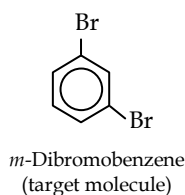
- | | | |
|------------------|--------------------------|--------------|
| i. Chlorobenzene | ii. Iodobenzene | iii. Benzene |
| iv. Phenol | v. 1,3,5-Tribromobenzene | |

Synthesis using Benzene Diazonium Chloride

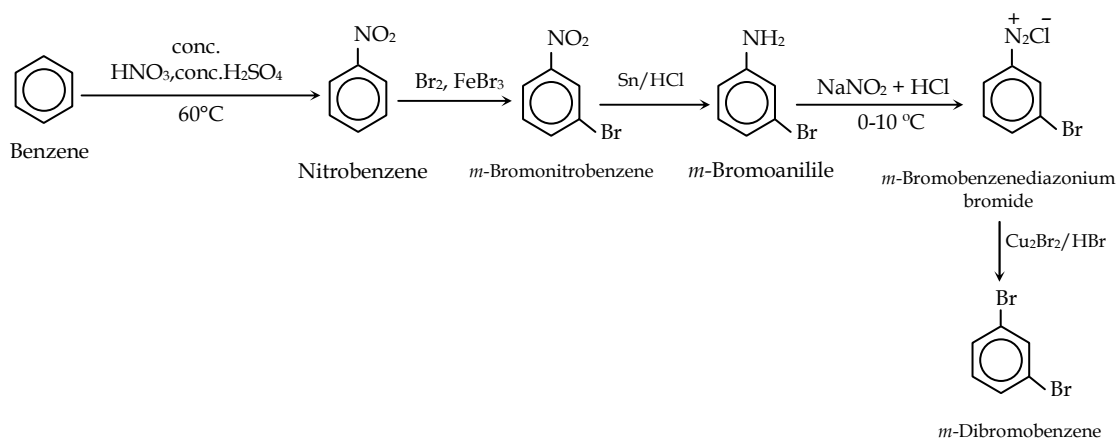
Diazonium salts also offers a means of preparing substituted aromatic compounds in which substituents are in positions which are not possible by direct substitution. For example,

1. Synthesis of *m*-Bromophenol*m*-Bromophenol

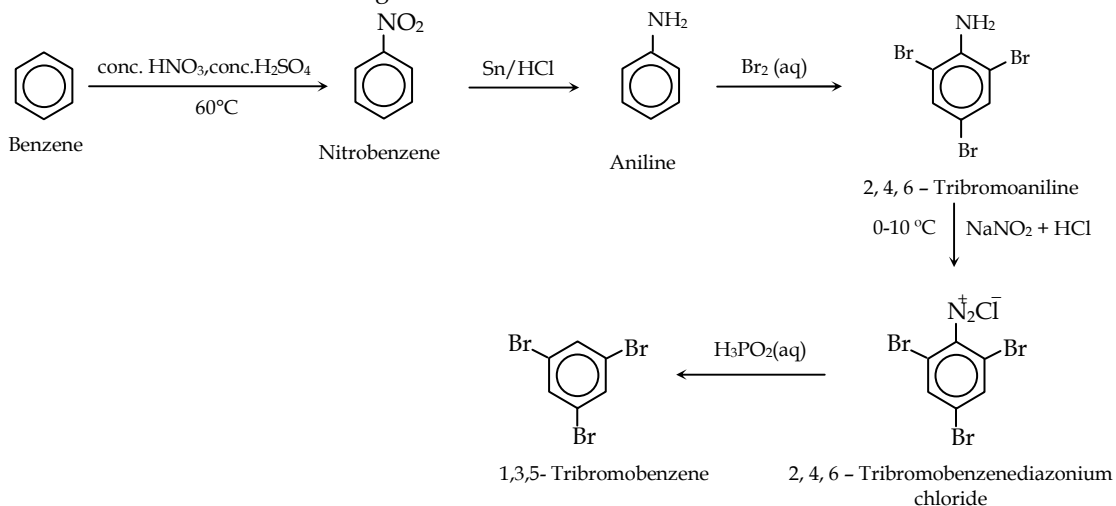
Here, the problem is two ortho para directing groups (-OH and -Br) are situated meta to each other. To solve the problem, bromination of nitrobenzene is carried out to get *m*-Bromonitrobenzene then -NO₂ is converted into -OH through the formation of diazonium salt.

**2. Synthesis of *m*-Dibromobenzene**

Here, the problem is same i.e. two ortho para directing groups (-Br) are situated meta to each other. To solve the problem the same technique is applied; bromination of nitrobenzene is carried out to get *m*-Bromonitrobenzene then -NO₂ is converted into -Br through the formation of diazonium salt.



3. **Synthesis of 1,3,5-Tribromobenzene:** Bromination of aniline gives its tribromoderivative. -NH_2 is then converted to -H through the formation of diazonium salt.

**Self Test**

15.24 Convert benzene into (i) *m*-Dichlorobenzene (ii) *m*-Chloriodobenzene

Solved Questions

1. Write the isomeric amines of molecular formula $\text{C}_3\text{H}_9\text{N}$ and give their IUPAC names.



i. $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
Propanamine

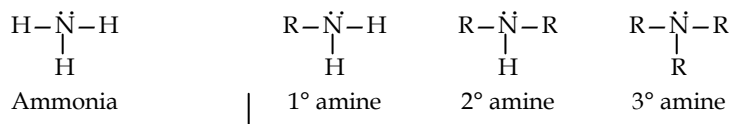
ii. $\text{CH}_3\text{CH}_2\text{NHCH}_3$
N-methylethanamine

iii. $\text{CH}_3\text{N}(\text{CH}_3)_2$
N,N-dimethylmethanamine

iv. $(\text{CH}_3)_2\text{CHNH}_2$
1-Methylethanamine

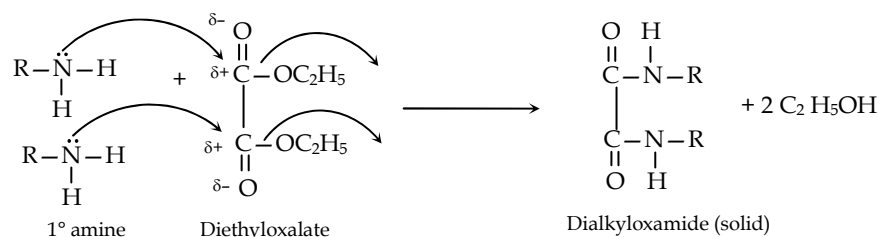
2. What are amines? How do different classes of amines react with diethyloxalate?

When one, two or all three hydrogen atoms in ammonia molecule are replaced by alkyl or aryl groups, the resulting organic compounds are called amines.

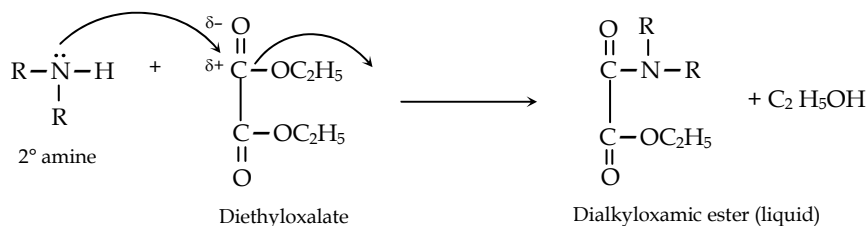


Different classes of amines react with diethyloxalate differently.

i. Primary amine reacts with diethyloxalate to form dialkyl oxamide which is solid.

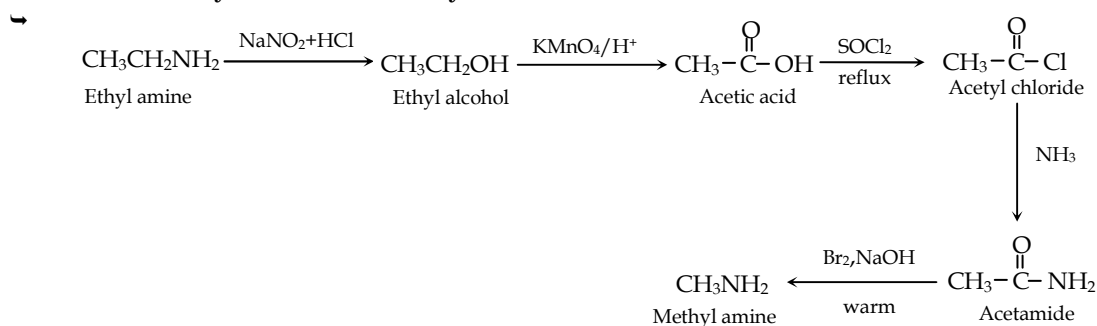
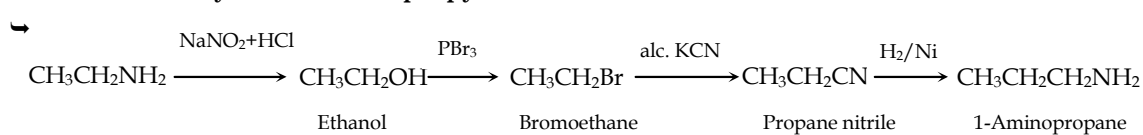


ii. Secondary amine reacts with diethyloxalate to form dialkyl oxamic ester which is generally liquid.



Dialkyl oxamic ester does not react further with another molecule of 2° amine due to steric hinderance.

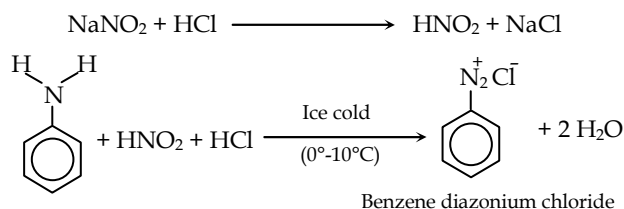
iii. Tertiary amine does not react with diethyloxalate presumely due to lack of hydrogen with nitrogen.

3. Convert ethylamine into methylamine.**4. Convert ethylamine into n-propyl amine.**

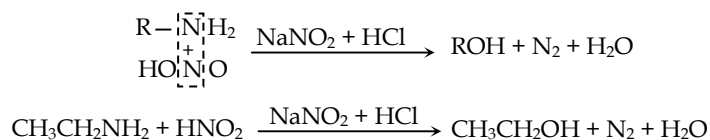
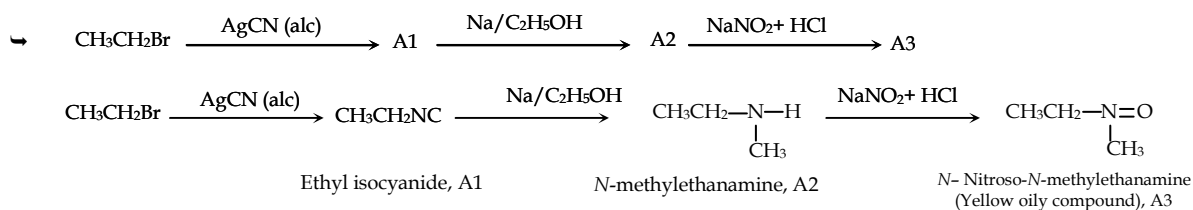
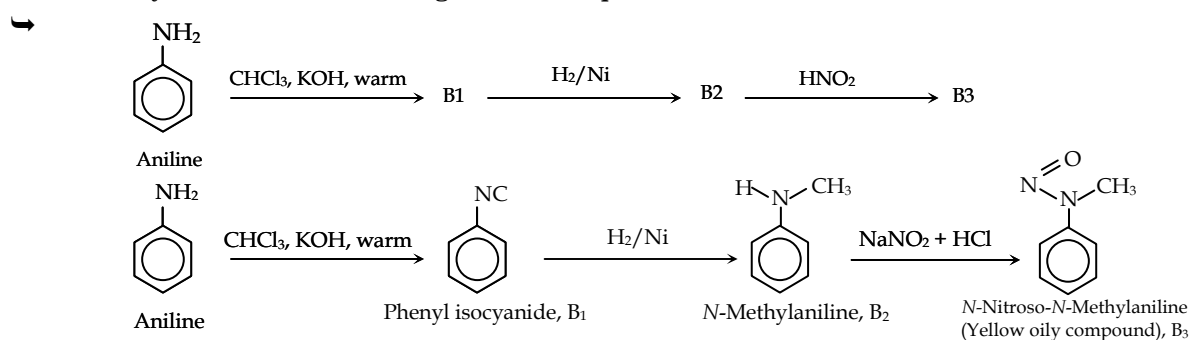
5. How do primary aliphatic and aromatic amine react with nitrous acid?

→ Reactions of primary amines

- i. **Primary aromatic amine** reacts with nitrous acid in ice cold solution to form diazonium salt. The reaction is called diazotization.

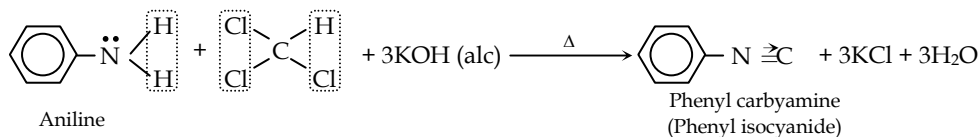


- ii. **Primary aliphatic amine** reacts with nitrous to give alcohol. N₂ (as bubbles) is evolved quantitatively. Hence this reaction can be used as a test for primary aliphatic amine since none of other classes of amines evolve N₂.

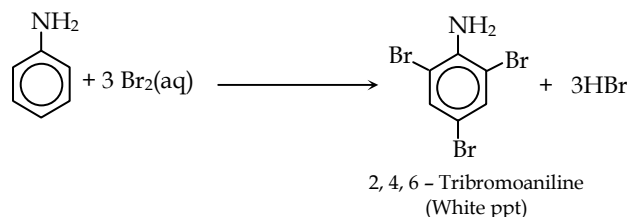
**6. Identify A and in the following reaction sequence.****7. Identify B and in the following reaction sequence****8. What happens when**

- Aniline is warmed with chloroform and alcoholic caustic alkali?
- Aniline is shaken with aqueous bromine?

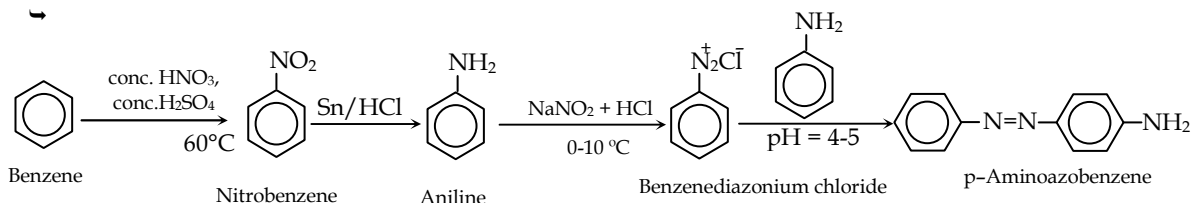
- a. Offensive odor is obtained due to formation of phenyl isocyanide.



- b. White precipitate is obtained due to formation of 2,4,6-Tribromoaniline.



9. How do you get *p*-Aminoazobenzene from benzene?



Questions

Very Short Answer Questions:

- How can you distinguish primary and secondary amines?
- What happens when primary amines are treated with diethyl oxalate?
- What happens when secondary amines are treated with diethyl oxalate?
- What do you mean by diazotization process?
- How can you distinguish aniline and ethyl amine?
- What happens when primary and secondary amines are separately treated with Hinsberg reagent + aq. KOH?
- Why is $-\text{NH}_2$ group of aniline protected before nitration?
- How are suphanilic acid and acetanilide prepared from aniline?
- Write isomers of $\text{C}_3\text{H}_9\text{N}$ with their IUPAC and common names.
- What happens when:
 - Aniline is shaken with aq. Br_2 ?
 - Aniline is coupled with benzene diazonium chloride?
 - Aniline is heated with methyl iodide?
 - Benzamide is allowed to react with bromine in KOH solution?
 - Aniline is warmed with chloroform and alcoholic KOH?
- Why is aniline less reactive than ethanamine?

Short /Long Answer Questions:

- What are amines? How are they classified? Indicate the functional groups of different classes of amines. How do different classes of amines react with diethyl oxalate?
- Briefly discuss the Hofmann's method for the separation of primary, secondary and tertiary amines from their mixture.

- How is aniline prepared in laboratory in pure and dry state? Convert aniline into benzamide and vice versa.
- How would you convert:
 - methanamine into ethanamine and vice-versa?
 - ethanamine into propanamine?
 - aniline into (i) azodye (ii) N-methyl aniline (iii) *o*-nitroaniline (iv) picric acid (v) phenol (vi) phenyl isocyanide (vii) sulphanilic acid (viii) *p*- benzoquinone (ix) benzenediazonium chloride
- Explain with an example of each of the following:
 - Diazotization reaction
 - Diazo coupling reaction
 - Acylation of aniline
 - Carbylamine test
 - Hofmann bromamide reaction
- How can you distinguish:
 - Ethnamine and aniline
 - Ethanamine and N-methylethanamine
 - Nitrobenzene and aniline
 - N-methylethanamine and N, N-dimethylethanamine.

Multiple Choice Questions

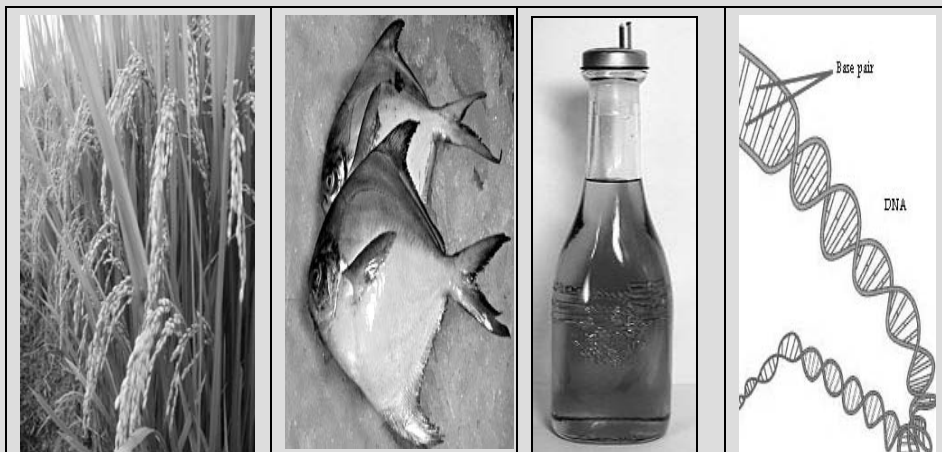
- Hofmann's ammonolysis of alkyl halides gives
 - Primary amine
 - Secondary amines
 - Tertiary amines
 - Mixture of amines
- In Hofmann's ammonolysis of alkyl halides, the good yield is obtained in
 - Primary alkyl halide
 - Secondary alkyl halide
 - Tertiary alkyl halide
 - Haloarene
- Which of the followings is not involved in degradation of carbon chain?
 - Hofmann degradation
 - Hofmann elimination
 - Hofmann bromamide reaction
 - Hofmann rearrangement
- Gabriel Phthalimide synthesis gives
 - Primary amine
 - Secondary amines
 - Tertiary amines
 - Mixture of amines
- Boiling point is highest
 - Ethyl amine
 - Ethyl alcohol
 - Formic acid
 - Trimethyl amine
- Primary aliphatic amine and primary aromatic amine can be distinguished by
 - Carbylamine test
 - Nitrous acid test
 - Victor Mayer test
 - Lucas test
- Which of the following exists as zwitter ion?
 - Suifanilic acid
 - Salicylic acid
 - Succinic acid
 - Urea
- Which of the followings does not give CH_4 when treated with CH_3MgI ?
 - Trimethyl amine
 - Ethyl amine
 - Dimethyl amine
 - Ammonia
- Primary and secondary amines can not be distinguished by
 - Nitrous acid test
 - Hinsberg test
 - Isocyanide test
 - DNP test
- n-Propyl amine and Ethylmethyl amine are
 - Chain isomers
 - Functional isomers
 - Positional isomers
 - Metamers

Answer Key

1	2	3	4	5	6	7	8	9	10
d	a	b	a	c	b	a	a	d	b

Molecules of Life

16



The complex lifeless molecules which built up body of organisms and essential for growth, development, regulation and maintenance of life are called biomolecules or molecules of life. In general, biomolecules are larger and more complicated than other organic molecules. The principal classes of biomolecules are carbohydrates, proteins, nucleic acids and lipids which are briefly discussed here.

16.1 Carbohydrates



Rice (starch)



Sugarcane(sucrose)



Cotton (cellulose)

16.1.1 Introduction

Carbohydrates are naturally occurring organic compounds which provide us the necessities of life: food, shelter, and clothing. For example, glucose, sucrose, starch, cellulose etc. Glucose is the key source of energy in human body which is obtained by enzymatic hydrolysis of starch mainly from rice, potatoes etc. Sucrose is obtained mainly from sugarcane. Similarly, cotton is cellulose and wood is chiefly cellulose which provides us clothing and shelter.

Chemically, carbohydrates are organic compounds which are composed of C, H and O. The name carbohydrate was given originally to signify 'hydrates of carbon' since carbohydrates satisfy formula $C_x(H_2O)_y$. For example, glucose ($C_6H_{12}O_6$ or $C_6(H_2O)_6$), sucrose ($C_{12}H_{22}O_{11}$ or $C_{12}(H_2O)_{11}$) etc. However, this concept is not correct in all the cases. There are some compounds which satisfy the given formula but they are not carbohydrates *e.g.* acetic acid (CH_3COOH or $C_2(H_2O)_2$), formaldehyde ($HCHO$) etc., while there are some compounds which do not satisfy the given formula but they are carbohydrates *e.g.* deoxyribose ($C_5H_{10}O_4$), rhamnose ($C_6H_{12}O_5$). Modern theories show, structurally carbohydrates are not hydrates of carbon.

Carbohydrates are **polyhydroxyaldehydes** or **polyhydroxyketones** or any organic compounds that produce these compounds on hydrolysis. Carbohydrates are optically active compounds.

16.1.2 Classification of Carbohydrates

Carbohydrates can be classified by different ways;

- A. Based on hydrolysis
- B. Based on solubility and taste
- C. Based on reducing behavior

A. Based on hydrolysis

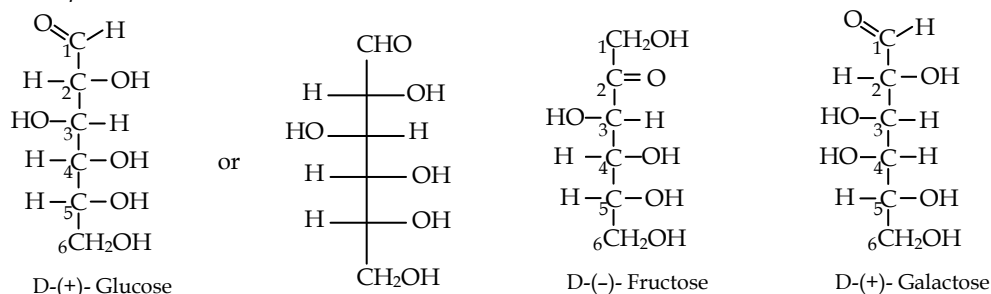
Based on hydrolysis, carbohydrates are classified into three classes:

- 1. Monosaccharide:** Carbohydrates which can not be hydrolyzed into simpler compounds are called monosaccharides. They are the simplest carbohydrates. For example, glucose, fructose, galactose etc.

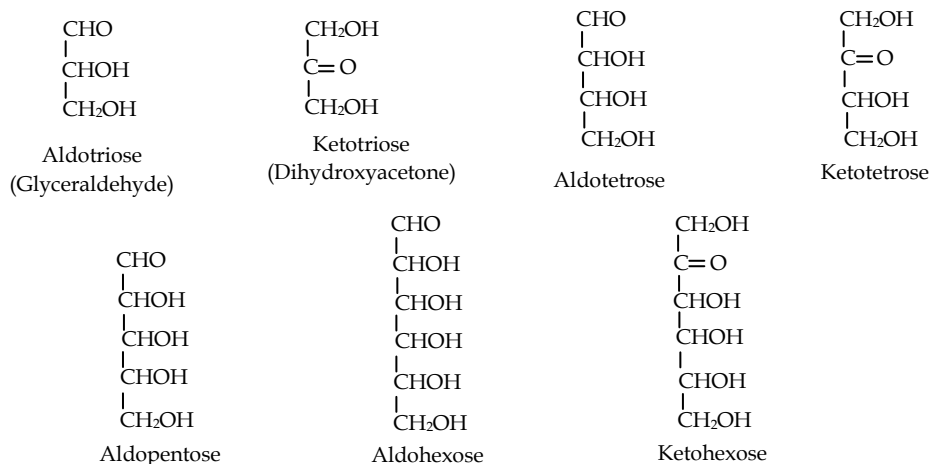
Monosaccharides are further classified as follows;

- On the basis of functional group in the molecule:** On the basis of functional group present in the molecule, monosaccharides are classified as aldoses and ketoses. A monosaccharide having aldehyde (-CHO) group is called an *aldose* and a monosaccharide having a keto group (>C=O) is called a *ketose*.

Examples:



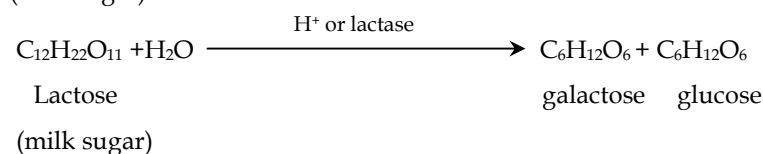
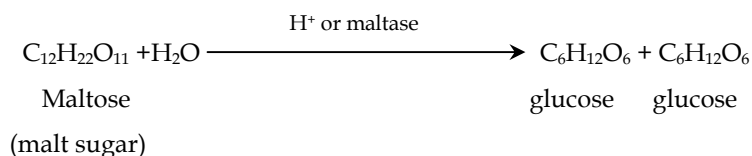
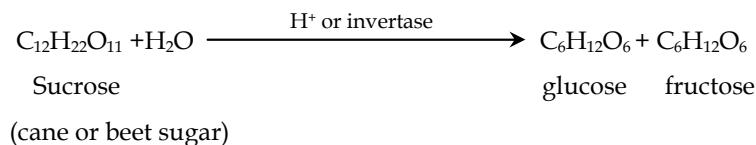
- On the basis of total number of carbons in the molecule:** On the basis of total number of carbons present in a molecule, monosaccharides are further classified as triose, tetrose, pentose, hexose and heptose for the carbohydrates containing three, four, five, six and seven carbon atoms respectively. Pentoses and hexoses are most common naturally occurring monosaccharides.



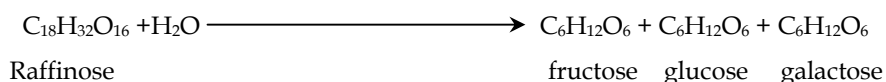
- 2. Oligosaccharide:** Carbohydrate which on hydrolysis gives two to ten monosaccharides is called oligosaccharide (Greek; oligos = few, sacchar = sugar). Carbohydrate which produces two monosaccharides on hydrolysis is called disaccharide and so on.

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Examples: Sucrose, maltose, lactose etc are naturally occurring disaccharides.



Raffinose is a trisaccharide, which is mostly found in beet molasses and on hydrolysis gives fructose, glucose and galactose.



3. **Polysaccharide:** Carbohydrate which on hydrolysis gives more than ten monosaccharides is called polysaccharide. Polysaccharides are represented by general formula $(\text{C}_6\text{H}_{10}\text{O}_5)_n$ where $n=100$ to 4000. For example, cellulose, starch etc. They are natural polymers of monosaccharides. Cellulose and starch are polymers of D-(+)-glucose.

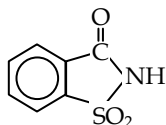
B. Based on solubility and taste

Based on solubility and taste, carbohydrates are classified as sugars and non sugars.

1. **Sugars:** Carbohydrates which are soluble in water and sweet in taste are called sugars. They are crystalline. All monosaccharides and oligosaccharides are sugars. Fructose is the sweetest of all carbohydrates.

Sweetening power of common sugars: fructose > invert sugar > sucrose > glucose > galactose > maltose > lactose

2. **Non sugars:** Carbohydrates which are insoluble in water and not sweet in taste are called non sugars. They are amorphous. All polysaccharides are non-sugars. For example, cellulose, starch etc.



o-Sulphobenzoic imide (Saccharin)

Saccharin is not a carbohydrate. It is about 550 times sweeter than sugar. It is almost insoluble in water and hence commercially available in its sodium salt. It is used instead of sugar mainly for diabetics and obese persons.

C. Based on reducing behavior

Carbohydrates which reduce Fehling's solution, Benedict's solution or Tollen's reagent are called reducing sugars. All monosaccharides and most disaccharides are reducing sugars. Exceptionally, **sucrose**, a disaccharide, is a non-reducing sugar. Polysaccharides are non-reducing sugars.

Test	Reducing sugar	Non reducing sugar
Tollen's test	positive	negative
Fehling test	positive	negative
Benedict test	positive	negative
Mutarotation	Yes	No

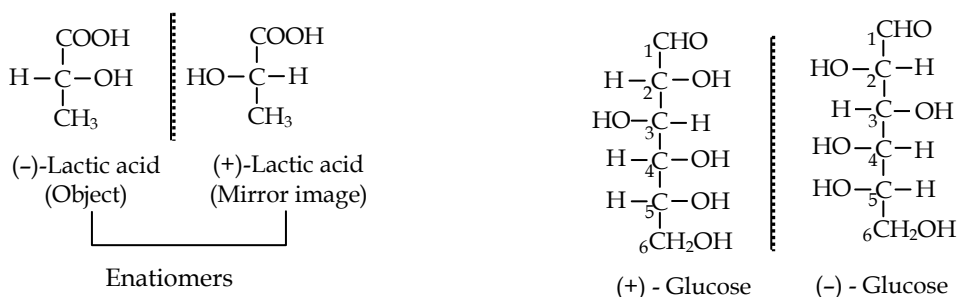
Self Test

- 16.1 What are biomolecules? Give examples.
- 16.2 What are carbohydrates? Give the examples of monosaccharides, oligosaccharides and polysaccharides.
- 16.3 How does a sugar differ from a non-sugar? Justify with example.
- 16.4 Differentiate a reducing sugar from a non-reducing sugar.
- 16.5 What are disaccharides? Write the open chain structure of monosaccharides produced on hydrolysis of table sugar.
- 16.6 Draw the structure of (i) aldotriose, (ii) ketohexose, (iii) aldopentose, (iv) ketotetrose.

16.1.3 Meaning of (+) and (-): Optically Active Compounds

A compound which rotates plane of polarized light is called *optically active* compound and the phenomenon is called optical activity. Optical activity is detected and measured by an instrument called *polarimeter*.

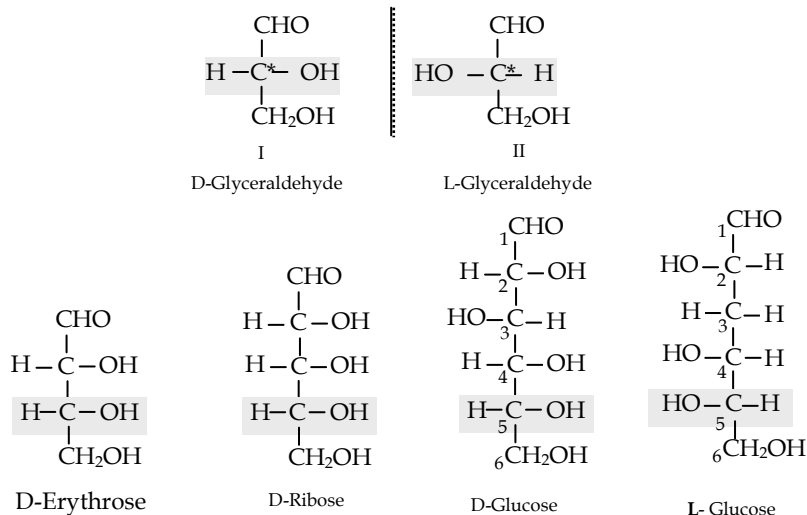
An optically active compound rotates plane of polarized light either in clockwise direction or in anticlockwise direction. If it rotates the plane of polarized light in clockwise direction, the compound is called **dextrorotatory** (Latin: *dexter*, = right) and represented by *d*- or (+). Similarly, if it rotates plane of polarized light in anticlockwise direction, the compound is called **laevorotatory** (Latin: *laevo*, = left) and represented by *l*- or (-). Mostly, naturally occurring substances are dextrorotatory. For example, naturally occurring glucose rotates plane of polarized light in clockwise direction. Hence it is called (+)-glucose or *d*-glucose. Lactic acid obtained from muscle is (+)-Lactic acid.

**16.1.4 D and L Designations: Relative Configuration**

All monosaccharides are divided into two families; D-family and L-family, by taking the configuration of glyceraldehyde, the simplest carbohydrate, as standard. Configuration I was arbitrarily assigned D-glyceraldehyde and II was assigned L-glyceraldehyde (proposed by American chemist Rosanoff in 1906 then universally accepted). The carbohydrates having the same configuration to that of D-glyceraldehyde at the **chiral carbon**¹ farthest away from the carbonyl

¹ Carbon which is bonded to four different atoms or group of atoms are called chiral carbon or asymmetric carbon. An aldohexose has four chiral carbons hence 16 configurations are possible. One of them is D-(+)-Glucose. (Total optical isomers = 2ⁿ, n is number of chiral carbon if two ends of compounds are not same.) Chiral carbon is indicated by asterisk(*) in glyceraldehyde.

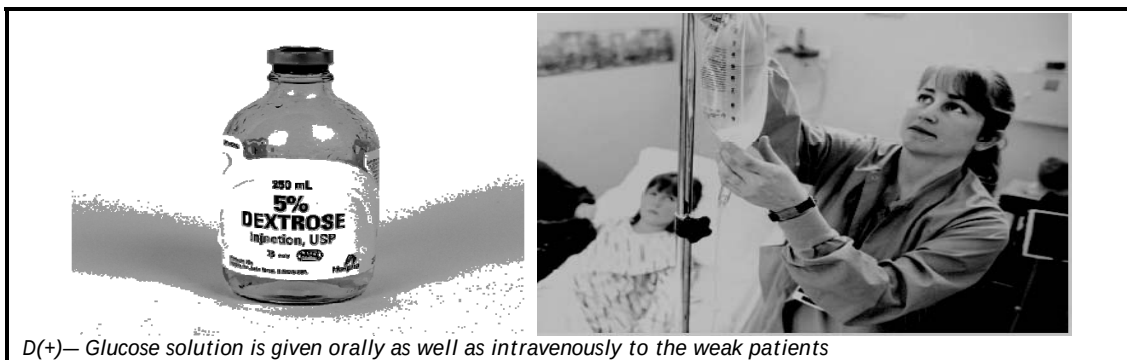
group (-OH on right hand side) are assigned D and those having configuration opposite to D (-OH on left hand side) are assigned L. It should be noted that D- and L- designations do not indicate the sign of rotation (*d*- and *l*-) of the plane polarized light.



Self Test

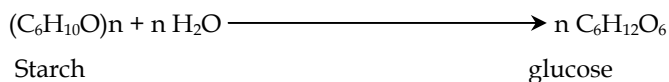
16.7 Indicate chiral carbon and predict the number of optical isomers in aldotriose, aldotetrose and aldopentose. Hint: 2, 4, 9 optical isomers respectively.

16.1.5 (+)-Glucose (Dextrose, Grape Sugar, Blood Sugar)



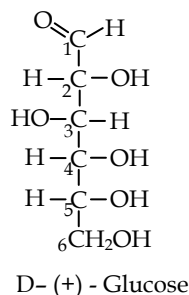
D(+)- Glucose solution is given orally as well as intravenously to the weak patients

D-(+)-Glucose (grape sugar) is the most common monosaccharide. It occurs in honey and most sweet fruits especially in ripen grapes (20 – 30% in free state). It is also an essential constituent of human blood hence it is known as **blood sugar**. Naturally occurring glucose is dextrorotary hence it is also called **dextrose**. Commercially, it is obtained by hydrolysis of starch at high temperature and pressure.



In glucose, chiral carbons are C₂, C₃, C₄ and C₅.

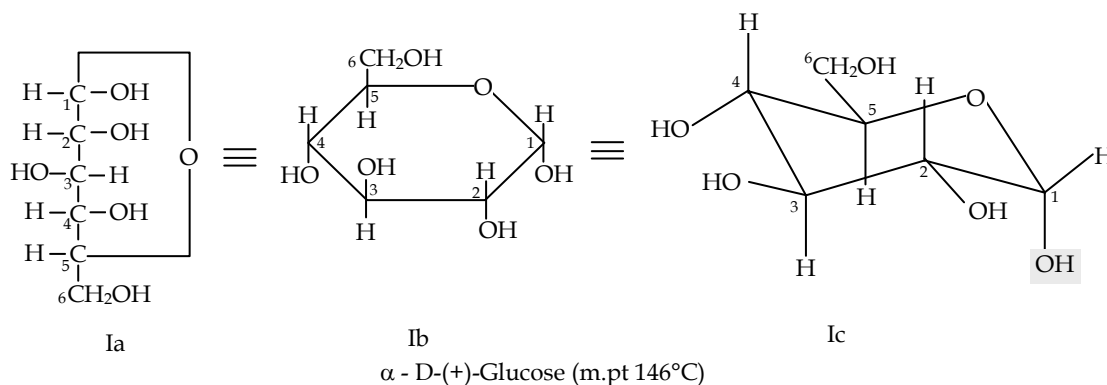
Open chain structure of glucose is



Open chain structure can not explain some of the properties of glucose.

- D-(+) Glucose does not give some typical reactions of aldehydes. It gives positive Tollen's test, Fehling test but does not form bisulphite addition product.
- D-(+)- Glucose exists in two crystalline forms; one with melting point 146°C (α - form which is obtained by crystallization in aqueous solution at 30°C) and another with melting point 150°C (β - form which is obtained by crystallization in acetic acid above 98°C).

To account these facts, cyclic structure² was proposed. The cycle structure of glucose is represented crudely by Fischer formula (Ia and IIa) and Haworth³ pyranose ring structure (Ib and IIb) and more accurately by structures Ic and IIc.

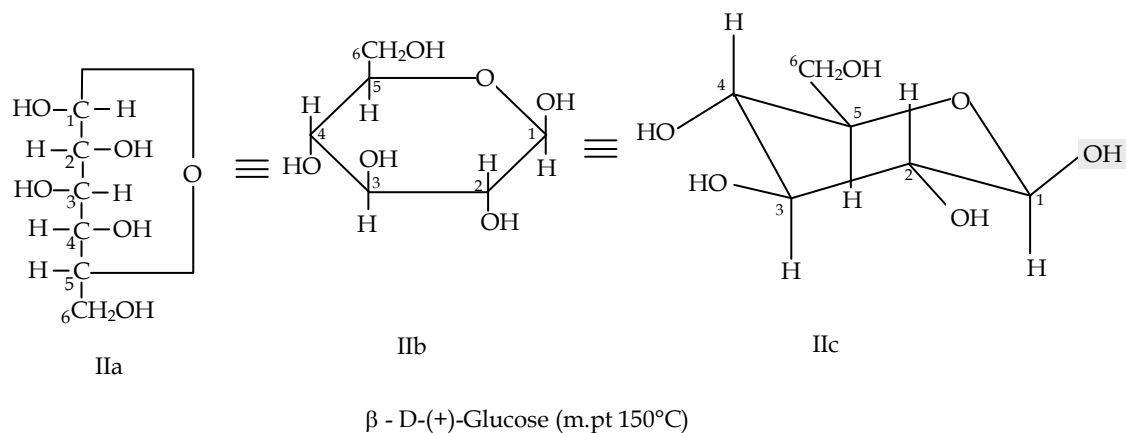


² In 1883 Tollens proposed the cyclic structure of glucose. However, he suggested five membered ring structure for it instead of six membered.

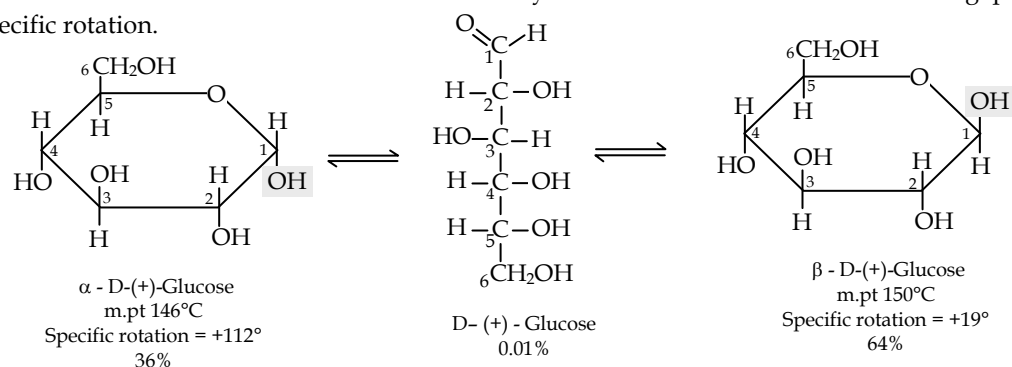
³ In 1926, Haworth established the six membered ring of glucose (aldohexose) and introduced hexagonal ring structure resembling pyran. α - and β - forms of D-(+)- glucose are represented by Haworth formula as Ib (α -D-Glucopyranose) and IIb (β -D-Glucopyranose) respectively. Their name is also given in term of pyran to indicate their ring size.



Pyran (a six membered of heterocyclic compound)



The two forms of glucose (α - and β -) differ in the orientation of -OH group at C_1 . Hence they are **anomers**. These two forms exist in different crystalline forms with different melting point and specific rotation.



When any of these forms is dissolved in water, an equilibrium mixture is obtained having about 36% of α - form, about 64% of β - form and very low percentage (0.01%) of open chain form. This low concentration of open chain form (aldehyde) is sufficient to give positive Tollen's and Fehling test but not sufficient to give the bisulphite addition product which is reversible.

When any of these forms is dissolved in water, the specific rotation gradually changes to $+52.7^{\circ}$. The change in specific rotation to the equilibrium value is called **mutarotation**.

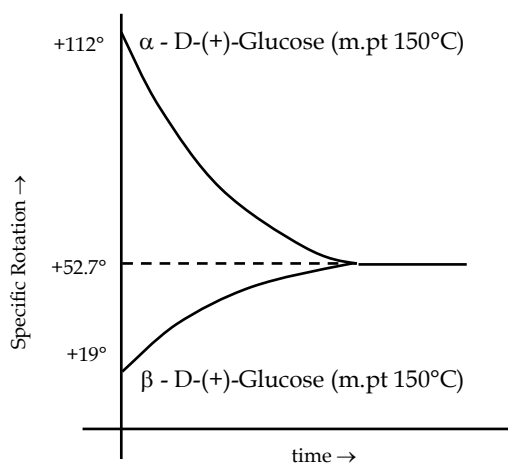
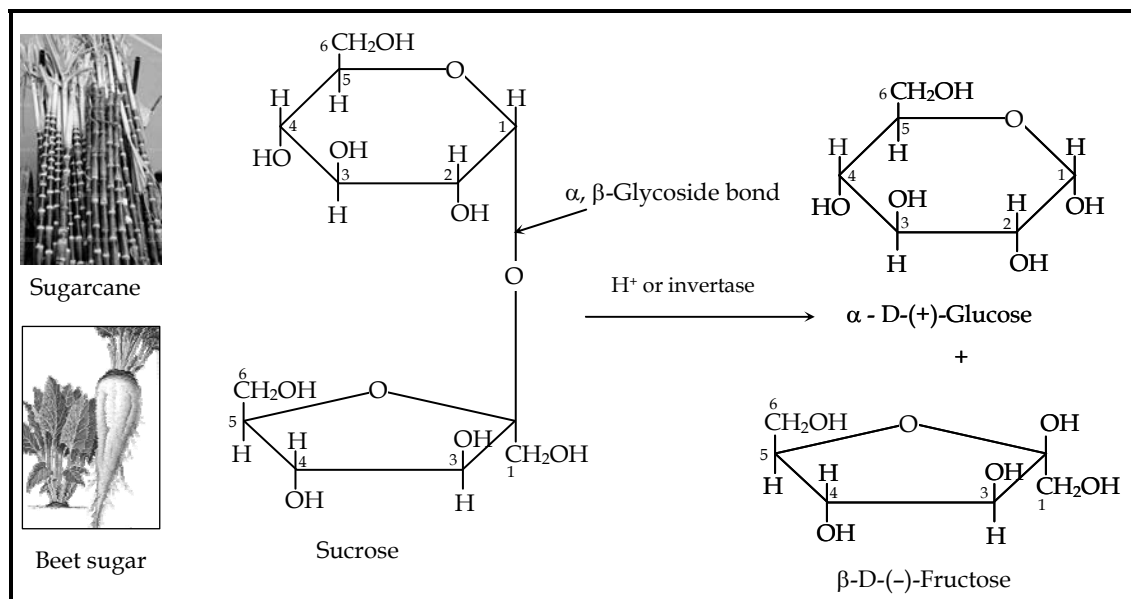


Fig. 16.1: Mutarotation of D - (+) - glucose

16.1.6 (-)-Fructose (Laevulose, Fruit Sugar)

D-(-)-Fructose is the most common ketohexose. It occurs in honey and most sweet fruits. Naturally occurring fructose is levorotatory hence it is also called **laevulose**. Commercially, it is obtained by hydrolysis of inulin.

**Self Test**

16.8 Aqueous solution of sucrose is called invert sugar, give reasons.

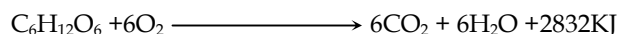
Starch

It is a polysaccharide and forms reserve food of plants. It contains 20% of water soluble fraction called **amylose** and 80% of water insoluble fraction called **amylopectin**. On complete hydrolysis, it gives D-(+)- glucose.

16.1.8 Functions of Carbohydrates

Carbohydrates provide us the necessities of life: food, shelter and clothing. The roles of carbohydrates categorized as;

1. **Energy source (bio-fuels):** Carbohydrates (all except cellulose) are the ultimate source of most of our food. Starch is the reserve food in plants and glycogen in animals.



2. **Structural role:** Cellulose is the chief structural component in plant cells. Cellulose in the form of wood is used for houses and furniture, cellulose in the form of cotton, linen, rayon is used for clothing and cellulose in the form of paper is used for books, newspapers, money, bags, boxes, sheets and rolls etc. Chitin forms exoskeleton of arthropods and crustaceans. Cellulose can be used to prepare explosive like gun cotton.
3. Carbohydrates are also the constituent of DNA and RNA (deoxyribose sugar in DNA and ribose sugar in RNA).

16.1.9 Test for Carbohydrates: Molisch's Test

Carbohydrates produce deep purple solution with Molisch's reagent (10% α -Naphthol in alcohol) in presence of concentrated H_2SO_4 .

16.2 Proteins



16.2.1 Introduction

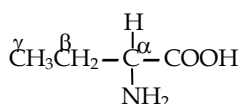
Proteins are complex nitrogenous organic compounds of high molecular mass. They make up a large part of animal body and essential for growth, development and maintenance of life. They are found in all living cells. The name protein is given to signify their prime importance to life (Greek; *proteios* = first). **Chemically, proteins are high molecular mass polymers of α -amino acids.**

Cereals, milk, fishes, meat, eggs are some major sources of proteins.

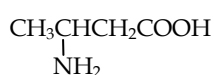
Source	Protein	Source	Protein
Soya bean	42%	Cashuenut	21%
Pea	28%	Almonds	21%
Milk powder (whole)	25%	Mutton	18%
Fish	22%		

16.2.2 Amino Acids

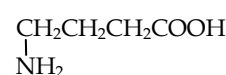
In chemistry, amino acids are organic compounds having amino group ($-\text{NH}_2$) and carboxyl group ($-\text{COOH}$) in the same molecule. The position of amino group varies. For example,



α -Aminocarboxylic acid

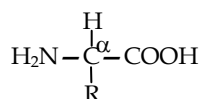


β -Aminocarboxylic acid



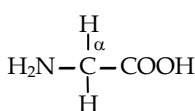
γ -Aminocarboxylic acid

Amino acids found in proteins are only **α -aminocarboxylic acid**. The general formula is

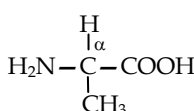


General formula of α -amino acid

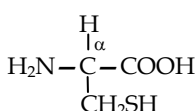
Examples:



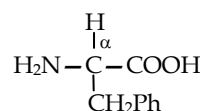
Glycine
Gly



Alanine
Ala



Cysteine
Cys

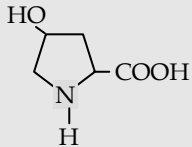
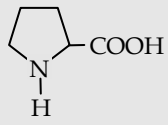


Phenylalanine
Phe

There are altogether 26 α - amino acids known, out of them only 23 amino acids are found in the proteins (used by cells in the protein biosynthesis), remaining are found in the special tissues.

List of natural amino acids

Name	Abbreviation	Structure
(+)-Alanine	Ala or A	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}^{\alpha}-\text{COOH} \\ \\ \text{CH}_3 \end{array}$
(+)-Arginine ^e	Arg or R	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}^{\alpha}-\text{COOH} \\ \\ \text{CH}_2\text{CH}_2\text{CH}_2\text{NH}-\text{C}-\text{NH}_2 \\ \\ \text{NH} \end{array}$
(-)-Asparagine	Asn or N	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{CONH}_2 \end{array}$
(+)-Aspartic acid	Asp or D	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{COOH} \end{array}$
(-)-Cysteine	Cys or C	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}^{\alpha}-\text{COOH} \\ \\ \text{CH}_2\text{SH} \end{array}$
(-)-Cystine	Cys-Cys	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{S}-\text{SCH}_2-\text{CHCOOH} \\ \\ \text{NH}_2 \end{array}$
(+)-Glutamic acid	Glu or E	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}^{\alpha}-\text{COOH} \\ \\ \text{CH}_2\text{CH}_2\text{COOH} \end{array}$
(+)-Glutamine	Gln or Q	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{CH}_2\text{CONH}_2 \end{array}$
Glycine	Gly or G	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{H} \end{array}$
(-)-Histidine ^e	His or H	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}^{\alpha}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{Imidazole ring} \end{array}$

(-)-Hydroxylysine	Hyl	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{CH}_2\underset{\text{OH}}{\text{CH}}\text{CH}_2\text{NH}_2 \end{array} $
(-)-Hydroxyproline	Hyp	
(+)-Isoleucine ^e	Ile or I	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CHCH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $
(-)-Leucine ^e	Leu or L	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_3 \\ \\ \text{H} \end{array} $
(+)-Lysine ^e	Lys or K	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2 \end{array} $
(-)-Methionine ^e	Met or M	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{CH}_2-\text{S}-\text{CH}_3 \end{array} $
(-)-Phenylalanine ^e	Phe or F	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}^{\alpha}-\text{COOH} \\ \\ \text{CH}_2-\text{C}_6\text{H}_5 \end{array} $
(-)-Proline	Pro or P	
(-)-Serine	Ser or S	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2\text{OH} \end{array} $
(-)-Threonine ^e	Thr or T	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}-\text{CH}_3 \\ \\ \text{OH} \end{array} $

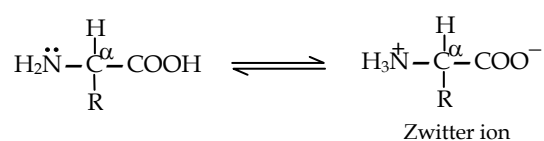
(-)-Tryptophane ^e	Trp or W	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{Indole ring} \end{array} $
(-)-Tyrosine	Tyr or Y	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{Benzene ring with OH} \end{array} $
(+)-Valine ^e	Val or V	$ \begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH} \\ / \quad \backslash \\ \text{CH}_3 \quad \text{CH}_3 \end{array} $
<i>e = essential amino acid</i>		

Out of 23 α - amino acids found in proteins, 10 α - amino acids cannot be synthesized by animals in their body; therefore they must be ingested in the form of food for the proper growth. These α - amino acids are called **essential or indispensable amino acids**. The remaining α - amino acids are synthesized in the body and are called **non essential or dispensable amino acids**.

All amino acids found in proteins are optically active compounds except glycine. Glycine is optically inactive since it has no chiral carbon.

Some amino acids contain second acidic group (carboxyl group) which are called acidic amino acids. For example, Aspartic acid, Glutamic acid. Some amino acids contain another basic group which are called basic amino acids. For example, Arginine, Histidine, Lysine.

Amino acids are insoluble in non-polar solvents like petroleum ether, ether, benzene but soluble in water. They are crystalline solids with high melting point. Various experiments show that amino acids exist as *zwitter* ions.

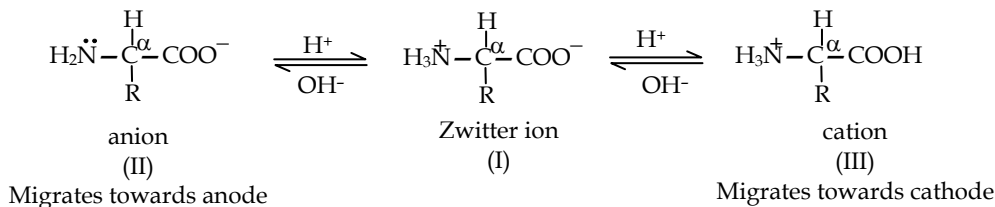


In an amino acid, since acidic group (-COOH) and basic group (-NH₂) are in the same molecule, -COOH loses a proton and -NH₂ accepts a proton forming dipolar ion or *zwitter* ion. (Greek: *Zwitter*= hermaphrodite).

Self Test

- 16.9 What are proteins? Name major proteins found in milk and egg.
- 16.10 What are amino acids? How do essential amino acids differ from non-essential amino acids? Give any two examples of each.
- 16.11 Give the name of
- | | |
|-------------------------------------|----------------------------------|
| (i) All essential amino acids | (ii) Acidic amino acid (any two) |
| (iii) Basic amino acid (any two) | (iv) Amino acid having sulphur |
| (v) Amino acid having aromatic ring | |

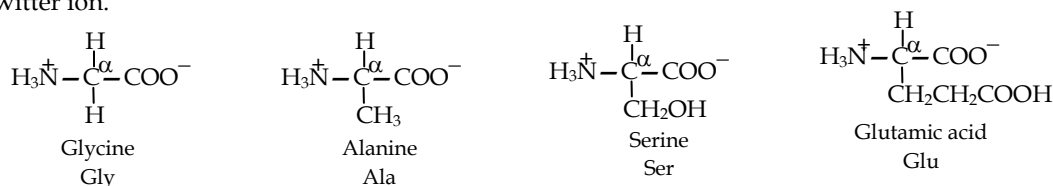
16.2.3 Isoelectric Point of Amino Acids



Amino acid exists as dipolar ion (zwitter ion). In alkaline medium, amino acids exist mostly in the anion (II) form, which migrates towards anode in the electric field. While in acidic medium, cation (III) exists in excess and migrates towards cathode. At certain *pH*, concentration of anion (II) becomes equal to that of cation (III), in which, there is no net migration of amino acid towards any electrode in the electric field. Such *pH* at which amino acid does not migrate is called **isoelectric point**. It is characteristics of all amino acids and each amino acid has its own characteristics isoelectric point.

Example: Isoelectric point of glycine, phenylalanine and glutamic acid are 6.1, 5.5 and 3.2 respectively.

An amino acid shows minimum solubility at its isoelectric point due to highest concentration of its zwitter ion.

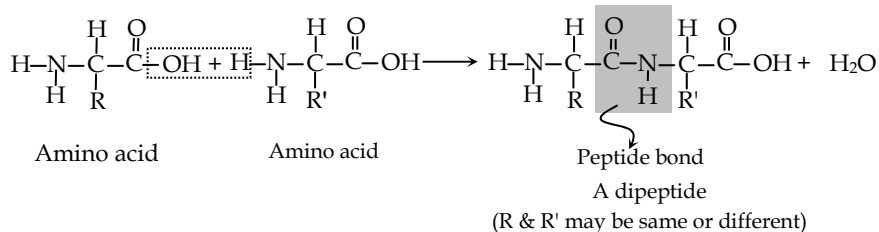


Self Test

- 16.12 What is a zwitter ion? Illustrate with an example of α amino acid.
 16.13 What is isoelectric point of an amino acid?

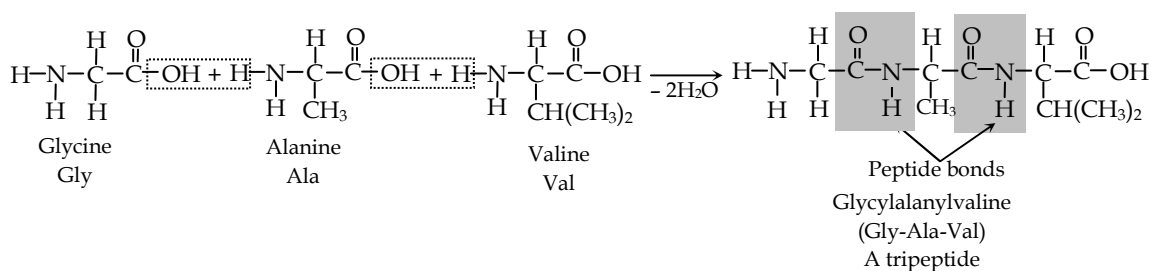
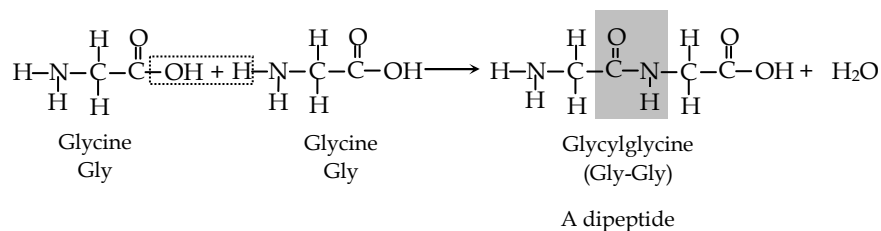
16.2.4 Peptides and Peptide Bond

Acidic carboxyl group ($-\text{COOH}$) of an amino acid reacts with the basic amino group ($-\text{NH}_2$) of other amino acid to give $-\text{NHCO}-$ group, which is called peptide bond or peptide linkage. The compound formed by the condensation reaction between two amino acids is called dipeptide.



Depending upon number of amino acid residue per molecule, the molecules are called dipeptide, tripeptide and so on, and finally polypeptide.

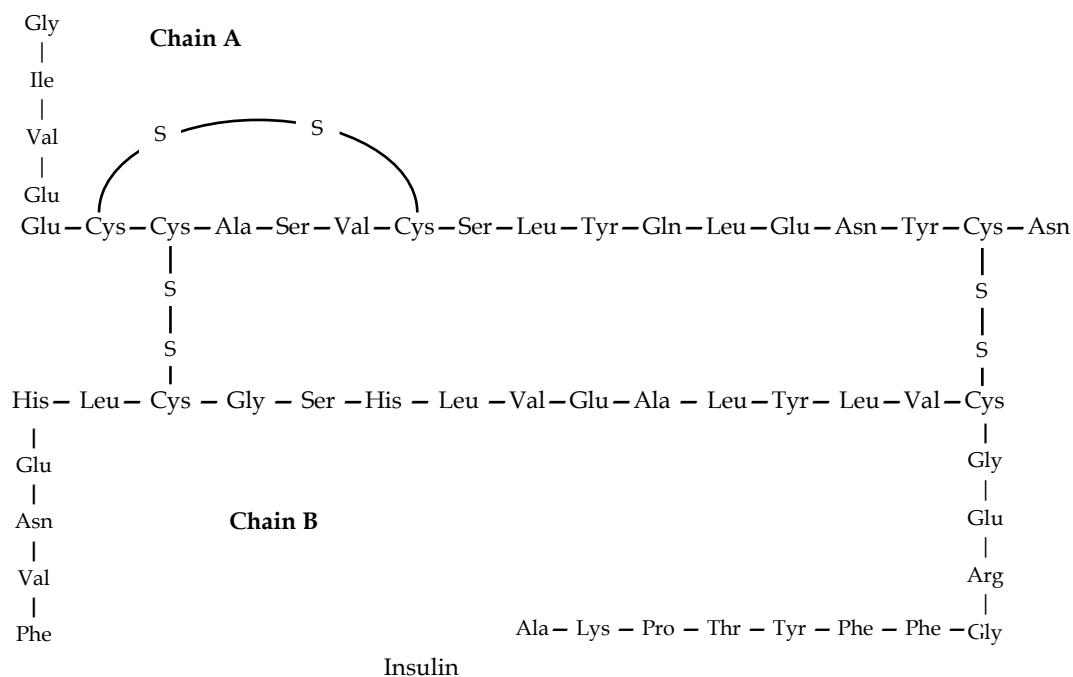
Examples:



According to convention, the **N-terminal amino acid residue** is written at the left end and the **C-terminal amino acid residue** at the right end in polypeptides.

Oxytocin is nonapeptide (a polypeptide having nine amino acid residues) is a pituitary hormone concerned with contraction of uterus.

Insulin is a polypeptide. It consists of two polypeptide chains; chain A and chain B which are hold together by disulphide bond. Chain A consists of 21 amino acids and chain B consists of 30 amino acids. Insulin is a hormone produced by β -cells of pancreas and has a key role in the metabolism of carbohydrates, proteins and fats. Deficiency of insulin causes the disease called **diabetes mellitus**.





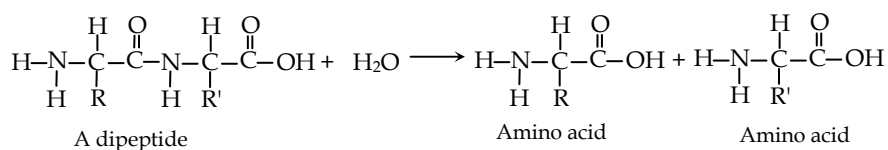
Human insulin is the synthetic insulin which is laboratory grown to mimic the insulin in humans and approved for pharmaceutical use in 1982.

By convention, proteins are polypeptides of high molecular masses above 10,000 are called proteins and hence they are macromolecules.

Some proteins	molecular mass
Insulin	12,000
Egg albumin	45,000
Haemoglobin	70,000
Casein	1,80,000

Hydrolysis of proteins

Proteins are hydrolyzed by dilute mineral acids, alkalis and by enzymes. The final products of hydrolysis are α -amino acids. For example,



Self Test

- 16.14 What is a peptide bond? Give an example of a dipeptide.
 16.15 Give a chemical reaction for hydrolysis of dipeptide.
 16.16 What happens when a protein is completely hydrolyzed?

16.2.5 Classification of Proteins

1. On the basis of molecular structure, proteins are broadly classified into two classes;
 - i. Fibrous proteins
 - ii. Globular proteins
- i. **Fibrous proteins:** Molecules of fibrous proteins are long threadlike and tend to lie side by side to form fibers. They are insoluble in water. Fibrous proteins serve as the *chief structural materials* of animal tissues, a function to which their insolubility and fiber forming tendency suit them. For example, *keratin* in skin, hair, nails, wool, horn, and feathers; *myosin* in muscle, *fibroin* in silk etc.
- ii. **Globular proteins:** Molecules of globular proteins are folded into compact units of spheroidal shapes. Globular proteins are soluble in water and aqueous solutions of acids, bases and salts. Globular proteins serve a variety of functions related to the maintenance and regulation of the life processes, functions that require mobility and solubility. For example, all enzymes, many hormones, albumin, hemoglobin etc.

2. **On the basis of constituents in the molecule**, proteins are classified into two classes;
- Simple proteins:** Protein which gives only α -amino acids on hydrolysis are called simple proteins. For example, albumin, globulin, histone etc.
 - Conjugated proteins:** Protein which gives non protein part in addition to α -amino acids are called conjugated proteins. For example, glycoprotein, lipoprotein nucleoprotein, hemoglobin etc.

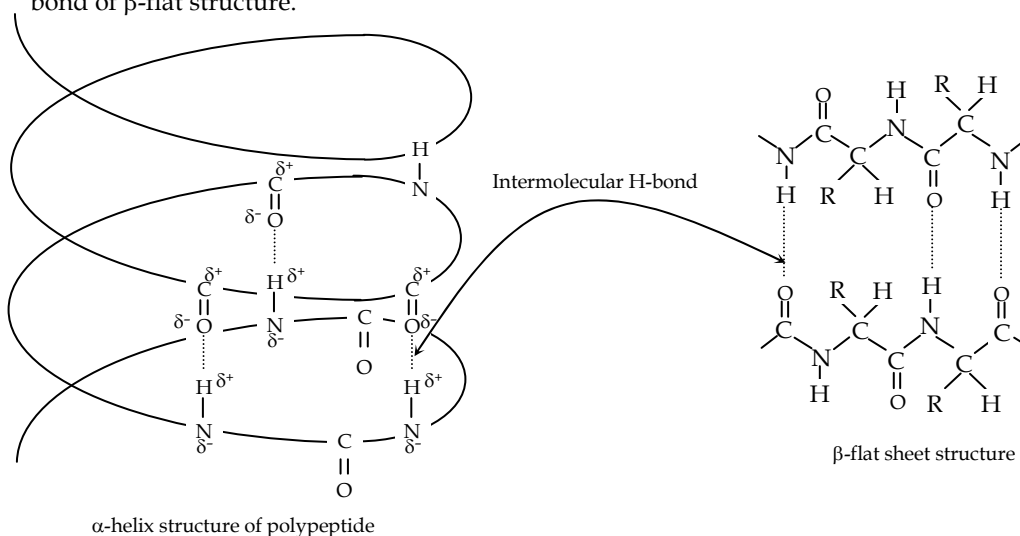
Self Test

16.17 How do fibrous proteins differ from globular proteins?

16.2.6 Structure of Proteins

Proteins are the high molecular mass polymers of α - amino acids. The structure is complex and can be divided into following levels of organization.

- Primary structure:** It is the sequence of amino acids in the protein. A protein molecule has its own sequence of amino acids. Just as the letters of the alphabet combined in a different ways to form an almost endless variety of words, amino acids linked together on varying sequence form a huge variety of proteins.
- Secondary structure:** It tells us about the spatial arrangement of polypeptide in a protein. The arrangement may provide twisting of polypeptide chain which is mainly maintained by H-bond of β -flat structure.



- Tertiary structure:** It is three dimensional arrangement of protein structure in native environment. It is compact structure with H-bond, S-S-bond, ionic interaction, and hydrophobic interaction to produce a specific shape which is most stable called native state.

16.2.7 Denaturation of Proteins

Denaturation is the disorganization of native protein structure. During denaturation, tertiary and secondary structures of protein are destroyed whereas primary structure remains the same. During denaturation, the biological activities of protein are lost and its physical properties are changed, but chemical properties remain same. In

**Fig. 16.2 Denaturation of Protein of Egg**

denaturation, soluble globular protein gets converted into insoluble fibrous protein. Proteins can be denatured by various agents such as;

- i. physical agents like heat
- ii. radiations
- iii. chemicals agents like strong solutions of acids, bases or salts

Examples:

1. Coagulation of egg- white (albumin) on heating: Soluble globular albumin changes into insoluble fibrous albumin.
2. Conversion of milk (casein) into cheese: Soluble globular casein changes into insoluble fibrous casein.

Self Test

16.18 What is denaturation of protein? What physical, chemical and biological changes take place during denaturation?

16.2.8 Enzymes

Enzymes are a group of globular proteins that catalyze the biochemical reactions in cells. Hence they are also called biological catalysts.

Each cell produces particular enzymes to speed up the reactions. Even a primitive cell of bacteria contains about 3000 different enzymes. Without enzymes, food materials like carbohydrates, proteins cannot be digested.

Characteristics of enzymes

1. Enzymes catalyze the reactions up to ten millions times faster than unanalyzed reactions.
2. Enzymes are highly specific in their actions.
3. Enzymes are highly efficient i.e. small amount of enzyme can bring a large extent of reactions.
4. Enzymes function at moderate temperature and pH range in living cells. If conditions are changed catalytic activities are decreased or lost.
5. Catalytic activities of enzymes are enhanced by some non-proteins which are called **cofactors**. It may be some metal cations like K^+ , Na^+ , Cu^+ , Mg^{++} , Zn^{++} , Ca^{++} , Ni^{++} , Fe^{++} , CO^{++} or other smaller organic compounds.

If the cofactors are organic compounds, they are called **coenzymes**. The peptide portion of enzyme without coenzyme is called **apoenzyme** and the enzyme with its cofactor is called **holoenzyme**.

Self Test

16.19 What are enzymes? Give any two characteristics of them.

16.2.9 Functions of Proteins

1. Enzymes: All enzymes are proteins.
2. Structural: Proteins are the major structural materials of animal tissues. E.g. *keratin*
3. Transport proteins: *Globin* in hemoglobin of blood transport oxygen
4. Storage proteins: Some proteins are food storage. E.g. *albumin* of egg, *casein* of milk
5. Hormones: Many hormones are proteins e.g. *Insulin* which regulates blood sugar.

16.2.10 Some Visual Tests of Proteins

1. Ninhydrin test: This test is given by amino acid as well as proteins.
2. Biuret test
3. Xantho-proteic test
4. Millon's test
5. Hopkins-Cole test

16.3 Nucleic Acids (DNA and RNA)

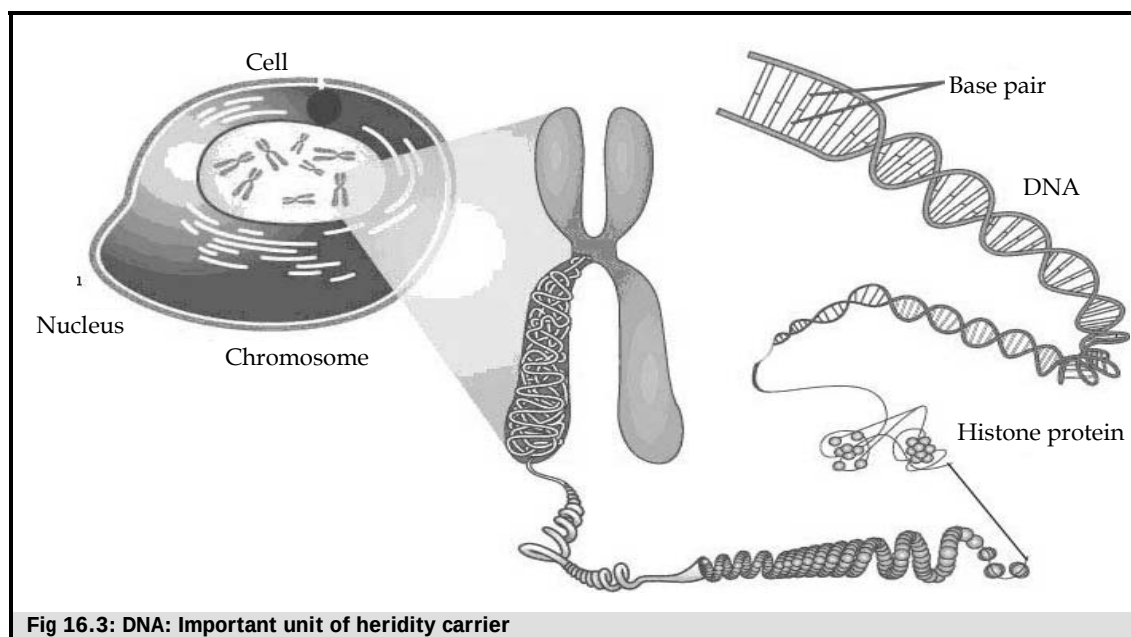


Fig 16.3: DNA: Important unit of heridity carrier

16.3.1 Introduction

Nucleic acids are non-protein, nitrogenous substances which are polymers of nucleotides hence they are also called polynucleotides. Nucleic acids are genetic materials of the cells which are responsible for transmission of heredity character and also carry out biosynthesis of proteins.

There are two different types of nucleic acids found in living organizations.

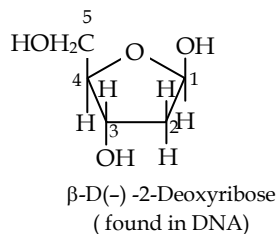
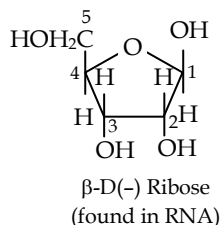
1. RNA (Ribonucleic acid)
2. DNA (Deoxyribonucleic acid)

16.3.2 Components of Nucleic Acids

Nucleic acids are made up of three components;

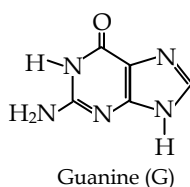
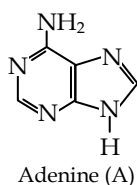
1. Pentose sugar
2. Nitrogenous bases
3. Phosphate group

1. **Pentose sugar:** There are two types of pentose sugars (Carbohydrate having five carbon atoms) in nucleic acids; deoxyribose sugar in DNA and ribose sugar in RNA.

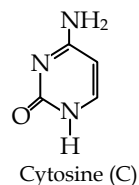
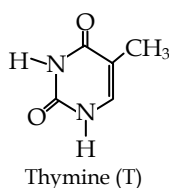
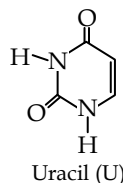


2. **Nitrogenous bases:** These are heterocyclic organic compounds having nitrogen atoms. There are two types of nitrogenous bases in nucleic acid; purine bases and pyrimidine bases.

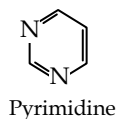
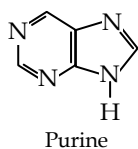
- i. **Purine bases:** It includes adenine and guanine.



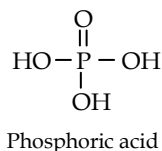
- ii. **Pyrimidine bases:** It includes uracil, thymine and cytosine.



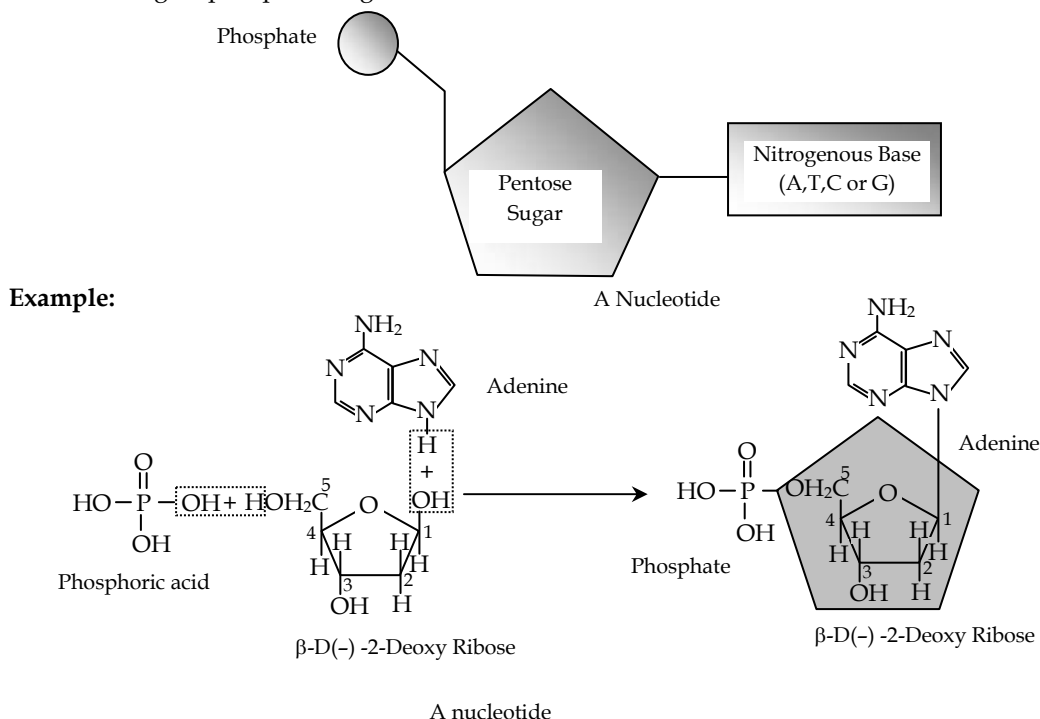
Nucleic acid	Purine bases	Pyrimidine bases
DNA	Adenine, Guanine	Cytosine, Thymine Uracil is absent
RNA	Adenine, Guanine	Cytosine, Uracil Thymine is absent



3. **Phosphoric acid unit**



Nucleotide: Pentose sugar combines with nitrogenous base at its C₁ and with phosphoric acid at C₅ to give phosphate-sugar-base unit which is called nucleotide.

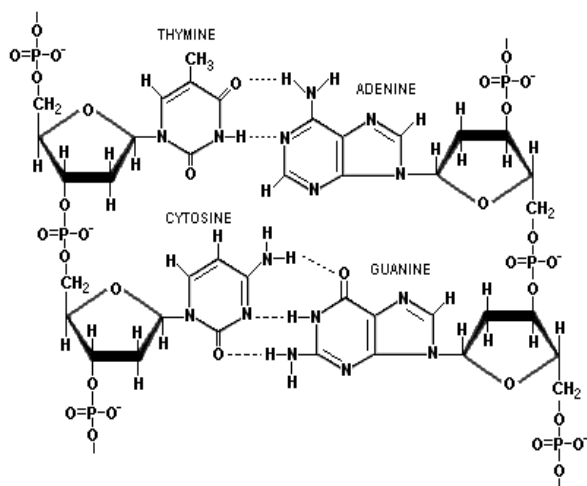


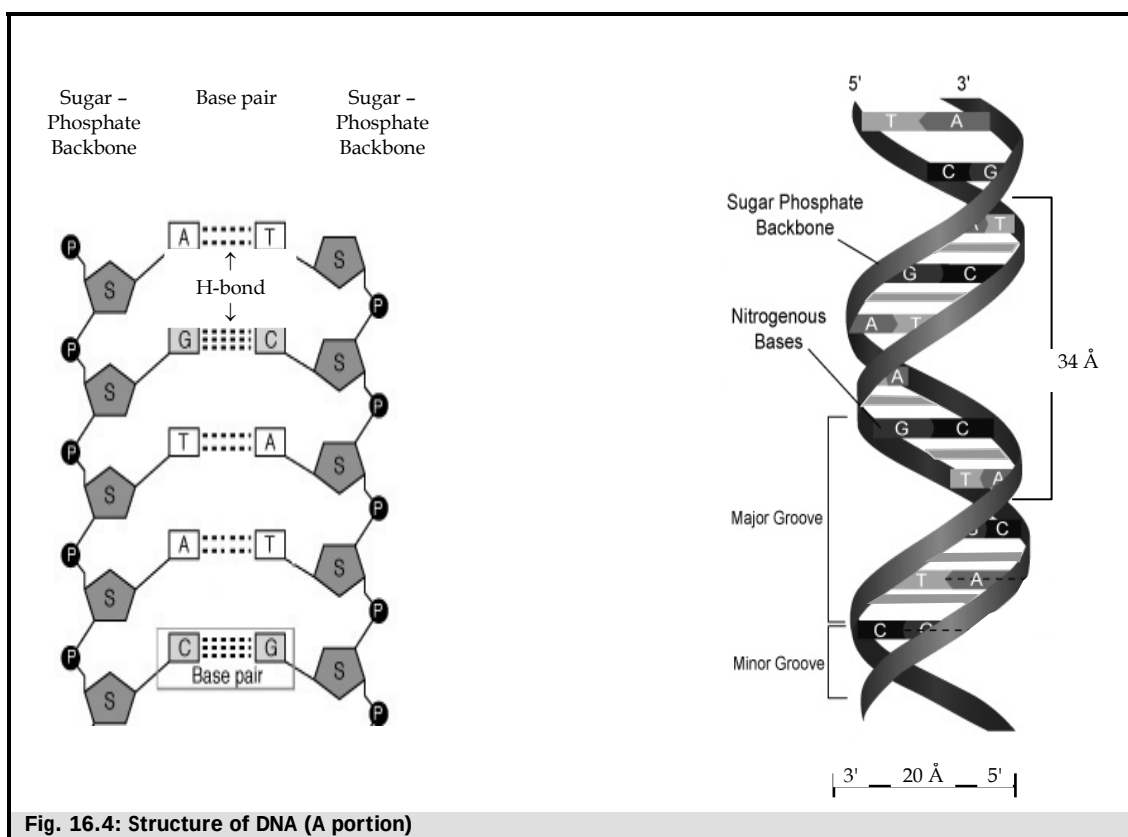
Polynucleotide: Phosphoric acid unit of one nucleotide combines with sugar unit of other nucleotide at C₃ of sugar (ester linkage) to give finally polynucleotide which is nucleic acid.

16.3.3 Double Helix Structure of DNA

In 1953, James Watson and Francis Crick reported the double helix structure of DNA and received the Nobel Prize in 1962 for this work. According to this model, DNA is made up of two polynucleotide chains wound as right handed helix round a common axis forming a double helix but heading in opposite direction so that the sequence is C-3, C-5 in one chain and C-5, C-3 in the other chain.

The two polynucleotide chains are held together at intervals by hydrogen bonds between purine and pyrimidine base pairs. Thymine is always bonded to adenine by two H-bonds and cytosine is always bonded to guanine by three H-bonds. Hence two chains are not identical but complementary. A double helix is 20Å(2nm) in diameter and each complete turns is 34Å(3.4nm) along the axis and has ten nucleotide units.





16.3.4 Differences between RNA and DNA

Some of the important difference between DNA and RNA are listed in the table:

DNA	RNA
1. It is double stranded and helical. (except in paraovins)	1. It is single stranded and non-helical.
2. The pentose sugar is deoxyribose .	2. The pentose sugar is ribose.
3. Purine nitrogenous bases found in DNA are Adenine (A) and Guanine (G). Pyrimidine nitrogenous bases found are Thymine (T) and Cytosine (C). Uracil (U) is absent.	3. Purine nitrogenous bases found in RNA are Adenine (A), Guanine (G). Pyrimidine nitrogenous bases found are Uracil (U) and Cytosine (C). Thymine (T) is absent.
4. DNA can undergo replication.	4. It does not replicate.
5. It transmits heredity characters.	5. It plays main role in protein synthesis.

16.3.5 Biological Functions of Nucleic Acids

1. The nucleic acid DNA carry genetic information from parent to offspring.
2. DNA synthesizes RNA by the process of transcription.
3. DNA controls all biological activities of cell.
4. The nucleic acid RNA plays major role on protein biosynthesis.
5. In some of the virus like HIV and TMV viruses, RNA acts as genetic material.

Self Test

- 16.20 What are the nucleic acids? Point out their constituents.
 16.21 Name purine and pyrimidine bases of DNA and RNA.
 16.22 Write the differences of DNA and RNA.

16.4 Lipids

16.4.1 Introduction

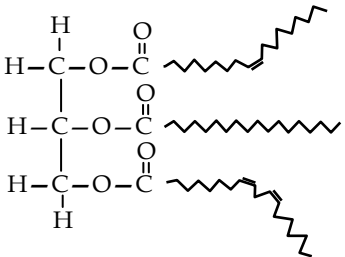
The term lipid is derived from the Greek word *Lipos*; meaning fat. Water insoluble compounds like fats and oils and their derivatives occurring in the living organisms are called lipids. Lipids are a group of bio-molecules which are greasy, insoluble in water (hydrophobic) and soluble in organic solvents like ether, chloroform etc. Lipids are the constituents of all cell membranes.

Lipids are classified as;

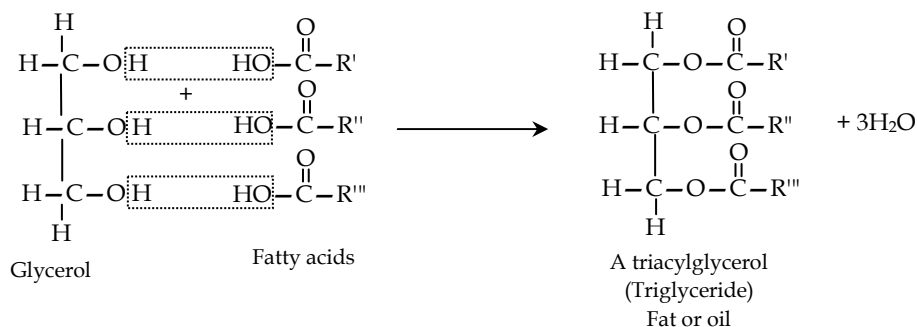
1. Simple lipids (fats and oils)
2. Phospholipids
3. Waxes

16.4.2 Simple Lipids (Fats and Oils)

Fats are the major constituents of fat cells in animals and plants, and are the important food reserves of organisms. They are found mainly under skin, around intestine in animals and in seeds and fruits in plants.

		
Olive (<i>Olea europaea</i>) branch with fruits	Olive oil	Triglyceride (Oil molecule)

Chemically, a fat or an oil molecule is an **ester** of glycerol (a trihydric alcohol) with fatty acids (long chain mono carboxylic acids) ie. *Triacylglycerol* or *triglyceride*.

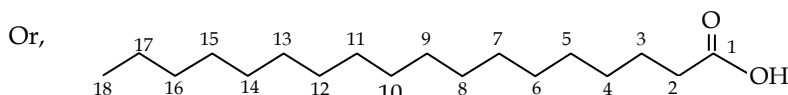
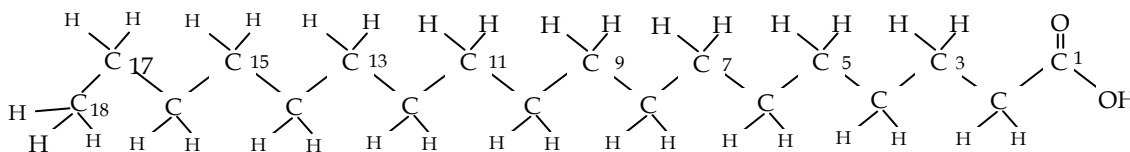


If $R' = R'' = R'''$, the triglyceride is called simple triglyceride otherwise called mixed triglyceride. All natural fats and oils are a mixture of mixed triglycerides.

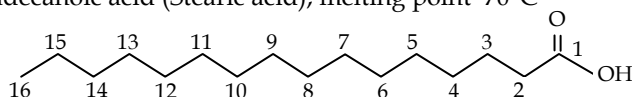
Fatty acids

Fatty acids are open chain monocarboxylic acids having carbon number 3 to 18. Except for C_3 and C_5 acids, only acids with even number of carbons are present in substantial amount. Fatty acids are saturated as well as unsaturated.

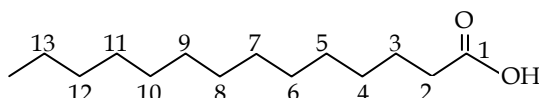
Some saturated fatty acids



Octadecanoic acid (Stearic acid), melting point 70°C



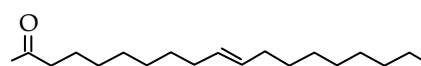
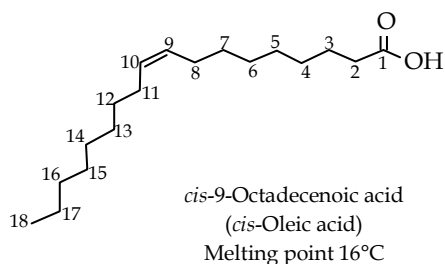
Hexadecanoic acid (Palmitic acid), melting point 63°C



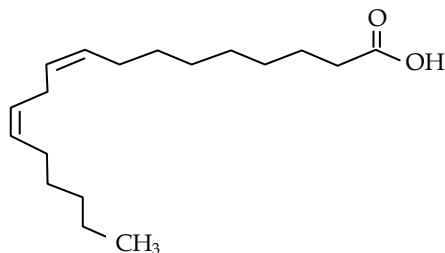
Tetradecanoic acid (Myristic acid), melting point 55°C

Saturated fatty acids are linear and fit well together hence have high melting point.

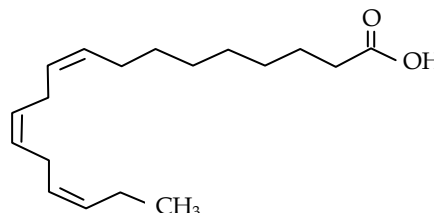
Some unsaturated fatty acids



Trans -9-Octadecenoic acid



cis,cis-9,12-Octadecadienoic acid
(Linoleic acid), melting point -5°C



cis,cis,cis-9,12,15-Octadecatrienoic acid
(Linolenic acid), melting point -11°C

cis-unsaturated fatty acid has a bent at the double bond i.e. the molecule is nonlinear and fits poorly hence has a low melting point.

Self Test

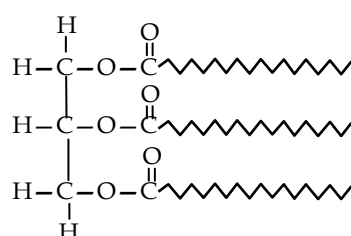
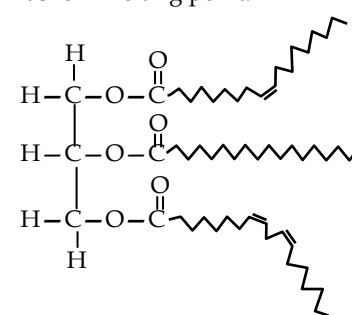
16.23 How do you account for the fact that octadecanoic acid is solid but cis-9-octadecanoic acid is liquid at room temperature though both have nearly equal molecular weight?

Hint: Compare surface area and molecular geometry

Table 16.1: Fatty acid composition of some fats and oils

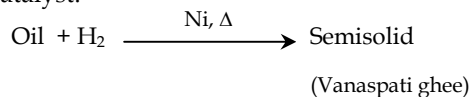
fats and oils	saturated fatty acids	unsaturated fatty acids
Butter (नौनी घ्यू)	Myristic acid (8-13%) Palmitic acid (25-32%) Stearic acid (8-13%)	Oleic acid (22-29%) Linoleic acid (3%) Linolenic acid (2-10%)
Lard (pig fat)	Myristic acid (1%) Palmitic acid (25-30%) Stearic acid (12-16%)	Oleic acid (41-51%) Linoleic acid (3-8%)
Soybean oil	Palmitic acid (7-11%) Stearic acid (2-5%)	Oleic acid (22-34%) Linoleic acid (50-60%) Linolenic acid (2-10%)
Olive oil (जैतूनको तेल)	Palmitic acid (7-20%) Stearic acid (1-3%)	Oleic acid (53-86%) Linoleic acid (4-22%)

16.4.3 Differences between Fats and Oils

Fats	Oils
Natural triglycerides that are solid at room temperature are called fats.	Natural triglycerides that are liquid at room temperature are called oils.
Fat contains a large proportion of saturated fatty acid residues like palmitic acid (mpt.63°C), stearic acid (mpt.70°C). The molecules of fats fit well together since the acid residues are saturated and hence linear due to which intermolecular force of attraction is stronger thus has high melting point.  Saturated triglyceride Fat molecule	Oils contain large proportions of <i>cis</i>-unsaturated fatty acid residues like oleic acid (mpt.16°C), linoleic acid (mpt.-5°C), linolenic acid (mpt.-11°C). The molecules do not fit each other well since the acid residues are unsaturated and hence nonlinear due to which intermolecular force of attraction is weak thus has low melting point.  Unsaturated triglyceride Oil molecule

16.4.4 Hydrogenation of Oil

For hydrogenation, oil is heated to 150-200°C and hydrogen is passed in under pressure in presence of finely divided nickel catalyst.



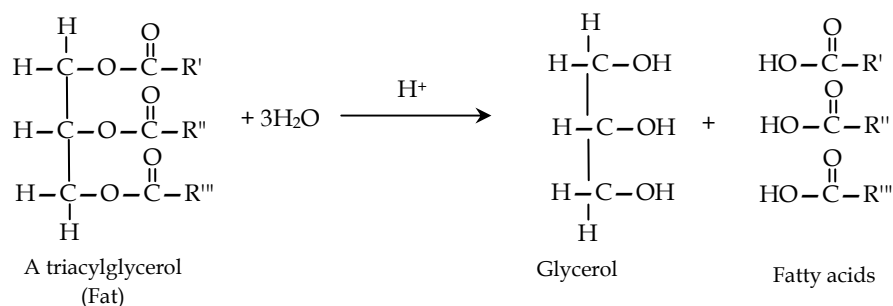
Hydrogenation of oil changes its physical as well as chemical properties. It hardens the oil and lowers rancidity.

- 1. Hardening of oils:** Oils contain large proportions of *cis*-unsaturated fatty acid residues. The molecules do not fit each other well since the acid residues are *cis*-unsaturated and hence nonlinear. This makes intermolecular force of attraction weak thus low melting point i.e. liquid at room temperature. On hydrogenation, the molecules become saturated and hence linear due to which intermolecular force of attraction increases i.e. melting point increases. Hence oils are converted into solid form. The process is also called hardening of oil. Only partial hydrogenation is carried out otherwise a fat as hard as tallow is formed.
- 2. Rancidity of fats and oils:** when fats and oils are exposed to moist air for a long time, bad smell and sour taste are developed and fats and oils are said to be rancid. Rancidity is due to presence of volatile, bad smelling acids and aldehydes which result at least from attack of oxygen at double bond positions (allylic positions). Hydrogenation slows down the rancidity by decreasing the number of double bonds.

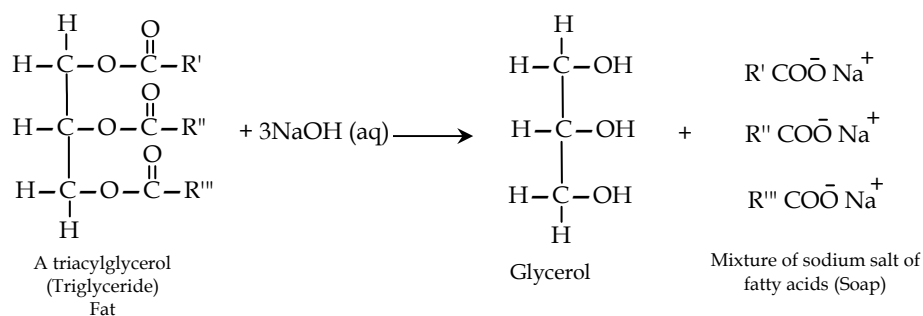
Drying oils: Oils with high degree of unsaturation like linseed oil (oil obtained from the ripe seeds of the flax plant *Linum usitatissimum*), tung oil (oil obtained from the seeds of the tung tree, *Aleurites fordii*.) polymerize in presence of air to form a thin tough waterproof film which protect the surface and gives good finish. Such oils are called drying oils. Drying means not the evaporation of solvent. Drying oils are used as vehicles in paints, varnishes, and printing inks.

16.4.5 Hydrolysis of Fats

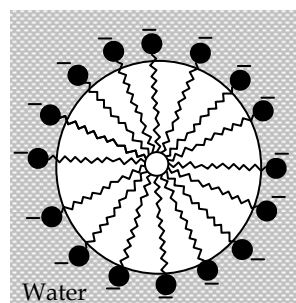
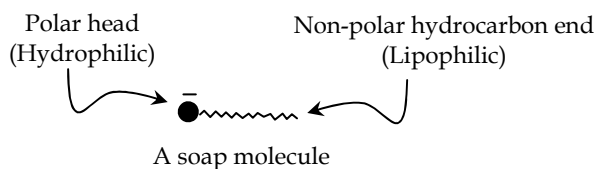
Fat (ester) can be hydrolyzed in acidic as well as in alkaline medium. Hydrolysis of fat with mineral acid gives glycerol and a mixture of carboxylic acids. They can be separated by fractional distillation of over 90% purity. Fats are thus the source of monocarboxylic acids of even carbon number ranging from six to eighteen carbons.



Alkaline hydrolysis of fat gives a mixture salt of fatty acids which is called soap and the process is called *saponification*.



A soap molecule is **amphipathic** i.e. has a polar end (hydrophilic) and nonpolar hydrocarbon end (lipophilic). In water, nonpolar ends huddle together in the center projecting polar ends outwards in the solvent forming a spherical cluster called **micelle**.



A soap micelle

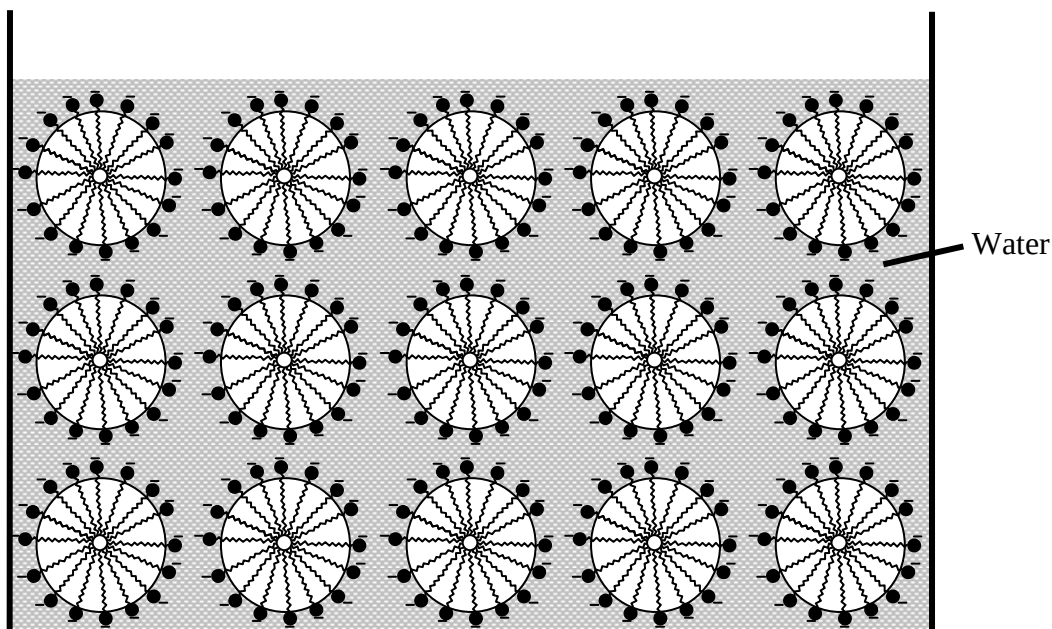
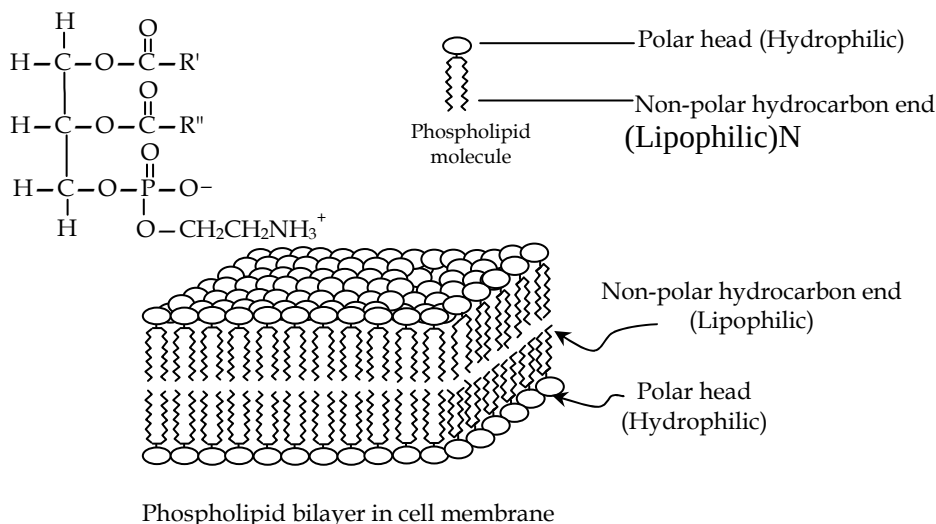


Fig 16.5: Soap micelles dispersed in water

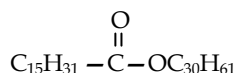
16.4.6 Phospholipids

Phospholipids are triglycerides in which two hydroxyl groups of glycerol are esterified with fatty acids and the third hydroxyl group is esterified with derivatives of phosphoric acid having nitrogen.



16.4.7 Waxes

Waxes are esters of higher fatty acids with long chain monohydric alcohols. For example, beeswax is largely myricyl palmitate.



Triacontanyl palmitate (myricyl palmitate)

16.4.8 Functions of Lipids

1. Lipids are good source of energy. They provide energy on oxidation.
2. Body stores food as reserve food in the forms of lipid because they are insoluble in water.
3. They are structural components of cell membrane.
4. They act as carrier of fat soluble vitamins A, D and E.
5. They surround vital organs and protect from mechanical shock and injury.
6. Fats are present below skin acts as insulator which reduces heat loss.
7. Drying oils are used as vehicles in paints, varnishes, and printing inks.

Self Test

16.24 What are lipids? Mention their role in a human body.

16.25 What are fats and oils chemically?

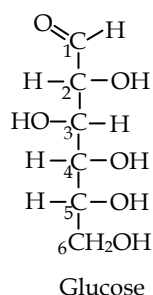
16.26 What are physical and chemical difference of fat and oil?

16.27 What is saponification?

Solved Questions

1. What are carbohydrates chemically?

- Carbohydrates are **polyhydroxyaldehydes** or **polyhydroxyketones** or any organic compounds that produce these compounds on hydrolysis. For example,



2. What are fibrous and globular proteins?

- On the basis of molecular structure, proteins are broadly classified into two classes;
- Fibrous proteins
 - Globular proteins
- a. Fibrous proteins:** Molecules of fibrous proteins are long thread like and tend to lie side by side to form fibers. They are insoluble in water. Fibrous proteins serve as the *chief structural materials* of animal tissues, a function to which their insolubility and fiber forming tendency suit them. For example, *keratin* in skin, hair, nails, wool, horn and feathers; *myosin* in muscle, *fibroin* in silk, etc.
- b. Globular proteins:** Molecules of globular proteins are folded into compact units of spheroidal shapes. Globular proteins are soluble in water and aqueous solutions of acids, bases and salts. Globular proteins serve a variety of functions related to the maintenance and regulation of the life processes, functions that require mobility and solubility. For example, All enzymes, many hormones, albumin, hemoglobin etc.

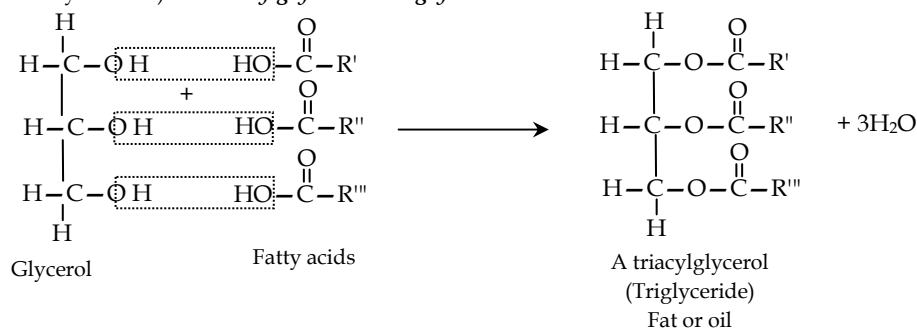
3. What is denaturation of protein?

- Denaturation is the disorganization of native protein structure. During denaturation, tertiary and secondary structures of protein are destroyed whereas primary structure remains same. During denaturation, the biological activities of protein are lost and its physical properties are changed, but chemical properties remain same. In denaturation, soluble globular protein gets converted into insoluble fibrous protein. Proteins can be denatured by various agents such as;
- physical agents like heat,
 - radiations,
 - chemicals agents like strong solutions of acids, bases or salts

For example, coagulation of egg- white (albumin) on heating: Soluble globular albumin changes into insoluble fibrous albumin.

4. What are fats and oils chemically? Why do they exist in different physical states?

- Chemically, a fat or an oil molecule is an **ester** of glycerol (a trihydric alcohol) with fatty acids (long chain mono carboxylic acids) i.e. **Triacylglycerol** or **triglyceride**.



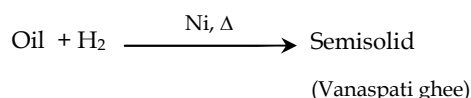
Natural triglycerides that are solid at room temperature are called fats otherwise oils. The difference in their physical states is due to differences in their molecular geometry.

Fat contains a large proportion of saturated fatty acid residues like palmitic acid (mpt. 63°C), stearic acid (m.pt.70°C). The molecules of fats fit well together since the acid residues are saturated and hence linear due to which intermolecular force of attraction is stronger thus has high melting point.

Oils contain large proportions of **cis-unsaturated** fatty acid residues like oleic acid (mpt.16°C), linoleic acid (m.pt.-5°C), linolenic acid (m.pt.-11°C). The molecules do not fit each other well since the acid residues are unsaturated and hence nonlinear due to which intermolecular force of attraction is weak thus has low melting point.

5. What happens when oil is hydrogenated?

- For hydrogenation, oil is heated to 150-200°C and hydrogen is passed in under pressure in presence of finely divided nickel catalyst.



Hydrogenation of oil changes its physical as well as chemical properties. It hardens the oil and lowers rancidity.

- i. **Hardening of oils:** Oils contain large proportions of **cis-unsaturated** fatty acid residues. The molecules do not fit each other well since the acid residues are *cis*-unsaturated and hence nonlinear. This makes intermolecular force of attraction is weak thus low melting point i.e. liquid at room temperature. On hydrogenation, the molecules become saturated and hence linear due to which intermolecular force of attraction increases i.e. melting point increases. Hence oils are converted into solid form. The process is also called hardening of oil. Only partial hydrogenation is carried out otherwise a fat as hard as tallow is formed.
- ii. **Rancidity of fats and oils:** when fats and oils are exposed to moist air for a long time, bad smell and sour taste are developed and fats and oils are said to be rancid. Rancidity is due to presence of volatile, bad smelling acids and aldehydes which result at least from attack of oxygen at double bond positions (allylic positions). Hydrogenation slows down the rancidity by decreasing the number of double bonds.

Questions

Very Short Answer Questions:

1. Write an example of each of the following
 - i. reducing sugar
 - ii. simple lipid
 - iii. non-sugar
 - iv. disaccharide
2. What is meant by peptide bond? Write an example of dipeptide.
3. Write molecular formula of each of the following:
 - i. Simple lipid
 - ii. Dipeptide
4. What are disaccharides? What happens when they get hydrolysed?
5. Distinguish between sugars and non-sugars.
6. What is denaturation of protein? Give an example:
7. What are lipids? Give one example of it.
8. What happens when;
 - i. fats get hydrolysed
 - ii. Protein is heated
9. How would you obtain glucose from cane sugar?
10. What are the constituent of nucleic acid?
11. What is different between essential and non-essential amino acids?
12. Name the pyrimidine bases found in RNA.
13. Point out any two differences between RNA and DNA molecules.

14. Distinguish both chemically and physically between a fat and a vegetable oil.
15. Write the open chain structure of glucose and fructose.
16. What happens when fats are hydrolysed using aq. NaOH?
17. Write important functions of lipids.
18. Write the structures of the compounds obtained by the hydrolysis of sucrose.
19. How can the presence of sugar be tested in urine?
20. Write any two important functions of carbohydrates.
21. Write any two examples each for sugars and non-sugars.

Multiple Choice Questions

1. Which of the followings is a carbohydrate?
 - a. Glyceraldehyde
 - b. Casein
 - c. Insulin
 - d. Saccharin
2. Which of the followings is a disaccharide?
 - a. Galactose
 - b. Lactose
 - c. Deoxyribose
 - d. Ribose
3. Which of the followings is a disaccharide?
 - a. Cellulose
 - b. Starch
 - c. Cellobiose
 - d. Glycogen
4. Water soluble component of starch is
 - a. Amylose
 - b. Amylopectin
 - c. Sucrose
 - d. Inulin
5. Glucose is
 - a. A hexose
 - b. A non sugar
 - c. Non reducing sugar
 - d. invert sugar
6. Milk sugar is
 - a. Lactose
 - b. Maltose
 - c. Glucose
 - d. Casein
7. Which of the followings is non sugar?
 - a. Glucose
 - b. Cellulose
 - c. Sucrose
 - d. Galactose
8. Which of the followings is non reducing sugar?
 - a. Maltose
 - b. Sucrose
 - c. Fructose
 - d. Lactose
9. Invert sugar is
 - a. Sucrose
 - b. Equimolar mixture of D (+)-Glucose and D (-)-Fructose
 - c. Equimolar mixture of D (+)-Glucose and D (+)-Fructose
 - d. Equimolar mixture of D (-)-Glucose and D (+)-Fructose
10. α -D (+)-Glucose and β -D (+)-Glucose differ in
 - a. Configuration at C-1
 - b. Configuration at C-2
 - c. Configuration at C-5
 - d. Configuration at C-1 and C-2
11. α -D (+)-Glucose and β -D (+)-Glucose are
 - a. Enantiomers
 - b. Epimers
 - c. Mirror image of each other
 - d. Anomers
12. Cotton is
 - a. Protein
 - b. Starch
 - c. Polysaccharide
 - d. Cellulose
13. Carbohydrates are tested by
 - a. Ninhydrin test
 - b. Molisch's test
 - c. Millon's test
 - d. Biuret test
14. Charring of sugar is
 - a. Oxidation
 - b. Dehydration
 - c. Reduction
 - d. Reduction and dehydration
15. In our body glucose is stored as
 - a. Protein
 - b. Fat
 - c. Glycogen
 - d. Lipid
16. The carbohydrate present in plant cell wall is
 - a. Cellulose
 - b. Maltose
 - c. Sucrose
 - d. Starch
17. Amino acids found in proteins are
 - a. α -amino acids
 - b. β -amino acids
 - c. γ -amino acids
 - d. Mixture of all
18. An example of Zwitter ion is
 - a. Alanine
 - b. Glycine hydrochloride
 - c. Urea
 - d. Ammonium acetate
19. Which of the followings is an essential amino acid?
 - a. Serine
 - b. Glycine
 - c. Alanine
 - d. Histidine
20. Albumin is
 - a. Simple protein
 - b. Conjugated protein
 - c. Derived protein
 - d. Lipoprotein
21. Protein responsible for regulation of body function is
 - a. Denatured protein
 - b. Fibrous protein
 - c. Globular protein
 - d. Derived protein
22. Which of the following test is given by proteins?
 - a. Tollen's test
 - b. Molisch's test
 - c. Ninhydrin test
 - d. Benedict's test
23. Primary structure of protein refers to
 - a. Amino acid sequence
 - b. Arrangement of polypeptides
 - c. Orientation of polypeptides
 - d. α -helix form
24. Denaturation of protein destroys its
 - a. Primary structure
 - b. Chemical properties
 - c. Secondary structure
 - d. Amino acid sequence
25. An enzyme is
 - a. Carbohydrate
 - b. Fibrous protein
 - c. Globular protein
 - d. Metal ion
26. Protein part of enzyme after removal of prosthetic group is called
 - a. Apoenzyme
 - b. Coenzyme
 - c. Holoenzyme
 - d. Cofactor

27. DNA is
 a. Simple protein b. Nucleoprotein
 c. Nonprotein d. Lipid
28. Nucleic acid is polymer of
 a. Nucleoside
 b. not a polymer
 c. Base, sugar and phosphate
 d. Nucleotide
29. DNA is polymer of
 a. Nucleotide b. Ribonucleotide
 c. Deoxyribonucleotide d. Nucleoside
30. In nucleic acid, sugar-phosphate unit forms the backbone. Here, sugar is bonded to phosphate unit by
 a. H-bond b. Ether bond
 c. Ester bond d. Glycoside bond
31. In DNA, two strands of polydeoxyrionucleotides are held by
 a. Peptide bond b. Covalent bond
 c. Ester bond d. H-bond
32. The sugar present in DNA is
 a. Hexose sugar b. Pentose sugar
 c. D- Ribose d. Deoxyribose
33. Which of the followings base is not present in DNA?
 a. Adenine b. Thymine
 c. Uracil d. Cytosine
34. Which of the followings base is a purine base ?
 a. Adenine b. Thymine
 c. Uracil d. Cytosine
35. In DNA, cytosine pairs with
 a. Adenine b. Thymine
 c. Uracil d. Guanine
36. The number of hydrogen bond between adenine and thymine is
 a. 1 b. 2
 c. 3 d. 4
37. Which of the followings base is not present in RNA?
 a. Adenine b. Thymine
 c. Uracil d. Cytosine
38. Which of the following base pair is not present in DNA?
 a. A-T b. G-C
 c. T-A d. U-C
39. Which of the followings is hydrocarbon?
 a. Animal fat b. Vegetable oil
 c. Beewax d. Paraffin oil
40. Chemically, fats and oils are
 a. Esters of glycerol with fatty acids
 b. Esters of monohydric alcohols with fatty acids
 c. Esters of glycols with fatty acids
 d. Esters of glycols with stearic acids
41. Oil contains a large proportions of
 a. Saturated fatty acid residues
 b. Cis unsaturated fatty acid residues
 c. Trans unsaturated fatty acid residues
 d. Unsaturated fatty acid residue
42. Drying oils must have
 a. Carboxyl group b. High saturation
 c. High unsaturation d. High m.pt.
43. Biuret test is done for
 a. Oils b. Sugars
 c. Fats d. Proteins
44. The calorific values is in the order
 a. Fats > carbohydrates > proteins
 b. Fats > proteins > carbohydrates
 c. Proteins > carbohydrates > fats
 d. Carbohydrates > fats > proteins
45. Soap molecule is
 a. Hydrophilic b. Hydrophobic
 c. Amphipathic d. Lipophilic

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
a	b	ca	a	a	a	b	b	b	a	d	d	b	b	c
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
a	a	a	d	a	c	c	a	c	c	a	c	d	c	c
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
d	d	c	a	d	b	b	d	d	a	b	c	d	b	c

Chemistry in the Service of Mankind

17



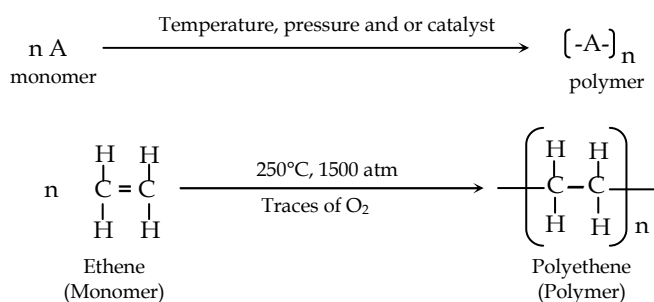
Chemistry has been in the service of mankind since the first birth. Food, clothing, shelter, diseases and employment are the major challenges of our society. Chemistry not only tells all about natural substances but also enables us to design and make new products. Chemistry has helped in getting more food by providing chemical fertilizers, pesticides for agriculture. Polymers like nylon, orlon, dacron, rayon etc. which are synthesized in laboratories are major fabrics of clothing. Chemistry has helped in the construction of houses by providing cement, steel, plastics etc. Synthetic dyes make them colourful. Modern synthetic drugs have played wonderful role to cure and eradicate many deadly diseases. Many industries are based on principles of chemistry which provide employments for people. Some important classes of organic compounds in service to mankind are polymers, dyes, drugs, chemical fertilizers, pesticides which are briefly discussed here.

17.1 Polymers

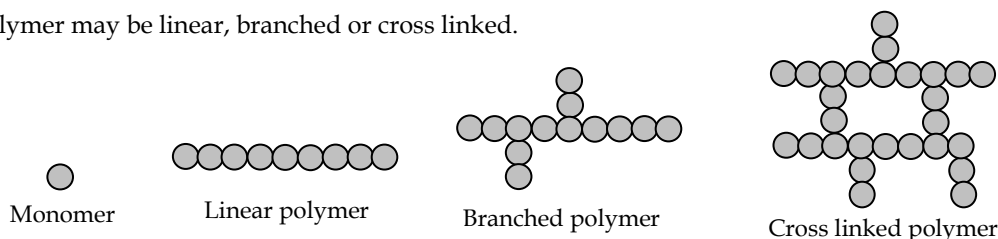
17.1.1 Monomer, Polymer and Polymerization

A *polymer* (Greek: *poly* = many, *meros* = parts) is a **giant molecule** (macromolecule) with **high molecular mass** which is made up of a number of smaller repeating units. The repeating units are usually obtained from low molecular mass simple molecules which are called *monomers*. The process of forming a polymer from monomers under the suitable conditions of temperature, pressure and or catalyst is called *polymerization*.

Example:



A polymer may be linear, branched or cross linked.



17.1.2 Classification of Polymers

Polymers can be classified by different ways;

17.1.3 Addition and Condensation Polymers

Based on *mode of synthesis*, polymers can be classified into addition polymers and condensation polymers.

A. Condensation polymers or step growth polymers: A polymer formed by addition of monomers with elimination of smaller molecules (condensation reaction) is called condensation polymer and the process is called condensation polymerization. Monomers in this type of polymerization have at least two functional groups. Polyamides and polyesters are two important types of condensation polymers.

Some examples,

1. **Nylon-6,6:** It is the polymer having many amide linkage hence belongs to polyamide. It is formed by condensation of two types of monomers: adipic acid (dicarboxylic acid) and hexamethylenediamine (diamine). The name nylon-6,6 is given since each type of monomer has six carbon atoms.

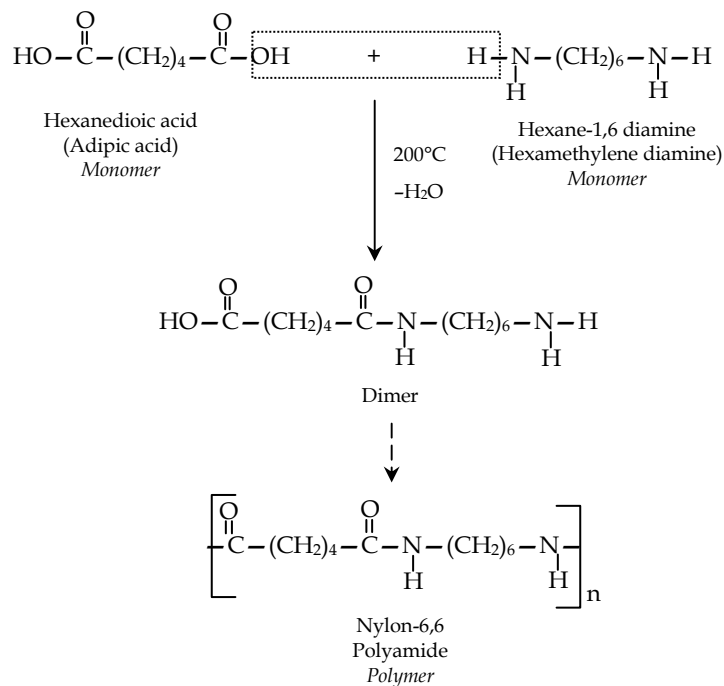
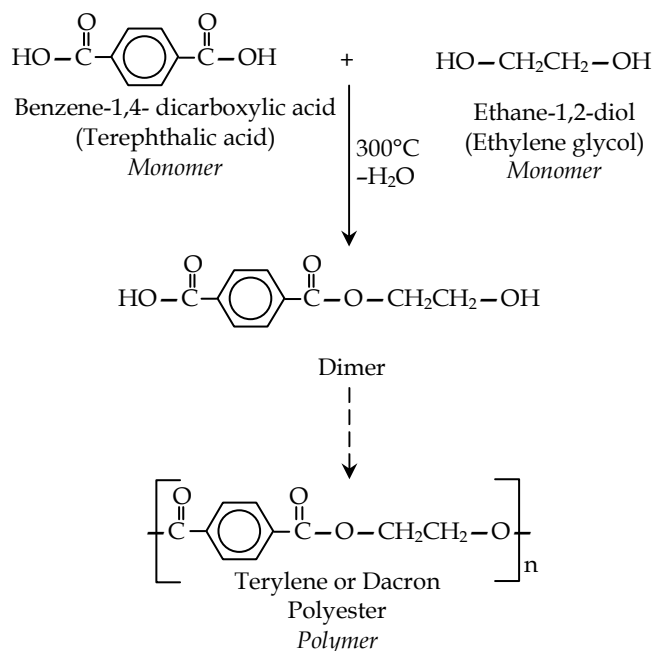
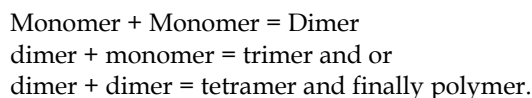


Fig 17.1 Nylon rope has high tensile strength hence used in rock climbing, mountain climbing and parachute

2. **Terylene or dacron** is the polymer having many *ester linkage* hence belongs to polyester. It is formed by condensation of two types of monomers; terephthalic acid (dicarboxylic acid) and ethylene glycol (dihydric alcohol).



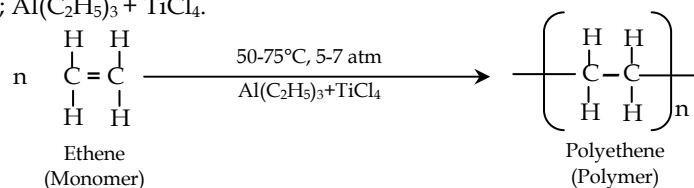
Condensation polymers are also called step growth polymers since the polymers are formed in stepwise manner. i.e.



- B. Addition polymers or chain polymers:** A polymer formed by addition of monomers without elimination of any molecule is called addition polymer and the process is called addition polymerization. Monomers in this type of polymerization are unsaturated. High molecular mass polymer is formed at once but not step by step.

Examples:

- 1. Polyethene:** Commercially, it is prepared from ethene in presence of Ziegler-Natta catalyst; $\text{Al}(\text{C}_2\text{H}_5)_3 + \text{TiCl}_4$.



Ziegler and Natta were awarded Nobel Prize in 1963 for this work. It is resistant to the action of water, acids, alkalis, salts, can be molded, folded. It is used to prepare house wares like buckets, dustbins, carpet backing, packaging materials, cable insulation, etc.

- 2. Polyvinyl chloride (PVC):** It is obtained by polymerization of vinyl chloride. It is used for pipes and fittings, imitation leather, floor covering, containers, window and door frames, gramophone recorders, blood bags etc.



Fig 17.2 PVC in different purposes (a) pipes (b) cosmetic bottles (c) blood bag and saline bottle and pipes

3. **Polytetrafluoroethene (Teflon or PTFE):** It is obtained by polymerization of tetrafluoroethylene. It is inert to chemical reagents, even to boiling aqua regia, resistant to heat and electricity and used as nonstick coating especially for cooking utensils, waterproof clothing, for insulation of electrical items, for the manufacture of gaskets, valves etc.



Fig 17.3 Polytetrafluoroethene in (a) nonstick coating on frying pan and waterproof jacket (b) gaskets and valves

4. **Polyvinylcyanide (Orlon):** It is obtained by polymerization of vinyl cyanide (acrylonitrile). It is used as synthetic textiles fiber and in clothing and carpeting.



5. **Polypropylene:** It is obtained by polymerization of propylene. It is used in the manufacture of house wares, toys, medical equipments, laboratory equipments etc.



Fig 17.4: Different uses of polypropylene

6. **Polystyrene:** It is obtained by polymerization of styrene. It is used for the manufacture of safety cabinets for electrical and electronic items, food containers etc.

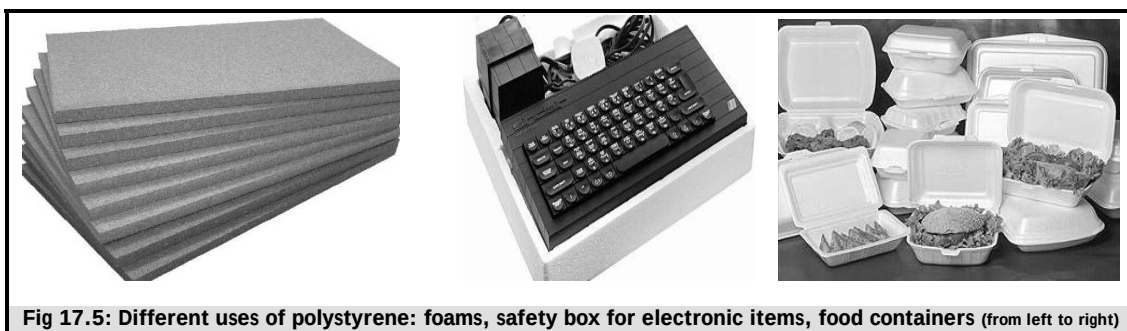
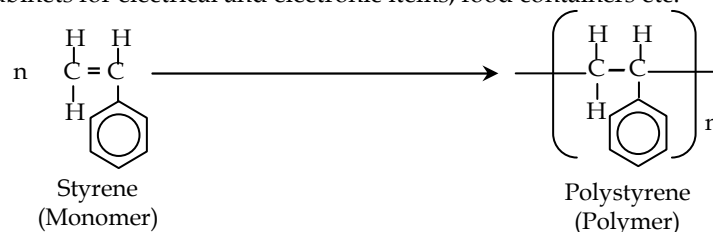
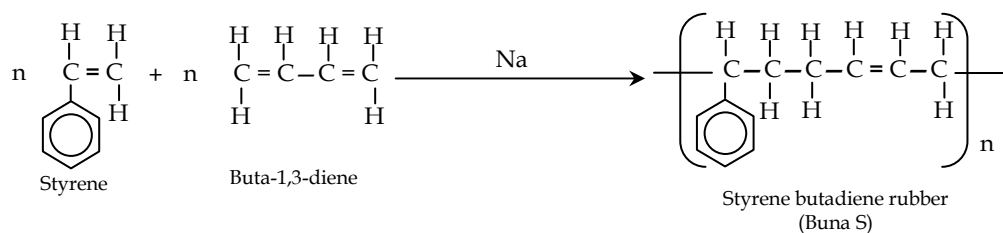


Fig 17.5: Different uses of polystyrene: foams, safety box for electronic items, food containers (from left to right)

7. **Styrene butadiene rubber (SBR):** It is obtained by polymerization of styrene and butadiene in presence of sodium metal. Hence the name is given Buna S. It is rubber and used in the manufacture of tubes and tyres.



8. **Neoprene:** It is obtained by polymerization of chloroprene.

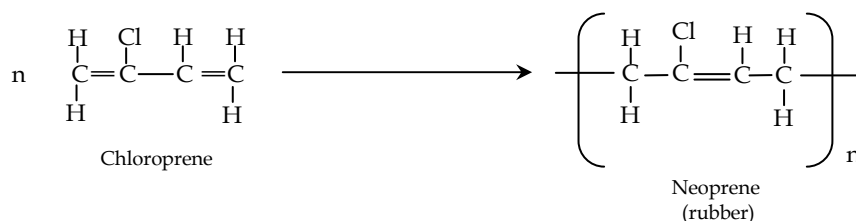


Fig 17.6 Different uses of rubber

17.1.4 Homopolymers and copolymers

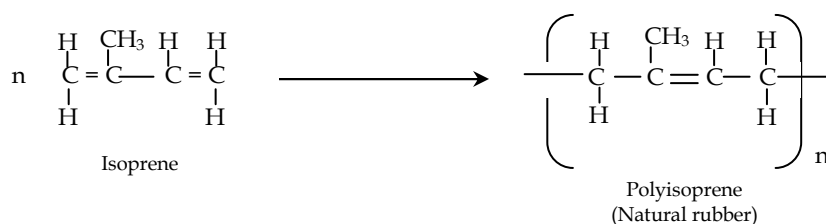
Based on the types of monomers, polymers can be classified as homopolymer and copolymers.

1. **Homopolymers:** A polymer formed by only one type of monomers is called homopolymer. Polyvinyl chloride (PVC) is made up of only one type of monomer, vinyl chloride. Hence it is an example of homopolymer. Some examples are; polyethene, teflon, orlon etc.
2. **Copolymers:** A polymer formed by more than one type of monomers is called copolymer. Nylon-6,6 is made up of two types of monomers; adipic acid (dicarboxylic acid) and hexamethylenediamine (diamine). Hence it is an example of copolymer. Other examples are; nylon, dacron, SBR etc.

17.1.5 Natural and Synthetic Polymers

On the basis of source or origin, polymers can be classified as natural and synthetic polymers.

1. **Natural polymers:** The polymers formed in the body of plants and animals are natural polymers. They are
 - i. **Polysaccharides:** Starch and cellulose are common polysaccharides. They are polymers of **glucose** (monomer). Starch is the chief food reserve in plants and cellulose is chief structural material of plants.
 - ii. **Proteins:** They are the polymers of **α-amino acids**. Protein is chief material of animal body. Wool, natural silk, leather, etc. are proteins.
 - iii. **Nucleic acids:** They are the polymers of **nucleotides**. Nucleic acids include DNA and RNA. DNA is the polymer of deoxyribonucleotide and RNA is the polymer of ribonucleotide.
 - iv. **Natural rubbers:** They are the polymers of **isoprene** (2-Methyl-1,3-butadiene). It is obtained from rubber tree.



Raw rubber is weak (tensile strength is 200 Kg/cm²), non resistant to vegetable and mineral oils, and other organic solvents, soft at high temperature and brittle at low temperature so can be used in the range of only 10-60°C, has high water absorption capacity. To improve the quality of rubber, raw rubber is heated with sulphur. The process is called **vulcanization** of rubber.



Fig.17.7: Rubber being collected from rubber tree ig. 17.8: Cotton (cellulose) is obtained from cotton plant

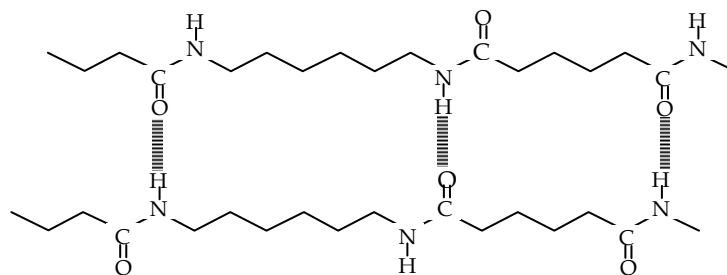
17.1.6 Fibers, Elastomers, Thermoplastics and Thermosetting Polymers

The utilities of polymers depend upon their mechanical properties like tensile strength, elasticity, toughness etc which is governed by intermolecular force of attraction. According to intermolecular force of attraction, polymers can be divided as;

1. Fibers
2. Elastomers
3. Thermoplastics
4. Thermosetting polymers

1. Fibers

Fibers are long thread like material with high tensile (pulling) strength. The molecules of fiber are long, thin and **linear** with polar groups which permits side by side alignment and **strong intermolecular force** to maintain this alignment and prevent slipping of molecules. Hence fibers have high tensile strength. Fibers are twisted into threads which can be woven into cloth or embedded in plastics material to impart strength. For example, nylon-6,6,



Intermolecular hydrogen bonding in nylon -6,6

2. Elastomers

An elastomer has a high degree of elasticity i.e. can be deformed (for example: can be stretched up to eight times) but retains its original shape. Molecules of elastomer are long but remain in **random nonlinear forms** due to unsaturation and have no polar groups. Hence fibers **lack of strong intermolecular force** to hold them linear except occasional cross links and can be stretched upto certain limit.

3. Thermoplastics

Polymers which becomes soft on heating and hard on cooling reversibly are called thermoplastics. Molecules in thermoplastics are long and linear which are held together by weak Van der Waal force of attraction. There are no covalent cross links among molecules. Hence intermolecular force of attraction can be overcome by heating easily. Therefore, thermoplastics can be moulded, remoulded and reused. Example: polyethylene, PVC, polystyrene, etc.

4. Thermosetting polymers

Polymers which once hardened on heating can not be soften are called thermosetting polymers. Molecules in thermoplastics are cross linked by covalent bonds forming three dimensional network. Hence intermolecular force of attraction can not be overcome by heating. Therefore, thermosetting polymers can not be remoulded and reused. Example: bakelite. (See phenols for details.)

Differences between thermoplastics and thermosetting polymers

Thermoplastics	Thermosetting polymers
1. They becomes soft on heating and hard on cooling reversibly. Hence they can be reshaped, reused and reclaimed from wastes.	1. Once hardened, they can not be soften on heating. Hence they can not be reshaped, reused and reclaimed from wastes.
2. They consists of long, linear molecules held together by weak Van der Waal force of attraction.	2. Their molecules are cross linked by covalent bonds forming three dimensional networks.
3. They are formed by addition polymerization.	3. They are formed by condensation polymerization.
4. They are generally soft, fusible, weak and less brittle.	4. They are generally hard, infusible, strong and brittle.

Some polymers in biomedical uses

Polymers	Applications
Polyurethane	Artificial heart, heart valves etc
Polydimethyl siloxane(silicone rubber)	Heart valves
Polypropylene	Heart valves, disposable syringes
Polymethyl methacrylate	Contact lenses
Polyethylene, PVC	disposable syringes

Self Test

17.1 Define the following terms with suitable examples.

- a. Monomer b. polymer c. polymerization

17.2 How do condensation polymers differ from addition polymers?

17.3 What is copolymerization? Give an example.

17.4 Write main differences between thermoplastics and thermosetting polymers.

17.5 Write the structure of monomer and polymer of

- a. polystyrene b. teflon c. polypropylene
d. bakelite e. nylon 66 d. dacron

17.6 How do fibres differ from elastomers?

17.2 Dyes



Fig. 17.8: Dyes

17.2.1 Introduction

Dyes are chemical substances, generally organic compounds, used in imparting color to textiles, leather, wool, plastics, foodstuffs etc.

All coloured substances are not necessarily dyes. For a substance to act as a dye, certain conditions must be fulfilled;

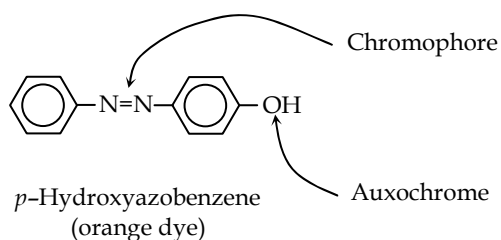
- It must have a suitable color,
- It must be able to fix itself or be capable of being fixed to the fabric,
- The fixed dye must not have fastness to light, water, dilute acids, dilute alkalis, various organic solvents used in dry cleaning etc.

17.2.2 Structural features

A dye possesses two main constituents;

- Chromophores:** These are certain unsaturated groups responsible for the colour of dye (Greek: *Chroma* = colour, *phoros* = bearer). For example, $(>C=C<)_n$, $>C=O$ (carbonyl group), $-N=N-$ (azo group), $-NO_2$ (nitro group), p -quinoid group etc.
- Auxochromes:** These are some functional groups which do not produce colour themselves but when present along with chromophore increase the intensity of colour of dye. (Greek: *auxo* = increase, *chroma* = colour). e.g. $-OH$, $-SO_3H$, $-COOH$, $-NH_2$, $-NHR$, $-NR_2$ etc. Polar auxochromes make the dye water soluble and bind the dye to the fabric.

Example:



17.2.3 Classification of Dyes

Dyes can be classified by different ways. Some of them are as follows;

1. Classification based on the source
2. Classification based on the chemical constitution
3. Classification based on the mode of application

17.2.4 Classification Based on the Source

On the basis of source, dyes can be classified into natural dyes and synthetic dyes.

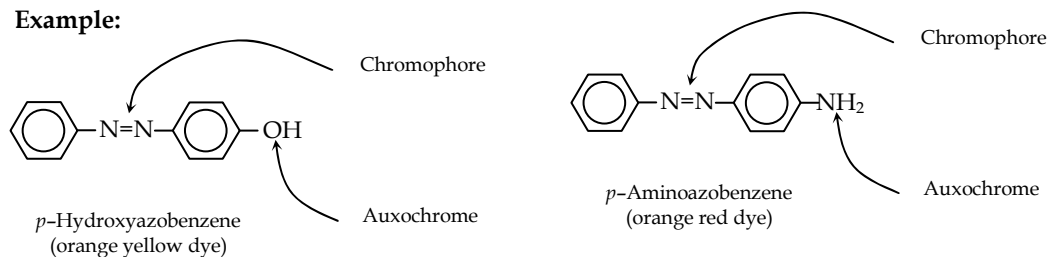
1. **Natural dyes:** Dyes obtained from plants and animals are called natural dyes. In ancient times, people used various plant and animal extracts for coloring household materials, clothes etc. For example, indigo (indigotin) is obtained from plants of the *indigofera* group. It is a dark blue solid with coppered lustre. It is the oldest known dye. Mummy clothes of 5000 years old have been found dyed with indigo. The name of the dye is from Greek word *indikon*, meaning substance from India. Evidences show that, indigo was used in India since ancient times. Even today, people in different regions use plant extracts to get colour. Today, however, all dyes are obtained by synthesis.
2. **Synthetic dyes:** Dyes prepared in laboratories are called synthetic dyes.

17.2.5 Classification Based on the Chemical Constitution

On the basis of chemical constitution, dyes are classified into different classes;

Azo dyes: Dyes having *azo group* ($-N=N-$) joined to aromatic system as chromophore are called azo dyes.

Example:



Azo dyes are the largest class of synthetic dyes. They are prepared by coupling of diazonium salt with phenols or aromatic amines. Some other types are tabulated as follows;

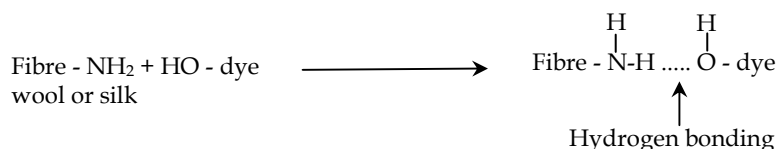
Types of Dye	Structural unit	Examples	Uses
1. Phthalein dyes		 Phenolphthalein	Indicator in acid and base titration

2. Nitro dye		 Martius yellow Picric acid	dye used in wool
3. Indigo dye		 Indigo	for making printing inks
4. Anthraquinone dye		 Alizarin	used for dyeing cotton
5. Triaryl methane dye		 Malachite green	

17.2.5 Classification Based on the Mode of Application

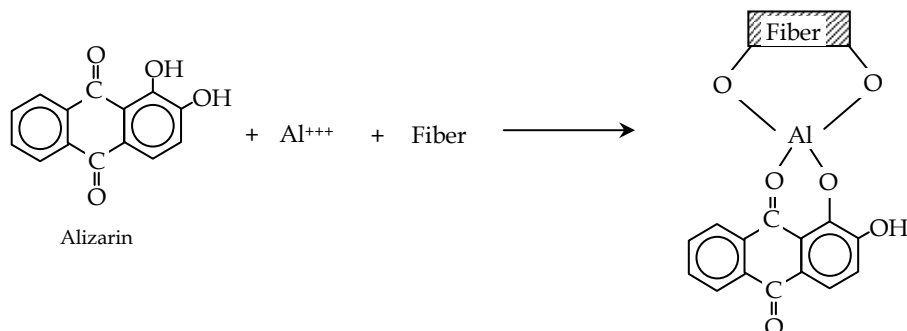
A dye molecule may be bounded to the fabric by covalent bond, hydrogen bond, ionic bond as well as Van der Waal force. The mode of application of a dye depends upon the nature of the dye as well as the nature of the fiber. On the basis of mode of application, dyes are classified into different classes;

- Vat dyes:** These dyes are insoluble in water. It is reduced into colourless soluble form (*leuco* form) when agitated with alkaline reagents like sodium hyposulphite then applied to the fabric and oxidized in air. When exposed in air, insoluble dye is regenerated in the fabric. For example, Indigo. Such dyeing was done in wooden vats hence the name vat dyes.
- Direct dyes:** These dyes are directly applied to the fabric in hot aqueous solution. The acidic or basic auxochrome of the dye combine directly with the fibre. These dyes are suited to the fabric which can form hydrogen bonds and used for silk, nylon, rayon etc. For example, Martius yellow.



- Mordant dyes:** These dyes do not dye the fabric directly and fixed to the fabric with the help of basic salts of metals like Al, Cr, Fe, Ba etc which are called mordants. The mordant forms an insoluble coordination complex between the fabric and the dye. If the dye is basic, acidic

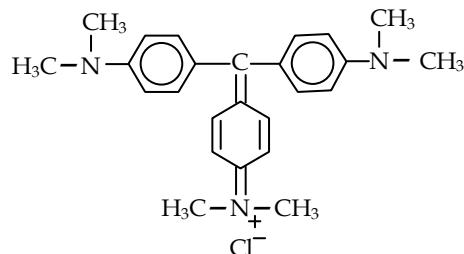
mordant like tannic acid is used and vice versa. Mordant dyes are used for dyeing wool. The same dye may give different colour depending upon the mordant used. For example, Alizarin gives red with aluminum, violet black with iron, brown violet with chromium, blue with barium.



4. **Disperse dyes:** Disperse dyes are insoluble in water and dispersed in water in a colloidal form. When fabric is immersed in the colloidal form of dye, fine particles are absorbed in the crystal structure of the fabrics. These dyes are used for dyeing synthetic fibres like nylons, polyesters and hence these dyes are also called *synthetic fiber dyes*.
5. **Azoic dyes:** (ingrain dyes): These are water insoluble azo-dyes which are formed on the fibre. The fabric is immersed in the solution of the coupling reagents like phenol, aniline, naphthol, etc. and then immersed in appropriate diazonium salts.

Dyes in medical field

Dyes have values in medical field too. Gram staining method (developed by Hans Christian Gram) uses the dye, crystal violet, to classify pathogenic bacteria into Gram positive and Gram negative. Bacteria which turn purple on staining with the dye are called Gram positive and bacteria which turn pink on staining with the dye are called Gram negative bacteria.



Crystal Violet

Certain dyes stain and paralyse micro organisms hence can be used as chemotherapeutic agent. For example, Prontosil is antibacterial. Crystal violet has antibacterial, antifungal, and anthelmintic properties.

Self Test

17.7 What are dyes? Give examples indicating chromophore and auxochrome.

17.8 Define the followings with suitable example.

- a. Azo dyes b. Vat dyes c. Mordant dyes d. Synthetic fibre dyes

17.3 Drugs

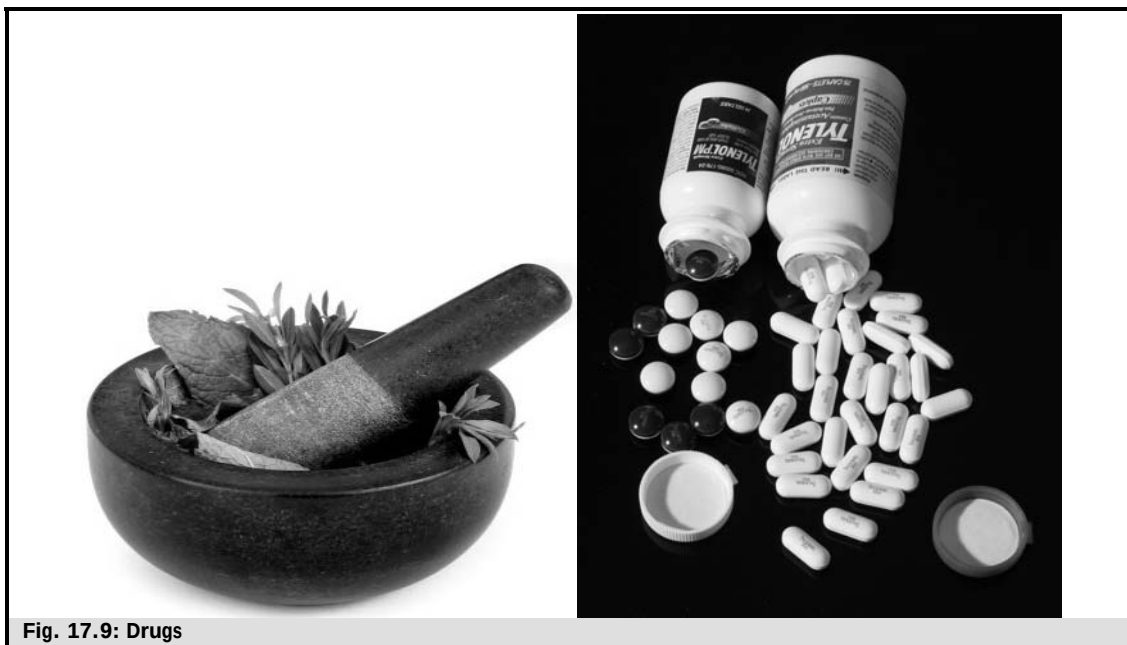


Fig. 17.9: Drugs

17.3.1 Introduction

The word 'drug' is derived from the French word '**drogue**' which means a **dry herb**. In ancient times, people used many herbs for the treatment of different diseases. In Nepal, a collection of many medicinal plants along with their diagram and medicinal value can be found in a book called '**Chandra Nigantu**'.

Drug is a substance used in prevention, diagnosis and treatment of diseases in human beings. According to World Health Organization (WHO), '*a drug is defined as any product (substance) that is used to modify or explore physiological systems or pathological states for the benefit of recipient*'.

The process of treating diseases by using chemicals (drugs) is called chemotherapy. **Paul Ehrlich**, in 1904, indicated that chemical can interfere with the proliferation of invading microorganisms at a concentration that is tolerated by the host. This is the principle of chemotherapy. Hence Paul Ehrlich is called as **Father of Chemotherapy**.

Characteristics of Drug

An ideal drug must fulfill the following characteristics.

1. The action of drug should be localized at the site where it is desired to act.
2. It should be efficient.
3. It should have minimum side effects.
4. It should not injure host tissues.
5. It should not make the host cell resistant to the drug after its use for a long time.

17.3.2 Classification of Drugs

On the basis of the therapeutic action, drugs are classified as chemotherapeutic agents and pharmacodynamic agents.

Chemotherapeutic agents

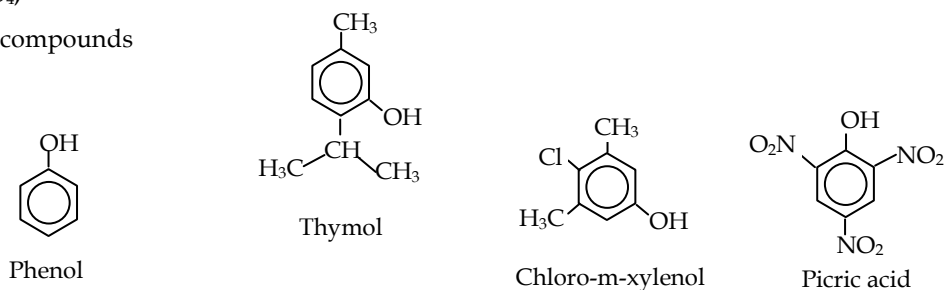
These drugs are used to treat infectious diseases. These are made in such a way that they kill the invading organism selectively without any harmful effects on host tissues. For example, antibacterial, antiviral, antifungal, antiprotozoal, antimalarial etc.

1. **Antiseptics:** These are drugs used to inhibit the growth of microorganism either by inhibiting their reproduction or causing their death and safe to host tissues.

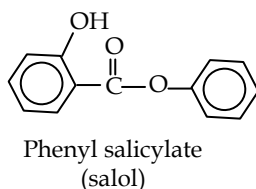
Examples:

(i) 1% KMnO_4 ,

(ii) Phenolic compounds



(iii) Benzoic acid derivatives

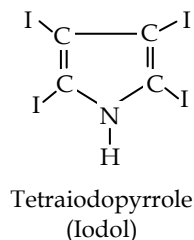


(iv) Iodine and its compounds:

Tincture of iodine ($\text{I}_2 + \text{KI}$ dissolved in ethanol),

Iodoform (CHI_3)

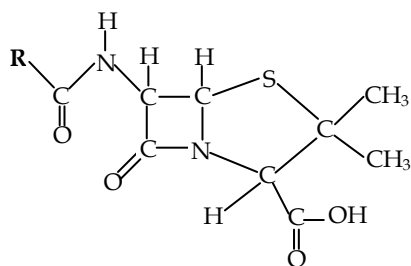
Iodol etc.



Alcoholic solution of chloro-m-xyleneol and terpene is commercially available as dettol.

Disinfectants are similar to antiseptics but harmful to host cells. So, they are applied only on inanimate objects such as floor, instruments, swimming pool etc. For example, chlorine used for the sterilization of water, sulphurdioxide used in bakery.

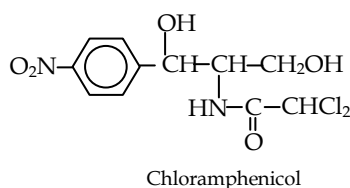
2. **Antibiotics:** Antibiotics are substances originally produced by various species of microorganisms (and later synthesized in laboratory) which inhibit the growth of other microorganisms and ultimately destroy them even given in small concentrations. For example, penicillin, streptomycin, chloramphenicol, tetracyclines, erythromycin etc. Penicillin is the first antibiotic which was discovered by Alexander Fleming in 1928.



Penicillin

Antibiotics can be classified into two classes.

- i. **Broad spectrum antibiotics:** Antibiotics which are effective against a wide range of microorganisms are called broad spectrum antibiotics. For example, chloramphenicol, tetracycline etc.

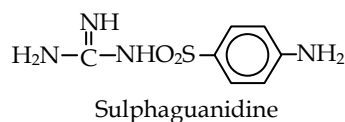


Chloramphenicol

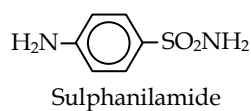
It is originally obtained from bacterium *streptomyces venezuelae*. It was introduced into clinical practice in 1949 under the trade name chloromycetin.

- ii. **Narrow spectrum antibiotics:** Antibiotics which are effective against a few microorganisms are called narrow spectrum antibiotics. For example, Bacitracin, Penicillin, Streptomycin, etc.
3. **Sulpha drugs:** They are first chemotherapeutic agents which are derivatives of sulphanilamide. They are used against bacteria.

Examples:



Sulphaguanidine

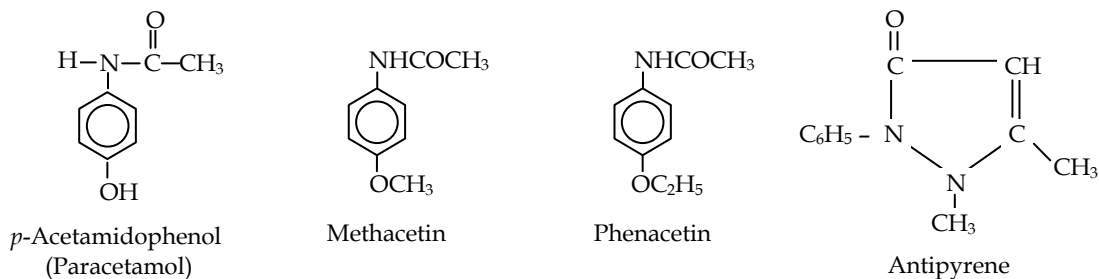


Sulphanilamide

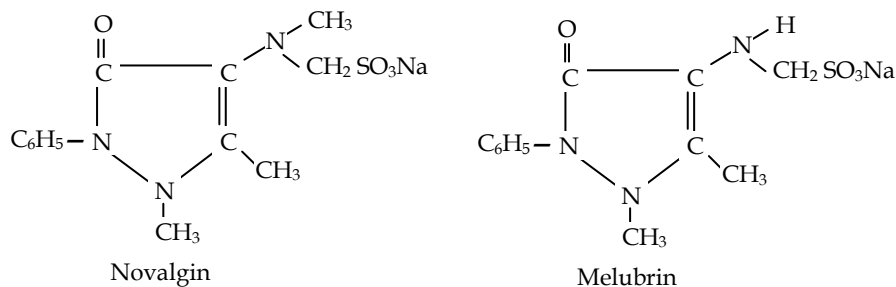
Pharmacodynamic agents

Many disease conditions in human beings appear due to functional disorder or degradation of body but not due to invading microorganisms. For example, anemia, cardiovascular disorders, cancer, depressions, diabetes, erectile dysfunction etc. The chemicals used to reduce such disease conditions are called pharmacodynamic agents.

1. **Antipyretics:** The drugs which are used to lower body temperature to normal in high fever are called as antipyretics.

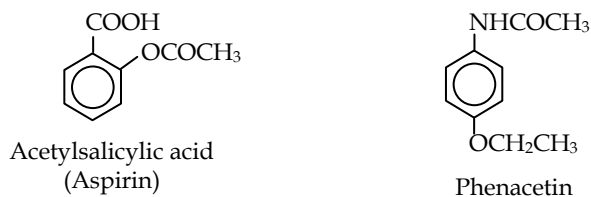
Examples:

2. **Analgesics:** The drugs which are used to relieve pain without loss of consciousness are called analgesics. They are commonly called as pain killers.

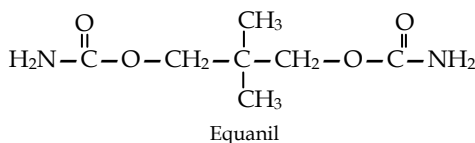
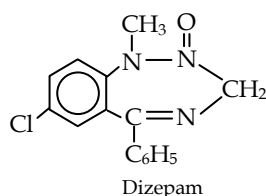
Examples:

Analgesics may be further classified as

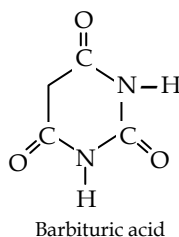
- i. **narcotic** which are habit forming. *For example, morphine, heroin etc*
 - ii. **non narcotic** which are not habit forming. *For example, aspirin, novalgin etc.*
3. **Analgesics and antipyretics:** The drugs which are used to lower body temperature as well as relief pain are called analgesics and antipyretics.

Examples:

4. **Psychotropic drugs:** The chemical substance which are used for the treatment of mental disorder like psychosis, neurosis, schizophrenia are called psychotropic drugs. These act on the central nervous system to reduce anxiety. People used opium, alcohol to be relief from mental anxiety in ancient times. They are of different types;
- i. **Sedatives:** Sedatives are central nervous system depressants that reduce nervous tension and promote relaxation without inducing sleep. *For example, dizepam, equinil etc.*



- ii. **Hypnotics:** Hypnotics are central nervous system depressants that induce sleep. For example, Barbituric acid.



- iii. **Antidepressants:** These drugs provide a feeling of well being in depressed mood. For example, cocaine.

5. **Anaesthetics:** The drugs which bring about the loss of sensation for a certain period of time are called anaesthetics. Use of anaesthetic has made surgery a more comfortable experience for patients, and allowed surgeons to take more time in which they can work carefully and more effectively. Hence use of anaesthetic has opened new advances in surgical technology and medical history.

Generally, there are following types of anaesthetics.

- i. **General or central anaesthetics:** They cause unconsciousness all over the body. For example, Diethyl ether, Ethylene, cyclopropane, Methyl n-propyl ether, Nitrous oxide (N_2O), chloroform, etc.
- ii. **Local anaesthetics:** The anaesthetic which causes unconsciousness to a limited region is called local anaesthetics. Examples: benzocaine, procaine, ethyl chloride, etc.

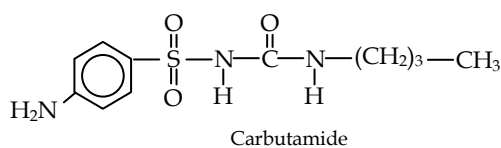
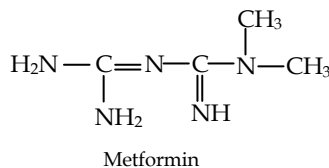
6. **Antacid:** The chemical substances which neutralize excess acid in gastric juice are called antacids. For example, NaHCO_3 , $\text{Al}(\text{OH})_3$, $\text{Mg}(\text{OH})_2$, etc. A commercial antacid called 'eno' contains a mixture of sodium bicarbonate, citric acid and anhydrous sodium carbonate.



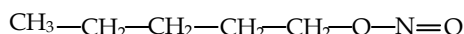
7. **Anti diabetic drugs:** Diabetes is a disease caused by defects in the metabolism of carbohydrates, proteins and fats. The disease brings many complications like degeneration of blood cells, blindness, renal and neurological disorders etc.

The cause of diabetes is not microorganisms but deficiency of insulin due to disorder in body function. β -cells of pancreas produce a hormone called insulin which has a key role in the metabolism of carbohydrates, proteins and fats. In some cases, insulin is not produced at all and causes type I diabetes commonly in juveniles. In other cases, insulin is secreted insufficiently and causes type II diabetes in adults.

Administration of insulin is only the therapy for type I diabetes. For type II diabetes, many drugs have been developed to lower the sugar level. For example, pheformin, metformin, carbutamide etc.



8. **Drugs for cardiovascular diseases:** Disease of heart and blood vessels are called cardiovascular diseases. Hypertension, high blood pressure caused by narrowing down of blood vessels etc. For example, Amylnitrite, glyceroltrinitrate, etc.



Amyl nitrite

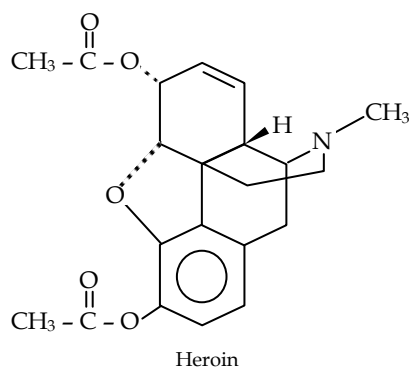
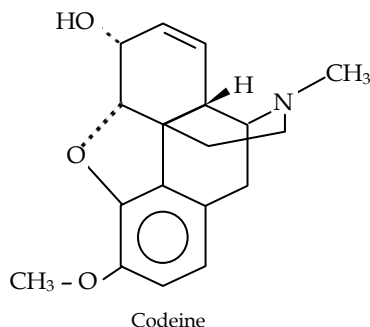
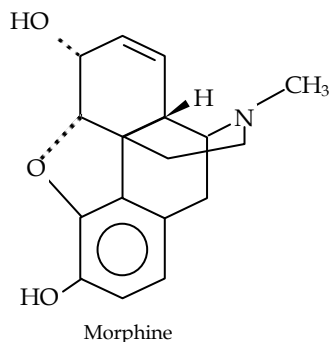
9. **Antihistamines:** Histamine is compound which is formed in the body due to biodegradation of amino acid, histidine. It is responsible for causing many physiological conditions like itching, hyper secretion of stomach acids etc. and allergy. Drugs used against histamine are called antihistamine. For example, avil, allegra etc.

17.3.3 Drugs Addiction

Drug addiction is a compulsive desire and dependence on an illegal drug or a medication despite the negative consequences. Once addiction develops, gene expression alters, alters the way the brain looks and functions which in turn affect human behavior. It includes wild mood swings, depression, anxiety, paranoia, violence, decrease in pleasure in everyday life, complication of mental illness, hallucinations, confusion, desire to engage in risky behavior, etc. These brain changes interfere with an individual's ability to make voluntary decisions, leading to compulsive drug craving, seeking and use. It creates many physical, physiological and psychological effects leading to serious problems in individual's life, in family and society. Many narcotic drugs like morphine, cocaine, heroin, etc which were once used as analgesics are responsible for drug addiction.

Drugs addiction can be treated

- By applying withdrawal of drug under proper medical supervision with the help of counselor in rehabilitation centre.
- By substituting the other less effective type of drug instead of using drug, like methadone instead of heroin



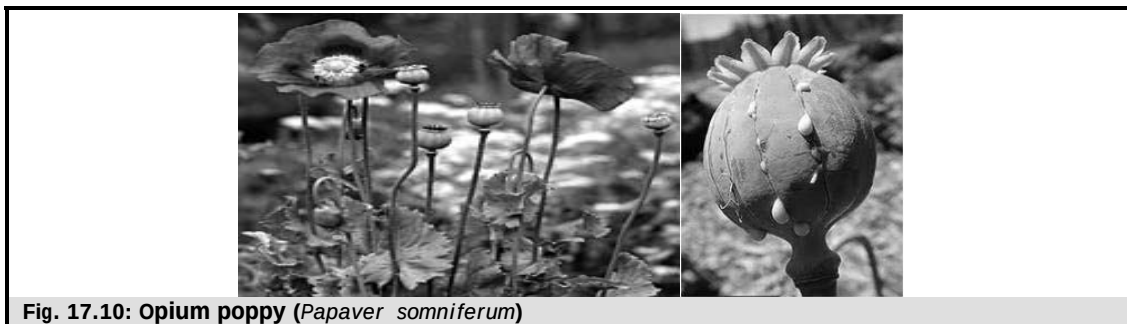


Fig. 17.10: Opium poppy (*Papaver somniferum*)

Morphine is the most abundantly found in opium, the dried latex from unripe seedpods of *Papaver somniferum* (the opium poppy). It is frequently processed chemically to produce **heroin** which is its diacetate derivative. The latex also includes the alkaloid **codeine** which is its methyl derivative. Morphine and its derivatives, such as heroin and codeine, remain important pain relievers.

Besides their analgesic properties, morphine and other potential analgesic alter mood and produce sensation of euphoria that can cause then to be habituating. Prolonged use of opium alkaloids leads to physiological and psychological need for the drug that result addiction.

Self Test

17.9 What are chemotherapeutic drugs ? How do they differ from pharmacodynamic drugs?

17.10 How do antiseptics differ from disinfectants?

17.11 Define with two examples of each:

- | | | |
|-----------------|----------------|------------------|
| a. antiseptics | b. analgesics | c. disinfectants |
| d. antipyretics | e. antibiotics | |

17.4 Chemical Fertilizers



Fig. 17.11: Chemical Fertilizers

17.4.1 Introduction

To fulfill the needs of society in terms of productivity from the land, people have been using fertilizers since long. Materials added to soil to provide necessary nutrients for better growth of plants or for increasing productivity are called **fertilizers**.

Plants require the following elements for the soil for their proper growth and better productivity.

1. **Primary nutrients:** Nitrogen, phosphorus and potash are primary nutrient.

Nitrogen: Plants require nitrogen for growth, protein formation and preparation of colouring materials.

Phosphorus: Plants require phosphorus for the stimulation of root growth, lipid formation and maturity of crops.

Potash: They require potash for healthy root system, ripening a fruits, formation of chlorophyll and to fight against the diseases.

2. **Secondary nutrients:** The elements like Mg and Ca are also required to the plants for metabolism.
3. **Micro nutrients:** Elements like Zn, Cu, Mo, Fe, Co, Cl, etc. are required in a very small quantity.

Criteria for an ideal fertilizer: Though perfectly ideal fertilizer is impossible the basic criteria for an ideal fertilizer are:

- i. It should not undergo chemical change with time.
- ii. It should be cheap.
- iii. It must be easily absorbable in soil or must be soluble in water.
- iv. It should not injure the host cells of the plant.

17.4.2 Classification of Fertilizers

Fertilizers are mainly classified in two categories based upon the source.

1. **Natural/ organic fertilizers:** The fertilizers which are directly obtained by decayed material of plants and animals are called natural/organic fertilizers. Examples: Cow dung, animal bone powders, etc.
2. **Artificial/ chemical fertilizers:** Chemical substances obtained directly for the earth or prepared artificially are **chemical fertilizers**. e.g. NPK fertilizers, mixed fertilizers.

17.4.3 Nitrogen Fertilizers

The fertilizer value of nitrogen fertilizer is indirectly measured by "available nitrogen" in the fertilizers.

Examples:

1. Basic calcium nitrate or Norwegian saltpetre, $\text{Ca}(\text{NO}_3)_2$ or $\text{Ca}(\text{NO}_3)_2 \cdot \text{CaO}$
2. Ammonium sulphate, $(\text{NH}_4)_2 \text{SO}_4$. Excessive use of this fertilizer makes the soil acidic.
3. Calcium ammonium nitrate, $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$
4. Urea or carbamide, H_2NCONH_2 . Has high % nitrogen and leaves only CO_2 in soil. So, it is considered to be a good nitrogen fertilizer.
5. Calcium cyanamide, CaCN_2 : It is second (to urea) preferred fertilizer. [Note: nitrolim = $\text{CaCN}_2 + \text{C}$]
6. Sodium nitrate or Chile saltpeter, NaNO_3 .

17.4.4 Phosphorus Fertilizers

The fertilizer value here is measured by available P_2O_5 .

Examples:

1. Super phosphate of lime or calcium super-phosphate,
2. $\text{Ca}(\text{H}_2\text{PO}_4)_2 + 2 (\text{CaSO}_4 \cdot 2 \text{H}_2\text{O})$. It has 16–20% P_2O_5 .
3. Triple super-phosphate, $\text{Ca}(\text{H}_2\text{PO}_4)_2$. It has 42–46% P_2O_5 . It is called triple super-phosphate as % P_2O_5 in this fertilizer is three times that in calcium super-phosphate.
4. Phosphoric slag or Thomas slag, $\text{Ca}_3(\text{PO}_4)_2$. It contains production of more hay.

17.4.5 Potash Fertilizers

Potash fertilizers are important for tobacco, cotton, madow grass, coffee, corn, etc.

The fertilizers value here is measured as available K_2O .

Examples:

1. Inuriate of potash or syluine, KCl
2. Potassium nitrate or nitre or saltpeter, KNO_3 .
3. It contains 50% K_2O .
4. Potassium sulphate, K_2SO_4 .
5. Potassium carbonate, K_2CO_3 .

17.4.6. Mixed (NPK) Fertilizers

They are compound or complex fertilizers or the mixture of nitrogen, phosphorus and potash fertilizers. Their compositions are variable.

17.4.7 Fertilizers as Pollutants

Today, fertilizer has become essential to modern agriculture to feed the growing population. However, it has created concern in health and environment. The biggest issue facing the use of chemical fertilizers is water pollution.

Fertilizers especially nitrates and phosphates are soluble in water hence washed and leached from soil into water bodies like lakes, rivers and oceans making them too rich in nutrients. As a result, aquatic plants like algae population grow rapidly covering the surface of water. This is called **algal bloom**. Visually, water becomes cloudy, typically green, yellow, brown or red. Algal bloom limits the sunlight available to aquatic plants. This restricts photosynthesis and causes reduction of dissolved oxygen level in water. As algae die, they are decomposed by bacteria. This process uses oxygen hence dissolved oxygen level is further decreased. When dissolved oxygen level decline, aquatic flora and fauna especially bottom dwellers die. Nitrogen groundwater contamination also responsible to marine "dead zones" which are hypoxic(low oxygen)areas in large lakes and the oceans due to overabundance of nutrients that deplete the oxygen required to sustain the marine life in the bottom and near bottom water. This response of ecosystem to overabundance of nutrients level is called **eutrophication**. Eutrophication also reduces the value of rivers and lakes for aesthetic enjoyment.

Nitrates are difficult to remove from drinking water supplies since they can not be precipitated. High nitrate levels above 100 ppm in water may cause 'blue baby syndrome' in which oxygen carrying capacity of haemoglobin is decreased in babies leading to death. WHO has recommended maximum concentration of NO_3^- in water is 10 ppm.



Fig. 17.12: A river suffering from eutrophication

Self Test

17.12 What are fertilizers? Give characteristics of fertilizers.

17.13 Give examples of nitrogenous, phosphate and potash fertilizers.

17.14 What is effect of fertilizer in environment?

17.5 Pesticides



Fig. 17.13: Pesticides

17.5.1 Introduction

A pest is any organism, either a plant or animal, harmful to human beings or human concerns especially in agriculture or livestock production. Any substance or mixture of substances used for the purpose of controlling pests is called pesticide.

The most common use of pesticides is in agriculture to protect crop plants and plant products from damaging influences of weeds, plant diseases, insects etc. Hence they are also called crop protection products. Pesticides are also used for non-agricultural purposes such as to control insects which damage structures, cause nuisance, or spread deadly diseases, in houses and recreational areas (lawns, gardens, public parks, playing fields, lakes and ponds for public enjoyment). It is also used to control vegetation that clogs waterways or obstruct highways. Pesticides have become major weapons for combating pests since as early as 1950.

The use of chemical pesticides began largely after the discovery of DDT for the control of agricultural pests and vector born diseases. History of chemical pesticide use in Nepal is not old. In 1950, DDT and pyrethrum was introduced for the first time. In 1955, Paris green /copper(II) acetate triarsenite, gammexane and nicotine sulfate were imported. All these pesticides were provided by USAID through grant for malaria control. Chronologically, the use of pesticides in Nepal is organochlorines- 1950s, organophosphates-1960s, carbamates- 1970s and synthetic pyrethroids- 1980s.

17.5.2 Classification of Pesticides

Pesticides are often referred and classified to according to the type of pest (target organism) they control. There are various types of pesticides.

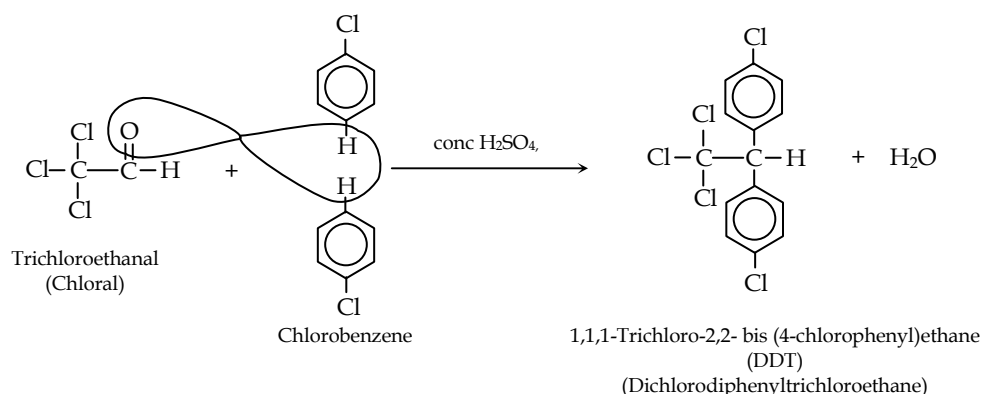
A. Insecticides

The chemical substance or mixture of substances that are used to kill and control insects are called insecticides. Many insects such as grasshoppers destroy cultivated plants. Many insects like bedbugs destroy fruits and food grains during processing and storage. Many insects such as termites can damage wood structures. Many insects cause nuisance in houses and recreational areas and spread

simple to deadly diseases such as mosquitoes are disease vector for malaria, filaria etc. Insecticides increase agricultural productivity by preventing crop losses to insects. Simple to deadly diseases caused by insect vectors are under control or at the stage of eradication due to use of insecticides. Some of commonly used insecticides are;

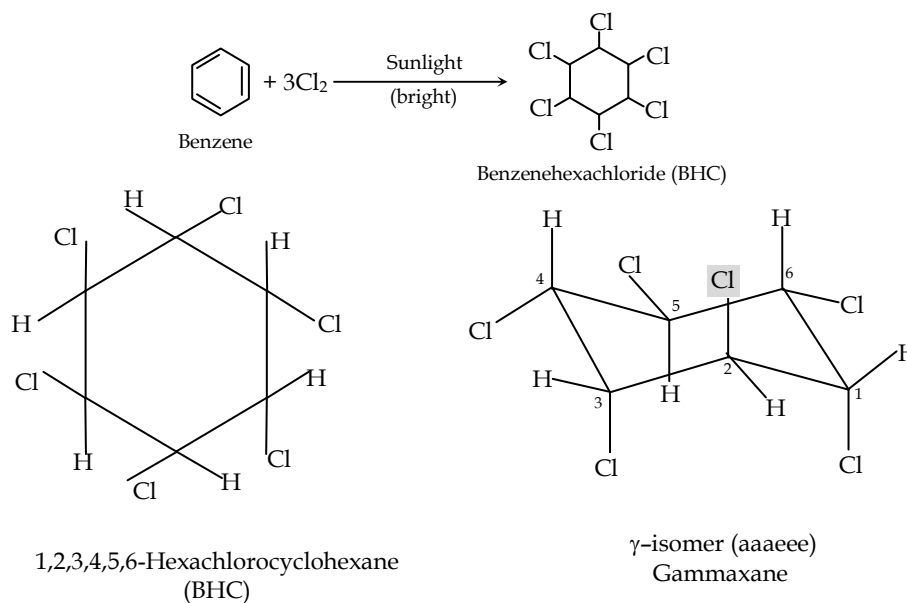
1. Organochlorine insecticides

- i. **DDT:** DDT is a contact poison against several arthropods. DDT is commercially prepared by heating chlorobenzene with chloral in presence of dehydrating agent. It is a colorless, tasteless crystalline compound.

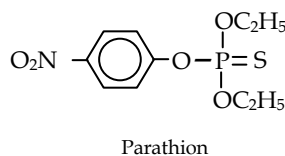
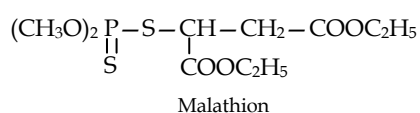


In the second half of World War II, it was used with great success to control malaria and typhus among civilians and troops. After the war, DDT was made available for use as an agricultural insecticide and extensively used in agriculture. DDT is toxic to a wide range of living organisms including human beings. Because of its hazards in health and environment, the use of DDT as agricultural insecticide was outlawed in 2004 under *Stockholm Convention* and restricted to disease vector control.

- ii. **BHC:** Many stereoisomers are possible for the formula C₆H₆Cl₆. out of which γ - isomer is most effective insecticide therefore BHC is also known as *gammexane*. Since it contains six carbon, six hydrogen and six chlorine atoms, it is known as "666". Commercially it is also called "lindane". It was first prepared by Michael Faraday in 1825 but its pesticidal action was discovered only in 1942. It has been used to treat food crops and to forestry products and to treat livestock and pets. It has also been used as pharmaceutical treatment for lice and scabies. It is very stable and acts more quickly than DDT. It is a persistent organic pollutant and has health hazards. Now, its use in agriculture has been banned internationally (the Stockholm Convention on Persistent Organic Pollutants, 2009) except in second-line treatments for the head lice and scabies; meaning they can only be prescribed when other "first-line" treatments have failed or cannot be used. It is prepared as



2. **Organophosphate insecticides:** Parathion and malathion are organic phosphate insecticides.



Organochlorine insecticides are persistent organic compounds. However, organophosphates usually are not persistent in the environment. Use of organochlorine insecticides in agriculture has been restricted except in second line vector control.

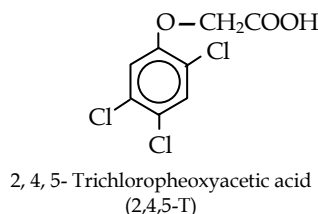
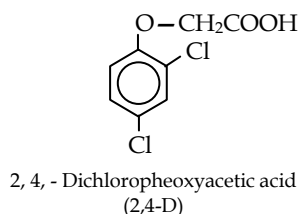
Pyrethroids now constitute the majority of commercial household insecticides against flies, mosquitoes and cockroaches. *Prallethrin*, *Cyfluthrin*, *Transfluthrin* which are active ingredients of Baygon.

Nicotine, pyrethrum and neem extracts are used as natural insecticides. Elemental sulfur powder is used as an insecticide against ticks and mites.

Pheromones are hormones secreted by certain female insects to attract their males. They are sex attractants. They are specific to attract their males. Pheromones can be used to attract male insects (which comprises the major insect population) which are then trapped.

B. Weedicides or herbicides

Weeds are undesired plant species grown in an area (plant pests). The chemicals which are used to kill and control weeds are called weedicides or herbicides. Herbicides are widely used in agriculture to control plant pests.

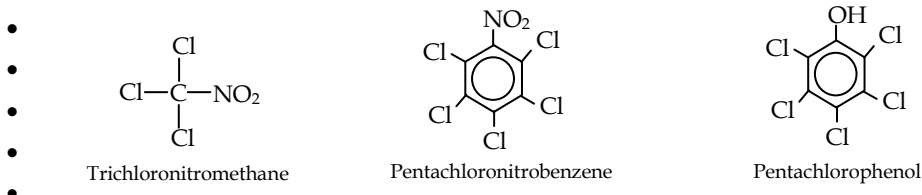
Example:

One of the most commonly used herbicides in the world is 2,4-D. It is a selective herbicide and used to control weed in wheat, maize, rice, and similar cereal grass crops, because it kills dicots (broadleaf plants), but not most monocots (grasses). Herbicides are also used to control vegetation in highways and railways, in landscape turf management, clear waste ground, industrial sites, etc.

C. Fungicides

The chemicals used to kill and control fungi are called fungicides. Fungi can cause serious damage in agriculture, resulting in critical losses of yield as well as quality. Many fungi can damage structures such as furniture, houses. Fungicides are used both in agriculture and to fight fungal infections in animals. Some examples are

1. Copper sulphate solution (CuSO_4)
2. Aluminum phosphide (A/P)
3. Bordeaux mixture (mixture of CuSO_4 and lime)

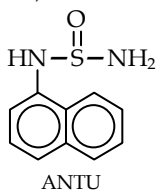


Elemental sulfur is one of the oldest fungicides. "Dusting sulfur," elemental sulfur in powdered form, is a common fungicide for grapes, strawberry, many vegetables and several other crops. In organic production, sulfur is the most important fungicide. Dusting sulfur is applied to crops with a sulfur duster or from a dusting plane.

D. Rodenticides

The chemicals which kill and control rodents are called as rodenticides. Some example of rodenticides are;

1. $\text{ClCH}_2\text{COONa}$ (Sodium monochloroacetate)
2. Zn_3P_2 (Zinc phosphide)
3. Alphanaphthylthiourea (ANTU)



17.5.3 Pesticides, Environmental Pollution and Human Health

Many pesticides are toxic to living organisms. Sprayed pesticides reach to non-target species, air, water and soil. Hence, pesticide use reduces biodiversity, contributes to pollinator decline, destroys habitat and threatens endangered species. Pesticides are one of the causes of water pollution, and some pesticides are persistent organic pollutants and contribute to soil contamination. Since chlorinated hydrocarbon pesticides dissolve in fats and are not excreted, organisms tend to retain them almost indefinitely.

Problems related to the use of synthetic pesticides also arises due to misuse of pesticides which includes use of pesticides even when it is not necessary, improper selection of pesticides, use of highly toxic and broad spectrum pesticides, not following pre harvest waiting period, dipping vegetables in pesticides, use of pesticides in fishing and low public awareness about pesticides and their safety concern.

Environmental and human health problems related to the use of synthetic pesticides have created an increasing pressure against their use. Hence, safer and more specific pesticides are to be developed and technical developments of the application equipment have also improved to enable their proper application. The non-chemical alternatives are also in use which include **methods of cultivation** like *polyculture* (growing multiple types of plants), *crop rotation*, (planting crops in areas where the pests that damage them do not live, *timing planting* according to when pests will be least problematic), and *use of trap crops* that attract pests away from the real crop, choice of resistant varieties, creation of an environment favorable for natural enemies of pests, and use of biological products and agents, including beneficial insects.

Self Test

17.15 What are pesticides? Indicate their role in agriculture.

17.16 Define insecticide. Give the structure and IUPAC name of DDT and BHC.

17.17 Give two examples along with their structure.

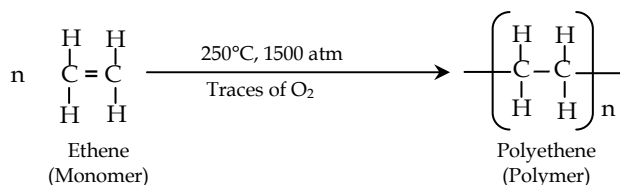
a. fungicide b. herbicide c. rodenticide

■ Solved Questions

1. Define polymer and polymerisation.

- A polymer (Greek: poly = many, meros = parts) is a giant molecule (macromolecule) with high molecular mass which is made up of a number of smaller repeating units. The repeating units are usually obtained from low molecular mass simple molecules which are called monomers. The process of forming a polymer from monomers under the suitable conditions of temperature pressure and or catalyst is called polymerization.

For example,

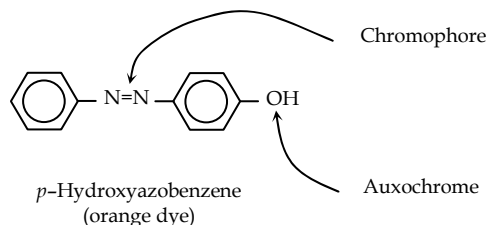


2. What are dyes?

- Dyes are chemical substances, generally organic compounds, used in imparting color to textiles, leather, wool, plastics, foodstuffs etc. A dye must following characteristics;
- It must have a suitable color,

- ii. It must be able to fix itself or be capable of being fixed to the fabric,
- iii. The fixed dye must have fastness to light, water, dilute acids, dilute alkalis, various organic solvents used in dry cleaning etc.

For example,



3. What are antibiotics? Give example of broad spectrum and narrow spectrum antibiotics.

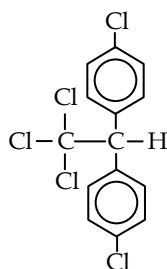
- Antibiotics are substances originally produced by various species of microorganisms (and later synthesized in laboratory) which inhibit the growth of other microorganisms and ultimately destroy them even given in small concentrations (dose). For example, penicillin, streptomycin, chloramphenicol, tetracyclines, erythromycin etc

Antibiotics can be classified into two classes.

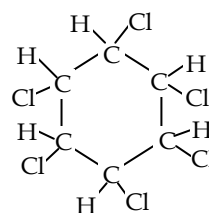
- i. **Broad spectrum antibiotics:** Antibiotics which are effective against a wide range of microorganisms are called broad spectrum antibiotics. For example, chloramphenicol, tetracycline etc.
- ii. **Narrow spectrum antibiotics:** Antibiotics which are effective against a few microorganisms are called narrow spectrum antibiotics. For example, Penicillin, Streptomycin, etc.

4. What are insecticides? Indicate their importance. Give any two example.

- The chemical substance or mixture of substances that are used to kill and control insects are called insecticides. Many insects such as grasshoppers destroy cultivated plants. Many insects like bedbugs destroy fruits and food grains during processing and storage. Many insects such as termites can damage wood structures. Many insects cause nuisance in houses and recreational areas and spread simple to deadly diseases such as mosquitoes are disease vector for malaria, filaria etc. Insecticides increase agricultural productivity by preventing crop losses to insects. Simple to deadly diseases caused by insect vectors are under control or at the stage of eradication due to use of insecticides.



1,1,1-Trichloro-2,2- bis (4-chlorophenyl)ethane
(DDT)



1,2,3,4,5,6- Hexachlorocyclohexane
(BHC)

Questions

Very Short Answer Questions:

1. What is the difference between an antiseptic and an antipyretic drug?
2. Write an example of each of the following with their formula.
(i) A synthetic fertilizer (ii) An insecticide
3. What are natural and synthetic dyes? Give one example of each.
4. Write any two examples of azo-dyes with their formulae.

5. Classify polymers on the basis of monomer unit.
6. What are the starting materials for preparing the followings:
(i) Polyvinyl chloride (ii) Bakelite
7. Name the monomers of (i) bakelite (ii) Nylon-6,6
8. Name any one condensation polymer and addition polymer. What are their monomers?
9. What are chemical fertilizers? Give two examples.
10. Write a method of preparation of each of the following polymers:
a. Bakelite b. PVC
11. What are insecticides? Give an example.
12. What is meant by copolymer? Give an example.
13. Distinguish between:
a. Addition polymers and condensation polymer
b. Thermoplastics and thermosetting plastics.
c. Natural and synthetic polymers
14. What do you mean by fungicides. Write an example.
15. Define the following terms writing one example each of:
a. Herbicides b. Insecticides
c. Analgesic drug d. Phosphatic fertilizers
e. Germicides f. Phthalic dye
g. Antacids h. Antibiotic drugs
i. Elastomers j. Thermoplastics
k. Potash fertilizers

Multiple Choice Questions

1. Which of the followings is a natural polymer?
a. Polystyrene b. Nylon-66
c. Neoprene d. Protein
2. Polymers have
a. Absolute melting point
b. Average molecular weight
c. Low molecular weight
d. Absolute molecular weight
3. $F_2C=CF_2$ is monomer of
a. Buna-S b. Orlon
c. Nylon-66 d. Teflon
4. Which of the following is an addition polymer?
a. Polypropylene b. Nylon-66
c. Terylene d. Polyamide
5. Polymer formed by the addition of two or more different monomers is
a. Addition polymer b. Copolymer
c. Homo polymer d. Chain polymer
6. Polymer used in making heart valve is
a. Silicone b. PVC
c. Teflon d. Polystyrene
7. Polymer used in making tyres of fighter aircrafts is
a. Silicone b. Natural rubber
c. Isoprene d. Polystyrene
8. The natural rubber is the polymer of
a. Cis-isoprene b. Trans-isoprene
c. Chloroprene d. Styrene
9. The synthetic polymer which resembles the natural rubber is
a. Neoprene b. Chloroprene
c. Galyptal d. Isoprene
10. Nonstick coating of cooking utensils is
a. Polypropylene b. PTFE
c. Bakelite d. Polystyrene
11. Terylene is the condensation polymer of ethylene glycol and
a. Benzoic acid b. Phthalic acid
c. Terephthalic acid d. Salicylic acid
12. The hexamethylene diamine and adipic acid are the monomer of
a. Dacron b. Nylon-66
c. Rayon d. Teflon
13. Cellulose is the polymer of
a. Maltose b. α -glucose
c. Fructose d. β -glucose
14. Bakelite is the polymer of
a. Methanal and alcohol b. Aniline and urea
c. Phenol and methanal d. Phenol and chloroform
15. Which of the following is thermosetting polymer?
a. Polyethylene b. Neoprene
c. PVC d. Bakelite
16. The strongest intermolecular forces are present in
a. Elastomers b. Fibers
c. Thermoplastics d. Natural rubber

17. Of the following which one is the step growth polymer
- a. Bakelite b. Polyethylene
c. Teflon d. PVC
18. Which of the following is used in contact lense?
- a. Polypropylene b. Polymethylmethacrylate
c. PVC d. Chloroprene
19. Which is the correct statement about dye?
- a. Every colored substance can act as dye
b. Presence of chromophore is necessary
c. Presence of chromophore and auxochrome are necessary
d. Presence of auxochromophore is necessary
20. Which of the following is an example of chromophore?
- a. $-C=C-$ b. $-NO_2$
c. Methyl isocyanide d. N_2
21. Bismarck is an example of
- a. Phthalein dye b. Azo dye
c. Anthraquinone dye d. Nitro dye
22. Moth balls are
- a. Napthalene b. Anthracene
c. Benzoic acid d. Benzophenone
23. Bordeaux mixture contains
- a. Copper sulphate b. Zinc sulphate
c. Zinc phosphide d. DDT
24. Oil of wintergreen is
- a. Methylacetate b. Methyl salicylate
c. Ethyl benzoate d. Sodium salicylate

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12
d	b	d	a	b	a	a	a	a	b	c	
13	14	15	16	17	18	19	20	21	22	23	24
b	c	d	b	a	b	c	b	c	a	a	b

Heavy Metals

18



The photo is of suspension bridge (झोलुङ्गे पुल) which connects Kushma to Gyandi, i.e. two cliffs separated by Modi river situated in Parbat, Nepal. It is at the height of 135 meters from the base and is considered as suspension bridge highest height. It is one of the longest bridges with length of 344 meters. It reduces a tough 1 hour walk to about 5 minutes. The bridge has just been completed and in use now. Many people cling to the safety wall of bridge due to the height and length in fear.

18.1 Introduction to Transition Metals

Elements of group 3 to 12 (IIB, IIIB, IVB, VB, VIB, VIIB, VIII, IB, and IIB) of periodic table are called d-block elements as the differentiating electron enters into d orbitals in their electronic configurations. Here, d-orbitals are incompletely filled. These elements are also called transition elements. On simply saying, transition elements are called so, because their properties lie in between s-block and p-block elements. In other words, while moving from IIA to IIIA group in the long form of periodic table, we have to transit through these elements. For example, atomic number of calcium is 20. Its electronic configuration is $[\text{Ar}]4s^2$. Therefore, it is a s-block element. Scandium ($Z = 21$), has an electronic configuration of $[\text{Ar}]4s^23d^1$. Now this is the first member of 3d series in period 4. It should be placed in group 3 (IIIB). The ten d-block elements of 3d series are scandium ($[\text{Ar}] 4s^23d^1$) to zinc ($[\text{Ar}] 4s^23d^{10}$). Similar pattern is found in 4d, and 5d elements also.

Periodic Table of the Elements

Legend:

- Metal Metal
- Semimetal Semimetal
- Nonmetal Nonmetal
- Alkali metal Alkali metal
- Alkaline earth metal Alkaline earth metal
- Transition metals Transition metals
- Lanthanide Lanthanide
- Actinide Actinide
- Chalcogens element Chalcogens element
- Halogens element Halogens element
- Noble gas Noble gas

Standard State (25 °C; 101 kPa):

- Ne - gas Ne - gas
- Fe - solid Fe - solid
- Hg - liquid Hg - liquid
- Ts - synthetic Ts - synthetic

Lanthanide Series:

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
LANTHANUM	CERUM	PRASEODYMIUM	NEODYMIUM	PROMETHIUM	SAMARIUM	EUROPIUM	GADOLINIUM	TERBIUM	DYSPROSIUM	HOLMIUM	ERBIUM	THULIUM	YTTTERBIUM	LUTETIUM

Actinide Series:

89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
ACTINIUM	THORIUM	PROTACTINIUM	URANIUM	NEPTUNIUM	PLUTONIUM	AMERICIUM	CURIUM	BERKELIUM	CALIFORNIUM	EINSTEINIUM	FERMILIUM	MENDELEVIUM	NOBELIUM	LAWRENCIUM

(1) Pure Appl. Chem., 81, No. 11, 2131-2150 (2009)
Relative atomic masses are expressed with five significant figures. For elements that have no stable nuclides, the value enclosed in brackets indicates the mass number of the longest-lived isotope of the element. However three such elements (Tl, Pa and U) do have a characteristic terrestrial isotopic composition, and for these an atomic weight is tabulated.

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General Characteristics of Transition Metals (d-block Elements)

1. In transition metals, differentiating electron enters in d orbital. Therefore, their general outer electronic configuration is $ns^{0-2} (n-1) d^{1-10}$, where n is the principal quantum number of the valance shell.
2. Their two outermost shells are incompletely filled with electron. They show variable valency as electrons of both valence shell and penultimate shell take part in bonding

Example: Electronic configurations of



3. Their d-orbitals are incompletely filled with electron. Therefore, they allow d-d electronic transitions. Hence, most of their compounds are coloured.

When a transition ion is isolated, $(n-1)$ d orbitals are degenerate. But when the ion is surrounded by anions or ligands, this degeneracy is lost and d-orbitals split-up into e_g and t_{2g} sets. Jumping of electrons in between these sets of d-orbital is associated with absorption and emission of energy of definite wave length, which corresponds to a particular colour in visible range. So, the compound of transition metals are coloured.

4. The ionization potential and electronegativity values of d-block elements lie between that of s-block and p-block elements.
5. Presence of unpaired d-electrons make them hard.
6. These metals or their compounds can act as catalyst in various processes. In some cases, the metals provide the surface for all reaction (eg Ni in hydrogenation) and in some cases they can provide alternate path owing to their variable valency (eg TiCl_4 in Ziegler-Natta catalysis)
7. Since they have unpaired electrons in d-orbitals, they exhibit paramagnetic property.
8. Their atoms are small sized. Therefore, they can form alloys. eg. Brass (Zn + Cu), Zinc-amalgam (Zn-Hg), etc.

18.2 Copper

18.2.1 Introduction

Symbol	Cu	Atomic number	29
Atomic mass	63.57 (amu)	Electronic configuration	$[\text{Ar}] 4s^1 3d^{10}$
Melting point	1083°C	Specific gravity	8.85
Boiling point	2350°C		

Copper has been known since long time. Romans and Greeks extracted this metal from island of cuprus, where it was named "*Cuprum*". The Nepali name of copper is Tama (तामा). Copper, silver and gold are commonly known as coinage metals. This is because, they had been used in past of making coins for currency.



Fig. 18.1: Copper wire

18.2.2 Position of Copper in Periodic Table

Copper lies in group 11 (IB) of the periodic table. The group members are $_{29}\text{Cu}$, $_{47}\text{Ag}$ and $_{79}\text{Au}$. Their general electronic configuration is $ns^1 (n-1)d^{10}$. They all are hard, malleable and ductile metals. In combined state, common state of Cu is Cu_2^{2+} and Cu^{2+} , Ag is Ag^+ and Au is Au^{3+} . Ag^{2+} and Au^+ are not common, as Au^+ is unstable form and Ag^{2+} is rare. They all lie below hydrogen in electrochemical series. Therefore, they don't liberate H_2 from dilute acids. Their reactivity order as reducing agents is, $\text{Cu} > \text{Ag} > \text{Au}$.

18.2.3 Occurrence

Copper exists in nature in native as well as combined state. Copper has been found in native form near superior lake of USA and in Sweden. Copper exists in combined form as its sulphides, oxides and carbonates. Some major ores of copper are given below:

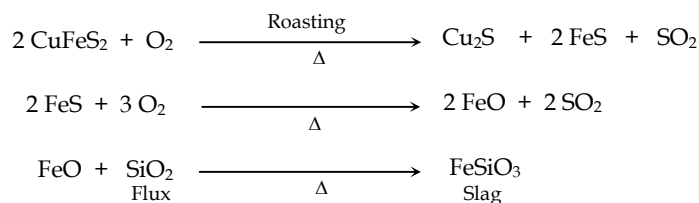
1. Copper pyrites or chalcopyrites	$\text{Cu}_2\text{S} \cdot \text{Fe}_2\text{S}_3$ or CuFeS_2	2. Malachite	$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
3. Copper glance or chalcocite	Cu_2S	4. Cuprite	Cu_2O
5. Azurite	$2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$	6. Bornite	Cu_3FeS_3

18.2.4 Extraction of Copper from Copper Pyrites

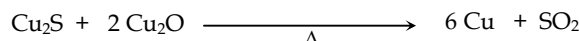
Main ore of copper is copper pyrites. Mostly, copper is extracted from copper pyrite using dry process (pyrometallurgy).

Theory

Copper pyrite is the main ore of copper. Cuprous and ferrous sulphides are obtained after roasting of copper pyrites. FeS thus obtained is oxidized to FeO and then removed as slag by the action of flux.



Cu_2S is oxidized partially to Cu_2O which undergoes reaction by unreacted Cu_2S to give metallic copper.



Steps involved in the metallurgy

- Crushing and pulverization:** Big lumps of ore are crushed using jaw crushers to get crushed ore which are pulverized (powdered) using pulverisers or ball mills to get powdered ore.
- Concentration of ore by froth-floatation process:** Powdered ore is taken into a tank containing water and small quantity of pine oil or eucalyptus oil. The mixture is agitated by using blast of air. Gangue particles are wetted by water and get collected at the bottom of the tank. The ore particles being wetted by oil come on the surface in froth. The froth is skimmed off to collect the concentrated ore.

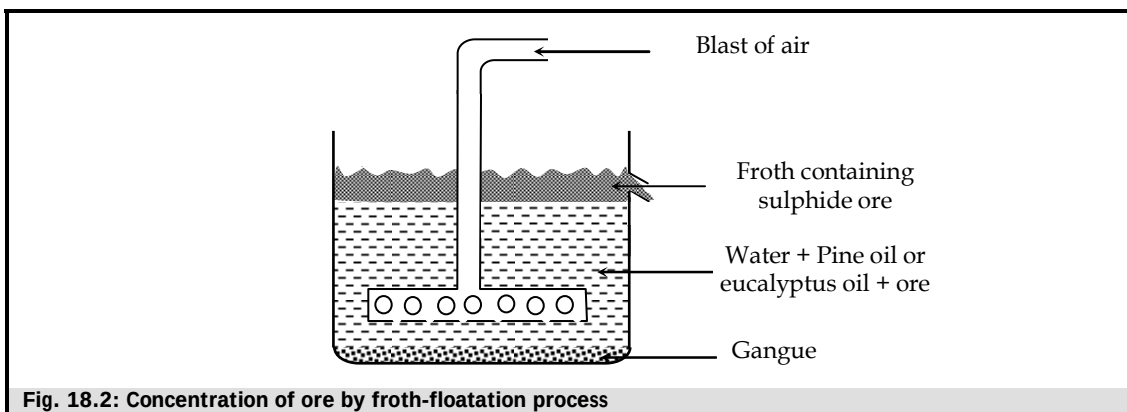


Fig. 18.2: Concentration of ore by froth-floatation process

3. **Roasting:** The concentrated ore is heated in a reverberatory furnace in presence of air, at a temperature insufficient to melt the ore.

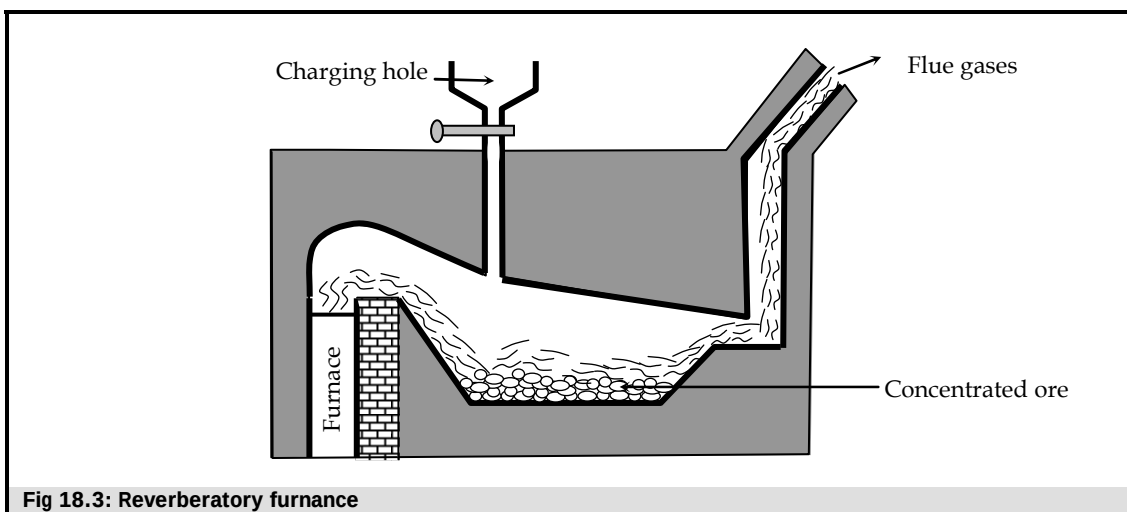
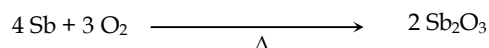
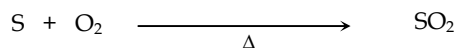
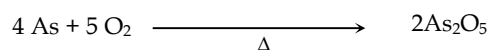
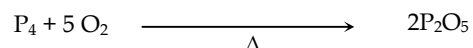


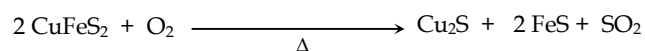
Fig 18.3: Reverberatory furnace

Following change occurs during roasting process.

- i. Volatile impurities and moistures are removed.
- ii. Impurities like P_4 , As, S, etc., are oxidized.

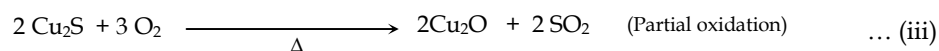
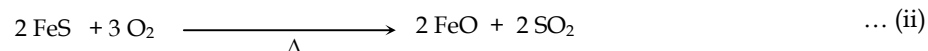


- iii. Iron pyrite is decomposed into cuprous sulphide and ferrous sulphide.



... (i)

iv. There is conversion of FeS into FeO and partial oxidation of Cu_2S into Cu_2O .



4. **Smelting:** Roasted ore is mixed with sand and coke followed by feeding into a blast furnace. The modern blast furnace is made of steel lined inside with refractory bricks. The furnace is of about 15 to 20 feet in height and about 6 feet in diameter. Hot air of about 800°C temperature is introduced through tuyeres. Coke burns with hot air and temperature inside the furnace reaches to about 1000°C . The following changes take place here.

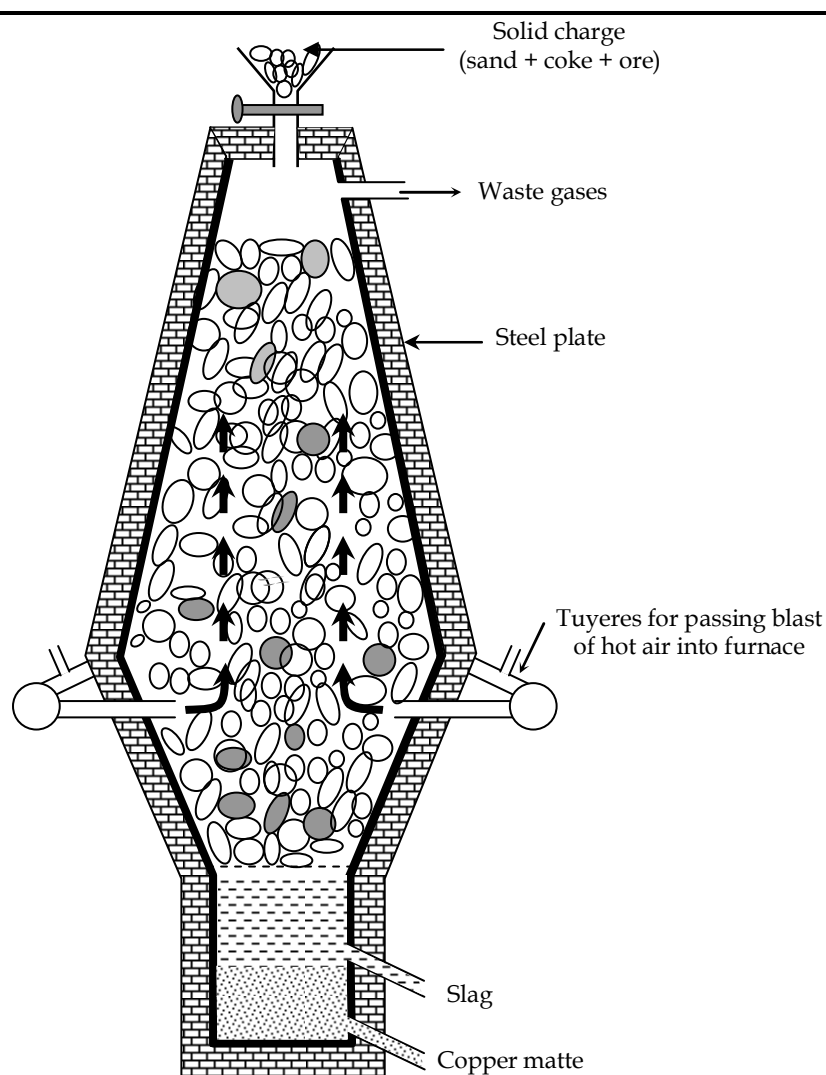
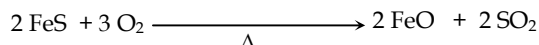
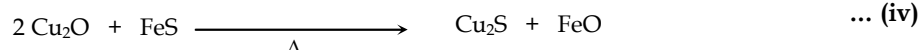
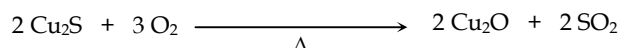


Fig 18.4: Blast furnace for smelting of copper ore

- i. Most of unreacted FeS gets oxidized to FeO.



- ii. Cu_2O formed by the oxidation Cu_2S reacts with unreacted FeS to get Cu_2S again. Hence complete removal of FeS is most important step in the extraction of copper.



- iii. FeO is converted into slag by the action of sand (silica is used as flux).



Slag being light and molten which can be removed from the upper layer.

At the hearth of furnace, molten mass is obtained which is known as "Matte". This mass mainly contains cuprous sulphide and little ferrous sulphide.

Self Test

18.1 Cu_2O is not formed till FeS is present in blast furnace. Given reason.

5. **Bessemerisation:** Metallic copper is recovered from copper matte by bessemerisation process which is carried out in a vessel called Bessemer converter.

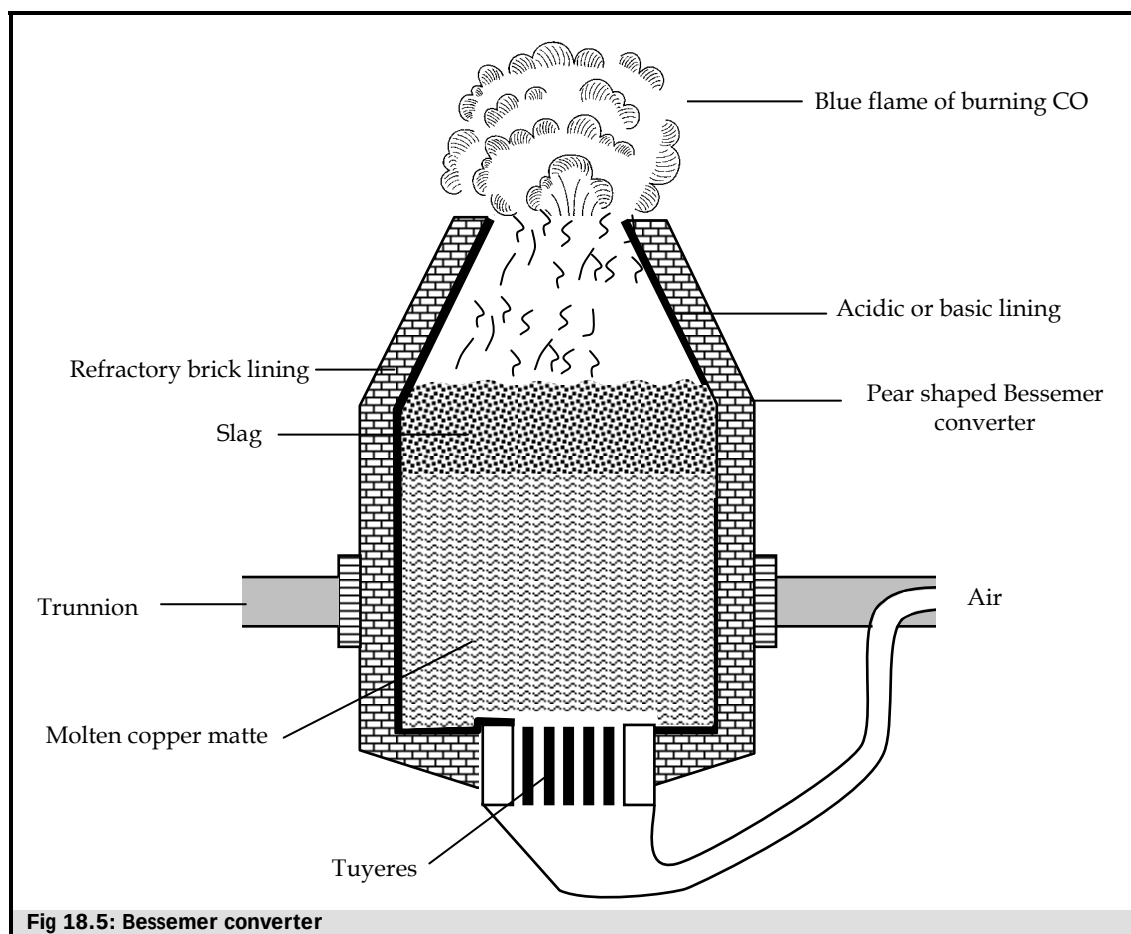


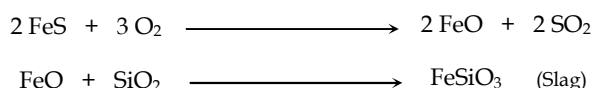
Fig 18.5: Bessemer converter

Bessemer converter is a pear-shaped steel vessel lined with magnetite and quartz. This is special Bessemer converter where tuyeres are present in the sides mounted in such away that the converter can be tilted in the desired direction. Copper matte thus obtained after smelting is transferred to Bessemer converter. Small amount of sand is also added.

Blast of hot air is passed through tuyeres. Air is blown through the molten matte.

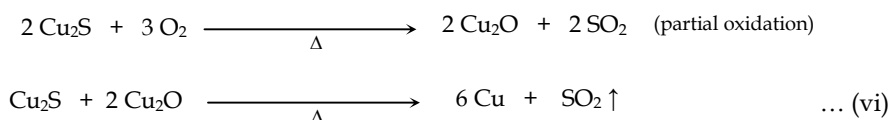
The following changes take place here.

- a. Ferrous sulphide left behind in the above mentioned steps is oxidized to ferrous oxide, which reacts with silica to give light fusible mass, called slag.



Slag being light is drained out from the top of molten mass at regular intervals by tilting the vessel. SO_2 escapes out from the converter as waste gas.

- b. After entire iron is removed as slag, cuprous sulphide gets partially oxidized to cuprous oxide. The Cu_2O thus formed reacts with remaining Cu_2S to give free copper metal.



The reactions from conversion of copper pyrite into cuprous sulphide and ferrous sulphide and self reduction to copper complete in the Bessemer converter (reaction (i) to (vi)).

The metallic copper is collected from the base of Bessemer converter by tilting it.

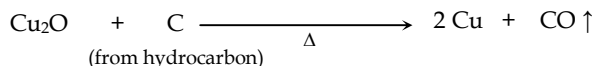
The metal thus obtained is allowed to cool, dissolved SO_2 escapes out forming large blisters on the surface of metal. Therefore, this metal thus formed is called **blister copper**. Blister copper contains 98% pure copper.

Self Test

18.1 What is blister copper?

6. **Purification or Refining:** Blister copper is purified mainly by two method.

- a. **Poling:** Blister copper is heated in a reverberatory furnace lined with silica. Through which molten mass is obtained. This molten mass is stirred with green wood poles. The hydrocarbons present in green wood poles reduce the cuprous oxide (as impurity) into metallic copper. This poling reduces Cu_2O into metallic copper.



Copper thus obtained after poling is about 99.5% pure. This copper is tough pitch copper.

- b. **Electro-refining:** The copper thus obtained may still contains impurities like Ag, Au, Ni, Zn, etc. which can be purified further by electrolytic refining. The impure copper (to be purified) is suspended as anode and thin sheet of pure copper as suspended as cathode in a large tank. Copper sulphate solution (15%) acidified with sulphuric acid (5%) is taken as electrolyte. When D.C. is supplied, copper is dissolved from anode and is deposited as pure copper at cathode. The impurities are left behind near anode as anode mud or anode slime.

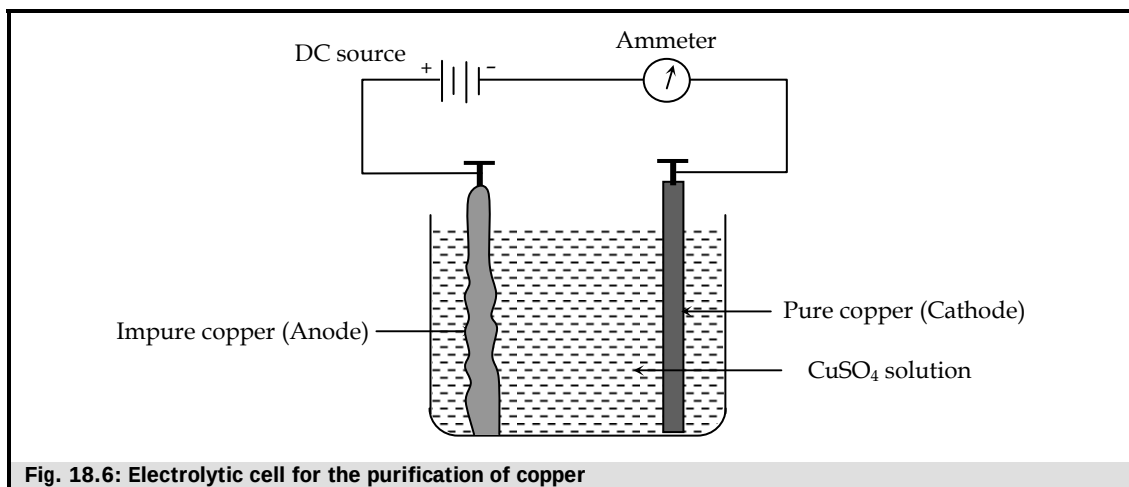
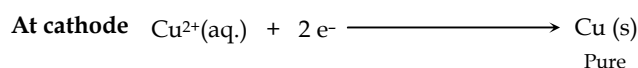
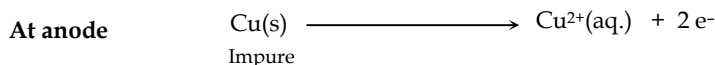
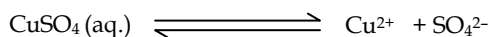


Fig. 18.6: Electrolytic cell for the purification of copper

Following reactions take place during the process.



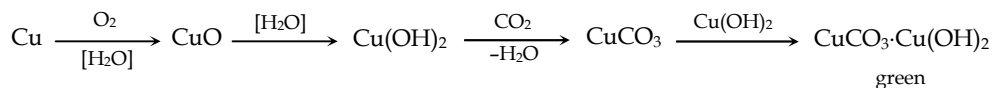
Copper thus obtained is about 99.95% pure.

18.2.5 Physical Properties

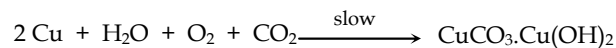
1. Copper is reddish brown coloured solid.
2. It is good conductor of heat and electricity.
3. It is highly malleable and ductile.
4. Its melting point is 1083°C , boiling point is 2350°C and specific gravity is 8.85.

18.2.6 Chemical Properties

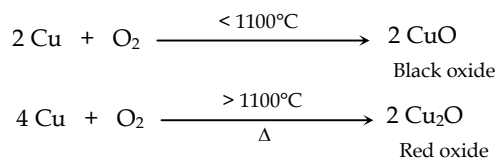
1. **Action with air:** Dry air has no effect on copper. However, if copper is exposed to moist air, it forms thin green film of basic copper carbonate, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$.



OR

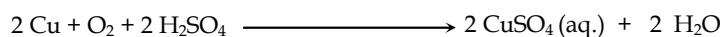


On heating copper with air (O_2), it forms cupric oxide (black oxide) which changes to cuprous oxide (red oxide) if temperature exceeds 1100°C .



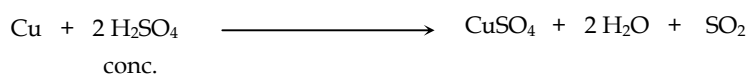
2. **Action with acids:** Copper lies below hydrogen in electrochemical series and cannot displace hydrogen gas from dilute mineral and non-oxidising acids.

- a. Non oxidizing warm and dilute acids (dil. HCl, dil. H₂SO₄) dissolve copper only in presence of air.

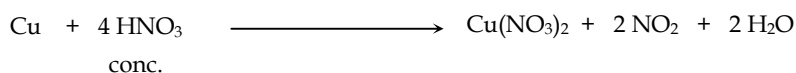


In absence of air, these acids do not react with copper.

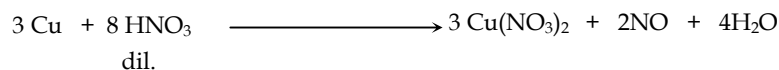
- b. With conc. H₂SO₄, it gives CuSO₄ and SO₂.



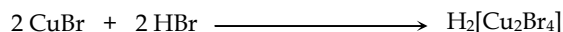
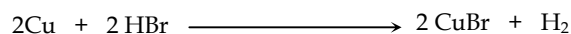
- c. With nitric acid copper is oxidized to Cu(NO₃)₂. However, the nature of nitrogen oxide depends upon the strength of HNO₃. Copper reacts with concentrated nitric acid to yield nitrogen peroxide as one of the products.



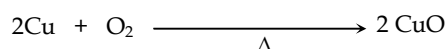
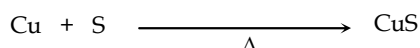
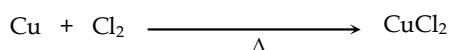
when copper reacts with dilute nitric acid, nitric oxide is produced.



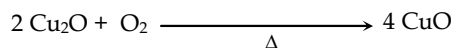
Copper reacts with hydrobromic acid and hydro-iodic acid to form complexes.



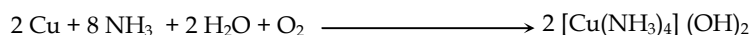
3. **Action with non-metals:** Copper reacts with non metal.



and



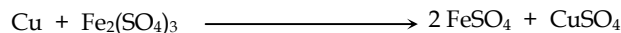
4. **Action with Ammonia:** Copper is dissolved in aqueous ammonia in presence of air, giving a complex compound.



5. **Displacement reaction:** Copper can displace the metals below it in electrochemical series.



6. **Reducing property of copper.** Copper acts as reducing agent. In the above reaction mentioned, copper is oxidized and behaves as reducing agent. It also reduces Ferric salt to ferrous salt.



The itching of copper plates using ferric chloride solution is based on this principle.

18.2.7 Uses of Copper

1. Copper is used in manufacture of electrical wires, cables, etc. owing to its good conducting power.
2. It is used to prepare household utensils, calorimeters, etc.
3. Copper is used in electroplating and electrotyping.
4. It can be used to prepare coins, ornaments and jewellery.
5. It is used to prepare various types of alloys.

Some of the important alloys have been listed below:

Alloy	Composition	Used in
Brass	Cu(70%) + Zn ($\approx$ 30%)	Domestic utensils
Bronze	Cu($\approx$ 80%) + Tin ($\approx$ 20%)	Coins, utensils, statues, bells, etc.
Gun metal	Cu(87%) + Sn ($\approx$ 10%) + Zn($\approx$ 3%)	Machine parts, guns
Rolled gold	Cu(95%) + zinc (20%) + Nickel (20%)	Utensils, ornaments, resistance coils.
German silver or Nickel silver	Cu (50%) + Zn(30%) + Ni(20%)	Utensils, plates, ornaments.



Different types of statues are being made since ancient period. Most of them are alloys of copper.

Fig. 18.7: Statues made of copper alloys

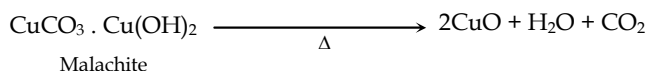
18.2.8 Compounds of Copper

A. Cupric Oxide or Black Oxide of Copper, (CuO)

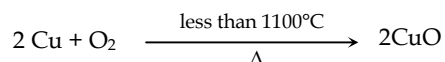
Cupric oxide is black oxide of copper which is insoluble in water.

Preparation

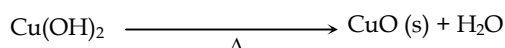
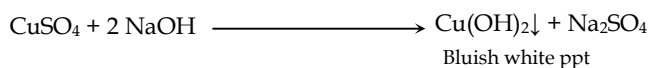
1. **From malachite:** Cupric oxide can be prepared in commercial scale, by heating malachite. Malachite is a naturally occurring compound of copper.



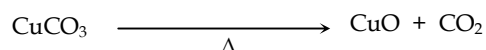
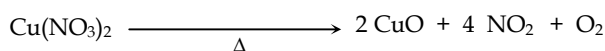
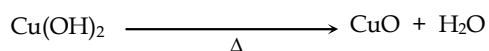
2. **From copper wire:** Cupric oxide can be obtained by burning copper with air.



3. **From water soluble copper salt:** Cupric oxide can be obtained by the precipitation reaction of water soluble of copper salts .



4. **From copper hydroxide, copper nitrate or copper carbonate:** Cupric oxide can be obtained by heating copper hydroxide, copper nitrate or copper carbonate.

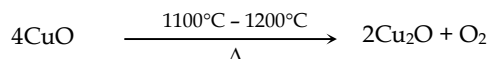


Physical properties

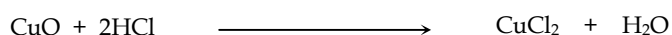
Cupric oxide is black, amorphous powder, which is insoluble in water but soluble in mineral acids.

Chemical properties

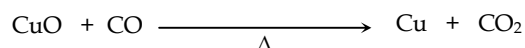
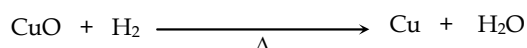
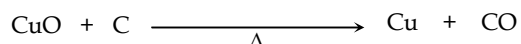
1. **Action of heat:** Cupric oxide is not easily decomposed under ordinary conditions, however, it is decomposed at $1100^\circ - 1200^\circ\text{C}$ to give Cu_2O



2. **Action of acids:** Cupric oxide reacts with mineral acids giving cupric salts.



3. **Reduction:** Cupric oxide can be reduced by H_2 , C, CO into copper.



Uses of Cupric Oxide

- Cupric oxide is used for coloring glass in blue or black colour.
- It is used for the estimation of C and H percentage in organic compounds. (Quantitative analysis of organic compounds).

B. Cuprous Oxide or Red Oxide of Copper (Cu₂O)

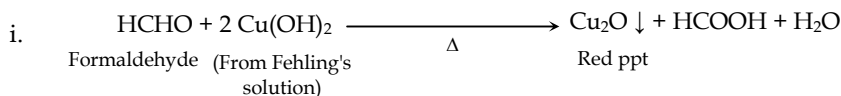
Cuprous oxide is red oxide of copper which is insoluble in water.

Preparation

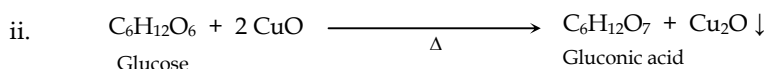
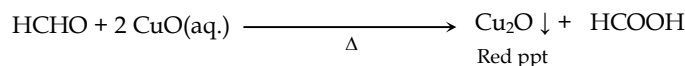
1. Cupric oxide can be prepared by the reduction of Fehling's solution with the help of glucose or an aldehyde.

Fehling's solution A contains aqueous CuSO₄ solution acidified with dil. H₂SO₄ and Fehling's solution B contains sodium potassium tartarate (Rochelle salt) made alkaline with NaOH.

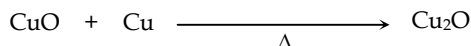
On mixing Fehling solution A and Fehling solution B, blue coloured solution of Cu(OH)₂ or CuO is obtained.



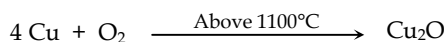
or,



2. Cuprous oxide can be obtained by heating CuO with Cu



3. When copper is heated at about 1100°C then red oxide of copper is obtained

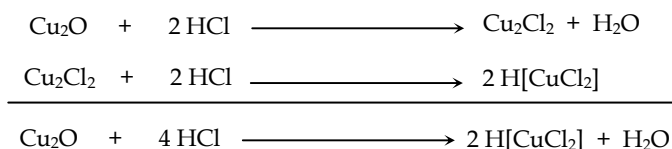


Physical properties

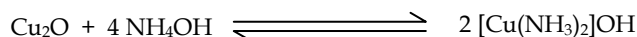
1. Cuprous oxide is a red coloured powder.
2. Cuprous oxide is insoluble in water.

Chemical properties

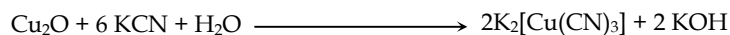
1. **Action of Acids:** Cuprous oxide reacts with dil. HCl to give cuprous chloride which further reacts with HCl to give a complex hydro-cupro-chloric acid. Usually lower and less stable oxidation states of metal are stabilized by complex formation.



2. **Action of NH₃:** Cuprous oxide reacts with ammonia solution to form amine complex.



3. **Action of Alkali cyanide:** Cuprous oxide forms a colourless complex with alkali cyanide.



Uses of Cuprous Oxide

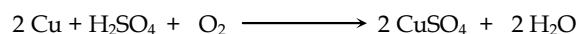
1. Cuprous oxide is used for the preparation of ruby red glass.
2. It is used as enamel and pigment.
3. It is used for the preparation of antirust paint.
4. It is used as pesticide.

C. Copper Sulphate or Blue Vitriol ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)

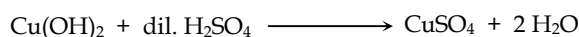
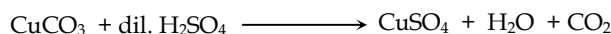
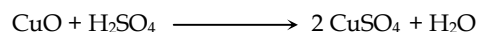
Copper sulphate pentahydrate is also called blue vitriol. In Nepali it is called *Nilo Tootho*. Blue vitriol is triclinic crystal ($a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$). It is soluble in water.

Preparation

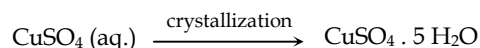
1. **From Cu or CuO or CuCO_3 or Cu(OH)_2 etc.:** When CuO or Cu(OH)_2 or CuCO_3 is treated with dil. H_2SO_4 and the resulting solution is crystallized, blue vitriol is obtained.



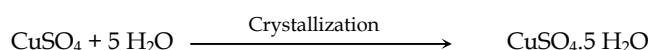
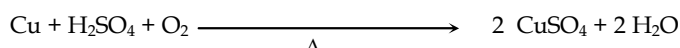
Or



Thus obtained copper sulphate solution is heated upto crystallization point where by triclinic crystals of blue vitriols are obtained. During the preparation of blue vitriol by this method, it is to be noted that sulphuric acid should not be excess which otherwise makes blue vitriol crystals dehydrated.

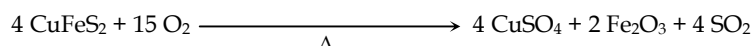


2. **From scrap copper:** When scrap copper is treated with dil. H_2SO_4 in presence of air, CuSO_4 is obtained which on crystallization gives blue vitriol.



On larger scale copper sulphate is manufactured by this method.

3. **From copper pyrites:** Copper pyrite is roasted whereby copper sulphide and iron sulphides are formed. If temperature is carefully regulated, copper sulphide is converted into copper sulphate.



CuSO_4 thus obtained is extracted with water and crystallized to get blue vitriol.

Physical properties

1. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is blue in colour.
2. Anhydrous CuSO_4 is white.

Chemical properties

1. Action of heat:

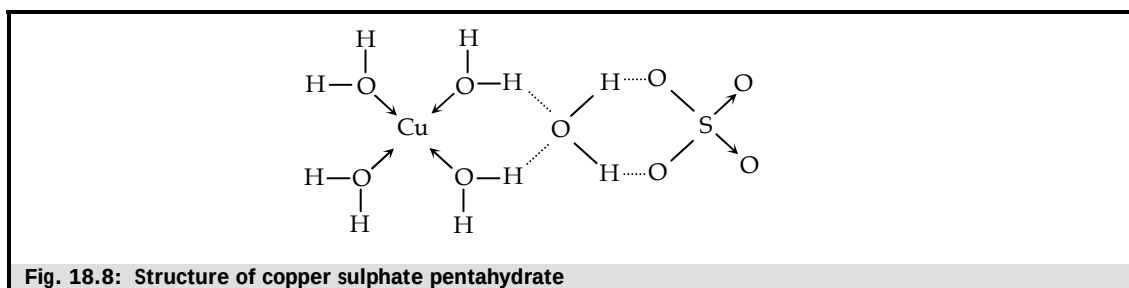
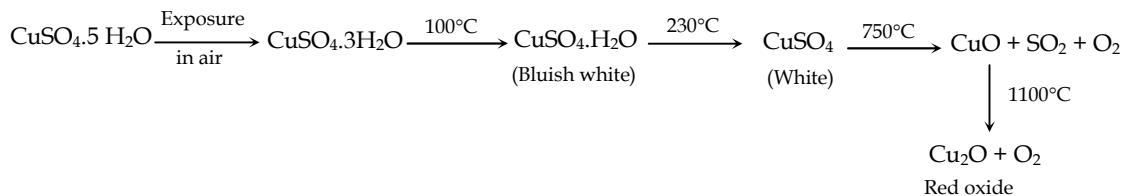
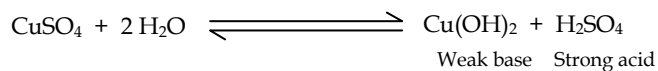


Fig. 18.8: Structure of copper sulphate pentahydrate

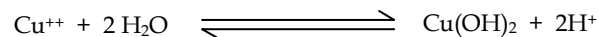
2. **Formation of double salts:** Copper sulphate forms double salts like $\text{K}_2\text{SO}_4 \cdot \text{CuSO}_4 \cdot 6\text{H}_2\text{O}$; $(\text{NH}_4)_2\text{SO}_4 \cdot \text{CuSO}_4 \cdot 6\text{H}_2\text{O}$.



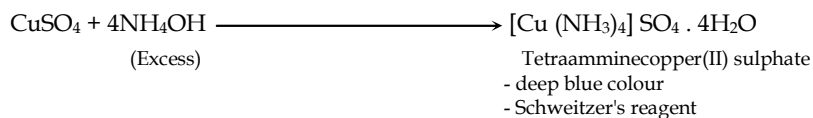
3. **Acidic nature:** Aqueous copper sulphate solution is acidic due to cationic hydrolysis.



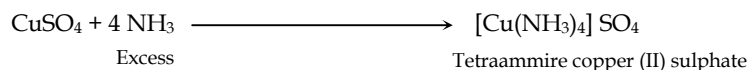
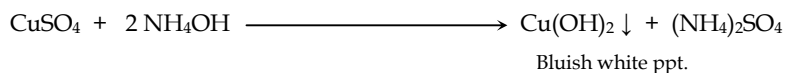
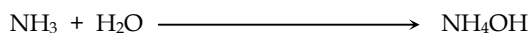
OR



4. Complex Formation



When ammonium hydroxide is added dropwise into copper sulphate solution, initially a bluish white precipitate of copper hydroxide is formed which dissolves to give deep blue solution on adding excess of ammonia solution. The deep blue colour is due to the formation of tetra-amine copper (II) sulphate.



The deep blue solution of tetraamine copper(II) sulphate is also known as **Schweitzer's reagent**. It has capacity to dissolve cellulose.

- Name two important ores of copper with formulae.
- Give any one method of conversion of cupric sulphate into cuprous oxide with chemical equations.
- When copper wire is put into a solution of silver nitrate beautiful crystals are deposited on the surface of the copper. What are these crystals chemically? Write the reaction involved.
- What is the effect of heat on copper sulphate penta-hydrate?
- Mention the important uses of the copper metal.

■ Short/Long Answer Questions:

- Write the preparations, properties and uses of Blue vitriol.
- How is blister copper obtained from copper pyrites?
- How would you convert copper to blue vitriol? Name any one alloy of each of:
 - Copper and Zinc.
 - Copper and tin.
- How is metal copper extracted from its pyrite ore? How is it refined? Write the action of it upon: (a) moist air and (b) Conc. H_2SO_4
Ans: (a) $2\text{Cu} + \text{H}_2\text{O} + \text{CO}_2 + \text{O}_2 \rightarrow \text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ (green basic carbonate) (b) $\text{Cu} + \text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + 2\text{H}_2\text{O} + \text{SO}_2$
- Starting from copper pyrite, how would you obtain pure copper? Explain the steps involved in the process with necessary diagram for it. Give the chemical reaction for the followings:
 - Copper is exposed to moist air.
 - Conversion of copper into blue vitriol.
- How does copper metal react with:
 - moist air
 - conc. H_2SO_4
 - conc. HNO_3
- Write the pertinent reaction
 - Copper sulphate crystal turns white powder on heating.
 - A brown gas is obtained when copper is treated with conc. nitric acid.
 - Copper becomes green when exposed to air for long time.
- Describe the action of blue vitriol with
 - ammonia solution till excess
 - aqueous potassium iodide
- What happens when?
 - Ammonium hydroxide is added to a solution of copper sulphate.
 - A copper coin is dropped into concentrated nitric acid in a test tube.
 - Copper turning is treated with conc. HNO_3 .
- Write short notes on (a) Chemistry of black oxide of copper (b) Chemistry of blue vitriol

■ Multiple Choice Questions

- Ruby copper is
 - Cu_2S
 - Cu_2O
 - $\text{Cu}(\text{OH})_2$
 - $\text{Cu}(\text{OH})_2\text{CuCO}_3$
- Blister copper consists . . . percent copper.
 - 90
 - 95
 - 98
 - 99.5
- Solution of Schweitzer's reagent gives the positive test of
 - Cu^{++}
 - SO_4^{--}
 - NH_3
 - Cu^{++} , SO_4^{--} and NH_3
- Bordeaux mixture is . . .
 - insecticide
 - fungicide
 - CuSO_4 solution
 - lime in aqueous CuSO_4
- Malachite is . . .
 - Cu_2S
 - $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
 - Cu_2O
 - CuCO_3
- Copper matter is a
 - CuS containing little amount of FeS .
 - CuS containing little amount of FeO
 - FeS containing little amount of CuS
 - Nearly equal amount of CuS and FeS

Answer Key

1	2	3	4	5	6
a	c	b	d	b	a

18.3 Zinc

18.3.1 Introduction

Symbol:	Zn	Melting point:	429.4°C
Atomic number:	30	Boiling point:	907°C
Atomic mass:	65.37 (amu)	Electronic configuration:	[Ar] 3d ¹⁰ 4s ²
Density:	7.14 g/cm ³		

Zinc has been known since long. Its Nepali name is '*Jasta*'. It has been used for a long time in the manufacture of brass (an alloy of copper and zinc).



Fig. 18.9: Zinc Sheet (जस्ता पाता) for roofing house

18.3.2 Position of Zinc in Periodic Table

Zinc lies in group 12 (group IIB) of the periodic table. The members of this group are Zinc ($_{30}\text{Zn}$), Cadmium ($_{48}\text{Cd}$) and Mercury ($_{80}\text{Hg}$).

1. These elements are known as non-typical transition metals. The reason is that they do not show typical properties of the transition metals like variable oxidation state, formation of coloured compounds, etc. They have completely filled d-orbitals. Therefore, they cannot give coloured compounds. Their d-electrons cannot participate in bond formation. Hence, they do not exhibit variable oxidation state in general.
2. Their common oxidation state in combined state is +2.
3. All of these elements are metals. Zinc and cadmium are solid and mercury is liquid at room temperature.

18.3.3 Occurrence of Zinc

Zinc does not occur in free state. In combined form, it exists as sulphides, oxides, carbonates, etc. Some ores of zinc are:

1. Zinc blende (ZnS)
2. Calamine (ZnCO_3)
3. Zincite or red zinc oxide (ZnO)
4. Franklinite ($\text{ZnO} \cdot \text{Fe}_2\text{O}_3$)
5. Willemite (Zn_2SiO_4)

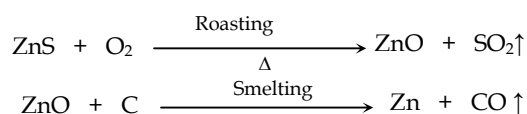
Zinc blende is the major ore of zinc. In Nepal, zinc is found in the form of its ores in Ganesh Himal, Phulchowki, etc. Zinc is found in the form of organometallic compounds in snake poison also.

18.3.4 Extraction of Zinc

Theory

Zinc blende (ZnS) is the main ore of zinc. Zinc blende after concentration and roasting gives zinc oxide in dry form. This oxide form of ore when heated strongly with powdered coke at about 1500°C gives zinc vapors which on condensation gives molten zinc (97.8% pure). This liquid form is called zinc spelter. Spelter on purification by fractional distillation followed by electrolytic refining gives pure zinc.

Reactions:



Steps involved in this metallurgy

1. **Crushing and pulverization:** Big lumps of ore are crushed using jaw crushers to get crushed ore which are pulverized (powdered) using pulverisers or ball mills to get powdered ore.
2. **Concentration of ore by froth-floatation process:** Powdered ore is dipped into a tank containing water and small quantity of pine oil. The mixture is agitated by using blast of air. Gangue particles are wetted by water and get collected at the bottom of the tank. The sulphide particles being preferentially wetted by oil come on the surface in the form of froth. The froth is skimmed off to collect the concentrated ore (figure 18.10).

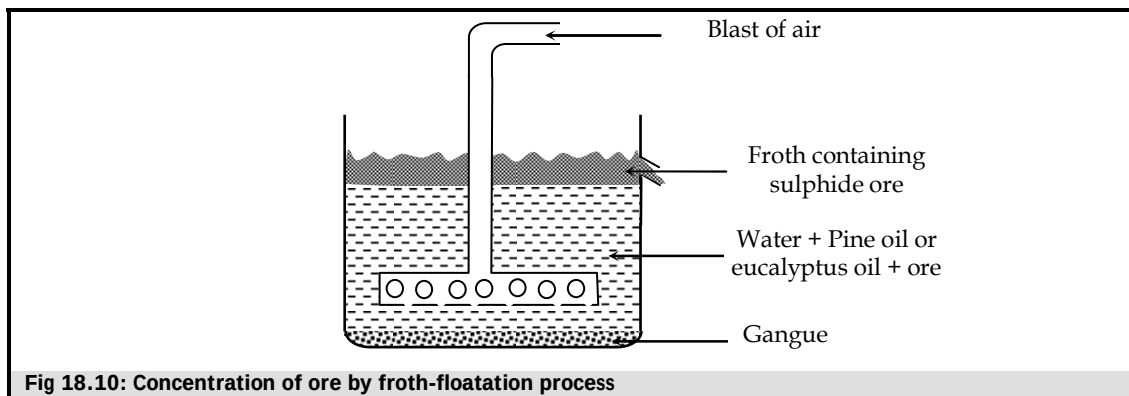
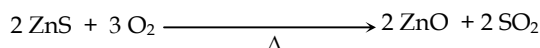
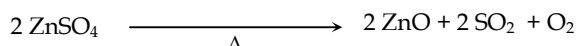
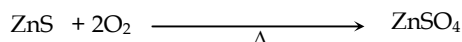


Fig 18.10: Concentration of ore by froth-floatation process

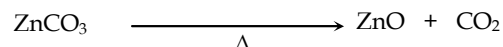
3. **Roasting:** The concentrated ore is heated in a reverberatory furnace in presence of air at a high temperature of about 900°C (to decompose ZnSO_4 , if formed) (figure 10.11). On doing so following changes take place.
 - i. Zinc sulphide present in the ore is oxidized to its oxide



- ii. If zinc sulphate is formed, it gets decomposed to give zinc oxide.



iii. If ore contains calamine, it is decomposed to zinc oxide.



iv. Volatile impurities are removed.

v. Impurities like sulphur, phosphorus, antimony, arsenic etc. are removed as their oxide.

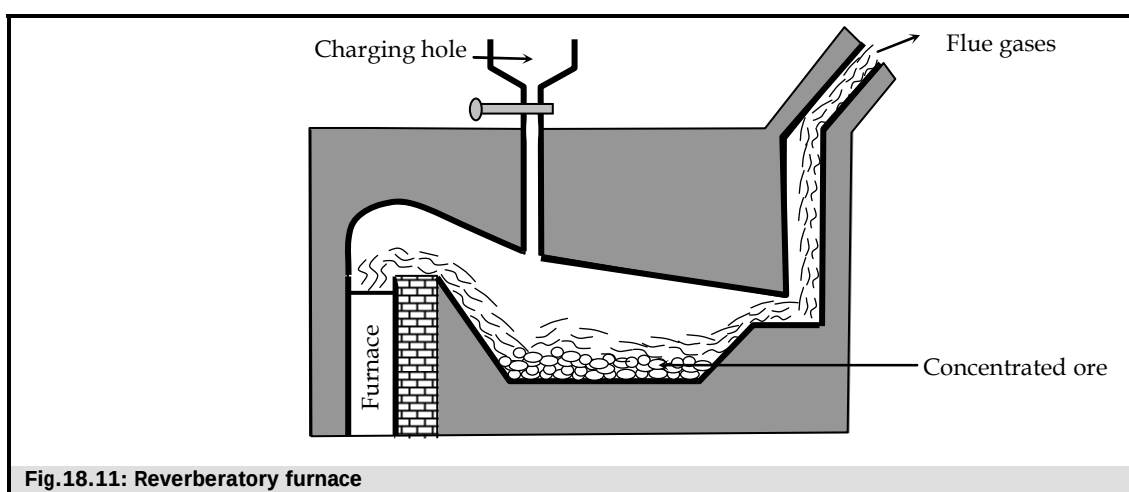
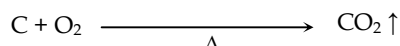
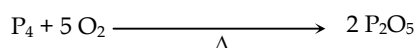
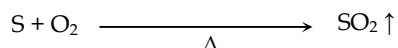


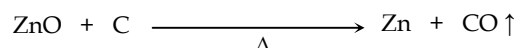
Fig.18.11: Reverberatory furnace

4. **Reduction.** Zinc oxide formed after roasting can be subjected to reduction by any of the following methods.

- Continuous vertical retort process.
- Belgian method (uses circular or elliptical retorts)
- Silesian process (reverberatory furnace is used)

Vertical retort process is modern and highly economical process (figure 18.12). This process will be discussed here.

The powdered oxide form of the ore (ZnO) obtained from roasting process is mixed with coke (C) in a ratio of 2:1 to make small briquets. The briquets are fed into the retort furnace through charging door. This feeding is a continuous process. The retort is heated externally by burning the producer gas ($\text{CO} + \text{N}_2$) at around $1400\text{--}1500^\circ\text{C}$. On doing so, zinc oxide gets reduced by coke (C) to the zinc as:



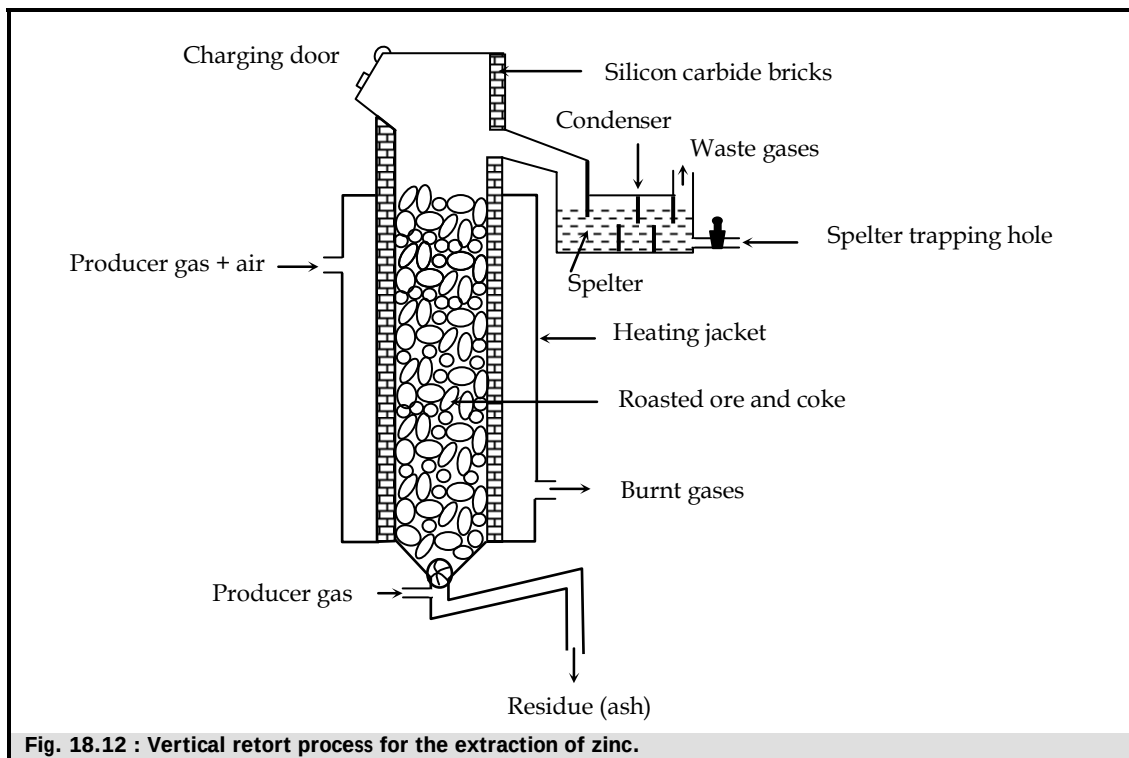
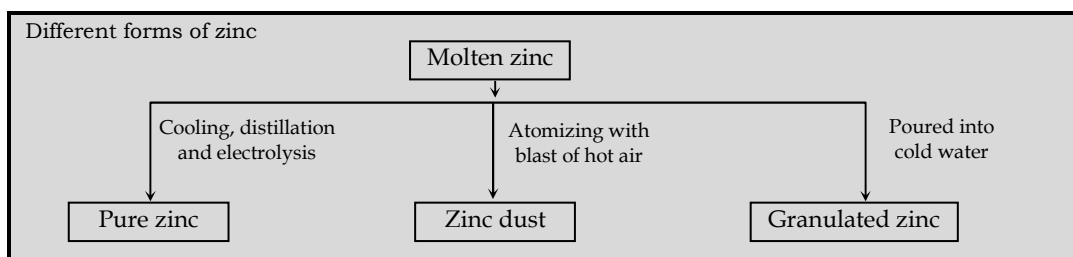


Fig. 18.12 : Vertical retort process for the extraction of zinc.

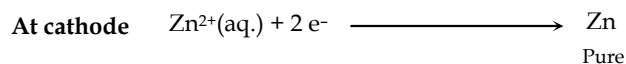
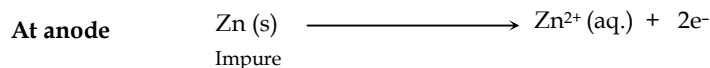
The zinc vapours and carbon-monoxide gas thus obtained are expelled out by the mild current of producer gas. The mixture of gases pass through a condenser where carbon-monoxide gas is expelled out as waste gases. Zinc vapours get condensed into a condenser or receiver and collected in condenser as molten zinc which is called **spelter zinc**. The spelter zinc is tapped out from the condenser from time to time. The purity of zinc obtained here is 97.8%.



5. **Purification or Refining of Zinc:** Possible impurities of zinc in zinc spelter are iron, lead, cadmium, arsenic etc. The methods of purification of the zinc thus obtained are discussed below:

- Distillation:** The zinc spelter is subjected to distillation at about 1000°C . Where impurities like iron (B.pt. = 3000°C) and lead (B.pt. = 1620°C) are left behind. Zinc (B.pt. = 907°C) and cadmium (B.pt. = 767°C) are collected as distillate. The distillate is distilled again at a temperature of 800°C , where cadmium is distilled off and zinc is left behind. Thus obtained zinc is about 99% pure.
- Electrolysis:** Thin sheet of pure zinc is made cathode and the zinc to be purified is made anode. Both electrodes are dipped in a solution containing ZnSO_4 solution containing dil.

H_2SO_4 (as electrolyte). When D.C. is supplied, zinc is dissolved from anode and gets collected at cathode leaving the impurities at anode. Thus pure zinc is collected at cathode.



Zinc thus obtained is 99.5% pure.

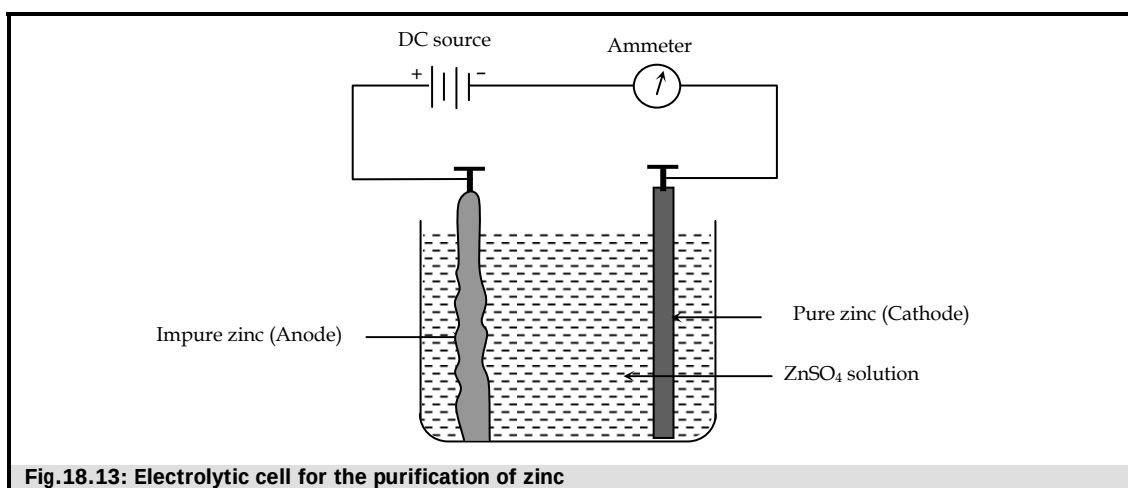
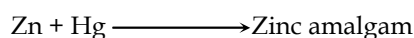
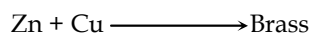


Fig.18.13: Electrolytic cell for the purification of zinc

18.3.5 Properties of Zinc

Physical properties

1. Zinc is bluish white solid. It is tarnished to grey colour when exposed on moist air.
2. Zinc is a good conductor of heat and electricity.
3. Zinc is malleable and ductile at about 100-150°C, where as it is brittle at about 200°C.
4. Melting point of zinc is 429.4°C and boiling point is 907°C.
5. Zinc forms homogeneous mixture with metal to form alloys.



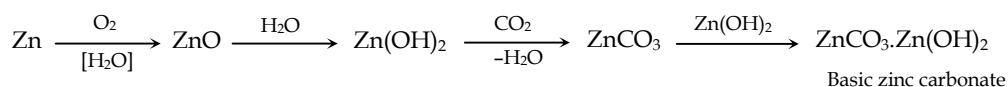
Chemical Properties of Zinc

Note:

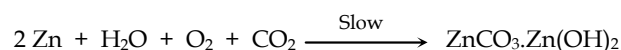
$$E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V} \quad E^\circ_{\text{Hg}^+/\text{Hg}} = +0.79 \text{ V} \quad E^\circ_{\text{Hg}^{2+}/\text{Hg}} = +0.85 \text{ V}$$

The above data show that general reactivity of Zn >>> Hg. (as reducing agent)

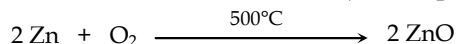
1. **Action of air:** Zinc is not affected by dry air, however it gets tarnished when exposed to moist air giving basic zinc carbonate.



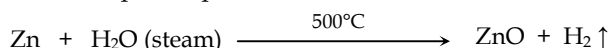
Overall reaction:



When zinc is heated with air at 500°C, it burns with bluish white flame giving white woolly flocks of ZnO (dense white smoke of zinc oxide) called **philosophers' wool** or **pompholyx**.

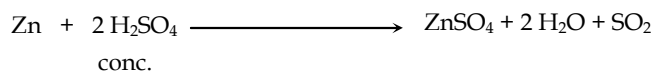


2. **Action of water:** Pure zinc does not react with water. Impure zinc (commercial or granulated) decomposes steam to produce H₂. Zinc in presence of Cu forms a Zn-Cu couple, which acts as electrochemical couple. So presence of Cu increases electron donating character of zinc.

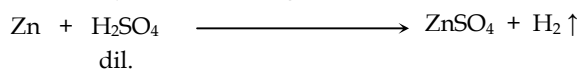


3. **Action of acids:** Zinc lies above hydrogen in electrochemical series. Therefore, it produces H₂ gas from dilute, mineral and non-oxidizing acid. Zinc shows amphoteric nature. Hence, it reacts with acids as well as bases to give salts.

- a. **Action of sulphuric acid:** Zinc is oxidized by hot conc. H₂SO₄ as

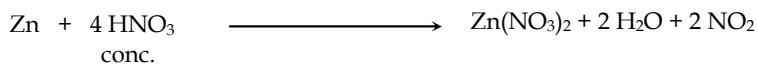


Zinc is oxidized by dil. H₂SO₄ to give ZnSO₄. Zn displaces H₂ from dil. H₂SO₄ as:



- b. **Action of nitric acid (HNO₃):** Zn is oxidized by nitric acid to Zn(NO₃)₂. HNO₃ gets reduced to various products like NO₂ or NO or N₂O or NH₄NO₃ depending upon the concentration of nitric acid:

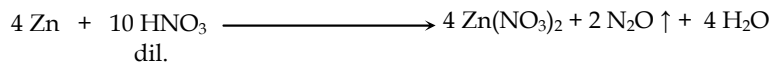
- i. **With conc. HNO₃**



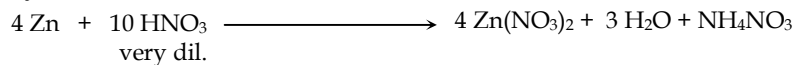
- ii. **With moderately concentrated (1:1) HNO₃**



- iii. **With dilute HNO₃**



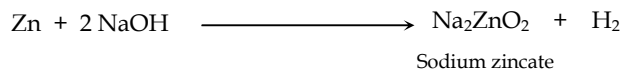
- iv. **With very dilute HNO₃**



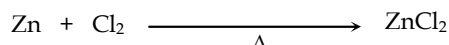
- c. **Action of HCl:** With dilute HCl, Zn displaces H₂ gas.



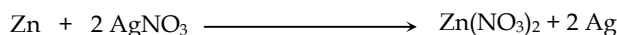
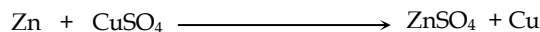
4. **Action of alkali:** Zinc reacts with alkali (eg. NaOH) to give sodium zincate.



5. **Action of non-metals.** Zn reacts with non-metals such as S, Cl₂, etc. to give corresponding salts.



6. **Displacement reaction:** Zinc can displace less reactive metals (metals below Zn in electrochemical series) from their salts.



18.3.6 Uses of Zinc

1. Zinc is used in making alloys (Examples: Brass (Cu + Zn), German silver (Cu + Zn + Ni)).
2. Zinc is used as a reducing agent (disilverization) during the extraction of Ag and Au.
3. It is used as a lab reagent. Examples: Zinc dust in reduction of phenol, Granulated zinc in the preparation of H_2 , etc.
4. It is used as anode in dry cells.
5. It is used in galvanization of iron.

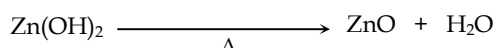
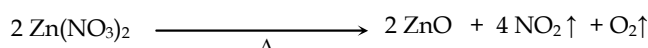
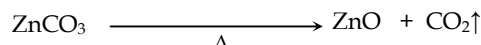
18.3.7 Zinc White or Zinc Oxide (ZnO)

Zinc oxide is known as philosopher wool or zinc white or Chinese white. When zinc is heated with air at 500°C , it burns with bluish white flame giving white woolly flocks of zinc oxide which is called **Philosopher's wool or pompholyx**.

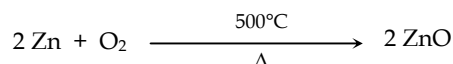
Preparation:

Zinc oxide can be prepared by the following method:

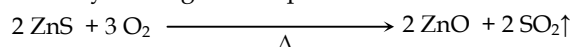
1. When zinc carbonate or zinc nitrate or zinc hydroxide is heated, ZnO is obtained



2. It can be obtained by heating zinc with air.



3. It can be obtained by heating zinc sulphide with air.

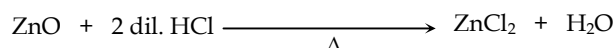


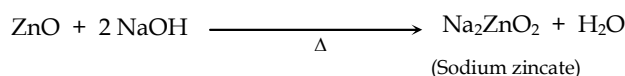
Physical properties

1. It is a white amorphous powder insoluble in water.
2. It becomes yellow on heating and white on cooling. On heating O^{2-} ions leave the crystal lattice as O and electrons are trapped by holes or crystal lattice giving yellow colour. On cooling the reverse process takes place.
3. It sublimes at 400°C .
4. Its melting point is high (1975°C).

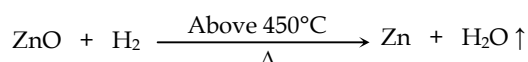
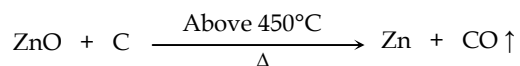
Chemical properties

1. **Amphoteric nature:** Zinc oxide reacts with acid as well as base giving salt i.e. it shows amphoteric nature.

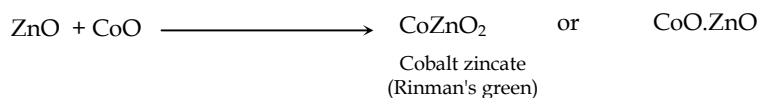
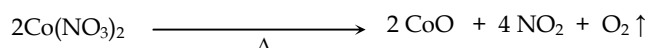




2. **Oxidizing nature:** Zinc oxide is reduced by C or H₂ into metallic zinc.



3. **Action of cobalt nitrate:** When zinc oxide is heated with cobalt nitrate, a green coloured mass of cobalt zincate is obtained (Rinman's green). In laboratory this reaction is utilized for the identification of Zn²⁺ in the salt solution, which is known as "Filter Ash Test".



Uses

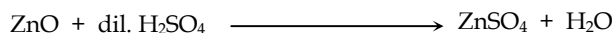
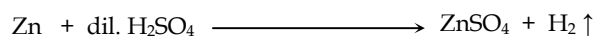
1. Zinc oxide is used as white pigment in paint under the name zinc white or Chinese white. It is preferred over the pigment of lead as it does not get blackened by atmospheric gases like H₂S.
2. It is used in preparing green pigment called Rinmann's green.
3. It is used as glaze in ceramics.
4. It is used as filler of soft rubber.
5. It is used in cosmetic powders or creams.

18.3.8 White Vitriol, ZnSO₄·7H₂O

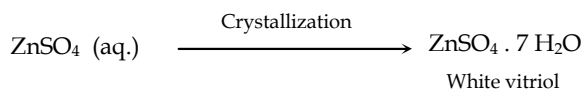
Heptahydrated zinc sulphate is called white vitriol. In nature, it exists as zincosite. It is soluble in water.

Preparation

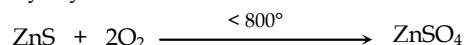
1. When zinc or ZnO or ZnCO₃ is treated with dil. H₂SO₄, aqueous solution of ZnSO₄ is obtained. This solution on crystallization gives white vitriol.

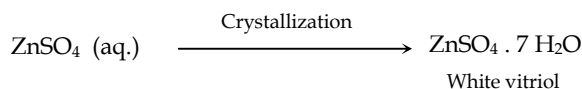


and



2. It can be prepared by the controlled oxidation between zinc blende (ZnS) and oxygen (Roasting), followed by crystallization.



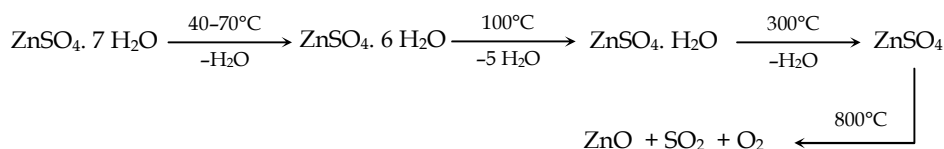


Physical properties

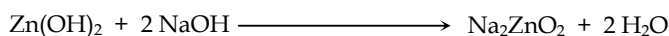
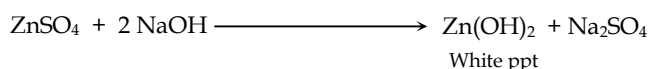
1. It is crystalline white solid.
2. It is efflorescent solid (loses water of crystallization when exposed to atmosphere).
3. It is highly soluble in water.

Chemical properties

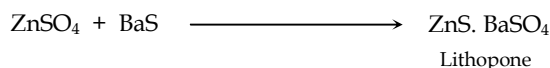
1. **Action of heat:** When white vitriol is heated to about 70°C, it loses one water molecule to give hexahydrated zinc sulphate. When temperature reaches to 100°C, it loses 5 more water molecules giving monohydrated ZnSO₄. On heating the monohydrated ZnSO₄ at 300°C gives anhydrous ZnSO₄. Anhydrous ZnSO₄ gets decomposed at 800°C to give zinc oxide.



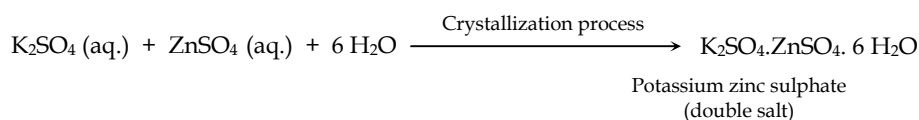
2. **Action of sodium hydroxide:** It reacts with NaOH to give white precipitate of Zn(OH)₂, which dissolves in excess NaOH to give sodium zincate.



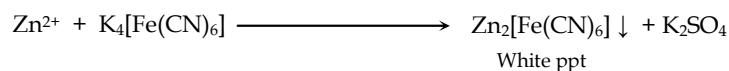
3. **Reaction with barium sulphide:** When zinc sulphate is treated with barium sulphide, double salt, *lithopone* is obtained, which is used as white pigment.



4. **Formation of double salts:** White vitriol solution on crystallization with alkali metal sulphates gives double salts.



5. **Action of potassium ferrocyanide:** Zn²⁺ reacts with potassium ferrocyanide giving white precipitate of zinc ferrocyanide.



Uses

1. It is used as electrolyte during electro-galvanization.
2. Its solution is used as eye lotion. (0.1 – 0.2% solution).
3. It is used as mordant for dyeing.
4. It is used for the preparation of lithopone, which is a white pigment.

18.3.9 Galvanization

Introduction

A large amount of iron used as a structural material is being destroyed due to rusting. To protect iron from rusting, it is coated with a layer of zinc. The process of coating a layer of zinc over the surface of iron is called **galvanisation**.

The Galvanisation process

Galvanization of iron is carried out in the following steps:

Step 1: Cleaning: The article to be galvanized is first cleaned by sand blasting.

Step 2: Pickling: After cleaning the article (to be galvanized) using sand blast, it is dipped into dil. HCl (or HNO_3) solution so as to remove the oxide films formed on the iron surface. This process is called pickling process.

Step 3: Galvanising the article: Coating of Zn on the surface of iron can be achieved by the following methods.

- Hot dip Galvanising:** In this method, cleaned article is kept in molten zinc, such that thin layer of Zn is coated. Then a gentle press is given to make the surface smooth and homogeneous.
- Sherardising:** In this method, cleaned article is heated in rotating drum with zinc dust at 800°C for several hours (At this temperature alloy is not formed). Uniform layer of zinc is coated. But the process is costly (is being used for galvanizing nuts, bolts, etc.).
- Metallizing:** In this method, molten zinc is sprayed over the article. This method is suitable for galvanization of iron in bridges, ships, etc.
- Electroplating:** In this method, iron is made cathode zinc is made anode. Acidified zinc sulphate is used as electrolyte. When D.C. is supplied, Zn gets dissolved from anode and is deposited at cathode.

Step 4: Rolling: Galvanized iron is pressed through rollers to make the surface smooth.

Mode of protection

- Barrier protection:** Zinc acts as barrier between corroding atmosphere and iron. In this way galvanization process protects from rusting.
- Cathodic protection (sacrificial protection):** Even if small amount of iron is exposed, iron acts as cathode and zinc acts as anode. Thus oxidation takes place at zinc. Here zinc is sacrificed for protecting iron.

■ Solved Questions

1. What is galvanization?

- ↪ The process of coating a thin layer of zinc on the surface of iron so as to protect it from rusting is called galvanization.

2. What is sacrificial protection?

- ↪ If iron surface is exposed to atmosphere from galvanized iron, zinc acts as anode and iron acts as cathode. A galvanic cell like arrangement is set up, where Zn dissolved from anode end and gets deposited at cathode end. The iron does not rust so long as all the zinc has not disappeared. Here cathode is protected by sacrifice of anode. This process is called sacrificial protection or cathodic protection.

3. What is granulated zinc? Why is it more reactive than pure zinc?

- When molten zinc spelter is dropped in cold water irregular shape of solid zinc is obtained. This impure form of zinc is called granulated zinc. It may consist of copper as impurity and forms Zn-Cu couple. These form a galvanic cell where zinc acts as anode and copper acts as cathode. Zinc loses electron and copper gains electron. Presence of copper increases the electron donating character of zinc. That is why granulated zinc is more reactive than pure zinc.

Questions

Very Short Answer Questions:

- Give a correct balanced chemical reaction for the preparation of
 - White vitriol
 - Philosopher's wool
- How is Rinman's green prepared? Give its one important use.
- Write action of heat on white vitriol.
- What happens when zinc white is heated?
- Write the molecular formulae of:

i. Philosopher's wool	ii. Rinman's green
iii. Calamine	iv. Zinc white
- How would you obtain white vitriol from zinc?
- What is the action of heat on white vitriol?
- Why does zinc become dull in contact of moist air?
- Write the name of two ores of zinc with their molecular formulae.
- What action takes place when aqueous sodium hydroxide is added to zinc sulphate solution drop by drop till excess?
- Why does zinc displace copper from copper sulphate solution?
- Why is zinc not considered as a transition element?
- How will you obtain crystals of white vitriol? Give a chemical reaction.

Short /Long Answer Questions:

- Give the chemistry of zinc white.
- Write the chemistry of blue vitriol.
- Starting from zinc blende, how would you obtain pure zinc? What is galvanization?
- Explain the principle and process sketching a well-labelled diagram for the extraction of zinc from its ore. What happens when zinc is exposed to moist air?
- Write two chemical reactions for the preparation of Zinc white, and justify its amphoteric nature by giving suitable chemical reactions. Write its one important use.
- What happens when zinc white is:
 - Heated with cobalt nitrate
 - Dissolved with caustic alkali
 - Zinc white is heated with cobalt nitrate
 - Metallic zinc is dissolved in hot conc. NaOH
 - Zinc carbonate is heated

Multiple Choice Questions

- Rinman's green is
 - Mixture of ZnO and CO
 - Mixture of ZnO and COO
 - Double salt of ZnO and CoO
 - Green pigment
- Phiosopher's wool is . . .
 - Zincite
 - Red zinc oxide
 - Precipitate of zinc oxide
 - Light mass of ZnO
- Lithopone can be used as
 - eye lotion
 - antiseptic
 - paint
 - Rinnmann's green
- Galvanisation is carried out to protect
 - Zinc
 - Magnesium
 - Iron
 - Barium
- . . . is used as flux in soldering
 - dil. solution of ZnCl_2
 - conc. solution of ZnCl_2
 - Mixture of ZnCl_2 and ZnSO_4
 - Mixture of ZnCl_2 and PbCl_2

Answer Key

1	2	3	4	5
c	d	c	c	b

18.4 Mercury

18.4.1 Introduction

Symbol	Hg	Density	13.6 g/cm ³
Atomic number	80	Boiling point	357°C
Atomic mass	200 (amu)	Melting point	- 38.85°C
Electronic configuration	[Xe] 5d ¹⁰ 6s ²		

Mercury has been known since long time. The symbol Hg is taken from Latin name *hydrargyrum* [Greek: *Hydro* = water, *Argyros* = Silver]. In Nepali it is called *paro* (पारो). Mercury preparation finds their applications in "Aayurveda". Aristotle named it as *quick silver* or liquid silver. It is only the metal which is found in liquid state at room temperature.

18.4.2 Occurrence of Mercury

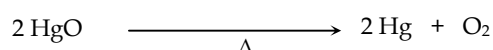
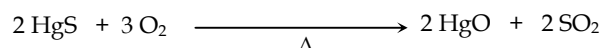
Very small amount of mercury is found in free state. In combined form, it exists as its halides, amalgams, sulphides, etc. Main ore of mercury is cinnabar (HgS).

18.4.3 Extraction of Mercury from Cinnabar

Theory

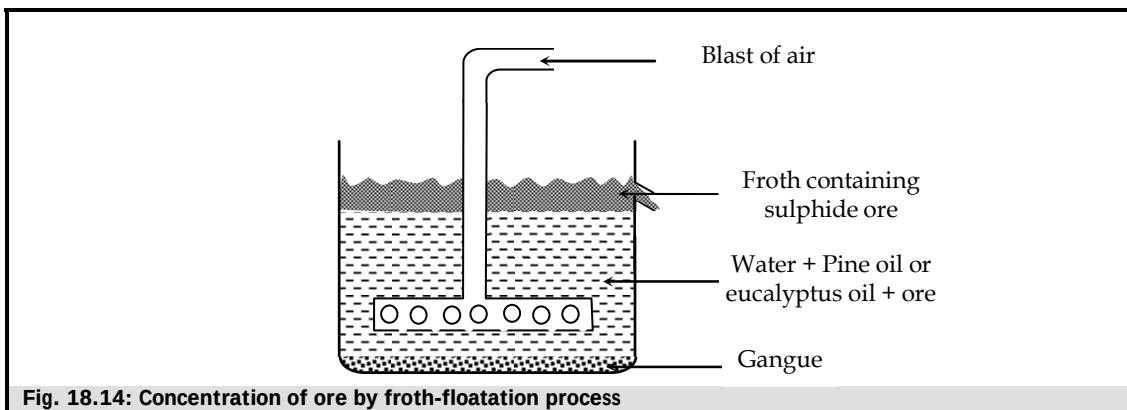
Cinnabar after concentration by froth-floatation process is subjected to roasting followed by distillation. Mercury vapour is condensed to get liquid mercury, which on purification by filtration, treatment with 5% HNO₃ and vacuum distillation gives pure mercury.

Main reactions are:



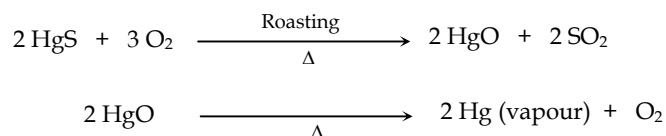
Steps involved in the metallurgy

- Crushing and pulverization:** Big lumps of ore are crushed using jaw crushers to get crushed ore which are pulverized (powdered) using pulverisers or ball mills to get powdered ore.
- Concentration of ore by froth-floatation process:** Powdered ore is taken into a tank containing water and small quantity of pine oil or eucalyptus oil. The mixture is agitated by using blast of air. Gangue particles are wetted by water and get collected at the bottom of the tank. The ore particles being wetted by oil come on the surface in froth. The froth is skimmed off to collect the concentrated ore.

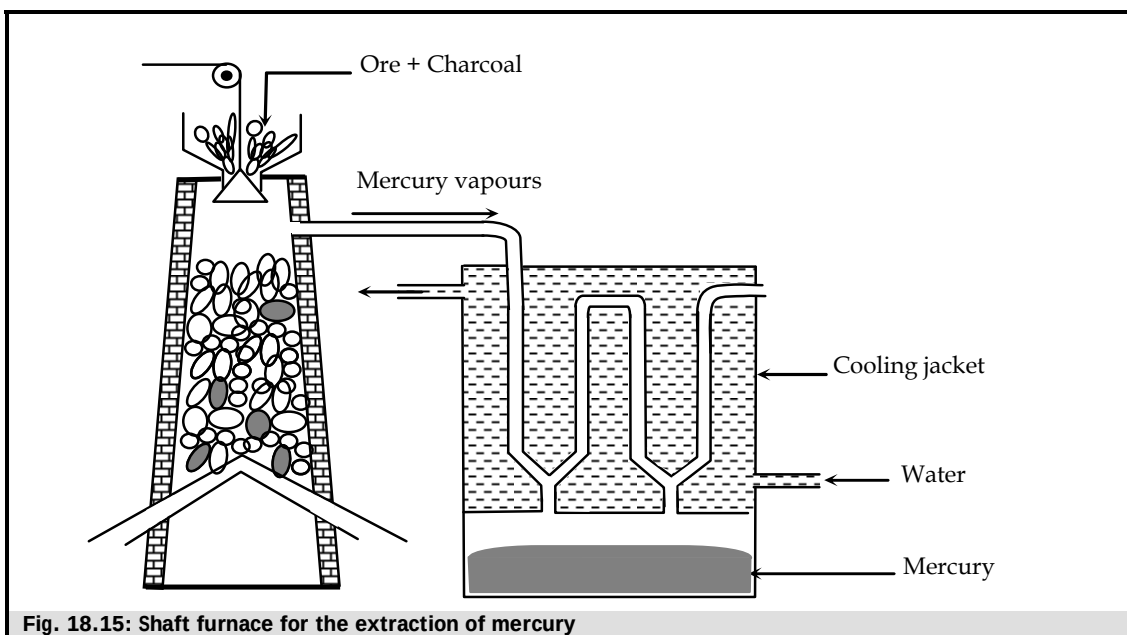


3. **Roasting and distillation in shaft furnace:** Shaft furnace is long cylindrical body having cup and cone arrangement as feed-end. This cup and cone arrangement allows the charge (ore + coke + lime) to enter through the furnace but prevents mercury vapours from escaping. The shaft-furnace contains a Y-shaped side tubes which are passed through cooler.

The concentrated ore, coke (2% only as a fuel) and lime are mixed together and fed through cup and cone arrangement to the shaft furnace. The furnace is heated by burning fuel (air is passed along with the flames). The furnace is heated upto red hot. On doing so, mercuric sulphide gets oxidized to mercuric oxide which subsequently decomposes in to mercury vapours.

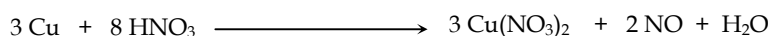


Mercury vapour thus obtained, is passed through the water-cooled Y-shaped iron pipes. These vapour gets condensed to give *liquid mercury*. The spent ore is taken out from the bottom of the furnace. The waste gases escape through the outlet of the receiver or condenser.



5. **Purification of mercury:** Main impurities present in the liquid mercury thus obtained are copper, zinc, bismuth, lead and suspended impurities of metal oxide. The following steps are employed to obtain pure mercury.

- i. **Filtration:** Mercury thus obtained is filtered (actually squeezed) through chamois leather or thick canvas. On doing so, suspended oxide impurities are removed.
- ii. **Treatment with 5% nitric acid solution:** Mercury obtained after filtration is passed through a long tube containing 5% HNO_3 solution. Base metals remain in aqueous solution.



Small amount of $\text{Hg}_2(\text{NO}_3)_2$ which is formed by the reaction of Hg with HNO_3 , react with other more electropositive metals to displace Hg.

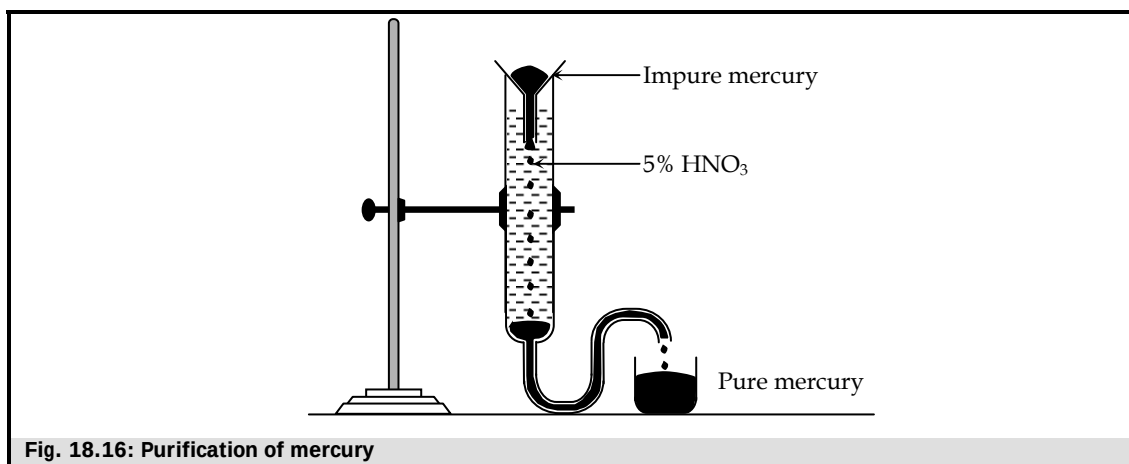
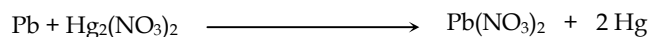
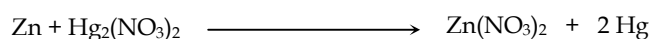
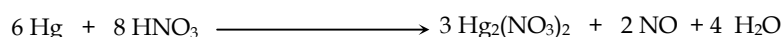


Fig. 18.16: Purification of mercury

- iii. **Vacuum distillation:** Mercury obtained from the above process still contains some impurities like Au, Ag, etc. So mercury thus obtained is subjected to distillation under reduced pressure (vacuum distillation), where mercury gets distilled off leaving behind the impurities.

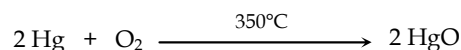
18.4.4 Physical Properties of Mercury

- Mercury is a liquid metal at room temperature.
- It is silvery-white in colour.
- It solidifies at -38.85°C and boils at 357°C .
- Mercury vapours are deadly poisonous.
- It is a poor conductor of heat and good conductor of electricity.

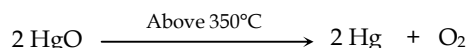
18.4.5 Chemical Properties of Mercury

1. Action of air:

- i. Under ordinary conditions mercury does not react with air.
- ii. When it is heated with air at about 350°C, it gives mercuric oxide. It forms red scum of HgO on the surface of liquid mercury.

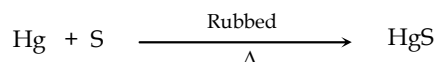
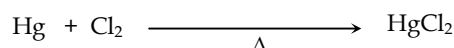


But if temperature exceeds 350°C, the oxide is decomposed into metallic mercury.



2. Action of water and alkali: It does not react with water and alkali (Difference with zinc)

3. Action of non-metals: Mercury reacts with non metals like sulphur and halogens to give their salts.

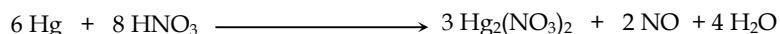


4. Action of acids: Mercury does not displace H₂ from dilute acids because it lies below hydrogen in electrochemical series.

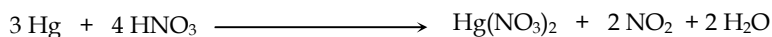
- i. With hot conc. H₂SO₄, mercury is oxidized to HgSO₄.



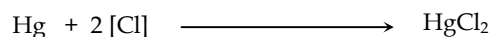
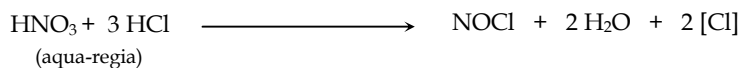
- ii. With dilute HNO₃, mercury is oxidized to mercurous nitrate.



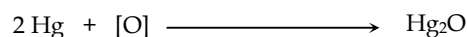
- iii. With conc. HNO₃, mercury is oxidized to mercuric nitrate.



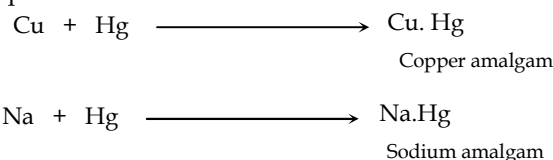
- iv. With aqua-regia, mercury gives mercuric chloride.



5. Action of O₃: When mercury comes in contact with O₃, mercurous oxide is formed. So, the meniscus of mercury gets depressed (lowered) and it leaves a tail when allowed to run through an inclined surface. This is called **tailing of mercury**.



6. **Formation of amalgams:** Mercury can form amalgams with almost all metals. In fact these metals dissolve in mercury. Iron does not form amalgam directly with mercury. Therefore, mercury is transported in iron vessels.



18.4.6 Uses of Mercury

1. Mercury is used as thermometric and barometric liquid.
2. It is used in the manufacture of caustic soda (eg, Castner-Kellner process).
3. It is used in the metallurgy of silver and gold (amalgamations process)
4. It is used in the mercury vapour lamps. The lamps impart bluish light rich in UV rays.
5. It is used in the preparation of various amalgams.
6. It is used in making dental plastic.
7. Mercury fulminate is used as detonator.

18.4.7 Mercury Poisoning: Health Hazards and Environmental Pollution

Mercury and many of its compounds are poisonous substances. They cause instant as well as long-term effects.

1. If mercury vapors are inhaled, it causes brain and lungs damage.
2. Organometallic compounds have been used as fungicides in agricultural fields or in paper industries. But exposure of these compounds may cause the possibility of its penetration in brain via gastrointestinal path. This causes brain damage, loss of vision or even death. A person may suffer from psychotic behaviour or even death.
3. Hundreds of Japanese in Mina Mata where killed due to the consumption of fishes which suffered from mercury poisoning. Severe health problems were observed in newly born babies. This disease was later on named as "Minamata disease".
4. Mercurous pesticides have been mostly banned these days.
5. Penetration of mercury can take place through skin also.

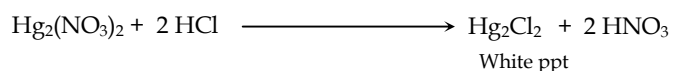
18.4.8 Calomel (Mercurous chloride or Mercury(I) chloride), Hg_2Cl_2

Calomel is a non-poisonous mercurous compound. It contains Hg in dimer form as $[\text{Hg}^+-\text{Hg}^+]$ ion. There is a covalent bond between two atoms of Hg. So, its molecular formula is Hg_2Cl_2 rather than HgCl .

Preparation

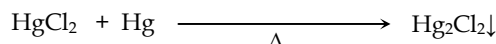
1. From mercurous nitrate

Mercurous chloride is prepared by treating mercurous nitrate with water soluble chloride such as hydrochloric acid. This is the laboratory method of preparation of calomel.

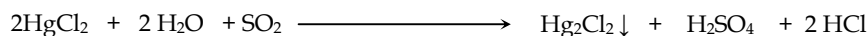
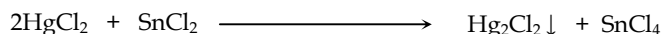


2. From mercuric chloride:

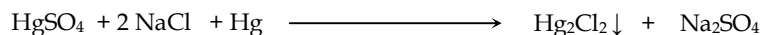
- a. It can be prepared by heating mercuric chloride with mercury. This method has been used to prepare calomel for medical purpose.



- b. When mercuric chloride is reduced with SnCl_2 or SO_2 , calomel is obtained.

**3. From mercuric sulphate**

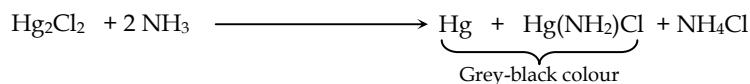
It can be prepared by grinding a mixture of mercuric sulphate, sodium chloride and mercury.

**Physical properties**

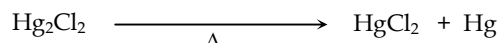
1. Calomel is a white powder insoluble in water.
2. It sublimes at a temperature more than 350°C .
3. It is non-poisonous in nature.

Chemical properties

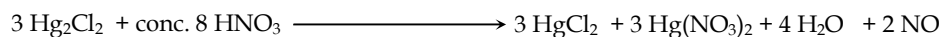
1. **Action of ammonia:** Calomel reacts with ammonia, to form grey black product due to the formation of mixture of finely divided Hg (black) and amido-mercuric chloride (white).



2. **Action of heat:** When strongly heated, calomel is decomposed to mercuric chloride.



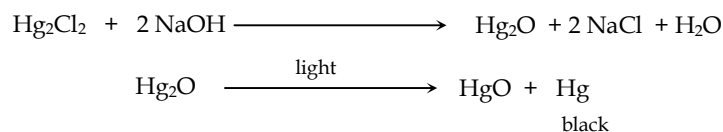
3. **Oxidation:** Calomel is oxidized by the oxidizing agents like Cl_2 , KClO_3 , HNO_3 , etc. to give mercuric chloride.



4. **Reduction:** Calomel is reduced to metallic mercury using reducing agent like stannous chloride.



5. **Action of sodium hydroxide:** Calomel reacts with aqueous sodium hydroxide solution to give grey black precipitate of mercurous oxide which gets decomposed to mercuric oxide and mercury in presence of sunlight.



Uses of Calomel

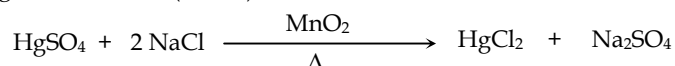
1. Calomel is used as purgative drug.
2. It is used as an antiseptic drug also.
3. It is used as a bed pill along with Na_2CO_3 .
4. It is used to make calomel electrode which is a secondary reference electrode.

18.4.9 Corrosive Sublimate (Mercuric Chloride or Mercury (II) Chloride), HgCl_2

Mercuric chloride is called corrosive sublimate. Corrosive sublimate is poisonous compound of mercury (II) state. It can easily sublime which is highly corrosive in nature. Therefore, it is called corrosive sublimate.

Preparation

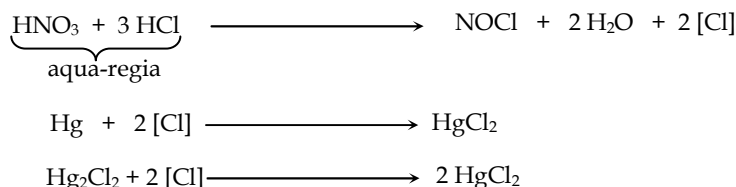
1. **From mercuric sulphate:** When mercuric sulphate is heated with common salt (NaCl) in presence of manganese dioxide (MnO_2), corrosive sublimate is obtained.



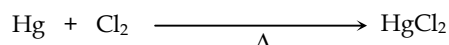
Here, MnO_2 prevents the formation of Hg_2Cl_2 . Thus obtained HgCl_2 sublimes and is collected at the cooler parts of the apparatus. This is commercial method of its preparation.

2. **From mercuric chloride:**

- a. When mercury or Hg_2Cl_2 is treated with aqua-regia, corrosive sublimate is obtained.

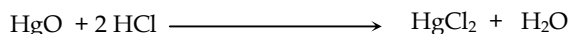


- b. When mercury is heated with excess chlorine, corrosive sublimate is obtained.



3. **From mercuric oxide:**

Corrosive sublimate can be obtained by dissolving mercuric oxide with dil. HCl . Mercuric oxide is obtained by the reaction of Hg^{2+} compound with NaOH .

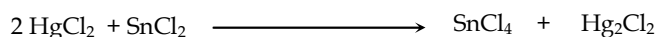


Physical properties

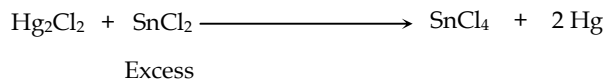
1. Corrosive sublimate is white crystalline solid and sparingly soluble in cold water.
2. It is soluble in hot water.
3. It is extremely poisonous. Its antidote is "white of egg" (white of egg is coagulated by HgCl_2 and is eliminated with HgCl_2 on it).

Chemical properties

1. **Reduction:** Corrosive sublimate is reduced by excess SnCl_2 to calomel and then to Hg.

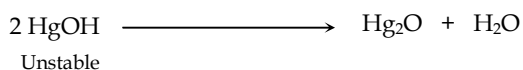
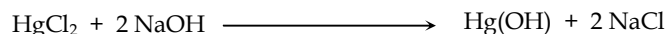


and

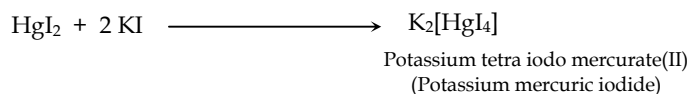
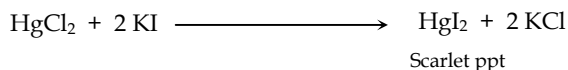


Mercuric chloride can also be reduced by SO_2

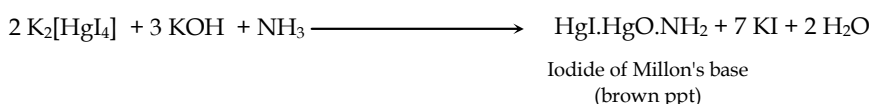
2. **Action of alkali:** When HgCl_2 is treated with sodium hydroxide, mercuric oxide is formed.



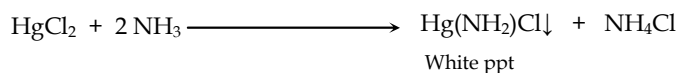
3. **Action of potassium iodide (KI):** When corrosive sublimate is treated with potassium iodide, scarlet precipitate of mercuric iodide (HgI_2) is obtained which dissolves in excess KI to give colourless water soluble complex of potassium mercuric iodide.



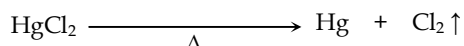
Alkaline solution of $\text{K}_2[\text{HgI}_4]$ is called **Nessler's reagent**. This reagent is used to detect NH_4^+ or NH_3 in a solution.



4. **Action of ammonia:** When corrosive sublimate is treated with ammonia, white precipitate of mercuric amino chloride is obtained.



5. **Action of heat:** On strong heating mercuric chloride gets decomposed into free mercury and chlorine.



Uses of Corrosive Sublimate

1. Corrosive sublimate is used as fungicide in agricultural fields.
2. It is used as preservative of timber and leather.
3. It is used for the preparation of Nessler's reagent and calomel.
4. Its very dilute solution is used to sterilize the surgical instruments.
5. It is used in the preparation of dry cells.



Questions

Very Short Answer Questions:

- Write an important use of corrosive sublimate
- Write an important use of calomel
- Given a balanced chemical equation for the preparation of corrosive sublimate. What is its action with KI solution?
- What is Nessler's reagent? How is it prepared?
- How is corrosive sublimate converted into calomel and vice-versa?
- How is Nessler's reagent prepared? Give its one use.
- What is meant by mercury poisoning?
- Give the molecular formula and one use of the following compounds:
(a) Nessler's reagent (b) Calomel (c) Corrosive sublimate
- Write the action of (i) Mercury with aqua regia (ii) tin with caustic soda
- What happens when HgCl_2 is allowed to react with SnCl_2 ?
- What happens when mercuric chloride reacts with excess of KI solution?

Short /Long Answer Questions:

- How is mercury extracted from Cinnabar?
- Write one method for the preparation of HgCl_2 and Hg_2Cl_2 each. What happens when an aqueous solution of HgCl_2 is treated with:
 - an aqueous solution of KI
 - an aqueous solution of SnCl_2
- Write short notes on chemistry of calomel
- Write two methods for the preparation of corrosive sublimate. Give its action upon.
(a) excess of KI (b) NH_3 .
- How is calomel prepared? Give its important properties & uses.
- Write Short Notes on
 - Extraction of mercury.
 - Chemistry of Calomel.
 - Purification of mercury.



Multiple Choice Questions

- Mercury can be stored in
 - Iron
 - Copper
 - Zinc
 - Tin
- Mercury tree consists . . .
 - Hg
 - HgO
 - Ag-Hg
 - AgNO_3
- Nessler's reagent is
 - $\text{K}_2[\text{HgI}_4]$
 - acidic $\text{K}_2[\text{HgI}_4]$
 - Alkaline $\text{K}_2[\text{HgI}_4]$
 - Neutral $\text{K}_2[\text{HgI}_4]$
- The sublimation product of mercuric sulphate, sodium chloride and mercury is . . .
 - Mercurous sulphate
 - Mercurous chloride
 - Mercuric chloride
 - Mercurous chloride and sodium sulphate
- Antidote of . . . is white of egg
 - Hg_2Cl_2
 - HgCl_2
 - Hg
 - Hg and HgCl_2

Answer Key

1	2	3	4	5
a	c	c	b	b

18.5 Iron

18.5.1 Introduction

Iron has been known since long time. The word iron is derived from scandinavian word. The symbol Fe is derived from the latin word *ferrum*. In Nepali iron is called *Falam* (फलाम). The use of iron and steel by human being is from immemorial time.

Symbol	Fe
Atomic number	26
Atomic weight	55.85 amu

²⁶ Fe	²⁷ Co	²⁸ Ni
⁴⁴ Ru	⁴⁵ Rh	⁴⁶ Pd
⁷⁶ Os	⁷⁷ Ir	⁷⁸ Pt

Position in the periodic table,

Period: 4

Group: 8 or VIII

Block : d-block (3d series)

Electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ or $[\text{Ar}] 4s^2 3d^6$

18.5.2 Occurrence

Iron is widely distributed in the earth's crust. It is easily attacked by moisture and air which makes it difficult to find in native state. It is mostly found in the combined states, such as oxides, sulphides and carbonates.

Oxide ore	(i)	Haematite	Fe_2O_3
	(ii)	Limonite (Brown haematite)	$2 \text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$
	(iii)	Magnetite	Fe_3O_4
Sulphide ore	(iv)	Iron pyrite	FeS_2
	(v)	Copper pyrite	CuFeS_2
Carbonate ore	(vi)	Siderite (spathic ore)	FeCO_3

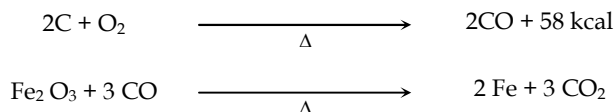
Iron forms essential constituent in the haemoglobin of blood.

18.5.3 Extraction of Iron

Haematite is the main ore of iron. The following steps are employed to extract iron from its main ore, haematite.

Theory

Ferric oxide present in the concentrated and calcined ore is reduced with the help of coke (C) to get iron in molten state.



The molten iron thus obtained is cooled to get pig iron. The pig iron is remelted and poured into moulds to obtain iron in desired shapes. Such iron is called cast iron.

Steps involved in the metallurgy

Extraction of iron from its ore such as oxides or carbonates is carried out in the following steps.

- Crushing and pulverisation:** The big lumps of iron ore are broken down into small pieces by using jaw crusher or hammer. Then, the pieces are further converted into fine powder with the help of ball mill or stamp mill.
- Dressing and concentration:** The pulverised ore is washed with water and concentrated by gravity separation method.

This process is based on the difference in the density of the ores and its impurities. The pulverised ore is washed with running stream of water where by heavy particles settle down washing the lighter impurities. If iron is to be extracted from its haematite ore, gravity separation process applied for concentration (fig. 18.17).

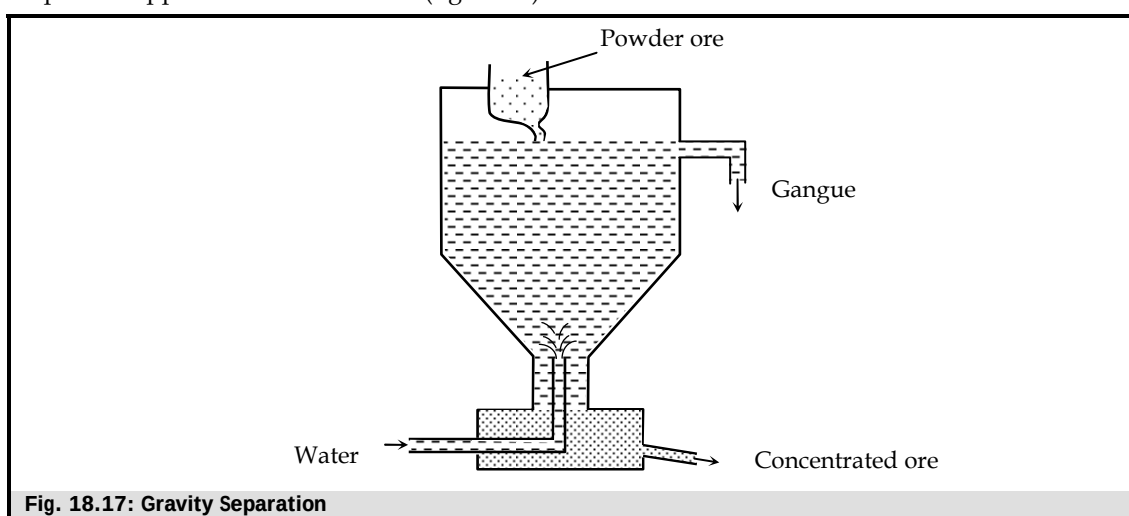
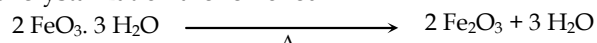


Fig. 18.17: Gravity Separation

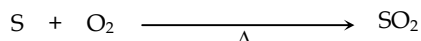
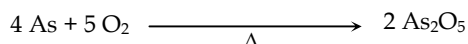
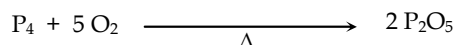
Oxide ores which contain little impurities can be used directly without concentration.

- Calcinations:** The concentrated ore is heated with some amount of coal till redness in a shallow kiln. Excess of air is admitted to oxidise volatile impurities. During this process, following changes take place.

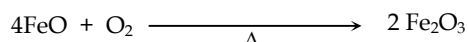
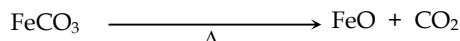
- Moisture and water of crystalization are removed



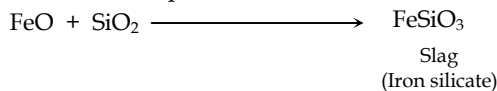
- Volatile impurities such as P, As, S etc are oxidised to their corresponding oxides and these are expelled.



- If iron carbonate is present, it decomposes to form ferrous oxide. The ferrous oxide may be converted into ferric oxide.



Here, it is to be noted that there is probability of combination of ferrous oxide with silica to form a slag of ferrous silicate and loss of iron takes place.



Fortunately conversion of ferrous oxide into ferric oxide prevents the formation of ferrous silicate (slag) and thus preventing loss of iron.

During calcination, water vapour and carbon dioxide make the whole mass porous which accelerate, the reduction of iron in successive step.

4. **Extraction of crude iron:** The calcined or roasted ore is reduced to iron in a blast furnace by **smelting or carbon reduction process**.

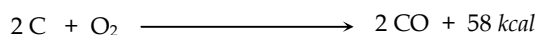
The blast furnace is cylindrical tower with 30 - 40 meter in height (fig. 18.18). The broadest part of the furnace is called **bosh** and upper part is called body. It is about 6 meter wide. The blast furnace is externally lined with fire-bricks. Hot blast of air is admitted under pressure into the furnace through water cooled narrow tubes called tuyeres. These tuyeres are fitted above 2 - 3 meter from the base. Since, blast of air is passed into the furnace, it is called blast furnace. The waste gas can be recirculated for heating purpose.

The lime-stone, desulphurised coke and calcined or roasted ore is mixed in 1:4:8 ratio. The mixture, also called charge, is fed from the top of furnace through the cup-cone arrangement. Here, coke acts as reducing agent and CaCO_3 acts as flux.

Different zones of the blast furnace possess different temperature and different chemical reactions take place accordingly. On the basis of range of temperature and nature of chemical reactions, the region of the blast furnace can be classified into the following zones.

a. **Zone of combustion (1300 - 1500 °C)**

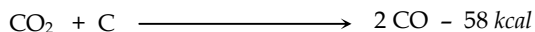
This region lies near the hearth of furnace. Combustion of carbon takes place in this zone and being exothermic, large amount of heat is produced.



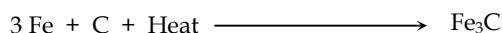
Temperature reaches upto 1500°C in this zone. This is the hottest region of blast furnace because heat energy is produced due to the exothermic reaction.

b. **Zone of fusion (1200 - 1300 °C)**

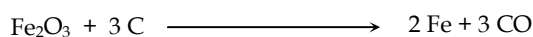
This region lies above the zone of combustion. Carbondioxide gas produced in the zone of combustion rises up and meets with hot coke and carbondioxide is reduced to carbon monoxide as

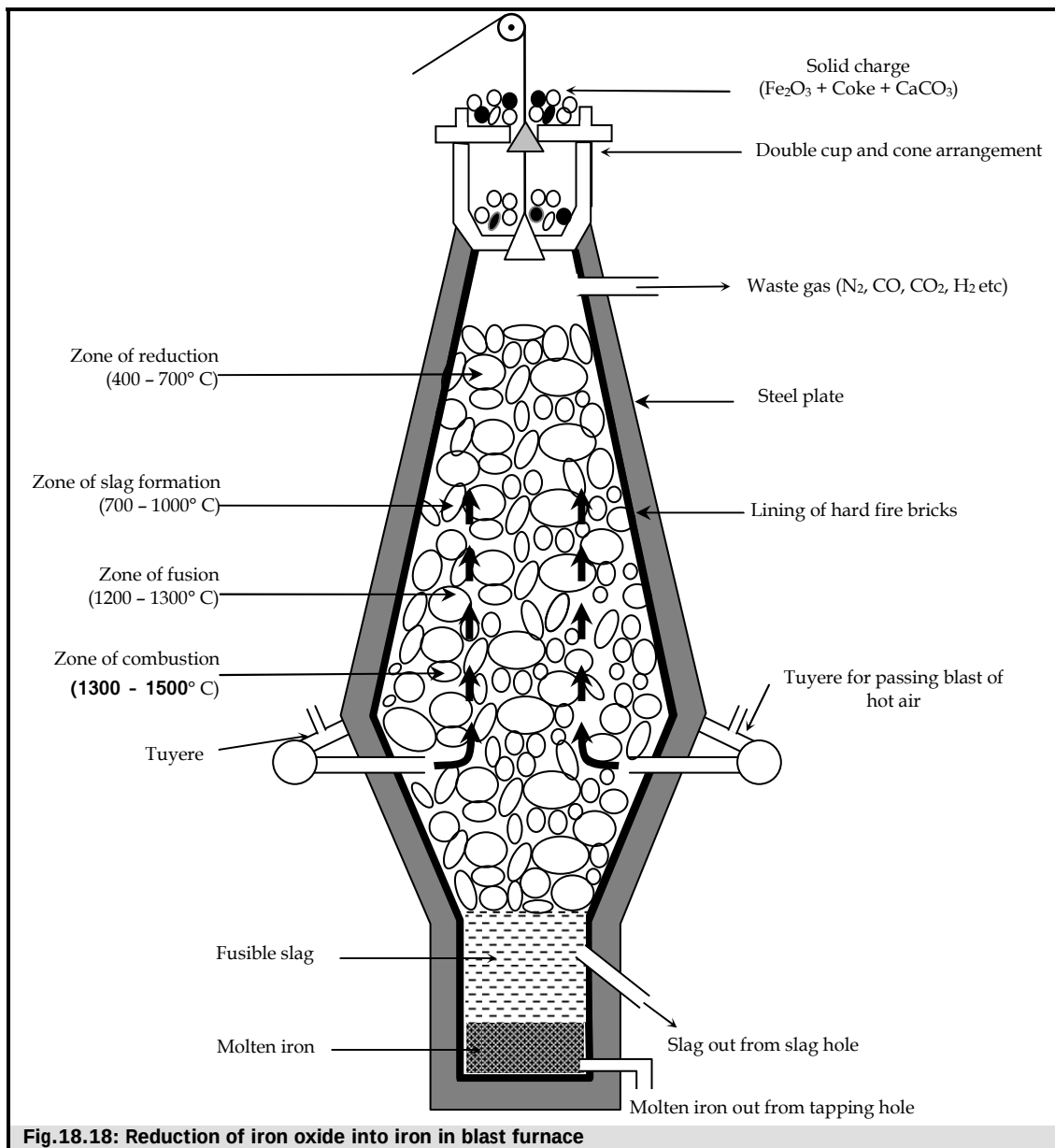


Since the reaction is endothermic, the temperature of this zone falls within the range of 1200°C to 1300°C. The spongy (porous) iron sliding from the top of the blast furnace is fused in this region. Therefore, this region is called zone of fusion. The molten iron may absorb carbon, phosphorus, silicon, sulphur at the region. Iron combines with carbon to form cementite.



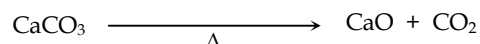
If ferric oxide (Fe_2O_3) is present, it is reduced in this zone to give iron.



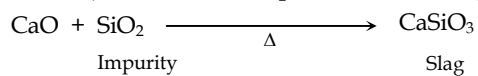


c. **Zone of slag formation ($700 - 1000^\circ \text{C}$)**

This region lies almost in the middle part of furnace. Limestone present here is decomposed according to following reaction.

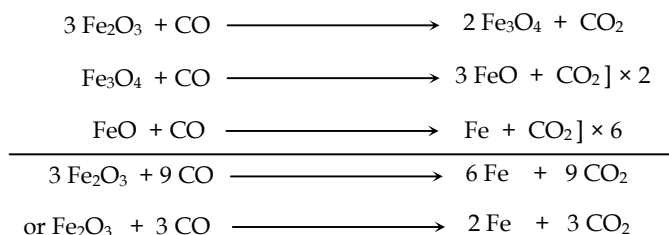


The temperature of this zone is about 700°C to 1000°C . The lime produced by the decomposition of CaCO_3 combines with silica (which acts as impurities in the ore) to form slag.

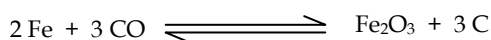


d. Zone of reduction (400 – 700°C)

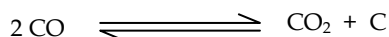
This region lies near about the top of the furnace. As the charge slides down, the temperature rises. The temperature of this zone is about 400°C to 700°C. Carbon monoxide coming from the bottom of the furnace reduces ferric oxide into iron. Thus obtained iron is spongy iron because the available temperature of this zone is insufficient to melt the iron. Here, it is to be noted that carbon monoxide is effective reducing agent rather than carbon under the condition. The reduction of ferric oxide takes place as follows:



In this reaction, decomposition of carbon monoxide may be carried out by spongy iron to some extent.



Besides this, decomposition of CO takes place at bright red heat.



Therefore, there exist equilibrium between carbon and its oxide.

Due to high temperature, calcination of ore may take place whereby moisture and water of crystallization are removed from the top of the furnace.

In this way, spongy iron is obtained from the zone of reduction and at the zone of fusion. It may contain carbon, phosphorus, silica as impurities.

The fused iron and molten slag fall into the lower part of the furnace or hearth whereby these form separate layers. Molten iron being heavier, forms lower layer and slag floats over the molten iron. These two liquids are collected separately. The molten iron is allowed to solidify in a big lump which is called **pig iron**.

The pig iron is remelted and poured into mould to obtain in desired shape which is called **cast iron**. Cast iron is not pure form of iron. It may consist 2.5 to 4.5 % carbon along with little quantity of Si, P, S, Mn etc. as impurities.

18.5.4 Different forms of Iron

On the basis of carbon content as the major impurity, iron is classified into following types.

- i. Cast iron
- ii. Steel
- iii. Wrought iron

Out of these varieties of iron, wrought iron is purest commercial form. It consists up to 0.25% of carbon while cast iron consists 2 to 4.5% carbon. Steel is an intermediate variety between cast iron and wrought iron. The comparison of these three forms of iron is given below:

Property	Cast Iron	Steel	Wrought Iron
1. Carbon content	2 – 4.5 %	0.2 – 2 %	up to 0.25%
2. Melting point	1200°C	1300 – 1400°C	1500°C
3. Malleability	Brittle	Malleable and brittle	Malleable
4. Magnetization	Can't be magnetised permanently	Can be magnetised permanently	Can be magnetised temporarily
5. Rusting	Does not rust easily	Depend upon the quality of steel	Rusts easily
6. Welding	Cannot be welded easily	Can be welded	Can be welded
7. Hardness	Hard	Depends upon carbon content	Soft
8. Tempering	Can't be tempered	Can be tempered	Can't be tempered
9. Uses	Casting metal objects like pipes, heavier parts of machinery	To prepare knives, swords, spring etc	To prepare chains and core of magnet

In cast iron, carbon is present in the form of graphite or in the form of cementite (Fe_3C). The cast iron may be white or grey in colour.

If molten iron is suddenly cooled most of the carbon will be present in cementite form. This variety of iron is very hard and brittle which is known as **white cast iron**. On the other hand, if the molten iron is cooled slowly, most of the carbon will be present in the form of graphite. This variety of iron is relatively soft and is known as **grey cast iron**. Cast iron is most fusible form of iron having melting point around 1200°C. The molten cast iron solidifies with slight expansion and it is used for casting metal objects.

18.5.5 Properties of Iron

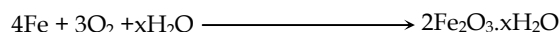
Physical properties

1. Pure form of iron is soft and white solid. Presence of carbon makes iron hard. Ordinary iron is grey and crystalline. This is due to presence of carbon.
2. Specific gravity of iron is 7.86.
3. Its melting point is 1530°C and boiling point is 2750°C.
4. Iron is magnetic substance and it losses this property if heated at about 760°C.
5. Iron is good conductor of heat and electricity.
6. It is malleable and ductile. It has high tensile strength.

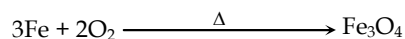
Chemical properties

1. Action of air:

Pure iron is not affected by dry air. When impure iron is exposed to moist air, iron is oxidised to its hydrated oxide or hydroxide or both. It forms a brown layer on the surface of iron which is called rust.

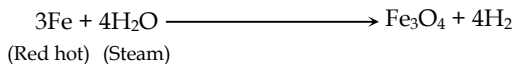


Finely divided iron when heated with oxygen yields, ferrosferric oxide.



2. Action of water:

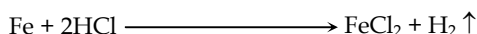
Iron is not affected by cold water. Red hot iron decomposes steam to give hydrogen gas.

**3. Action of acid:**

Iron lies above hydrogen in electrochemical series. It liberates hydrogen gas from dilute, mineral and non-oxidising acid.

(i) Action with HCl

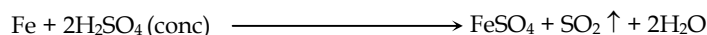
Iron reacts with hydrochloric acid to liberate hydrogen gas

**(ii) Action with H₂SO₄**

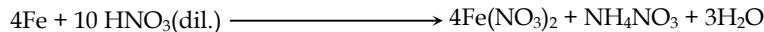
- ♦ Iron reacts with dilute sulphuric acid to liberate hydrogen gas



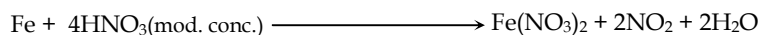
- ♦ But iron reacts with conc sulphuric acid to liberate sulphurdioxide gas

**(iii) Action with HNO₃**

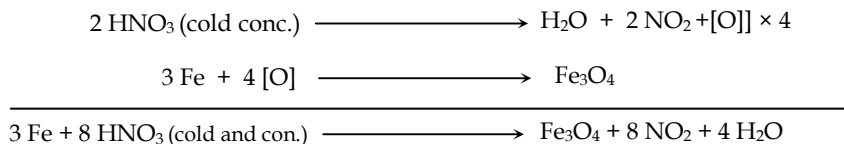
- ♦ Iron reacts with dilute nitric acid to produce ammonium nitrate as one of the products.



- ♦ Iron reacts with moderately conc. nitric acid to produce nitrogen dioxide gas as one of the products



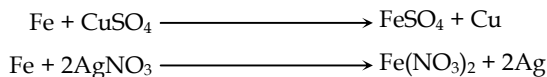
- ♦ Hot and conc nitric acid makes iron passive. When Fe reacts with cold and conc. HNO₃ an extremely thin (almost invisible) film of ferroso-ferric oxide (Fe₃O₄) is formed on the surface of iron and iron becomes passive (exhibit internees). This phenomenon is called as **passivity of iron**. The inertness exhibited by a metal under the condition of expected chemical activity is called passivity. Besides conc nitric acid, oxidising agents such as KMnO₄, K₂Cr₂O₇ also make iron passive.

**(iv) Reaction with alkali**

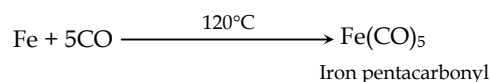
Iron does not react with alkali

(v) Displacement reaction

Iron can displace the metal lying below it in electrochemical series from their salts.

**(vi) Reaction with carbon monoxide**

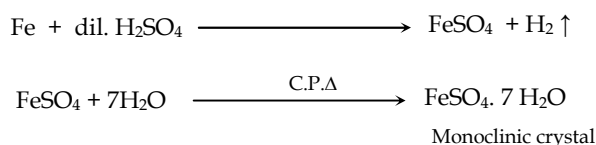
When finely divided iron is heated with carbon monoxide at about 120°C, iron pentacarbonyl is formed.

**18.5.6 Some Compounds of Iron****A. Ferrous Sulphate or Green Vitriol ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)**

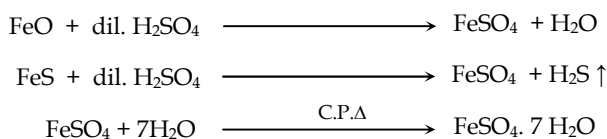
It is one of the most important compounds of iron. Ferrous sulphate heptahydrate is also called green vitriol. The compound is also called **copperas** or melanterite, which is obtained as oxidation product of pyrites in presence of air and water.

Methods of Preparation

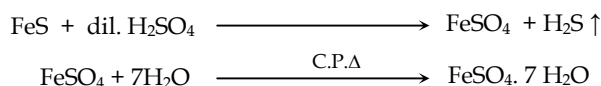
1. **From Iron:** When iron is treated with dilute sulphuric acid, ferrous sulphate solution is obtained which when heated up to crystallization point (C.P.) yields crystals of green vitriol.



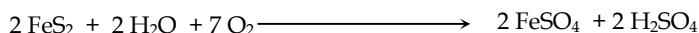
2. **From ferrous oxide or iron carbonate or iron (II) sulphides:** When iron oxide or iron carbonate or iron sulphides are treated with dilute sulphuric acid, iron(II) sulphate solution is formed which when heated up to crystallization point yields crystals of green vitriol.



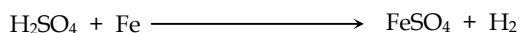
3. **From the waste of Kipps apparatus:** The waste solution obtained from the Kipps apparatus during the preparation of H_2S gas is taken, filtered and evaporated to crystallization which produces monoclinic crystals of green vitriol.



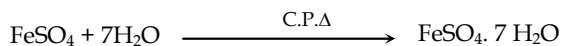
4. **From iron pyrites:** Green vitriol is manufactured from iron pyrite (FeS_2). The iron pyrite is exposed to moist air and it is allowed to undergo slow oxidation. Ferrous sulphate is formed along with free sulphuric acid.



The free acid is neutralized by adding some amount of scrap iron.



The ferrous sulphate solution is heated up to crystallization point to obtain monoclinic crystals of green vitriol.



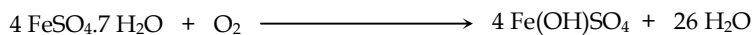
Properties of Green Vitriol

A. Physical Properties

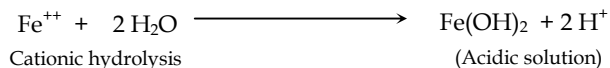
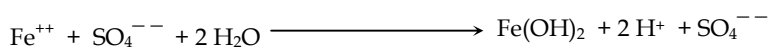
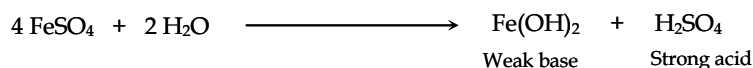
1. It is pale green crystal isomorphous with $\text{MSO}_4 \cdot 7\text{H}_2\text{O}$ where $\text{M} = \text{Co}, \text{Ni}, \text{Cr}, \text{Mn}, \text{Mg}, \text{Zn}$ etc.
2. It effloresces on exposure to air.
3. It has monoclinic crystal structure i.e. $a \neq b \neq c$, $\alpha = \gamma = 90^\circ$ and $\beta \neq 90^\circ$

B. Chemical properties

1. When green vitriol is exposed to air, the light green colour changes to brownish due to its oxidation.



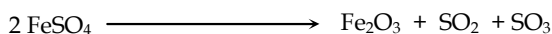
2. Green vitriol crystals are soluble in water and its aqueous solution is acidic in nature due to cationic hydrolysis.



3. **Action of heat:** When green vitriol is heated below 300°C , six molecules of water are lost at 300°C to give anhydrous FeSO_4 , which is white in colour.

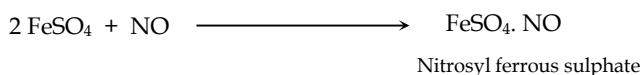


At red hot in vacuum at 140°C , ferrous sulphate decomposes to ferric oxide.

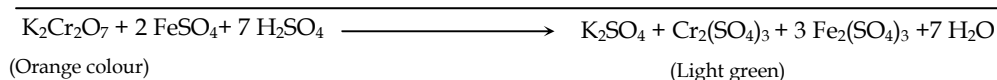
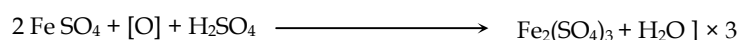
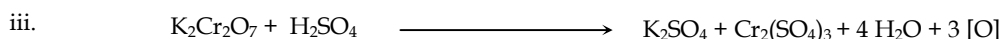
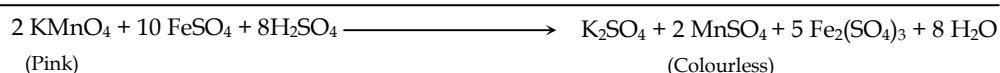
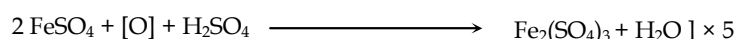
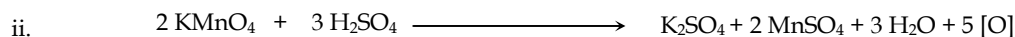
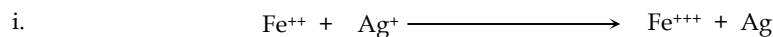


Fe_2O_3 thus formed is used as a pigment under the name of **Rouge** or **Venetian Red**.

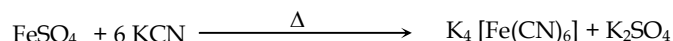
4. **Action with NO:** FeSO_4 solution absorbs NO to give brown colour compound.



5. **Reducing behaviour:** Ferrous sulphate is good reducing agent. In acidic medium, it reduces silver ion, acidified KMnO_4 solution and acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution.



6. **Action of KCN:** (KCN is deadly poisonous!) When excess of KCN is added to ferrous sulphate solution, potassium ferrocyanide is obtained.



Uses of Green Vitriol

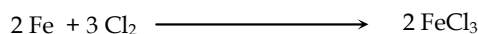
1. It is used in dyeing industry as mordant.
2. It is used in the preparation of writing ink (Mohr's salt).
3. It is used in tanning industries.
4. It is used as weedicide in agriculture.
5. It is used as reducing agent.

B. Ferric Chloride (FeCl_3)

Ferric chloride is one of the most important compounds of iron. Its chemical name is Iron (III) chloride. In vapour state, it exists as dimer Fe_2Cl_6 . Ferric chloride exists in anhydrous as well as hydrated form.

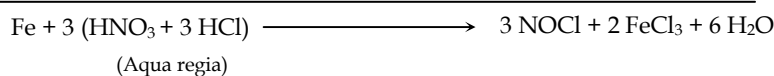
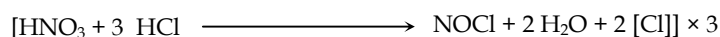
Preparation of anhydrous ferric chloride:

When dry chlorine gas is passed into heated iron in a hard glass tube, anhydrous ferric chloride is sublimed off at 315°C and is collected inside the bottle.

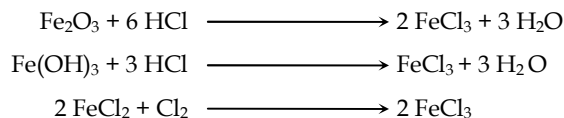


Preparation of hydrated ferric chloride

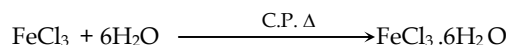
a. **From iron:** When iron is dissolved in aqua-regia then ferric chloride is formed



- b. **From ferric oxide or ferric hydroxide:** When ferric oxide or ferric hydroxide is reacted with dilute hydrochloric acid, ferric chloride is obtained



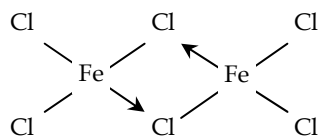
When ferric chloride solution is heated upto crystallization point then yellow crystals of ferric chloride are formed.



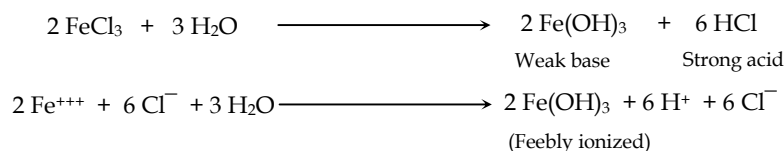
Properties of Ferric Chloride

A. Physical properties

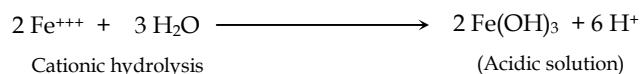
1. Anhydrous ferric chloride is black while hydrated ferric chloride is yellow in colour.
2. Ferric chloride is deliquescent crystalline solid.
3. At lower temperature ferric chloride exists as Fe_2Cl_6 while at 700°C the dimer dissociates to monomer.



4. Ferric chloride is highly soluble in water. Its aqueous solution is acidic in nature due to cationic hydrolysis



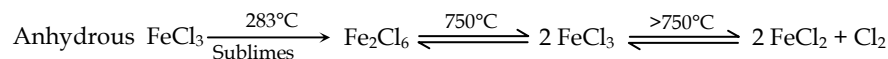
Omitting the common ion from both sides



Ferric chloride is also soluble in alcohol and ether because it is not ionic compound (it is covalent compound).

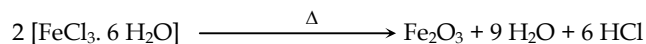
Chemical Properties

1. Action of heat:



At high temperature Fe_2Cl_6 decomposed into FeCl_3 . Above 750°C ferric chloride decomposes to FeCl_2 and Cl_2 .

On heating $\text{FeCl}_3 \cdot 6 \text{H}_2\text{O}$ it is changed into Fe_2O_3 , H_2O and HCl .

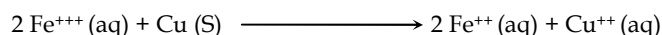


$$\text{FeCl}_3 + [\text{H}] \xrightarrow{\text{Zn} + \text{HCl}} \text{FeCl}_2 + \text{HCl}$$
$$2 \text{FeCl}_3 + \text{H}_2\text{S} \longrightarrow 2 \text{FeCl}_2 + 2 \text{HCl} + \text{S}$$
$$\text{FeCl}_3 + 6 \text{C}_6\text{H}_5\text{OH} \longrightarrow [\text{Fe}(\text{C}_6\text{H}_5\text{O})_6]^{--} + 3\text{H}^+ + 3\text{HCl}$$

Iron hexaphenoxide
(Violet colour)

$$2 \text{FeCl}_3 + 2 \text{KI} \longrightarrow 2 \text{KCl} + \text{I}_2 + 2 \text{FeCl}_2$$

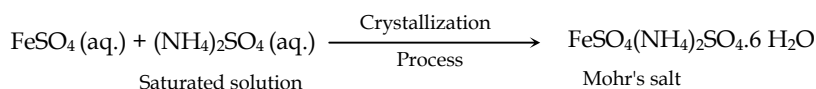
1. Ferric chloride is used as mordant in dyeing.
2. It is used as disinfectant.
3. Concentrated solution of ferric chloride is used for etching copper to prepare circuit board.



- ### C. Ferrous ammonium sulphate $[\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6 \text{H}_2\text{O}]$

Chemical name of Mohr's salt is ferrous ammonium sulphate hexahydrate.

When solution containing equimolar proportion of ferrous sulphate (with little dilute sulphuric acid) and ammonium sulphate is heated upto crystallization point (at about 210°C), monoclinic crystals of Mohr's salt are obtained.



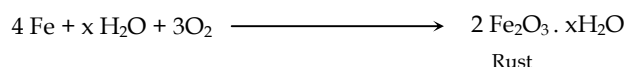
1. It is used to prepare standard solution of ferrous ion in volumetric analysis instead of FeSO_4 because Mohr's salt is readily soluble in water and does not effloresce. It is less readily oxidised by atmospheric air than does ferrous sulphate. Therefore, Mohr's salt is used as primary standard substance but ferrous sulphate is not.
2. It is used to standardise KMnO_4 solution.

18.5.7 Rusting of Iron

Metals are extracted from their ores at the expense of energy and pure metals can be regarded in higher energy state in comparison to their corresponding ores. When metals are used in various forms, they are likely to be attacked under certain atmospheric condition and tend to revert back into combined state. This forms the basis of corrosion.

The phenomenon of deterioration and consequent loss of solid metallic substance through unwanted chemical reaction or electrochemical attack under the set of environmental condition is called corrosion. If corrosion takes place in iron, it is specially called rusting.

When iron is exposed to moist air for a sufficient period, the surface of iron gets converted into a mixture of hydrated ferric oxides and ferric hydroxide which is called rusting and the amorphous brown scale obtained is called rust.



Factors affecting the rusting

Some of the factors which affect rusting are as follows.

- a. Presence of impurities of other metals
- b. Moisture and air
- c. Electrolytes
- d. Presence of certain gases such as O_2 , CO_2 , SO_2 , H_2S , Cl_2 etc .
- e. Strain in metal
- f. Small layer of rust (A small layer of rust further promotes rusting by auto catalysis)

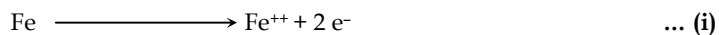
Theories of Rusting

There are different theories about rusting. Out of these theories, two important theories are explained below.

1. **Electrochemical theory:** This is modern theory. Commercial forms of iron are not in pure form. These have non-uniform surfaces with strains or impurities. Due to this, some portion of iron surface is at lower reduction potential and acts as anode while another portion of iron surface at higher reduction potential acts as cathode. In this way a kind of tiny voltaic cell is set up. Gases such as carbon dioxide, sulphur dioxide, hydrogen sulphides etc. dissolved in water and the solution is believed to act as electrolytic solution. Following electrode reactions take place on iron and produce rust.

Reaction at anodic area:

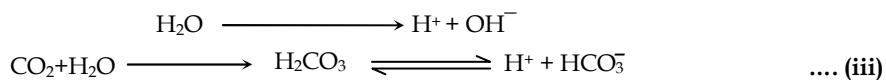
Oxidation of iron takes place at anodic area



Reaction at cathodic area: The electrons liberated at anodic area travel towards the cathodic area through conducting medium. These electrons are consumed at cathode either by the evolution of hydrogen or by the absorption of oxygen depending upon the pH of environment in acidic medium,



The hydrogen necessary for the reaction is obtained by the ionization of water or carbonic acid.

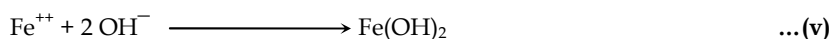


634 Modern Approach to Chemistry-II

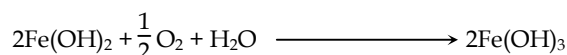
In neutral medium, reduction of oxygen takes place.



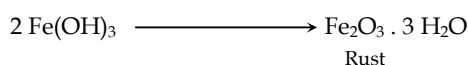
The Fe^{++} obtained from anode and OH^- obtained at cathode combine to form ferrous hydroxide.



In excess of oxygen and water the ferrous hydroxide is oxidized to ferric hydroxide.

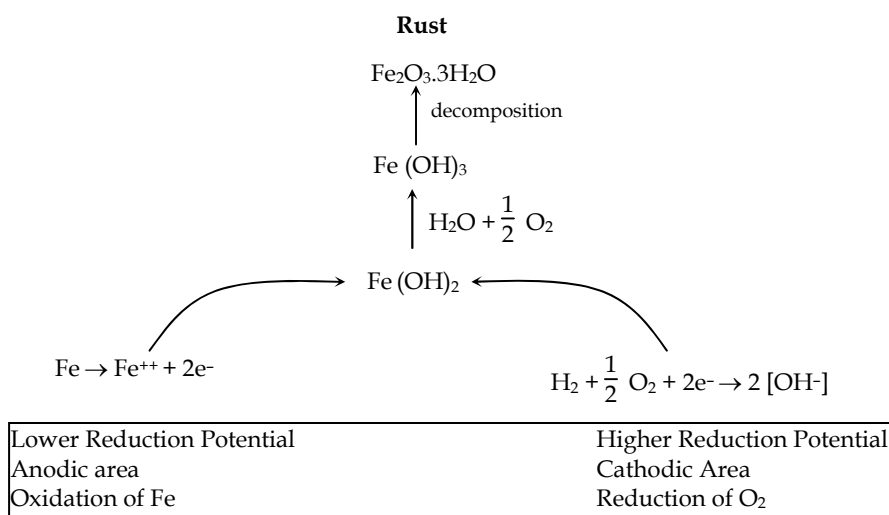


Being less stable, the ferric hydroxide undergoes decomposition to give hydrated ferric oxides, called rust.



However, in the limited presence of oxygen the corrosion product may be anhydrous black powder of composition of ferrosferric oxide (Fe_3O_4).

Summary of Rusting

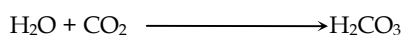


Evidence in the favour of the theory

Presence of Fe^{++} ions at anode and presence of OH^- ions at cathode have been detected by chemical tests.

2. **Chemical theory of rusting:** This theory was given by Calvert and Crum Brown in 1888. According to this theory, rusting of iron takes place in presence of water and carbon-dioxide on the surface of iron. The rusting of iron is considered to take place in the following steps.

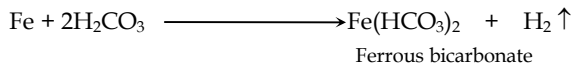
i. Formation of carbonic acid



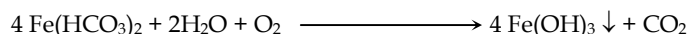
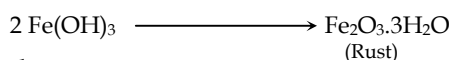
The atmospheric carbon-dioxide combines with moisture to form carbonic acid.

ii. Formation of ferrous bicarbonate

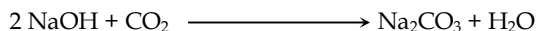
The carbonic acid attacks on the iron surface to form ferrous bicarbonate.

**iii. Oxidation of ferrous bicarbonate to ferric hydroxides.**

The ferrous bicarbonate gives precipitate in the form of ferric hydroxide in presence of water and oxygen.

**iv. Decomposition of ferric hydroxide into hydrated ferric oxide.****Evidence in the favour of this theory**

Alkali prevents the rusting of iron. Alkali combines with carbon-dioxide and prevents the formation of bicarbonate and hence rusting does not take place.



However, this theory fails to explain the rusting of iron in the absence of CO_2 .

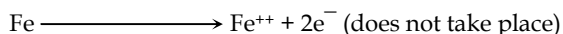
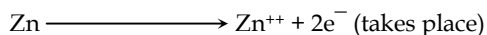
Prevention of Rusting

Iron forms the major constituent of building and industrial constituent material. Rusting of iron is a rampant problem in household as well as industrial material. Loss of iron due to rusting or corrosion can't be measured only in terms of cost of metal, it should be evaluated in terms of cost of maintenance and time factor too. To avoid the rusting any of the following processes may be adopted.

a. Cathodic protection or sacrificial protection: It is the best method for the prevention of rusting. In this method, the iron is kept in contact with more electropositive metal such as Zn, Al, Mg etc.

If iron is likely to be rusted, there should be the formation of electrochemical cell in which the more electropositive metal forms anode and corrodes while less electropositive iron does not corrode or rust. Here, the anode sacrifices itself and prevents cathode from being rusted. Therefore, the process is called sacrificial protection or cathodic protection.

A better way of sacrificial protection of iron is carried out by galvanization process in which iron surface is coated with thin layer of zinc. The galvanization process protects from rusting by following two ways.



- i. Zinc layer forms a barrier between iron surface and atmosphere.
 - ii. If the rusting environment is set up, it is not the iron but zinc which undergoes corrosion. This is due to lower reduction potential of zinc.
- b. By the formation of protective barrier:** Moisture and air are inevitable factors for rusting. If a coating is made on the surface of iron then it prevents the contact of iron and moist air. Hence, it prevents iron from being rusted. The following methods can be applied to develop protective barrier.
- i. Coating of oil and grease may be used on tools and machinery parts to prevent from rusting.
 - ii. Thin coating of paint, enamel, lacquer, varnishes may be used to prevent from rusting. (Barrier protection)

- iii. By passing steam on the red hot iron (Barf's protection).
- c. **By the formation of alloys:** If iron forms an alloys with nickel and chromium, the alloy becomes resistant to corrosion. Chromium is the best suitable alloying metal for iron because its film is self healing. The alloy has homogenous surface.
- d. **Environmental medication:** The rusting environment can be reduced by:
 - i. removing harmful constituents e.g. dehumidification
 - ii. adding specific substance which neutralize the effect of corrosive constituents of environment e.g. addition of alkaline substance to remove acidic character of corrosive environment.
- e. **Uses of corrosion inhibitors:** The chemical substance which when added to aqueous corrosive environment decrease the corrosion of metal by altering the surface reaction is called corrosion inhibitor. Chromates (e.g. K_2CrO_4), phosphates, tungsten, amine, heavy metal, soap etc are used as inhibitors.

Self Test

- 18.1 Do you agree with the statement that rusting is reverse process of extraction of metal?
- 18.2 Losses occurring due to corrosion can't be measured only in terms of cost of metal. Comment the statement.

18.5.8 Steel**Introduction****Fig. 18.19: Uses of steel for construction**

Steel is an alloy of iron containing 0.2 to 2% carbon as one of the constituents and traces of sulphur, phosphorous along with some metals. Plain-carbon steel has limited use. Hence, certain special steel alloys can be prepared by adding metals such as V, Cr, Mn, Co, Ni etc. The reasons for alloying steel are given below.

- i. To improve tensile strength, ductility, toughness, elasticity, heat and corrosion resistance.
This is usually obtained by adding calculated quantity of nickel.
- ii. To improve hardness
 - Chromium improves hardness and toughness both.
 - Addition of 1.0 to 1.5% manganese (Mn) increases strength, toughness and brittleness.
- iii. To improve resistance to abrasion, acid and chemical resisting property
 - Molybdenum increase corrosion resistance, abrasion resistance.
- iv. To improve shock-resistance.
 - Tungsten increase toughness, abrasion resistance and shock-resistance.

Some of the applications of special steels are given below:

- i. Steel containing 3.5% Mo is used as heat resisting steels.

- ii. Chromium is a universal constituent of all heat resisting steel because it increases corrosion resistance at high temperature. Heat resisting steel contains more than 12% chromium.

The amount of carbon in steel is intermediate to cast iron and wrought iron. Therefore, steel can be manufactured either by decreasing the carbon content of cast iron or by increasing the carbon content of wrought iron.

Manufacture of Steel

Steel can be manufactured by any of the following methods.

- Bessemer process. (From cast iron)
- Open hearth process (from cast and wrought iron)
- Cementation process (from wrought iron)
- Crucible process (from wrought iron)
- Electrical process (From pure ore and carbon)

Here, we discuss only Bessemer process and open hearth process:

a. Bessemer process

This process was developed by Henry Bessemer in 1856. The cast iron is taken in a pear shaped vessel called Bessemer converter. A number of holes are provided at the bottom of the converter to pass the blast of pre-heated air. Depending upon the nature of impurities the internal lining of Bessemer converter is made of acidic material such as silica (SiO_2) or basic materials such as dolomite. Accordingly, Bessemer process is two types.

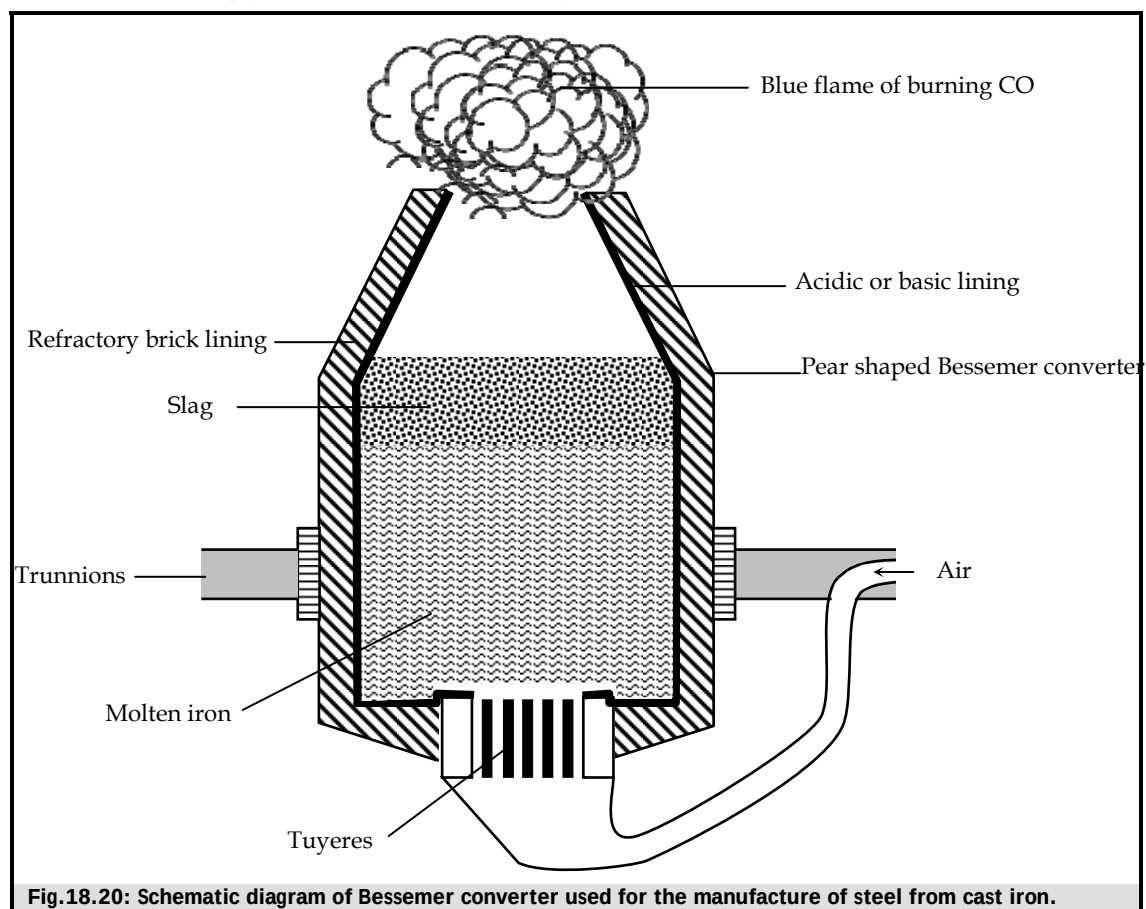
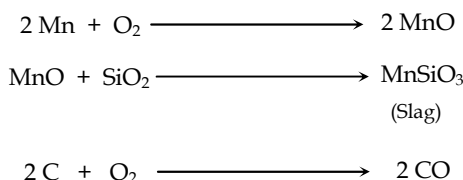


Fig.18.20: Schematic diagram of Bessemer converter used for the manufacture of steel from cast iron.

i. Acidic Bessemer Process

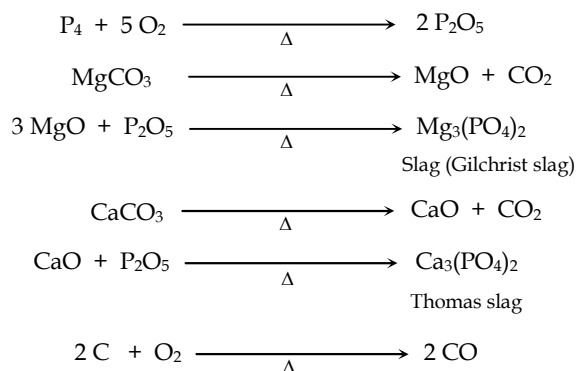
If cast iron used for the manufacture of steel consists basic impurities such as Mn and Si or the phosphorous content is low, acidic Bessemer process is adopted. In this process internal lining of converter is carried out by an acidic material such as SiO_2 . Cast iron obtained from the blast furnace is put into the converter and blast of pre-heated air is admitted through tuyeres at the bottom. The impurities of cast iron are oxidised and combine with SiO_2 to give slag.



The last to be oxidized is carbon which is oxidized to carbon monoxide with the production of blue flame at the mouth of the converter. When the flame dies out, it indicates the completion of oxidation of impurities. When the blue flame dies, calculated quantity of spiegeleisen¹ (Mn + Fe + C) is added and blast of air is continued so as to mix it thoroughly. Little aluminium or ferrosilicon is used to remove entrapped O_2 , NO_2 , CO_2 etc. Steel is poured out by tilting the converter.

ii. Basic Bessemer process

If cast iron used for the manufacture of steel consists acidic impurities or phosphorus content is high, basic Bessemer process is adopted. In this process, internal lining of Bessemer converter is made by basic material such as dolomite ($\text{CaCO}_3 \cdot \text{MgCO}_3$). Cast iron obtained from the blast furnace is put into Bessemer converter and blast of pre-heated air is admitted through tuyeres at the bottom. The phosphorus is oxidised to P_2O_5 which combines with basic lining to form slag.



The Gilchrist slag and Thomas slag can be used as phosphate fertilizer. If Si and Mn are present these are also oxidized. Carbon is last to be oxidized. When the impurities are completely combusted, calculated quantity of spiegeleisen (Mn + Fe + C) is added and a blast of air is passed for some time to ensure thorough mixing. Then steel is poured out by tilting the converter.

¹ Mn and C act as reducing agent and reduce FeO if present. Besides this, carbon is the essential part of steel while Mn increases hardness and tensile strength of steel.

Advantage of Bessemer process

- i. The process is quick to run and it completes within 15 to 20 minute.
- ii. It is relatively inexpensive.

Disadvantage of Bessemer process

- i. Quality of steel obtained by this process varies from batch to batch.
- ii. It can't remove the impurities completely. Thus produced steel rusts easily.
- iii. The process is too rapid to carry out analytical test of the product from time to time.

b. Open Hearth Process

This process was developed by Siemen and Martin. This is the modern process for the manufacture of steel. For the manufacture of steel, 70 - 80% cast iron, 20 - 30% scrap iron and little haematite is used. The charge is heated by passing producer gas ($\text{CO} + \text{N}_2$ in 1 : 2) and it works on the regenerative system of heat economy. In this process, oxidation of impurities is carried out by using haematite. Depending upon the nature of impurities, the open hearth process may be acidic process or basic process.

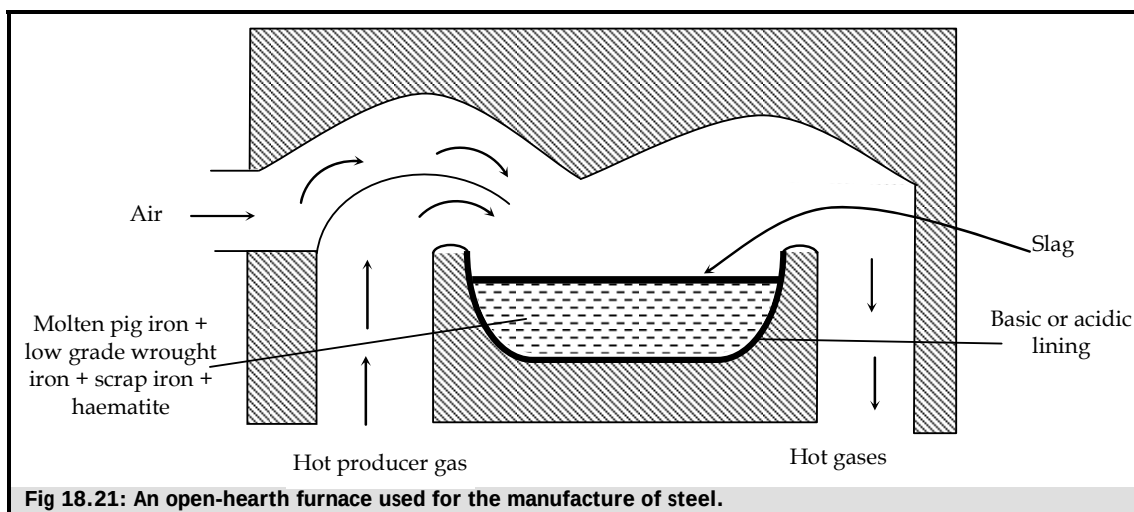
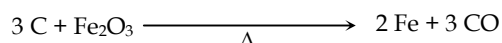
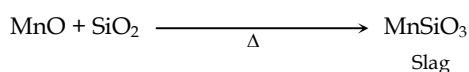
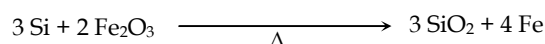
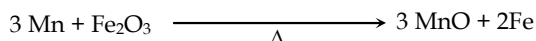


Fig 18.21: An open-hearth furnace used for the manufacture of steel.

Acidic Process

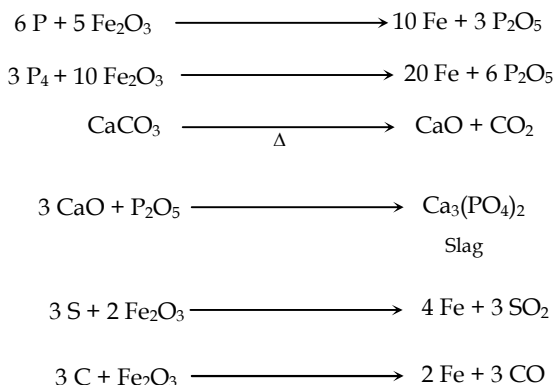
If impurities present in the iron are basic in nature such as Mn, Si etc or the phosphorus content is low, acid process is applied. In acid process, the hearth of the furnace is lined with acidic material such as SiO_2 . The impurities are oxidised by haematite ore.



After the impurities are removed, calculated quantity of spiegeleisen (Mn + Fe + C) is added to obtain desire quality of steel.

Basic Process

If impurities present in cast iron are acidic in nature and consist more quantity of phosphorus then basic process is adopted. In this process, the hearth of the furnace is lined with basic materials such as dolomite ($\text{CaCO}_3 \cdot \text{MgCO}_3$). The impurities are oxidised by haematite ores.



The samples are drawn from time to time and chemical test is carried out. When the test shows completion of process, the calculated quantity of spiegeleisen ($\text{Mn} + \text{Fe} + \text{C}$) is added and stirred thoroughly. The molten steel is taken out and poured into mould to prepare steel of desired shapes.

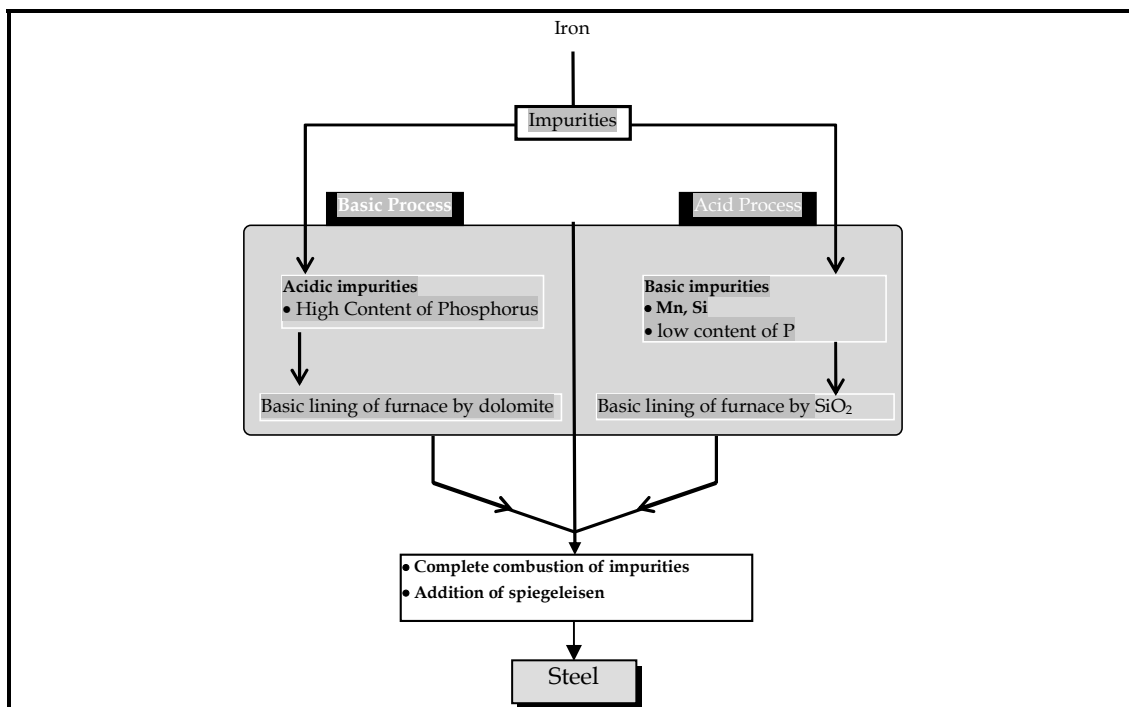
The open-hearth process has following advantages over Bessemer process:

1. It is modern process.
2. Loss of iron as slag is very little. (Loss of iron as slag in open hearth process is about 4% while that in Bessemer process is about 15%)
3. Temperature can be easily controlled because heat is supplied from external sources.
4. The overall process is carried out smoothly within about 8 to 10 hours. Therefore, the product obtained is remarkably uniform.
5. Scrap iron or steel can be used for the preparation of steel. Open hearth furnace works on the regenerative principle of heat economy so the overall process is economical.

Self Test

18.3 What are the basic differences between Bessemer process and open heart process?

18.4 What are the points of similarities between Bessemer process and open hearth process?



Heat Treatment of Steel

Properties of steel are mostly determined by carbon content, presence of other metals and heat treatment. Here heat treatment of steel is described in brief.

The properties of steel are largely determined by degree of heating or cooling and mode of casting.

The phenomenon of heating and cooling solid form of steel under controlled conditions so as to modify certain physical properties without changing chemical composition is called heat treatment of steel. Particularly, heat treatment of steel develops following changes in the steel.

1. Refinement of grain structure
2. Removal of imprisoned gases
3. Change in internal stresses.

Some heat treatment processes are given below:

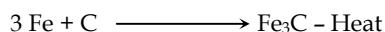
- a. **Hardening (Quenching):** The process of heating steel beyond critical temperature and cooling it suddenly (quickly) either plunging in oil or in water is called **hardening** or **quenching**. The steel thus obtained is called **hardened** or **quenched steel**.
 - Faster the rate of cooling, harder will be the steel produced.
 - Low carbon steel² cannot be hardened but medium and high carbon steel can be hardened.

² Steel can be classified as carbon steel and alloy steel. However alloy steel also consists carbon. On the basis of carbon content, carbon steel is further classified into low carbon -steel, medium carbon - steel and high carbon - steel.

Type of steel	Carbon content	Nature	Used in
A. Carbon Steel			
a. Low carbon-steel	Less than 0.2%	Malleable and ductile	Nail and Chain
b. Medium carbon-steel	0.2 – 0.6 %	Harder	Rail, steel beam
c. High carbon-steel	0.6 – 1.5 %	Hard and tough	Spring, tools and cutlery
B. Alloy steel	Alloy steel is that kind of steel which consists at least one other element besides carbon to give special properties. Some examples of alloy steels are stainless steel, invar, nickel steel, stainless steel, chrome steel etc.		

Hardening increases the resistance to wear, ability to cut other metals and brittleness of steel. Such types of steels are used for cutting tools.

- b. **Annealing (softening):** The process of heating steel to bright redness (200 – 350°C) followed by slow cooling is called **annealing**. The steel thus obtained is soft, weak and ductile. It removes strain in the steel. At high temperature iron forms cementite (Fe_3C) with carbon



On cooling slowly, Fe_3C is partly decomposed and carbon is spreaded uniformly in steel. Therefore, steel becomes soft and ductile.

- c. **Tempering of steel:** If hardened or quenched steel is heated to a temperature much below redness and then cooled slowly, it neither becomes too hard nor too brittle. The process is called **tempering of steel** and steel thus obtained is called **tempered steel**. Tempering is quenching followed by annealing of steel.

In the process of tempering, a thin film of oxide is formed on the surface of hardened steel. The colour of oxide film depends on temperature at which the hardened steel is heated.

Temperature range (°C)	200-230	225-270	300
Colour of oxide film	yellow	brown	blue

- d. **Case-hardening of steel:** When mild steel (steel containing low % of carbon) is heated with charcoal and then plunged into oil, a thin coating of hardened or quenched steel is produced on the surface of the steel, it is called **case hardening** and steel obtained is called **case-hardened steel**. It is resistant to tear and wear and is largely used in the manufacture of **armour plates, locomotive axles** etc.
- e. **Nitriding of steel:** When steel is heated in the atmosphere of NH_3 in sealed boxes for about 3-4 days, hard coating of iron nitride is produced on the surface of the steel. The process is called **nitriding of steel**.

Stainless Steel

It is a special type of alloy steel. It contains (12-14%) Cr and (1-2%) Ni. It resists corrosion, heat and action of **organic acids**.

Uses of stainless steel

1. It is used to make cutting tools (cutlery) knives, forks etc.
2. It is used to make cycle and automobile parts.
3. It is used to make food processing machinery.
4. It is used to make surgical instruments, ball bearings etc.



Questions

Very Short Answer Questions:

1. Why cannot anhydrous ferric chloride be prepared by heating hydrated ferric chloride?
2. What are corrosion and rusting?
3. What is the composition of spiegeleisen? Write its one use.
4. What is meant by quenching of steel?
5. What is the composition of stainless steel? Write its one use.
6. Differentiate between cast iron and wrought iron.
7. Why is open-hearth process more advantageous than Bessemer process for the manufacture of steel?

8. Write the molecular formula of:
 - (i) Mohr's salt
 - (ii) Green vitriol
9. Write chemical reaction occurring in zone of reduction in the extraction of iron.
10. How is red oxide of copper converted into black oxide and vice versa.
11. What is meant by tempering of steel? Mention one important use of tempered steel.
12. What reaction occurs in the combustion zone in the blast furnace during extraction of iron?
13. What is meant by plumbosolvency?
14. Mention the function of Spiegeleisen in the manufacture of Steel.
15. Name two important ore of iron.
16. What happens when: Few drops of potassium ferrocyanide are added to aqueous solution of FeCl_3 .
17. What is meant by galvanization?
18. Write an important use of wrought iron.
19. Write two methods by which rusting of iron can be prevented.
20. What is tempering of steel? What is its use?
21. What is stainless steel?
22. What is the function of limestone in the smelting of during the extraction of iron?
23. What is a transition element? Mention two of its important characteristics?
24. Write the principle involved in manufacture of steel by open hearth process.

■ Short /Long Answer Questions:

1. How is cast iron extracted from Iron pyrites?
2. Write short notes on manufacture of steel by open hearth process.
3. Draw a neat labelled sketch of blast furnace for the extraction of pig iron and write down the chemical reactions involved in different zones.
4. What is rusting? Describe electrochemical theory for the rusting of iron. List any two methods of prevention of rusting.
5. How is steel manufactured by open hearth process? Mention its advantage over Bessemer process.
6. Explain any four methods for preventing rusting of iron.
7. Write short notes on:
 - a. Theory of corrosion.
 - b. Rusting of iron and its prevention.
 - c. General properties of Transition elements.

■ Multiple Choice Questions

1. Which of the following consists maximum content of iron?
 - a. Cast iron
 - b. pig iron
 - c. wrought iron
 - d. steel
2. During the extraction of iron spongy iron is formed in
 - a. Zone of combustion
 - b. Zone of fusion
 - c. Zone of slag formation
 - d. Zone of reduction
3. Alloy of iron with carbon is heated below its melting point and left it for cooling slowly. The process is
 - a. Annealing
 - b. Tempering
 - c. Quenching
 - d. Nitriding
4. Iron can be protected by keeping it in . . . with minimum loss.
 - a. conc. HNO_3
 - b. dil. HNO_3
 - c. dil. HCl
 - d. dil. H_2SO_4
5. Don't touch the Kipp's waste because you can get . . . from it.
 - a. Green vitriol
 - b. Blue vitriol
 - c. White vitriol
 - d. Oil of vitriol

Answer Key

1	2	3	4	5
c	d	a	a	a

18.6 Silver

The Latin name of silver is *Argentum* which means white. Silver is very important metal having ornamental value.



Fig. 18.22: Different uses of silver

Symbol	Ag	Atomic number	47
Atomic mass	107.9 (amu)	Electronic configuration	[Kr] 4d ¹⁰ 5s ¹

18.6.1 Occurrence

Silver is found in native state as well as in combined state. In native state, it is found in USA, Mexico, Canada, etc. It is found in combined state as sulphide and chloride.

Ores of Silver

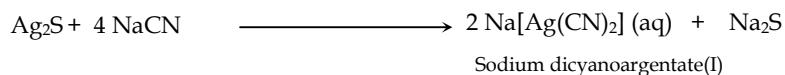
1. Argentite or silver glance (Ag_2S)
2. Horn silver (AgCl)
3. Silver copper glance ($(\text{CuAg})_2\text{S}_2$)
4. Argentiferous galena (PbS , Ag_2S)
5. Pyrargyrite or ruby silver $3\text{Ag}_2\text{S} \cdot \text{Sb}_2\text{S}_3$

18.6.2 Extraction of Silver from Argentite ore Using Cyanide Process

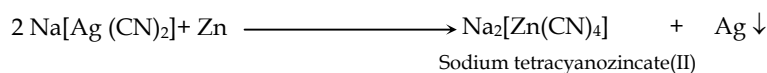
This method is also called Mac-Arthur-Forrest process.

Theory

Ag_2S present in the argentite ore is converted into solution of sodium argentocyanide (Sodium dicyanoargentate(I)) by treating ore with dilute NaCN solution



Sodium argentocyanide solution thus obtained on the treatment with zinc gives metallic silver as precipitate.



Steps involved in the metallurgy

1. **Crushing and pulverization:** Big lumps of ore are crushed using jaw crushers to get crushed ore which are pulverized (powdered) using pulverisers or ball mills to get powdered ore.

2. **Concentration of ore by froth-floatation process:** Powdered ore is dipped into a tank containing water and small quantity of pine oil. The mixture is agitated by using blast of air. Gangue particles are wetted by water and get collected at the bottom of the tank. The ore particles being preferentially wetted by oil come on the surface in the form of froth. The froth is skimmed out to collect the concentrated ore.

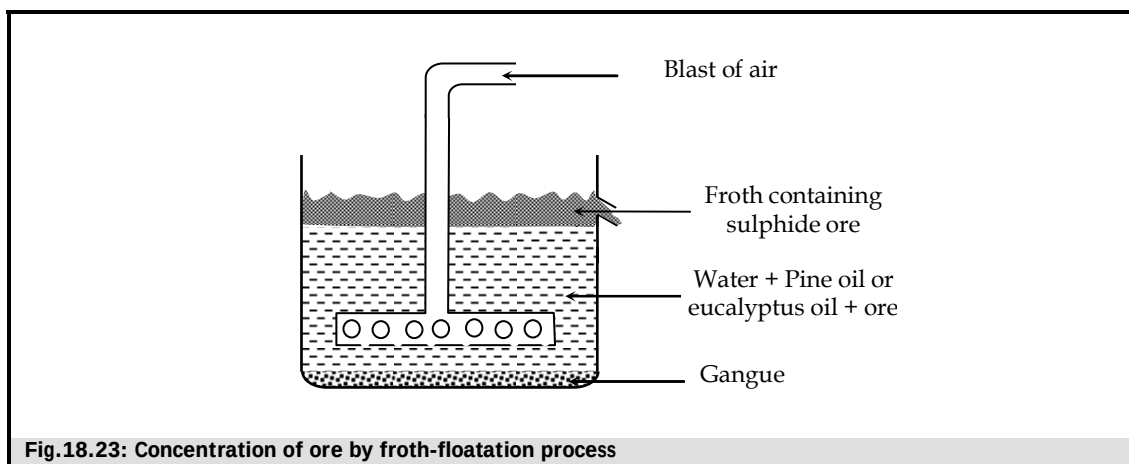
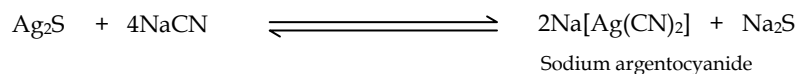
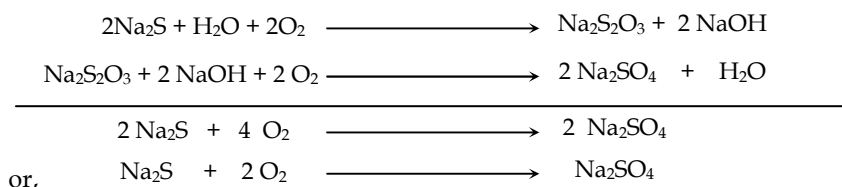


Fig.18.23: Concentration of ore by froth-floatation process

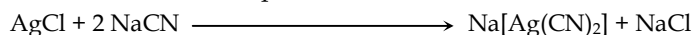
3. **Formation of complex cyanide:** Concentrated ore is agitated with dilute sodium cyanide solution (0.5%) using a current of air. Then solution of sodium argentocyanide is obtained.



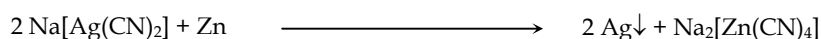
Due to the case that reaction is reversible, air is blown, which removes Na_2S from equilibrium mixture as Na_2SO_4 . So reaction proceeds in forward direction.



Even if ore is horn silver, same process can be carried out.



4. **Reduction by precipitation method:** The solution obtained from the above method is filtered off and treated with finely divided Zn where Ag will be precipitated in the form of dark amorphous powder.



This precipitate of silver is removed by filtration process and the precipitate is dried out.

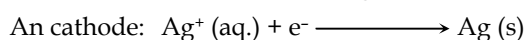
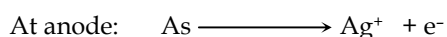
5. Purification of silver

Steps:

- Dried silver is heated with KNO_3 solid in crucible to get a compact mass.
The impurities are oxidised on the surface. Liquid silver forms compact mass on cooling.
- Then impure silver (containing Cu, Al, etc.) is purified by electrolysis.
Impure silver is made anode and pure silver is made cathode. 6% AgNO_3 solution acidified with 1% HNO_3 is used as electrolyte.

On passing D.C. silver dissolves from anode and gets collected at cathode.

The reactions take place as follows:



Cupellation may also be used if Pb is suspected. In this process, Pb is blown away by air in the form of PbO

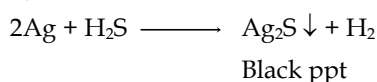
Properties of Silver

Physical properties

- Silver metal is white in colour. Alchemist used the term *luna* for silver which means moon.
- Silver is good conductor of heat and electricity. Due to its good conductance, it is used to prepare electrode.
- Silver is malleable and ductile.
- The density of silver is 10.52 gram per cc.
- Melting point of silver is 956°C and boiling point is 1955°C .

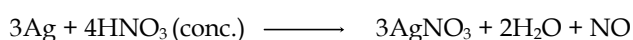
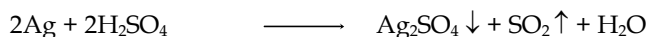
Chemical Properties

- Action with air:** Silver is not chemically active metal. Therefore, under ordinary condition, it does not react with air. However, it is tarnished by air containing even traces quantity of hydrogen sulphide.

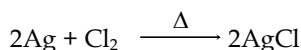
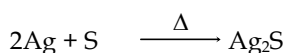


Due to hydrogen sulphide pollution, silver monuments are being tarnished. This demands regular cleaning of silver wares.

- Action with acids:** Silver lies below hydrogen in electrochemical series. So it cannot displace hydrogen gas from dilute, mineral and non-oxidizing acids such as HCl , H_2SO_4 . However, silver reacts with concentrated nitric acid and concentrated sulphuric acid as follows:



- Action with non-metals:** Some non-metals such as sulphur, chlorine, etc. react with silver on heating to form corresponding compounds.



Uses of silver

1. In the early days silver were used to prepare coins. An alloy of silver and copper is used to prepare coins. Hence, it is regarded as coinage metal.
2. It is used to prepare electrodes.
3. It is used in photography.
4. It is used for electrical purposes.
5. It is used to prepare jewellery.
6. It is used to prepare alloy.

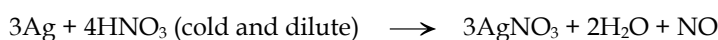
18.6.3 Silver Nitrate AgNO_3

Chemical formula:	AgNO_3	Melting point:	209.7°C
Name:	Silver (I) nitrate	Boiling point:	440°C
Molecular mass:	169.87	Solubility :	256 gram per 100mL at 25°C
Colour:	White	Hazards:	Corrosive, explosively reacts with ethanol

Silver nitrate is an inorganic salt with chemical formula AgNO_3 . It is also called lunar caustic because (*luna* means moon, *caustic* = corrosive) it can be cast into sticks and it exhibits strong caustic action.

Preparation

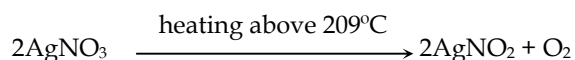
Silver nitrate can be prepared by the action of nitric acid on silver.



These reactions should be carried out in fume cupboard or fumehood to expell the toxic nitrogen oxide gases.

Physical properties

1. Silver nitrate forms colourless, transparent rhombic crystal ($a \neq b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$).
2. Its density is 4.35 and melting point is 209.7°C
3. The fused state of silver nitrate can be casted into sticks due to its profound caustic action. That is why silver nitrate is called lunar caustic.
4. Silver nitrate is highly soluble in water. Silver nitrate solution darkens slowly in presence of light. That is why both silver nitrate crystals and silver nitrate solutions are stored in dark bottles.
5. Silver nitrate is also soluble in alcohol, pyridine, acetone, etc.
6. **Decomposition by heat:** On heating, silver nitrate above its melting point, it gets decomposed to silver nitrite and oxygen.

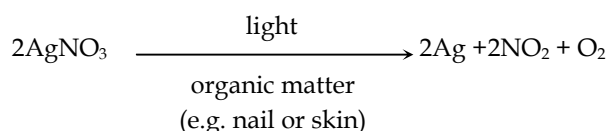


If silver nitrate is heated upto red, it decomposes to metallic silver.



Silver nitrate decomposes completely at about 440°C . In fact, most of the metal nitrates decompose to corresponding metal oxides and so does the silver nitrate. Silver oxides decompose at even lower temperature than does the silver nitrate. Hence, decomposition of silver nitrate produces metallic silver.

7. **Decomposition by organic matter:** Some organic compounds such as skin, paper, cloth, etc. decomposes silver nitrate into finely divided metallic silver (black) in presence of light.



This property of silver nitrate is used to prepare indelible ink which is used to mark.

This indelible ink is used in nail and skin after casting vote (fig. 18.24).

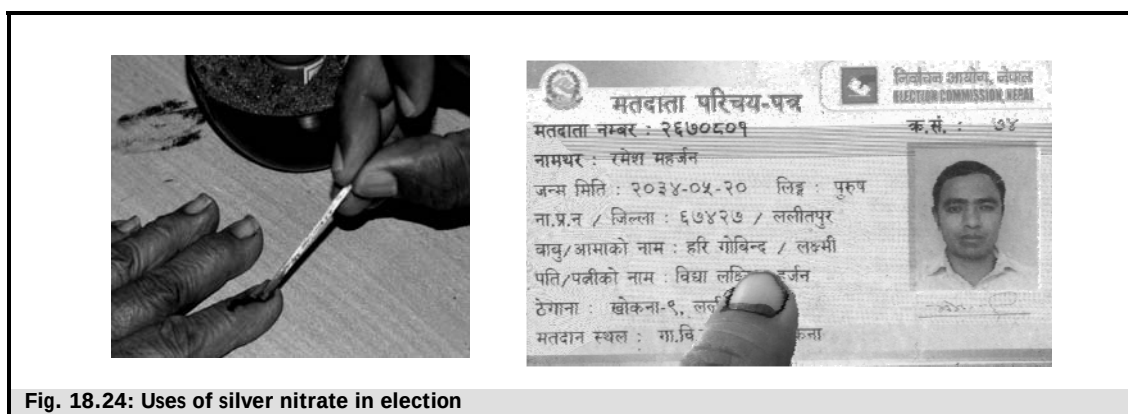


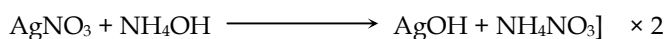
Fig. 18.24: Uses of silver nitrate in election

Chemical Properties

1. **Action with NaOH:** Silver nitrate reacts with sodium hydroxide to give brown precipitate of silver oxide with the production of sodium nitrate and water.

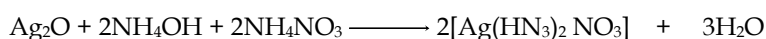


2. **Action with NH_4OH :** When ammonium hydroxide is added into silver nitrate solution, a brown precipitate of silver oxide is formed which gets dissolved to form a complex in excess of ammonium hydroxide.



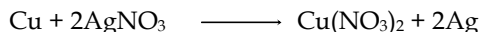
(Brown ppt)

With excess of NH_4OH

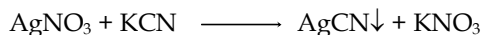


Diamminesilver (I) nitrate

3. **Displacement reaction:** Silver lies below metals such as Na, Mg, Al, Zn, Sn, Pb, Cu, Hg, etc. and above Pt, Au in electrochemical series. Hence, metals lying above it in electro-chemical series can displace it from its salt.



4. **Action with KCN:** Silver nitrate gives white precipitate with potassium cyanide.



5. **Action with halide, sulphate, thiosulphate, thiocyanate, phosphate, sulphide, chromate:**
Aqueous solution of halide, phosphate, sulphide, sulphate, thiosulphate, form precipitate of corresponding silver salts with silver nitrate solution.

- | | | | | |
|-------|--|-------------------|---|-----------------|
| i. | $\text{AgNO}_3 + \text{Cl}^-$ | $\longrightarrow$ | $\text{AgCl}\downarrow + \text{NO}_3^-$ | white ppt |
| ii. | $2\text{AgNO}_3 + \text{SO}_4^{--}$ | $\longrightarrow$ | $\text{Ag}_2\text{SO}_4\downarrow + 2\text{NO}_3^-$ | White ppt |
| iii. | $2\text{AgNO}_3 + \text{S}_2\text{O}_3^{--}$ | $\longrightarrow$ | $\text{Ag}_2\text{S}_2\text{O}_3 + 2\text{NO}_3^-$ | White ppt |
| iv. | $\text{AgNO}_3 + \text{CNS}^-$ | $\longrightarrow$ | $\text{AgCNS}\downarrow + \text{NO}_3^-$ | White ppt |
| v. | $3\text{AgNO}_3 + \text{PO}_4^{--}$ | $\longrightarrow$ | $\text{Ag}_3\text{PO}_4\downarrow + 3\text{Ag}$ | Yellow ppt |
| vi. | $2\text{AgNO}_3 + \text{S}^{--}$ | $\longrightarrow$ | $\text{Ag}_2\text{S}\downarrow + 2\text{NO}_3^-$ | Black ppt |
| vii. | $2\text{AgNO}_3 + \text{CrO}_4^{--}$ | $\longrightarrow$ | $\text{Ag}_2\text{CrO}_4 + 2\text{NO}_3^-$ | Brick red ppt |
| viii. | $\text{AgNO}_3 + \text{Br}^-$ | $\longrightarrow$ | $\text{AgBr}\downarrow + \text{NO}_3^-$ | Pale yellow ppt |
| ix. | $\text{AgNO}_3 + \text{I}^-$ | $\longrightarrow$ | $\text{AgI}\downarrow + \text{NO}_3^-$ | Yellow ppt |

Self Test

18.5 A little quantity of water was observed in a sea cargo. Some argued that the water appeared from condensation of humid air while some other argued it is from sea water leaking from pores. What test would you prefer to identify it?

Uses

1. It is used as laboratory reagent. For example, it is used for halide test (Cl^- , Br^- , I^-)



- It is used in silvering of mirror.
- It is used to prepare indelible ink.
- It is used to prepare some silver based explosives such as silver acetylide, silver fulminate, silver azide, etc.
- It is used in the preparation of photographic film where silver nitrate is reacted with sodium or potassium salts.
- It is used to prepare Tollen's reagent.



Questions

Very Short Answer Questions:

1. Why does Silver nitrate solution produce black-stain on the skin in presence of sun light?
2. What is the action of heat on silver nitrate?
3. What do you mean by noble metals? Give any two examples.
4. What happens when AgNO_3 is heated at 450°C .
5. Give two uses of silver nitrate.
6. How is presence of halide ions are tested with silver nitrate solution?
7. Why silver ores are leached with metal cyanides for the extraction of silver?
8. What do you mean by noble metals? Name at least three noble metals you know.

Short /Long Answer Questions:

1. Give the various steps involved in the extraction of silver from silver glance.



Multiple Choice Questions

1. Mac-Arthur-Forrest process is concerned with extraction of
 - a. Cu
 - b. Ag
 - c. Au
 - d. Pt
2. Lunar caustic is
 - a. AgNO_3
 - b. AgCl
 - c. Ag_2C
 - d. $\text{AgNO}_3 + \text{KOH}$
3. Ruby silver is
 - a. Ag_2S
 - b. Sb_2S_3
 - c. AgCl
 - d. Ag_2SbS_3
4. In the extraction of silver, sodium cyanide is reacted with silver sulphide to yield sodium argentocyanide and sodium sulphide. Air oxidation of sodium sulphide is carried out to make sodium sulphate because . . .
 - a. Sodium sulphate is less harmful waste product.
 - b. Sodium sulphate can be used as useful reagent.
 - c. Air should be passed during the reaction.
 - d. The transformation drives the reaction to forward direction.
5. Black stain is production when . . . comes into contact of skin in presence of light.
 - a. AgNO_3
 - b. NaNO_3
 - c. AgCl
 - d. AgBr

Answer Key

1	2	3	4	5
b	a	d	d	a